

FULL SPECIFICATION TCC8132

Digital Multimedia Processor

TCC8132_FULL_SPEC

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Telechips

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1 Introduction

TCC8132 is a Digital Multimedia Processor based on the ARM926EJ-S, ARM's proprietary 32-bit RISC CPU core. It can decode and encode MP3 or other types of audio/video compression/decompression standards by software based architecture. The on-chip USB Host Controller and other flexible features provide cost effective solutions for hosting player applications such as car audio, home audio, and etc.

1.1 Features

- ARM926EJ-S CPU
 - 400MHz @1.1V (500MHz @TBD V)
 - 16KB instruction, 16KB data cache
- Stacked 32MB SDRAM
- 32KB internal SRAM
- 16KB Boot ROM for various boot modes
 - Serial-NOR, NAND, SD Card, eMMC, USB
- 1 KB OTPROM
- External Bus Controller supports
 - Serial-NOR / NAND Flash, and etc.
 - 16/32-bit SDR SDRAM up to 166MHz clock
- Two USB 2.0 Interface Ports shared with two Host Controller and one Device controller
 - Port 0 (USB0DP/DM) is dedicated to USB2.0 Host Controller 0
 - Port 1 (USB1DP/DM) is shared with Device Controller and USB2.0 Host Controller 1
 - Host Controller supports high-speed , full-speed , and low-speed
 - Device Controller supports high-speed, and full-speed (Not support low-speed)
 - Battery Charge 1.2 (Support SDP, CDP, DCP)
- ECC generator supports
 - SLC ECC
 - MLC 4/6/12/16/24 bit ECC
- Three I2S interfaces
- SPDIF interface (TX / RX)
- ASRC (Sampling Rate Conversion)
 - Two Stereo Channel
 - Asynchronous or Synchronous Sampling Rate Conversion
 - Sampling Rate : input/output [kHz] : 1 to 192
 - + Up Conversion - 8:1
 - + Down Conversion - 1:7.5
 - + Format: 16 or 24 bits
 - + Memory to Memory transfer is possible
 - + Memory to I2S transfer is possible
 - + I2S to memory transfer is possible
 - + I2S to I2S transfer is possible
 - Mixer
 - + Two sources can be mixed
 - + Only Memory / I2S to I2S Interface
 - Digital Volume
 - + Range : -120 to 18dB
 - + Support Fade-in and Fade-out
- UART (Five channels)
- SPI compatible Master/Slave (Six channels)
- I2C compatible serial bus controllers (Four masters and two slaves)
- SD/SDIO/MMC controller
 - Two channels (with up-to 8 bit-width)
 - Compliant to SD3.0, SDIO3.0, MMC4.41
- Serial NOR Flash controller
 - Single, Dual, Quad I/O support
 - Single transfer rate support
- Three three-channel DMA controllers (Total 9 channels)
- Four 16-bit timer/counters, two 20-bit timers and 32-bit 1Hz counter
- Three Channel PWM generator
- Three Channel Input Capture Timer
 - 8bit Pre-Scaling Count Clock
 - 256 Capture Input
 - Interrupt : Time-Out, Frequency Detect, Duty Detect, Fail Edge Full.
 - Noise Filter : 16 Mode (Immediate, Up/Down, Hold, Reset, Mixed Mode)
 - 16bit Falling Edge Count of Capture Input
 - 28bit Duty Capture, Frequency Capture Counter.

- Eight external interrupt inputs
- General purpose ADC with 8 inputs
 - 12-bit resolution
 - 1MSPS conversion rate
- Four PLLs (one of them is dithered PLL)
- Real Time Clock
- I/Os alive with 1KBytes Backup RAM.
- Cipher
 - AES : key length 128 / 192 / 256 bits
 - DES : single / double / triple (2key, 3key)
 - CBC-MAC
- CAN controller
 - Two channels
 - Supports CAN protocol version 2.0 part A, B
 - Bit rate up to 1MBit/s
 - 32 Message Objects
- Display Controller
 - One Display Controller
 - Controller can overlay up to 4 image channels
 - 4 Channel Overlay / Chroma-Keying / Alpha-blending
 - Gamma Correction with LUT
 - Look-up table for Indexed or RGB Color
 - Contrast, Brightness, Hue Function Supported
 - De-interlacer and Scaler can be plugged in to each video or graphic data path
 - TFT-LCD, CPU Interface
- Camera Interface
 - 8-bit one channels (with time-demultiplexor. only support 2-channel demultiplexor)
 - CCIR-601/656 Interface
 - Camera Input Supported
 - 1 Channel Overlay / Chroma-Keying
 - Up-scaling (x4), Down-scaling(x1/1024: X/32, Y/32)
- 2D Accelerator
 - 8bpp(RGB332)
 - RGB (444, 454, 555, 565, 666, 888)
 - Alpha-RGB (4444, 3454, 1555, 6666, 8888)
 - Packed RGB(888)
 - Sequential YUV (444,422)
 - Separate YUV (444,440,422,420,411,410)
 - Interleaved YUV(422, 420)
 - BitBLT (16 Raster Operations)
 - 4-Channel Source Mirror, Flip, 90°, 180°, 270° rotate (1 Channel for Packed-RGB only)
 - 1-Channel Destination Mirror, Flip, 90°, 180°, 270° rotate
 - 3-Channel Arithmetic Operation
 - 3-Channel YCbCr-to-RGB color space converting
 - Overlay and alphablending (2 overlay, 256-level alphablending)
 - Color LUT
 - Dithering
- Security Feature
 - Chip Boot-Code JTAG Debug Disable
 - Permanent JTAG Debug Disable
 - Password JTAG Mode
 - Secure Boot (Authentication Algorithm for Serial Flash Boot)
- Four Boot modes
 - NAND BOOT (8bits parallel mode, page size 512, 2048, 4096)
 - SDMMC BOOT (SD3.0, MMC, eMMC4.41)
 - Serial Flash BOOT (3Byte, 4Byte mode, Authentication Algorithm for secure boot)
 - USB BOOT
- 45nm low-power CMOS process
- 324-ball BGA

1.2 Applications

- Car Audio
- USB Host Player

1.3 Block Diagram

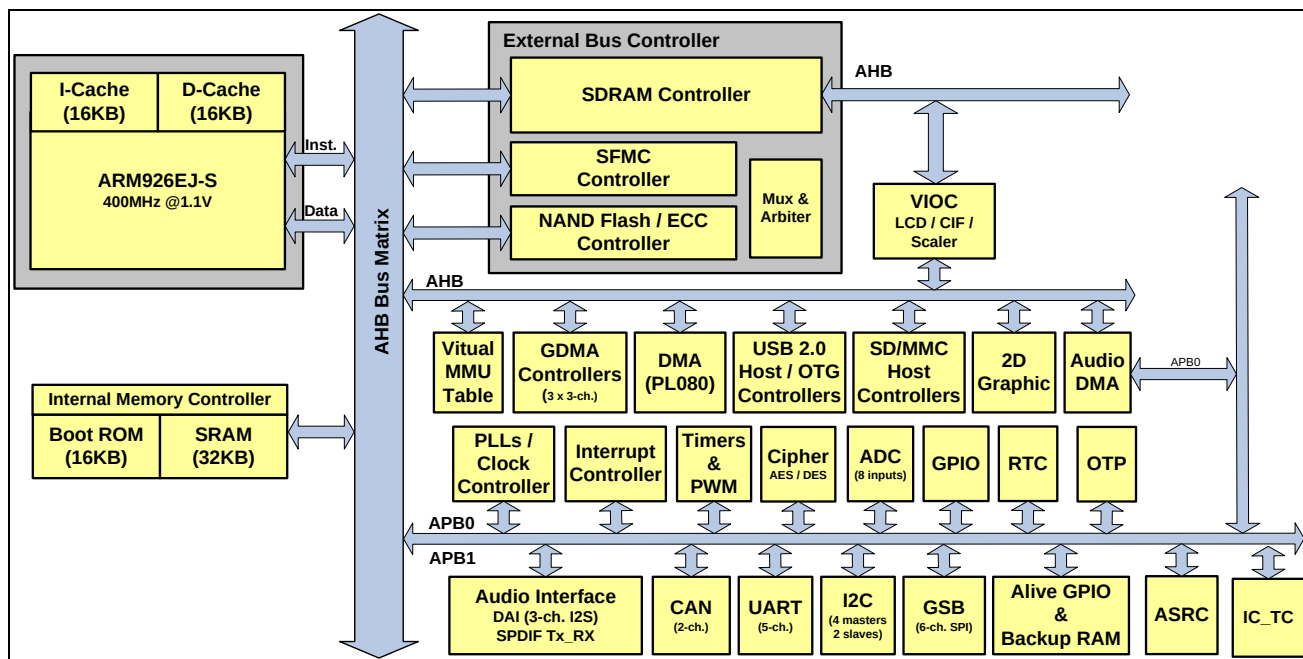


Figure 1.1 TCC8132 Block Diagram

1.4 IP Specification

Table 1.1 TCC8132 Pin Description

IP	Description
ARM926EJS	- 400MHz @1.1V $\pm$ 5% - 500MHZ @ TBD V - 16 KB instruction , 16KB data cache - JTAG I/F + Chip Boot-Code JTAG Debug Disable + Permanent JTAG Debug Disable + Password JTAG Mode + Debug Interface + Separate AHB Port (Instructure & Data) + AHB Clock is 1/n of ARM9 Core
Memory Organization	- 0x00000000 ~ 0x0FFFFFFF : Depending on Remap Register Value + Internal BootRom / Internal SRAM / External SDRAM / Serial Flash - 0x10000000 ~ 0x1FFFFFFF : Internal SRAM (32KB) - 0x20000000 ~ 0x2FFFFFFF : External SDR-SDRAM - 0x30000000 ~ 0x3FFFFFFF : Internal SRAM (32KB), (Mirror 0x10000000) - 0x40000000 ~ 0x7FFFFFFF : Not used - 0x80000000 ~ 0x8FFFFFFF : AHB Peripherals (Refer to Chapter 1.6) - 0x90000000 ~ 0x9FFFFFFF : APB Peripherals (Refer to Chapter 1.6) - 0xA0000000 ~ 0xAFFFFFFF : Serial Flash - 0xB0000000 ~ 0xBFFFFFFF : External SDR-SDRAM (Mirror 0x20000000) - 0xC0000000 ~ 0xDFFFFFFF : Not used - 0xE0000000 ~ 0xEFFFFFFF : Internal BOOTROM (16KB) - 0xF0000000 ~ 0xFFFFFFFF : External memory controller configuration register space
Internal SRAM Controller	- Memory Size :32Kbyte
SDRAM Controller	- Density : 256Mbit (Stacked SDRAM) - BusWidth : 32bits - CASWidth : 9bits - RASWidth : 12bits
Internal BootROM	- NAND BOOT + 24bits ECC + DataWidth : 8 bits + Page size : 512, 2048, 4096 - SDMMC BOOT + SD, MMC Boot mode + eMMC4.3 Boot mode - Serial Flash BOOT + 3 Byte Boot mode + 4 Byte Boot mode + using Authentication Algorithm for secure boot - USB BOOT - Operating Frequency is PLL/CPUCCLK/HCLK (196 / 92 / 48 MHz)
OTPROM	- Size : 1 KB - Vender : Kilopass - Lock Read/Write function for JTAG Password
Serial Flash	- Support serial peripheral interface – mode0 and mode3 - Protocol Support :Single, Dual, Quad I/O (Not support DDR mode)
ECC	- SLC ECC - MCL 4/6/12/16/24 bit ECC data / 512 Bytes user data
NFC	- 8 bit I/O bus width supportable - 1 page burst program and read (Maximum page size : 8192Bytes + ECC size) - Configurable read data capture timing to reduce read cycle period - Support 1 or 4 ready - NFC dedicated ECC encoding / decoding capability - Support SLC / ECC
I2C	- Master + 4-channel master + Multi master operation + Programmable clock frequency + Fast Mode (400kbps) + Wait state by clock stretching - Slave + 2-channel slave

IP	Description
	+ Fast Mode (400Kbps)
SPI	- 6-channel SPI + 3 channels with DMA and 3 channels without DMA - Support master and slave mode - Configurable data bit width 8~32 bit
UART	- 5-channel UART - Support hardware flow control (CTS / RTS) - Generate baudrate with 16 bits clock divider - Offers similar functionality to the industry-standard 16C650 - Baud Rate Clock (3MHz <- 48MHz/16) - Support SmartCard IF + RXD : SmartCard CLK + TXD : SmartCard Data line - Support UART transfer with general DMA
SDMMC	- 2-channel SDMMC - 2-channel with up-to 8-bit width + SDSC : up to and include 2GB + SDHC : more than 2GB and up to and include 32GB + SDXC : more than 32GB and up to and include 2TB - Complaint to SD3.0, SDIO3.0, eMMC4.41 + Default speed : 3.3V signaling, Frequency up to 25MHz + High speed : 3.3V signaling, Frequency up to 50MHz + SDR50 mode : 1.8V signaling, Frequency up to 80MHz (this mode is only using SD1CH (GPIO_FC)) + SDR104 mode : 1.8V signaling, Frequency up to 120MHz (this mode is only using SD1CH(GPIO_FC)) - Automatic CRC generation & Checking the Data / Command - Data transmit / receive FIFO (32bit x 8) - Supported SD/MMC mode + 1bit serial or 4bit parallel SD + 1bit serial or 4/8bit parallel transfer for MMC
USB pico PHY	- Integrated HS and FS termination and signalling switching - 480Mbps High Speed / 12Mbps Full Speed / 1.5Mbps Low speed - On-Chip PLL for 480MHz - 45Ohm termination / 1.5K pull-up / 15k pull-down integrated - Support 8/16bit unidirectional parallel interface - Support FS/HS dual mode operation - Data Recovery from Serial data on the usb connector - Sync/End-of-Packet Generation and Checking - Bit stuffing and unstuffing, bit stuffing error detection. - NRZI encoding / decoding - Support of suspend, sleep, resume, remote wakeup operation - Support Battery charge 1.2V. (SDP, CDP, DCP)
USB DEVICE LINK	- Compliant to USB2.0 Specification - 480Mbps High Speed / 12Mbps Full Speed - Support interrupt, isochronous, bulk transfer - Support FS/HS dual mode operation - 16 bidirectional endpoint, including control endpoint 0 - Internal DMA controller - On-Chip UTMI PHY parallel interface - 30MHz with 16bits or 60MHz with 8bit, parallel interface
USB 2.0 HOST LINK	- Compliant to USB2.0 Specification - 480Mbps High Speed / 12Mbps Full Speed / 1.5Mbps Low speed - Compliant to EHCI Specification and OHCI Specification - Descriptor and data prefetching
AUDIO	- I2S + 3 channels + Support 16 / 24 bit mode + Range of sampling Frequency (8, 16, 24, 32, 44.1, 48, 64, 96, 128, 192kHz) + Support MSB/LSB justified mode + Each 32bit x 16 depth FIFO + Available to use general DMA or specialized DMA - SPDIF TX/RX + Each 1 channel + Support 16, 24 bit mode + Available to use general DMA or specialized DMA

IP	Description
	- Digital Volume + -96dB to 0dB (Step : 6dB)
ASRC	- Two Stereo Channel - Asynchronous or Synchronous Sampling Rate Conversion - Sampling Rate : input/output [kHz] : 1 to 192 + Up conversion : 8:1 + Down conversion : 1:7.5 + Format : 16 or 24 bits + Memory to memry transfer + Memory to i2s transfer + I2c to memory transfer - Mixer + Two sources can be mixed + Only memory / I2S to I2S interface - Digital volume + Range : -120 to 18dB + Support fade-in and fade-out
CKC	- 6 primary clock sources from PLL0~PLL3, XI, XTI + PLL0,1,3 is normal PLL (Max. 2.0GHz) + PLL2 is dithered PLL (Max. 660MHz)\ - All clocks are generated by SCLK.
Interrupt Controller	- 64 interrupt source. + There are two types of interrupt in ARM926EJ-S: IRQ type, FIQ type. - Various kinds of method + Rising / falling edge and level high / low
PL080 DMA	- ASRC spcialized DMA - Internal four-word FIFO per channel - Support for 8, 16, and 32 bit wide transaction. - Memory to Memory, Memory to peri., and peri. to memory transfers - Audio pair support 2CH
GDMA	- 3-channel DMA (x3) - Perform transfer between external memory / on-chip memory / external device / peripheral moule
Watch-dog Timer	- 16bit counter - Counter overflows (0xffff->0x0000) , reset / interrupt generated.
Timer	-16 bit timer - 20 bit timer - 32 bit 1HZ Counter
PDM(PWM)	- 3 channels + Each channel has enable control bit - Phase mode + Mode 1 (2 phase) : Width of each phase is controllable by using 2 registers + Mode 2 (4 phase) : Width of each phase is controllable by using 4 registers - Registter mode + In each PWM cclock cycle, PWM controller generates PWM output signal by reading all bits of 1 register + In each PWM cclock cycle, PWM controller generates PWM output signal by reading all bits of 4 register + In each PWM cclock cycle, PWM controller generates PWM output signal by reading configurable size up to 4 - Supports continuous mode (infinite loop)
Input Capture Timer	- 3 channels - Interrupt : time-out, frequency detect, duty detect fall edge full - Noise filter : 16 mode (immediate, up/down, hold, reset, mixed mode) - 16 bit falling edge count of capture input - 28 bit duty capture frequency capture counter
Ext Interrupt	- 8 source external interrupt controller + GPIO PK[7:0] is mapped to external interrupt source.
ADC	- 8-channel input general propose ADC. - 12bit resolution - 1MSPS conversion rate (with 5MHz clock) - Differential nonlinearity (± 1 LSB (Maximum.)) - Integral nonlinearity (± 4 LSB (Maximum.)) - Signal to noise & distortion ratir : 54dB
PLL	- Normal PLL (PLL0, 1, 3) + 32MHz ~ 2000MHz + Duty : 40:60 (1~2GHz), 45:55 (under 1GHz)

IP	Description
	- Dithered PLL (PLL2) + 11MHz ~ 660MHz + Duty : 40:60 (S=0), 45:55 (S>0) + Modulation Frequency : 30kHz ~ 150kHz + Modulation rate : 0~5%
RTC	- General Features + Sub Oscillator included + Clock and Calender function (BCD display) : Sec / Min / Hour / Date/ Week / month / year + Leap year generation - Interrupt & round reset + Alarm interrupt in normal operation mode + Cyclic interrupts 1/256, 1/64, 1/16, 1/4, 1/2, 1 second interrupts + Round-reset function 30-, 40-, 50- second - Wakeup Fucntion - OSC Frequency + 32.76kHz, 4.19MHz
PMIO	- Alive IO (Independent by normal IO power) - 1Kbytes Backup RAM
CIPHER	- AES : key length 128 / 192 / 256 bits - DES : signgle / double / tripple (2key, 3key) - CBC-MAC
CAN	- 2 channels - Support CAN protocol version 2.0 part A, B - Bit rate up to 1Mbit/s - 32 Message Object
Display Controller	- One display Controller - 4 overlay image channel + Overlay, chroma-keying / alpha-blending (level is 256) - Gamma correction with LUT - Look-up Table for indexed or RGB color - Contrast, Birghness, Hue function supported + This mode is used only when R2Y of DISP module is enabled. - Deintelacer and scaler can be plugged in to each video or graphic data path - TFT-LCD, CPU interface + Maximum resolution is 800x480(TBD) (TFT-LCD)
VIDEOIN	- 8 bit one channel (wih time-demultiplexor. only support 2ch demultiplexor) - Support CCIR 601, 656 Interface - VIDEOIN supported + YUV422 8bit mode, time-multiplex mode (TW2867 device for Techwell Inc.) - 1 channel overlay / chroma-keying - Up-scaling (x4), down-scaling (x1/1024)
2D Accelerator	- 8 bpp (RGB332), - RGB / alphaRGB + 444 / 454 / 555 / 565 / 666 / 888 - Packed RGB888 - Sequential YUV + 444 / 422 - Separate YUV + 444 / 440 / 422 / 420 / 411 / 410 - Interleaved YUV + 422 / 420 - BitBLT (16 raster operation) - 4-channel source mirror, flip 90°, 180°, 270° rotate (1 Channel for Packed-RGB only) - 1-channel destination mirror, Flip, 90°, 180°, 270° rotate - 3-channel arithmetic operation - 3-channel YUV-to-RGB color space converting - Overlay and alphablending (2 overlay, 256-level alphablending) - Color LUT - Dithering
PKG	- TCC8130 + Single product - TCC8131 + Wafer + 16MB SDRAM MCP - TCC8132 + Wafer + 32MB SDRAM MCP

1.5 Boot Mode

Table 1.2 TCC8132 Boot Mode

BM	Boot Mode	Port	Option	Description
4'b0000	Not used			
4'b0001				
4'b0010				
4'b0011				
4'b0100	NAND	PS[13:00]	8 bit	- Only use 8 bit nand - Page size is 512, 2048, 4096.
4'b0101	SDMMC	PS[11:00]	eMMC	- Support MMC4.3 - Only use 3.3V signaling.
4'b0110	Serial Flash	PS[27:22]	3B	- 3 byte mode - Support Authentication Algorithm for secure boot
4'b0111	Serial Flash	PS[27:22]	4B	- 4 byte mode - Support Authentication Algorithm for secure boot
4'b1000	Not used			
4'b1001				
4'b1010	SDMMC	FC[11:0]	SD	- SD mode - Use 1.8V/3.3V signaling
4'b1011	SDMMC	FC[11:0]	eMMC	- Support MMC4.3 - Use 1.8V/3.3V signaling
4'b1100	Not used			
4'b1101	SDMMC	PS[11:0]	SD	- SD mode - only use 3.3V signaling.
4'b1110	USB	USB1DP/DM	-	USB BOOT
4'b1111	Not used			

Address	Description			
		M8	SCFG	0x13000 ~ 0x13FFF
		M9	ECID	0x14000 ~ 0x14FFF
		M10	CIPHER	0x18000 ~ 0x18FFF
		M11	PM	0x1C000 ~ 0x1CFFF
		M13	ASRC	0x20000 ~ 0x2FFFF
0xA0000000 ~ 0xFFFFFFFF	Serial Flash			
0xB0000000 ~ 0xDFFFFFFF	Not used			
0xE0000000 ~ 0xEFFFFFFF	Internal BOOTROM (16KB)			
0xF0000000 ~ 0xFFFFFFFF	External memory controller configuration register space			

2 Pin Description

The TCC8132 is a CMOS device. If any unused input pin is left open, it may cause a malfunction or abnormal current consumption. Pull-up or pull-down resistors must be used appropriately for unused inputs or bidirectional pins in input state.

Do not tie bidirectional pins directly to power or ground. They must be connected via resistors. During power-on/off, there is certain period of time in which all the I/O pins are in an undefined state. If bidirectional pins are tied to power or ground directly and their direction happens to be in output mode during this period, short circuit current flows through the I/O cells. This may damage the device or cause a system malfunction.

Notation:

I:	Input
IU:	Input with Pull-Up Enabled
ID:	Input with Pull-Down Enabled
O:	Output
OL:	Output Low
OH:	Output High
OT:	Tri-state Output
B:	Bidirectional
AI:	Analog Input
AO:	Analog Output
DP:	Digital Power
DG:	Digital Ground
AP:	Analog Power
AG:	Analog Ground
PU:	Pull-Up
PD:	Pull-Down
Z:	High Impedance

2.1 TCC8132 Pin Description

Table 2.1 Power/Ground Information

Group	Ball Name	# of Balls	Ball#	MIN	TYP	MAX	Description
PWRCORE	VDDI_CORE	13	G9 M9 K10 H9 F10 L10 J9 G10 M10 K9 H10 L9 J10	1.045	1.1	1.155	Digital Power for Internal Core (Typical 1.1V)
PWROSC_33D	VDD_OSC	1	E7	3.135	3.3	3.465	Digital Power for Oscillator I/O (XI/XO)
PWRUSB20DO_11D	VDDDO_USB11	1	F9	1.045	1.1	1.155	Digital Power for USB2.0 Device or Host Internal Core (Typical 1.1V)
PWRUSB20DH_11D	VDDDH_USB11	1	F8	1.045	1.1	1.155	Digital Power for USB2.0 Host Internal Core (Typical 1.1V)
PWRUSB20DO_33A	VDDAO_USB33	1	E11	3.135	3.3	3.465	Analog Power for USB2.0 Device or Host PHY (Typical 3.3V)
PWRUSB20DH_33A	VDDAH_USB33	1	D10	3.135	3.3	3.465	Analog Power for USB2.0 Host PHY (Typical 3.3V)
PWRADC_33A	VDD_ADC	1	K7	3.0	3.3	3.6	Analog Power for ADC and ADIN[7:0] (Typical 3.3V)
PWRPLL_11A	VDD_PLL	2	N10 N11	1.045	1.1	1.155	Analog Power for PLL (Typical 1.1V)
PWRRTC_33D	VDD_RTC	1	N7	1.65	3.3	3.6	Digital Power for RTC (XTI/XTO, RTCRST#)
PWRPMP_33D	VDD_RTC_PM	1	P7	1.65	3.3	3.6	Digital Power for PMIO core
PWROTP_25D	VDDD_OTP25	1	K12	2.38	2.5	2.62	Digital Power for OTP I/O(Typical 2.5V)
PWRMEMIO_33D	VDD_XA	1	H7	1.65	3.3	3.6	Digital Power for I/O (GPIOXA[12:0], XA[16:13], XDCLK, XDCKE, XWEn, XDSCn0)
	VDD_XD	1	G12	1.65	3.3	3.6	Digital Power for I/O (XD[31:00],XA[21:18])
PWRSDRAM_33D	VDD_MEM	4	J7 G16 H16 H15	2.7	3.3	3.6	Digital Power for SDRAM Core (Typical 3.3V)
PWRIOPK_33D	VDD_PK	2	N8 P8	1.65	3.3	3.6	Digital Power for alive 3.3V I/O (GPIOPK[34:0], TESTM, PRSTn, KRSTn, TJDBG)
PWRIOPS_33D	VDD_PS	1	H12	1.65	3.3	3.6	Digital Power for I/O (GPIOPS[31:0], TDO, RTCK, TCK, TMS, TRSTn, TDI, PKG[1:0])
PWRIOPC_33D	VDD_PC	1	E12	1.65	3.3	3.6	Digital Power for I/O (GPIOPC[27:0])
PWRIOPD_33D	VDD_PD	1	F12	1.65	3.3	3.6	Digital Power for I/O (GPIOPD[23:0], BM[3:0])
PWRIOFC_33D	VDD_FC	1	H13	1.65	3.3	3.6	Digital Power for I/O (GPIOFC[11:0])
PWRIOLD_33D	VDD_LD	1	G7	1.65	3.3	3.6	Digital Power for I/O (GPIOLC[5:0], GPIOLD[23:0])
GNDCORE	VSS_CORE	27	A1 M8 K11 V1 E9 L11 F7 N9 M11 M7 P9 P11 E8 E10 N12 G8 P10 A18 H8 F11 V18 J8 G11 K8 H11 L8 J11	-	-	-	Digital / Analog Ground for Core
GNDADC_33A	VSS_ADC	1	L7				Analog Ground for ADC
FSOURCE_ECID	FSOURCE1	1	M3				

Table 2.2 PWRIOPS Group I/O Pin Description

PIN NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
PKG0	PKG0	V5	B	I	3.6V	X	CMOS
RTCK	RTCK	V7	B	O	3.6V	X	CMOS
TDO	TDO	U8	B	O	3.6V	X	CMOS
TCK	TCK	V8	B	IU	3.6V	X	CMOS
TDI	TDI	T8	B	IU	3.6V	X	CMOS
TRSTN	TRSTN	U9	B	ID	3.6V	X	CMOS
TMS	TMS	R8	B	IU	3.6V	X	CMOS
GPIO_PS0	PS0	R9	B	I	3.6V	X	CMOS
GPIO_PS1	PS1	R10	B	I	3.6V	X	CMOS
GPIO_PS2	PS2	T9	B	I	3.6V	X	CMOS
GPIO_PS3	PS3	V11	B	I	3.6V	X	CMOS
GPIO_PS4	PS4	V9	B	I	3.6V	X	CMOS
GPIO_PS5	PS5	V10	B	I	3.6V	X	CMOS
GPIO_PS6	PS6	U10	B	I	3.6V	X	CMOS
GPIO_PS7	PS7	T11	B	I	3.6V	X	CMOS
GPIO_PS8	PS8	T10	B	I	3.6V	X	CMOS
GPIO_PS9	PS9	U12	B	I	3.6V	X	CMOS
GPIO_PS10	PS10	U11	B	I	3.6V	X	CMOS
GPIO_PS11	PS11	V12	B	I	3.6V	X	CMOS
GPIO_PS12	PS12	V14	B	I	3.6V	X	CMOS
GPIO_PS13	PS13	V13	B	I	3.6V	X	CMOS
GPIO_PS14	PS14	V15	B	I	3.6V	X	CMOS
GPIO_PS15	PS15	U13	B	I	3.6V	X	CMOS
GPIO_PS16	PS16	U14	B	I	3.6V	X	CMOS
GPIO_PS17	PS17	R11	B	I	3.6V	X	CMOS
GPIO_PS18	PS18	T13	B	I	3.6V	X	CMOS
GPIO_PS19	PS19	T12	B	I	3.6V	X	CMOS
GPIO_PS20	PS20	R12	B	I	3.6V	X	CMOS
GPIO_PS21	PS21	V17	B	I	3.6V	X	CMOS
GPIO_PS22	PS22	V16	B	I	3.6V	X	CMOS
GPIO_PS23	PS23	U17	B	I	3.6V	X	CMOS
GPIO_PS24	PS24	U15	B	I	3.6V	X	CMOS
GPIO_PS25	PS25	T15	B	I	3.6V	X	CMOS
GPIO_PS26	PS26	U16	B	I	3.6V	X	CMOS
GPIO_PS27	PS27	U18	B	I	3.6V	X	CMOS
GPIO_PS28	PS28	T14	B	I	3.6V	X	CMOS
GPIO_PS29	PS29	T16	B	I	3.6V	X	CMOS
GPIO_PS30	PS30	R13	B	I	3.6V	X	CMOS
GPIO_PS31	PS31	R14	B	I	3.6V	X	CMOS

Table 2.3 GPIO_PS Function Map

PIN NAME	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
PS0	GPIO_PS[00]	SD0D[0]	-	-	EDI_XD[0]	GBUS[0]
PS1	GPIO_PS[01]	SD0D[1]	-	-	EDI_XD[1]	GBUS[1]
PS2	GPIO_PS[02]	SD0D[2]	-	-	EDI_XD[2]	GBUS[2]
PS3	GPIO_PS[03]	SD0D[3]	-	-	EDI_XD[3]	GBUS[3]
PS4	GPIO_PS[04]	SD0CMD	-	-	FALE	GBUS[4]
PS5	GPIO_PS[05]	SD0CLK	-	-	FCLE	GBUS[5]
PS6	GPIO_PS[06]	SD0DET	-	-	FWEn	GBUS[6]
PS7	GPIO_PS[07]	SD0WP	-	-	FOEn	GBUS[7]
PS8	GPIO_PS[08]	SD0D[4]			EDI_XD[4]	GBUS[8]
PS9	GPIO_PS[09]	SD0D[5]			EDI_XD[5]	GBUS[9]
PS10	GPIO_PS[10]	SD0D[6]			EDI_XD[6]	GBUS[10]
PS11	GPIO_PS[11]	SD0D[7]			EDI_XD[7]	GBUS[11]
PS12	GPIO_PS[12]				FCSn[0]	GBUS[12]
PS13	GPIO_PS[13]				FRDY[0]	GBUS[13]
PS14	GPIO_PS[14]				-	GBUS[14]
PS15	GPIO_PS[15]				-	GBUS[15]
PS16	GPIO_PS[16]				-	GBUS[16]
PS17	GPIO_PS[17]				-	GBUS[17]
PS18	GPIO_PS[18]				-	GBUS[18]
PS19	GPIO_PS[19]				-	GBUS[19]
PS20	GPIO_PS[20]				-	GBUS[20]
PS21	GPIO_PS[21]				QSPI_RST	GBUS[21]
PS22	GPIO_PS[22]				QSPI_CS	GBUS[22]
PS23	GPIO_PS[23]				QSPI_CLK	GBUS[23]
PS24	GPIO_PS[24]	-			QSPI_IO0	GBUS[24]
PS25	GPIO_PS[25]	-			QSPI_IO1	GBUS[25]
PS26	GPIO_PS[26]	-			QSPI_IO2	GBUS[26]
PS27	GPIO_PS[27]	-			QSPI_IO3	GBUS[27]
PS28	GPIO_PS[28]	-			-	GBUS[28]
PS29	GPIO_PS[29]	-			-	GBUS[29]
PS30	GPIO_PS[30]	-			-	GBUS[30]
PS31	GPIO_PS[31]	-			-	GBUS[31]

Table 2.4 PWRIOPK Group I/O Pin Description

PIN NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
TESTM	TESTM	T7	B	I	TBD	TBD	SCHMITT
TJDBG	TJDBG	V6	B	I	TBD	TBD	SCHMITT
PRSTN	PRSTN	U6	B	I	TBD	TBD	SCHMITT
KRSTN	KRSTN	U7	B	I	TBD	TBD	SCHMITT
GPIO_PK0	PK0	L4	B	I	TBD	TBD	SCHMITT
GPIO_PK1	PK1	K4	B	I	TBD	TBD	SCHMITT
GPIO_PK2	PK2	N1	B	I	TBD	TBD	SCHMITT
GPIO_PK3	PK3	L3	B	I	TBD	TBD	SCHMITT
GPIO_PK4	PK4	N2	B	I	TBD	TBD	SCHMITT
GPIO_PK5	PK5	L2	B	I	TBD	TBD	SCHMITT
GPIO_PK6	PK6	N3	B	I	TBD	TBD	SCHMITT
GPIO_PK7	PK7	L1	B	I	TBD	TBD	SCHMITT
GPIO_PK8	PK8	M4	B	I	TBD	TBD	SCHMITT
GPIO_PK9	PK9	N4	B	I	TBD	TBD	SCHMITT
GPIO_PK10	PK10	P1	B	I	TBD	TBD	SCHMITT
GPIO_PK11	PK11	R3	B	I	TBD	TBD	SCHMITT
GPIO_PK12	PK12	P2	B	I	TBD	TBD	SCHMITT
GPIO_PK13	PK13	R2	B	I	TBD	TBD	SCHMITT
GPIO_PK14	PK14	P3	B	I	TBD	TBD	SCHMITT
GPIO_PK15	PK15	T2	B	I	TBD	TBD	SCHMITT
GPIO_PK16	PK16	R1	B	I	TBD	TBD	SCHMITT
GPIO_PK17	PK17	T1	B	I	TBD	TBD	SCHMITT
GPIO_PK18	PK18	R5	B	I	TBD	TBD	SCHMITT
GPIO_PK19	PK19	P4	B	I	TBD	TBD	SCHMITT
GPIO_PK20	PK20	T5	B	I	TBD	TBD	SCHMITT
GPIO_PK21	PK21	T3	B	I	TBD	TBD	SCHMITT
GPIO_PK22	PK22	T4	B	I	TBD	TBD	SCHMITT
GPIO_PK23	PK23	U2	B	I	TBD	TBD	SCHMITT
GPIO_PK24	PK24	R6	B	I	TBD	TBD	SCHMITT
GPIO_PK25	PK25	U1	B	I	TBD	TBD	SCHMITT
GPIO_PK26	PK26	V4	B	I	TBD	TBD	SCHMITT
GPIO_PK27	PK27	R4	B	I	TBD	TBD	SCHMITT
GPIO_PK28	PK28	U5	B	I	TBD	TBD	SCHMITT
GPIO_PK29	PK29	U3	B	I	TBD	TBD	SCHMITT
GPIO_PK30	PK30	U4	B	I	TBD	TBD	SCHMITT
GPIO_PK31	PK31	V2	B	I	TBD	TBD	SCHMITT
GPIO_PK32	PK32	T6	B	I	TBD	TBD	SCHMITT
GPIO_PK33	PK33	V3	B	I	TBD	TBD	SCHMITT
GPIO_PK34	PK34	R7	B	I	TBD	TBD	SCHMITT

Table 2.5 GPIO_PK Function Map

PIN NAME	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
PK0	GPIO_PK[00]	EXTINT[0]	-	-	-	-
PK1	GPIO_PK[01]	EXTINT[1]	-	-	-	-
PK2	GPIO_PK[02]	EXTINT[2]	-	-	-	-
PK3	GPIO_PK[03]	EXTINT[3]	-	-	-	-
PK4	GPIO_PK[04]	EXTINT[4]	-	-	-	GBUS[164]
PK5	GPIO_PK[05]	EXTINT[5]	-	-	-	GBUS[165]
PK6	GPIO_PK[06]	EXTINT[6]	-	-	-	GBUS[166]
PK7	GPIO_PK[07]	EXTINT[7]	-	-	-	GBUS[167]
PK8	GPIO_PK[08]	-	-	-	-	GBUS[168]
PK9	GPIO_PK[09]	-	-	-	-	GBUS[169]
PK10	GPIO_PK[10]	-	-	-	-	GBUS[170]
PK11	GPIO_PK[11]	-	-	-	-	GBUS[171]
PK12	GPIO_PK[12]	-	-	-	-	GBUS[172]
PK13	GPIO_PK[13]	-	-	-	-	GBUS[173]
PK14	GPIO_PK[14]	-	-	-	-	GBUS[174]
PK15	GPIO_PK[15]	-	-	-	-	GBUS[175]
PK16	GPIO_PK[16]	-	-	-	TCO[0]	GBUS[176]
PK17	GPIO_PK[17]	-	-	-	TCO[1]	GBUS[177]
PK18	GPIO_PK[18]	-	-	-	TCO[2]	GBUS[178]
PK19	GPIO_PK[19]	-	-	-	TCO[3]	GBUS[179]
PK20	GPIO_PK[20]	-	-	-	TCO[4]	GBUS[180]
PK21	GPIO_PK[21]	-	-	-	TCO[5]	GBUS[181]
PK22	GPIO_PK[22]	CAN0TXD	-	-	-	GBUS[182]
PK23	GPIO_PK[23]	CAN0RXD	-	-	-	GBUS[183]
PK24	GPIO_PK[24]	CAN1TXD	-	-	-	GBUS[184]
PK25	GPIO_PK[25]	CAN1RXD	-	-	-	GBUS[185]
PK26	GPIO_PK[26]		-	-	TCO[3]	GBUS[186]
PK27	GPIO_PK[27]		-	-	TCO[4]	GBUS[187]
PK28	GPIO_PK[28]	-	-	-	-	GBUS[188]
PK29	GPIO_PK[29]	-	-	-	-	GBUS[189]
PK30	GPIO_PK[30]	-	-	-	-	GBUS[190]
PK31	GPIO_PK[31]	-	-	-	-	GBUS[191]
PK32	GPIO_PK[32]/ PWRCTL[0]	-	-	-	-	-
PK33	GPIO_PK[33]/ PWRCTL[1]	HRESETn	-	-	-	-
PK34	GPIO_PK[34]	HRESETn	-	-	-	-

Table 2.6 PWRIOPC Group I/O Pin Description

PIN NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
GPIO_PC0	PC0	D13	B	I	3.6V	X	CMOS
GPIO_PC1	PC1	D15	B	I	3.6V	X	CMOS
GPIO_PC2	PC2	C14	B	I	3.6V	X	CMOS
GPIO_PC3	PC3	D14	B	I	3.6V	X	CMOS
GPIO_PC4	PC4	C15	B	I	3.6V	X	CMOS
GPIO_PC5	PC5	B13	B	I	3.6V	X	CMOS
GPIO_PC6	PC6	C13	B	I	3.6V	X	CMOS
GPIO_PC7	PC7	D11	B	I	3.6V	X	CMOS
GPIO_PC8	PC8	D12	B	I	3.6V	X	CMOS
GPIO_PC9	PC9	C12	B	I	3.6V	X	CMOS
GPIO_PC10	PC10	B15	B	I	3.6V	X	CMOS
GPIO_PC11	PC11	B12	B	I	3.6V	X	CMOS
GPIO_PC12	PC12	B14	B	I	3.6V	X	CMOS
GPIO_PC13	PC13	A12	B	I	3.6V	X	CMOS
GPIO_PC14	PC14	A14	B	I	3.6V	X	CMOS
GPIO_PC15	PC15	C11	B	I	3.6V	X	CMOS
GPIO_PC16	PC16	A13	B	I	3.6V	X	CMOS
GPIO_PC17	PC17	B11	B	I	3.6V	X	CMOS
GPIO_PC18	PC18	A9	B	I	3.6V	X	CMOS
GPIO_PC19	PC19	A11	B	I	3.6V	X	CMOS
GPIO_PC20	PC20	B9	B	I	3.6V	X	CMOS
GPIO_PC21	PC21	C10	B	I	3.6V	X	CMOS
GPIO_PC22	PC22	B7	B	I	3.6V	X	CMOS
GPIO_PC23	PC23	B10	B	I	3.6V	X	CMOS
GPIO_PC24	PC24	C7	B	I	3.6V	X	CMOS
GPIO_PC25	PC25	A10	B	I	3.6V	X	CMOS
GPIO_PC26	PC26	D9	B	I	3.6V	X	CMOS
GPIO_PC27	PC27	A7	B	I	3.6V	X	CMOS

Table 2.7 GPIO_PC Function Map

PIN NAME	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
PC0	GPIO_PC[00]	CIF0PD[0]				GBUS[80]
PC1	GPIO_PC[01]	CIF0PD[1]				GBUS[81]
PC2	GPIO_PC[02]	CIF0PD[2]	-	-	-	GBUS[82]
PC3	GPIO_PC[03]	CIF0PD[3]	-	-	-	GBUS[83]
PC4	GPIO_PC[04]	CIF0PD[4]	-	-	-	GBUS[84]
PC5	GPIO_PC[05]	CIF0PD[5]	-	-	-	GBUS[85]
PC6	GPIO_PC[06]	CIF0PD[6]	-	-	-	GBUS[86]
PC7	GPIO_PC[07]	CIF0PD[7]	-	-	-	GBUS[87]
PC8	GPIO_PC[08]	CIF0CKI	-	-	-	GBUS[88]
PC9	GPIO_PC[09]	CIF0VS	-	-	-	GBUS[89]
PC10	GPIO_PC[10]	CIF0HS	-	-	-	GBUS[90]
PC11	GPIO_PC[11]	CIF0CKO	-	-	-	GBUS[91]
PC12	GPIO_PC[12]	CIF0CFIELD	-	-	-	GBUS[92]
PC13	GPIO_PC[13]	CIF0CDE	-	-	-	GBUS[93]
PC14	GPIO_PC[14]	CIF0PD[8]	CIF1PD[0]	-	-	GBUS[94]
PC15	GPIO_PC[15]	CIF0PD[9]	CIF1PD[1]	-	-	GBUS[95]
PC16	GPIO_PC[16]	CIF0PD[10]	CIF1PD[2]	-	-	GBUS[96]
PC17	GPIO_PC[17]	CIF0PD[11]	CIF1PD[3]	-	-	GBUS[97]
PC18	GPIO_PC[18]	CIF0PD[12]	CIF1PD[4]	-	-	GBUS[98]
PC19	GPIO_PC[19]	CIF0PD[13]	CIF1PD[5]	-	-	GBUS[99]
PC20	GPIO_PC[20]	CIF0PD[14]	CIF1PD[6]	-	-	GBUS[100]
PC21	GPIO_PC[21]	CIF0PD[15]	CIF1PD[7]	-	-	GBUS[101]
PC22	GPIO_PC[22]	-	CIF1CKI	-	-	GBUS[102]
PC23	GPIO_PC[23]	-	CIF1VS	-	-	GBUS[103]
PC24	GPIO_PC[24]	-	CIF1HS	-	-	GBUS[104]
PC25	GPIO_PC[25]	-	CIF1CKO	-	-	GBUS[105]
PC26	GPIO_PC[26]	-	CIF1CFIELD	-	-	GBUS[106]
PC27	GPIO_PC[27]	-	CIF1CDE	-	-	GBUS[107]

Table 2.8 PWRIOPD Group I/O Pin Description

PIN NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
BM0	BM0	E17	B	I	3.6V	X	CMOS
BM1	BM1	D17	B	I	3.6V	X	CMOS
BM2	BM2	E16	B	I	3.6V	X	CMOS
BM3	BM3	D16	B	I	3.6V	X	CMOS
GPIO_PD0	PD0	F17	B	I	3.6V	X	CMOS
GPIO_PD1	PD1	A15	B	I	3.6V	X	CMOS
GPIO_PD2	PD2	G17	B	I	3.6V	X	CMOS
GPIO_PD3	PD3	A16	B	I	3.6V	X	CMOS
GPIO_PD4	PD4	H17	B	I	3.6V	X	CMOS
GPIO_PD5	PD5	A17	B	I	3.6V	X	CMOS
GPIO_PD6	PD6	J17	B	I	3.6V	X	CMOS
GPIO_PD7	PD7	B16	B	I	3.6V	X	CMOS
GPIO_PD8	PD8	K17	B	I	3.6V	X	CMOS
GPIO_PD9	PD9	B17	B	I	3.6V	X	CMOS
GPIO_PD10	PD10	L17	B	I	3.6V	X	CMOS
GPIO_PD11	PD11	B18	B	I	3.6V	X	CMOS
GPIO_PD12	PD12	M18	B	I	3.6V	X	CMOS
GPIO_PD13	PD13	C16	B	I	3.6V	X	CMOS
GPIO_PD14	PD14	G18	B	I	3.6V	X	CMOS
GPIO_PD15	PD15	F18	B	I	3.6V	X	CMOS
GPIO_PD16	PD16	H18	B	I	3.6V	X	CMOS
GPIO_PD17	PD17	E18	B	I	3.6V	X	CMOS
GPIO_PD18	PD18	J18	B	I	3.6V	X	CMOS
GPIO_PD19	PD19	D18	B	I	3.6V	X	CMOS
GPIO_PD20	PD20	K18	B	I	3.6V	X	CMOS
GPIO_PD21	PD21	C18	B	I	3.6V	X	CMOS
GPIO_PD22	PD22	L18	B	I	3.6V	X	CMOS
GPIO_PD23	PD23	C17	B	I	3.6V	X	CMOS

Table 2.9 GPIO_PD Function Map

PIN NAME	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
PD0	GPIO_PD[00]	-	-	-	-	GBUS[56]
PD1	GPIO_PD[01]	-	-	-	-	GBUS[57]
PD2	GPIO_PD[02]	-	-	-	-	GBUS[58]
PD3	GPIO_PD[03]	-	-	-	-	GBUS[59]
PD4	GPIO_PD[04]	-	-	-	-	GBUS[60]
PD5	GPIO_PD[05]	-	-	-	-	GBUS[61]
PD6	GPIO_PD[06]	-	-	-	-	GBUS[62]
PD7	GPIO_PD[07]	-	-	-	-	GBUS[63]
PD8	GPIO_PD[08]	-	-	-	-	GBUS[64]
PD9	GPIO_PD[09]	-	-	-	-	GBUS[65]
PD10	GPIO_PD[10]	-	-	-	-	GBUS[66]
PD11	GPIO_PD[11]	-	-	-	-	GBUS[67]
PD12	GPIO_PD[12]	-	-	-	-	GBUS[68]
PD13	GPIO_PD[13]	-	-	-	-	GBUS[69]
PD14	GPIO_PD[14]	-	-	-	-	GBUS[70]
PD15	GPIO_PD[15]	-	-	-	-	GBUS[71]
PD16	GPIO_PD[16]	-	-	-	-	GBUS[72]
PD17	GPIO_PD[17]	-	-	-	-	GBUS[73]
PD18	GPIO_PD[18]	-	-	-	-	GBUS[74]
PD19	GPIO_PD[19]	-	-	-	-	GBUS[75]
PD20	GPIO_PD[20]	-	-	-	-	GBUS[76]
PD21	GPIO_PD[21]	-	-	-	-	GBUS[77]
PD22	GPIO_PD[22]	DEV_VBUS_P WR_CTL	-	-	-	GBUS[78]
PD23	GPIO_PD[23]	HVBUS_PWR CTL	-	-	-	GBUS[79]

Table 2.10 PWRIOFC Group I/O Pin Description

PIN NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
GPIO_FC0	FC0	P18	B	I	3.6V	X	CMOS
GPIO_FC1	FC1	R15	B	I	3.6V	X	CMOS
GPIO_FC2	FC2	P16	B	I	3.6V	X	CMOS
GPIO_FC3	FC3	R16	B	I	3.6V	X	CMOS
GPIO_FC4	FC4	P17	B	I	3.6V	X	CMOS
GPIO_FC5	FC5	R17	B	I	3.6V	X	CMOS
GPIO_FC6	FC6	N18	B	I	3.6V	X	CMOS
GPIO_FC7	FC7	T17	B	I	3.6V	X	CMOS
GPIO_FC8	FC8	N17	B	I	3.6V	X	CMOS
GPIO_FC9	FC9	T18	B	I	3.6V	X	CMOS
GPIO_FC10	FC10	M17	B	I	3.6V	X	CMOS
GPIO_FC11	FC11	R18	B	I	3.6V	X	CMOS

Table 2.11 GPIO_FC Function Map

PIN NAME	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
FC0	GPIO_FC[00]	SD1D[0]	TCO[0]	CKMO[0]	-	GBUS[116]
FC1	GPIO_FC[01]	SD1D[1]	TCO[1]	CKMO[1]		GBUS[117]
FC2	GPIO_FC[02]	SD1D[2]	TCO[2]	CKMO[0]		GBUS[118]
FC3	GPIO_FC[03]	SD1D[3]	TCO[3]	CKMO[1]		GBUS[119]
FC4	GPIO_FC[04]	SD1CMD	TCO[4]	CKMO[0]		GBUS[120]
FC5	GPIO_FC[05]	SD1CLK	TCO[5]	CKMO[1]		GBUS[121]
FC6	GPIO_FC[06]	SD1DET	TCO[0]	CKMO[0]		GBUS[122]
FC7	GPIO_FC[07]	SD1WP	TCO[1]	CKMO[1]		GBUS[123]
FC8	GPIO_FC[08]	SD1D[4]	-	-		GBUS[124]
FC9	GPIO_FC[09]	SD1D[5]	-	-		GBUS[125]
FC10	GPIO_FC[10]	SD1D[6]	-	-		GBUS[126]
FC11	GPIO_FC[11]	SD1D[7]	-	-		GBUS[127]

Table 2.12 PWRIOLC Group I/O Pin Description

PIN NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
GPIO_LC0	LC0	C6	B	I	3.6V	X	CMOS
GPIO_LC1	LC1	D5	B	I	3.6V	X	CMOS
GPIO_LC2	LC2	C5	B	I	3.6V	X	CMOS
GPIO_LC3	LC3	C4	B	I	3.6V	X	CMOS
GPIO_LC4	LC4	D6	B	I	3.6V	X	CMOS
GPIO_LC5	LC5	D4	B	I	3.6V	X	CMOS
GPIO_LD0	LD0	A3	B	I	3.6V	X	CMOS
GPIO_LD1	LD1	B3	B	I	3.6V	X	CMOS
GPIO_LD2	LD2	A2	B	I	3.6V	X	CMOS
GPIO_LD3	LD3	E3	B	I	3.6V	X	CMOS
GPIO_LD4	LD4	B1	B	I	3.6V	X	CMOS
GPIO_LD5	LD5	E2	B	I	3.6V	X	CMOS
GPIO_LD6	LD6	B2	B	I	3.6V	X	CMOS
GPIO_LD7	LD7	F4	B	I	3.6V	X	CMOS
GPIO_LD8	LD8	C3	B	I	3.6V	X	CMOS
GPIO_LD9	LD9	F1	B	I	3.6V	X	CMOS
GPIO_LD10	LD10	C2	B	I	3.6V	X	CMOS
GPIO_LD11	LD11	F3	B	I	3.6V	X	CMOS
GPIO_LD12	LD12	C1	B	I	3.6V	X	CMOS
GPIO_LD13	LD13	F2	B	I	3.6V	X	CMOS
GPIO_LD14	LD14	D3	B	I	3.6V	X	CMOS
GPIO_LD15	LD15	G1	B	I	3.6V	X	CMOS
GPIO_LD16	LD16	D2	B	I	3.6V	X	CMOS
GPIO_LD17	LD17	G2	B	I	3.6V	X	CMOS
GPIO_LD18	LD18	E1	B	I	3.6V	X	CMOS
GPIO_LD19	LD19	G3	B	I	3.6V	X	CMOS
GPIO_LD20	LD20	D1	B	I	3.6V	X	CMOS
GPIO_LD21	LD21	H4	B	I	3.6V	X	CMOS
GPIO_LD22	LD22	E4	B	I	3.6V	X	CMOS
GPIO_LD23	LD23	G4	B	I	3.6V	X	CMOS

Table 2.13 GPIO_LC / GPIO_LD Function Map

PIN NAME	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
LC0	GPIO_LC[00]	LVS	LXCSn[0]	-	-	-
LC1	GPIO_LC[01]	LHS	LXCSn[1]	-	-	-
LC2	GPIO_LC[02]	LAC	LXOEn	-	-	-
LC3	GPIO_LC[03]	LCLK	LXWEn	-	-	-
LC4	GPIO_LC[04]	-	LXA[0]			GBUS[114]
LC5	GPIO_LC[05]	LIRQ	LXA[1]	-	-	GBUS[115]
LD0	GPIO_LD[00]	LPD[0]	LXD[0]	-	-	-
LD1	GPIO_LD[01]	LPD[1]	LXD[1]	-	-	-
LD2	GPIO_LD[02]	LPD[2]	LXD[2]	-	-	-
LD3	GPIO_LD[03]	LPD[3]	LXD[3]	-	-	-
LD4	GPIO_LD[04]	LPD[4]	LXD[4]	-	-	-
LD5	GPIO_LD[05]	LPD[5]	LXD[5]	-	-	-
LD6	GPIO_LD[06]	LPD[6]	LXD[6]	-	-	-
LD7	GPIO_LD[07]	LPD[7]	LXD[7]	-	-	-
LD8	GPIO_LD[08]	LPD[8]	LXD[8]	-	-	-
LD9	GPIO_LD[09]	LPD[9]	LXD[9]	-	-	-
LD10	GPIO_LD[10]	LPD[10]	LXD[10]	-	-	-
LD11	GPIO_LD[11]	LPD[11]	LXD[11]	-	-	-
LD12	GPIO_LD[12]	LPD[12]	LXD[12]	-	-	-
LD13	GPIO_LD[13]	LPD[13]	LXD[13]	-	-	-
LD14	GPIO_LD[14]	LPD[14]	LXD[14]	-	-	-
LD15	GPIO_LD[15]	LPD[15]	LXD[15]	-	-	-
LD16	GPIO_LD[16]	LPD[16]	LXD[16]	-	-	-
LD17	GPIO_LD[17]	LPD[17]	LXD[17]	-	-	-
LD18	GPIO_LD[18]	LPD[18]	LXD[18]	-	-	GBUS[108]
LD19	GPIO_LD[19]	LPD[19]	LXD[19]	-	-	GBUS[109]
LD20	GPIO_LD[20]	LPD[20]	LXD[20]	-	-	GBUS[110]
LD21	GPIO_LD[21]	LPD[21]	LXD[21]	-	-	GBUS[111]
LD22	GPIO_LD[22]	LPD[22]	LXD[22]	-	-	GBUS[112]
LD23	GPIO_LD[23]	LPD[23]	LXD[23]	-	-	GBUS[113]

Table 2.14 PWRMEMIO Group I/O Pin Description

PIN NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
XDCLK	XDCLK	J6	B	I	3.6V	X	CMOS
XDCKE	XDCKE	L5	B	I	3.6V	X	CMOS
XDCSN0	XDCSN0	K5	B	I	3.6V	X	CMOS
XWEN	XWEN	K6	B	I	3.6V	X	CMOS
XA0	XA0	H6	B	I	3.6V	X	CMOS
XA1	XA1	E5	B	I	3.6V	X	CMOS
XA2	XA2	E6	B	I	3.6V	X	CMOS
XA3	XA3	F5	B	I	3.6V	X	CMOS
XA4	XA4	L6	B	I	3.6V	X	CMOS
XA5	XA5	M5	B	I	3.6V	X	CMOS
XA6	XA6	G5	B	I	3.6V	X	CMOS
XA7	XA7	M6	B	I	3.6V	X	CMOS
XA8	XA8	P6	B	I	3.6V	X	CMOS
XA9	XA9	P5	B	I	3.6V	X	CMOS
XA10	XA10	F6	B	I	3.6V	X	CMOS
XA11	XA11	N6	B	I	3.6V	X	CMOS
XA12	XA12	K2	B	I	3.6V	X	CMOS
XA13	XA13	G6	B	I	3.6V	X	CMOS
XA14	XA14	N5	B	I	3.6V	X	CMOS
XA15	XA15	H5	B	I	3.6V	X	CMOS
XA16	XA16	J5	B	I	3.6V	X	CMOS
XA18	XA18	K13	B	I	3.6V	X	CMOS
XA19	XA19	L12	B	I	3.6V	X	CMOS
XA20	XA20	F15	B	I	3.6V	X	CMOS
XA21	XA21	J13	B	I	3.6V	X	CMOS
XD0	XD0	P15	B	I	3.6V	X	CMOS
XD1	XD1	N15	B	I	3.6V	X	CMOS
XD2	XD2	N14	B	I	3.6V	X	CMOS
XD3	XD3	L14	B	I	3.6V	X	CMOS
XD4	XD4	P14	B	I	3.6V	X	CMOS
XD5	XD5	M14	B	I	3.6V	X	CMOS
XD6	XD6	N16	B	I	3.6V	X	CMOS
XD7	XD7	M15	B	I	3.6V	X	CMOS
XD8	XD8	M16	B	I	3.6V	X	CMOS
XD9	XD9	L16	B	I	3.6V	X	CMOS
XD10	XD10	L15	B	I	3.6V	X	CMOS
XD11	XD11	K14	B	I	3.6V	X	CMOS
XD12	XD12	P13	B	I	3.6V	X	CMOS
XD13	XD13	N13	B	I	3.6V	X	CMOS
XD14	XD14	M13	B	I	3.6V	X	CMOS
XD15	XD15	L13	B	I	3.6V	X	CMOS
XD16	XD16	K15	B	I	3.6V	X	CMOS
XD17	XD17	J14	B	I	3.6V	X	CMOS
XD18	XD18	J15	B	I	3.6V	X	CMOS
XD19	XD19	E15	B	I	3.6V	X	CMOS
XD20	XD20	J16	B	I	3.6V	X	CMOS
XD21	XD21	K16	B	I	3.6V	X	CMOS
XD22	XD22	E14	B	I	3.6V	X	CMOS
XD23	XD23	G15	B	I	3.6V	X	CMOS
XD24	XD24	E13	B	I	3.6V	X	CMOS
XD25	XD25	H14	B	I	3.6V	X	CMOS
XD26	XD26	G13	B	I	3.6V	X	CMOS
XD27	XD27	G14	B	I	3.6V	X	CMOS
XD28	XD28	F16	B	I	3.6V	X	CMOS
XD29	XD29	M12	B	I	3.6V	X	CMOS
XD30	XD30	F14	B	I	3.6V	X	CMOS
XD31	XD31	F13	B	I	3.6V	X	CMOS

Table 2.15 GPIO_LC / GPIO_LD Function Map

PIN NAME	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
XDCLK	XDCLK	XDCLK	-	-	-	-
XDCKE	XDCKE	XDCKE	-	-	-	-
XDCS0#	XDCSn[0]	XDCSn[0]	-	-	-	-
XWE#	XWEn	XWEn	-	-	-	-
XA0	XA[0]	XA[0]	-	-	-	-
XA1	XA[1]	XA[1]	-	-	-	-
XA2	XA[2]	XA[2]	-	-	-	-
XA3	XA[3]	XA[3]	-	-	-	-
XA4	XA[4]	XA[4]	-	-	-	GBUS[144]
XA5	XA[5]	XA[5]	-	-	-	GBUS[145]
XA6	XA[6]	XA[6]	-	-	-	GBUS[146]
XA7	XA[7]	XA[7]	-	-	-	GBUS[147]
XA8	XA[8]	XA[8]	-	-	-	GBUS[148]
XA9	XA[9]	XA[9]	-	-	-	GBUS[149]
XA10	XA[10]	XA[10]	-	-	-	GBUS[150]
XA11	XA[11]	XA[11]	-	-	-	GBUS[151]
XA12	XA[12]	XA[12]	-	-	-	GBUS[152]
XA13	XA[13]	XA[13]	-	-	-	GBUS[153]
XA14	XA[14]	XA[14]	-	-	-	GBUS[154]
XA15	XA[15]	XA[15]	-	-	-	GBUS[155]
XA16	XA[16]	XA[16]	-	-	-	GBUS[156]
XA18	XA[18]	XA[18]	-	-	-	GBUS[158]
XA19	XA[19]	XA[19]	-	-	-	GBUS[159]
XA20	XA[20]	XA[20]	-	-	-	GBUS[160]
XA21	XA[21]	XA[21]	-	-	-	GBUS[161]
XA18	XA[18]	XA[18]	-	-	-	GBUS[158]
XA19	XA[19]	XA[19]	-	-	-	GBUS[159]
XA20	XA[20]	XA[20]	-	-	-	GBUS[160]
XA21	XA[21]	XA[21]	-	-	-	GBUS[161]
XD0	XD[0]	XD[0]	-	-	-	-
XD1	XD[1]	XD[1]	-	-	-	-
XD2	XD[2]	XD[2]	-	-	-	-
XD3	XD[3]	XD[3]	-	-	-	-
XD4	XD[4]	XD[4]	-	-	-	-
XD5	XD[5]	XD[5]	-	-	-	-
XD6	XD[6]	XD[6]	-	-	-	-
XD7	XD[7]	XD[7]	-	-	-	-
XD8	XD[8]	XD[8]	-	-	-	-
XD9	XD[9]	XD[9]	-	-	-	-
XD10	XD[10]	XD[10]	-	-	-	-
XD11	XD[11]	XD[11]	-	-	-	-
XD12	XD[12]	XD[12]	-	-	-	-
XD13	XD[13]	XD[13]	-	-	-	-
XD14	XD[14]	XD[14]	-	-	-	-
XD15	XD[15]	XD[15]	-	-	-	-
XD16	GPIOXD[16]	XD[16]	-	-	-	GBUS[128]
XD17	GPIOXD[17]	XD[17]	-	-	-	GBUS[129]
XD18	GPIOXD[18]	XD[18]	-	-	-	GBUS[130]
XD19	GPIOXD[19]	XD[19]	-	-	-	GBUS[131]
XD20	GPIOXD[20]	XD[20]	-	-	-	GBUS[132]
XD21	GPIOXD[21]	XD[21]	-	-	-	GBUS[133]
XD22	GPIOXD[22]	XD[22]	-	-	-	GBUS[134]
XD23	GPIOXD[23]	XD[23]	-	-	-	GBUS[135]
XD24	GPIOXD[24]	XD[24]	-	-	-	GBUS[136]
XD25	GPIOXD[25]	XD[25]	-	-	-	GBUS[137]
XD26	GPIOXD[26]	XD[26]	-	-	-	GBUS[138]
XD27	GPIOXD[27]	XD[27]	-	-	-	GBUS[139]
XD28	GPIOXD[28]	XD[28]	-	-	-	GBUS[140]
XD29	GPIOXD[29]	XD[29]	-	-	-	GBUS[141]
XD30	GPIOXD[30]	XD[30]	-	-	-	GBUS[142]
XD31	GPIOXD[31]	XD[31]	-	-	-	GBUS[143]

Table 2.16 PWROSC Group I/O Pin Description

PAD NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
XIN	XI	B6	I	I	X	X	OSC
XOUT	XO	A6	O	O			

Table 2.17 PWRRTC Group I/O Pin Description

PAD NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
XTI	XTI	M1	I	I	X	X	OSC
XTO	XTO	M2	O	O	X	X	OSC
RTCRSTN	RTCRSTN	K1	I	I	TBD	TBD	SCHMITT

Table 2.18 PWRUSB20H Group I/O Pin Description

PAD NAME	BALL NAME	BALL #	TYPE	I/O	FAILSAFE	TOLERANT
USB20H_DM	USB20H_DM	A4		AI	X	5V
USB20H_DP	USB20H_DP	B4		AI	X	5V
USB20H_ID	USB20H_ID	B5		AI	X	TBD
USB20H_REXT	USB20H_REXT	A5		AI	X	TBD
USB20H_VBUS	USB20H_VBUS	D7		AI	X	5V

Note: USB20H_TXRTUNE: External resistor (TBD Ohm) is connected to GND through REXT.

Table 2.19 PWRUSB20DH Group I/O Pin Description

PAD NAME	BALL NAME	BALL #	TYPE	I/O	FAILSAFE	TOLERANT
USB20DH_DM	USB20DH_DM	A8		AI	X	5V
USB20DH_DP	USB20DH_DP	B8		AI	X	5V
USB20DH_ID	USB20DH_ID	C9		AI	X	TBD
USB20DH_REXT	USB20DH_REXT	D8		AI	X	TBD
USB20DH_VBUS	USB20DH_VBUS	C8		AI	X	5V

Note: USB20DH_TXRTUNE: External resistor (TBD Ohm) is connected to GND through REXT

Table 2.20 PWRADC Group I/O Pin Description

PAD NAME	BALL NAME	BALL #	TYPE	I/O	FAILSAFE	TOLERANT
ADIN0	ADIN0	H3	0 Ohm	AI	X	3.6V
ADIN1	ADIN1	J4	0 Ohm	AI	X	3.6V
ADIN2	ADIN2	H2	0 Ohm	AI	X	3.6V
ADIN3	ADIN3	K3	0 Ohm	AI	X	3.6V
ADIN9	ADIN9	H1	0 Ohm	AI	X	3.6V
ADIN10	ADIN10	J3	0 Ohm	AI	X	3.6V
ADIN11	ADIN11	J1	0 Ohm	AI	X	3.6V
ADIN12	ADIN12	J2	0 Ohm	AI	X	3.6V

Table 2.21 PWRSSPLL Group I/O Pin Description

PAD NAME	BALL NAME	BALL #	TYPE	I/O	FAILSAFE	TOLERANT
FILTER	FILTER	P12	0 Ohm	AI	X	X

Table 2.22 PWROTP25 Group I/O Pin Description

PIN NAME	BALL NAME	BALL Num	I/O	INIT	TOLERANT	FAILSAFE	IN TYPE
OTPROM_WP	OTPROM_WP	J12	I	I	3.6V	X	CMOS

2.2 TCC8132 Ball Map

Table 2.23 TCC8132 Ball Map (324-BGA-1818 Top View)

	1	2	3	4	5	6	7	8	9	10	11
A	VSS_CORE	LD2	LD0	USB20H_DM	USB20H_REXT	XO	PC27	USB20DH_DM	PC18	PC25	PC19
B	LD4	LD6	LD1	USB20H_DP	USB20H_ID	XI	PC22	USB20DH_DP	PC20	PC23	PC17
C	LD12	LD10	LD8	LC3	LC2	LC0	PC24	USB20DH_VBUS	USB20DH_ID	PC21	PC15
D	LD20	LD16	LD14	LC5	LC1	LC4	USB20H_VBUS	USB20DH_REXT	PC26	VDDAH_USB33	PC7
E	LD18	LD5	LD3	LD22	XA1	XA2	VDD_OSC	VSS_CORE	VSS_CORE	VSS_CORE	VDDAO_USB33
F	LD9	LD13	LD11	LD7	XA3	XA10	VSS_CORE	VDDDH_USB11	VDDDO_USB11	VDDI_CORE	VSS_CORE
G	LD15	LD17	LD19	LD23	XA6	XA13	VDD_LD	VSS_CORE	VDDI_CORE	VDDI_CORE	VSS_CORE
H	ADIN9	ADIN2	ADIN0	LD21	XA15	XA0	VDD_XA	VSS_CORE	VDDI_CORE	VDDI_CORE	VSS_CORE
J	ADIN11	ADIN12	ADIN10	ADIN1	XA16	XDCLK	VDD_MEM	VSS_CORE	VDDI_CORE	VDDI_CORE	VSS_CORE
K	RTCRSTN	XA12	ADIN3	PK1	XDCSN0	XWEN	VDD_ADC	VSS_CORE	VDDI_CORE	VDDI_CORE	VSS_CORE
L	PK7	PK5	PK3	PK0	XDCKE	XA4	VSS_ADC	VSS_CORE	VDDI_CORE	VDDI_CORE	VSS_CORE
M	XTI	XTO	FSOURCE1	PK8	XA5	XA7	VSS_CORE	VSS_CORE	VDDI_CORE	VDDI_CORE	VSS_CORE
N	PK2	PK4	PK6	PK9	XA14	XA11	VDD_RTC	VDD_PK	VSS_CORE	VDD_PLL	VDD_PLL
P	PK10	PK12	PK14	PK19	XA9	XA8	VDD_RTC_PM	VDD_PK	VSS_CORE	VSS_CORE	VSS_CORE
R	PK16	PK13	PK11	PK27	PK18	PK24	PK34	TMS	PS0	PS1	PS17
T	PK17	PK15	PK21	PK22	PK20	PK32	TESTM	TDI	PS2	PS8	PS7
U	PK25	PK23	PK29	PK30	PK28	PRSTN	KRSTN	TDO	TRSTN	PS6	PS10
V	VSS_CORE	PK31	PK33	PK26	PKG0	TJDBG	RTCK	TCK	PS4	PS5	PS3

	12	13	14	15	16	17	18
A	PC13	PC16	PC14	PD1	PD3	PD5	VSS_CORE
B	PC11	PC5	PC12	PC10	PD7	PD9	PD11
C	PC9	PC6	PC2	PC4	PD13	PD23	PD21
D	PC8	PC0	PC3	PC1	BM3	BM1	PD19
E	VDD_PC	XD24	XD22	XD19	BM2	BM0	PD17
F	VDD_PD	XD31	XD30	XA20	XD28	PD0	PD15
G	VDD_XD	XD26	XD27	XD23	VDD_MEM	PD2	PD14
H	VDD_PS	VDD_FC	XD25	VDD_MEM	VDD_MEM	PD4	PD16
J	OTPROM_WP	XA21	XD17	XD18	XD20	PD6	PD18
K	VDDD_OTP25	XA18	XD11	XD16	XD21	PD8	PD20
L	XA19	XD15	XD3	XD10	XD9	PD10	PD22
M	XD29	XD14	XD5	XD7	XD8	FC10	PD12
N	VSS_CORE	XD13	XD2	XD1	XD6	FC8	FC6
P	FILTER	XD12	XD4	XD0	FC2	FC4	FC0
R	PS20	PS30	PS31	FC1	FC3	FC5	FC11
T	PS19	PS18	PS28	PS25	PS29	FC7	FC9
U	PS9	PS15	PS16	PS24	PS26	PS23	PS27
V	PS11	PS13	PS12	PS14	PS22	PS21	VSS_CORE

2.3 I/O Type

The following table illustrates the I/O types used in the TCC8132.

Table 2.24 I/O Type Description

I/O Cell Type	Power	Signal Name	Diagram(Logical Equivalent)
Oscillator	VDDOSC	XI, XO	
Oscillator for RTC	VDDRTC	XTI, XTO	
Wide range IO supply, Programmable bi-direction I/O with cmos/schmitt trigger input, input disable, pull-up/down and 4-step strength output	VDDE_XD	XD[31:0], XDCLK	
	VDDE_XA	XA[23:0], XDCKE, XRCS#, XWE#, XDCS[0]#,	
	VDDE_LD	LC[5:0], LD[23:0]	
	VDDE_PC	PC[27:0]	
	VDDE_PD	PD[23:0], BM[3:0]	
	VDDE_PS	PS[31:0], PKG[1:0], TCK, TDI, TMS, TRST#, TDO, RTCK,	
	VDDE_FC	FC[11:0]	
3.3V IO supply, Programmable bi-direction I/O with cmos/schmitt trigger input, input disable, pull-up/down and 4-step strength output	VDDK	PK[34:0], PRST#, KRST#, TESTM, TJDBG	
	VDDRTC	RTCRST#	
Analog and digital mixed PAD	VDDADC	ADIN[7:0] (ADC Inputs)	

Note: USB signals are connected directly to the internal USB2.0 PHY modules.

Table 2.25 GPIO_XD/XA/LD/PC/PD/PS/FC Drive Strength

VDD	STRENGTH	MIN	TYP	MAX	UNIT
3.0V(MIN) 3.3V(TYP) 3.6V(MAX)	0	6.4	10.3	14.8	mA
	1	10.6	16.4	23.9	mA
	2	15.9	24.7	35.9	mA
	3	23.9	37.1	53.9	mA
2.3V(MIN) 2.5V(TYP) 2.7V(MAX)	0	4.1	6.5	9.8	mA
	1	6.5	10.5	15.8	mA
	2	9.8	15.8	23.7	mA
	3	14.8	23.7	35.6	mA
1.65V(MIN)	0	2.1	3.6	5.8	mA

1.8V(TYP)	1	3.4	5.8	9.3	mA
1.95V(MAX)	2	5.1	8.7	13.9	mA
	3	7.7	13.1	20.9	mA

Note:

MIN = T=125℃, Process=Slow, Isink at VDD*0.2V, Isource at VDD*0.8V

TYP = T=25℃, Process=Nominal, Isink at VDD*0.2V, Isource at VDD*0.8V

MAX= T=-40℃, Process=Fast, Isink at VDD*0.2V, Isource at VDD*0.8V

Table 2.26 GPIO_XD/XA/LD/PC/PD/PS/FC Pull-Up/Down Resistance

VDD	TYPE	MIN	TYP	MAX	UNIT
3.0V(MIN)	PULL-UP	33.9	24.8	18.1	KOhm
3.3V(TYP)	PULL-DOWN	34.6	21.8	15.2	KOhm
3.6V(MAX)					
2.3V(MIN)	PULL-UP	43.6	32.2	21.6	KOhm
2.5V(TYP)	PULL-DOWN	44.6	27.5	18.4	KOhm
2.7V(MAX)					
1.65V(MIN)	PULL-UP	64.3	43.7	18.1	KOhm
1.8V(TYP)	PULL-DOWN	69.4	41.7	15.2	KOhm
1.95V(MAX)					

Note:

MIN = T=125℃, Process=Slow

TYP = T=25℃, Process=Nominal

MAX= T=-40℃, Process=Fast

2.4 Multiplexed Functions

The following table lists the functions selectable for each pin. Refer to Chapter 18 for GPIO function selection registers and Graphic BUS.

Pin Name	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
ADIN0	ADIN[0]	ADIN[0]	-	-	-	-
ADIN1	ADIN[1]	ADIN[1]	-	-	-	-
ADIN2	ADIN[2]	ADIN[2]	-	-	-	-
ADIN3	ADIN[3]	ADIN[3]	-	-	-	-
ADIN4	ADIN[4]	ADIN[4]	-	-	-	-
ADIN5	ADIN[5]	ADIN[5]	-	-	-	-
ADIN6	ADIN[6]	ADIN[6]	-	-	-	-
ADIN7	ADIN[7]	ADIN[7]	-	-	-	-
FC0	GPIOFC[0]	SD1D[0]	TCO[0]	CKMO[0]	-	GBUS[116]
FC1	GPIOFC[1]	SD1D[1]	TCO[1]	CKMO[1]	-	GBUS[117]
FC2	GPIOFC[2]	SD1D[2]	TCO[2]	CKMO[0]	-	GBUS[118]
FC3	GPIOFC[3]	SD1D[3]	TCO[3]	CKMO[1]	-	GBUS[119]
FC4	GPIOFC[4]	SD1CMD	TCO[4]	CKMO[0]	-	GBUS[120]
FC5	GPIOFC[5]	SD1CLK	TCO[5]	CKMO[1]	-	GBUS[121]
FC6	GPIOFC[6]	SD1DET	TCO[0]	CKMO[0]	-	GBUS[122]
FC7	GPIOFC[7]	SD1WP	TCO[1]	CKMO[1]	-	GBUS[123]
FC8	GPIOFC[8]	SD1D[4]	-	-	-	GBUS[124]
FC9	GPIOFC[9]	SD1D[5]	-	-	-	GBUS[125]
FC10	GPIOFC[10]	SD1D[6]	-	-	-	GBUS[126]
FC11	GPIOFC[11]	SD1D[7]	-	-	-	GBUS[127]
LC0	GPIOLC[0]	LVS	LXCSn[0]	-	-	-
LC1	GPIOLC[1]	LHS	LXCSn[1]	-	-	-
LC2	GPIOLC[2]	LAC	LXOEn	-	-	-
LC3	GPIOLC[3]	LCLK	LXWEn	-	-	-
LC4	GPIOLC[4]	-	LXA[0]	-	-	GBUS[114]
LC5	GPIOLC[5]	LIRQ	LXA[1]	-	-	GBUS[115]
LD0	GPIOLD[0]	LPD[0]	LXD[0]	-	-	-
LD1	GPIOLD[1]	LPD[1]	LXD[1]	-	-	-
LD2	GPIOLD[2]	LPD[2]	LXD[2]	-	-	-
LD3	GPIOLD[3]	LPD[3]	LXD[3]	-	-	-
LD4	GPIOLD[4]	LPD[4]	LXD[4]	-	-	-
LD5	GPIOLD[5]	LPD[5]	LXD[5]	-	-	-
LD6	GPIOLD[6]	LPD[6]	LXD[6]	-	-	-
LD7	GPIOLD[7]	LPD[7]	LXD[7]	-	-	-
LD8	GPIOLD[8]	LPD[8]	LXD[8]	-	-	-
LD9	GPIOLD[9]	LPD[9]	LXD[9]	-	-	-
LD10	GPIOLD[10]	LPD[10]	LXD[10]	-	-	-
LD11	GPIOLD[11]	LPD[11]	LXD[11]	-	-	-
LD12	GPIOLD[12]	LPD[12]	LXD[12]	-	-	-
LD13	GPIOLD[13]	LPD[13]	LXD[13]	-	-	-
LD14	GPIOLD[14]	LPD[14]	LXD[14]	-	-	-
LD15	GPIOLD[15]	LPD[15]	LXD[15]	-	-	-
LD16	GPIOLD[16]	LPD[16]	LXD[16]	-	-	-
LD17	GPIOLD[17]	LPD[17]	LXD[17]	-	-	-
LD18	GPIOLD[18]	LPD[18]	LXD[18]	-	-	GBUS[108]
LD19	GPIOLD[19]	LPD[19]	LXD[19]	-	-	GBUS[109]
LD20	GPIOLD[20]	LPD[20]	LXD[20]	-	-	GBUS[110]
LD21	GPIOLD[21]	LPD[21]	LXD[21]	-	-	GBUS[111]
LD22	GPIOLD[22]	LPD[22]	LXD[22]	-	-	GBUS[112]
LD23	GPIOLD[23]	LPD[23]	LXD[23]	-	-	GBUS[113]
PC0	GPIOPC[0]	CIF0PD[0]	-	-	-	GBUS[80]
PC1	GPIOPC[1]	CIF0PD[1]	-	-	-	GBUS[81]
PC2	GPIOPC[2]	CIF0PD[2]	xRSC# (for test)	xSCS0# (for test)	xSCS1# (for test)	GBUS[82]
PC3	GPIOPC[3]	CIF0PD[3]	xOE# (for test)	-	-	GBUS[83]
PC4	GPIOPC[4]	CIF0PD[4]	XA[12] (for test)	-	-	GBUS[84]
PC5	GPIOPC[5]	CIF0PD[5]	XA[17] (for test)	-	-	GBUS[85]
PC6	GPIOPC[6]	CIF0PD[6]	-	-	-	GBUS[86]
PC7	GPIOPC[7]	CIF0PD[7]	-	-	-	GBUS[87]
PC8	GPIOPC[8]	CIF0CKI	-	-	-	GBUS[88]
PC9	GPIOPC[9]	CIF0VS	-	-	-	GBUS[89]
PC10	GPIOPC[10]	CIF0HS	-	-	-	GBUS[90]
PC11	GPIOPC[11]	CIF0CKO	-	-	-	GBUS[91]
PC12	GPIOPC[12]	CIF0CFIELD	-	-	-	GBUS[92]
PC13	GPIOPC[13]	CIF0CDE	-	-	-	GBUS[93]

Pin Name	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
PC14	GPIOPC[14]	CIF0PD[8]	CIF1PD[0]	-	-	GBUS[94]
PC15	GPIOPC[15]	CIF0PD[9]	CIF1PD[1]	-	-	GBUS[95]
PC16	GPIOPC[16]	CIF0PD[10]	CIF1PD[2]	-	-	GBUS[96]
PC17	GPIOPC[17]	CIF0PD[11]	CIF1PD[3]	-	-	GBUS[97]
PC18	GPIOPC[18]	CIF0PD[12]	CIF1PD[4]	-	-	GBUS[98]
PC19	GPIOPC[19]	CIF0PD[13]	CIF1PD[5]	-	-	GBUS[99]
PC20	GPIOPC[20]	CIF0PD[14]	CIF1PD[6]	-	-	GBUS[100]
PC21	GPIOPC[21]	CIF0PD[15]	CIF1PD[7]	-	-	GBUS[101]
PC22	GPIOPC[22]	-	CIF1CKI	-	-	GBUS[102]
PC23	GPIOPC[23]	-	CIF1VS	-	-	GBUS[103]
PC24	GPIOPC[24]	-	CIF1HS	-	-	GBUS[104]
PC25	GPIOPC[25]	-	CIF1CKO	-	-	GBUS[105]
PC26	GPIOPC[26]	-	CIF1CFIELD	-	-	GBUS[106]
PC27	GPIOPC[27]	-	CIF1CDE	-	-	GBUS[107]
PD0	GPIOPD[0]	-	-	-	-	GBUS[56]
PD1	GPIOPD[1]	-	-	-	-	GBUS[57]
PD2	GPIOPD[2]	-	-	-	-	GBUS[58]
PD3	GPIOPD[3]	-	-	-	-	GBUS[59]
PD4	GPIOPD[4]	-	-	-	-	GBUS[60]
PD5	GPIOPD[5]	-	-	-	-	GBUS[61]
PD6	GPIOPD[6]	-	-	-	-	GBUS[62]
PD7	GPIOPD[7]	-	-	-	-	GBUS[63]
PD8	GPIOPD[8]	-	-	-	-	GBUS[64]
PD9	GPIOPD[9]	-	-	-	-	GBUS[65]
PD10	GPIOPD[10]	-	-	-	-	GBUS[66]
PD11	GPIOPD[11]	-	-	-	-	GBUS[67]
PD12	GPIOPD[12]	-	-	-	-	GBUS[68]
PD13	GPIOPD[13]	-	-	-	-	GBUS[69]
PD14	GPIOPD[14]	-	-	-	-	GBUS[70]
PD15	GPIOPD[15]	-	-	-	-	GBUS[71]
PD16	GPIOPD[16]	-	-	-	-	GBUS[72]
PD17	GPIOPD[17]	-	-	-	-	GBUS[73]
PD18	GPIOPD[18]	-	-	-	-	GBUS[74]
PD19	GPIOPD[19]	-	-	-	-	GBUS[75]
PD20	GPIOPD[20]	-	-	-	-	GBUS[76]
PD21	GPIOPD[21]	-	-	-	-	GBUS[77]
PD22	GPIOPD[22]	DEV_VBUS_PW R_CTL	-	-	-	GBUS[78]
PD23	GPIOPD[23]	HVBUS_PWR_ CTL	-	-	-	GBUS[79]
PK0	GPIOPK[0]	EXTINT[0]	-	-	-	-
PK1	GPIOPK[1]	EXTINT[1]	-	-	-	-
PK2	GPIOPK[2]	EXTINT[2]	-	-	-	-
PK3	GPIOPK[3]	EXTINT[3]	-	-	-	-
PK4	GPIOPK[4]	EXTINT[4]	-	-	-	GBUS[164]
PK5	GPIOPK[5]	EXTINT[5]	-	-	-	GBUS[165]
PK6	GPIOPK[6]	EXTINT[6]	-	-	-	GBUS[166]
PK7	GPIOPK[7]	EXTINT[7]	-	-	-	GBUS[167]
PK8	GPIOPK[8]	-	-	-	-	GBUS[168]
PK9	GPIOPK[9]	-	-	-	-	GBUS[169]
PK10	GPIOPK[10]	-	-	-	-	GBUS[170]
PK11	GPIOPK[11]	-	-	-	-	GBUS[171]
PK12	GPIOPK[12]	-	-	-	-	GBUS[172]
PK13	GPIOPK[13]	-	-	-	-	GBUS[173]
PK14	GPIOPK[14]	-	-	-	-	GBUS[174]
PK15	GPIOPK[15]	-	-	-	-	GBUS[175]
PK16	GPIOPK[16]	-	-	-	TCO[0]	GBUS[176]
PK17	GPIOPK[17]	-	-	-	TCO[1]	GBUS[177]
PK18	GPIOPK[18]	-	-	-	TCO[2]	GBUS[178]
PK19	GPIOPK[19]	-	-	-	TCO[3]	GBUS[179]
PK20	GPIOPK[20]	-	-	-	TCO[4]	GBUS[180]
PK21	GPIOPK[21]	-	-	-	TCO[5]	GBUS[181]
PK22	GPIOPK[22]	CAN0TXD	-	-	-	GBUS[182]
PK23	GPIOPK[23]	CAN0RXD	-	-	-	GBUS[183]
PK24	GPIOPK[24]	CAN1TXD	-	-	-	GBUS[184]
PK25	GPIOPK[25]	CAN1RXD	-	-	-	GBUS[185]
PK26	GPIOPK[26]	-	-	-	TCO[3]	GBUS[186]
PK27	GPIOPK[27]	-	-	-	TCO[4]	GBUS[187]
PK28	GPIOPK[28]	-	-	-	-	GBUS[188]
PK29	GPIOPK[29]	-	-	-	-	GBUS[189]
PK30	GPIOPK[30]	-	-	-	-	GBUS[190]

Pin Name	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
PK31	GPIOPK[31]	-	-	-	-	GBUS[191]
PK32	GPIOPK[32]/PWRCTL[0]	-	-	-	-	-
PK33	GPIOPK[33]/PWRCTL[1]	HRESETn	-	-	-	-
PK34	GPIOPK[34]	HRESETn	-	-	-	-
PS0	GPIOPS[0]	SD0D[0]	-	-	EDI_XD[0]	GBUS[0]
PS1	GPIOPS[1]	SD0D[1]	-	-	EDI_XD[1]	GBUS[1]
PS2	GPIOPS[2]	SD0D[2]	-	-	EDI_XD[2]	GBUS[2]
PS3	GPIOPS[3]	SD0D[3]	-	-	EDI_XD[3]	GBUS[3]
PS4	GPIOPS[4]	SD0CMD	-	-	FALE	GBUS[4]
PS5	GPIOPS[5]	SD0CLK	-	-	FCLE	GBUS[5]
PS6	GPIOPS[6]	SD0DET	-	-	FWEn	GBUS[6]
PS7	GPIOPS[7]	SD0WP	-	-	FOEn	GBUS[7]
PS8	GPIOPS[8]	SD0D[4]	-	-	EDI_XD[4]	GBUS[8]
PS9	GPIOPS[9]	SD0D[5]	-	-	EDI_XD[5]	GBUS[9]
PS10	GPIOPS[10]	SD0D[6]	-	-	EDI_XD[6]	GBUS[10]
PS11	GPIOPS[11]	SD0D[7]	-	-	EDI_XD[7]	GBUS[11]
PS12	GPIOPS[12]	-	-	-	FCSn[0]	GBUS[12]
PS13	GPIOPS[13]	-	-	-	FRDY[0]	GBUS[13]
PS14	GPIOPS[14]	-	-	-	-	GBUS[14]
PS15	GPIOPS[15]	-	-	-	-	GBUS[15]
PS16	GPIOPS[16]	-	-	-	-	GBUS[16]
PS17	GPIOPS[17]	-	-	-	-	GBUS[17]
PS18	GPIOPS[18]	-	-	-	-	GBUS[18]
PS19	GPIOPS[19]	-	-	-	-	GBUS[19]
PS20	GPIOPS[20]	-	-	-	-	GBUS[20]
PS21	GPIOPS[21]	-	-	-	QSPI_RST	GBUS[21]
PS22	GPIOPS[22]	-	-	-	QSPI_CS	GBUS[22]
PS23	GPIOPS[23]	-	-	-	QSPI_CLK	GBUS[23]
PS24	GPIOPS[24]	-	-	-	QSPI_IO0	GBUS[24]
PS25	GPIOPS[25]	-	-	-	QSPI_IO1	GBUS[25]
PS26	GPIOPS[26]	-	-	-	QSPI_IO2	GBUS[26]
PS27	GPIOPS[27]	-	-	-	QSPI_IO3	GBUS[27]
PS28	GPIOPS[28]	-	-	-	-	GBUS[28]
PS29	GPIOPS[29]	-	-	-	-	GBUS[29]
PS30	GPIOPS[30]	-	-	-	-	GBUS[30]
PS31	GPIOPS[31]	-	-	-	-	GBUS[31]
XA0	XA[0]	XA[0]	-	-	-	-
XA1	XA[1]	XA[1]	-	-	-	-
XA2	XA[2]	XA[2]	-	-	-	-
XA3	XA[3]	XA[3]	-	-	-	-
XA4	XA[4]	XA[4]	-	-	-	GBUS[144]
XA5	XA[5]	XA[5]	-	-	-	GBUS[145]
XA6	XA[6]	XA[6]	-	-	-	GBUS[146]
XA7	XA[7]	XA[7]	-	-	-	GBUS[147]
XA8	XA[8]	XA[8]	-	-	-	GBUS[148]
XA9	XA[9]	XA[9]	-	-	-	GBUS[149]
XA10	XA[10]	XA[10]	-	-	-	GBUS[150]
XA11	XA[11]	XA[11]	-	-	-	GBUS[151]
XA12	XA[12]	XA[12]	-	-	-	GBUS[152]
XA13	XA[13]	XA[13]	-	-	-	GBUS[153]
XA14	XA[14]	XA[14]	-	-	-	GBUS[154]
XA15	XA[15]	XA[15]	-	-	-	GBUS[155]
XA16	XA[16]	XA[16]	-	-	-	GBUS[156]
XA18	XA[18]	XA[18]	-	-	-	GBUS[158]
XA19	XA[19]	XA[19]	-	-	-	GBUS[159]
XA20	XA[20]	XA[20]	-	-	-	GBUS[160]
XA21	XA[21]	XA[21]	-	-	-	GBUS[161]
XD0	XD[0]	XD[0]	-	-	-	-
XD1	XD[1]	XD[1]	-	-	-	-
XD2	XD[2]	XD[2]	-	-	-	-
XD3	XD[3]	XD[3]	-	-	-	-
XD4	XD[4]	XD[4]	-	-	-	-
XD5	XD[5]	XD[5]	-	-	-	-
XD6	XD[6]	XD[6]	-	-	-	-
XD7	XD[7]	XD[7]	-	-	-	-
XD8	XD[8]	XD[8]	-	-	-	-
XD9	XD[9]	XD[9]	-	-	-	-
XD10	XD[10]	XD[10]	-	-	-	-
XD11	XD[11]	XD[11]	-	-	-	-
XD12	XD[12]	XD[12]	-	-	-	-
XD13	XD[13]	XD[13]	-	-	-	-

Pin Name	Reset Default	Functions Selected by GPIO Function Selection Registers				
	All FS* Bits=0	FS0=1	FS1=1	FS2=1	FS3=1	FS4=1
XD14	XD[14]	XD[14]	-	-	-	-
XD15	XD[15]	XD[15]	-	-	-	-
XD16	GPIOXD[16]	XD[16]	-	-	-	GBUS[128]
XD17	GPIOXD[17]	XD[17]	-	-	-	GBUS[129]
XD18	GPIOXD[18]	XD[18]	-	-	-	GBUS[130]
XD19	GPIOXD[19]	XD[19]	-	-	-	GBUS[131]
XD20	GPIOXD[20]	XD[20]	-	-	-	GBUS[132]
XD21	GPIOXD[21]	XD[21]	-	-	-	GBUS[133]
XD22	GPIOXD[22]	XD[22]	-	-	-	GBUS[134]
XD23	GPIOXD[23]	XD[23]	-	-	-	GBUS[135]
XD24	GPIOXD[24]	XD[24]	-	-	-	GBUS[136]
XD25	GPIOXD[25]	XD[25]	-	-	-	GBUS[137]
XD26	GPIOXD[26]	XD[26]	-	-	-	GBUS[138]
XD27	GPIOXD[27]	XD[27]	-	-	-	GBUS[139]
XD28	GPIOXD[28]	XD[28]	-	-	-	GBUS[140]
XD29	GPIOXD[29]	XD[29]	-	-	-	GBUS[141]
XD30	GPIOXD[30]	XD[30]	-	-	-	GBUS[142]
XD31	GPIOXD[31]	XD[31]	-	-	-	GBUS[143]
XDCKE	XDCKE	XDCKE	-	-	-	-
XDCLK	XDCLK	XDCLK	-	-	-	-
XDCS0#	XDCSn[0]	XDCSn[0]	-	-	-	-
XWE#	XWEn	XWEn	-	-	-	-

Note: The reset default state of XA[23:4] is output and they may be toggled during reset and boot code execution. Care must be taken if XA[23:4] is used as GPIO.

3 Package Dimensions

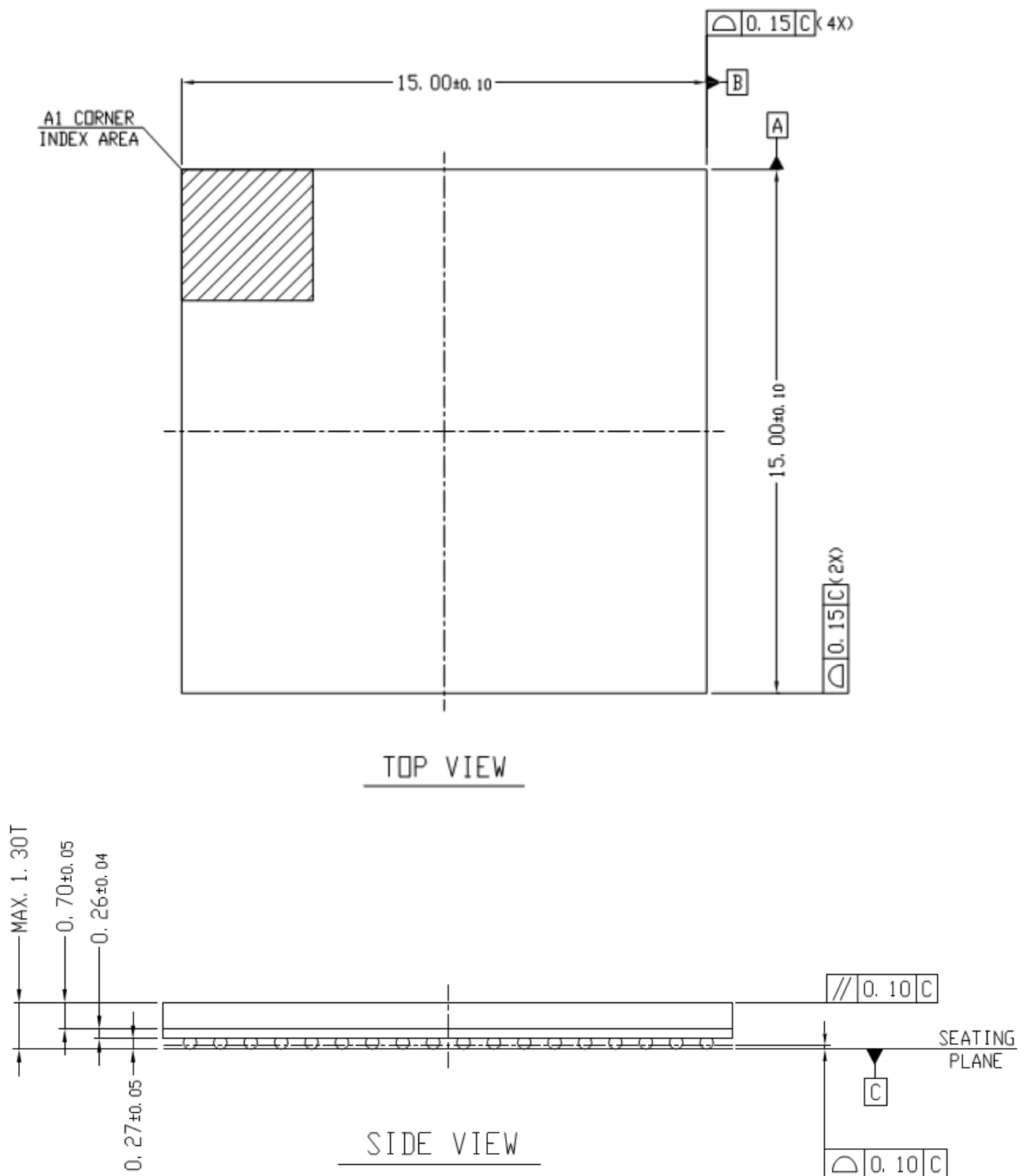


Figure 3.1 TCC8132 Package Dimension (324-BGA-1818)

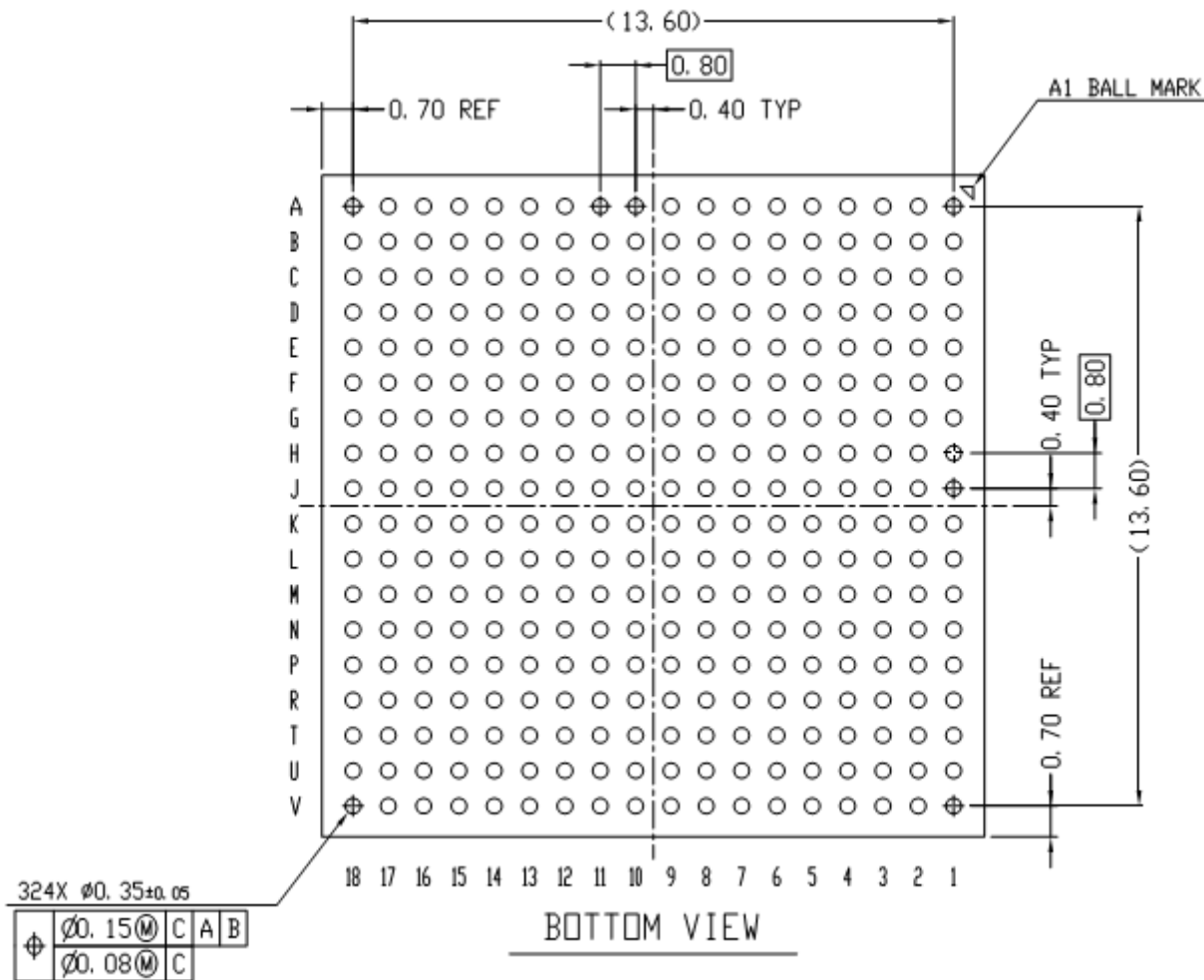


Figure 3.2 TCC8132 Package Dimension (Bottom View)

4 Electrical Specification

4.1 Absolute Maximum Ratings

Table 4.1 Absolute Maximum Ratings

Parameter	Symbol	Rating	Unit
DC Supply Voltage for I/O (VDD_P*,L*,FS)	V _{DDIO}	4.6	V
DC Supply Voltage for Backup Power (VDD_PK)	V _{DDK}	4.6	V
DC Supply Voltage for Internal Logic (VDDI, VDDD* USB11)	V _{DDI}	1.8	V
DC Supply Voltage for PLL (VDD_PLL*)	V _{DDPLL}	1.8	V
DC Supply Voltage for ADC (VDD_ADC)	V _{DDADC}	4.6	V
DC Supply Voltage for USB (VDD* USB33)	V _{DDUSB}	4.6	V
DC Supply Voltage for OSC (XI/XO)	V _{DDOSC}	4.6	V
DC Supply Voltage for RTC (XTI/XTO)	V _{DDRTC}	4.6	V
Digital Input Voltage for Input Buffer	V _{IN}	V _{DDIO} + 0.3	V
Digital Output Voltage for Output Buffer	V _{OUT}	V _{DDIO} + 0.3	V
Digital Input Voltage for USB20DH_VBUS and USB20H_VBUS	V _{USB_VBUS}	6.0	V
Analog Input Voltage for ADC	V _{INA}	0 ~ V _{DDADC}	V
DC Input Current	I _{IN}	± 20	mA
Storage Temperature	T _{STG}	-55 ~ 150	°C

Note:

1. Absolute maximum ratings specify the values beyond which the device may be damaged permanently. Exposure to absolute maximum rating conditions for extended periods may affect reliability. Each condition value is applied with the other values kept within the following operating conditions and functional operation under any of these conditions is not implied.
2. All voltages are measured with respect to VSS unless otherwise specified.
3. V_{DDI} must always be less than or equal to V_{DDIO}
4. The voltage difference between analog and digital grounds must always be within 0.3V.

4.2 Recommended Operating Conditions

Table 4.2 Recommended Operating Conditions

Parameter		Symbol	MIN	TYP	MAX	Unit
DC Supply Voltage for Internal Logic (VDDI, VDD12USB*, VDD11_OTP)	F _{CPU} = 400MHz, F _{BUS} = 166MHz	V _{DDI}	1.045	1.1	1.155	V
DC Supply Voltage for PLL (VDD_PLL*)		V _{DDPLL}	V _{DDI}	V _{DDI}	V _{DDI}	V
DC Supply Voltage for I/O (VDD_P*,L*,FS)		V _{DDIO}	1.65	3.3	3.6	V
DC Supply Voltage for Backup Power (VDD_PK)		V _{DDK}	1.65	3.3	3.6	V
DC Supply Voltage for ADC (VDD_ADC)		V _{DDADC}	3.0	3.3	3.6	V
DC Supply Voltage for USB (VDD* USB33)		V _{DDUSB}	3.135	3.3	3.465	V
DC Supply Voltage for OSC (XI/XO)		V _{DDOSC}	3.135	3.3	3.465	V
DC Supply Voltage for RTC (XTI/XTO)		V _{DDRTC}	1.65	3.3	3.6	V
DC Supply Voltage for OTP (VDD_OTP25)		V _{DDOTP}	2.38	2.5	2.62	V
Digital Input Voltage for Input Buffer		V _{IN}	-0.1	-	V _{DDIO} +0.15	V
Digital Output Voltage		V _{OUT}	-0.1	-	V _{DDIO} +0.15	V
Analog Input Voltage for ADC		V _{INA}	0.0	-	V _{DDADC}	V
Operating Temperature		T _A	-40	-	85	°C

Note: F_{BUS} is limited by the characteristics of SDRAM and the loading on the external bus.

The following table shows the speed characteristics of the CKC (Clock Controller) dividers and each peripheral. All the Clock Controller registers must be programmed to operate within the limit specified.

Table 4.3 Maximum Frequency Specification for CKC Dividers

Peripheral Clock Generator / Clock Divider	Peripherals	Maximum frequency of the Clock Divider Input	Maximum frequency of the Clock Output to each Peripheral
		VDDI=1.10V	VDDI=1.10V
ACLKDAI0 ~ 2	I2S Interface 0 ~ 2	840MHz	Dependent on the external component
ACLKREF	SDRAM Controller	840MHz	Dependent on the SDRAM refresh rate
ACLKI2C	I2C Controller	840MHz	(HCLK frequency) / 4
ACLKUART0 ~ 5	UART Interface	840MHz	100MHz
ACLKTCT/X/Z	Timer/Counters	840MHz	200MHz
ACLKADC	ADC Interface	840MHz	66MHz
ACLKLCD*	LCD Controller	840MHz	125MHz
ACLKSPDIF	SPDIF Tx/Rx Controller	840MHz	40MHz
ACLKSDH0 ~ 1	SD/MMC Host	840MHz	208MHz
ACLKEXT	External Output Pin	840MHz	Dependent on the external component
ACLKCAN0 ~ 1	CAN Controller	840MHz	66MHz
ACLKGSB0 ~ 5	GSB Master	840MHz	100MHz
ACLKPDM	PDM	840MHz	300MHz
ACLKCIF0 ~ 1	Camera Interface 0 ~ 1	840MHz	Dependent on the external component
PLL0~3 Divider	Clock Generators	1200MHz	-
SYS CLK Divider	CPU	1200MHz	≤ Maximum F _{CPU}

4.3 Electrical Characteristics for General I/O

Table 4.4 DC Electrical Characteristics for General I/O

Parameter	Symbol	Test Conditions	MIN	TYP	MAX	Unit
High Level Input Voltage	V _{IH}		0.7V _{DDIO}		V _{DDIO} +0.0	V
Low Level Input Voltage	V _{IL}		0.0		0.30V _{DDIO}	V
Hysteresis Voltage	ΔV		0.1V _{DDIO}			V
High Level Input Current	I _{IH}	V _{DDIO} = 3.3V, pull-down disabled	-10		10	μA
		V _{DDIO} = 3.3V, pull-down enabled	79		260	μA
		V _{DDIO} = 2.5V, pull-down enabled	46		131	μA
		V _{DDIO} = 1.8V, pull-down enabled	21		82	μA
Low Level Input Current	I _{IL}	V _{DDIO} = 3.3V, pull-up disabled	-10		10	μA
		V _{DDIO} = 3.3V, pull-up enabled	-220		-70	μA
		V _{DDIO} = 2.5V, pull-up enabled	-112		-43	μA
		V _{DDIO} = 1.8V, pull-up enabled	-76		-10	μA
High Level Output Voltage	V _{OH}	I _{OH} = -2mA I _{OH} = -4mA I _{OH} = -8mA I _{OH} = -12mA	0.8V _{DDIO}			V
Low Level Output Voltage	V _{OL}	I _{OL} = +2mA I _{OL} = +4mA I _{OL} = +8mA I _{OL} = +12mA			0.2 V _{DDIO}	V
High Level Output Voltage (GPIO Port)	V _{OH}	I _{OH} = -2mA I _{OH} = -4mA I _{OH} = -12mA I _{OH} = -20mA	0.8V _{DDIO}			V
Low Level Output Voltage (GPIO Port)	V _{OL}	I _{OL} = +2mA I _{OL} = +4mA I _{OL} = +12mA I _{OL} = +20mA			0.2 V _{DDIO}	V
Tri-state Output Leakage Current	I _{OZ}	V _{OUT} = V _{SS} or V _{DDIO}	-10		10	μA
XI/XTI High Level Input Voltage	V _{IH_osc}	V _{DDOSC} = V _{DDRTC} = 3.3V ± 0.3V	0.7V _{DDOSC}			V
XI/XTI Low Level Input Voltage	V _{IL_osc}	V _{DDOSC} = V _{DDRTC} = 3.3V ± 0.3V			0.3V _{DDOSC}	V
XI/XO Frequency	F _{OSC1}	V _{DDOSC} = 3.3V ± 0.3V		12		MHz
XTI/XTO Frequency	F _{OSC2}	V _{DDRTC} = 3.3V ± 0.3V		32.768		kHz
		V _{DDRTC} = 3.3V ± 0.3V		4.1943		MHz
Backup Power Static Current	I _{DS_K}	V _{DDK} = 3.3V, No load		40	40	μA
		V _{IN} = V _{SS} or V _{DDK}		260	260	μA
RTC Static Current	I _{DS_RTC}	V _{DDRTC} = 3.3V, No load		10	10	μA
		V _{IN} = V _{DDRTC}		60	60	μA
RTC Dynamic Current @32.768kHz	I _{DD_RTC}	V _{DDRTC} = 3.3V, C _L = 30pF V _{IN} = V _{SS} or V _{DDRTC}		20	20	μA
				20	20	μA
				70	70	μA
RTC Dynamic Current @4.1943MHz	I _{DD_RTC1}			690	690	μA
				650	650	μA
				660	660	μA

Note:

1. $T_A = -40 \sim 85^\circ C$, $V_{DDIO} = V_{DDOSC} = V_{DDADC} = 3.3V \pm 0.3V$, $V_{DDI} = V_{DDPLL} = 1.1V$, $V_{SS} = V_{SSPLL} = V_{SSADC} = 0.0V$ unless otherwise specified.
2. The cycle unit of PLL Lock Time is "1 cycle = $1/F_{ref} = 1(F_{in}/p)$

Note that not all parameters are tested. Guaranteed by design characterization.

4.4 Electrical Characteristics for Main Oscillator

Table 4.5 AC Electrical Characteristics for Main Oscillator

Parameter	Symbol	Test Conditions	MIN	TYP	MAX	Unit
Oscillator Frequency	F_R	PWROSC=3.3V	—	12		MHz
Frequency Tolerance	F_{TOL}		-50		50	ppm
Duty Cycle	D_C		40		60	%
Jitter	J_{CLKI}	Peak-to-Peak Jitter			50	ps
		RMS Jitter			3.5	ps

4.5 Electrical Characteristics for PLL

Table 4.6 AC Electrical Characteristics for PLL

Parameter	Symbol	Test Conditions	MIN	TYP	MAX	Unit
Power Down Current	I_{PD}	$V_{DDPLL} = 1.1V$			120	μA
Current Consumption	I_{DD}	$V_{DDPLL} = 1.1V$			2	mA

Table 4.7 DC Electrical Characteristics for PLL

Parameter	Symbol	Test Conditions		MIN	TYP	MAX	Unit
PLL Input Frequency	F_{IN_PLL}	$V_{DDPLL} = V_{DDI} = 1.1V$		10		300	MHz
PLL0,1,3 Output Frequency	F_{OUT_PLL}			32		2000	MHz
PLL2 Output Frequency	F_{OUT_PLL2}			11		660	MHz
PLL0,1,3 VCO Frequency	F_{VCO_PLL}			1000		2000	MHz
PLL2 VCO Frequency	F_{VCO_PLL2}			330		660	MHz
PLL Locking Time (PLL0,1,3)	T_{LT}					400	Cycle
PLL Locking Time (PLL2)	T_{LT}					1500	Cycle
PLL Dynamic Current (PLL0,1,3)	I_{DD_PLL}	$V_{DDPLL} = 1.1V$ $T_A = 25^\circ C$	All PLLs running			2	mA
PLL Power Down Current (PLL0,1,3)	I_{PD_PLL}		All PLLs stopped			120	μA
PLL Dynamic Current (PLL2)	I_{DD_PLL}	$V_{DDPLL} = 1.1V$ $T_A = 25^\circ C$	All PLLs running			3	mA
PLL Power Down Current(PLL2)	I_{PD_PLL}		All PLLs stopped			300	μA

4.6 Electrical Characteristics for ADC

Table 4.8 DC Electrical Characteristics for ADC

Parameter	Symbol	Test Conditions	MIN	TYP	MAX	Unit
Resolution	Bit		-	-	12	Bits
Differential Non-Linearity	DNL	$V_{REF}=3.3V, GND=0V$ $f_{STC} = 0.5MHz, f_{CKIN} = 2.5MHz$	-	± 0.5	± 1	LSB
Integral Non-Linearity	INL		-	± 2	± 4	LSB
Offset Voltage	TOPOFF BOTOFF		-	± 50	± 85	LSB

(Converter Specifications: $V_{DDADC}=3.3V, V_{SSADC}=0V$, Operating Ambient Temperature/ $T_A=25^\circ C$, $V_{REF}=3.3V, GND=0.0V$ unless otherwise specified)

Table 4.9 AC Electrical Characteristics for ADC

Parameter	Symbol	Test Conditions	MIN	TYP	MAX	Unit
Maximum conversion rate	f_c	$f_{CKIN} = 5MHz$	—	—	1	MSPS
Standby supply current	—	STBY = VDD	—	10	50	μA
Dynamic supply current	I_{VDD}	$f_{CKIN} = 5MHz$ (without system load)	—	4.5	8	mA
Power-Down Off Time	I_{PD_OFF}	STBY : High-to-Low		0.5	1	μS
Signal-to-noise & distortion ratio	SNDR	$f_{CKIN} = 5MHz$ AINT = 50kHz	54	-	—	dB

(Converter Specifications: $V_{DDADC}=3.3V, V_{SSADC}=0V$, Operating Ambient Temperature/ $T_A=25^\circ C$, $V_{REF}=3.3V, GND=0.0V$ unless otherwise specified, $V_{DDIO}=3.3V \pm 0.3V$)

4.7 Electrical Characteristics for LCD interface

The following figure shows the timing diagram for TFT-LCD with RGB interface. All the timing parameters can be configured by LHTIME1, LHTIME2, LVTIME1 ~ 4 registers.

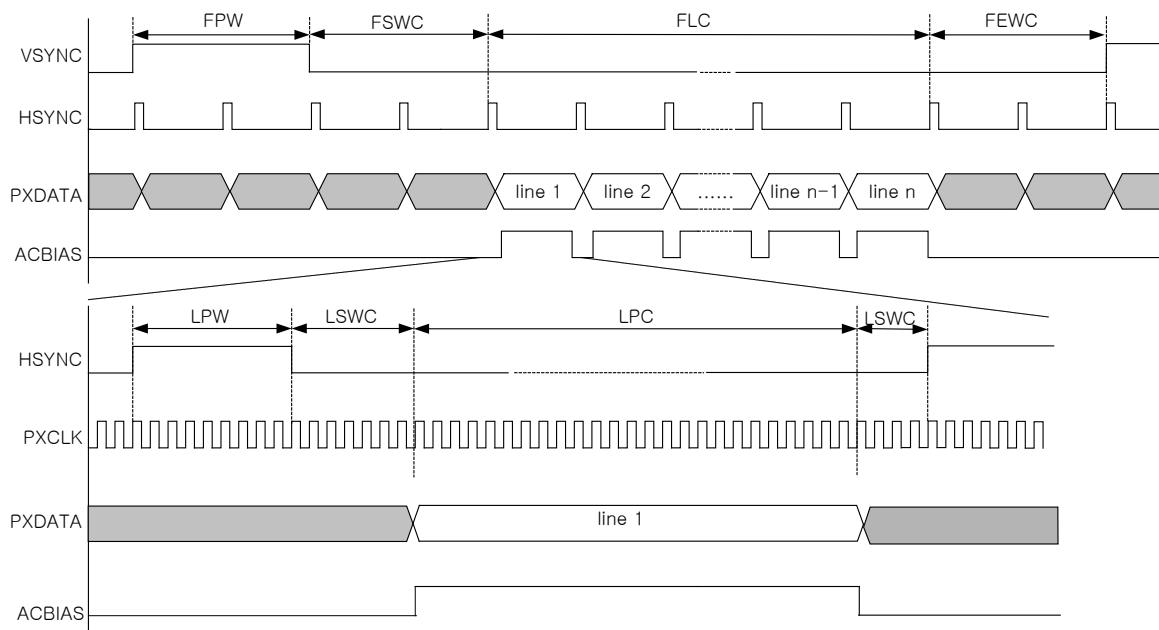


Figure 4.1 Timing Diagram for LCD Controller

The LHS (HSYNC), LVS (VSYNC), LBIAS (ACBIAS, Data Enable), and LPD[17:0] (PXDATA[17:0]) signals are referenced by LCK (PXCLK). Each minimum and maximum timing for the output delay are shown in the following figure.

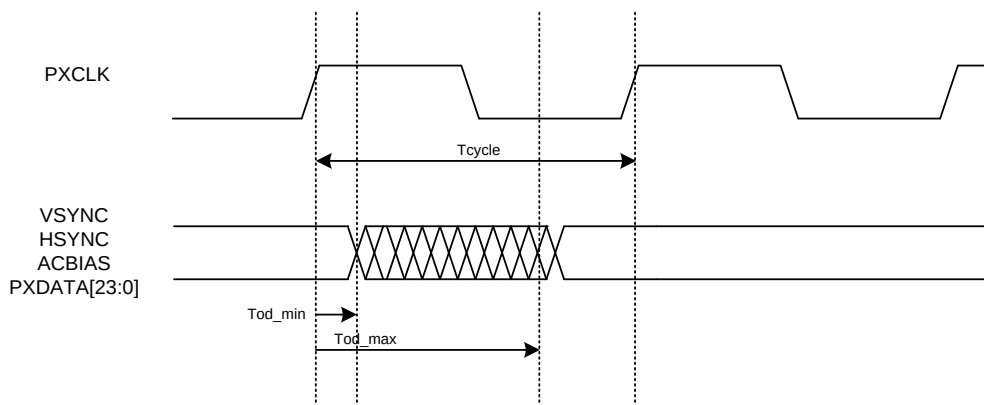


Figure 4.2 Timing Diagram Data Output Referenced to PXCLK

Table 4.10 Timing Parameters for Each Symbol

Parameter	Symbol	MIN	MAX	Unit	Remark
Clock Cycle	T_{CYCLE}	10	-	Ns	
Output Delay	T_{OD}	0	7	Ns	

Table 4.11 I/O Function Name for Corresponding Signal Name

Signal Name	I/O Function Name
VSYNC	L0_LVS
HSYNC	L0_LHS
ACBIAS	L0_LDE
PXDATA[23:0]	L0_LPD[23:0],
PXCLK	L0_LCK

The following figure shows the timing diagram of bus interface to CPU interface LCD device. The reference clock is used internally and the cycle time is defined as the register value written by software.

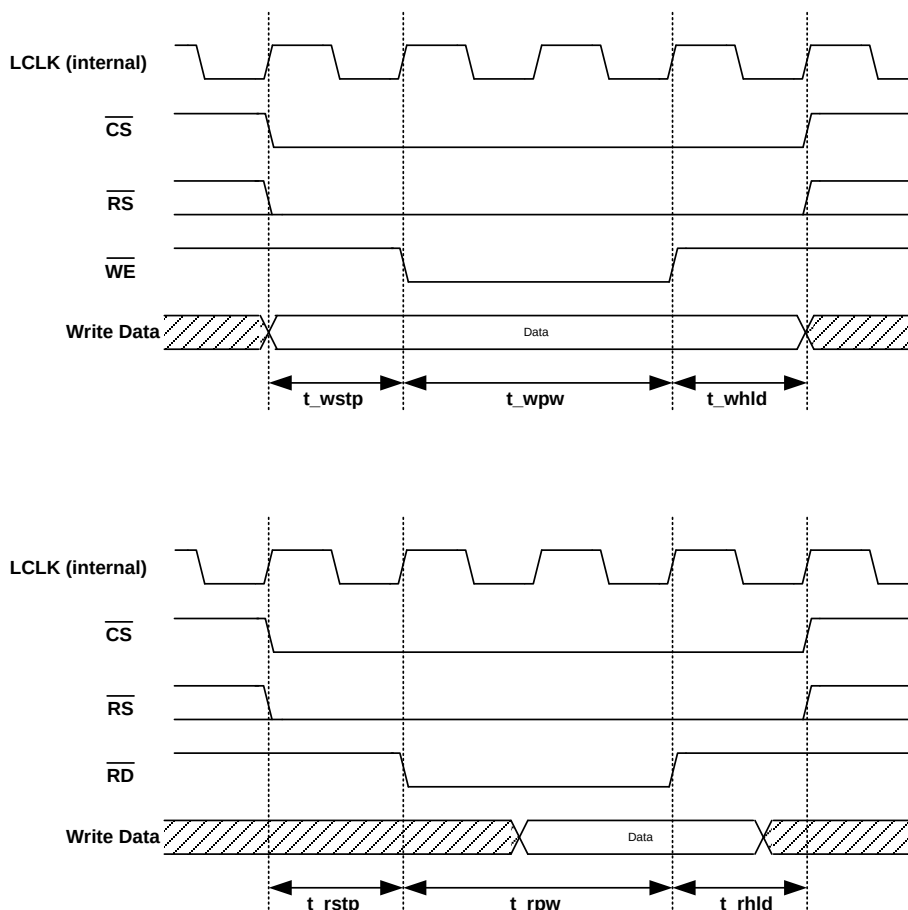


Figure 4.3 Timing Diagram Data Output Referenced to LCDSI

Table 4.12 Timing Parameters for Each Symbols

Parameter	Symbol	MIN	MAX	Unit	Remark
LCLK Clock Period (LCDSI Clock)	tCLK	18		ns	
RD Setup Time Referenced to LCLK	T_rstp	0 * tCLK	7 * tCLK	ns	
RD Pulse Width Referenced to LCLK	T_rpw	1 * tCLK	256 * tCLK	ns	
RD Hold Time Referenced to LCLK	T_rhld	0 * tCLK	7 * tCLK	ns	
WE Setup Time Referenced to LCLK	T_wstp	0 * tCLK	7 * tCLK	ns	
WE Pulse Width Referenced to LCLK	T_wpw	1 * tCLK	256 * tCLK	ns	
WE Hold Time Referenced to LCLK	T_whld	0 * tCLK	7 * tCLK	ns	

Table 4.13 I/O Function Name for Corresponding Signal Name

Signal Name	I/O Function Name
#CS	LCSN0, LCSN1
#RS	LXA[0]
#RD	LOEN
#WE	LWEN
Write Data[17:0]	LXD[17:0]

4.8 Electrical Characteristics for Camera Interface

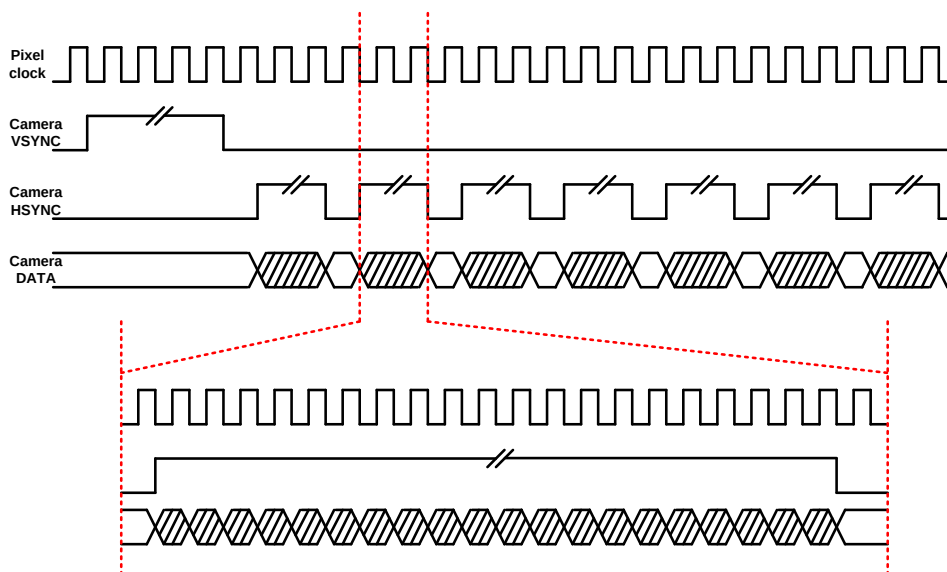


Figure 4.4 Timing Diagram for Camera Interface

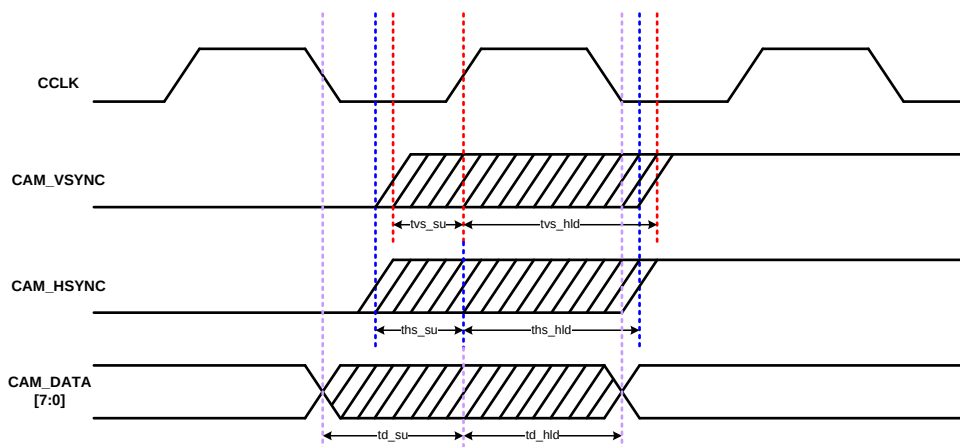


Figure 4.5 Timing Diagram Data Output Referenced to CCLK

Table 4.14 Timing Parameters for Each Symbol

Parameter	Symbol	MIN	MAX	Unit	Remark
Clock Frequency	TCLK		125	MHz	
Setup time for CVS(CAM_VSYNC)	Tvs_su	2		ns	
Hold time for CVS(CAM_VSYNC)	Tvs_hld	2		ns	
Setup time for CHS(CAM_HSYNC)	Ths_su	2		ns	
Hold time for CHS(CAM_HSYNC)	Ths_hld	2		ns	
Setup time for CPD[7:0](CAM_DATA)	Td_su	2		ns	
Hold time for CPD[7:0](CAM_DATA)	Td_hld	2		ns	

Table 4.15 I/O Function Name for Corresponding Signal Name

Signal Name	I/O Function Name
CCLK	CCKI
CAM_VSYNC	CVS
CAM_HSYNC	CHS
CAM_DATA[7:0]	CPD[7:0]

4.9 Electrical Characteristics for I2C Controller

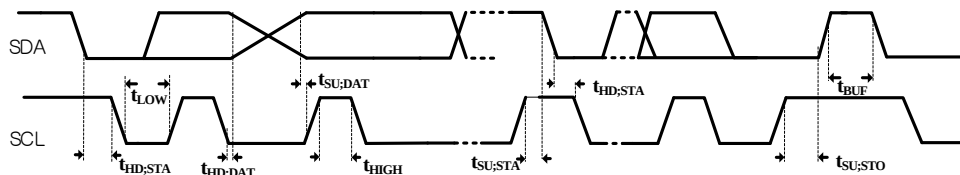


Figure 4.6 Timing Diagram for I2C Controller

Table 4.16 Timing Parameters for Each Symbol (Master)

Parameter	Symbol	STANDARD-MODE		FAST-MODE		Unit
		MIN	MAX	MIN	MAX	
SCL clock frequency			100		400	kHz
Hold time(repeated) START condition After this period, the first clock pulse is generated	$t_{HD,STA}$	4.0		0.6		μs
LOW period of the SCL clock	t_{LOW}	4.7		1.3		μs
HIGH period of the SCL clock	t_{HIGH}	4.0		0.6		μs
Set-up time for a repeated START condition *	$t_{SU,STA}$	2.0		0.5		μs
Data hold time	$t_{HD,DAT}$	20		20		ns
Data set-up time	$t_{SU,DAT}$	250		100		ns
Set-up time for STOP condition	$t_{SU,STO}$	4.0		1		μs
Bus free time between a STOP and START condition	t_{BUF}	4.7		1.3		μs

* Parameter $t_{SU,STA}$ is not used in TCC8132 I2C Master.

If some devices may not work with STANDARD or FAST MODE, it is recommended to follow the directions below.

1. Change Clock (ACLK12C DIV) under 40kHz.
2. Write START command.
3. Read TIP of Status Register transfer.
4. Return to the desired clock frequency and complete the transfer.

Table 4.17 Timing Parameters for Each Symbol (Slave)

Parameter	Symbol	STANDARD-MODE		FAST-MODE		Unit
		MIN	MAX	MIN	MAX	
SCL clock frequency			100		400	kHz
Hold time(repeated) START condition After this period, the first clock pulse is generated	$t_{HD,STA}$	4.0		0.6		μs
LOW period of the SCL clock	t_{LOW}	4.7		1.3		μs
HIGH period of the SCL clock	t_{HIGH}	4.0		0.6		μs
Set-up time for a repeated START condition	$t_{SU,STA}$	4.7		0.6		μs
Data hold time	$t_{HD,DAT}$	20		20		ns
Data set-up time	$t_{SU,DAT}$	250		100		ns
Set-up time for STOP condition	$t_{SU,STO}$	4.0		0.6		μs
Bus free time between STOP and START condition	t_{BUF}	4.7		1.3		μs

4.10 Electrical Characteristics for SPDIF Transmitter

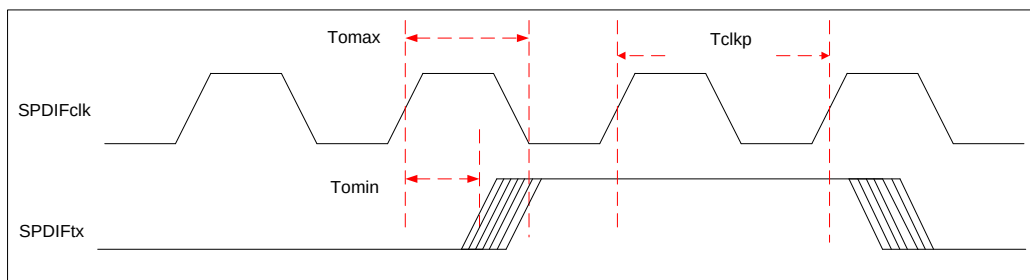


Figure 4.7 Timing Diagram for SPDIF Transmitter

Table 4.18 Timing Parameters for Each Symbol

Parameter	Symbol	MIN	MAX	Unit	Remark
SPDIFclk Clock Cycle Time	Tclkp	110		ns	
SPDIFclk Data Output Time Referenced to SPDIFclk	Tomin/Tomax	1	10	ns	CL = 50pF

4.11 Electrical Characteristics for DAI

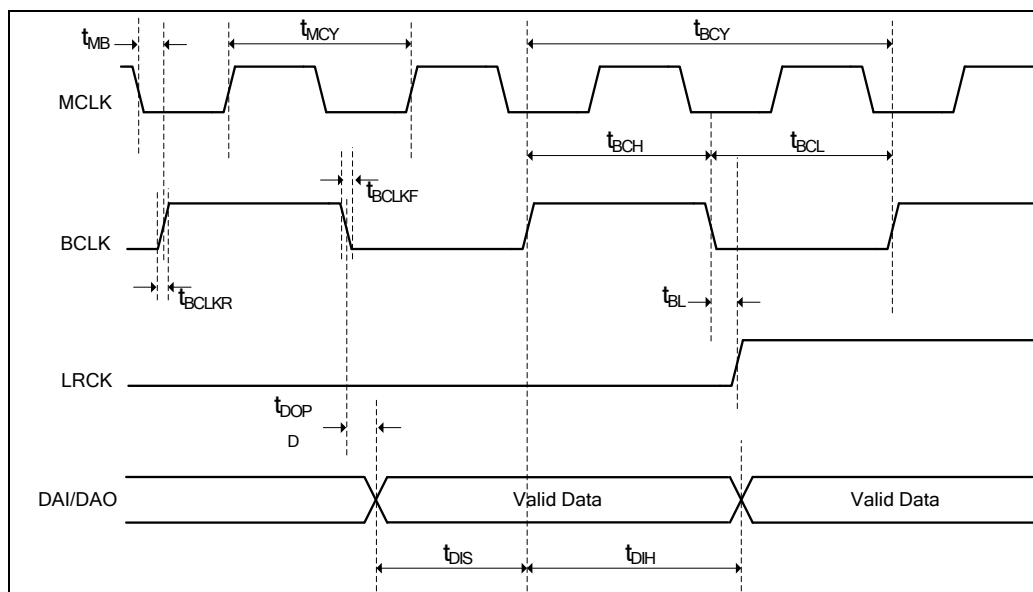


Figure 4.8 Timing Diagram for DAI

Test Condition:

MODE = I2S, fs = 48kHz, MCLK = 256fs, BCLK = 64fs, (Polarity: DAMR[16] = '0', DAMR[3] = '0')

Table 4.19 Timing for DAI

Parameter	Symbol	MIN	TYP	MAX	Unit	Remark
MCLK cycle time	tMCY	19.40	81.40		ns	
BCLK cycle time	tBCY	4 * tMCY	4 * tMCY		ns	
BCLK pulse width high	tBCH	39	163		ns	
BCLK pulse width low	tBCL	38	162		ns	
BCLK / CBCLK rise time	tBCLKR			10	ns	
BCLK/ CBCLK fall time	tBCLKF			10	ns	
MCLK to BCLK	tMB	-3	3		ns	
BCLK to LRCK	tBL	-3	3		ns	
DAI setup time to BCLK rising edge	tDIS	1	1		ns	
DAI hold time from BCLK rising edge	tDIH	1	1		ns	
DAO Output Timing Referenced to BCLK	tDOPD			1	ns	

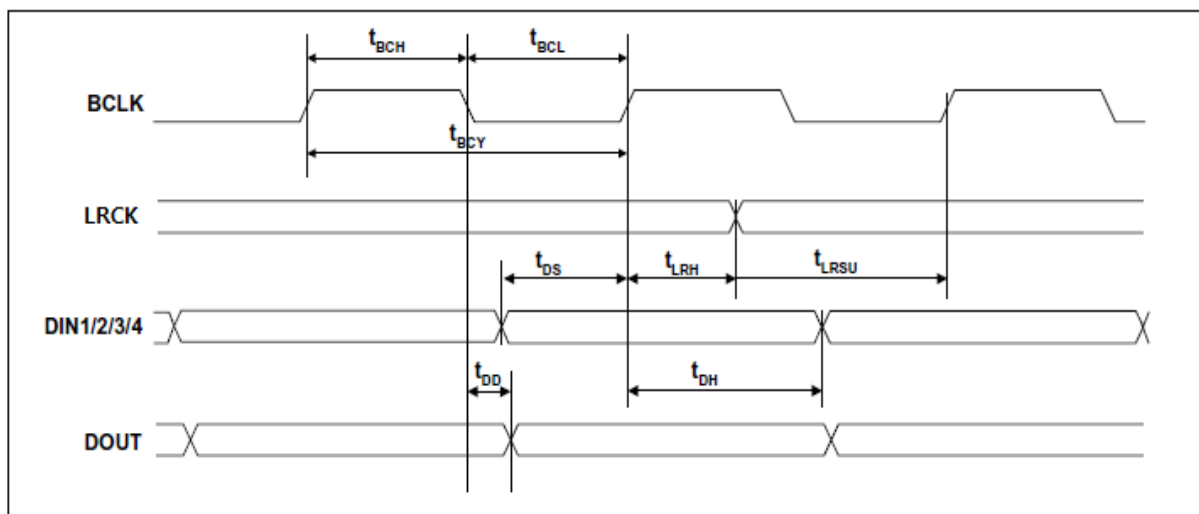


Figure 4.9 Slave Timing Diagram for DAI / CDIF

Table 4.20 Slave Timing for DAI / CDIF

Parameter	Symbol	MIN	MAX	Unit	Remark
BCLK cycle time	tBCY	50		ns	
BCLK pulse width high	tBCH	-		ns	Audio uses BCLK rising edge only
BCLK pulse width low	tBCL	-		ns	Audio uses BCLK rising edge only
LRCK set-up time to BCLK rising edge	tLRSU	0.3		ns	
LRCK hold time from BCLK rising edge	tLRH	0		ns	
DIN set-up time to BCLK rising edge	tDS	2.4		ns	
DIN hold time from BCLK rising edge	tDH	0		ns	
DOUT propagation delay from BCLK falling edge	tDD	0	9.98	ns	

4.12 Electrical Characteristics for NAND Flash Controller

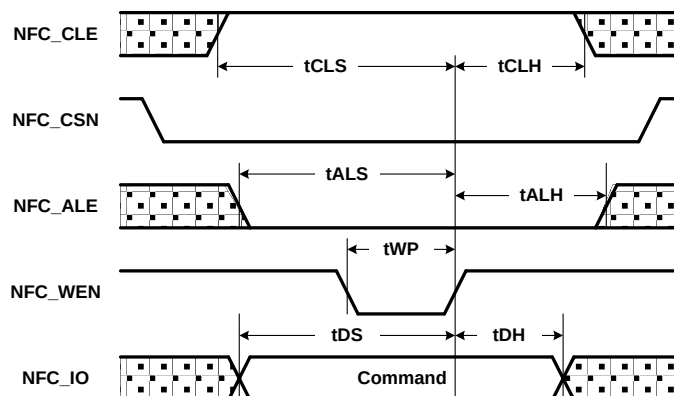


Figure 4.10 Timing Diagram for Command Latch Enable Cycle

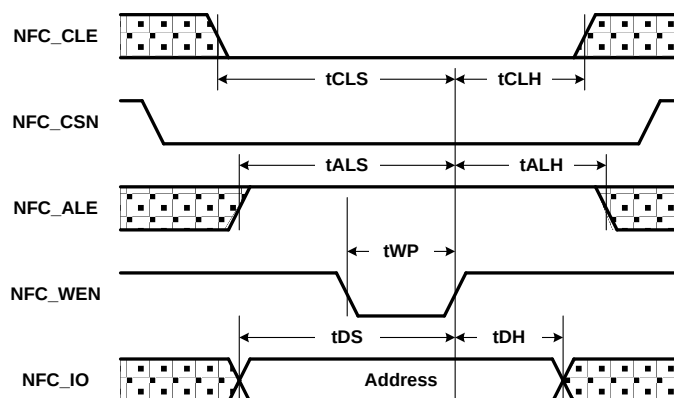


Figure 4.11 Timing Diagram for Single Address Latch Cycle

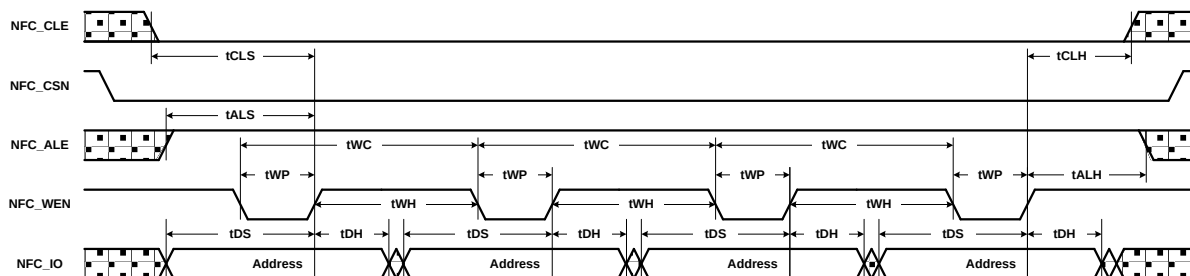


Figure 4.12 Timing Diagram for Linear Address Latch Cycle

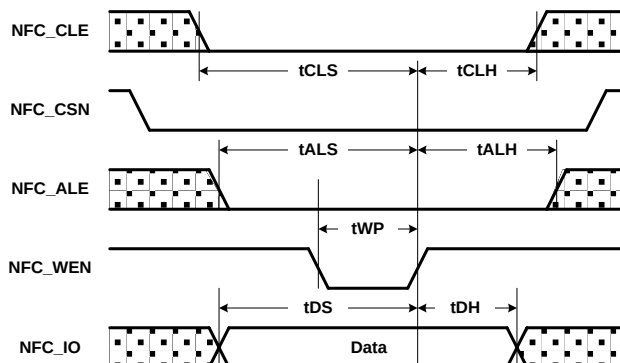


Figure 4.13 Timing Diagram for Single Data Write Cycle

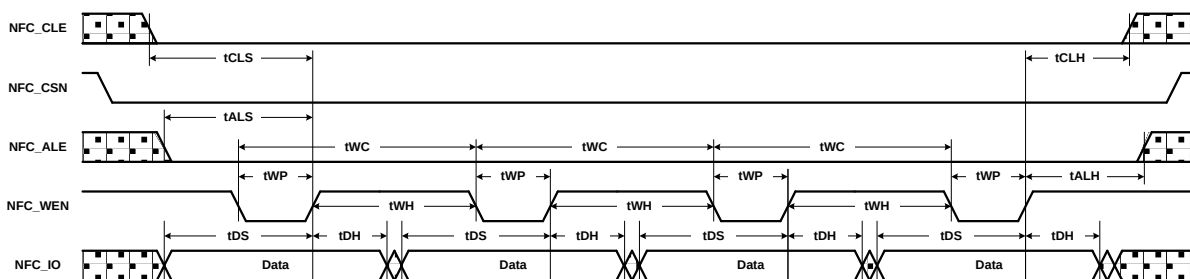


Figure 4.14 Timing Diagram for Linear Data Write Cycle

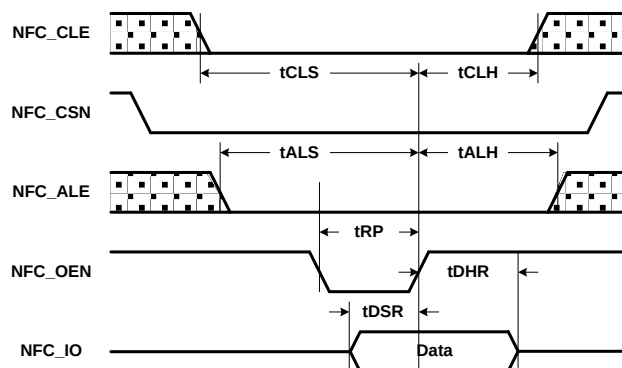


Figure 4.15 Timing Diagram for Single Data Read Cycle

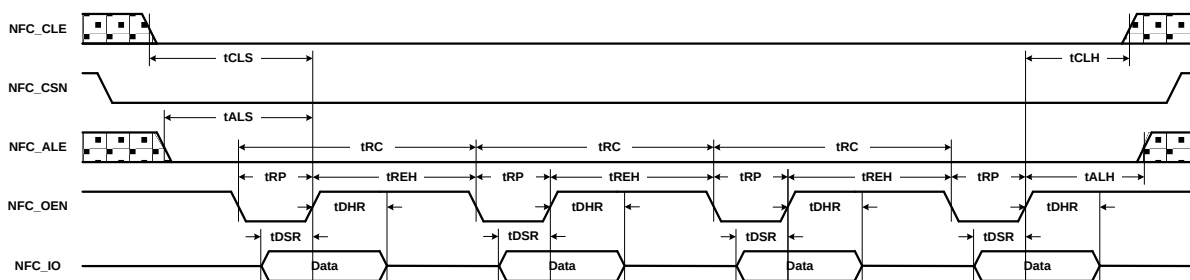


Figure 4.16 Timing Diagram for Linear Data Read Cycle

Table 4.21 Timing Parameters for Each Symbol

Parameter	Symbol	MIN	MAX	Unit
Clock Period	tHCLK	5.11		ns
CLE Set-Up Time	tCLS	$(STP + PW) \times tHCLK + 1.0$	$(STP + PW) \times tHCLK + 2$	ns
CLE Hold Time	tCLH	$HLD \times tHCLK - 2.0$	$HLD \times tHCLK - 1.0$	ns
WEN Pulse Width	tWP	$PW \times tHCLK$	$PW \times tHCLK$	ns
WEN High Hold Time	tWH	$(STP + HLD) \times tHCLK$	$(STP + HLD) \times tHCLK$	ns
Write Cycle Time	tWC	$(STP + PW + HLD) \times tHCLK$	$(STP + PW + HLD) \times tHCLK$	ns
OEN Pulse Width	tRP	$PW \times tHCLK$	$PW \times tHCLK$	ns
OEN High Hold Time	tREH	$(STP + HLD) \times tHCLK$	$(STP + HLD) \times tHCLK$	ns
Read Cycle Time	tRC	$(STP + PW + HLD) \times tHCLK$	$(STP + PW + HLD) \times tHCLK$	ns
ALE Set-Up Time	tALS	$(STP + PW) \times tHCLK - 1.00$	$(STP + PW) \times tHCLK + 2.00$	ns
ALE Hold Time	tALH	$HLD \times tHCLK - 2.00$	$HLD \times tHCLK + 1.00$	ns
Data Set-Up Time	tDS	$(STP + PW) \times tHCLK - 7.00$	$(STP + PW) \times tHCLK - 1.00$	ns
Data Hold Time	tDH	$HLD \times tHCLK - 1.00$	$HLD \times tHCLK + 1.00$	ns
Data Set-Up Time in READ	tDSR	5.00	15.0	ns
Data Hold Time in READ	tDHR	0	0	ns

Note: NFC Clock source, tHCLK is same as IOBUS Clock.

Table 4.22 I/O Function Name for Corresponding Signal Name

Signal Name	I/O Function Name
NFC_CSN	NAND_CSN0, NAND_CSN1
NFC_ALE	NAND_ALE
NFC_CLE	NAND_CLE
NFC_OEN	NAND_OEN
NFC_WEN	NAND_WEN
NFC_IO[15:0]	NANDXD[15:0]

4.13 Electrical Characteristics for UART Controller

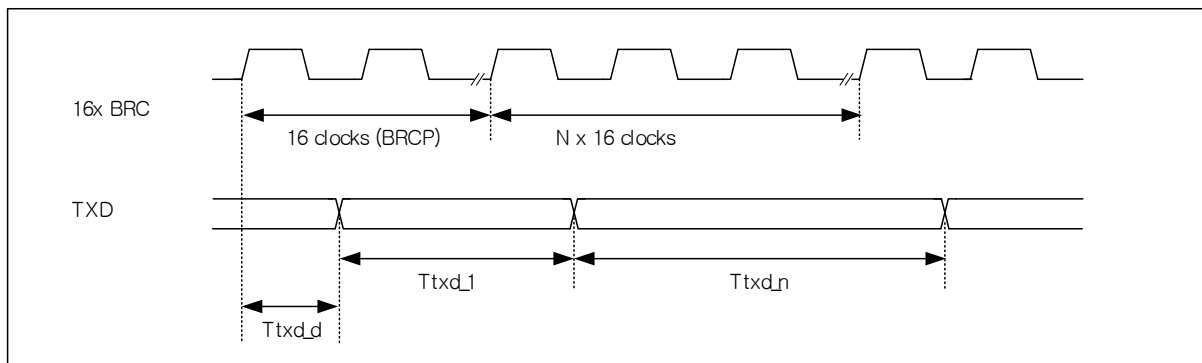


Figure 4.17 Timing Diagram for TXD

Table 4.23 Timing Parameters for Each Symbol

Parameter	Symbol	MIN	MAX	Unit	Remark
Pulse duration of 1bit TXD	Ttxd_1	BRCP -BRCP/28	BRCP +BRCP/28	ns	3.3V
Pulse duration of nbit TXD	Ttxd_n	N (BRCP) -BRCP/28	N(BRCP) +BRCP/28	ns	3.3V
TXD output delay time	Ttxd_d	BRCP/28	BRCP/28	ns	3.3V

- BRC : Baud-Rate Clock
- BRCP : Baud-Rate Clock Period
- CL : 71pF

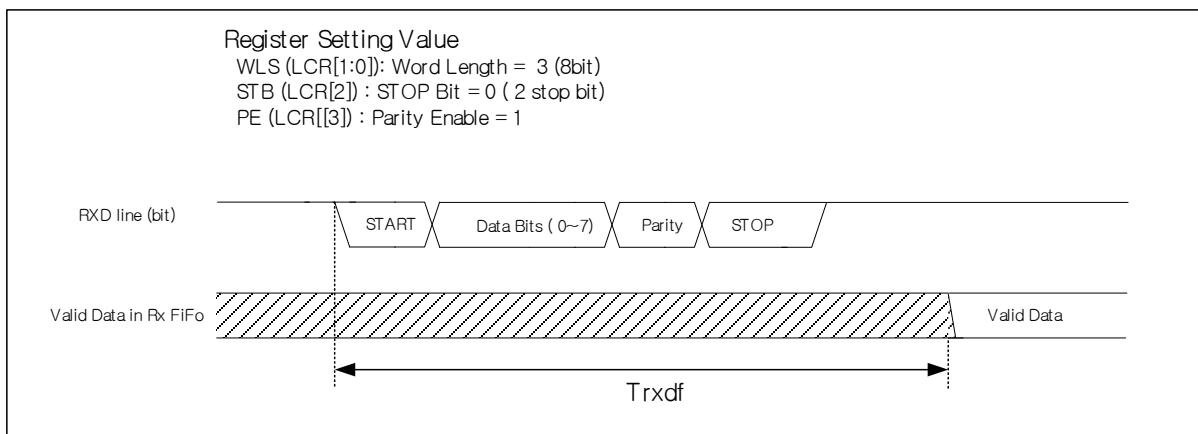


Figure 4.18 Timing Diagram for RXD

Table 4.24 Timing Parameters for Each Symbol

Parameter	Symbol	MIN	MAX	Unit	Remark
RXD Start to Rx FiFo.	Trxdf	10.5 x BRCP	11 x BRCP	ns	3.3V

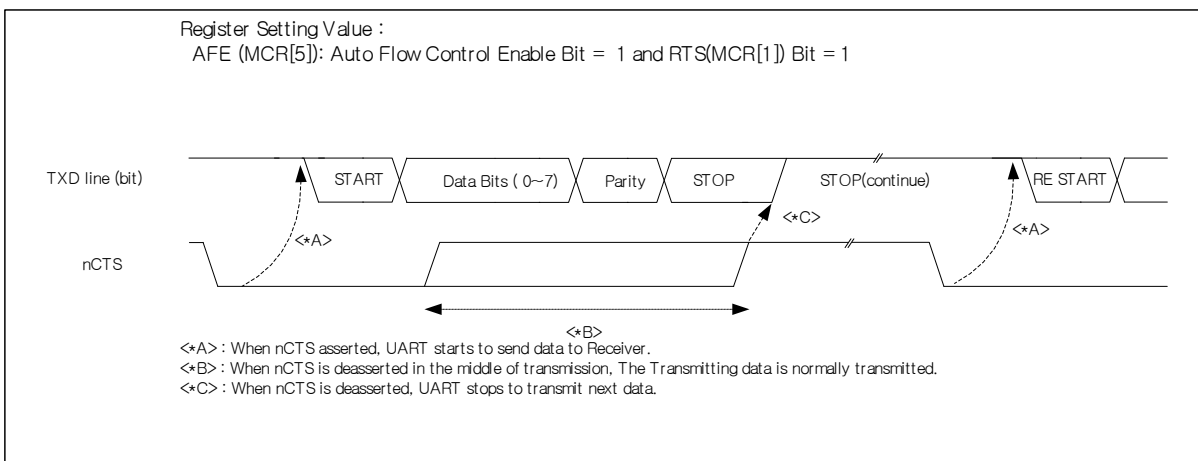


Figure 4.19 Timing Diagram for TX Operation with Hardware Flow Control

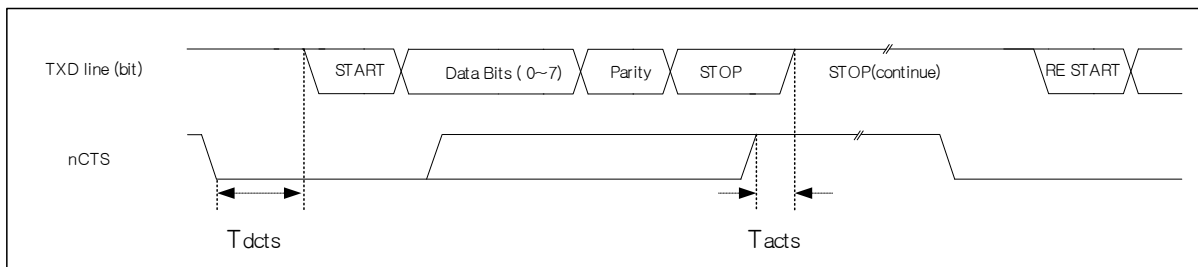


Figure 4.20 Timing Diagram for nCTS Timing Diagram

Table 4.25 Timing Parameters for Each Symbol

Parameter	Symbol	MIN	MAX	Unit	Remark
Deasserted nCTS to Tx Start	Tdcts	-	BRCP	ns	3.3V
Deasserted nCTS to Tx Stop :to stop next transmission (setup time)	Tacts	4 x BRCP/16	-	ns	3.3V

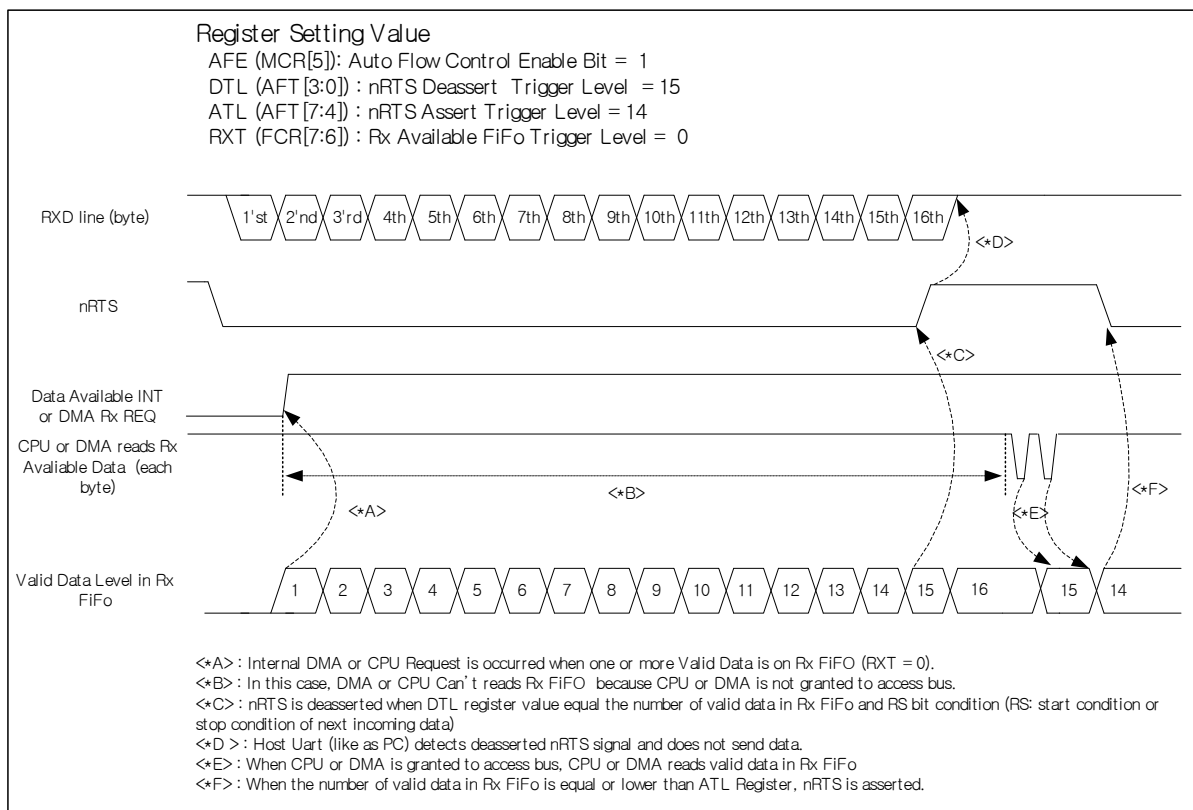


Figure 4.21 Timing Diagram for RX Operation with Hardware Flow Control

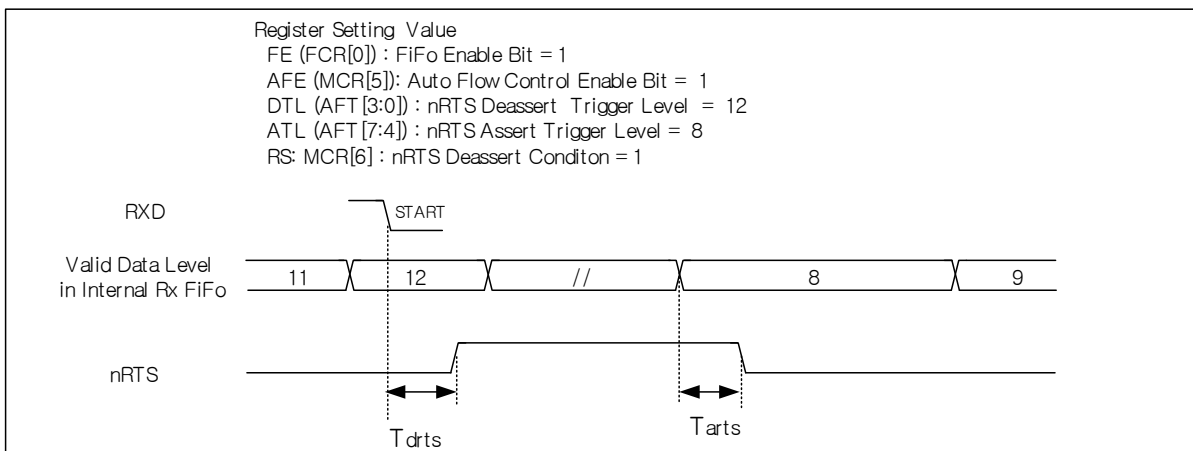


Figure 4.22 Timing Diagram for nRTS Timing Diagram

Table 4.26 Timing Parameters for Each Symbols

Parameter	Symbol	MIN	MAX	Unit	Remark
DTL (and RXD start condition) to deasserted nRTS	Tdrts	-	BRCP	ns	3.3V
ATL to asserted nRTS	Tarts	-	BRCP/16 + 8	ns	3.3V

4.14 Electrical Characteristics for OTPROM

Table 4.27 DC Electrical Characteristics for OTP ROM

Parameter	Symbol	Test Condition	MIN	TYP	MAX	Unit
Standby Current	I _{dd}	CEB = High		7	70	uA
Read Operation						
Input Leakage	I _{li}	V _{in} = 0 to VDD			2.5	uA
Output Leakage	I _{lo}	V _{out} = 0 to VDD			2.5	uA
Active Current	I _{dd}	f = 10MHz		2	3	mA
Programming Operation						
Input Leakage	I _{li}	V _{in} = 0 to VDD			2.5	uA
Output Leakage	I _{lo}	V _{out} = 0 to VDD			2.5	uA
Active Current	I _{dd}	CEB = Low		13.5	23	mA

T_A = 25°C unless otherwise specified.

4.15 Electrical Characteristics for SD/MMC Controller

The SD/MMC host controller is designed to supports high-speed mode (SD rev.1.10, up to 50 MHz Clock) as well as default speed mode (SD rev.1.01, up to 25 MHz Clock). A user does not need to set the SD/MMC host controller differently for mode change between default mode and high speed mode. If you want to change to high-speed mode or to default mode by using switch-function command (CMD6), the SD/MMC cards are set to such mode. The timing diagram shows the input/output timing criterion referenced to SD/MMC clock.

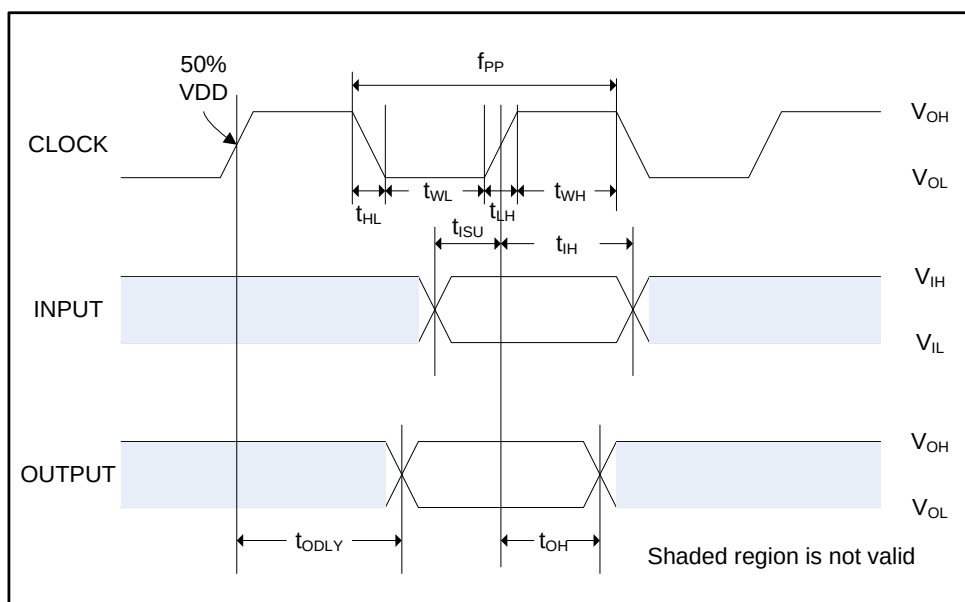


Figure 4.23 Timing Diagram for SD/MMC Controller

Table 4.28 Timing Parameters for Each Symbol

Parameter	Symbol	MIN	MAX	Unit	Remark
Clock frequency Data Transfer Mode	f _{PP}	0	TBD	MHz	Ccard ≤ 30pF
Clock low time	t _{WL}	2		ns	Ccard ≤ 30pF
Clock high time	t _{WH}	2		ns	Ccard ≤ 30pF
Clock rise time	t _{LH}		1	ns	Ccard ≤ 30pF
Clock fall time	t _{HL}		1	ns	Ccard ≤ 30pF
Input set-up time	t _{ISU}	1.8		ns	Ccard ≤ 30pF
Input hold time	t _{IH}	1		ns	Ccard ≤ 30pF
Output delay time	t _{ODLY}		2.7	ns	Ccard ≤ 30pF
Output hold time	t _{OH}	1.4		ns	Ccard ≤ 30pF

4.16 Electrical Characteristics for SDRAM Interface Controller

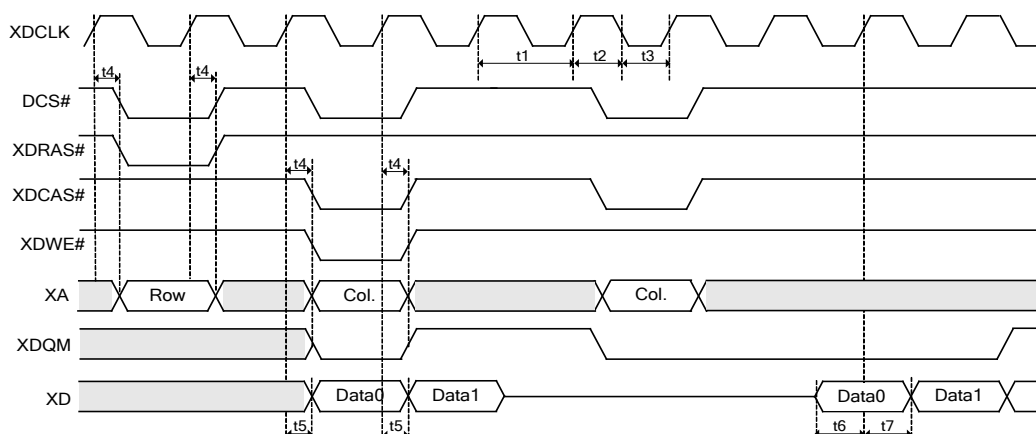


Figure 4.24 SDRAM Interface Timing

Table 4.29 SDRAM Interface Timing Parameters

Parameter	Symbol	Test Conditions	MIN	MAX	Unit
Clock cycle time	t_1	$V_{DDI} = 1.2V, V_{DDIO} = 3.3V, C_L = 30pF$	6.02	-	ns
Clock high pulse width	t_2	$V_{DDI} = 1.2V, V_{DDIO} = 3.3V, C_L = 30pF$	2.5	-	ns
Clock low pulse width	t_3	$V_{DDI} = 1.2V, V_{DDIO} = 3.3V, C_L = 30pF$	2.5	-	ns
Clock to address/command delay	t_4	$V_{DDI} = 1.2V, V_{DDIO} = 3.3V, C_L = 30pF$	1	7	ns
Clock to data output delay	t_5	$V_{DDI} = 1.2V, V_{DDIO} = 3.3V, C_L = 30pF$	1	7	ns
Input data setup time to clock	t_6	$V_{DDI} = 1.2V, V_{DDIO} = 3.3V$	1.5	-	ns
Input data hold time from clock	t_7	$V_{DDI} = 1.2V, V_{DDIO} = 3.3V$	1.5	-	ns

Note: Clock cycle time is limited by the internal HCLK frequency.

5 Address & Register Map

5.1 Address Map

The TCC8132 has a fixed address map. The address space is separated by the MSB 4bits of address bus. The following table represents the address space of TCC8132 system.

Table 5.1 Address Allocation Map

Address Space	Device Name
0x00000000 ~ 0x0FFFFFFF	Depending on REMAP register value - Internal Boot ROM - Internal SRAM - External SDRAM
0x10000000 ~ 0x1FFFFFFF	Internal SRAM (32 KB)
0x20000000 ~ 0x2FFFFFFF	External SDR SDRAM
0x30000000 ~ 0x3FFFFFFF	Internal SRAM (32 KB)
0x40000000 ~ 0x4FFFFFFF	Not used
0x50000000 ~ 0x5FFFFFFF	Not used
0x60000000 ~ 0x6FFFFFFF	Not used
0x70000000 ~ 0x7FFFFFFF	Not used
0x80000000 ~ 0x8FFFFFFF	AHB Peripherals
0x90000000 ~ 0x9FFFFFFF	APB Peripherals
0xA0000000 ~ 0xAFFFFFFF	Serial Flash
0xB0000000 ~ 0xBFFFFFFF	External SDR SDRAM
0xC0000000 ~ 0xCFFFFFFF	Not used
0xD0000000 ~ 0xDFFFFFFF	Not used
0xE0000000 ~ 0xEFFFFFFF	Internal Boot ROM
0xF0000000 ~ 0xFFFFFFFF	External Memory Controller Configuration Register Space

The address space (0x00000000 ~ 0x0FFFFFFF) is initially allocated to internal or external PROM for booting procedure, and a special flag exists in system configuration unit for remapping this space to other types of memories. Refer to the description of system configuration module for detailed operation.

The TCC8132 has several peripherals for various multimedia functions. These peripherals can be configured appropriately by their own registers that can be accessed through specially allocated address. The base addresses of these peripherals are described in the following table.

Refer to corresponding sections for detailed information of each peripheral.

Table 5.2 Address Allocations for Internal Peripherals (Base = 0x80000000)

Offset Address Space	Peripheral
0x00000 ~ 0x007FFF	External Memory Controller
0x00800 ~ 0x00FFFF	Reserved
0x01000 ~ 0x01FFFF	G2D (Graphic Engine)
0x02000 ~ 0x03FFFF	Reserved
0x04000 ~ 0x043FFF	GDMA0 (General DMA0)
0x04400 ~ 0x047FFF	GDMA1 (General DMA1)
0x04800 ~ 0x04BFFF	GDMA2 (General DMA2)
0x05000 ~ 0x05FFFF	Serial Flash (Manual)
0x06000 ~ 0x06FFFF	PL080 (DMA)
0x07000 ~ 0x07FFFF	TTCAN (CH0)
0x08000 ~ 0x08FFFF	TTCAN (CH1)
0x09000 ~ 0x09FFFF	TTCAN (CFG)
0x0A000 ~ 0x0A7FFF	GDMA1 (General DMA1)
0x0A800 ~ 0x0BFFFF	Reserved
0x0C000 ~ 0x0CFFFF	MMU Table
0x0D000 ~ 0x0DFFFF	NFC (Nand Flash Controller)
0x0E000 ~ 0x0E1FFF	SD/MMC 0 (SD/MMC Controller0)
0x0E200 ~ 0x0E3FFF	SD/MMC 1 (SD/MMC Controller1)
0x0E400 ~ 0x0FFFFF	Reserved
0x20000 ~ 0x2FFFFF	VIOC (LCD)
0x30000 ~ 0x3FFFFF	LCD SI
0x40000 ~ 0x7FFFFF	USB 20DH
0x80000 ~ 0x8FFFFF	USB Host 0 EHCI
0x90000 ~ 0x9FFFFF	USB Host 0 OHCI
0xA0000 ~ 0xAFFFFF	USB Host 1 EHCI
0xB0000 ~ 0xBFFFFF	USB Host 1 OHCI
0xC0000 ~ 0xCFFFFF	USB Configuration Registers

Table 5.3 Address Allocations for Internal Peripherals (Base = 0x90000000)

Offset Address Space	Peripheral
0x00000 ~ 0x00FFFF	DAI0 ~ 3 (Digital Audio Interface 0 ~ 3)
0x01000 ~ 0x01FFFF	SPDIF
0x02000 ~ 0x06FFFF	Reserved
0x07000 ~ 0x07FFFF	UART0
0x08000 ~ 0x08FFFF	UART1
0x09000 ~ 0x09FFFF	UART2
0x0A000 ~ 0x0AFFFF	UART3
0x0B000 ~ 0x0BFFFF	I2C Compatible Interface
0x0C000 ~ 0x0EFFFF	IC_TC
0x0F000 ~ 0x0FFFFF	GSB0 ~ 5 (SPI Master/Slave 0 ~ 5)
0x10000 ~ 0x14FFFF	Reserved
0x15000 ~ 0x15FFFF	UART4
0x16000 ~ 0x18FFFF	Reserved
0x19000 ~ 0x19FFFF	Reserved
0x1A000 ~ 0x1AFFFF	Reserved
0x1B000 ~ 0x1BFFFF	Reserved
0x1C000 ~ 0x1CFFFF	Reserved
0x1D000 ~ 0x1DFFFF	Reserved

Table 5.4 Address Allocations for Internal Peripherals (Base = 0x98000000)

Offset Address Space	Peripheral
0x00000 ~ 0x01FFF	Reserved
0x02000 ~ 0x02FFF	ADMA0 ~ 2 (Audio DMA 0 ~ 2)
0x03000 ~ 0x03FFF	PIC (Interrupt Controller)
0x04000 ~ 0x040FF	Timer
0x04100 ~ 0x041FF	PDM
0x05000 ~ 0x05FFF	GPIO
0x06000 ~ 0x06FFF	CKC (Clock Controller)
0x07000 ~ 0x07FFF	Reserved
0x08000 ~ 0x08FFF	Reserved
0x09000 ~ 0x09FFF	Reserved
0x0A000 ~ 0x0AFFF	Reserved
0x0B000 ~ 0x0BFFF	Reserved
0x0C000 ~ 0x0CFFF	Reserved
0x0D000 ~ 0x0DFFF	Reserved
0x0E000 ~ 0x0EFFF	Reserved
0x0F000 ~ 0x0FFFF	Reserved
0x10000 ~ 0x1003F	PMRAM (Backup RAM)
0x10800 ~ 0x10FFF	PMIO (Alive I/O Control)
0x11000 ~ 0x11FFF	ADC
0x12000 ~ 0x12FFF	RTC
0x13000 ~ 0x13FFF	SCFG (System Configuration Register)
0x14000 ~ 0x14FFF	CID
0x15000 ~ 0x15FFF	Reserved
0x16000 ~ 0x16FFF	Reserved
0x17000 ~ 0x17FFF	Reserved
0x18000 ~ 0x18FFF	Cipher
0x19000 ~ 0x19FFF	Reserved
0x1A000 ~ 0x1AFFF	Reserved
0x1B000 ~ 0x1BFFF	Reserved
0x1C000 ~ 0x1CFFF	GPIOPK (GPIO Controller for PK[34:0])
0x1D000 ~ 0x1DFFF	Reserved
0x1E000 ~ 0x1EFFF	Reserved
0x1F000 ~ 0x1FFFF	Reserved
0x20000 ~ 0x3FFFF	Reserved

Address decoding logic only monitors base address (For example, 0x8xxxxxxx, 0x9xxxxxxx, and etc.), and bit16~bit12 of accessing address bus. There can be a lot of mirror images of address space that are repeated at every 4Kbyte boundary, a user can access certain registers with these mirror images also, so care must be taken not to modify these registers unintentionally.

5.2 Address Remapping

When the CPU is reset, it starts instruction fetch by accessing to 0 address. The TCC8132 provides address remapping function in order for various memories to be mapped to 0 address on system address space.

When reset is asserted, 0 address of the TCC8132 can be mapped according to boot mode.

Table 5.5 Address Mapping in Reset State

Boot Mode	0 Address
External NOR Boot Mode	Not used
Other Boot Mode	Internal Boot ROM

For more detailed information about Boot Mode, refer to Chapter 1.5 .

After reset, remapping function can be carried out through REMAP register (4bit) of system configuration module (0x98013000). Memory mapping according to REMAP register setting is as follows.

Table 5.6 Address Mapping According to REMAP Register

REMAP[3:0]	0 Address
0x0	0x1xxxxxxx - Internal SRAM
0x1	0x1xxxxxxx - Internal SRAM
0x2	0x2xxxxxxx - External SDRAM
0x3	0x3xxxxxxx - Internal SRAM
0x4	Reserved
0x5	Reserved
0x7	0x7xxxxxxx - External NOR, CS3 (XRCS#)
0xB	0x2xxxxxxx - External SDRAM
0xC	Reserved
0xD	Reserved
0xE	0xExxxxxxx - Internal Boot ROM

6 CPU

6.1 Overview

The TCC8132 adopted the ARM926EJ-S core for controlling system and processing various kinds of digital signals. The ARM926EJ-S is a Harvard architecture cached processor with separate 16Kbyte data and 16Kbytes instruction caches, each with 8-word of line length.

The followings are key features of the TCC8132 CPU core.

- CPU ARM926EJ-S
- Cache 16KB for Data / 16KB for Instruction
- MMU Memory Management Unit

In this chapter, the key functions of the ARM926EJ-S core are described briefly. Refer to the *ARM Technical Reference Manual version of r0p5* for detailed information of the ARM926EJ-S core. [1]

6.2 Functional Description

6.2.1 Memory Formats

The ARM926EJ-S views memory as a linear collection of bytes numbered upwards from 0. Bytes 0 to 3 hold the first word, bytes 4 to 7 holds the second word, and so on. The ARM926EJ-S itself can treat words in memory as being stored either in big-endian or little-endian, but in the TCC8132, there is only little-endian supported by the memory controller. The following figure illustrates the structure of little-endian type of memories.

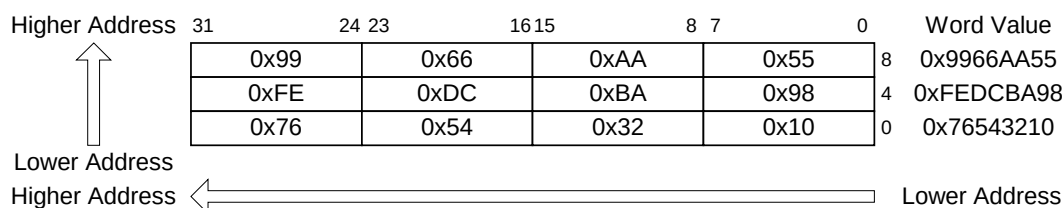


Figure 6.1 Little-Endian Addresses of Bytes-Words

6.2.2 Operating Modes

The ARM926EJ-S supports seven modes of operation:

- USER (usr) The normal ARM program execution mode
- FIQ (fiq) Designed to support a data transfer or channel process
- IRQ (irq) Used for general purpose interrupt handling
- Supervisor (svc) Protected mode for the operating system
- Abort (abt) Entered after a data or instruction prefetch abort
- System (sys) A privileged user mode for the operating system
- Undefined (und) Entered when an undefined instruction is executed

Switching between these modes may be made under software control, or may be brought about by interrupts or exception processing. Most application programs will be executed in user mode. The non-user modes known as privileged modes are entered in order to service interrupts or exceptions, or to access protected resources.

6.2.3 Instruction Set Features

ARM Instruction Set

The ARM instruction set contains the following six main instruction types.

- Branch instructions
- Data processing instructions
- Status register transfer instructions
- Load/store instructions
- Cache/Protection instructions
- Exception generation instructions

Most data processing instructions update the four condition code flags (N, Z, C, V) in the current program status register (CPSR).

6.2.4 Addressing Modes

Load/Store Instructions

Load/store instructions provide three basic types of addressing modes, each specifying a base register and an offset.

- Offset addressing mode: The memory address consists of the offset added to or subtracted from the base register contents
- Preindex addressing mode: The memory address is same as the offset addressing mode. This address then overwrites the base register contents.
- Postindex addressing mode: The memory address is the base register contents. The hardware then updates the base register by adding or subtracting the offset.

The offset is either an immediate value or the contents of an index register. There are options for shifting the contents of an index register before the addition or subtraction.

Multiple Load/Store Instructions

Multiple load/store instructions load/store two or more general-purpose registers from/to memory. These instructions offer four addressing modes.

- Increment After (IA): The base register contents are incremented after the transfer.
- Increment Before (IB): The base register contents are incremented before the transfer.
- Decrement After (DA): The base register contents are decremented after the transfer.
- Decrement Before (DB): The base register contents are decremented before the transfer.

6.2.5 Exceptions

Exceptions occur whenever the normal flow of a program has to be halted temporarily, for example to service an interrupt from a peripheral. Before an exception can be handled, the current processor state must be preserved so that the original program can resume when the handler routine has finished. It is possible for several exceptions to arise at the same time, and if this happens, they are dealt with in a fixed priority. This priority is illustrated in the table below.

Table 6.1 Exception Vector and Priorities

Address	Exception	Mode in Entry	Priority
0x00000000	Reset	Supervisor	1
0x00000004	Undefined Instruction	Undefined	6
0x00000008	Software Interrupt	Supervisor	6
0x0000000C	Abort (Prefetch)	Abort	5
0x00000010	Abort (Data)	Abort	2
0x00000014	Reserved	Reserved	-
0x00000018	IRQ	IRQ	4
0x0000001C	FIQ	FIQ	3

Note: The higher priority has the lower number.

6.2.5.1 Actions on Entering an Exception

- Preserves the address of the next instruction in the link register of appropriate mode. If the exception has been entered from ARM state, then the address of the next instruction is copied into the link register (current PC + 4, or +8 depending on the exception types). If the exception has been entered from THUMB state, then the value written into the link register is the current PC offset by a value such that the program resumes from the correct place on return from the exception. That is, the exception handler needs not to determine which state the exception was entered from. For example, in case of SWI exception, MOVS PC, R14_svc will always return to the next instruction regardless of whether the SWI was executed in ARM or THUMB state.
- Copies the CPSR into the SPSR of appropriate mode
- Forces the CPSR mode bits to a value that represents the mode of exception
- Forces the PC to fetch the next instruction from the relevant exception vector

It may also set the interrupt disable flags to prevent otherwise unmanageable nestings of exceptions. If the processor is in THUMB state when an exception occurs, it will automatically switch into ARM state when the PC is loaded with the exception vector address.

6.2.5.2 Actions on Leaving an Exception

- Forces the PC to fetch the next instruction from the relevant exception vector
- Moves the link register, minus an offset where appropriate, to the PC. (The offset will vary depending on the type of exception.)
- Copies back the SPSR to the CPSR

An explicit switch back to THUMB state is never needed, since restoring to the CPSR from the SPSR automatically sets the T bit to the value it held immediately prior to the exception.

6.2.5.3 Exception Entry/Exit Summary

The following table summarizes the PC value preserved in the relevant R14 on exception entry, and the recommended instruction for exiting the exception handler.

Table 6.2 Exception Entry/Exit

Entry	Return Instruction	Previous State		Note
		ARM	THUMB	
BL	MOV PC, R14	PC + 4	PC + 2	1
SWI	MOVS PC, R14	PC + 4	PC + 2	1
UNDEF	MOVS PC, R14	PC + 4	PC + 2	1
FIQ	SUBS PC, R14, #4	PC + 4	PC + 4	2
IRQ	SUBS PC, R14, #4	PC + 4	PC + 4	2
PABT	SUBS PC, R14, #4	PC + 4	PC + 4	1
DABT	SUBS PC, R14, #8	PC + 8	PC + 8	3
RESET	-			4

Note:

1. Where PC is the address of the BL or SWI instruction or Undefined Instruction which had the prefetch abort.
2. Where PC is the address of the instruction which did not get executed since the FIQ or IRQ interrupt took priority.
3. Where PC is the address of the Load or Store instruction which generated the data abort.
4. The value saved in R14 upon reset is unpredictable.

6.2.5.4 FIQ

The FIQ (Fast Interrupt Request) exception is designed to support a data transfer or channel process, and in ARM state, it has sufficient private registers to remove the need for saving registers (thus minimizing the overhead of context switching). The FIQ is generated by nFIQ signal that is controlled by internal or external interrupts defined as FIQ type. Irrespective of whether the exception is entered from ARM state or THUMB state, a FIQ handler should leave the interrupt by executing

```
SUBS PC, R14, #4
```

The FIQ may be disabled by setting the CPSR's F flag (This is not possible in user mode). If the F flag is cleared, ARM926EJ-S checks for a low level on the nFIQ signal at the end of every instruction.

6.2.5.5 IRQ

The IRQ (Interrupt Request) exception is a normal interrupt caused by a low level on the nIRQ signal that is controlled by internal or external interrupt defined as IRQ type. The IRQ has lower priority than FIQ and is masked out when a FIQ sequence is entered. It may be disabled at any time by setting the I bit in the CPSR, though this can only be done in a privileged (non-user) mode.

Irrespective of whether the exception is entered from ARM state or THUMB state, an IRQ handler should return from the interrupt by executing

```
SUBS PC, R14, #4
```

6.2.5.6 Abort

An abort indicates that the current memory access cannot be completed. ARM926EJ-S checks for the abort condition during memory access cycles.

There are two types of abort:

- Prefetch abort: Occurs during an instruction prefetch
- Data abort: Occurs during a data access

If a prefetch abort occurs, the prefetched instruction is marked as invalid, but the exception will not be taken until the

instruction reaches the thread of the pipeline. If the instruction is not executed, the abort does not take place.

If a data abort occurs, the action taken depends on the instruction type:

The swap instruction (SWP) is aborted as if it had not been executed.

Block data transfer instructions (LDM and STM) are completed. If write-back is set, the base is updated. If the instruction has overwritten the base with data, the overwriting is prevented. All register overwriting are prevented after an abort is indicated, which means in particular that R15 is preserved in an aborted LDM instruction.

After fixing the reason for the abort, the handler should execute the following irrespective of the state (ARM or THUMB):

```
SUBS    PC, R14, #4      ; For a prefetch abort
SUBS    PC, R14, #8      ; For a data abort
```

This restores both the PC and the CPSR, and retries the aborted instruction.

6.2.5.7 Software Interrupt

The Software Interrupt Instruction (SWI) is used for entering Supervisor mode, usually to request a particular supervisor function. A SWI handler should return by executing the following irrespective of the state (ARM or THUMB):

```
MOV     PC, R14
```

This restores the PC and CPSR, and returns to the instruction following the SWI instruction.

6.2.5.8 Undefined Instruction

When ARM926EJ-S comes across an instruction which it cannot handle, it takes the undefined instruction trap. This mechanism is used to extend either the THUMB or ARM instruction set by software emulation. After emulating the failed instruction, the trap handler should execute the following instruction irrespective of the state (ARM or THUMB):

```
MOVS    PC, R14
```

This restores the CPSR and returns to the instruction following the undefined instruction.

6.3 Coprocessor CP15

Refer to ARM's technical reference manual. [1]

7 Virtual MMU Table

7.1 Overview

MMU table module sets access properties by region such as cacheable/bufferable by setting maximum 8 regions in applications where ARM926EJ-S does not use MMU.

7.2 Register Description

Table 7.1 MMU Register (Base Address = 0x8000C000)

Name	Address	Type	Reset	Description
REGIONx	0x00	R/W	0x00	Region Configuration Register

Table 7.2 MMU Descriptor Register (Base Address = 0x8001C000)

Name	Address	Type	Reset	Description
DESCRIPTOR	0x00	R	0x00	MMU section descriptor

It provides section descriptor (1Mbyte). Base Address, 0x8001C000 should be registered in TTB register of ARM926EJS MMU.

Example:

nBase = 0x8001C000

mcr p15, 0, nBase, c2, c0, 0

Note: "R" in register map table represents **Reserved**.

Region Configuration Register (REGIONx)

0x8000C000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SA												SZ			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SZ				AP		EN	DO				R	CA	BU	R	

SA	[31:20]	Section Base Address
N		Section base address is N

Defined same as ARM926EJ-S

SZ	[19:12]	Region size
0x13		1MB Region
0x14		2MB Region
0x15		4MB Region
...		...
0x1E		2GB Region
0x1F		4GB Region

AP	[11:10]	Access Permission Register
00	No Access	Any access generates a domain fault. Priviledged : No access User : No access
01	Client	Accesses are checked against the access permission bits in the section or page descriptor. Priviledged : Read/write User : No access
10	Reserved	Reserved. Currently behaves like the no access mode. Priviledged : Read/write User : Read only
11	Manager	Accesses are not checked against the access permission bits so a permission fault cannot be generated. Priviledged : Read/write User : Read/write

Defined same as ARM926EJ-S

EN	[9]	Region Enable Register
0		Disable the region
1		Enable the region

DO	[8:5]	Domain Register
N	There are 16 domains from 0 to 15	

Defined same as ARM926EJ-S

CA	[3]	Cacheable Register
0	Not cacheable	
1	Cacheable	

Defined same as ARM926EJ-S

BU	[2]	Bufferable Register
0	CA == 0 : Not Bufferable CA == 1 : Write through	
1	CA == 0 : Bufferable CA == 1 : Write back	

Defined same as ARM926EJ-S

Note: All the bit field except for SZ and EN are defined same as in ARM926EJ-S technical reference manual. [1]
The REGION7 has the highest priority and the REGION0 has the lowest priority.
After setting register, the address of TABBASE should be written into TLB base address for ARM926EJ-S.

8 Booting Procedure

8.1 Overview

In the TCC8132, there is an internal boot ROM for system initialization process. It contains the fundamental routines for system initialization or boot procedure through various devices or interfaces.

The figure below illustrates the timing of reset sequence at power-up. During this process, the configuration value including boot mode is selected by the state of {BM3, BM2, BM1, BM0} at the PRST# rising edge. The 4-bit configuration value (BM[3:0]) determines the boot mode as listed in the table below.

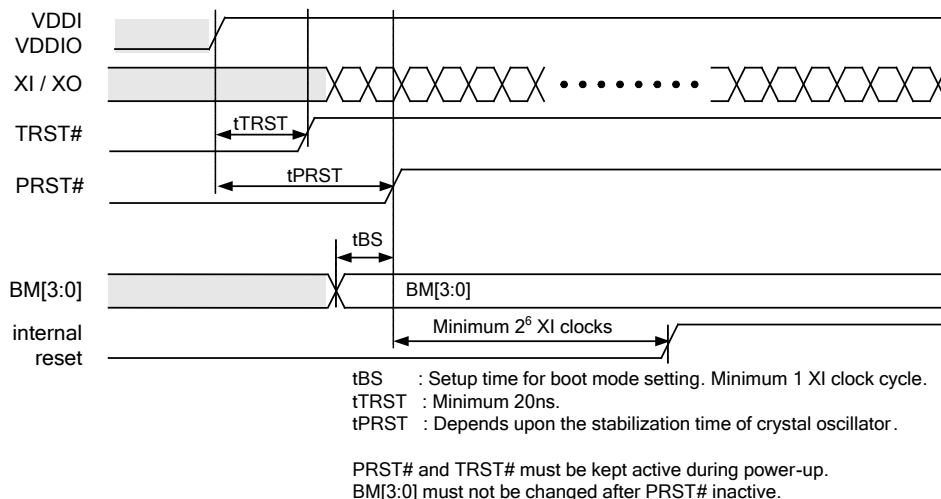


Figure 8.1 Reset Sequence

Table 8.1 Boot Mode & Configuration Value

BOOT MODE	BM[3]	BM[2]	BM[1]	BM[0]	Ports	Option
Reserved	0	0	0	0	-	-
	0	0	0	1	-	-
Reserved	0	0	1	0	-	-
	0	0	1	1	-	-
NAND	0	1	0	0	PS[13:0]	8bit
SDMMC	0	1	0	1	PS[11:0]	eMMC
Serial Flash	0	1	1	0	PS[27:22]	3B-
	0	1	1	1		4B
Reserved	1	0	0	0	-	-
	1	0	0	1	-	-
SD	1	0	1	0	FC[11:0]	SD
	1	0	1	1		eMMC
Reserved	1	1	0	0	-	-
SDMMC	1	1	0	1	PS[11:0]	SD
USB	1	1	1	0	USB1DP/DM	-
Reserved	1	1	1	1	-	-

8.2 Dual CRC Checking

Except the external NOR boot mode, the boot code checks the CRC for the downloaded image according to the algorithm described below.

- 1) Calculate CRC on the downloaded image up to 128Kbytes. In calculation, the word at the offset of 0x10 and 0x14 is skipped, because the good CRC code is contained at the offset of 0x10, and the word at the offset of 0x14 means the address of user-defined CRC routine.
- 2) After the first CRC calculation is completed, and the result is same as the value at the offset address of 0x10, the boot code checks if the address of user-defined CRC routine is not 0.
- 3) If there is user-defined CRC routine, that routine is called.
The user-defined CRC routine checks its own CRC code and returns Equal condition flag if the CRC codes are equal. If not equal, it returns Not Equal condition flag.
- 4) If there is a failure on the CRC checking in the above steps, the CPU goes to pre-defined boot mode automatically.

The following code shows same CRC generation algorithm used in the first CRC checking routine.

```
word calc_crc (word *base, word length, word *crctable)
{
    word    crcout = 0;
    word    cnt, i, code, tmp;

    length >>= 2;                // convert into word unit.
    for (cnt = 0; cnt < length; cnt++) {
        if (cnt == 0x04 or cnt == 0x05)    // skip offset of 0x10 and 0x14
            continue;
        code = base[cnt];
        for (i = 0; i < 4; i++) {
            tmp    = code ^ crcout;
            crcout = (crcout >> 8) ^ crctable[tmp & 0xFF];
            code   = code >> 8;
        }
    }
    return crcout;
}

void gen_crc (word *crctable)
{
    word    crc, cnt, i;
    // Polynomial = x32 + x31 + x16 + x15 + x4 + x3 + x + 1

    for (cnt = 0; cnt < 0x100; cnt++) {
        crc = cnt;
        for (i = 0; i < 8; i++) {
            if (crc & 1)    crc    = (crc >> 1) ^ 0xD8018001;
            else            crc    = (crc >> 1);
        }
        crctable[cnt]      = crc;
    }
}
```

8.3 NAND Boot

In this mode, the firmware code is read from NAND flash attached at the PS[13:0]. NAND chip enable is controlled by PS[13:0]. To make use of this mode, the predefined structure such as Master Block and Master Cluster should be fused in the NAND flash. The detailed construction of this structure is illustrated in Figure 8.3. The supportable configuration of NAND flash is as follows.

Case1) One NAND flash of 8bit bus-width (BM[3:0] = 4)

NAND flash I/O connection of the TCC8132 is as follows

1. 8 bits I/O : The configuration of Case1 can be possible
 - nWE : PS[6]
 - nOE : PS[7]
 - nCS : PS[12]
 - ALE, CLE : PS[4], PS[5]
 - Data I/O[7:0] : PS[11:8], PS[3:0]

To read the firmware code from NAND flash, first of all, should know the type of attached NAND flash such as I/O band width, connection type (single/double/quad) and so on.

The boot sequence of this mode is as follows. If there is any problem hard to recover during this sequence, it automatically goes to USB boot mode. It assumes the contents are stored as little endian format.

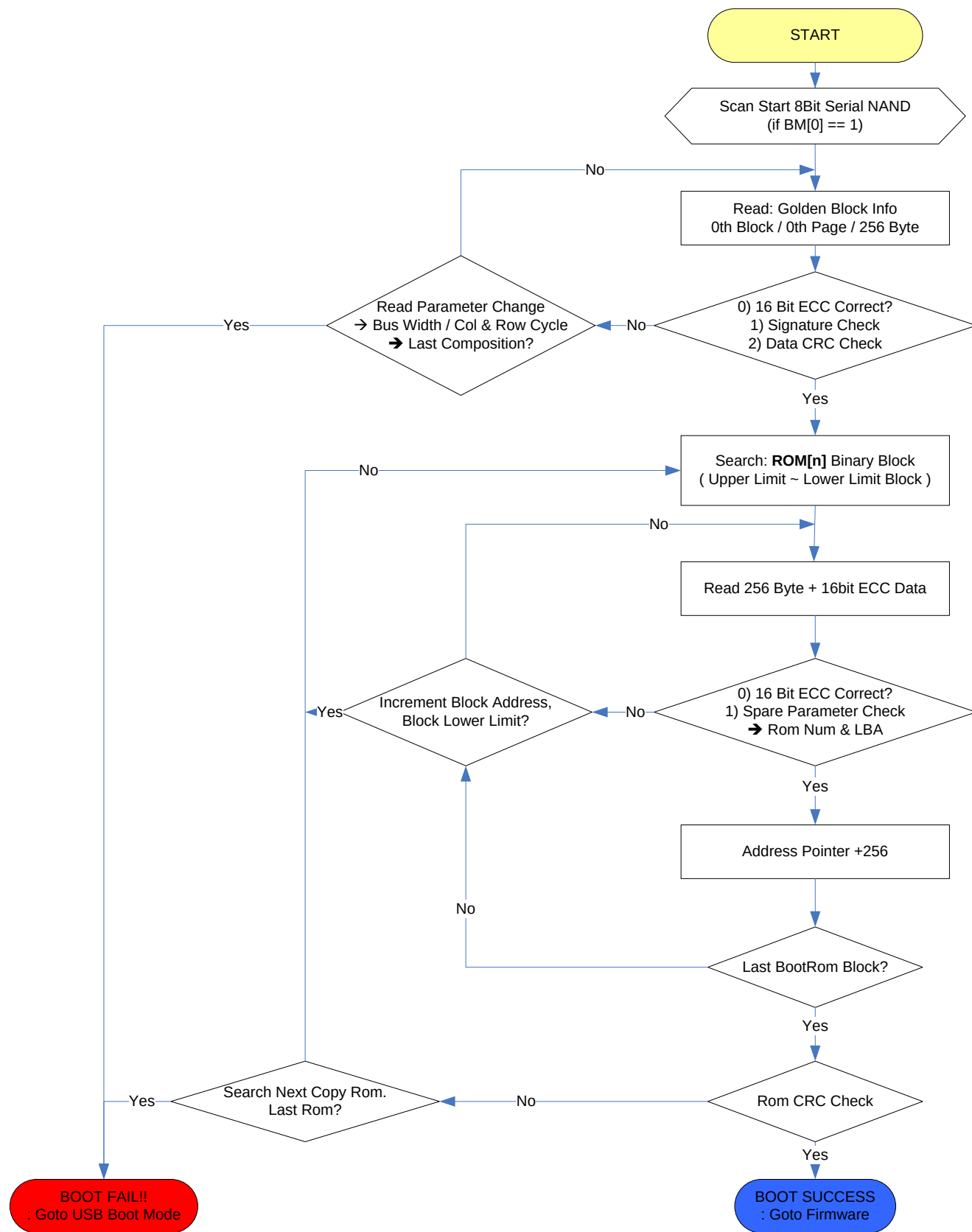


Figure 8.2 NAND_V2 Boot Procedure

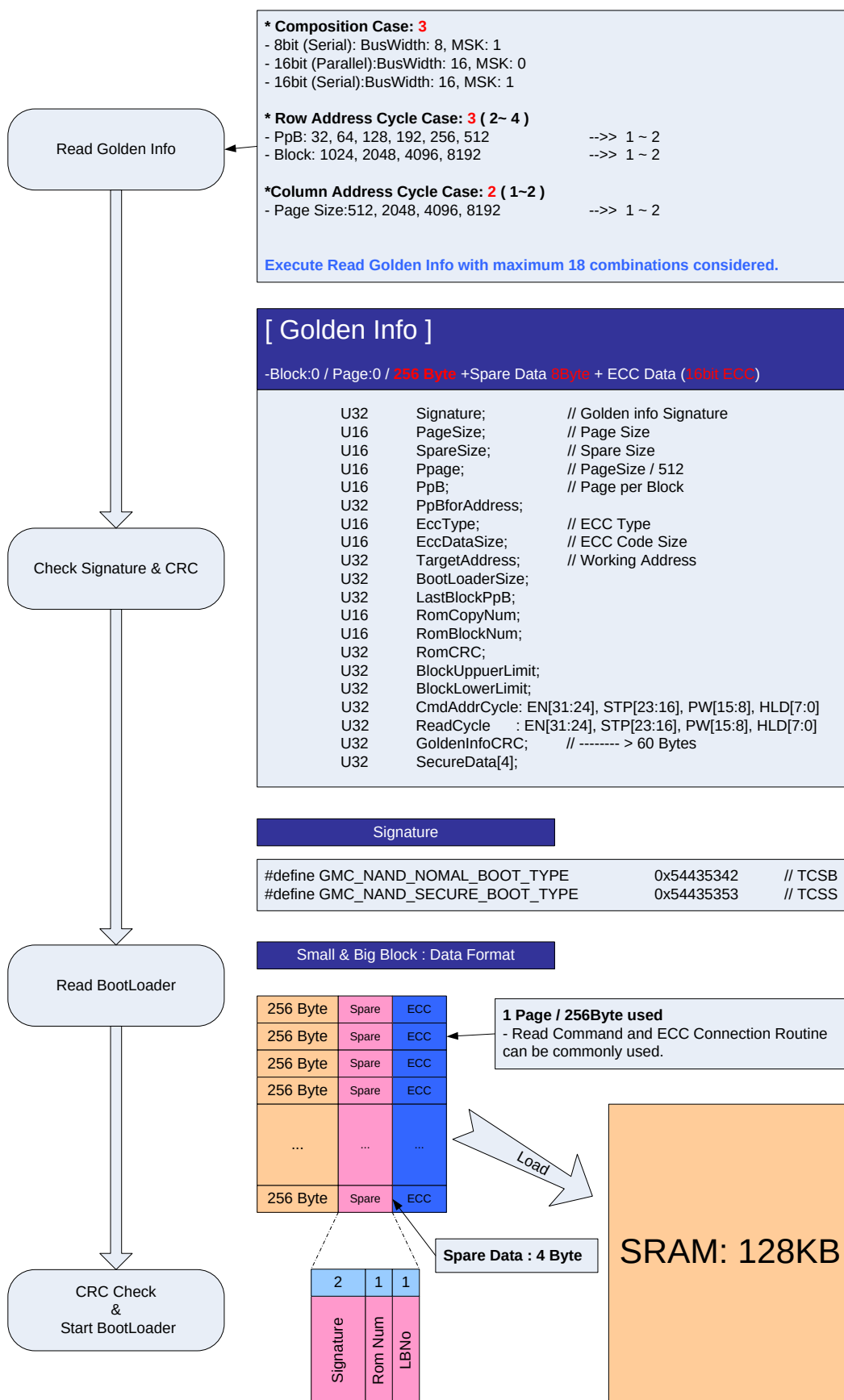


Figure 8.3 NAND_V2 Golden Information Structure

8.4 Serial NOR Flash Boot

The BM=0x6 is 3B Address NOR Serial flashboot mode and the BM=0x7 is 4B Address NOR Serial flash boot mode. The GPIO_PS[21:27] are used for interfacing the serial NOR Flash. The boot mode changes to USB boot after failed.

```
typedef struct {
unsigned long ulSignature;
unsigned long ulRomCodeSize_cm4;
unsigned long ulRomCodeOffset_cm4;
unsigned long ulRomTargetAddress_cm4;
unsigned long ulRomCodeCRC_cm4;

unsigned long ulReserved[4];

unsigned long ulRomCodeSize_dram;
unsigned long ulRomCodeOffset_dram;
unsigned long ulRomTargetAddress_dram;
unsigned long ulRomCodeCRC_dram;

unsigned long ulRomCodeSize_boot;
unsigned long ulRomCodeOffset_boot;
unsigned long ulRomTargetAddress_boot;
unsigned long ulRomCodeCRC_boot;
unsigned long ulReserved[110];
unsigned long uCRC;
} sSFBootHeader;
```

Figure 8.4 Data Structure for Header Information Stored In Serial Flash

In the figure above, the data structure required for booting from serial flash is described. The variables named as ulRomCodeSize, ulRomCodeOffset, ulRomTargetAddress, ulRomCodeCRC, and uCRC are used in the boot code. The ulRomCodeSize field is total number of bytes to store in the serial flash and should be aligned to word (4bytes).

Even if the first Header fails, the second header tries again.

OFFSET	DESCRIPTION	SIZE
	1 st SFMC Init Table	64*4 Byte
0x100	Cm4_CodeSize	4 Byte
0x104	Cm4_CodeOffset	4 Byte
0x108	Cm4_TargetAddress	4 Byte
0x10C	Cm4_CRC	4 Byte
:	Reserved	4*4 Byte
0x120	DRAM_CodeSize	4 Byte
0x124	DRAM_CodeOffset	4 Byte
0x128	DRAM_TargetAddress	4 Byte
0x12C	DRAM_CRC	4 Byte
0x130	BootLoader_CodeSize	4 Byte
0x134	BootLoader_CodeOffset	4 Byte
0x138	BootLoader_TargetAddress	4 Byte
0x13C	BootLoader_CRC	4 Byte
:	Reserved	16305*4Byte
0x10000	2 nd SFMC Init Table	64*4 Byte
0x10100	Cm4_CodeSize	4 Byte
0x10104	Cm4_CodeOffset	4 Byte
0x10108	Cm4_TargetAddress	4 Byte
0x1010C	Cm4_CRC	4 Byte
:	Reserved	4*4 Byte
0x10120	DRAM_CodeSize	4 Byte
0x10124	DRAM_CodeOffset	4 Byte
0x10128	DRAM_TargetAddress	4 Byte
0x1012C	DRAM_CRC	4 Byte
0x10130	BootLoader_CodeSize	4 Byte
0x10134	BootLoader_CodeOffset	4 Byte
0x10138	BootLoader_TargetAddress	4 Byte
0x1013C	BootLoader_CRC	4 Byte

Figure 8.5 SFMC Boot Parameter

The overall procedure for serial flash boot is shown in the following figure.

```
// Function to read from serial flash
// "data" is destination pointer
// "nwords" is the total number of words to read

Void StartSFlashBoot ()
{
    SFLASH_QPI_PortConfig (); // Port Configuration for Serial Flash Interface
    SFLASH_QPI_Init (); // Initialization for SFMC controller

    for(i=0; i<2; i++)
    {
        If( i== 0 ) {
            nPS = SBIF_IMG_PS_PRIMARY;
            uHeaderOffset = SFQPI_1st_HEADER_OFFSET;
        } else {
            nPS = SBIF_IMG_PS_SECONDARY;
            uHeaderOffset = SFQPI_2nd_HEADR_OFFSET;
        }
    }
}
```

```
OTP_READ_QUAD_SPI_OCTA_ORNOT();
PMU_SATUS_READ();
Compatible_recovery_time_wait();

If(nSfmcMODE==3Bmode)
{
    3B_MODE_Set();
}
else
{
    4B_MODE_Set();
}

SFLASH_QPI_ReadData ((unsigned
int*)pHeader,uHeaderOffset,sizeof(sSFQPIBootHeader));

if ((pHeader->u1RomCodeSize_cm4 | pHeader->u1RomCodeOffset_cm4)==0xFFFFFFFF)
StartUSBBoot ();// Booting Failed

SFLASH_QPI_ReadData ((unsigned*)pHeader->u1RomTargetAddrss_cm4,
                    pHeader->u1RomCodeOffset_cm4,
                    pHeader->u1RomCodeSize_cm4));

if(IsSecureBootEnabled()){
    if(DecryptAndVerifyCM4Code(pHeader->u1RomTargetAddrss_cm4,
                             pHeader->u1RomCodeSize_cm4))
        goto BOOT_FAIL;
}

else{
    if(IO_UTIL_CheckCODE((unsigned*)pHeader->u1RomTargetAddrss_cm4,
                        pHeader->u1RomCodeSize_cm4, 0) !=0)
        goto BOOT_FAIL;
}

CM4_PostProcess();

SFLASH_QPI_ReadData ((unsigned*)pHeader->u1RomTargetAddrss_dram,
                    pHeader->u1RomCodeOffset_dram,
                    pHeader->u1RomCodeSize_dram));

if(IsSecureBootEnabled()){
    if(DecryptAndVerifySB1(nPS, pHeader->u1RomTargetAddrss_dram,
                          pHeader->u1RomCodeSize_dram))
        goto BOOT_FAIL;}

else{
    if(IO_UTIL_CheckCODE((unsigned*)pHeader->u1RomTargetAddrss_dram,
                        pHeader->u1RomCodeSize_dram, 0) !=0)
        goto BOOT_FAIL;}

    if(fDRAMInit == 0 {
        CallSafe(pHeader->u1RomTargetAddrss_dram);
        fDRAMInit =1;}

SFLASH_QPI_ReadData ((unsigned*)pHeader->u1RomTargetAddrss_boot,
                    pHeader->u1RomCodeOffset_boot,
                    pHeader->u1RomCodeSize_boot));

if(IsSecureBootEnabled()){
    if(DecryptAndVerifySB2(nPS, pHeader->u1RomTargetAddrss_boot,
                          pHeader->u1RomCodeSize_boot))
        goto BOOT_FAIL;}

else{
    if(IO_UTIL_CheckCODE((unsigned*)pHeader->u1RomTargetAddrss_boot,
                        pHeader->u1RomCodeSize_boot, 0) !=0)
        goto BOOT_FAIL;}

goto_firmware (u1EntryPoint);}
}
```

Figure 8.6 Pseudo Code for Serial Flash Booting Procedure

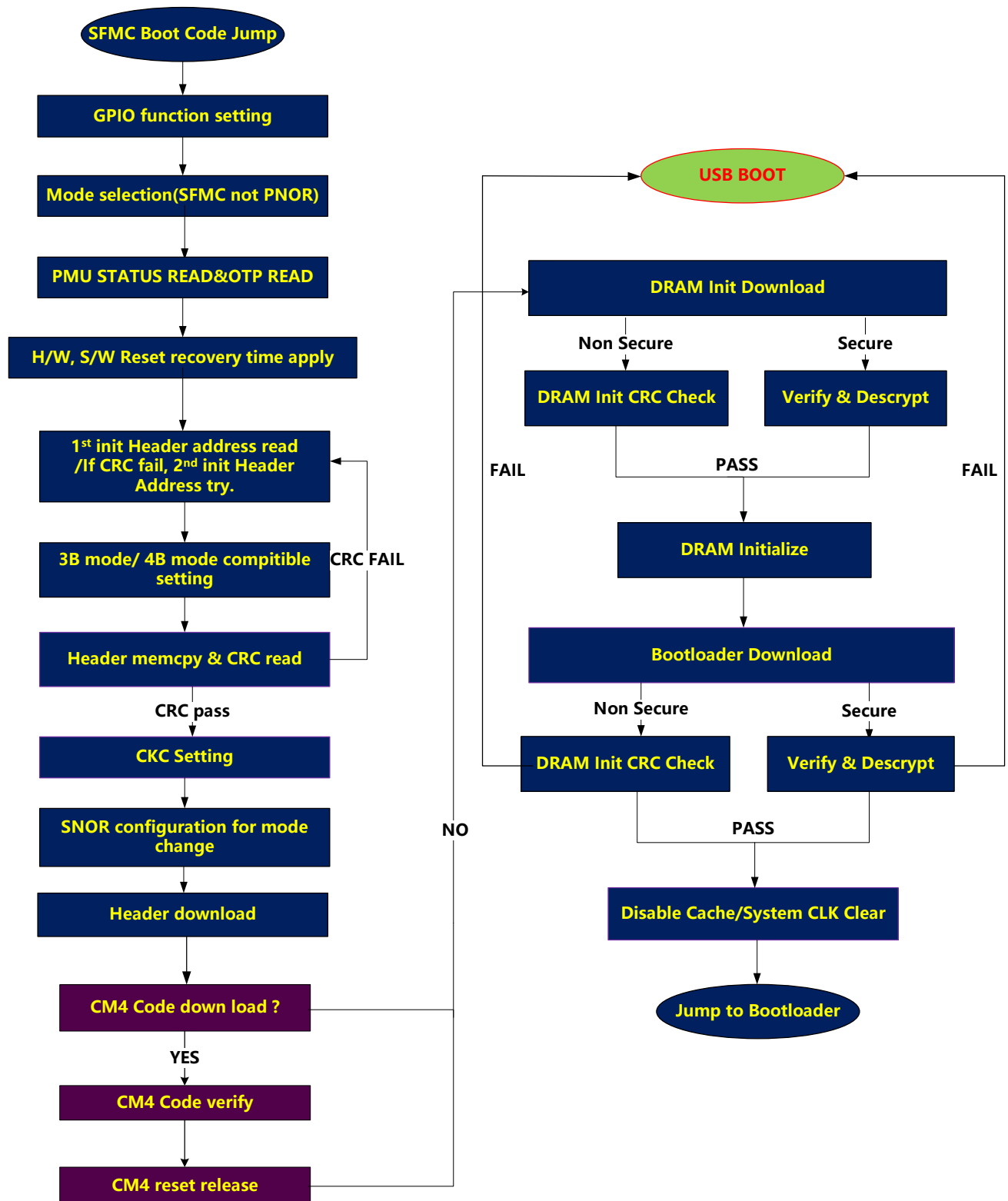


Figure 8.7 Flow Chart of Serial Flash Boot Procedure

8.5 SD Boot

The boot code supports booting from SD (Secure Digital) cards or eMMC (embedded Multimedia Card) which contains boot image with specific-format. The SD Host Controller 0 (0x8000E000) is used for booting. The PS[5:0] and FC[5:0] are mapped as SDCLK, SDCMD, DATA3, DATA2, DATA1, and DATA0 respectively. The SD host controller reads data through 1 or 4-bit bus under normal speed mode.

The SD and eMMC boot mode are processed by the following procedure.

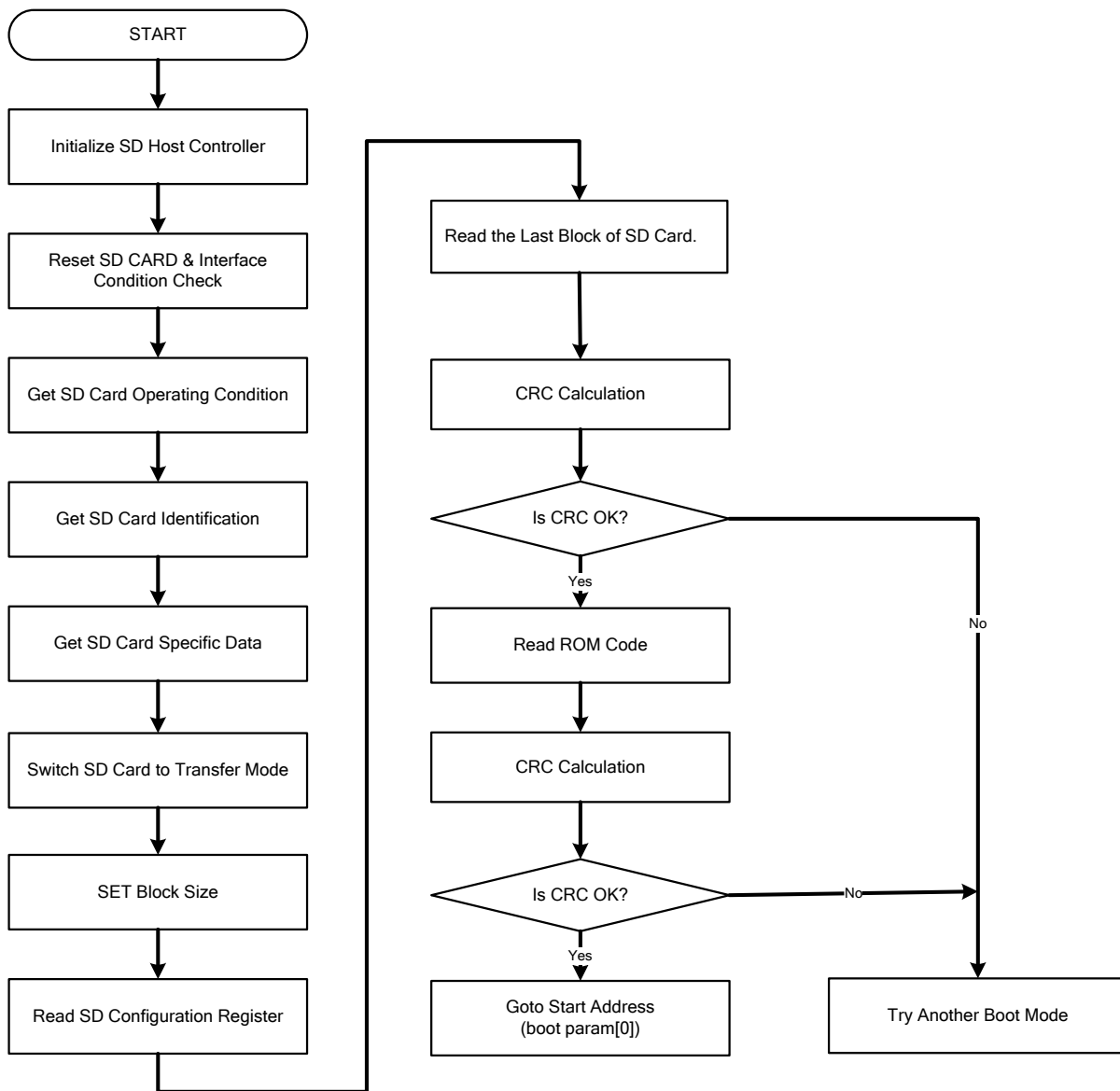


Figure 8.8 SD Boot Procedure

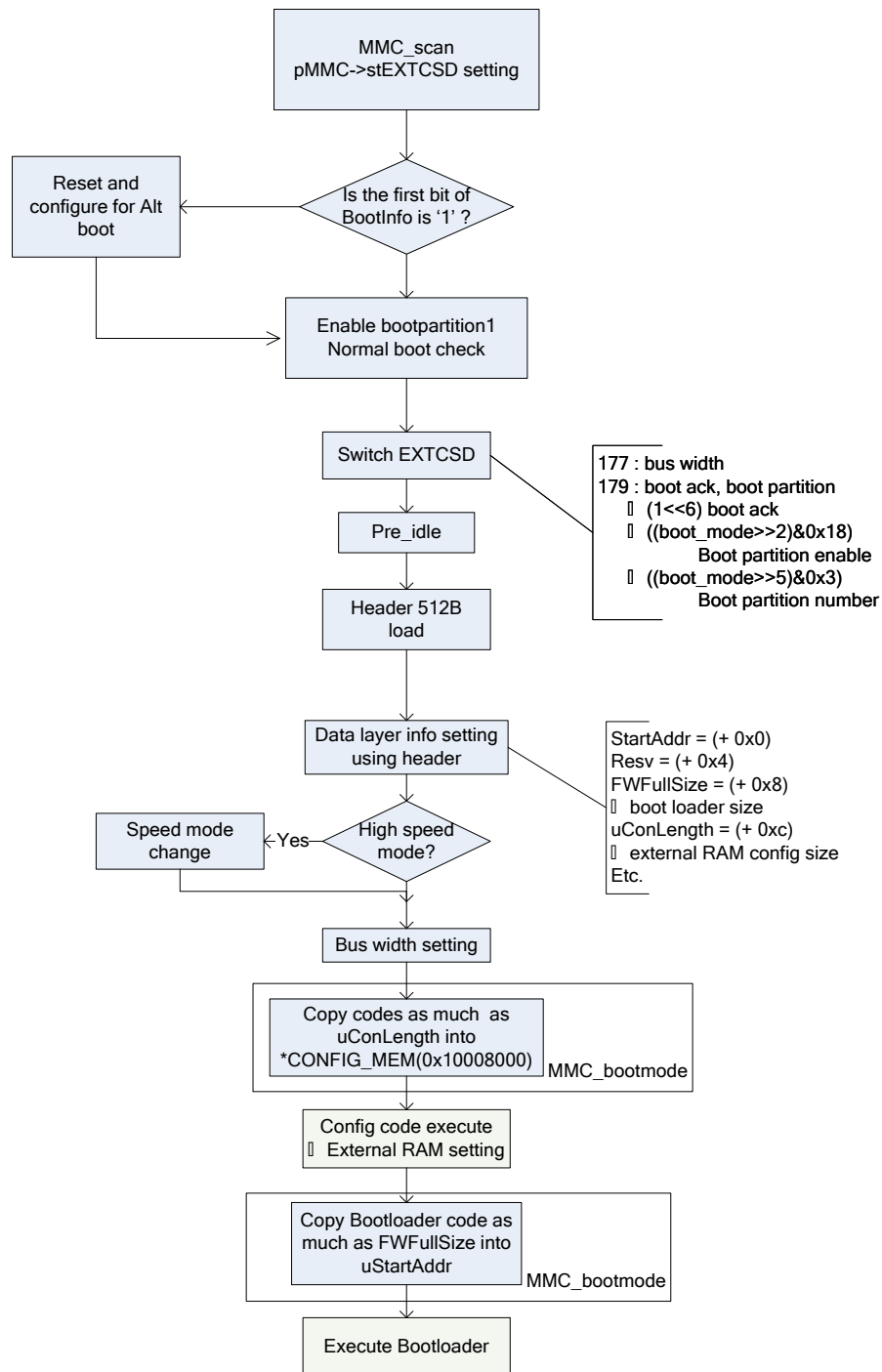


Figure 8.9 eMMC Boot Procedure

The header Information of SD card and eMMC is shown in the figure below.

0	Entry Point(4B)	Base Address(4B)	ConfigCode Size(4B)	Bootloader Size(4B)
16	BootConfig(4B)	BootPartition Size(4B)		
32				
96	Serial Number (64B)			
128	Hidden Area Start Addr	Hidden0's Size	Hidden1's Size	Hidden2's Size
	Hidden3's Size	Hidden4's Size	Hidden5's Size	Hidden6's Size
	Hidden7's Size	Hidden8's Size	Hidden9's Size	
256				
384				
			"HEAD"	CRC(4B)

Figure 8.10 Header Information of SD and eMMC cards

The structure of SD boot partition is shown in the figure below.

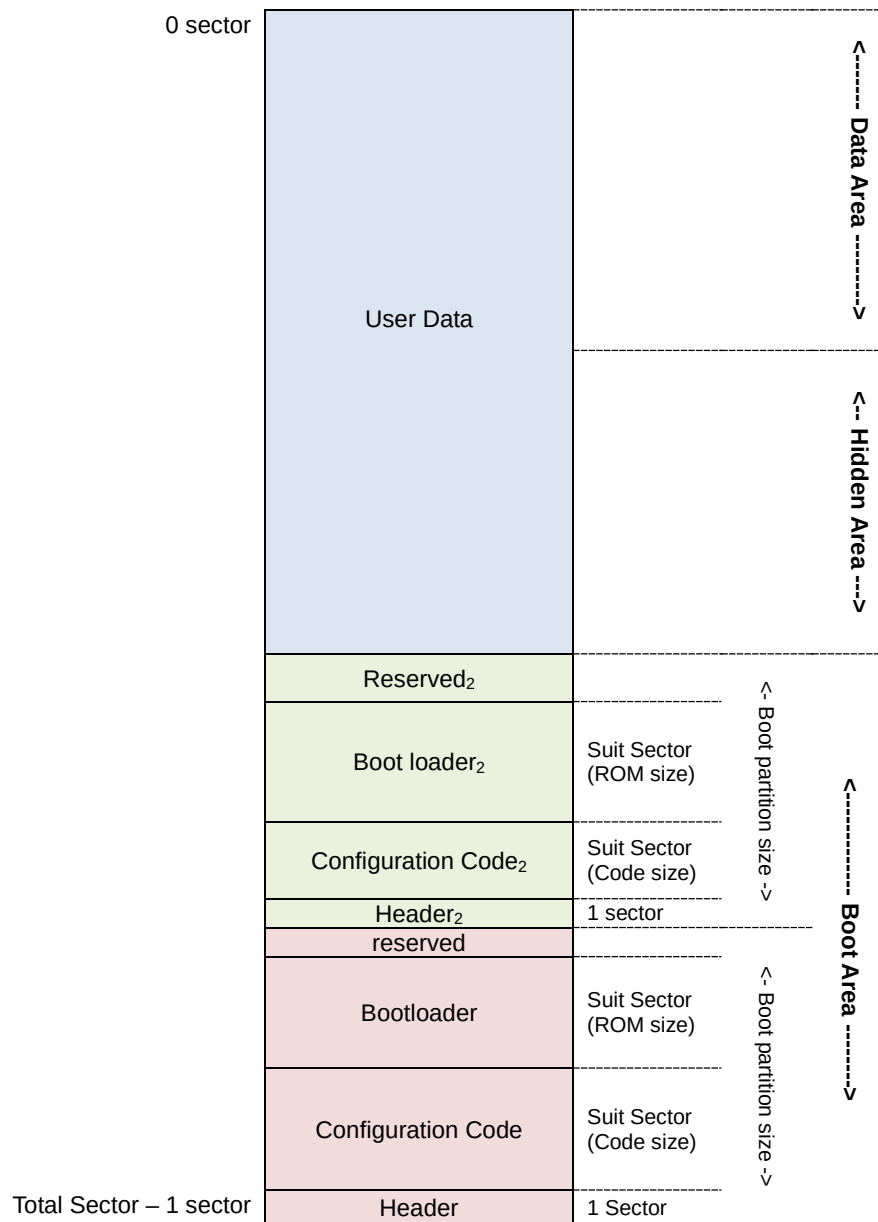


Figure 8.11 Structure of SD Boot Partition

The structure of eMMC boot partition is shown in the figure below.

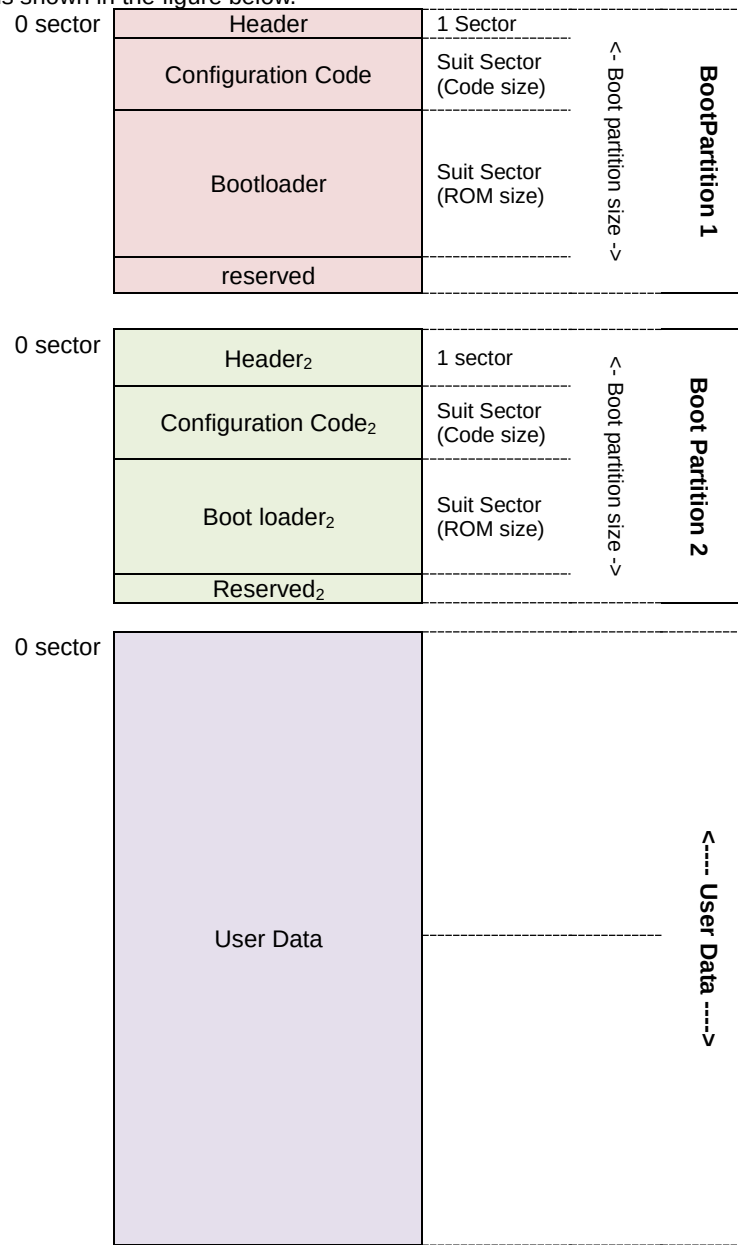


Figure 8.12 Structure of eMMC Boot Partition

8.6 USB Boot

This mode is mainly used for firmware upgrade. In this mode, a user can download a program into the user defined area.

The procedure of this mode is as follows.

- It waits until USB connection is established.
- Once it is connected, host transfers first the parameter for USB loader routine including destination address and the amount of data to be transferred (with a unit of packet).
- The CPU starts communicating fixed amount of data which is called packet with a host PC.
- At every successful reception of packet, it copies them where the destination address pointed, and after all amounts of data have been copied, it starts program where the start address pointed.

Normally, the program downloaded is for writing user system firmware to non-volatile memory like NOR or NAND flash.

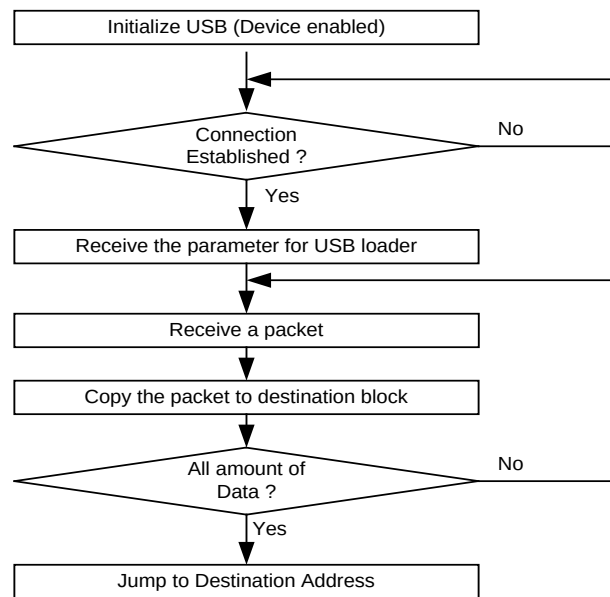


Figure 8.13 USB Boot Procedure

9 JTAG Interface

The JTAG Interface of the TCC8132 supports two independent modes:

- Multi-ICE interface for the CPU debug mode
- IEEE1149.1 Boundary Scan mode

As illustrated in the figure below, each mode has its own TAP controller and can be selected by TJDBG pin. The TAP controller in the ARM926EJ-S core is not compliant to the IEEE1149.1 Boundary Scan standard.

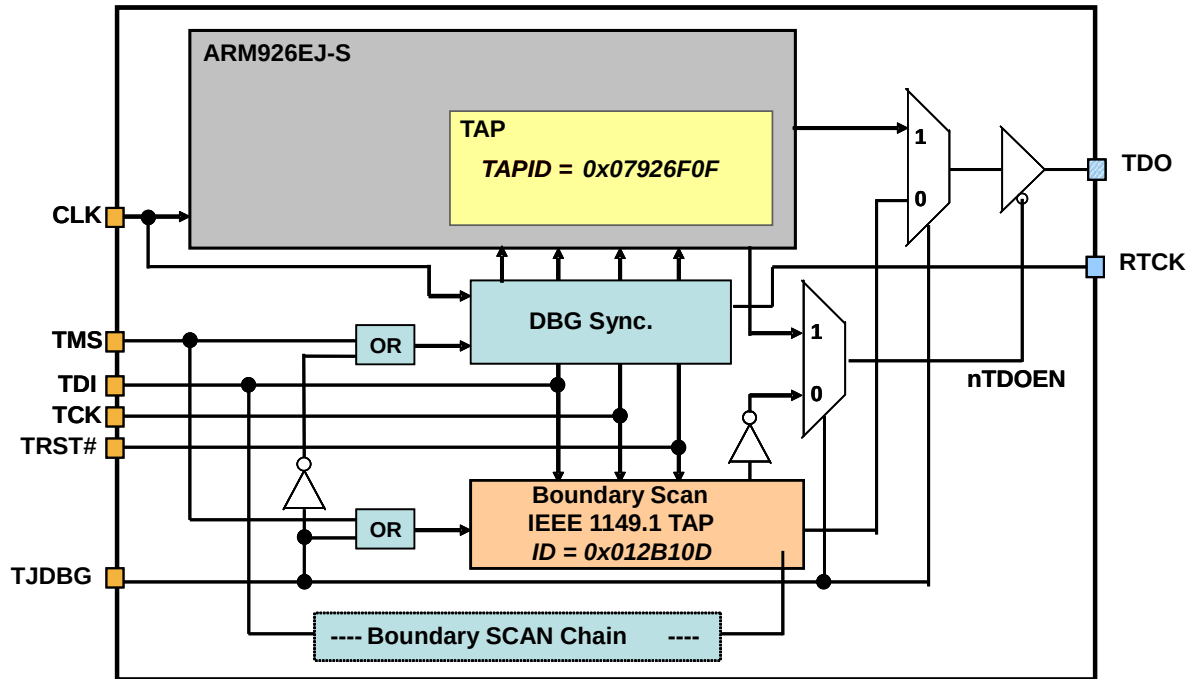


Figure 9.1 JTAG Interface Logic Connections

9.1 JTAG Debug Mode

Figure 9.2 shows the application circuit for JTAG Debug interface. Care must be taken not to combine system reset with JTAG reset signal. If you want to design the application board without the pull-up resistors, the interface circuits between JTAG connector and the TCC8132 should be designed to the following guidance. The interface circuit between the TCC8132 and JTAG connector works as level-shifter between two different powers. If you can design with the single power, this circuit does not need to work as level-shifter. The output drivers to the TCC8132 in the interface circuit have the operating power as VDDK and the output drivers to the JTAG connector have the operating power as VDD_JTAG. The grounds for each power can be shorted.

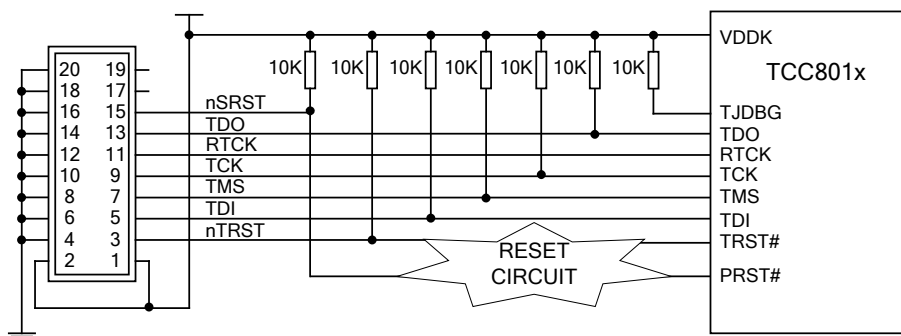


Figure 9.2 JTAG Debug Interface Circuit Diagram

Note that TJDBG pin must be pulled-up to select JTAG debug mode.

9.2 JTAG Boundary Scan Mode

The TCC8132 supports JTAG Boundary Scan compliant to IEEE Standard 1149.1. To enable JTAG Boundary Scan Mode, the TJDBG and PKG[1:0] pins must be tied to ground.

9.2.1 Supported Instructions

The TCC8132 supports only the mandatory instructions of IEEE Standard 1149.1. These instructions are used to interconnect testing in board-level, but not for internal logic test.

- BYPASS
- SAMPLE/PRELOAD
- EXTEST
- IDCODE

9.2.2 ID Code

The ID code of the TCC8132 is 0x012B10D (00000001001010110001000010011101). It can be read by IDCODE command of TAP controller.

Bit	Name	Value
31:28	Version (4 bit)	0000
27:12	Part Number (16 bit)	0001001010110001
11:8	Continuation (4 bit)	0000
7:1	JEDEC Code (7 bit)	1001110
0	Header (1 bit)	1

9.2.3 Boundary Scan Chain

The figure below shows an example structure of boundary scan design that complies with IEEE Standard 1149.1.

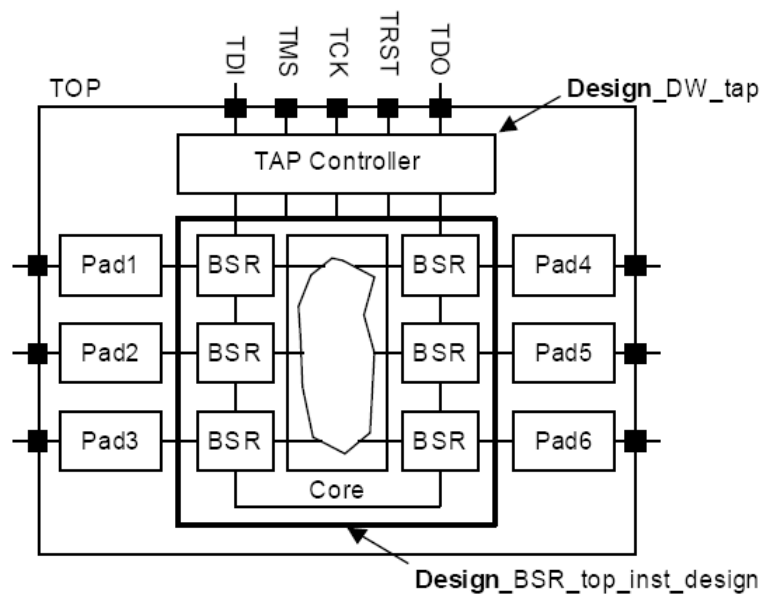


Figure 9.3 Structure of Boundary SCAN Design

The boundary scan logic of the TCC8132 has three types of BSR(Boundary Scan Register), that is, BSR1, BSR2, and BSR4. The BSR1 is used for output pins and output pins of bidirectional pads. The BSR2 is used for control pins of bidirectional pads And the BSR4 is used for input pins and input pins of bidirectional pads. The Figure 9.4, Figure 9.5, Figure 9.6, and Table 9.1 show the internal logic diagram of BSRs and scan chain order.

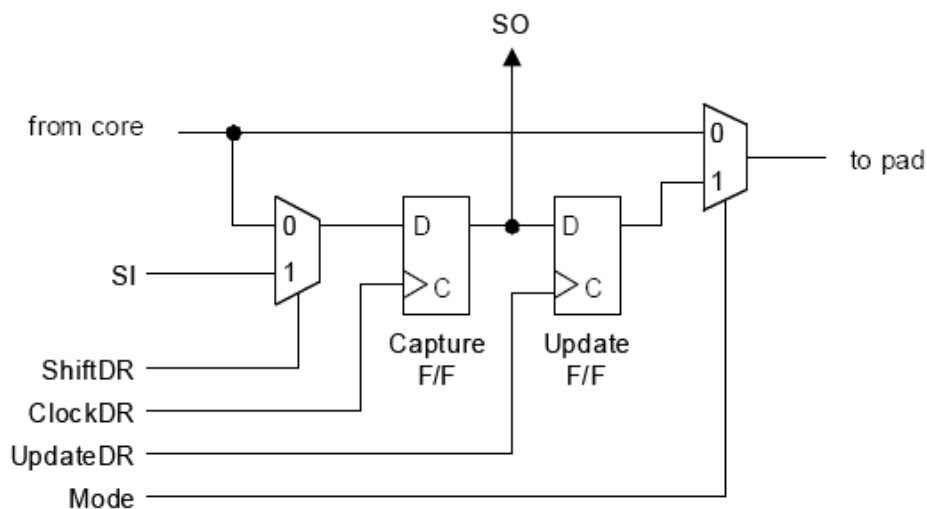


Figure 9.4 BSR1

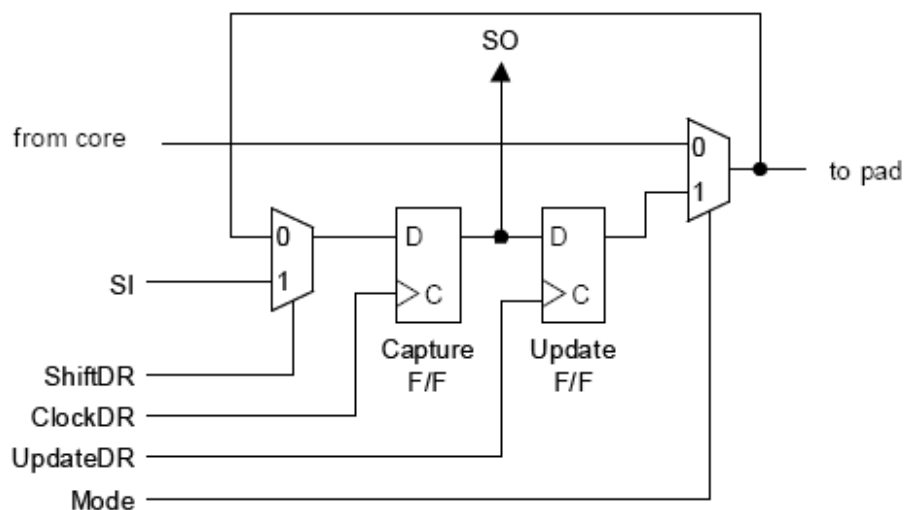


Figure 9.5 BSR2

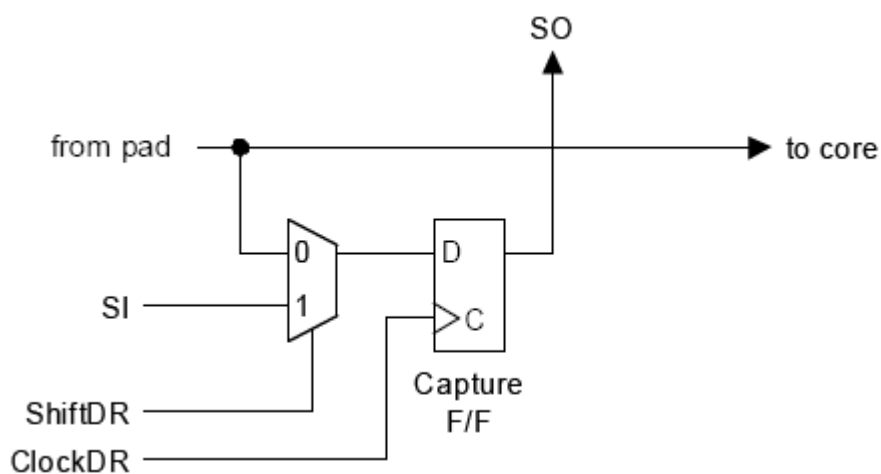


Figure 9.6 BSR4

Table 9.1 Boundary Scan Chain Pin List

Order	Signal	Used BSR	Order	Signal	Used BSR	Order	Signal	Used BSR	Order	Signal	Used BSR
1	XD[16]	4, 2, 1	67	XA[13]	4, 2, 1	133	PC[25]	4, 2, 1	199	PD[23]	4, 2, 1
2	XD[17]	4, 2, 1	68	XA[14]	4, 2, 1	134	PC[26]	4, 2, 1	200	PU[0]	4, 2, 1
3	XD[18]	4, 2, 1	69	XA[15]	4, 2, 1	135	PC[27]	4, 2, 1	201	PU[1]	4, 2, 1
4	XD[19]	4, 2, 1	70	XA[16]	4, 2, 1	136	ADIN[0]	4, 2, 1	202	PU[2]	4, 2, 1
5	XD[20]	4, 2, 1	71	XA[17]	4, 2, 1	137	ADIN[1]	4, 2, 1	203	PU[3]	4, 2, 1
6	XD[21]	4, 2, 1	72	XA[18]	4, 2, 1	138	ADIN[2]	4, 2, 1	204	PU[4]	4, 2, 1
7	XD[22]	4, 2, 1	73	XA[19]	4, 2, 1	139	ADIN[3]	4, 2, 1	205	PU[5]	4, 2, 1
8	XD[23]	4, 2, 1	74	XA[20]	4, 2, 1	140	ADIN[4]	4, 2, 1	206	PU[6]	4, 2, 1
9	XD[24]	4, 2, 1	75	XA[21]	4, 2, 1	141	ADIN[5]	4, 2, 1	207	PU[7]	4, 2, 1
10	XD[25]	4, 2, 1	76	XA[22]	4, 2, 1	142	ADIN[6]	4, 2, 1	208	PU[8]	4, 2, 1
11	XD[26]	4, 2, 1	77	XA[23]	4, 2, 1	143	ADIN[7]	4, 2, 1	209	PU[9]	4, 2, 1
12	XD[27]	4, 2, 1	78	LC[0]	4, 2, 1	144	PS[0]	4, 2, 1	210	PU[10]	4, 2, 1
13	XD[28]	4, 2, 1	79	LC[1]	4, 2, 1	145	PS[1]	4, 2, 1	211	PU[11]	4, 2, 1
14	XD[29]	4, 2, 1	80	LC[2]	4, 2, 1	146	PS[2]	4, 2, 1	212	PU[12]	4, 2, 1
15	XD[30]	4, 2, 1	81	LC[3]	4, 2, 1	147	PS[3]	4, 2, 1	213	PU[13]	4, 2, 1
16	XD[31]	4, 2, 1	82	LC[4]	4, 2, 1	148	PS[4]	4, 2, 1	214	PU[14]	4, 2, 1
17	XD[0]	4, 2, 1	83	LC[5]	4, 2, 1	149	PS[5]	4, 2, 1	215	PU[15]	4, 2, 1
18	XD[1]	4, 2, 1	84	LD[0]	4, 2, 1	150	PS[6]	4, 2, 1	216	PU[16]	4, 2, 1
19	XD[2]	4, 2, 1	85	LD[1]	4, 2, 1	151	PS[7]	4, 2, 1	217	PU[17]	4, 2, 1
20	XD[3]	4, 2, 1	86	LD[2]	4, 2, 1	152	PS[8]	4, 2, 1	218	PU[18]	4, 2, 1
21	XD[4]	4, 2, 1	87	LD[3]	4, 2, 1	153	PS[9]	4, 2, 1	219	PU[19]	4, 2, 1
22	XD[5]	4, 2, 1	88	LD[4]	4, 2, 1	154	PS[10]	4, 2, 1	220	PU[20]	4, 2, 1
23	XD[6]	4, 2, 1	89	LD[5]	4, 2, 1	155	PS[11]	4, 2, 1	221	PU[21]	4, 2, 1
24	XD[7]	4, 2, 1	90	LD[6]	4, 2, 1	156	PS[12]	4, 2, 1	222	PU[22]	4, 2, 1
25	XD[8]	4, 2, 1	91	LD[7]	4, 2, 1	157	PS[13]	4, 2, 1	223	PU[23]	4, 2, 1
26	XD[9]	4, 2, 1	92	LD[8]	4, 2, 1	158	PS[14]	4, 2, 1	224	BM[0]	4
27	XD[10]	4, 2, 1	93	LD[9]	4, 2, 1	159	PS[15]	4, 2, 1	225	BM[1]	4
28	XD[11]	4, 2, 1	94	LD[10]	4, 2, 1	160	PS[16]	4, 2, 1	226	BM[2]	4
29	XD[12]	4, 2, 1	95	LD[11]	4, 2, 1	161	PS[17]	4, 2, 1	227	BM[3]	4
30	XD[13]	4, 2, 1	96	LD[12]	4, 2, 1	162	PS[18]	4, 2, 1	228	PK[0]	4, 2, 1
31	XD[14]	4, 2, 1	97	LD[13]	4, 2, 1	163	PS[19]	4, 2, 1	229	PK[1]	4, 2, 1
32	XD[15]	4, 2, 1	98	LD[14]	4, 2, 1	164	PS[20]	4, 2, 1	230	PK[2]	4, 2, 1
33	XWE#	4, 2, 1	99	LD[15]	4, 2, 1	165	PS[21]	4, 2, 1	231	PK[3]	4, 2, 1
34	XOE#	4, 2, 1	100	LD[16]	4, 2, 1	166	PS[22]	4, 2, 1	232	PK[4]	4, 2, 1
35	XSCS[0]#	4, 2, 1	101	LD[17]	4, 2, 1	167	PS[23]	4, 2, 1	233	PK[5]	4, 2, 1
36	XSCS[1]#	4, 2, 1	102	LD[18]	4, 2, 1	168	PS[24]	4, 2, 1	234	PK[6]	4, 2, 1
37	XRCS#	4, 2, 1	103	LD[19]	4, 2, 1	169	PS[25]	4, 2, 1	235	PK[7]	4, 2, 1
38	XDCS[0]#	4, 2, 1	104	LD[20]	4, 2, 1	170	PS[26]	4, 2, 1	236	PK[8]	4, 2, 1
39	XDCS[1]#	4, 2, 1	105	LD[21]	4, 2, 1	171	PS[27]	4, 2, 1	237	PK[9]	4, 2, 1
40	XDCLK	4, 2, 1	106	LD[22]	4, 2, 1	172	PS[28]	4, 2, 1	238	PK[10]	4, 2, 1
41	XDCKE	4, 2, 1	107	LD[23]	4, 2, 1	173	PS[29]	4, 2, 1	239	PK[11]	4, 2, 1
42	FC[0]	4, 2, 1	108	PC[0]	4, 2, 1	174	PS[30]	4, 2, 1	240	PK[12]	4, 2, 1
43	FC[1]	4, 2, 1	109	PC[1]	4, 2, 1	175	PS[31]	4, 2, 1	241	PK[13]	4, 2, 1
44	FC[2]	4, 2, 1	110	PC[2]	4, 2, 1	176	PD[0]	4, 2, 1	242	PK[14]	4, 2, 1
45	FC[3]	4, 2, 1	111	PC[3]	4, 2, 1	177	PD[1]	4, 2, 1	243	PK[15]	4, 2, 1
46	FC[4]	4, 2, 1	112	PC[4]	4, 2, 1	178	PD[2]	4, 2, 1	244	PK[16]	4, 2, 1
47	FC[5]	4, 2, 1	113	PC[5]	4, 2, 1	179	PD[3]	4, 2, 1	245	PK[17]	4, 2, 1
48	FC[6]	4, 2, 1	114	PC[6]	4, 2, 1	180	PD[4]	4, 2, 1	246	PK[18]	4, 2, 1
49	FC[7]	4, 2, 1	115	PC[7]	4, 2, 1	181	PD[5]	4, 2, 1	247	PK[19]	4, 2, 1
50	FC[8]	4, 2, 1	116	PC[8]	4, 2, 1	182	PD[6]	4, 2, 1	248	PK[20]	4, 2, 1
51	FC[9]	4, 2, 1	117	PC[9]	4, 2, 1	183	PD[7]	4, 2, 1	249	PK[21]	4, 2, 1
52	FC[10]	4, 2, 1	118	PC[10]	4, 2, 1	184	PD[8]	4, 2, 1	250	PK[22]	4, 2, 1
53	FC[11]	4, 2, 1	119	PC[11]	4, 2, 1	185	PD[9]	4, 2, 1	251	PK[23]	4, 2, 1
54	XA[0]	4, 2, 1	120	PC[12]	4, 2, 1	186	PD[10]	4, 2, 1	252	PK[24]	4, 2, 1
55	XA[1]	4, 2, 1	121	PC[13]	4, 2, 1	187	PD[11]	4, 2, 1	253	PK[25]	4, 2, 1
56	XA[2]	4, 2, 1	122	PC[14]	4, 2, 1	188	PD[12]	4, 2, 1	254	PK[26]	4, 2, 1
57	XA[3]	4, 2, 1	123	PC[15]	4, 2, 1	189	PD[13]	4, 2, 1	255	PK[27]	4, 2, 1
58	XA[4]	4, 2, 1	124	PC[16]	4, 2, 1	190	PD[14]	4, 2, 1	256	PK[28]	4, 2, 1
59	XA[5]	4, 2, 1	125	PC[17]	4, 2, 1	191	PD[15]	4, 2, 1	257	PK[29]	4, 2, 1
60	XA[6]	4, 2, 1	126	PC[18]	4, 2, 1	192	PD[16]	4, 2, 1	258	PK[30]	4, 2, 1
61	XA[7]	4, 2, 1	127	PC[19]	4, 2, 1	193	PD[17]	4, 2, 1	259	PK[31]	4, 2, 1
62	XA[8]	4, 2, 1	128	PC[20]	4, 2, 1	194	PD[18]	4, 2, 1	260	PK[32]	4, 2, 1

Order	Signal	Used BSR	Order	Signal	Used BSR	Order	Signal	Used BSR	Order	Signal	Used BSR
63	XA[9]	4, 2, 1	129	PC[21]	4, 2, 1	195	PD[19]	4, 2, 1	261	PK[33]	4, 2, 1
64	XA[10]	4, 2, 1	130	PC[22]	4, 2, 1	196	PD[20]	4, 2, 1	262	PK[34]	4, 2, 1
65	XA[11]	4, 2, 1	131	PC[23]	4, 2, 1	197	PD[21]	4, 2, 1	263	RTCK	4, 2, 1
66	XA[12]	4, 2, 1	132	PC[24]	4, 2, 1	198	PD[22]	4, 2, 1			

9.3 Treatment of JTAG Ports in Mass Production Board

The TCC8132 is a CMOS device. If the JTAG interface is not used, JTAG input mode pins must not be left unconnected. Treat the JTAG related pins according to the table below.

Pin Name	Description
TRST#	N.C.
TCK	N.C.
TMS	N.C.
TDI	N.C.
TJDBG	Must be tied to ground via an external resistor.

10 Clock Controller

10.1 Overview

The CKC (Clock Controller) block has 6 primary clock sources from PLL0, PLL1, PLL2, PLL3, XI and XTI. The PLL clock sources, XI and XTI can generate divided clocks. These 6 primary clock sources and 6 divided clock sources are used to generate the CPU clock, bus clock, and peripheral clocks.

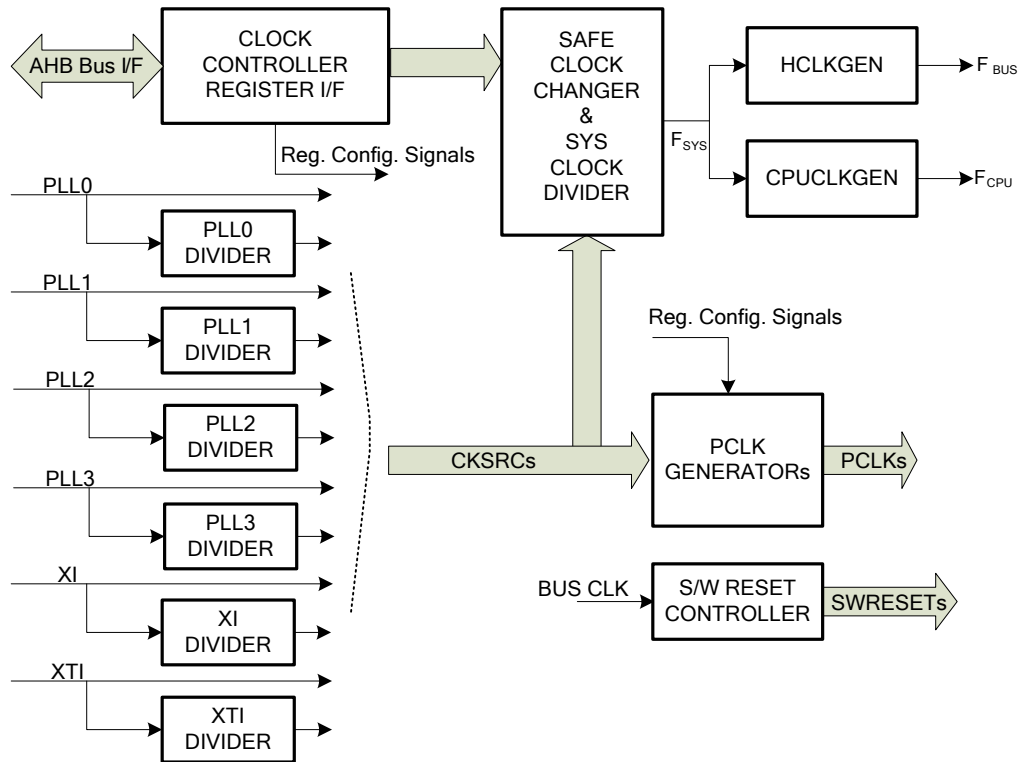


Figure 10.1 CKC Block Diagram

10.2 Register Description

Table 10.1 CKC Register Map (Base Address = 0x98006000)

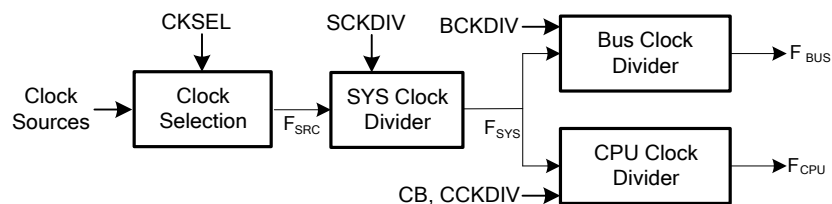
Name	Address	Type	Reset	Description
CLKCTRL	0x000	R/W	0x80000004	CPU & Bus Clock Control Register
PLL0CFG	0x004	R/W	0x80016003	PLL0 Configuration Register
PLL1CFG	0x008	R/W	0x80016003	PLL1 Configuration Register
CLKDIVC0	0x00C	R/W	0x03030303	Divided Clock Configuration Register 0
MODECTR	0x010	R/W	0x00000000	Reserved
BCLKCTR0	0x014	R/W	0x5BFFFFFF	Bus Clock Enable Register 0
SWRESET0	0x018	R/W	0x00000000	Software Reset Register 1
PCLKCFGx	0x01C ~ 0x05C	R/W	-	Reserved (TCC813x Peri-Clock Registers)
BCLKCTR1	0x060	R/W	0x001FFFFFF	Bus Clock Enable Register 1
SWRESET1	0x064	R/W	0xFFE00000	Software Reset Register 1
PWDCTL	0x068	R/W	0x00000000	Power Down Control Register
PLL2CFG	0x06C	R/W	0x80016003	PLL2 Configuration Register
CLKDIVC1	0x070	R/W	0x00000003	Divided Clock Configuration Register 1
PLL3CFG	0x074	R/W	0x80219003	PLL3 Configuration Register
ACLKREF	0x080	R/W	0x140000BA	SDRAM Refresh Clock Register
ACLKI2C	0x084	R/W	0x00000000	I2C Clock Register
ACLKSFMC	0x088	R/W	0x00000000	Serial Flash Clock Register
ACLKRSVD0	0x08C	R/W	0x00000000	Reserved
ACLKUART0	0x090	R/W	0x00000000	UART0 Clock Register
ACLKUART1	0x094	R/W	0x00000000	UART1 Clock Register
ACLKUART2	0x098	R/W	0x00000000	UART2 Clock Register
ACLKUART3	0x09C	R/W	0x00000000	UART3 Clock Register
ACLKUART4	0x0A0	R/W	0x00000000	UART4 Clock Register
ACLKTCT	0x0A4	R/W	0x00000000	Timer T-Clock Register
ACLKTCX	0x0A8	R/W	0x14000000	Timer X-Clock (WDT) Register
ACLKTCZ	0x0AC	R/W	0x00000000	Timer Z-Clock (TC32) Register
ACLKADC	0x0B0	R/W	0x00000000	ADC Clock Register
ACLKDAI0	0x0B4	R/W	0x00000000	DAI0 Clock Register
ACLKDAI1	0x0B8	R/W	0x00000000	DAI1 Clock Register
ACLKLCD	0x0BC	R/W	0x00000000	LCD Clock Register
ACLKSPDIF	0x0C0	R/W	0x00000000	SPDIF Clock Register
ACLKRSVD1	0x0C4	R/W	0x00000000	Reserved
ACLKSDH0	0x0C8	R/W	0x00000000	SD Host 0 Clock Register
ACLKSDH1	0x0CC	R/W	0x00000000	SD Host 1 Clock Register
ACLKLCDTC	0x0D0	R/W	0x00000000	LCDC Timer Clock Register
ACLKICTC0	0x0D4	R/W	0x00000000	IC_TC 0 Clock Register
ACLKCAN0	0x0D8	R/W	0x00000000	CAN0 Clock Register
ACLKCAN1	0x0DC	R/W	0x00000000	CAN1 Clock Register
ACLKGSB0	0x0E0	R/W	0x00000000	GSB0 Clock Register
ACLKGSB1	0x0E4	R/W	0x00000000	GSB1 Clock Register
ACLKGSB2	0x0E8	R/W	0x00000000	GSB2 Clock Register
ACLKGSB3	0x0EC	R/W	0x00000000	GSB3 Clock Register
ACLKGSB4	0x0F0	R/W	0x00000000	GSB4 Clock Register
ACLKGSB5	0x0F4	R/W	0x00000000	GSB5 Clock Register
ACLKDAI2	0x0F8	R/W	0x00000000	DAI2 Clock Register
ACLKICTC1	0x0FC	R/W	0x00000000	IC_TC 1 Clock Register
ACLKICTC2	0x100	R/W	0x00000000	IC_TC 2 Clock Register
ACLKASRC2	0x104	R/W	0x00000000	ASRC_REF2 Clock Register
ACLKPDM	0x108	R/W	0x00000000	PDM Clock Register
ACLKCIF0	0x10C	R/W	0x00000000	CIF0 Clock Register
ACLKCIF1	0x110	R/W	0x00000000	CIF1 Clock Register
ACLKASRC0	0x114	R/W	0x00000000	ASRC_REF0 Clock Register
ACLKASRC1	0x118	R/W	0x00000000	ASRC_REF1 Clock Register
ACLKLCDSI	0x11C	R/W	0x00000000	LCDSI Clock Register
PLL0_MISC	0x1A8	R/W	0x00000000	PLL0_MISC Configuration Register
PLL1_MISC	0x1AC	R/W	0x00000000	PLL1_MISC Configuration Register
PLL3_MISC	0x1B0	R/W	0x00000000	PLL3_MISC Configuration Register
PLL2_MISC0	0x1B4	R/W	0x00000000	PLL2_MISC Configuration Register 0
PLL2_MISC1	0x1B8	R/W	0x00000000	PLL2_MISC Configuration Register 1

BUS Clock Control Register (CLKCTRL)

0x98006000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
XE	0	CB		0				SCKDIV[5:0]					CCKDIV[3:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0				BCKDIV[7:0]								CKSEL[3:0]			

Bit	Name	Type	Default	Description
31	XE	R/W	1	XI Oscillator Enable Bit 0 = Disable 1 = Enable
30	-	R	0	Reserved
29	CB	R/W	0	CPU Clock(F_{CPU}) Selection Bit If set to “1”, $F_{CPU} = F_{BUS}$
28:26	-	R	0	Reserved
25:20	SCKDIV[5:0]	R/W	0	SYS Clock(F_{SYS}) Divider $F_{SYS} = F_{SRC} (SCKDIV + 1)$ If non-zero value is written, Source Clock is divided before fed to CPU & Bus clock generation logic. - Value of (SCKDIV+1) is even.
19:16	CCKDIV[3:0]	R/W	0xF	CPU Clock(F_{CPU}) Divider $F_{CPU} = F_{BUS} + ((F_{SYS} - F_{BUS}) * (CCKDIV + 1)) / 16$
15:12	-	R	0	Reserved
11:4	BCKDIV[7:0]	R/W	0	Bus Clock(F_{BUS}) Divider $F_{BUS} = F_{SYS} / (BCLKDIV + 1)$
3:0	CKSEL	R/W	0x4	Clock Source(F_{SRC}) Selection Bits



Note: Regardless of the CCKDIV value, F_{SYS} should not exceed the maximum frequency of the CPU.

Clock Change Sequence

If BCKDIV and CKSEL/SCLKDIV are written at the same time, F_{SYS} may output to F_{BUS} 2 ~ 3 clock period. It means that the intermediate time interval can exist which the bus clock exceeds the maximum operating frequency. To avoid malfunction caused by this abnormal behavior, make sure to follow the clock change sequence below.

When targeting F_{SYS} is higher than the maximum bus frequency (F_{BUS}), modify BCKDIV first and then modify CKSEL and/or SCLKDIV.

If targeting F_{SYS} is less than or equal to the maximum F_{BUS} , all the bits in CLKCTRL register can be written at the same time.

PLL0 Configuration Register (PLL0CFG)

0x98006004

PLL1 Configuration Register (PLL1CFG)

0x98006008

PLL2 Configuration Register (PLL2CFG)

0x9800606C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PD	0					FIN	BP	0					S[2:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
M[9:0]										P[5:0]					

Bit	Name	Type	Default	Description
31	PD	R/W	1	PLL Power Down. 0=Power Up 1=Power Down
30:26	-	R	0	Reserved
25	FIN	R/W	0	Reserved. This bit must be written as 0 for normal operation.
24	BP	R/W	0	Bypass enable. This bit must be written as 0 for normal operation.
23:19	-	R	0	Reserved
18:16	S	R/W	0	PLL Post-Divisor (0 ≤ S ≤ 5)
15:6	M	R/W	0xXX	PLL Main-Divisor (PLL0,1 : 16 ≤ M ≤ 1023) (PLL2 : 23 ≤ M ≤ 511)
5:0	P	R/W	0xXX	PLL Pre-Divisor (1 ≤ P ≤ 63)

Note:

- PLL0, 1 : $F_{VCO} = (M * FIN) / (P * 2^S)$: 32MHz ~ 2000MHz
- PLL0, 1 : $F_{REF} = (FIN) / (P)$: 1MHz ~ 10MHz
- PLL2 : $F_{VCO} = ((M+k/1024) * FIN) / (P * 2^S)$: 11MHz ~ 660MHz
- PLL2 : $F_{REF} = (FIN) / (P)$: 2MHz ~ 30MHz

PLL3 Configuration Register (PLL3CFG)

0x98006074

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PD	0					FIN	BP	0	S[2:0]			0		M[9:8]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
M[9:0]								0		P[5:0]					

Bit	Name	Type	Default	Description
31	PD	R/W	1	PLL Power Down 0=Power Up 1=Power Down
30:26	-	R	0	Reserved
25	FIN	R/W	0	Reserved. This bit must be written as 0 for normal operation.
24	BP	R/W	0	Bypass enable. This bit must be written as 0 for normal operation.
23	-	R	0	Reserved
22:20	S	R/W	0	PLL Post-Divisor (0 ≤ S ≤ 5)
17:8	M	R/W	0xXXX	PLL Main-Divisor (64 ≤ M ≤ 1023)
5:0	P	R/W	0xXX	PLL Pre-Divisor (1 ≤ P ≤ 63)

Note:

- $F_{VCO} = (M * FIN) / (P * 2^S)$: 32MHz ~ 2000MHz
- $F_{REF} = (FIN) / (P)$: 1MHz ~ 10MHz

Caution:

Not all the P/M/S values derived from the equations above are usable.

For guaranteed P/M/S values, refer to chapter 10.4 Recommended PLL Parameter Values. Using any P/M/S value not listed in the chapter may cause system malfunction.

PLL Locking Time

Whenever any register value is updated, PLL needs a period called “Locking Time” to generate a target frequency. Actually PLL generates an unknown frequency during the “Locking Time”. The PLL clock path should be disabled by the CPU during “Locking Time” to avoid malfunction caused by the abnormal frequency.

The PLL Locking time is 300μs.

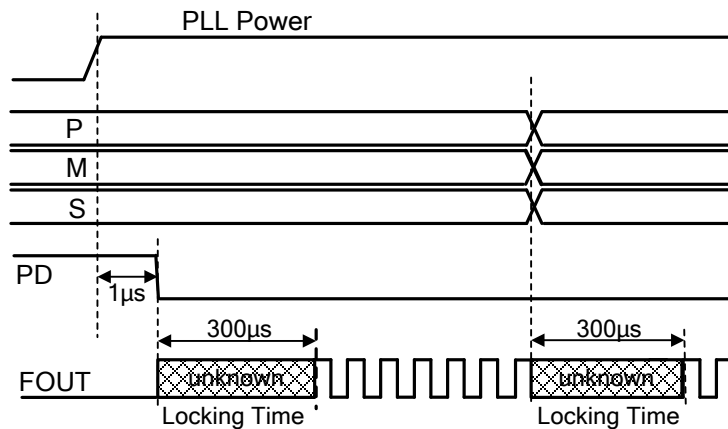


Figure 10.2 PLL Timing Diagram

PLL0 MISC Register (PLL0MISC)

0x980061A8

PLL1 MISC Register (PLL1MISC)

0x980061AC

PLL3 MISC Register (PLL3MISC)

0x980061B0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								0							
15								0							
0								0							

Bit	Name	Type	Default	Description
31:25	-	R	0	Reserved
24	LOCK_EN	R/W	0	Enable pin for lock detector (Active-high)
23:17	-	R	0	Reserved
16	FVCO_EN	R/W	0	Enable pin for FVCO_OUT (Active-high) Default setting is 0
15:9	-	R	0	Reserved
8	FSEL	R/W	0	Pin selection for monitoring purposes and others. Default setting is FSEL = 0.
7:1	-	R	0	Reserved
0	VSEL	R/W	0xX	- VSEL = 0 for lower VCO range- VSEL = 1 for higher VCO range. 0 : FVCO0 : 1.0GHz ~ 1.4GHz 1 : FVCO1 : 1.4GHz ~ 2.0GHz

PLL2 MISC0 Register (PLL2MISC0)

0x980061B4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MFR						0	MRR					0	SEL PF		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
K												0	SSCG VSEL		

Bit	Name	Type	Default	Description
31:26	MFR	R/W	6'h0	Value of 6-bit modulation frequency control
25	-	R	0	Reserved
24:20	MRR	R/W	5'h15	Value of 5-bit modulation rate control
19:18	-	R	0	Reserved
17:16	SEL_PF	R	2'h2	Value of 2-bit modulation method control 00: down spread 01: up spread 1x: center spread
15:4	K	R/W	12'h2AB	Value of 12-bit DSM (Delta-Sigma Modulator)
3	-	R	0	Reserved
1	SSCG_EN	R/W	0	If SSCG_EN = 1, dithered mode is enabled If SSCG_EN = 0, dithered mode is disabled.
0	VSEL	R/W	1	VSEL selects VCO frequency range. - VSEL = 0 for low VCO range Range : 330MHz ~ 460MHz - VSEL = 1 for high VCO range. Range : 460MHz ~ 660MHz

PLL2 MISC1 Register (PLL2MISC1)

0x980061B8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0													LOCK		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0							EXTAFC					0		AFC_ENB	

Bit	Name	Type	Default	Description
31:19	-	R	0	Reserved
18:16	LOCK	R	0x0	PLL LOCK monitor Bit[0] : PLL0 Bit[1] : PLL1 Bit[2] : PLL3
3:1	-	R	0	Reserved
8:4	EXTAFC	R/W	0	- If AFC_ENB is equal to 0 (mandatory), AFC is enabled and VCO is automatically calibrated. - If AFC_ENB is equal to 1, AFC is disabled and VCO is calibrated manually by EXTAFC[4:0] for test of VCO range. AFC disabled mode is not allowed for the product applications.
0	AFC_ENB	R/W	0	

Divider Control Register 0 (CLKDIVC0)

0x9800600C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
P0E	0	P0DIV[5:0]						P1E	0	P1DIV[5:0]					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XE	0	XDIV[5:0]						XTE	0	XTDIV[5:0]					

P0E/P1E	[31, 23]	PLL 0/1 Divider Enable
0		PLL 0/1 Divider Disabled
1		PLL 0/1 Divider Enabled

P0DIV[5:0] P1DIV[5:0]	[29:24] [21:16]	PLL 0 Divisor PLL 1 Divisor
0		Divider Disabled
1 ~ 31		$F_{P0DIV} = F_{PLL0} / (P0DIV+1)$ $F_{P1DIV} = F_{PLL1} / (P1DIV+1)$

XE	[15]	XIN Divider Enable
0		XI Divider Disabled
1		XI Divider Enabled

XDIV[5:0]	[13:8]	XTIN Divisor
0		Divider Disabled
1 ~ 31		$F_{XDIV} = F_{XI} / (XDIV+1)$

XTE	[7]	XTIN Divider Enable
0		XTI Divider Disabled
1		XTI Divider Enabled

XTDIV[5:0]	[5:0]	XTIN Divisor
0		Divider Disabled
1 ~ 31		$F_{XTDIV} = F_{XTI} / (XTDIV+1)$

Divider Control Register 1 (CLKDIVC1)

0x98006070

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
P3E	0	P3DIV[5:0]						P2E	0	P2DIV[5:0]					

P3E/P2E	[15, 7]	PLL 3/2 Divider Enable
0	PLL 3/2 Divider Disabled	
1	PLL 3/2 Divider Enabled	

P3DIV[5:0] P2DIV[5:0]	[13:8] [5:0]	PLL 3 Divisor PLL 2 Divisor
0	Divider Disabled	
1 ~ 31	$F_{P3DIV} = F_{PLL3} / (P3DIV+1)$ $F_{P2DIV} = F_{PLL2} / (P2DIV+1)$	

MODE Control Register (MODECTR)

0x98006010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0															

Note: This register is reserved in the TCC813x. Do not write to this register.

Bus Clock Enable Register 0 (BCLKCTR0)

0x98006014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICTC1	ICTC0					SFMC	UART1	GDMA1		ASDMA		G2D	SD0	NFC
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RTC		LCD	GDMA0	USBH0	ADC		UART3	I2C	UART2	UART0	USBO	GPIO	TC	PIC	DAI0

Bus Clock Enable Register 1 (BCLKCTR1)

0x98006060

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					ADMA2		DAI2	GSB5	GSB4					GDMA2	GSB3
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GSB2	GSB1	GSB0		CAN1	CAN0	CIPHER	ADMA1	DAI1	UART4	SD1	ECID	SCFG	SPDIF	ADMA0	ICTC2

BCLKEN0 BCLKEN1	[31:0] [31:0]	Bus Clock Enable Bits
0 (for each bit)		Disable the Bus Clock for Corresponding Controller
1 (for each bit)		Enable the Bus Clock for Corresponding Controller

BCLKCTR0			BCLKCTR1		
Bit	Name	Block	Bit	Name	Block
0	DAI0	DAI Controller 0	0	IC TC2	IC TC 2
1	PIC	Interrupt Controller	1	ADMA0	Audio DMA Controller 0
2	TC	Timer/Counter	2	SPDIF	SPDIF Tx Controller
3	GPIO	GPIO Controller	3	SCFG	System CFG
4	USBO	USB 20DH Controller	4	ECID	ECID Interface
5	UART0	UART Controller 0	5	SD1	SD/MMC Controller 1
6	UART2	UART Controller 2	6	UART4	UART Controller 4
7	I2C	I2C Controller	7	DAI1	DAI Controller 1
8	UART3	UART Controller 3	8	ADMA1	Audio DMA Controller 1
9	-	Reserved	9	CIPHER	Cipher Controller
10	ADC	ADC Interface Controller	10	CAN0	CAN Controller 0
11	USBH0	USB Host Controller 0	11	CAN1	CAN Controller 1
12	GDMA0	GDMA Controller 0	12	-	Reserved
13	LCD	LCD Controller	13	GSB0	GSB Controller 0
14	-	Reserved	14	GSB1	GSB Controller 1
15	RTC	RTC Interface	15	GSB2	GSB Controller 2
16	NFC	NAND Flash Controller	16	GSB3	GSB Controller 3
17	SD0	SD/MMC Controller 0	17	GDMA2	GDMA Controller 2
18	G2D	G2D	18	-	Reserved
19	-	Reserved	19	-	Reserved
20	ASDMA	ASRC DMA	20	-	Reserved
21	-	Reserved	21	-	Reserved
22	GDMA1	GDMA Controller 1	22	GSB4	GSB Controller 4
23	UART1	UART Controller 1	23	GSB5	GSB Controller 5
24	SFMC	Serial Flash controller	24	DAI2	DAI Controller 2
25	-	Reserved	25	-	Reserved
26	-	Reserved	26	ADMA2	Audio DMA Controller 2
27	-	Reserved	27	-	Reserved
28	-	Reserved	28	-	Reserved
29	ICTC0	IC TC 0	29	-	Reserved
30	ICTC1	IC TC 1	30	-	Reserved
31	-	Reserved	31	-	Reserved

SoftWare Reset Register 0 (SWRESET0)

0x98006018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICTC1	ICTC0					SFMC	UART1	GDMA1		ASDMA		G2D	SD0	NFC
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RTC		LCD	GDMA0	USBH0	ADC		UART3	I2C	UART2	UART0	USBO	GPIO	TC	PIC	DAI0

SoftWare Reset Register 1 (SWRESET1)

0x98006064

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					ADMA2		DAI2	GSB5	GSB4					GDMA2	GSB3
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GSB2	GSB1	GSB0		CAN1	CAN0	CIPHER	ADMA1	DAI1	UART4	SD1	ECID	SCFG	SPDIF	ADMA0	ICTC2

SWRESET0 SWRESET1	[31:0] [31:0]	Software Reset Control Bits
0 (for each bit)	Not Reset	
1 (for each bit)	Reset	

These bits are not automatically cleared.

SWRESET0			SWRESET1		
Bit	Name	Block	Bit	Name	Block
0	DAI0	DAI Controller 0	0	IC_TC2	IC_TC 2
1	PIC	Interrupt Controller	1	ADMA0	Audio DMA Controller 0
2	TC	Timer/Counter	2	SPDIF	SPDIF Tx Controller
3	GPIO	GPIO Controller	3	SCFG	System CFG
4	USBO	USB 20DH Controller	4	ECID	ECID Interface
5	UART0	UART Controller 0	5	SD1	SD/MMC Controller 1
6	UART2	UART Controller 2	6	UART4	UART Controller 4
7	I2C	I2C Controller	7	DAI1	DAI Controller 1
8	UART3	UART Controller 3	8	ADMA1	Audio DMA Controller 1
9	-	Reserved	9	CIPHER	Cipher Controller
10	ADC	ADC Interface Controller	10	CAN0	CAN Controller 0
11	USBH0	USB Host Controller 0	11	CAN1	CAN Controller 1
12	GDMA0	GDMA Controller 0	12	-	Reserved
13	LCD	LCD Controller	13	GSB0	GSB Controller 0
14	-	Reserved	14	GSB1	GSB Controller 1
15	RTC	RTC Interface	15	GSB2	GSB Controller 2
16	NFC	NAND Flash Controller	16	GSB3	GSB Controller 3
17	SD0	SD/MMC Controller 0	17	GDMA2	GDMA Controller 2
18	G2D	G2D	18	-	Reserved
19	-	Reserved	19	-	Reserved
20	ASDMA	ASRC DMA	20	-	Reserved
21	-	Reserved	21	-	Reserved
22	GDMA1	GDMA Controller 1	22	GSB4	GSB Controller 4
23	UART1	UART Controller 1	23	GSB5	GSB Controller 5
24	SFMC	Serial Flash controller	24	DAI2	DAI Controller 2
25	-	Reserved	25	-	Reserved
26	-	Reserved	26	ADMA2	Audio DMA Controller 2
27	-	Reserved	27	-	Reserved
28	-	Reserved	28	-	Reserved
29	ICTC0	IC_TC 0	29	-	Reserved
30	ICTC1	IC_TC 1	30	-	Reserved
31	-	Reserved	31	-	Reserved

Power Down Control Register (PWDCTL)

0x98006068

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CKOP0	CKOE0	CKOSEL0[5:0]						CKOP1	CKOE1	CKOSEL1[5:0]					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								PDACK	PDREQ	PDXTI	PDXI	PDXIWS			

Bit	Name	Type	Default	Description
31	CKOP0	R	0	Clock Output Polarity 0
30	CKOE0	R/W	0	Clock Output Enable 0
29:24	CKOSEL0	R/W	0	Clock Output Select 0
23	CKOP1	R/W	0	Clock Output Polarity 1
22	CKOE1	R/W	0	Clock Output Enable 1
21:16	CKOSEL1	R/W	0	Clock Output Select 1
15:8	-	R	0	Reserved
7	PDACK	R	0	Power Down Acknowledge from SDR SDRAM controller. "1" indicates the SDR SDRAM entered self-refresh mode.
6	PDREQ	R/W	0	Power Down Request to SDR SDRAM controller. If power down acknowledge is asserted and the SDR SDRAM controller is in idle state, SDR SDRAM enters self-refresh mode.
5	PDXTI	R/W	0	Reserved
4	PDXI	R/W	0	Reserved
3:0	PDXIWS	R/W	0	Reserved

Note: Clocks selected by CKOSEL0 and CKOSEL1 are sent to external component via GPIO pins. Refer to GPIOFC_FS2 Registers in Chapter 18.3 for available pins.

CKOSEL0/1	Selected Clock	CKOSEL0/1	Selected Clock
0	ACLKREF	20	ACLKLCDC
1	ACLKI2C	21	ACLKICTC0
2	ACLKSFMC	22	ACLKCAN0
3	-	23	ACLKCAN1
4	ACLKUART0	24	ACLKGSB0
5	ACLKUART1	25	ACLKGSB1
6	ACLKUART2	26	ACLKGSB2
7	ACLKUART3	27	ACLKGSB3
8	ACLKUART4	28	ACLKGSB4
9	ACLKTCT	29	ACLKGSB5
10	ACLKTCX	30	ACLKDAI2
11	ACLKTCZ	31	ACLKICTC1
12	ACLKADC	32	ACLKICTC2
13	ACLKDAI0	33	ACLKASRC0
14	ACLKDAI1	34	ACLKPDM
15	ACLKLCD	35	ACLKCIF0
16	ACLKSPDIF	36	ACLKCIF1
17	ACLKRSVD1	37	ACLKASRC1
18	ACLKSDH0	38	ACLKASRC2
19	ACLKSDH1	39	ACLKLCDI

SDRAM Refresh Clock Register (ACLKREF)	0x98006080
I2C Clock Register (ACLKI2C)	0x98006084
SFMC Clock Register 0 (ACLKSFMC)	0x98006088
RSVD0 Register (ACLKRSVD0)	0x9800608C
UART0 Clock Register (ACLKUART0)	0x98006090
UART1 Clock Register (ACLKUART1)	0x98006094
UART2 Clock Register (ACLKUART2)	0x98006098
UART3 Clock Register (ACLKUART3)	0x9800609C
UART4 Clock Register (ACLKUART4)	0x980060A0
Timer T-Clock Register (ACLKTCT)	0x980060A4
Timer X-Clock (WDT) Register (ACLKTCX)	0x980060A8
Timer Z-Clock (TC32) Register (ACLKTCZ)	0x980060AC
ADC Clock Register (ACLKADC)	0x980060B0
LCD Clock Register (ACLKLCD)	0x980060BC
SPDIF Clock Register (ACLKSPDIF)	0x980060C0
Reserved Clock Register 1 (ACLKRSVD1)	0x980060C4
SD Host 0 Clock Register (CLKSDH0)	0x980060C8
SD Host 1 Clock Register (CLKSDH1)	0x980060CC
LCDC Timer Clock Register (CLKLCDTC)	0x980060D0
ICTC0 Register (CLKICTC0)	0x980060D4
CAN0 Clock Register (CLKCAN0)	0x980060D8
CAN1 Clock Register (CLKCAN1)	0x980060DC
GSB0 Clock Register (CLKGSB0)	0x980060E0
GSB1 Clock Register (CLKGSB1)	0x980060E4
GSB2 Clock Register (CLKGSB2)	0x980060E8
GSB3 Clock Register (CLKGSB3)	0x980060EC
GSB4 Clock Register (CLKGSB4)	0x980060F0
GSB5 Clock Register (CLKGSB5)	0x980060F4
ICTC1 Clock Register (CLKICTC1)	0x980060FC
ICTC2 Clock Register (CLKICTC2)	0x98006100
PDM Clock Register (CLKPDM)	0x98006108
CIF0 Clock Register (CLKCIF0)	0x9800610C
CIF1 Clock Register (CLKCIF1)	0x98006110
LCDSI Clock Register (CLKLCDSI)	0x9800611C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0			EN	SEL				0							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0				DIV											

Bit	Name	Type	Default	Description																																				
31:29	-	R	0	Reserved																																				
28	EN	R/W	ACLKREF=1 ACLKTCX=1 All the others=0	Enable Clock Output 0=Disable 1=Enable																																				
27:24	SEL	R/W	ACLKREF=4 ACLKTCX=4 All the others=0	Clock Source Selection.																																				
				<table><tr><th>SEL</th><th>Clock Source Selected</th><th>SEL</th><th>Clock Source Selected</th></tr><tr><td>0</td><td>PLL0</td><td>8</td><td>PLL2</td></tr><tr><td>1</td><td>PLL1</td><td>9</td><td>PLL2 Divider</td></tr><tr><td>2</td><td>PLL0 Divider</td><td>10</td><td>PLL3</td></tr><tr><td>3</td><td>PLL1 Divider</td><td>11</td><td>PLL3 Divider</td></tr><tr><td>4</td><td>XI (main oscillator)</td><td>12</td><td>Reserved</td></tr><tr><td>5</td><td>XI Divider</td><td>13</td><td>Reserved</td></tr><tr><td>6</td><td>XTI (sub-oscillator)</td><td>14</td><td>Reserved</td></tr><tr><td>7</td><td>XTI Divider</td><td>15</td><td>Reserved</td></tr></table>	SEL	Clock Source Selected	SEL	Clock Source Selected	0	PLL0	8	PLL2	1	PLL1	9	PLL2 Divider	2	PLL0 Divider	10	PLL3	3	PLL1 Divider	11	PLL3 Divider	4	XI (main oscillator)	12	Reserved	5	XI Divider	13	Reserved	6	XTI (sub-oscillator)	14	Reserved	7	XTI Divider	15	Reserved
				SEL	Clock Source Selected	SEL	Clock Source Selected																																	
				0	PLL0	8	PLL2																																	
				1	PLL1	9	PLL2 Divider																																	
				2	PLL0 Divider	10	PLL3																																	
				3	PLL1 Divider	11	PLL3 Divider																																	
				4	XI (main oscillator)	12	Reserved																																	
				5	XI Divider	13	Reserved																																	
6	XTI (sub-oscillator)	14	Reserved																																					
7	XTI Divider	15	Reserved																																					
23:12	-	R	0	Reserved																																				
11:0	DIV[11:0]	R/W	ACLKREF=0x0BA All the others=0x000	Clock Divisor FOUT = FSELECTED / (DIV + 1)																																				

Note: The maximum operating frequency of the divider is 840MHz @1.2V.

DAI0 Clock Register (ACLKDAI0)	0x980060B4
DAI1 Clock Register (ACLKDAI1)	0x980060B8
DAI2 Clock Register (ACLKDAI2)	0x980060F8
ASRC2 Clock Register (ACLKASRC2)	0x98006104
ASRC0 Clock Register (ACLKASRC2)	0x98006114
ASRC1 Clock Register (ACLKASRC2)	0x98006118

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DVM	0		EN	SEL				0							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV															

Bit	Name	Type	Default	Description																																				
31	DVM	R/W	0	Divider / DCO Mode Select. (0=DCO, 1=Divider)																																				
30:29	-	R	0	Reserved																																				
28	EN	R/W	0	Enable Clock Output (0=Disable, 1=Enable)																																				
27:24	SEL	R/W	0	Clock Source Selection.																																				
				<table><tr><th>SEL</th><th>Clock Source Selected</th><th>SEL</th><th>Clock Source Selected</th></tr><tr><td>0</td><td>PLL0</td><td>8</td><td>PLL2</td></tr><tr><td>1</td><td>PLL1</td><td>9</td><td>PLL2 Divider</td></tr><tr><td>2</td><td>PLL0 Divider</td><td>10</td><td>PLL3</td></tr><tr><td>3</td><td>PLL1 Divider</td><td>11</td><td>PLL3 Divider</td></tr><tr><td>4</td><td>XI (main oscillator)</td><td>12</td><td>Reserved</td></tr><tr><td>5</td><td>XI Divider</td><td>13</td><td>Reserved</td></tr><tr><td>6</td><td>XTI (sub-oscillator)</td><td>14</td><td>Reserved</td></tr><tr><td>7</td><td>XTI Divider</td><td>15</td><td>Reserved</td></tr></table>	SEL	Clock Source Selected	SEL	Clock Source Selected	0	PLL0	8	PLL2	1	PLL1	9	PLL2 Divider	2	PLL0 Divider	10	PLL3	3	PLL1 Divider	11	PLL3 Divider	4	XI (main oscillator)	12	Reserved	5	XI Divider	13	Reserved	6	XTI (sub-oscillator)	14	Reserved	7	XTI Divider	15	Reserved
				SEL	Clock Source Selected	SEL	Clock Source Selected																																	
				0	PLL0	8	PLL2																																	
				1	PLL1	9	PLL2 Divider																																	
				2	PLL0 Divider	10	PLL3																																	
				3	PLL1 Divider	11	PLL3 Divider																																	
				4	XI (main oscillator)	12	Reserved																																	
				5	XI Divider	13	Reserved																																	
				6	XTI (sub-oscillator)	14	Reserved																																	
7	XTI Divider	15	Reserved																																					
23:16	-	R	0	Reserved																																				
15:0	DIV[15:0]	R/W	0	Clock Divisor																																				
				<table><tr><th>DVM</th><th>Clock Output Frequency</th></tr><tr><td>0</td><td>$F_{OUT} = F_{SELECTED} * (DIV + 1) / 2^{16}$</td></tr><tr><td>1</td><td>$F_{OUT} = F_{SELECTED} / (DIV + 1)$</td></tr></table>	DVM	Clock Output Frequency	0	$F_{OUT} = F_{SELECTED} * (DIV + 1) / 2^{16}$	1	$F_{OUT} = F_{SELECTED} / (DIV + 1)$																														
				DVM	Clock Output Frequency																																			
0	$F_{OUT} = F_{SELECTED} * (DIV + 1) / 2^{16}$																																							
1	$F_{OUT} = F_{SELECTED} / (DIV + 1)$																																							

Note: The maximum operating frequency of the DCO/divider is 840MHz @1.2V.

10.3 Configuration of Peripheral

Care must be taken in configuring the clock(s) for peripherals. (e.g., I2C, DAI, and etc.)

The peripheral controller should be in reset state by setting the SWRESET bit before the clock and the corresponding ports are configured. If you configure the clock and ports in non-reset state, which is not-initialized state, an unexpected operation can occur.

If you want to stop the peripheral controller operation, you should make the controller into reset-state by SWRESET bit and stop the clock and then release the port.

Note that if the controller is in reset-state (SWRESET='1'), the CPU cannot access any control register inside the controller.

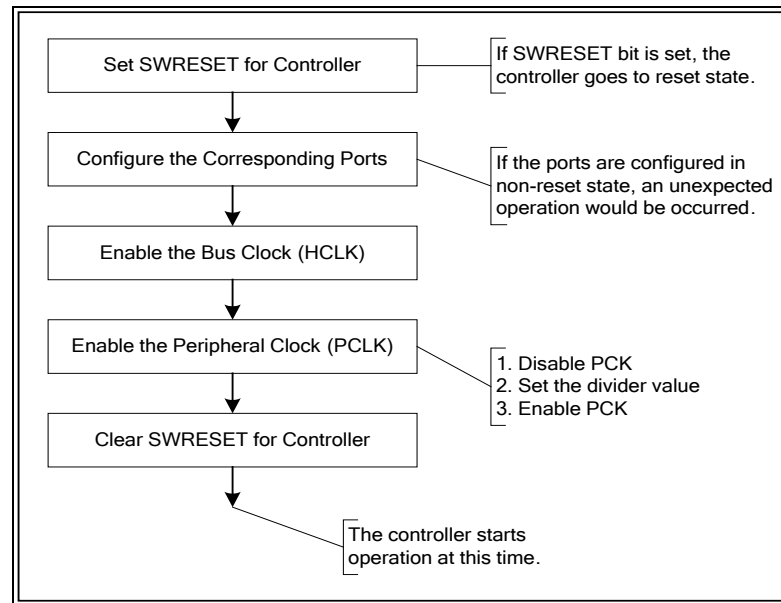


Figure 10.3 Peripheral Clock Configuration Procedure

10.4 Recommended PLL Parameter Values

All PLLs in TCC8132 are integer PLLs. It means that there must be a mismatch between actual PLL output frequency and the target frequency when a target frequency is not an integer value. Make sure to check whether the difference is affordable or not before using the parameter values.

PLL0, PLL1, PLL3					
FIN(MHz)	P	M	S	VSEL	FOUT(MHz)
12	3	280	4	0	70
12	3	284	4	0	71
12	3	288	4	0	72
12	3	292	4	0	73
12	3	296	4	0	74
12	3	300	4	0	75
12	3	304	4	0	76
12	3	308	4	0	77
12	3	312	4	0	78
12	3	316	4	0	79
12	3	320	4	0	80
12	3	324	4	0	81
12	3	328	4	0	82
12	3	332	4	0	83
12	3	336	4	0	84
12	3	340	4	0	85
12	3	344	4	0	86
12	3	348	4	0	87
12	3	352	4	1	88
12	3	356	4	1	89
12	3	360	4	1	90
12	3	364	4	1	91
12	3	368	4	1	92
12	3	372	4	1	93
12	3	376	4	1	94
12	3	380	4	1	95
12	3	384	4	1	96
12	3	388	4	1	97
12	3	392	4	1	98
12	3	396	4	1	99
12	3	400	4	1	100
12	3	404	4	1	101
12	3	408	4	1	102
12	3	412	4	1	103
12	3	416	4	1	104
12	3	420	4	1	105
12	3	424	4	1	106
12	3	428	4	1	107
12	3	432	4	1	108
12	3	436	4	1	109
12	3	440	4	1	110
12	3	444	4	1	111
12	3	448	4	1	112
12	3	452	4	1	113
12	3	456	4	1	114
12	3	460	4	1	115
12	3	464	4	1	116
12	3	468	4	1	117
12	3	472	4	1	118
12	3	476	4	1	119
12	3	480	4	1	120
12	3	484	4	1	121
12	3	488	4	1	122
12	3	492	4	1	123
12	3	496	4	1	124
12	3	500	4	1	125

PLL0, PLL1, PLL3					
FIN(MHz)	P	M	S	VSEL	FOUT(MHz)
12	3	252	3	0	126
12	3	254	3	0	127
12	3	256	3	0	128
12	3	258	3	0	129
12	3	260	3	0	130
12	3	262	3	0	131
12	3	264	3	0	132
12	3	266	3	0	133
12	3	268	3	0	134
12	3	270	3	0	135
12	3	272	3	0	136
12	3	274	3	0	137
12	3	276	3	0	138
12	3	278	3	0	139
12	3	280	3	0	140
12	3	282	3	0	141
12	3	284	3	0	142
12	3	286	3	0	143
12	3	288	3	0	144
12	3	290	3	0	145
12	3	292	3	0	146
12	3	294	3	0	147
12	3	296	3	0	148
12	3	298	3	0	149
12	3	300	3	0	150
12	3	302	3	0	151
12	3	304	3	0	152
12	3	306	3	0	153
12	3	308	3	0	154
12	3	310	3	0	155
12	3	312	3	0	156
12	3	314	3	0	157
12	3	316	3	0	158
12	3	318	3	0	159
12	3	320	3	0	160
12	3	322	3	0	161
12	3	324	3	0	162
12	3	326	3	0	163
12	3	328	3	0	164
12	3	330	3	0	165
12	3	332	3	0	166
12	3	334	3	0	167
12	3	336	3	0	168
12	3	338	3	0	169
12	3	340	3	0	170
12	3	342	3	0	171
12	3	344	3	0	172
12	3	346	3	0	173
12	3	348	3	0	174
12	3	350	3	0	175
12	3	352	3	1	176
12	3	354	3	1	177
12	3	356	3	1	178
12	3	358	3	1	179
12	3	360	3	1	180
12	3	362	3	1	181
12	3	364	3	1	182
12	3	366	3	1	183
12	3	368	3	1	184
12	3	370	3	1	185
12	3	372	3	1	186
12	3	374	3	1	187
12	3	376	3	1	188

PLL0, PLL1, PLL3					
FIN(MHz)	P	M	S	VSEL	FOUT(MHz)
12	3	378	3	1	189
12	3	380	3	1	190
12	3	382	3	1	191
12	3	384	3	1	192
12	3	386	3	1	193
12	3	388	3	1	194
12	3	390	3	1	195
12	3	392	3	1	196
12	3	394	3	1	197
12	3	396	3	1	198
12	3	398	3	1	199
12	3	400	3	1	200
12	3	402	3	1	201
12	3	404	3	1	202
12	3	406	3	1	203
12	3	408	3	1	204
12	3	410	3	1	205
12	3	412	3	1	206
12	3	414	3	1	207
12	3	416	3	1	208
12	3	418	3	1	209
12	3	420	3	1	210
12	3	422	3	1	211
12	3	424	3	1	212
12	3	426	3	1	213
12	3	428	3	1	214
12	3	430	3	1	215
12	3	432	3	1	216
12	3	434	3	1	217
12	3	436	3	1	218
12	3	438	3	1	219
12	3	440	3	1	220
12	3	442	3	1	221
12	3	444	3	1	222
12	3	446	3	1	223
12	3	448	3	1	224
12	3	450	3	1	225
12	3	452	3	1	226
12	3	454	3	1	227
12	3	456	3	1	228
12	3	458	3	1	229
12	3	460	3	1	230
12	3	462	3	1	231
12	3	464	3	1	232
12	3	466	3	1	233
12	3	468	3	1	234
12	3	470	3	1	235
12	3	472	3	1	236
12	3	474	3	1	237
12	3	476	3	1	238
12	3	478	3	1	239
12	3	480	3	1	240
12	3	482	3	1	241
12	3	484	3	1	242
12	3	486	3	1	243
12	3	488	3	1	244
12	3	490	3	1	245
12	3	492	3	1	246
12	3	494	3	1	247
12	3	496	3	1	248
12	3	498	3	1	249
12	3	500	3	1	250
12	3	251	2	0	251

PLL0, PLL1, PLL3					
FIN(MHz)	P	M	S	VSEL	FOUT(MHz)
12	3	252	2	0	252
12	3	253	2	0	253
12	3	254	2	0	254
12	3	255	2	0	255
12	3	256	2	0	256
12	3	257	2	0	257
12	3	258	2	0	258
12	3	259	2	0	259
12	3	260	2	0	260
12	3	261	2	0	261
12	3	262	2	0	262
12	3	263	2	0	263
12	3	264	2	0	264
12	3	265	2	0	265
12	3	266	2	0	266
12	3	267	2	0	267
12	3	268	2	0	268
12	3	269	2	0	269
12	3	270	2	0	270
12	3	271	2	0	271
12	3	272	2	0	272
12	3	273	2	0	273
12	3	274	2	0	274
12	3	275	2	0	275
12	3	276	2	0	276
12	3	277	2	0	277
12	3	278	2	0	278
12	3	279	2	0	279
12	3	280	2	0	280
12	3	281	2	0	281
12	3	282	2	0	282
12	3	283	2	0	283
12	3	284	2	0	284
12	3	285	2	0	285
12	3	286	2	0	286
12	3	287	2	0	287
12	3	288	2	0	288
12	3	289	2	0	289
12	3	290	2	0	290
12	3	291	2	0	291
12	3	292	2	0	292
12	3	293	2	0	293
12	3	294	2	0	294
12	3	295	2	0	295
12	3	296	2	0	296
12	3	297	2	0	297
12	3	298	2	0	298
12	3	299	2	0	299
12	3	300	2	0	300
12	3	302	2	0	302
12	3	304	2	0	304
12	3	306	2	0	306
12	3	308	2	0	308
12	3	310	2	0	310
12	3	312	2	0	312
12	3	314	2	0	314
12	3	316	2	0	316
12	3	318	2	0	318
12	3	320	2	0	320
12	3	322	2	0	322
12	3	324	2	0	324
12	3	326	2	0	326
12	3	328	2	0	328

PLL0, PLL1, PLL3					
FIN(MHz)	P	M	S	VSEL	FOUT(MHz)
12	3	330	2	0	330
12	3	332	2	0	332
12	3	334	2	0	334
12	3	336	2	0	336
12	3	338	2	0	338
12	3	340	2	0	340
12	3	342	2	0	342
12	3	344	2	0	344
12	3	346	2	0	346
12	3	348	2	0	348
12	3	350	2	0	350
12	3	352	2	1	352
12	3	354	2	1	354
12	3	356	2	1	356
12	3	358	2	1	358
12	3	360	2	1	360
12	3	362	2	1	362
12	3	364	2	1	364
12	3	368	2	1	368
12	3	369	2	1	369
12	3	370	2	1	370
12	3	372	2	1	372
12	3	374	2	1	374
12	3	376	2	1	376
12	3	378	2	1	378
12	3	380	2	1	380
12	3	381	2	1	381
12	3	384	2	1	384
12	3	386	2	1	386
12	3	388	2	1	388
12	3	390	2	1	390
12	3	392	2	1	392
12	3	393	2	1	393
12	3	396	2	1	396
12	3	399	2	1	399
12	3	400	2	1	400
12	3	404	2	1	404
12	5	676	2	1	405.6
12	3	406	2	1	406
12	3	408	2	1	408
12	3	410	2	1	410
12	3	412	2	1	412
12	3	416	2	1	416
12	5	696	2	1	417.6
12	3	418	2	1	418
12	3	420	2	1	420
12	3	422	2	1	422
12	3	424	2	1	424
12	3	428	2	1	428
12	3	429	2	1	429
12	3	430	2	1	430
12	3	432	2	1	432
12	3	434	2	1	434
12	3	436	2	1	436
12	3	438	2	1	438
12	3	440	2	1	440
12	3	442	2	1	442
12	3	444	2	1	444
12	3	446	2	1	446
12	3	448	2	1	448
12	3	450	2	1	450
12	3	452	2	1	452
12	3	454	2	1	454

PLL0, PLL1, PLL3					
FIN(MHz)	P	M	S	VSEL	FOUT(MHz)
12	3	456	2	1	456
12	5	772	2	1	463.2
12	3	464	2	1	464
12	3	466	2	1	466
12	3	468	2	1	468
12	3	470	2	1	470
12	3	472	2	1	472
12	3	474	2	1	474
12	3	476	2	1	476
12	3	478	2	1	478
12	3	480	2	1	480
12	3	482	2	1	482
12	3	484	2	1	484
12	3	486	2	1	486
12	3	488	2	1	488
12	3	490	2	1	490
12	3	492	2	1	492
12	3	496	2	1	496
12	5	828	2	1	496.8
12	3	498	2	1	498
12	3	500	2	1	500
12	3	251	1	0	502
12	3	252	1	0	504
12	3	253	1	0	506
12	3	254	1	0	508
12	3	255	1	0	510
12	3	256	1	0	512
12	3	257	1	0	514
12	3	258	1	0	516
12	3	259	1	0	518
12	3	260	1	0	520
12	3	261	1	0	522
12	3	262	1	0	524
12	3	263	1	0	526
12	3	264	1	0	528
12	3	265	1	0	530
12	3	266	1	0	532
12	3	267	1	0	534
12	3	268	1	0	536
12	3	269	1	0	538
12	3	270	1	0	540
12	3	271	1	0	542
12	3	272	1	0	544
12	3	273	1	0	546
12	3	274	1	0	548
12	3	275	1	0	550
12	3	276	1	0	552
12	3	277	1	0	554
12	3	278	1	0	556
12	3	279	1	0	558
12	3	280	1	0	560
12	3	281	1	0	562
12	3	282	1	0	564
12	3	283	1	0	566
12	3	284	1	0	568
12	3	285	1	0	570
12	3	286	1	0	572
12	3	287	1	0	574
12	3	288	1	0	576
12	3	289	1	0	578
12	3	290	1	0	580
12	3	291	1	0	582
12	3	292	1	0	584

PLL0, PLL1, PLL3					
FIN(MHz)	P	M	S	VSEL	FOUT(MHz)
12	3	293	1	0	586
12	3	294	1	0	588
12	3	295	1	0	590
12	3	296	1	0	592
12	3	297	1	0	594
12	3	298	1	0	596
12	3	299	1	0	598
12	3	300	1	0	600
12	3	304	1	0	608
12	3	308	1	0	616
12	3	312	1	0	624
12	3	316	1	0	632
12	3	320	1	0	640
12	3	324	1	0	648
12	3	328	1	0	656
12	3	330	1	0	660
12	3	332	1	0	664
12	3	336	1	0	672
12	3	340	1	0	680
12	3	348	1	0	696
12	3	352	1	1	704
12	3	356	1	1	712
12	3	360	1	1	720
12	3	362	1	1	724
12	3	364	1	1	728
12	3	369	1	1	738
12	3	370	1	1	740
12	3	372	1	1	744
12	3	380	1	1	760
12	3	381	1	1	762
12	3	384	1	1	768
12	3	388	1	1	776
12	3	390	1	1	780
12	3	392	1	1	784
12	3	393	1	1	786
12	3	396	1	1	792
12	3	399	1	1	798
12	3	400	1	1	800
12	3	404	1	1	808
12	5	676	1	1	811.2
12	3	406	1	1	812
12	3	408	1	1	816
12	3	410	1	1	820
12	3	412	1	1	824
12	3	416	1	1	832
12	5	696	1	1	835.2
12	3	420	1	1	840
12	3	424	1	1	848
12	3	428	1	1	856
12	3	429	1	1	858
12	3	430	1	1	860
12	3	432	1	1	864
12	3	436	1	1	872
12	3	440	1	1	880
12	3	442	1	1	884
12	3	444	1	1	888
12	3	448	1	1	896
12	3	450	1	1	900
12	3	452	1	1	904
12	3	454	1	1	908
12	3	456	1	1	912
12	3	460	1	1	920
12	5	772	1	1	926.4

PLL0, PLL1, PLL3					
FIN(MHz)	P	M	S	VSEL	FOUT(MHz)
12	3	464	1	1	928
12	3	466	1	1	932
12	3	468	1	1	936
12	3	470	1	1	940
12	3	472	1	1	944
12	3	474	1	1	948
12	3	476	1	1	952
12	3	480	1	1	960
12	3	484	1	1	968
12	3	486	1	1	972
12	3	488	1	1	976
12	3	490	1	1	980
12	3	492	1	1	984
12	3	496	1	1	992
12	5	828	1	1	993.6
12	3	498	1	1	996
12	3	500	1	1	1000
12	3	252	0	0	1008
12	3	254	0	0	1016
12	3	256	0	0	1024
12	3	258	0	0	1032
12	3	260	0	0	1040
12	3	262	0	0	1048
12	3	264	0	0	1056
12	3	266	0	0	1064
12	3	268	0	0	1072
12	3	270	0	0	1080
12	3	272	0	0	1088
12	3	274	0	0	1096
12	3	276	0	0	1104
12	3	278	0	0	1112
12	3	280	0	0	1120
12	3	282	0	0	1128
12	3	284	0	0	1136
12	3	286	0	0	1144
12	3	288	0	0	1152
12	3	290	0	0	1160
12	3	292	0	0	1168
12	3	294	0	0	1176
12	3	296	0	0	1184
12	3	298	0	0	1192
12	3	300	0	0	1200

Note: P/M/S value is decimal

PLL2														
FIN (MHz)	P	M	S	K	VSEL	FMR	MRR	SEL PF	FOUT (MHz)	DUTY (%)	MF (kHz)	MF ERR (MHz)	MR (%)	MR_ER R (%)
12	1	32	4	1024	0	2	4	2	24.75	49~51	31.25	1.25	2.27	0.27
12	1	47	3	1024	1	2	5	2	72	49~51	31.25	1.25	1.95	-0.05
12	1	49	3	1024	1	2	5	2	75	49~51	31.25	1.25	1.88	-0.13
12	1	51	3	1024	1	2	6	2	78	49~51	31.25	1.25	2.16	0.16
12	1	53	3	1024	1	2	6	2	81	49~51	31.25	1.25	2.08	0.08
12	1	27	2	1024	0	2	3	2	84	49~51	31.25	1.25	2.01	0.01
12	1	28	2	1024	0	2	3	2	87	49~51	31.25	1.25	1.94	-0.06
12	1	29	2	1024	0	2	3	2	90	49~51	31.25	1.25	1.88	-0.13
12	1	30	2	1024	0	2	3	2	93	49~51	31.25	1.25	1.81	-0.19
12	1	31	2	1024	0	2	3	2	96	49~51	31.25	1.25	1.76	-0.24
12	1	32	2	1024	0	2	4	2	99	49~51	31.25	1.25	2.27	0.27
12	1	33	2	1024	0	2	4	2	102	49~51	31.25	1.25	2.21	0.21
12	1	34	2	1024	0	2	4	2	105	49~51	31.25	1.25	2.14	0.14
12	1	35	2	1024	0	2	4	2	108	49~51	31.25	1.25	2.08	0.08
12	1	36	2	1024	0	2	4	2	111	49~51	31.25	1.25	2.03	0.03
12	1	37	2	1024	0	2	4	2	114	49~51	31.25	1.25	1.97	-0.03
12	1	38	2	1024	1	2	4	2	117	49~51	31.25	1.25	1.92	-0.08
12	1	39	2	1024	1	2	4	2	120	49~51	31.25	1.25	1.88	-0.13
12	1	40	2	1024	1	2	4	2	123	49~51	31.25	1.25	1.83	-0.17
12	1	41	2	1024	1	2	4	2	126	49~51	31.25	1.25	1.79	-0.21
12	1	42	2	1024	1	2	5	2	129	49~51	31.25	1.25	2.18	0.18
12	1	43	2	1024	1	2	5	2	132	49~51	31.25	1.25	2.13	0.13
12	1	44	2	1024	1	2	5	2	135	49~51	31.25	1.25	2.08	0.08
12	1	45	2	1024	1	2	5	2	138	49~51	31.25	1.25	2.04	0.04
12	1	46	2	1024	1	2	5	2	141	49~51	31.25	1.25	1.99	-0.01
12	1	47	2	1024	1	2	5	2	144	49~51	31.25	1.25	1.95	-0.05
12	1	48	2	1024	1	2	5	2	147	49~51	31.25	1.25	1.91	-0.09
12	1	49	2	1024	1	2	5	2	150	49~51	31.25	1.25	1.88	-0.13
12	1	50	2	1024	1	2	5	2	153	49~51	31.25	1.25	1.84	-0.16
12	1	51	2	1024	1	2	6	2	156	49~51	31.25	1.25	2.16	0.16
12	1	52	2	1024	1	2	6	2	159	49~51	31.25	1.25	2.12	0.12
12	1	53	2	1024	1	2	6	2	162	49~51	31.25	1.25	2.08	0.08
12	1	54	2	1024	1	2	6	2	165	49~51	31.25	1.25	2.05	0.05
12	1	27	1	1024	0	2	3	2	168	49~51	31.25	1.25	2.01	0.01
12	1	28	1	512	0	2	3	2	171	49~51	31.25	1.25	1.97	-0.03
12	1	28	1	1024	0	2	3	2	174	49~51	31.25	1.25	1.94	-0.06
12	1	29	1	512	0	2	3	2	177	49~51	31.25	1.25	1.91	-0.09
12	1	29	1	1024	0	2	3	2	180	49~51	31.25	1.25	1.88	-0.13
12	1	30	1	512	0	2	3	2	183	49~51	31.25	1.25	1.84	-0.16
12	1	30	1	1024	0	2	3	2	186	49~51	31.25	1.25	1.81	-0.19
12	1	31	1	512	0	2	3	2	189	49~51	31.25	1.25	1.79	-0.21
12	1	31	1	1024	0	2	3	2	192	49~51	31.25	1.25	1.76	-0.24
12	1	32	1	512	0	2	3	2	195	49~51	31.25	1.25	1.73	-0.27
12	1	32	1	1024	0	2	4	2	198	49~51	31.25	1.25	2.27	0.27
12	1	33	1	512	0	2	4	2	201	49~51	31.25	1.25	2.24	0.24
12	1	33	1	1024	0	2	4	2	204	49~51	31.25	1.25	2.21	0.21
12	1	34	1	512	0	2	4	2	207	49~51	31.25	1.25	2.17	0.17
12	1	34	1	1024	0	2	4	2	210	49~51	31.25	1.25	2.14	0.14
12	1	35	1	512	0	2	4	2	213	49~51	31.25	1.25	2.11	0.11
12	1	35	1	1024	0	2	4	2	216	49~51	31.25	1.25	2.08	0.08
12	1	36	1	512	0	2	4	2	219	49~51	31.25	1.25	2.05	0.05
12	1	36	1	1024	0	2	4	2	222	49~51	31.25	1.25	2.03	0.03
12	1	37	1	512	0	2	4	2	225	49~51	31.25	1.25	2.00	0.00
12	1	37	1	1024	0	2	4	2	228	49~51	31.25	1.25	1.97	-0.03
12	1	38	1	512	1	2	4	2	231	49~51	31.25	1.25	1.95	-0.05
12	1	38	1	1024	1	2	4	2	234	49~51	31.25	1.25	1.92	-0.08
12	1	39	1	512	1	2	4	2	237	49~51	31.25	1.25	1.90	-0.10
12	1	39	1	1024	1	2	4	2	240	49~51	31.25	1.25	1.88	-0.13
12	1	40	1	512	1	2	4	2	243	49~51	31.25	1.25	1.85	-0.15
12	1	40	1	1024	1	2	4	2	246	49~51	31.25	1.25	1.83	-0.17

PLL2														
FIN (MHz)	P	M	S	K	VSEL	FMR	MRR	SEL_ PF	FOUT (MHz)	DUTY (%)	MF (kHz)	MF_ ERR (MHz)	MR (%)	MR_ER R (%)
12	1	41	1	512	1	2	4	2	249	49~51	31.25	1.25	1.81	-0.19
12	1	41	1	1024	1	2	4	2	252	49~51	31.25	1.25	1.79	-0.21
12	1	42	1	512	1	2	5	2	255	49~51	31.25	1.25	2.21	0.21
12	1	42	1	1024	1	2	5	2	258	49~51	31.25	1.25	2.18	0.18
12	1	43	1	512	1	2	5	2	261	49~51	31.25	1.25	2.16	0.16
12	1	43	1	1024	1	2	5	2	264	49~51	31.25	1.25	2.13	0.13
12	1	44	1	512	1	2	5	2	267	49~51	31.25	1.25	2.11	0.11
12	1	44	1	1024	1	2	5	2	270	49~51	31.25	1.25	2.08	0.08
12	1	45	1	512	1	2	5	2	273	49~51	31.25	1.25	2.06	0.06
12	1	45	1	1024	1	2	5	2	276	49~51	31.25	1.25	2.04	0.04
12	1	46	1	512	1	2	5	2	279	49~51	31.25	1.25	2.02	0.02
12	1	46	1	1024	1	2	5	2	282	49~51	31.25	1.25	1.99	-0.01
12	1	47	1	512	1	2	5	2	285	49~51	31.25	1.25	1.97	-0.03
12	1	47	1	1024	1	2	5	2	288	49~51	31.25	1.25	1.95	-0.05
12	1	48	1	512	1	2	5	2	291	49~51	31.25	1.25	1.93	-0.07
12	1	48	1	1024	1	2	5	2	294	49~51	31.25	1.25	1.91	-0.09
12	1	49	1	512	1	2	5	2	297	49~51	31.25	1.25	1.89	-0.11
12	1	49	1	1024	1	2	5	2	300	49~51	31.25	1.25	1.88	-0.13
12	1	50	1	1024	1	2	5	2	306	49~51	31.25	1.25	1.84	-0.16
12	1	51	1	1024	1	2	6	2	312	49~51	31.25	1.25	2.16	0.16
12	1	52	1	1024	1	2	6	2	318	49~51	31.25	1.25	2.12	0.12
12	1	53	1	1024	1	2	6	2	324	49~51	31.25	1.25	2.08	0.08
12	1	54	1	1024	1	2	6	2	330	49~51	31.25	1.25	2.05	0.05
12	1	27	0	1024	0	2	3	2	336	40~60	31.25	1.25	2.01	0.01
12	1	28	0	512	0	2	3	2	342	40~60	31.25	1.25	1.97	-0.03
12	1	28	0	1024	0	2	3	2	348	40~60	31.25	1.25	1.94	-0.06
12	1	29	0	512	0	2	3	2	354	40~60	31.25	1.25	1.91	-0.09
12	1	29	0	1024	0	2	3	2	360	40~60	31.25	1.25	1.88	-0.13
12	1	30	0	768	0	2	3	2	369	40~60	31.25	1.25	1.83	-0.17
12	1	30	0	1024	0	2	3	2	372	40~60	31.25	1.25	1.81	-0.19
12	1	31	0	512	0	2	3	2	378	40~60	31.25	1.25	1.79	-0.21
12	1	31	0	768	0	2	3	2	381	40~60	31.25	1.25	1.77	-0.23
12	1	31	0	1024	0	2	3	2	384	40~60	31.25	1.25	1.76	-0.24
12	1	32	0	512	0	2	3	2	390	40~60	31.25	1.25	1.73	-0.27
12	1	32	0	768	0	2	3	2	393	40~60	31.25	1.25	1.72	-0.28
12	1	32	0	1024	0	2	4	2	396	40~60	31.25	1.25	2.27	0.27
12	1	32	0	1280	0	2	4	2	399	40~60	31.25	1.25	2.26	0.26
12	1	33	0	1024	0	2	4	2	408	40~60	31.25	1.25	2.21	0.21
12	1	34	0	1024	0	2	4	2	420	40~60	31.25	1.25	2.14	0.14
12	1	35	0	768	0	2	4	2	429	40~60	31.25	1.25	2.10	0.10
12	1	35	0	1024	0	2	4	2	432	40~60	31.25	1.25	2.08	0.08
12	1	36	0	512	0	2	4	2	438	40~60	31.25	1.25	2.05	0.05
12	1	36	0	1024	0	2	4	2	444	40~60	31.25	1.25	2.03	0.03
12	1	37	0	512	0	2	4	2	450	40~60	31.25	1.25	2.00	0.00
12	1	37	0	1024	0	2	4	2	456	40~60	31.25	1.25	1.97	-0.03
12	1	38	0	1024	1	2	4	2	468	40~60	31.25	1.25	1.92	-0.08
12	1	39	0	512	1	2	4	2	474	40~60	31.25	1.25	1.90	-0.10
12	1	39	0	1024	1	2	4	2	480	40~60	31.25	1.25	1.88	-0.13
12	1	40	0	512	1	2	4	2	486	40~60	31.25	1.25	1.85	-0.15
12	1	40	0	1024	1	2	4	2	492	40~60	31.25	1.25	1.83	-0.17
12	1	41	0	512	1	2	4	2	498	40~60	31.25	1.25	1.81	-0.19
12	1	41	0	1024	1	2	4	2	504	40~60	31.25	1.25	1.79	-0.21
12	1	42	0	512	1	2	5	2	510	40~60	31.25	1.25	2.21	0.21
12	1	42	0	1024	1	2	5	2	516	40~60	31.25	1.25	2.18	0.18
12	1	43	0	512	1	2	5	2	522	40~60	31.25	1.25	2.16	0.16
12	1	43	0	1024	1	2	5	2	528	40~60	31.25	1.25	2.13	0.13
12	1	44	0	512	1	2	5	2	534	40~60	31.25	1.25	2.11	0.11
12	1	44	0	1024	1	2	5	2	540	40~60	31.25	1.25	2.08	0.08
12	1	45	0	512	1	2	5	2	546	40~60	31.25	1.25	2.06	0.06
12	1	45	0	1024	1	2	5	2	552	40~60	31.25	1.25	2.04	0.04
12	1	46	0	512	1	2	5	2	558	40~60	31.25	1.25	2.02	0.02

PLL2														
FIN (MHz)	P	M	S	K	VSEL	FMR	MRR	SEL_ PF	FOUT (MHz)	DUTY (%)	MF (kHz)	MF_ ERR (MHz)	MR (%)	MR_ER R (%)
12	1	46	0	1024	1	2	5	2	564	40~60	31.25	1.25	1.99	-0.01
12	1	47	0	512	1	2	5	2	570	40~60	31.25	1.25	1.97	-0.03
12	1	47	0	1024	1	2	5	2	576	40~60	31.25	1.25	1.95	-0.05
12	1	48	0	512	1	2	5	2	582	40~60	31.25	1.25	1.93	-0.07
12	1	48	0	1024	1	2	5	2	588	40~60	31.25	1.25	1.91	-0.09
12	1	49	0	512	1	2	5	2	594	40~60	31.25	1.25	1.89	-0.11
12	1	49	0	1024	1	2	5	2	600	40~60	31.25	1.25	1.88	-0.13

Note: P/M/S value is decimal

10.5 Port Multiplexing for Clock Outputs

Some of the clock divider outputs can be monitored via GBUS ports.

Refer to GBUS[191:0] found in GPIO Function Selection Registers (GPIO*_FS* of GPIO block).

Any clock output signal can be mapped to any bit position of GBUS[191:0] by writing an index number to the registers below.

Table 10.2 GSB Port Selection Register Map (Base Address = 0x98006180)

Name	Address	Type	Reset	Description
CKOPORTSEL0	0x00	R/W	0xFFFFFFFF	Clock Output Port Selection Register0
CKOPORTSEL3	0x0C	R/W	0x00FFFF00	Clock Output Port Selection Register3
CKOPORTSEL5	0x14	R/W	0x0000FF00	Clock Output Port Selection Register5
CKOPORTSEL7	0x1C	R/W	0xFFFF0000	Clock Output Port Selection Register7
CKOPORTSEL8	0x20	R/W	0xFF000000	Clock Output Port Selection Register8
CKOPORTSEL9	0x24	R/W	0x000000FF	Clock Output Port Selection Register9

Clock Output Port Selection Register0 (CKOPORTSEL0)

0x98006180

Bit	Name	R/W	Reset	Description
31:24	IrDACLK	R/W	0xFF	Port number for IrDA
23:16	-	R/W	0xFF	Reserved
15:8	I2CCLK	R/W	0xFF	Port number for I2CCLK
7:0	REFCLK	R/W	0xFF	Port number for REFCLK

Clock Output Port Selection Register3 (CKOPORTSEL3)

0x9800618C

Bit	Name	R/W	Reset	Description
31:24	-	R	0	Reserved
23:16	DAI1CLK	R/W	0xFF	Port number for DAI1CLK
15:8	DAI0CLK	R/W	0xFF	Port number for DAI0CLK
7:0	-	R	0	Reserved

Clock Output Port Selection Register5 (CKOPORTSEL5)

0x98006194

Bit	Name	R/W	Reset	Description
31:16	-	R	0	Reserved
15:8	EXTCLK	R/W	0xFF	Port number for EXTCLK
7:0	-	R	0	Reserved

Clock Output Port Selection Register7 (CKOPORTSEL7)

0x9800619C

Bit	Name	R/W	Reset	Description
31:24	DAI3CLK	R/W	0xFF	Port number for DAI3CLK
23:16	DAI2CLK	R/W	0xFF	Port number for DAI2CLK
15:0	-	R	0	Reserved

Clock Output Port Selection Register8 (CKOPORTSEL8)

0x980061A0

Bit	Name	R/W	Reset	Description
31:24	CIF0CLK	R/W	0xFF	Port number for CIF0CLK
23:0	-	R	0	Reserved

Clock Output Port Selection Register9 (CKOPORTSEL9)

0x980061A4

Bit	Name	R/W	Reset	Description
31:8	-	R	0	Reserved
7:0	CIF1CLK	R/W	0xFF	Port number for CIF1CLK

11 Interrupt Controller

11.1 Overview

The following figure represents the block diagram of the interrupt controller. The Interrupt controller can manage up to 64 interrupt sources. In TCC8132, there are eight external interrupt sources that can be detected by various kinds of method, which is a rising edge / falling edge / level high / level low can be detected from external interrupt sources.

There are two types of interrupt in ARM926EJ-S; IRQ type, and FIQ type.
Interrupt controller can select these two types for each interrupt sources separately.

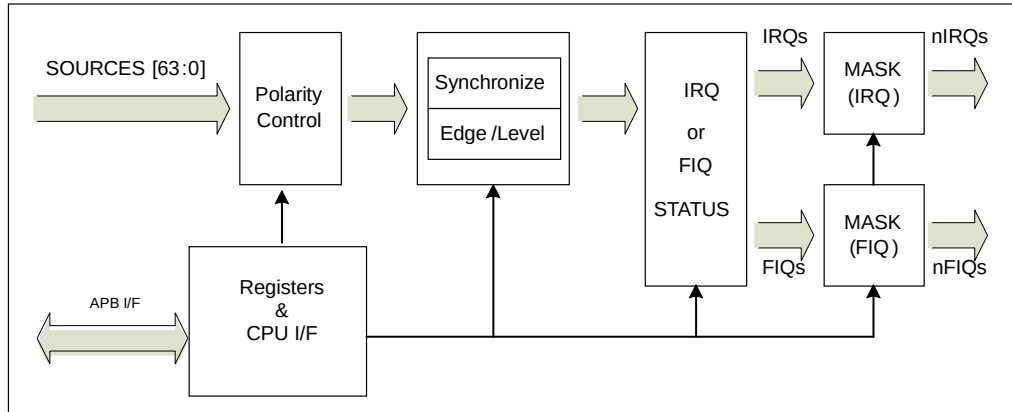


Figure 11.1 Interrupt Controller Block Diagram

11.2 Register Description

Table 11.1 Interrupt Controller Register Map (Base Address = 0x98003000)

Name	Address	Type	Reset	Description
IEN_A	0x00	R/W	0x00000000	PIC0 Interrupt Enable Register
CREQ_A	0x04	W	-	PIC0 Clear Interrupt Request Register
IREQ_A	0x08	R/W	-	PIC0 Interrupt Request Register
IRQSEL_A	0x0C	R/W	0x00000000	PIC0 IRQ / FIQ Select Register
SRC_A	0x10	R/W	0x00000000	PIC0 Interrupt Source Status
MREQ_A	0x14	R/W	0x00000000	PIC0 Masked Interrupt Request Register
TSTREQ_A	0x18	W	0x00000000	PIC0 Test Mode Register (must be remained as 0)
POL_A	0x1C	R/W	0x00000000	PIC0 Polarity control
IRQ_A	0x20	R/W	-	PIC0 IRQ Raw Status (IREQ & IRQSEL)
FIQ_A	0x24	R/W	-	PIC0 FIQ Raw Status (IREQ & ~IRQSEL)
MIRQ_A	0x28	R/W	-	PIC0 Masked IRQ Status (IRQ & IEN)
MFIQ_A	0x2C	R/W	-	PIC0 Masked FIQ Status (FIQ & IEN)
TMODE_A	0x30	R/W	0x00000000	PIC0 Trigger Mode 0: Edge 1: Level
SYNC_A	0x34	R/W	0x00000000	PIC0 Synchronizer Control
WKUP_A	0x38	R/W	0x00000000	PIC0 Wakeup Control
TMODEA_A	0x3C	R/W	0x00000000	PIC0 Trigger Mode 0: Single edge 1: Both edge
INTOEN_A	0x40	R/W	0xFFFFFFFF	PIC0 Interrupt Output Enable
MEN_A	0x44	R/W	0x00000003	PIC0 Master Enable
MEN	0x48	R/W	0x00000003	Master Enable for both controllers
IEN_B	0x80	R/W	0x00000000	PIC1 Interrupt Enable Register
CREQ_B	0x84	W	-	PIC1 Clear Interrupt Request Register
IREQ_B	0x88	R/W	-	PIC1 Interrupt Request Register
IRQSEL_B	0x8C	R/W	0x00000000	PIC1 IRQ / FIQ Select Register
SRC_B	0x10	R/W	0x00000000	PIC1 Interrupt Source Status
MREQ_B	0x94	R/W	0x00000000	PIC1 Masked Interrupt Request Register
TSTREQ_B	0x98	W	0x00000000	PIC1 Test Mode Register (must be remained as 0)
POL_B	0x9C	R/W	0x00000000	PIC1 Polarity control
IRQ_B	0xA0	R/W	-	PIC1 IRQ Raw Status (IREQ & IRQSEL)
FIQ_B	0xA4	R/W	-	PIC1 FIQ Raw Status (IREQ & ~IRQSEL)
MIRQ_B	0xA8	R/W	-	PIC1 Masked IRQ Status (IRQ & IEN)
MFIQ_B	0xAC	R/W	-	PIC1 Masked FIQ Status (FIQ & IEN)
TMODE_B	0xB0	R/W	0x00000000	PIC1 Trigger Mode 0: Edge 1: Level
SYNC_B	0xB4	R/W	0x00000000	PIC1 Synchronizer Control
WKUP_B	0xB8	R/W	0x00000000	PIC1 Wakeup Control
TMODEA_B	0xBC	R/W	0x00000000	PIC1 Trigger Mode 0: Single edge 1: Both edge
INTOEN_B	0xC0	R/W	0xFFFFFFFF	PIC1 Interrupt Output Enable
MEN_B	0xC4	R/W	0x00000003	PIC1 Master Enable
MEN	0xC8	R/W	0x00000003	Master Enable for both controllers

Caution:

Some peripherals have their own request flags as well as the flag in the interrupt controller, so in the interrupt service routine, a user should clear their own request flags in the peripherals ahead of clearing the flag in the interrupt controller.

Interrupt controller registers are divided into A and B group. Both are composed of the registers with the same functions. Their interrupt output signals are OR'ed together before sent to the CPU.

Watch out for the default value of Trigger Mode registers. By default, all the interrupt sources are treated as "Edge Trigger" which may not be suitable for some internal peripherals.

PIC0 Interrupt Enable Register (IEN_A)

0x98003000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
GSB1	GSDMA	UART1	-	CIPHER	G2D	-	USBH0	PK	GSB0	SD0	NFC	RTCA	RTCP	I2C	ADC
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LCD	TC32	GDMA0	GDMA2	LIRQ	SV	-	USBO	UART0	TC	ADCWK	ADCSD	GDMA1	AUDIO	UHC	UHF

PIC1 Interrupt Enable Register (IEN_B)

0x98003080

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADCSM	UOC	UOF	SD1DET	SD0DET	SFMC	KRST	GSB3	-	CAN	ASRC	ICTC2	ICTC1	UART4	SD1	VBOFF
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VBON	ASDMA	ICTC0	GSB2	GSB5	GSB4	UART3	UART2	E7	E6	E5	E4	E3	E2	E1	E0

These registers are used to enable/disable interrupt individually.

0: Disable,

1: Enable

For master enable control, refer to "[Master Enable Register](#)" description.

Table 11.2 Interrupt Assignment

Interrupt Controller 0 (A)			Interrupt Controller 1 (B)		
Bit	Name	Source	Bit	Name	Source
[31]	GSB1	GSB Controller 1	[31]	ADCSM	ADC Stop Mode
[30]	GSDMA	GSB DMA Controller	[30]	UOC	USB 20DH CHGDET
[29]	UART1	UART Controller 1	[29]	UOF	USB 20DH FSVPLUS
[28]	-	-	[28]	SD1DET	SD1DET (FS[6] External Pin)
[27]	CIPHER	Cipher Controller	[27]	SD0DET	SD0DET (PS[6] External Pin)
[26]	G2D	Graphic 2D Engine	[26]	SFMC	Serial Flash Controller
[25]	-	-	[25]	KRST	KRST# (External Pin)
[24]	USBH0	USB Host Controller 0	[24]	GSB3	GSB Controller 3
[23]	PK	GPIOPK Interrupt Controller	[23]	-	-
[22]	GSB0	GSB Controller 0	[22]	CAN0	CAN Controller
[21]	SD0	SD/MMC Controller 0	[21]	ASRC	ASRC
[20]	NFC	NAND Flash Controller	[20]	ICTC2	ICTC 2
[19]	RTCA	RTC Alarm	[19]	ICTC1	ICTC 1
[18]	RTCP	RTC Power-Down Wakeup	[18]	UART4	UART Controller 4
[17]	I2C	I2C Controller	[17]	SD1	SD/MMC Controller 1
[16]	ADC	ADC Interface Controller	[16]	VBOFF	USB 20DH VBUS Off Detection
[15]	LCD	LCD Controller	[15]	VBON	USB 20DH VBUS On Detection
[14]	TC32	32-bit Timer	[14]	ASDMA	ASRC_DMA
[13]	GDMA0	General DMA Controller 0	[13]	ICTC0	ICTC 0
[12]	GDMA2	General DMA Controller 2	[12]	GSB2	GSB Controller 2
[11]	LIRQ	LC[5] (External Pin)	[11]	GSB5	GSB Controller 5
[10]	SV	HSIZE Violation	[10]	GSB4	GSB Controller 4
[9]	-	-	[9]	UART3	UART Controller 3
[8]	USBO	USB 20DH Controller	[8]	UART2	UART Controller 2
[7]	UART0	UART Controller 0	[7]	E7	PK[7] External Request 7
[6]	TC	Timer/Counter	[6]	E6	PK[6] External Request 6
[5]	ADCWK	ADC Wake Up	[5]	E5	PK[5] External Request 5
[4]	ADCSD	ADC Stylus Down	[4]	E4	PK[4] External Request 4
[3]	GDMA1	General DMA Controller 1	[3]	E3	PK[3] External Request 3
[2]	AUDIO	Audio (DAI/CDIF/SPDIFTx)	[2]	E2	PK[2] External Request 2
[1]	UHC	USB Host CHGDET	[1]	E1	PK[1] External Request 1
[0]	UHF	USB Host FSVPLUS	[0]	E0	PK[0] External Request 0

PIC0 Clear Interrupt Request Register (CREQ_A)	0x98003004
PIC1 Clear Interrupt Request Register (CREQ_B)	0x98003084

By writing “1” to each field, the interrupt request flag of corresponding interrupt is cleared. Writing “0” does not mean anything and the corresponding flag remains its previous state.

PIC0 Interrupt Request Register (IREQ_A)	0x98003008
PIC1 Interrupt Request Register (IREQ_B)	0x98003088

If each field is “1”, it means that the corresponding interrupt has been requested. If each peripheral has its own request flags, it means at least one of those flags has been set.

By writing “1” to each field, the interrupt request flag of corresponding interrupt is cleared. Writing “0” does not mean anything and the corresponding flag remains its previous state.

PIC0 IRQ Interrupt Select Register (IRQSEL_A)	0x9800300C
PIC1 IRQ Interrupt Select Register (IRQSEL_B)	0x9800308C

If each field is “1”, it means that the corresponding interrupt is considered as IRQ interrupt, if each field is ‘0’ it means that its interrupt is considered as FIQ interrupt.

PIC0 Interrupt Source Status Register (SRC_A)	0x98003010
PIC1 Interrupt Source Status Register (SRC_B)	0x98003090

Interrupt source line raw status (Polarity controlled).

PIC0 Masked Interrupt Request Register (MREQ_A)	0x98003014
PIC1 Masked Interrupt Request Register (MREQ_B)	0x98003094

Same meaning as IREQ except that it represents only that of the enabled interrupts. Only the flags of enabled interrupts can be checked by this register. It is recommended to use MREQ register instead of IREQ in the interrupt handler routine.

MREQ_A = IREQ_A & IEN_A

MREQ_B = IREQ_B & IEN_B

By writing “1” to each field, the corresponding bit is cleared. Writing “0” has no effect.

PIC0 Test Mode Register (TSTREQ_A)	0x98003018
PIC1 Test Mode Register (TSTREQ_B)	0x98003098

These registers can be used to generate an interrupt (1 clock pulse) by writing 1 to the corresponding bit of the internal interrupt source. These registers are for testing purpose only. It must be remained ‘0’ during normal operation.

PIC0 Polarity Control Register (POL_A)	0x9800301C
PIC1 Polarity Control Register (POL_B)	0x9800309C

By default, all interrupt source lines are high active

0: Active high

1: Active low

PIC0 IRQ Raw Status Register (IRQ_A)**0x98003020****PIC1 IRQ Raw Status Register (IRQ_B)****0x980030A0**

These registers reflect IREQ bits selected when the corresponding IRQSEL bit is high.

IRQ_A = IREQ_A & IRQSEL_A

IRQ_B = IREQ_B & IRQSEL_B

By writing "1" to each field, the corresponding bit is cleared. Writing "0" has no effect.

PIC0 FIQ Raw Status Register (FIQ_A)**0x98003024****PIC1 FIQ Raw Status Register (FIQ_B)****0x980030A4**

These registers reflect IREQ bits selected when corresponding IRQSEL bit is low.

FIQ_A = IREQ_A & ~IRQSEL_A

FIQ_B = IREQ_B & ~IRQSEL_B

By writing "1" to each field, the corresponding bit is cleared. Writing "0" has no effect.

PIC0 Masked IRQ Raw Status Register (MIRQ_A)**0x98003028****PIC1 Masked IRQ Raw Status Register (MIRQ_B)****0x980030A8**

MIRQ_A = IRQ_A & IEN_A & ((MEN_A[0] & MEN[0]) ? (IRQSEL_A & INTOEN_A) : 0)

MIRQ_B = IRQ_B & IEN_B & ((MEN_B[0] & MEN[0]) ? (IRQSEL_B & INTOEN_B) : 0)

By writing "1" to each field, the corresponding bit is cleared. Writing "0" has no effect.

PIC0 Masked FIQ Raw Status Register (MFIQ_A)**0x9800302C****PIC1 Masked FIQ Raw Status Register (MFIQ_B)****0x980030AC**

MFIQ_A = FIQ_A & IEN_A & ((MEN_A[1] & MEN[1]) ? (~IRQSEL_A & INTOEN_A) : 0)

MFIQ_B = FIQ_B & IEN_B & ((MEN_B[1] & MEN[1]) ? (~IRQSEL_B & INTOEN_B) : 0)

By writing "1" to each field, the corresponding bit is cleared. Writing "0" has no effect.

PIC0 Trigger Mode Register (TMODE_A)**0x98003030****PIC1 Trigger Mode Register (TMODE_B)****0x980030B0**

This register selects trigger mode (0: edge, 1: level) for each internal interrupt source.

PIC0 Synchronization Control Register (SYNC_A)**0x98003034****PIC1 Synchronization Control Register (SYNC_B)****0x980030B4**

By default, all internal interrupt source lines are synchronized to HCLK. This register disables synchronization registers (0: sync enabled, 1: sync disabled). Do not disable synchronization if an interrupt source is asynchronous to HCLK

PIC0 Wakeup Control Register (WKUP_A)**0x980003038****PIC1 Wakeup Control Register (WKUP_B)****0x9800030B8**

By default, all interrupt source lines are used for system wakeup events in power saving mode. (0: Enabled, 1: Disabled). Appropriate bits must be enabled before the system enters power saving mode (clock stop mode). Otherwise, system cannot wakeup.

PIC0 Trigger Mode A Register (TMODEA_A)
PIC1 Trigger Mode A Register (TMODEA_B)

0x9800303C
0x980030BC

Along with TMODE register, this register selects both edge trigger mode (0: Single edge, 1: Both edge) for each internal interrupt source.

PIC0 Interrupt Output Enable Register (INTOEN_A)
PIC1 Interrupt Output Enable Register (INTOEN_B)

0x98003040
0x980030C0

These registers have the same function as IEN registers except that only the MIRQ and MFIQ status is affected (0: Disables interrupt, 1: Enables interrupt).

PIC0 Master Enable Register (MEN_A)

0x98003044

Bit	Name	R/W	Reset	Description
31:2	-	R	0	Reserved
1	MENFIQA	R/W	1	FIQ Master Enable for Controller 0 (A)
0	MENIRQA	R/W	1	IRQ Master Enable for Controller 0 (A)

PIC1 Master Enable Register (MEN_B)

0x980030C4

Bit	Name	R/W	Reset	Description
31:2	-	R	0	Reserved
1	MENFIQB	R/W	1	FIQ Master Enable for Controller 1 (B)
0	MENIRQB	R/W	1	IRQ Master Enable for Controller 1 (B)

Master Enable Register (MEN)

0x98003048, 0x980030C8

Bit	Name	R/W	Reset	Description
31:2	-	R	0	Reserved
1	MENFIQ	R/W	1	FIQ Master Enable
0	MENIRQ	R/W	1	IRQ Master Enable

12 DMA Controller

12.1 Overview

The TCC8132 has three 3-channel GDMA (General DMA) controllers for data transfer. The GDMA can be used to perform high-speed transfers between external memory, on-chip memory, memory-mapped external devices, and on-chip peripheral module. It is possible to select channel priority levels with fixed priority or round-robin priority.

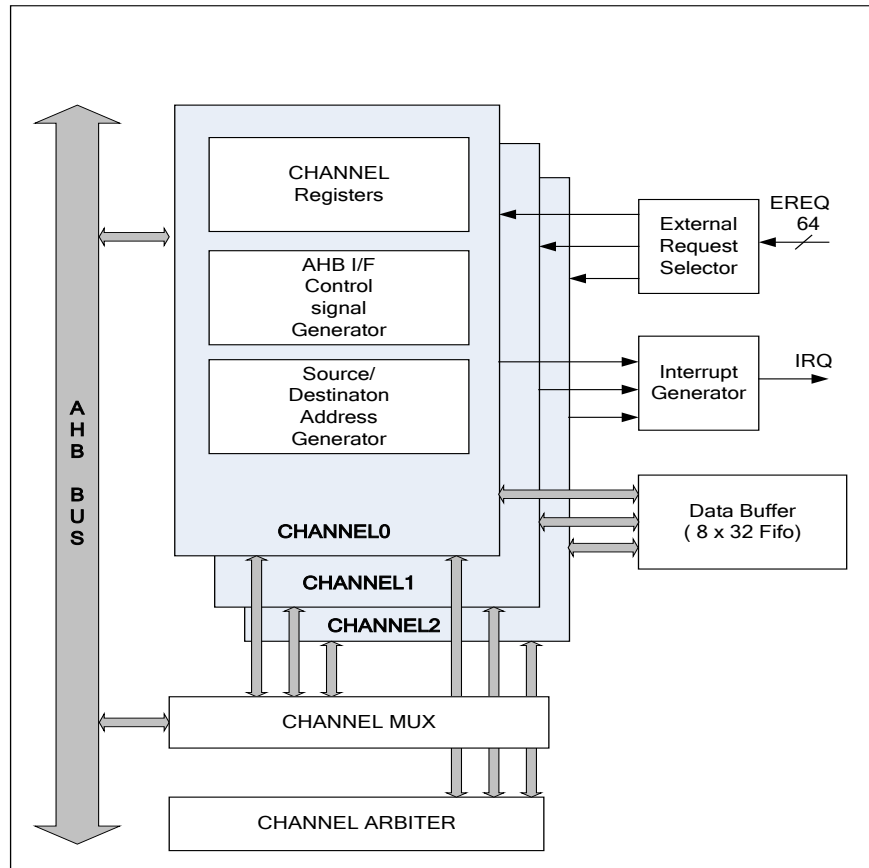


Figure 12.1 GDMA Controller Block Diagram

12.2 Register Description

Table 12.1 General DMA Controller Register Map

DMA Controller 0 Base Address = 0x80004000

DMA Controller 1 Base Address = 0x80004400

DMA Controller 2 Base Address = 0x80004800

	Name	Address	Type	Reset	Description
C H A N N E L 0	ST_SADR0	0x00	R/W	0x00000000	Start Address of Source Block
	SPARAM0	0x04	R/W	0x00000000	Parameter of Source Block
	C_SADR0	0x0C	R	0x00000000	Current Address of Source Block
	ST_DADR0	0x10	R/W	0x00000000	Start Address of Destination Block
	DPARAM0	0x14	R/W	0x00000000	Parameter of Destination Block
	C_DADR0	0x1C	R	0x00000000	Current Address of Destination Block
	HCOUNT0	0x20	R/W	0x00000000	Initial and Current Hop count
	CHCTRL0	0x24	R/W	0x00000000	Channel Control Register
	RPTCTRL0	0x28	R/W	0x00000000	Repeat Control Register
	EXTREQ0_A	0x2C	R/W	0x00000000	External DMA Request Register A
C H A N N E L 1	EXTREQ0_B	0xA0	R/W	0x00000000	External DMA Request Register B
	ST_SADR1	0x30	R/W	0x00000000	Start Address of Source Block
	SPARAM1	0x34	R/W	0x00000000	Parameter of Source Block
	C_SADR1	0x3C	R	0x00000000	Current Address of Source Block
	ST_DADR1	0x40	R/W	0x00000000	Start Address of Destination Block
	DPARAM1	0x44	R/W	0x00000000	Parameter of Destination Block
	C_DADR1	0x4C	R	0x00000000	Current Address of Destination Block
	HCOUNT1	0x50	R/W	0x00000000	Initial and Current Hop count
	CHCTRL1	0x54	R/W	0x00000000	Channel Control Register
	RPTCTRL1	0x58	R/W	0x00000000	Repeat Control Register
C H A N N E L 2	EXTREQ1_A	0x5C	R/W	0x00000000	External DMA Request Register A
	EXTREQ1_B	0xA4	R/W	0x00000000	External DMA Request Register B
	ST_SADR2	0x60	R/W	0x00000000	Start Address of Source Block
	SPARAM2	0x64/0x68	R/W	0x00000000	Parameter of Source Block
	C_SADR2	0x6C	R	0x00000000	Current Address of Source Block
	ST_DADR2	0x70	R/W	0x00000000	Start Address of Destination Block
	DPARAM2	0x74/0x78	R/W	0x00000000	Parameter of Destination Block
	C_DADR2	0x7C	R	0x00000000	Current Address of Destination Block
	HCOUNT2	0x80	R/W	0x00000000	Initial and Current Hop count
	CHCTRL2	0x84	R/W	0x00000000	Channel Control Register
	RPTCTRL2	0x88	R/W	0x00000000	Repeat Control Register
	EXTREQ2_A	0x8C	R/W	0x00000000	External DMA Request Register A
	EXTREQ2_B	0xA8	R/W	0x00000000	External DMA Request Register B
	CHCONFIG	0x90	R/W	0x00000000	Channel Configuration Register

$$[\text{nnnn} = 0x4000/0x4400/0x4800 + 0x30 \times N, N=0,1,2]$$

Start Source Address Register (ST_SADR)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ST_SADR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ST_SADR[15:0]															

This register contains the start address of source memory block for DMA transfer. The transfer begins reading data from this address.

$$[\text{nnnn} = 0x4010/0x4410/0x4810 + 0x30 \times N, N=0,1,2]$$

Start Destination Address Register (ST_DADR)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ST_DADR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ST_DADR[15:0]															

This register contains the start address of destination memory block for DMA transfer.

$$[\text{nnnn} = 0x4004/0x4404/0x4804 + 0x30 \times N, N=0,1,2]$$

Source Block Parameter Register (SPARAM)

0x8000nnnn

Source Mask Parameters Register (S.Mask)															Source Mask		
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16		
								SMASK[23:8]									
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
SMASK[7:0]								SINC[7:0]									

SMASK[23:0]	[31:8]	Source Address Mask Register
0	Non-masked	
1	Masked so that source address bit does not change during DMA transfer	

Each bit field controls the dedicated bit of source address field. That is, if SMASK[23] is set to 1, the 28th bit of source address is masked, and if SMASK[22] is set to 1, the 27th bit of source address is masked, and so on. If a bit is masked, the corresponding bit of address bus does not change during DMA transfer. This function can be used to generate circular buffer address.

SINC[7:0]	[7:0]	Source Address Increment Register
sinc	Source address is added by amount of sinc at every write cycles. sinc is represented as 2's complement, so if SINC[7] is 1, the source address is decremented.	

The addresses of DMA transfer have 32bit wide, but the upper 4bit of them are not affected during DMA transfer. If the source or destination address reaches its maximum address space like 0x7FFFFFFF or 0x2FFFFFFF, the next transfer starts from 0x70000000 or 0x20000000 not from 0x80000000 or 0x30000000.

[nnnn = 0x4014/0x4414/0x4814 + 0x30×N, N=0,1,2]

Destination Block Parameter Register (DPARAM)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMASK[23:8]								7	6	5	4	3	2	1	0
DMASK[7:0]								DINC[7:0]							

DMASK[28:0]	[31:8]	Destination Address Mask Register
0		Non-masked
1		Masked so that destination address bit does not change during DMA transfer

Each bit field controls the corresponding bit of source address field. That is, if DMASK[23] is set to 1, the 28th bit of source address is masked, and if DMASK[22] is set to 1, the 27th bit of source address is masked, and so on. If a bit is masked, the corresponding bit of address bus does not change during DMA transfer. This function can be used to generate circular buffer address.

DINC[7:0]	[7:0]	Destination Address Increment Register
dinc		Destination address is added by amount of dinc at every write cycles. dinc is represented as 2's complement, so if DINC[7] is 1, the destination address is decremented.

The addresses of DMA transfer have 32bit wide, but the upper 4bit of them are not affected during DMA transfer. If the source or destination address reaches its maximum address space like 0x7FFFFFFF or 0x2FFFFFFF, the next transfer starts from 0x70000000 or 0x20000000, not from 0x80000000 or 0x30000000.

[nnnn = 0x400C/0x440C/0x480C + 0x30×N, N=0,1,2]

Current Source Address Register (C_SADR)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
C_SADR[31:16]								7	6	5	4	3	2	1	0
C_SADR[15:0]															

This register contains the current source address of DMA transfer. It represents the address which the DMA currently reads. This is read only register.

[nnnn = 0x401C/0x441C/0x481C + 0x30×N, N=0,1,2]

Current Destination Address Register (C_DADR)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
C_DADR[31:16]								7	6	5	4	3	2	1	0
C_DADR[15:0]															

This register contains current destination address of DMA transfer. It represents the address which the DMA currently writes.

$$[\text{nnnn} = 0x4020/0x4420/0x4820 + 0x30 \times N, N=0,1,2]$$

HOP Count Register (HCOUNT)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
C_HCNT[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ST_HCNT[15:0]															

C_HCNT[15:0]	[31:16]	Current Hop Count
cn	Represent cn number of Hop transfer remains	

ST_HCNT[15:0]	[15:0]	Start Hop Count
sn	DMA transfers data by amount of sn Hop transfers	

At the beginning of transfer, the C_HCNT is updated by ST_HCNT register. At the end of every hop transfer, this is decremented by 1 until it reaches zero. When this reaches zero, the DMA finishes its transfer and may generate its interrupt according to IEN flag of CHCTRL register.

$$[\text{nnnn} = 0x4024/0x4424/0x4824 + 0x30 \times N, N=0,1,2]$$

Channel Control Register (CHCTRL)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CONT	DTM	SYNC	HRD	LOCK	BST	TYPE		BSIZE		WSIZE		FLG	IEN	REP	EN

CONT	[15]	Issue Continuous Transfer
0	DMA transfer begins from ST_SADR / ST_DADR address	
1	DMA transfer begins from C_SADR / C_DADR address. It must be used after the former transfer has been executed, so that C_SADR and C_DADR contain a meaningful value.	

DTM	[14]	Differential Transfer Mode
0	Differential Transfer Mode Disable	
1	Differential Transfer Mode Enable for WSIZE = 10 and BSIZE = 11: 32bit to 16bit transfer 4 Read(Word Unit) / 8 Write(Half-Word Unit) for WSIZE = 11 and BSIZE = 11: 16bit to 32bit transfer. 8 Read(Half-Word Unit) / 4 Write(Word Unit)	

SYNC	[13]	Hardware Request Synchronization
0	Do not Synchronize Hardware Request.	
1	Synchronize Hardware Request. This bit must be set as "1" for the following requests. UT0_TX, UT0_RX, UT1_TX, UT1_RX, UT2_TX, UT2_RX, UT3_TX, UT3_RX, UT4_TX, UT4_RX, UT5_TX, UT5_RX, CD_RX, I2S0_TX, I2S0_RX, I2S1_TX, I2S1_RX, I2S2_TX, I2S2_RX, I2S3_TX, I2S3_RX, CAN0, CAN1, CAN2	

HRD	[12]	Hardware Request Direction
0	ACK/EOT signals are issued when DMA-Read Operation.	
1	ACK/EOT signals are issued When DMA-Write Operation.	

LOCK	[11]	Issue Locked Transfer
1	DMA transfer executed with lock transfer	

Lock field controls the LOCK signal (refer to AHB specification). When the LOCK is set to 1, the DMA transfer is not bothered by other AHB masters like LCD controller, ARM etc. This field is only meaningful in case of non-burst type transfers.

BST	[10]	BURST Transfer
0	DMA transfer executed with arbitration.	
1	DMA transfer executed with no arbitration. (burst operation)	

Arbitration means that at the end of every HOP transfer, the AHB bus is released from DMA channel so other master can occupy the bus when that master has requested the bus.

Burst means that once the DMA request occurs, all of transfers are executed without further DMA requests.

TYPE[1:0]	[9:8]	Transfer Type
00		SINGLE transfer with edge-triggered detection
11		SINGLE transfer with level-sensitive detection
01		Hardware transfer
10		Software transfer

In SINGLE Type, after one Hop data transferring DMA checks External DMA Request (DREQ) and then if it is active, DMA transfers next hop data. DREQ is detected either level-sensitive or edge-triggered by SINGLE transfer TYPE. The 1 Hop of transfer means 1 burst of read followed by 1 burst of write. 1 burst means 1, 2 or 4 consecutive read or write cycles defined by BSIZE field of CHCTRL register. The figure below illustrates the relation between the transfers above.

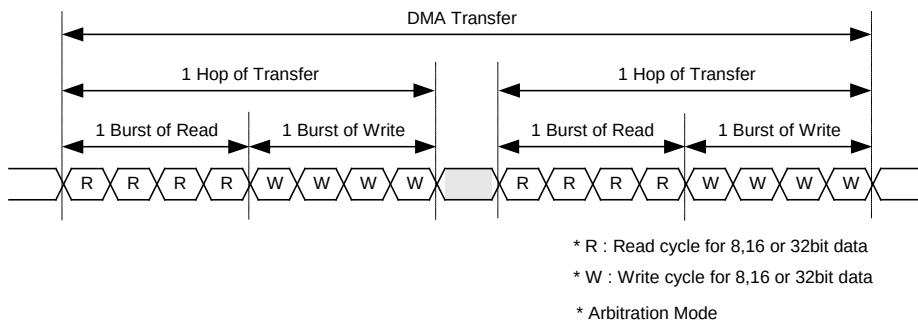


Figure 12.2 Relation between Hop and Burst Transfers (If burst size is 4)

Hardware type transfer means that the DMA transfer triggered by external or internal hardware blocks selected by DMASEL field in CHCTRL register.

Software type transfer means the DMA transfer triggered by EN bit of CHCTRL Register. When this is set to 1, the transfer request signal is generated internally and then the transfer begins immediately.

Figure 12.3 below shows the example of various types of transfer.

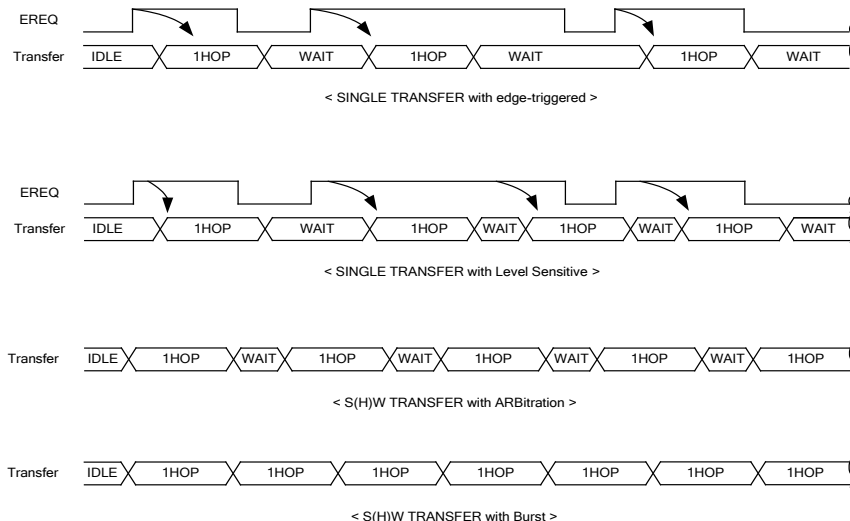


Figure 12.3 Example of Various Types of Transfer.

BSIZE[1:0]	[7:6]	Burst Size
0		1 Burst transfer consists of 1 read or write cycle.
1		1 Burst transfer consists of 2 read or write cycles
2		1 Burst transfer consists of 4 read or write cycles
3		1 Burst transfer consists of 8 read or write cycles

WSIZE[1:0]	[5:4]	Word Size
0		Each cycle read or write 8bit data
1		Each cycle read or write 16bit data
2, 3		Each cycle read or write 32bit data

FLAG	[3]	DMA Done Flag
1		Represents that all hop of transfers are fulfilled.
1		Clears FLAG to 0

It is not automatically cleared by another transfer start, so a user must clear this flag to 0 to check DMA status correctly before starting any other DMA transfer.

IEN	[2]	Interrupt Enable
1		At the same time the FLAG goes to 1, DMA interrupt request is generated.

To generate IRQ or FIQ interrupt, the DMA flag of IEN register in the interrupt controller must be set to 1 in advance.

REP	[1]	Repeat Mode Control
0		After all of hop transfer has executed, the DMA channel is disabled
1		The DMA channel remains enabled. When another DMA request occurred, the DMA channel starts transferring data again with the same manner (type, address, increment, mask) as the latest transfer of that channel.

EN	[0]	DMA Channel Enable
0		DMA channel is terminated and disabled. It does not affect the HCOUNT register, so if the current hop counter is not zero when channel is disabled, it is possible that the transfer illegally starts right after channel is re-enabled. Make sure that HCOUNT is zero not to continue transfer after channel is re-enabled.
1		DMA channel is enabled. If software type transfer is selected, this bit generates DMA request directly, or if hardware type transfer is used, the selected interrupt request flag generates DMA request.

[nnnn = 0x4028/0x4428/0x4828 + 0x30×N, N=0,1,2]

Repeat Control Register (RPTCTRL)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DRI	EOT	0						RPTCNT[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RPTCNT[15:0]															

DRI	[31]	Disable Repeat Interrupt
0		DMA Interrupt occurs at the end of each Repeated DMA operation.
1		DMA Interrupt occurs at the last DMA Repeated DMA operation

This bit is meaningful when the Repeat Mode is enabled.

EOT	[30]	EOT
0		EOT signal occurs at the end of each Repeated DMA operation in hardware (including Single) transfer Mode.
1		EOT Signal occurs at the last Repeated DMA operation in hardware (including Single) transfer Mode.

This bit is meaningful when the Repeat Mode is enabled.

RPTCNT[23:0]	[23:0]	Repeat Count
0		DMA transfer data endlessly.
None zero		DMA transfer the number of (N + 1) * HCOUNT data

This bit is meaningful when the Repeat Mode is enabled. When this bit is cleared in repeat mode, DMA runs endlessly. To exit endless repeat mode, clear EN bit of DMACTRL or disable Repeat Mode. The circular transfer is available if using repeat count.

[nnnn = 0x402C/0x442C/0x482C+ 0x30×N, N=0,1,2]

External DMA Request Register A (EXTREQn_A)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UT4_RX	I2CS0_TX	I2CS0_RX	EINT7	EINT6	EINT5	EINT4	UT4_TX	SF_USR	SF_PKT	UT3_RX	UT3_TX	UT2_RX	UT2_TX	UT1_TX	UT0_TX
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TC	UT1_RX	UT0_RX	GSB5_RX	GSB5_TX	TSIF	GSB4_RX	NFC	CD_RX	GSB4_TX	I2S0_TX	I2S0_RX	EINT3	EINT2	EINT1	EINT0

[nnnn = 0x40A0/0x44A0/0x48A0+ 0x4×N, N=0,1,2]

External DMA Request Register B (EXTREQn_B)

0x8000nnnn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EOT2[2:0]			EOT1[2:0]			EOT0[2:0]			0	I2CS1_TX	I2CS1_RX	ICTC2	ICTC1	I2S2_TX	I2S2_RX
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ICTC0	CAN2	CAN1	CAN0	UT5_TX	ADC	I2S1_TX	I2S1_RX	GSB3_RX	GSB2_RX	GSB1_RX	GSB0_RX	GSB3_TX	GSB2_TX	GSB1_TX	GSB0_TX

External DMA Request Register A						External DMA Request Register B					
NAME	BIT FIELD	Description				NAME	BIT FIELD	Description			
UT4_RX	EXTREQ[31]	UART4 Receive				EOT2[2]	EXTREQB[31]	GDMA2 EOT[2]			
I2CS0_TX	EXTREQ[30]	I2C Slave 0 TX DREQ				EOT2[1]	EXTREQB[30]	GDMA2 EOT[1]			
I2CS0_RX	EXTREQ[29]	I2C Slave 0 RX DREQ				EOT2[0]	EXTREQB[29]	GDMA2 EOT[0]			
EINT7	EXTREQ[28]	External Interrupt 7				EOT2[2]	EXTREQB[28]	GDMA1 EOT[2]			
EINT6	EXTREQ[27]	External Interrupt 6				EOT2[1]	EXTREQB[27]	GDMA1 EOT[1]			
EINT5	EXTREQ[26]	External Interrupt 5				EOT2[0]	EXTREQB[26]	GDMA1 EOT[0]			
EINT4	EXTREQ[25]	External Interrupt 4				EOT2[2]	EXTREQB[25]	GDMA0 EOT[2]			
UT4_TX	EXTREQ[24]	UART4 Transmit				EOT2[1]	EXTREQB[24]	GDMA0 EOT[1]			
SF_USR	EXTREQ[23]	SPDIF USR DREQ				EOT2[0]	EXTREQB[23]	GDMA0 EOT[0]			
SF_PKT	EXTREQ[22]	SPDIF PKT DREQ				-	EXTREQB[22]	Reserved			
UT3_RX	EXTREQ[21]	UART3 Receive				I2CS1_TX	EXTREQB[21]	I2C Slave 1 TX DREQ			
UT3_TX	EXTREQ[20]	UART3 Transmit				I2CS1_RX	EXTREQB[20]	I2C Slave 1 RX DREQ			
UT2_RX	EXTREQ[19]	UART2 Receive				I2S3_TX	EXTREQB[19]	ICTC Controller 2			
UT2_TX	EXTREQ[18]	UART2 Transmit				I2S3_RX	EXTREQB[18]	ICTC Controller 1			
UT1_TX	EXTREQ[17]	UART1 Transmit				I2S2_TX	EXTREQB[17]	I2S2 Transmit			
UT0_TX	EXTREQ[16]	UART0 Transmit				I2S2_RX	EXTREQB[16]	I2S2 Receive			
TC	EXTREQ[15]	TC DMA Request				UT5_TX	EXTREQB[15]	ICTC Controller 0			
UT1_RX	EXTREQ[14]	UART1 Receive				CAN2	EXTREQB[14]	CAN Controller 2			
UT0_RX	EXTREQ[13]	UART0 Receive				CAN1	EXTREQB[13]	CAN Controller 1			
GSB5_RX	EXTREQ[12]	GSB5 Receive				CAN0	EXTREQB[12]	CAN Controller 0			
GSB5_TX	EXTREQ[11]	GSB5 Transmit				UT5_TX	EXTREQB[11]	UART5 Transmit			
TSIF	EXTREQ[10]	TSIF Controller				ADC	EXTREQB[10]	ADC Controller			
GSB4_RX	EXTREQ[09]	GSB4 Receive				I2S1_RX	EXTREQB[09]	I2S1 Receive			
NFC	EXTREQ[08]	NFC Controller				I2S1_TX	EXTREQB[08]	I2S1 Transmit			
CD_RX	EXTREQ[07]	CD Receive				GSB3_RX	EXTREQB[07]	GSB3 Receive			
GSB4_TX	EXTREQ[06]	GSB4 Transmit				GSB2_RX	EXTREQB[06]	GSB2 Receive			
I2S0_TX	EXTREQ[05]	I2S0 Transmit				GSB1_RX	EXTREQB[05]	GSB1 Receive			
I2S0_RX	EXTREQ[04]	I2S0 Receive				GSB0_RX	EXTREQB[04]	GSB0 Receive			
EINT3	EXTREQ[03]	External Interrupt 3				GSB3_TX	EXTREQB[03]	GSB3 Transmit			
EINT2	EXTREQ[02]	External Interrupt 2				GSB2_TX	EXTREQB[02]	GSB2 Transmit			
EINT1	EXTREQ[01]	External Interrupt 1				GSB1_TX	EXTREQB[01]	GSB1 Transmit			
EINT0	EXTREQ[00]	External Interrupt 0				GSB0_TX	EXTREQB[00]	GSB0 Transmit			

Note:

The following DMA requests are Asynchronous to the DMA controller clock. The SYNC bit of CHCTRL Register must be set to "1" for these requests.

UT0_TX, UT0_RX, UT1_TX, UT1_RX, UT2_TX, UT2_RX, UT3_TX, UT3_RX, UT4_TX, UT4_RX,
 UT5_TX, UT5_RX, CD_RX, I2S0_TX, I2S0_RX, I2S1_TX, I2S1_RX, I2S2_TX, I2S2_RX,
 I2S1_TX, I2S1_RX, CAN0, CAN1, CAN2

Channel Configuration Register (CHCONFIG)

0x80004090/4490/4890

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0									IS2	IS1	IS0	0	MIS2	MIS1	MIS0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0					SWP2	SWP1	SWP0	0	PRI[2:0]			0			FIX

IS2	[22]	Channel 2 Alternate interrupt status
IS1	[21]	Channel 1 Alternate interrupt status
IS0	[20]	Channel 0 Alternate interrupt status
0	No interrupt in channel	
1	Interrupt occurs	

Regardless of the Interrupt enable bit (IEN) of each channel, these bits indicate the interrupt status of the corresponding channel. This bit is automatically cleared when FLAG bit of the corresponding channel is cleared. These bits are read only.

MIS2	[18]	Channel 2 Masked Interrupt Status
MIS1	[17]	Channel 1 Masked Interrupt Status
MIS0	[16]	Channel 0 Masked Interrupt Status
0	Masked interrupt does not occur	
1	Masked Interrupt occurs	

These bits are set when an interrupt occurred and the interrupt enable bit (IEN) of the corresponding channel is set. These are automatically cleared when the FLAG bit of the corresponding bit is cleared. These bits are read only.

SWP2	[10]	Channel 2 SWAP Enable bit
SWP1	[9]	Channel 1 SWAP Enable bit
SWP0	[8]	Channel 0 SWAP Enable bit
0	Do not Swap Channel Data.	
1	Swap Channel Data. Data to be written to destination address is swapped.	

For example, the 32bit source data which consists of 4bytes {D3,D2,D1,D0} is stored {D0,D1,D2,D3} in destination address. The 16bit source data which consists of 2bytes {D1,D0} is stored {D0,D1} in destination address.

PRI[2:0]	[5:4]	Channel Priority
000	CH0 > CH1 > CH2	
001	CH0 > CH2 > CH1	
010	CH1 > CH0 > CH2	
011	CH1 > CH2 > CH0	
100	CH2 > CH1 > CH0	
101	CH2 > CH0 > CH1	

PRI bits are meaningful when fix[0] bit is enabled.

FIX	[0]	Fixed Priority Operation
0	Round-Robin (Cyclic) Mode. Each channel is enabled one by one every one hop transferring.	
1	Fixed Priority Mode. According to PRI bit, the highest priority channel is serviced first and lower priority channel is serviced after higher priority channel operation is finished.	

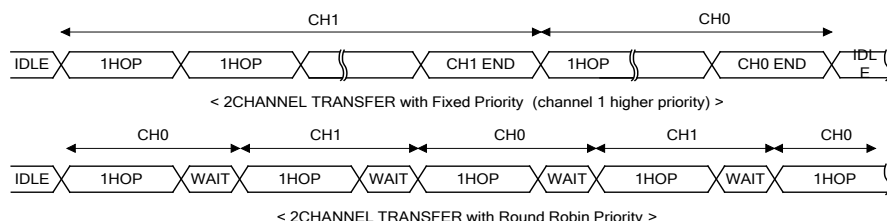


Figure 12.4 Transfer with Two Channels Enabled

13 Timer

13.1 Overview

The Timer block has four 16-bit timers (TC16), two 20-bit timers (TC20), one watchdog timer (WDT) and a 32-bit 1Hz timer (TC32).

The main clock frequency of timers can be configured by the Clock Controller. With the internal basic counter (prescaler), the timer/counter can generate various intervals from microseconds to seconds unit.

13.2 Register Description

Table 13.1 Timer Register Map (Base Address = 0x98004000)

Name	Address	Type	Reset	Description
TCFG0	0x00	R/W	0x000	Timer/Counter 0 Configuration Register
TCNT0	0x04	R/W	0x0000	Timer/Counter 0 Counter Register
TREF0	0x08	R/W	0xFFFF	Timer/Counter 0 Reference Register
TMREF0	0x0C	R/W	0x0000	Timer/Counter 0 Middle Reference Register
TCFG1	0x10	R/W	0x000	Timer/Counter 1 Configuration Register
TCNT1	0x14	R/W	0x0000	Timer/Counter 1 Counter Register
TREF1	0x18	R/W	0xFFFF	Timer/Counter 1 Reference Register
TMREF1	0x1C	R/W	0x0000	Timer/Counter 1 Middle Reference Register
TCFG2	0x20	R/W	0x000	Timer/Counter 2 Configuration Register
TCNT2	0x24	R/W	0x0000	Timer/Counter 2 Counter Register
TREF2	0x28	R/W	0xFFFF	Timer/Counter 2 Reference Register
TMREF2	0x2C	R/W	0x0000	Timer/Counter 2 Middle Reference Register
TCFG3	0x30	R/W	0x000	Timer/Counter 3 Configuration Register
TCNT3	0x34	R/W	0x0000	Timer/Counter 3 Counter Register
TREF3	0x38	R/W	0xFFFF	Timer/Counter 3 Reference Register
TMREF3	0x3C	R/W	0x0000	Timer/Counter 3 Middle Reference Register
TCFG4	0x40	R/W	0x000	Timer/Counter 4 Configuration Register
TCNT4	0x44	R/W	0x00000	Timer/Counter 4 Counter Register
TREF4	0x48	R/W	0xFFFFF	Timer/Counter 4 Reference Register
TCFG5	0x50	R/W	0x000	Timer/Counter 5 Configuration Register
TCNT5	0x54	R/W	0x00000	Timer/Counter 5 Counter Register
TREF5	0x58	R/W	0xFFFFF	Timer/Counter 5 Reference Register
TIREQ	0x60	R/W	0x0000	Timer/Counter n Interrupt Request Register
TWD CFG	0x70	R/W	0x00	Watchdog Timer Configuration Register
TWD CLR	0x74	W	-	Watchdog Timer Clear Register
TWD CNT	0x78	R/W	0xFFE0	Watchdog Timer Counter Register
TC32EN	0x80	R/W	0x00007FFF	32-bit Counter Enable / Pre-scale Value
TC32LDV	0x84	R/W	0x00000000	32-bit Counter Load Value
TC32CMP0	0x88	R/W	0x00000000	32-bit Counter Match Value 0
TC32CMP1	0x8C	R/W	0x00000000	32-bit Counter Match Value 1
TC32PCNT	0x90	R/W	-	32-bit Counter Current Value (pre-scale counter)
TC32MCNT	0x94	R/W	-	32-bit Counter Current Value (main counter)
TC32IRQ	0x98	R/W	0x0000----	32-bit Counter Interrupt Control
PWM	0x100~0x1FF	R/W	-	PDM(PWM) Registers
TCOPORTSEL0	0x800	R/W	0xFFFFFFFF	TCO Port Selection Register 0
TCOPORTSEL1	0x804	R/W	0xFFFFFFFF	TCO Port Selection Register 1
PDMPORTSEL	0x808	R/W	0xFFFFFFFF	PDM(PWM) Port Selection Register

The address of each timer/counter is 16byte-aligned.

The number n represents for each timer/counter in case of timer/counter 4, 5 (that is n = 4 or 5)

13.2.1 Registers for TC16/20

Each timer/counter has three registers for basic operation modes. When operating in counter modes, the external interrupt pin is used as counting clock for that counter.

The output of Timer/Counters (TCO[5:0]) can be monitored through GPIO ports. (Refer to Chapter 18).

The following figures show the block and timing diagram for the 16/20-bit timer/counters.

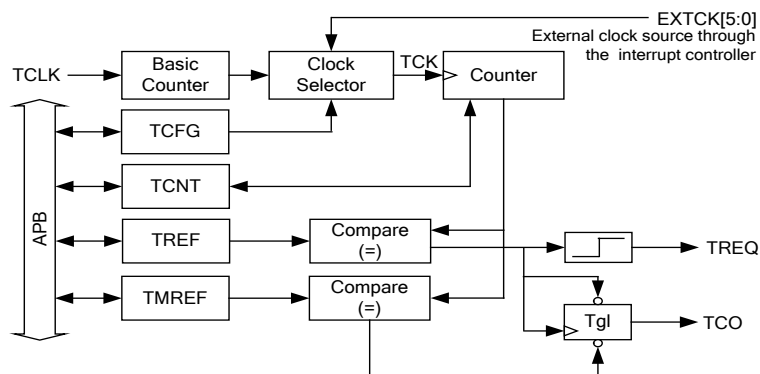


Figure 13.1 Timer Block Diagram

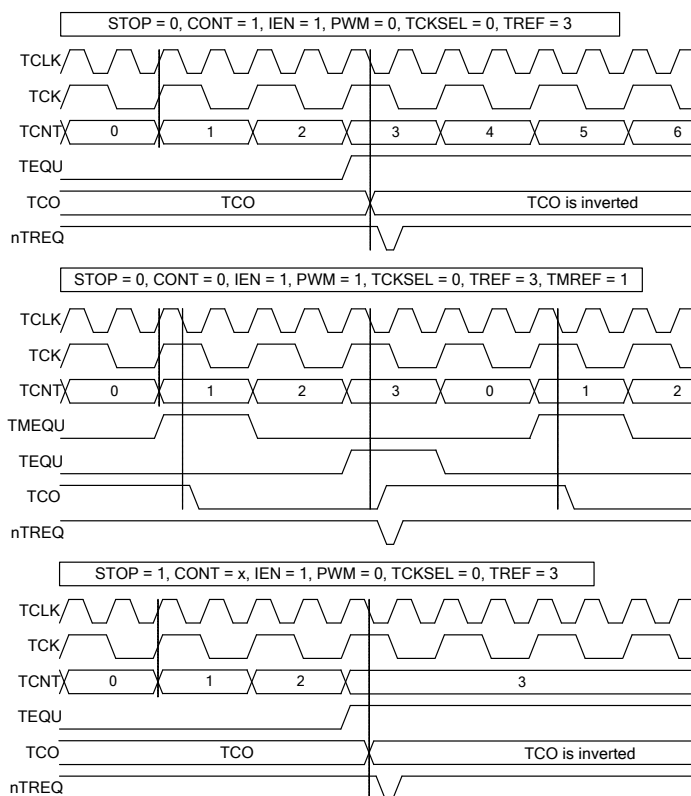


Figure 13.2 Timing Diagram of Timer/Counter

Timer/Counter n Configuration Register (TCFGn)

0x980040n0(n=0~5)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0						STOP	CC	POL	TCKSEL			IEN	PWM	CON	EN

Bit	Name	R/W	Reset	Description
9	STOP	R/W	0	STOP Mode 0 Continuous counting mode. 1 If TCNTn is equal to the TREFn, the TCNTn counter stops to increment.
8	CC	R/W	0	Clear Count 0 TCNTn is not cleared. 1 TCNTn is cleared to zero (1 clock reset pulse is sent to the counter).
7	POL	R/W	0	TCK Polarity 0 TCNTn is incremented at rising edge of the selected counting clock 1 TCNTn is incremented at falling edge of the selected counting clock
6:4	TCKSEL	R/W	0	TCK Select k = 0 ~ 4 TCK is internally generated from divider circuit. It is driven by TCLK, and this value determines the division factor of this circuit. Division factor is $2^{(k+1)}$. k = 5, 6 TCK is internally generated from divider circuit. It is driven by TCLK, and this value determines the division factor of this circuit. Division factor is 2^{2k}. k = 7 TCK is the external pin shared by external interrupt signal.
3	IEN	R/W	0	Interrupt Enable (1 = Enable, 0 = Disable)
2	PWM	R/W	0	PWM Mode Enable 1 Timer/Counter output is changed at every time the TCNTn is equal to TREFn and TMREFn value. It can be used to generate PWM waveform, by changing TMREFn while TREFn fixed. (where, TREFn > TMREFn) 0 Timer/Counter output is changed only when the TCNTn is equal to TREFn. It can be used to generate a rectangular pulse of variable frequency.
1	CON	R/W	0	Continue Counting (If the STOP bit is set, this bit is meaningless). 0 When the TCNTn is reached to TREFn, TCNTn restarts counting from 0 at the next pulse of selected clock source. 1 The TCNTn continues counting from the TREFn.
0	EN	R/W	0	Timer/Counter Enable (1 = Enable, 0 = Disable)

Timer/Counter n Counter Register (TCNTn)

0x980040n4(n=0~5)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0												TCNTn[19:16] (n=4,5)			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TCNTn[15:0] (n=0~5)															

TCNTn is increased by 1 at every pulse of selected clock source. TCNTn can be set to any value by writing to this register. In case of timer 4 and timer 5, it has 20 bits, otherwise it has 16 bits.

Timer/Counter n Reference Register (TREFn)

0x980040n8(n=0~5)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0												TREFn[19:16] (n=4,5)			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TREFn[15:0] (n=0~5)															

When TCNTn is reached at TREFn and the CON flag of TCFGn register is set to 1, the TCNTn is cleared to 0 at the next pulse of selected clock source. According to the TCFGn settings, various kinds of operations may be done. In case of timer 4 and timer 5, it has 20 bit, otherwise it has 16 bit.

Timer/Counter n Middle Reference Register (TMREFn)**0x980040nC(n=0~3)**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TMREFn[15:0] (n=0~3)															

When TCNTn reaches at TMREFn and the PWM of TCFGn register is set to 1, the timer output (TCOn) is cleared to 0 at the negative edge of selected clock source. The TCOn is set to 1 when the TCNTn is reached at TREFn. Therefore, you can generate PWM signal by modifying TMREFn between 0 ~ (TREFn-1).

TMREFn is valid only for timer 0 through 3 (TMREF4 and TMREF5 do not exist).

Timer/Counter Interrupt Request Register (TIREQ)**0x98004060**

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	TWF	TF5	TF4	TF3	TF2	TF1	TF0	0	TWI	TI5	TI4	TI3	TI2	TI1	TI0

Bit	Name	R/W	Reset	Description
14	TWF	R/W	0	Watchdog Timer Flag
13:8	TFn	R/W	0	Timer/Counter n Flag
6	TWI	R/W	0	Watchdog Timer Interrupt Request Flag = TWF & TWDCFG[3]
5:0	TIn	R/W	0	Timer/Counter n Interrupt Request Flag = TFn & TCFGn[3]

If a timer n has reached its reference value, the TFn is set. (Bit n represents for Timer n). If its interrupt request is enabled by bit 3 of TCFGn register, the TIn is set. Each bit can be cleared by writing "1".

13.2.3 Registers for TC32

As illustrated in the figure below, TC32 consists of a pre-scale counter, a main counter and two comparators. The pre-scale counter is a simple 24-bit up-counter which always counts from zero to PRESCALE value programmed in TC32EN register. The 32-bit main counter is incremented only when the prescale counter reaches PRESCALE value. The clock input of TC32 module is generated by the Clock Controller. (Refer to Chapter 10).

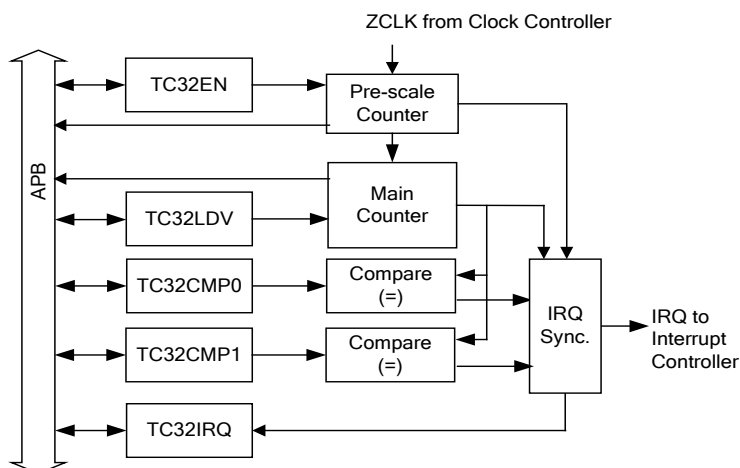


Figure 13.4 32-bit Counter Block Diagram

The available counter modes are described in the table below.

Table 13.2 TC32 Count Mode

Mode	TC32EN Register Bits			Main Counter Operation	
	LOADZERO	LDM1	LDM0	Start Count Value	End Count Value
0	0	0	0	LOADVAL	0xFFFFFFFF
1	0	0	1	LOADVAL	CMP0 (if LOADVAL < CMP0)
2	0	1	0	LOADVAL	CMP1 (if LOADVAL < CMP1)
3	0	1	1	LOADVAL	CMP0 (if LOADVAL < CMP0 ≤ CMP1) or CMP1 (if LOADVAL < CMP1 ≤ CMP0)
4	1	0	0	0	LOADVAL – 1
5	1	0	1	0	CMP0 (if LOADVAL > CMP0)
6	1	1	0	0	CMP1 (if LOADVAL > CMP1)
7	1	1	1	0	CMP0 (if LOADVAL > CMP1 ≥ CMP0) or CMP1 (if LOADVAL > CMP0 ≥ CMP1)

Refer to register descriptions below for CMP0, CMP1 and LOADVAL.

Mode 0 can be used as 1Hz counter mode, if PRESCALE = 0x007FFF, STOPMODE = 0, ZCLK = XTI (32.768kHz)

TC32 Enable / Pre-scale Value Register (TC32EN)

0x98004080

Bit	Name	Default	R/W	Description
31:30	-	0	R	Reserved
29	LDM1	0	R/W	Re-load counter when the counter value matched with CMP1. LOADZERO bit below selects the counter load (start) value.
28	LDM0	0	R/W	Re-load counter when the counter value matched with CMP0. LOADZERO bit below selects the counter load (start) value.
27	-	0	R	Reserved
26	STOPMODE	0	R/W	0 = Free Running Mode 1 = Stop Mode.
25	LOADZERO	0	R/W	By default, counter starts from LOADVAL. When this bit is enabled (1), the counter is forced to count from "0" to "LOADVAL – 1".
24	ENABLE	0	R/W	Counter Enable
23:0	PRESCALE	0x007FFF	R/W	Pre-scale counter load value. The pre-scale counter always runs from "0" up to PRESCALE. The default value is for 1Hz counter when ZCLK = XTI (32.768kHz).

TC32 Load Value Register (TC32LDV)**0x98004084**

Bit	Name	Default	R/W	Description
31: 0	LOADVAL	0x00000000	R/W	Counter Load Value.

The counter restarts whenever one of the TC32En and TC32LDV is written (Disabled and then enabled).

TC32 Match Value 0 Register (TC32CMP0)**0x98004088**

Bit	Name	Default	R/W	Description
31: 0	CMP0	0x00000000	R/W	Counter Match Value

TC32 Match Value 1 Register (TC32CMP1)**0x9800408C**

Bit	Name	Default	R/W	Description
31: 0	CMP1	0x00000000	R/W	Counter Match Value

TC32 Pre-scale Counter Current Value Register (TC32PCNT)**0x98004090**

Bit	Name	Default	R/W	Description
31:24	-	0x00	R	Reserved
23: 0	PCNT	0x00000000	R	Pre-scale counter current value. The AHB system clock must be three times faster than the frequency of ZCLK to read valid value.

TC32 Main Counter Current Value Register (TC32MCNT)**0x98004094**

Bit	Name	Default	R/W	Description
31: 0	MCNT	0x00000000	R	Main counter current value. When RSYNC is enabled, the AHB system clock must be faster than the frequency calculated below. $(ZCLK \text{ frequency}) / (PRESCALE + 1) * 3$

TC32 Interrupt Control Register (TC32IRQ)**0x98004098**

Bit	Name	Default	R/W	Description
31	IRQCLR	0	R/W	Interrupt Clear Control. When this bit is 0, interrupt status bits (IRQRSTAT) are cleared by reading this register. When this bit is set, IRQSTAT bits are cleared only if written with non-zero value.
30	RSYNC	0	R/W	Synchronization control for Counter Current Value Registers (TC32PCNT and TC32MCNT) 0 = Enable 1 = Disable
29:24	BITSEL	0x00	R/W	Counter bit selection value for interrupt generation. Any one of the counter bits {MCNT[31:0], PCNT[23:0]} selected by BITSEL is used to generate an interrupt. 0x00 ~ 0x17 : PCNT[0] ~ PCNT[23] 0x18 ~ 0x38: MCNT[0] ~ MCNT[31]
23:21	-	0	R/W	Reserved
20	IRQEN[4]	0	R/W	Enable Interrupt at the rising edge of a counter bit selected by BITSEL.
19	IRQEN[3]	0	R/W	Enable Interrupt at the end of pre-scale count
18	IRQEN[2]	0	R/W	Enable Interrupt at the end of count
17	IRQEN[1]	0	R/W	Enable Interrupt when the counter value matched with CMP1
16	IRQEN[0]	0	R/W	Enable Interrupt when the counter value matched with CMP0
15:13	-	0	R/W	Reserved
12:8	IRQRSTAT	0x00	R/W	Interrupt Raw Status. Refer to the description for IRQEN above.
7:5	-	0	R/W	Reserved
4:0	IRQMSTAT	0x00	R/W	Masked Interrupt Status = IRQRSTAT & IRQEN

13.2.4 Registers for PWM (Pulse Width Modulation)

PWM module has four separate PWM blocks (PWM A/B/C/D). Each PWM block can be programmed for actuator driver or other devices controlled by PWM/PDM pulse.

Table 13.3 PWM Register Map (Base Address = 0x98004100)

Name	Address	Type	Reset	Description
PwmSt	0x0000	R	0x00000000	PWM block status
PwmEn	0x0004	R/W	0x0000000F	PWM block enable & PWM Trigger
PwmMode	0x0008	R/W	0x000000AA	PWM output mode & PWMCLKDiv
PwmLoop	0x000C	R/W	0x00000000	PWM block operation Steps
PwmAPstn12	0x0010	R/W	0x00200010	PWM block A position 1 & 2
PwmAPstn34	0x0014	R/W	0x00400030	PWM block A position 3 & 4
PwmBPstn12	0x0018	R/W	0x00200010	PWM block B position 1 & 2
PwmBPstn34	0x001C	R/W	0x00400030	PWM block B position 3 & 4
PwmCPstn12	0x0020	R/W	0x00200010	PWM block C position 1 & 2
PwmCPstn34	0x0024	R/W	0x00400030	PWM block C position 3 & 4
PwmDPstn12	0x0028	R/W	0x00200010	PWM block D position 1 & 2
PwmDPstn34	0x002C	R/W	0x00400030	PWM block D position 3 & 4
PwmOutRegA1	0x0030	R/W	0xAA0000AA	PWM A1 block Reg mode Output value
PwmOutRegA2	0x0034	R/W	0x0AAAAA00	PWM A2 block Reg mode Output value
PwmOutRegA3	0x0038	R/W	0x0AAAAA00	PWM A3 block Reg mode Output value
PwmOutRegA4	0x003C	R/W	0xAA0000AA	PWM A4 block Reg mode Output value
PwmOutRegB1	0x0040	R/W	0x55000000	PWM B1 block Reg mode Output value
PwmOutRegB2	0x0044	R/W	0x00550000	PWM B2 block Reg mode Output value
PwmOutRegB3	0x0048	R/W	0x00005500	PWM B3 block Reg mode Output value
PwmOutRegB4	0x004C	R/W	0x00000055	PWM B4 block Reg mode Output value
PwmOutRegC1	0x0050	R/W	0xAA000000	PWM C1 block Reg mode Output value
PwmOutRegC2	0x0054	R/W	0x00AA0000	PWM C2 block Reg mode Output value
PwmOutRegC3	0x0058	R/W	0x0000AA00	PWM C3 block Reg mode Output value
PwmOutRegC4	0x005C	R/W	0x000000AA	PWM C4 block Reg mode Output value
PwmOutRegD1	0x0060	R/W	0x55000000	PWM D1 block Reg mode Output value
PwmOutRegD2	0x0064	R/W	0x00550000	PWM D2 block Reg mode Output value
PwmOutRegD3	0x0068	R/W	0x00005500	PWM D3 block Reg mode Output value
PwmOutRegD4	0x006C	R/W	0x00000055	PWM D4 block Reg mode Output value

PWM Status (PwmSt)

0x98004100

Bit	Name	R/W	Reset	Description
31:4	-	R	0x0	Reserved
3	BUSY_D	R	0x0	PWM_D is Busy
2	BUSY_C	R	0x0	PWM_C is Busy
1	BUSY_B	R	0x0	PWM_B is Busy
0	BUSY_A	R	0x0	PWM_A is Busy

PWM Enable (PwmEn)

0x98004104

Bit	Name	R/W	Reset	Description
31:20	-	R/W	0xFF	Reserved
19	TRIG_D	R/W	0x0	PWM_D Output Generation Triggering
18	TRIG_C	R/W	0x0	PWM_C Output Generation Triggering
17	TRIG_B	R/W	0x0	PWM_B Output Generation Triggering
16	TRIG_A	R/W	0x0	PWM_A Output Generation Triggering
15:4	-	R	0x0	Reserved
3	EN_D	R/W	0x0	PWM_D Enable
2	EN_C	R/W	0x0	PWM_C Enable
1	EN_B	R/W	0x0	PWM_B Enable
0	EN_A	R/W	0x0	PWM_A Enable

PWM Mode (PwmMode)

0x98004108

Bit	Name	R/W	Reset	Description
31:30	DIV_D	R/W	0x0	PWM D Block input Clock divide 3: Divide by 16 2: Divide by 8 1: Divide by 4 0: Divide by 2
29:28	DIV_C	R/W	0x0	PWM C Block input Clock divide 3: Divide by 16 2: Divide by 8 1: Divide by 4 0: Divide by 2
27:26	DIV_B	R/W	0x0	PWM B Block input Clock divide 3: Divide by 16 2: Divide by 8 1: Divide by 4 0: Divide by 2
25:24	DIV_A	R/W	0x0	PWM A Block input Clock divide 3: Divide by 16 2: Divide by 8 1: Divide by 4 0: Divide by 2
23:20	-	R	0x0	Reserved
19	INV_D	R/W	0x0	PWM D Output Signal Inverse
18	INV_C	R/W	0x0	PWM C Output Signal Inverse
17	INV_B	R/W	0x0	PWM B Output Signal Inverse
16	INV_A	R/W	0x0	PWM A Output Signal Inverse
15:12	MODE_D	R/W	0x2	PWM D block mode 4'b0001: Phase Mode 4'b0010: Register Out Mode 4'b0100: Register Out Mode2 4'b1001: Phase Mode2 Others: Reserved
11:8	MODE_C	R/W	0x2	PWM C block mode 4'b0001: Phase Mode 4'b0010: Register Out Mode 4'b0100: Register Out Mode2 4'b1001: Phase Mode2 Others: Reserved
7:4	MODE_B	R/W	0x2	PWM B block mode 4'b0001: Phase Mode 4'b0010: Register Out Mode 4'b0100: Register Out Mode2 4'b1001: Phase Mode2 Others: Reserved
3:0	MODE_A	R/W	0x2	PWM A block mode 4'b0001: Phase Mode 4'b0010: Register Out Mode 4'b0100: Register Out Mode2 4'b1001: Phase Mode2 Others: Reserved

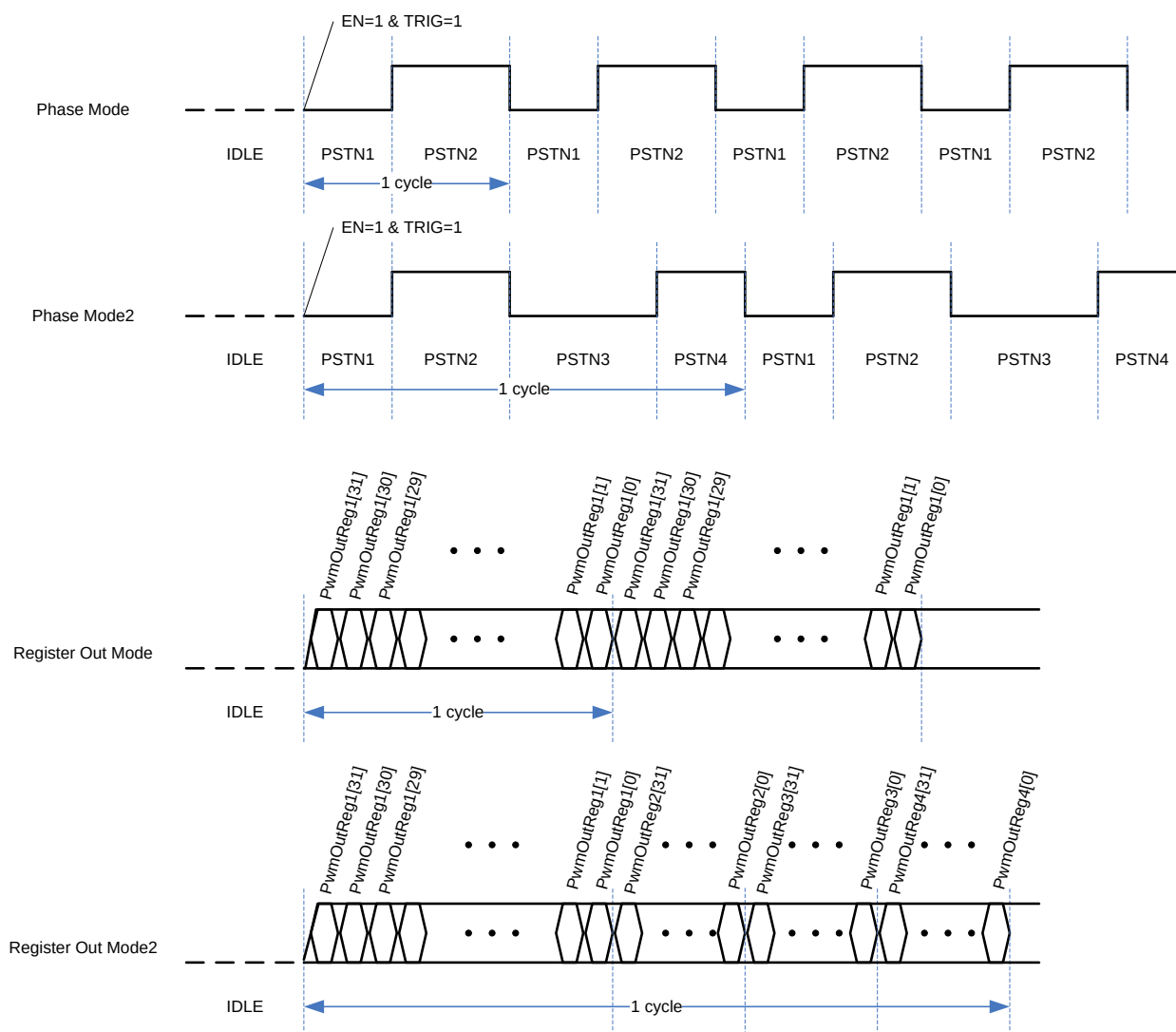


Figure 13.5 PWM Operation Modes

PWM Loop (PwmLoop)

0x9800410C

Bit	Name	R/W	Reset	Description
31:16	-	-	0x0	Reserved
15:12	Loop_D	R/W	0x0	PWM block D step repetition value (if 0, infinite value mode)
11:8	Loop_C	R/W	0x0	PWM block C step repetition value (if 0, infinite value mode)
7:4	Loop_B	R/W	0x0	PWM block B step repetition value (if 0, infinite value mode)
3:0	Loop_A	R/W	0x0	PWM block A step repetition value (if 0, infinite value mode)

PWM PSTN12 n (PwmAPstn12/PwmBPstn12/PwmCPstn12/PwmDPstn12)

0x98004110/18/20/28

Bit	Name	R/W	Reset	Description
31:16	PwmPstn2_N	R/W	0x20	Input Clock (PWMClk N) count value2 (N=A,B,C,D)
15:0	PwmPstn1_N	R/W	0x10	Input Clock (PWMClk N) count value1 (N=A,B,C,D)

PWM PSTN34 n (PwmAPstn34/PwmBPstn34/PwmCPstn34/PwmDPstn34)

0x98004114/1C/24/2C

Bit	Name	R/W	Reset	Description
31:16	PwmPstn4_N	R/W	0x40	Input Clock (PWMClk N) count value4 (N=A,B,C,D)
15:0	PwmPstn3_N	R/W	0x30	Input Clock (PWMClk N) count value3 (N=A,B,C,D)

PWMOUT n1 (PwmOutRegA1/PwmOutRegB1/PwmOutRegC1/PwmOutRegD1)

0x98004130/40/50/60

Bit	Name	R/W	Reset	Description
31:0	PwmOutReg_N1	R/W	A1: 0xAA0000AA B1: 0x55000000 C1: 0xAA000000 D1: 0x55000000	PWM output data pattern value 1

PWMOUT n2 (PwmOutRegA2/PwmOutRegB2/PwmOutRegC2/PwmOutRegD2) 0x98004134/44/54/64

Bit	Name	R/W	Reset	Description
31:0	PwmOutReg_N2	R/W	A2: 0x00AAAA00 B2: 0x00550000 C2: 0x00AA0000 D2: 0x00550000	PWM output data pattern value 2

PWMOUT n3 (PwmOutRegA3/PwmOutRegB3/PwmOutRegC3/PwmOutRegD3) 0x98004138/48/58/68

Bit	Name	R/W	Reset	Description
31:0	PwmOutReg_N3	R/W	A3: 0x00AAAA00 B3: 0x00005500 C3: 0x0000AA00 D3: 0x00005500	PWM output data pattern value 3

PWMOUT n4 (PwmOutRegA4/PwmOutRegB4/PwmOutRegC4/PwmOutRegD4) 0x9800413C/4C/5C/6C

Bit	Name	R/W	Reset	Description
31:0	PwmOutReg_N4	R/W	A4: 0xAA0000AA B4: 0x00000055 C4: 0x000000AA D4: 0x00000055	PWM output data pattern value 4

13.2.5 Port Multiplexing

The signal name GBUS[191:0] found in GPIO Function Selection Registers (GPIO*_FS* of GPIO block) is the final output of the port multiplexers in this module.

Any TCO/PWM signal can be mapped to any bit position of GBUS[191:0] by writing an index number to the registers below in TCC8132. The pre-mapped bit position of GBUS[191:0] is shown in the table below.

Table 13.4 Pre-mapped bit position for PWM Port

GBUS	GPIO
GBUS[181]	PK[21]
GBUS[111:110]	GPIOLD[21:20]

Table 13.5 TCO Port Selection Register Map (Base Address = 0x98004800)

Name	Address	Type	Reset	Description
TCOPORTSEL0	0x00	R/W	0xFFFFFFFF	TCO Port Selection Register 0
TCOPORTSEL1	0x04	R/W	0xFFFFFFFF	TCO Port Selection Register 1
PDMPORTSEL	0x08	R/W	0xFFFFFFFF	PDM(PWM) Port Selection Register

TCO Port Selection Register 0 (TCOPORTSEL0) 0x98004800

Bit	Name	R/W	Reset	Description
31:24	TCO[3]	R/W	0xFF	Port number for TCO[3]
23:16	TCO[2]	R/W	0xFF	Port number for TCO[2]
15:8	TCO[1]	R/W	0xFF	Port number for TCO[1]
7:0	TCO[0]	R/W	0xFF	Port number for TCO[0]

TCO Port Selection Register 1 (TCOPORTSEL1) 0x98004804

Bit	Name	R/W	Reset	Description
31:16	-	R/W	0xFF	Reserved
15:8	TCO[5]	R/W	0xFF	Port number for TCO[5]
7:0	TCO[4]	R/W	0xFF	Port number for TCO[4]

PDM (PWM) Port Selection Register (PDMPORTSEL) 0x98004808

Bit	Name	R/W	Reset	Description
31:24	PDM[3]	R/W	0xFF	Port number for PDM[3]
23:16	PDM[2]	R/W	0xFF	Port number for PDM[2]
15:8	PDM[1]	R/W	0xFF	Port number for PDM[1]
7:0	PDM[0]	R/W	0xFF	Port number for PDM[0]

14 IC_TC (Input Capture Timer)

14.1 Feature

- 8bit Pre-Scaling Count Clock
- 256 Capture Input
- Interrupt: Time-Out, Frequency Change Detect, Duty Change Detect, Filter Count Full, Fall Edge Count Full, PD Count Full, Rise Edge Notify, Fall Edge Notify.
- Noise Filter: 16 Mode (Immediate, Up/Down, Up/Hold, Up/Reset, Mixed Mode)
- 16bit Falling Edge Count of Capture Input
- 28bit Duty Capture, Frequency Capture Counter.
- Free TimeStamp Counter

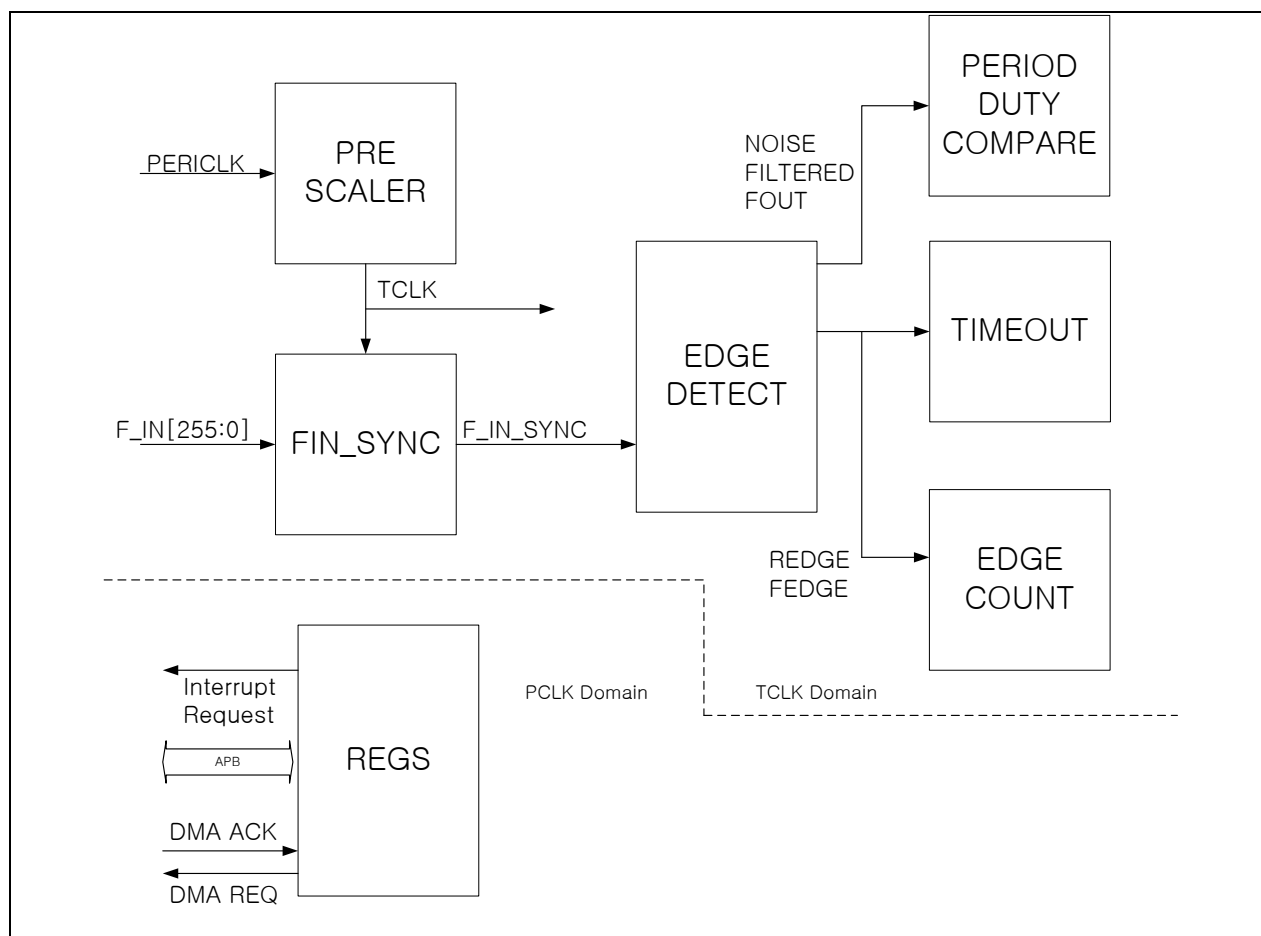


Figure 14.1 IC_TC Block Diagram

14.2 Function Description

14.2.1 Time-Out Unit

You can check if input signal for capturing is propagated or not. 28-bit time-out counter is counted until rise or fall edge is detected. If there are no edge detection, interrupt is generated.

14.2.2 Noise Filter & Edge Detect Unit

IC_TC provides 16 modes for noise filtering. There are Immediate, Up/Down, Up/Hold, Up/Reset mode for detecting valid both rise and fall edge. You can also use mixed mode for noise filtering such as Up/Down for rise edge and Up/Reset for fall edge. Noise filtered signal is propagated to Frequency & Duty Compare Unit, and edge detected signal is propagated to Time-Out Unit and Fall edge Count Unit.

14.2.3 Fall Edge Count Unit

16bit Fall Edge Counter detects and counts fall edge from Noise Filter & Edge Detect Unit. When the counter becomes full, interrupt is generated.

14.2.4 Frequency & Duty Compare Unit

Noise filtered signal is measured for comparing frequency and duty. When you detect frequency or duty is changed, interrupt is generated. Frequency change means that PD counter value changes between rise edge and rise edge. Duty change means PD counter value changes between high level of previous and current signal.

14.3 Signal Descriptions

14.3.1 Global Signals

Name	Direction	Description	Related Blocks
PORn	Input	Reset Signal	

14.3.2 Clock Source Signals

Name	Direction	Description	Related Blocks
PERICK	Input	Clock Signal from CKC	CKC

14.3.3 Interrupt Signals

Name	Direction	Description	Related Blocks
IREQ_IC_TC	Output	Output Interrupt	

14.3.4 DMA Request Signals

Name	Direction	Description	Related Blocks
DMA_REQ	Output	DMA Request	
DMA_ACK	Input	DMA Acknowledge	

14.3.5 Edge Detection Related Signals

Name	Direction	Description	Related Blocks
F_IN	Input	Input Signal for Detection	Edge Detection Unit

14.4 Register Description

Table 14.1 IC_TC Register Map (Base Address = 0x9800C000, 0x9800C100, 0x9800C200)

Name	Address	Type	Reset	Description
IC_TC_EN	0x000	R/W	0x0	IC_TC_EN Register
IC_TC_CTRL	0x004	R/W	0x00004800	IC_TC Ctrl Register
IC_TC_IRQ	0x008	R/W	0x0	IC_TC IRQ Register
IC_TC_TOVL	0x00c	R/W	0x0ffffff	IC_TC TimeOut Value Setting Register
IC_TC_REDGE_VL	0x010	R/W	0x0ffffff	IC_TC REDGE Matching Value Setting Register
IC_TC_FEDGE_VL	0x014	R/W	0x0ffffff	IC_TC FEDGE Matching Value Setting Register
IC_TC_PRD_RND_VL	0x018	R/W	0x0ffffff	IC_TC Period Compare Rounding Value Setting Register
IC_TC_DT_RND_VL	0x01c	R/W	0x0ffffff	IC_TC Duty Compare Rounding Value Setting Register
IC_TC_ECNT_VL	0x020	R/W	0xffff	IC_TC Fall Edge Counter Matching Value Setting Register
IC_TC_PRD_CNT_PVL	0x030	R	0x0	Counted Previous Period Value
IC_TC_PRD_CNT_PPVL	0x034	R	0x0	Counted 1 cycle before-previous Period Value
IC_TC_DT_CNT_PVL	0x038	R	0x0	Counted Previous Duty Value
IC_TC_DT_CNT_PPVL	0x03c	R	0x0	Counted 1 cycle before-previous Duty Value
IC_TC_FEDGE_DET_CNT_CVL	0x040	R	0x0	Counted Current Falling Edge Count Value
IC_TC_FEDGE_DET_CNT_PVL	0x044	R	0x0	Counted Previous Falling Edge Count Value
IC_TC_REDGE_TSCNT_VL	0x048	R	0x0	Counted Rising Edge TimeStamp Count Value
IC_TC_FEDGE_TSCNT_VL	0x04c	R	0x0	Counted Falling Edge TimeStamp Count Value

IC_TC_EN Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IC_TC_EN										TCLK_EN	TS_CNT_EN	FLT_CNT_EN	TO_CNT_EN	FEDGE_CNT_EN	PD_CMP_CNT_EN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
											TS_CNT_CLEAR	FLT_CNT_CLEAR	TO_CNT_CLEAR	FEDGE_CNT_CLEAR	PD_CMP_CNT_CLEAR

Field	Name	RW	Reset	Description
[31]	IC_TC_EN	R/W	0	IC_TC Enable
[30:22]		R/W	0	Reserved
[21]	TCLK_EN	R/W	1'b0	Pre-Scaler Counter Enable
[20]	TS_CNT_EN	R/W	0	TimeStamp Counter Enable
[19]	FLT_CNT_EN	R/W	1'b0	Filter Counter Enable
[18]	TO_CNT_EN	R/W	1'b0	TimeOut Counter Enable
[17]	FEDGE_CNT_EN	R/W	1'b0	Falling Edge Counter Enable
[16]	PD_CMP_CNT_EN	R/W	1'b0	Period/Duty CMP Counter Enable
[15:5]		R/W	0	Reserved
[4]	TS_CNT_CLEAR	R/W	0	TimeStamp Counter CLEAR
[3]	FLT_CNT_CLEAR	R/W	0	Filter Counter CLEAR
[2]	TO_CNT_CLEAR	R/W	0	TimeOut Counter CLEAR
[1]	FEDGE_CNT_CLEAR	R/W	0	Falling Edge Counter CLEAR
[0]	PD_CMP_CNT_CLEAR	R/W	0	Period/Duty CMP Counter CLEAR

IC_TC_CTRL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCKEN					TCLK_POL	TCLK_SEL				FLT_F_MODE		FLT_R_MODE		CMP_ERR_SEL	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMA_EN	DMA_REQ_SEL								F_IN_SEL						

Field	Name	RW	Reset	Description																																								
[31]	LOCKEN	W	1'b0	Disable to change external input for capture																																								
[26]	TCLK_POL	R/W	1'b0	Pre-Scaler Output Clock Polarity Change Enable																																								
[25:22]	TCLK_SEL	R/W	4'b0	Pre-Scaler Output Clock Select TCLK_SEL[2:0]: select divided clock (2,4,8,16, 32, 64, 128, 256) TCLK_SEL[3] = 1'b0: counter out select for output clock TCLK_SEL[3] = 1'b1: bypass clock select for output clock																																								
[21:20]	FLT_F_MODE	R/W	2'b0	Filter mode for Falling Edge 2'b00 : RST mode 2'b01 : HD mode 2'b10 : UD mode 2'b11 : IMM mode																																								
[19:18]	FLT_R_MODE	R/W	2'b0	Filter mode for Rising Edge 2'b00 : RST mode 2'b01 : HD mode 2'b10 : UD mode 2'b11 : IMM mode																																								
[17:16]	CMP_ERR_SEL	R/W	2'b10	Interrupt Selection between period/duty compare result for counting falling edge 2'b00 : Select period compare intr. 2'b01 : Select duty compare intr. 2'b1x : Select period & duty compare intr.																																								
[15:]	DMA_EN	R/W	1'b0	DMA_EN																																								
[14:12]	DMA_REQ_SEL	R/W	3'b0	DMA_REQ_SEL 3'b000 : PRD_PVL 3'b001 : PRD_PPVL 3'b010 : DT_PVL 3'b011 : DT_PPVL 3'b100 : FEDGE_CNT_CVL 3'b101 : FEDGE_CNT_PVL 3'b110 : REDGE_TS_CNT_VL 3'b111 : FEDGE_TS_CNT_VL																																								
[11:8]		R/W	0	Reserved																																								
[7:0]	F_IN_SEL	R/W	8'b0	Select F_IN among 256ea External Inputs for Detect <table><tr><th>BIT</th><th>GPIO</th></tr><tr><td>[011:000]</td><td>FC[11:00]</td></tr><tr><td>[031:012]</td><td>0</td></tr><tr><td>[037:032]</td><td>LC[05:00]</td></tr><tr><td>[063:038]</td><td>0</td></tr><tr><td>[087:064]</td><td>LD[23:00]</td></tr><tr><td>[095:088]</td><td>0</td></tr><tr><td>[123:096]</td><td>PC[27:00]</td></tr><tr><td>[127:124]</td><td>0</td></tr><tr><td>[151:128]</td><td>PD[23:00]</td></tr><tr><td>[159:152]</td><td>0</td></tr><tr><td>[191:160]</td><td>PS[31:00]</td></tr><tr><td>[223:192]</td><td>PK[31:00]</td></tr><tr><td>[226:224]</td><td>PK[34:32]</td></tr><tr><td>[227:227]</td><td>0</td></tr><tr><td>[228:228]</td><td>0</td></tr><tr><td>[229:229]</td><td>IREQ_ADCSTOP</td></tr><tr><td>[230:230]</td><td>IREQ_ADCWKUP</td></tr><tr><td>[231:231]</td><td>IREQ_RTCWKUP</td></tr><tr><td>[234:232]</td><td>0</td></tr></table>	BIT	GPIO	[011:000]	FC[11:00]	[031:012]	0	[037:032]	LC[05:00]	[063:038]	0	[087:064]	LD[23:00]	[095:088]	0	[123:096]	PC[27:00]	[127:124]	0	[151:128]	PD[23:00]	[159:152]	0	[191:160]	PS[31:00]	[223:192]	PK[31:00]	[226:224]	PK[34:32]	[227:227]	0	[228:228]	0	[229:229]	IREQ_ADCSTOP	[230:230]	IREQ_ADCWKUP	[231:231]	IREQ_RTCWKUP	[234:232]	0
BIT	GPIO																																											
[011:000]	FC[11:00]																																											
[031:012]	0																																											
[037:032]	LC[05:00]																																											
[063:038]	0																																											
[087:064]	LD[23:00]																																											
[095:088]	0																																											
[123:096]	PC[27:00]																																											
[127:124]	0																																											
[151:128]	PD[23:00]																																											
[159:152]	0																																											
[191:160]	PS[31:00]																																											
[223:192]	PK[31:00]																																											
[226:224]	PK[34:32]																																											
[227:227]	0																																											
[228:228]	0																																											
[229:229]	IREQ_ADCSTOP																																											
[230:230]	IREQ_ADCWKUP																																											
[231:231]	IREQ_RTCWKUP																																											
[234:232]	0																																											

				[242:235]	EXTINT[7:0]
				[255:243]	0

Note : You can use the maximum IC TC clock frequency by 200MHz, if you do not use Time-Out Unit.

If you want to use Time-Out Unit, the maximum IC TC clock frequency is dependent on the frequency of input signal for capturing and Time-Out Counter Matching Value. For more details, refer to IC_TC_TOVL Register description below.

IC_TC_IRQ Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IEN7	IEN6	IEN5	IEN4	IEN3	IEN2	IEN1	IEN0	FLG7	FLG6	FLG5	FLG4	FLG3	FLG2	FLG1	FLG0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ICLR7	ICLR6	ICLR5	ICLR4	ICLR3	ICLR2	ICLR1	ICLR0	IRQ7	IRQ6	IRQ5	IRQ4	IRQ3	IRQ2	IRQ1	IRQ0

Field	Name	RW	Reset	Description
[31]	IEN7	R/W	0	FEDGE_NOTIFY_IRQ_EN
[30]	IEN6	R/W	0	REDGE_NOTIFY_IRQ_EN
[29]	IEN5	R/W	0	PD_CMP_CNT_FULL_IRQ_EN
[28]	IEN4	R/W	0	CMP_PRD_ERR_IRQ_EN
[27]	IEN3	R/W	0	CMP_DT_ERR_IRQ_EN
[26]	IEN2	R/W	0	FEDGE_CNT_FULL_IRQ_EN
[25]	IEN1	R/W	0	TO_CNT_FULL_IRQ_EN
[24]	IEN0	R/W	0	FLT_CNT_FULL_IRQ_EN
[23]	FLG7	R	0	FEDGE_NOTIFY_FLAG
[22]	FLG6	R	0	REDGE_NOTIFY_FLAG
[21]	FLG5	R	0	PD_CMP_CNT_FULL_FLAG
[20]	FLG4	R	0	CMP_PRD_ERR_FLAG
[19]	FLG3	R	0	CMP_DT_ERR_FLAG
[18]	FLG2	R	0	FEDGE_CNT_FULL_FLAG
[17]	FLG1	R	0	TO_CNT_FULL_FLAG
[16]	FLG0	R	0	FLT_CNT_FULL_FLAG
[15]	ICLR7	R/W	0	FEDGE_NOTIFY_IRQ(FLAG)_CLEAR
[14]	ICLR6	R/W	0	REDGE_NOTIFY_IRQ(FLAG)_CLEAR
[13]	ICLR5	R/W	0	PD_CMP_CNT_FULL_IRQ(FLAG)_CLEAR
[12]	ICLR4	R/W	0	CMP_PRD_ERR_IRQ(FLAG)_CLEAR
[11]	ICLR3	R/W	0	CMP_DT_ERR_IRQ(FLAG)_CLEAR
[10]	ICLR2	R/W	0	FEDGE_CNT_FULL_IRQ(FLAG)_CLEAR
[9]	ICLR1	R/W	0	TO_CNT_FULL_IRQ(FLAG)_CLEAR
[8]	ICLR0	R/W	0	FLT_CNT_FULL_IRQ(FLAG)_CLEAR
[7]	IRQ7	R	0	FEDGE_NOTIFY_IRQ
[6]	IRQ6	R	0	REDGE_NOTIFY_IRQ
[5]	IRQ5	R	0	PD_CMP_CNT_FULL_IRQ
[4]	IRQ4	R	0	CMP_PRD_ERR_IRQ
[3]	IRQ3	R	0	CMP_DT_ERR_IRQ
[2]	IRQ2	R	0	FEDGE_CNT_FULL_IRQ
[1]	IRQ1	R	0	TO_CNT_FULL_IRQ
[0]	IRQ0	R	0	FLT_CNT_FULL_IRQ

Note 1: FLT_CNT_FULL means that Counter for Noise Filter and Edge Detection becomes full.

Note 2: TO_CNT_FULL means that Time-Out Counter becomes full.

Note 3: FEDGE_CNT_FULL means that Counter for Fall edge at Fall Edge Count Unit becomes full.

Note 4: CMP_DT_ERR means that Duty is changed between previous and current signal.

Note 5: CMP_PRD_ERR means that Frequency is changed between previous and current signal

Note 6: PD_CMP_CNT_FULL means that counter for comparing duty and frequency becomes full.

IC_TC_TOVL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TO_VL															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TO_VL															

Field	Name	RW	Reset	Description
[27:0]	TO_VL	R/W	0x0ffffff	IC_TC TimeOut Value Setting Register

Note : For example, when IC TC clock is 100MHz(10ns) and TO_VL is 0xA, Time-Out counter is full after 100ns. In this case, if input signal is slower than 5MHz (because Time-Out counter is cleared at rise and fall edge detect), Time-Out interrupt is generated. Therefore, you should consider IC TC clock frequency depending on the input signal frequency and TO_VL.

IC_TC_REDGE_VL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REDGE_VL															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REDGE_VL															

Field	Name	RW	Reset	Description
[27:0]	REDGE_VL	R/W	0x0ffffff	IC_TC REDGE Matching Value Setting Register

Note: You can configure how long period is treated as noise. For example, when you configure this register to 3, if the input signal does not maintain during 3-cycle after rise edge, this rise edge is treated as noise.

IC_TC_FEDGE_VL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FEDGE_VL															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FEDGE_VL															

Field	Name	RW	Reset	Description
[27:0]	FEDGE_VL	R/W	0x0ffffff	IC_TC FEDGE Matching Value Setting Register

Note: You can configure how long period is treated as noise. For example, when you configure this register to 3, if the input signal does not maintain during 3-cycle after fall edge, this fall edge is treated as noise.

IC_TC_PRD_RND_VL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PRD_RND_VL															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRD_RND_VL															

Field	Name	RW	Reset	Description
[27:0]	PRD_RND_VL	R/W	0x0	IC_TC Period Compare Rounding Value Setting Register

Note: For example, when the previous frequency counter value is 0xff33 and the current frequency counter value is 0xff34, the CMP_PRD_ERR interrupt is generated. which is undesirable. In this case, you can mask LSB bits by configuring 0xf to this register. Then, the previous frequency counter value becomes 0xff30 and the current frequency counter value becomes 0xff30, and no interrupt is generated finally.

IC_TC_DT_RND_VL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
									DT_RND_VL						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DT_RND_VL															

Field	Name	RW	Reset	Description
[27:0]	DT_RND_VL	R/W	0x0	IC_TC Duty Compare Rounding Value Setting Register

Note: For example, when the previous duty counter value is 0xff33 and the current duty counter value is 0xff34, the CMP_DT_ERR interrupt is generated which is undesirable. In this case, you can mask LSB bits by configuring 0xf to this register. Then, the previous duty counter value becomes 0xff30 and the current duty counter value becomes 0xff30, and no interrupt is generated finally.

IC_TC_ECNT_VL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ECNT_VL															

Field	Name	RW	Reset	Description
[15:0]	ECNT_VL	R/W	0x0000ffff	IC_TC Fall Edge Counter Match Value Setting Register

IC_TC_PRD_CNT_PVL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
									PRD_CNT_PVL						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRD_CNT_PVL															

Field	Name	RW	Reset	Description
[27:0]	PRD_CNT_PVL	R	0x0	IC_TC Counted Previous Period Value This value is used for comparing frequency change with PRD_CNT_PPVL

IC_TC_PRD_CNT_PPVL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
									PRD_CNT_PPVL						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRD_CNT_PPVL															

Field	Name	RW	Reset	Description
[27:0]	PRD_CNT_PPVL	R	0x0	IC_TC Counted 1 cycle before-previous Period Value This value is used for comparing frequency change with PRD_CNT_PVL

IC_TC_DT_CNT_PVL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

DT_CNT_PVL

Field	Name	RW	Reset	Description
[27:0]	DT_CNT_PVL	R	0x0	IC_TC Counted Previous Duty Value This value is used for comparing duty change with DT_CNT_PPVL

IC_TC_DT_CNT_PPVL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

DT_CNT_PPVL

Field	Name	RW	Reset	Description
[27:0]	DT_CNT_PPVL	R	0x0	IC_TC Counted 1 cycle before-previous Duty Value This value is used for comparing duty change with DT_CNT_PVL

IC_TC_FEDGE_DET_CNT_CVL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

FEDGE_DET_CNT_CVL

Field	Name	RW	Reset	Description
[15:0]	FEDGE_DET_CNT_CVL	R	0x0	IC_TC Counted Current Falling Edge Count Value

IC_TC_FEDGE_DET_CNT_PVL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x44

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

FEDGE_DET_CNT_PVL

Field	Name	RW	Reset	Description
[15:0]	FEDGE_DET_CNT_PVL	R	0x0	IC_TC Counted Previous Falling Edge Count Value

IC_TC_REDGE_TSCNT_VL Register

0x9800C000, 0x9800C100, 0x9800C200 + 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

REDGE_TSCNT_VL

Field	Name	RW	Reset	Description
[15:0]	REDGE_TSCNT_VL	R	0x0	IC_TC Counted Rising Edge TimeStamp Count Value

IC_TC_FEDGE_TSCNT_VL Register**0x9800C000, 0x9800C100, 0x9800C200 + 0x4C**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FEDGE_TSCNT_VL															

Field	Name	RW	Reset	Description
[15:0]	FEDGE_TSCNT_VL	R	0x0	IC_TC Counted Falling Edge TimeStamp Count Value

15 RTC (Real Time Clock)

15.1 Overview

The Real Time Clock (RTC) unit can operate by the backup battery if the system power is turned off. The RTC provides time keeping data to CPU as BCD (Binary Coded Decimal) values. The data includes second, minute, hour, date, day of the week, month, and year. The RTC unit also can perform the alarm function and the block diagram is shown in the figure below.

15.2 Feature

- Clock and calendar functions (BCD display): seconds, minutes, hours, date, day of week, month, year
- Leap year generator
- Wake-up (PMWKUP) signal generation: supported in the power down mode
- Alarm interrupt (ALMINT) in normal operation mode
- Supports 32.768kHz or 4.194304MHz XTAL input

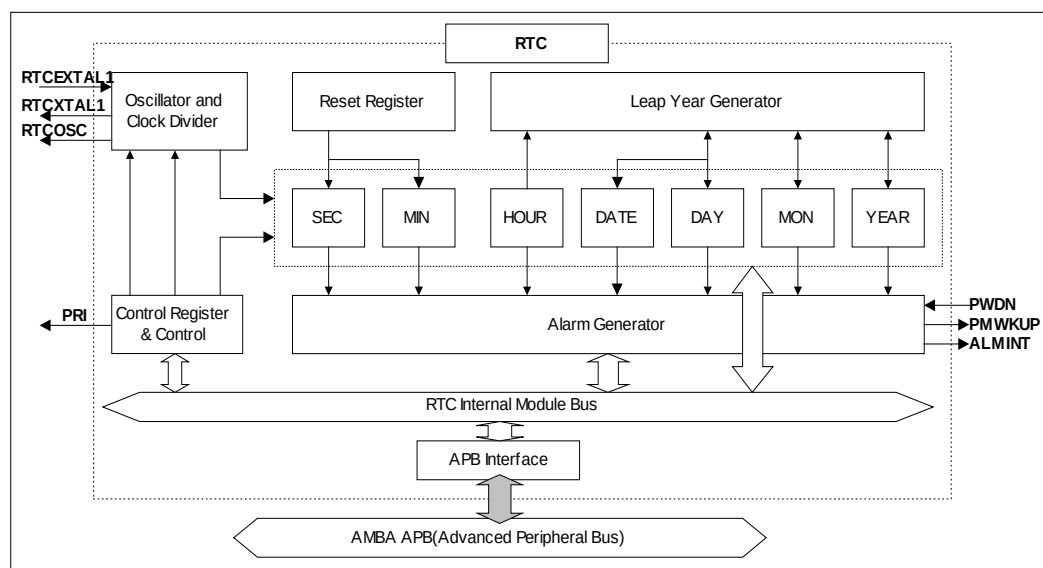


Figure 15.1 RTC Block Diagram

Note: In TCC8132, the PMWKUP can be output to PK[33] or PK[32]. It is masked and controlled by PMGPIO (Alive GPIO module). Refer to PMGPIO_CTL[7:6] register description in Chapter 18 . The polarity of PMWKUP must be programmed as "high active".

15.3 Function Description

15.3.1 System Clock Frequency Control

The leap year generator calculates whether the last date of each month is 28,29,30 or 31 based on data from BCDDAY, BCDMON, and BCDYEAR. This also considers leap years in deciding the last date. A 16bit counter can just represent four BCD digits, so it can decide whether any year is a leap year or not.

15.3.2 System Power Operation

It is required to set bit 1 of the RTCCON register for interfacing between CPU and RTC logic. A one second error can occur when the CPU reads or writes data into BCD counters and this can cause the change of the higher time units. When the CPU reads/writes data to/from the BCD counters, another time unit may be changed if BCDSEC register is overflowed. To avoid this problem, the CPU should reset BCDSEC register to 00h. The reading sequence of the BCD counters is BCDYEAR, BCDMON, BCDDATE, BCDDAY, BCDHOUR, BCDMIN, and BCDSEC. It is required to read it again from BCDYEAR to BCDSEC if BCDSEC is zero.

15.3.3 Backup Battery Operation

The RTC logic is driven by backup battery if the system power is turned off. The interface between the CPU and RTC logic is blocked and the battery only drives the oscillation circuit and the BCD counters to minimize power dissipation.

15.3.4 Alarm Function

The RTC generates alarm signal at specified time in the power down mode or normal operation mode. In normal operation mode, the alarm interrupt (ALMINT) is activated and in the power down mode the power management wake up (PMWKUP) signal is activated. The RTC alarm register, RTCALM, determines the alarm enable and the condition of the alarm time setting. The PWDN signal determines one of the normal operation mode and power down mode.

15.4 RTC Operation

15.4.1 Boot-Up Sequence

The figure below shows how to initialize register of RTCINT block (RTCALM, RTCIM, RTCPEND).

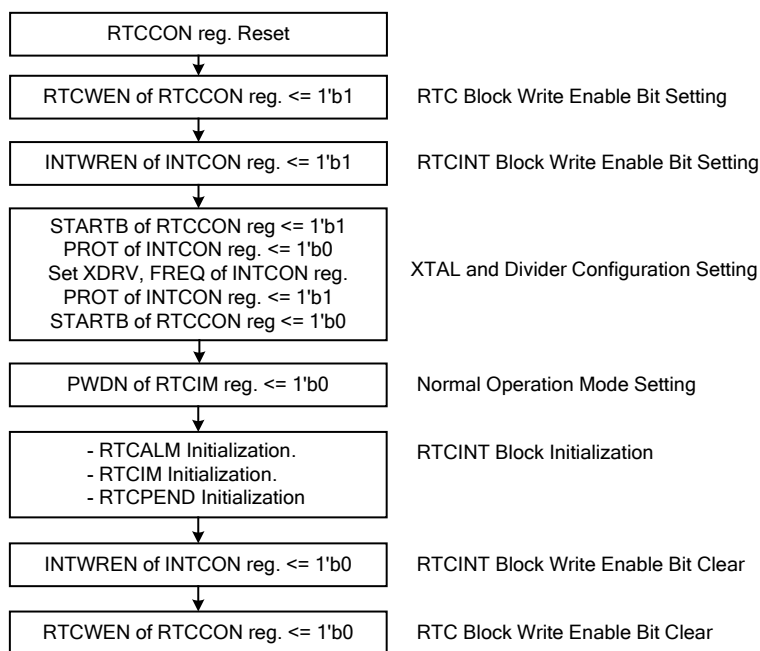


Figure 15.2 Boot-Up Sequence

15.4.2 RTC Time Setting

Figure 15.3 shows how to set the time when the clock stops. This works when the entire calendar or clock is to be set.

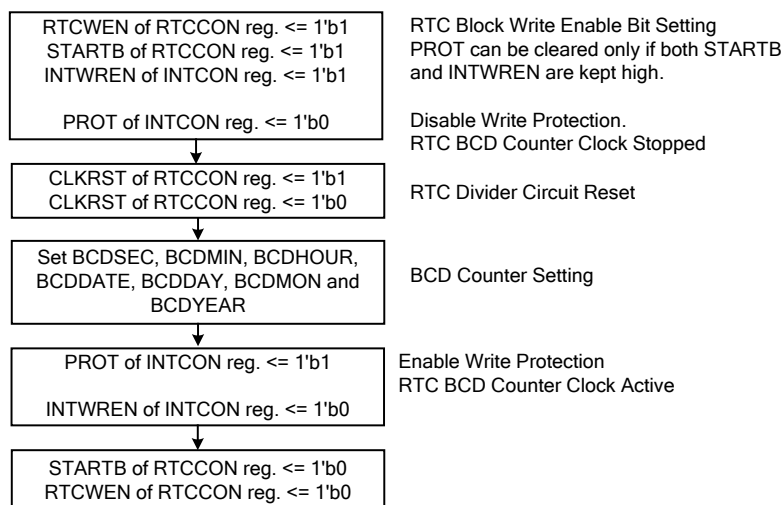


Figure 15.3 RTC Time Setting Sequence

15.4.3 RTC Alarm Time Setting

Figure 15.4 shows how to set Alarm Time.

The alarms can be generated using the seconds, minutes, hours, day of week, date, month, year or any combination of these.

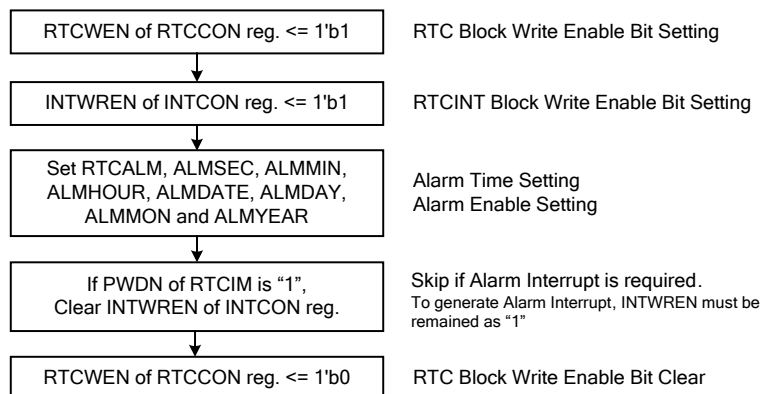


Figure 15.4 RTC Alarm Time Setting Sequence

15.4.4 Clearing Pending Interrupt

Figure 15.5 shows how to Clear RTC interrupt. There are two types of interrupt.

- Alarm Interrupt (ALMINT)
- Wake-Up Interrupt (WKUPINT)

Alarm Interrupt can be cleared immediately by writing "0" to RTCPEND register.

To clear Wake-Up Interrupt and PMWKUP signal, PWDN bit of RTCIM register must be written as "0" and remained "0" until WKUPINT is actually cleared.

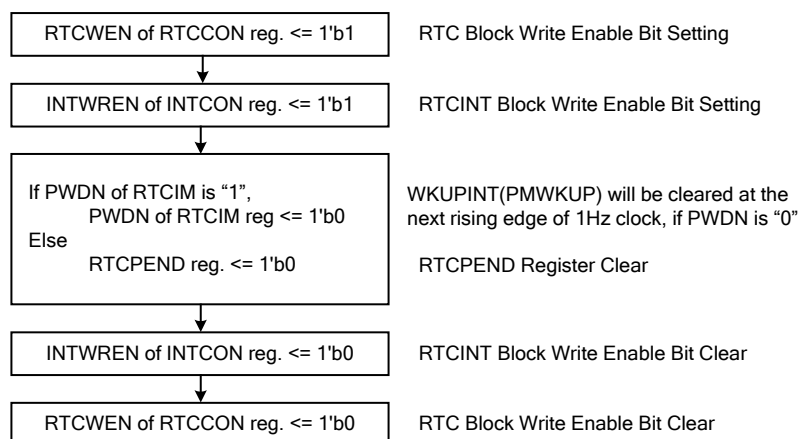


Figure 15.5 PEND Clear Sequence

15.4.5 RTC Operation

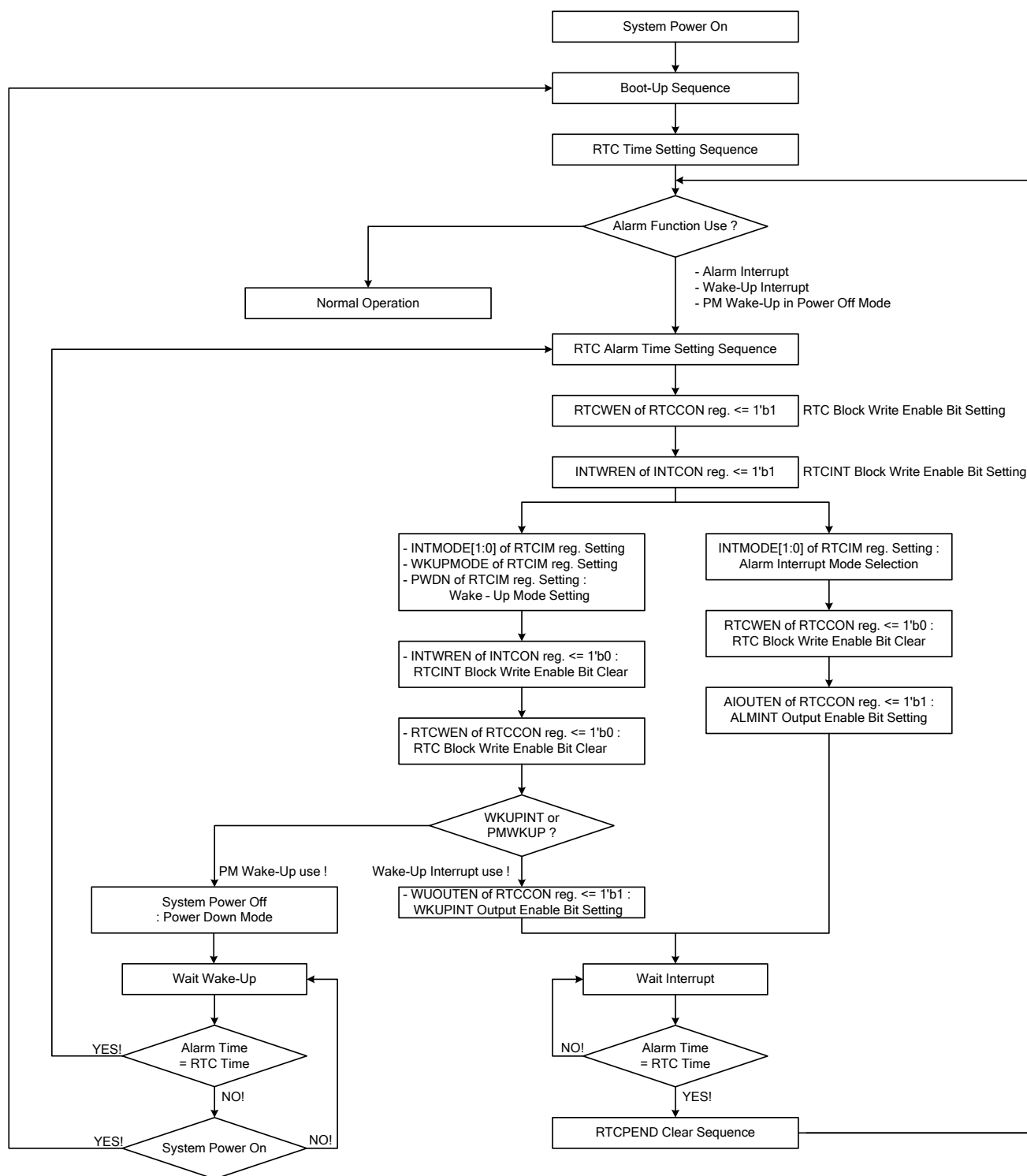


Figure 15.6 RTC Operation Process Flow Chart.

15.4.6 Crystal Oscillator Circuit

Crystal oscillator circuit constants (recommended values) are shown in Table 15.1, and the RTC crystal oscillator circuit in Figure 15.7.

Table 15.1 Recommended Oscillator Circuit Constants

fosc	Cin	Cout
32.768 kHz	from 10 pF to 22 pF	from 10 pF to 22 pF
4.194304 MHz	from 10 pF to 22 pF	from 10 pF to 22 pF

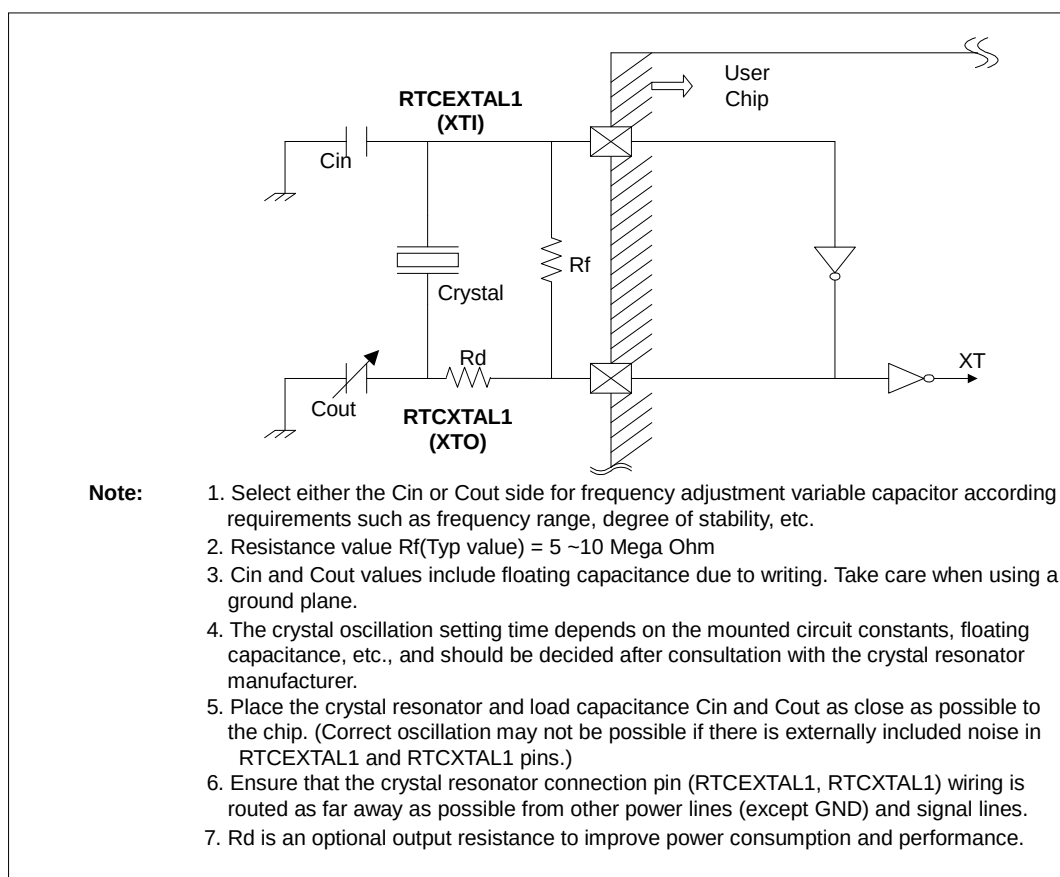


Figure 15.7 Example of Crystal Oscillator Circuit Connection

15.5 Register Description

Table 15.2 RTC Register Map (Base Address = 0x98012000)

Name	Address	Type	Reset	Description
RTCCON	0x00	R/W	0x00	RTC Control Register
INTCON	0x04	R/W	-	RTC Interrupt Control Register
RTCALM	0x08	R/W	-	RTC Alarm Control Register
ALMSEC	0x0C	R/W	-	Alarm Second Data Register
ALMMIN	0x10	R/W	-	Alarm Minute Data Register
ALMHOUR	0x14	R/W	-	Alarm Hour Data Register
ALMDATE	0x18	R/W	-	Alarm Date Data Register
ALMDAY	0x1C	R/W	-	Alarm Day of Week Data Register
ALMMON	0x20	R/W	-	Alarm Month Data Register
ALMYEAR	0x24	R/W	-	Alarm Year Data Register
BCDSEC	0x28	R/W	-	BCD Second Register
BCDMIN	0x2C	R/W	-	BCD Minute Register
BCDHOURL	0x30	R/W	-	BCD Hour Register
BCDDATE	0x34	R/W	-	BCD Date Register
BCDDAY	0x38	R/W	-	BCD Day of Week Register
BCDMON	0x3C	R/W	-	BCD Month Register
BCDYEAR	0x40	R/W	-	BCD Year Register
RTCIM	0x44	R/W	-	RTC Interrupt Mode Register
RTCPEND	0x48	R/W	-	RTC Interrupt Pending Register
RTCAPB	0x100	R/W	0x1001	RTC APB Interface Timing Register

RTC Control Register (RTCCON)

0x98012000

Bit	Name	Type	Default	Description	
31:8	-	R	0	Reserved	
7	WUOUTEN	R/W	0	Wake Up Interrupt Output Enable to the CPU	
				0	Disable
				1	Enable if (RTCIM.PWDN == 1)
6	AIOUTEN	R/W	0	Alarm Interrupt Output Enable to the CPU	
				0	Disable
				1	Enable if (RTCIM.PWDN == 0)
5	OSCEN	R/W	0	Oscillator and Divider Circuit Test Enable	
				0	Disable
				1	Enable
4	CLKRST	R/W	0	RTC Clock Count(Divider) Reset	
				This bit resets the frequency divider logic in the RTC.	
				0	No reset (Normal operation)
3	CNTSEL	R/W	0	BCD Count Test Type Select	
				This bit forces BCD counters into test mode.	
				0	Test Disable (Normal operation)
2	CLKSEL	R/W	0	BCD Counter Test Clock Select	
				0	1Hz clock (Normal operation)
				1	XTI divided by 2 if (~PROT & INTWREN & RTCWEN).
1	RTCWEN	R/W	0	RTC Write Enable	
				This bit controls all interfaces between the CPU and the RTC, so it should be set to '1' to enable data transfer.	
				0	Disable
0	STARTB	R/W	0	RTC Start Bit	
				0	Run (Normal operation)
				1	Halt
Note: If PROT bit of INTCON register is set as “1”, the BCD counters start normal operation with 1Hz clock regardless of STARTB status.					

Note: CLKRST, CNTSEL and CLKSEL are affected by PROT and INTWREN of INTCON register. They are valid only if (~PROT & INTWREN & RTCWEN).

RTC Interrupt Control Register (INTCON)

0x98012004

Bit	Name	Type	Default	Description								
15	PROT	R/W	0	Protection Enable Bit. (0 = Disable, 1 = Enable) This bit must be set as “1” before the core power is turned off. To clear this bit, RTCWEN, STARTB and INTWREN must be set first. To enable this bit, RTCWEN and STARTB must be set first. If enabled, the following fields are affected. - XDRV, FSEL of INTCON register are write protected - All the BCD Data Registers are write protected - CLKRST, CNTSEL and CLKSEL bits of RTCCON register are disabled.								
13:12	XDRV[1:0]	R/W	X	XTAL I/O Driver Strength Select <table><tr><td>00</td><td>32.768kHz XTAL @1.8V</td></tr><tr><td>01</td><td>32.768kHz XTAL @3.3V</td></tr><tr><td>10</td><td>4.194304MHz XTAL @1.8V</td></tr><tr><td>11</td><td>4.194304MHz XTAL @3.3V</td></tr></table> These bits can be written only if (RTCWEN & STARTB & INTWREN & ~PROT)	00	32.768kHz XTAL @1.8V	01	32.768kHz XTAL @3.3V	10	4.194304MHz XTAL @1.8V	11	4.194304MHz XTAL @3.3V
00	32.768kHz XTAL @1.8V											
01	32.768kHz XTAL @3.3V											
10	4.194304MHz XTAL @1.8V											
11	4.194304MHz XTAL @3.3V											
10:8	FSEL[2:0]	R/W	X	Divider Output Select <table><tr><td>000</td><td>32.768kHz XTAL</td></tr><tr><td>111</td><td>4.194304MHz XTAL</td></tr><tr><td>Others</td><td>Reserved</td></tr></table> These bits can be written only if (RTCWEN & STARTB & INTWREN & ~PROT)	000	32.768kHz XTAL	111	4.194304MHz XTAL	Others	Reserved		
000	32.768kHz XTAL											
111	4.194304MHz XTAL											
Others	Reserved											
2:1	STATUS	R/W	0	Scratch bits for user defined status.								
0	INTWREN	R/W	0	Interrupt Block Write Enable Bit. (0 = Disable, 1 = Enable)								

RTC Alarm Control Register (RTCALM)

0x98012008

Bit	Name	Type	Default	Description (1 = Enable, 0 = Disable)
31:8	-	R	0	Reserved
7	ALMEN	R/W	0	Alarm Global Enable Bit To generate ALMINT or PMWKUP, this bit must be set as "1" and at least one of the bit[6:0] below must be enabled.
6	YEAREN	R/W	0	Year Alarm Enable Bit
5	MONEN	R/W	0	Month Alarm Enable Bit
4	DAYEN	R/W	0	Day of week Alarm Enable Bit
3	DATEEN	R/W	0	Date Alarm Enable Bit
2	HOUREN	R/W	0	Hour Alarm Enable Bit
1	MINEN	R/W	0	Minute Alarm Enable Bit
0	SECEN	R/W	0	Second Alarm Enable Bit

RTCALM register controls alarm enable and the condition of the alarm time setting. Note that RTCALM register generates the alarm signal through ALMINT in normal operation mode.

Alarms can be generated by using the seconds, minutes, hours, day of week, date, month, year or any combination of them.

Alarm Second Data Register (ALMSEC)

0x9801200C

Bit	Name	Type	Default	Description	
31:7	-	R	0	Reserved	
6:0	SECDATA	R/W	X	Alarm Value for Second Bits 0 ~ 59 second	
				[6:4]	Alarm BCD value from 0 to 5
				[3:0]	Alarm BCD value from 0 to 9

You must set RTCWEN (RTCCON[1]) and INTWREN (INTCON[0]) to Write ALMSEC register.

You must set INTWREN (INTCON[0]) to Read ALMSEC register.

Alarm Minute Data Register (ALMMIN)**0x98012010**

Bit	Name	Type	Default	Description
31:7	-	R	0	Reserved
6:0	MINDATA	R/W	X	Alarm Value for Minute Bits 0 ~ 59 minute
				[6:4] Alarm BCD value from 0 to 5
				[3:0] Alarm BCD value from 0 to 9

You must set RTCWEN (RTCCON[1]) and INTWREN (INTCON[0]) to Write ALMMIN register.

You must set INTWREN (INTCON[0]) to Read ALMMIN register.

Alarm Hour Data Register (ALMHOUR)**0x98012014**

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	HOURLDATA	R/W	X	Alarm Value for Hour Bits 0 ~ 23 hour
				[5:4] Alarm BCD value from 0 to 2
				[3:0] Alarm BCD value from 0 to 9

You must set RTCWEN (RTCCON[1]) and INTWREN (INTCON[0]) to Write ALMHOUR register.

You must set INTWREN (INTCON[0]) to Read ALMHOUR register.

Alarm Date Data Register (ALMDATE)**0x98012018**

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	DATEDATA	R/W	X	Alarm Value for Date Bits 1 ~ 31 date
				[5:4] Alarm BCD value from 0 to 3
				[3:0] Alarm BCD value from 0 to 9

You must set RTCWEN (RTCCON[1]) and INTWREN (INTCON[0]) to Write ALMDATE register.

You must set INTWREN (INTCON[0]) to Read ALMDATE register.

Alarm Day of Week Data Register (ALMDAY)**0x9801201C**

Bit	Name	Type	Default	Description
31:3	-	R	0	Reserved
2:0	DAYDATA	R/W	X	Alarm Value for Day Bits 0 ~ 6 day (SUN. ~ SAT.)
				000 Sunday
				001 Monday
				010 Tuesday
				011 Wednesday
				100 Thursday
				101 Friday
				110 Saturday
				111 Reserved

You must set RTCWEN (RTCCON[1]) and INTWREN (INTCON[0]) to Write ALMDAY register.

You must set INTWREN (INTCON[0]) to Read ALMDAY register.

Alarm Month Data Register (ALMMON)

0x98012020

Bit	Name	Type	Default	Description
31:5	-	R	0	Reserved
4:0	MONDATA	R/W	X	Alarm Value for Month Bits 1 ~ 12 month (JAN. ~ DEC.)
				[4] Alarm BCD value from 0 to 1
				[3:0] Alarm BCD value from 0 to 9

You must set RTCWEN (RTCCON[1]) and INTWREN (INTCON[0]) to Write ALMMON register.
You must set INTWREN (INTCON[0]) to Read ALMMON register.

Alarm Year Data Register (ALMYEAR)

0x98012024

Bit	Name	Type	Default	Description
31:16	-	R	0	Reserved
15:0	YEARDATA	R/W	X	Alarm Value for Year Bits 0 ~ 9999 year
				[15:8] Alarm BCD value from 0 to 99
				[7:0] Alarm BCD value from 0 to 99

You must set RTCWEN (RTCCON[1]) and INTWREN (INTCON [0]) to Write ALMYEAR register.
You must set INTWREN (INTCON[0]) to Read ALMYEAR register.

BCD Second Data Register (BCDSEC)

0x98012028

Bit	Name	Type	Default	Description
31:7	-	R	0	Reserved
6:0	SECDATA	R/W	X	BCD Value for Second Bits 0 ~ 59 second
				[6:4] BCD value from 0 to 5
				[3:0] BCD value from 0 to 9

To Write BCDSEC register, you must set STARTB (RTCCON[0]) and RTCWEN (RTCCON [1]) and clear PROT (NTCON[15]).
To Read BCDSEC register, you must set INTWREN (INTCON[0]).

BCD Minute Data Register (BCDMIN)

0x9801202C

Bit	Name	Type	Default	Description
31:7	-	R	0	Reserved
6:0	MINDATA	R/W	X	BCD Value for Minute Bits 0 ~ 59 minute
				[6:4] BCD value from 0 to 5
				[3:0] BCD value from 0 to 9

To Write BCDMIN register, you must set STARTB (RTCCON[0]) and RTCWEN (RTCCON[1]) and clear PROT (NTCON[15]).
To Read BCDMIN register, you must set INTWREN (INTCON[0]).

BCD Hour Data Register (BCDHOURL)

0x98012030

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	HOURLDATA	R/W	X	BCD Value for Hour Bits 0 ~ 23 hour
				[5:4] BCD value from 0 to 2
				[3:0] BCD value from 0 to 9

To Write BCDHOURL register, you must set STARTB (RTCCON[0]) and RTCWEN (RTCCON[1]) and clear PROT (NTCON[15]).
To Read BCDHOURL register, you must set INTWREN (INTCON[0]).

BCD Date Data Register (BCDDATE)**0x98012034**

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	DATEDATA	R/W	X	BCD Value for Date Bits 1 ~ 31 date
				[5:4] BCD value from 0 to 3
				[3:0] BCD value from 0 to 9

To Write BCDDATE register, you must set STARTB (RTCCON[0]) and RTCWEN (RTCCON[1]) and clear PROT (INTCON[15]).

To Read BCDDATE register, you must set INTWREN (INTCON[0]).

BCD Day of Week Data Register (BCDDAY)**0x98012038**

Bit	Name	Type	Default	Description
31:3	-	R	0	Reserved
2:0	DAYDATA	R/W	X	BCD Value for Day Bits 0 ~ 6 day (SUN. ~ SAT.)
				000 Sunday
				001 Monday
				010 Tuesday
				011 Wednesday
				100 Thursday
				101 Friday
				110 Saturday
				111 Reserved

To Write BCDDAY register, you must set STARTB (RTCCON[0]) and RTCWEN (RTCCON[1]) and clear PROT (INTCON[15]).

To Read BCDDAY register, you must set INTWREN (INTCON[0]).

BCD Month Data Register (BCDMON)**0x9801203C**

Bit	Name	Type	Default	Description
31:5	-	R	0	Reserved
4:0	MONDATA	R/W	X	BCD Value for Month Bits 1 ~ 12 month (JAN. ~ DEC.)
				[4] BCD value from 0 to 1
				[3:0] BCD value from 0 to 9

To Write BCDMON register, you must set STARTB (RTCCON[0]) and RTCWEN (RTCCON[1]) and clear PROT (INTCON[15]).

To Read BCDMON register, you must set INTWREN (INTCON[0]).

BCD Year Data Register (BCDYEAR)**0x98012040**

Bit	Name	Type	Default	Description
31:16	-	R	0	Reserved
15:0	YEARDATA	R/W	X	BCD Value for Year Bits 0 ~ 9999 year
				[15:8] BCD value from 0 to 99
				[7:0] BCD value from 0 to 99

To Write BCDYEAR register, you must set STARTB (RTCCON[0]) and RTCWEN (RTCCON[1]) and clear PROT (INTCON[15]).

To Read BCDYEAR register, you must set INTWREN (INTCON[0]).

RTC Interrupt Mode Register (RTCIM)

0x98012044

Bit	Name	Type	Default	Description
31:4	-	R	0	Reserved
3	PWDN	R/W	0	Operation Mode Selection Bit
				0 Normal operation mode
				1 Power down mode (PMWKUP output enabled)
2	WKUPMD	R/W	1	Wakeup Mode Selection Bit
				0 PMWKUP active low
				1 PMWKUP active high
1:0	INTMODE	R/W	0	Interrupt Mode Selection Bit
				x0 Disable alarm interrupt mode
				x1 Enable alarm interrupt mode
				01 Supports on the edge alarm interrupt
				11 Supports on the level alarm interrupt

To Write RTCIM register, you must set RTCWEN (RTCCON[1]) and INTWREN (INTCON[0])

To Read RTCIM register, you must set INTWREN (INTCON[0]).

When the INTMODE bit of RTCIM register is high, and the clock and alarm times match, "1" is set in the PEND bit of RTCPEND register. The detection of alarm can be checked with reading the PEND bit.

To generate an alarm interrupt, you must set INTMODE(RTCIM[0] = 1), clear PWDN(RTCIM [3] = 0) and RTCALM register should be set properly. Also, INTWREN bit of INTCON register must be kept as "1".

To generate PMWKUP signal, PWDN bit should be set to "1". Once PMWKUP signal is asserted, it maintains asserted state until PWDN bit is cleared. Note that PWDN must be kept as "0" for one second to disable PMWKUP output.

RTC Interrupt Pending Register (RTCPEND)

0x98012048

Bit	Name	Type	Default	Description	
31:1	-	R	0	Reserved	
0	PEND	R/W	0	Interrupt Pending Enable Bit	
				0	PEND bit is cleared
				1	PEND bit is pending
PEND bit is for Alarm Interrupt only. To clear PMWKUP interrupt, PWDN bit of RTCIM should be written as "0" and kept as "0" at least one second.					

To Clear RTCPEND register, you must set RTCWEN(RTCCON[1]) and INTWREN(INTCON[0]).

To Read RTCPEND register, you must set INTWREN(INTCON[0]).

RTC APB Timing Register (RTCAPB)

0x98012100

Bit	Name	Type	Default	Description
15:12	DIV	R/W	1	Clock Divisor. $RTC\ APB\ CLK = HCLK / (2 * (DIV + 1))$
11:8	HOLD	R/W	0	Additional Hold Time in HCLK count
7:4	-	R/W	0	Reserved
3:0	SETUP	R/W	1	Additional Setup Time in HCLK count. Should not be zero.

The default value of this register is for HCLK(Bus Clock) frequency up to 133MHz.

For HCLK frequency higher than 133MHz, 0x2101 is recommended.

16 ADC Interface

16.1 Overview

The TCC8132 has 8-channel general purpose low-power ADC for battery level detection, remote control interface, and etc. It is a CMOS type 10/12-bit A/D converter with 8-channel analog input multiplexer. It converts the analog input signal into 10/12-bit binary digital codes at a maximum conversion rate of 1 MSPS with 5MHz ADC (Analog to Digital Converter) clock (CKIN). The ADC operates with on-chip sample-and-hold function. The power down mode is supported.

- Resolution: 10-bit/12-bit
- Maximum Conversion Rate: 1MSPS
- Main Clock: 5MHz (Maximum)
- Input Range: 0.0V ~ ADC Power
- Normal Conversion mode
- Waiting for Interrupt mode
- STOP mode wakeup source

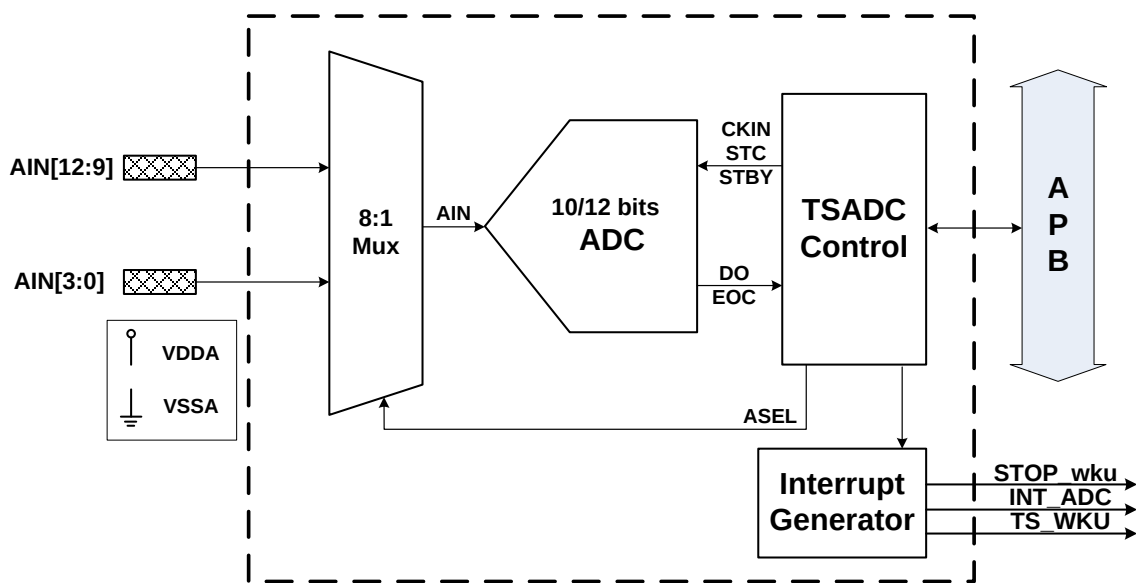


Figure 16.1 ADC Controller Block Diagram

The ADC interface has 8 normal analog inputs. The maximum operating frequency of APB interface (PCLK) is 50 MHz. The ADC clock (CKIN) is generated by PCLK. The maximum operating frequency of CKIN is 5 MHz (1 MSPS). (CKIN should be set less than PCLK by 5 times)

Also, the ADC control signal is directly controlled by GPIO.

16.1.1 A/D Conversion Time

When the PCLK frequency is 50 MHz and the prescaler value is 49, total 10-bit/12-bit conversion time is as follows.

- ADC frequency = 50 MHz / (49 + 1) = 1 MHz
- Conversion time = 1 / (1 MHz / 5 cycles) = 1 / 200 kHz = 5 us

Note 1: Maximum frequency of PCLK is 50MHz.

Note 2: This ADC is designed to operate at maximum 5 MHz clock, so the conversion rate can go up to 1 MSPS.

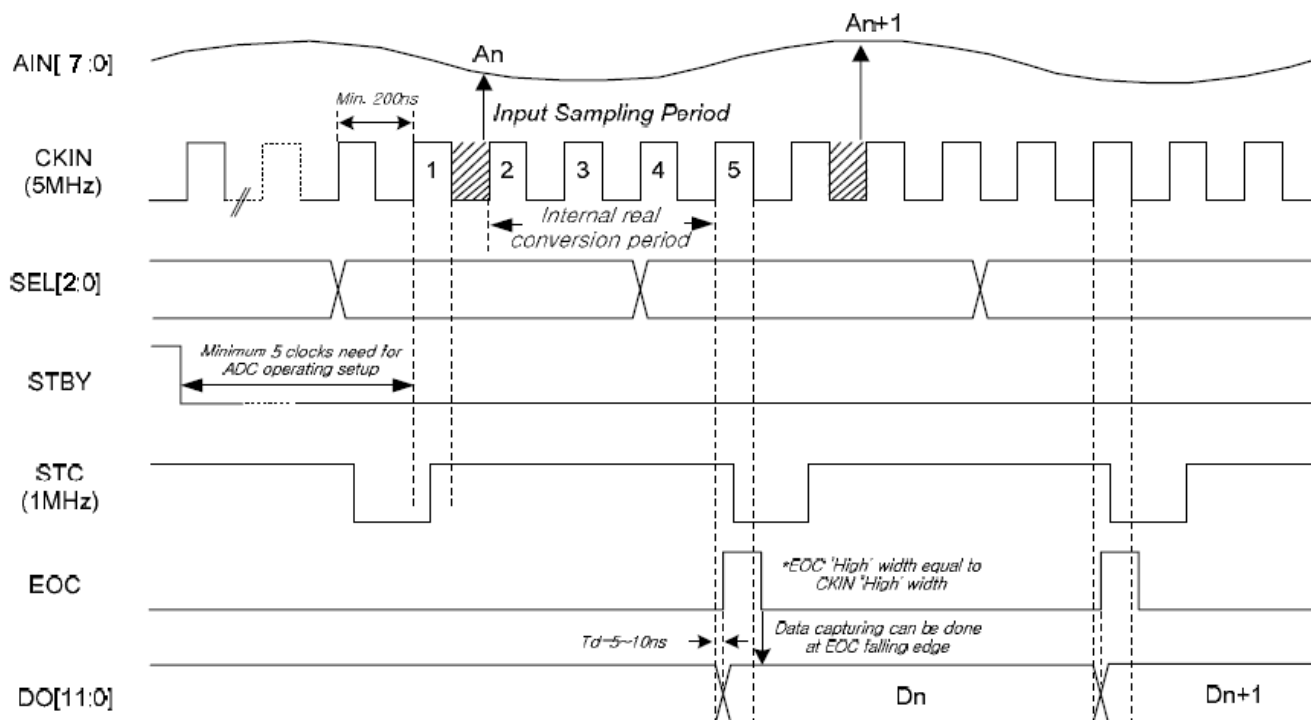


Figure 16.2 Main waveform

16.1.1.1 Stand-By Mode

Stand-by mode is activated when ADCCON[2] (STBY) is set to '1'. In this mode, A/D Conversion operation is halted and [ADCDAT0](#) and [ADCDAT1](#) registers contain the previous converted data.

16.2 ADC Controller Register Description

Table 16.1 ADC Controller Register Map (Base Address = 0x98011000)

Name	Address	Type	Reset	Description
-	0x000~0x090	R/W	-	Reserved
ADCCON	0x100	R/W	0x00003FC4	ADC Control Register
ADCTSC	0x104	R/W	0x00000058	ADC Touch Screen Control Register
ADCDLY	0x108	R/W	0x000000FF	ADC Start or Interval Delay Register
ADCDAT0	0x10C	R	0x00008000	ADC Conversion Data Register
ADCUPDN	0x114	R/W	0x00000000	Stylus Up or Down Interrupt Register
ADCCLRINT	0x118	W	0x00000000	Clear ADC Interrupt
-	0x11C	-	-	Reserved
ADCCLRUPDN	0x120	W	0x00000000	Clear Pen UP/DOWN Interrupt
PCFG	0x200	R/W	0x00000000	Miscellaneous Configuration

ADC Control Register (ADCCON)

0x98011100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														RES	E_FLG
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PS_EN	PS_VAL								ASEL				STBY	RD_ST	EN_ST

Bit	Name	RW	Reset	Description
17	RES	R/W	0x0	ADC Resolution Selection 0 = 10 bits A/D Conversion 1 = 12 bits A/D Conversion
16	E_FLG	R	0x0	End of Conversion (EOC) Flag 0 = A/D Conversion in Progress 1 = End of A/D Conversion
15	PS_EN	R/W	0x0	Enable Prescaler 0 = Disable 1 = Enable
14:7	PS_VAL	R/W	0xFF	Value of Prescaler Data value : 4 ~ 255 Note that division factor is (N+1) when the prescaler value is N. Note: ADC frequency should be set less than PCLK by 5 times. e.g. PCLK=10MHz, ADC Freq<2MHz)
6:3	ASEL	R/W	0x0	Analog Input Channel Select 4'b0010 = AIN0 4'b0011 = AIN1 4'b0100 = AIN2 4'b0101 = AIN3 4'b1011 = AIN9 4'b1100 = AIN10 4'b1101 = AIN11 4'b1110 = AIN12
2	STBY	R/W	0x1	Stand-By mode selection 0 = Operation mode 1 = Stand-By mode
1	RD_ST	R/W	0x0	A/D Conversion Start by data read 0 = Disable 1 = Enable
0	EN_ST	R/W	0x0	A/D Conversion Start by Enable signal 0 = Disable 1 = Enable

ADC Touch Screen Register (ADCTSC)

0x98011104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							UDEN	YMEN	YPEN	XMEN	XPEN	P_UP	AUTO	XY_PST	

Caution:

Do not use this ADCTSC register.

While waiting for stylus up/down interrupt, XPEN bit must be set to '1', namely 'XP output disable' and P_UP bit must be set to '0', namely 'XP Pull-Up Enable'.

AUTO_PST bit should be set '1' only in Automatic and Sequential X/Y Position Conversion.

If you do not use touch screen, you should not tie AIN[6] (XM) and AIN[4] (YM) to VSSA. Because Pull-Up switch, XP and YP switch are enabled in SLEEP mode.

Bit	Name	RW	Reset	Description
8	UDEN	R/W	0x0	Detect Stylus UP / DOWN status 0 = Detect Stylus-DOWN status 1 = Detect Stylus-UP status
7	YMEN	R/W	0x0	YM to GND Switch Enable 0 = Disable (YM = AIN4, Hi-z) 1 = Enable (YM = VSSA)
6	YPEN	R/W	0x1	YP to VDD Switch Enable 0 = Enable (YP = VDDA) 1 = Disable (YP = AIN5, Hi-z)
5	XMEN	R/W	0x0	XM to GND Switch Enable 0 = Disable (XM = AIN6, Hi-z) 1 = Enable (XM = VSSA)
4	XPEN	R/W	0x1	XP to VDD Switch Enable 0 = Enable (XP = VDDA) 1 = Disable (XP = AIN7, Hi-z)
3	P_UP	R/W	0x1	Pull-Up Switch Enable 0 = XP Pull-up Enable 1 = XP Pull-up Disable
2	AUTO	R/W	0x0	Automatic (sequential) conversion of X-pos. and Y-pos 0 = Normal A/D conversion 1 = Automatic (sequential) conversion of X-pos. and Y-pos
1:0	XY_PST	R/W	0x0	Select conversion of X-pos or Y-pos 2'b00 = No operation 2'b01 = X-pos conversion 2'b10 = Y-pos conversion 2'b11 = Waiting Interrupt mode

ADC Start Delay Register (ADCDLY)

0x98011108

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															CLK src
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DELAY															

Before A/D Conversion, touch screen uses X-tal clock (3.68 MHz).

During A/D Conversion, PCLK (Maximum 50 MHz) is used.

Bit	Name	RW	Reset	Description
16	CLKsrc	R/W	0	ADCDLY clock source In waiting for interrupt mode, CLKsrc is used as delay filter clock source. 0 = External input clock 1 = RTC clock
15:0	DELAY	R/W	0xFF	1. In case of A/D conversion mode (normal, separate, auto): A/D conversion is delayed by counting clock is PCLK. -> A/D conversion delay value 2. In case of waiting interrupt mode: when stylus down occurs in waiting interrupt mode, it generates interrupt signal (INT_PENUPDN) at interval of several ms for AUTO X/Y-pos conversion. If this interrupt occurs in STOP mode, it generates Wake-Up signal, having interval (several ms), for exiting STOP MODE.

ADC Data0 Register (ADCDAT0)

0x9801110C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UPDN	AUTO	XYPST		XPDATA_12		XPDATA									

Bit	Name	RW	Reset	Description
15	UPDN	R	1	UP/DOWN state of Stylus in Waiting Interrupt Mode. 0 = Stylus-DOWN 1 = Stylus-UP
14	AUTO	R	0	Automatic sequencing conversion of X-pos. and Y-pos. 0 = Normal A/D Conversion 1 = Automatic (sequential) conversion of X-pos, and Y-pos
13:12	XYPST	R	0	Select conversion of X-pos. or Y-pos. 2'b00 = No operation. 2'b01 = X-pos conversion. 2'b10 = Y-pos conversion. 2'b11 = Waiting Interrupt mode Caution: Do not use this bit.
11:10	XPDATA_12	R	0	When A/D resolution is 12bits, this is X-pos. conversion data[11:0] value.
9:0	XPDATA	R	0	X-pos. conversion data[9:0] value (Includes normal A/D Conversion data value) Data value : 0x00 ~ 0x3FF

ADC Touch Screen UP/DOWN Register (ADCUPDN)

0x98011114

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														UP	DOWN

Bit	Name	RW	Reset	Description
1	UP	R/W	0	Stylus-UP Interrupt history (After check, this bit must be cleared manually) 0 = No Stylus-UP Interrupt. 1 = Stylus-UP Interrupt has been occurred.
0	DOWN	R/W	0	Stylus-DOWN Interrupt history (After check, this bit should be cleared manually) 0 = No Stylus-DOWN Interrupt 1 = Stylus-DOWN Interrupt has occurred.

ADC Touch Screen Interrupt Clear Register (ADCCLRINT)

0x98011118

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															INT_CLR

ISR (Interrupt Service Routine) is supposed to clear interrupts after the interrupt service is completed. Writing any values on this register will clear up the relevant interrupts asserted.

Bit	Name	RW	Reset	Description
0	INT_CLR	W	0	Clear the INT_ADC Interrupt

ADC Touch Screen Interrupt Clear Register (ADCCLRUPDN)

0x98011120

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															UPDN_CLR

ISR is supposed to clear interrupts after the interrupt service is completed. Writing any values on this register will clear up the relevant interrupts asserted.

Bit	Name	RW	Reset	Description
0	UPDN_CLR	W	0x00000000	Clear the Stylus UP/DOWN Interrupt

Miscellaneous Configuration Register (PCFG)

0x98011200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CSEL	PWD	STOP	0

Bit	Name	RW	Reset	Description
3	CSEL	R/W	0	Controller Selection 0 = Touch Screen Controller 1 = TCC800x Compatible Controller Caution: Do not use this bit.
2	PWD	R/W	0	ADC Internal level shifter control 0 = Enable 1 = Disable
1	STOP	R/W	0	ADC Clock Stop Mode 0 = Clock Enabled 1 = Clock Stopped

16.3 Programming Note

The converted data can be accessed by means of interrupt or polling method. With interrupt method, the overall conversion time – from A/D converter start to convert data read – may be delayed because of the return time of interrupt service routine and data access time. With polling method, by checking the ADCCON[15] – end of conversion flag – bit, the read time from ADCDAT register can be determined.

A/D conversion can be activated in different way. After ADCCON[1] – A/D conversion start-by-read mode – is set to 1. The A/D conversion starts simultaneously when converted data is read.

If pen up/down interrupt is used as an wake-up source in STOP mode, XY_PST bit(ADCTSC[1:0]) should be set to 2'b11(waiting interrupt mode). To choose stylus pen up/down wake-up, UD_SEN bit (ADCTSC[8]) is used.

16.3.1 Input range

The analog input is single-ended type and the range is from VREF (ADC Power) to AGND (ADC Ground). This analog input voltage follows reference voltage range fundamentally. Therefore, you should change the voltage value of VREF (ADC Power) if you want to alter into another input range.

Table 16.2 I/O Chart

Index	AIN input	Digital output	YPEN
0	~ 0.0008056	0000 0000 0000	1 LSB = 0.805 mV VREF = 3.3V AGND = 0.0V (at 3.3V typical supply voltage)
1	0.0008056 ~ 0.0016112	0000 0000 0001	
~	~	~	
2047	1.6491944 ~ 1.65000	0111 1111 1111	
2048	1.6500000 ~ 1.6508056	1000 0000 0000	
2049	1.6508056 ~ 1.6516112	1000 0000 0001	
~	~	~	
	3.2983888 ~ 3.2991944	1111 1111 1110	
	3.2991944 ~ 3.3	1111 1111 1111	

17 System Configuration Register (SCFG)

17.1 Overview

SCFG sets miscellaneous operation mode of the TCC8132 and its internal function blocks.

17.2 Register Description

Table 17.1 SCFG Register Map (Base Address = 0x98013000)

Name	Addr	Type	Reset	Description
Configuration (BMI)	0x00	R	-	Boot Mode Status Register
APBPWE	0x08	R/W	0x00000000	Reserved. (APB Posted Write Enable Register)
TCMWAIT	0x0C	R/W	0x0000003F	TCM Interface Wait Control Register
AHBCON1	0x14	R/W	0x000F0000	AHB Control Register 1
REMAP	0x20	R/W	-	Address REMAP Register
LCDSIAE	0x24	R/W	0x00000000	VIOC Configuration Register
PAD_IS0	0xA8	R/W	0x00000000	PAD IS Control Register0
PAD_IS1	0xAC	R/W	0x00000000	PAD IS Control Register1
PAD_IS2	0xB0	R/W	0x00000000	PAD IS Control Register2
PAD_IS3	0xB4	R/W	0x00000000	PAD IS Control Register3
PAD_IS4	0xB8	R/W	0x00000000	PAD IS Control Register4
PAD_IS5	0xBC	R/W	0x00000000	PAD IS Control Register5
PAD_IS6	0xC0	R/W	0x00000000	PAD IS Control Register6
PAD_SR0	0xC4	R/W	0x00000000	PAD SR Control Register0
PAD_SR1	0xC8	R/W	0x00000000	PAD SR Control Register1
PAD_SR2	0xCC	R/W	0x00000000	PAD SR Control Register2
PAD_SR3	0xD0	R/W	0x00000000	PAD SR Control Register3
PAD_SR4	0xD4	R/W	0x00000000	PAD SR Control Register4
PAD_SR5	0xD8	R/W	0x00000000	PAD SR Control Register5
PAD_SR6	0xDC	R/W	0x00000000	PAD SR Control Register6
XMCCFG	0xE0	R/W	-	External Memory Control Configuration Register
IMCCFG	0xE4	R/W	0x0000007F	Internal Memory Control Configuration Register
JTAG_SECTRL	0xE8	R/W	0x00000003	JTAG Secure Control Register
SEMA_CTRL	0xEC	R/W	0x00003041	Single RAM EMA Control Register
DEMA_CTRL	0xF0	R/W	0x000006DB	Dual RAM EMA Control Register
VEMA_CTRL	0xF4	R/W	0x3	VROM EMA Control Register

Boot Mode Configuration (BMI)

0x98013000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						PKG[1:0]		BM[7:0]							

Bit	Name	Type	Default	Description
31:10	-	R	0	Reserved
9:8	PKG[1:0]	R	0	Reflects the status of PKG[1:0] pins
7:4	BM[7:4]	R	0	Reserved
3:0	BM[3:0]	R	0	Reflects the status of BM[3:0] pins

TCM Wait Control Register (TCMWAIT)

0x9801300C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									DDRA	TCMARB		TCMWS			

Bit	Name	Type	Default	Description
31:7	-	R	0	Reserved
6	DDRA	R/W	0	Bypass AHB Bus Matrix for DDR SDRAM access. If this bit is set as "1", all the CPU accesses to address range 0x20000000 ~ 0x5FFFFFFF will be routed directly to the DDR SDRAM Controller.
5	TCMARB[1]	R/W	1	CPU ITCM Interface Arbitration Enable.
4	TCMARB[0]	R/W	1	CPU DTCM Interface Arbitration Enable.
3	TCMWS[3]	R/W	1	AHB ITCM Interface Wait State Enable.
2	TCMWS[2]	R/W	1	AHB DTCM Interface Wait State Enable.
1	TCMWS[1]	R/W	1	CPU ITCM Interface Wait State Enable.
0	TCMWS[0]	R/W	1	CPU DTCM Interface Wait State Enable.

Caution: Do not change the default value of TCMARB and TCMWS.

AHB Control Register1 (AHBCON1)

0x98013014

Bit	Name	Type	Default	Description
31:24	BMARBFP[7:0]	R/W	0	AHB Bus Matrix Arbitration Mode for Fixed Priority. If it is set to "1", the highest priority will be given to the CPU.
23:20	-	R	0	Reserved
19:16	HBURST	R/W	0xF	AHB Burst Mode Configuration. Do not change the default value.
15:8	BMARBRR[7:0]	R/W	0	AHB Bus Matrix Arbitration Mode - Round Robin Priority Enable Do not change the default value.
7:0	BMINCRB[7:0]	R/W	0	Arbitration Mode for INCR Cycle. Do not change the default value.

The VIOC always has the highest arbitration priority regardless of the value written in this register.

Each bit of BMARBFP and BMARBRR controls corresponding output port (OP0~7) of the AHB Bus Matrix shown in Figure 17.1 respectively. (OP8 is controlled together with OP0).

The input port numbers of the AHB Bus Matrix shown in Figure 17.1 represent priority order (IP0 is the highest priority master).

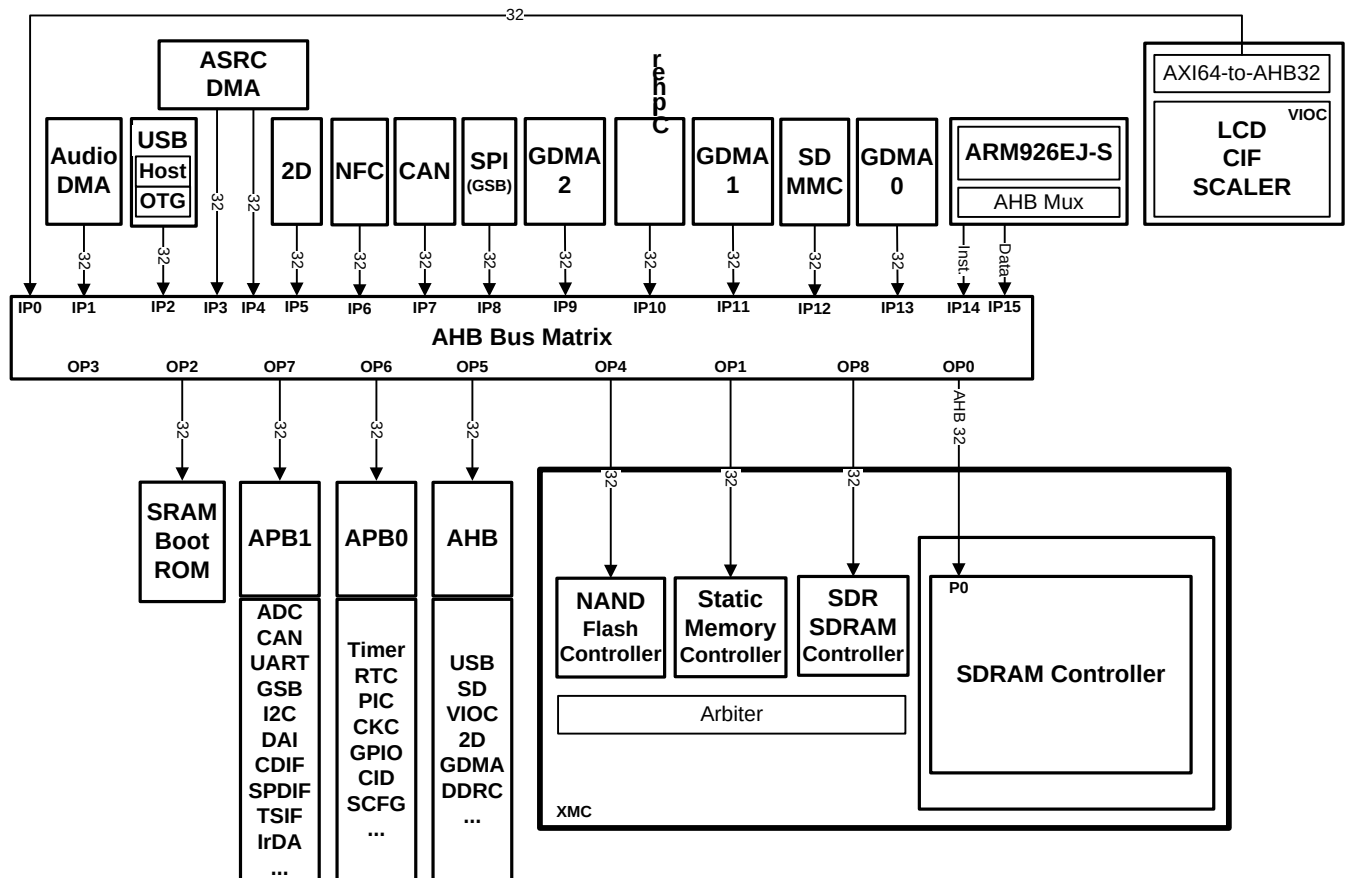


Figure 17.1 Internal Bus Block Diagram

REMAP Register (REMAP)

0x98013020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												REMAP[3:0]			

REMAP[3:0]	REMAP(When ADDI[31:28] == 0x0)
0x0	ADDRO[31:28] = 0x1 (Internal SRAM)
0x1	0x10000000 is remapped to 0x00000000.
0x2	ADDRO[31:28] = 0x2 (External SDRAM)
0x3	0x20000000 is remapped to 0x00000000.
0x4	ADDRO[31:28] = 0x1 (Internal SRAM)
0x5	0x10000000 is remapped to 0x00000000.
0x6	ADDRO[31:28] = 0x4 (External SRAM)
0x7	0x40000000 is remapped to 0x00000000.
0x8	ADDRO[31:28] = 0x7 (External CS3)
0x9	0x70000000 is remapped to 0x00000000.
0xA	ADDRO[31:28] = 0xA (Serial Flash Controller)
0xB	0xA0000000 is remapped to 0x00000000.
0xC	ADDRO[31:28] = 0xB (External SDRAM)
0xD	0xB0000000 is remapped to 0x00000000.
0xE	ADDRO[31:28] = 0xC (External SRAM)
0xF	0xC0000000 is remapped to 0x00000000.
0x0E	ADDRO[31:28] = 0xE (Internal Boot ROM)
0x0F	0xE0000000 is remapped to 0x00000000.

Note: At RESET, if boot mode is external NOR boot, REMAP[3:0] == 0x7, otherwise REMAP[3:0] == 0xE.

LCDSI Access Enable (LCDSIAE)

0x98013024

Bit	Name	Type	Default	Description
31:12	-	R	0	Reserved.
11	LCDFB	R/W	0	Reserved for test. Must be written as 0.
10	CDEMUX[1]	R/W	0	VIOC CIF DEMUX Enable.
9	CDEMUX[0]	R/W	0	VIOC CIF DEMUX Input Selection. 0=CIF0 1=CIF1
8	LCDAXI	R/W	0	VIOC Master Bus Selection.
				0 AHB. If SDR SDRAM is used, this bit must be set as 0.
				1 AXI. If DDR SDRAM is used, this bit must be set as 1.
7:0	-	R	0	Reserved

PAD_IS0 Register (PAD IS Control Register)

0x980130A8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IS[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IS[15:00]															

PAD_IS1 Register (PAD IS Control Register)

0x980130AC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IS[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IS[47:32]															

PAD_IS2 Register (PAD IS Control Register)

0x980130B0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IS[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IS[79:64]															

PAD_IS3 Register (PAD IS Control Register)

0x980130B4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IS[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IS[111:96]															

PAD_IS4 Register (PAD IS Control Register)

0x980130B8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IS[159:144]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IS[143:128]															

PAD_IS5 Register (PAD IS Control Register)

0x980130BC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IS[191:176]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IS[175:160]															

PAD_IS6 Register (PAD IS Control Register)

0x980130C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IS[223:208]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IS[207:192]															

Bit	Name	Type	Default	Description
[223:0]	IS	R/W	0x60000000_00000000_00000000_00000000_00000000_00000000_00000000_00000000_	Schmitt Trigger mode enable 0: CMOS 1: Schmitt

Table 17.2 PAD_IS Assignment

Bit	IS0	IS1	IS2	IS3	IS4	IS5	IS6
[31]	XA[10]	PS[10]	FC[10]	XD[30]	PC[1]	LC[5]	-
[30]	XA[9]	PS[9]	FC[9]	XD[29]	PC[0]	LC[4]	-
[29]	XA[8]	PS[8]	FC[8]	XD[28]	PD[23]	LC[3]	-
[28]	XA[7]	PS[7]	FC[7]	XD[27]	PD[22]	LC[2]	-
[27]	XA[6]	PS[6]	FC[6]	XD[26]	PD[21]	LC[1]	-
[26]	XA[5]	PS[5]	FC[5]	XD[25]	PD[20]	LC[0]	-
[25]	XA[4]	PS[4]	FC[4]	XD[24]	PD[19]	PC[27]	-
[24]	XA[3]	PS[3]	FC[3]	XD[23]	PD[18]	PC[26]	-
[23]	XA[2]	PS[2]	FC[2]	XD[22]	PD[17]	PC[25]	-
[22]	XA[1]	PS[1]	FC[1]	XD[21]	PD[16]	PC[24]	-
[21]	XA[0]	PS[0]	FC[0]	XD[20]	PD[15]	PC[23]	-
[20]	LD[23]	RTCK	PS[31]	XD[19]	PD[14]	PC[22]	-
[19]	LD[22]	TDO	PS[30]	XD[18]	PD[13]	PC[21]	-
[18]	LD[21]	-	PS[29]	XD[17]	PD[12]	PC[20]	-
[17]	LD[20]	ADIN[12]	PS[28]	XD[16]	PD[11]	PC[19]	-
[16]	LD[19]	ADIN[11]	PS[27]	XD[15]	PD[10]	PC[18]	-
[15]	LD[18]	ADIN[10]	PS[26]	XD[14]	PD[9]	PC[17]	-
[14]	LD[17]	ADIN[9]	PS[25]	XD[13]	PD[8]	PC[16]	-
[13]	LD[16]	ADIN[3]	PS[24]	XD[12]	PD[7]	PC[15]	-
[12]	LD[15]	ADIN[2]	PS[23]	XD[11]	PD[6]	PC[14]	-
[11]	LD[14]	ADIN[1]	PS[22]	XD[10]	PD[5]	PC[13]	-
[10]	LD[13]	ADIN[0]	PS[21]	XD[9]	PD[4]	PC[12]	-
[9]	LD[12]	XDCSn	PS[20]	XD[8]	PD[3]	PC[11]	-
[8]	LD[11]	XWEn	PS[19]	XD[7]	PD[2]	PC[10]	-
[7]	LD[10]	XDCKE	PS[18]	XD[6]	PD[1]	PC[9]	-
[6]	LD[9]	XA[16]	PS[17]	XD[5]	PD[0]	PC[8]	-
[5]	LD[8]	XA[15]	PS[16]	XD[4]	OTP_WP	PC[7]	-
[4]	LD[7]	XDCLK	PS[15]	XD[3]	XA[21]	PC[6]	-
[3]	LD[6]	XA[14]	PS[14]	XD[2]	XA[20]	PC[5]	-
[2]	LD[5]	XA[13]	PS[13]	XD[1]	XA[19]	PC[4]	LD[2]
[1]	LD[4]	XA[12]	PS[12]	XD[0]	XA[18]	PC[3]	LD[1]
[0]	LD[3]	XA[11]	PS[11]	FC[11]	XD[3]	PC[2]	LD[0]

The IS of GPIO_PK is controlled by PMGPIO register.

PAD_SR0 Register (PAD SR Control Register)

0x980130C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SR [15:00]															

PAD_SR1 Register (PAD SR Control Register)

0x980130C8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SR [63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SR [47:32]															

PAD_SR2 Register (PAD SR Control Register)

0x980130CC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SR [95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SR [79:64]															

PAD_SR3 Register (PAD SR Control Register)

0x980130D0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SR [127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SR [111:96]															

PAD_SR4 Register (PAD SR Control Register)

0x980130D4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SR [159:144]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SR [143:128]															

PAD_SR5 Register (PAD SR Control Register)

0x980130D8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SR [191:176]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SR [175:160]															

PAD_SR6 Register (PAD SR Control Register)

0x980130DC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SR [223:208]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SR [207:192]															

Bit	Name	Type	Default	Description
[223:0]	SR	R/W	0x00000000_00000000_00000000_00000000_00000000_00000000_00000000_00000000_	Control Slew-rate 0: Fast Slew 1: Slow Slew

Table 17.3 PAD_SR Assignment

Bit	SR0	SR1	SR2	SR3	SR4	SR5	SR6
[31]	XA[10]	PS[10]	FC[10]	XD[30]	PC[1]	LC[5]	-
[30]	XA[9]	PS[9]	FC[9]	XD[29]	PC[0]	LC[4]	-
[29]	XA[8]	PS[8]	FC[8]	XD[28]	PD[23]	LC[3]	-
[28]	XA[7]	PS[7]	FC[7]	XD[27]	PD[22]	LC[2]	-
[27]	XA[6]	PS[6]	FC[6]	XD[26]	PD[21]	LC[1]	-
[26]	XA[5]	PS[5]	FC[5]	XD[25]	PD[20]	LC[0]	-
[25]	XA[4]	PS[4]	FC[4]	XD[24]	PD[19]	PC[27]	-
[24]	XA[3]	PS[3]	FC[3]	XD[23]	PD[18]	PC[26]	-
[23]	XA[2]	PS[2]	FC[2]	XD[22]	PD[17]	PC[25]	-
[22]	XA[1]	PS[1]	FC[1]	XD[21]	PD[16]	PC[24]	-
[21]	XA[0]	PS[0]	FC[0]	XD[20]	PD[15]	PC[23]	-
[20]	LD[23]	RTCK	PS[31]	XD[19]	PD[14]	PC[22]	-
[19]	LD[22]	TDO	PS[30]	XD[18]	PD[13]	PC[21]	-
[18]	LD[21]	-	PS[29]	XD[17]	PD[12]	PC[20]	-
[17]	LD[20]	ADIN[12]	PS[28]	XD[16]	PD[11]	PC[19]	-
[16]	LD[19]	ADIN[11]	PS[27]	XD[15]	PD[10]	PC[18]	-
[15]	LD[18]	ADIN[10]	PS[26]	XD[14]	PD[9]	PC[17]	-
[14]	LD[17]	ADIN[9]	PS[25]	XD[13]	PD[8]	PC[16]	-
[13]	LD[16]	ADIN[3]	PS[24]	XD[12]	PD[7]	PC[15]	-
[12]	LD[15]	ADIN[2]	PS[23]	XD[11]	PD[6]	PC[14]	-
[11]	LD[14]	ADIN[1]	PS[22]	XD[10]	PD[5]	PC[13]	-
[10]	LD[13]	ADIN[0]	PS[21]	XD[9]	PD[4]	PC[12]	-
[9]	LD[12]	XDCSn	PS[20]	XD[8]	PD[3]	PC[11]	-
[8]	LD[11]	XWEn	PS[19]	XD[7]	PD[2]	PC[10]	-
[7]	LD[10]	XDCKE	PS[18]	XD[6]	PD[1]	PC[9]	-
[6]	LD[9]	XA[16]	PS[17]	XD[5]	PD[0]	PC[8]	-
[5]	LD[8]	XA[15]	PS[16]	XD[4]	OTP_WP	PC[7]	-
[4]	LD[7]	XDCLK	PS[15]	XD[3]	XA[21]	PC[6]	-
[3]	LD[6]	XA[14]	PS[14]	XD[2]	XA[20]	PC[5]	-
[2]	LD[5]	XA[13]	PS[13]	XD[1]	XA[19]	PC[4]	LD[2]
[1]	LD[4]	XA[12]	PS[12]	XD[0]	XA[18]	PC[3]	LD[1]
[0]	LD[3]	XA[11]	PS[11]	FC[11]	XD[3]	PC[2]	LD[0]

The SR of GPIO_PK is controlled by PMGPIO register.

External Memory Control Configuration Register (XMCCFG)

0x980130E0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
												BMD[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
				XDSMC	XDNFC	BW		XAW	PUI	DDXD	XDFD	0	FDXD	NFCE	DDRE

Bit	Name	Type	Default	Description				
31:10	-	R	0	Reserved				
19:16	BMD	R/W	0	Bus Matrix Decoder Control for DDR SDRAM. “0” enables DDR SDRAM on to the corresponding address region below. These bits are effective only when “DDRE” bit is set (DDR SDRAM Controller enabled) and “DDRA” bit of TCMWAIT register described above is “0”. BMD[0] : 0x20000000 ~ 0x2FFFFFFF BMD[1] : 0x30000000 ~ 0x3FFFFFFF BMD[2] : 0x40000000 ~ 0x4FFFFFFF BMD[3] : 0x50000000 ~ 0x5FFFFFFF				
11	XDSMC	R/W	0	XD Bus Configuration for EMC (SDR SDRAM / Static Memory Controller) <table><tr><td>0</td><td>XD[31:16] ⇔ EMC_DATA[31:16]</td></tr><tr><td>1</td><td>XD[31:16] ⇔ EMC_DATA[15:00]</td></tr></table>	0	XD[31:16] ⇔ EMC_DATA[31:16]	1	XD[31:16] ⇔ EMC_DATA[15:00]
0	XD[31:16] ⇔ EMC_DATA[31:16]							
1	XD[31:16] ⇔ EMC_DATA[15:00]							
10	XDNFC	R/W	0	XD Bus Configuration for NFC (NAND Flash Controller) <table><tr><td>0</td><td>XD[31:16] ⇔ NFC_DATA[31:16]</td></tr><tr><td>1</td><td>XD[31:16] ⇔ NFC_DATA[15:00]</td></tr></table>	0	XD[31:16] ⇔ NFC_DATA[31:16]	1	XD[31:16] ⇔ NFC_DATA[15:00]
0	XD[31:16] ⇔ NFC_DATA[31:16]							
1	XD[31:16] ⇔ NFC_DATA[15:00]							
9:8	BW	R/W	Boot	CS3(XRCS#) Bus Width 10: 16bit 11: 8bit				
7	XAW	R/W	0	Reserved. Must be written as 0				
6	PUI	R/W	Boot	SDR SDRAM PowerUp Initialization Control at controller reset. <table><tr><td>0</td><td>SDRAM Power up initialization is enabled at reset.</td></tr><tr><td>1</td><td>SDRAM Power up initialization is disabled (skipped) at reset.</td></tr></table>	0	SDRAM Power up initialization is enabled at reset.	1	SDRAM Power up initialization is disabled (skipped) at reset.
0	SDRAM Power up initialization is enabled at reset.							
1	SDRAM Power up initialization is disabled (skipped) at reset.							
5	DDXD	R/W	Boot	Reserved				
4	XDFD	R/W	Boot	Reserved Must be written as 0				
3	-	R/W	0	Reserved				
2	FDXD	R/W	Boot	Reserved Must be written as 1				
1	NFCE	R/W	Boot	NAND Flash Controller Enable 0=Disable 1=Enable				
0	DDRE	R/W	Boot	DDR SDRAM Controller Enable 0=Disable 1=Enable Must be written as 0.				

The default value is determined by BM[3:0] configuration and boot code.

Internal Memory Control Configuration Register (IMCCFG)

0x980130E4

Bit	Name	Type	Default	Description
31:7	-	R	0	Reserved
11	SDH	R/W	1	4KB – SD/MMC Controller 0/1 FIFOs 4KB – USB Host 0 FIFO 4KB – USB Host 1 FIFO 16KB – USB20DH FIFO Reserved 20KB – G2D Memories 0: CPU access disable 1: CPU access enable
10	USBH0	R/W	1	
9	USBH1	R/W	1	
8	USB20DH	R/W	1	
7:5	-	R/W	0x7	
4	BMEM	R/W	1	
3	FMEM0	R/W	1	
2	FMEM1	R/W	1	
1	FMEM2	R/W	1	
0	FMEM3	R/W	1	

These bits must be cleared before each function block is enabled for normal operation.
 The CPU can access the SRAMs through 0x10008000 ~ 0x1000FFFF.

Caution: Bit[3:0] is always set to "0"

JTAG_SECTRL Register (JTAG Secure Control Register)

0x980130E8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
														T2I	JS

Bit	Name	Type	Default	Description
1	T2I	R/W	0x1	Type2Indicator bit 1: Can access ROMCODE 0: Cannot access ROMCODE This bit is only change from 1 to 0. (Cannot change 0 to 1) This value changes by bootcode.
0	JS	R/W	0x1	JTAG SELECT *for Debug

SEMA CTRL Register (Single port memory EMA control Register)

0x980130EC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
														SEMA[17:16]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SEMA[15:00]															

Bit	Name	Type	Default	Description
[17:0]	SEMA	R/W	0x3041	Single Port Memory EMA Control Register Caution: Do not change this value.

DEMA CTRL Register (Dual port memory EMA control Register)

0x980130F0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DEMA[11:00]															

Bit	Name	Type	Default	Description
[11:0]	DEMA	R/W	0x06DB	Dual Port Memory EMA Control Register Caution: Do not change this value.

VEMA CTRL Register (VROM memory EMA control Register)

0x980130F4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
														VEMA[01:00]	

Bit	Name	Type	Default	Description
[1: 0]	VEMA	R/W	0x3	VROM Memory EMA Control Register Caution: Do not change this value.

18 GPIO Port

18.1 Overview

The GPIO controller provides the following features.

- Internal pull-up/pull-down resistor
- Output driver strength control
- I/O direction control
- Programmable Interrupt (GPIOPK only)

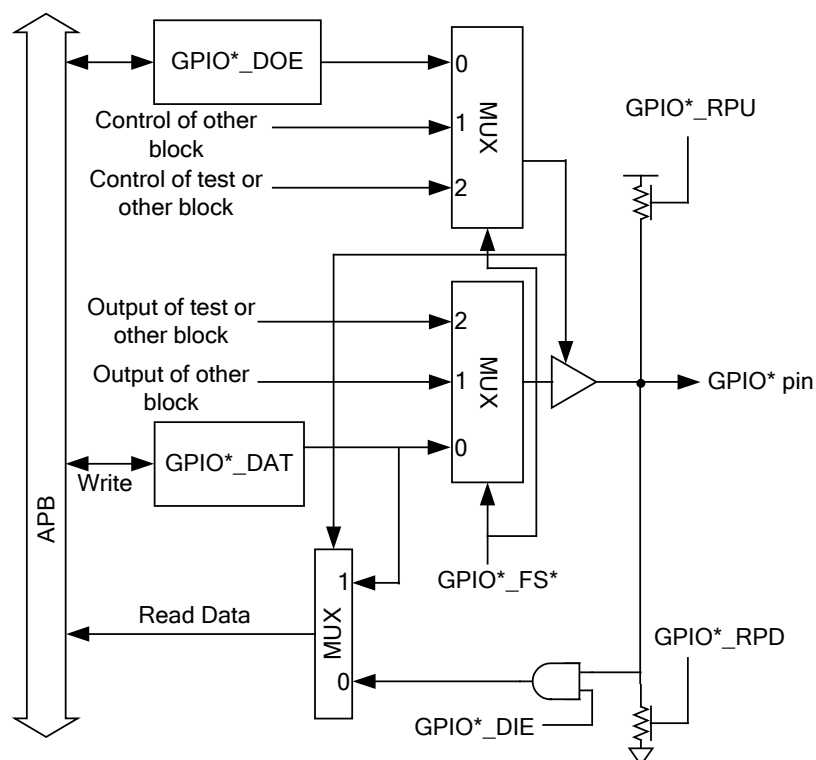


Figure 18.1 GPIO Block Diagram

The I/O mode can be set by the state of GPIO*_DOE register. If a bit of GPIO*_DOE register is written as 1, the corresponding GPIO pin is set to output mode, and if written as 0, which is the default state of GPIO*_DOE register, the corresponding GPIO pin is set to input mode.

If GPIO pin is in input mode, the state of GPIO pin can be read by the CPU through GPIO*_DAT register. In output mode, the state of GPIO pin can be controlled by the value of the corresponding bit of GPIO*_DAT register.

In TCC8132, most of the GPIO pins are shared and multiplexed with the signals from/to internal function blocks. The multiplexing between GPIO mode and normal function mode is controlled by the state of the GPIO*_FS* registers. If a bit of GPIO*_FS* is 1, the corresponding GPIO pin is configured in function mode, and can be used by peripherals instead of GPIO block. The direction control method of GPIO pins in the function mode is determined case by case.

Caution:

1. Must Not enable both pull-up and pull-down resistors together for the same pin. If the internal pull-up resistor is enabled for a pin, then the pull-down resistor for that pin must be disabled and vice versa.
2. Do not change the default reset values for those bits or registers without external pins.

18.2 Alive GPIO (PMGPIO)

The following figure shows the internal logic circuit for PMGPIO controller for PK[31:0].
For ports PK[34:32], refer to the Figure 18.3, Figure 18.4, and Figure 18.5..

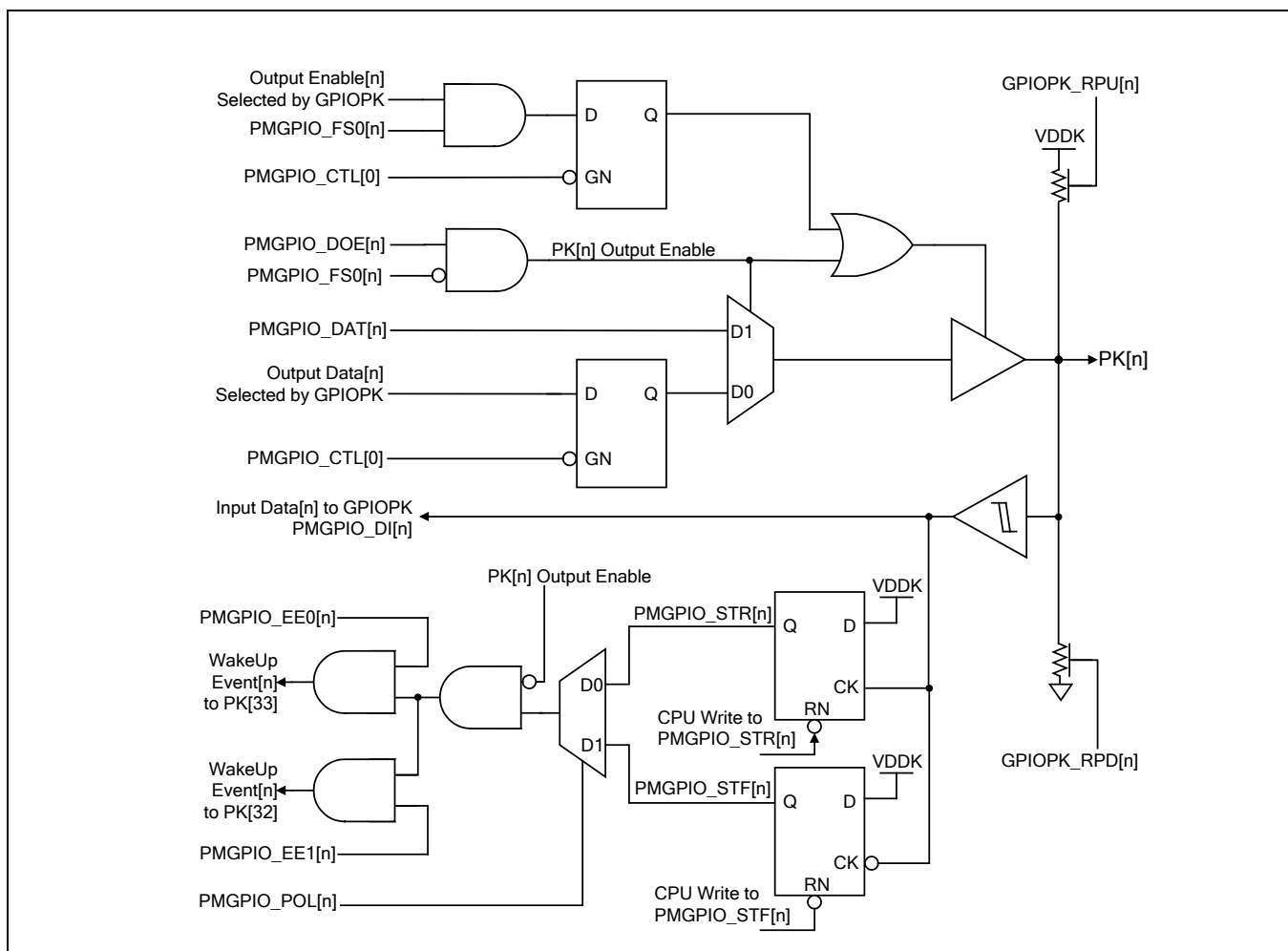


Figure 18.2 PMGPIO Logic Diagram – PK[31:0]

Note:

1. All the logic circuits in the PMGPIO controller operate with VDDK power (1.8V ~ 3.3V).
2. In case of polling the external pin state of PK[31:0], using PMGPIO_DI port consumes more power and wait states than using GPIOPK_DAT.
3. Pay attention not to access PMGPIO and backup RAM if VDDK is supplied from the backup battery. The dynamic current consumption may affect the backup battery life.

The following figures show the internal logic circuit for PK[34:32] output generation. Note that unlike other PK ports, the reset default state of PK[34:32] is output low and there is no wakeup event generation logic for input mode.

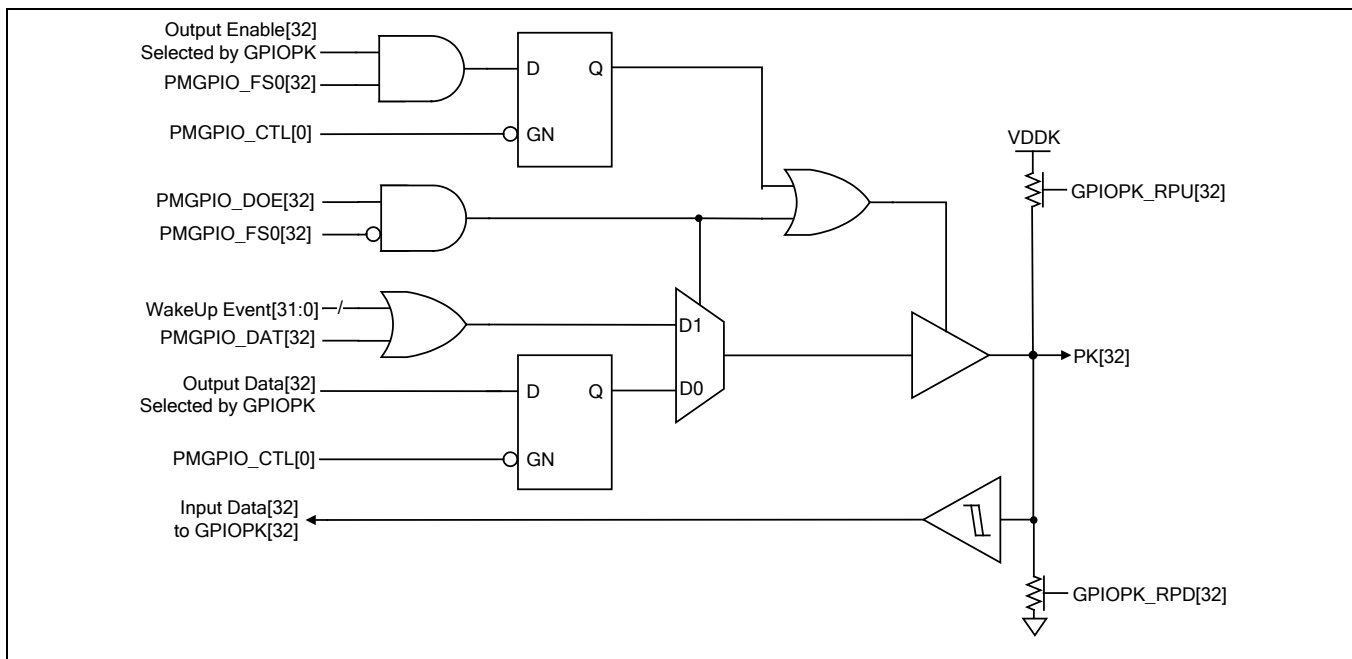


Figure 18.3 PMGPIO Logic Diagram – PK[32]

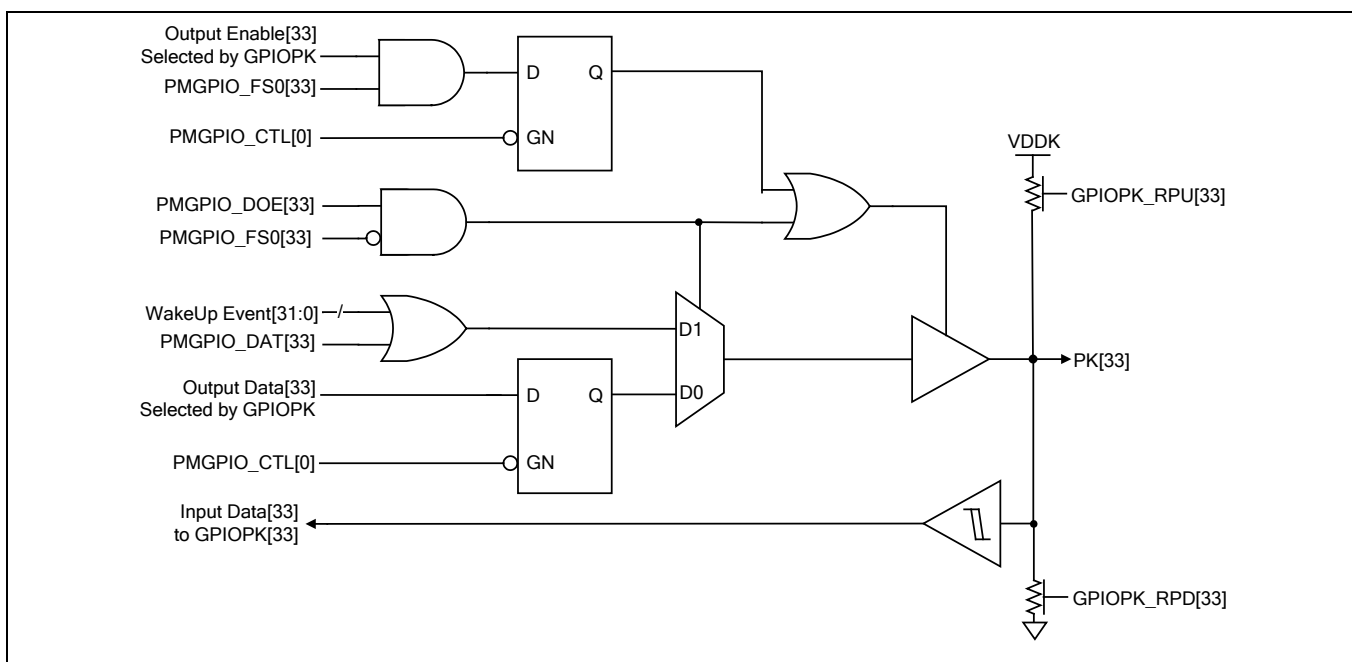


Figure 18.4 PMGPIO Logic Diagram – PK[33]

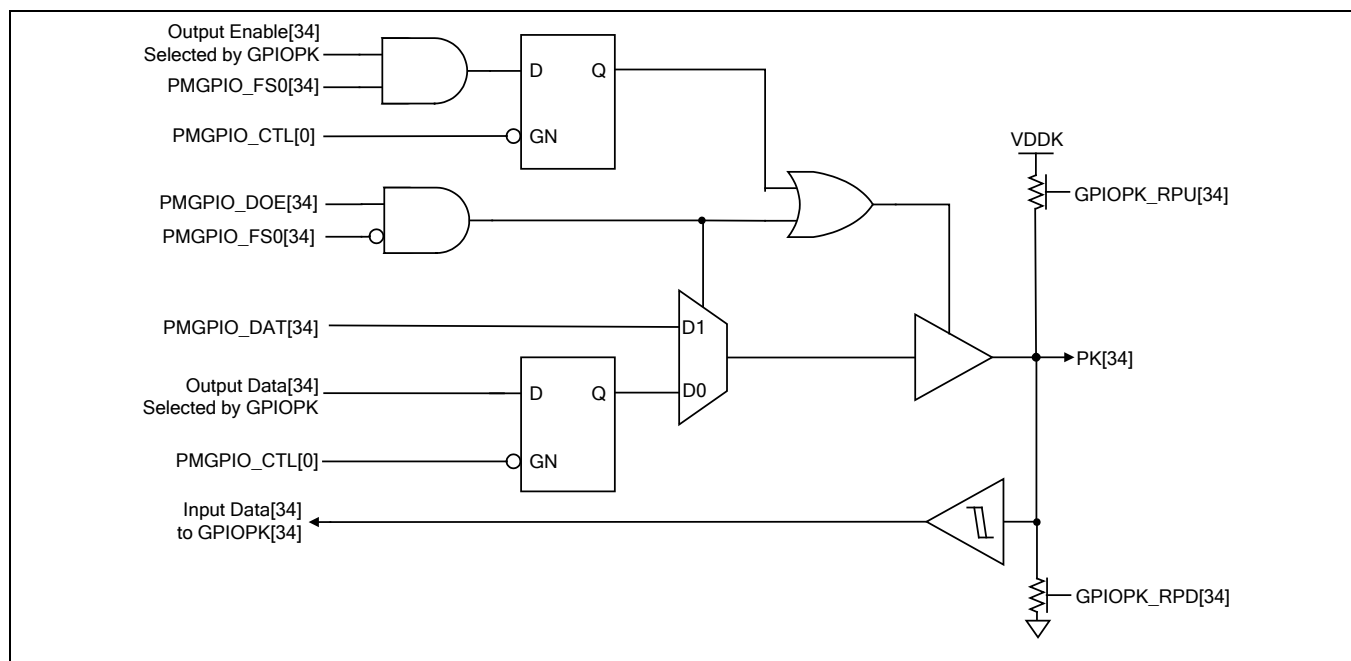


Figure 18.5 PMGPIO Logic Diagram – PK[34]

18.3 Register Description

Table 18.1 GPIOPD Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
PD[23:0]	GPIOPD_DAT	0x000	R/W	0x00000000	GPIOPD Data Register
	GPIOPD_DOE	0x004	R/W	0x00000000	GPIOPD Direction Control Register
	GPIOPD_FS4	0x018	R/W	0x00000000	GPIOPD Function Select 4 Register
	GPIOPD_DIE	0x02C	R/W	0x00000000	GPIOPD Input Enable Register
	GPIOPD_RPE	0x030	R/W	0x00000000	GPIOPD Pull-Up / Down Enable Register
	GPIOPD_RPS	0x034	R/W	0x00000000	GPIOPD Pull-Up / Down Select Register
	GPIOPD_DV0	0x038	R/W	0x00000000	GPIOPD Driver Strength 0 Register
	GPIOPD_DV1	0x03C	R/W	0x00000000	GPIOPD Driver Strength 1 Register

Table 18.2 GPIOPS Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
PS[31:0]	GPIOPS_DAT	0x040	R/W	0x00000000	GPIOPS Data Register
	GPIOPS_DOE	0x044	R/W	0x00000000	GPIOPS Direction Control Register
	GPIOPS_FS0	0x048	R/W	0x00000000	GPIOPS Function Select 0 Register
	GPIOPS_FS1	0x04C	R/W	0x00000000	GPIOPS Function Select 1 Register
	GPIOPS_FS2	0x050	R/W	0x00000000	GPIOPS Function Select 2 Register
	GPIOPS_FS3	0x054	R/W	0x00000000	GPIOPS Function Select 3 Register
	GPIOPS_FS4	0x058	R/W	0x00000000	GPIOPS Function Select 4 Register
	GPIOPS_DIE	0x06C	R/W	0xFFFFFFFF	GPIOPS Input Enable Register
	GPIOPS_RPE	0x070	R/W	0x00000000	GPIOPS Pull-Up / Down Enable Register
	GPIOPS_RPS	0x074	R/W	0x00000000	GPIOPS Pull-Up / Down Select Register
	GPIOPS_DV0	0x078	R/W	0x00000000	GPIOPS Driver Strength 0 Register
	GPIOPS_DV1	0x07C	R/W	0x00000000	GPIOPS Driver Strength 1 Register

Table 18.3 GPIOFC Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
FC[11:0]	GPIOFC_DAT	0x0C0	R/W	0x00000000	GPIOFC Data Register
	GPIOFC_DOE	0x0C4	R/W	0x00000000	GPIOFC Direction Control Register
	GPIOFC_FS0	0x0C8	R/W	0x00000000	GPIOFC Function Select 0 Register
	GPIOFC_FS1	0x0CC	R/W	0x00000000	GPIOFC Function Select 1 Register
	GPIOFC_FS2	0x0D0	R/W	0x00000000	GPIOFC Function Select 2 Register
	GPIOFC_FS3	0x0D4	R/W	0x00000000	GPIOFC Function Select 3 Register
	GPIOFC_FS4	0x0D8	R/W	0x00000000	GPIOFC Function Select 4 Register
	GPIOFC_DIE	0x0EC	R/W	0x00000FFF	GPIOFC Input Enable Register
	GPIOFC_RPE	0x0F0	R/W	0x00000003	GPIOFC Pull-Up / Down Enable Register
	GPIOFC_RPS	0x0F4	R/W	0x00000000	GPIOFC Pull-Up / Down Select Register
	GPIOFC_DV0	0x0F8	R/W	0x00000FFF	GPIOFC Driver Strength 0 Register
	GPIOFC_DV1	0x0FC	R/W	0x00000000	GPIOFC Driver Strength 1 Register

Table 18.4 GPIOXD Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
XD[31:0]	GPIOXD_DAT	0x100	R/W	0x00000000	GPIOXD Data Register
	GPIOXD_DOE	0x104	R/W	0x00000000	GPIOXD Direction Control Register
	GPIOXD_FS0	0x108	R/W	0x00000000	GPIOXD Function Select 0 Register
	GPIOXD_FS4	0x118	R/W	0x00000000	GPIOXD Function Select 4 Register
	GPIOXD_DIE	0x12C	R/W	0xFFFFFFFF	GPIOXD Input Enable Register
	GPIOXD_RPE	0x130	R/W	0x00000000	GPIOXD Pull-Up / Down Enable Register
	GPIOXD_RPS	0x134	R/W	0x00000000	GPIOXD Pull-Up / Down Select Register
	GPIOXD_DV0	0x138	R/W	0xFFFFFFFF	GPIOXD Driver Strength 0 Register
	GPIOXD_DV1	0x13C	R/W	0x00000000	GPIOXD Driver Strength 1 Register

Table 18.5 GPIOLC Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
LC[5:0]	GPIOLC_DAT	0x140	R/W	0x00000000	GPIOLC Data Register
	GPIOLC_DOE	0x144	R/W	0x00000000	GPIOLC Direction Control Register
	GPIOLC_FS0	0x148	R/W	0x00000000	GPIOLC Function Select 0 Register
	GPIOLC_FS1	0x14C	R/W	0x00000000	GPIOLC Function Select 1 Register
	GPIOLC_FS4	0x158	R/W	0x00000000	GPIOLC Function Select 4 Register
	GPIOLC_DIE	0x16C	R/W	0x0000003F	GPIOLC Input Enable Register
	GPIOLC_RPE	0x170	R/W	0x0000000F	GPIOLC Pull-Up / Down Enable Register

Table 18.5 GPIOLC Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
	GPIOLC_RPS	0x174	R/W	0x00000000	GPIOLC Pull-Up / Down Select Register
	GPIOLC_DV0	0x178	R/W	0x00000000	GPIOLC Driver Strength 0 Register
	GPIOLC_DV1	0x17C	R/W	0x00000000	GPIOLC Driver Strength 1 Register

Table 18.6 GPIOLD Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
LD[23:0]	GPIOLD_DAT	0x180	R/W	0x00000000	GPIOLD Data Register
	GPIOLD_DOE	0x184	R/W	0x00000000	GPIOLD Direction Control Register
	GPIOLD_FS0	0x188	R/W	0x00000000	GPIOLD Function Select 0 Register
	GPIOLD_FS1	0x18C	R/W	0x00000000	GPIOLD Function Select 1 Register
	GPIOLD_FS4	0x198	R/W	0x00000000	GPIOLD Function Select 4 Register
	GPIOLD_DIE	0x1AC	R/W	0xFFFFFFFF	GPIOLD Input Enable Register
	GPIOLD_RPE	0x1B0	R/W	0x00000000	GPIOLD Pull-Up / Down Enable Register
	GPIOLD_RPS	0x1B4	R/W	0x00000000	GPIOLD Pull-Up / Down Select Register
	GPIOLD_DV0	0x1B8	R/W	0x00000000	GPIOLD Driver Strength 0 Register
	GPIOLD_DV1	0x1BC	R/W	0x00000000	GPIOLD Driver Strength 1 Register

Table 18.7 GPIOAD Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
ADIN[12,9][3:0]	GPIOAD_DAT	0x1C0	R/W	0x00000000	GPIOAD Data Register
	GPIOAD_DOE	0x1C4	R/W	0x00000000	GPIOAD Direction Control Register
	GPIOAD_FS0	0x1C8	R/W	0x000000FF	GPIOAD Function Select 0 Register
	GPIOAD_DIE	0x1EC	R/W	0x00000000	GPIOAD Input Enable Register
	GPIOAD_RPE	0x1F0	R/W	0x00000000	GPIOAD Pull-Up / Down Enable Register
	GPIOAD_RPS	0x1F4	R/W	0x00000000	GPIOAD Pull-Up / Down Select Register
	GPIOAD_DV0	0x1F8	R/W	0x00000000	GPIOAD Driver Strength 0 Register
	GPIOAD_DV1	0x1FC	R/W	0x00000000	GPIOAD Driver Strength 1 Register

Table 18.8 GPIOXC Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
XDCKE XDSCS#[1:0] XRCS# XSCS#[1:0] XOE# XWE# XA[23:0]	GPIOXC_DAT	0x200	R/W	0x00000000	GPIOXC Data Register
	GPIOXC_DOE	0x204	R/W	0x00000000	GPIOXC Direction Control Register
	GPIOXC_FS0	0x208	R/W	0xFFFFFFFF	GPIOXC Function Select 0 Register
	GPIOXC_FS4	0x218	R/W	0x00000000	GPIOXC Function Select 4 Register
	GPIOXC_DIE	0x22C	R/W	0xFFFFFFFF	GPIOXC Input Enable Register
	GPIOXC_RPE	0x230	R/W	0x00000000	GPIOXC Pull-Up / Down Enable Register
	GPIOXC_RPS	0x234	R/W	0x00000000	GPIOXC Pull-Up / Down Select Register
	GPIOXC_DV0	0x238	R/W	0xFFFFFFFF	GPIOXC Driver Strength 0 Register
	GPIOXC_DV1	0x23C	R/W	0x00000000	GPIOXC Driver Strength 1 Register

Table 18.9 GPIODD Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
XDCLK	GPIODD_DAT	0x240	R/W	0x00000000	GPIODD Data Register
	GPIODD_FS0	0x248	R/W	0xFFFFFFFF	GPIODD Function Select 0 Register
	GPIODD_DIE	0x26C	R/W	0xFFFFFFFF	GPIODD Input Enable Register
	GPIODD_RPE	0x270	R/W	0x00000000	GPIODD Pull-Up / Down Enable Register
	GPIODD_RPS	0x274	R/W	0x00000000	GPIODD Pull-Up / Down Select Register
	GPIODD_DV0	0x278	R/W	0xFFFFFFFF	GPIODD Driver Strength 0 Register
	GPIODD_DV1	0x27C	R/W	0x00000000	GPIODD Driver Strength 1 Register

Table 18.10 GPIOPC Register Map (Base Address =0x98005000)

Ports	Register	Addr	Type	Reset	Description
PC[27:0]	GPIOPC_DAT	0x280	R/W	0x00000000	GPIOPC Data Register
	GPIOPC_DOE	0x284	R/W	0x00000000	GPIOPC Direction Control Register
	GPIOPC_FS0	0x288	R/W	0x00000000	GPIOPC Function Select 0 Register
	GPIOPC_FS1	0x28C	R/W	0x00000000	GPIOPC Function Select 1 Register
	GPIOPC_FS4	0x298	R/W	0x00000000	GPIOPC Function Select 4 Register
	GPIOPC_DIE	0x2AC	R/W	0x0FFFFFFF	GPIOPC Input Enable Register
	GPIOPC_RPE	0x2B0	R/W	0x00000000	GPIOPC Pull-Up / Down Enable Register
	GPIOPC_RPS	0x2B4	R/W	0x00000000	GPIOPC Pull-Up / Down Select Register
	GPIOPC_DV0	0x2B8	R/W	0x00000000	GPIOPC Driver Strength 0 Register
	GPIOPC_DV1	0x2BC	R/W	0x00000000	GPIOPC Driver Strength 1 Register

Table 18.11 GPIOPK Register Map (Base Address =0x9801C000)

Ports	Register	Addr	Type	Reset	Description
PK[34:0]	GPIOPK_RST	0x008	R/W	0x00000000	GPIOPK Reset Control Register
	GPIOPK_DAT	0x100	R/W	0x00000000	GPIOPK Data Register
	GPIOPK_DOE	0x104	R/W	0x00000000	GPIOPK Direction Control Register
	GPIOPK_FS0	0x108	R/W	0x00000000	GPIOPK Function Select 0 Register
	GPIOPK_FS1	0x10C	R/W	0x00000000	GPIOPK Function Select 1 Register
	GPIOPK_FS2	0x110	R/W	0x00000000	GPIOPK Function Select 2 Register
	GPIOPK_FS3	0x114	R/W	0x00000000	GPIOPK Function Select 3 Register
	GPIOPK_FS4	0x118	R/W	0x00000000	GPIOPK Function Select 4 Register
	GPIOPK_DIE	0x12C	R/W	0xFFFFFFFF	GPIOPK Input Enable Register
	GPIOPK_RPE	0x130	R/W	0x00000000	GPIOPK Pull-Up / Down Enable Register
	GPIOPK_RPS	0x134	R/W	0x00000000	GPIOPK Pull-Up / Down Select Register
	GPIOPK_DV0	0x138	R/W	0x00000000	GPIOPK Driver Strength 0 Register
	GPIOPK_DV1	0x13C	R/W	0x00000000	GPIOPK Driver Strength 1 Register
	GPIOPK_DAT_H	0x140	R/W	0x00000000	GPIOPK Data Register
	GPIOPK_DOE_H	0x144	R/W	0x00000007	GPIOPK Direction Control Register
	GPIOPK_FS0_H	0x148	R/W	0x00000000	GPIOPK Function Select 0 Register
	GPIOPK_DIE_H	0x16C	R/W	0x00000007	GPIOPK Input Enable Register
	GPIOPK_RPE_H	0x170	R/W	0x00000000	GPIOPK Pull-Up / Down Enable Register
	GPIOPK_RPS_H	0x174	R/W	0x00000000	GPIOPK Pull-Up / Down Select Register
	GPIOPK_DV0_H	0x178	R/W	0x00000000	GPIOPK Driver Strength 0 Register
	GPIOPK_DV1_H	0x17C	R/W	0x00000000	GPIOPK Driver Strength 1 Register
	GPIOPK_FDAT	0x200	R/W	0x00000000	GPIOPK Filtered Data Port
	GPIOPK_IRQST	0x210	R/W	0x00000000	GPIOPK Interrupt Status Register
	GPIOPK_IRQEN	0x214	R/W	0x00000000	GPIOPK Interrupt Enable Register
	GPIOPK_IRQPOL	0x218	R/W	0x00000000	GPIOPK Interrupt Polarity Register
	GPIOPK_IRQTM0	0x21C	R/W	0x00000000	GPIOPK Interrupt Trigger Mode 0
	GPIOPK_IRQTM1	0x220	R/W	0x00000000	GPIOPK Interrupt Trigger Mode 1
	GPIOPK_FCK	0x224	R/W	0x00000000	GPIOPK Filter Clock Control
	GPIOPK_FBP	0x228	R/W	0x00000000	GPIOPK Filter Bypass
	GPIOPK_CTL	0x22C	R/W	0x00000000	GPIOPK Miscellaneous Control

Note: PMGPIO registers must be programmed correctly before GPIOPK registers are used to control PK[34:0].

Table 18.12 Backup RAM & Alive GPIO (PMGPIO) Register Map (Base Address =0x98010000)

Ports	Register	Addr	Type	Reset	Description
	Backup RAM	0x000	R/W	-	1Kbytes Backup RAM.
PK[31:0]	PMGPIO_DAT	0x800	R/W	0x00000000	PMGPIO Data Register
	PMGPIO_DOE	0x804	R/W	0x00000000	PMGPIO Direction Control Register
	PMGPIO_FS0	0x808	R/W	0xFFFFFFFF	PMGPIO Function Select Register
	PMGPIO_DIE	0x80C	R/W	0xFFFFFFFF	PMGPIO Input Enable Register
	PMGPIO_RPE	0x810	R/W	0x00000000	PMGPIO Pull-Up / Down Enable Register
	PMGPIO_RPS	0x814	R/W	0x00000000	PMGPIO Pull-Up / Down Select Register
	PMGPIO_DV0	0x818	R/W	0x00000000	PMGPIO Driver Strength 0 Register
	PMGPIO_DV1	0x81C	R/W	0x00000000	PMGPIO Driver Strength 1 Register
	PMGPIO_EE0	0x820	R/W	0x00000000	PMGPIO Event Enable 0 Register
	PMGPIO_EE1	0x824	R/W	0x00000000	PMGPIO Event Enable 1 Register
	PMGPIO_CTL	0x828	R/W	0x00011F00	PMGPIO Control Register
	PMGPIO_DI	0x82C	R/W	0x00000000	PMGPIO Input Raw Status Register
	PMGPIO_STR	0x830	R/W	0x00000000	PMGPIO Rising Event Status Register
	PMGPIO_STF	0x834	R/W	0x00000000	PMGPIO Falling Event Status Register
	PMGPIO_POL	0x838	R/W	0x00000000	PMGPIO Event Polarity Register
	PMGPIO_PROT	0x83C	R/W	0x00000000	PMGPIO Protect Register
	PMGPIO_IS	0x860	R/W	0xFFFFFFFF	PMGPIO Schmitt Trigger Control Register
	PMGPIO_SR	0x864	R/W	0x00000000	PMGPIO Slew-rate Control Register
PK[34:32]	PMGPIO_DAT_H	0x840	R/W	0x00000000	PMGPIO Data Register
	PMGPIO_DOE_H	0x844	R/W	0x00000007	PMGPIO Direction Control Register
	PMGPIO_FS0_H	0x848	R/W	0x00000004	PMGPIO Function Select Register
	PMGPIO_DIE_H	0x84C	R/W	0xFFFFFFFF	PMGPIO Input Enable Register
	PMGPIO_RPE_H	0x850	R/W	0x00000000	PMGPIO Pull-Up / Down Enable Register
	PMGPIO_RPS_H	0x854	R/W	0x00000000	PMGPIO Pull-Up / Down Select Register
	PMGPIO_DV0_H	0x858	R/W	0x00000000	PMGPIO Driver Strength 0 Register
	PMGPIO_DV1_H	0x85C	R/W	0x00000000	PMGPIO Driver Strength 1 Register
	PMGPIO_IS_H	0x868	R/W	0xFFFFFFFF	PMGPIO Schmitt Trigger Control Register
	PMGPIO_SR_H	0x86C	R/W	0x00000000	PMGPIO Slew-rate Control Register
	PMGPIO_APB	0x900	R/W	0x00000001	PMGPIO & PMRAM APB Timing Register

Note: Backup RAM does not support byte or half word write.

Note:

1. Read-modify-write sequence is recommended for all the GPIO registers.
2. Reset values for GPIO*_DAT and PMGPIO_DAT registers represent write data value (data for output). They cannot be read correctly unless the corresponding GPIO*_DOE or PMGPIO_DOE register bits are set as output mode.
3. If a GPIO* pin is in output mode and the corresponding bits of GPIO*_FS* or PMGPIO_FS are all 0, the corresponding bit of GPIO*_DAT or PMGPIO_DAT register controls the status of the GPIO* pin. If it is in input mode and acts as GPIO, the corresponding bit represents the status of the GPIO* pin.
4. GPIO*_DOE and PMGPIO_DOE register bits are effective only if the corresponding bit(s) of GPIO*_FS* or PMGPIO_FS* are all 0.
5. If any bit of PMGPIO_FS0 is "0", the corresponding bit of GPIOPK register cannot control PK ports. Also, note that some bits of PMGPIO_CTL register can block all the GPIOPK outputs.

GPIOPD Data Register (GPIOPD_DAT)**0x98005000**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOPD_DAT	R/W	0	Data Port.

GPIOPD Direction Control Register (GPIOPD_DOE)**0x98005004**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOPD_DOE	R/W	0	Direction Control 0 = Input 1 = Output

GPIOPD Function Select 4 Register (GPIOPD_FS4)**0x98005018**

Bit	Name	Type	Default	Description (Function Selected)
31:24	-	R	0	Reserved.
23:0	GBUS[79:56]	R/W	0	GBUS[79:56]

GPIOPD Input Enable Register (GPIOPD_DIE)**0x9800502C**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOPD_DIE	R/W	0	Input Enable 0 = Disable 1 = Enable

GPIOPD Pull-Up / Down Enable Register (GPIOPD_RPE)**0x98005030**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOPD_RPE	R/W	0	Pull-Up / Down Resistor Enable 0 = Disable 1 = Enable

GPIOPD Pull-Up / Down Select Register (GPIOPD_RPS)**0x98005034**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOPD_RPS	R/W	0	Pull-Up / Down Resistor Select 0 = Pull-down 1 = Pull-up

GPIOPD Driver Strength 0 Register (GPIOPD_DV0)**0x98005038**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOPD_DV0	R/W	0	Output Driver Strength Control 0

GPIOPD Driver Strength 1 Register (GPIOPD_DV1)**0x9800503C**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOPD_DV1	R/W	0	Output Driver Strength Control 1

GPIOPD_DV0	GPIOPD_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOPS Data Register (GPIOPS_DAT)

0x98005040

Bit	Name	Type	Default	Description
31:0	GPIOPS_DAT	R/W	0	Data Port.

GPIOPS Direction Control Register (GPIOPS_DOE)

0x98005044

Bit	Name	Type	Default	Description
31:0	GPIOPS_DOE	R/W	0	Direction Control 0 = Input 1 = Output

GPIOPS Function Select 0 Register (GPIOPS_FS0)

0x98005048

Bit	Name	Type	Default	Description (Function Selected)
31:24	-	R/W	0	Reserved
23:20	SD1D[7:4]	R/W	0	SD/MMC Controller 1 Data [7:4]
19	SD1WP	R/W	0	SD/MMC Controller 1 Write Protect Input
18	SD1DET	R/W	0	SD/MMC Controller 1 Card Detect Input
17	SD1CLK	R/W	0	SD/MMC Controller 1 Clock
16	SD1CMD	R/W	0	SD/MMC Controller 1 Command
15:12	SD1D[3:0]	R/W	0	SD/MMC Controller 1 Data [3:0]
11:8	SD0D[7:4]	R/W	0	SD/MMC Controller 0 Data [7:4]
7	SD0WP	R/W	0	SD/MMC Controller 0 Write Protect Input
6	SD0DET	R/W	0	SD/MMC Controller 0 Card Detect Input
5	SD0CLK	R/W	0	SD/MMC Controller 0 Clock
4	SD0CMD	R/W	0	SD/MMC Controller 0 Command
3:0	SD0D[3:0]	R/W	0	SD/MMC Controller 0 Data [3:0]

GPIOPS Function Select 1 Register (GPIOPS_FS1)

0x9800504C

Bit	Name	Type	Default	Description (Function Selected)
31:20	CIF1PD[15:4]	R/W	0	CAM. Interface 1 Data [15:4]
19:12	-	R/W	0	Reserved
11:8	CIF1PD[3:0]	R/W	0	CAM. Interface 1 Data [3:0]
7:0	-	R/W	0	Reserved

GPIOPS Function Select 2 Register (GPIOPS_FS2)

0x98005050

Bit	Name	Type	Default	Description (Function Selected)
31:24	CIF1PD[7:0]	R/W	0	CAM. Interface 1 Data [7:0]
23:20	CIF1PD[15:12]	R/W	0	CAM. Interface 1 Data [15:12]
19:12	-	R/W	0	Reserved
11:8	CIF1PD[11:8]	R/W	0	CAM. Interface 1 Data [11:8]
7:0	-	R/W	0	Reserved

GPIOPS Function Select 3 Register (GPIOPS_FS3)

0x98005054

Bit	Name	Type	Default	Description (Function Selected)
31:28	-	R/W	0	Reserved
27:24	QSPI_IO[3:0]	R/W	0	Serial Flash Data [3:0]
23	QSPI_CLK	R/W	0	Serial Flash Clock
22	QSPI_CS	R/W	0	Serial Flash CS
21	QSPI_RST	R/W	0	Serial Flash Reset
20:14	-	R/W	0	Reserved
13	FRDY[0]	R/W	0	NAND Flash READY
12	FCSn[0]	R/W	0	NAND Flash CS
11:8	EDI_XD[7:4]	R/W	0	NAND Flash DATA [7:4]
7	FOEn	R/W	0	NAND Flash OEn
6	FWEn	R/W	0	NAND Flash WEn
5	FCLE	R/W	0	NAND Flash CLE
4	FALE	R/W	0	NAND Flash ALE
3:0	EDI_XD[3:0]	R/W	0	NAND Flash DATA[3:0]

GPIOPS Function Select 4 Register (GPIOPS_FS4)

0x98005058

Bit	Name	Type	Default	Description (Function Selected)
31:0	GBUS[31:0]	R/W	0	GBUS[31:0]

GPIOPS Input Enable Register (GPIOPS_DIE)

0x9800506C

Bit	Name	Type	Default	Description
31:0	GPIOPS_DIE	R/W	0xFFFFFFFF	Input Enable. 0 = Disable 1 = Enable

GPIOPS Pull-Up / Down Enable Register (GPIOPS_RPE)

0x98005070

Bit	Name	Type	Default	Description
31:0	GPIOPS_RPE	R/W	0	Pull-Up / Down Resistor Enable. 0 = Disable 1 = Enable

GPIOPS Pull-Up / Down Select Register (GPIOPS_RPS)

0x98005074

Bit	Name	Type	Default	Description
31:0	GPIOPS_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

GPIOPS Driver Strength 0 Register (GPIOPS_DV0)

0x98005078

Bit	Name	Type	Default	Description
31:0	GPIOPS_DV0	R/W	0	Output Driver Strength Control 0

GPIOPS Driver Strength 1 Register (GPIOPS_DV1)

0x9800507C

Bit	Name	Type	Default	Description
31:0	GPIOPS_DV1	R/W	0	Output Driver Strength Control 1

GPIOPS_DV0	GPIOPS_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOFC Data Register (GPIOFC_DAT)

0x980050C0

Bit	Name	Type	Default	Description
11:0	GPIOFC_DAT	R/W	0	Data Port

GPIOFC Direction Control Register (GPIOFC_DOE)

0x980050C4

Bit	Name	Type	Default	Description
11:0	GPIOFC_DOE	R/W	0	Direction Control 0 = Input 1 = Output

GPIOFC Function Select 0 Register (GPIOFC_FS0)

0x980050C8

Bit	Name	Type	Default	Description (Function Selected)
11:10	FCSn[3:2]	R/W	0	Nand Flash Controller CSn[3:2] Output
9:6	FRDY[3:0]	R/W	0	Nand Flash Controller READY[3:0] Input
5	FALE	R/W	0	Nand Flash Controller ALE Output
4	FCLE	R/W	0	Nand Flash Controller CLE Output
3	FOEn	R/W	0	Nand Flash Controller OEn Output
2	FWEn	R/W	0	Nand Flash Controller WEn Output
1:0	FCSn[1:0]	R/W	0	Nand Flash Controller CSn[1:0] Output

GPIOFC Function Select 1 Register (GPIOFC_FS1)

0x980050CC

Bit	Name	Type	Default	Description (Function Selected)
11:8	-	R/W	0	Reserved
7:6	TCO[1:0]	R/W	0	Timer TCO[1:0] Output
5:0	TCO[5:0]	R/W	0	Timer TCO[5:0] Output

GPIOFC Function Select 2 Register (GPIOFC_FS2)

0x980050D0

Bit	Name	Type	Default	Description (Function Selected)
11:8	-	R/W	0	Reserved
7:6	CKMO[1:0]	R/W	0	Clock Output [1:0] from CKC
5:4	CKMO[1:0]	R/W	0	Clock Output [1:0] from CKC
3:2	CKMO[1:0]	R/W	0	Clock Output [1:0] from CKC
1:0	CKMO[1:0]	R/W	0	Clock Output [1:0] from CKC

GPIOFC Function Select 3 Register (GPIOFC_FS3)

0x980050D4

Bit	Name	Type	Default	Description (Function Selected)
11	XICS _n	R/W	0	IDE Chip Select Output
10	XSCSn[2]	R/W	0	Static Memory Controller Chip Select 2.
9:7	XIRDY	R/W	0	READY Input to EMC
6:2	-	R/W	0	Reserved.
1	XICS _n	R/W	0	IDE Chip Select Output
0	-	R/W	0	Reserved.

GPIOFC Function Select 4 Register (GPIOFC_FS4)

0x980050D8

Bit	Name	Type	Default	Description (Function Selected)
11:0	GBUS[127:116]	R/W	0	GBUS[127:116]

GPIOFC Input Enable Register (GPIOFC_DIE)

0x980050EC

Bit	Name	Type	Default	Description
11:0	GPIOFC_DIE	R/W	0xFFF	Input Enable. 0 = Disable 1 = Enable

GPIOFC Pull-Up / Down Enable Register (GPIOFC_RPE)

0x980050F0

Bit	Name	Type	Default	Description
11:0	GPIOFC_RPE	R/W	0x03	Pull-Up / Down Resistor Enable. 0 = Disable 1 = Enable

GPIOFC Pull-Up / Down Select Register (GPIOFC_RPS)

0x980050F4

Bit	Name	Type	Default	Description
11:0	GPIOFC_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

GPIOFC Driver Strength 0 Register (GPIOFC_DV0)

0x980050F8

Bit	Name	Type	Default	Description
11:0	GPIOFC_DV0	R/W	0xFFF	Output Driver Strength Control 0

GPIOFC Driver Strength 1 Register (GPIOFC_DV1)

0x980050FC

Bit	Name	Type	Default	Description
11:0	GPIOFC_DV1	R/W	0	Output Driver Strength Control 1

GPIOFC_DV0	GPIOFC_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOXD Data Register (GPIOXD_DAT)

0x98005100

Bit	Name	Type	Default	Description
31:0	GPIOXD_DAT	R/W	0	Data Port.

GPIOXD Direction Control Register (GPIOXD_DOE)

0x98005104

Bit	Name	Type	Default	Description
31:0	GPIOXD_DOE	R/W	0	Direction Control. 0 = Input 1 = Output

GPIOXD Function Select 0 Register (GPIOXD_FS0)**0x98005108**

Bit	Name	Type	Default	Description (Function Selected)
31:0	XD[31:0]	R/W	0	External Bus Data [31:0]

GPIOXD Function Select 4 Register (GPIOXD_FS4)**0x98005118**

Bit	Name	Type	Default	Description (Function Selected)
31:16	GBUS[143:128]	R/W	0	GBUS[143:128]
15:0	-	R/W	0	Reserved.

GPIOXD Input Enable Register (GPIOXD_DIE)**0x9800512C**

Bit	Name	Type	Default	Description
31:0	GPIOXD_DIE	R/W	0xFFFFFFFF	Input Enable. 0 = Disable 1 = Enable

GPIOXD Pull-Up / Down Enable Register (GPIOXD_RPE)**0x98005130**

Bit	Name	Type	Default	Description
31:0	GPIOXD_RPE	R/W	0	Pull-Up / Down Resistor Enable. 0 = Disable 1 = Enable

GPIOXD Pull-Up / Down Select Register (GPIOXD_RPS)**0x98005134**

Bit	Name	Type	Default	Description
31:0	GPIOXD_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

GPIOXD Driver Strength 0 Register (GPIOXD_DV0)**0x98005138**

Bit	Name	Type	Default	Description
31:0	GPIOXD_DV0	R/W	0xFFFFFFFF	Output Driver Strength Control 0

GPIOXD Driver Strength 1 Register (GPIOXD_DV1)**0x9800513C**

Bit	Name	Type	Default	Description
31:0	GPIOXD_DV1	R/W	0	Output Driver Strength Control 1

GPIOXD_DV0	GPIOXD_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOLC Data Register (GPIOLC_DAT)**0x98005140**

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	GPIOLC_DAT	R/W	0	Data Port.

GPIOLC Direction Control Register (GPIOLC_DOE)**0x98005144**

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	GPIOLC_DOE	R/W	0	Direction Control. 0 = Input 1 = Output

GPIOLC Function Select 0 Register (GPIOLC_FS0)**0x98005148**

Bit	Name	Type	Default	Description (Function Selected)
31:6	-	R	0	Reserved
5	LIRQ	R/W	0	Interrupt Input to LCD Controller
4	-	R/W	0	Reserved
3	LCLK	R/W	0	LCD Pixel Clock Output
2	LAC	R/W	0	LCD AC Bias Output
1	LHS	R/W	0	LCD HSync Output
0	LVS	R/W	0	LCD VSync Output

GPIOLC Function Select 1 Register (GPIOLC_FS1)

0x9800514C

Bit	Name	Type	Default	Description (Function Selected)
31:6	-	R	0	Reserved
5:4	LXA[1:0]	R/W	0	LCD XA Output
3	LXWEn	R/W	0	LCD Write Enable Output
2	LXOEn	R/W	0	LCD Read Enable Output
1:0	LXCsn[1:0]	R/W	0	LCD Chip Select [1:0] Output

GPIOLC Function Select 4 Register (GPIOLC_FS4)

0x98005158

Bit	Name	Type	Default	Description (Function Selected)
31:6	-	R	0	Reserved.
1:0	GBUS[115:114]	R/W	0	GBUS[115:114]

GPIOLC Input Enable Register (GPIOLC_DIE)

0x9800516C

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	GPIOLC_DIE	R/W	0x3F	Input Enable 0 = Disable 1 = Enable

GPIOLC Pull-Up / Down Enable Register (GPIOLC_RPE)

0x98005170

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	GPIOLC_RPE	R/W	0x0F	Pull-Up / Down Resistor Enable 0 = Disable 1 = Enable

GPIOLC Pull-Up / Down Select Register (GPIOLC_RPS)

0x98005174

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	GPIOLC_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

GPIOLC Driver Strength 0 Register (GPIOLC_DV0)

0x98005178

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	GPIOLC_DV0	R/W	0	Output Driver Strength Control 0

GPIOLC Driver Strength 1 Register (GPIOLC_DV1)

0x9800517C

Bit	Name	Type	Default	Description
31:6	-	R	0	Reserved
5:0	GPIOLC_DV1	R/W	0	Output Driver Strength Control 1

GPIOPD_LC0	GPIOPD_LC1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOLD Data Register (GPIOLD_DAT)

0x98005180

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOLD_DAT	R/W	0	Data Port.

GPIOLD Direction Control Register (GPIOLD_DOE)

0x98005184

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOLD_DOE	R/W	0	Direction Control. 0 = Input 1 = Output

GPIOLD Function Select 0 Register (GPIOLD_FS0)**0x98005188**

Bit	Name	Type	Default	Description (Function Selected)
31:24	-	R	0	Reserved.
23:0	LPD[23:0]	R/W	0	LCD Panel Data [23:0] Output

GPIOLD Function Select 1 Register (GPIOLD_FS1)**0x9800518C**

Bit	Name	Type	Default	Description (Function Selected)
31:24	-	R	0	Reserved.
23:0	LXD[23:0]	R/W	0	LCD Data Bus

GPIOLD Function Select 4 Register (GPIOLD_FS4)**0x98005198**

Bit	Name	Type	Default	Description (Function Selected)
31:24	-	R	0	Reserved.
23:18	GBUG[113:108]	R/W	0	GBUG[113:108]
17:0	-	R/W	0	Reserved.

GPIOLD Input Enable Register (GPIOLD_DIE)**0x980051AC**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOLD_DIE	R/W	0xFFFFFFFF	Input Enable. 0 = Disable 1 = Enable

GPIOLD Pull-Up / Down Enable Register (GPIOLD_RPE)**0x980051B0**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOLD_RPE	R/W	0	Pull-Up / Down Resistor Enable. 0 = Disable, 1 = Enable

GPIOLD Pull-Up / Down Select Register (GPIOLD_RPS)**0x980051B4**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOLD_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

GPIOLD Driver Strength 0 Register (GPIOLD_DV0)**0x980051B8**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOLD_DV0	R/W	0	Output Driver Strength Control 0

GPIOLD Driver Strength 1 Register (GPIOLD_DV1)**0x980051BC**

Bit	Name	Type	Default	Description
31:24	-	R	0	Reserved
23:0	GPIOLD_DV1	R/W	0	Output Driver Strength Control 1

GPIOLD_DV0	GPIOLD_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOAD Data Register (GPIOAD_DAT)**0x980051C0**

Bit	Name	Type	Default	Description
31:8	-	R	0	Reserved
7:0	GPIOAD_DAT	R/W	0	Data Port.

GPIOAD Direction Control Register (GPIOAD_DOE)**0x980051C4**

Bit	Name	Type	Default	Description
31:8	-	R	0	Reserved
7:0	GPIOAD_DOE	R/W	0	Direction Control 0 = Input 1 = Output

GPIOAD Function Select 0 Register (GPIOAD_FS0)**0x980051C8**

Bit	Name	Type	Default	Description (Function Selected)
31:8	-	R	0	Reserved.
7:0	ADIN[7:0]	R/W	0xFF	ADC Inputs [7:0]

GPIOAD Input Enable Register (GPIOAD_DIE)**0x980051EC**

Bit	Name	Type	Default	Description
31:8	-	R	0	Reserved
7:0	GPIOAD_DIE	R/W	0	Input Enable. 0 = Disable 1 = Enable

GPIOAD Pull-Up / Down Enable Register (GPIOAD_RPE)**0x980051F0**

Bit	Name	Type	Default	Description
31:8	-	R	0	Reserved
7:0	GPIOAD_RPE	R/W	0	Pull-Up / Down Resistor Enable. 0 = Disable 1 = Enable

GPIOAD Pull-Up / Down Select Register (GPIOAD_RPS)**0x980051F4**

Bit	Name	Type	Default	Description
31:8	-	R	0	Reserved
7:0	GPIOAD_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

GPIOAD Driver Strength 0 Register (GPIOAD_DV0)**0x980051F8**

Bit	Name	Type	Default	Description
31:8	-	R	0	Reserved
7:0	GPIOAD_DV0	R/W	0	Output Driver Strength Control 0

GPIOAD Driver Strength 1 Register (GPIOAD_DV1)**0x980051FC**

Bit	Name	Type	Default	Description
31:8	-	R	0	Reserved
7:0	GPIOAD_DV1	R/W	0	Output Driver Strength Control 1

GPIOAD_DV0	GPIOAD_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOXC Data Register (GPIOXC_DAT)**0x98005200**

Bit	Name	Type	Default	Description
31:0	GPIOXC_DAT	R/W	0	Data Port.

GPIOXC Direction Control Register (GPIOXC_DOE)**0x98005204**

Bit	Name	Type	Default	Description
31:0	GPIOXC_DOE	R/W	0	Direction Control. 0 = Input 1 = Output

GPIOXC Function Select 0 Register (GPIOXC_FS0)**0x98005208**

Bit	Name	Type	Default	Description (Function Selected)
31	XDCKE	R/W	1	SDR SDRAM CKE
31:29	XDSCn[1:0]	R/W	0x3	SDR SDRAM Chip Select [1:0]
28	XRCSn	R/W	1	External Bus ROM Chip Select
27:26	XSCSn[1:0]	R/W	0x3	External Bus Chip Select [1:0]
25	XOEn	R/W	1	External Bus Output Enable
24	XWEn	R/W	1	External Bus Write Enable
23:0	XA[23:0]	R/W	0xFFFFFFFF	External Bus Address [23:0]

GPIOXC Function Select 4 Register (GPIOXC_FS4)**0x98005218**

Bit	Name	Type	Default	Description (Function Selected)
31:24	-	R/W	0	Reserved
23:4	GBUS[163:144]	R/W	0	GBUS[163:144]
3:0	-	R/W	0	Reserved

GPIOXC Input Enable Register (GPIOXC_DIE)**0x9800522C**

Bit	Name	Type	Default	Description
31:0	GPIOXC_DIE	R/W	0xFFFFFFFF	Input Enable. 0 = Disable 1 = Enable

GPIOXC Pull-Up / Down Enable Register (GPIOXC_RPE)**0x98005230**

Bit	Name	Type	Default	Description
31:0	GPIOXC_RPE	R/W	0	Pull-Up / Down Resistor Enable. 0 = Disable 1 = Enable

GPIOXC Pull-Up / Down Select Register (GPIOXC_RPS)**0x98005234**

Bit	Name	Type	Default	Description
31:0	GPIOXC_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

GPIOXC Driver Strength 0 Register (GPIOXC_DV0)**0x98005238**

Bit	Name	Type	Default	Description
31:0	GPIOXC_DV0	R/W	0xFFFFFFFF	Output Driver Strength Control 0

Refer to GPIOXC_FS0 Register for port controlled by each bit.

GPIOXC Driver Strength 1 Register (GPIOXC_DV1)**0x9800523C**

Bit	Name	Type	Default	Description
31:0	GPIOXC_DV1	R/W	0	Output Driver Strength Control 1

Refer to GPIOXC_FS0 Register for port controlled by each bit.

GPIOXC_DV0	GPIOXC_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIODD Data Register (GPIODD_DAT)

0x98005240

Bit	Name	Type	Default	Description
31:17	-	R/W	0	Reserved
16	XDCLK_DAT	R/W	0	Data Port for XDCLK
15:0	-	R/W	0	Reserved

GPIODD Function Select 0 Register (GPIODD_FS0)

0x98005248

Bit	Name	Type	Default	Description (Function Selected)
31:17	-	R/W	0x7FFF	Reserved
16	XDCLK	R/W	1	SDR SDRAM Clock
15:0	-	R/W	0xFFFF	Reserved

GPIODD Pull-Up / Down Select Register (GPIODD_RPE)

0x98005270

Bit	Name	Type	Default	Description
31:0	-	R/W	0	Reserved

GPIODD Pull-Up / Down Enable Register (GPIODD_RPS)

0x98005274

Bit	Name	Type	Default	Description
31:0	-	R/W	0	Reserved

GPIODD Driver Strength 0 Register (GPIODD_DV0)

0x98005278

Bit	Name	Type	Default	Description
31:0	GPIODD_DV0	R/W	0xFFFFFFFF	Output Driver Strength Control 0

Refer to GPIODD_FS0 Register for port controlled by each bit.

GPIODD Driver Strength 1 Register (GPIODD_DV1)

0x9800527C

Bit	Name	Type	Default	Description
31:0	GPIODD_DV1	R/W	0	Output Driver Strength Control 1

Refer to GPIODD_FS0 Register for port controlled by each bit.

GPIODD_DV0	GPIODD_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOPC Data Register (GPIOPC_DAT)

0x98005280

Bit	Name	Type	Default	Description
27:0	GPIOPC_DAT	R/W	0	Data Port.

GPIOPC Direction Control Register (GPIOPC_DOE)

0x98005284

Bit	Name	Type	Default	Description
27:0	GPIOPC_DOE	R/W	0	Direction Control. 0 = Input 1 = Output

GPIOPC Function Select 0 Register (GPIOPC_FS0)

0x98005288

Bit	Name	Type	Default	Description (Function Selected)
27:22	-	R/W	0	Reserved.
21:14	CIF0PD[15:8]	R/W	0	CAM Interface 0 Data [15:8]
13	CIF0CDE	R/W	0	CAM Interface 0 Data Enable Signal
12	CIF0CFIELD	R/W	0	CAM Interface 0 Field Signal
11	CIF0CKO	R/W	0	CAM Interface 0 Clock Output
10	CIF0HS	R/W	0	CAM Interface 0 Horizontal Sync
9	CIF0VS	R/W	0	CAM Interface 0 Vertical Sync
8	CIF0CKI	R/W	0	CAM Interface 0 Clock Input
7:0	CIF0PD[7:0]	R/W	0	CAM Interface 0 Data [7:0]

GPIOC Function Select 1 Register (GPIOC_FS1)**0x9800528C**

Bit	Name	Type	Default	Description (Function Selected)
27	CIF1CDE	R/W	0	CAM Interface 1 Data Enable Signal
26	CIF1CFIELD	R/W	0	CAM Interface 1 Field Signal
25	CIF1CKO	R/W	0	CAM Interface 1 Clock Output
24	CIF1HS	R/W	0	CAM Interface 1 Horizontal Sync
23	CIF1VS	R/W	0	CAM Interface 1 Vertical Sync
22	CIF1CKI	R/W	0	CAM Interface 1 Clock Input
21:14	CIF1PD[7:0]	R/W	0	CAM Interface 1 Data [7:0]
13:0	-	R/W	0	Reserved.

GPIOC Function Select 4 Register (GPIOC_FS4)**0x98005298**

Bit	Name	Type	Default	Description (Function Selected)
27:0	GBUS[107:80]	R/W	0	GBUS[107:80]

GPIOC Input Enable Register (GPIOC_DIE)**0x980052AC**

Bit	Name	Type	Default	Description
27:0	GPIOC_DIE	R/W	0x0FFFFFFF	Input Enable. 0 = Disable 1 = Enable

GPIOC Pull-Up / Down Enable Register (GPIOC_RPE)**0x980052B0**

Bit	Name	Type	Default	Description
27:0	GPIOC_RPE	R/W	0	Pull-Up / Down Resistor Enable. 0 = Disable 1 = Enable

GPIOC Pull-Up / Down Select Register (GPIOC_RPS)**0x980052B4**

Bit	Name	Type	Default	Description
27:0	GPIOC_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

GPIOC Driver Strength 0 Register (GPIOC_DV0)**0x980052B8**

Bit	Name	Type	Default	Description
27:0	GPIOC_DV0	R/W	0	Output Driver Strength Control 0

GPIOC Driver Strength 1 Register (GPIOC_DV1)**0x980052BC**

Bit	Name	Type	Default	Description
27:0	GPIOC_DV1	R/W	0	Output Driver Strength Control 1

GPIOC_DV0	GPIOC_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOPK Reset Control Register (GPIOPK_RST)**0x9801C008**

Bit	Name	Type	Default	Description
31:2	-	R	0	Reserved
1	RST_PIC	R/W	0	Mask soft-reset for PK Interrupt Control Logic.
0	RST_GPC	R/W	0	Mask soft-reset for PK GPIO Control Logic.

When enabled (written as "1"), the corresponding block can be reset by PRST# only.

GPIOPK Data Register (GPIOPK_DAT)**0x9801C100**

Bit	Name	Type	Default	Description
31:0	GPIOPK_DAT	R/W	0	Data Port.

GPIOPK Direction Control Register (GPIOPK_DOE)**0x9801C104**

Bit	Name	Type	Default	Description
31:0	GPIOPK_DOE	R/W	0	Direction Control. 0 = Input 1 = Output

GPIOPK Function Select 0 Register (GPIOPK_FS0)

0x9801C108

Bit	Name	Type	Default	Description (Function Selected)
31:28	-	R/W	0	Reserved
27	C2RXD	R/W	0	CAN Controller 2 RXD
26	C2TXD	R/W	0	CAN Controller 2 TXD
25	C1RXD	R/W	0	CAN Controller 1 RXD
24	C1TXD	R/W	0	CAN Controller 1 TXD
23	C0RXD	R/W	0	CAN Controller 0 RXD
22	C0TXD	R/W	0	CAN Controller 0 TXD
21:8	-	R/W	0	Reserved
7:0	EXTINT[7:0]	R/W	0	External Interrupt Inputs [7:0] to Interrupt Controller

GPIOPK Function Select 1 Register (GPIOPK_FS1)

0x9801C10C

Bit	Name	Type	Default	Description (Function Selected)
31:22	-	R/W	0	Reserved
21	HRESETn	R/W	0	HRESETn Output. The bit 0 of GPIOPK_RST register must be written as "1", to protect these bits from reset.
20	HRESETn	R/W	0	
19	HRESETn	R/W	0	
18:0	-	R/W	0	Reserved.

GPIOPK Function Select 3 Register (GPIOPK_FS3)

0x9801C114

Bit	Name	Type	Default	Description (Function Selected)
31:28	-	R/W	0	Reserved
27:26	TCO[4:3]	R/W	0	TCO Output [4:3]
21:16	TCO[5:0]	R/W	0	TCO Output [5:0]
15:0	-	R/W	0	Reserved.

GPIOPK Function Select 4 Register (GPIOPK_FS4)

0x9801C118

Bit	Name	Type	Default	Description (Function Selected)
31:4	GBUS[191:164]	R/W	0	GBUS[191:164]
3:0	-	R/W	0	Reserved

GPIOPK Input Enable Register (GPIOPK_DIE)

0x9801C12C

Bit	Name	Type	Default	Description
31:0	GPIOPK_DIE	R/W	0xFFFFFFFF	Input Enable 0 = Disable 1 = Enable

Caution: Do not use this register. If you want to use, use PMGPIO Input Enable Register.

GPIOPK Pull-Up / Down Enable Register (GPIOPK_RPE)

0x9801C130

Bit	Name	Type	Default	Description
31:0	GPIOPK_RPE	R/W	0	Pull-Up / Down Resistor Enable 0 = Disable 1 = Enable

Caution: Do not use this register. If you want to use, use PMGPIO Pull-Up Enable Register.

GPIOPK Pull-Up / Down Select Register (GPIOPK_RPS)

0x9801C134

Bit	Name	Type	Default	Description
31:0	GPIOPK_RPS	R/W	0	Pull-Up / Down Resistor Select 0 = Pull-down 1 = Pull-up

Caution: Do not use this register. If you want to use, use PMGPIO Pull-Down Enable Register.

GPIOPK Driver Strength 0 Register (GPIOPK_DV0)

0x9801C138

Bit	Name	Type	Default	Description
31:0	GPIOPK_DV0	R/W	0	Output Driver Strength Control 0

Caution: Do not use this register. If you want to use, use PMGPIO Driver Strength 0 Register.

GPIOPK Driver Strength 1 Register (GPIOPK_DV1)**0x9801C13C**

Bit	Name	Type	Default	Description
31:0	GPIOPK_DV1	R/W	0	Output Driver Strength Control 1

Caution: Do not use this register. If you want to use, use PMGPIO Driver Strength 1 Register.

GPIOPK_DV0	GPIOPK_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOPK Data Register (GPIOPK_DAT_H)**0x9801C140**

Bit	Name	Type	Default	Description
31:3	-	R	0	Reserved
2:0	GPIOPK_DAT_H	R/W	0	Data Port [34:32]

GPIOPK Direction Control Register (GPIOPK_DOE_H)**0x9801C144**

Bit	Name	Type	Default	Description
31:3	-	R	0	Reserved
2:0	GPIOPK_DOE_H	R/W	0x7	Direction Control [34:32] 0 = Input 1 = Output

GPIOPK Function Select 0 Register (GPIOPK_FS0_H)**0x9801C148**

Bit	Name	Type	Default	Description (Function Selected)
31:3	-	R/W	0	Reserved
2	HRESETn	R/W	0	HRESETn Output to PK[34:32] The bit 0 of GPIOPK_RST register must be written as "1" to protect these bits from reset.
1	HRESETn	R/W	0	
0	HRESETn	R/W	0	

GPIOPK Input Enable Register (GPIOPK_DIE_H)**0x9801C16C**

Bit	Name	Type	Default	Description
31:3	-	R/W	0	Reserved
2:0	GPIOPK_DIE_H	R/W	0x7	Input Enable. 0 = Disable 1 = Enable

Caution: Do not use this register. If you want to use, use PMGPIO Input Enable Register

GPIOPK Pull-Up / Down Enable Register (GPIOPK_RPE_H)**0x9801C170**

Bit	Name	Type	Default	Description
31:0	GPIOPK_RPE_H	R/W	0	Pull-Up / Down Resistor Enable 0 = Disable 1 = Enable

Caution: Do not use this register.. If you want to use, use PMGPIO Pull-Up Enable Register

GPIOPK Pull-Up / Down Select Register (GPIOPK_RPS_H)**0x9801C174**

Bit	Name	Type	Default	Description
31:0	GPIOPK_RPS_H	R/W	0	Pull-Up / Down Resistor Select. (0 = Pull-down, 1 = Pull-up)

Caution: Do not use this register. If you want to use, use PMGPIO Pull-Down Enable Register.

GPIOPK Driver Strength 0 Register (GPIOPK_DV0_H)**0x9801C178**

Bit	Name	Type	Default	Description
31:0	GPIOPK_DV0_H	R/W	0	Output Driver Strength Control 0

Caution: Do not use this register. If you want to use, use PMGPIO Driver Strength 0 Register.

GPIOPK Driver Strength 1 Register (GPIOPK_DV1_H)

0x9801C17C

Bit	Name	Type	Default	Description
31:0	GPIOPK_DV1_H	R/W	0	Output Driver Strength Control 1

Caution: Do not use this register. If you want to use, use PMGPIO Driver Strength 1 Register.

GPIOPK_DV0	GPIOPK_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

GPIOPK Filtered Data Port (GPIOPK_FDAT)

0x9801C200

Bit	Name	Type	Default	Description
31:16	-	R/W	-	Reserved
15:0	GPIOPK_FDAT	R/W	-	Filtered PK[15:0] input data. The corresponding bit of GPIOPK_FBP register should be "0" (filter enabled).

GPIOPK Interrupt Status Register (GPIOPK_IRQST)

0x9801C210

Bit	Name	Type	Default	Description
31:0	PK_IRQST	R/W	-	Interrupt Status for PK[31:0] (GPIOPK_uIRQST & GPIOPK_IREQN)

GPIOPK Interrupt Status Register (GPIOPK_uIRQST)

0x9801C210 + 0x400

Bit	Name	Type	Default	Description
31:0	PK_uIRQST	R/W	-	Interrupt Status for PK[31:0] (unmasked)

GPIOPK Interrupt Enable Register (GPIOPK_IRQEN)

0x9801C214

Bit	Name	Type	Default	Description
31:0	PK_IRQEN	R/W	0	Interrupt Enable for PK[31:0] 0 = Disable 1 = Enable

GPIOPK Interrupt Polarity Register (GPIOPK_IRQPOL)

0x9801C218

Bit	Name	Type	Default	Description
31:0	PK_IRQPOL	R/W	0	Interrupt Polarity for PK[31:0] 0 = Non-inverted 1 = Inverted

GPIOPK Interrupt Trigger Mode 0 Register (GPIOPK_IRQTM0)

0x9801C21C

Bit	Name	Type	Default	Description
31:0	PK_IRQTM0	R/W	0	0 = Edge Trigger, 1 = Level Trigger

GPIOPK Interrupt Trigger Mode 1 Register (GPIOPK_IRQTM1)

0x9801C220

Bit	Name	Type	Default	Description
31:0	PK_IRQTM1	R/W	0	This bit is effective only when the corresponding bit of GPIOPK_IRQTM0 is "0". 0 = Single Edge 1 = Both Edge.

GPIOPK Filter Clock Configuration Register (GPIOPK_FCK)

0x9801C224

Bit	Name	Type	Default	Description
31:16	LDV	R/W	0	Filter Count Load Value. PK[15:0] input signals are sent to GPIOPK_IRQST[15:0] after the filter counter is expired.
15:0	DIV	R/W	0	Filter Clock Frequency = fFCK / (DIV+1)

GPIOPK Filter Bypass Register (GPIOPK_FBP)

0x9801C228

Bit	Name	Type	Default	Description
31:16	MEN	R/W	0	PK[31:16] Interrupt Source Enable 0: Input Disabled 1: Input Enabled
15:0	FEN	R/W	0	PK[15:0] Interrupt Source Filter Enable 0: Filter Enabled 1: Filter Disabled

There is no noise filter for PK[31:16]. To generate interrupt with PK[31:16] inputs, bit[31:16] must be set to "1".

GPIOPK Miscellaneous Control Register (GPIOPK_CTL)

0x9801C22C

Bit	Name	Type	Default	Description
31	RCLR	R/W	0	0: Interrupt status is retained until CPU writes. 1: Interrupt status is cleared by CPU read.
30	UIRQ	R/W	0	This bit is used to bypass external interrupt events to the PIC when the system clock is stopped. 0: Registered version of IRQ is sent to the main PIC. 1: Un-registered version of IRQ is sent to the main PIC.
29	FCKSEL	R/W	0	This bit is used to select clock source for the input filter logic. 0: XTI input 1: PCLK
28:0	-	R/W	0	Reserved

GPIOPK Group Interrupt Controller

There is an interrupt controller dedicated to PK[31:0] inputs. The final interrupt output signal of this interrupt controller is connected to channel 23 of Main Interrupt Controller A. Refer to Table 11.2.

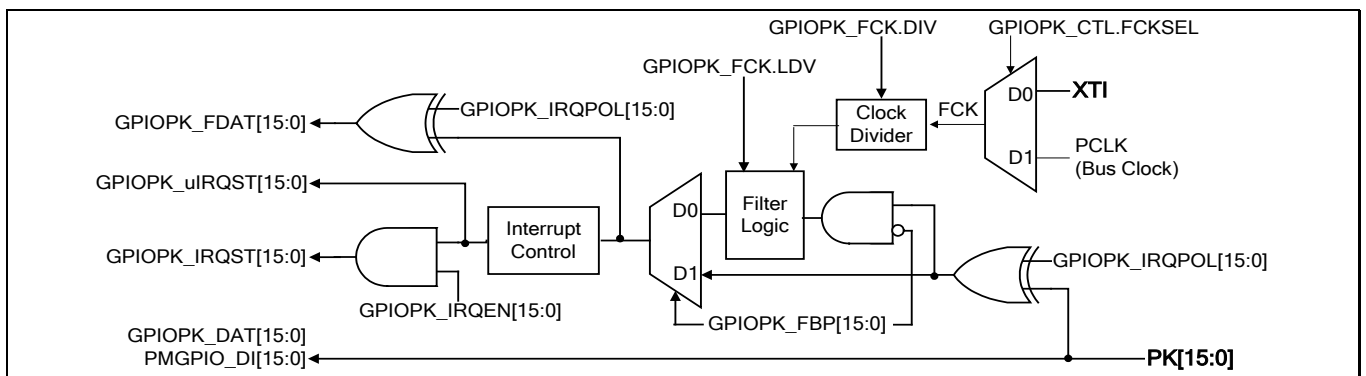


Figure 18.6 Interrupt Control for PK[15:0]

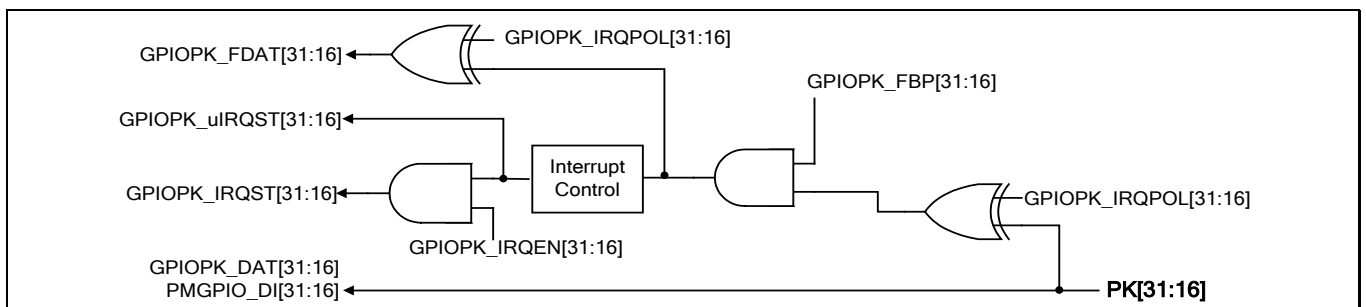


Figure 18.7 Interrupt Control for PK[31:16]

Note:

1. The noise filter is available for PK[15:0] inputs only.
2. Bitwise noise filtering is not supported. The filter logic monitors and compares all PK[15:0] inputs together at the same time. Any bit transition will restart the counter in the filter logic.
3. The default value of GPIOPK_FBP[31:16] is "0x0000" (Input Disabled). To generate interrupt with PK[31:16] inputs, GPIOPK_FBP[31:16] must be set to "1".

PMGPIO Data Register (PMGPIO_DAT) 0x98010800

Bit	Name	Type	Default	Description
31:0	PMGPIO_DAT	R/W	0	Data Port.

PMGPIO Direction Control Register (PMGPIO_DOE) 0x98010804

Bit	Name	Type	Default	Description
31:0	PMGPIO_DOE	R/W	0	Direction Control. 0 = Input 1 = Output

PMGPIO Function Select 0 Register (PMGPIO_FS0) 0x98010808

Bit	Name	Type	Default	Description
31:0	PMGPIO_FS0	R/W	0xFFFFFFFF	0: PK pin controlled by PMGPIO_DAT/DOE 1: PK pin controlled by GPIOPK_DAT/DOE/FS0~4

PMGPIO Input Enable Register (PMGPIO_DIE) 0x9801080C

Bit	Name	Type	Default	Description
31:0	GPIOPK_DIE	R/W	0xFFFFFFFF	Input Enable 0 = Disable 1 = Enable

PMGPIO Pull-Up / Down Enable Register (PMGPIO_RPE) 0x98010810

Bit	Name	Type	Default	Description
31:0	GPIOPK_RPE	R/W	0	Pull-Up / Down Resistor Enable. 0 = Disable 1 = Enable

PMGPIO Pull-Up / Down Select Register (GPIOPK_RPS) 0x98010814

Bit	Name	Type	Default	Description
31:0	GPIOPK_RPS	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

PMGPIO Driver Strength 0 Register (GPIOPK_DV0) 0x98010818

Bit	Name	Type	Default	Description
31:0	GPIOPK_DV0	R/W	0	Output Driver Strength Control 0

PMGPIO Driver Strength 1 Register (GPIOPK_DV1) 0x9801081C

Bit	Name	Type	Default	Description
31:0	GPIOPK_DV1	R/W	0	Output Driver Strength Control 1

GPIOPK_DV0	GPIOPK_DV1	Current Driving Strength of Output Driver
0	0	X1 Driving Strength
0	1	X2 Driving Strength
1	0	X4 Driving Strength
1	1	X8 Driving Strength

PMGPIO Event Enable 0 Register (PMGPIO_EE0) 0x98010820

Bit	Name	Type	Default	Description
31:0	PMGPIO_EE0	R/W	0	Enabled pin can change PK[33] output from low to high. 0 = Disable 1 = Enable

Note: Once PK[33] goes to high, PK[33] maintains the high level until the corresponding bit of PMGPIO_STR or PMGPIO_STF register is cleared.

PMGPIO Event Enable 1 Register (PMGPIO_EE1) 0x98010824

Bit	Name	Type	Default	Description
31:0	PMGPIO_EE1	R/W	0	Enabled pin can change PK[32] output from low to high. 0 = Disable 1 = Enable

Note: Once PK[32] goes to high, PK[32] maintains the high level until the corresponding bit of PMGPIO_STR or PMGPIO_STF register is cleared.

PMGPIO Control Register (PMGPIO_CTL)

0x98010828

Bit	Name	Type	Default	Description
31:17	-	R/W	0	Reserved
16	DCKERTO	R/W	1	Retention Control for DCKE, DRESETn and DOT1. 0 = Retention Enable 1 = Retention Disable
15	SKIPCRC	R/W	0	If this bit is set as "1", the CRC check is skipped during boot code execution.
14:13	-	R/W	0	Reserved
12	CKCWDR	R/W	1	0 = CKC is NOT reset by WDT 1 = CKC is reset by WDT
11:8	XIWAIT	R/W	0xF	XI XTAL oscillator wait time control (for CKC reset or POR). 0x0 ~ 0xB: XI Clock Period x 2^(6+XIWAIT) 0xC ~ 0xF: XI Clock Period x 2^18
7:6	PMWKUPEN	R/W	0	PMWKUP Output Enable. If written as "1", PMWKUP signal from RTC can be output to PK[33:32] respectively. Before any of these bits is enabled, the polarity of PMWKUP signal must be programmed as "active high" in the RTC module.
5:4	OFFCTL[1:0]	R/W	0	Once PK[33:32] output change from high to low state, keep low until PRST# active.
3	-	R/W	0	Reserved
2	RAMWP	R/W	0	Write Protect Bit for backup RAM address region. 0 = Write Enabled 1 = Write Protected
1	LVSCTL	R/W	0	0 = Enable LVS 1 = Disable LVS
0	LATCTL	R/W	0	0 = Enable Latch 1 = Disable Latch <i>This bit must be set to "1" before core power (VDDI) is off.</i>

PMGPIO Input Raw Status Register (PMGPIO_DI)

0x9801082C

Bit	Name	Type	Default	Description
31:0	PMGPIO_DI	R	0	External PK[31:0] states can be read from this port

PMGPIO Rising Event Status Register (PMGPIO_STR)

0x98010830

Bit	Name	Type	Default	Description
31:0	PMGPIO_STR	R/C	0	Set by PK pin rising edge. Cleared by writing "1"

PMGPIO Falling Event Status Register (PMGPIO_STF)

0x98010834

Bit	Name	Type	Default	Description
31:0	PMGPIO_STF	R/C	0	Set by PK falling edge. Cleared by writing "1"

PMGPIO Event Polarity Register (PMGPIO_POL)

0x98010838

Bit	Name	Type	Default	Description
31:0	PMGPIO_POL	R/W	0	Selects which event is used to force PK[33:32] high level. 0 = PMGPIO_STR 1 = PMGPIO_STF

PMGPIO Protect Register (PMGPIO_PROT)

0x9801083C

Bit	Name	Type	Default	Description
31:8	-	R	0	Reserved
7:1	-	R/W	0	Reserved
0	PMGPIO_WP	R/W	0	Write Protect Bit for PMGPIO register address region. 0 = Write Enabled 1 = Write Protected

PMGPIO Data Register (PMGPIO_DAT_H)

0x98010840

Bit	Name	Type	Default	Description
31:3	-	R	0	Reserved
2:0	PMGPIO_DAT_H	R/W	0	Data Port [34:32]

PMGPIO Direction Control Register (PMGPIO_DOE_H)

0x98010844

Bit	Name	Type	Default	Description
31:3	-	R	0	Reserved
2:0	PMGPIO_DOE_H	R/W	0x7	Direction Control. 0 = Input 1 = Output

PMGPIO Function Select 0 Register (PMGPIO_FS0_H)

0x98010848

Bit	Name	Type	Default	Description
31:3	-	R	0	Reserved.
2:0	PMGPIO_FS0_H	R/W	0x4	0: PK[34:32] pin controlled by PMGPIO_DAT_H/DOE_H 1: PK[34:32] pin controlled by GPIOPK_DAT_H/DOE_H/FS0_H

PMGPIO Input Enable Register (PMGPIO_DIE_H)

0x9801084C

Bit	Name	Type	Default	Description
2:0	GPIOPK_DIE_H	R/W	0x7	Input Enable. 0 = Disable 1 = Enable

PMGPIO Pull-Up / Down Enable Register (PMGPIO_RPE_H)

0x98010850

Bit	Name	Type	Default	Description
2:0	GPIOPK_RPE_H	R/W	0	Pull-Up / Down Resistor Enable. 0 = Disable 1 = Enable

PMGPIO Pull-Up / Down Select Register (GPIOPK_RPS_H)

0x98010854

Bit	Name	Type	Default	Description
2:0	GPIOPK_RPS_H	R/W	0	Pull-Up / Down Resistor Select. 0 = Pull-down 1 = Pull-up

PMGPIO Driver Strength 0 Register (GPIOPK_DV0_H)

0x98010858

Bit	Name	Type	Default	Description
2:0	GPIOPK_DV0_H	R/W	0	Output Driver Strength Control 0

PMGPIO Driver Strength 1 Register (GPIOPK_DV1_H)

0x9801085C

Bit	Name	Type	Default	Description
2:0	GPIOPK_DV1_H	R/W	0	Output Driver Strength Control 1

GPIOPK_DV0	GPIOPK_DV1	Current Driver Strength of Output Driver
0	0	X1 Driver Strength
0	1	X2 Driver Strength
1	0	X4 Driver Strength
1	1	X8 Driver Strength

PMGPIO IS Enable Register (PMGPIO_IS)

0x98010860

Bit	Name	Type	Default	Description
31:0	GPIOPK_IS	R/W	0xFFFFFFFF	Schmitt trigger Resistor Enable. 0 : CMOS 1 : Schmitt trigger

PMGPIO SR Select Register (GPIOPK_SR)

0x98010864

Bit	Name	Type	Default	Description
31:0	GPIOPK_SR	R/W	0	Control slew-rate 0 : Fast Slew 1 : Slow Slew

PMGPIO IS Enable_H Register (GPIOPK_IS_H)

0x98010868

Bit	Name	Type	Default	Description
2:0	GPIOPK_IS_H	R/W	0x7	Schmitt trigger Resistor Enable. 0 : CMOS 1 : Schmitt trigger

PMGPIO SR Select_H Register (GPIOPK_SR_H)

0x9801086C

Bit	Name	Type	Default	Description
2:0	GPIOPK_SR_H	R/W	0	Control slew-rate 0 : Fast Slew 1 : Slow Slew

PMGPIO & PMRAM APB Timing Register (PMGPIO_APB)

0x98010900

Bit	Name	Type	Default	Description
10:8	HOLD	R/W	0	Hold Time (HCLK period) * (HOLD+1) ≥ 4ns
6:4	CMDW	R/W	0	Command Pulse Width (HCLK period) * (CMDW+1) ≥ 4ns
2:0	SETUP	R/W	1	Setup Time. (HCLK period) * (SETUP+1) > 15ns

This register controls the APB interface timing to the PMGPIO controller and the Backup RAM. The default value of this register is for HCLK (Bus Clock) frequency up to 133MHz.

Note: 0x113 is recommended for HCLK frequency higher than 133MHz.

18.4 GBUS Port Multiplexing

In this section, all the port selection registers for the GBUS are listed.

Table 18.13 GBUS Port Selection Registers

Name	Address	Type	Reset	Description
GSB0PORTSEL	0x9000F800	R/W	0x03020100	GSB0 Port Selection Register
GSB1PORTSEL	0x9000F804	R/W	0x07060504	GSB1 Port Selection Register
GSB2PORTSEL	0x9000F808	R/W	0x0B0A0908	GSB2 Port Selection Register
GSB3PORTSEL	0x9000F80C	R/W	0x0F0E0D0C	GSB3 Port Selection Register
GSB4PORTSEL	0x9000F810	R/W	0x13121110	GSB4 Port Selection Register
GSB5PORTSEL	0x9000F814	R/W	0x17161514	GSB5 Port Selection Register
UART0PORTSEL	0x90007800	R/W	0x23222120	UART0 Port Selection Register
UART1PORTSEL	0x90007804	R/W	0x27262524	UART1 Port Selection Register
UART2PORTSEL	0x90007808	R/W	0x2B2A2928	UART2 Port Selection Register
UART3PORTSEL	0x9000780C	R/W	0x2F2E2D2C	UART3 Port Selection Register
UART4PORTSEL	0x90007810	R/W	0x33323130	UART4 Port Selection Register
DAI0PORTSEL	0x98002800	R/W	0x3B3A3938	DAI0 Port Selection Register
DAI1PORTSEL	0x98002804	R/W	0x3F3E3D3C	DAI1 Port Selection Register
DAI2PORTSEL	0x98002808	R/W	0x43424140	DAI2 Port Selection Register
DAIMPORTSEL	0x98002810	R/W	0x4B4A4948	DAI MCLK Port Selection Register
DAICPORTSEL	0x98002814	R/W	0x4F4E4D4C	DAI CDIF/SPDIF Port Selection Register
I2CM0/1PORTSEL	0x9000B200	R/W	0xFFFFFFFF	I2C Master 0/1 Port Selection Register
I2CM2/3PORTSEL	0x9000B204	R/W	0xFFFFFFFF	I2C Master 2/3 Port Selection Register
I2CS0/1PORTSEL	0x9000B208	R/W	0xFFFFFFFF	I2C Slave 0/1 Port Selection Register
TCOPORTSEL0	0x98004800	R/W	0xFFFFFFFF	TCO Port Selection Register 0
TCOPORTSEL1	0x98004804	R/W	0xFFFFFFFF	TCO Port Selection Register 1
PDMPORTSEL	0x98004808	R/W	0xFFFFFFFF	PDM Port Selection Register

Note: There is no priority control for port selection. Care must be taken to prevent contention.

Table 18.14 GBUS Map – Port Name Order

Port	GBUS	Port	GBUS	Port	GBUS	Port	GBUS	Port	GBUS
FC[00]	GBUS[116]	PC[20]	GBUS[100]	PK[12]	GBUS[172]	PS[20]	GBUS[020]	XA[08]	GBUS[148]
FC[01]	GBUS[117]	PC[21]	GBUS[101]	PK[13]	GBUS[173]	PS[21]	GBUS[021]	XA[09]	GBUS[149]
FC[02]	GBUS[118]	PC[22]	GBUS[102]	PK[14]	GBUS[174]	PS[22]	GBUS[022]	XA[10]	GBUS[150]
FC[03]	GBUS[119]	PC[23]	GBUS[103]	PK[15]	GBUS[175]	PS[23]	GBUS[023]	XA[11]	GBUS[151]
FC[04]	GBUS[120]	PC[24]	GBUS[104]	PK[16]	GBUS[176]	PS[24]	GBUS[024]	XA[12]	GBUS[152]
FC[05]	GBUS[121]	PC[25]	GBUS[105]	PK[17]	GBUS[177]	PS[25]	GBUS[025]	XA[13]	GBUS[153]
FC[06]	GBUS[122]	PC[26]	GBUS[106]	PK[18]	GBUS[178]	PS[26]	GBUS[026]	XA[14]	GBUS[154]
FC[07]	GBUS[123]	PC[27]	GBUS[107]	PK[19]	GBUS[179]	PS[27]	GBUS[027]	XA[15]	GBUS[155]
FC[08]	GBUS[124]	PD[00]	GBUS[056]	PK[20]	GBUS[180]	PS[28]	GBUS[028]	XA[16]	GBUS[156]
FC[09]	GBUS[125]	PD[01]	GBUS[057]	PK[21]	GBUS[181]	PS[29]	GBUS[029]	XA[17]	GBUS[157]
FC[10]	GBUS[126]	PD[02]	GBUS[058]	PK[22]	GBUS[182]	PS[30]	GBUS[030]	XA[18]	GBUS[158]
FC[11]	GBUS[127]	PD[03]	GBUS[059]	PK[23]	GBUS[183]	PS[31]	GBUS[031]	XA[19]	GBUS[159]
LC[04]	GBUS[114]	PD[04]	GBUS[060]	PK[24]	GBUS[184]	PU[00]	GBUS[032]	XA[20]	GBUS[160]
LC[05]	GBUS[115]	PD[05]	GBUS[061]	PK[25]	GBUS[185]	PU[01]	GBUS[033]	XA[21]	GBUS[161]
LD[18]	GBUS[108]	PD[06]	GBUS[062]	PK[26]	GBUS[186]	PU[02]	GBUS[034]	XA[22]	GBUS[162]
LD[19]	GBUS[109]	PD[07]	GBUS[063]	PK[27]	GBUS[187]	PU[03]	GBUS[035]	XA[23]	GBUS[163]
LD[20]	GBUS[110]	PD[08]	GBUS[064]	PK[28]	GBUS[188]	PU[04]	GBUS[036]	XD[16]	GBUS[128]
LD[21]	GBUS[111]	PD[09]	GBUS[065]	PK[29]	GBUS[189]	PU[05]	GBUS[037]	XD[17]	GBUS[129]
LD[22]	GBUS[112]	PD[10]	GBUS[066]	PK[30]	GBUS[190]	PU[06]	GBUS[038]	XD[18]	GBUS[130]
LD[23]	GBUS[113]	PD[11]	GBUS[067]	PK[31]	GBUS[191]	PU[07]	GBUS[039]	XD[19]	GBUS[131]
PC[00]	GBUS[080]	PD[12]	GBUS[068]	PS[00]	GBUS[000]	PU[08]	GBUS[040]	XD[20]	GBUS[132]
PC[01]	GBUS[081]	PD[13]	GBUS[069]	PS[01]	GBUS[001]	PU[09]	GBUS[041]	XD[21]	GBUS[133]
PC[02]	GBUS[082]	PD[14]	GBUS[070]	PS[02]	GBUS[002]	PU[10]	GBUS[042]	XD[22]	GBUS[134]
PC[03]	GBUS[083]	PD[15]	GBUS[071]	PS[03]	GBUS[003]	PU[11]	GBUS[043]	XD[23]	GBUS[135]
PC[04]	GBUS[084]	PD[16]	GBUS[072]	PS[04]	GBUS[004]	PU[12]	GBUS[044]	XD[24]	GBUS[136]
PC[05]	GBUS[085]	PD[17]	GBUS[073]	PS[05]	GBUS[005]	PU[13]	GBUS[045]	XD[25]	GBUS[137]
PC[06]	GBUS[086]	PD[18]	GBUS[074]	PS[06]	GBUS[006]	PU[14]	GBUS[046]	XD[26]	GBUS[138]
PC[07]	GBUS[087]	PD[19]	GBUS[075]	PS[07]	GBUS[007]	PU[15]	GBUS[047]	XD[27]	GBUS[139]
PC[08]	GBUS[088]	PD[20]	GBUS[076]	PS[08]	GBUS[008]	PU[16]	GBUS[048]	XD[28]	GBUS[140]
PC[09]	GBUS[089]	PD[21]	GBUS[077]	PS[09]	GBUS[009]	PU[17]	GBUS[049]	XD[29]	GBUS[141]
PC[10]	GBUS[090]	PD[22]	GBUS[078]	PS[10]	GBUS[010]	PU[18]	GBUS[050]	XD[30]	GBUS[142]
PC[11]	GBUS[091]	PD[23]	GBUS[079]	PS[11]	GBUS[011]	PU[19]	GBUS[051]	XD[31]	GBUS[143]
PC[12]	GBUS[092]	PK[04]	GBUS[164]	PS[12]	GBUS[012]	PU[20]	GBUS[052]		
PC[13]	GBUS[093]	PK[05]	GBUS[165]	PS[13]	GBUS[013]	PU[21]	GBUS[053]		
PC[14]	GBUS[094]	PK[06]	GBUS[166]	PS[14]	GBUS[014]	PU[22]	GBUS[054]		
PC[15]	GBUS[095]	PK[07]	GBUS[167]	PS[15]	GBUS[015]	PU[23]	GBUS[055]		
PC[16]	GBUS[096]	PK[08]	GBUS[168]	PS[16]	GBUS[016]	XA[04]	GBUS[144]		
PC[17]	GBUS[097]	PK[09]	GBUS[169]	PS[17]	GBUS[017]	XA[05]	GBUS[145]		
PC[18]	GBUS[098]	PK[10]	GBUS[170]	PS[18]	GBUS[018]	XA[06]	GBUS[146]		
PC[19]	GBUS[099]	PK[11]	GBUS[171]	PS[19]	GBUS[019]	XA[07]	GBUS[147]		

Table 18.15 GBUS Map – GBUS Index Order

GBUS	Port	GBUS	Port	GBUS	Port	GBUS	Port	GBUS	Port
GBUS[000]	PS[00]	GBUS[040]	PU[08]	GBUS[080]	PC[00]	GBUS[120]	FC[04]	GBUS[160]	XA[20]
GBUS[001]	PS[01]	GBUS[041]	PU[09]	GBUS[081]	PC[01]	GBUS[121]	FC[05]	GBUS[161]	XA[21]
GBUS[002]	PS[02]	GBUS[042]	PU[10]	GBUS[082]	PC[02]	GBUS[122]	FC[06]	GBUS[162]	XA[22]
GBUS[003]	PS[03]	GBUS[043]	PU[11]	GBUS[083]	PC[03]	GBUS[123]	FC[07]	GBUS[163]	XA[23]
GBUS[004]	PS[04]	GBUS[044]	PU[12]	GBUS[084]	PC[04]	GBUS[124]	FC[08]	GBUS[164]	PK[04]
GBUS[005]	PS[05]	GBUS[045]	PU[13]	GBUS[085]	PC[05]	GBUS[125]	FC[09]	GBUS[165]	PK[05]
GBUS[006]	PS[06]	GBUS[046]	PU[14]	GBUS[086]	PC[06]	GBUS[126]	FC[10]	GBUS[166]	PK[06]
GBUS[007]	PS[07]	GBUS[047]	PU[15]	GBUS[087]	PC[07]	GBUS[127]	FC[11]	GBUS[167]	PK[07]
GBUS[008]	PS[08]	GBUS[048]	PU[16]	GBUS[088]	PC[08]	GBUS[128]	XD[16]	GBUS[168]	PK[08]
GBUS[009]	PS[09]	GBUS[049]	PU[17]	GBUS[089]	PC[09]	GBUS[129]	XD[17]	GBUS[169]	PK[09]
GBUS[010]	PS[10]	GBUS[050]	PU[18]	GBUS[090]	PC[10]	GBUS[130]	XD[18]	GBUS[170]	PK[10]
GBUS[011]	PS[11]	GBUS[051]	PU[19]	GBUS[091]	PC[11]	GBUS[131]	XD[19]	GBUS[171]	PK[11]
GBUS[012]	PS[12]	GBUS[052]	PU[20]	GBUS[092]	PC[12]	GBUS[132]	XD[20]	GBUS[172]	PK[12]
GBUS[013]	PS[13]	GBUS[053]	PU[21]	GBUS[093]	PC[13]	GBUS[133]	XD[21]	GBUS[173]	PK[13]
GBUS[014]	PS[14]	GBUS[054]	PU[22]	GBUS[094]	PC[14]	GBUS[134]	XD[22]	GBUS[174]	PK[14]
GBUS[015]	PS[15]	GBUS[055]	PU[23]	GBUS[095]	PC[15]	GBUS[135]	XD[23]	GBUS[175]	PK[15]
GBUS[016]	PS[16]	GBUS[056]	PD[00]	GBUS[096]	PC[16]	GBUS[136]	XD[24]	GBUS[176]	PK[16]
GBUS[017]	PS[17]	GBUS[057]	PD[01]	GBUS[097]	PC[17]	GBUS[137]	XD[25]	GBUS[177]	PK[17]
GBUS[018]	PS[18]	GBUS[058]	PD[02]	GBUS[098]	PC[18]	GBUS[138]	XD[26]	GBUS[178]	PK[18]
GBUS[019]	PS[19]	GBUS[059]	PD[03]	GBUS[099]	PC[19]	GBUS[139]	XD[27]	GBUS[179]	PK[19]
GBUS[020]	PS[20]	GBUS[060]	PD[04]	GBUS[100]	PC[20]	GBUS[140]	XD[28]	GBUS[180]	PK[20]
GBUS[021]	PS[21]	GBUS[061]	PD[05]	GBUS[101]	PC[21]	GBUS[141]	XD[29]	GBUS[181]	PK[21]
GBUS[022]	PS[22]	GBUS[062]	PD[06]	GBUS[102]	PC[22]	GBUS[142]	XD[30]	GBUS[182]	PK[22]
GBUS[023]	PS[23]	GBUS[063]	PD[07]	GBUS[103]	PC[23]	GBUS[143]	XD[31]	GBUS[183]	PK[23]
GBUS[024]	PS[24]	GBUS[064]	PD[08]	GBUS[104]	PC[24]	GBUS[144]	XA[04]	GBUS[184]	PK[24]
GBUS[025]	PS[25]	GBUS[065]	PD[09]	GBUS[105]	PC[25]	GBUS[145]	XA[05]	GBUS[185]	PK[25]
GBUS[026]	PS[26]	GBUS[066]	PD[10]	GBUS[106]	PC[26]	GBUS[146]	XA[06]	GBUS[186]	PK[26]
GBUS[027]	PS[27]	GBUS[067]	PD[11]	GBUS[107]	PC[27]	GBUS[147]	XA[07]	GBUS[187]	PK[27]
GBUS[028]	PS[28]	GBUS[068]	PD[12]	GBUS[108]	LD[18]	GBUS[148]	XA[08]	GBUS[188]	PK[28]
GBUS[029]	PS[29]	GBUS[069]	PD[13]	GBUS[109]	LD[19]	GBUS[149]	XA[09]	GBUS[189]	PK[29]
GBUS[030]	PS[30]	GBUS[070]	PD[14]	GBUS[110]	LD[20]	GBUS[150]	XA[10]	GBUS[190]	PK[30]
GBUS[031]	PS[31]	GBUS[071]	PD[15]	GBUS[111]	LD[21]	GBUS[151]	XA[11]	GBUS[191]	PK[31]
GBUS[032]	PU[00]	GBUS[072]	PD[16]	GBUS[112]	LD[22]	GBUS[152]	XA[12]		
GBUS[033]	PU[01]	GBUS[073]	PD[17]	GBUS[113]	LD[23]	GBUS[153]	XA[13]		
GBUS[034]	PU[02]	GBUS[074]	PD[18]	GBUS[114]	LC[04]	GBUS[154]	XA[14]		
GBUS[035]	PU[03]	GBUS[075]	PD[19]	GBUS[115]	LC[05]	GBUS[155]	XA[15]		
GBUS[036]	PU[04]	GBUS[076]	PD[20]	GBUS[116]	FC[00]	GBUS[156]	XA[16]		
GBUS[037]	PU[05]	GBUS[077]	PD[21]	GBUS[117]	FC[01]	GBUS[157]	XA[17]		
GBUS[038]	PU[06]	GBUS[078]	PD[22]	GBUS[118]	FC[02]	GBUS[158]	XA[18]		
GBUS[039]	PU[07]	GBUS[079]	PD[23]	GBUS[119]	FC[03]	GBUS[159]	XA[19]		

Table 18.16 GSB Port Selection Register Map (Base Address = 0x9000F800)

Name	Address	Type	Reset	Description
GSB0PORTSEL	0x00	R/W	0x03020100	GSB0 Port Selection Register
GSB1PORTSEL	0x04	R/W	0x07060504	GSB1 Port Selection Register
GSB2PORTSEL	0x08	R/W	0x0B0A0908	GSB2 Port Selection Register
GSB3PORTSEL	0x0C	R/W	0x0F0E0D0C	GSB3 Port Selection Register
GSB4PORTSEL	0x10	R/W	0x13121110	GSB4 Port Selection Register
GSB5PORTSEL	0x14	R/W	0x17161514	GSB5 Port Selection Register

GSB0 Port Selection Register (GSB0PORTSEL)

0x9000F800

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x03	Port number for SDO
23:16	FRM	R/W	0x02	Port number for FRM
15:8	SCK	R/W	0x01	Port number for SCK
7:0	SDI	R/W	0x00	Port number for SDI

GSB1 Port Selection Register (GSB1PORTSEL)

0x9000F804

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x07	Port number for SDO
23:16	FRM	R/W	0x06	Port number for FRM
15:8	SCK	R/W	0x05	Port number for SCK
7:0	SDI	R/W	0x04	Port number for SDI

GSB2 Port Selection Register (GSB2PORTSEL)

0x9000F808

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x0B	Port number for SDO
23:16	FRM	R/W	0x0A	Port number for FRM
15:8	SCK	R/W	0x09	Port number for SCK
7:0	SDI	R/W	0x08	Port number for SDI

GSB3 Port Selection Register (GSB3PORTSEL)

0x9000F80C

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x0F	Port number for SDO
23:16	FRM	R/W	0x0E	Port number for FRM
15:8	SCK	R/W	0x0D	Port number for SCK
7:0	SDI	R/W	0x0C	Port number for SDI

GSB4 Port Selection Register (GSB4PORTSEL)

0x9000F810

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x13	Port number for SDO
23:16	FRM	R/W	0x12	Port number for FRM
15:8	SCK	R/W	0x11	Port number for SCK
7:0	SDI	R/W	0x10	Port number for SDI

GSB5 Port Selection Register (GSB5PORTSEL)

0x9000F814

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x17	Port number for SDO
23:16	FRM	R/W	0x16	Port number for FRM
15:8	SCK	R/W	0x15	Port number for SCK
7:0	SDI	R/W	0x14	Port number for SDI

Table 18.17 UART Port Selection Register Map (Base Address = 0x90007800)

Name	Address	Type	Reset	Description
UART0PORTSEL	0x00	R/W	0x23222120	UART0 Port Selection Register
UART1PORTSEL	0x04	R/W	0x27262524	UART1 Port Selection Register
UART2PORTSEL	0x08	R/W	0x2B2A2928	UART2 Port Selection Register
UART3PORTSEL	0x0C	R/W	0x2F2E2D2C	UART3 Port Selection Register
UART4PORTSEL	0x10	R/W	0x33323130	UART4 Port Selection Register
UART5PORTSEL	0x14	R/W	0x37363534	UART5 Port Selection Register

UART0 Port Selection Register (UART0PORTSEL)**0x90007800**

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x23	Port number for RTS
23:16	CTS	R/W	0x22	Port number for CTS
15:8	RXD	R/W	0x21	Port number for RXD
7:0	TXD	R/W	0x20	Port number for TXD

UART1 Port Selection Register (UART1PORTSEL)**0x90007804**

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x27	Port number for RTS
23:16	CTS	R/W	0x26	Port number for CTS
15:8	RXD	R/W	0x25	Port number for RXD
7:0	TXD	R/W	0x24	Port number for TXD

UART2 Port Selection Register (UART2PORTSEL)**0x90007808**

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x2B	Port number for RTS
23:16	CTS	R/W	0x2A	Port number for CTS
15:8	RXD	R/W	0x29	Port number for RXD
7:0	TXD	R/W	0x28	Port number for TXD

UART3 Port Selection Register (UART3PORTSEL)**0x9000780C**

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x2F	Port number for RTS
23:16	CTS	R/W	0x2E	Port number for CTS
15:8	RXD	R/W	0x2D	Port number for RXD
7:0	TXD	R/W	0x2C	Port number for TXD

UART4 Port Selection Register (UART4PORTSEL)**0x90007810**

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x33	Port number for RTS
23:16	CTS	R/W	0x32	Port number for CTS
15:8	RXD	R/W	0x31	Port number for RXD
7:0	TXD	R/W	0x30	Port number for TXD

UART5 Port Selection Register (UART5PORTSEL)**0x90007814**

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x37	Port number for RTS
23:16	CTS	R/W	0x36	Port number for CTS
15:8	RXD	R/W	0x35	Port number for RXD
7:0	TXD	R/W	0x34	Port number for TXD

Table 18.18 DAI Port Selection Register Map (Base Address = 0x98002800)

Name	Address	Type	Reset	Description
DAI0PORTSEL	0x00	R/W	0x3B3A3938	DAI0 Port Selection Register
DAI1PORTSEL	0x04	R/W	0x3F3E3D3C	DAI1 Port Selection Register
DAI2PORTSEL	0x08	R/W	0x43424140	DAI2 Port Selection Register
DAI3PORTSEL	0x0C	R/W	0x47464544	DAI3 Port Selection Register
DAIMPORTSEL	0x10	R/W	0x4B4A4948	DAI MCLK Port Selection Register
DAICPORTSEL	0x14	R/W	0x4F4E4D4C	DAI CDIF Port Selection Register

DAI0 Port Selection Register (DAI0PORTSEL)

0x98002800

Bit	Name	R/W	Reset	Description
31:24	DO	R/W	0x3B	Port number for DO
23:16	LRCK	R/W	0x3A	Port number for LRCK
15:8	BCLK	R/W	0x39	Port number for BCLK
7:0	DI	R/W	0x38	Port number for DI

DAI1 Port Selection Register (DAI1PORTSEL)

0x98002804

Bit	Name	R/W	Reset	Description
31:24	DO	R/W	0x3F	Port number for DO
23:16	LRCK	R/W	0x3E	Port number for LRCK
15:8	BCLK	R/W	0x3D	Port number for BCLK
7:0	DI	R/W	0x3C	Port number for DI

DAI2 Port Selection Register (DAI2PORTSEL)

0x98002808

Bit	Name	R/W	Reset	Description
31:24	DO	R/W	0x43	Port number for DO
23:16	LRCK	R/W	0x42	Port number for LRCK
15:8	BCLK	R/W	0x41	Port number for BCLK
7:0	DI	R/W	0x40	Port number for DI

DAI3 Port Selection Register (DAI3PORTSEL)

0x9800280C

Bit	Name	R/W	Reset	Description
31:24	DO	R/W	0x47	Port number for DO
23:16	LRCK	R/W	0x46	Port number for LRCK
15:8	BCLK	R/W	0x45	Port number for BCLK
7:0	DI	R/W	0x44	Port number for DI

DAI MCLK Port Selection Register (DAIMPORTSEL)

0x98002810

Bit	Name	R/W	Reset	Description
31:24	DAI3MCLK	R/W	0x4B	Port number for DAI3MCLK
23:16	DAI2MCLK	R/W	0x4A	Port number for DAI2MCLK
15:8	DAI1MCLK	R/W	0x49	Port number for DAI1MCLK
7:0	DAI0MCLK	R/W	0x48	Port number for DAI0MCLK

DAI CDIF/SPDIF Port Selection Register (DAICPORTSEL)

0x98002814

Bit	Name	R/W	Reset	Description
31:24	SPDIFTX	R/W	0x4B	Port number for SPDIFTX
23:16	CDLRCK	R/W	0x4A	Port number for CDLRCK
15:8	CDBCLK	R/W	0x49	Port number for CDBCLK
7:0	CDDI	R/W	0x48	Port number for CDDI

Table 18.19 I2C Port Selection Register Map (Base Address = 0x9000B200)

Name	Address	Type	Reset	Description
I2CM0/1PORTSEL	0x00	R/W	0xFFFFFFFF	I2C Master 0/1 Port Selection Register
I2CM2/3PORTSEL	0x04	R/W	0xFFFFFFFF	I2C Master 2/3 Port Selection Register
I2CS0/1PORTSEL	0x08	R/W	0xFFFFFFFF	I2C Slave 0/1 Port Selection Register

I2C Master 0/1 Port Selection Register (I2CM0/1PORTSEL)

0x9000B200

Bit	Name	R/W	Reset	Description
31:24	M1SDA	R/W	0xFF	Port number for I2C Master 1 SDA
23:16	M1SCL	R/W	0xFF	Port number for I2C Master 1 SCL
15:8	M0SDA	R/W	0xFF	Port number for I2C Master 0 SDA
7:0	M0SCL	R/W	0xFF	Port number for I2C Master 0 SCL

I2C Master 2/3 Port Selection Register (I2CM2/3PORTSEL)

0x9000B204

Bit	Name	R/W	Reset	Description
31:24	M3SDA	R/W	0xFF	Port number for I2C Master 3 SDA
23:16	M3SCL	R/W	0xFF	Port number for I2C Master 3 SCL
15:8	M2SDA	R/W	0xFF	Port number for I2C Master 2 SDA
7:0	M2SCL	R/W	0xFF	Port number for I2C Master 2 SCL

I2C Slave 0/1 Port Selection Register (I2CS0/1PORTSEL)

0x9000B208

Bit	Name	R/W	Reset	Description
31:24	S1SDA	R/W	0xFF	Port number for I2C Slave 1 SDA
23:16	S1SCL	R/W	0xFF	Port number for I2C Slave 1 SCL
15:8	S0SDA	R/W	0xFF	Port number for I2C Slave 0 SDA
7:0	S0SCL	R/W	0xFF	Port number for I2C Slave 0 SCL

Table 18.20 TCO Port Selection Register Map (Base Address = 0x98004800)

Name	Address	Type	Reset	Description
TCOPORTSEL0	0x00	R/W	0xFFFFFFFF	TCO Port Selection Register 0
TCOPORTSEL1	0x04	R/W	0xFFFFFFFF	TCO Port Selection Register 1
PDMPORTSEL	0x08	R/W	0xFFFFFFFF	PDM Port Selection Register

TCO Port Selection Register 0 (TCOPORTSEL0)

0x98004800

Bit	Name	R/W	Reset	Description
31:24	TCO[3]	R/W	0xFF	Port number for TCO[3]
23:16	TCO[2]	R/W	0xFF	Port number for TCO[2]
15:8	TCO[1]	R/W	0xFF	Port number for TCO[1]
7:0	TCO[0]	R/W	0xFF	Port number for TCO[0]

TCO Port Selection Register 1 (TCOPORTSEL1)

0x98004804

Bit	Name	R/W	Reset	Description
31:16	-	R/W	0xFF	Reserved
15:8	TCO[5]	R/W	0xFF	Port number for TCO[5]
7:0	TCO[4]	R/W	0xFF	Port number for TCO[4]

PDM Port Selection Register (PDMPORTSEL)

0x98004808

Bit	Name	R/W	Reset	Description
31:24	PDM[3]	R/W	0xFF	Port number for PDM[3]
23:16	PDM[2]	R/W	0xFF	Port number for PDM[2]
15:8	PDM[1]	R/W	0xFF	Port number for PDM[1]
7:0	PDM[0]	R/W	0xFF	Port number for PDM[0]

Table 18.21 IRDA Port Selection Register Map (Base Address = 0x9801E080)

Name	Address	Type	Reset	Description
IRDAPORTSEL	0x00	R/W	0x0000008F	IRDA Port Selection Register

IRDA Port Selection Register (IRDAPORTSEL)

0x9801E080

Bit	Name	R/W	Reset	Description
31:8	-	R/W	0x00	Reserved
7:0	IRDAI	R/W	0x8F	Port number for IrDA Input.

Table 18.22 TSIF Port Selection Register Map (Base Address = 0x80006600)

Name	Address	Type	Reset	Description
TSIFPORTSEL0	0x00	R/W	0xFFFFFFFF	TSIF Port Selection Register 0
TSIFPORTSEL1	0x04	R/W	0xFFFFFFFF	TSIF Port Selection Register 1
TSIFPORTSEL2	0x08	R/W	0xFFFFFFFF	TSIF Port Selection Register 2
TSIFPORTOE	0x0C	R/W	0x00000000	TSIF Port Direction Control Register

TSIF Port Selection Register 0 (TSIFPORTSEL0)

0x80006600

Bit	Name	R/W	Reset	Description
31:24	TSD[3]	R/W	0xFF	Port number for TSD[3]
23:16	TSD[2]	R/W	0xFF	Port number for TSD[2]
15:8	TSD[1]	R/W	0xFF	Port number for TSD[1]
7:0	TSD[0]	R/W	0xFF	Port number for TSD[0]

TSIF Port Selection Register 1 (TSIFPORTSEL1)

0x80006604

Bit	Name	R/W	Reset	Description
31:24	TSD[7]	R/W	0xFF	Port number for TSD[7]
23:16	TSD[6]	R/W	0xFF	Port number for TSD[6]
15:8	TSD[5]	R/W	0xFF	Port number for TSD[5]
7:0	TSD[4]	R/W	0xFF	Port number for TSD[4]

TSIF Port Selection Register 2 (TSIFPORTSEL2)

0x80006608

Bit	Name	R/W	Reset	Description
31:24	-	R	0	Reserved
23:16	TSVALID	R/W	0xFF	Port number for TSVALID
15:8	TSSYNC	R/W	0xFF	Port number for TSSYNC
7:0	TSCLK	R/W	0xFF	Port number for TSCLK

TSIF Port Direction Control Register (TSIFPORTOE)

0x8000660C

Bit	Name	R/W	Reset	Description
31:11	-	R	0	Reserved
10	TSVALIDOE	R/W	0	TSVALID Output Enable. (0=Input, 1=Output)
9	TSSYNCOE	R/W	0	TSSYNC Output Enable. (0=Input, 1=Output)
8	TSCLKOE	R/W	0	TSCLK Output Enable. (0=Input, 1=Output)
7:0	TSDOE	R/W	0	TSD[7:0] Output Enable. (0=Input, 1=Output)

Table 18.23 GSB Port Selection Register Map (Base Address = 0x98006180)

Name	Address	Type	Reset	Description
CKOPORTSEL0	0x00	R/W	0xFFFFFFFF	Clock Output Port Selection Register0
CKOPORTSEL3	0x0C	R/W	0x00FFFF00	Clock Output Port Selection Register3
CKOPORTSEL5	0x14	R/W	0x0000FF00	Clock Output Port Selection Register5
CKOPORTSEL7	0x1C	R/W	0xFFFF0000	Clock Output Port Selection Register7
CKOPORTSEL8	0x20	R/W	0xFF000000	Clock Output Port Selection Register8
CKOPORTSEL9	0x24	R/W	0x000000FF	Clock Output Port Selection Register9

Clock Output Port Selection Register0 (CKOPORTSEL0) 0x98006180

Bit	Name	R/W	Reset	Description
31:24	IrDACLK	R/W	0xFF	Port number for IrDA
23:16	-	R/W	0xFF	Reserved
15:8	I2CCLK	R/W	0xFF	Port number for I2CCLK
7:0	REFCLK	R/W	0xFF	Port number for REFCLK

Clock Output Port Selection Register3 (CKOPORTSEL3) 0x9800618C

Bit	Name	R/W	Reset	Description
31:24	-	R	0	Reserved
23:16	DAI1CLK	R/W	0xFF	Port number for DAI1CLK
15:8	DAI0CLK	R/W	0xFF	Port number for DAI0CLK
7:0	-	R	0	Reserved

Clock Output Port Selection Register5 (CKOPORTSEL5) 0x98006194

Bit	Name	R/W	Reset	Description
31:16	-	R	0	Reserved
15:8	EXTCLK	R/W	0xFF	Port number for EXTCLK
7:0	-	R	0	Reserved

Clock Output Port Selection Register7 (CKOPORTSEL7) 0x9800619C

Bit	Name	R/W	Reset	Description
31:24	DAI3CLK	R/W	0xFF	Port number for DAI3CLK
23:16	DAI2CLK	R/W	0xFF	Port number for DAI2CLK
15:0	-	R	0	Reserved

Clock Output Port Selection Register8 (CKOPORTSEL8) 0x980061A0

Bit	Name	R/W	Reset	Description
31:24	CIF0CLK	R/W	0xFF	Port number for CIF0CLK
23:0	-	R	0	Reserved

Clock Output Port Selection Register9 (CKOPORTSEL9) 0x980061A4

Bit	Name	R/W	Reset	Description
31:8	-	R	0	Reserved
7:0	CIF1CLK	R/W	0xFF	Port number for CIF1CLK

19 Memory Controller

19.1 Overview

19.1.1 Internal Memory

In TCC8132, there is 32Kbytes of internal SRAM. The SRAM area is dedicated to area 1 (0x10000000 ~ 0x1FFFFFFF), and accessed by area 0 (0x00000000 ~ 0x0FFFFFFF) when there is no device assigned to area 0. The Boot Code area is dedicated to area E (0xE0000000 ~ 0xEFFFFFFF), and accessed by area 0 (0x00000000 ~ 0x0FFFFFFF) when REMAP flag of SCFG register is cleared to 0.

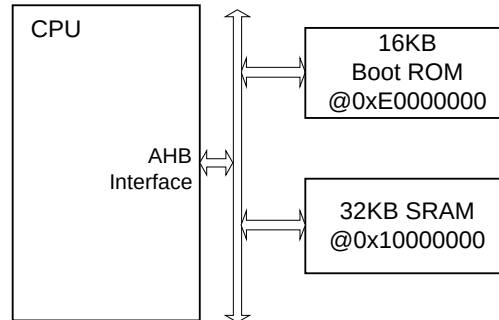


Figure 19.1 Internal Memories

19.1.2 SDR SDRAM Memory Controller

19.1.2.1 Feature

- Supports 16/32bit SDR SDRAM
- Supports 16Mbit up to 512Mbit density

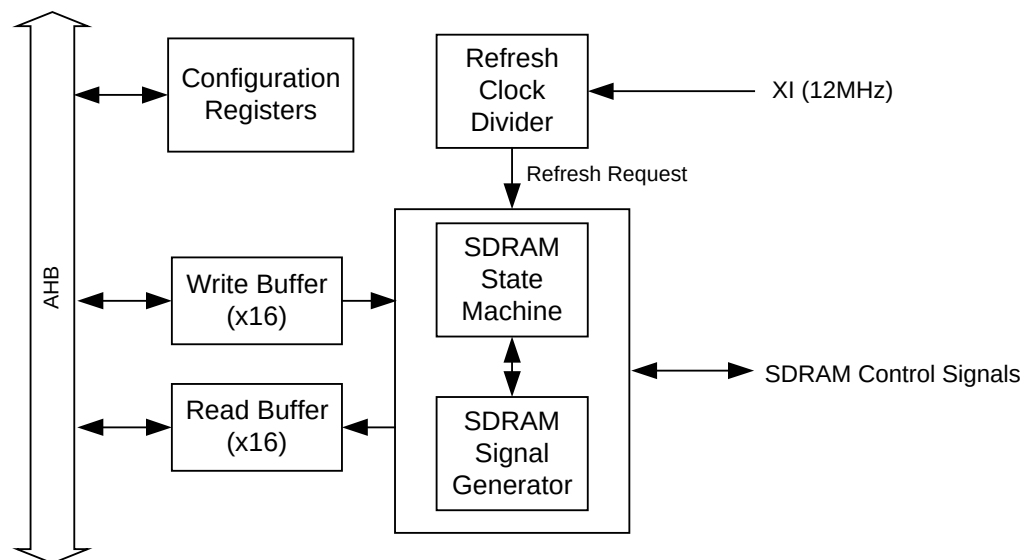


Figure 19.2 SDR SDRAM Controller Block Diagram

19.2 SDR SDRAM Memory Controller

19.2.1 Register Description

Table 19.1 SDR SDRAM Memory Controller Register Map (Base Address = 0xF0000000)

Name	Address	Type	Reset	Description
SDCFG	0x00	R/W	0x06101400	SDRAM Configuration Register
SDFSM	0x04	R	-	SDRAM FSM Status Register
MCFG	0x08	R/W	0xFFFFFFFF42	Miscellaneous Configuration Register
CLKCFG	0x20	R/W	0x15110200	Periodic Clock Enable Count Register
SDCMD	0x24	R/W	0x0000F000	SDRAM Command Write Register
MCKDV	0x28	R/W	0x00000000	Memory Clock Divider Load Value Register
RFDV	0x2C	R/W	0x001B005C	Refresh Clock Divider Load Value Register
AMCFG0	0x40	R/W	0x00010082	SDRAM Controller Configuration 0
AMCFG1	0x44	R/W	0x00100007	SDRAM Controller Configuration 1
AMCFG2	0x48	R/W	0x0000001F	SDRAM Controller Configuration 2
AMCFG3	0x4C	R/W	0x00000000	SDRAM Controller Configuration 3
DFREQ	0xF0	R/W	0x00000060	SDRAM Controller Frequency Register
DTIME	0xF4	R/W	0x4614142D	SDRAM Controller Time Register
DTCFG	0xF8	R/W	0x00116114	SDRAM Controller Timing Configuration
DTYPE	0xFC	R/W	0x00000010	SDRAM Controller Type Configuration

Note:

1. Most of the reset default values are subject to change by the boot code, debug interface, and etc. Some registers are automatically updated when the other registers are written. Do not rely on the default value of the registers above.
2. SDCFG and MCFG registers are preserved for backward compatibility. It is recommended to use AMCFG0, DTCFG and DTYPE registers for additional parameter settings.

Table 19.2 NAND Flash Register Map (Base Address = N × 0x1000000)

Name	Address	Type	Reset	Description
NDCMD	0x00	R/W	-	Command Cycle Register
NDLADR	0x04	W	-	Linear Address Cycle Register
NDRADR	0x08	W	-	Row Address Cycle Register
NDIADR	0x0C	W	-	Single Address Cycle Register
NDDATA	0x10	R/W	-	Data Access Cycle Register

Note 1: N represents BASE field of configuration register (CSCFGx) for each chip select.

Note 2: This table is used for reading/writing through memory port.

19.2.2 SDR SDRAM Control Registers

The SDRAM controller can control from 16Mbit up to 512Mbit density SDRAM. The SDRAM parameter such as size, refresh period, RAS to CAS delay, refresh to idle delay can be programmed by internal register.

SDRAM Configuration Register (SDCFG)

0xF0000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CL	BW	CW		RFC				RCD				WR		0	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RP				RW		0						AM	0	PPD	SR

Note: The reset value changes by the boot code.

Bit	Name	R/W	Reset	Description		
31	CL	R/W	0	CAS Latency $t_{CL} = HCLK * (CL+2)$		
30	BW	R/W	0	Bus Width		
				0	32 bit	
				1	16 bit	
29:28	CW	R/W	0	Column Address Width		
				CW	Column Address Width	SDRAM Density
				0, 1	8bit is used for column address	16Mbit, 64Mbit
				2	9bit is used for column address	128Mbit, 256Mbit
				3	10bit is used for column address	512Mbit
27:24	RFC	R/W	6	Auto Refresh Cycle Time $t_{RFC} = (HCLK \text{ period}) * (RFC + 1)$ Refresh-to-Idle cycle parameter is also controlled by this value		
23:20	RCD	R/W	1	RAS to CAS Delay $t_{RCD} = (HCLK \text{ period}) * (RCD + 1)$		
19:18	WR	R/W	0	Write to Read Wait $t_{WR} = (HCLK \text{ period}) * WR$		
17:16	-	R	0	Reserved		
15:12	RP	R/W	1	Row Precharge Time $t_{RP} = (HCLK \text{ period}) * (RP + 1)$ Controls delay time from “Row Precharge Command” to a next command.		
11:10	RW	R/W	1	Row Address Width		
				RW	Row Address Width	SDRAM Density
				0	12bit is used for row address (Default)	64Mbit, 128Mbit
				2	13bit is used for row address	256Mbit, 512Mbit
				1, 3	11bit is used for row address	16Mbit
9:4	-	R	0	Reserved		
3	AM	R/W	0	Address Matching Configuration selects BA address mapping.		
				0	BA-RAS-CAS address order	
				1	RAS-BA-CAS address order	
2	-	R/W	0	Reserved. Must be written as “0”		
1	PPD	R/W	0	Precharge/Active Power-Down Mode Enable. When enabled and SDRAM is in idle state, CKE signal is disabled. Whenever the CKE is re-enabled, one wait cycle penalty is incurred.		
0	SR	R/W	0	Self-Refresh Mode Enable. Writing “1” enables self-refresh mode.		
				0	Disable (Exit from the self-refresh mode)	
				1	Enable (Enter the self-refresh mode)	

Except for Bit 0 (SR), the sequence below must be followed whenever this register is written.

1. Clear SDEN bit of MCFG register (SDRAM controller is disabled)
2. Update SDCFG register value
3. Set SDEN bit of MCFG register (SDRAM controller is enabled and MRS cycle is issued to the SDRAM).

Caution:

1. BW, CW and RW are written by DTYPE register. (Do not use SDCFG register)
2. CR, RC, RCD, WR, and RP are written by DFREQ or DTCFG register (Do not use SDCFG register)
3. AM, PPD, and SR are written by AMCFG0 register (Do not use SDCFG register)
4. It is not recommended to use the yellow box fields above. In case of using them, must use AMCFG0, DTCFG, DTYPE Registers.

SDRAM FSM Status Register (SDFSM)

0xF0000004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0				SDFSM											

This register is read only and represents current status of finite state machine in the SDRAM controller. This can be used for test purpose only.

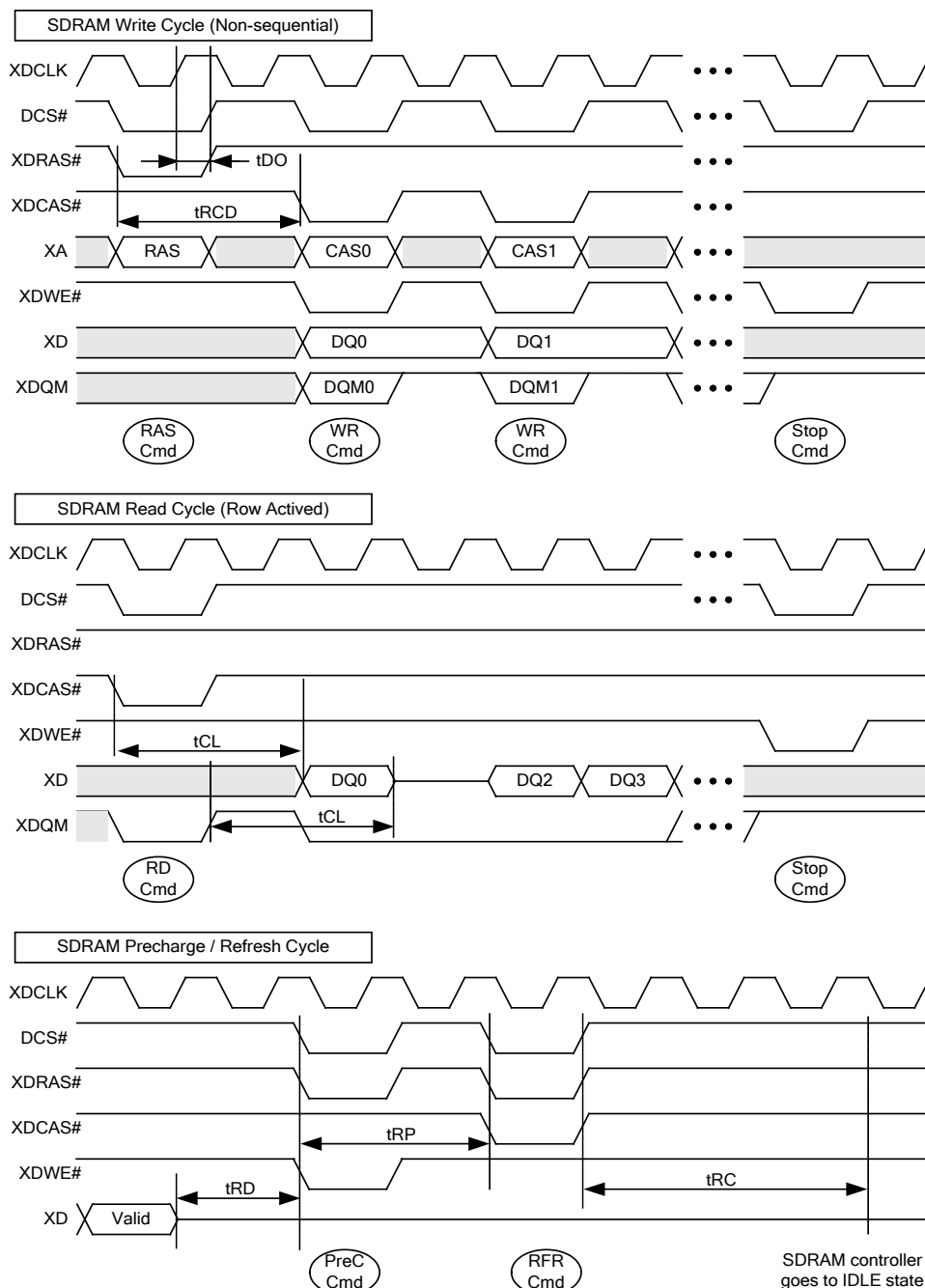


Figure 19.3 SDRAM Cycle Diagram

Miscellaneous Configuration Register (MCFG)

0xF0000008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
X (Don't care – read only area)															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								SDW	ENCK	ECKE	SDEN	-	-	CKM	-

Bit	Name	R/W	Reset	Description	
31:16	-	R	X	Reserved	
15:8	-	R	X	Reserved	
7	SDW	R/W	0	SDRAM Wait State Control	
				0	Zero wait
				1	One wait
				Normally this should be zero. This can be used in case of very high speed SDRAM interface or too high load capacitance for memory bus.	
6	ENCK	R/W	1	SDRAM Clock dynamic control enable bit	
				0	Direct output
				1	Gating controlled
				If enabled, the memory controller masks the CLK to reduce power consumption. The ratio of masked clock period is determined by CLKCFG register.	
5	ECKE	R/W	0	CKE Enable bit	
				0	CKE disabled
				1	CKE enabled (controlled by the SDRAM controller)
4	SDEN	R/W	0	SDRAM Controller Enable Whenever this bit goes from low to high, SDRAM controller is enabled and SDRAM MRS cycle is generated immediately after auto refresh cycles. Before enabled, the SDRAM Refresh Request (clock) register of the Clock controller must be enabled with valid value. This bit must be cleared before CL bit of SDCFG register is changed.	
3:2	-	R/W	0	Reserved.	
1	CKM	R/W	1	SDRAM Clock Masking bit.	
				0	Masked (Disabled)
				1	Unmasked
0	-	R/W	0	Reserved.	

Caution:

1. SDW is updated whenever WAIT field of DTCFG register is written.
2. ENCK, SDEN, and CKM are written by AMCFG0 register (Do not use MCFG register)
3. It is not recommended to access the yellow box field above. In case of using them, must use AMCFG0 Register.

SDRAM Clock Control Register (CLKCFG)

0xF0000020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved (Read as 0x1511)															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved (Read as 0x02)								PCLKCNT							

This register determines the period of SDRAM clock masked. In the active period, the HCLK is out with the same frequency. In the in-active period, this register generates divided HCLK and masks the clock output. This can lower the power consumption in idle mode. If 0, the HCLK is out directly with the same frequency. (default)

PCLKCNT	[7:0]	Masked Period of SDCLK
M	This determines the period of SDCLK masked. If 0, the HCLK is out directly with same frequency.	

SDRAM Command Register (SDCMD)

0xF0000024

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
							-								A10
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CS#	RAS#	CAS#	WE#	BA[1:0]		A[9:0]									

Any value written to this register must be a valid SDRAM command (MRS or EMRS, etc).

Supported commands: MRS, EMRS, Auto-refresh.

Memory Clock Divider Load Value Register (MCKDV)

0xF0000028

Bit	Name	R/W	Reset	Description
31:4	-	R	0	Reserved
3:0	MCKDV	R/W	0	Memory clock divider load value. SDRAM clock period = (HCLK period) * (MCKDV + 1)

Refresh Clock Divider Load Value Register (RFDV)

0xF000002C

Bit	Name	R/W	Reset	Description	
31:24	-	R	0	Reserved	
23:16	PUD	R/W	0x1B	Power Up Initialization Delay.	tPUD = (Refresh period) * PUD Default delay = 209.25us
15	CKSEL	R/W	0	Divider clock source	
				0	12MHz clock (XI main oscillator)
				1	HCLK (Bus clock)
14:12	-	R	0	Reserved	
12:0	RFDV	R/W	0x05C	Divider load value.	Refresh period = (Refresh clock period) * (RFDV + 1). Default refresh period = 7.75us

SDRAM Controller Configuration Register 0 (AMCFG0)

0xF0000040

SDRAM Controller Configuration Register (PMR 0)				0x000014					
Bit	Name	R/W	Reset	Description					
31	PD0	R/W	0	Power Down and Force 0 Level on to the SDRAM interface signals. When this bit is set, the SDRAM controller enters self-refresh mode and then, all the SDRAM interface signals are driven with “0” level. SDRAM can be powered off in this mode. If SDRAM is re-powered, the DPD bit must be set before this bit is cleared.					
30	-	R/W	0	Reserved					
29	DPD	R/W	0	Deep Power Down selects command when SR bit or PD0 bit is enabled. <table><tr><td>0</td><td>Self Refresh command</td></tr><tr><td>1</td><td>Deep Power Down command</td></tr></table>		0	Self Refresh command	1	Deep Power Down command
0	Self Refresh command								
1	Deep Power Down command								
28:16	RFTH	R/W	0x0001	Refresh Threshold					
15	CSEL	R/W	0	Reserved					
14:13	RMAX	R/W	0	Row Active Time tRASmax = (Refresh period) * 3 * (RMAX + 1). If any row is opened over tRASmax, pre-charge cycle is forced.					
12:8	-	R	0	Reserved					
7	ENCK	R/W	1	SDRAM Clock dynamic control enable bit <table><tr><td>0</td><td>Direct output</td></tr><tr><td>1</td><td>Gating controlled</td></tr></table> If enabled, the memory controller masks the CLK to reduce power consumption. The ratio of masked clock period is determined by CLKCFG register.		0	Direct output	1	Gating controlled
0	Direct output								
1	Gating controlled								
6	MARF	R/W	0	Mask (Disable) Auto Refresh <table><tr><td>0</td><td>Auto Refresh Enabled</td></tr><tr><td>1</td><td>Auto Refresh Disabled</td></tr></table>		0	Auto Refresh Enabled	1	Auto Refresh Disabled
0	Auto Refresh Enabled								
1	Auto Refresh Disabled								
5	-	R/W	0	Reserved					
4	SDEN	R/W	0	SDRAM Controller Enable.					
3	AM	R/W	0	Address Matching Configuration selects BA address mapping. <table><tr><td>0</td><td>BA-RAS-CAS address order</td></tr><tr><td>1</td><td>RAS-BA-CAS address order</td></tr></table>		0	BA-RAS-CAS address order	1	RAS-BA-CAS address order
0	BA-RAS-CAS address order								
1	RAS-BA-CAS address order								
2	PPD	R/W	0	Precharge/Active Power-Down Mode Enable. When enabled and SDRAM controller is in idle state, CKE signal is disabled. Whenever the CKE is re-enabled, one wait cycle penalty is incurred.					
1	CKM	R/W	1	SDRAM Clock Masking bit. <table><tr><td>0</td><td>Masked (Disabled)</td></tr><tr><td>1</td><td>Unmasked</td></tr></table>		0	Masked (Disabled)	1	Unmasked
0	Masked (Disabled)								
1	Unmasked								
0	SR	R/W	0	Self-Refresh Mode Enable. Writing “1” enables self-refresh mode. <table><tr><td>0</td><td>Disable (Exit self-refresh mode)</td></tr><tr><td>1</td><td>Enable (Enter self-refresh mode)</td></tr></table>		0	Disable (Exit self-refresh mode)	1	Enable (Enter self-refresh mode)
0	Disable (Exit self-refresh mode)								
1	Enable (Enter self-refresh mode)								

ENCK, SDEN, and CKM are duplicated bits of MCFG register.

AM, PPD, and SR are duplicated bits of SDCFG register.

SDRAM Controller Configuration Register 1 (AMCFG1)

0xF0000044

Bit	Name	R/W	Reset	Description	
31	HOLD	R/W	0	Reserved (for test purpose)	
30	HLDA	R	0	Reserved	
29:21	-	R	0	Reserved	
20:16	-	R/W	0x10	Reserved	
15:12	-	R/W	0	Reserved	
11:8	WBTH	R/W	0	Write Buffer Threshold Control	
7:4	-	R/W	0	Reserved Must be written as "0".	
3:0	PFE	R/W	0x7	Prefetch Enable for AHB INCRx burst cycles.	
				PFE[3:0]	Description
				1xxx	8 words (INCR8 word access only)
				x1xx	4 Cycles
				x01x	2 Cycles
x001	1 Cycle				

SDRAM Controller Configuration Register 2 (AMCFG2)

0xF0000048

Bit	Name	R/W	Reset	Description						
31:28	-	R	0	Reserved						
27:10	WBA0	R/W	0	Prefetch Window 0 Base Address [27:10]						
9:6	-	R	0	Reserved						
5	WSZ0	R/W	0	Prefetch Window 0 Inversion						
				<table><tr><th>WSZ0[5]</th><th>Description</th></tr><tr><td>0</td><td>Prefetch enabled for the area inside the window</td></tr><tr><td>1</td><td>Prefetch enabled for the area outside the window</td></tr></table>	WSZ0[5]	Description	0	Prefetch enabled for the area inside the window	1	Prefetch enabled for the area outside the window
				WSZ0[5]	Description					
				0	Prefetch enabled for the area inside the window					
1	Prefetch enabled for the area outside the window									
4:0	0x1F	Prefetch Window 0 Size								
		<table><tr><th>WSZ0[4:0]</th><th>Description</th></tr><tr><td>0</td><td>Prefetch disabled</td></tr><tr><td>Others</td><td>Size = (1 << WSZ0[4:0]) * 1KBytes</td></tr></table>	WSZ0[4:0]	Description	0	Prefetch disabled	Others	Size = (1 << WSZ0[4:0]) * 1KBytes		
		WSZ0[4:0]	Description							
0	Prefetch disabled									
Others	Size = (1 << WSZ0[4:0]) * 1KBytes									

SDRAM Controller Configuration Register 3 (AMCFG3)

0xF000004C

Bit	Name	R/W	Reset	Description						
31:28	-	R	0	Reserved						
27:10	WBA1	R/W	0	Prefetch Window 1 Base Address [27:10]						
9:6	-	R	0	Reserved						
5	WSZ1	R/W	0	Prefetch Window 1 Inversion						
				<table><tr><th>WSZ1[5]</th><th>Description</th></tr><tr><td>0</td><td>Prefetch enabled for the area inside the window</td></tr><tr><td>1</td><td>Prefetch enabled for the area outside the window</td></tr></table>	WSZ1[5]	Description	0	Prefetch enabled for the area inside the window	1	Prefetch enabled for the area outside the window
				WSZ1[5]	Description					
				0	Prefetch enabled for the area inside the window					
1	Prefetch enabled for the area outside the window									
4:0	0	Prefetch Window 1 Size								
		<table><tr><th>WSZ1[4:0]</th><th>Description</th></tr><tr><td>0</td><td>Prefetch disabled</td></tr><tr><td>Others</td><td>Size = (1 << WSZ1[4:0]) * 1KBytes</td></tr></table>	WSZ1[4:0]	Description	0	Prefetch disabled	Others	Size = (1 << WSZ1[4:0]) * 1KBytes		
WSZ1[4:0]	Description									
0	Prefetch disabled									
Others	Size = (1 << WSZ1[4:0]) * 1KBytes									

SDRAM Controller Frequency Register (DFREQ)

0xF00000F0

Bit	Name	R/W	Reset	Description
31:8	-	R	0	Reserved
7:0	FREQ	R/W	0x60	Frequency of SDRAM Clock (Default value is 96MHz) Right after this register is written, DTCFG Register is updated with the value calculated with DTIME Register.

Caution: Do not use this register (yellow box).

SDRAM Controller Time Register (DTIME)

0xF00000F4

Bit	Name	R/W	Reset	Description
31:24	tRFC	R/W	0x46	Auto Refresh Cycle Time Default = 70ns
23:16	tRCD	R/W	0x14	RAS to CAS Delay Time Default = 20ns
15:8	tRP	R/W	0x14	Row Precharge Time Default = 20ns
7:0	tRASmin	R/W	0x2D	Minimum Row Active Time Default = 45ns

DFREQ register must be written after DTIME register is modified.

Caution: Do not use this register (yellow box).

SDRAM Controller Timing Configuration Register (DTCFG)

0xF00000F8

Bit	Name	R/W	Reset	Description	
31	CL	R/W	0	CAS Latency tCL = HCLK * (CL+2)	
30	-	R/W	0	Reserved	
29:28	WAIT	R/W	0	Wait State Control	
				00	Zero wait
				01, 10	Reserved
				11	One wait
27:26	RWD	R/W	0	Read to Write Delay tRWD = (HCLK period) * (RWD + 1)	
25:24	WRD	R/W	0	Write to Read Delay tWRD = (HCLK period) * (WRD + 1)	
23:20	RRD	R/W	0x1	RAS to RAS Bank Active Delay tRRD = (HCLK period) * (RRD + 1)	
19:16	RDL	R/W	0x1	Last Data In to Row Pre-Charge Command Delay tRDL = (HCLK period) * (RDL + 1)	
15:12	RFC	R/W	0x6	Auto Refresh Cycle Time tRFC = (HCLK period) * (RFC + 1)	
11:8	RCD	R/W	0x1	RAS to CAS Delay tRCD = (HCLK period) * (RCD + 1)	
7:4	RP	R/W	0x1	Row Precharge Time tRP = (HCLK period) * (RP + 1)	
3:0	RMIN	R/W	0x4	Minimum Row Active Time tRASmin= (HCLK period) * (RMIN + 1)	

This register is updated whenever SDCFG or DFREQ is written.

SDRAM Controller Type Configuration Register (DTYPE)

0xF00000FC

Bit	Name	R/W	Reset	Description	
30:11	-	R	0	Reserved	
10	F8	R/W	0	Reserved	
9:8	BW	R/W	0	Bus Width	
				00	32 bit
				10	16 bit
				x1	Reserved
7	-	R/W	0	Reserved	
6	CSW	R/W	0	Reserved	
5	-	R/W	0	Reserved	
4	BAW	R/W	1	Bank Address Width	
				0	1 bit
				1	2 bit
3:2	RAW	R/W	0	Row Address Width	
				00	11 bit
				01	12 bit
				10	13 bit
				11	Reserved
1:0	CAW	R/W	0	Column Address Width	
				00	8 bit
				01	9 bit
				10	10 bit
				11	Reserved

This register is updated whenever SDCFG is written.

19.2.3 Sub-registers of NAND Type Memory

In case of NAND flash type memories, there are several sub-registers for generating command, address, and data cycles. Followings are these sub-registers. (M is base field of CSCFGx register)

Except the data register (NDDATA), the sub-register has implicit size of 32bit, so the bus-width of CSCFGx register does not affect the cycle of command and address registers. It only affects the cycle of data register.

Command Cycle Register (NDCMD)

0x10000000 * M

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NDCMD3								NDCMD2							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NDCMD1								NDCMD0							

Note: If bus width of NAND flash is more than 8bit, the NDCMD1 ~ 3 may be used as command register, otherwise only NDCMD0 is used as command register. The following values are example commands for NAND flash of SAMSUNG. Refer to corresponding datasheet of NAND flash chip for more detailed list of commands.

0x00/0x01 : Page Read Command

0x80 : Page Program Command

0x60 : Block Erase Command

0x70 : Status Read Command (generated by reading from 0xM0000700 address)

Linear Address Cycle Register (NDLADR)

0x10000000 * M + 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								NDLADR[31:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NDLADR[15:0]															

Note: By writing to this register, the memory controller generates linear address cycle for NAND flash.

Row Address Cycle Register (NDRADR)

0x10000000 * M + 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								NDRADR[31:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NDRADR[15:0]															

Note: By writing to this register, the memory controller generates row address cycle for NAND flash.

The table below represents the relation between each cycle and address generation. A user must set this information appropriately to PSIZE and CADDR field of CSCFGx register ahead of accessing NAND data.

Table 19.3 Page Size of NAND Flash

# of Cycle	Address Generation			
	PSIZE = 0	PSIZE = 1	PSIZE = 2	PSIZE = 3
1 st	ADR[7:0]	ADR[7:0]	ADR[7:0]	ADR[7:0]
2 nd	ADR[15:8]	ADR[16:9]	ADR[10:8]	ADR[11:8]
3 rd	ADR[23:16]	ADR[24:17]	ADR[18:11]	ADR[19:12]
4 th	ADR[31:24]	ADR[31:25]	ADR[26:19]	ADR[27:20]
5 th	-	-	ADR[31:27]	ADR[31:28]

Note: ADR means address value that is written to NDLADR or NDRADR register. The shaded cycles represent row address cycles. That is, NAND address cycles start from there when NDRADR register is accessed.

Single Address Cycle Register (NDIADR)

0x10000000 * M + 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								NDIADR							

Note: When CPU writes to this register, one cycle of address cycle is generated.

Data Register (NDDATA)

0x10000000 * M + 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NDDATA3								NDDATA2							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NDDATA1								NDDATA0							

Note: NDDATA3~1 may be used as the value of data register, otherwise only NDDATA0 is used as data register. It is dependent on the bus-width of CSCFGx register in NAND flash.

19.2.4 SDRAM Interface Timing

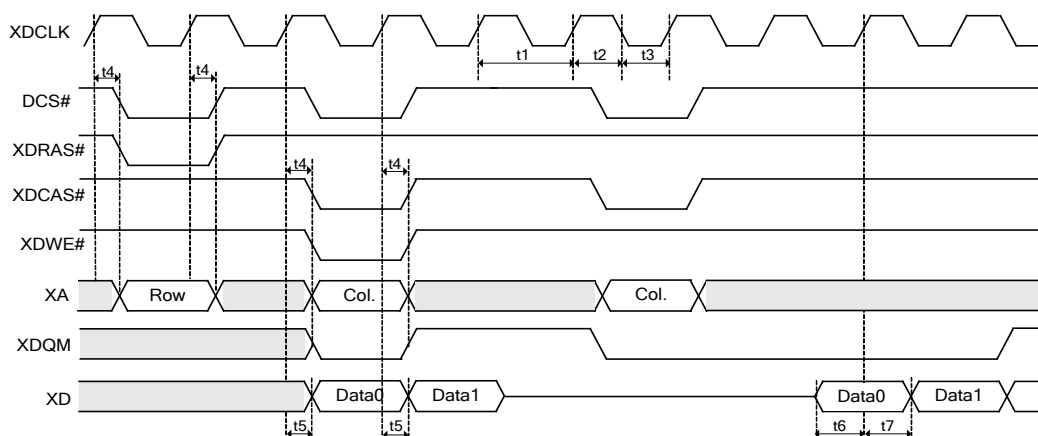


Figure 19.4 SDRAM Interface Timing

Table 19.4 SDRAM Interface Timing Parameters

Parameter	Symbol	Test Conditions	MIN	MAX	Unit
Clock cycle time	t1	V _{DDI} = 1.1V, V _{DDIO} = 3.3V, C _L = 30pF	6.02	-	ns
Clock high pulse width	t2	V _{DDI} = 1.1V, V _{DDIO} = 3.3V, C _L = 30pF	2.5	-	ns
Clock low pulse width	t3	V _{DDI} = 1.1V, V _{DDIO} = 3.3V, C _L = 30pF	2.5	-	ns
Clock to address/command delay	t4	V _{DDI} = 1.1V, V _{DDIO} = 3.3V, C _L = 30pF	1	7	ns
Clock to data output delay	t5	V _{DDI} = 1.1V, V _{DDIO} = 3.3V, C _L = 30pF	1	7	ns
Input data setup time to clock	t6	V _{DDI} = 1.1V, V _{DDIO} = 3.3V	1.5	-	ns
Input data hold time from clock	t7	V _{DDI} = 1.1V, V _{DDIO} = 3.3V	1.5	-	ns

Note: Clock cycle time is limited by the internal HCLK frequency.

20 OTPROM

20.1 Register Description

Table 20.1 OTP-ROM Controller Register Map (Base Address = 0x9001B000)

Name	Address	Type	Reset	Description
OTPCTRL	0x00	R/W	0x0	OTP-ROM Control Register
OTPDIRECT	0x04	R/W	0x0	OTP-ROM Direct Control Register
OTPTIME0	0x08	R/W	0x0	OTP-ROM Timing Information Register 0
OTPTIME1	0x0C	R/W	0x0	OTP-ROM Timing Information Register 1
OTPTIME2	0x10	R/W	0x0	OTP-ROM Timing Information Register 2
OTPTIME3	0x14	R/W	0x0	OTP-ROM Timing Information Register 3
OTPTIME4	0x18	R/W	0x0	OTP-ROM Timing Information Register 4
OTPTIME5	0x1C	R/W	0x0	OTP-ROM Timing Information Register 5
OTPTIME6	0x20	R/W	0x0	OTP-ROM Timing Information Register 6
OTPTIME7	0x24	R/W	0x0	OTP-ROM Timing Information Register 7
OTPSECRD0	0x28	R/W	0x0	OTP-ROM Secure Read Information Register 0
OTPSECRD1	0x2C	R/W	0x0	OTP-ROM Secure Read Information Register 1
OTPSECADDR	0x30	R/W	0x0	OTP-ROM Secure Address Register

OTP-ROM Control Register (OTPCTRL)

0x9001B000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DIRECT	EXT_WP														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
									MAX_NWR			WP		RETRY	CEB

Field	Name	RW	Reset	Description
31	DIRECT	R/W	0x0	OTP-ROM Direct Control Enable Register 0b : Accessed through controller 1b : Accessed through OTPDIRECT register.
30	EXT_WP	R	0x0	Status of External Write Protect Pin 0 : Can program 1 : Cannot program
6:4	MAX_NWR	R/W	0x4	Number of Maximum Overwriting Note: 4 is strongly recommended.
3	WP	R/W	0x1	Software Write Protect Signal 0 : Can program 1 : Cannot program Note: The OTP-ROM can be written when both EXT_WP and WP are in programmable state.
1	RETRY	R/W	0x1	Number of maximum retry 0 : Up to 16 1 : Up to 32
0	CEB	R/W	0x0	CEB (Chip Enable Signal) Mode Control Register 0 : CEB de-asserted to high after long idle cycle. 1 : CEB de-asserted to high after every 8 bytes transfer.

OTP-ROM Direct Control Register (OTPDIRECT)

0x9001B004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
							WEB	RSTB	PGMEN	CPUMPEN	DIN	READEN	DLE	CLE	CEB
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
A															

Field	Name	RW	Reset	Description
24	WEB	R/W	0x0	WEB signal
23	RSTB	R/W	0x0	RSTB signal
22	PGMEN	R/W	0x0	PGMEN signal
21	CPUMPEN	R/W	0x0	CPUMPEN signal
20	DIN	R/W	0x0	DIN signal
19	READEN	R/W	0x0	READEN signal
18	DLE	R/W	0x0	DLE signal
17	CLE	R/W	0x0	CLE signal
16	CEB	R/W	0x0	CEB signal
13:0	A	R/W	0x0	A[13:0] signal

OTP-ROM Timing Configuration Register 0 (OTPTIME0)

0x9001B008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
tRW				tREW								tDS			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
tDH				tD2PGM				tCEW							

Field	Name	RW	Reset	Description
31:28	tRW	R/W	0x0	Should be greater than 20ns * Time = tCYC * (tRW+1)
27:24	tREW	R/W	0x0	Should be greater than 20ns * Time = tCYC * (tREW+1)
19:16	tDS	R/W	0x0	Should be greater than 20ns * Time = tCYC * (tDS+1)
15:12	tDH	R/W	0x0	Should be greater than 20ns * Time = tCYC * (tDH+1)
11:8	tD2PGM	R/W	0x0	Should be greater than 30ns * Time = tCYC * (tD2PGM+1)
7:4	tCEW	R/W	0x0	Should be greater than 20ns * Time = tCYC * (tCEW+1)

Note: tCYC is the memory bus clock period.

OTP-ROM Timing Configuration Register 1 (OTPTIME1)

0x9001B00C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
tREADEN								tACC							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
tPGMAS												tWWL			

Field	Name	RW	Reset	Description
31:24	tREADEN	R/W	0x0	Should be greater than 70ns * Time = tCYC * (tREADEN+1)
23:16	tACC	R/W	0x0	Should be greater than 70ns * Time = tCYC * (tACC+1)
15:8	tPGMAS	R/W	0x0	Should be greater than 100ns * Time = tCYC * (tPGMAS+1)
3:0	tWWL	R/W	0x0	Should be greater than 20ns * Time = tCYC * (tWWL+1)

OTP-ROM Timing Configuration Register 2 (OTPTIME2)

0x9001B010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
tCSRST															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
tCHRST															

Field	Name	RW	Reset	Description
31:16	tCSRST	R/W	0x0	Should be greater than 1090ns * Time = tCYC * (tCSRST+1)
15:0	tCHRST	R/W	0x0	Should be greater than 1000ns * Time = tCYC * (tCHRST+1)

OTP-ROM Timing Configuration Register 3 (OTPTIME3)

0x9001B014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
tPES															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
tPEH															

Field	Name	RW	Reset	Description
31:16	tPES	R/W	0x0	Should be greater than 1000ns * Time = tCYC * (tPES+1)
15:0	tPEH	R/W	0x0	Should be greater than 1000ns * Time = tCYC * (tPEH+1)

OTP-ROM Timing Configuration Register 4 (OTPTIME4)

0x73820018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
tRSRD															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
tRETRY															

Field	Name	RW	Reset	Description
31:16	tRSRD	R/W	0x0	Should be greater than 1000ns * Time = tCYC * (tRSRD+1)
15:0	tRETRY	R/W	0x0	Should be greater than 30ns * Time = tCYC * (tRETRY+1)

OTP-ROM Timing Configuration Register 5 (OTPTIME5)

0x9001B01C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
tCPS															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
tCPS															

Field	Name	RW	Reset	Description
17:0	tCPS	R/W	0x0	Should be greater than 3000ns * Time = tCYC * (tCPS+1) * "tCPS + tPW + tCPH" should be in 9500ns ~ 10500ns

OTP-ROM Timing Configuration Register 6 (OTPTIME6)

0x9001B020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
														tPW	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
tPW															

Field	Name	RW	Reset	Description
17:0	tPW	R/W	0x0	Should be in 4500ns ~ 5500ns * Time = tCYC * (tCPS+1) * "tCPS + tPW + tCPH" should be in 9500ns ~ 10500ns

OTP-ROM Timing Configuration Register 7 (OTPTIME7)

0x9001B024

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
														tCPH	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
tCPH															

Field	Name	RW	Reset	Description
17:0	tCPH	R/W	0x0	Should be greater than 2000ns * Time = tCYC * (tCPS+1) * "tCPS + tPW + tCPH" should be in 9500ns ~ 10500ns

OTP-ROM Secure Read Information Register 0 (OTPSECRD0)

0x9001B028

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SECRD[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECRD[15:0]															

Field	Name	RW	Reset	Description
31:0	SECRD[31:0]	W	0x0	Secure read data key value – the least significant word

OTP-ROM Secure Read Information Register 1 (OTPSECRD1)

0x9001B02C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SECRD[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECRD[47:32]															

Field	Name	RW	Reset	Description
31:0	SECRD[63:32]	W	0x0	Secure read data key value – the most significant word

The SECRD is used for key for secure region. The data in secure region will be read with pre-defined function with SECRD data as a key.

OTP-ROM Secure Address Register 1(OTPSECADDR)

0x9001B030

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IND															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR															

Field	Name	RW	Reset	Description
31	IND	R	0x0	Indirect Access for Secure Region 0 : direct access 1 : indirect access
8:0	ADDR	R/W	0x0	Indirect Address for Secure Region

21 Serial Flash Controller

21.1 Feature

- Serial Peripheral Interface support – Mode 0 and Mode 3
- Protocol Support
 - Single I/O, Dual I/O, Quad I/O
- Transfer Rate Support
 - Single Transfer Rate support
- Bus Interface
 - 1 x AHB Slave Bus Interface : Read Only I/F for Code Memory. 32bit data bus width
 - 1 x AHB Slave Bus Interface : Control Erase/Write by configuring Register.
 - 1 x Interrupt Source
- Dual-Quad SPI mode support
 - Two Quad SPI memory
- Micro Code Architecture
 - Flexible and easy-to-use
 - Serial Flash Memory Control
 - Interrupt & I/O Delay Control
 - Use SRAM for Code Memory
- I/O Delay configuration register
 - SIO/SCLK/RCLK delay control
- Maximum Operating Frequency.
 - STR : 166MHz

21.2 Block Diagram

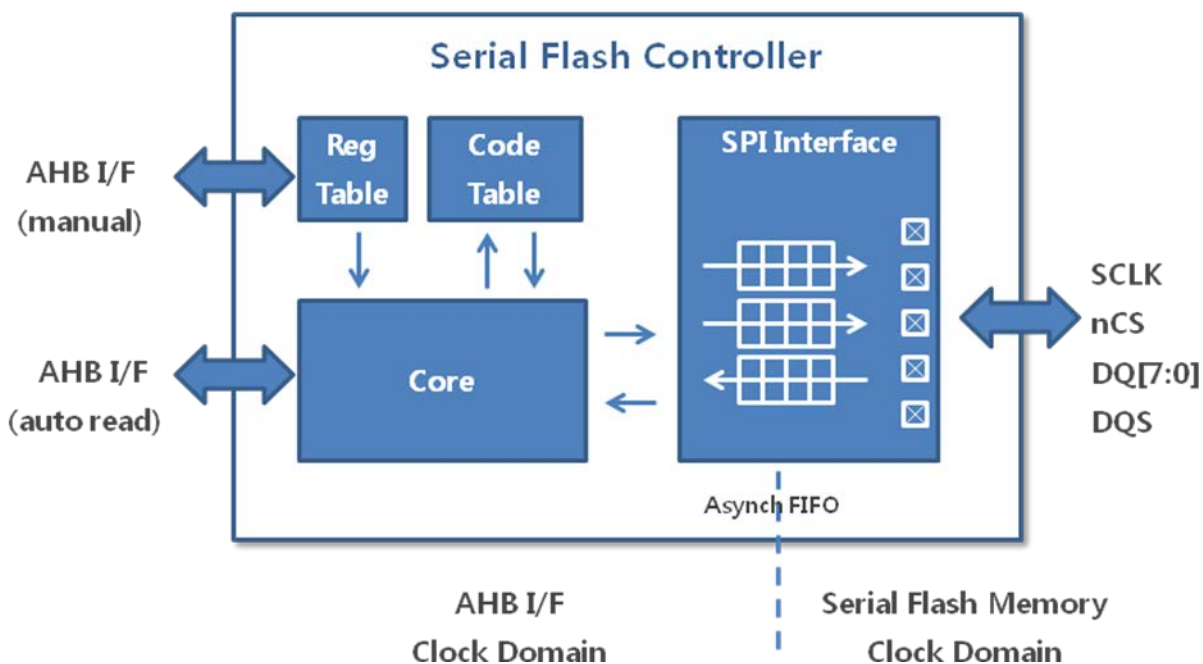


Figure 21.1 Serial NOR Flash Controller Block Diagram

21.3 Interfaces

21.3.1 Auto Read Interface

- AMBA 3 AHB-Lite Protocol support
- 32-bit data on AHB Slave Interface
- When write task, Error Response is generated.

- When RUN_AUTO register is "0", Read value is "DEADCODE".
- About INCR Burst protocol, the hardware changes it to single burst protocol internally.
- 8/16bit Burst Transaction
- The AHB slave interface does not support following protocol:
HTRANS.BUSY, HMASTLOCK, HPROT[3:0], HWDATA[31:0]

21.3.2 Manual Interface

- AMBA 3 AHB-Lite Protocol support
- 32-bit data on AHB Slave Interface
- About 8bit/16bit Write task, the hardware changes it to 32bit write task
- When write at out of Code Table, Error Response is generated.
- The AHB slave interface does not support following protocol:
HMASTLOCK, HPROT[3:0]

21.3.3 Serial Flash Memory Interface

- Serial Peripheral Interface Mode 0, Mode 3 support
- Single I/O, Dual I/O and Quad I/O support
- STR (Single Transfer Rate) Mode support

21.4 Operation

21.4.1 Instruction Set

31	30	29	28	27	26	25	24	23	12		11	4		3	0	
Cmd0	La	D	AR	R	IO	BS				Addr						Read
Cmd0	La	D	AA	Du	IO	BS				Addr						Write
Cmd0	La	D	BS		IO	Data(3/2/1 Byte)										Write Byte
Cmd0	Cmd1	OP		Cond	Addr1				Addr0						Operation	
Cmd0	Cmd1	R		Cond	Addr1				Addr0						Compare	
Cmd0	Cmd1	Cmd2	Cond		R				Addr						Branch	
Cmd0	Cmd1	Cmd2	Cond		R								Int		Interrupt	
Cmd0	Cmd1	Cmd2	Cond		Delay										Wait	
Cmd0	Cmd1	Cmd2	Cond		R										Stop	

Figure 21.2 Instruction Set Formats

Table 21.1 Keys about Instruction Set

Type	Description
Cmd0	Command 0. Classify SFM Read/Write command and other commands. 00b : Read command 01b : Write command 10b : Write Byte command 11b : Other commands
Cmd1	Command 1. Classify Op/CMP command and other commands 00b : Operation command 01b : Compare command 10b : Branch/Interrupt command 11b : Wait/Stop command
Cmd2	Command 2. Classify rest commands Ex) Cmd1_Cmd2 : Command type 10_00b : Branch command. 10_01b : Interrupt command 11_00b : Wait command 11_01b : Stop command
La	Last transaction. Determine CS# value. 0b : Keep CS# value 1b : Deassert CS#
D	Double Transfer Rate(DTR) mode enable 0b : DTR mode is disabled 1b : DTR mode is enabled (Not supported in TCC8132)
AR	AHB bus data bypassing enable 0b : Register table write

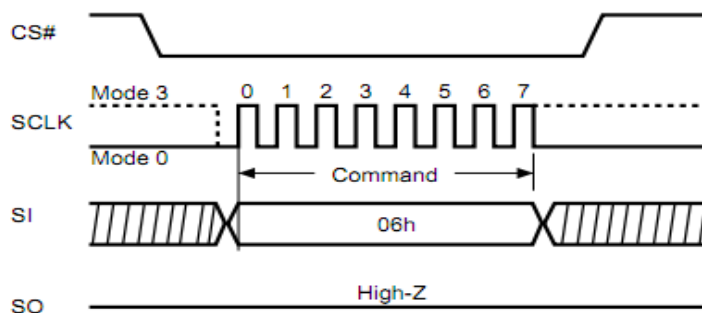
Type	Description
	1b : Bypass serial flash memory data to AHB bus
R	Reserved
IO	Using number of IO for transaction 00b : Single SPI 01b : Dual SPI 10b : Quad SPI 11b : Octa SPI (Not supported in TCC8132)
BS	Byte Size(Byte unit)
Addr	16-bit Address for code table
AA	AHB bus address bypassing enable 0b : Use register table data as address for serial flash memory 1b : Bypass AHB bus address to serial flash memory
Du	Dummy Flag. 0b : Writing data is actual real data. 1b : Writing data is dummy data.
Data	Write data
OP	Operations. 000b : Logical AND 001b : Logical OR 010b : Logical NOT 011b : Logical XOR 100b : Add 101b : Subtraction 110b : Multiplication 111b : Reserved
Cond	Conditional execution flags 00b : Always execute 01b : Execute in an equal case 10b : Execute in a not-equal case. 11b : Reserved
Addr1/0	12-bit Address for code table
Int	Interrupt Flags. 00b : Transaction is Finished 01b : Reserved 10b : Reserved 11b : Error
Delay	Clock cycle delay for wait mode

Table 21.2 Instruction

Mnemonic	Command	Action
RD	Read	Read byte size of data from flash memory to code table
WR	Write	Write byte size of data from code table to flash memory
WRB	Write Byte	Write (1/2/3)-Byte to flash memory
B	Branch	Branch to addr of code table
CMP	Compare	Compare data from addr1/addr0 of code table
INT	Interrupt	Generate an interrupt request to AHB bus master
OP	Operation	Arithmetic/logical operation between data from addr1/addr0 of code table
STOP	Stop	Stop the operation and wait for the next event

21.4.2 Instruction Set Examples

- Write Enable Sequence (SPI Mode)



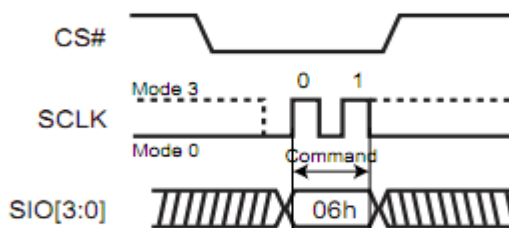
Code:

```

0 : WRB      LAST      1IO      1(Byte)      0x06
1 : STOP

```

- Write Enable Sequence (QPI Mode)



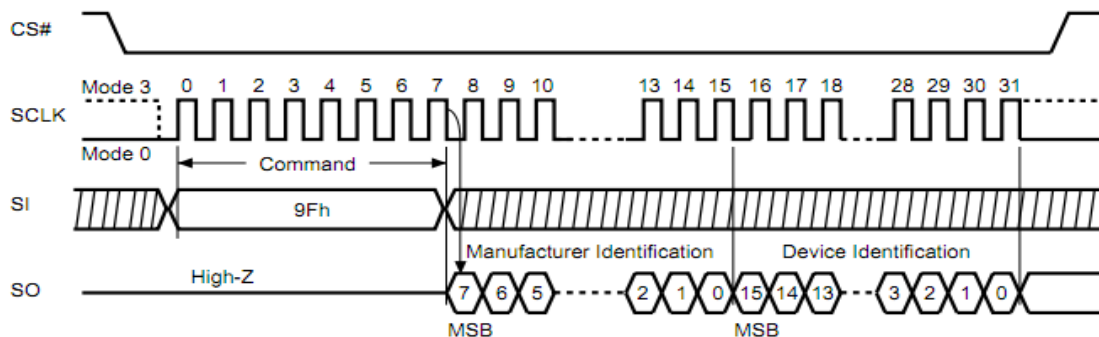
Code:

```

0 : WRB      LAST      4IO      1(Byte)      0x06
1 : STOP

```

- Read Identification(RDID) Sequence (SPI Mode only)



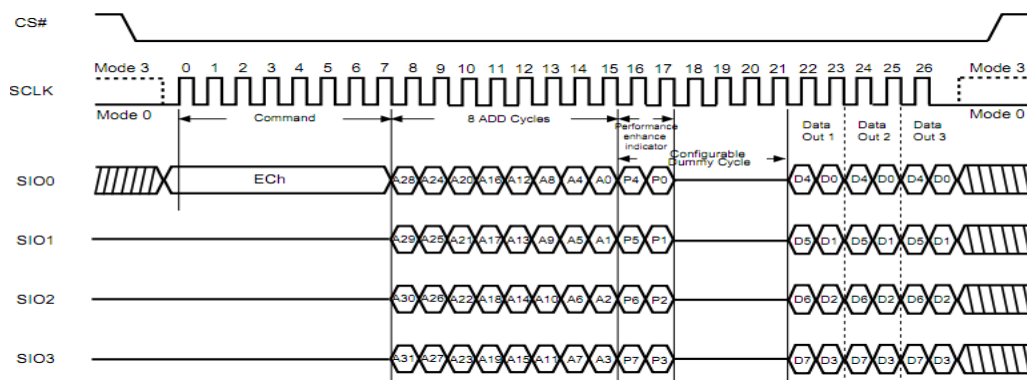
Code:

```

0 : WRB      1IO      1(Byte)      0x9F
1 : RD      LAST      1IO      2(Byte)      0x10      // read data is stored at 0x10
2 : STOP

```

- 4-I/O Fast Read using 4-Byte Address Sequence (4READ4B)



Code:

0 : WRB	1IO	1(Byte)	0xEC	
1 : WR	4IO	4(Byte)	0x40	// Address of code table
2 : WRB	4IO	1(Byte)	0xA5	// performance enhance indicator
3 : WR_dummy	4IO	2(Byte)		
4 : RD	LAST	4IO	3(Byte)	0x50 // read data is stored at 0x50
5 : STOP				

21.4.3 Interrupts

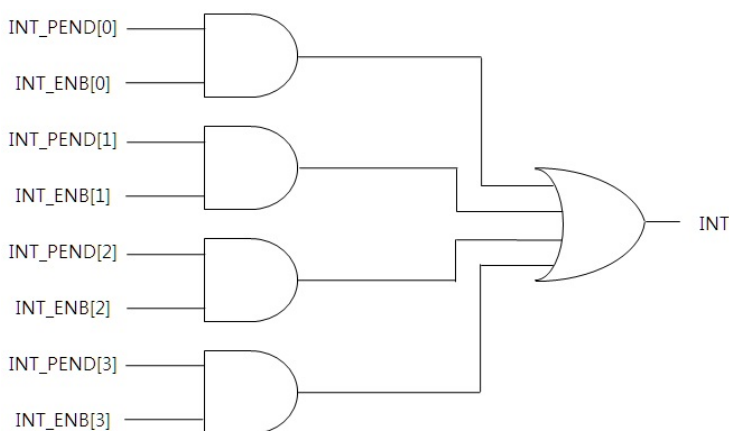


Figure 21.3 Block Diagram of Interrupts in Serial Flash Controller

Originally, you can use User-Interrupt 0~3 & Timeout Interrupt & Core-Error Interrupt, by using Interrupt Instruction as example below. But you can only use User-Interrupt 3 & Timeout Interrupt after revision for support 3-Byte mode read task. Refer to this hardware limitation for Interrupt using.

addr(Code_Table)	Command	Value
0x100	Interrupt	0x1

21.4.4 4B-mode vs. 3B-mode

	Read	Write	Write Byte	Op	Compare	Branch	irq	Wait	Stop
31	0x0	0x1	0x2	0x3	0x3	0x3	0x3	0x3	0x3
30	LAST	LAST	LAST	0x0	0x1	0x2	0x2	0x3	0x3
29	DTR	DTR	DTR						
28	AHB rdata	AHB addr	Byte Size	OP	-	0x0	0x1	0x0	0x1
27	-	Dummy	IO	Cond	Cond	Cond	Cond	Cond	Cond
26	IO	IO							
25	Byte Size	Byte Size	Data (3/2/1Byte)	Addr1	Addr1	-	-	Delay	-
24									
23									
22									
21	Addr	Addr		Addr0	Addr0	Addr	Int		
20									
19									
18									
17	3B/4B Mode								
16									
15									
14									
13									
12									
11									
10									
9									
8									
7									
6									
5									
4									
3									
2									
1									
0									

For using 3B-mode (24bit address), you should control [0]-bit of Write Instruction as the table above.
You can control 32bit address mode or 24bit address mode as the reference code below.

QREAD4B vs. QREAD3B
<pre> If (4B_MODE == 1) { *(volatile unsigned int *) (SFMC_REG_CODE_TABLE + 0x14) = 0x8400006C; //QREAD4B command *(volatile unsigned int *) (SFMC_REG_CODE_TABLE + 0x18) = 0x48000000; //4B-mode (32bit address) } else { // 3B-mode *(volatile unsigned int *) (SFMC_REG_CODE_TABLE + 0x14) = 0x8400006B; //QREAD3B command *(volatile unsigned int *) (SFMC_REG_CODE_TABLE + 0x18) = 0x48000001; //3B-mode (24bit address) } *(volatile unsigned int *) (SFMC_REG_CODE_TABLE + 0x1c) = 0x44001000; *(volatile unsigned int *) (SFMC_REG_CODE_TABLE + 0x20) = 0x2A000000; *(volatile unsigned int *) (SFMC_REG_CODE_TABLE + 0x24) = 0xF4000000 </pre>

21.5 Register Description

Table 21.3 Serial Flash Controller Register Map (Base Address = 0x80005000)

Name	Address	RW	Reset	Description
MAGIC	0x00	R	parameter	IP Number
VERSION	0x04	R	parameter	Version Number
PARAM	0x08	R	parameter	Num of Code Register
STATUS	0x0C	R	0x0	FIFO Full, Empty Status Register
RUN	0x10	R/W	0x0	Manual Mode, Auto Mode Run Register
INT_PEND	0x14	R/W	0x0	Interrupt Pending Register
INT_ENB	0x18	R/W	0x0	Interrupt Enable Register
BASE_ADDR_MAN	0x1C	R/W	0x0	Manual Mode Start Address Register
BASE_ADDR_AUTO	0x20	R/W	0x0	Auto Mode Start Address Register
MODE	0x24	R/W	0x0	Mode Control Register
TIMING	0x28	R/W	0x0	Timing Control Register
DELAY_SO	0x2C	R/W	0x0	Delay Control0 Register
DC_CLK	0x30	R/W	0x0	Delay Control1 Register
DC_WBD0	0x34	R/W	0x0	SO Write Bit Delay Control Register
DC_WBD1	0x38	R/W	0x0	SO Write Bit Delay Control Register
DC_RBD0	0x3C	R/W	0x0	SI Read Bit Delay Control Register
DC_RBD1	0x40	R/W	0x0	SI Read Bit Delay Control Register
DC_WOEBD0	0x44	R/W	0x0	SO Write Output Enable Bit Delay Control Register
DC_WOEBD1	0x48	R/W	0x0	SO Write Output Enable Bit Delay Control Register
TIMEOUT	0x4C	R/W	0x0	Timeout Control Register
CODE_TABLE	0x800 ~ 0x9FC	R/W	0x0	Code Memory Area, size : 32bit x 128ea

IP Number (MAGIC)

0x80005000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MAGIC															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MAGIC															

Field	Name	RW	Reset	Description
31:0	MAGIC	R	parameter	IP Number

Version Number (VERSION)

0x80005004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VERSION															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VERSION															

Field	Name	RW	Reset	Description
31:0	VERSION	R	parameter	Version Number

Num of Code Register (PARAM)

0x80005008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PARAM															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PARAM															

Field	Name	RW	Reset	Description
31:0	PARAM	R	Parameter	Number of Code Register

FIFO Full, Empty Status Register (STATUS)

0x8000500C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
												FIFO_FULL			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								FIFO_EMPTY							
												PC			

Field	Name	RW	Reset	Description
31:21	-	-	-	Reserved
20:16	FIFO_FULL	R	0x00	FIFO Full Status [20]: CORE_FIFO full [19]: RD_FIFO full [18]: WR_FIFO full [17]: CMD_FIFO full [16]: RUN_FIFO full
15:13	-	-	-	Reserved
12:8	FIFO_EMPTY	R	0x0	FIFO Empty Status [12]: CORE_FIFO empty [11]: RD_FIFO empty [10]: WR_FIFO empty [9]: CMD_FIFO empty [8]: RUN_FIFO empty
7:0	PC	R	0x00	Current Program Counter

Manual Mode, Auto Mode Run Register (RUN)

0x80005010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
							SOFT_RESET			RUN_AUTO				RUN_MAN	

Field	Name	RW	Reset	Description
31:9	-	-	-	Reserved
8	SOFT_RESET	R/W	0	Soft Reset 0 : Soft Reset Clear 1 : Soft Reset
7:6	-	-	-	Reserved
5:4	RUN_AUTO	R/W	0x0	Auto Read Mode Run - Read 0 : Idle 1 : Run 3 : Stop (auto clear) - Write 0, 2 : Don't care 1 : Run 3 : Stop
3:2	-	-	-	Reserved
1:0	RUN_MAN	R/W	0x0	Manual Mode Run - Read 0 : Idle 1 : Run 3 : Stop (auto clear) - Write 0, 2 : Don't care 1 : Run 3 : Stop

Interrupt Pending Register (INT_PEND)

0x80005014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
										INT_ERROR		INT_PEND			

Field	Name	RW	Reset	Description
31:6	-	-	-	Reserved
5:4	INT_ERROR	R/W	0x0	Error Pending Interrupt Register [5] : Timeout Interrupt [4] : Core Error Interrupt
3:0	INT_PEND	R/W	0x0	Code Interrupt Pending Register [3] : User Interrupt Pending 3 [2] : User Interrupt Pending 2 [1] : User Interrupt Pending 1 [0] : User Interrupt Pending 0

Interrupt Enable Register (INT_ENB)

0x80005018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
										INT_ERR_ENB		INT_ENB			

Field	Name	RW	Reset	Description
31:6	-	-	-	Reserved
5:4	INT_ERR_ENB	R/W	0x0	Error Interrupt Enable Register [5] : Timeout Interrupt Enable [4] : Core Error Interrupt Enable
3:0	INT_ENB	R/W	0x0	Interrupt Enable Register [3] : User Interrupt Enable 3 [2] : User Interrupt Enable 2 [1] : User Interrupt Enable 1 [0] : User Interrupt Enable 0

Manual Mode Start Address Register (BASE_ADDR_MAN)

0x8000501C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

BASE_ADDR_MAN

Field	Name	RW	Reset	Description
31:12	-	-	-	Reserved
11:0	BASE_ADDR_MAN	R/W	0x0	Manual Mode Start Address

Auto Mode Start Address Register (BASE_ADDR_AUTO)

0x80005020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

BASE_ADDR_AUTO

Field	Name	RW	Reset	Description
31:12	-	-	-	Reserved
11:0	BASE_ADDR_AUTO	R/W	0x0	Auto Mode Start Address

Mode Control Register (MODE)

0x80005024

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SIO_OFF								CROSS_RD	CROSS_WR	MODE_SERIAL			FIFO_CTRL	MODE_DQUAD	FLASH_RESET

Field	Name	RW	Reset	Description
31:16	-	-	-	Reserved
15:8	SIO_OFF	R/W	0x00	SIO Data Off 0xF0 : Quad Memory 0x00 : Octa or Dual Quad Memory (Not supported in TCC8132)
7	CROSS_RD	R/W	0x0	Read Data Cross Enable (only Octa DTR Mode) 0 : Disable 1 : Cross RDATA[31:0] -> {RDATA[23:16],RDATA[31:24],RDATA[7:0],RDATA[15:8]}; (Not supported in TCC8132)
6	CROSS_WR	R/W	0x0	Write Data Cross Enable (only Octa DTR Mode) 0 : Disable 1 : Cross WDATA[31:0] -> {WDATA[23:16],WDATA[31:24],WDATA[7:0],WDATA[15:8]}; (Not supported in TCC8132)
5:4	MODE_SERIAL	R/W	0x0	Serial Mode 0 : Disable clock gating 1 : Serial Mode 0 2 : Serial Mode 3
3	-	-	-	Reserved
2	FIFO_CTRL	R/W	0x0	FIFO_CTRL 0 : Disable FIFO_CTRL 1 : Enable FIFO_CTRL
1	MODE_DQUAD	R/W	0x0	Dual Quad Mode Enable (Active High)
0	FLASH_RESET	R/W	0x0	Serial Flash Memory Reset Control

Timing Control Register (TIMING)

0x80005028

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
										CS_EXTEND		CS_TO_CS			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
				READ_LATENCY						SEL_DQS		SEL_PN		SEL_N_180	SEL_P_180

Field	Name	RW	Reset	Description
31:22	-	-	-	Reserved
21:20	CS_EXTEND	R/W	0x0	Chip Select Extension - Extend CS inactive timing - FCLK Clock cycle based 0 : Not extend 1 : 1 tFCLK Extend 2 : 2 tFCLK Extend 3 : 3 tFCLK Extend
19:16	CS_TO_CS	R/W	0x0	CS to CS Timing (minimum timing) 0 : 1 clock 1 : 2 clock ~ 15 : 16 clock
15:12	-	-	-	Reserved
11:8	READ_LATENCY	R/W	0x0	Read Latency 0 : No latency 1 : 1 clock latency ~ 7 : 7 clock latency
7:6	-	-	-	Reserved
5:4	SEL_DQS	R/W	0x0	Select DQS Clock 0: fclk 1: fclk (pad input) 2: i_dqs (Octa only) 3: ~fclk
3:2	SEL_PN	R/W	0x0	Select p data - must be set to 1 in DTR mode (Not supported in TCC8132)
1	SEL_N_180	R/W	0x0	Select rd_n_180 data
0	SEL_P_180	R/W	0x0	Select rd_p_180 data

Delay Control0 Register (DELAY_SO)

0x8000502C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					INV_SCLK	SLCH			SLDH						

Field	Name	RW	Reset	Description
31:11	-	-	-	Reserved
10	INV_SCLK	R/W	0x0	Invert SCLK
9:8	SLCH	R/W	0x0	CS Low to SCLK Latency 0~2
7:0	SLDH	R/W	0x0	CS Low to SO Latency [7] : SO[7] delay (Not supported in TCC8132) [6] : SO[6] delay (Not supported in TCC8132) [5] : SO[5] delay (Not supported in TCC8132) [4] : SO[4] delay (Not supported in TCC8132) [3] : SO[3] delay [2] : SO[2] delay [1] : SO[1] delay [0] : SO[0] delay

Delay Control1 Register (DC_CLK)

0x80005030

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
									CLK_RD_TAB	CLK_WR_TAB	CSN_BD				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CLK_RD								CLK_WD							

Field	Name	RW	Reset	Description
31:23	-	-	-	Reserved
22	CLK_RD_TAB	R/W	0x0	CLK_RD tab_delay_chg
21	CLK_WR_TAB	R/W	0x0	CLK_WD tab_delay_chg
20:16	CSN_BD	R/W	0x0	Flash Memory CSN Signal Delay
15:8	CLK_RD	R/W	0x0	Read Clock Delay [15:14] : delay_ctrl [13:8] : clk_buf_sel
7:0	CLK_WD	R/W	0x0	Flash Memory SCLK Clock Delay (0xE7) [7:6] : delay_ctrl [5:0] : clk_buf_sel

SO Write Bit Delay Control Register (DC_WBD0)

0x80005034

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
S3WBD								S2WBD							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
S1WBD								S0WBD							

Field	Name	RW	Reset	Description
31:24	S3WBD	R/W	0x0	Flash Memory SO[3] Write Bit Delay
23:16	S2WBD	R/W	0x0	Flash Memory SO[2] Write Bit Delay
15:8	S1WBD	R/W	0x0	Flash Memory SO[1] Write Bit Delay
7:0	S0WBD	R/W	0x0	Flash Memory SO[0] Write Bit Delay

SO Write Bit Delay Control Register (DC_WBD1)

0x80005038

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
S7WBD								S6WBD							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
S5WBD								S4WBD							

Field	Name	RW	Reset	Description
31:24	S7WBD	R/W	0x0	Flash Memory SO[7] Write Bit Delay (Not supported in TCC8132)
23:16	S6WBD	R/W	0x0	Flash Memory SO[6] Write Bit Delay (Not supported in TCC8132)
15:8	S5WBD	R/W	0x0	Flash Memory SO[5] Write Bit Delay (Not supported in TCC8132)
7:0	S4WBD	R/W	0x0	Flash Memory SO[4] Write Bit Delay (Not supported in TCC8132)

SI Read Bit Delay Control Register (DC_RBD0)

0x1B00003C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
S3RBD								S2RBD							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
S1RBD								S0RBD							

Field	Name	RW	Reset	Description
31:24	S3RBD	R/W	0x0	Flash Memory SI[3] Read Bit Delay
23:16	S2RBD	R/W	0x0	Flash Memory SI[2] Read Bit Delay
15:8	S1RBD	R/W	0x0	Flash Memory SI[1] Read Bit Delay
7:0	S0RBD	R/W	0x0	Flash Memory SI[0] Read Bit Delay

SI Read Bit Delay Control Register (DC_RBD1)

0x80005040

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
S7RBD								S6RBD							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
S5RBD								S4RBD							

Field	Name	RW	Reset	Description
31:24	S7RBD	R/W	0x0	Flash Memory SI[7] Read Bit Delay (Not supported in TCC8132)
23:16	S6RBD	R/W	0x0	Flash Memory SI[6] Read Bit Delay (Not supported in TCC8132)
15:8	S5RBD	R/W	0x0	Flash Memory SI[5] Read Bit Delay (Not supported in TCC8132)
7:0	S4RBD	R/W	0x0	Flash Memory SI[4] Read Bit Delay (Not supported in TCC8132)

SO Write Output Enable Bit Delay Control Register (DC_WOEBD0)

0x80005044

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
S3WOEBD								S2WOEBD							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
S1WOEBD								S0WOEBD							

Field	Name	RW	Reset	Description
31:24	S3WOEBD	R/W	0x0	Flash Memory SO[3] Write Output Enable Bit Delay
23:16	S2WOEBD	R/W	0x0	Flash Memory SO[2] Write Output Enable Bit Delay
15:8	S1WOEBD	R/W	0x0	Flash Memory SO[1] Write Output Enable Bit Delay
7:0	S0WOEBD	R/W	0x0	Flash Memory SO[0] Write Output Enable Bit Delay

SO Write Output Enable Bit Delay Control Register (DC_WOEBD1)

0x80005048

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
S7WOEBD								S6WOEBD							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
S5WOEBD								S4WOEBD							

Field	Name	RW	Reset	Description
31:24	S7WOEBD	R/W	0x0	Flash Memory SO[7] Write Output Enable Bit Delay (Not supported in TCC8132)
23:16	S6WOEBD	R/W	0x0	Flash Memory SO[6] Write Output Enable Bit Delay (Not supported in TCC8132)
15:8	S5WOEBD	R/W	0x0	Flash Memory SO[5] Write Output Enable Bit Delay (Not supported in TCC8132)
7:0	S4WOEBD	R/W	0x0	Flash Memory SO[4] Write Output Enable Bit Delay (Not supported in TCC8132)

Timeout Control Register (TIMEOUT)

0x8000504C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
															ENB
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMEOUT_CNT															

Field	Name	RW	Reset	Description
31:17	-	-	-	Reserved
16	TIMEOUT_ENB	R/W	0x0	Timeout Enable
15:0	TIMEOUT_CNT	R/W	0x0	Timeout Counter [15:0]

Code Memory Area (CODE_TABLE)

0x80005800~9FC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CODE_TABLE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CODE_TABLE															

Field	Name	RW	Reset	Description
31:0	CODE_TABLE	R/W	0x0	Code Memory Area - Size : 32bit x 128 depth

22 NFC (NAND Flash Controller)

22.1 Function Description

The TCC8132 has a NAND flash controller which can support various type of NAND flash such as 532(512 + 16) bytes/page, 2112(2048+64) bytes/page, 4314(4096 + 218) bytes/page NAND, 8644(8192 +448) bytes/page NAND, and so on.

The supportable configuration for external NAND flash is as follows.

Case1) single NAND flash of 8-bit bus-width. (8bit I/O BUS configuration)

The supportable main features are as follows

- 8 bit I/O bus width is supportable
- AHB Master I/F with DMA Controller for burst data transfer
- AHB Slave I/F for command, address and single/burst data transfer.
- 1 page burst program and read (Maximum page size : 8192Bytes + ECC size)
- Configurable read data capture timing to reduce read cycle period
- NFC dedicated memory (2048Bytes) for spare data (ECC or system related data)
- NAND 1 Ready is supportable
- NFC dedicated ECC encoding/decoding capability
 - SLC ECC
 - MLC 4/6/12/16/24 bit ECC
 - MLC ECC Error Correction Pass

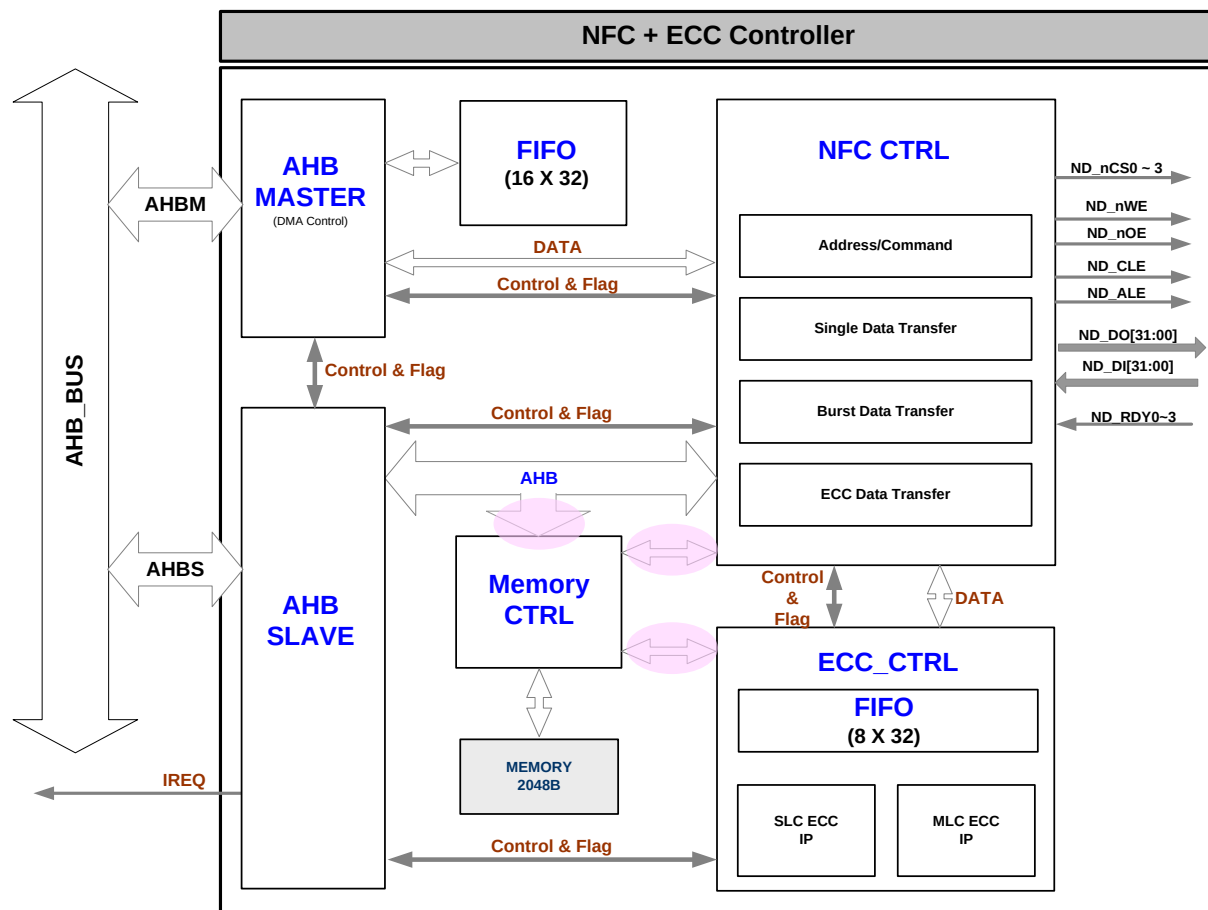


Figure 22.1 NAND Flash Controller Block Diagram

22.1.1 NAND Burst Program / Read By DMA

The TCC8132 NFC (NAND Flash Controller) provides two modes of NAND Burst Program/Read. One is General DMA (GDMA) and the other is NFC dedicated DMA (NDMA).

1) Burst Program / Read by GDMA (General DMA) with ECC

In this case, NAND burst program and NAND burst read operation can be executed by access to NAND Linear Data Register (NFC_LDATA) through GDMA. Maximum transfer data size is limited to 1024 Bytes (Maximum ECC Data Size). If 1 page data size of NAND is larger than 1024 Bytes, the data is divided into less than 1024 Bytes in order to transmit DMA and to execute ECC.

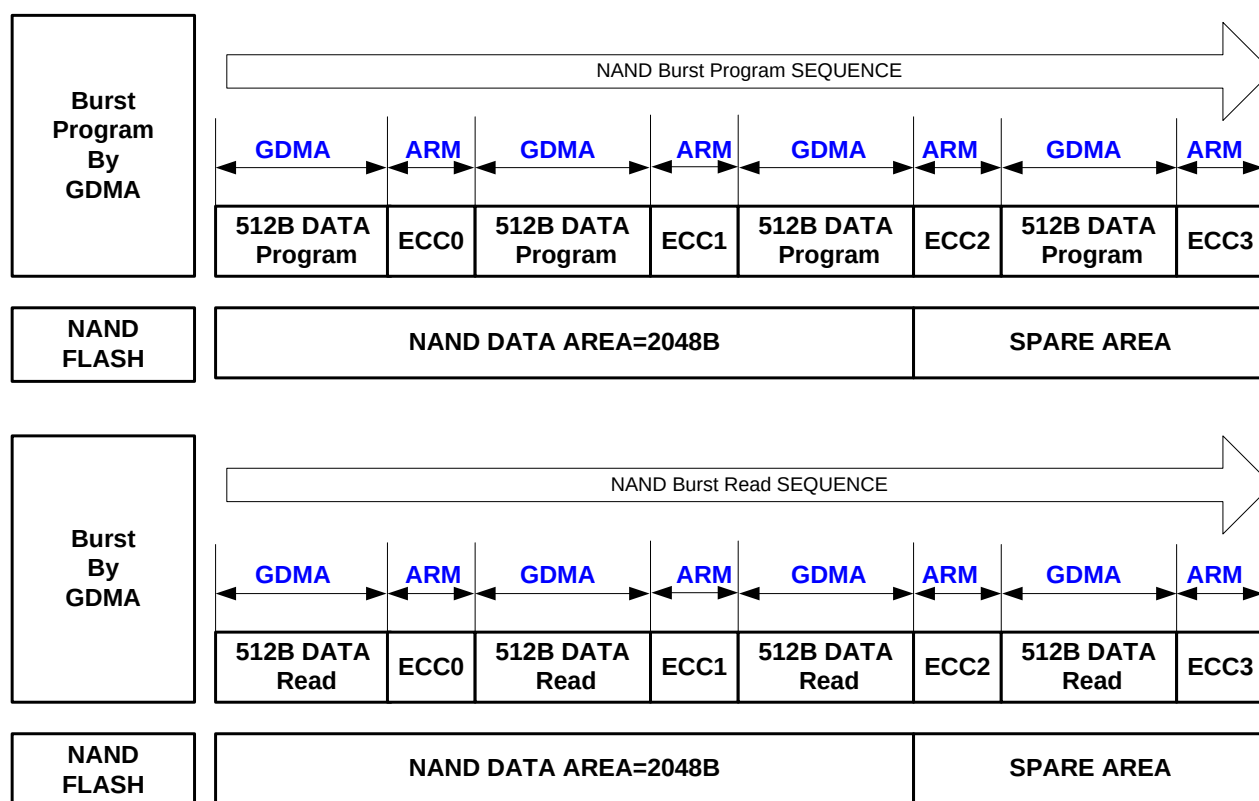


Figure 22.2 Structure of NAND Flash for Burst Program/Read By GDMA

2) Burst Program/Read by NDMA (NFC DMA) with ECC

NAND burst program and read can be possible by NFC dedicated DMA (NDMA). In this case, even though a page size is larger than 1024 Bytes, 1 page burst program and read operation are possible. However, data transfer and ECC are internally executed per each subpage and each subpage should be configured by related register. The number of subpage can be set up to 32. The maximum data size of the subpage is 1024 bytes.

Encoded/decoded ECC of each subpage data is orderly saved to NFC dedicated memory. For the saved ECC data, burst transfer is automatically executed after all of the subpaged data (subpage X N) is transferred. (NFC_CTRL[BMODE] == 1) For example, if 1 page data area of NAND is 2048 bytes and ECC requirement is 12bit/512bytes, the number of sub-pages should be set to 4 and the data size of each subpage should be set 512 bytes in order to execute 1 page burst program and read operation via NDMA. The following figure is the page structure of NAND flash when burst read or program by NDMA is executed.

As a result, the NDMA can provide better bandwidth than GDMA and reduce bus traffic.

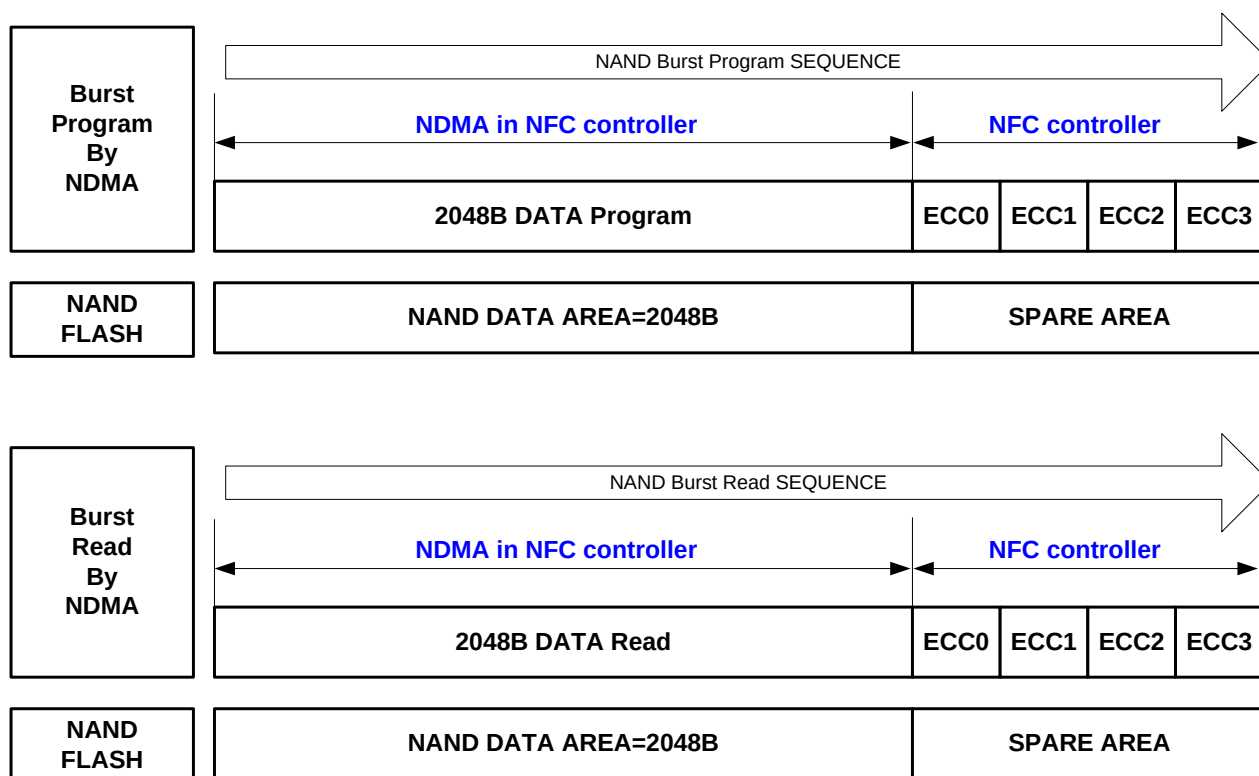


Figure 22.3 Structure of NAND Flash for Burst Program/Read By NDMA

22.1.2 NAND Ready Configuration

In TCC8132, the READY input can be configured to 1 ready or 4 ready input of NAND Flash by NFC_CTRL[RDYSEL]. In case of 4 ready, ND_nCS[N](N=0,...,3) can mask its own ready input.

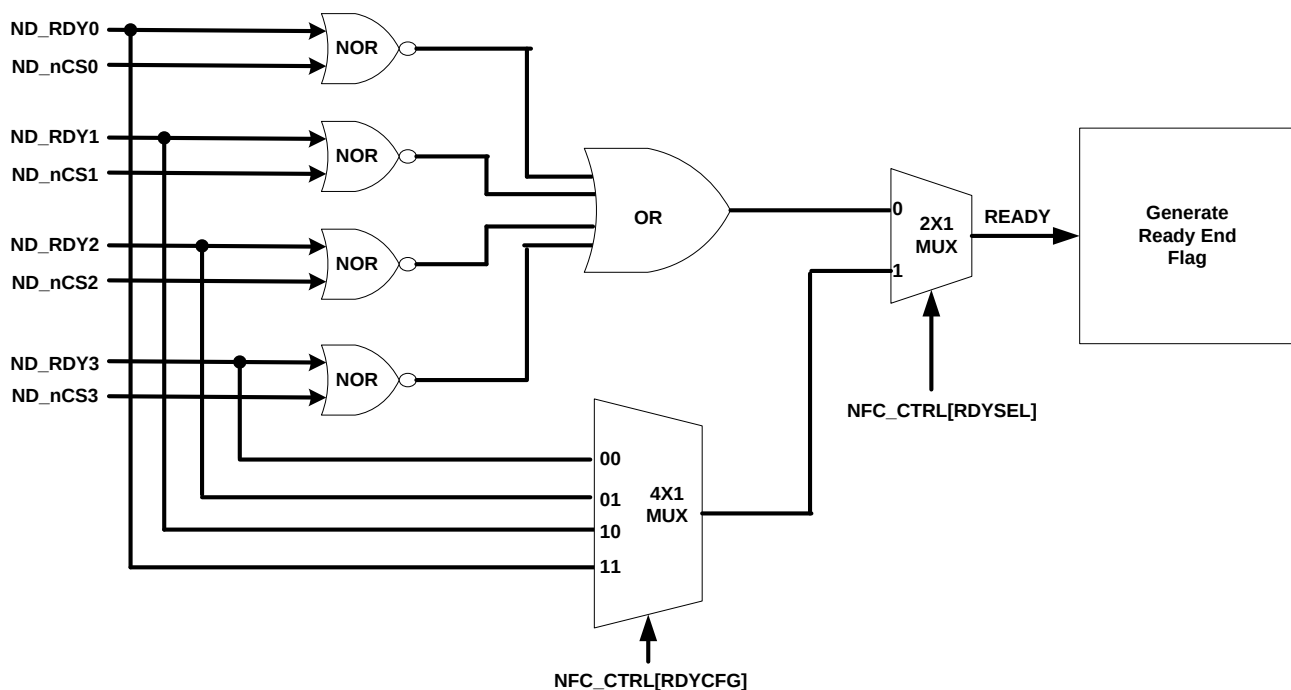


Figure 22.4 Ready

22.2 Register Description

Table 22.1 NAND Flash Controller Register Map (Base Address=0x8000D000)

Name	Address	Type	Reset	Description
NFC_CMD	0x00	W	-	NAND Flash Command Register
NFC_LADDR	0x04	W	-	NAND Flash Linear Address Register
NFC_SADDR	0x08	W	-	NAND Flash Signal Address Register
NFC_SDAT	0x0C	R/W	-	NAND Flash Single Data Register
NFC_WDAT	0x1x	R/W	Unknown	NAND Flash Word Data Register
NFC_LDAT	0x2x/3x	R/W	Unknown	NAND Flash Linear Data Register
NFC_MDAT	0x40	R/W	Unknown	NAND Flash Multiple Data Register
NFC_CACYC	0x44	R/W	0x00010101	NAND Flash Command & Address Cycle Register
NFC_WRCYC	0x48	R/W	0x00010101	NAND Flash Write Cycle Register
NFC_RECYC	0x4C	R/W	0x00010101	NAND Flash Read Cycle Register
NFC_CTRL	0x50	R/W	0xF006C800	NFC Control Register
NFC_DSIZ	0x54	R/W	0x0000FFFF	NFC Data Size Register
NFC_SPARE	0x58	R/W	0x00000200	NFC Spare Burst Transfer Control Register
NFC_PRSTRT	0x5C	R/W	0x00000000	NFC Burst Program/Read Start
NFC_RST	0x60	R/W	0x00000000	NFC Reset Register
NFC_IRQCFG	0x64	R/W	0x00000000	NFC Interrupt Request Control Register
NFC_IRQ	0x68	R/W	0x00000000	NFC Interrupt Request Register
NFC_STATUS	0x6C	R/W	-	NFC Status Register
NFC_RFWBASE	0x70	R/W	0x00000000	NFC Read Flag Wait Base Word Register
NDMA_ADDR	0x100	R/W	0x00000000	NFC DMA Source/Destination Register
NDMA_PARAM	0x104	R/W	0x00000000	NFC DMA Parameter Register
NDMA_CADDR	0x108	R	0x00000000	NFC DMA Current Address Register
NDMA_CTRL	0x10C	R/W	0x00000000	NFC DMA Control Register
NDMA_SPCTRL	0x110	R/W	0x00000000	NFC DMA SUBPAGE Control Register
NDMA_STATUS	0x114	R/W	0x00000000	NFC DMA Status Register
SP_CFG0	0x120	R/W	0x00000000	NFC Sub Page Configuration Register 0
SP_CFG1	0x124	R/W	0x00000000	NFC Sub Page Configuration Register 1
SP_CFG2	0x128	R/W	0x00000000	NFC Sub Page Configuration Register 2
SP_CFG3	0x12C	R/W	0x00000000	NFC Sub Page Configuration Register 3
SP_CFG4	0x130	R/W	0x00000000	NFC Sub Page Configuration Register 4
SP_CFG5	0x134	R/W	0x00000000	NFC Sub Page Configuration Register 5
SP_CFG6	0x138	R/W	0x00000000	NFC Sub Page Configuration Register 6
SP_CFG7	0x13C	R/W	0x00000000	NFC Sub Page Configuration Register 7
SP_CFG8	0x140	R/W	0x00000000	NFC Sub Page Configuration Register 8
SP_CFG9	0x144	R/W	0x00000000	NFC Sub Page Configuration Register 9
SP_CFG10	0x148	R/W	0x00000000	NFC Sub Page Configuration Register 10
SP_CFG11	0x14C	R/W	0x00000000	NFC Sub Page Configuration Register 11
SP_CFG12	0x150	R/W	0x00000000	NFC Sub Page Configuration Register 12
SP_CFG13	0x154	R/W	0x00000000	NFC Sub Page Configuration Register 13
SP_CFG14	0x158	R/W	0x00000000	NFC Sub Page Configuration Register 14
SP_CFG15	0x15C	R/W	0x00000000	NFC Sub Page Configuration Register 15
SP_CFG16	0x160	R/W	0x00000000	NFC Sub Page Configuration Register 16
SP_CFG17	0x164	R/W	0x00000000	NFC Sub Page Configuration Register 17
SP_CFG18	0x168	R/W	0x00000000	NFC Sub Page Configuration Register 18
SP_CFG19	0x16C	R/W	0x00000000	NFC Sub Page Configuration Register 19
SP_CFG20	0x170	R/W	0x00000000	NFC Sub Page Configuration Register 20
SP_CFG21	0x174	R/W	0x00000000	NFC Sub Page Configuration Register 21
SP_CFG22	0x178	R/W	0x00000000	NFC Sub Page Configuration Register 22
SP_CFG23	0x17C	R/W	0x00000000	NFC Sub Page Configuration Register 23
SP_CFG24	0x180	R/W	0x00000000	NFC Sub Page Configuration Register 24
SP_CFG25	0x184	R/W	0x00000000	NFC Sub Page Configuration Register 25
SP_CFG26	0x188	R/W	0x00000000	NFC Sub Page Configuration Register 26
SP_CFG27	0x18C	R/W	0x00000000	NFC Sub Page Configuration Register 27
SP_CFG28	0x190	R/W	0x00000000	NFC Sub Page Configuration Register 28
SP_CFG29	0x194	R/W	0x00000000	NFC Sub Page Configuration Register 29
SP_CFG30	0x198	R/W	0x00000000	NFC Sub Page Configuration Register 30
SP_CFG31	0x19C	R/W	0x00000000	NFC Sub Page Configuration Register 31
ECC_CTRL	0x200	R/W	0x00000000	ECC Control Register
SECC_ERRNUM	0x204	R/W	-	SLC ECC Error Number Register
MECC_ERRNUM	0x204	R/W	-	MLC ECC Error Number Register

Name	Address	Type	Reset	Description
ECC_BASE	0x208	R/W	0x8000D800	ECC Base Address
ECC_MASK	0x20C	R/W	0x00008000	ECC Mask Address
ECC_CLEAR	0x210	R/W	0x00000000	NFC ECC Register Clear
ECC_RSTART	0x214	R/W	-	NFC ECC Decoding Start
ECC_STLDCTL	0x218	R/W	-	NFC ECC Store Status and Load Control Register
ECC_STATUS	0x21C	R/W	-	NFC ECC STATUS Register
SECC_0	0x220	R/W	0x00000000	1st SLC ECC Code Register
SECC_1	0x224	R/W	0x00000000	2nd SLC ECC Code Register
SECC_2	0x228	R/W	0x00000000	3rd SLC ECC Code Register
SECC_3	0x22C	R/W	0x00000000	4th SLC ECC Code Register
SECC_4	0x230	R/W	0x00000000	5th SLC ECC Code Register
SECC_5	0x234	R/W	0x00000000	6th SLC ECC Code Register
SECC_6	0x238	R/W	0x00000000	7th SLC ECC Code Register
SECC_7	0x23C	R/W	0x00000000	8th SLC ECC Code Register
SECC_8	0x240	R/W	0x00000000	9th SLC ECC Code Register
SECC_9	0x244	R/W	0x00000000	10th SLC ECC Code Register
SECC_10	0x248	R/W	0x00000000	11th SLC ECC Code Register
SECC_11	0x24C	R/W	0x00000000	12th SLC ECC Code Register
SECC_12	0x250	R/W	0x00000000	13th SLC ECC Code Register
SECC_13	0x254	R/W	0x00000000	14th SLC ECC Code Register
SECC_14	0x258	R/W	0x00000000	15th SLC ECC Code Register
SECC_15	0x25C	R/W	0x00000000	16th SLC ECC Code Register
MECC4_0	0x220	R/W	0x00000000	1st MLC ECC4 Code Register
MECC4_1	0x224	R/W	0x00000000	2nd MLC ECC4 Code Register
MECC6_0	0x220	R/W	0x00000000	1st MLC ECC6 Code Register
MECC6_1	0x224	R/W	0x00000000	2nd MLC ECC6 Code Register
MECC6_2	0x228	R/W	0x00000000	3rd MLC ECC6 Code Register
MECC12_0	0x220	R/W	0x00000000	1st MLC ECC12 Code Register
MECC12_1	0x224	R/W	0x00000000	2nd MLC ECC12 Code Register
MECC12_2	0x228	R/W	0x00000000	3rd MLC ECC12 Code Register
MECC12_3	0x22C	R/W	0x00000000	4th MLC ECC12 Code Register
MECC12_4	0x230	R/W	0x00000000	5 th MLC ECC12 Code Register
MECC12_5	0x234	R/W	0x00000000	6 th MLC ECC12 Code Register
MECC16_0	0x220	R/W	0x00000000	1st MLC ECC14 Code Register
MECC16_1	0x224	R/W	0x00000000	2nd MLC ECC14 Code Register
MECC16_2	0x228	R/W	0x00000000	3rd MLC ECC14 Code Register
MECC16_3	0x22C	R/W	0x00000000	4th MLC ECC14 Code Register
MECC16_4	0x230	R/W	0x00000000	5th MLC ECC14 Code Register
MECC16_5	0x234	R/W	0x00000000	6th MLC ECC14 Code Register
MECC16_6	0x238	R/W	0x00000000	7th MLC ECC14 Code Register
MECC16_7	0x23C	R/W	0x00000000	8th MLC ECC14 Code Register
MECC24_0	0x220	R/W	0x00000000	1st MLC ECC16 Code Register
MECC24_1	0x224	R/W	0x00000000	2nd MLC ECC16 Code Register
MECC24_2	0x228	R/W	0x00000000	3rd MLC ECC16 Code Register
MECC24_3	0x22C	R/W	0x00000000	4th MLC ECC16 Code Register
MECC24_4	0x230	R/W	0x00000000	5th MLC ECC16 Code Register
MECC24_5	0x234	R/W	0x00000000	6th MLC ECC16 Code Register
MECC24_6	0x238	R/W	0x00000000	7th MLC ECC16 Code Register
MECC24_7	0x23C	R/W	0x00000000	8th MLC ECC16 Code Register
MECC24_8	0x240	R/W	0x00000000	9th MLC ECC16 Code Register
MECC24_9	0x244	R/W	0x00000000	10th MLC ECC16 Code Register
MECC24_10	0x248	R/W	0x00000000	11th MLC ECC16 Code Register
SECC_EADDR0	0x260	R	0x00000000	SLC ECC Error Address Register 0
SECC_EADDR1	0x264	R	0x00000000	SLC ECC Error Address Register 1
SECC_EADDR2	0x268	R	0x00000000	SLC ECC Error Address Register 2
SECC_EADDR3	0x26C	R	0x00000000	SLC ECC Error Address Register 3
SECC_EADDR4	0x270	R	0x00000000	SLC ECC Error Address Register 4
SECC_EADDR5	0x274	R	0x00000000	SLC ECC Error Address Register 5
SECC_EADDR6	0x278	R	0x00000000	SLC ECC Error Address Register 6
SECC_EADDR7	0x27C	R	0x00000000	SLC ECC Error Address Register 7
SECC_EADDR8	0x280	R	0x00000000	SLC ECC Error Address Register 8
SECC_EADDR9	0x284	R	0x00000000	SLC ECC Error Address Register 9
SECC_EADDR10	0x288	R	0x00000000	SLC ECC Error Address Register 10
SECC_EADDR11	0x28C	R	0x00000000	SLC ECC Error Address Register 11

Name	Address	Type	Reset	Description
SECC_EADDR12	0x290	R	0x00000000	SLC ECC Error Address Register 12
SECC_EADDR13	0x294	R	0x00000000	SLC ECC Error Address Register 13
SECC_EADDR14	0x298	R	0x00000000	SLC ECC Error Address Register 14
SECC_EADDR15	0x29C	R	0x00000000	SLC ECC Error Address Register 15
MECC4_EADDR0	0x260	R	0x00000000	MLC ECC4 Error Address Register 0
MECC4_EADDR1	0x264	R	0x00000000	MLC ECC4 Error Address Register 1
MECC4_EADDR2	0x268	R	0x00000000	MLC ECC4 Error Address Register 2
MECC4_EADDR3	0x26C	R	0x00000000	MLC ECC4 Error Address Register 3
MECC6_EADDR0	0x260	R	0x00000000	MLC ECC6 Error Address Register 0
MECC6_EADDR1	0x264	R	0x00000000	MLC ECC6 Error Address Register 1
MECC6_EADDR2	0x268	R	0x00000000	MLC ECC6 Error Address Register 2
MECC6_EADDR3	0x26C	R	0x00000000	MLC ECC6 Error Address Register 3
MECC6_EADDR4	0x270	R	0x00000000	MLC ECC6 Error Address Register 4
MECC6_EADDR5	0x27C	R	0x00000000	MLC ECC6 Error Address Register 5
MECC12_EADDR0	0x260	R	0x00000000	MLC ECC12 Error Address Register 0
MECC12_EADDR1	0x264	R	0x00000000	MLC ECC12 Error Address Register 1
MECC12_EADDR2	0x268	R	0x00000000	MLC ECC12 Error Address Register 2
MECC12_EADDR3	0x26C	R	0x00000000	MLC ECC12 Error Address Register 3
MECC12_EADDR4	0x270	R	0x00000000	MLC ECC12 Error Address Register 4
MECC12_EADDR5	0x274	R	0x00000000	MLC ECC12 Error Address Register 5
MECC12_EADDR6	0x278	R	0x00000000	MLC ECC12 Error Address Register 6
MECC12_EADDR7	0x27C	R	0x00000000	MLC ECC12 Error Address Register 7
MECC12_EADDR8	0x280	R	0x00000000	MLC ECC12 Error Address Register 8
MECC12_EADDR9	0x284	R	0x00000000	MLC ECC12 Error Address Register 9
MECC12_EADDR10	0x288	R	0x00000000	MLC ECC12 Error Address Register 10
MECC12_EADDR11	0x28C	R	0x00000000	MLC ECC12 Error Address Register 11
MECC16_EADDR0	0x260	R	0x00000000	MLC ECC16 Error Address Register 0
MECC16_EADDR1	0x264	R	0x00000000	MLC ECC16 Error Address Register 1
MECC16_EADDR2	0x268	R	0x00000000	MLC ECC16 Error Address Register 2
MECC16_EADDR3	0x26C	R	0x00000000	MLC ECC16 Error Address Register 3
MECC16_EADDR4	0x270	R	0x00000000	MLC ECC16 Error Address Register 4
MECC16_EADDR5	0x274	R	0x00000000	MLC ECC16 Error Address Register 5
MECC16_EADDR6	0x278	R	0x00000000	MLC ECC16 Error Address Register 6
MECC16_EADDR7	0x27C	R	0x00000000	MLC ECC16 Error Address Register 7
MECC16_EADDR8	0x280	R	0x00000000	MLC ECC16 Error Address Register 8
MECC16_EADDR9	0x284	R	0x00000000	MLC ECC16 Error Address Register 9
MECC16_EADDR10	0x288	R	0x00000000	MLC ECC16 Error Address Register 10
MECC16_EADDR11	0x28C	R	0x00000000	MLC ECC16 Error Address Register 11
MECC16_EADDR12	0x290	R	0x00000000	MLC ECC16 Error Address Register 12
MECC16_EADDR13	0x294	R	0x00000000	MLC ECC16 Error Address Register 13
MECC16_EADDR14	0x298	R	0x00000000	MLC ECC16 Error Address Register 14
MECC16_EADDR15	0x29C	R	0x00000000	MLC ECC16 Error Address Register 15
MECC24_EADDR0	0x260	R	0x00000000	MLC ECC24 Error Address Register 0
MECC24_EADDR1	0x264	R	0x00000000	MLC ECC24 Error Address Register 1
MECC24_EADDR2	0x268	R	0x00000000	MLC ECC24 Error Address Register 2
MECC24_EADDR3	0x26C	R	0x00000000	MLC ECC24 Error Address Register 3
MECC24_EADDR4	0x270	R	0x00000000	MLC ECC24 Error Address Register 4
MECC24_EADDR5	0x274	R	0x00000000	MLC ECC24 Error Address Register 5
MECC24_EADDR6	0x278	R	0x00000000	MLC ECC24 Error Address Register 6
MECC24_EADDR7	0x27C	R	0x00000000	MLC ECC24 Error Address Register 7
MECC24_EADDR8	0x280	R	0x00000000	MLC ECC24 Error Address Register 8
MECC24_EADDR9	0x284	R	0x00000000	MLC ECC24 Error Address Register 9
MECC24_EADDR10	0x288	R	0x00000000	MLC ECC24 Error Address Register 10
MECC24_EADDR11	0x28C	R	0x00000000	MLC ECC24 Error Address Register 11
MECC24_EADDR12	0x290	R	0x00000000	MLC ECC24 Error Address Register 12
MECC24_EADDR13	0x294	R	0x00000000	MLC ECC24 Error Address Register 13
MECC24_EADDR14	0x298	R	0x00000000	MLC ECC24 Error Address Register 14
MECC24_EADDR15	0x29C	R	0x00000000	MLC ECC24 Error Address Register 15
MECC24_EADDR16	0x2A0	R	0x00000000	MLC ECC24 Error Address Register 16
MECC24_EADDR17	0x2A4	R	0x00000000	MLC ECC24 Error Address Register 17
MECC24_EADDR18	0x2A8	R	0x00000000	MLC ECC24 Error Address Register 18
MECC24_EADDR19	0x2AC	R	0x00000000	MLC ECC24 Error Address Register 19
MECC24_EADDR20	0x2B0	R	0x00000000	MLC ECC24 Error Address Register 20
MECC24_EADDR21	0x2B4	R	0x00000000	MLC ECC24 Error Address Register 21

Name	Address	Type	Reset	Description
MECC24_EADDR22	0x2B8	R	0x00000000	MLC ECC24 Error Address Register 22
MECC24_EADDR23	0x2BC	R	0x00000000	MLC ECC24 Error Address Register 23
NFC_MEMORY	0x800~0xDFF	R/W		NFC MEMORY for SPARE Area

Command Register (NFC_CMD)

0x8000D000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CMD3[7:0]								CMD2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMD1[7:0]								CMD0[7:0]							

Bit	Name	RW	Reset	Description
31:24	CMD3	W	-	NAND Flash Command Register3
23:16	CMD2	W	-	NAND Flash Command Register2
15:8	CMD1	W	-	NAND Flash Command Register1
7:0	CMD0	W	-	NAND Flash Command Register0

For 32bit bus width configuration of 8bit NAND flash or 16bit NAND flash, the NFC_CMD[CMD3] down to NFC_CMD[CMD0] is used as command register. (NFC_CTRL[BW] = 2'b10)

For 16bit bus width configuration of 8bit NAND flash or 16bit NAND flash, the NFC_CMD[CMD1] down to NFC_CMD[CMD0] is used as command register. NFC_CMD[CMD3] and NFC_CMD[CMD2] is ignored. (NFC_CTRL[BW] = 2'b01)

For 8bit bus width of NAND flash, the NFC_CMD[CMD0] is used as command register. NFC_CMD[CMD3] down to NFC_CMD[CMD1] is ignored. (NFC_CTRL[BW] = 2'b00)

If MSK of NFC_CTRL is high, the NFC_CMD[CMD3] down to NFC_CMD[CMD1] is masked.

1. NFC_CTRL[BW]=2'01 (16bit BW)

NFC_CMD	MSK=1	MSK=0	I/O[15:00]	
			MSK=1	MSK=0
0x00000000	0x00000000	0x00000000	0x0000	0x0000
0x80808080	0x00000080	0x00008080	0x0080	0x8080
0x60606060	0x00000060	0x00006060	0x0060	0x6060
0x70707070	0x00000070	0x00007070	0x0070	0x7070
...	...	...	...	...

2. NFC_CTRL[BW]=2'10 (32bit BW)

NFC_CMD	MSK=1	MSK=0	I/O[31:00]	
			MSK=1	MSK=0
0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x80808080	0x00000080	0x80808080	0x00000080	0x80808080
0x60606060	0x00000060	0x60606060	0x00000060	0x60606060
0x70707070	0x00000070	0x70707070	0x00000070	0x70707070
...	...	...	...	...

Linear Address Register (NFC_LADDR)

0x8000D004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LADDR[15:0]															

Bit	Name	RW	Reset	Description
31:0	LADDR	W	-	NAND Flash Linear Address Register

By writing to this register, memory controller generates linear address shown in Table 22.2.

For NFC_LADDR = 0x12345600, CADDR NFC_CTRL[14:12] = 2 and PSIZE(NFC_CTRL[17:16]) = 1 (512 bytes)

- MSK(NFC_CTRL[15]) = 1 : ND_IO[15:0] = 0x0000 – 0x002B – 0x001A
- MSK(NFC_CTRL[15]) = 0 : ND_IO[15:0] = 0x0000 – 0x2B2B – 0x1A1A

Single Address Cycle Register (NFC_SADDR)

0x8000D008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR3[7:0]								SADDR2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR1[7:0]								SADDR0[7:0]							

Bit	Name	RW	Reset	Description
31:24	SADDR3	W	-	NAND Flash Signal Address Register3
23:16	SADDR2	W	-	NAND Flash Signal Address Register2
15:8	SADDR1	W	-	NAND Flash Signal Address Register1
7:0	SADDR0	W	-	NAND Flash Signal Address Register0

For 32bit bus width configuration of 8 bit NAND flash or 16 bit NAND flash, NFC_SADDR[SADDR3] down to NFC_SADDR[SADDR0] is used as single address register. (NFC_CTRL[BW] = 2'b10)

For 16bit bus width configuration of 8 bit NAND flash or 16 bit NAND flash, the NFC_SADDR[SADDR1] down to NFC_SADDR[SADDR0] is used as single address register. NFC_SADDR[SADDR3] and NFC_SADDR[SADDR2] is ignored. (NFC_CTRL[BW] = 2'b01)

For 8bit bus width of NAND flash, the NFC_SADDR[SADDR0] is used as single address register. NFC_SADDR[SADDR3] down to NFC_SADDR[SADDR1] is ignored. (NFC_CTRL[BW] = 2'b00)

If MSK of NFC_CTRL is high, the NFC_SADDR[SADDR3] down to NFC_SADDR[SADDR1] is masked.

For example,

1. NFC_CTRL[BW]=2'01 (16bit BW)

NFC_SADDR	MSK=1	MSK=0	I/O[15:00]	
			MSK=1	MSK=0
0xFFFFFFFF	0x000000FF	0x0000FFFF	0x00FF	0xFFFF
0xAAAAAAAA	0x000000AA	0x0000AAAA	0x00AA	0xAAAA
...	...	...	...	...

2. NFC_CTRL[BW]=2'10 (32bit BW)

NFC_SADDR	MSK=1	MSK=0	I/O[31:00]	
			MSK=1	MSK=0
0xFFFFFFFF	0x000000FF	0xFFFFFFFF	0x000000FF	0xFFFFFFFF
0xAAAAAAAA	0x000000AA	0xAAAAAAAA	0x000000AA	0xAAAAAAAA
...	...	...	...	...

Table 22.2 Page Size of NAND Flash

# of Cycle (CADDR)	PSIZE = 0	PSIZE = 1	PSIZE = 2	PSIZE = 3	PSIZE > 3
1st	ADDR[7:0]	ADDR[7:0]	ADDR[7:0]	ADDR[7:0]	ADDR[7:0]
2nd	ADDR[16:9]	ADDR[16:9]	ADDR[10:8]	ADDR[11:8]	ADDR[12:8]
3rd	ADDR[24:17]	ADDR[24:17]	ADDR[18:11]	ADDR[19:12]	ADDR[20:13]
4th	ADDR[31:25]	ADDR[31:25]	ADDR[26:19]	ADDR[27:20]	ADDR[28:21]
5th	-	-	ADDR[31:27]	ADDR[31:28]	ADDR[31:29]

Figure 22.5 below represents the relation between each cycle and address generation.

A user must set this information appropriately to PSIZE and CADDR field of NFC_CTRL register ahead of accessing NAND data.

ADDR means address value that is written to NFC_LADDR or NFC_BADDR register. The shaded cycles represent block address cycles. That is, NAND address cycles start from there when NFC_BADDR register is accessed.

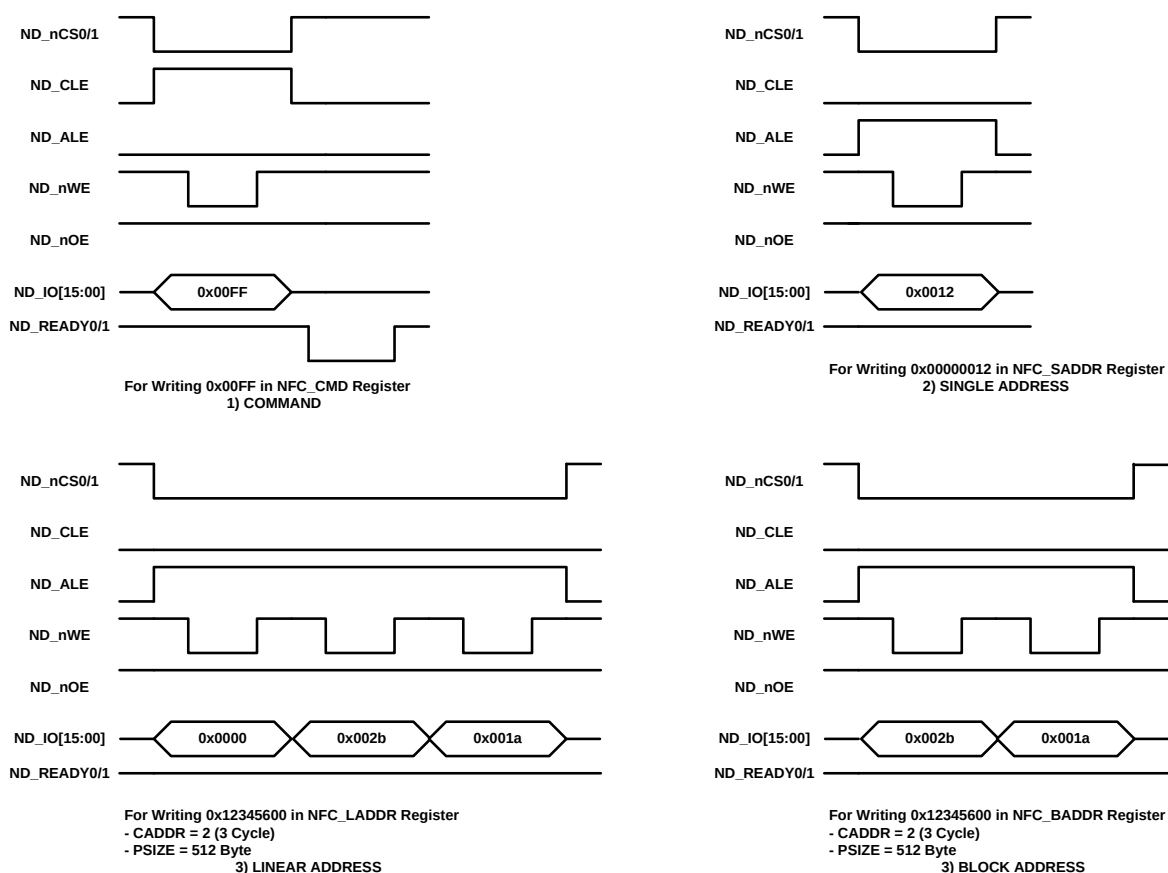


Figure 22.5 Example of Address/Command Writing Operation

Word Data Register (NFC_WDATA)

0x8000D01x

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NFC_WDATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NFC_WDATA[15:0]															

Bit	Name	RW	Reset	Description
31:0	NFC_WDATA	R/W	Unknown	NAND Flash Word Data Register

This register is used for single word data transfer by CPU and it is useful in reading and writing NAND Flash data of spare area.

Linear Data Register (NFC_LDATA)

0x8000D02x/0x8000D03x

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NFC_LDATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NFC_LDATA[15:00]															

Bit	Name	RW	Reset	Description
31:0	NFC_LDATA	R/W	Unknown	NAND Flash Linear Data Register

This register is used for burst data transfer by DMA/CPU.

To start burst data transfer, a user must write any value to NFC_PSTART or NFC_RSTART.

Multiple Data Register (NFC_MDATA)

0x8000D040

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NFC_MDATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NFC_MDATA[15:0]															

Bit	Name	RW	Reset	Description
31:0	NFC_MDATA	R/W	Unknown	NAND Flash Multiple Data Register

This register is used for reading and writing multiple 8bit or 16bit data according to NAND Flash Bus Size.
NFC_CTRL.DNUM[01:00] register defines the number of multiple 8 bit or 16 bit data for data transfer.

CMD/ADDRESS CYCLE

0x8000D044

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								STP							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PW								HLD[7:0]							

Note: Command / Address Cycle Configuration Register.

Bit	Name	RW	Reset	Description
23:16	STP	R/W	0x01	Setup Time Cycles
15:8	PW	R/W	0x01	Pulse Width Cycles
7:0	HLD	R/W	0x01	Hold Time Cycles

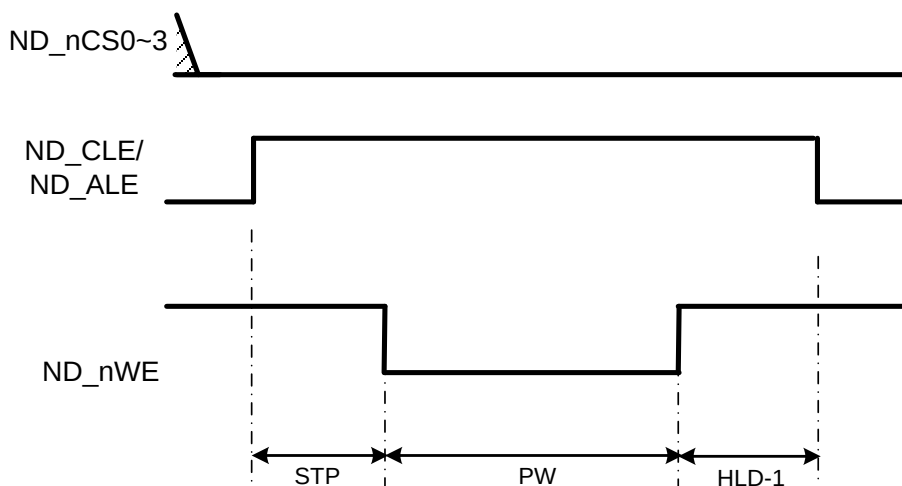


Figure 22.6 Timing Diagram of Read/Write Enable Signal for Command or Address Cycle

WRITE CYCLE

0x8000D048

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								STP							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PW								HLD[7:0]							

Note: Write Cycle Configuration Register. Refer to CMD/ADDR Cycle Configuration Register.

Bit	Name	RW	Reset	Description
23:16	STP	R/W	0x01	Setup Time Cycles
15:8	PW	R/W	0x01	Pulse Width Cycles
7:0	HLD	R/W	0x01	Hold Time Cycles

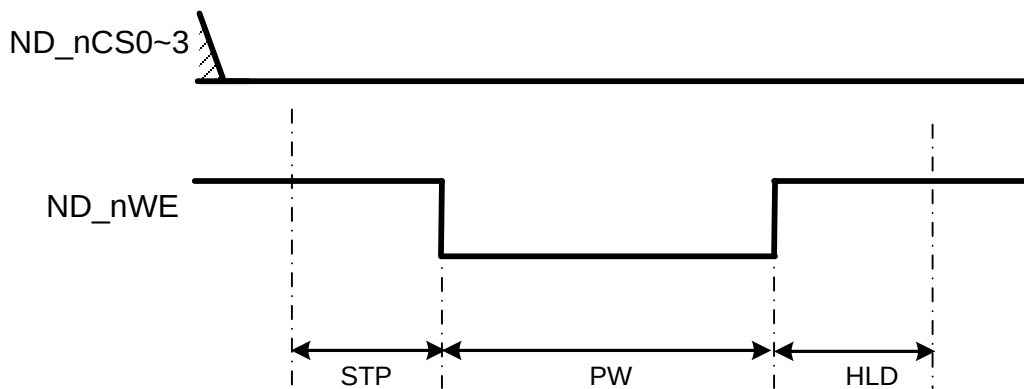


Figure 22.7 Timing Diagram of Write Enable Signal for Write cycle

READ CYCLE												0x8000D04C			
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								STP							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PW								HLD[7:0]							

Note: Read Cycle Configuration Register.

Bit	Name	RW	Reset	Description
23:16	STP	R/W	0x01	Setup Time Cycles
15:8	PW	R/W	0x01	Pulse Width Cycles
7:0	HLD	R/W	0x01	Hold Time Cycles

For Single / Word data read (SDATA/WDATA/MDATA): The pulse width is expanded as much as NFC_CTRL[RDNDLY] value.

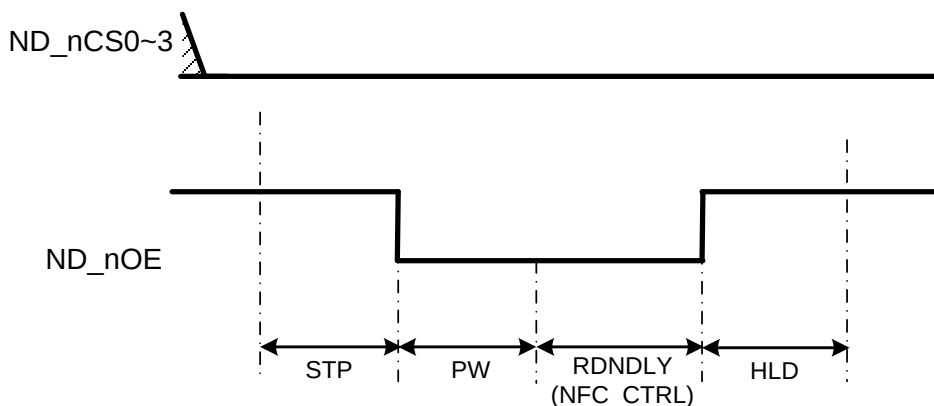


Figure 22.8 Timing Diagram of Read Enable Signal for Single Read cycle

For Burst data read (LDATA or NDMA):

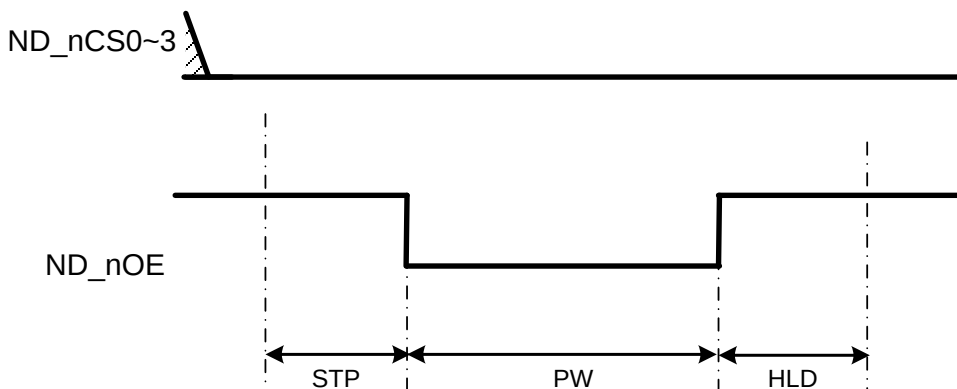


Figure 22.9 Timing Diagram of Read Enable Signal for Burst Read cycle

NFC CTRL

0x8000D050

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CS3	CS2	CS1	CS0	RDNDLY			ARFW	RDYCFG			DACK	DMODE	BMODE	ECCON	PRSEL
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DNUM[01:00]			RDYSEL	RM	MSK	BW		BSIZE		PSIZE			CADDR[2:0]		

Bit	Name	RW	Reset	Description
31	CS3SEL	R/W	0x1	NAND Flash CS3 Selection 0 : NAND Flash nCS3 is Enable 1 : NAND Flash nCS3 is Disable
30	CS2SEL	R/W	0x1	NAND Flash CS2 Selection 0 : NAND Flash nCS2 is Enable 1 : NAND Flash nCS2 is Disable
29	CS1SEL	R/W	0x1	NAND Flash CS1 Selection 0 : NAND Flash nCS1 is Enable 1 : NAND Flash nCS1 is Disable
28	CS0SEL	R/W	0x1	NAND Flash CS0 Selection 0 : NAND Flash nCS0 is Enable 1 : NAND Flash nCS0 is Disable

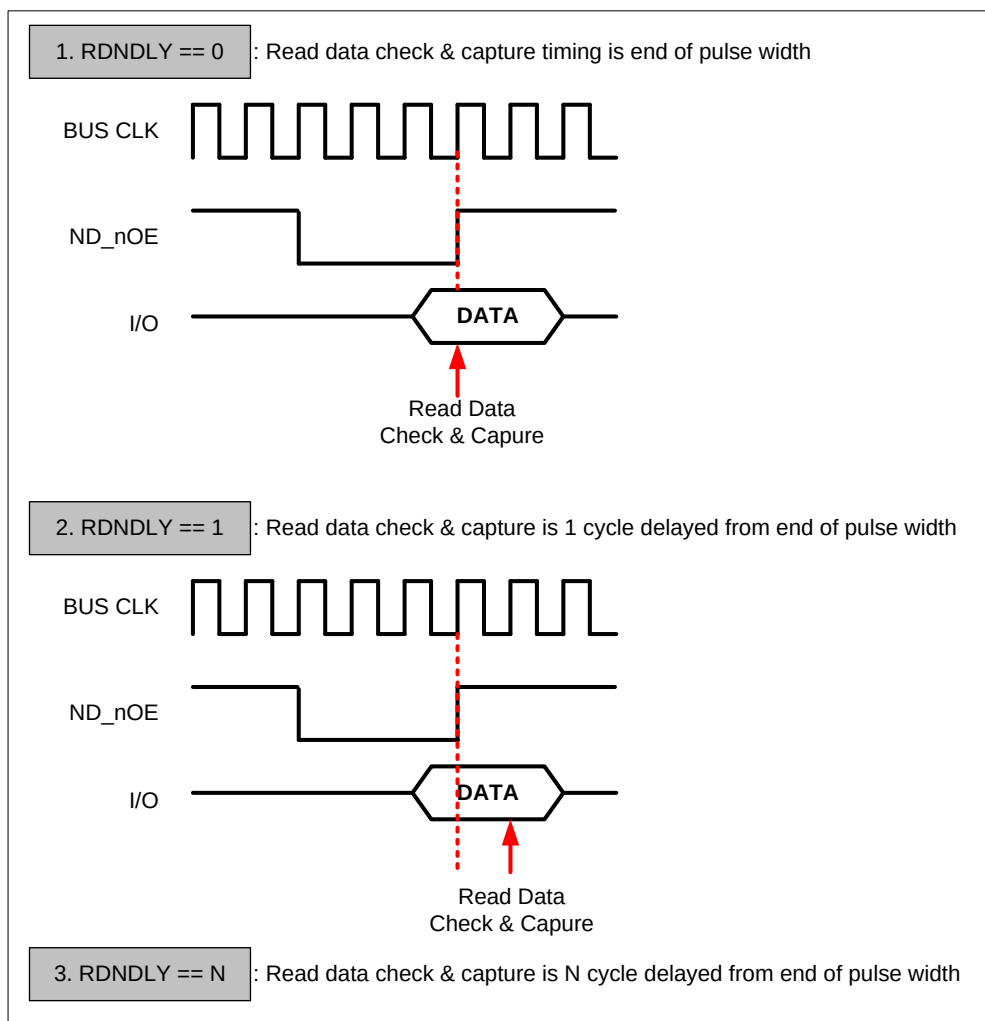
Bit	Name	RW	Reset	Description
27:25	RDNDLY	R/W	0x0	NAND Read Data Capture Delay Control 000 : None 001 : 1 bus clock delay 010 : 2 bus clock delay 011 : 3 bus clock delay 100 : 4 bus clock delay 101 : 5 bus clock delay 110 : 6 bus clock delay 111 : 7 bus clock delay

Note: Read data capture timing is adjusted in burst data read. If RDNDLY is 0, at the end of the pulse width, read data is checked and captured.

If RDNDLY is N, the timing of checking and capturing read data is as below.

Available read data timing = end of pulse width + (Bus clock X N)

RDNDLY is useful for compensating read data cycle increase caused by I/O delay.



Bit	Name	RW	Reset	Description
24	ARFW	R/W	0x0	Automatically Ready Flag Wait Enable 0 : Disable 1 : Enable

Note: Once ARFW is set, in burst data read mode ready flag is automatically checked after Read Command. After this, the next procedure (burst data read) can be automatically executed. However, this is available only when there is the 2nd read command.

Example:

NAND Flash of SAMSUNG
1st READ_CMD (ex. 0x00) – ADDRESS –2nd READ_CMD (e.g. 0x30)

In order to automatically check ready flag, the NFC_RFWBASE Register value should be same as the 2nd Read Command value.

Example:

If 2nd READ_CMD (0x00000030) → set ARFWBASE (0x00000030)
If 2nd READ_CMD (0x30303030) → set ARFWBASE (0x30303030)

Bit	Name	RW	Reset	Description
23:22	RDYCFG	R/W	0x0	This bit field is used to select one of four external ready input of NAND Flash. (In case that NFC_CTRL[RDYSEL] is '1') The selected ready input is used for generating ready end flag/interrupt. 2'b00 : External ready input 0 is selected 2'b01 : External ready input 1 is selected 2'b10 : External ready input 2 is selected 3'b11 : External ready input 3 is selected

Bit	Name	RW	Reset	Description
20	DACK	R/W	0x0	DMA Acknowledge Enable 0 : Disable 1 : Enable

Note: In case of burst transfer with CPU used, it should be set to '0'.

Bit	Name	RW	Reset	Description
19	DMODE	R/W	0x0	DMA Mode Selection 0 : General DMA (GDMA) 1 : NFC Dedicate DMA (NDMA)

Bit	Name	RW	Reset	Description
18	BMODE	R/W	0x1	Burst Operation Mode Selection for NDMA 0 : Only user data is burst-transferred 1 : User data + spare (ECC and etc.) is burst-transferred

Note: This function is only available for burst data transfer by NDMA.

The detailed descriptions are as follows.

1) If BMODE is '1'

- Program:

In this mode, both user data and spare data are burst-written to the NAND flash.

1st STEP, burst program is orderly executed in NAND as much as data size set in SPCFGn register. At this time, ECC data is generated whenever transmitting data of each subpage is completed. The generated ECC data is automatically saved to a designated address of SPCFGn[ECC_ADDR] in NFC dedicated memory.

2nd STEP, if transmitting all subpage data as many as subpages which have been set is completed, data as much as the data size which has been set in NFC_SPARE is read from NFC dedicated memory and transmitted to the NAND flash.

3rd STEP, if the transmission is completed, program interrupt or program end flag is occurred.

- Read:

In this mode, both user data and spare data are burst-read from the NAND flash.

1st STEP, data size as much as set in SPCFGn register is orderly burst-read from the NAND.

2nd STEP, if all subpage data read is completed, spare data as big as data size of NFC_SPARE is read from the NAND flash and saved to NFC dedicated memory.

3rd STEP, once spare data read is completed, by referring to SP_CFGn register per subpage ECC decoding is executed. ECC data of each subpage is read from SP_CFGn[ECC_ADDR] of NFC dedicated memory for error detecting. If an error occurs, an error flag with a read flag or read interrupt occurs. If ECC error does not occur in all subpages, the error flag does not occur and only read flag or read interrupt occurs.

If an error flag occurs, a user can correct the error by referring to an error number and error address register. Once error correction is completed, the number of subpage where and error occurs must be checked by referring to MECC_STATUS[C_SUBPAGE] register. If it is not the last subpage, ECC decoding should be restarted by setting ECC Decoding Register (ECC_RSTART) to decode ECC for the rest of subpage.

2) If BMODE is '0'

- Program

Burst program is orderly executed in NAND flash as much as data size set in SPCFGn register. At this time, ECC data is generated whenever transmitting data of each subpage is completed. The generated ECC data is automatically saved to a designated address of SPCFGn[ECC_ADDR] in NFC dedicated memory. Once all subpage data is written, a flag or write interrupt occurs.

After data write is completed, a user should write the encoded ECC data in dedicated memory to NAND flash using SPW or CPU.

- Read:

1st STEP, data size as much as set in SPCFGn register is orderly burst-read from the NAND flash.

Once all subpage data is read, a flag or read interrupt occurs. In order for ECC decoding, ECC data of each subpage should be single or burst-read from the spare area of the NAND Flash and after this ECC decoding should be manually executed per each subpage by using ECC Store Status & Load Control Register.

Bit	Name	RW	Reset	Description
17	ECCON	R/W	0x0	NAND Flash ECC Encoding/Decoding Enable 0 : Disable 1 : Enable

If this bit is set, the ECC encoding/decoding can be possible for burst transfer, single word data (WDATA) and multiple data.

Bit	Name	RW	Reset	Description
16	PRSEL	R/W	0x0	1) Program/Read Mode Selection for Burst Transfer 0 : Read 1 : Program This register should be set before the burst transfer starts. 2) Randomizer direction selection 0 : Read 1 : Program When the randomizer enable register (NFC_RAND_CTRL) is set, the PRSEL is used to select the randomizer direction regardless of burst transfer

Bit	Name	RW	Reset	Description
15:14	DNUM	R/W	0x0	The Number of Multiple Data Configuration N : NFC MDATA Access count

The number of transfer byte/half words of multiple bytes/half word data transfer (NFC_MDATA) is set.

For 8 bit bus width of NAND flash, transferred bytes = DNUM + 1

For 16 bit bus width of NAND flash, transferred half word = DNUM + 1 (must be DNUM<2)

If the 32 bit bus width of NAND flash is only used, ignore NFC_CTRL[DNUM].

Bit	Name	RW	Reset	Description
12	RDYSEL	R/W	0x0	NAND Flash External Ready Input Selection 0: External four ready inputs can be used for generation of ready end flag/interrupt 1: One of external four ready inputs is only available for generation of ready end flag/interrupt. Refer to NFC_CTRL[RDYCFG].
11	RM	R/W	0x1	NAND Flash Ready Mask Disable 0: READY input from NAND Flash is masked. If it is '0', READY flag or interrupt is not generated. 1: READY input from NAND Flash is unmasked.
10	MSK	R/W	0x0	NAND Flash IO Mask Enable 0: NAND Flash High Bytes(MSB 31bit down to 8bit) is not masked 1: NAND Flash High Bytes(MSB 31bit down to 8bit) is masked
9:8	BW	R/W	0x0	NAND Flash Bus Width Select 00: Bus width = 8bit 01: Bus width = 16bit 10: Bus width = 32bit 11: Reserved.

Bit	Name	RW	Reset	Description
7:6	BSIZE	R/W	0x0	NAND Flash Burst Size 00: 1 burst 01: 2 burst 10: 4 burst 11: 8 burst

Note: This function is only available for burst transfer by GDMA or CPU. A burst size must be same as that of GDMA or that of CPU.

Bit	Name	RW	Reset	Description
5:3	PSIZE	R/W	0x0	Page Size of NAND Flash 000 : 1 page = 256 Half-Word 001 : 1 page = 512 Bytes 010 : 1 page = 1024 Half-Word 011 : 1 page = 2048 Bytes 100 : 1 page = 4096 Bytes 101~111 : 1 page = 8192 Bytes

A linear address for external NAND is affected by CTRL[PSIZE]. If the single address mode is only used, ignore CTRL[PSIZE].

Bit	Name	RW	Reset	Description
2:0	CADDR	R/W	0x0	The Number of Address Cycles N: The number of address cycle for NAND flash. (N+1) cycle is used for generating address cycle command

NFC_DSIZE

0x8000D054

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA_SIZE[15:00]															

Bit	Name	RW	Reset	Description
15:0	DATA_SIZE	R/W	0xFFFF	NAND Flash Data Size N : N is transferred byte data size This is only available in burst transfer by GDMA or CPU. In case of burst-transfer by NDMA, ignore NFC_DSIZE register. For Samsung 258 Half-word/512 Byte small block, N is 512(byte). For Samsung 1024 Half-Word/2048 Byte big block, N is 2048(byte).

NFC_SPARE

0x8000D058

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
														SPW	SPR
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPARE_DSIZE[11:00]															

Bit	Name	RW	Reset	Description
17	SPW	R/W	0x0	0 : Read-only 1 : Spare Area Burst Program If this bit is set, the data of NFC dedicated memory is burst-transferred to NAND Flash as many as SPARE_DSIZE. This bit is automatically cleared and Spare Area Burst Write Flag (NFC_IRQ[SPWF]) occurs.
16	SPR	R/W	0x0	0 : Read-only 1 : Spare Area Burst Read If this bit is set, data as many as SPARE_DSIZE is burst-read from the NAND Flash and saved to NFC dedicated memory. Once this operation is completed, this bit is automatically cleared and Spare Area Burst Read Flag occurs.
11:0	SPARE_DSIZE	R/W	0x200	NAND Flash Spare Area Data Size N : The byte size of burst transfer data between NFC dedicated memory and NAND Flash It is available for burst transfer by NDMA or spare area burst program/read (by SPW and SPR).

NFC Burst Program/Read Start Register (NFC_PRSTART)

0x8000D05C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Don't care															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Don't care															

Bit	Name	RW	Reset	Description
31:0	Don't Care	R/W	0x00000000	Don't care

When this register is written by any value, data transfer is started. That is, Programming operation to NAND flash or Reading Operation from NAND starts as NFC_CTRL[PRSEL].

For burst data transfer by GDMA, you must set EN flag of DMA_CHCTRL register first, and then write any value to NFC_BSTART ahead of accessing NFC_LDATA.

For burst data transfer by NDMA, you must set EN flag of NDMA_CHCTRL register first, and then write any value to NFC_BSTART.

NFC RESET

0x8000D060

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Don't care															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Don't care															

If any value is written, the NFC controller is internally reset.

Bit	Name	RW	Reset	Description
31:0	Don't Care	R/W	0x00000000	Don't Care

NFC_IRQCFG

0x8000D064

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDYIEN	PIEN	RDYIEN		SPWIEEN	SPRIEN			MEIEN							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bit	Name	RW	Reset	Description
31	RDYIEN	R/W	0x0	NAND Flash Ready Interrupt Enable 0 : Ready Interrupt Disable 1 : Ready Interrupt Enable
30	PIEN	R/W	0x0	NAND Flash Program Interrupt Enable 0 : Program Interrupt Disable 1 : Program Interrupt Enable Program interrupt occurs when all of data is programmed to NAND flash by burst transfer. The data size or burst-transfer is decided by NFC_DSIZE and NFC_SPCFGn[DSIZE]
29	RDYIEN	R/W	0x0	NAND Flash Read Interrupt Enable 0 : Read Interrupt Disable 1 : Read Interrupt Enable Read interrupt occurs when all of data is read from NAND flash by burst transfer. The data size or burst-transfer is decided by NFC_DSIZE or NFC_SPCFGn[DSIZE]
27	SPWIEEN	R/W	0x0	NAND Flash Spare Area Burst Write Interrupt Enable 0 : Disable 1 : Enable
26	SPRIEN	R/W	0x0	NAND Flash Spare Area Burst Read Interrupt Enable 0 : Disable 1 : Enable
23	MEIEN	R/W	0x0	MLC ECC Decoding Interrupt Enable 0 : Disable 1 : Enable

NFC_IRQ

0x8000D068

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDYIRQ	PIRQ	RDIRQ		SPWIRQ	SPRIRQ			MEDIRQ							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDYFG	PFG	RDFG		SPWFG	SPRFG	Reserve	Reserve	MEDFG	MEEFG	SEDFG	SEEFG	ERRFG			

Bit	Name	RW	Reset	Description
31	RDYIRQ	R/W	0x0	NAND Flash Ready End Interrupt 0 (Read) : None 1 (Read) : Ready interrupt occurred 1 (Write) : Ready Interrupt clear
30	PIRQ	R/W	0x0	NAND Flash Program End Interrupt 0 (Read) : None 1 (Read) : Program interrupt occurred 1 (Write) : Program Interrupt clear
29	RDIRQ	R/W	0x0	NAND Flash Read End Interrupt 0 (Read) : None 1 (Read) : Read interrupt occurred 1 (Write) : Read Interrupt clear
27	SPWIRQ	R/W	0x0	NAND Flash Spare Burst Write End Interrupt 0 (Read) : None 1 (Read) : Spare burst write interrupt occurred 1 (Write) : Spare burst write Interrupt clear
26	SPRIRQ	R/W	0x0	NAND Flash Spare Burst Read End Interrupt 0 (Read) : None 1 (Read) : Spare burst read interrupt occurred 1 (Write) : Spare burst read Interrupt clear
23	MEDIRQ	R/W	0x0	MLC ECC Decoding End Interrupt 0 (Read) : None 1 (Read) : MLC ECC decoding interrupt occurred 1 (Write) : MLC ECC decoding Interrupt clear
15	RDYFG	R/W	0x0	NAND Flash Ready End Flag 0 (Read) : None 1 (Read) : Ready flag occurred 1 (Write) : Ready flag clear
14	PFG	R/W	0x0	NAND Flash Program End Flag 0 (Read) : None 1 (Read) : Program flag occurred 1 (Write) : Program flag clear
13	RDFG	R/W	0x0	NAND Flash Read End Flag 0 (Read) : None 1 (Read) : Read flag occurred 1 (Write) : Read flag clear
11	SPWFG	R/W	0x0	NAND Flash Spare Burst Write End Flag 0 (Read) : None 1 (Read) : Spare burst write flag occurred 1 (Write) : Spare burst write flag clear
10	SPRFG	R/W	0x0	NAND Flash Spare Burst Read End Flag 0 (Read) : None 1 (Read) : Spare burst read flag occurred 1 (Write) : Spare burst read flag clear
7	MEDFG	R/W	0x0	MLC ECC Decoding End Flag 0 (Read) : None 1 (Read) : MLC ECC decoding flag occurred 1 (Write) : MLC ECC decoding flag clear
6	MEEFG	R/W	0x0	MLC ECC Encoding End Flag 0 (Read) : None 1 (Read) : MLC ECC encoding flag occurred 1 (Write) : MLC ECC encoding flag clear
5	SEDFG	R/W	0x0	SLC ECC Decoding End Flag 0 (Read) : None 1 (Read) : SLC ECC decoding flag occurred 1 (Write) : SLC ECC decoding flag clear
4	SEEFG	R/W	0x0	SLC ECC Encoding End Flag 0 (Read) : None 1 (Read) : SLC ECC encoding flag occurred 1 (Write) : SLC ECC encoding flag clear

3	ERRFG	R/W	0x0	MLC/SLC Error Flag 0 (Read) : None 1 (Read) : Error flag occurred 1 (Write) : Error flag clear
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NFC_STATUS

0x8000D06C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDY	FS											CURR_SUBPAGE[4:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
												NFC_STATE			

Bit	Name	RW	Reset	Description
31	RDY	R	0x0	NAND Flash Ready Flag Status 0 : External ready flag of dedicated NAND Flash is low 1 : External ready flag of dedicated NAND Flash is high

Bit	Name	RW	Reset	Description
30	FS	R	0x0	NAND Flash FIFO Status 0 : FIFO is not ready to write and read for burst transfer 1 : FIFO is ready to write and read for burst transfer

For burst data transfer by ARM, a user must check whether this bit is high before accessing NFC_LDData.

Bit	Name	RW	Reset	Description
20:16	CURR_SUBPAGE	R	0x0	Current Sub-Page Indicator N : Nth Sub-page data is transferred from/to NAND Flash

Bit	Name	RW	Reset	Description
5:0	NFC_STATE	R	0x0	NFC FSM Current State N : Reserved

NFC_RFWBASE

0x8000D070

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RFWBASE[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RFWBASE[15:0]															

Refer to NFC_CTRL[RFWEN].

Bit	Name	RW	Reset	Description
31:0	RFWBASE	R/W	-	NFC Read Flag Wait Base Word Register

NFC DMA Source/Destination Address (NDMA_ADDR)

0x8000D100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ST_ADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ST_ADDR[15:0]															

Bit	Name	RW	Reset	Description
31:0	ST_ADDR	R/W		NFC DMA Source/Destination Register

This register contains the start address of source memory block for NFC program operation or the start address of destination memory block for NFC Read operation. This register field is only available in NDMA mode enable of NFC_CTRL.

NFC DMA Parameter Register (NDMA_PARAM)

0x8000D104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MASK[23:8]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MASK[7:0]								INC[7:0]							

Bit	Name	RW	Reset	Description
31:8	MASK	R/W	0x000000	Source/Destination Address mask Register 0 : Non-masked 1 : Masked so that source address bit does not change during DMA transfer

Each bit field controls the dedicated bit of source/destination address field. That is, if MASK[23] is set to 1, the 28th bit of source/destination address is masked, and if MASK[22] is set to 1, the 27th bit of source/destination address is masked, and so on. If a bit is masked, a corresponding bit of address bus does not change during DMA transfer. This function can be used to generate circular buffer address.

Bit	Name	RW	Reset	Description
7:0	INC	R/W	0x00	Source/Destination address is added by amount of sinc at every write cycles. sinc is represented as 2's complement, so if INC[7] is 1, the source address is decremented

The addresses of DMA transfer are 32bit wide, but the upper 4bit of them are not affected during DMA transfer. If the source or destination address reaches its maximum address space like 0x7FFFFFFF or 0x2FFFFFFF, the next transfer starts from 0x70000000 or 0x20000000 not from 0x80000000 or 0x30000000.

NFC DMA Current Address (NDMA_CADDR)

0x8000D108

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
C_ADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C_ADDR[15:0]															

Bit	Name	RW	Reset	Description
31:0	C_ADDR	R	0x00000000	NFC DMA Current Address Register

This register contains the current source/destination address of DMA transfer. It represents that the current transfer read data from this address. This is read only register.

NFC DMA Control Register (NDMA_CTRL)

0x8000D10C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		Reserved	LOCK	ARB				BSIZE		WSIZE		FLAG		SEQ	EN

Bit	Name	RW	Reset	Description
11	LOCK	R/W	0x0	Issue Locked Transfer 0 : Non-locked transfer mode 1: NDMA transfer executed with lock transfer.

Lock field controls the LOCK signal (refer to AHB specification [1]). When the LOCK is set to 1, the DMA transfer is not bothered by other AHB masters like LCD controller, ARM, and etc. This field is only meaningful in case of non-burst type transfers.

Bit	Name	RW	Reset	Description
10	ARB	R/W	0x0	Arbitration Transfer 0 : NDMA transfer executed with arbitration 1: NDMA transfer executed with no arbitration.

Arbitration means that at the end of every HOP transfer, the AHB bus is released from DMA channel so other master can occupy the bus when that master has requested the bus. Burst means that once the DMA request occurs, all of transfers are executed without further DMA requests.

Bit	Name	RW	Reset	Description
7:6	BSIZE	R/W	0x0	Burst Size 00 : 1 burst consists of 1 read or write cycle 01 : 1 burst consists of 2 read or write cycle 10 : 1 burst consists of 4 read or write cycle 11 : 1 burst consists of 8 read or write cycle
5:4	WSIZE	R/W	0x0	Word Size 00 : Read or write 8bit data for each cycle 01 : Read or write 16bit data for each cycle 10,11 : Read or write 32bit data for each cycle
3	FLAG	R/W	0x0	DMA Done Flag Represents that all hop of transfers are fulfilled. When writing 1 to this bit, it is cleared to 0
2	SEQ	R/W	0x0	Sequential mode control If the number of subpage is equal and above '2', this bit must be set to '1' 0 : After all of hop transfer has executed, the DMA channel is disabled 1: The DMA channel remains enabled. When another DMA request has occurred, the DMA channel starts transferring data again with the same manner (address, increment, mask) as the latest transfer of that channel.
1	EN	R/W	0x0	NFC DMA Enable 0 : NDMA Channel is disabled 1 : NDMA Channel is enabled, before NDMA transfer starts by NFC_CTRL[BSTART], this bit must be set to '1'.

NFC DMA SUBPAGE Control (NDMA_SPCTRL)

0x8000D110

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
												SUBPAGE CNT[04:00]			

Bit	Name	RW	Reset	Description
4:0	SUBPAGE_CNT	R/W	0x00	Set the Number of Subpage N+1 : The number of subpage

NFC DMA STATUS (NDMA_STATUS)

0x8000D114

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
C_HOP_COUNT[13:00]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C_STATE[05:00]															

Bit	Name	RW	Reset	Description
29:16	C_HOP_COUNT	R	0x0000	Current Hop Count
5:0	C_STATE	R	0x00	Current NDMA State

NFC SUB-PAGE Configuration Register (SP_CFGn, n=0,...,31)

0x8000D120+n*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ECC_ADDR[10:00]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DSIZE[15:00]															

Bit	Name	RW	Reset	Description
26:16	ECC_ADDR	R/W	0x000	ECC offset address in NFC dedicate memory

It represents the offset address of ECC data in NFC dedicated memory. n (n=0,...,31) is the number of subpage.

In case of burst program by NDMA, after ECC encoding of each subpage is completed. The generated ECC data is automatically saved to the NFC dedicated memory where ECC_ADDR is.

In case of burst read by NDMA, if NFC_CTRL[BMODE] equal '1', the controller reads ECC data from the address and ECC decoding for each subpage is automatically processed after all of the data is transferred from NAND Flash to the destination memory (payload data) and NFC dedicated memory (ECC or information data as much as NFC_SPARE[SPARE_DSIZE])

Bit	Name	RW	Reset	Description
15:0	DSIZE	R/W	0x0000	The number of data for each subpage[n]

Note: In case of burst transfer by NDMA, it represents ECC encoding / decoding data size and transferred data size from/to NAND Flash. Its size must be set.

NFC ECC Control Register (ECC_CTRL)

0x8000D200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ECC_DSIZE[26:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
reserved										CP	MODE	ECC_EN[03:00]			

Bit	Name	RW	Reset	Description
26:16	ECC_DSIZE	R/W	0x000	N : ECC data size It is only useful for the following case 1) NAND Flash burst transfer by GDMA & CPU and ECC_CTRL[MODE] = 0 2) ECC_CTRL[MODE] = 1
5	CP	R/W	0x0	MLC ECC Correction Pass Enable 0 : Correction pass disable 1 : Correction pass enable
4	MODE	R/W	0x0	ECC Mode Selection 0: ECC is calculated for I/O data transferred to NAND Flash. 1 : ECC is calculated for internal data written to NFC dedicated memory or read from NFC dedicated memory Note: To access NFC memory, MODE should be '1'
3:0	ECC_EN	R/W	0x0	ECC Enable Configuration 00xx : Disable 0100 : SLC ECC encoding enable 0101 : SLC ECC decoding enable 0110 : MLC ECC4 encoding enable 0111 : MLC ECC4 decoding enable 1000 : MLC ECC6 encoding enable 1001 : MLC ECC6 decoding enable 1010 : MLC ECC12 encoding enable 1011 : MLC ECC12 decoding enable 1100 : MLC ECC16 encoding enable 1101 : MLC ECC16 decoding enable 1110 : MLC ECC24 encoding enable 1111 : MLC ECC24 decoding enable

NFC SECC Error Number Register (SECC_STATUS)

0x8000D204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ENUM15		ENUM14		ENUM13		ENUM12		ENUM11		ENUM10		ENUM09		ENUM08	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ENUM07		ENUM06		ENUM05		ENUM04		ENUM03		ENUM02		ENUM01		ENUM00	

(n = 0, ..., 15)

Bit	Name	RW	Reset	Description
[2*n+1:2*n]	ENUM[n]	R/W	0x0	SLC ECC Error Status 00 : No error 01 : 2-bit ECC error is detected (uncorrectable) 10 : 1-bit ECC error is detected (correctable) 11 : Uncorrectable error

NFC MLC ECC Error Status Register (MECC_STATUS)

0x8000D204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
											C_SUBPAGE				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
											ERRNUM[04:00]				

Table 22.3 Error Status Table for MLC ECC4/6/12/16/24 bit

ERRNUM	MLC ECC4	MLC ECC6	MLC ECC12	MLC ECC16	MLC ECC24
[4:0]	DECODE	DECODE	DECODE	DECODE	DECODE
00000	No Error	No Error	No Error	No Error	No Error
00001	1 BIT Error	1 BIT Error	1 BIT Error	1 BIT Error	1 BIT Error
00010	2 BIT Error	2 BIT Error	2 BIT Error	2 BIT Error	2 BIT Error
00011	3 BIT Error	3 BIT Error	3 BIT Error	3 BIT Error	3 BIT Error
00100	4 BIT Error	4 BIT Error	4 BIT Error	4 BIT Error	4 BIT Error
00101	-unknown	5 BIT Error	5 BIT Error	5 BIT Error	5 BIT Error
00110		6 BIT Error	6 BIT Error	6 BIT Error	6 BIT Error
00111		-unknown	7 BIT Error	7 BIT Error	7 BIT Error
01000			8 BIT Error	8 BIT Error	8 BIT Error
01001			9 BIT Error	9 BIT Error	9 BIT Error
01010			10 BIT Error	10 BIT Error	10 BIT Error
01011			11 BIT Error	11 BIT Error	11 BIT Error
01100			12 BIT Error	12 BIT Error	12 BIT Error
01101		-unknown	13 BIT Error	13 BIT Error	
01110			14 BIT Error	14 BIT Error	
01111			15 BIT Error	15 BIT Error	
10000			16 BIT Error	16 BIT Error	
10001		-unknown	-unknown	-unknown	17 BIT Error
10010					18 BIT Error
10011					19 BIT Error
10100					20 BIT Error
10101	21 BIT Error				
10110	22 BIT Error				
10111	23 BIT Error				
11000	24 BIT Error				
11001~				unknown	
11111	Uncorrectable error				

NFC ECC Clear Register (ECC_BASE)

0x8000D208

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE_ADDR															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE_ADDR															

Bit	Name	RW	Reset	Description
31:0	BASE_ADDR	R/W	0x8000D800	When ECC is enabled and MODE of ECC is AHB, this register is used for Base address to calculate ECC.

NFC ECC Clear Register (ECC_MASK)

0x8000D20C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MASK_ADDR															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MASK_ADDR															

Bit	Name	RW	Reset	Description
31:0	MASK_ADDR	R/W	0x1000	When ECC is enabled and MODE of ECC is AHB, this register is used for Mask Address to calculate ECC. Note: The ECC is calculated whenever the specified region of memory is accessed. The region for ECC calculating is determined by ECC_BASE & ECC_MASK Register. The real base address is determined by following formula. Real base address = ECC_BASE & ~(ECC_MASK[27:2] << 2) (The real base address is assumed to be word aligned, so the least 2 bits are always 0.) The size of region is also determined by ECC_MASK Register. If ECC_MASK Register has N concatenated 0 from LSB, the region size is set to 2^N byte

NFC ECC Clear Register (ECC_CLEAR)

0x8000D210

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Don't care															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Don't care															

Bit	Name	RW	Reset	Description
31:0	Don't Care	R/W	0x00000000	Don't Care

If any bit is written, some of internal logic is cleared. For ECC encoding or decoding, this register must be set before and after ECC_CTRL[ECC_EN] is set.

NFC ECC Decoding Restart (ECC_RSTART)

0x8000D214

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Don't care															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Don't care															

Bit	Name	RW	Reset	Description
31:0	Don't Care	R/W	-	Don't Care

This register is used to restart ECC decoding procedure for the next subpage.

In case of burst read by NDMA, the ECC decoding of each subpage is subsequently processed by the controller. When the ECC error occurs during ECC decoding for some of subpages, the controller generates an error flag and holds the ECC decoding procedure for the rest of subpages.

To start ECC [decoding] decoding for the next subpages, this register should be written with any value.

NFC ECC Store Status & Load Control Register (ECC_STLDCTL)

0x8000D218

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STLD31	STLD30	STLD29	STLD28	STLD27	STLD26	STLD25	STLD24	STLD23	STLD22	STLD21	STLD20	STLD19	STLD18	STLD17	STLD16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STLD15	STLD14	STLD13	STLD12	STLD11	STLD10	STLD9	STLD8	STLD7	STLD6	STLD5	STLD4	STLD3	STLD2	STLD1	STLD0

Bit	Name	RW	Reset	Description
31:0	STLD0~STLD31	R/W	0x00000000	ECC Store Status and Load Enable Control Each bit filed is synchronized with subpage N. 0 (read) : Others 1 (read): ECC of subpage N is stored in internal store register. 1 (write): ECC of subpage N is loaded in internal LFSR (Linear Feedback Shift Register) for ECC decoding of the subpage when NFC_CTRL[BMODE] is '0'. Once any of bit filed which value is '1' is written to '1', it is cleared (1->0). If all of the bit field is cleared (32'b0), there is no subpage for decoding.

NFC ECC Current State Register (ECC_CSTATE)

0x8000D21C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ECC_STATE[29:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ECC_STATE[15:00]															

Bit	Name	RW	Reset	Description
29:0	ECC_STATE	R/W	0x00000001	0x00000001: ECC_IDLE. 0x00000002: SLC ECC data wait 0x00000004: SLC ECC encode or decode 0x00000008: SLC ECC End 0x00000010: MLC ECC encode data wait 0x00000020: MLC ECC decode data wait 0x00000040: MLC ECC encode 0x00000080: MLC ECC decode 0x00000100: MLC ECC4 encoded ecc input wait 0x00000200: MLC ECC4 decoded ecc input wait 0x00000400: MLC ECC4 error calculation is on process. 0x00000800: MLC ECC6 encoded ecc input wait 0x00001000: MLC ECC6 decoded ecc input wait 0x00002000: MLC ECC6 error calculation is on process. 0x00004000: MLC ECC12 encoded ecc input wait 0x00008000: MLC ECC12 decoded ecc input wait 0x00010000: MLC ECC12 error calculation is on process. 0x00020000: MLC ECC16 encoded ecc input wait 0x00040000: MLC ECC16 decoded ecc input wait 0x00080000: MLC ECC16 error calculation is on process. 0x00020000: MLC ECC24 encoded ecc input wait 0x00040000: MLC ECC24 decoded ecc input wait 0x00400000: MLC ECC24 error calculation is on process. 0x08000000: MLC ECC Error Occur in NDMA mode 0x08000000: MLC ECC Error and Holt State (MECC_CORR) 0x20000000: MLC ECC next subpage decode wait (MECC_PWAIT) Others: Reserved If burst read by NDMA and BMODE=0, should check if ECC_STATE is the followings at the end of every subpage decoding because MECC decoding end flag does not occur. 1) ECC_IDLE: Represents all of the subpage decoding completed and last subpage decoding has no error. Controller is ready. 2) MECC_CORR: Error occurs in current subpage decoding. 3) MECC_PWAIT: There is no error in current subpage decoding and controller is ready for next subpage decoding.

MLC ECC4 Code Register 0 (MECC4_0)

0x8000D220

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC4_CODE1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC4_CODE0[15:00]															

Bit	Name	RW	Reset	Description
31:16	MECC4_CODE1	R/W	0x0000	MLC ECC4 Code Register1
15:0	MECC4_CODE0	R/W	0x0000	MLC ECC4 Code Register0

MLC ECC4 Code Register 1 (MECC4_1)

0x8000D224

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								MECC4_CODE4[56:48]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC4_CODE3[47:32]															

Bit	Name	RW	Reset	Description
23:16	MECC4_CODE4	R/W	0x00	MLC ECC4 Code Register4
15:0	MECC4_CODE3	R/W	0x0000	MLC ECC4 Code Register3

Note: Real ECC Code Size for MLC ECC4 mode is 52 bits. For the remainder bits of MLC ECC4 Code Register 1, zeros are padded.

Note: MLC ECC4 Code Register must be written and read by word access.

MLC ECC6 Code Register 0 (MECC6_0)

0x8000D220

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC6_CODE1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC6_CODE0[15:00]															

Bit	Name	RW	Reset	Description
31:16	MECC6_CODE1	R/W	0x0000	MLC ECC6 Code Register1
15:0	MECC6_CODE0	R/W	0x0000	MLC ECC6 Code Register0

MLC ECC6 Code Register 1 (MECC6_1)

0x8000D224

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC6_CODE3[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC6_CODE2[47:32]															

Bit	Name	RW	Reset	Description
31:16	MECC6_CODE3	R/W	0x0000	MLC ECC6 Code Register3
15:0	MECC6_CODE2	R/W	0x0000	MLC ECC6 Code Register2

MLC ECC6 Code Register 2 (MECC6_2)

0x8000D228

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								MECC6_CODE5[83:80]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC6_CODE4[79:64]															

Bit	Name	RW	Reset	Description
19:16	MECC6_CODE5	R/W	0x0	MLC ECC6 Code Register5
15:0	MECC6_CODE4	R/W	0x0000	MLC ECC6 Code Register4

Note: Real ECC Code Size for MLC ECC8 mode is 104 bits. For the remainder bits of MLC ECC6 Code Register 3, zeros are padded.

Note: MLC ECC8 Code Register must be written and read by word access.

MLC ECC12 Code Register 0 (MECC12_0)

0x8000D220

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC12_CODE1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC12_CODE0[15:00]															

Bit	Name	RW	Reset	Description
31:16	MECC12_CODE1	R/W	0x0000	MLC ECC12 Code Register1
15:0	MECC12_CODE0	R/W	0x0000	MLC ECC12 Code Register0

MLC ECC12 Code Register 1 (MECC12_1)

0x8000D224

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC12_CODE3[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC12_CODE2[47:32]															

Bit	Name	RW	Reset	Description
31:16	MECC12_CODE3	R/W	0x0000	MLC ECC12 Code Register3
15:0	MECC12_CODE2	R/W	0x0000	MLC ECC12 Code Register2

MLC ECC12 Code Register 2 (MECC12_2)

0x8000D228

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC12_CODE5[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC12_CODE4[79:64]															

Bit	Name	RW	Reset	Description
31:16	MECC12_CODE5	R/W	0x0000	MLC ECC12 Code Register5
15:0	MECC12_CODE4	R/W	0x0000	MLC ECC12 Code Register4

MLC ECC12 Code Register 3 (MECC12_3)

0x8000D22C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC12_CODE7[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC12_CODE6[111:96]															

Bit	Name	RW	Reset	Description
31:16	MECC12_CODE7	R/W	0x0000	MLC ECC12 Code Register7
15:0	MECC12_CODE6	R/W	0x0000	MLC ECC12 Code Register6

MLC ECC12 Code Register 4 (MECC12_4)

0x8000D230

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC12_CODE9[159:144]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC12_CODE8[143:128]															

Bit	Name	RW	Reset	Description
31:16	MECC12_CODE9	R/W	0x0000	MLC ECC12 Code Register9
15:0	MECC12_CODE8	R/W	0x0000	MLC ECC12 Code Register8

MLC ECC12 Code Register 4 (MECC12_5)

0x8000D234

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0															
MECC12_CODE10[168:160]															

Bit	Name	RW	Reset	Description
7:0	MECC12_CODE10	R/W	0x00	MLC ECC12 Code Register12

Note: Real ECC Code Size for MLC ECC12 mode is 156 bits. For the remainder bits of MLC ECC12 Code Register 4, zeros are padded.

Note: MLC ECC12 Code Register must be written and read by word access.

MLC ECC16 Code Register 0 (MECC16_0)

0x8000D220

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC16_CODE1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC16_CODE0[15:00]															

Bit	Name	RW	Reset	Description
31:16	MECC16_CODE1	R/W	0x0000	MLC ECC16 Code Register1
15:0	MECC16_CODE0	R/W	0x0000	MLC ECC16 Code Register0

MLC ECC16 Code Register 1 (MECC16_1)

0x8000D224

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC16_CODE3[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC16_CODE2[47:32]															

Bit	Name	RW	Reset	Description
31:16	MECC16_CODE3	R/W	0x0000	MLC ECC16 Code Register3
15:0	MECC16_CODE2	R/W	0x0000	MLC ECC16 Code Register2

MLC ECC16 Code Register 2 (MECC16_2)

0x8000D228

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC16_CODE5[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC16_CODE4[79:64]															

Bit	Name	RW	Reset	Description
31:16	MECC16_CODE5	R/W	0x0000	MLC ECC16 Code Register5
15:0	MECC16_CODE4	R/W	0x0000	MLC ECC16 Code Register4

MLC ECC16 Code Register 3 (MECC16_3)

0x8000D22C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC16_CODE7[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC16_CODE6[111:96]															

Bit	Name	RW	Reset	Description
31:16	MECC16_CODE7	R/W	0x0000	MLC ECC16 Code Register7
15:0	MECC16_CODE6	R/W	0x0000	MLC ECC16 Code Register6

MLC ECC16 Code Register 4 (MECC16_4)

0x8000D230

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC16_CODE9[159:144]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC16_CODE8[143:128]															

Bit	Name	RW	Reset	Description
31:16	MECC16_CODE9	R/W	0x0000	MLC ECC16 Code Register9
15:0	MECC16_CODE8	R/W	0x0000	MLC ECC16 Code Register8

MLC ECC16 Code Register 5 (MECC16_5)

0x8000D234

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC16_CODE11[191:176]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC16_CODE10[175:160]															

Bit	Name	RW	Reset	Description
31:16	MECC16_CODE11	R/W	0x0000	MLC ECC16 Code Register11
15:0	MECC16_CODE10	R/W	0x0000	MLC ECC16 Code Register10

MLC ECC16 Code Register 6 (MECC16_6)

0x8000D238

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC16_CODE13[223:208]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC16_CODE12[207:192]															

Bit	Name	RW	Reset	Description
31:16	MECC16_CODE13	R/W	0x0000	MLC ECC16 Code Register13
15:0	MECC16_CODE12	R/W	0x0000	MLC ECC16 Code Register12

Note: Real ECC Code Size for MLC ECC16 mode is 207 bits. For the remainder bits of MLC ECC16 Code Register 6, zeros are padded.

Note: MLC ECC16 Code Register must be written and read by word access.

MLC ECC24 Code Register 0 (MECC24_0)

0x8000D220

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE0[15:00]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE1	R/W	0x0000	MLC ECC24 Code Register1
15:0	MECC24_CODE0	R/W	0x0000	MLC ECC24 Code Register0

MLC ECC24 Code Register 1 (MECC24_1)

0x8000D224

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE3[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE2[47:32]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE3	R/W	0x0000	MLC ECC24 Code Register3
15:0	MECC24_CODE2	R/W	0x0000	MLC ECC24 Code Register2

MLC ECC24 Code Register 2 (MECC24_2)

0x8000D228

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE5[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE4[79:64]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE5	R/W	0x0000	MLC ECC24 Code Register5
15:0	MECC24_CODE4	R/W	0x0000	MLC ECC24 Code Register4

MLC ECC24 Code Register 3 (MECC24_3)

0x8000D22C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE7[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE6[111:96]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE7	R/W	0x0000	MLC ECC24 Code Register7
15:0	MECC24_CODE6	R/W	0x0000	MLC ECC24 Code Register6

MLC ECC24 Code Register 4 (MECC24_4)

0x8000D230

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE9[159:144]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE8[143:128]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE9	R/W	0x0000	MLC ECC24 Code Register9
15:0	MECC24_CODE8	R/W	0x0000	MLC ECC24 Code Register8

MLC ECC24 Code Register 5 (MECC24_5)

0x8000D234

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE11[191:176]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE10[175:160]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE11	R/W	0x0000	MLC ECC24 Code Register11
15:0	MECC24_CODE10	R/W	0x0000	MLC ECC24 Code Register10

MLC ECC24 Code Register 6 (MECC24_6)

0x8000D238

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE13[223:208]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE12[207:192]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE13	R/W	0x0000	MLC ECC24 Code Register13
15:0	MECC24_CODE12	R/W	0x0000	MLC ECC24 Code Register12

MLC ECC24 Code Register 7 (MECC24_7)

0x8000D23C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE15[255:240]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE14[239:224]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE15	R/W	0x0000	MLC ECC24 Code Register15
15:0	MECC24_CODE14	R/W	0x0000	MLC ECC24 Code Register14

MLC ECC24 Code Register 8 (MECC24_8)

0x8000D240

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE17[287:272]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE16[271:256]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE17	R/W	0x0000	MLC ECC24 Code Register17
15:0	MECC24_CODE16	R/W	0x0000	MLC ECC24 Code Register16

MLC ECC24 Code Register 9 (MECC24_9)

0x8000D244

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MECC24_CODE19[319:304]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE18[303:288]															

Bit	Name	RW	Reset	Description
31:16	MECC24_CODE19	R/W	0x0000	MLC ECC24 Code Register19
15:0	MECC24_CODE18	R/W	0x0000	MLC ECC24 Code Register18

MLC ECC24 Code Register 10 (MECC24_10)

0x8000D248

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MECC24_CODE20[335:320]															

Bit	Name	RW	Reset	Description
15:0	MECC24_CODE20	R/W	0x0000	MLC ECC24 Code Register20

SLC ECC Error Address (SECC_EADDR_x, x = 0,...,15)

0x8000D260+n*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECC_EADDR _x [14:0]															

Note: To correct errors for SLC ECC

- SECC_EADDR[14:3]: Byte Error Address

- SECC_EADDR[2:0]: Error Bit Plane

Corrected DATA = ADDR[SECC_EADDR[14:3]] ^ (1 << SECC_EADDR[2:0])

(Byte)

(Byte)

Bit	Name	RW	Reset	Description
14:0	SECC_EADDR	R	0x0000	SLC ECC Error Address Register

MLC ECC4 Error Address Register (MECC4_EADDR_n, n = 0,1,2,3)

0x8000D260+n*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EADDR[13:00]															

Bit	Name	RW	Reset	Description
13:0	EADDR	R	0x0000	N is bit address of ECC ERROR

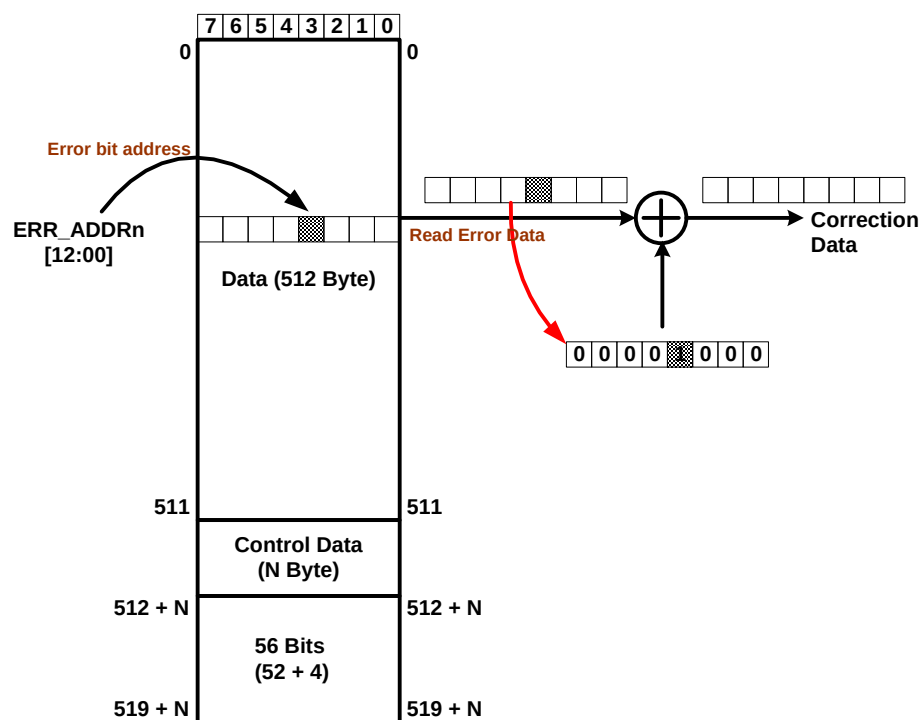


Figure 22.10 Example of MLC ECC4 error correction

MLC ECC6 Error Address Register (MECC8_EADDR_n, n = 0,...,5)

0x8000D260+n*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EADDR[13:00]															

Note: Refer to MECC4_EADDR_n.

Bit	Name	RW	Reset	Description
13:0	EADDR	R	0x0000	N is bit address of ECC6 ERROR

MLC ECC12 Error Address Register (MECC12_EADDRn, n = 0,...,11) 0x8000D260+n*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EADDR[13:00]															

Note: Refer to MECC4_EADDRn.

Bit	Name	RW	Reset	Description
13:0	EADDR	R	0x0000	N is bit address of ECC12 ERROR

MLC ECC16 Error Address Register (MECC16_EADDRn, n = 0,...,15) 0x8000D260+n*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EADDR[13:00]															

Note: Refer to MECC4_EADDRn.

Bit	Name	RW	Reset	Description
13:0	EADDR	R	0x0000	N is bit address of ECC16 ERROR

MLC ECC24 Error Address Register (MECC24_EADDRn, n = 0,...,23) 0x8000D260+ n*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EADDR[13:00]															

Note: Refer to MECC4_EADDRn.

Bit	Name	RW	Reset	Description
13:0	EADDR	R	0x0000	N is bit address of ECC24 ERROR

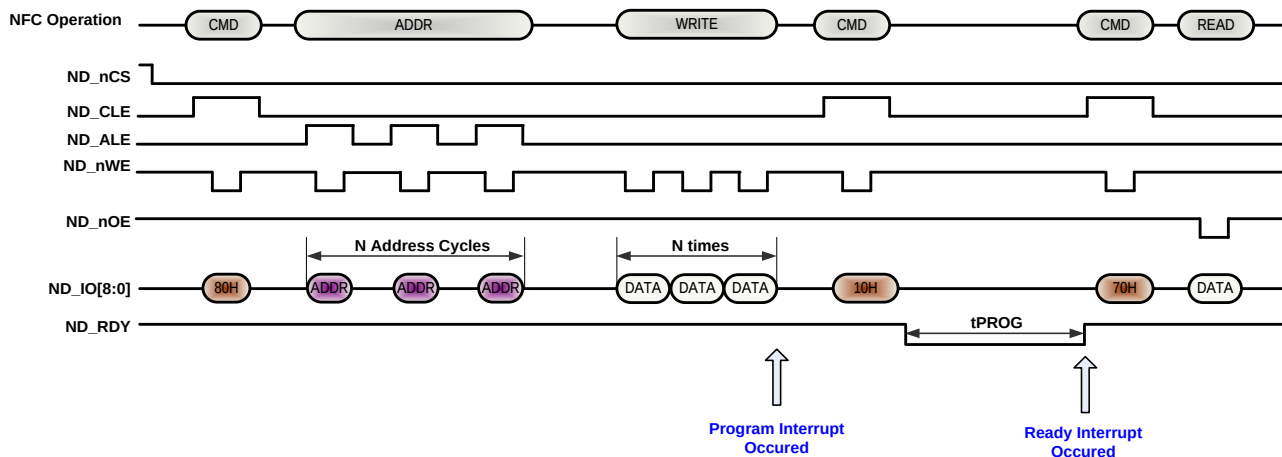


Figure 22.11 Timing Diagram of NAND Page Program

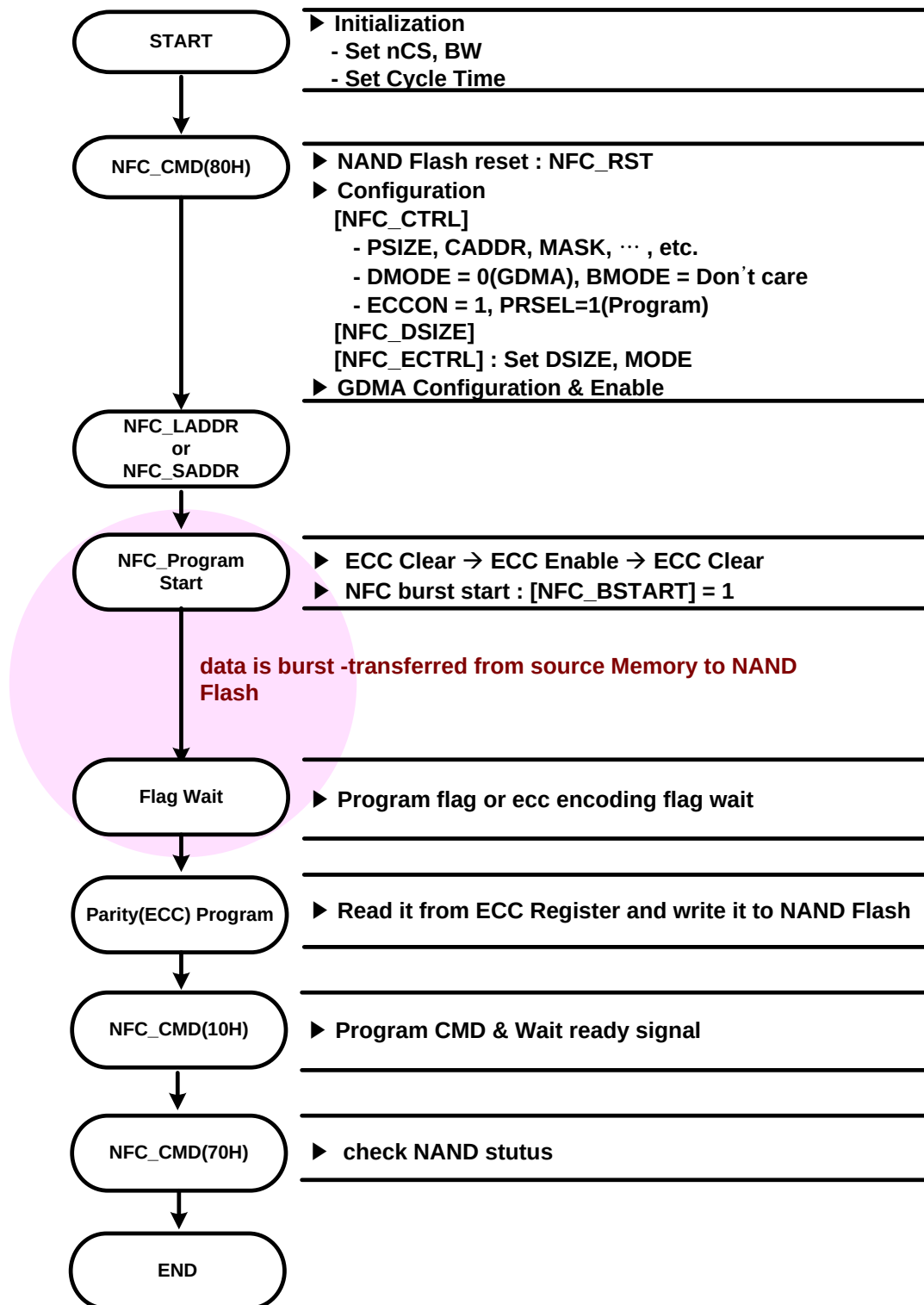


Figure 22.12 NAND Burst Program Sequence by GDMA

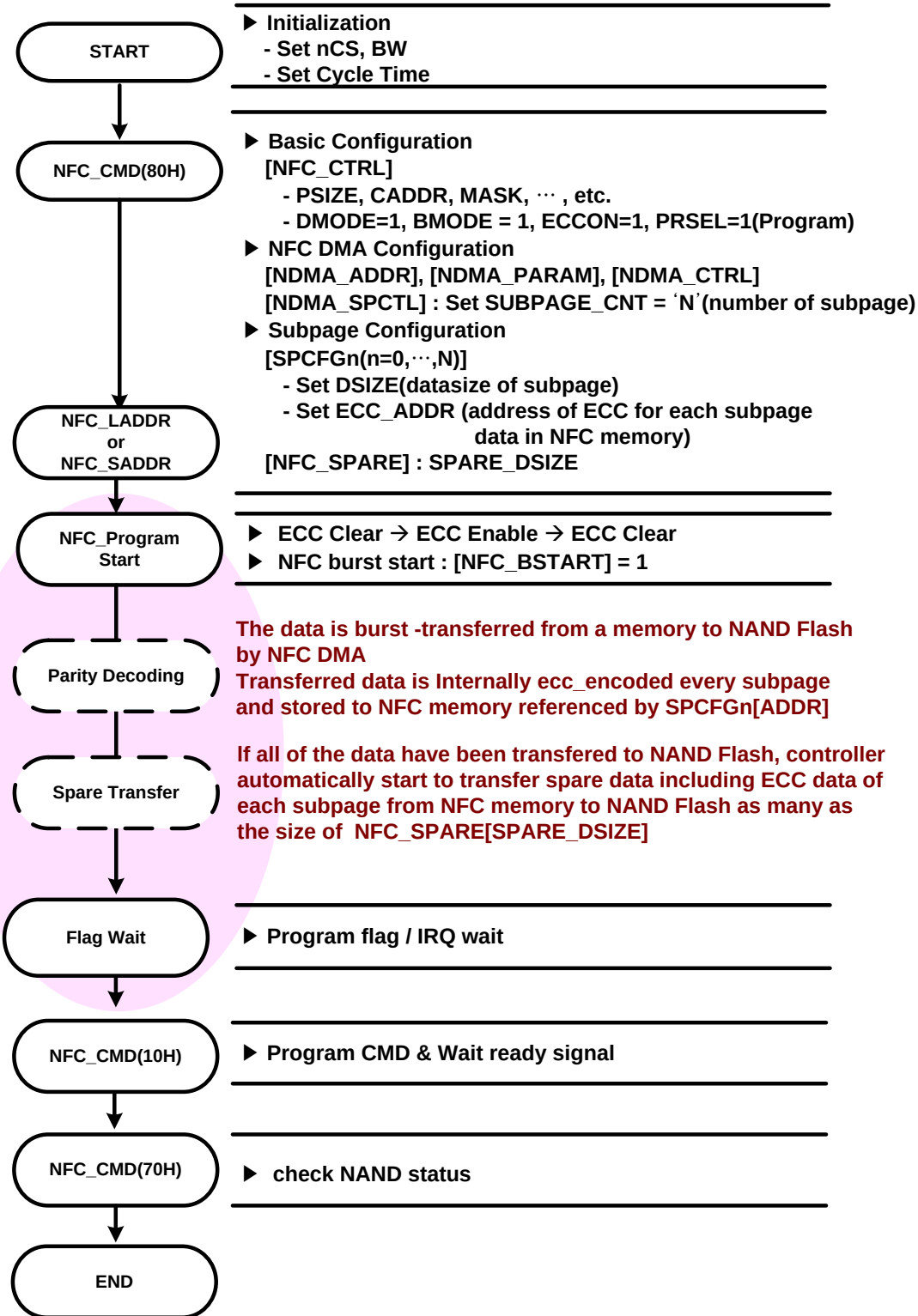


Figure 22.13 NAND Burst Program Sequence by NDMA (BMODE=1)

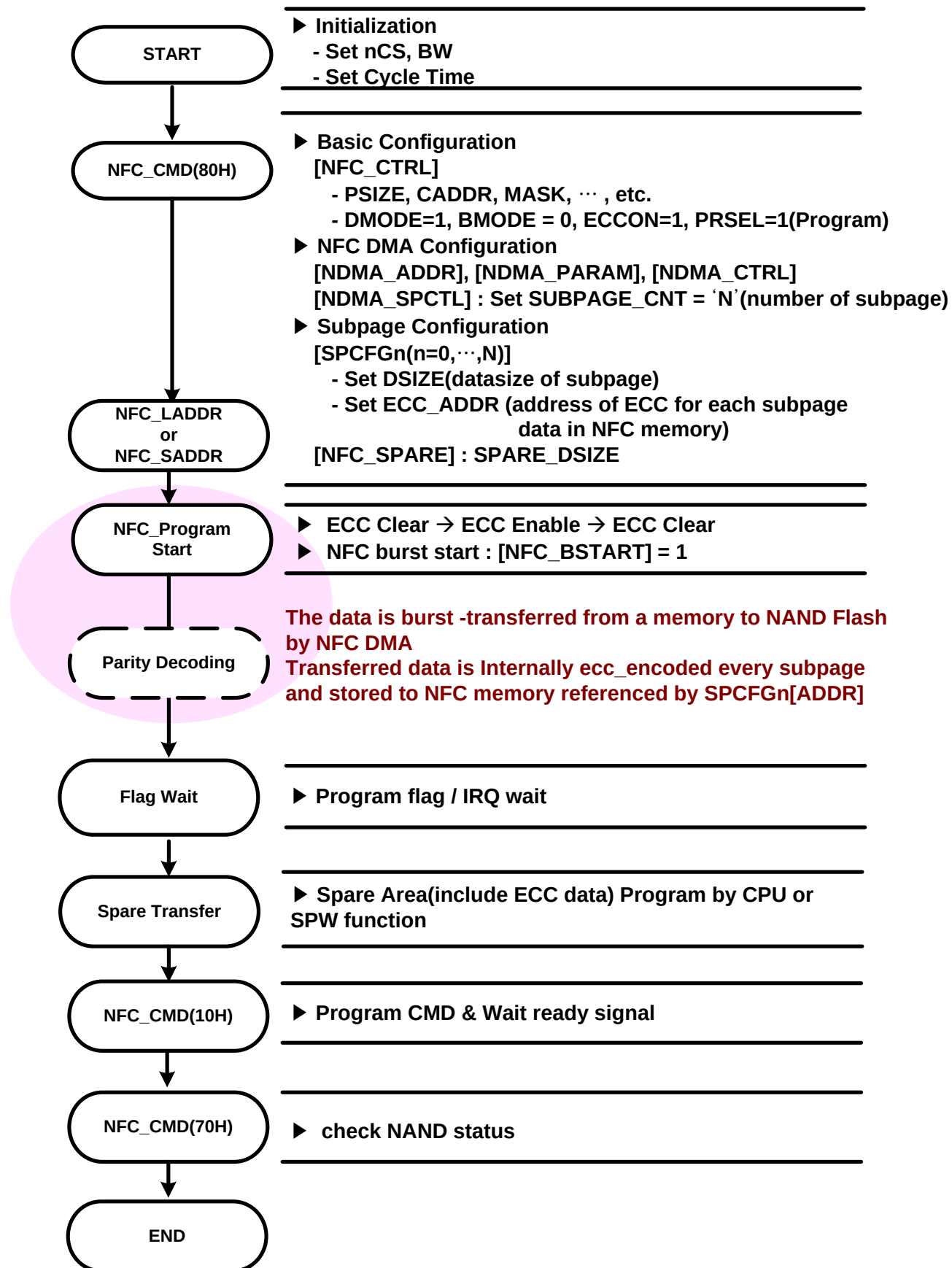


Figure 22.14 NAND Burst Program Sequence by NDMA (BMODE=0)

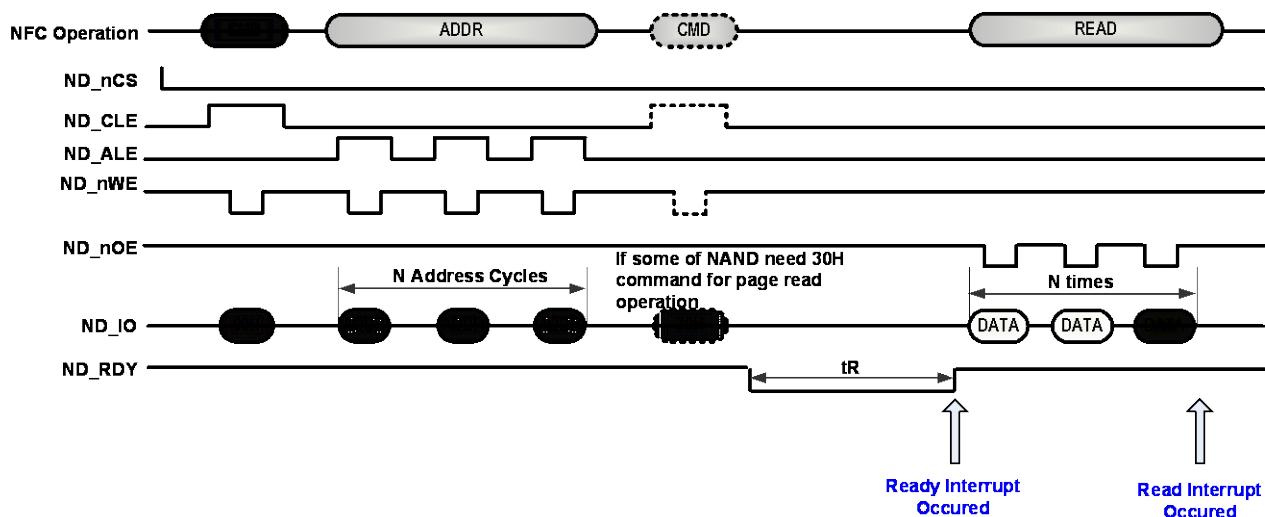


Figure 22.15 Timing Diagram of NAND Page Read

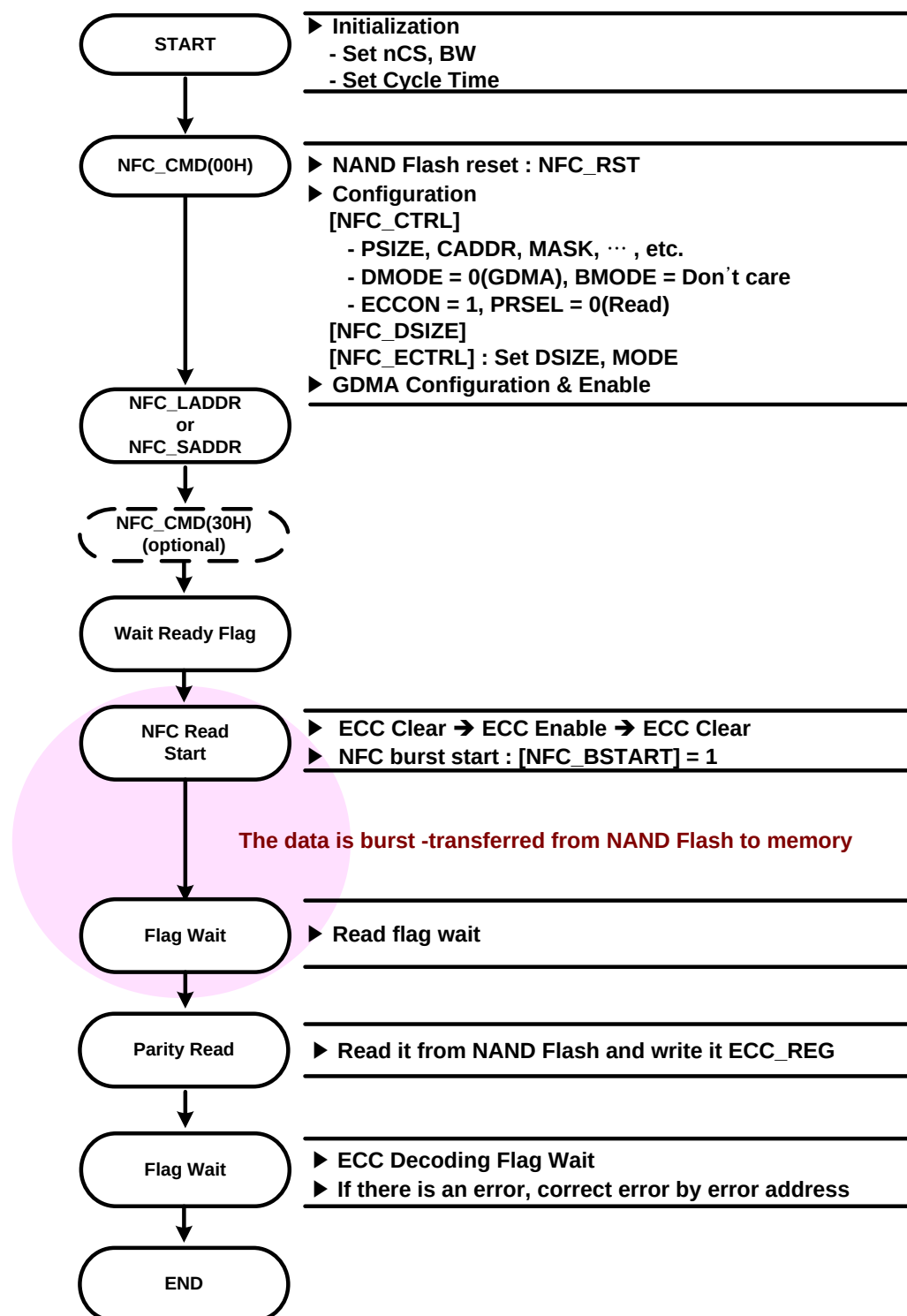


Figure 22.16 NAND Burst Read Sequence by GDMA

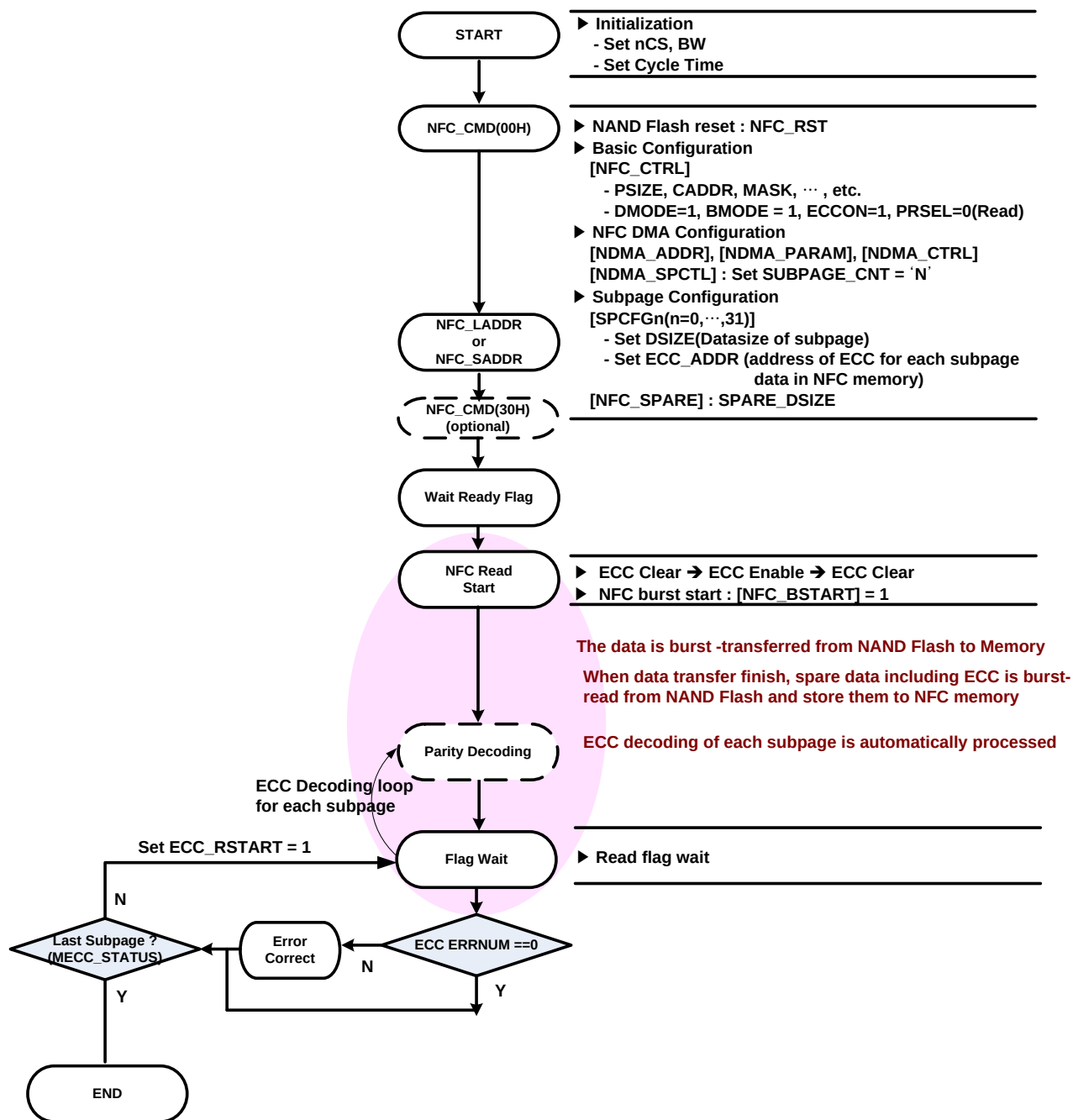


Figure 22.17 NAND Burst Read Sequence by NDMA (BMODE=1)

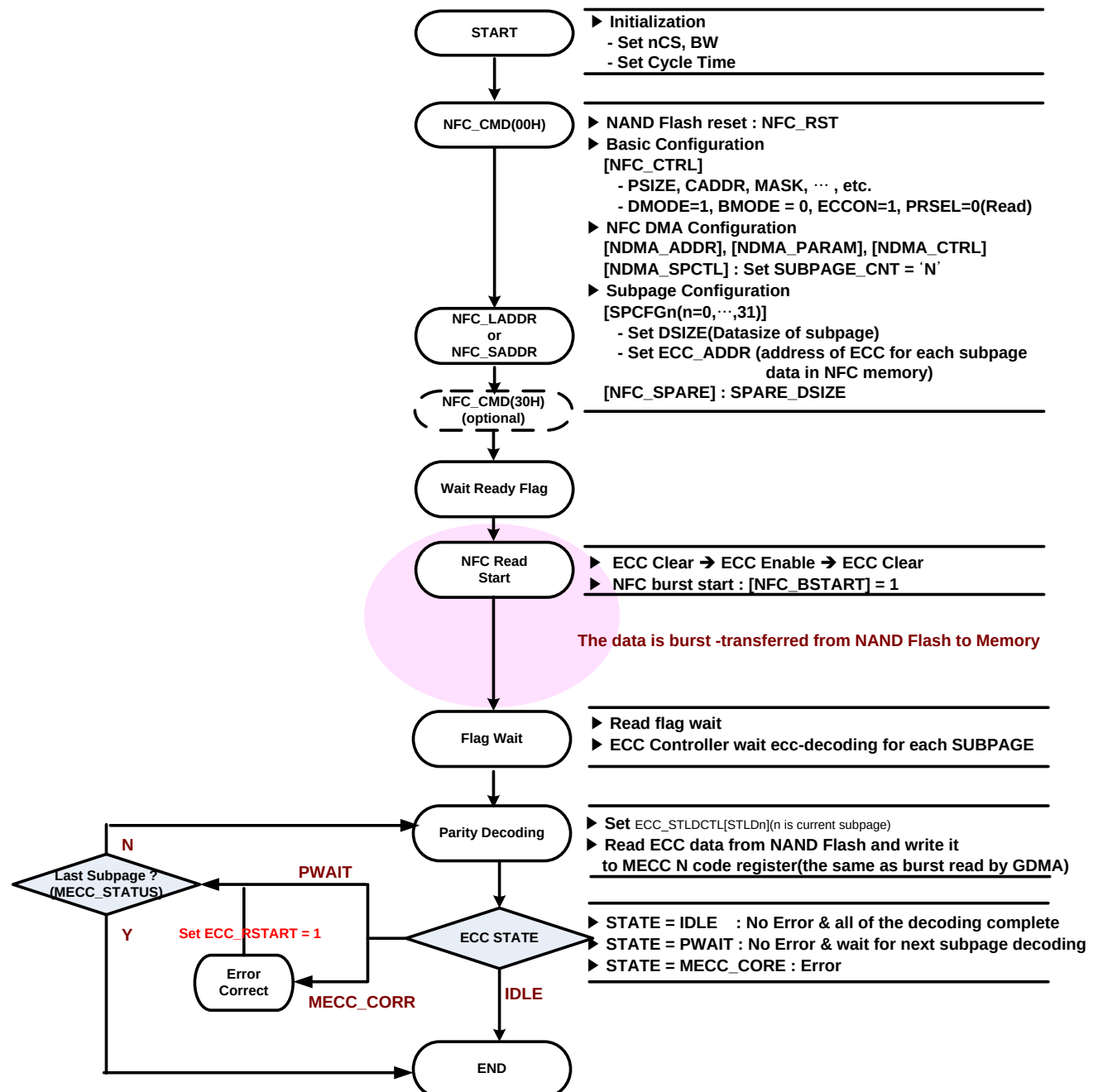


Figure 22.18 NAND Burst Read Sequence by NDMA (BMODE=0)

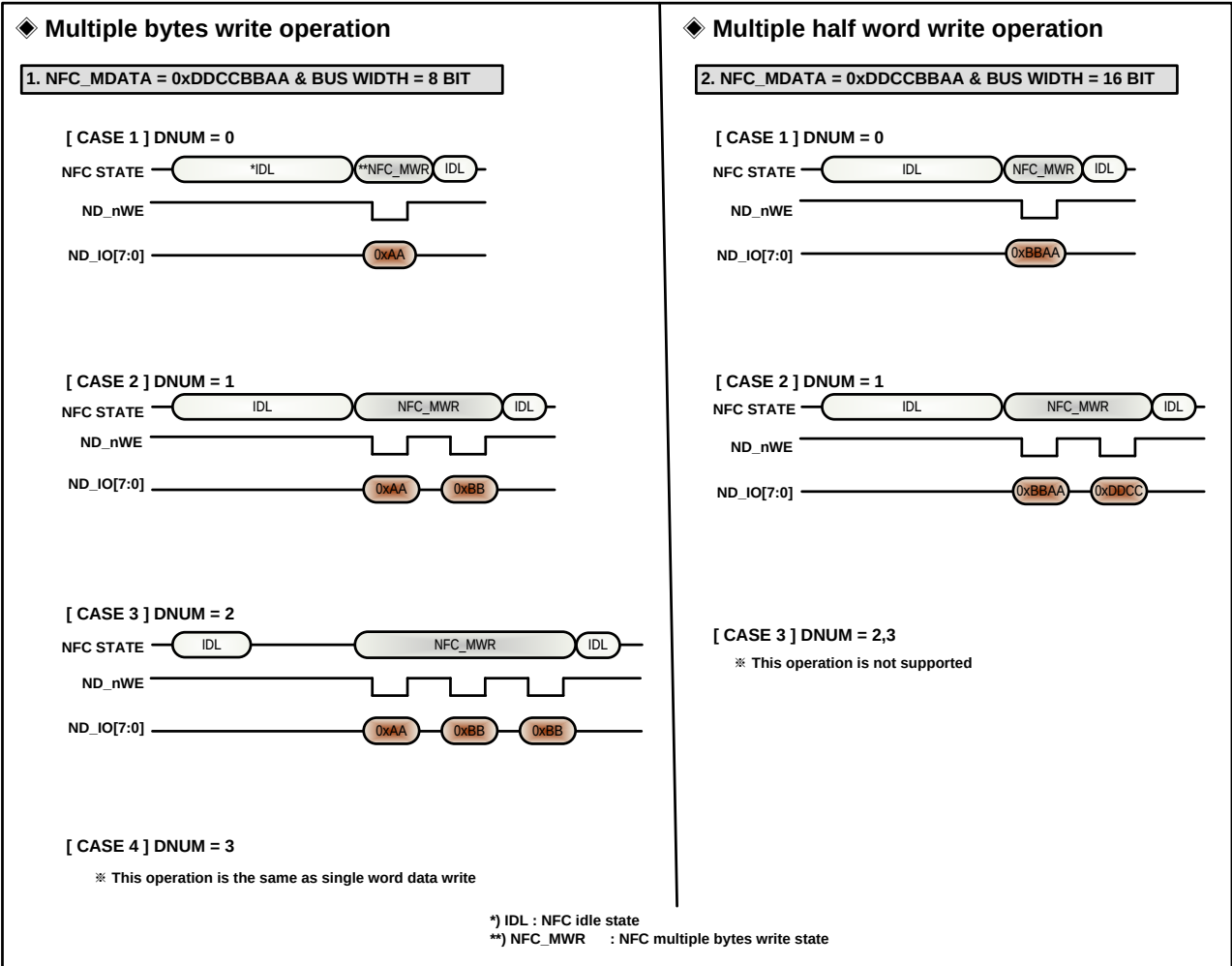


Figure 22.19 Example of Multiple bytes/half Word Write Operation

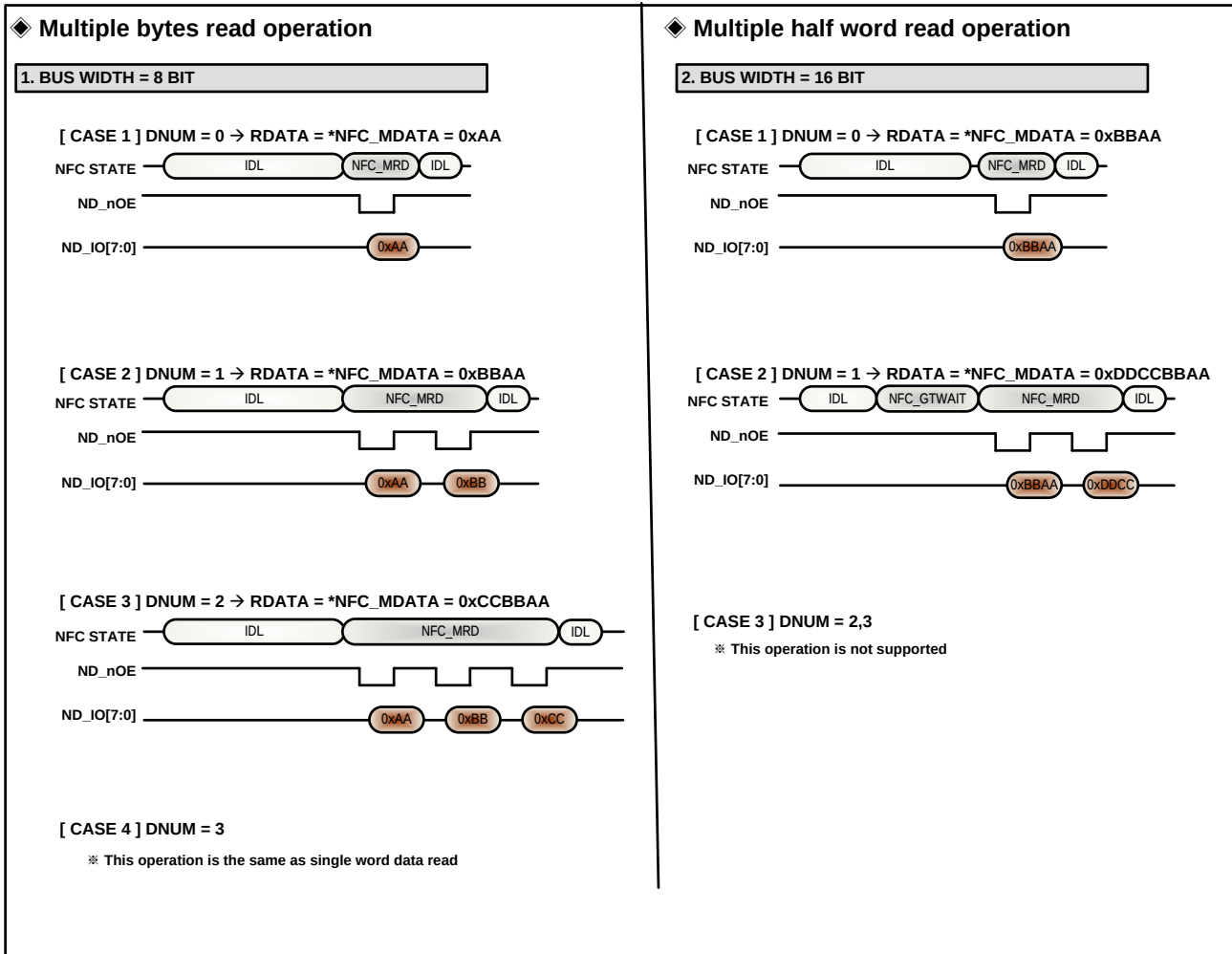


Figure 22.20 Example of Multiple bytes/half Word Read Operation

22.3 Nand Flash Controller Timing

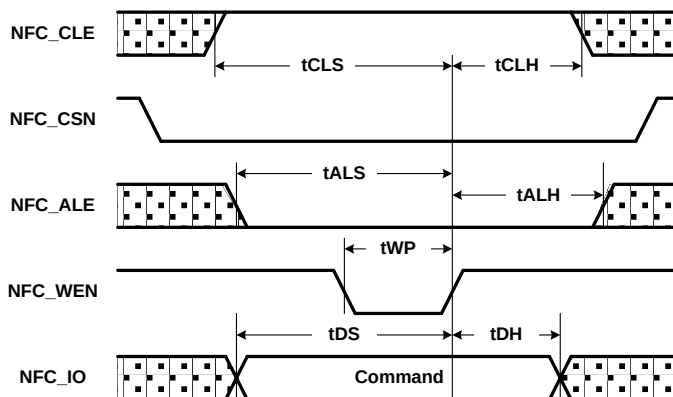


Figure 22.21 Timing Diagram for Command Latch Enable Cycle

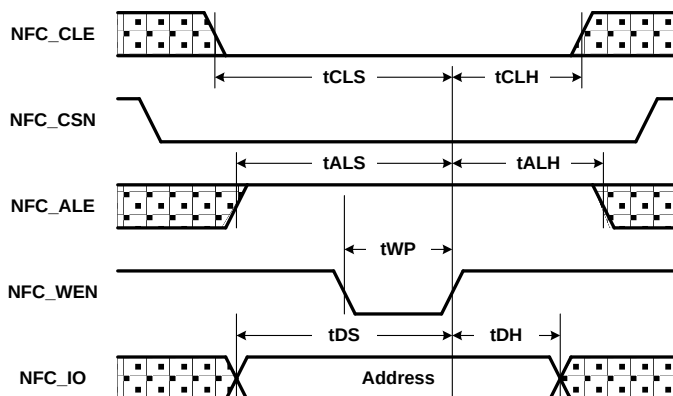


Figure 22.22 Timing Diagram for Single Address Latch Cycle

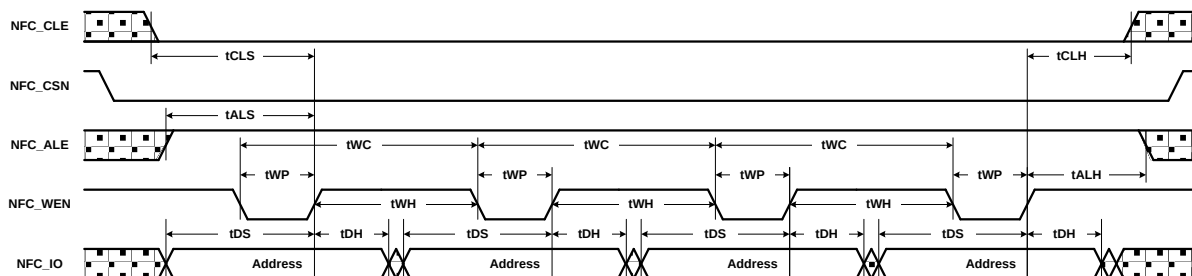


Figure 22.23 Timing Diagram for Linear Address Latch Cycle

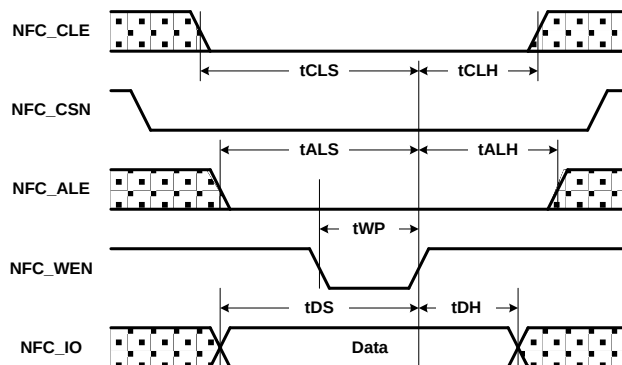


Figure 22.24 Timing Diagram for Single Data Write Cycle

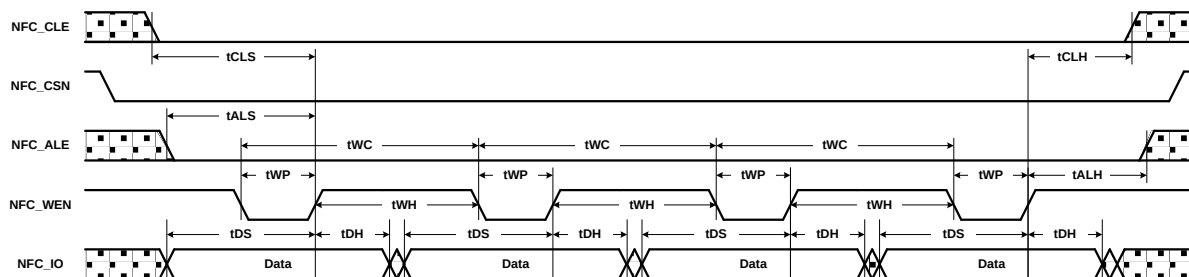


Figure 22.25 Timing Diagram for Linear Data Write Cycle

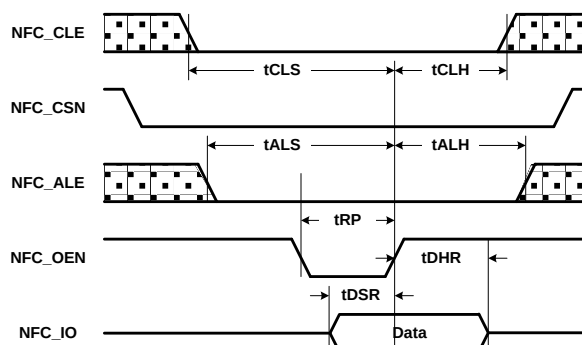


Figure 22.26 Timing Diagram for Single Data Read Cycle

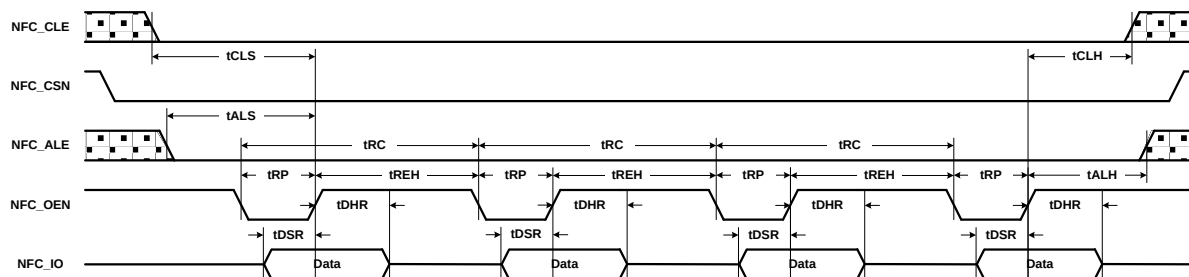


Figure 22.27 Timing Diagram for Linear Data Read Cycle

Table 22.4 Timing Parameters for NAND Flash Controller

Parameter	Symbol	Test Condition	MIN	MAX	Unit
Clock Period	t_{HCLK}	$V_{DDI} = 1.1V$	5.00		ns
Clock Frequency	f_{HCLK}	$V_{DDI} = 1.1V$		110	MHz
CLE Set-Up Time	t_{CLS}		$(STP + PW) \times t_{HCLK}$		ns
CLE Hold Time	t_{CLH}		$HLD \times t_{HCLK}$		ns
WEN Pulse Width	t_{WP}		$PW \times t_{HCLK}$		ns
WEN High Hold Time	t_{WH}		$(STP + HLD) \times t_{HCLK}$		ns
Write Cycle Time	t_{WC}		$(STP + PW + HLD) \times t_{HCLK}$		ns
OEN Pulse Width	t_{RP}		$PW \times t_{HCLK}$		ns
OEN High Hold Time	t_{REH}		$(STP + HLD) \times t_{HCLK}$		ns
Read Cycle Time	t_{RC}		$(STP + PW + HLD) \times t_{HCLK}$		ns
ALE Set-Up Time	t_{ALS}		$(STP + PW) \times t_{HCLK}$		ns
ALE Hold Time	t_{ALH}		$HLD \times t_{HCLK}$		ns
Data Set-Up Time	t_{DS}		$(STP + PW) \times t_{HCLK}$		ns
Data Hold Time	t_{DH}		$HLD \times t_{HCLK}$		ns
Output Valid Delay from HCLK	t_{OV}	$V_{DDI} = 1.1V, V_{DDIO} = 3.3V, C_L = 30pF$	1	10	ns
Data Set-Up Time to HCLK	t_{DSU}	$V_{DDI} = 1.1V, V_{DDIO} = 3.3V, C_L = 30pF$	4		ns
Data Set-Up Time in READ	t_{DSR}	$V_{DDI} = 1.1V, V_{DDIO} = 3.3V, C_L = 30pF$	$t_{DSU} + t_{OV}$		ns
Data Hold Time in READ	t_{DHR}	$V_{DDI} = 1.1V, V_{DDIO} = 3.3V$	0		ns

Note: Refer to NFC_CTRL register for STP, PW, and HLD.

23 USB 2.0 Host Controller

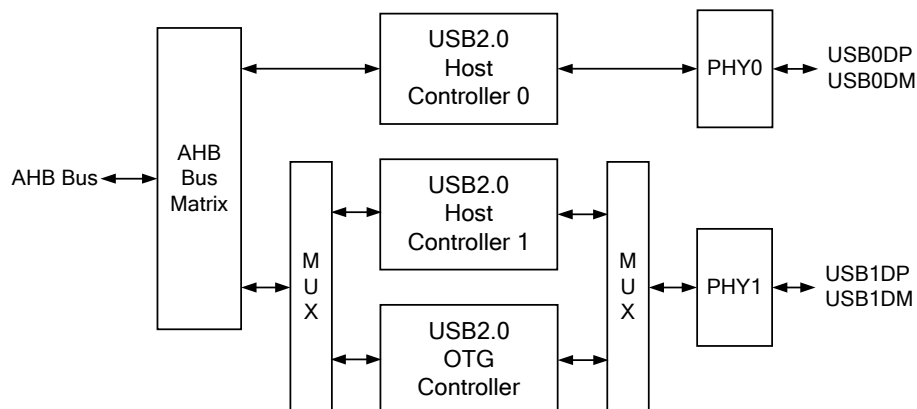


Figure 23.1 USB Interface Block Diagram

Table 23.1 USB Interface Address Map

Address	Description
0x80040000 ~ 0x8007FFFF	USB 20DH
0x80080000 ~ 0x8008FFFF	USB Host 0 EHCI
0x80090000 ~ 0x8009FFFF	USB Host 0 OHCI
0x800A0000 ~ 0x800AFFFF	USB Host 1 EHCI
0x800B0000 ~ 0x800BFFFF	USB Host 1 OHCI
0x800C0000 ~ 0x800C001F	USB (PHY) Configuration Registers – USB 20DH
0x800C0020 ~ 0x800C003F	USB (PHY) Configuration Registers – USB Host 0
0x800C0040 ~ 0x800C005F	USB (PHY) Configuration Registers – USB Host 1
0x800C0060	USB Controller Selection Register

23.1 Overview

The main features of the TCC8132 USB 2.0 Host controller are as follows.

- Complies with the USB 2.0 High-speed, Full-speed, and Low-speed (Host Mode only) Specification
- Supports ping and split transactions
- Complies with Enhanced Host Controller Interface (EHCI) Specification, Revision 1.0, and the Open Host Controller Interface (OHCI) Specification Release 1.0a. (full-speed only)
- Supports high-speed, 480-Mbps transfers (40 times faster than USB 1.1 full-speed mode) using an EHCI Host Controller, as well as full speed through an integrated OHCI Host Controller.
- At the USB 2.0 physical interface, the subsystem provides a USB 2.0 Transceiver Macrocell Interface (UTMI), which is fully compliant to the UTMI Specification, Revision 1.05.[2]

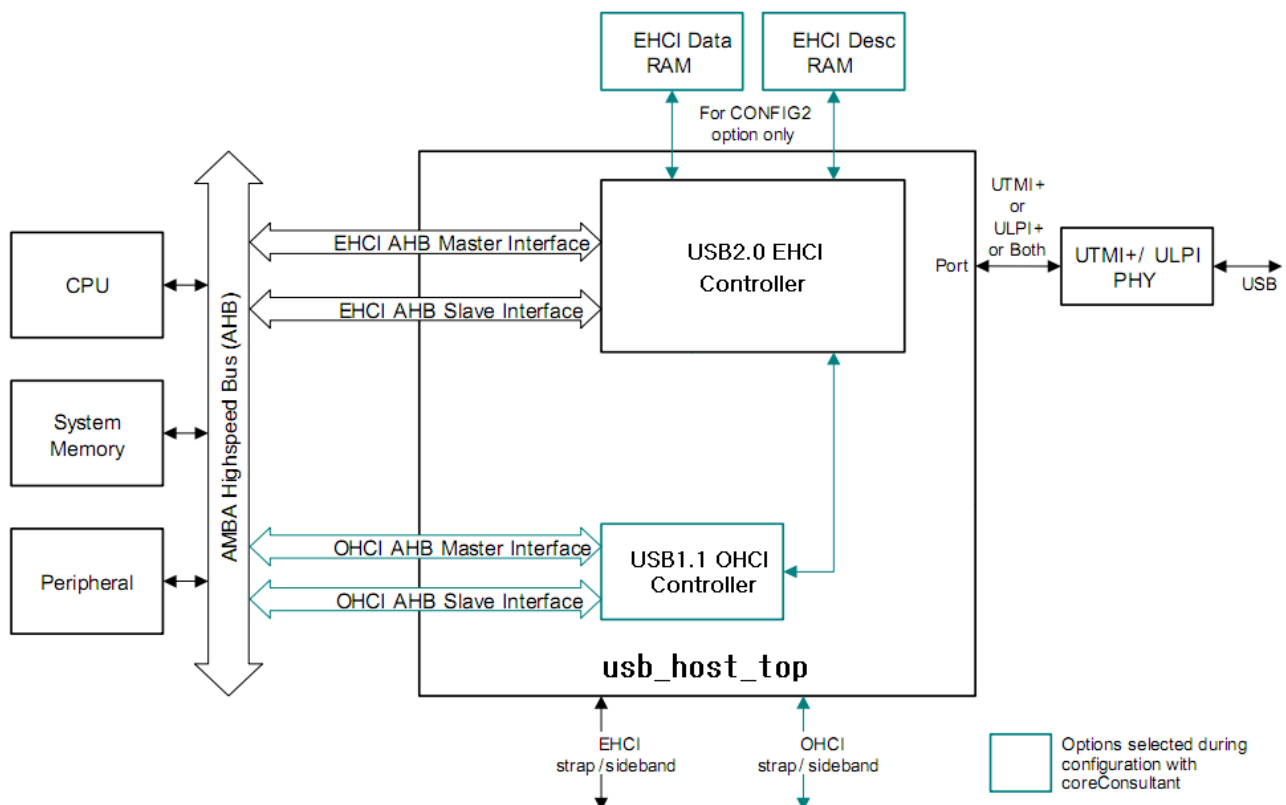


Figure 23.2 USB Controller Block Diagram

23.2 Functional Description for USB 2.0 Host Controller

The Slave Bus Interface Unit (BIU) provides the application interface to the EHCI/OHCI Operational registers, where the application can program the Capability and Operational registers or read back the registers for status.

In case of EHCI, when the Run/Stop bit is set to 1, the List Processor begins to execute the schedule by fetching the descriptors from the system memory through the Master BIU, which in turn generates the transaction on the AHB.

The List Processor stores the fetched descriptors in the Data Structure Cache register and triggers the corresponding descriptor servicer (ITD, siTD, or QH) to service the descriptor. The List Processor then initiates the transaction to the Root Hub.

The Host Parallel Interface Engine (HPiE) in the Root Hub performs all front-end functions, such as token packet decoding, CRC checking and generation, and byte packing and unpacking, before it transmits this data to, or receives it from, the external UTMI PHY.

The UTMI block is implemented with port routing logic, which routes the peripheral attached to either the EHCI or OHCI Host Controller. (For information of the OHCI, refer to the *DesignWare USB 1.1 OHCI Host Controller with AHB/VCI Databook* [2]).

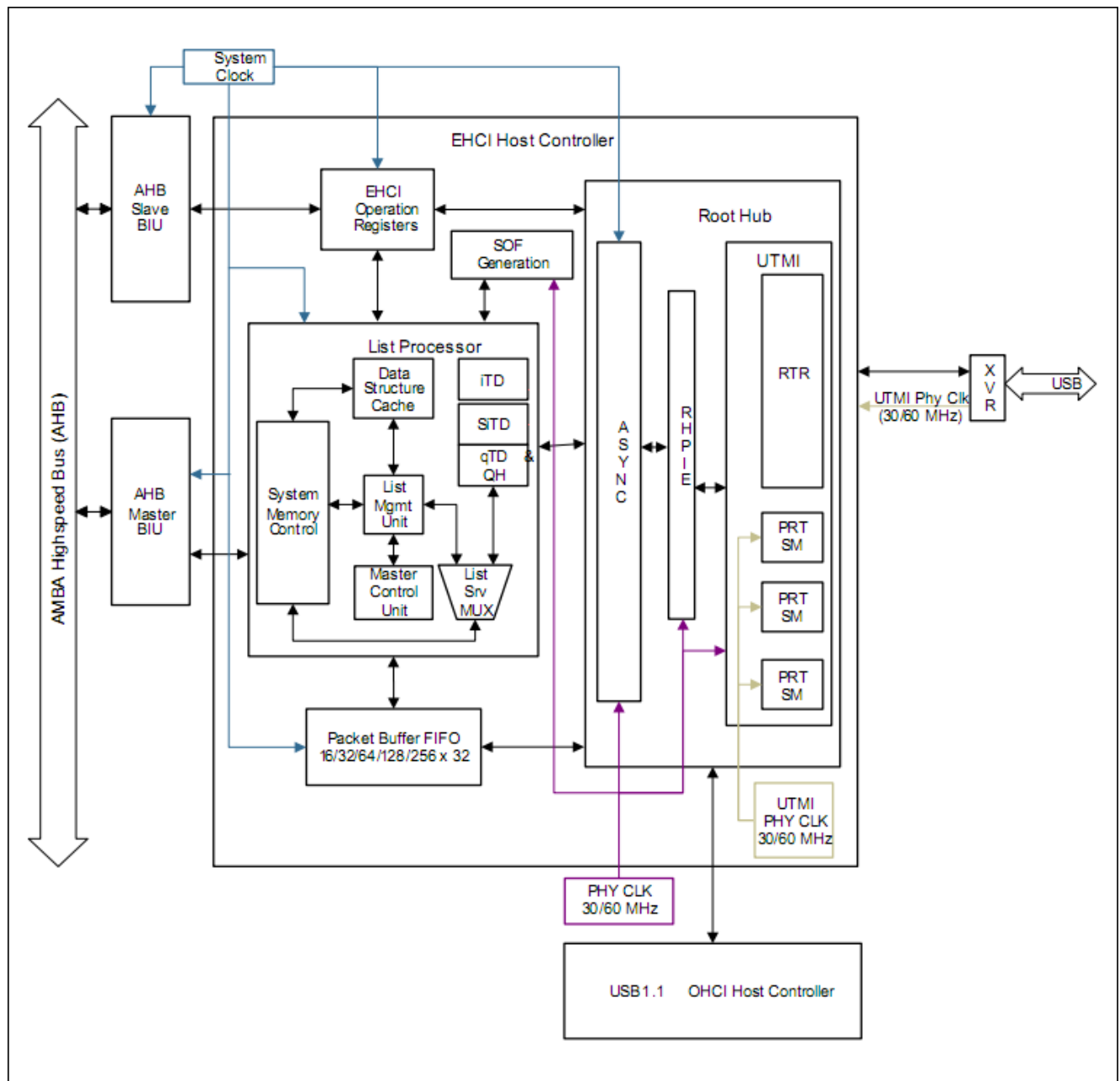


Figure 23.3 usb_host_top with EHCI Detail

23.3 Register Description for USB 2.0 Host Controller

The following table describes the address mapping of EHCI Rev1.0 registers.

❖ *Enhanced Host Controller Interface Specification for Universal Serial Bus, Revision 1.0, Intel Corporation [3]*

Table 23.2 USB EHCI Register (Base Address = 0x80080000/0x800A0000)

Name	Address	Type	Reset	Description
HCCAPBASE	0x00	R	0x01000010	Capability Registers
HCSPARAMS	0x04	R	0x00001116	
HCCPARAMS	0x08	R/W	0x0000A010	
USBCMD	0x10	R/W	0x00080000 or 0x00080B00	Operational Registers
USBSTS	0x14	R/W	0x00001000	
USBINTR	0x18	R/W	0x0	
FRINDEX	0x1C	R/W	0x0	
CTRLDSSEGMENT	0x20	R/W	0x0	
PERIODICLISTBASE	0x24	R/W	0x0	
ASYNCLISTADDR	0x28	R/W	0x0	
CONFIGFLAG	0x50	R/W	0x0	Auxiliary Power Well Registers
PORTSC_1 to PORTSC_15	0x54 to 0x8C	R/W	0x00002000	
INSNREG00	0x90	R/W	0x0	Synopsys-Specific Registers
INSNREG01	0x94	R/W	0x00100010	
INSNREG02	0x98	R/W	0x00000100	
INSNREG03	0x9C	R/W	0x0	
INSNREG04	0x100	R/W	0x0	
INSNREG05	0x104	R/W	0x00001000	

The following table describes the address mapping of OHCI Rev1.0 registers.

❖ *Open Host Controller Interface Specification for USB, Release 1.0a, Compaq, Inc., Microsoft Corporation, National Semiconductor Corporation October 1996 [4]*

Table 23.3 USB OHCI Register (Base Address = 0x80090000/0x800B0000)

Name	Address	Type	Reset	Description
HcRevision	0x00	R	0x00000010	Control and status registers
HcControl	0x04	R/W	0x0	
HcCommandStatus	0x08	R	0x0	
HcInterruptStatus	0x0C	R	0x0	
HcInterruptEnable	0x10	R/W	0x0	
HcInterruptDisable	0x14	W	0x0	
HcHCCA	0x18	R/W	0x0	Memory pointer registers
HcPeriodCurrentED	0x1C	R	0x0	
HcControlHeadED	0x20	R/W	0x0	
HcControlCurrentED	0x24	R/W	0x0	
HcBulkHeadED	0x28	R/W	0x0	
HcBulkCurrentED	0x2C	R/W	0x0	
HcDoneHead	0x30	R	0x0	
HcRmInterval	0x34	R/W	0x00002EDF	Frame counter registers
HcFmRemaining	0x38	R/W	0x0	
HcFmNumber	0x3C	R/W	0x0	
HcPeriodStart	0x40	R/W	0x0	
HcLSThreshold	0x44	R/W	0x00000628	Root hub registers
HcRhDescriptorA	0x48	R/W	0x02001202	
HcRhDescriptorB	0x4C	R/W	0x0	
HcRhStatus	0x50	R/W	0x0	
HcRhPortStatus1	0x54	R/W	0x00000100	
HcRhPortStatus2	0x58	R/W	0x00000100	

Table 23.4 Register Description for UTMI (USB PHY) (Base Address = 0x800C0000)

Name	Address	Type	Reset	Description
UPCR0 ~ 3	0x00 ~ 0x1F	R/W	-	USB200 PHY Configuration Register0 ~ 3
USB20H_PCFG0	0x24/0x44	R/W	0x00000100	USB20H PHY Configuration Register0
USB20H_PCFG1	0x28/0x48	R/W	0x80000000	USB20H PHY Configuration Register1
USB20H_PCFG2	0x2C/0x4C	R/W	0x00000000	USB20H PHY Configuration Register2
USB20H_STS	0x30/0x50	R	0x00000000	USB20H Status Register
USB20HAHB	0x34/0x54	R/W	0x00000000	USB20H AHB Configuration Register
USB20H_PCFG3	0x38/0x58	R/W	0x00000100	USB20H PHY Configuration Register3
USB20H_PCFG4	0x3C/0x5C	R/W	0x80000000	USB20H PHY Configuration Register4
USB20H_PCFG5	0x40	R/W	0x00000000	USB20H PHY Configuration Register5 for HOST0
USB20H_PCFG5	0x60	R/W	0x00000000	USB20H PHY Configuration Register5 for HOST1
USBCSEL	0x64	R/W	0x00000001	USB Controller Selection Register

Refer to Chapter 24 for UPCR 0 ~ 3 description.

USB20H PHY Configuration Register0 (USB20H_PCFG0)

0x800C0024/44

CODEDHT: Configuration Register (CODEDHT: C: 0)																CODEDHT: 17			
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16				
TXFSLST				-	SQRXT			-	USD20DHT			-	CDT						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
-	PR	CM	RCS		RCD		SDI	-	VBDS	DMPD	DPPD	TBSH	TBS	VBD	LBE				

Bit	Name	RW	Reset	Description																		
31:28	TXFSLST	RW	0	TXFSLSTUNE[3:0] These bits have no effect on the host operation. Reserved.																		
27	-	RW	0	Reserved																		
26:24	SQRXT	RW	0	SQRXTUNE[2:0] Squelch Threshold Tune. This bus adjusts the voltage level for the threshold used to detect valid high-speed data. <table><tr><td>111</td><td>-20%</td></tr><tr><td>110</td><td>-15%</td></tr><tr><td>101</td><td>-10%</td></tr><tr><td>100</td><td>-5%</td></tr><tr><td>011</td><td>Design default</td></tr><tr><td>010</td><td>+5%</td></tr><tr><td>001</td><td>+10%</td></tr><tr><td>000</td><td>+15%</td></tr><tr><td>Others</td><td>Reserved</td></tr></table>	111	-20%	110	-15%	101	-10%	100	-5%	011	Design default	010	+5%	001	+10%	000	+15%	Others	Reserved
111	-20%																					
110	-15%																					
101	-10%																					
100	-5%																					
011	Design default																					
010	+5%																					
001	+10%																					
000	+15%																					
Others	Reserved																					
23	-	RW	0	Reserved																		
22:20	USD20DHT	RW	0	USB20DHTUNE[2:0] These bits have no effect on the host operation. Reserved.																		
19	-	RW	0	Reserved																		
18:16	CDT	RW	0	COMPDISTUNE[2:0] Disconnect Threshold Adjustment. This bus adjusts the voltage level for the threshold used to detect a disconnect event at the host.																		
15	-	RW	0	Reserved																		
14	PR	RW	0	PORTRESET When asserted, this signal resets the port's transmit and receive logic without disabling the clocks within the USB 2.0 nanoPHY. <table><tr><td>0</td><td>The transmit and receive FSMs are operational, and the line_state logic becomes sequential after 11 PHYCLOCK cycles.</td></tr><tr><td>1</td><td>The transmit and receive finite state machines are reset, and the line_state logic combinationally reflects the state of the single-ended receivers.</td></tr></table>	0	The transmit and receive FSMs are operational, and the line_state logic becomes sequential after 11 PHYCLOCK cycles.	1	The transmit and receive finite state machines are reset, and the line_state logic combinationally reflects the state of the single-ended receivers.														
0	The transmit and receive FSMs are operational, and the line_state logic becomes sequential after 11 PHYCLOCK cycles.																					
1	The transmit and receive finite state machines are reset, and the line_state logic combinationally reflects the state of the single-ended receivers.																					
13	CM	RW	0	COMMONONN Common Block Power-Down Control. This signal controls the power-down signals in the XO, Bias, and PLL blocks when the USB 2.0 nanoPHY is suspended. <table><tr><td>0</td><td>The XO, Bias, and PLL Blocks remain powered in suspend mode.</td></tr><tr><td>1</td><td>The XO, Bias, and PLL Blocks are powered down in suspend mode.</td></tr></table> If COMMONONN is set to high, clocks to the USB Host Controllers are disabled. Accessing Host Controllers in suspend mode may cause system hangup.	0	The XO, Bias, and PLL Blocks remain powered in suspend mode.	1	The XO, Bias, and PLL Blocks are powered down in suspend mode.														
0	The XO, Bias, and PLL Blocks remain powered in suspend mode.																					
1	The XO, Bias, and PLL Blocks are powered down in suspend mode.																					
12:11	RCS	RW	0	REFCLKSEL[1:0] These bits select the reference clock source for the PLL block. <table><tr><th>RCS</th><th>Reference Clock Select for PLL Block</th></tr><tr><td>01</td><td>The XO block uses an external clock supplied on the XO pin(default)</td></tr><tr><td>Others</td><td>Reserved</td></tr></table>	RCS	Reference Clock Select for PLL Block	01	The XO block uses an external clock supplied on the XO pin(default)	Others	Reserved												
RCS	Reference Clock Select for PLL Block																					
01	The XO block uses an external clock supplied on the XO pin(default)																					
Others	Reserved																					
10:9	RCD	RW	0	REFCLKDIV[1:0] These bits select the reference clock frequency. <table><tr><th>RCD</th><th>Reference Clock Frequency Select</th></tr><tr><td>00</td><td>12 MHz</td></tr><tr><td>Others</td><td>Reserved</td></tr></table>	RCD	Reference Clock Frequency Select	00	12 MHz	Others	Reserved												
RCD	Reference Clock Frequency Select																					
00	12 MHz																					
Others	Reserved																					

8	SDI	W	1	SIDDQ This test signal enables you to perform IDDQ testing by powering down all analog blocks. <table><tr><td>SDI</td><td>IDDQ Test Enable</td></tr><tr><td>1</td><td>The analog blocks are powered down</td></tr><tr><td>0</td><td>The analog blocks are not powered down.</td></tr></table>	SDI	IDDQ Test Enable	1	The analog blocks are powered down	0	The analog blocks are not powered down.
SDI	IDDQ Test Enable									
1	The analog blocks are powered down									
0	The analog blocks are not powered down.									
7	-	RW	0	Reserved						
6	VBDS	RW	0	Reserved						
5	DMPD	RW	0	DMPULLDOWN This controller signal enables or disables the pull-down resistance on the D-line. This bit must be enabled for host operation. <table><tr><td>DMPD</td><td>D- Pull-Down Resistor Enable</td></tr><tr><td>1</td><td>The pull-down resistance on D- is enabled.</td></tr><tr><td>0</td><td>The pull-down resistance on D- is disabled.</td></tr></table>	DMPD	D- Pull-Down Resistor Enable	1	The pull-down resistance on D- is enabled.	0	The pull-down resistance on D- is disabled.
DMPD	D- Pull-Down Resistor Enable									
1	The pull-down resistance on D- is enabled.									
0	The pull-down resistance on D- is disabled.									
4	DPPD	RW	0	DPPULLDOWN This controller signal enables or disables the pull-down resistance on the D+line. This bit must be enabled for host operation. <table><tr><td>DPPD</td><td>D+ Pull-Down Resistor Enable</td></tr><tr><td>1</td><td>The pull-down resistance on D+ is enabled</td></tr><tr><td>0</td><td>The pull-down resistance on D+ is disabled</td></tr></table>	DPPD	D+ Pull-Down Resistor Enable	1	The pull-down resistance on D+ is enabled	0	The pull-down resistance on D+ is disabled
DPPD	D+ Pull-Down Resistor Enable									
1	The pull-down resistance on D+ is enabled									
0	The pull-down resistance on D+ is disabled									
3	TBSH	RW	0	TXBITSTUFFENH This controller signal enables or disables bits stuffing on DATAINH[7:0] when OPMODE[1:0] =2'b11 <table><tr><td>TBSH</td><td>High-Byte Transmit Bit-Stuffing Enable</td></tr><tr><td>1</td><td>Bit stuffing is enabled</td></tr><tr><td>0</td><td>Bit stuffing is disabled.</td></tr></table>	TBSH	High-Byte Transmit Bit-Stuffing Enable	1	Bit stuffing is enabled	0	Bit stuffing is disabled.
TBSH	High-Byte Transmit Bit-Stuffing Enable									
1	Bit stuffing is enabled									
0	Bit stuffing is disabled.									
2	TBS	RW	0	TXBITSTUFFEN This controller signal enables or disables bits stuffing on DATAIN[7:0] when OPMODE[1:0] =2'b11 <table><tr><td>TBS</td><td>Low-Byte Transmit Bit-Stuffing Enable</td></tr><tr><td>1</td><td>Bit stuffing is enabled</td></tr><tr><td>0</td><td>Bit stuffing is disabled</td></tr></table>	TBS	Low-Byte Transmit Bit-Stuffing Enable	1	Bit stuffing is enabled	0	Bit stuffing is disabled
TBS	Low-Byte Transmit Bit-Stuffing Enable									
1	Bit stuffing is enabled									
0	Bit stuffing is disabled									
1	VBD	RW	0	Reserved						
0	LBE	RW	0	LOOPBACKENB This controller signal places the USB 2.0 nanoPHY in Loopback mode, which enables the receive and transmit logic concurrently. Loopback mode is for test purpose only and cannot be used for normal operation. <table><tr><td>LBE</td><td>Loopback Test Enable</td></tr><tr><td>1</td><td>During data transmission, the receive logic is enabled.</td></tr><tr><td>0</td><td>During data transmission, the receive logic is disabled.</td></tr></table>	LBE	Loopback Test Enable	1	During data transmission, the receive logic is enabled.	0	During data transmission, the receive logic is disabled.
LBE	Loopback Test Enable									
1	During data transmission, the receive logic is enabled.									
0	During data transmission, the receive logic is disabled.									

USB20H PHY Configuration Register1 (USB20H_PCFG1)

0x800C0028/48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SR	-	SM	U20DH	IDPU	DVB	CVB	DCV	-				TXHSXVT		WI	ABE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-							TXVRT				-		TXRT	-	TP

Bit	Name	RW	Reset	Description	
31	SR	RW	1	PHY Soft Reset.	
				SR	Soft Reset Enable
				0	Reset PHY. Once enabled, this bit must be maintained at least 10us. Do not access the host controllers when this bit is low (system hangup can happen).
1	Normal operation enabled				
30	-	RW	0	Reserved	
29	SM	RW	0	SLEEPM Reserved.	
28	U20DH	RW	0	USB20DH DISABLE This bit must be written as “1” to save power in the host operation.	
27	IDPU	RW	0	IDPULLUP Reserved	
26	DVB	RW	0	DRVVBUS Reserved.	
25	CVB	RW	0	CHRGVBUS Reserved.	
24	DCV	RW	0	DISCHRGVBUS Reserved.	
23:20	-	RW	0	Reserved	
19:18	TXHSXVT	RW	0	TXHSXVTUNE[1:0] Transmitter High Speed Crossover Adjustment.	
				3	Default setting
				2	The crossover voltage is increased by 15mV
				1	The crossover voltage is decreased by 15mV
0	Reserved				
TXHSXVT adjusts the voltage at which the DP and DM signals cross while transmitting in HS mode					
17	WI	RW	0	WORDINTERFACE UTMI Data Bus Width and Clock Select.	
				1	16-bit data interface
0	8-bit data interface				
The USB 2.0 nanoPHY supports 8/16-bit data interfaces. • WORDINTERFACE = 1'b1 (16-bit operation): PHYCLOCK = 30 MHz • WORDINTERFACE = 1'b0 (8-bit operation): PHYCLOCK = 60 MHz					
16	ABE	RW	0	ATEBISTERROR TX Jitter Adjustment.	
				1	TX Jitter Fix enable (default)
0	TX Jitter Fix disable				
15:9	-	RW	0	Reserved	
8:5	TXVRT	RW	0	TXVREFTUNE[3:0] HS DC Voltage Level Adjustment.	
				1111	+24%
				1110	+22%
				1101	+20%
				1100	+18%
				1011	+16%
				1010	+14%
				1001	+12%
				1000	+10%
				0111	+8%
				0110	+6%
				0101	+4%

				<table><tr><td>0100</td><td>+2%</td></tr><tr><td>0011</td><td>Design default</td></tr><tr><td>0010</td><td>-2%</td></tr><tr><td>0001</td><td>-4%</td></tr><tr><td>0000</td><td>-6%</td></tr></table>	0100	+2%	0011	Design default	0010	-2%	0001	-4%	0000	-6%
0100	+2%													
0011	Design default													
0010	-2%													
0001	-4%													
0000	-6%													
This bus adjusts the voltage to which the high speed DC level is tuned.														
4	-	RW	0	Reserved										
3:2	TXRT	RW	0	TXRISSETUNE. HS Transmitter Rise/Fall Time Adjustment. <table><tr><td>3</td><td>-30%</td></tr><tr><td>2</td><td>-20%</td></tr><tr><td>1</td><td>Design default (default)</td></tr><tr><td>0</td><td>+10%</td></tr></table> This bus adjusts the rise/fall times the high speed waveform	3	-30%	2	-20%	1	Design default (default)	0	+10%		
3	-30%													
2	-20%													
1	Design default (default)													
0	+10%													
1:0	-	RW	0	Reserved										

USB20H PHY Configuration Register2 (USB20H_PCFG2)

0x800C002C/4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLAH						OCS	POA	APO	SM	FLA					

Bit	Name	RW	Reset	Description																
31:30	-	RW	0	Reserved																
29:16	-	RW	0	Reserved for test.																
15:10	FLAH	RW	0	Frame Length Adjustment Register This feature adjusts any offset from the clock source that drives the μSOF counter. The μSOF cycle time (number of μSOF counter clock periods to generate a μSOF microframe length) is equal to 59,488 plus this value. The default value is decimal 32 (0x20), which gives an SOF cycle time of 60,000 (each microframe has 60,000 bit times). Frame Length (decimal) FLADJ Value (decimal) <table><tr><td>59488</td><td>0 (0x00)</td></tr><tr><td>59504</td><td>1 (0x01)</td></tr><tr><td>59520</td><td>2 (0x02)</td></tr><tr><td>...</td><td>...</td></tr><tr><td>59984</td><td>31 (0x1F)</td></tr><tr><td>60000</td><td>32 (0x20)</td></tr><tr><td>...</td><td>...</td></tr><tr><td>60496</td><td>63 (0x3F)</td></tr></table> Note that this register must be modified only when the HCHalted bit in the USBSTS register is set to 1; otherwise, the EHCI yields undefined results. The register must not be reprogrammed by the USB system software unless the default or BIOS programmed values are incorrect, or the system is restoring the register while returning from a suspended state. Connect this signal to value 0x20 (32 decimal) for no offset.	59488	0 (0x00)	59504	1 (0x01)	59520	2 (0x02)	...	...	59984	31 (0x1F)	60000	32 (0x20)	...	...	60496	63 (0x3F)
59488	0 (0x00)																			
59504	1 (0x01)																			
59520	2 (0x02)																			
...	...																			
59984	31 (0x1F)																			
60000	32 (0x20)																			
...	...																			
60496	63 (0x3F)																			
9	OCS	RW	0	OHCI Count Select Selects the counter value for simulation or real time for 1 ms <table><tr><td>1</td><td>Simulation time</td></tr><tr><td>0</td><td>Count full 1 ms</td></tr></table>	1	Simulation time	0	Count full 1 ms												
1	Simulation time																			
0	Count full 1 ms																			
8	POA	RW	0	Port Overcurrent Indication From Application. When asserted by the application, the corresponding port enters Disable state. This signal controls both EHCI and OHCI controller port state machines. Depending on ownership of the port, the corresponding EHCI or OHCI controller generates an Overcurrent Detect interrupt.																
7	APO	RW	0	Auto Port Power Disable on Overcurrent This strap signal enables automatic port power disable in the host controller. When this signal is active, if an over-current condition is detected on a powered port and PPC is 1, the PP bit in each affected port is automatically transitioned by the host controller from 1 to 0, removing power from the port.																
6	SM	RW	0	Simulation Mode When set to 1'b1, this bit sets the PHY in a non-driving mode so the EHCI can detect device connection. This signal is used only for simulation.																
5:0	FLA	RW	0	Frame Length Adjustment Register per Port This feature adjusts the frame length of the microframe per port. This value must be same as that of FLAH, or the EHCI yields undefined results. See FLAH above for the adjustment value.																

USB20H Status Register (USB20H_STS)

0x800C0030/50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TDOUT				-						USBSTS					

Bit	Name	RW	Reset	Description
15:12	TDOUT	R	0x0	USB PHY TEST DATA OUT TESTDATAOUT[3:0] status.
5:0	USBSTS	R	0x0	USB Status EHCI USBSTS register[5:0] bits. This signal indicates pending interrupts and various host controller statuses. These 6 bits reflect the value in the USBSTS[5:0] register.

USB20H AHB Configuration Register (USB20HAHB)

0x800C0034/54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-								BST[3:0]				FGNTV[1:0]		FGNTS[1:0]	

Bit	Name	RW	Reset	Description
31:8	-	R	0	Reserved
7	BST[3]	R/W	0	AHB INCR16 Burst (0=enable, 1=disable)
6	BST[2]	R/W	0	AHB INCR8 Burst (0=enable, 1=disable)
5	BST[1]	R/W	0	AHB INCR4 Burst (0=enable, 1=disable)
4	BST[0]	R/W	0	AHB INCRx Burst Align (0=enable, 1=disable)
3:2	FGNTV	R/W	0	AHB Bus Grant signal control.
1:0	FGNTS	R/W	0	

USB20H PHY Configuration Register3 (USB20H_PCFG3)

0x800C0038/58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-								PHY_VSTATUS							

Bit	Name	RW	Reset	Description
7:0	PHY_VSTATUS	R/W	0	phy_vstatus

USB20H PHY Configuration Register4 (USB20H_PCFG4)

0x800C003C/5C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					TXRESTUNE1				TXRESTUNE0	TXPREEMPULSETUNE	TXPREEMPAMPTUNE		FSEL		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTREFCLK	RETENABLEN	RESREQIN	TESTBURNIN	VATESTENB0	ATERESE	UARTSE		UARTBP		VATESTENB1	ACAENB	DCDENB	VDATDENB	VDATSR	CHRGSE

Bit	Name	RW	Reset	Description
26	TXRESTUNE1	R/W	0	USB Source Impedance Adjustment In some applications, there can be significant series resistance on the D+ and D– paths between the transceiver and cable. This bus adjusts the driver source impedance to compensate for added series resistance on the USB. Note: Any setting other than the default can result in source impedance variation across process, voltage, and temperature conditions that does not meet USB 2.0 specification limits. ■ 11: Source impedance is decreased by approximately 4 Ω. ■ 10: Source impedance is decreased by approximately 2 Ω. ■ 01: Design default ■ 00: Source impedance is increased by approximately 1.5 Ω. If this bus is not used, leave it at the default setting.
22	TXRESTUNE0	R/W	0	
21	TXPREEMPULSETUNE	R/W	0	HS Transmitter Pre-Emphasis Duration Control This signal controls the duration for which the HS pre-emphasis current is sourced onto DP0 or DM0. The HS Transmitter pre-emphasis duration is defined in terms of unit amounts. One unit of pre-emphasis duration is approximately 580 ps and is defined as 1X pre-emphasis duration. This signal is valid only if either TXPREEMPAMPTUNE0[1] or TXPREEMPAMPTUNE0[0] is set to 1'b1. ■ 1: 1X, short pre-emphasis current duration ■ 0 (design default): 2X, long pre-emphasis current duration If TXPREEMPULSETUNE0 is not used, set it to 1'b0.
20:19	TXPREEMPAMPTUNE	R/W	0	HS Transmitter Pre-Emphasis Current Control This signal controls the amount of current sourced to DP0 and DM0 after a J-to-K or K-to-J transition. The HS Transmitter pre-emphasis current is defined in terms of unit amounts. One unit amount is approximately 600 μA and is defined as 1X pre-emphasis current. ■ 11: HS Transmitter pre-emphasis circuit sources 3X pre-emphasis current. ■ 10: HS Transmitter pre-emphasis circuit sources 2X pre-emphasis current. ■ 01: HS Transmitter pre-emphasis circuit sources 1X pre-emphasis current. ■ 00 (design default): HS Transmitter pre-emphasis is disabled. If these signals are not used, set them to 2'b00.
18:16	FSEL	R/W	0	Reference Clock Frequency Select Selects the USB 2.0 picoPHY reference clock frequency. ■ 111: 50 MHz ■ 110: Reserved ■ 101: 24 MHz ■ 100: 20 MHz ■ 011: 19.2 MHz ■ 010: 12 MHz ■ 001: 10 MHz ■ 000: 9.6 MHz
15	EXTREFCLK	R/W	0	Reserved If this signal is not used, tie it to ground.
14	RETENABLEN	R/W	0	Retention Mode Enable Reserved Tie this signal high and keep it high.
13	RESREQIN	R/W	0	Reserved Tie this pin to 1'b0.
12	TESTBURNIN	R/W	0	Burn-in Test Enable This input enables the built-in self burn-in testing function of the PHY.

				At assertion of this signal, the PHY performs a power-on-reset sequence and then enters HS Loopback mode. In HS Loopback mode, the PHY continues to transmit alternating 1's and 0's at the maximum bit rate until TESTBURNIN is deasserted. This input has precedence over the SUSPENDM0, SIDDQ, PORTRESET0, ATERESET, and SLEEPM0 signals. Therefore, these signals are internally overridden when TESTBURNIN is asserted. This function requires a 50-MHz reference clock source on the CLKCORE input. If this signal is not used, tie it low.
11	VATESTENB0	R/W	0	Analog Test Pin Select Enables analog test voltages to be placed on either the ANALOGTEST or ID0 pin. ■ 11: Reserved. Invalid setting. ■ 10: Analog test voltages can be viewed or applied on ANALOGTEST. ■ 01: Analog test voltages can be viewed or applied on ID0. ■ 00: Analog test voltages cannot be viewed or applied on either ANALOGTEST or ID0. If this bus is not used, tie these inputs low.
10	ATERESET	R/W	0	Reset Input from Automatic Test Equipment When the USB 2.0 picoPHY is powered up (not in Suspend or Sleep mode), an automatic tester can use this test signal to disable PHYCLOCK0 and FREECLK, then re-enable them with an aligned phase. ■ 1: The PHYCLOCK0 and FREECLK outputs are disabled. ■ 0: The PHYCLOCK0 and FREECLK outputs are available within a specific period after ATERESET is deasserted. If this signal is not used, tie it low.
9	UARTSEL	R/W	0	UARTSEL signals
7	UARTBYP	R/W	0	UART Data Value
5	VATESTENB1	R/W	0	11bit
4	ACAENB	R/W	0	ACA ID_USB20DH Pin Resistance Detection Enable Enables detection of resistance on the ID_USB20DH pin of an ACA. ■ 1: Enables detection of resistance on the ID_USB20DH pin of an ACA. ■ 0: Disables detection of resistance on the ID_USB20DH pin of an ACA. If this signal is not used, tie this input to 1'b0.
3	DCDENB	R/W	0	Data Contact Detection Enable Enables current sourcing on the D+ line and pull-down resistance on the D- line for Data Contact Detect (DCD). ■ 1: IDP_SRC current is sourced onto DP0, pull-down resistance on DM0 is enabled. ■ 0: IDP_SRC current is disabled, pull-down resistance on DM0 is disabled. Note: During USB 2.0 picoPHY normal operation, set this signal low. If this signal is not used, tie this input to 1'b0.
2	VDATDETENB	R/W	0	Battery Charging Attach/Connect Detection Enable Enables or disables attach/connect detection. ■ 1: Data detect voltage (CHG_DET) is enabled. ■ 0: Data detect voltage (CHG_DET) is disabled. During USB 2.0 picoPHY normal operation, set this signal low.
1	VDATSRCEB	R/W	0	Battery Charging Sourcing Select Enables or disables sourcing for battery charging. ■ 1: Data source voltage (VDAT_SRC) is enabled. ■ 0: Data source voltage (VDAT_SRC) is disabled. During USB 2.0 picoPHY normal operation, set this signal low.
0	CHRGSEL	R/W	0	Battery Charging Source Select Determines whether current is sourced onto or sunk from DP0 or DM0. ■ 1: Data source voltage (VDAT_SRC) is sourced onto DM0 and sunk from DP0. ■ 0: Data source voltage (VDAT_SRC) is sourced onto DP0 and sunk from DM0. For information about battery charging operation, refer to <i>USB Battery Charging 1.2</i>. [6]

USB20H PHY Configuration Register5 (USB20H_PCFG5)-HOST0

0x800C0040

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			CFG_OP MODE	OPMODE		CFG_SU SPENDM	SUSPEN DM	SELO_H OSTDIS CONNEC T			TO_HOS TDISCO NNECT	SELH_H OSTDIS CONNEC T			TH_HOS TDISCO NNECT
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	TO_VDA TSRCEN B	TH1_VD ATSRCE NB	TH0_VD ATSRCE NB	TO_CHG SEL	TO_SUS PENDM	TO_OPMODE		TH1_CH GSEL	TH1_SU SPENDM	TH1_OPMODE		TH0_CH GSEL	TH0_SU SPENDM	TH0_OPMODE	

Bit	Name	RW	Reset	Description
28	CFG_OPMODE	R/W	0	OPMDE select register 1:OPMODE 0:utmi_opmode
27:26	OPMODE	R/W	0	User Setting OPMODE
25	CFG_SUSPENDM	R/W	0	SUSPENDM select register 1:SUSPENDM 0:utmi_suspendm
24	SUSPENDM	R/W	0	User Setting SUSPENDM
23	SELO_HOSTDISCONNECT	R/W	0	USB20DH PHY HOSTDISCONNECT select register
20	TO_HOSTDISCONNECT	R/W	0	User Setting USB20DH HOSTDISCONNECT
19	SELH_HOSTDISCONNECT	R/W	0	HOSTPHY HOSTDISCONNECT select register
16	TH_HOSTDISCONNECT	R/W	0	User Setting USB20DH HOSTDISCONNECT
14	TO_VDATSRCEB	R/W	0	User Setting USB20DHPHY Battery Charging Sourcing Select
13	TH1_VDATSRCEB	R/W	0	User Setting HOST0PHY Battery Charging Sourcing Select
12	TH0_VDATSRCEB	R/W	0	User Setting HOST1PHY Battery Charging Sourcing Select
11	TO_CHGSEL	R/W	0	User Setting USB20DHPHY Battery Charging Sourcing Select
10	TO_SUSPENDM	R/W	0	User Setting USB20DHPHY Suspend Assertion
9:8	TO_OPMODE	R/W	0	User Setting USB20DHPHY OPMODE
7	TH1_CHGSEL	R/W	0	User Setting HOST0PHY Battery Charging Sourcing Select
6	TH1_SUSPENDM	R/W	0	User Setting HOST0PHY Suspend Assertion
5:4	TH1_OPMODE	R/W	0	User Setting HOST0PHY OPMODE
3	TH0_CHGSEL	R/W	0	User Setting HOST1PHY Battery Charging Sourcing Select
2	TH0_SUSPENDM	R/W	0	User Setting HOST1PHY Suspend Assertion
1:0	TH0_OPMODE	R/W	0	User Setting HOST1PHY OPMODE

USB20H PHY Configuration Register5 (USB20H_PCFG5)-HOST1

0x800C0060

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			CFG_O PMODE	OPMODE		CFG_S USPEN DM	SUSPE NDM								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bit	Name	RW	Reset	Description
28	CFG_OPMODE	R/W	0	OPMDE select register 1:OPMODE 0:utmi_opmode
27:26	OPMODE	R/W	0	User Setting OPMODE
25	CFG_SUSPENDM	R/W	0	SUSPENDM select register 1:SUSPENDM 0:utmi_suspendm
24	SUSPENDM	R/W	0	User Setting SUSPENDM

USB Controller Selection Register (USBCSEL)

0x800C0064

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-															SEL

Bit	Name	RW	Reset	Description
31:1	-	R	0x0	Reserved
0	SEL	R	0x1	This bit selects USB controller for PHY1. 0: HOST Controller 1 1: USB20 Device Controller

24 USB 2.0DH (USB2.0 Host & Device) Controller

24.1 Overview

The TCC8132 supports Dual-Role Device (DRD) controller, which supports both device and host functions and is fully compliant to the On-The-Go Supplement to the USB 2.0 Specification, Revision 1.0a.

It can also be configured as a Host-only or Device-only controller, fully compliant with the USB 2.0 Specification. The USB 2.0 configurations support high-speed (HS, 480-Mbps), full-speed (FS, 12-Mbps), and low-speed (LS, 1.5-Mbps) transfers. Additionally, it can be configured as a USB 1.1 full-speed/low-speed DRD.

The main features of the TCC8132 USB 2.0DH controller are as follows.

[GENERAL FEATURES]

- Supports Slave, External DMA Controller Interface
- Includes USB power management features
- Includes power-saving features (clock gating, two power rails for advanced power management)
- Supports packet-based, Dynamic FIFO memory allocation for endpoints for small FIFOs and flexible, efficient use of RAM
- Uses single-port RAM
- Provides support to change an endpoint's FIFO memory size during transfers
- Supports endpoint FIFO sizes that are not powers of 2, to allow the use of contiguous memory locations
- Supports the Keep-Alive in Low-Speed mode and SOFs in High/Full-Speed modes
- Optional support for Transmit and Receive thresholding in DMA mode when dedicated Tx FIFO is selected in Device mode. Thresholding and threshold length selectable through global registers. For supporting thresholding, the AHB must be run at 60 MHz or higher.

[SOFTWARE FEATURES]

- Software handles USB commands (SETUP transactions are detected and their command payloads are forwarded to the application for decoding).
- Software handles USB errors.

[APPLICATION FEATURES]

- Interfaces for the application via the AHB:
- AHB Slave interface for accessing Control and Status Registers (CSRs), the Data FIFO, and queues
- Optional AHB Master interface for Data FIFO access when Internal DMA is enabled
- Supports all AHB burst types in AHB Slave interface
- Software-selectable AHB burst type on AHB Master interface
- Takes care of the 1KB boundary breakup.
- Optional support for a dedicated transmit FIFO for each of the device IN endpoints in Slave and DMA modes. Each FIFO can hold multiple packets.

[USB 2.0 SUPPORTED FEATURES]

- Complies with the On-The-Go Supplement to the USB 2.0 Specification (Revision 1.0a)
- Operates in High-Speed (HS, 480-Mbps), Full-Speed (FS, 12-Mbps) and Low-Speed (LS, 1.5-Mbps) modes
- **Caution:** In host role controller, Low-Speed device connection through Full-Speed Hub is not supported.
- Supports the UTMI+ Level 3 interface (Revision 1.0, February 25th, 2004). 8-, 16-, and 8/16-bit data buses are supported.
- Supports Session Request Protocol (SRP)
- Supports Host Negotiation Protocol (HNP)
- Supports up to 16 bidirectional endpoints, including control endpoint 0
- Supports up to 16 host channels. In Host mode, when the number of device endpoints to be supported is more than the number of host channels, software can reprogram the channels to support up to 127 devices, each having 32 endpoints (IN + OUT), for a maximum of 4, 064 endpoints.
- Supports a generic root hub
- Includes automatic ping capabilities

[POWER FEATURES]

- PHY clock gating support during USB Suspend mode and Session-Off mode
- AHB clock gating support during USB Suspend mode and Session-Off mode

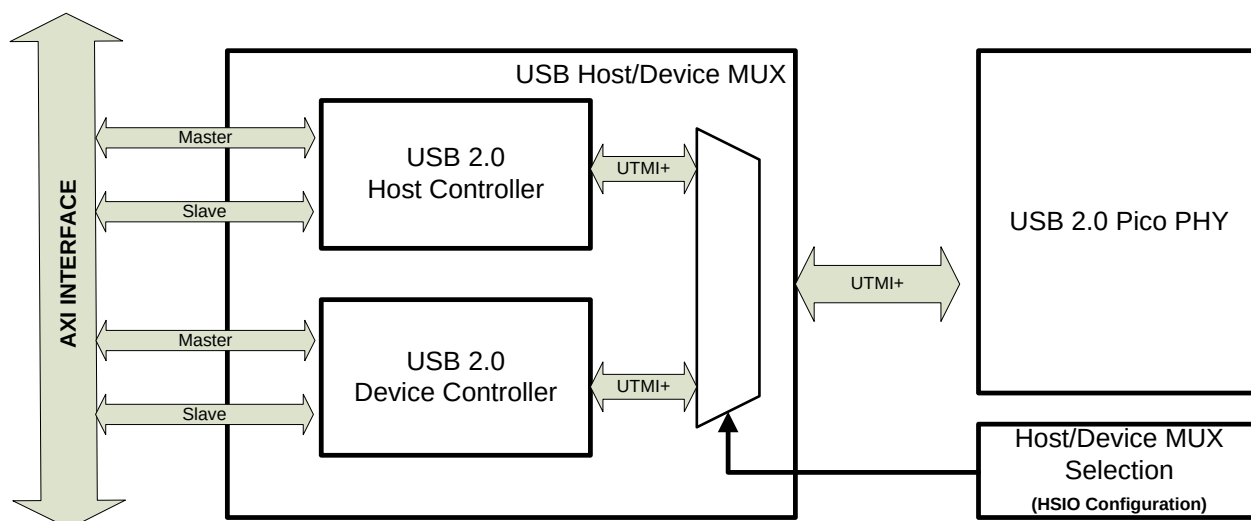


Figure 24.1 USB Controller Block Diagram

24.2 Register Description for UTMI (USB PHY)

Table 24.1 Register Description for UTMI (USB PHY) (Base Address = 0x800C0000)

Name	Address	Type	Reset	Description
UPCR0	0x08	R/W	0x00002530	USB PHY Configuration Register0
UPCR1	0x0C	R/W	0x00007343	USB PHY Configuration Register1
UPCR2	0x10	R/W	0x00005100	USB PHY Configuration Register2
UPCR3	0x14	R/W	0x00008005	USB PHY Configuration Register3
UPCR4	0x18	R/W	0x004A8081	USB PHY Configuration Register1
UPCR5	0x1C	R/W	0x00000020	USB PHY Configuration Register2
UPCR6	0x20	R/W	0x00008000	USB PHY Configuration Register3

USB PHY Configuration Register0 (UPCR0)

0x800C0008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODE	PR	CM	RCS		RCD		SDI	FO	VBDS	DMPD	DPPD	TBSH	TBS	VBD	LBE

Bit	Name	R/W	Reset	Description						
15	MODE	R/W	0	Opmode select signal 1: Register setting value 0: utmi_opmode output signal						
14	PR	R/W	0	When asserted, this signal resets the port's transmit and receive logic without disabling the clocks within the USB 2.0 nanoPHY. <table><tr><td>0</td><td>The transmit and receive FSMs are operational, and the line_state logic becomes sequential after 11 PHYCLOCK cycles.</td></tr><tr><td>1</td><td>The transmit and receive finite state machines are reset, and the line_state logic combinationally reflects the state of the single-ended receivers.</td></tr></table>	0	The transmit and receive FSMs are operational, and the line_state logic becomes sequential after 11 PHYCLOCK cycles.	1	The transmit and receive finite state machines are reset, and the line_state logic combinationally reflects the state of the single-ended receivers.		
0	The transmit and receive FSMs are operational, and the line_state logic becomes sequential after 11 PHYCLOCK cycles.									
1	The transmit and receive finite state machines are reset, and the line_state logic combinationally reflects the state of the single-ended receivers.									
13	CM	R/W	1	This signal controls the power-down signals in the XO, Bias, and PLL blocks when the USB 2.0 nanoPHY is suspended. <table><tr><td>0</td><td>The XO, Bias, and PLL Blocks remains powered in suspend mode.</td></tr><tr><td>1</td><td>The XO, Bias, and PLL Blocks are powered down in suspend mode.</td></tr></table>	0	The XO, Bias, and PLL Blocks remains powered in suspend mode.	1	The XO, Bias, and PLL Blocks are powered down in suspend mode.		
0	The XO, Bias, and PLL Blocks remains powered in suspend mode.									
1	The XO, Bias, and PLL Blocks are powered down in suspend mode.									
12:11	RCS	R/W	1	This signal selects the reference clock source for the PLL block. <table><tr><th>RCS</th><th>Reference Clock Select</th></tr><tr><td>01</td><td>The XO block uses an external clock supplied on the XO pin(default)</td></tr><tr><td>Others</td><td>Reserved</td></tr></table>	RCS	Reference Clock Select	01	The XO block uses an external clock supplied on the XO pin(default)	Others	Reserved
RCS	Reference Clock Select									
01	The XO block uses an external clock supplied on the XO pin(default)									
Others	Reserved									
10:9	RCD	R/W	1	Reference Clock Divider						
8	SDI	R/W	0	This test signal enables you to perform IDDQ testing by powering down all analog blocks. <table><tr><th>SID</th><th>IDDQ Test Enable</th></tr><tr><td>1</td><td>The analog blocks are powered down</td></tr><tr><td>0</td><td>The analog blocks are not powered down.</td></tr></table>	SID	IDDQ Test Enable	1	The analog blocks are powered down	0	The analog blocks are not powered down.
SID	IDDQ Test Enable									
1	The analog blocks are powered down									
0	The analog blocks are not powered down.									
7	FO	R/W	0	This controller signal enables the UTMI or serial interface. Note: It is effective only when UPCR0.MODE is set and you should not clear this bit when UPCR0.MODE is set <table><tr><th>FO</th><th>UTMI/Serial Interface Select</th></tr><tr><td>1</td><td>The TXENABLEN, FSDATAEXT, and FSSE0EXT inputs drive USB 2.0 nanoPHY data output onto the D+ and D- lines</td></tr><tr><td>0</td><td>Data on the D+ and D- lines is transmitted and received through the UTMI.</td></tr></table>	FO	UTMI/Serial Interface Select	1	The TXENABLEN, FSDATAEXT, and FSSE0EXT inputs drive USB 2.0 nanoPHY data output onto the D+ and D- lines	0	Data on the D+ and D- lines is transmitted and received through the UTMI.
FO	UTMI/Serial Interface Select									
1	The TXENABLEN, FSDATAEXT, and FSSE0EXT inputs drive USB 2.0 nanoPHY data output onto the D+ and D- lines									
0	Data on the D+ and D- lines is transmitted and received through the UTMI.									
6	VBDS	R/W	0	This signal selects the VBUSVLDEXT input or the internal Session Valid comparator to indicate when the VBUS signal on the USB cable is valid <table><tr><th>VBDS</th><th>External VBUS Valid Select</th></tr><tr><td>1</td><td>The VBUSVLDEXT input is used</td></tr><tr><td>0</td><td>The internal Session Valid comparator is used</td></tr></table>	VBDS	External VBUS Valid Select	1	The VBUSVLDEXT input is used	0	The internal Session Valid comparator is used
VBDS	External VBUS Valid Select									
1	The VBUSVLDEXT input is used									
0	The internal Session Valid comparator is used									
5	DMPD	R/W	0	This controller signal enables or disables the pull-down resistance on the D-line. When an A/B device is acting as a host (Downstream-facing port), DPPULLDOWN and DMPULLDOWN are enabled. The UTMI+/USB20DH-compliant controllers are not allowed to toggle DPPULLDOWN and DMPULLDOWN during normal operation. UPCR2.OPMODE =						

				2'b01(non-driving) has precedence over the DPPULLDOWN and DMPULLDOWN controls. Note: It is effective only when UPCR0.MODE is set. <table><tr><td>DMPD</td><td>D- Pull-Down Resistor Enable</td></tr><tr><td>1</td><td>The pull-down resistance on D- is enabled.</td></tr><tr><td>0</td><td>The pull-down resistance on D- is disabled.</td></tr></table>	DMPD	D- Pull-Down Resistor Enable	1	The pull-down resistance on D- is enabled.	0	The pull-down resistance on D- is disabled.
DMPD	D- Pull-Down Resistor Enable									
1	The pull-down resistance on D- is enabled.									
0	The pull-down resistance on D- is disabled.									
4	DPPD	R/W	0	This controller signal enables or disables the pull-down resistance on the D+line. Note: It is effective only when UPCR0.MODE is set. <table><tr><td>DPPD</td><td>D+ Pull-Down Resistor Enable</td></tr><tr><td>1</td><td>The pull-down resistance on D+ is enabled</td></tr><tr><td>0</td><td>The pull-down resistance on D+ is disabled</td></tr></table>	DPPD	D+ Pull-Down Resistor Enable	1	The pull-down resistance on D+ is enabled	0	The pull-down resistance on D+ is disabled
DPPD	D+ Pull-Down Resistor Enable									
1	The pull-down resistance on D+ is enabled									
0	The pull-down resistance on D+ is disabled									
3	TBSH	R/W	0	This controller signal enables or disables bits stuffing on DATAINH[7:0] when OPMODE[1:0] =2'b11 <table><tr><td>TBSH</td><td>High-Byte Transmit Bit-Stuffing Enable</td></tr><tr><td>1</td><td>Bit stuffing is enabled</td></tr><tr><td>0</td><td>Bit stuffing is disabled.</td></tr></table>	TBSH	High-Byte Transmit Bit-Stuffing Enable	1	Bit stuffing is enabled	0	Bit stuffing is disabled.
TBSH	High-Byte Transmit Bit-Stuffing Enable									
1	Bit stuffing is enabled									
0	Bit stuffing is disabled.									
2	TBS	R/W	0	This controller signal enables or disables bits stuffing on DATAIN[7:0] when OPMODE[1:0] =2'b11 <table><tr><td>TBS</td><td>Low-Byte Transmit Bit-Stuffing Enable</td></tr><tr><td>1</td><td>Bit stuffing is enabled</td></tr><tr><td>0</td><td>Bit stuffing is disabled</td></tr></table>	TBS	Low-Byte Transmit Bit-Stuffing Enable	1	Bit stuffing is enabled	0	Bit stuffing is disabled
TBS	Low-Byte Transmit Bit-Stuffing Enable									
1	Bit stuffing is enabled									
0	Bit stuffing is disabled									
1	VBD	R/W	0	This signal is valid in device mode only when the VBUSVLDEXTSEL signal is high. VBUSVLDEXTSEL indicates whether the VBUS signal on the USB cable is valid. In addition, VBUSVLDEXTSEL enables the pull-up resistor on the D+ line. <table><tr><td>VBD</td><td>External VBUS Valid Indicator</td></tr><tr><td>1</td><td>The VBUS signal is valid, and the pull-up resistor on D+ is enabled</td></tr><tr><td>0</td><td>The VBUS signal is not valid, and the pull-up resistor on D+ is disabled.</td></tr></table>	VBD	External VBUS Valid Indicator	1	The VBUS signal is valid, and the pull-up resistor on D+ is enabled	0	The VBUS signal is not valid, and the pull-up resistor on D+ is disabled.
VBD	External VBUS Valid Indicator									
1	The VBUS signal is valid, and the pull-up resistor on D+ is enabled									
0	The VBUS signal is not valid, and the pull-up resistor on D+ is disabled.									
0	LBE	R/W	0	This controller signal places the USB 2.0 nanoPHY in Loopback mode, which enables the receive and transmit logic concurrently. <table><tr><td>LBE</td><td>Loopback Test Enable</td></tr><tr><td>1</td><td>During data transmission, the receive logic is enabled.</td></tr><tr><td>0</td><td>During data transmission, the receive logic is disabled.</td></tr></table>	LBE	Loopback Test Enable	1	During data transmission, the receive logic is enabled.	0	During data transmission, the receive logic is disabled.
LBE	Loopback Test Enable									
1	During data transmission, the receive logic is enabled.									
0	During data transmission, the receive logic is disabled.									

[VBUS Valid and End Interrupt]

VBUS is the USB power supply pin. In device mode, an off-chip charge pump must provide power to VBUS pin. Alternatively the VBUS can be set internally by VBUSVLDEXT and VBUSVLDEXTSEL in USB PHY Configuration Register 0(UPCR0). If VBUSVLDEXTSEL is set to 1, VBUSVLDEXT value can be supplied as VBUS data of USB PHY.

There are two interrupts related to VBUS and these interrupts, which indicate the state of VBUS, are generated from B_VALID and SESSEND signals. The B_VALID signal is set to 1 when VBUS is 1 and the SESSEND signal is set to 1 when VBUS is 0. Therefore, when VBUS value changes, one of two interrupts definitely occurs.

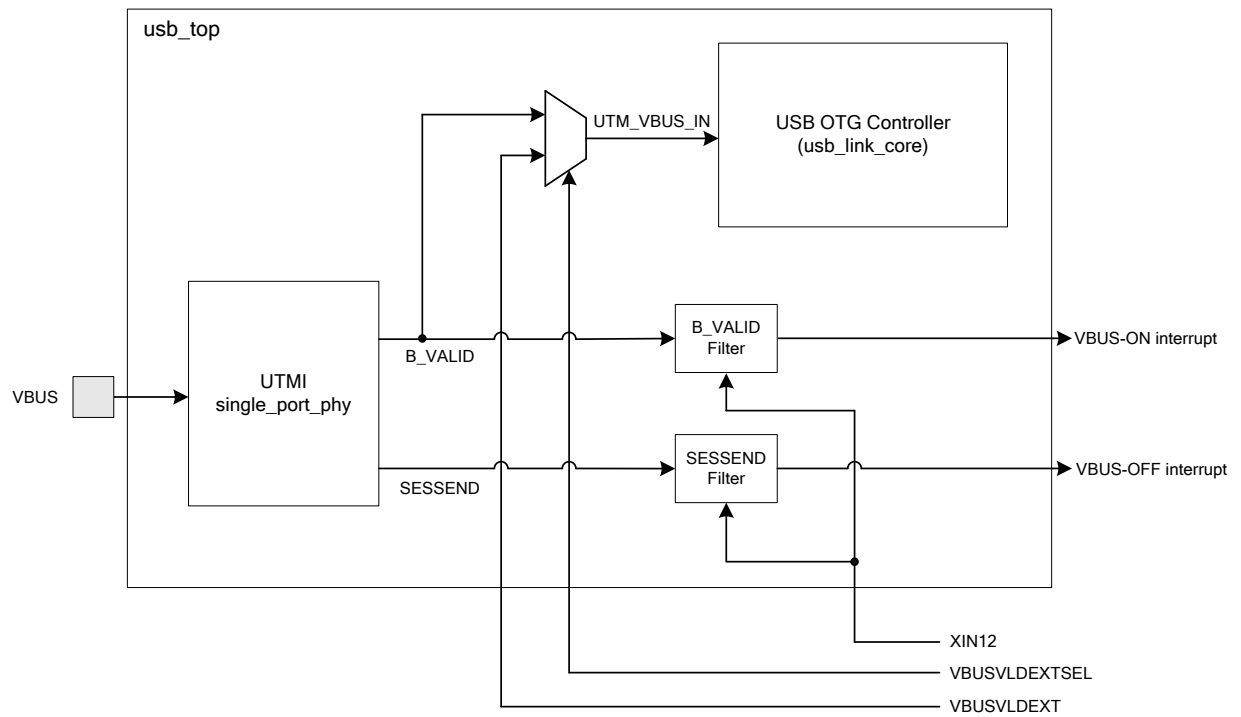


Figure 24.2 VBUS Control and Interrupts

USB PHY Configuration Register1 (UPCR1)

0x800C000C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TXFSLST					SQRXT				OTGT				CDT		

Bit	Name	R/W	Reset	Description	
15:12	TXFSLST	R/W	0x3	FS/LS Pull-Up Resistance Adjustment. This bus adjusts the low- and full-speed pull-up resistance, based on nominal power, voltage, and temperature.	
				1111	-5%
				0111	-2.5%
				0011	Design default (default)
				0001	+2.5%
				0000	+5%
				Others	Reserved
10:8	SQRXT	R/W	0x3	Squelch Threshold Tune. This bus adjusts the voltage level for the threshold used to detect valid high-speed data	
				111	-20%
				110	-15%
				101	-10%
				100	-5%
				011	Design default (default)
				010	+5%
				001	+10%
				000	+15%
6:4	OTGT	R/W	0x4	VBUS Valid Threshold Adjustment. This bus adjusts the voltage level for the threshold used to detect valid high-speed data.	
				111	+9%
				110	+6%
				101	+3%
				100	Design default(default)
				011	-3%
				010	-6%
				001	-7%
				000	-12%
2:0	CDT	R/W	0x4	Disconnect Threshold Adjustment. This bus adjusts the voltage level for the threshold used to detect a disconnect event at the host.	

USB PHY Configuration Register2 (UPCR2)

0x800C0010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	TM	XCVRSEL			OPMODE		TXVRT					TXRT			TP

Bit	Name	R/W	Reset	Description	
14	TM	R/W	0	USB Termination Select	
				1	Full Speed termination is enabled.
				0	High Speed termination is enabled.
				This controller signal selects FS or HS termination.	
				Note: It is effective only when UPCR0.MODE is set	
13:12	XCVRSEL	R/W	0x1	FS/LS Pull-Up Resistance Adjustment	
				11	Sends an LS packet on an FS bus or receives an Ls packet
				10	LS transceiver
				01	FS transceiver (default)
				00	HS transceiver
				This controller bus selects the HS, FS, or LS transceiver.	
				Note: It is effective only when UPCR0.MODE is set	
10:9	OPMODE	R/W	0x1	UTMI Operational Mode	
				11	Normal operation SYNC or EOP generation
				10	Disable bit stuffing and NRZI encoding
				01	Non driving
				00	Normal(default)
				This controller bus selects the UTMI operational mode.	
				Note: It is effective only when UPCR0.MODE is set.	
8:5	TXVRT	R/W	0x8	HS DC Voltage Level Adjustment	
				1111	+24%
				1110	+22%
				1101	+20%
				1100	+18%
				1011	+16%
				1010	+14%
				1001	+12%
				1000	+10%
				0111	+8%
				0110	+6%
				0101	+4%
				0100	+2%
				0011	Design default
				0010	-2%
				0001	-4%
0000	-6%				
				This bus adjusts the voltage to which the high speed DC level is tuned.	
3:2	TXRT	R/W	1	HS Transmitter Rise/Fall Time Adjustment	
				3	-30%
				2	-20%
				1	Design default (default)
				0	+10%
				This bus adjusts the rise/fall times the high speed waveform	
0	TP	R/W	0	HS transmitter Pre-Emphasis Enable	
				1	The HS Transmitter pre-emphasis is enabled
				0	The HS transmitter pre-emphasis is disabled. (default)
				This signal enables or disables the pre-emphasis for a J-K or K-J state transition in HS mode	

USB PHY Configuration Register3 (UPCR3)

0x800C0014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SR	SCALE		U20DH	IDPULLUP	DRVVBUS	CHRGVBUS	DISCHRGVBUS				IDDIG	TXHSXVT			ATERESET

Bit	Name	R/W	Reset	Description	
31	-	R	0	Reserved.	
30:16	-	R/W	0	Reserved. Must be written as “0”.	
15	SR	R/W	0	Soft Reset	
				1	Soft Reset Disabled (Normal)
				0	Soft Reset Enabled. Once enabled, this bit must be maintained at least 10us.
14:13	SCALE	R/W	0	Scale down mode for Simulation. For normal operation, these bits must be disabled.	
				11	Enables bit 0 and bit 1 scale down timing values.
				10	Enables scale-down of Device mode suspend and resume timing values only.
				01	Enables scale-down of all timing values except Device mode suspend and resume
			00	Disable all scale-downs. Actual timing values are used.	
12	U20DH	R/W	0	USB20DH Disable	
				1	The USB20DH block is powered down.
				0	The USB20DH block is not powered down(default)
				This controller signal powers down the USB20DH block, including the VBUS Valid comparator. If the application does not use USB20DH functionality, you can set this input high to save power.	
11	IDPULLUP	R/W	1	Analog ID Input Sample Enable This controller signal controls ID line sampling. ■ 1: ID pin sampling is enabled, and the IDDIG0 output is valid. ■ 0: ID pin sampling is disabled, and the IDDIG0 output is not valid.	
10	DRVVBUS	R/W	0	Drive VBUS This controller signal controls the VBUS Valid comparator. This signal drives 5 V on VBUS through an external charge pump. When USB20DH DISABLE0 is set to 1'b0 and DRVVBUS0 is asserted, the Bandgap circuitry and VBUS Valid comparator are powered, even in Suspend or Sleep mode. ■ 1: The VBUS Valid comparator is enabled. ■ 0: The VBUS Valid comparator is disabled.	
9	CHRGVBUS	R/W	0	VBUS Input Charge Enable Directs the PHY to charge VBUS. ■ 1'b0: Do not charge VBUS through a resistor. ■ 1'b1: Charge VBUS through a resistor (must be active for at least 30 ms).	
8	DISCHRGVBUS	R/W	0	VBUS Input Discharge Enable Directs the PHY to discharge VBUS ■ 1'b0: Do not discharge VBUS through a resistor. ■ 1'b1: Discharge VBUS through a resistor (must be active for at least 50 ms).	
4	IDDIG	R/W	0	Mini A/B Plug Indicator This controller signal indicates whether the connected plug is a mini-A or mini-B plug. This signal must be valid within 20 ms after the IDPULLUP0 signal is set to 1'b1. ■ 1: The connected plug is a mini-B plug. ■ 0: The connected plug is a mini-A plug.	
3:2	TXHSXVT	R/W	1	Transmitter High Speed Crossover Adjustment	
				3	The crossover voltage is increased by 15mV
				2	The crossover voltage is increased by 30mV
				1	Default setting
			0	Reserved	
				TXHSXVT adjusts the voltage at which the DP and DM signals cross while transmitting in HS mode	
0	ATERESET	R/W	0	Reset Input from Automatic Test Equipment	

				When the USB 2.0 picoPHY is powered up (not in Suspend or Sleep mode), an automatic tester can use this test signal to disable PHYCLOCK0 and FREECLK, then re-enable them with an aligned phase. ■ 1: The PHYCLOCK0 and FREECLK outputs are disabled. ■ 0: The PHYCLOCK0 and FREECLK outputs are available within a specific period after ATERESET is deasserted. If this signal is not used, tie it low.
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USB PHY Configuration Register4 (UPCR4)

0x800C0018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CFG_SUSPENDM			SUSPENDM					TXRESTUNE		TXPREEMPULSETUNE	TXPREEMPAMPTUNE				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RETENABLEN	RESREQIN	TESTBURNIN	BYPASSSEL	BYPASSDMEN	BYPASSDPEN	BYPASSDMDATA	BYPASSDPDATA	ACAENB	DCDENB	VDATADATENB	VDATASRCENB	CHRGSEL	FSSE0EXT	FSDATAEXT	TXENABLEN

Bit	Name	R/W	Reset	Description
31	CFG_SUSPENDM	R/W	0	SUSPENDM select register 1:user_setting 0:utmi_SUSPENDM
28	SUSPENDM	R/W	0	User setting SUSPENDM
23:22	TXRESTUNE	R/W	1	USB Source Impedance Adjustment In some applications, there can be significant series resistance on the D+ and D- paths between the transceiver and cable. This bus adjusts the driver source impedance to compensate for added series resistance on the USB. Note: Any setting other than the default can result in source impedance variation across process, voltage, and temperature conditions that does not meet USB 2.0 specification limits. ■ 11: Source impedance is decreased by approximately 4 Ω. ■ 10: Source impedance is decreased by approximately 2 Ω. ■ 01: Design default ■ 00: Source impedance is increased by approximately 1.5 Ω. If this bus is not used, leave it at the default setting.
21	TXPREEMPULSETUNE	R/W	0	HS Transmitter Pre-Emphasis Duration Control This signal controls the duration for which the HS pre-emphasis current is sourced onto DP0 or DM0. The HS Transmitter pre-emphasis duration is defined in terms of unit amounts. One unit of pre-emphasis duration is approximately 580 ps and is defined as 1X pre-emphasis duration. This signal is valid only if either TXPREEMPAMPTUNE0[1] or TXPREEMPAMPTUNE0[0] is set to 1'b1. ■ 1: 1X, short pre-emphasis current duration ■ 0 (design default): 2X, long pre-emphasis current duration If TXPREEMPULSETUNE0 is not used, set it to 1'b0.
20:19	TXPREEMPAMPTUNE	R/W	1	HS Transmitter Pre-Emphasis Current Control This signal controls the amount of current sourced to DP0 and DM0 after a J-to-K or K-to-J transition. The HS Transmitter pre-emphasis current is defined in terms of unit amounts. One unit amount is approximately 600 μA and is defined as 1X pre-emphasis current. ■ 11: HS Transmitter pre-emphasis circuit sources 3X pre-emphasis current. ■ 10: HS Transmitter pre-emphasis circuit sources 2X pre-emphasis current. ■ 01: HS Transmitter pre-emphasis circuit sources 1X pre-emphasis current. ■ 00 (design default): HS Transmitter pre-emphasis is disabled. If these signals are not used, set them to 2'b00.
18:16	FSEL	R/W	2	Reference Clock Frequency Select Selects the USB 2.0 picoPHY reference clock frequency. ■ 111: 50 MHz ■ 110: Reserved ■ 101: 24 MHz ■ 100: 20 MHz ■ 011: 19.2 MHz ■ 010: 12 MHz ■ 001: 10 MHz ■ 000: 9.6 MHz
15	RETENABLEN	R/W	1	Retention Mode Enable Reserved Tie this signal high and keep it high.
14	RESREQIN	R/W	0	Reserved Tie this pin to 1'b0.
13	TESTBURNIN	R/W	0	Burn-in Test Enable This input enables the built-in self burn-in testing function of the PHY. At assertion of this signal, the PHY performs a power-on-reset sequence and then enters HS Loopback mode. In HS Loopback mode, the PHY continues to transmit alternating 1's and 0's at the maximum bit rate until TESTBURNIN is deasserted. This input has precedence over the SUSPENDM0, SIDDQ, PORTRESET0, ATERASET, and SLEEPM0 signals. Therefore, these signals are internally

				overridden when TESTBURNIN is asserted. This function requires a 50-MHz reference clock source on the CLKCORE input. If this signal is not used, tie it low.
12	BYPASSSEL	R/W	0	Transmitter Digital Bypass Select Enables/disables Transmitter Digital Bypass mode. ■ 1: Transmitter Digital Bypass mode is enabled. ■ 0: Transmitter Digital Bypass mode is disabled. If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.
11	BYPASSDMEN	R/W	0	DM0 Transmitter Digital Bypass Enable Enables/disables the DM0 FS/LS driver in Transmitter Digital Bypass mode. ■ 1: DM0 FS/LS driver is enabled and driven with the BYPASSDPDATA0 signal. ■ 0: DM0 FS/LS driver is disabled in Transmitter Digital Bypass mode. If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.
10	BYPASSDPEN	R/W	0	DP0 Transmitter Digital Bypass Enable Enables/disables the DP0 FS/LS driver in Transmitter Digital Bypass mode. ■ 1: DP0 FS/LS driver is enabled and driven with the BYPASSDPDATA0 signal. ■ 0: DP0 FS/LS driver is disabled in Transmitter Digital Bypass mode. If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.
9	BYPASSDMDATA	R/W	0	Data for DM0 Transmitter Digital Bypass In Transmitter Digital Bypass mode, this signal provides the data that is transmitted on DM0. ■ 1: DM0 FS/LS driver drives to a high state. ■ 0: DM0 FS/LS driver drives to a low state. If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.
8	BYPASSDPDATA	R/W	0	Data for DP0 Transmitter Digital Bypass In Transmitter Digital Bypass mode, this signal provides the data that is transmitted on DP0. ■ 1: DP0 FS/LS driver drives to a high state. ■ 0: DP0 FS/LS driver drives to a low state. If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.
7	ACAENB	R/W	1	ACA ID_USB20DH Pin Resistance Detection Enable Enables detection of resistance on the ID_USB20DH pin of an ACA. ■ 1: Enables detection of resistance on the ID_USB20DH pin of an ACA. ■ 0: Disables detection of resistance on the ID_USB20DH pin of an ACA. If this signal is not used, tie this input to 1'b0.
6	DCDENB	R/W	0	Data Contact Detection Enable Enables current sourcing on the D+ line and pull-down resistance on the D– line for Data Contact Detect (DCD). ■ 1: IDP_SRC current is sourced onto DP0, pull-down resistance on DM0 is enabled. ■ 0: IDP_SRC current is disabled, pull-down resistance on DM0 is disabled. Note: During USB 2.0 picoPHY normal operation, set this signal low. If this signal is not used, tie this input to 1'b0.
5	VDATDETENB	R/W	0	Battery Charging Attach/Connect Detection Enable Enables or disables attach/connect detection. ■ 1: Data detect voltage (CHG_DET) is enabled. ■ 0: Data detect voltage (CHG_DET) is disabled. During USB 2.0 picoPHY normal operation, set this signal low.
4	VDATSRCEB	R/W	0	Battery Charging Sourcing Select Enables or disables sourcing for battery charging. ■ 1: Data source voltage (VDAT_SRC) is enabled. ■ 0: Data source voltage (VDAT_SRC) is disabled. During USB 2.0 picoPHY normal operation, set this signal low.
3	CHRGSEL	R/W	0	Battery Charging Source Select Determines whether current is sourced onto or sunk from DP0 or DM0. ■ 1: Data source voltage (VDAT_SRC) is sourced onto DM0 and sunk from DP0. ■ 0: Data source voltage (VDAT_SRC) is sourced onto DP0 and sunk from DM0.
2	FSSE0EXT	R/W	0	USB 1.1 SE0 Generation This controller signal sets the USB to an SE0 state. This signal is valid only if FSXCVROWNER0 is set to 1'b1 and TXENABLEN0 is set to 1'b0. ■ 1: The D+ and D– lines are driven to an SE0 state. ■ 0: The D+ and D– line states are determined by FSDATAEXT0. If FSSE0EXT0 is not used, set it to 1'b0.

1	FSDATAEXT	R/W	0	USB 1.1 Transmit Data This controller signal sets the USB to either a J or K state. This signal is valid only if FSXCVROWNER0 is set to 1'b1, TXENABLEN0 is set to 1'b0, and FSSE0EXT0 is set to 1'b0. ■ 1: The D+ and D– lines are driven to a Differential 1. ■ 0: The D+ and D– lines are driven to a Differential 0. If FSDATAEXT0 is not used, set it to 1'b0.
0	TXENABLEN	R/W	1	USB 1.1 Data Enable This controller signal enables the FSDATAEXT0 and FSSE0EXT0 inputs. TXENABLEN0 is valid only when the FSXCVROWNER0 signal is set to 1'b1. ■ 1: FSDATAEXT0 and FSSE0EXT0 are disabled. ■ 0: FSDATAEXT0 and FSSE0EXT0 are enabled. If TXENABLEN0 is not used, set it to 1'b1.

USB PHY Configuration Register5 (UPCR5)

0x800C001C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
GPIN															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	CFG_FSXCVROWNER	CFG_TXVALIDH	CFG_TXVALID	CFG_DMPULLDOWN	CFG_DPPULLDOWN	CFG_XCVRSEL	CFG_TERMSEL			SYNHRESET	ANDHRESET		SSSCALEDOWNMODE	ADPRESETN	

Bit	Name	R/W	Reset	Description
31:16	GPIN	R/W	0	For debug
14	CFG_FSXCVROWNER	R/W	0	FSXCVROWNER select register 1: user_setting 0: utmi_FSXCVROWNER
13	CFG_TXVALIDH	R/W	0	TXVALIDH select register 1: fixed 0: utmi_TXVALIDH
12	CFG_TXVALID	R/W	0	TXVALID select register 1: fixed 0: utmi_TXVALID
11	CFG_DMPULLDOWN	R/W	0	DMPULLDOWN select register 1: user_setting 0: utmi_DMPULLDOWN
10	CFG_DPPULLDOWN	R/W	0	DPPULLDOWN select register 1: user_setting 0: utmi_DPPULLDOWN
9	CFG_XCVRSEL	R/W	0	XCVRSEL select register 1: user setting 0: utmi_XCVRSEL
8	CFG_TERMSEL	R/W	0	TERMSEL select register 1: user setting 0: utmi_TERMSEL
5	SYNHRESET	R/W	0	Sync HRESETn AND select register
4	ANDHRESET	R/W	0	HRESETn AND select register
2:1	SSSCALEDOWNMODE	R/W	1	Scale-Down Mode ■ When this signal is enabled during simulation, the core uses scaled-down timing values, resulting in faster simulations. This value is connected to invert the USB Link - Write mode 0 : Simulation mode 1 : Normal mode - Read mode 0 : Normal mode 1 : Simulation mode Caution: You must not set "read & bit write operation" but must write the word base.
0	ADPRESETN	R/W	0	ADP Reset signal

USB PHY Configuration Register6 (UPCR6)

0x800C0020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
											SESEND PMU	VBUSVA LIDPMU	BVALIDP M	HOSTDI SCONP MU	UTMIPM UCON
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DBNCEF LTRBYP ASS	SOF COUNT														SOFUPD ATETOG GLE

Bit	Name	R/W	Reset	Description
20	SESENDPMU	R/W	0	B-Device Session End for PMU Indicates whether the voltage on VBUS is below the B-device Session End threshold. The comparator thresholds are: ■ 1'b0: VBUS > 0.8 V ■ 1'b1: VBUS < 0.2 V
19	VBUSVALIDPMU	R/W	0	VBUS Valid for PMU Indicates whether the voltage on VBUS is valid for A/B-device/peripheral operation. The comparator thresholds are: ■ 1'b0: VBUS < 4.4 V ■ 1'b1: VBUS > 4.75 V
18	BVALIDPMU	R/W	0	B-Peripheral Session Valid Indicates whether the voltage on VBUS is at a valid level for a B-peripheral session. The comparator thresholds are: USB20DH Revision 1.3 and Revision 2.0. ■ 1'b0: VBUS < 0.8 V ■ 1'b1: VBUS = 0.8 V to 4 V Device Mode: ■ Asserting this signal In Device mode sets the Session Request / New Session Detected Interrupt bit in the Interrupt Register (GINTSTS) (GINTSTS.SessReqInt). ■ Deasserting this signal sets the Session End Detected bit in the Interrupt Register (GOTGINT) (GOTGINT.SesEndDet).
17	HOSTDISCONPMU	R/W	0	Peripheral Disconnect Indicator to Host for PMU Indicates that the USB transceiver has detected a disconnect condition on the cable. ■ This signal is valid only when utmiotg_dppulldown and utmiotg_dmpulldown are sampled asserted. ■ When utmiotg_dppulldown and utmiotg_dmpulldown are not sampled asserted, then the behavior of utmi_hostdisconnect is undefined. Connection: ■ No peripheral connected: signal remains asserted. ■ Peripheral connected: signal de-asserted.
16	UTMIPMUCON	R/W	0	UTMI+ PMU I/F select signal
15	DBNCEFLTRBYPASS	R/W	1	Mode: Host and Device Debounce Filter Bypass Bypass Debounce filters for avalid, bvalid, vbusvalid, sessend, and iddig signals when enabled. ■ 1'b0: Disabled ■ 1'b1: Enabled
14:1	SOF COUNT	R/W	0	SOF Input Count Wireless USB Wire Adapter (DWA) application ■ Wireless USB Device Wire Adapter (DWA) application: Input value to be loaded into Host Frame Number/Frame Time Remaining Register (HFNUM) field FrNum. ■ Non-Wireless USB application: Must be tied to 0. This signal is not present when parameter OTG_RM_OPT_FEATURES=Yes.
0	SOFUPDATETOGGLE	R/W	0	Update SOF Toggle input to update SOF value Host Frame Number/Frame Time Remaining Register (HFNUM) Field FrNum. ■ Wireless USB Device Wire Adapter (DWA) application: Can be used to synchronize the wired Host SOF to wireless USB time. ■ Non-wireless USB application: Must be tied to 0. This signal is not present when parameter OTG_RM_OPT_FEATURES=Yes.

25 Audio Interface

25.1 Overview

25.1.1 ADMA

ADMA is a dedicated DMA controller for Audio Interfaces such as DAI and SPDIF. For earlier TCC series, only general DMA is available. When a general DMA is used for transferring audio data, it has to read from memory and write to Audio Interface registers. It can be a redundant bus cycle time in time-critical applications so a dedicated DMA to Audio Interface is applied to reduce redundant bus cycle time. The ADMA enables users to utilize audio data for more flexible purpose. The structure of ADMA is similar to that of general DMA. Therefore, a user can use general DMA instead of ADMA. It is not recommended to use ADMA and general DMA at the same time.

The TCC8132 has three ADMA's;

ADMA 0 is dedicated to DAI0 and SPDIF.

ADMA 1 ~ 2 are dedicated to DAI1 ~ 2 respectively.

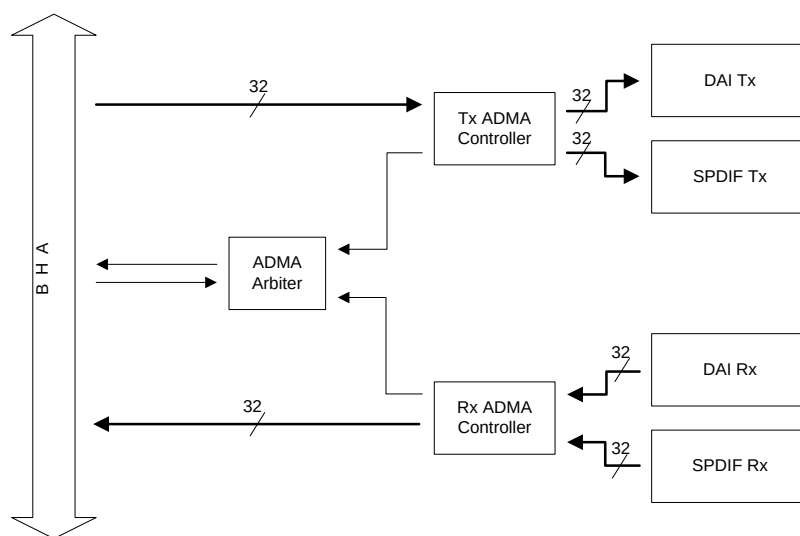


Figure 25.1 ADMA Top Block Diagram

25.1.2 DAI

The TCC8132 provides digital audio interface that complies with IIS (Inter-IC Sound). The DAI has five input/output pins for IIS interface; MCLK, BCLK, LRCK, DAI, and DAO. All DAI input/output pins are multiplexed with GPIO pins.

The MCLK is the system clock pin that is used for CODEC system clock. In master mode, the MCLK can be generated from clock generator in which that is known as a DCLK, or fed from the outside of chip in slave mode. The DAI can process 128fs, 192fs, 256fs, 288fs, 384fs and 512fs as a system clock. 256fs means that the system clock has 256 times of sampling frequency (fs).

The BCLK is the serial bit clock for IIS data exchange. The DAI can generate 64fs, 48fs and 32fs by dividing a system clock. The polarity of BCLK can be programmed. That is, the serial bit can be stable either rising edge of BCLK or falling edge of BCLK.

The LRCK is the frame clock for the stereo audio channel Left and Right. The frequency of LRCK is known as the "fs" – sampling frequency. Generally, for audio application – such as MP3 Player, CD player, the fs can be set to 8kHz ~ 192kHz. To support the wide range of sampling frequency in audio application, the DCO function is very useful to generate a system clock. Refer to the Chapter 10 Clock Controller for detailed information.

All three clocks (MCLK, BCLK, LRCK) are selectable as master or slave.

The DAI and DAO are the serial data input output pins respectively.

The DAI has two 8-word input/output buffers. It has a banked buffer structure so that one side of buffer is receiving/transmitting data while the other side of that can be read/written through the DADI_XX/DADO_XX registers. The maximum data word size is 24 bit. Data is justified to MSB of 32bits and zeros are padded to LSB.

There are 2 types of interrupt from IIS; transmit done interrupt, and receive done interrupt. The transmit-done interrupt is generated when the 8 words are transferred successfully in the output buffer. At this interrupt, a user should fill another 8 more words into the other part of the output buffer in the interrupt service routine (ISR). In this ISR routine, 8 consecutive stores of word data to the DADO registers are needed. The receive-done interrupt is generated when the 8 words are

received successfully in the input buffer. At this interrupt, a user should read 8 received words from the input buffer using 8 consecutive load instructions from the DADI registers.

The TCC8132 has three DAIs (DAI0, DAI1 and DAI2) and you can run three DAIs independently.

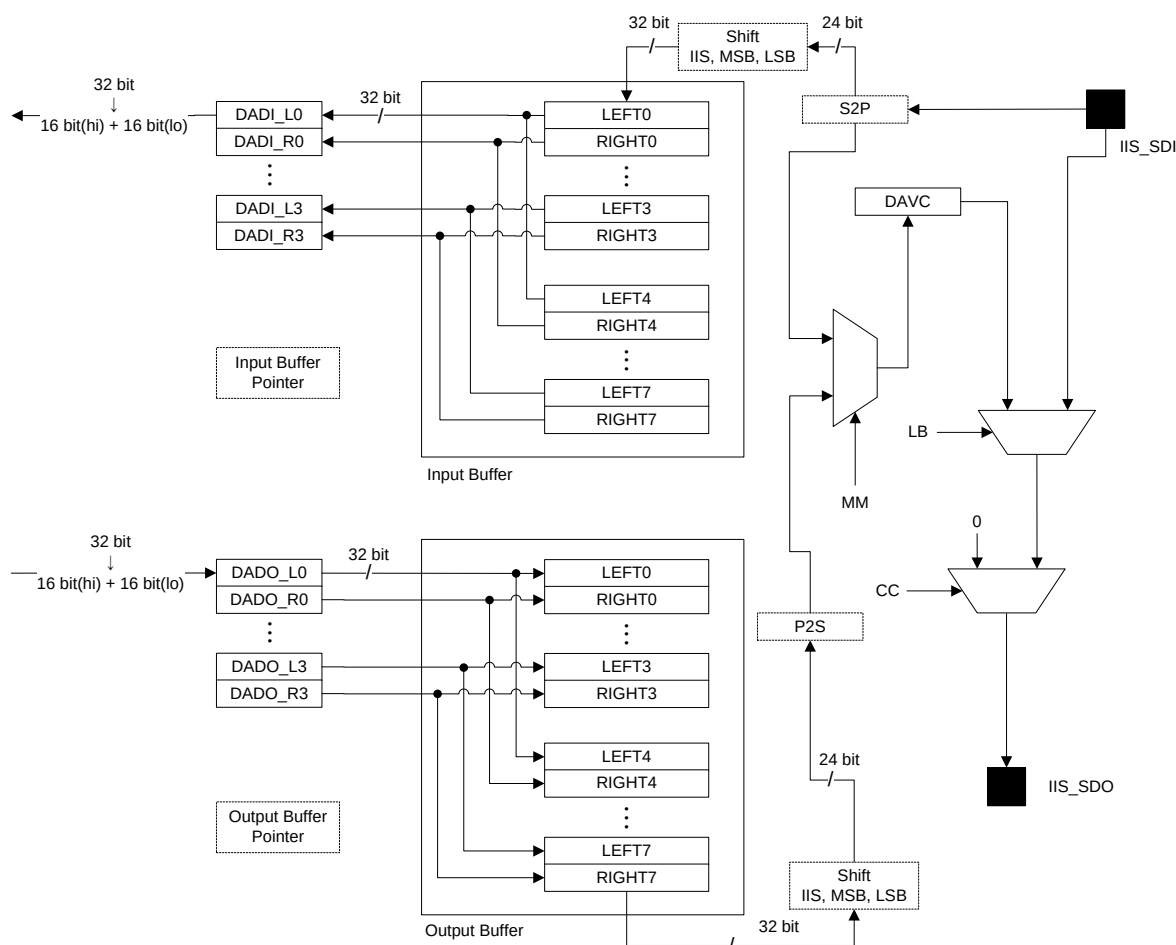


Figure 25.2 DAI Block Diagram

25.1.3 SPDIF

The SPDIF (or AES/EBU, IEC950 standards) is a point-to-point protocol for serial transmission of digital audio through a single transmission line. The transmission medium can be either electrical or optical (e.g. TosLink). It provides two channels for audio data, a method for communicating control information, and some error detection capabilities. The control information is transmitted as one bit per sample and accumulates in a block structure. The data is by-phase encoded, which enables the receiver to extract a clock from the data. Coding violations, defined as preambles, are used to identify sample and block boundaries.

The SPDIF interfaces are found on most CD/DVD players, audio equipment and computer sound cards.

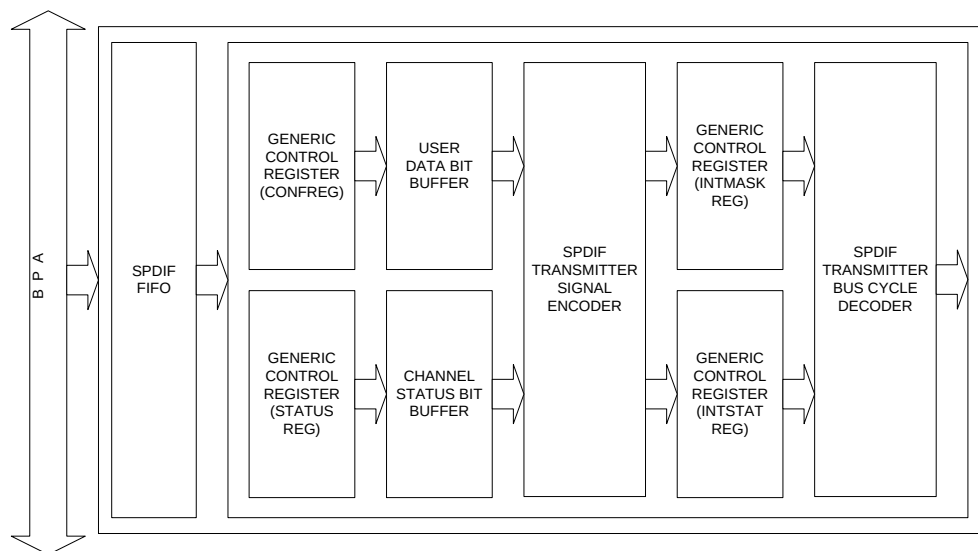


Figure 25.3 SPDIF Tx Block Diagram

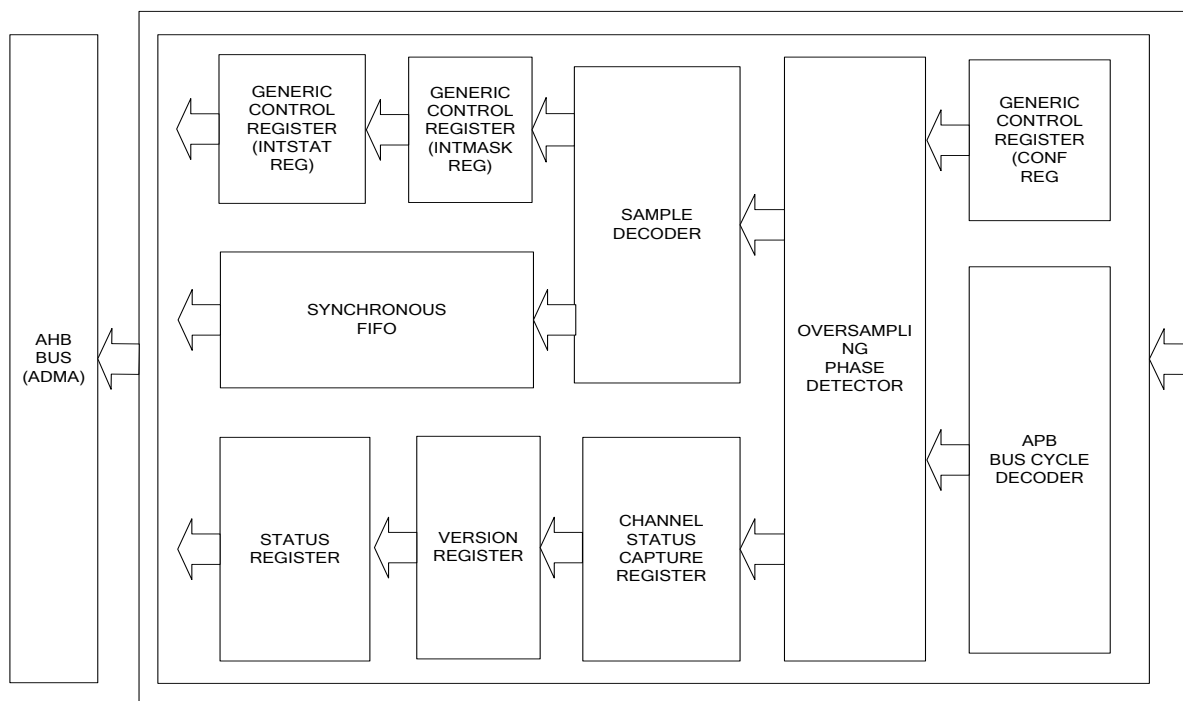


Figure 25.4 SPDIF Rx Block Diagram

25.2 Register Description

Table 25.1 Audio DMA0 Register Map (Base Address = 0x98002000)

Name	Addr	Type	Reset	Description
RxDaDar	0x00	R/W	0x00000000	DAI Rx (Right) Data Destination Address
RxDaParam	0x04	R/W	0x00000000	DAI Rx Parameters
RxDaTCnt	0x08	R/W	0x00000000	DAI Rx Transmission Counter Register
RxDaCdar	0x0C	R	0x00000000	DAI Rx (Right) Data Current Destination Address
RxCdDar	0x10	R/W	0x00000000	CDIF Rx (Right) Data Destination Address
RxCdParam	0x14	R/W	0x00000000	CDIF Rx Parameters
RxCdTCnt	0x18	R/W	0x00000000	CDIF Rx Transmission Counter Register
RxCdCdar	0x1C	R	0x00000000	CDIF Rx (Right) Data Current Destination Address
RxDaDarL	0x28	R/W	0x00000000	DAI Rx Left Data Destination Address
RxDaCdarL	0x2C	R	0x00000000	DAI Rx Left Data Current Destination Address
RxCdDarL	0x30	R/W	0x00000000	CDIF Rx Left Data Destination Address
RxCdCdarL	0x34	R	0x00000000	CDIF Rx Left Data Current Destination Address
TransCtrl	0x38	R/W	0x0000AA00	DMA Transfer Control Register
RptCtrl	0x3C	R/W	0x00000000	DMA Repeat Control Register
TxDaSar	0x40	R/W	0x00000000	DAI Tx (Right) Data Source Address
TxDaParam	0x44	R/W	0x00000000	DAI Tx Parameters
TxDaTCnt	0x48	R/W	0x00000000	DAI Tx Transmission Counter Register
TxDaCsar	0x4C	R	0x00000000	DAI Tx (Right) Data Current Source Address
TxSpSar	0x50	R/W	0x00000000	SPDIF Tx (Right) Data Source Address
TxSpParam	0x54	R/W	0x00000000	SPDIF Tx Parameters
TxSpTCnt	0x58	R/W	0x00000000	SPDIF Tx Transmission Counter Register
TxSpCsar	0x5C	R	0x00000000	SPDIF Tx (Right) Data Current Source Address
TxDaSarL	0x68	R/W	0x00000000	DAI Tx Left Data Source Address
TxDaCsarL	0x6C	R	0x00000000	DAI Tx Left Data Current Source Address
TxSpSarL	0x70	R/W	0x00000000	SPDIF Tx Left Data Source Address
TxSpCsarL	0x74	R	0x00000000	SPDIF Tx Left Data Current Source address
ChCtrl	0x78	R/W	0x00008000	DMA Channel Control Register
IntStatus	0x7C	R/W	0x00000000	DMA Interrupt Status Register

Table 25.2 Audio DMA1 ~ 2 Register Map
(ADMA1 Base Address = 0x98002100)
(ADMA2 Base Address = 0x98002200)

Name	Addr	Type	Reset	Description
RxDaDar	0x00	R/W	0x00000000	DAI Rx (Right) Data Destination Address
RxDaParam	0x04	R/W	0x00000000	DAI Rx Parameters
RxDaTCnt	0x08	R/W	0x00000000	DAI Rx Transmission Counter Register
RxDaCdar	0x0C	R	0x00000000	DAI Rx (Right) Data Current Destination Address
RxDaDarL	0x28	R/W	0x00000000	DAI Rx Left Data Destination Address
RxDaCdarL	0x2C	R	0x00000000	DAI Rx Left Data Current Destination Address
TransCtrl	0x38	R/W	0x0000AA00	DMA Transfer Control Register
RptCtrl	0x3C	R/W	0x00000000	DMA Repeat Control Register
TxDaSar	0x40	R/W	0x00000000	DAI Tx (Right) Data Source Address
TxDaParam	0x44	R/W	0x00000000	DAI Tx Parameters
TxDaTCnt	0x48	R/W	0x00000000	DAI Tx Transmission Counter Register
TxDaCsar	0x4C	R	0x00000000	DAI Tx (Right) Data Current Source Address
TxDaSarL	0x68	R/W	0x00000000	DAI Tx Left Data Source Address
TxDaCsarL	0x6C	R	0x00000000	DAI Tx Left Data Current Source Address
ChCtrl	0x78	R/W	0x00008000	DMA Channel Control Register
IntStatus	0x7C	R/W	0x00000000	DMA Interrupt Status Register

Table 25.3 DAI0 ~ 2 Register Map
(DAI0 Base Address = 0x90000000)
(DAI1 Base Address = 0x90000100)
(DAI2 Base Address = 0x90000200)

Name	Address	Type	Reset	Description
DADI_L0	0x00	R	-	Digital Audio Left Input Register 0
DADI_R0	0x04	R	-	Digital Audio Right Input Register 0
DADI_L1	0x08	R	-	Digital Audio Left Input Register 1
DADI_R1	0x0C	R	-	Digital Audio Right Input Register 1
DADI_L2	0x10	R	-	Digital Audio Left Input Register 2
DADI_R2	0x14	R	-	Digital Audio Right Input Register 2
DADI_L3	0x18	R	-	Digital Audio Left Input Register 3
DADI_R3	0x1C	R	-	Digital Audio Right Input Register 3
DADO_L0	0x20	R/W	-	Digital Audio Left Output Register 0
DADO_R0	0x24	R/W	-	Digital Audio Right Output Register 0
DADO_L1	0x28	R/W	-	Digital Audio Left Output Register 1
DADO_R1	0x2C	R/W	-	Digital Audio Right Output Register 1
DADO_L2	0x30	R/W	-	Digital Audio Left Output Register 2
DADO_R2	0x34	R/W	-	Digital Audio Right Output Register 2
DADO_L3	0x38	R/W	-	Digital Audio Left Output Register 3
DADO_R3	0x3C	R/W	-	Digital Audio Right Output Register 3
DAMR	0x40	R/W	0x00000000	Digital Audio Mode Register
DAVC	0x44	R/W	0x0000	Digital Audio Volume Control Register
ASRC_MUX	0x48	R/W	0x00000000	ASRC Path Control Register

Table 25.4 SPDIF TX Register Map (Base Address = 0x90001000)

Name	Address	Type	Reset	Description
TxVersion	0x00	R	0x00003111	Version Register
TxConfig	0x04	R/W	0x00000000	Configuration Register
TxChStat	0x08	R/W	0x00000000	Channel Status Control Register
TxIntMask	0x0C	R/W	0x00000000	Interrupt Mask Register
TxIntStat	0x10	R/W	0x00000000	Interrupt Status Register
UserData	0x80~0xDC	W	-	User Data Buffer
ChStatus	0x100~0x15C	W	-	Channel Status Buffer
TxBuffer	0x200~0x23C	W	-	Transmit Data Buffer
DMACFG	0x400	R/W	0x00000007	Additional Configuration for DMA
CSBUDB	0x680~0x6DC	W	-	Merged Window for CSB/UDB

Table 25.5 SPDIF Rx Register Map (Base Address = 0x90001800)

Name	Address	Type	Reset	Description
RxVersion	0x00	R	0x00080111	Version Register
RxConfig	0x04	R/W	0x00000000	Configuration Register
RxStatus	0x08	R	0x00000000	Signal Status Buffer
RxIntMask	0x0C	R/W	0x00000000	Interrupt Mask Register
RxIntStat	0x10	R/W	0x00000000	Interrupt Status Register
RxCapCtl [n]	0x40~0x7C(even)	RW	0x00000000	Channel Status Capture Control Register
RxCap [n]	0x40~0x7C(odd)	R	0x00000000	Captured Channel Status / user bit
RxBuffer	0xA00~0xA1C	R	-	Receive Data Buffer

25.2.1 Audio DMA Registers

DAI Rx / DAI(SPDIF)Tx (Right) Data Destination Address Register

0x98002000(0x10)/0x40(0x50)
0x98002100/0x40
0x98002200/0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RxDaDar[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RxDaDar[15:0]															

This register contains the start address of target memory block for DAI Rx DMA transfer. The transfer begins reading data from this address. When LRMode is asserted, this is used for Right Data of DAI Rx.

DAI Rx / DAI(SPDIF) Tx Parameters Register

0x98002004(0x14)/0x44(0x54)
0x98002104/0x44
0x98002204/0x44

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SMASK[23:8]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMASK[7:0]								SINC[7:0]							

SMASK[23:0]	[31:8]	Destination Address Mask Register
0	non-masked	
1	Masked so that source address bit does not change during DMA transfer	

Each bit field controls the dedicated bit of source address field. That is, if SMASK[23] is set to 1, the 28th bit of source address is masked, and if SMASK[22] is set to 1, the 27th bit of source address is masked, and so on. If a bit is masked, a corresponding bit of address bus is not changed during DMA transfer. This function can be used to generate circular buffer address.

SINC[7:0]	[7:0]	Destination Address Increment Register
sinc	Source address is added by amount of sinc at every write cycles. sinc is represented as 2's complement, so if SINC[7] is 1, the source address is decremented.	

The addresses of DMA transfer have 32bit wide, but the upper 4bit of them are not affected during DMA transfer. If the source or destination address reaches its maximum address space like 0x7FFFFFFF or 0x2FFFFFFF, the next transfer is starting from 0x70000000 or 0x20000000 not from 0x80000000 or 0x30000000.

DAI Rx / DAI(SPDIF) Tx Transmission Counter Register

0x98002008(0x18)/0x48(0x58)
0x98002108/0x48
0x98002208/0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
C_TCNT[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ST_TCNT[15:0]															

C_TCNT[15:0]	Type	[31:16]	Current Transmission Count
cn	R		Represent cn number of Hop transfer remains

ST_TCNT[15:0]	Type	[15:0]	Start Transmission Count
sn	R/W		DMA transfers data by amount of sn Hop transfers

At the beginning of transfer, the C_TCNT is updated by ST_TCNT register. At the end of every hop transfer, this is decremented by 1 until it reaches 0. When this reaches 0, the DMA finishes its transfer and may generate its interrupt according to IEN flag of CHCTRL register.

DAI Rx / DAI(SPDIF) Tx (Right) Data Current Destination Address Register

0x9800200C(0x1C)/0x4C(0x5C)

0x9800210C/0x4C

0x9800220C/0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RxDaCdar[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RxDaCdar[15:0]															

This register contains the current destination address of DAI Rx DMA transfer. It represents that the current transfer read data from this address. This is read only register. When LRMode is asserted, this is used for Right Data of DAI Rx.

DAI Rx / DAI(SPDIF) Tx Left Data Destination Address Register

0x98002028(0x30)/0x68(0x70)

0x98002128/0x68

0x98002228/0x68

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RxDaDarL[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RxDaDarL[15:0]															

This register contains the start address of target memory block for DAI Rx DMA transfer. The transfer begins reading data from this address. When LRMode is asserted, this is used for Left Data of DAI Rx, otherwise this is not available.

DAI Rx / DAI(SPDIF) Left Data Current Destination Address Register

0x9800202C(0x34)/0x6C(0x74)

0x9800212C/0x6C

0x9800222C/0x6C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RxDaCdarL[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RxDaCdarL[15:0]															

This register contains the current destination address of DAI Rx DMA transfer. It represents that the current transfer reads data from this address. This register is read only. When LRMode is asserted, this register is used for Left Data of DAI Rx, otherwise this is not available.

Transfer Control Register

0x98002038
0x98002138
0x98002238

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	RCN	TCN	CRLCK	DRLCK	STLCK	DTLCK	CRTRG	DRTRG	STTRG	DTTRG	CRRPT	DRRPT	STRPT	DTRPT	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CRBSIZE	DRBSIZE	STBSIZE	DTBSIZE	CRWSIZE	DRWSIZE	STWSIZE	DTWSIZE								

RCN	[29]	Issue Continuous Transfer of RX DMA
0		DMA transfer begins from source/destination address
1		DMA transfer begins from current source/destination address. It must be used after the former transfer has been executed, so that current source/destination address registers contain a meaningful value.

TCN	[28]	Issue Continuous Transfer of TX DMA
0		DMA transfer begins from source/destination address
1		DMA transfer begins from current source/destination address. It must be used after the former transfer has been executed, so that current source/destination address registers contain a meaningful value.

CRLCK	[27]	Issue Locked Transfer of CDIF RX or SPDIF RX DMA
1		DMA transfer executed with lock transfer (NOT USE CDIFRX)

Lock field controls the LOCK signal (Refer to AHB specification). When the LOCK is set to 1, the DMA transfer is not bothered by other AHB masters like LCD controller, ARM, and etc. This bit is not available on ADMA1, 2, 3.

DRLCK	[26]	Issue Locked Transfer of DAI RX DMA
1		DMA transfer executed with lock transfer

STLCK	[25]	Issue Locked Transfer of SPDIF TX DMA
1		DMA transfer executed with lock transfer

This bit is not available on ADMA1, 2.

DTLCK	[24]	Issue Locked Transfer of DAI TX DMA
1		DMA transfer executed with lock transfer

CRTRG	[23]	Trigger Type of CDIF RX or SPDIF RX DMA
0		SINGLE edge-triggered detection (NOT USE CDIFRX)
1		SINGLE level-sensitive detection (NOT USE CDIFRX)

In SINGLE Type, after one Hop data transferring DMA checks External DMA Request (DREQ) and then if its bit is active, DMA transfers next hop data. The DREQ is detected level-sensitive or edge-triggered by SINGLE transfer TYPE. This bit is not available on ADMA1, 2.

DRTRG	[22]	Trigger Type of DAI RX DMA
0		SINGLE edge-triggered detection
1		SINGLE level-sensitive detection

STTRG	[21]	Trigger Type of SPDIF TX DMA
0		SINGLE edge-triggered detection
1		SINGLE level-sensitive detection

This bit is not available on ADMA1, 2.

DTTRG	[20]	Trigger Type of DAI TX DMA
0		SINGLE edge-triggered detection
1		SINGLE level-sensitive detection

CRRPT	[19]	Repeat Mode Control of CDIF RX or SPDIF RX DMA
0		After all of hop transfer has executed, the DMA channel is disabled
1		The DMA channel remains enabled. When another DMA request has occurred, the DMA channel start transfer data again with the same manner (type, address, increment, mask) as the latest transfer of that channel.

This bit is not available on ADMA1, 2.
This package does not have CDIF RX.

DRRPT	[18]	Repeat Mode Control of DAI RX DMA
0		After all of hop transfer has executed, the DMA channel is disabled
1		The DMA channel remains enabled. When another DMA request has occurred, the DMA channel start transfer data again with the same manner (type, address, increment, mask) as the latest transfer of that channel.

STRPT	[17]	Repeat Mode Control of SPDIF TX DMA
0		After all of hop transfer has executed, the DMA channel is disabled
1		The DMA channel remains enabled. When another DMA request has occurred, the DMA channel start transfer data again with the same manner (type, address, increment, mask) as the latest transfer of that channel.

This bit is not available on ADMA1, 2.

DTRPT	[16]	Repeat Mode Control of DAI TX DMA
0		After all of hop transfer has executed, the DMA channel is disabled
1		The DMA channel remains enabled. When another DMA request has occurred, the DMA channel start transfer data again with the same manner (type, address, increment, mask) as the latest transfer of that channel.

The 1 Hop of transfer means 1 burst of read followed by 1 burst of write. 1 burst means 1, 2 or 4 consecutive read or write cycles defined by BSIZE field of CHCTRL register. The figure below illustrates 1 Hop of transfer

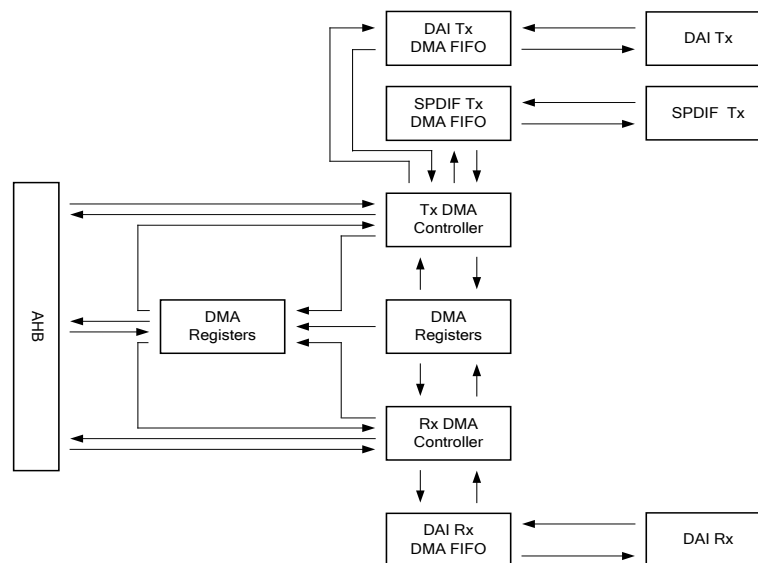


Figure 25.5 Relation between Hop and Burst Transfers

CRBSIZE[1:0]	[15:14]	Burst Size of CDIF RX or SPDIF RX DMA
0		1 Burst transfer consists of 1 read or write cycle.
1		1 Burst transfer consists of 2 read or write cycles
2		1 Burst transfer consists of 4 read or write cycles
3		1 Burst transfer consists of 8 read or write cycles

When LRMode is enabled, 8 Read/Write is not available. These bits are not available on ADMA1, 2.
This package does not have CDIF RX.

DRBSIZE[1:0]	[13:12]	Burst Size of DAI RX DMA
0		1 Burst transfer consists of 1 read or write cycle.
1		1 Burst transfer consists of 2 read or write cycles
2		1 Burst transfer consists of 4 read or write cycles
3		1 Burst transfer consists of 8 read or write cycles

When LRMode is enabled, 8 Read/Write is not available.

STBSIZE[1:0]	[11:10]	Burst Size of SPDIF TX DMA
0	1	Burst transfer consists of 1 read or write cycle.
1	1	Burst transfer consists of 2 read or write cycles
2	1	Burst transfer consists of 4 read or write cycles
3	1	Burst transfer consists of 8 read or write cycles

When LRMode is enabled, 8 Read/Write is not available. These bits are not available on ADMA1, 2.

DTBSIZE[1:0]	[9:8]	Burst Size of DAI TX DMA
0	1	Burst transfer consists of 1 read or write cycle.
1	1	Burst transfer consists of 2 read or write cycles
2	1	Burst transfer consists of 4 read or write cycles
3	1	Burst transfer consists of 8 read or write cycles

When LRMode is enabled, 8 Read/Write is not available.

CRWSIZE[1:0]	[7:6]	Word Size of CDIF RX or SPDIF RX DMA
0		Each cycle read or write 8bit data
1		Each cycle read or write 16bit data
2, 3		Each cycle read or write 32bit data

These bits are not available on ADMA1, 2.
This package does not have CDIF RX.

DRWSIZE[1:0]	[5:4]	Word Size of DAI RX DMA
0		Each cycle read or write 8bit data
1		Each cycle read or write 16bit data
2, 3		Each cycle read or write 32bit data

STWSIZE[1:0]	[3:2]	Word Size of SPDIF TX DMA
0		Each cycle read or write 8bit data
1		Each cycle read or write 16bit data
2, 3		Each cycle read or write 32bit data

These bits are not available on ADMA1, 2.

DTWSIZE[1:0]	[1:0]	Word Size of DAI TX DMA
0		Each cycle read or write 8bit data
1		Each cycle read or write 16bit data
2, 3		Each cycle read or write 32bit data

Repeat Control Register

0x9800203C

0x9800213C

0x9800223C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DRI	Reserved							RPTCNT[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RPTCNT[15:0]															

DRI	[31]	Disable Repeat Interrupt
0		DMA Interrupt is occurred at the end of each Repeated DMA operation.
1		DMA Interrupt is occurred at the last DMA Repeated DMA operation

This bit is meaningful when Repeat Mode is enabled.

RPTCNT[23:0]	[23:0]	Repeat Count
0		DMA transfer data endlessly.
None zero		DMA transfer the number of $(N + 1) * TCOUNT$ data

This bit is meaningful when repeat mode is enabled. When this bit is cleared in repeat mode, the DMA runs endlessly. To exit endless repeat mode, clear EN bit of ChCtrl or disable Repeat Mode. It is possible to circular transfer using repeat count.

Channel Control Register

0x98002078

0x98002178

0x98002278

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CREN	DREN	STEN	DTEN		SRSEL	Reserved					SPCLK	CRLR	DRLR	STLR	DTLR
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CRDW	DRDW	STDW	DTDW	CRSEN	DRSEN	STSEN	DTSEN	CRWB	DRWB	STWB	DTWB	CRIEN	DRIEN	STIEN	DTIEN

CREN	[31]	DMA Channel Enable of CDIF RX or SPDIF RX
0		DMA channel is terminated and disabled. It does not affect the TCOUNT register. If the current hop counter is not zero when channel is disabled, it is possible that the transfer illegally starts right after channel is re-enabled. Make sure that TCOUNT is zero not to continue transfer after channel is re-enabled.
1		DMA channel is enabled. Source is selected by SRSEL (bit [26])

This bit is not available on ADMA1, 2.
This package does not have CDIF RX.

DREN	[30]	DMA Channel Enable of DAI RX
0		DMA channel is terminated and disabled.
1		DMA channel is enabled.

STEN	[29]	DMA Channel Enable of SPDIF TX
0		DMA channel is terminated and disabled.
1		DMA channel is enabled.

This bit is not available on ADMA1, 2.

DTEN	[28]	DMA Channel Enable of DAI TX
0		DMA channel is terminated and disabled.
1		DMA channel is enabled.

SRSEL	[26]	Select CDIF or SPDIF RX
0		CDIF (Not used)
1		SPDIF_RX (Not used)

This package does not have CDIF RX.

SPCLK	[20]	SPDIF Clock Source
1		Provide SPDIF with DAI clock from CKC.
0		Provide SPDIF with SPDIF clock from CKC.

CRLR	[19]	Left/Right Data Mode of CDIF RX or SPDIF RX
1		Enable LRMode. When LRMode is enabled, audio data will be written to memory separately according to whether it is left or right one.
0		Disable LRMode.

This bit is not available on ADMA1, 2.
This package does not have CDIF RX.

DRLR	[18]	Left/Right Data Mode of DAI RX
1		Enable LRMode.
0		Disable LRMode.

STLR	[17]	Left/Right Data Mode of SPDIF TX
1		Enable LRMode.
0		Disable LRMode.

This bit is not available on ADMA1, 2.

DTLR	[16]	Left/Right Data Mode of DAI TX
1		Enable LRMode.
0		Disable LRMode.

CRDW	[15]	Width of Audio Data of CDIF RX or SPDIF RX
1		Assume width of audio data is 16bits. Audio DMA manipulates audio data as 16bit-length one. If it gets 24bit-length audio data from audio devices, it will cut out lsb-16bits and put those into DMA FIFO. For more details, refer to attached pictures.
0		Assume width of audio data is 24bits.

This bit is not available on ADMA1, 2.
This package does not have CDIF RX.

DRDW	[14]	Width of Audio Data of DAI RX
1		Assume width of audio data is 16bits.
0		Assume width of audio data is 24bits.

STDW	[13]	Width of Audio Data of SPDIF TX
1		Assume width of audio data is 16bits.
0		Assume width of audio data is 24bits.

This bit is not available on ADMA1, 2.

DTDW	[12]	Width of Audio Data of DAI TX
1		Assume width of audio data is 16bits.
0		Assume width of audio data is 24bits.

CRSEN	[11]	Swapping Half-word/Byte align of CDIF RX or SPDIF RX
0		Disable swapping
1		Swaps half-word or byte align according to WB bits

This bit is not available on ADMA1, 2.
This package does not have CDIF RX.

DRSEN	[10]	Swapping Half-word/Byte align of DAI RX
0		Disable swapping
1		Swaps half-word or byte align according to WB bits

STSEN	[9]	Swapping Half-word/Byte align of SPDIF TX
0		Disable swapping
1		Swaps half-word or byte align according to WB bits

This bit is not available on ADMA1, 2.

DTSSEN	[8]	Swapping Half-word/Byte align of DAI TX
0		Disable swapping
1		Swaps half-word or byte align according to WB bits

CRWB	[7]	Choose Half-word/Byte of CDIF RX or SPDIF RX
0		Byte swapping. This is available only when SWP is enabled.
1		Half-word swapping. This is available only when SWP is enabled.

This bit is not available on ADMA1, 2.
This package does not have CDIF RX.

DRWB	[6]	Choose Half-word/Byte of DAI RX
0		Byte swapping. This is available only when SWP is enabled.
1		Half-word swapping. This is available only when SWP is enabled.

STWB	[5]	Choose Half-word/Byte of SPDIF TX
0		Byte swapping. This is available only when SWP is enabled.
1		Half-word swapping. This is available only when SWP is enabled.

This bit is not available on ADMA1, 2.

DTWB	[4]	Choose Half-word/Byte of DAI TX
0		Byte swapping. This is available only when SWP is enabled.
1		Half-word swapping. This is available only when SWP is enabled.

CRIEN	[3]	Interrupt Enable of CDIF RX or SPDIF RX
1		At the same time the FLAG goes to 1, DMA interrupt request is generated.

To generate IRQ or FIQ interrupt, the DMA flag of IEN register in the interrupt controller must be set to 1 in advance. This bit is not available on ADMA1, 2.
This package does not have CDIF RX.

DRIEN	[2]	Interrupt Enable of DAI RX
1		At the same time the FLAG goes to 1, DMA interrupt request is generated.

STIEN	[1]	Interrupt Enable of SPDIF TX
1		At the same time the FLAG goes to 1, DMA interrupt request is generated.

This bit is not available on ADMA1, 2, 3.

DTIEN	[0]	Interrupt Enable of DAI TX
1		At the same time the FLAG goes to 1, DMA interrupt request is generated.

DMA Interrupt Status Register

0x9800207C

0x9800217C

0x9800227C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								CRI	DRI	STI	DTI	CRMI	DRMI	STMI	DTMI

CRI	[7]	DMA Interrupt Status of CDIF Rx or SPDIF RX
0		No interrupt occurred while CDIF Rx DMA transfer.
1		Interrupt occurred while CDIF Rx DMA transfer.

Without regard to Interrupt enable bit(CRIEN) of ChCtrl, this bit indicates the CDIF Rx DMA interrupt status.
This bit is read only and is automatically cleared when FLAG bit of ChCtrl is cleared. This bit is not available on ADMA1, and ADMA2.
This package does not have CDIF RX.

DRI	[6]	DMA Interrupt Status of DAI RX
0		No interrupt occurred while DAI Rx DMA transfer.
1		Interrupt occurred while DAI Rx DMA transfer.

Without regard to Interrupt enable bit(DRIEN) of ChCtrl, this bit indicates the DAI Rx DMA interrupt status. This bit is read only and is automatically cleared when FLAG bit of ChCtrl is cleared.

STI	[5]	DMA Interrupt Status of SPDIF TX
0		No interrupt occurred while SPDIF Tx DMA transfer.
1		Interrupt occurred while SPDIF Tx DMA transfer.

Without regard to Interrupt enable bit(STIEN) of ChCtrl, this bit indicates the SPDIF Tx DMA interrupt status. This bit is read only and is automatically cleared when FLAG bit of ChCtrl is cleared. This bit is not available on ADMA1, and ADMA 2.

DTI	[4]	DMA Interrupt Status of DAI TX
0		No interrupt occurred while DAI Tx DMA transfer.
1		Interrupt occurred while DAI Tx DMA transfer.

Without regard to Interrupt enable bit(DTIEN) of ChCtrl, this bit indicates the DAI Tx DMA interrupt status. This bit is read only and is automatically cleared when FLAG bit of ChCtrl is cleared.

CRMI	[3]	DMA Masked Interrupt Status of CDIF RX or SPDIF RX
0		No interrupt occurred while CDIF Rx DMA transfer.
1		Interrupt occurred while CDIF Rx DMA transfer.

This bit indicates the CDIF Rx DMA interrupt status when CRIEN is asserted. This bit is not available on ADMA1, and ADMA 2.

This package does not have CDIF RX.

DRMI	[2]	DMA Masked Interrupt Status of DAI RX
0		No interrupt occurred while DAI Rx DMA transfer.
1		Interrupt occurred while DAI Rx DMA transfer.

This bit indicates the DAI Rx DMA interrupt status when DRIEN is asserted.

STMI	[1]	DMA Masked Interrupt Status of SPDIF TX
0		No interrupt occurred while SPDIF Tx DMA transfer.
1		Interrupt occurred while SPDIF Tx DMA transfer.

This bit indicates the SPDIF Tx DMA interrupt status when STIEN is asserted. This bit is not available on ADMA1, and ADMA 2.

DTMI	[0]	DMA Masked Interrupt Status of DAI TX
0		No interrupt occurred while DAI Tx DMA transfer.
1		Interrupt occurred while DAI Tx DMA transfer.

This bit indicates the DAI Tx DMA interrupt status when DTIEN is asserted.

25.2.2 DAI Registers

Digital Audio Mode Register (DAMR)

0x90000040

0x90000140

0x90000240

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BPS	LPS		R	AFE		R	NMDR	NMDT	RXE	RXS<1:0>		TXS<1:0>		R	SP
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EN	TE	RE	MD	SM	BM	FM	CC	BD<1:0>		FD<1:0>		BP	CM	MM	LB

BPS	[31]	BCLK Pad Select
0		BCLK source select BCLK Pad
1		BCLK direct at master mode

LPS	[30]	LRCK Pad Select
0		LRCK source select LRCK Pad
1		LRCK direct at master mode

AFE	[27]	Audio Filter Enable
0		Audio filters will be disabled.
1		Audio filters will be enabled. When CD BCLK, CD LRCK, DAI MCLK, DAI BCLK and DAI LRCK are exposed to noises, you may suppress them by enable this bit. Audio Filter clock has to be enabled before enabling this bit and be at least four times faster than CD BCLK or DAI MCLK.

NMDR	[24]	Rx Justified Mode
0		Rx Left-justified Mode
1		Rx Right-justified Mode MD = '1' & DSP = '0' & NMDR = '0' → Left-justified Rx Mode MD = '1' & DSP = '0' & NMDR = '1' → Right-justified Rx Mode

NMDT	[23]	Tx Justified Mode
0		Tx Left-justified Mode
1		Tx Right-justified Mode MD = '1' & DSP = '0' & NMDT = '0' → Left-justified Tx Mode MD = '1' & DSP = '0' & NMDT = '1' → Right-justified Tx Mode

RXE	[22]	DAI RX Data Sign Extension
0		Disable (zero extension)
1		Enable (sign bit extension)

RXS	[21:20]	DAI RX Shift
00		Bit-pack MSB and 24bit mode
01		Bit-pack MSB and 16bit mode
10		Bit-pack LSB and 24bit mode
11		Bit-pack LSB and 16bit mode

31 8 7 0

RX DATA	0
---------	---

RXS = 0

RXE = Don't care

31 16 15 0

RX DATA	0
---------	---

RXS = 1

RXE = Don't care

31 24 23 0

0	RX DATA
---	---------

RXS = 2 or 3 RXE = 0

31 24 23 0

S	RX DATA
---	---------

RXS = 2 or 3 RXE = 1

TXS	[19:18]	DAI TX Shift
0x		Bit-pack MSB mode

10	Bit-pack LSB and 24bit mode
11	Bit-pack LSB and 16bit mode

SP	[16]	DAI System Clock Polarity Inversion
0		Disable
1		Enable

EN	[15]	DAI Master Enable
0		Disable
1		Enable

TE	[14]	DAI Transmitter Enable
0		Disable
1		Enable

RE	[13]	DAI Receiver Enable
0		Disable
1		Enable

MD	[12]	DAI Bus Mode
0		IIS mode
1		MSB justified mode

SM	[11]	DAI System Clock Master Select
0		DAI system clock from external pin
1		DAI system clock is generated by the clock generator block

BM	[10]	DAI Bit Clock Master Select
0		DAI bit clock from external pin
1		DAI bit clock is generated by dividing DAI system clock

FM	[9]	DAI Frame Clock Master Select
0		DAI frame clock from external pin
1		DAI frame clock is generated by dividing DAI bit clock

CC	[8]	CDIF Clock Select
BLCK and LRCK from CDIF under master mode. This is associated with CDIF Monitor Mode		

BD	[7:6]	DAI Bit Clock Divider Select
00		Select Div 4 (256fs->64fs)
01		Select Div 6 (384fs->64fs)
10		Select Div 8 (512fs->64fs, 384fs->48fs , 256fs->32fs)
11		Select Div16 (512fs->32fs)

FD	[5:4]	DAI Frame Clock Divider Select
00		Select Div 32 (32fs->fs)
01		Select Div 48 (48fs->fs)
10		Select Div 64 (64fs->fs)

The combination of BD & FD determines the ratio between main system clock and the sampling frequency. The multiplication between the division factor of BD and FD must be equal to this ratio.

BP	[3]	DAI Bit Clock Polarity
0		Positive
1		Negative

CM	[2]	CDIF Monitor Mode
0		Disable
1		Enable

MM	[1]	DAI Monitor Mode
0		Disable
1		Enable (TE bit should be enabled)

LB	[0]	DAI Loop-Back Mode
0	Disable	
1	Enable	

Digital Audio Volume Control Register (DAVC)

0x90000044

0x90000144

0x90000244

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											VC<4:0>				

The volume of audio output can be manipulated by this register. It has –6dB unit so the output volume can be set from 0 dB to –96 dB as the following table.

VC[4:0]	DAI Volume control
00000	0dB
00001	-6dB
00010	-12dB
00011	-18dB
00100	-24dB
00101	-30dB
00110	-36dB
00111	-42dB
01000	-48dB
01001	-54dB
01010	-60dB
01011	-66dB
01100	-72dB
01101	-78dB
01110	-84dB
01111	-90dB
10000	-96dB

ASRC MUX Register (ASRCMUX)

0x90000048

0x90000148

0x90000248

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
														PATH	LRP

PATH	[1]	PATH Select
0	DAI TX DATA path	
1	ASRC DATA path	

LRP	[0]	LRCK polarity for ASRC path
0	Change	
1	Not change	

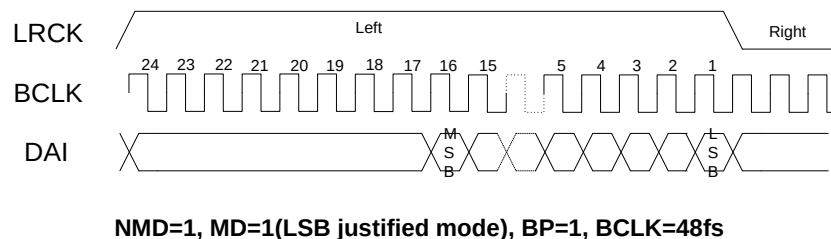
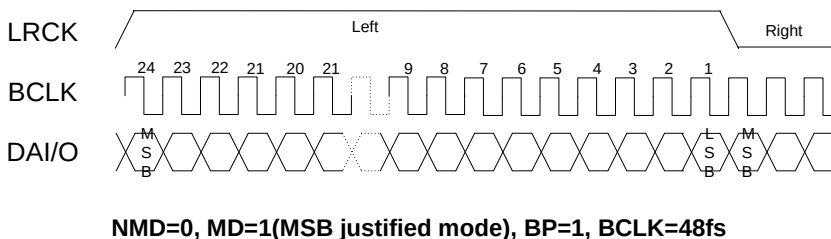
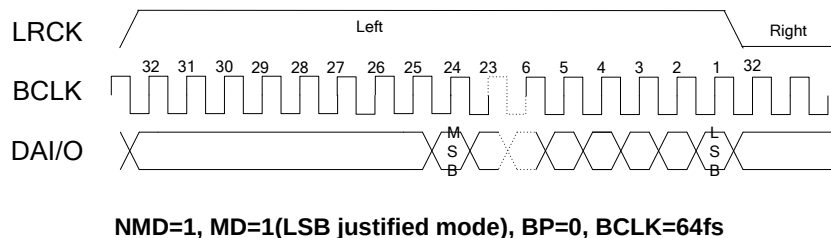
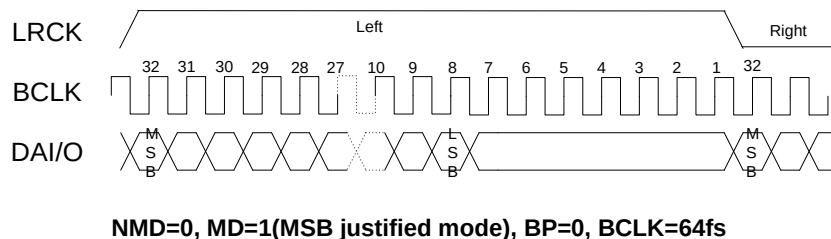
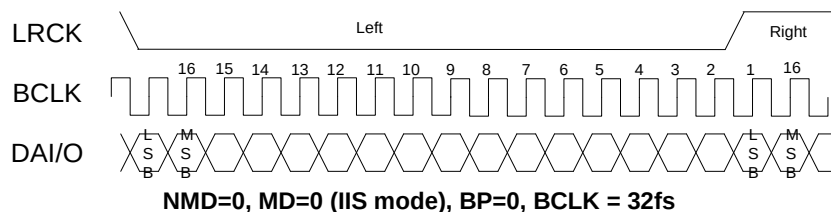


Figure 25.6 DAI Bus Timing Diagram

25.2.3 SPDIF Registers

SPDIF Transmitter Version Register

0x90001000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CSB	UDB	AW						DW	VER			

CSB[12]	Channel Status Buffer Available Bit
0	Channel status buffer is not available
1	Channel status buffer is available

UDB[11]	User Data Buffer Available Bit
0	User data buffer is not available
1	User data buffer is available

AW[10:5]	Value of Address Width
n	Value of Address Width

DW[4]	Value of Data Width
n	Value of Data Width

VER[3:0]	Design Version
n	Version Number

SPDIF Transmit Configuration Register

0x90001004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								MODE				Reserved			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RATIO								UDATEN		CHSTEN		0	IEN	TXD	TXEN

MODE[23:20]	Sample Format Mode Indicator
n	16 + n Bits Transmit Sample Format
9 ~ 15	Reserved

RATIO[15:8]	Clock Divider Ratio
n	Clock divider for the transmit frequency. The SPDIF clock is divided by a factor of (RATIO + 1) to generate the serial transmit clock.

UDATEN[7:6]	User Data Enable Bits
0	User data A & B set to 0.
1	User data A & B generated from UserData bit 7-0
2	User data A generated from UserData bit 7-0, B generated from UserData bit 15-8
3	Reserved

CHSTEN[5:4]	Channel Status Enable Bits
0	Channel status A & B generated from TxChStat
1	Channel status A & B generated from ChStat bit 7-0
2	Channel status A generated from ChStat bit 7-0, B generated from ChStat bit 15-8
3	Reserved

IEN[2]	Interrupt Output Enable Bit
n	'1' for enabling the interrupt output.

TXD[1]	Data Valid Bit
n	'1' for data being valid.

TXEN[0]	Transmitter Enable Bit
n	'1' for enabling the transmission.

SPDIF Transmit Channel Status Register

0x90001008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								FREQ		Reserved		GSTS	PRE	CPY	AU

FREQ[7:6]	Sampling Frequency
0	44.1kHz
1	48kHz
2	32kHz
3	Sample Rate Converter

GSTS[3]	Status Generation
0	No indication
1	Original/Commercially Pre-recorded data

PRE[2]	Pre-emphasis
0	None
1	50/15us

CPY[1]	Copyright
0	Copy inhibited
1	Copy permitted

AU[0]	Data Format
0	Audio Format
1	Data Format

SPDIF Transmit Interrupt Mask Register

0x9000100C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											HCSB	LCSB	HSB	LSB	R

HCSB[4]	Higher Channel Status/User Data Buffer Empty
n	'1' for enable for higher channel status/user data buffer empty interrupt

LCSB[3]	Lower Channel Status/User Data Buffer Empty
n	'1' for enable for lower channel status/user data buffer empty interrupt

HSB[2]	Higher Data Buffer Empty
n	'1' for enable for higher data buffer empty interrupt

LSB[1]	Lower Data Buffer Empty
n	'1' for enable for lower data buffer empty interrupt

SPDIF Transmit Interrupt Status Register

0x90001010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											HCSB	LCSB	HSB	LSB	R*

HCSB[4]	Higher Channel Status/User Data Buffer Empty
n	'1' for higher channel status/user data buffer empty interrupt activated

LCSB[3]	Lower Channel Status/User Data Buffer Empty
n	'1' for lower channel status/user data buffer empty interrupt activated

HSB[2]	Higher Data Buffer Empty
n	'1' for higher data buffer empty interrupt activated

LSB[1]	Lower Data Buffer Empty
n	'1' for lower data buffer empty interrupt activated

SPDIF Transmit User Data Buffer Register

0x90001080 ~ 0x900010DC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHBUD								CHAUD							

The user data is 24 bytes per sample block. Bit 0 of byte 0 is transmitted first. Bit 7 of byte 23 is transmitted last.

CHBUD[15:8]	User Data for Channel B
n	8 Bits User Data for Channel B

CHAUD[7:0]	User Data for Channel A
n	8 Bits User Data for Channel A

SPDIF Transmit Channel Status Buffer Register

0x90001100 ~ 0x9000115C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHBCS								CHACS							

The channel status data is 24 bytes per sample block. Bit 0 of byte 0 is transmitted first. Bit 7 of byte 23 is transmitted last.

CHBCS[15:8]	Channel Status for Channel B
n	8 Bits Channel Status for Channel B

CHACS[7:0]	Channel Status for Channel A
n	8 Bits Channel Status for Channel A

SPDIF Transmit Sample Data Buffer Register

0x90001200 ~ 0x900013FC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												DATH			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATL															

The format of data words in transmit sample buffer. Channel A is transmitted first, and must be stored on even addresses, while channel B is stored on odd addresses in the sample buffer.

DATH[19:16]	Upper 8 Bits for Sample Buffer Data
n	Audio data if > 16 bits resolution. Unused bits are 0

DATL[15:0]	Lower 16 Bits for Sample Buffer Data
n	Audio data (16 bits mode). Bits 0 is LSB

DMA Configuration (DMACFG)

0x90001400

Bit	Name	R/W	Reset	Description
31:21		R	0	Reserved
20:16	VCNT	R	X	FIFO Valid Entry Count
14	SS	R/W	0x0	Swap Sample 0 : Disable 1 : Enable
13:12	RALR	R/W	0x0	Read Address LR Mode 0 : Read Address 24bit Mode 1 : Read Address 24bit LR Mode 2 : Read Address 16bit Mode 3 : Read Address 16bit LR Mode
11	DRQEN1	R/W	0	DMA Request Enable for user data buffer
10	DRQEN0	R/W	0	DMA Request Enable for sample data buffer
9	AMODE1	R/W	0	Sample data buffer address mode Buffer is written with sequence below. 0, 2, 4, 6, 1, 3, 5, 7
8	AMODE0	R/W	0	0: Ignore address (FIFO mode) 1: Enable address (16 entries can be written with address).
7	FIFOCLR	R/W	0	Clear FIFO. Must be written as "0" for normal operation.
6:4		R	0	Reserved
3:0	FIFOTH	R/W	X	FIFO Threshold for DMA Request. DMA Request is asserted if (VCNT <= FIFOTH)

SPDIF Receiver Version Register (RxVersion)

0x90001800

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R												CSC			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R				AW								DW	VER		

The version register allows the software to read out all the parameter that was used to generate the receiver.

Field	Name	RW	Reset	Description
19:16	CSC	R	0x08	Channel Status Buffer Available Bit n : Value of Channel Status Capture
11:5	AW	R	0x08	Value of Address Width
4	DW	R	0x1	Value of Data Width 0 : Data Width is 16bit 1 : Data Width is 24bit
3:0	VER	R	0x1	Version Number

SPDIF Receiver Configuration Register (RxConfig)

0x90001804

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R*							BLKEN	MODE				PAREN	STATEN	USEREN	VALEN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R*											VALID	CHAS	RINTEN	SAMPLE	RXEN

Field	Name	RW	Reset	Description	
24	BLKEN	R/W	0x0	Block Boundary Marking 0: Do not use block boundary marking 1: Mark the first sample in each block with '1' in bit 27	
23:20	MODE	R/W	0x0	Store Samples as Bit	
				0	Store samples as 16bit
				1	Store samples as 17bit
				2~6	Store samples as 18, 19, 20, 21, 22bits
				7	Store samples as 23bit
				8	Store samples as 24bit
9~15	Reserved				
19	PAREN	R/W	0x0	Store Parity Bit 0: Do not use block boundary marking 1: Store parity bit 31 in sample buffer	
18	STATEN	R/W	0x0	Store Channel Status Bit 0: Do not store channel status bit 1: Store channel status bit in bit 30 in sample buffer	
17	USEREN	R/W	0x0	Store User Data Bit 0: Do not store user data bit 1: Store user data bit in bit 29 in sample buffer	
16	VALEN	R/W	0x0	Store Validity bit 0: Do not store validity bit 1: Store validity bit in bit 28 in sample buffer	
4	VALID	R/W	0x0	Sample Data Store 0: Sample data stored in buffers regardless of sub_frame Validity bit 1: Sample data stored only when sub_frame validity bit is 0	
3	CHAS	R/W	0x0	RxStatus Register Holds Channel 0: RxStatus register holds status from channel B 1: RxStatus register holds status from channel A	
2	RINTEN	R/W	0x0	Interrupt Output 0: Interrupt output is disabled 1: Interrupt output is enabled	
1	SAMPLE	R/W	0x0	Stored Data 0: No data is stored in the sample buffer 1: Data is stored in the sample buffer	
0	RXEN	R/W	0x0	Receiver Disable Enable 0: Receiver is disabled 1: Receiver is enabled	

SPDIF Receiver Status Register (RxStatus)

0x90001808

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R									COPY	EMPH			AUDIO	PRO	LOCK

The signal status register holds information about the received signal. When the receiver is disabled, all bits read 0. When lock bit is 0, all other bits are set to 0.

Field	Name	RW	Reset	Description
6	COPY	R	0x0	Copy Information 0: Copy inhibited 1: Copy permitted (copy bit only applicable in consumer mode)
5:3	EMPH*	R	0x0	Emphasis Code Emphasis code from the channel status block. Note that the interpretation of these bits is different in consumer and professional mode.
2	AUDIO	R	0x0	Signal Data 0: Signal is non-audio 1: Signal is audio
1	PRO	R	0x0	Signal format 0: Signal format is consumer 1: Signal format is professional
0	LOCK	R	0x0	Lock to SPDIF Signal 0: Receiver has no valid input signal 1: Receiver is locked to SPDIF signal

SPDIF Receiver Interrupt Mask Register (RxIntMask)

0x9000180C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R					DBS		CRD	CAP7	CAP6	CAP5	CAP4	CAP3	CAP2	CAP1	CAP0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R											PEB	PEA	HSBF	LSBF	LOCK

Field	Name	RW	Reset	Description
26:25	DBS	R/W	0x0	DMA Burst Select 0 : DMA 1 Burst 1 : DMA 2 Burst 2 : DMA 4 Burst 3 : DMA 8 Burst
24	CRD	R/W	0x0	Capture Register DMA Interrupt Mask 0 : Disable 1 : Enable
23	CAP7	R/W	0x0	Capture Register 7 Interrupt Mask 0 : Disable 1 : Enable
22	CAP6	R/W	0x0	Capture Register 6 Interrupt Mask 0 : Disable 1 : Enable
21	CAP5	R/W	0x0	Capture Register 5 Interrupt Mask 0 : Disable 1 : Enable
20	CAP4	R/W	0x0	Capture Register 4 Interrupt Mask 0 : Disable 1 : Enable
19	CAP3	R/W	0x0	Capture Register 3 Interrupt Mask 0 : Disable 1 : Enable
18	CAP2	R/W	0x0	Capture Register 2 Interrupt Mask 0 : Disable 1 : Enable
17	CAP1	R/W	0x0	Capture Register 1 Interrupt Mask 0 : Disable 1 : Enable
16	CAP0	R/W	0x0	Capture Register 0 Interrupt Mask 0 : Disable 1 : Enable

4	PEB	R/W	0x0	Parity Error Channel B Interrupt Mask 0 : Disable 1 : Enable
3	PEA	R/W	0x0	Parity Error Channel A Interrupt Mask 0 : Disable 1 : Enable
2	HSBF	R/W	0x0	Higher Sample Buffer Full Interrupt Mask 0 : Disable 1 : Enable
1	LSBF	R/W	0x0	Lower Sample Buffer Full Interrupt Mask 0 : Disable 1 : Enable
0	LOCK	R/W	0x0	Change in Lock Bit Interrupt Mask 0 : Disable 1 : Enable

SPDIF Receiver Interrupt Status Register (RxIntStat)

0x90001810

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R							CRD	CAP7	CAP6	CAP5	CAP4	CAP3	CAP2	CAP1	CAP0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R											PEB	PEA	HSBF	LSBF	LOCK

Field	Name	RW	Reset	Description
24	CRD	R/W	0x0	Capture Register DMA Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
23	CAP7	R/W	0x0	Capture Register 7 Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
22	CAP6	R/W	0x0	Capture Register 6 Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
21	CAP5	R/W	0x0	Capture Register 5 Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
20	CAP4	R/W	0x0	Capture Register 4 Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
19	CAP3	R/W	0x0	Capture Register 3 Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
18	CAP2	R/W	0x0	Capture Register 2 Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
17	CAP1	R/W	0x0	Capture Register 1 Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
16	CAP0	R/W	0x0	Capture Register 0 Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
4	PEB	R/W	0x0	Parity Error Channel B Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
3	PEA	R/W	0x0	Parity Error Channel A Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
2	HSBF	R/W	0x0	Higher Sample Buffer Full Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
1	LSBF	R/W	0x0	Lower Sample Buffer Full Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs
0	LOCK	R/W	0x0	Change in Lock Bit Interrupt 0 : Disable 1 : Enable A bit in this register is set to '1' when an event occurs

If the corresponding bit in RxIntMask is set to 1, an interrupt is generated (if enabled). Write 1 to a bit to clear the event. The interrupt signal goes inactive when all events are cleared.

SPDIF Receiver Channel Status Capture [n] Register (RxCapCtl [n]) 0x90001840 ~ 0x9000187C(even)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BITPOS								CDATA	CHID	BITLEN					

Field	Name	RW	Reset	Description
15:8	BITPOS	R/W	0x00	First Bit Position of Bit-Filed to be Captured n : Valid Range 0 to 191
7	CDATA	R/W	0x0	DATA Captured 0 : User data is captured 1 : Channel status data is captured
6	CHID	R/W	0x0	Source Channel 0 : Source is channel A 1 : Source is channel B
5:0	BITLEN	R/W	0x00	Bit Length 0 : Capture function disable 1~32 : Length of bit-field to be captured

The channel status capture register can be set up to capture a specified bit-field of the channel status frame or user data frame, and also generate an interrupt when the bit-field changes. The bit-field can be from 1 to 32 bits long. There can be up to eight sets of bit-field capture registers

SPDIF Receiver Channel Status Data [n] Register (RxCap [n]) 0x90001840~0x9000187C(odd)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA															

Field	Name	RW	Reset	Description
31:0	DATA	R	0x00000000	Captured Channel Status n : First bit captured is stored in bit 0 Captured channel status/user data bits

SPDIF Receiver Sample Data Buffer (RxBuffer) 0x90001A00 ~ 0x90001A1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PARITY	CHSTAT	USRDAT	VALID	BLKS	R				DATAH						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATAL															

Format of data words in receive sample buffer

Field	Name	RW	Reset	Description
31	PARITY	R	0x0	Parity Bit If enable, otherwise 0
30	CHSTAT	R	0x0	Channel Status Bit If enable, otherwise 0
29	USRDAT	R	0x0	User Data Bit If enable, otherwise 0
28	VALID	R	0x0	Validity Bit If enable, otherwise 0
27	BLKS	R	0x0	Start of Sample Block Bit If enable, otherwise 0
23:16	DATAH	R	0x00	DATA High n : Audio Data (if > 16bit Resolution) - Unused bit are 0
15:0	DATAL	R	0x0000	DATA Low n : Audio Data - Bit 0 is LSB

25.3 Audio Data Format

The following figures show the data formats supported between memory and buffer.

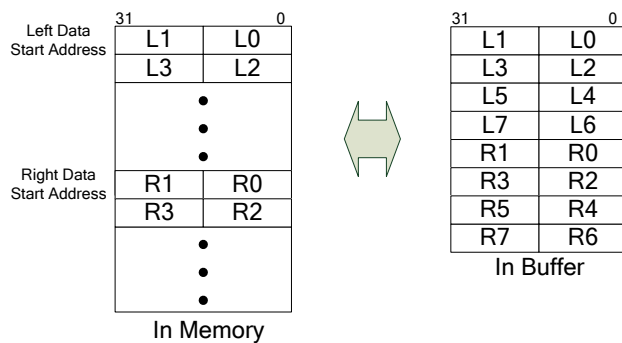


Figure 25.7 Data Format - LRMode = 1, 16bits

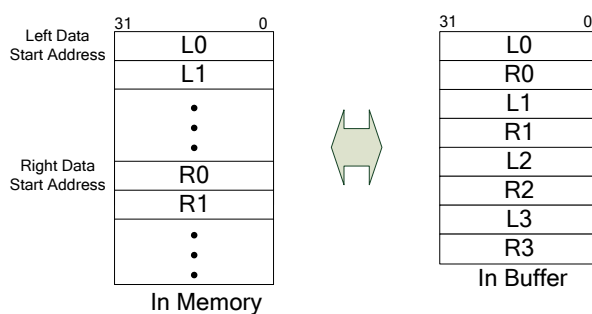


Figure 25.8 Data Format - LRMode = 1, 24bits

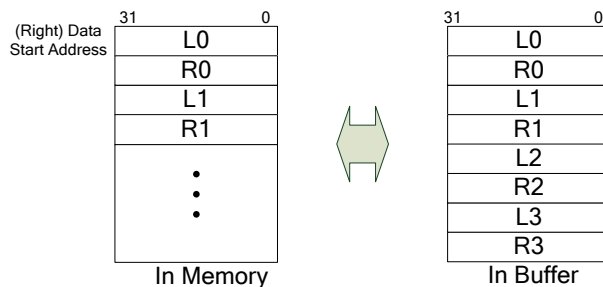


Figure 25.9 Data Format - LRMode = 0, 24bits

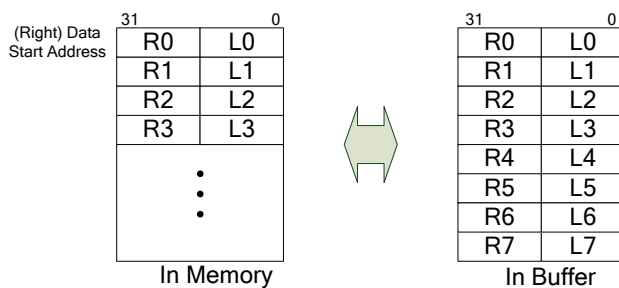


Figure 25.10 Data Format - LRMode = 0, 16bits

25.4 Port Multiplexing

Before the signals are sent to the GPIO block for pin multiplexing, they are internally rearranged to have more flexibility in port configuration. The signal name GBUS[191:0] found in GPIO Function Selection Registers (GPIO*_FS* of GPIO block) is the final output of the port multiplexers in this module.

Any DAI signal can be mapped to any bit position of GBUS[191:0] by writing an index number to the registers below in TCC8132. The pre-mapped bit position of GBUS[191:0] is shown in the table below.

Table 25.6 Pre-mapped bit position for DAI Port

GBUS	GPIO
GBUS[105:107]	PC[25:27]
GBUS[56:60]	PD[0:4]
GBUS[61:65]	PD[5:9]
GBUS[67:70]	PD[11:14]

Table 25.7 DAI Port Selection Register Map (Base Address = 0x98002800)

Name	Address	Type	Reset	Description
DAI0PORTSEL	0x00	R/W	0x3B3A3938	DAI0 Port Selection Register
DAI1PORTSEL	0x04	R/W	0x3F3E3D3C	DAI1 Port Selection Register
DAI2PORTSEL	0x08	R/W	0x43424140	DAI2 Port Selection Register
DAIMPORTSEL	0x10	R/W	0x4B4A4948	DAI MCLK Port Selection Register
DAICPORTSEL	0x14	R/W	0x4F4E4D4C	DAI CDIF Port Selection Register

DAI0 Port Selection Register (DAI0PORTSEL)

0x98002800

Bit	Name	R/W	Reset	Description
31:24	DO	R/W	0x3B	Port number for DO
23:16	LRCK	R/W	0x3A	Port number for LRCK
15:8	BCLK	R/W	0x39	Port number for BCLK
7:0	DI	R/W	0x38	Port number for DI

DAI1 Port Selection Register (DAI1PORTSEL)

0x98002804

Bit	Name	R/W	Reset	Description
31:24	DO	R/W	0x3F	Port number for DO
23:16	LRCK	R/W	0x3E	Port number for LRCK
15:8	BCLK	R/W	0x3D	Port number for BCLK
7:0	DI	R/W	0x3C	Port number for DI

DAI2 Port Selection Register (DAI2PORTSEL)

0x98002808

Bit	Name	R/W	Reset	Description
31:24	DO	R/W	0x43	Port number for DO
23:16	LRCK	R/W	0x42	Port number for LRCK
15:8	BCLK	R/W	0x41	Port number for BCLK
7:0	DI	R/W	0x40	Port number for DI

DAI MCLK Port Selection Register (DAIMPORTSEL)

0x98002810

Bit	Name	R/W	Reset	Description
31:24	DAI3MCLK	R/W	0x4B	Port number for DAI3MCLK
23:16	DAI2MCLK	R/W	0x4A	Port number for DAI2MCLK
15:8	DAI1MCLK	R/W	0x49	Port number for DAI1MCLK
7:0	DAI0MCLK	R/W	0x48	Port number for DAI0MCLK

DAI SPDIF_TX Port Selection Register (DAISTPORTSEL)

0x98002814

Bit	Name	R/W	Reset	Description
31:24	SPDIFTX	R/W	0x4F	Port number for SPDIFTX

DAI SPDIF_RX Port Selection Register (DAISRPORTSEL)

0x98002818

Bit	Name	R/W	Reset	Description
7:0	SPDIFRX	R/W	0x50	Port number for SPDIFRX

25.5 Audio Interrupt Request

The TCC8132 audio interface allows only one interrupt request to come out. Therefore, each interrupt source of audio interface channels has to be selected by configuring following registers.

Audio Interrupt Status Register (AUDIRQ)

0x98002820

Bit	Name	R/W	Reset	Description
14	SPDIF_RX	R/W	-	SPDIF Rx Interrupt Request of Channel0
13	SPDIF_TX	R/W	-	SPDIF Tx Interrupt Request of Channel0
12	CDIFRX	R/W	-	Reserved
11	DAITX3	R/W	-	Reserved
10	DAITX2	R/W	-	DAI Tx Interrupt Request of Channel2
9	DAITX1	R/W	-	DAI Tx Interrupt Request of Channel1
8	DAITX0	R/W	-	DAI Tx Interrupt Request of Channel0
7	DAIRX3	R/W	-	Reserved
6	DAIRX2	R/W	-	DAI Rx Interrupt Request of Channel2
5	DAIRX1	R/W	-	DAI Rx Interrupt Request of Channel1
4	DAIRX0	R/W	-	DAI Rx Interrupt Request of Channel0
3	ADMA3	R/W	-	Reserved
2	ADMA2	R/W	-	ADMA Interrupt Request of Channel2
1	ADMA1	R/W	-	ADMA Interrupt Request of Channel1
0	ADMA0	R/W	-	ADMA Interrupt Request of Channel0

Each bit of this register represents the status of interrupt requests regardless of the configuration bits of AUDIEN register. But configuration of AUDSYNC and AUDTMODE registers will work. Writing '1' will clear corresponding bits.

Audio Interrupt Register (AUDMIQ)

0x98002824

Bit	Name	R/W	Reset	Description
14	SPDIF_RX	R/W	-	SPDIF Rx Interrupt Request of Channel0
13	SPDIF_TX	R/W	-	SPDIF Tx Interrupt Request of Channel0
12	CDIFRX	R/W	-	Reserved
11	DAITX3	R/W	-	Reserved
10	DAITX2	R/W	-	DAI Tx Interrupt Request of Channel2
9	DAITX1	R/W	-	DAI Tx Interrupt Request of Channel1
8	DAITX0	R/W	-	DAI Tx Interrupt Request of Channel0
7	DAIRX3	R/W	-	Reserved
6	DAIRX2	R/W	-	DAI Rx Interrupt Request of Channel2
5	DAIRX1	R/W	-	DAI Rx Interrupt Request of Channel1
4	DAIRX0	R/W	-	DAI Rx Interrupt Request of Channel0
3	ADMA3	R/W	-	Reserved
2	ADMA2	R/W	-	ADMA Interrupt Request of Channel2
1	ADMA1	R/W	-	ADMA Interrupt Request of Channel1
0	ADMA0	R/W	-	ADMA Interrupt Request of Channel0

Each bit of this register represents the status of interrupt requests depending on the configuration of AUDIEN register. Writing '1' will clear corresponding bits.

Audio Interrupt Register (AUDIEN)

0x98002828

Bit	Name	R/W	Reset	Description
14	SPDIF_RX	R/W	1	SPDIF Rx Interrupt Request of Channel0
13	SPDIF_TX	R/W	1	SPDIF Tx Interrupt Request of Channel0
12	CDIFRX	R/W	1	Reserved
11	DAITX3	R/W	1	Reserved
10	DAITX2	R/W	1	DAI Tx Interrupt Request of Channel2
9	DAITX1	R/W	1	DAI Tx Interrupt Request of Channel1
8	DAITX0	R/W	1	DAI Tx Interrupt Request of Channel0
7	DAIRX3	R/W	1	Reserved
6	DAIRX2	R/W	1	DAI Rx Interrupt Request of Channel2
5	DAIRX1	R/W	1	DAI Rx Interrupt Request of Channel1
4	DAIRX0	R/W	1	DAI Rx Interrupt Request of Channel0
3	ADMA3	R/W	1	Reserved
2	ADMA2	R/W	1	ADMA Interrupt Request of Channel2
1	ADMA1	R/W	1	ADMA Interrupt Request of Channel1
0	ADMA0	R/W	1	ADMA Interrupt Request of Channel0

Each bit of this register enables or disables corresponding interrupt requests to come out as audio interface request.

Audio Interrupt Trigger Mode Register (AUDTMODE)**0x9800282C**

Bit	Name	R/W	Reset	Description
14	SPDIF_RX	R/W	0	SPDIF Rx Interrupt Request of Channel0
13	SPDIF_TX	R/W	0	SPDIF Tx Interrupt Request of Channel0
12	CDIFRX	R/W	0	Reserved
11	DAITX3	R/W	0	Reserved
10	DAITX2	R/W	0	DAI Tx Interrupt Request of Channel2
9	DAITX1	R/W	0	DAI Tx Interrupt Request of Channel1
8	DAITX0	R/W	0	DAI Tx Interrupt Request of Channel0
7	DAIRX3	R/W	0	Reserved
6	DAIRX2	R/W	0	DAI Rx Interrupt Request of Channel2
5	DAIRX1	R/W	0	DAI Rx Interrupt Request of Channel1
4	DAIRX0	R/W	0	DAI Rx Interrupt Request of Channel0
3	ADMA3	R/W	1	Reserved
2	ADMA2	R/W	1	ADMA Interrupt Request of Channel2
1	ADMA1	R/W	1	ADMA Interrupt Request of Channel1
0	ADMA0	R/W	1	ADMA Interrupt Request of Channel0

Each bit of this register represents how to trigger interrupt requests. If a bit is set, level-triggering is used, whereas edge-triggering is used if cleared.

Audio Interrupt Trigger Mode Register (AUDSYNC)**0x98002830**

Bit	Name	R/W	Reset	Description
14	SPDIF_RX	R/W	0	SPDIF Rx Interrupt Request of Channel0
13	SPDIF_TX	R/W	0	SPDIF Tx Interrupt Request of Channel0
12	CDIFRX	R/W	0	Reserved
11	DAITX3	R/W	0	Reserved
10	DAITX2	R/W	0	DAI Tx Interrupt Request of Channel2
9	DAITX1	R/W	0	DAI Tx Interrupt Request of Channel1
8	DAITX0	R/W	0	DAI Tx Interrupt Request of Channel0
7	DAIRX3	R/W	0	Reserved
6	DAIRX2	R/W	0	DAI Rx Interrupt Request of Channel2
5	DAIRX1	R/W	0	DAI Rx Interrupt Request of Channel1
4	DAIRX0	R/W	0	DAI Rx Interrupt Request of Channel0
3	ADMA3	R/W	1	Reserved
2	ADMA2	R/W	1	ADMA Interrupt Request of Channel2
1	ADMA1	R/W	1	ADMA Interrupt Request of Channel1
0	ADMA0	R/W	1	ADMA Interrupt Request of Channel0

Each bit of this register represents whether corresponding bit is synchronized with HCLK or not. If a bit is set, it is synchronized with HCLK, otherwise it is not synchronized.

Audio Interrupt Raw Status Register (AUDSTAT)**0x98002834**

Bit	Name	R/W	Reset	Description
14	SPDIF_RX	R/W	0	SPDIF Rx Interrupt Request of Channel0
13	SPDIF_TX	R/W	-	SPDIF Tx Interrupt Request of Channel0
12	CDIFRX	R/W	-	Reserved
11	DAITX3	R/W	-	Reserved
10	DAITX2	R/W	-	DAI Tx Interrupt Request of Channel2
9	DAITX1	R/W	-	DAI Tx Interrupt Request of Channel1
8	DAITX0	R/W	-	DAI Tx Interrupt Request of Channel0
7	DAIRX3	R/W	-	Reserved
6	DAIRX2	R/W	-	DAI Rx Interrupt Request of Channel2
5	DAIRX1	R/W	-	DAI Rx Interrupt Request of Channel1
4	DAIRX0	R/W	-	DAI Rx Interrupt Request of Channel0
3	ADMA3	R/W	-	Reserved
2	ADMA2	R/W	-	ADMA Interrupt Request of Channel2
1	ADMA1	R/W	-	ADMA Interrupt Request of Channel1
0	ADMA0	R/W	-	ADMA Interrupt Request of Channel0

Each bit of this register represents the status of interrupt requests regardless of configuration of AUDIEN, AUDTMODE and AUDSYNC registers.

25.6 DAI (I²S) Timing

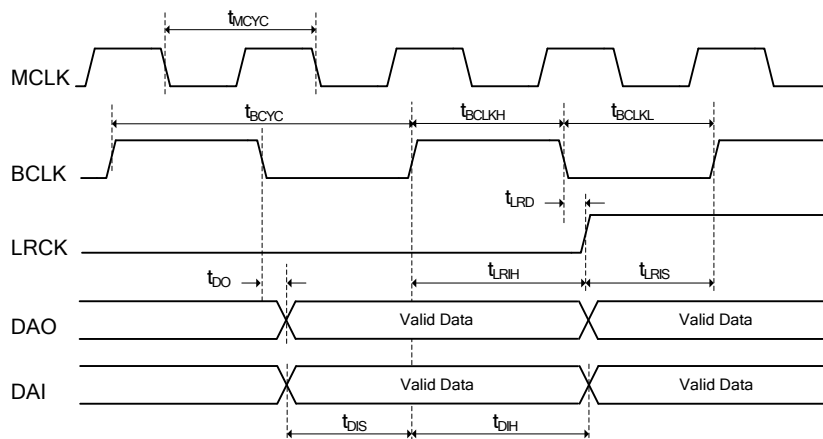


Figure 25.11 DAI / CDIF (I²S) Timing

Table 25.8 DAI Timing Parameters

Parameter	Symbol	MIN	MAX	Unit
MCLK cycle time	t_{MCLK}	20	-	ns
MCLK rise time	t_{MCLKR}	-	5	ns
MCLK fall time	t_{MCLKF}	-	5	ns
BCLK / CBCLK cycle time	t_{BCLK}	$4 * t_{MCLK}$	-	ns
BCLK / CBCLK pulse width high	t_{BCLKH}	t_{MCLK}	-	ns
BCLK / CBCLK pulse width low	t_{BCLKL}	t_{MCLK}	-	ns
BCLK / CBCLK rise time	t_{BCLKR}	-	10	ns
BCLK / CBCLK fall time	t_{BCLKF}	-	10	ns
BCLK to LRCK output	t_{LRD}	1	14	ns
DAO output delay from BCLK falling edge	t_{DO}	1	15	ns
DAI / CDAI setup time to BCLK / CBCLK rising edge	t_{DIS}	10	-	ns
DAI / CDAI hold time from BCLK / CBCLK rising edge	t_{DIH}	10	-	ns
LRCK / CLCK input setup time to BCLK / CBCLK rising edge	t_{LRIS}	10	-	ns
LRCK / CLCK input hold time from BCLK rising edge	t_{LRIH}	10	-	ns

Note: $C_L = 30pF$, $V_{DDI} = 1.2V$, $V_{DDIO} = 3.3V$

26 ASRC (Asynchronous Sample Rate converter)

26.1 Overview

Asynchronous Sample Rate Converter (ASRC) converts the sampling rate of a signal associated with an input clock into a signal associated with different output clock.

The ASRC supports up to two sampling rate pairs, volume control, and mixer function.

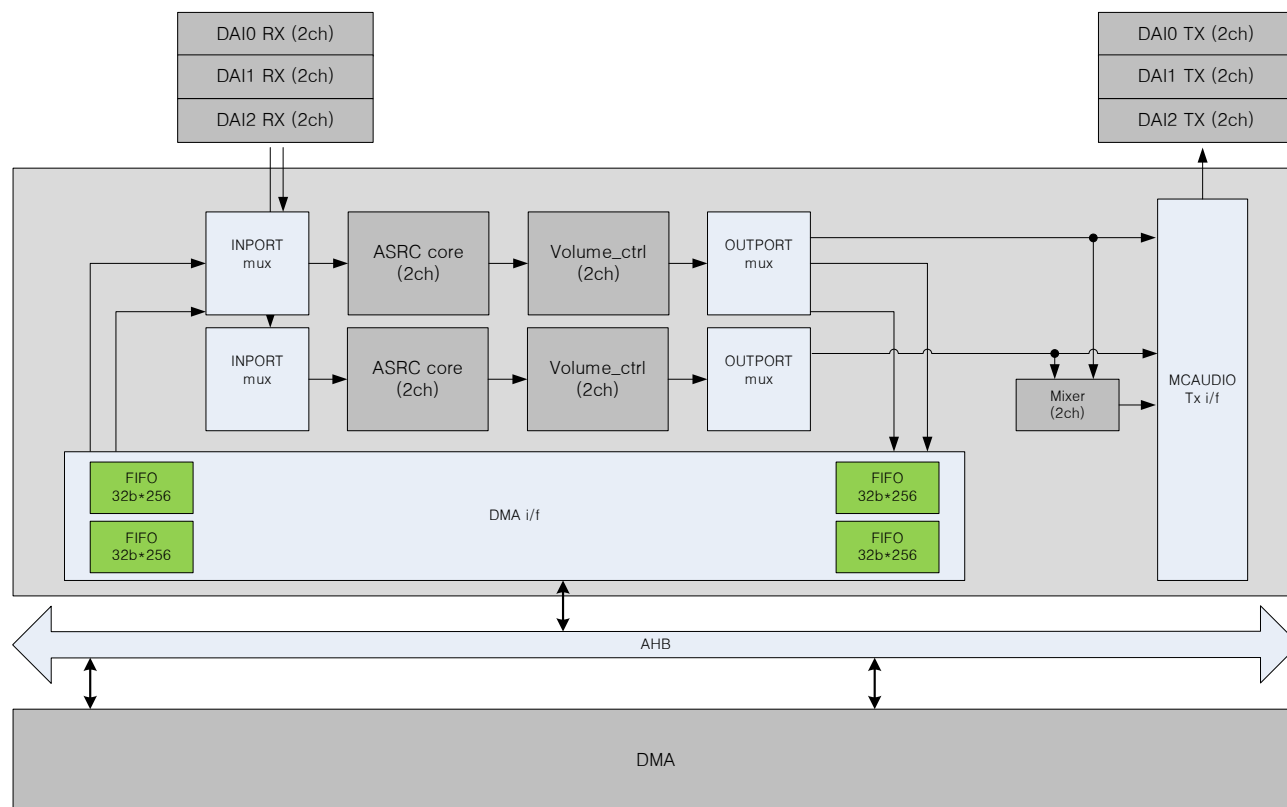


Figure 26.1 ASRC Block Diagram

26.2 Feature

[ASRC]

- SRC (Synchronous sample rate conversion) or ASRC (Asynchronous sample rate conversion) is supported.
- 2 pair supports 2channels.
- Up-Conversion, Down-Conversion, and 1:1 conversion are supported.
- Conversion ratio ranging from 1:7.5 (down-conversion) to 8:1 (up-conversion)
- Supported bit size are 16bit and 24bit.
- DMA transfer with memory and direct transfer with the DAI module are possible

[VOLUME CONTROL]

- Volume gain and volume ramp are supported.
- Volume gain : x0 ~ x8 (mute, -120dB to 18dB)
- Volume ramp can be used to perform soft mute, fade-in, fade-out (ramp-time can be changed and set within the sampling range of $2^0 \sim 2^{23}$)

[MIXER]

- Input source : Data of one to four system can be mixed
- The ratio to add the source is controlled by VOLUME_CONTROL's gain.
- Only direct transfer with the DAI module is possible

26.3 Function Description

26.3.1 Sampling Rate convertor

SRC (Synchronous sample rate conversion) or ASRC (Asynchronous sample rate conversion) is supported. In SRC mode, the rate is fixed and in ASRC mode, the rate is automatically calculated by external clock source.

26.3.2 Volume control

Digital Volume and Volume ramp (fade-in/out) are provided as functions to adjust the volume. The digital volume is set as a 24bit fixed-point value within the range from x0 to x8 (mute, -120dB to 18dB). The Volume ramp can be used to perform fade-in, fade-out and volume change as desired. The ramp time of volume ramp can be changed and set within the sampling range of 2^0 to 2^{23} .

26.3.3 Mixer

The mixer block is used to add the input audio data from VOLUME_CONTROL outputs. The mixer block can process data for 2 channels.

The mixer block does not have Gain control function. Therefore, the ratio to add the source can be set by VOLUME_CONTROL's gain.

Only direct transfer with the DAI is possible.

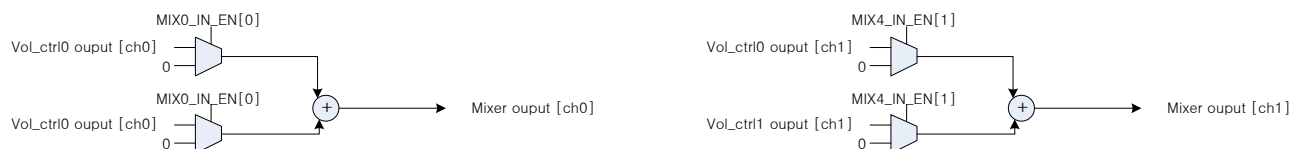


Figure 26.2 Mixer Block Diagram

26.3.4 MCAUDIO i/f

DAI0/1/2 path is available on 2 channels only (stereo).

26.3.5 ASRC_DMA

- Eight DMA channels. Each channel can support an unidirectional transfer.
- 16 DMA requests. The DMAC provides 16 peripheral DMA request lines.
- Single DMA and burst DMA request signals. Each peripheral connected to the DMAC can assert either a burst DMA request or a single DMA request. The DMA burst size is set by programming the DMAC.
- Memory-to-memory, memory-to-peripheral, peripheral-to-memory, and peripheral-to-peripheral transfers.
- Scatter or gather DMA support through the use of linked lists. This means that the source and destination areas do not have to occupy contiguous areas of memory.
- Hardware DMA channel priority. Each DMA channel has a specific hardware priority. DMA channel 0 has the highest priority and channel 1 has the lowest priority. If requests from two channels become active at the same time, the channel with the highest priority is serviced first.
- Incrementing or non-incrementing addressing for source and destination.
- Programmable DMA burst size. The DMA burst size can be programmed to transfer data more efficiently. Usually, the burst size is set to half the size of the FIFO in the peripheral.
- Internal four-word FIFO per channel.
- Support for 8, 16, and 32-bit wide transactions.
- Big-endian and little-endian are supported. The DMAC defaults to little-endian mode on reset.

26.3.6 Data transfer path

- Memory → ASRC → VOLUME CONTROL → Memory
- Memory → ASRC → VOLUME CONTROL → (MIXER) → DAI in
- DAO out → ASRC → VOLUME CONTROL → Memory
- DAO out → ASRC → VOLUME CONTROL → (MIXER) → DAI in

26.3.7 Word alignment

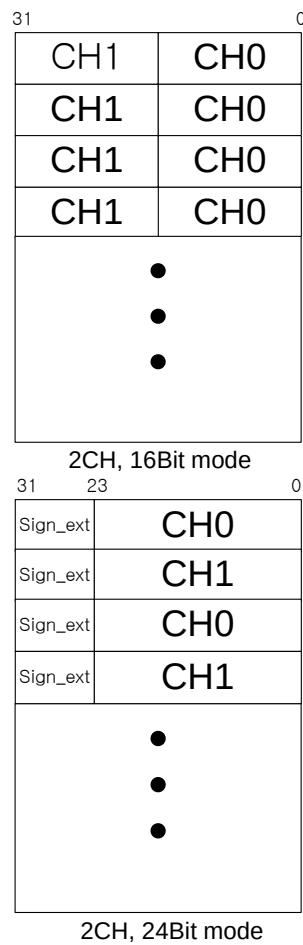


Figure 26.3 Input/Output Word Alignment

26.4 Register Description

Table 26.1 ASRC Register Map (Base Address = 0x9802000)

Name	Addr	Type	Reset	Description
ASRC_EN	0x00	R/W	0x00000000	ASRC Enable Register
ASRC_INIT	0x04	R/W	0x00000000	ASRC Initialization Register
ASRC_INPORT	0x08	R/W	0x00000000	ASRC Input Port Control Register
ASRC_OUTPORT	0x0C	R/W	0x00000000	ASRC Output Port Control Register
ASRC_INITZERO	0x10	R/W	0x00000000	ASRC Initialization Zero Register
ASRC_OPT_LEVEL	0x14	R/W	0x00000000	ASRC Optimum Buffer Level Register
ASRC_EXT_FMT	0x18	R/W	0x00000000	ASRC External I/O Format Register
	0x1C	R	0x00000000	Reserved
ASRC_SYNC_PER0	0x20	R/W	0x00000000	ASRC Sync Period Count Register 0
ASRC_RATE0	0x24	R/W	0x00000000	ASRC Rate Config Register 0
ASRC_RATE_MON0	0x28	R	0x00000000	ASRC Rate Monitor Register 0
ASRC_SRC_STAT0	0x2C	R	0x00000000	ASRC Source Status Register 0
ASRC_SYNC_PER1	0x30	R/W	0x00000000	ASRC Sync Period Count Register 1
ASRC_RATE1	0x34	R/W	0x00000000	ASRC Rate Config Register 1
ASRC_RATE_MON1	0x38	R	0x00000000	ASRC Rate Monitor Register 1
ASRC_SRC_STAT1	0x3C	R	0x00000000	ASRC Source Status Register 1
ASRC_MIX_IE	0x60	R/W	0x00000000	ASRC Mixer Input Enable Register
ASRC_TIME	0x64	R/W	0x00000000	ASRC Sample Timing Register
ASRC_VOL	0x70	R/W	0x00000000	ASRC Volume Control Register
ASRC_VOL_GAIN	0x80	R/W	0x00000000	ASRC Volume Gain Register
ASRC_VOL_RAMPDN	0x84	R/W	0x00000000	ASRC RAMP Down Volume Gain Register
ASRC_VOL_RAMP	0x88	R/W	0x00000000	ASRC RAMP Volume Gain Register
ASRC_VOL_RAMPUP	0x8C	R/W	0x00000000	ASRC RAMP Up Volume Gain Register
ASRC_IRQ_STAT	0xC0	R	0x00000000	ASRC IRQ Status Register
ASRC_MIRQ	0xC4	R	0x00000000	ASRC Mask-IRQ Register
ASRC_CIRQ	0xC8	R	0x00000000	ASRC Clear IRQ Register
ASRC_IREQEN	0xCC	R/W	0x00008000	ASRC IRQ Enable Register
ASRC_FIF0IN0	0x100	R/W	0x00000000	ASRC FIFO_IN Control Register 0
ASRC_FIF0IN_STAT0	0x104	R/W	0x00000000	ASRC FIFO_IN Status Control Register 0
ASRC_FIF0IN1	0x108	R/W	0x00000000	ASRC FIFO_IN Control Register 1
ASRC_FIF0IN_STAT1	0x10C	R/W	0x00000000	ASRC FIFO_IN Status Control Register 1
ASRC_FIF0OUT0	0x120	R/W	0x00000000	ASRC FIFO_OUT Control Register 0
ASRC_FIF0OUT_STAT0	0x124	R/W	0x00000000	ASRC FIFO_OUT Status Control Register 0
ASRC_FIF0OUT1	0x128	R/W	0x00000000	ASRC FIFO_OUT Control Register 1
ASRC_FIF0OUT_STAT1	0x12C	R/W	0x00000000	ASRC FIFO_OUT Status Control Register 1
ASRC_FIFO_MISC	0x140	R/W	0x00000000	ASRC FIFO MISC Control Register
ASRC_FIFO_IN_DATA0	0x200	R/W	0x00000000	ASRC FIFO DATAIN Register 0
ASRC_FIFO_IN_DATA1	0x240	R/W	0x00000000	ASRC FIFO DATAIN Register 1
ASRC_FIFO_OUT_DATA0	0x300	R/W	0x00000000	ASRC FIFO DATAOUT Register 0
ASRC_FIFO_OUT_DATA1	0x340	R/W	0x00000000	ASRC FIFO DATAOUT Register 1

ENABLE Register (ENABLE)

0x98020000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								EN_MIXER	EN_EXTIO	EN_OP	EN_IP	EN_ASRC			

Field	Name	RW	Reset	Description
7	EN_MIXER	R/W	0x0	MIXER Enable When this bit is set to 1, Mixer is enabled.
6	EN_EXTIO	R/W	0x0	External I/O Enable When this bit is set to 1, External I/O is enabled.
5	EN_OP	R/W	0x0	Output Port Enable When this bit is set to 1, Output Port is enabled.
4	EN_IP	R/W	0x0	Input Port Enable When this bit is set to 1, Input Port is enabled.
3:0	EN_ASRC	R/W	0x0	ASRC Enable When the bit [3] is always 0. When the bit [2] is always 0. When the bit [1] is set to 1, ASRC pair1 is enabled. When the bit [0] is set to 1, ASRC pair0 is enabled.

INIT Register (INIT)

0x98020004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								INIT_MIXER	INIT_EXTIO	INIT_OP	INIT_IP	INIT_ASRC			

Field	Name	RW	Reset	Description
7	INIT_MIXER	R/W	0x0	MIXER Initialization When this bit is set to 1, Mixer is initialized.
6	INIT_EXTIO	R/W	0x0	External I/O Initialization When this bit is set to 1, External I/O is initialized.
5	INIT_OP	R/W	0x0	Output Port Initialization When this bit is set to 1, Output Port is initialized.
4	INIT_IP	R/W	0x0	Input Port Initialization When this bit is set to 1, Input Port is initialized.
3:0	INIT_ASRC	R/W	0x0	ASRC Initialization When the bit [3] is always 0. When the bit [2] is always 0. When the bit [1] is set to 1, ASRC pair1 is initialized. When the bit [0] is set to 1, ASRC pair0 is initialized.

InPort Control Register (INPORT_CTRL)

0x98020008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IP3_ROUTE				IP2_ROUTE				IP1_ROUTE				IP0_ROUTE			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IP3_PATH		IP3_CLK_SEL		IP2_PATH		IP2_CLK_SEL		IP1_PATH		IP1_CLK_SEL		IP0_PATH		IP0_CLK_SEL	

Field	Name	RW	Reset	Description
31:28	IP3_ROUTE	R/W	0x0	Not used
27:24	IP2_ROUTE	R/W	0x0	Not used
23:20	IP1_ROUTE	R/W	0x0	Input source of ASRC pair1 0 : Ext. Input source is DAI0_out [1:0] 1 : Ext. Input source is DAI0_out [1:0] 2 : Ext. Input source is DAI0_out [1:0] 3 : Ext. Input source is DAI0_out [1:0] 4 : Ext. Input source is DAI1_out [1:0] Others : Not used Note: When IP1_PATH is 0, IP1_ROUTE is valid.
19:16	IP0_ROUTE	R/W	0x0	Input source of ASRC pair0 0 : Ext. Input source is MCAUDIO0_out [1:0] 4 : Ext. Input source is MCAUDIO1_out [1:0] Others : Not used Note: When IP0_PATH is 0, IP0_ROUTE is valid. Note: When the bits are set to 4/5/6, msb 6bit of input source are zero.
15	IP3_PATH	R/W	0x0	Not used
14:12	IP3_CLK_SEL	R/W	0x0	Not used
11	IP2_PATH	R/W	0x0	Not used
10:8	IP2_CLK_SEL	R/W	0x0	Not used
7	IP1_PATH	R/W	0x0	Input path of ASRC pair1 0 : Ext. I/O path (DAI → InPort → ASRC → VOL_CTRL) 1 : DMA path (DMA → InPort → ASRC → VOL_CTRL)
6:4	IP1_CLK_SEL	R/W	0x0	Input clock selection of ASRC pair1 0 : DAI0 L/R clock 1 : DAI1 L/R clock 2 : DAI2 L/R clock 3 : Not used 4 : AUXCLK0 clock 5 : AUXCLK1 clock 6 : AUXCLK2 clock 7 : Not used
3	IP0_PATH	R/W	0x0	Input path of ASRC pair0 0 : Ext. I/O path (DAI → InPort → ASRC → VOL_CTRL) 1 : DMA path (DMA → InPort → ASRC → VOL_CTRL)
2:0	IP0_CLK_SEL	R/W	0x0	Input clock selection of ASRC pair0 0 : DAI0 L/R clock 1 : DAI1 L/R clock 2 : DAI2 L/R clock 3 : Not used 4 : AUXCLK0 clock 5 : AUXCLK1 clock 6 : AUXCLK2 clock 7 : Not used

OutPort Control Register (OUTPORT_CTRL)

0x9802000C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OP3_ROUTE				OP2_ROUTE				OP1_ROUTE				OP0_ROUTE			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OP3_PATH		OP3_CLK_SEL		OP2_PATH		OP2_CLK_SEL		OP1_PATH		OP1_CLK_SEL		OP0_PATH		OP0_CLK_SEL	

Field	Name	RW	Reset	Description
31:28	OP3_ROUTE	R/W	0x0	Not used
27:24	OP2_ROUTE	R/W	0x0	Input source of DAI2_in 0 : VOL_CTRL of ASRC pair0[1:0] 1 : VOL_CTRL of ASRC pair0[1:0] 2 : VOL_CTRL of ASRC pair0[1:0] 3 : VOL_CTRL of ASRC pair0[1:0] 4 : VOL_CTRL of ASRC pair1[1:0] 5 : Not used 6 : Not used 8 : MIXER[1:0] 9 : MIXER[1:0] 10 : MIXER[1:0] 11 : MIXER[1:0] Note: When OP2_PATH is 0, OP2_ROUTE is valid
23:20	OP1_ROUTE	R/W	0x0	Input source of MCAUDIO1_in 0 : VOL_CTRL of ASRC pair0[1:0] 1 : VOL_CTRL of ASRC pair0[1:0] 2 : VOL_CTRL of ASRC pair0[1:0] 3 : VOL_CTRL of ASRC pair0[1:0] 4 : VOL_CTRL of ASRC pair1[1:0] 5 : Not used 6 : Not used 8 : MIXER[1:0] 9 : MIXER[1:0] 10 : MIXER[1:0] 11 : MIXER[1:0] Note: When OP1_PATH is 0, OP1_ROUTE is valid
19:16	OP0_ROUTE	R/W	0x0	Input source of MCAUDIO0_in 0 : VOL_CTRL of ASRC pair0[1:0] 4 : VOL_CTRL of ASRC pair1[1:0] 5 : VOL_CTRL of ASRC pair2[1:0] 6 : VOL_CTRL of ASRC pair3[1:0] 8 : MIXER[1:0] Note: When OP0_PATH is 0, OP0_ROUTE is valid Note: when the bit are set to 4/5/6, msb 6bit of input source are zero.
15	OP3_PATH	R/W	0x0	Not used
14:12	OP3_CLK_SEL	R/W	0x0	Not used
11	OP2_PATH	R/W	0x0	Input path of DAI2_in 0 : Ext. I/O path (ASRC → VOL_CTRL → OutPort → (MIXER) → DAI) 1 : DMA path (ASRC → VOL_CTRL → OutPort → DMA)
10:8	OP2_CLK_SEL	R/W	0x0	Input clock selection of DAI2_in 0 : DAI0 L/R clock 1 : DAI1 L/R clock 2 : DAI2 L/R clock 3 : Not used 4 : AUXCLK0 clock 5 : AUXCLK1 clock 6 : AUXCLK2 clock 7 : Not used
7	OP1_PATH	R/W	0x0	Input path of DAI1_in 0 : Ext. I/O path (ASRC → VOL_CTRL → OutPort → (MIXER) → MCAUDIO) 1 : DMA path (ASRC → VOL_CTRL → OutPort → DMA)
6:4	OP1_CLK_SEL	R/W	0x0	Input clock selection of DAI1_in 0 : DAI0 L/R clock 1 : DAI1 L/R clock 2 : DAI2 L/R clock

				3 : Not used 4 : AUXCLK0 clock 5 : AUXCLK1 clock 6 : AUXCLK2 clock 7 : Not used
3	OP0_PATH	R/W	0x0	Input path of DAI0_in 0 : Ext. I/O path (ASRC → VOL_CTRL → OutPort → (MIXER) → MCAUDIO) 1 : DMA path (ASRC → VOL_CTRL → OutPort → DMA)
2:0	OP0_CLK_SEL	R/W	0x0	Input clock selection of DAI0_in 0 : DAI0 L/R clock 1 : DAI1 L/R clock 2 : DAI2 L/R clock 3 : Not used 4 : AUXCLK0 clock 5 : AUXCLK1 clock 6 : AUXCLK2 clock 7 : Not used

INIT_ZERO_SIZE Register (SRC_INIT_ZERO_SZ)

0x98020010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SRC_INIT_ZERO_SZ1								SRC_INIT_ZERO_SZ0							

When ASRC is initialized, SRC_INIT_ZERO_SZ "0" data is automatically stored into the SRC ring buffer. SRC_INIT_ZERO_SZ is needed to prevent removing initial noisy sample caused FIR filter.

Recommend : filter_tap/2 (filter_tap : 32 @ oversample, 32 * downsample_rate @ downsample)

Example:

Input sample : 32K, output sample : 8K → downsample_rate = 4

SRC_INIT_ZERO = 32 * 4 = 128

Field	Name	RW	Reset	Description
15:8	SRC_INIT_ZERO_SZ1	R/W	0x0	Zero padding initializing size (ASRC pair1) to FIR filter
7:0	SRC_INIT_ZERO_SZ0	R/W	0x0	Zero padding initializing size (ASRC pair0) to FIR filter

OPTIMUM BUFFER LEVEL Register (SRC_OPT_BUF_LVL)

0x98020014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SRC_OPT_BUF_LVL1								SRC_OPT_BUF_LVL0							

Recommend value is '0x10'.

Field	Name	RW	Reset	Description
15:8	SRC_OPT_BUF_LVL1	R/W	0x0	Optimum buffer level for ASRC pair1
7:0	SRC_OPT_BUF_LVL0	R/W	0x0	Optimum buffer level for ASRC pair0

External I/O Format Register (EXT_IO_FMT)

0x98020018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	OP3_SWAP		OP3_FMT		OP2_SWAP		OP2_FMT		OP1_SWAP		OP1_FMT		OP0_SWAP		OP0_FMT
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	IP3_SWAP		IP3_FMT		IP2_SWAP		IP2_FMT		IP1_SWAP		IP1_FMT		IP0_SWAP		IP0_FMT

Field	Name	RW	Reset	Description
30	OP3_SWAP	R/W	0x0	Not used
29:28	OP3_FMT	R/W	0x0	Not used
26	OP2_SWAP	R/W	0x0	Swap control for OutPort2 0: No swap 1: Odd/even bit swap
25:24	OP2_FMT	R/W	0x0	Data format for OutPort2 0: No format changed. (24bit) 1: Left 8 bit shift, lsb 0 padding. 2: Right 8 bit shift, msb sign extension. Note: Internal format of ASRC is always 24bit. If external I/O format is 16bit, this register must be configured to 1 or 2.
22	OP1_SWAP	R/W	0x0	Swap control for OutPort1 0: No swap 1: Odd/even bit swap
21:20	OP1_FMT	R/W	0x0	Data format for OutPort1 0: No format changed. (24bit) 1: Left 8 bit shift, lsb 0 padding. 2: Right 8 bit shift, msb sign extension. Note: Internal format of ASRC is always 24bit. If external I/O format is 16bit, this register must be configured to 1 or 2.
18	OP0_SWAP	R/W	0x0	Swap control for OutPort0 0: No swap 1: Odd/even bit swap
17:16	OP0_FMT	R/W	0x0	Data format for OutPort0 0: No format changed. (24bit) 1: Left 8 bit shift, lsb 0 padding. 2: Right 8 bit shift, msb sign extension. Note: Internal format of ASRC is always 24bit. If external I/O format is 16bit, this register must be configured to 1 or 2.
14	IP3_SWAP	R/W	0x0	Not used
13:12	IP3_FMT	R/W	0x0	Not used
10	IP2_SWAP	R/W	0x0	Swap control for InPort2 0: No swap 1: Odd/even bit swap
9:8	IP2_FMT	R/W	0x0	Data format for InPort2 0: No format changed. (24bit) 1: 16bit → left 8 bit shift, lsb 0 padding. 2: 16bit → right 8 bit shift, msb sign extension. Note: Internal format of ASRC is always 24bit. If external I/O format is 16bit, this register must be configured to 1 or 2.
6	IP1_SWAP	R/W	0x0	Swap control for InPort1 0: No swap 1: Odd/even bit swap
5:4	IP1_FMT	R/W	0x0	Data format for InPort1 0: No format changed. (24bit) 1: 16bit → left 8 bit shift, lsb 0 padding. 2: 16bit → right 8 bit shift, msb sign extension. Note: Internal format of ASRC is always 24bit. If external I/O format is 16bit, this register must be configured to 1 or 2.
2	IP0_SWAP	R/W	0x0	Swap control for InPort0

				0: No swap 1: Odd/even bit swap
1:0	IP0_FMT	R/W	0x0	Data format for InPort0 0: No format changed. (24bit) 1: 16bit → left 8 bit shift, lsb 0 padding. 2: 16bit → right 8 bit shift, msb sign extension. Note: Internal format of ASRC is always 24bit. If external I/O format is 16bit, this register must be configured to 1 or 2.

SYNC PERIOD Register (PREDIOD_SYNC_CNT0/1)

0x98020020 + X*0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PERIOD_SYNC_CNT															

Field	Name	RW	Reset	Description
13:0	PERIOD_SYNC_CNT	R/W	0x0	Sync period count for ASRC pair X

Rate Config Register (SRC_RATE0/1)

0x98020024 + X*0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SRC_MODE						MANUAL_RATIO[25:16]									
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MANUAL_RATIO[15:0]															

Field	Name	RW	Reset	Description
31	SRC_MODE	R/W	0x0	SRC mode for pair X 0 : Async sample rate convertor (ratio calculated by external timing signal) 1 : Sync sample rate convertor (ratio fixed by MANUAL_RATIO)
25:0	MANUAL_RATIO	R/W	0x0	Manual ratio for pair X Format 4.22 (Bit[25:22] : integer, Bit[21:0] : fraction) When SRC_MODE is 0, MANUAL_RATIO is initial ratio When SRC_MODE is 1, MANUAL_RATIO is fixed ratio.

Rate monitor Register (SRC_CAL_RATE0/1)

0x98020028 + X*0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
						CAL_RATIO[25:16]									
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAL_RATIO[15:0]															

Field	Name	RW	Reset	Description
25:0	CAL_RATIO	R	0x0	Calculated ratio for pair X Format 4.22 (Bit[25:22] : integer, Bit[21:0] : fraction)

SRC STATUS Register (SRC_STATUS0/1)

0x9802002C + X*0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCKED	OVERFLOW	UNDERFLOW													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FIFO_LVL[15:0]															

Field	Name	RW	Reset	Description
31	LOCKED	R	0x0	CAL_RATIO Lock status for pair X 0 : Unlock 1 : Lock
30	OVERFLOW	R	0x0	Internal Ring Buffer OVERFLOW status for pair X 0 : No overflow 1 : Overflow
29	UNDERFLOW	R	0x0	Internal Ring Buffer UNDERFLOW status for pair X 0 : No underflow 1 : Underflow
15:0	FIFO_LVL	R	0x0	Internal Ring Buffer LEVEL for pair X

MIXER Input Enable Register (MIX_IN_EN)

0x98020060

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MIX7_IN_EN				MIX6_IN_EN				MIX5_IN_EN				MIX4_IN_EN			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MIX3_IN_EN				MIX2_IN_EN				MIX1_IN_EN				MIX0_IN_EN			

Field	Name	RW	Reset	Description
31	MIX7_IN_EN[3]	R/W	0x0	Not used
30	MIX7_IN_EN[2]	R/W	0x0	Not used
29	MIX7_IN_EN[1]	R/W	0x0	Input Source selection for MIXER CH1 0 : ASRC pair1 CH 1 disable 1 : ASRC pair1 CH 1 enable
28	MIX7_IN_EN[0]	R/W	0x0	Input Source selection for MIXER CH1 0 : ASRC pair0 CH 1 disable 1 : ASRC pair0 CH 1 enable
27	MIX6_IN_EN[3]	R/W	0x0	Not used
26	MIX6_IN_EN[2]	R/W	0x0	Not used
25	MIX6_IN_EN[1]	R/W	0x0	Input Source selection for MIXER CH0 0 : ASRC pair1 CH 0 disable 1 : ASRC pair1 CH 0 enable
24	MIX6_IN_EN[0]	R/W	0x0	Input Source selection for MIXER CH0 0 : ASRC pair0 CH 0 disable 1 : ASRC pair0 CH 0 enable
23	MIX5_IN_EN[3]	R/W	0x0	Not used
22	MIX5_IN_EN[2]	R/W	0x0	Not used
21	MIX5_IN_EN[1]	R/W	0x0	Input Source selection for MIXER CH1 0 : ASRC pair1 CH 1 disable 1 : ASRC pair1 CH 1 enable
20	MIX5_IN_EN[0]	R/W	0x0	Input Source selection for MIXER CH1 0 : ASRC pair0 CH 1 disable 1 : ASRC pair0 CH 1 enable
19	MIX4_IN_EN[3]	R/W	0x0	Not used
18	MIX4_IN_EN[2]	R/W	0x0	Not used
17	MIX4_IN_EN[1]	R/W	0x0	Input Source selection for MIXER CH0 0 : ASRC pair1 CH 0 disable 1 : ASRC pair1 CH 0 enable
16	MIX4_IN_EN[0]	R/W	0x0	Input Source selection for MIXER CH0 0 : ASRC pair0 CH 0 disable 1 : ASRC pair0 CH 0 enable
15	MIX3_IN_EN[3]	R/W	0x0	Not used
14	MIX3_IN_EN[2]	R/W	0x0	Not used
13	MIX3_IN_EN[1]	R/W	0x0	Input Source selection for MIXER CH1 0 : ASRC pair1 CH 1 disable 1 : ASRC pair1 CH 1 enable
12	MIX3_IN_EN[0]	R/W	0x0	Input Source selection for MIXER CH1 0 : ASRC pair0 CH 1 disable 1 : ASRC pair0 CH 1 enable
11	MIX2_IN_EN[3]	R/W	0x0	Not used
10	MIX2_IN_EN[2]	R/W	0x0	Not used
9	MIX2_IN_EN[1]	R/W	0x0	Input Source selection for MIXER CH0 0 : ASRC pair1 CH 0 disable 1 : ASRC pair1 CH 0 enable
8	MIX2_IN_EN[0]	R/W	0x0	Input Source selection for MIXER CH0 0 : ASRC pair0 CH 0 disable 1 : ASRC pair0 CH 0 enable
7	MIX1_IN_EN[3]	R/W	0x0	Not used
6	MIX1_IN_EN[2]	R/W	0x0	Not used
5	MIX1_IN_EN[1]	R/W	0x0	Input Source selection for MIXER CH1 0 : ASRC pair1 CH 1 disable 1 : ASRC pair1 CH 1 enable
4	MIX1_IN_EN[0]	R/W	0x0	Input Source selection for MIXER CH1 0 : ASRC pair0 CH 1 disable 1 : ASRC pair0 CH 1 enable
3	MIX0_IN_EN[3]	R/W	0x0	Not used
2	MIX0_IN_EN[2]	R/W	0x0	Not used
1	MIX0_IN_EN[1]	R/W	0x0	Input Source selection for MIXER CH0

				0 : ASRC pair1 CH 0 disable 1 : ASRC pair1 CH 0 enable
0	MIX0_IN_EN[0]	R/W	0x0	Input Source selection for MIXER CH0 0 : ASRC pair0 CH 0 disable 1 : ASRC pair0 CH 0 enable

SAMPLE TIMING Register (SAMP_TIMING)

0x98020064

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
									OP_TIMING				IP_TIMING		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
														MIX_TIMING	

Field	Name	RW	Reset	Description
23	OP_TIMING[3]	R/W	0x0	Not used
22	OP_TIMING[2]	R/W	0x0	Sample timing for OutPort2 0 : External clk 1 : Use the request which is automatically generated by ASRC
21	OP_TIMING[1]	R/W	0x0	Sample timing for OutPort1 0 : External clk 1 : Use the request which is automatically generated by ASRC
20	OP_TIMING[0]	R/W	0x0	Sample timing for OutPort0 0 : External clk 1 : Use the request which is automatically generated by ASRC
19	IP_TIMING[3]	R/W	0x0	Not used
18	IP_TIMING[2]	R/W	0x0	Sample timing for InPort2 0 : External clk 1 : Use the request which is automatically generated by ASRC
17	IP_TIMING[1]	R/W	0x0	Sample timing for InPort1 0 : External clk 1 : Use the request which is automatically generated by ASRC
16	IP_TIMING[0]	R/W	0x0	Sample timing for InPort0 0 : External clk 1 : Use the request which is automatically generated by ASRC
1:0	MIX_TIMING	R/W	0x0	Sample timing for Mixer 0 : ASRC pair0 output timing 1 : ASRC pair1 output timing 2 : Not used 3 : Not used

Note:

case 1) IP_TIMING=1, OP_TIMING=1

If Out_fifo is not full, out data is transferred. Input data is transferred automatically depending on buffer status.

case 2) IP_TIMING=1, OP_TIMING=0

Out data is transferred at ext_out clock's rising_edge. Input data is transferred automatically depending on buffer status.

case 3) IP_TIMING=0, OP_TIMING=0

case 4) IP_TIMING=0, OP_TIMING=1

VOLUME CONTROL Register (VOL_CTRL_EN)

0x98020070

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								RAMP EN				VOL EN			

Field	Name	RW	Reset	Description
7	RAMP_EN[3]	R/W	0x0	Not used
6	RAMP_EN[2]	R/W	0x0	Not used
5	RAMP_EN[1]	R/W	0x0	Volume ramp control for VOL_CTRL pair 1 0 : Disable 1 : Enable (auto clear)
4	RAMP_EN[0]	R/W	0x0	Volume ramp control for VOL_CTRL pair 0 0 : Disable 1 : Enable (auto clear)
3	VOL_EN[3]	R/W	0x0	Not used
2	VOL_EN[2]	R/W	0x0	Not used
1	VOL_EN[1]	R/W	0x0	Volume control for VOL_CTRL pair 1 0 : Disable 1 : Enable (auto clear)
0	VOL_EN[0]	R/W	0x0	Volume control for VOL_CTRL pair 0 0 : Disable 1 : Enable (auto clear)

VOLUME GAIN Register (VOL_GAIN0/1/2/3)

0x98020080+X*0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								VOL_GAIN[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VOL_GAIN[15:0]															

Field	Name	RW	Reset	Description
23:0	VOL_GAIN	R/W	0x0	Gain for VOL_CTRL pair X Bit [23]: sign Bit [22:20]: integer Bit [19:0]: fraction Range: 0x1 ~ 0x4ffff Example: 0x400000 : x4 (12dB) 0x100000 : x1 (0dB) 0x80000 : x0.5 (-6dB) 0x1 : x2 [^] (-20) (-120dB) Note: When VOL_GAIN is 0x0, malfunction occurs.

VOLUME RAMP GAIN Register (VOL_RAMP_GAIN0/1/2/3)

0x98020088+X*0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VOL_RAMP_GAIN															

Field	Name	RW	Reset	Description
9:0	VOL_RAMP_GAIN	R/W	0x0	Ramp Gain for VOL_CTRL pair X Gain = -0.125 x VOL_RAMP_GAIN (dB) Range : 0x0 ~ 0x2dc Example: 0x0 : x1 (0dB) 0x30 : x0.5 (-6dB) 0x60 : x0.25 (-12dB)

VOLUME RAMP-DN CONFIG Register (VOL_RAMPDN_CFG0/1/2/3)

0x98020084+X*0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VOL_RAMP_DN_TIME								VOL_RAMP_DN_WAIT[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VOL_RAMP_DN_WAIT[15:0]															

Field	Name	RW	Reset	Description
31:24	VOL_RAMP_DN_TIME	R/W	0x0	Ramp down (fade-out) complete time 0 : Maximum 128dB/1sample 1 : Maximum 64dB/ 1sample 2 : Maximum 32dB/1sample 3 : Maximum 16dB/1sample 4 : Maximum 8dB/1sample 5 : Maximum 4dB/1sample 6 : Maximum 2dB/1sample 7 : Maximum 1dB/1sample 8 : Maximum 0.5dB/1sample 9 : Maximum 0.25dB/1sample 10 : Maximum 0.125dB/1sample 11 : 0.125dB/2sample 12 : 0.125dB/4sample 13 : 0.125dB/8sample 14 : 0.125dB/16sample 15 : 0.125dB/32sample 16 : 0.125dB/64sample
23:0	VOL_RAM_DN_WAIT	R/W	0x0	VOL_RAMP_DN_WAIT sets the wait time before ramp down (fade-out) operation Wait time = output sampling rate * VOL_RAMP_DN_WAIT second

VOLUME RAMP-UP CONFIG Register (VOL_RAMPDN_CFG0/1/2/3)

0x9802008C+X*0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VOL_RAMP_UP_TIME								VOL_RAMP_UP_WAIT[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VOL_RAMP_UP_WAIT[15:0]															

Field	Name	RW	Reset	Description
31:24	VOL_RAMP_UP_TIME	R/W	0x0	Ramp up (fade-in) complete time 0 : Maximum 128dB/1sample 1 : Maximum 64dB/ 1sample 2 : Maximum 32dB/1sample 3 : Maximum 16dB/1sample 4 : Maximum 8dB/1sample 5 : Maximum 4dB/1sample 6 : Maximum 2dB/1sample 7 : Maximum 1dB/1sample 8 : Maximum 0.5dB/1sample 9 : Maximum 0.25dB/1sample 10 : Maximum 0.125dB/1sample 11 : 0.125dB/2sample 12 : 0.125dB/4sample 13 : 0.125dB/8sample 14 : 0.125dB/16sample 15 : 0.125dB/32sample 16 : 0.125dB/64sample
23:0	VOL_RAM_UP_WAIT	R/W	0x0	VOL_RAMP_UP_WAIT sets the wait time before ramp up (fade-in) operation Wait time = output sampling rate * VOL_RAMP_UP_WAIT second

IRQ RAW STATUS Register (IRQ_RAW_STATUS)

0x980200C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								IRQ_RAW_STATUS[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_RAW_STATUS[15:0]															

Field	Name	RW	Reset	Description
23	IRQ_RAW_STATUS[23]	R	0x0	Not used
22	IRQ_RAW_STATUS[22]	R	0x0	Not used
21	IRQ_RAW_STATUS[21]	R	0x0	ASRC pair1 ring buffer overflow
20	IRQ_RAW_STATUS[20]	R	0x0	ASRC pair0 ring buffer overflow
19	IRQ_RAW_STATUS[19]	R	0x0	Not used
18	IRQ_RAW_STATUS[18]	R	0x0	Not used
17	IRQ_RAW_STATUS[17]	R	0x0	ASRC pair1 ring buffer underflow
16	IRQ_RAW_STATUS[16]	R	0x0	ASRC pair0 ring buffer underflow
15	IRQ_RAW_STATUS[15]	R	0x0	Not used
14	IRQ_RAW_STATUS[14]	R	0x0	Not used
13	IRQ_RAW_STATUS[13]	R	0x0	FIFO_OUT1 Full
12	IRQ_RAW_STATUS[12]	R	0x0	FIFO_OUT0 Full
11	IRQ_RAW_STATUS[11]	R	0x0	Not used
10	IRQ_RAW_STATUS[10]	R	0x0	Not used
9	IRQ_RAW_STATUS[9]	R	0x0	FIFO_OUT1 Empty
8	IRQ_RAW_STATUS[8]	R	0x0	FIFO_OUT0 Empty
7	IRQ_RAW_STATUS[7]	R	0x0	Not used
6	IRQ_RAW_STATUS[6]	R	0x0	Not used
5	IRQ_RAW_STATUS[5]	R	0x0	FIFO_IN1 Full
4	IRQ_RAW_STATUS[4]	R	0x0	FIFO_IN0 Full
3	IRQ_RAW_STATUS[3]	R	0x0	Not used
2	IRQ_RAW_STATUS[2]	R	0x0	Not used
1	IRQ_RAW_STATUS[1]	R	0x0	FIFO_IN1 Empty
0	IRQ_RAW_STATUS[0]	R	0x0	FIFO_IN0 Empty

IRQ MASK STATUS Register (IRQ_MASK_STATUS)

0x980200C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								IRQ_MASK_STATUS[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_MASK_STATUS[15:0]															

Field	Name	RW	Reset	Description
23	IRQ_MASK_STATUS[23]	R	0x0	Not used
22	IRQ_MASK_STATUS[22]	R	0x0	Not used
21	IRQ_MASK_STATUS[21]	R	0x0	ASRC pair1 ring buffer overflow
20	IRQ_MASK_STATUS[20]	R	0x0	ASRC pair0 ring buffer overflow
19	IRQ_MASK_STATUS[19]	R	0x0	Not used
18	IRQ_MASK_STATUS[18]	R	0x0	Not used
17	IRQ_MASK_STATUS[17]	R	0x0	ASRC pair1 ring buffer underflow
16	IRQ_MASK_STATUS[16]	R	0x0	ASRC pair0 ring buffer underflow
15	IRQ_MASK_STATUS[15]	R	0x0	Not used
14	IRQ_MASK_STATUS[14]	R	0x0	Not used
13	IRQ_MASK_STATUS[13]	R	0x0	FIFO_OUT1 Full
12	IRQ_MASK_STATUS[12]	R	0x0	FIFO_OUT0 Full
11	IRQ_MASK_STATUS[11]	R	0x0	Not used
10	IRQ_MASK_STATUS[10]	R	0x0	Not used
9	IRQ_MASK_STATUS[9]	R	0x0	FIFO_OUT1 Empty
8	IRQ_MASK_STATUS[8]	R	0x0	FIFO_OUT0 Empty
7	IRQ_MASK_STATUS[7]	R	0x0	Not used
6	IRQ_MASK_STATUS[6]	R	0x0	Not used
5	IRQ_MASK_STATUS[5]	R	0x0	FIFO_IN1 Full
4	IRQ_MASK_STATUS[4]	R	0x0	FIFO_IN0 Full
3	IRQ_MASK_STATUS[3]	R	0x0	Not used
2	IRQ_MASK_STATUS[2]	R	0x0	Not used
1	IRQ_MASK_STATUS[1]	R	0x0	FIFO_IN1 Empty
0	IRQ_MASK_STATUS[0]	R	0x0	FIFO_IN0 Empty

IRQ CLEAR Register (IRQ_CLEAR)

0x980200C8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								IRQ_CLEAR[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_CLEAR[15:0]															

Field	Name	RW	Reset	Description
23	IRQ_CLEAR[23]	W	0x0	Not used
22	IRQ_CLEAR[22]	W	0x0	Not used
21	IRQ_CLEAR[21]	W	0x0	ASRC pair1 ring buffer overflow IRQ Clear
20	IRQ_CLEAR[20]	W	0x0	ASRC pair0 ring buffer overflow IRQ Clear
19	IRQ_CLEAR[19]	W	0x0	Not used
18	IRQ_CLEAR[18]	W	0x0	Not used
17	IRQ_CLEAR[17]	W	0x0	ASRC pair1 ring buffer underflow IRQ Clear
16	IRQ_CLEAR[16]	W	0x0	ASRC pair0 ring buffer underflow IRQ Clear
15	IRQ_CLEAR[15]	W	0x0	Not used
14	IRQ_CLEAR[14]	W	0x0	Not used
13	IRQ_CLEAR[13]	W	0x0	FIFO_OUT1 Full IRQ Clear
12	IRQ_CLEAR[12]	W	0x0	FIFO_OUT0 Full IRQ Clear
11	IRQ_CLEAR[11]	W	0x0	Not used
10	IRQ_CLEAR[10]	W	0x0	Not used
9	IRQ_CLEAR[9]	W	0x0	FIFO_OUT1 Empty IRQ Clear
8	IRQ_CLEAR[8]	W	0x0	FIFO_OUT0 Empty IRQ Clear
7	IRQ_CLEAR[7]	W	0x0	Not used
6	IRQ_CLEAR[6]	W	0x0	Not used
5	IRQ_CLEAR[5]	W	0x0	FIFO_IN1 Full IRQ Clear
4	IRQ_CLEAR[4]	W	0x0	FIFO_IN0 Full IRQ Clear
3	IRQ_CLEAR[3]	W	0x0	Not used
2	IRQ_CLEAR[2]	W	0x0	Not used
1	IRQ_CLEAR[1]	W	0x0	FIFO_IN1 Empty IRQ Clear
0	IRQ_CLEAR[0]	W	0x0	FIFO_IN0 Empty IRQ Clear

IRQ ENABLE Register (IRQ_EN)

0x980200CC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								IRQ_ENABLE[23:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_ENABLE[15:0]															

Field	Name	RW	Reset	Description
23	IRQ_ENABLE[23]	R/W	0x0	Not used
22	IRQ_ENABLE[22]	R/W	0x0	Not used
21	IRQ_ENABLE[21]	R/W	0x0	ASRC pair1 ring buffer overflow IRQ Enable
20	IRQ_ENABLE[20]	R/W	0x0	ASRC pair0 ring buffer overflow IRQ Enable
19	IRQ_ENABLE[19]	R/W	0x0	Not used
18	IRQ_ENABLE[18]	R/W	0x0	Not used
17	IRQ_ENABLE[17]	R/W	0x0	ASRC pair1 ring buffer underflow IRQ Enable
16	IRQ_ENABLE[16]	R/W	0x0	ASRC pair0 ring buffer underflow IRQ Enable
15	IRQ_ENABLE[15]	R/W	0x0	Not used
14	IRQ_ENABLE[14]	R/W	0x0	Not used
13	IRQ_ENABLE[13]	R/W	0x0	FIFO_OUT1 Full IRQ Enable
12	IRQ_ENABLE[12]	R/W	0x0	FIFO_OUT0 Full IRQ Enable
11	IRQ_ENABLE[11]	R/W	0x0	Not used
10	IRQ_ENABLE[10]	R/W	0x0	Not used
9	IRQ_ENABLE[9]	R/W	0x0	FIFO_OUT1 Empty IRQ Enable
8	IRQ_ENABLE[8]	R/W	0x0	FIFO_OUT0 Empty IRQ Enable
7	IRQ_ENABLE[7]	R/W	0x0	Not used
6	IRQ_ENABLE[6]	R/W	0x0	Not used
5	IRQ_ENABLE[5]	R/W	0x0	FIFO_IN1 Full IRQ Enable
4	IRQ_ENABLE[4]	R/W	0x0	FIFO_IN0 Full IRQ Enable
3	IRQ_ENABLE[3]	R/W	0x0	Not used
2	IRQ_ENABLE[2]	R/W	0x0	Not used
1	IRQ_ENABLE[1]	R/W	0x0	FIFO_IN1 Empty IRQ Enable
0	IRQ_ENABLE[0]	R/W	0x0	FIFO_IN0 Empty IRQ Enable

FIFO_IN Control Register (FIFO_IN0/1/2/3_CTRL)

0x98020100+X*0x8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMA_EN															FMT
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
THRESHOLD								MODE							

Field	Name	RW	Reset	Description
31	DMA_EN	R/W	0x0	DMA request enable 0: Disable 1: Enable
16	FMT	R/W	0x0	Data input format 0: 16bit - {16bit, 16bit} 1: 24bit - {sign_ext, 24bit}
10:8	THRESHOLD	R/W	0x00	0: DMA request at least 1-word available 1: DMA request at least 2-words available 2: DMA request at least 3-words available 3: DMA request at least 4-words available
3:0	MODE	R/W	0x0000	0: 2 channels 1: 4 channels 2: 6 channels 3: 8 channels

FIFO_IN_STATUS Register (FIFO_IN0/1/2/3_STATUS)

0x98020104+X*0x8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FULL	EMPTY														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LEVEL															

Field	Name	RW	Reset	Description
31	FULL	R	0x0	0: Not full 1: Full
30	EMPTY	R	0x0	0: Not empty 1: Empty
15:0	LEVEL	R	0x00	The number of available data

FIFO_OUT Control Register (FIFO_OUT0/1/2/3_CTRL)

0x98020120+X*0x8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMA_EN															FMT
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
THRESHOLD								MODE							

Field	Name	RW	Reset	Description
31	DMA_EN	R/W	0x0	DMA request enable 0: Disable 1: Enable
16	FMT	R/W	0x0	Data input format 0: 16bit - {16bit, 16bit} 1: 24bit - {sign_ext, 24bit}
10:8	THRESHOLD	R/W	0x00	0: DMA request at least 1-word available 1: DMA request at least 2-words available 2: DMA request at least 3-words available 3: DMA request at least 4-words available
3:0	MODE	R/W	0x0000	0: 2 channels 1: 4 channels 2: 6 channels 3: 8 channels

FIFO_OUT_STATUS Register (FIFO_OUT0/1/2/3_STATUS)

0x98020124+X*0x8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FULL	EMPTY														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LEVEL															

Field	Name	RW	Reset	Description
31	FULL	R	0x0	0: Not full 1: Full
30	EMPTY	R	0x0	0: Not empty 1: Empty
15:0	LEVEL	R	0x00	The number of available data

FIFO MISC CONTROL Register (FIFO_MISC_CTRL)

0x98020140

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
															DMA_ARB

Field	Name	RW	Reset	Description
0	DMA_ARB	R/W	0x0	DMA request arbitration 0: No arbitration 1: Round-robin arbitration

FIFO_IN DATA Register (FIFO_IN0/1/2/3_DATA)

0x98020200 +X*0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															

FIFO_OUT DATA Register (FIFO_OUT0/1/2/3_DATA)

0x98020300 +X*0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															

27 UART

27.1 Overview

The TCC8132 has 5 UART modules (UART0, UART1, UART2, UART3, and UART4). Each UART module has an automatic flow control and two DMA request outputs (Rx/Tx DMA request).

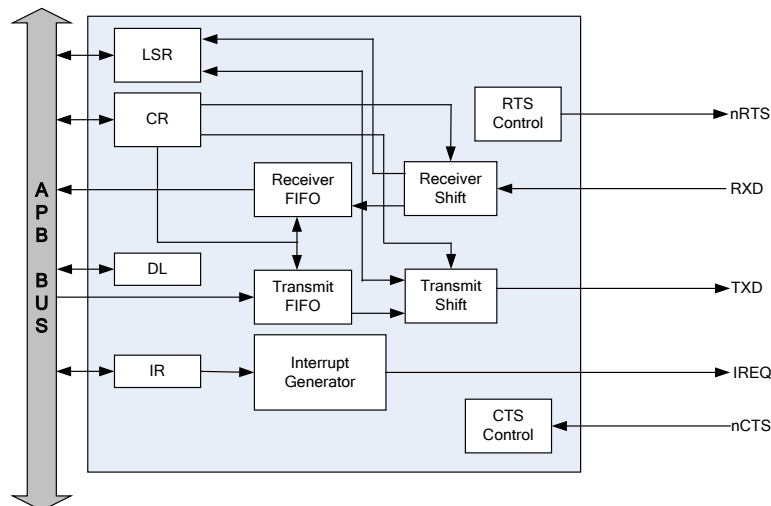


Figure 27.1 UART Block Diagram

Recommended Frequency:

The system bus (APB) interface of the UART has synchronization logic for the two different clock domains. Therefore, the bus access time from/to UART is quite longer than the other peripherals. If the frequency of the UART clock is too slow, significant performance degradation and latency problems can happen, especially when there are multiple bus masters running in parallel. It is recommended to set the UART clock as fast as possible.

Interrupt Trigger Mode:

The interrupt trigger mode for the UART must be set as level trigger. (Refer to Interrupt Controller TMODE register). The interrupt controller of the TCC8132 has default value of edge trigger mode. Make sure to program the interrupt controller before UART module is enabled.

DMA Configuration:

In case of UART interface with general DMA, the DMA controller configuration and the clock frequencies should be set according to the table below.

DMA.CHCTRL[13]	SYNC = 1							
DMA.CHCTRL[5:4]	WSIZE = 0 (1 Byte)							
DMA.CHCTRL[9:8]	TYPE=0x3 (Level Triggered)				TYPE=0x0 (Edge Triggered)			
UART Transfer	Tx		Rx		Tx		Rx	
DMA.CHCTRL[10]	BST=0	BST=1	BST=0	BST=1	BST=0	BST=1	BST=0	BST=1
Frequency Condition to avoid malfunction	$f_{UART} > f_{BUS} / 2.5$	N/A Do not use	$f_{UART} > f_{BUS} / 3.0$	$f_{UART} > f_{BUS} / 2.0$	$f_{UART} < f_{BUS} \times 16 \times \{DLM, DLL\}$ Bus clock must be faster than baud rate			

27.2 Register Description

Table 27.1 UART Register Map
(UART0 Base Address = 0x90007000)
(UART1 Base Address = 0x90008000)
(UART2 Base Address = 0x90009000)
(UART3 Base Address = 0x9000A000)
(UART4 Base Address = 0x90015000)

Name	Address	Type	Reset	Description
RBR	0x00	R	Unknown	Receiver Buffer Register (DLAB = 0)
THR	0x00	W	0x00	Transmitter Holding Register (DLAB=0)
DLL	0x00	R/W	0x00	Divisor Latch (LSB) (DLAB=1)
IER	0x04	R/W	0x00	Interrupt Enable Register (DLAB=0)
DLM	0x04	R/W	0x00	Divisor Latch (MSB) (DLAB=1)
IIR	0x08	R	Unknown	Interrupt Ident. Register (DLAB=0)
FCR	0x08	W	0xC0	FIFO Control Register (DLAB=1)
LCR	0x0C	R/W	0x03	Line Control Register
MCR	0x10	R/W	0x40	MODEM Control Register
LSR	0x14	R	Unknown	Line Status Register
MSR	0x18	R	Unknown	MODEM Status Register
SCR	0x1C	R/W	0x00	Scratch Register
AFC	0x20	R/W	0x00	AFC Trigger Level Register
UCR	0x24	R/W	0x00	UART Control Register
SRBR	0x40	R	Unknown	Rx Buffer Register
STHR	0x44	W	0x00	Transmitter Holding Register
SDLL	0x48	R/W	0x00	Divisor Latch (LSB)
SDLM	0x4C	R/W	0x00	Divisor Latch (MSB)
SIER	0x50	R/W	0x00	Interrupt Enable Register
SCCR	0x60	R/W	0x00	SmartCard Configuration Register
STC	0x64	R/W	0x00	SmartCard Tx Count
IRCFG	0x80	R/W	0x00	Reserved

UART Receiver Buffer Register (RBR)**0x900nn000(DLAB=0) (nn=07/08/09/0A/15)**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Received Data[7:0] (when read with DLAB=0)							

The RBR is an actual 16byte FIFO, a received data from external device is stored in the RBR and CPU (or DMA) can read this register by Rx interrupt (or Rx DMA Request).

UART Transmitter Holding Register (THR)**0x900nn000(DLAB=0) (nn=07/08/09/0A/15)**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Transmit Data[7:0] (when written with DLAB=0)							

The THR is an actual 16byte FIFO. To transmit data to external device, CPU (or DMA) should write data to the THR.

UART Divisor Latch Register (DLL)**0x900nn000(DLAB=1) (nn=07/08/09/0A/15)**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Divisor Latch LSB[7:0] (when accessed with DLAB=1)							

UART Divisor Latch Register (DLM)**0x900nn004(DLAB=1) (nn=07/08/09/0A/15)**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Divisor Latch MSB[7:0] (when accessed with DLAB=1)							

These registers are for generation of the desired baud rate clock. The value can be calculated as follows.

$$\{DLM, DLL\} = f_{UART} / (16 * \text{desired baud rate})$$

For example,

If UART clock frequency is 48MHz (from CKC) and the baud rate you want is 115,200 bps, the divisor value should be $26(48M / (115200 \times 16))$ in decimal.

UART Interrupt Register (IER)**0x900nn004(DLAB=0) (nn=07/08/09/0A/15)**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0												EMSI	ELSI	ETXI	ERXI

Bit	Name	Type	Reset	Description
3	EMSI	R/W	0	Enable MODEM status interrupt 1=Enable 0=Disable
2	ELSI	R/W	0	Enable receiver line status interrupt 1=Enable 0=Disable This field cannot detect parity error interrupt. (Just use checking the parity error status.)
1	ETXI	R/W	0	Enable transmitter holding register empty interrupt 1=Enable 0=Disable
0	ERXI	R/W	0	Enable received data available interrupt 1=Enable 0=Disable

UART Interrupt Ident. Register (IIR)

0x900nn008 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0				STF	0										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0												IID[2:0]		IPF	

Bit	Name	Type	Reset	Description			
27	STF	R	X	SmartCard Tx done Flag. This bit is effective when the smart card interface enabled.			
3:1	IID	R	X	IID	Interrupt ID	Interrupt Source	Interrupt Clear
				011	Receiver line status	Overrun / parity error / framing error / break error	Reading the line status register
				010	Received data available	Receiver data available or trigger level reached	Reading the receiver buffer register or the FIFO drops below the trigger level
				110	Character timeout indication	No characters have been removed from or input to the RX FIFO during the last 4 char. Times and there is at least 1 char in it during this time.	Reading the receiver buffer register
				001	Transmitter holding register empty	Transmitter holding register empty	Reading the IIR register(if source of interrupt) or writing into the transmitter holding register
				000	MODEM status	Clear to send data set ready or ring indicator or data carrier detect	Reading the MODEM status register
0	IPF	R	X	Interrupt Flag			
				0	Interrupt pending		
				1	Interrupt is not generated		

UART FIFO Control Register (FCR)

0x900nn008 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								RXT[1:0]		TXT[1:0]		DRTE	TXFR	RXFR	FE

Bit	Name	Type	Reset	Description															
7:6	RXT	W	0x3	RX FIFO Trigger Level															
				<table><tr><th>RXT</th><th>Bytes</th></tr><tr><td>00</td><td>1</td></tr><tr><td>01</td><td>4</td></tr><tr><td>10</td><td>8</td></tr><tr><td>11</td><td>14</td></tr></table>	RXT	Bytes	00	1	01	4	10	8	11	14					
				RXT	Bytes														
				00	1														
				01	4														
				10	8														
11	14																		
5:4	TXT	W	0	TX FIFO Trigger Level is used to set the FIFO threshold. When the Tx FIFO count decreases to the value selected below, the Tx interrupt or DMA Tx request is asserted and kept asserted until the Tx FIFO is filled above the threshold value.															
				<table><tr><th>TXT</th><th>Bytes (Threshold)</th><th>Number of bytes can be written</th></tr><tr><td>00</td><td>0 (when empty)</td><td>16</td></tr><tr><td>01</td><td>8</td><td>8</td></tr><tr><td>10</td><td>12</td><td>4</td></tr><tr><td>11</td><td>15</td><td>1</td></tr></table>	TXT	Bytes (Threshold)	Number of bytes can be written	00	0 (when empty)	16	01	8	8	10	12	4	11	15	1
				TXT	Bytes (Threshold)	Number of bytes can be written													
				00	0 (when empty)	16													
				01	8	8													
				10	12	4													
11	15	1																	
3	DRTE	W	0	DMA Rx/Tx Enable. "1" enables both TX and RX requests.															
				<table><tr><td>0</td><td>DMA transfer is dependent on RxDE or TxDE bits</td></tr><tr><td>1</td><td>Enable both Rx & Tx DMA transfer (Regardless of RxDE or TxDE status)</td></tr></table>	0	DMA transfer is dependent on RxDE or TxDE bits	1	Enable both Rx & Tx DMA transfer (Regardless of RxDE or TxDE status)											
				0	DMA transfer is dependent on RxDE or TxDE bits														
1	Enable both Rx & Tx DMA transfer (Regardless of RxDE or TxDE status)																		
2	TXFR	W	0	TX FIFO Reset. Reset TX FIFO counter and FIFO data															
1	RXFR	W	0	RX FIFO Reset. Reset RX FIFO counter and FIFO data															
0	FE	W	0	FIFO Enable. Enable TX and RX FIFOs.															

UART Line Control Register (LCR)

0x900nn00C (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								DLAB	SB	SP	EPS	PEN	STB	WLS[1:0]	

Bit	Name	Type	Reset	Description		
7	DLAB	R/W	0	Divisor Latch Access		
				0	Access the receiver buffer, the transmitter holding register, or the interrupt enable register	
				1	Access the divisor latches of the baud generator	
6	SB	R/W	0	Set Break		
				0	Disable the break	
				1	The serial output is forced to the spacing(logic 0) state	
5	SP	R/W	0	Sticky Parity Enable. 1=Enable 0=Disable If PEN=1 and this bit is enabled, inverted EPS value is used for parity bit value. This bit has no effect if PEN=0.		
4	EPS	R/W	0	Even Parity Select. 1=Even 0=Odd		
3	PEN	R/W	0	Parity Enable. 1=Enable 0=Disable		
2	STB	R/W	0	Number of Stop Bits		
				STB	Word Length	Selected
				0	X	One stop bit
				1	5 bits	One and a half stop bits
				1	6 bits	Two stop bits
				1	7 bits	
1	8 bits					
1:0	WLS	R/W	0	Word Length Select		
				00	5 bits	
				01	6 bits	
				10	7 bits	
				11	8 bits	

UART Modem Control Register (MCR)

0x900nn010 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0									RS	AFE	LOOP	0	0	RTS	0

Bit	Name	Type	Reset	Description		
6	RS	R/W	1	RTS Deassert Condition Control		
				0	RTS is de-asserted at the Rx Stop Condition	
				1	RTS is de-asserted at the Rx Start Condition (Reset value)	
5	AFE	R/W	0	Auto Flow Control Enable. 1=Enable 0=Disable		
4	LOOP	R/W	0	Loop Back Enable. 1=Enable 0=Disable		
3:2	-	R	0	Reserved		
1	RTS	R/W	0	Request To Send Line Control.		
				AFE	RTS	RTS / CTS Configuration
				0	0	RTS is kept inactive
				0	1	RTS is kept active
				1	0	Auto flow control enabled with CTS only. (RTS is kept inactive)
				1	1	Auto flow control enabled with both CTS and RTS.
0	-	R	0	Reserved		

UART Line Status Register (LSR)

0x900nn014 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								ERF	TEMT	THRE	BI	FE	PE	OE	DR

Bit	Name	Type	Reset	Description
7	ERF	R	X	Divisor Latch Access. This bit is set when there is at least one parity error, framing error or break indication in the FIFO.
6	TEMT	R	X	Transmitter Empty indicates that both the transmitter holding register and the transmitter shift register are empty.
5	THRE	R	X	Transmitter Holding Register Empty indicates UART is ready to accept a new char for transmission.
4	BI	R	X	Break Interrupt indicates that the received data input is held in the spacing (logic 0) state for longer than a full word transmission time.
3	FE	R	X	Framing Error indicates that the received character did not have a valid stop bit.
2	PE	R	X	Parity Error indicates that the received data character does not have the correct even or odd parity. This field cannot generate the interrupt source. Just use the status source.
1	OE	R	X	Overrun Error indicates that the receiver buffer register is not read by the CPU before the next character is transferred into the receiver buffer register.
0	DR	R	X	Data Ready indicates that the receiver data is ready.

UART Modem Status Register(MSR)

0x900nn018 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								0	0	0	CTS	0	0	0	DCTS

Bit	Name	Type	Reset	Description
4	CTS	R	0	Clear To Send This bit is the complement of the Clear to Send input
0	DCTS	R	0	Delta Clear to Send indicates CTS input change.

UART Scratch Register (SCR)

0x900nn01C (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Scratch Data[7:0]							

This 8-bit read/write register is for compatibility. It is intended to be used as a scratchpad register.

UART AFC Trigger Level Register (AFT)

0x900nn020 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								ATL[3:0]				DTL[3:0]			

Bit	Name	Type	Reset	Description
7:4	ATL	R/W	0	RTS Assert Trigger Level
3:0	DTL	R/W	0	RTS Deassert Trigger Level

UART Control Register (UCR)

0x900nn024 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0										POLC	POLR	RWA	TWA	RxDE	TxDE

Bit	Name	Type	Reset	Description
5	POLC	R/W	0	Polarity Control for CTS
				0 CTS is Active Low (Not inverted)
				1 CTS is Active High (Inverted)
4	POLR	R/W	0	Polarity Control for RTS
				0 RTS is Active Low (Not inverted)
				1 RTS is Active High (Inverted)
3	RWA	R/W	0	Rx Word Access enables word access (4bytes) to Rx FIFO
2	TWA	R/W	0	Tx Word Access enables word access (4bytes) to Tx FIFO
1	RxDE	R/W	0	Rx DMA Enable
0	TxDE	R/W	0	Tx DMA Enable

UART Receiver Buffer Register (SRBR)

0x900nn040 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Received Data (Read Only)							

UART Transmitter Holding Register (STHR)

0x900nn044 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Transmitting Data (Write Only)							

UART Divisor Latch Register (SDLL)

0x900nn048 (nn=07/08/09/0A/15/1D)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Divisor Latch LSB							

UART Divisor Latch Register (SDLM)

0x900nn04C (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								Divisor Latch MSB							

UART Interrupt Register (SIER)

0x900nn050 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0												EMSI	ELSI	ETXI	ERXI

SRBR, STHR, SDLL, SDLM, SIER registers are copy of the RBR, THR, DLL, DLM, IER registers. These registers can be accessed without concern of DLAB state.

SmartCard Configuration Register (SCCR)

0x900nn060 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SEN	STEN	DEN	P	STF	STE	0									
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV [15:0]															

Bit	Name	Type	Reset	Description
31	SEN	R/W	0	SmartCard Interface Enable
30	STEN	R/W	0	SmartCard Tx Enable
29	DEN	R/W	0	SmartCard Clock Divider Enable
28	P	R/W	0	SmartCard Clock Post Divider Enable
27	STF	R/W	0	SmartCard Tx done Flag Cleared by writing "1".
26	STE	R/W	0	SmartCard Tx done Interrupt Enable
25:16	-	R/W	0	Reserved
15:0	DIV	R/W	0	SmartCard Clock Divisor $F_{out} = F_{uart} / ((DIV+1) \times 2^P)$

SmartCard Tx Count (STC)

0x900nn064 (nn=07/08/09/0A/15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
C_CNT[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
S_CNT[15:0]															

Bit	Name	Type	Reset	Description
31:16	C_CNT	R	0	Current Tx Count When SmartCard Interface is enabled
15:0	S_CNT	R/W	0	Start Tx Count When SmartCard Interface is enabled

For smart card interface, the RXD signal is used as the smart card clock (SC_CLK) and the TXD signal is used as the smart card data (SC_DAT). All UART ports and channels are available for smart card interface.

27.3 Port Multiplexing

Before the UART signals are sent to the GPIO block for pin multiplexing, they are internally rearranged to have more flexibility in port configuration. The signal name GBUS[191:0] found in GPIO Function Selection Registers (GPIO*_FS* of GPIO block) is the final output of the UART port multiplexers in this module.

Any UART signal can be mapped to pre-mapped bit position of GBUS[191:0] by writing and index number to the register below in TCC8132. Pre-mapped bit position of GBUS[191:0] is shown in the table below.

Table 27.2 Pre-mapped bit position for UART Port

GBUS	GPIO
GBUS[112:113]	LD[22:23]
GBUS[71:74]	PD[15:18]
GBUS[190:191]	PK[30:31]

Table 27.3 UART Port Selection Register Map (Base Address = 0x90007800)

Name	Address	Type	Reset	Description
UART0PORTSEL	0x00	R/W	0x23222120	UART0 Port Selection Register
UART1PORTSEL	0x04	R/W	0x27262524	UART1 Port Selection Register
UART2PORTSEL	0x08	R/W	0x2B2A2928	UART2 Port Selection Register
UART3PORTSEL	0x0C	R/W	0x2F2E2D2C	UART3 Port Selection Register
UART4PORTSEL	0x10	R/W	0x33323130	UART4 Port Selection Register

UART0 Port Selection Register (UART0PORTSEL)

0x90007800

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x23	Port number for RTS
23:16	CTS	R/W	0x22	Port number for CTS
15:8	RXD	R/W	0x21	Port number for RXD
7:0	TXD	R/W	0x20	Port number for TXD

UART1 Port Selection Register (UART1PORTSEL)

0x90007804

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x27	Port number for RTS
23:16	CTS	R/W	0x26	Port number for CTS
15:8	RXD	R/W	0x25	Port number for RXD
7:0	TXD	R/W	0x24	Port number for TXD

UART2 Port Selection Register (UART2PORTSEL)

0x90007808

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x2B	Port number for RTS
23:16	CTS	R/W	0x2A	Port number for CTS
15:8	RXD	R/W	0x29	Port number for RXD
7:0	TXD	R/W	0x28	Port number for TXD

UART3 Port Selection Register (UART3PORTSEL)

0x9000780C

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x2F	Port number for RTS
23:16	CTS	R/W	0x2E	Port number for CTS
15:8	RXD	R/W	0x2D	Port number for RXD
7:0	TXD	R/W	0x2C	Port number for TXD

UART4 Port Selection Register (UART4PORTSEL)

0x90007810

Bit	Name	R/W	Reset	Description
31:24	RTS	R/W	0x33	Port number for RTS
23:16	CTS	R/W	0x32	Port number for CTS
15:8	RXD	R/W	0x31	Port number for RXD
7:0	TXD	R/W	0x30	Port number for TXD

27.4 UART Timing

TXD Timing

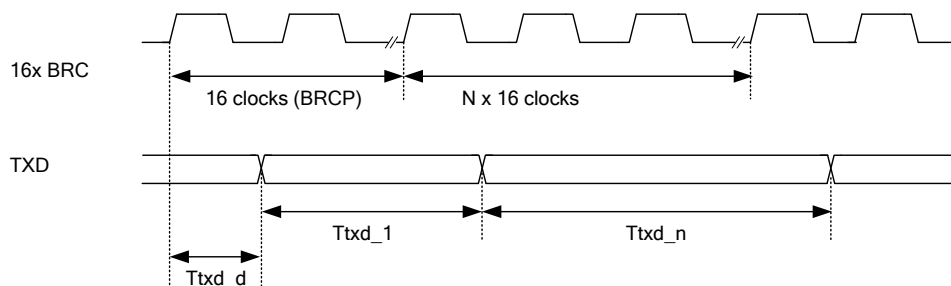


Figure 27.2 UART TXD Timing

RXD Timing

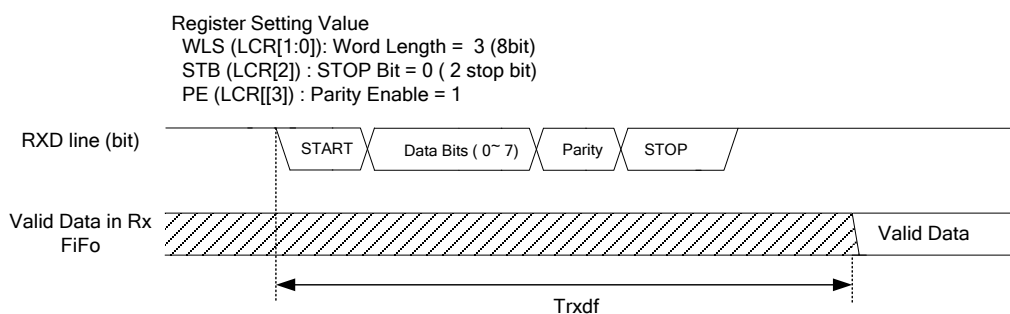


Figure 27.3 UART RXD Timing

Tx Operation with Hardware Auto Flow Control

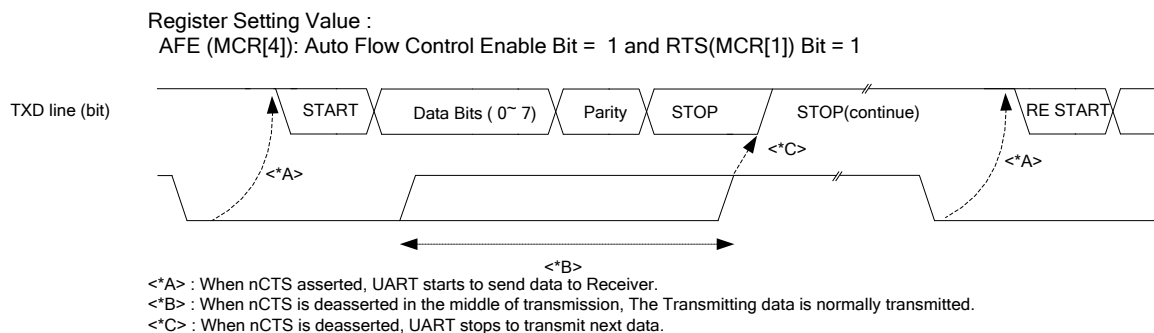


Figure 27.4 UART Tx WaveForm

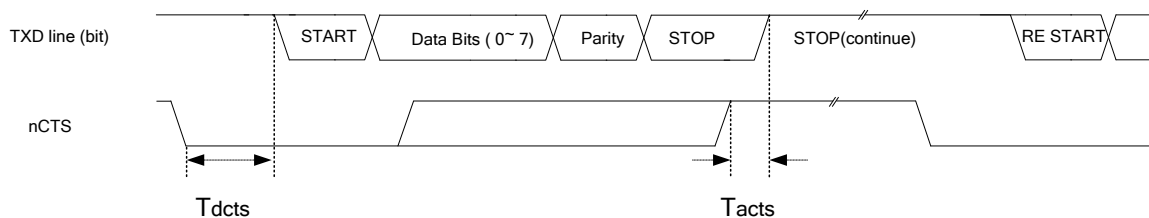
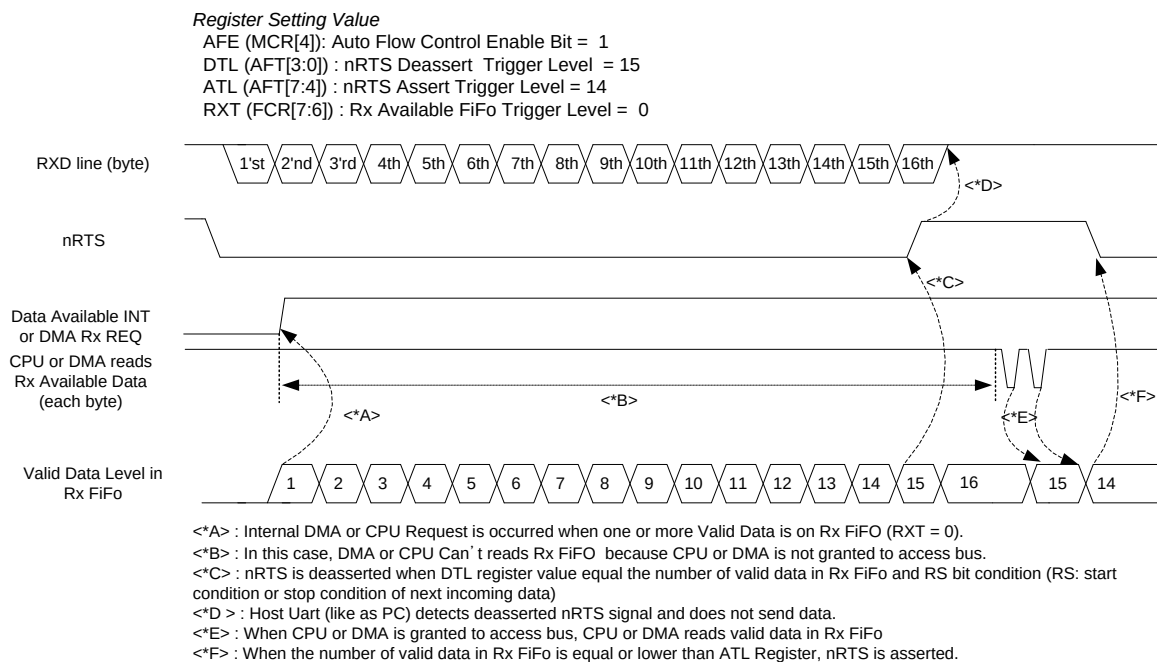
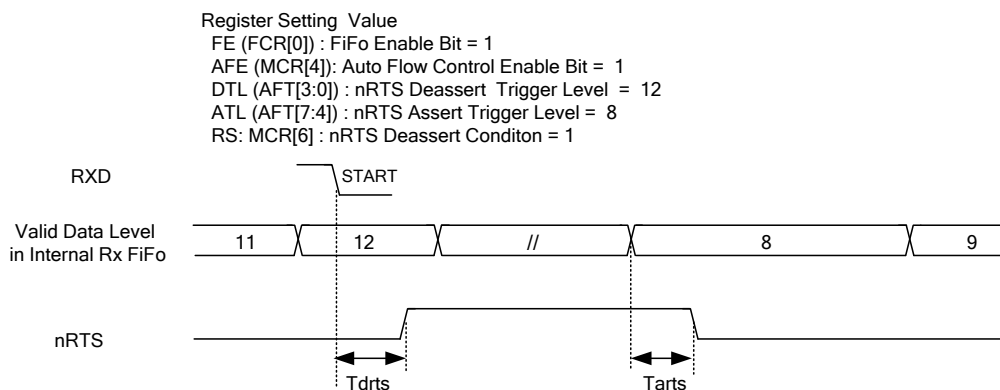


Figure 27.5 UART Tx nCTS Timing

Rx Operation with Hardware Auto Flow Control**Figure 27.6 UART Rx WaveForm****Figure 27.7 UART Rx nRTS Timing**

Parameter	Symbol	1.8V		3.3V		Unit
		MIN	MAX	MIN	MAX	
Pulse duration of 1bit TXD	Ttxd_1	BRCP -30	BRCP+30	BRCP-15	BRCP+15	ns
Pulse duration of nbit TXD	Ttxd_n	N x BRCP-30	N X BRCP+30	N x BRCP-15	N X BRCP+15	ns
TXD output delay time	Ttxd_d	0.5	30	0.5	15	ns
RXD Start to Rx FiFo	Trxdf	10.5*BRCP	10.5*BRCP	10.5*BRCP	11*BRCP	ns
Deasserted nCTS to Tx Start	Tdcts	-	BRCP	-	BRCP	ns
Deasserted nCTS to Tx Stop :to stop next transmission(setup time)	Tacts	4 x BRCP/16	-	4 x BRCP/16	-	ns
DTL(and RXD start condition) to deasserted nRTS	Tdrts	-	BRCP	-	BRCP	ns
ATL to asserted nRTS	Tarts	-	BRCP/16	-	BRCP/16	ns

Note:

BRC: Baud Rate Clock.
 BRCP: Baud Rate Clock Period.
 CL: 30 pF

28 CAN Controller

28.1 Overview

The CAN performs communication according to ISO 11898-1 (Identical to Bosch CAN protocol specification 2.0 part A,B) and according to ISO 11898-4 (Time triggered communication on CAN). In addition, the CAN supports communication according to CAN FD protocol specification 1.0.

All functions concerning the handling of messages are implemented by the Rx Handler and the Tx Handler. The Rx Handler manages message acceptance filtering, the transfer of received messages from the CAN Core to the Message RAM as well as providing receive message status information.

The Tx Handler is responsible for the transfer of transmit messages from the Message RAM to the CAN Core as well as providing transmit status information. It implements all functions concerning the time schedule and the global system time.

Acceptance filtering is implemented by a combination of up to 128 filter elements where each one can be configured as a range, as a bit mask, or as a dedicated ID filter.

The TCC8132 supports two CAN controllers that have the following features.

- Conform with CAN protocol version 2.0 part A, B and ISO 11898-1, -4
- CAN FD with up to 64 data bytes supported Wake-up (PMWKUP) signal generation: support on the power down mode (PWDN)
- CAN Error Logging
- AUTOSAR support
- SAE J1939 support
- Improved acceptance filtering
- Two configurable Receive FIFOs
- Separate signaling on reception of High Priority Messages
- Up to 64 dedicated Receive Buffers
- Up to 32 dedicated Transmit Buffers
- Configurable Transmit FIFO
- Configurable Transmit Queue
- Configurable Transmit Event FIFO
- Direct Message RAM access for Host CPU
- Multiple M_TTCANs may share the same Message RAM
- Programmable loop-back test mode
- Maskable module interrupts
- 8/16/32 bit Generic Slave Interface for connection customer-specific Host CPUs
- Two clock domains (CAN clock and Host clock)
- Power-down support
- Debug on CAN support

Caution:

Do not access 0x10000000. If you want to access SRAM, the address of SRAM is mapped to 0x30000000 (mirror address)

28.2 Function Description

28.2.1 Operation modes

28.2.1.1 Software Initialization

Software initialization is started by setting bit **CCCR.INIT**, either by software or by a hardware reset, when an uncorrected bit error was detected in the Message RAM, or by going *Bus_Off*. While **CCCR.INIT** is set, message transfer from and to the CAN bus is stopped, the status of the CAN bus output **m_can_tx** is *recessive* (HIGH). The counters of the Error Management Logic EML are unchanged. Setting **CCCR.INIT** does not change any configuration register. Resetting **CCCR.INIT** finishes the software initialization. Afterwards the Bit Stream Processor BSP synchronizes itself to the data transfer on the CAN bus by waiting for the occurrence of a sequence of 11 consecutive *recessive* bits (α *Bus_Idle*) before it can take part in bus activities and start the message transfer.

Access to the M_CAN configuration registers is only enabled when both bits **CCCR.INIT** and **CCCR.CCE** are set (protected write).

CCCR.CCE can only be set/reset while **CCCR.INIT** = '1'. **CCCR.CCE** is automatically reset when **CCCR.INIT** is reset.

The following registers are reset when **CCCR.CCE** is set

- **HPMS** - High Priority Message Status
- **RXF0S** - Rx FIFO 0 Status
- **RXF1S** - Rx FIFO 1 Status
- **TXFQS** - Tx FIFO/Queue Status
- **TXBRP** - Tx Buffer Request Pending
- **TXBTO** - Tx Buffer Transmission Occurred
- **TXBCF** - Tx Buffer Cancellation Finished
- **TXEFS** - Tx Event FIFO Status

The Timeout Counter value **TOCV.TOC** is preset to the value configured by **TOCC.TOP** when **CCCR.CCE** is set. In addition, the state machines of the Tx Handler and Rx Handler are held in idle state while **CCCR.CCE** = '1'.

The following registers are only writeable while **CCCR.CCE** = '0'

- **TXBAR** - Tx Buffer Add Request
- **TXBCR** - Tx Buffer Cancellation Request

CCCR.TEST and **CCCR.MON** can only be set by the Host while **CCCR.INIT** = '1' and **CCCR.CCE** = '1'. Both bits can be reset at any time. **CCCR.DAR** can be set/reset only while **CCCR.INIT** = '1' and **CCCR.CCE** = '1'.

28.2.1.2 Normal Operation

Once the M_CAN is initialized and **CCCR.INIT** is reset to *zero*, the M_CAN synchronizes itself to the CAN bus and is ready for communication.

After passing the acceptance filtering, received messages including Message ID and DLC are stored into a dedicated Rx Buffer or into Rx FIFO 0 or Rx FIFO 1.

For messages to be transmitted dedicated Tx Buffers and/or a Tx FIFO or a Tx Queue can be initialized or updated. Automated transmission on reception of remote frames is not implemented.

28.2.1.3 CAN FD Operation

There are two variants in the CAN FD frame transmission, the first variant is CAN FD frame without bit rate switching. The second variant is CAN FD frame where control field, data field, and CRC field are transmitted with a higher bit rate than the beginning and the end of the frame.

The previously reserved bit in CAN frames with 11-bit identifiers and the first previously reserved bit in CAN frames with 29-bit identifiers will now be decoded as **FDF** bit.

- **FDF** = *recessive* signifies a CAN FD frame
- **FDF** = *dominant* signifies a Classic CAN frame

In a CAN FD frame, the two bits following **FDF**, **res** and **BRS**, decide whether the bit rate inside of this CAN FD frame is switched. A CAN FD bit rate switch is signified by **res** = *dominant* and **BRS** = *recessive*. The coding of **res** = *recessive* is reserved for future expansion of the protocol.

In case that the M_CAN receives a frame with **FDF** = *recessive* and **res** = *recessive*, it will signal a Protocol Exception Event by setting bit **PSR.PXE**. When Protocol Exception Handling is enabled (**CCCR.PXHD** = '0'), this causes the operation state to change from Receiver (**PSR.ACT** = "10") to Integrating (**PSR.ACT** = "00") at the next sample point.

In case that Protocol Exception Handling is disabled (**CCCR.PXHD** = '1'), the M_CAN will treat a *recessive* **res** bit as an form error and will respond with an error frame.

CAN FD operation is enabled by programming **CCCR.FDOE**.

In case **CCCR.FDOE** = '1', transmission and reception of CAN FD frames are enabled. The transmission and reception of Classic CAN frames are always possible. Whether to transmit CAN FD frame or a Classic CAN frame can be configured via bit **FDF** in the respective Tx Buffer element.

With **CCCR.FDOE** = '0', received frames are interpreted as Classic CAN frames, which lead to the transmission of an error frame when receiving a CAN FD frame. When CAN FD operation is disabled, no CAN FD frames are transmitted even if bit **FDF** of a Tx Buffer element is set. **CCCR.FDOE** and **CCCR.BRSE** can change only while **CCCR.INIT** and **CCCR.CCE** are both set.

With **CCCR.FDOE** = '0', the setting of bits **FDF** and **BRS** is ignored and frames are transmitted in Classic CAN format. With **CCCR.FDOE** = '1' and **CCCR.BRSE** = '0', only bit **FDF** of a Tx Buffer element is evaluated. With **CCCR.FDOE** = '1' and **CCCR.BRSE** = '1', transmission of CAN FD frames with bit rate switching is enabled. All Tx Buffer elements with bits **FDF** and **BRS** set are transmitted in CAN FD format with bit rate switching.

A mode change during CAN operation is only recommended under the following conditions:

- The failure rate in the CAN FD data phase is significant higher than in the CAN FD arbitration phase. In this case disable

the CAN FD bit rate switching option for transmissions.

- During system startup all nodes are transmitting Classic CAN messages until it is verified that they are able to communicate in CAN FD format. If this is true, all nodes switch to CAN FD operation.
- Wake-up messages in CAN Partial Networking have to be transmitted in Classic CAN format.
- End-of-line programming in case that not all nodes are CAN FD capable. Non-CAN FD nodes are held in Silent mode until programming has completed. Then all nodes switch back to Classic CAN communication.

In the CAN FD format, the coding of the DLC differs from the standard CAN format. The DLC codes 0 to 8 have the same coding as in standard CAN, the codes 9 to 15, which in standard CAN all code a data field of 8 bytes, are coded according to Table 28.1 below.

Table 28.1 Coding of DLC in CAN FD

DLC	9	10	11	12	13	14	15
Number of Data Bytes	12	16	20	24	32	48	64

In CAN FD frames, the bit timing will be switched inside the frame, after the **BRS** (Bit Rate Switch) bit, if this bit is *recessive*. Before the **BRS** bit, in the CAN FD arbitration phase, the nominal CAN bit timing is used as defined by the Nominal Bit Timing & Prescaler Register **NBTP**. In the following CAN FD data phase, the data phase bit timing is used as defined by the Data Bit Timing & Prescaler Register **DBTP**. The bit timing is switched back from the data phase timing at the CRC delimiter or when an error is detected, whichever occurs first.

The maximum configurable bit rate in the CAN FD data phase depends on the CAN clock frequency (**m_can_cclk**).

Example:

With a CAN clock frequency of 20MHz and the shortest configurable bit time of 4 tq, the bit rate in the data phase is 5 Mbit/s.

In both data frame formats, CAN FD and CAN FD with bit rate switching, the value of the bit **ESI** (Error Status Indicator) is determined by the transmitter's error state at the start of the transmission. If the transmitter is error passive, **ESI** is transmitted *recessive*, else it is transmitted *dominant*.

28.2.1.4 Transmitter Delay Compensation

During the data phase of a CAN FD transmission only one node is transmitting, all others are receivers. The length of the bus line has no impact. When transmitting via pin **m_can_tx** the M_CAN receives the transmitted data from its local CAN transceiver via pin **m_can_rx**. The received data is delayed by the transmitter delay. In case that this delay is greater than TSEG1 (time segment before sample point), a bit error is detected. In order to enable a data phase bit time that is even shorter than the transmitter delay, the delay compensation is introduced. Without transmitter delay compensation, the bit rate in the data phase of a CAN FD frame is limited by the transmitter delay.

28.2.1.4.1 Description

The M_CAN's protocol unit has implemented a delay compensation mechanism to compensate the transmitter delay, thereby enabling transmission with higher bit rates during the CAN FD data phase independent of the delay of a specific CAN transceiver.

To check for bit errors during the data phase of transmitting nodes, the delayed transmit data is compared against the received data at the Secondary Sample Point SSP. If a bit error is detected, the transmitter will react on this bit error at the next following regular sample point. During arbitration phase the delay compensation is always disabled.

The transmitter delay compensation enables configurations where the data bit time is shorter than the transmitter delay, it is described in detail in ISO 11898-1:2015. It is enabled by setting bit **DBTP.TDC**.

The received bit is compared against the transmitted bit at the SSP. The SSP position is defined as the sum of the measured delay from the M_CAN's transmit output **m_can_tx** through the transceiver to the receive input **m_can_rx** plus the transmitter delay compensation offset as configured by **TDCR.TDCO**. The transmitter delay compensation offset is used to adjust the position of the SSP inside the received bit (e.g. half of the bit time in the data phase). The position of the secondary sample point is rounded down to the next integer number of mtq.

PSR.TDCV shows the actual transmitter delay compensation value. **PSR.TDCV** is cleared when **CCCR.INIT** is set and is updated at each transmission of an FD frame while **DBTP.TDC** is set.

28.2.1.4.2 Transmitter Delay Compensation Measurement

If transmitter delay compensation is enabled by programming **DBTP.TDC** = '1', the measurement is started within each transmitted CAN FD frame at the falling edge of bit **FDF** to bit **res**. The measurement is stopped when this edge is seen at the receive input **m_can_rx** of the transmitter. The resolution of this measurement is one mtq.

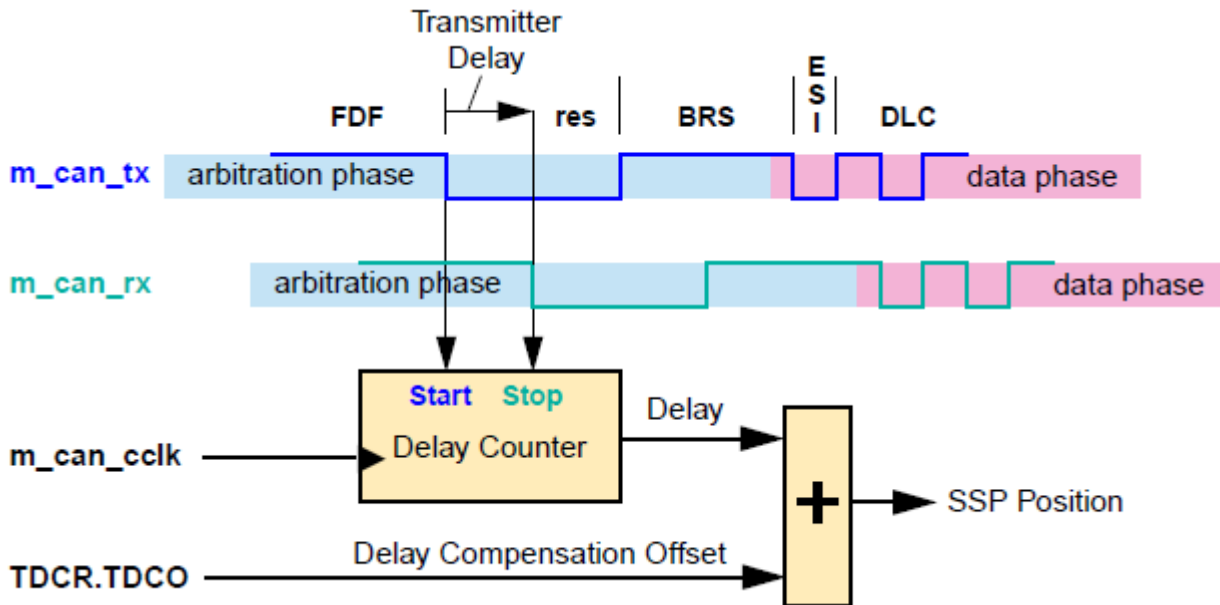


Figure 28.1 Transmitter Delay Measurement

To avoid that a dominant glitch inside the received **FDF** bit ends the delay compensation measurement before the falling edge of the received **res** bit, resulting in a too early SSP position, the use of a transmitter delay compensation filter window can be enabled by programming **TDCR.TDCF**. This defines a minimum value for the SSP position. Dominant edges on **m_can_rx**, that would result in an earlier SSP position are ignored for transmitter delay measurement. The measurement stops when the SSP position becomes low by arithmetic with **TDCR.TDCF** AND **m_can_rx** at least.

28.2.1.5 Restricted Operation Mode

In Restricted Operation Mode the node is able to receive data and remote frames and to give acknowledge to valid frames, but it does not send data frames, remote frames, active error frames, or overload frames. In case of an error condition or overload condition, it does not send dominant bits, instead it waits for the occurrence of bus idle condition to resynchronize itself to the CAN communication. The error counters (**ECR.REC**, **ECR.TEC**) are frozen while Error Logging (**ECR.CEL**) is active. The Host can set the **M_CAN** into Restricted Operation mode by setting bit **CCCR.ASM**. The bit can only be set by the Host when both **CCCR.CCE** and **CCCR.INIT** are set to '1'. The bit can be reset by the Host at any time. Restricted Operation Mode is automatically entered when the Tx Handler was not able to read data from the Message RAM in time. To leave Restricted Operation Mode, the Host CPU has to reset **CCCR.ASM**. The Restricted Operation Mode can be used in applications that adapt themselves to different CAN bit rates. In this case, the application tests different bit rates and leaves the Restricted Operation Mode after it has received a valid frame. If the **M_CAN** is connected to a Clock Calibration on CAN unit, **CCCR.ASM** is controlled by input **m_can_cok**. In case **m_can_cok** switches to '0', bit **CCCR.ASM** is set. When **m_can_cok** switches back to '1', bit **CCCR.ASM** returns to the previously written value. When there is no Clock Calibration on CAN unit, connected input **m_can_cok** is hardwired to '1'.

Note: The Restricted Operation Mode must not be combined with the Loop Back Mode (Internal or external).

28.2.1.6 Bus Monitoring Mode

The **M_CAN** is set in Bus Monitoring Mode by programming **CCCR.MON** to one. In Bus Monitoring Mode (See ISO 11898-1:2015, 10.14 Bus monitoring), the **M_CAN** is able to receive valid data frames and valid remote frames, but cannot start a transmission. In this mode, it sends only *recessive* bits on the CAN bus. If the **M_CAN** is required to send a *dominant* bit (ACK bit, overload flag, active error flag), the bit is rerouted internally so that the **M_CAN** monitors this *dominant* bit, although the CAN bus may remain in *recessive* state. In Bus Monitoring Mode register **TXBRP** is held in reset state. The Bus Monitoring Mode can be used to analyze the traffic on a CAN bus without affecting it by the transmission of *dominant* bits. Figure 28.2 function description shows the connection of signals **m_can_tx** and **m_can_rx** to the **M_CAN** in Bus Monitoring Mode.

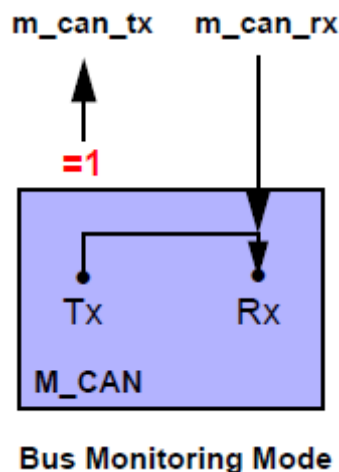


Figure 28.2 Pin control in Bus Monitoring Mode

28.2.1.7 Disabled Automatic Retransmission

According to the CAN Specification (see ISO 11898-1:2015, 8.3.4 Recovery Management), the M_CAN provides means for automatic retransmission of frames that have lost arbitration or that have been disturbed by errors during transmission. By default automatic retransmission is enabled. To support time-triggered communication as described in ISO 11898-1:2015, Chapter 28.2 Function Description may be disabled via **CCCR.DAR**.

28.2.1.7.1 Frame Transmission in DAR Mode

In DAR mode all transmissions are automatically cancelled after they started on the CAN bus. A Tx Buffer's Tx Request Pending bit **TXBRP.TRPx** is reset after successful transmission, when a transmission has not yet been started at the point of cancellation, has been aborted due to lost arbitration, or when an error occurred during frame transmission.

- Successful transmission:
 - Corresponding Tx Buffer Transmission Occurred bit **TXBTO.TOx** set
 - Corresponding Tx Buffer Cancellation Finished bit **TXBCF.CFx** not set
- Successful transmission in spite of cancellation:
 - Corresponding Tx Buffer Transmission Occurred bit **TXBTO.TOx** set
 - Corresponding Tx Buffer Cancellation Finished bit **TXBCF.CFx** set
- Arbitration lost or frame transmission disturbed:
 - Corresponding Tx Buffer Transmission Occurred bit **TXBTO.TOx** not set
 - Corresponding Tx Buffer Cancellation Finished bit **TXBCF.CFx** set

In case of a successful frame transmission, and if storage of Tx events is enabled, a Tx Event FIFO element is written with Event Type **ET** = "10" (transmission in spite of cancellation).

28.2.1.7.2 Power Down (Sleep Mode)

The M_CAN can be set into power down mode controlled by input signal **m_can_clkstop_req** or via CC Control Register **CCCR.CSR**. As long as the clock stop request signal **m_can_clkstop_req** is active, bit **CCCR.CSR** is read as *one*. When all pending transmission requests have completed, the M_CAN waits until bus idle state is detected. Then the M_CAN sets then **CCCR.INIT** to *one* to prevent any further CAN transfers. Now the M_CAN acknowledges that it is ready for power down by setting output signal **m_can_clkstop_ack** to *one* and **CCCR.CSA** to *one*. In this state, before the clocks are switched off, further register accesses can be made. A write access to **CCCR.INIT** has no effect. Now the module clock inputs **m_can_hclk** and **m_can_cclk** can be switched off. To leave power down mode, the application has to turn on the module clocks before resetting signal **m_can_clkstop_req** resp. CC Control Register flag **CCCR.CSR**. The M_CAN will acknowledge this by resetting output signal **m_can_clkstop_ack** and resetting **CCCR.CSA**. Afterwards, the application can restart CAN communication by resetting bit **CCCR.INIT**.

28.2.1.8 Test Modes

To enable write access to register **TEST**, bit **CCCR.TEST** has to be set to *one*. This allows the configuration of the test modes and test functions.

Four output functions are available for the CAN transmit pin **m_can_tx** by programming **TEST.TX**. Additionally to its default function – the serial data output – it can drive the CAN Sample Point signal to monitor the M_CAN's bit timing and it can drive constant dominant or recessive values. The actual value at pin **m_can_rx** can be read from **TEST.RX**. Both functions can be used to check the CAN bus' physical layer.

Due to the synchronization mechanism between CAN clock and Host clock domain, there may be a delay of several Host clock periods between writing to **TEST.TX** until the new configuration is visible at output pin **m_can_tx**. This applies also when reading input pin **m_can_rx** via **TEST.RX**.

Note: Test modes should be used for production tests or self test only. The software control for pin **m_can_tx** interferes with all CAN protocol functions. It is not recommended to use test modes for application.

28.2.1.8.1 External Loop Back Mode

The M_CAN can be set in External Loop Back Mode by programming **TEST.LBCK** to *one*. In Loop Back Mode, the M_CAN treats its own transmitted messages as received messages and stores them (if they pass acceptance filtering) into an Rx Buffer or an Rx FIFO. Figure 28.3 shows the connection of signals **m_can_tx** and **m_can_rx** to the M_CAN in External Loop Back Mode.

This mode is provided for hardware self-test. To be independent from external stimulation, the M_CAN ignores acknowledge errors (recessive bit sampled in the acknowledge slot of a data/remote frame) in Loop Back Mode. In this mode the M_CAN performs an internal feedback from its Tx output to its Rx input. The actual value of the **m_can_rx** input pin is disregarded by the M_CAN. The transmitted messages can be monitored at the **m_can_tx** pin.

28.2.1.8.2 Internal Loop Back Mode

Internal Loop Back Mode is entered by programming bits **TEST.LBCK** and **CCCR.MON** to *one*. This mode can be used for a "Hot Self test", meaning the M_CAN can be tested without affecting a running CAN system connected to the pins **m_can_tx** and **m_can_rx**. In this mode pin **m_can_rx** is disconnected from the M_CAN and pin **m_can_tx** is held *recessive*. Figure 28.3 shows the connection of **m_can_tx** and **m_can_rx** to the M_CAN in case of Internal Loop Back Mode.

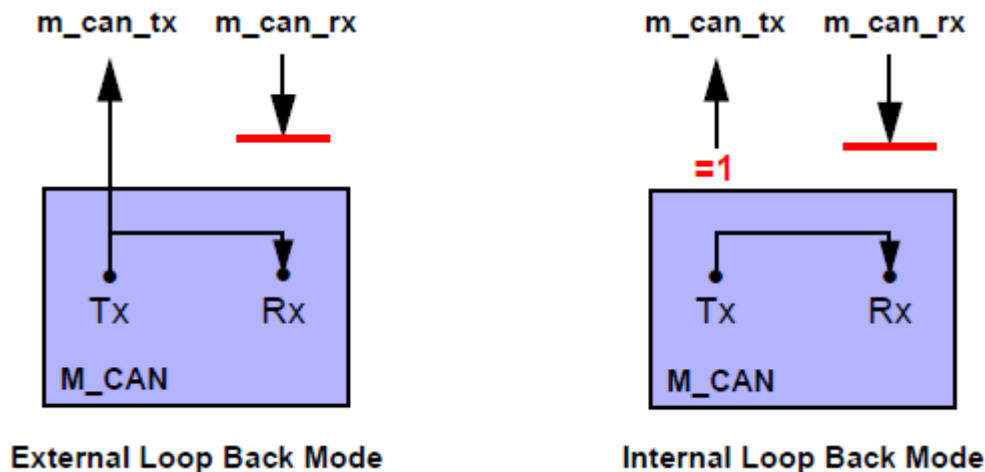


Figure 28.3 Pin Control in Loop Back Modes

28.2.2 Timestamp Generation

For timestamp generation the M_CAN supplies a 16-bit wrap-around counter. A prescaler **TSCC.TCP** can be configured to clock the counter in multiples of CAN bit times (1...16). The counter is readable via **TSCV.TSC**. A write access to register **TSCV** resets the counter to *zero*. When the timestamp counter wraps around interrupt flag **IR.TSW** is set.

On start of frame reception / transmission the counter value is captured and stored into the timestamp section of an Rx Buffer / Rx FIFO (**RXTS[15:0]**) or Tx Event FIFO (**TXTS[15:0]**) element.

By programming bit **TSCC.TSS** an external 16-bit timestamp can be used.

28.2.3 Timeout Counter

To signal timeout conditions for Rx FIFO 0, Rx FIFO 1, and the Tx Event FIFO the M_CAN supplies a 16-bit Timeout Counter. It operates as down-counter and uses the same prescaler controlled by **TSCC.TCP** as the Timestamp Counter.

The Timeout Counter is configured via register **TOCC**. The actual counter value can be read from **TOCV.TOC**. The Timeout Counter can start only while **CCCR.INIT** = '0'. It stops when **CCCR.INIT** = '1', e.g. when the M_CAN enters *Bus_Off* state. The operation mode is selected by **TOCC.TOS**. When operating in Continuous Mode, the counter starts when **CCCR.INIT** is reset. A write to **TOCV** presets the counter to the value configured by **TOCC.TOP** and continues down-counting. When the Timeout Counter is controlled by one of the FIFOs, an empty FIFO presets the counter to the value configured by **TOCC.TOP**. Down-counting starts when the first FIFO element is stored. Writing to **TOCV** has no effect. When the counter reaches *zero*, interrupt flag **IR.TOO** is set. In Continuous Mode, the counter is immediately restarted at **TOCC.TOP**.

Note: The clock signal for the Timeout Counter is derived from the CAN Core's sample point signal. Therefore the point in time where the Timeout Counter is decremented may vary due to the synchronization / re-synchronization mechanism of the CAN Core. If the bit rate switch feature in CAN FD is used, the timeout counter is clocked differently in arbitration and data field.

28.2.4 Rx Handling

The Rx Handler controls the acceptance filtering, the transfer of received messages to the Rx Buffers or to one of the two Rx FIFOs, as well as the Rx FIFO's Put and Get Indices.

28.2.4.1 Acceptance Filtering

The M_CAN offers the possibility to configure two sets of acceptance filters, one for standard identifiers and one for extended identifiers. These filters can be assigned to an Rx Buffer or to Rx FIFO 0,1. For acceptance filtering each list of filters is executed from element #0 until the first matching element. Acceptance filtering stops at the first matching element. The following filter elements are not evaluated for this message.

The main features are:

- Each filter element can be configured as
 - Range filter (from - to)
 - Filter for one or two dedicated IDs
 - Classic bit mask filter
- Each filter element is configurable for acceptance or rejection filtering
- Each filter element can be enabled / disabled individually
- Filters are checked sequentially, execution stops with the first matching filter element

Related configuration registers are:

- Global Filter Configuration **GFC**
- Standard ID Filter Configuration **SIDFC**
- Extended ID Filter Configuration **XIDFC**
- Extended ID AND Mask **XIDAM**

Depending on the configuration of the filter element (**SFEC/EFEC**) a match triggers one of the following actions:

- Store received frame in FIFO 0 or FIFO 1
- Store received frame in Rx Buffer
- Store received frame in Rx Buffer and generate pulse at filter event pin
- Reject received frame
- Set High Priority Message interrupt flag **IR.HPM**
- Set High Priority Message interrupt flag **IR.HPM** and store received frame in FIFO 0 or FIFO 1

Acceptance filtering is started after the complete identifier has been received. After acceptance filtering has completed, and if a matching Rx Buffer or Rx FIFO has been found, the Message Handler starts writing the received message data in portions of 32 bit to the matching Rx Buffer or Rx FIFO. If the CAN protocol controller has detected an error condition (e.g. CRC error), this message is discarded with the following impact on the affected Rx Buffer or Rx FIFO:

- Rx Buffer

New Data flag of matching Rx Buffer is not set, but Rx Buffer (partly) overwritten with received data. For error type, see **PSR.LEC** respectively **PSR.DLEC**.

- Rx FIFO

Put index of matching Rx FIFO is not updated, but related Rx FIFO element (partly) overwritten with received data. For error type see **PSR.LEC** respectively **PSR.DLEC**. In case the matching Rx FIFO is operated in overwrite mode, the boundary conditions described in Chapter 28.2.4.2.2 have to be considered.

Note: When an accepted message is written to one of the two Rx FIFOs, or into an Rx Buffer, the unmodified received identifier is stored independent of the filter(s) used. The result of the acceptance filter process is strongly depending on the sequence of configured filter elements.

28.2.4.1.1 Range Filter

The filter matches for all received frames with Message IDs in the range defined by **SF1ID/SF2ID** resp. **EF1ID/EF2ID**.

There are two possibilities when range filtering is used together with extended frames:

EFT = "00": The Message ID of received frames is ANDed with the Extended ID AND Mask (**XIDAM**) before the range filter is applied

EFT = "11": The Extended ID AND Mask (**XIDAM**) is not used for range filtering

28.2.4.1.2 Filter for specific IDs

A filter element can be configured to filter for one or two specific Message IDs. To filter for one specific Message ID, the filter element has to be configured with **SF1ID** = **SF2ID** resp. **EF1ID** = **EF2ID**.

28.2.4.1.3 Classic Bit Mask Filter

Classic bit mask filtering is intended to filter groups of Message IDs by masking single bits of a received Message ID. With classic bit mask filtering, the **SF1ID/EF1ID** is used as Message ID filter, while **SF2ID/EF2ID** is used as filter mask.

A zero bit at the filter mask will mask out the corresponding bit position of the configured ID filter, e.g. the value of the received Message ID at that bit position is not relevant for acceptance filtering. Only those bits of the received Message ID where the corresponding mask bits are one are relevant for acceptance filtering.

In case all mask bits are one, a match occurs only when the received Message ID and the Message ID filter are identical. If all mask bits are zero, all Message IDs match.

28.2.4.1.4 Standard Message ID Filtering

Figure 28.4 below shows the flow for standard Message ID (11-bit Identifier) filtering. The Standard Message ID Filter element is described in Chapter 28.3.3.5 .

Controlled by the Global Filter Configuration **GFC** and the Standard ID Filter Configuration **SIDFC** Message ID, Remote Transmission Request bit (RTR), and the Identifier Extension bit (IDE) of received frames are compared against the list of configured filter elements.

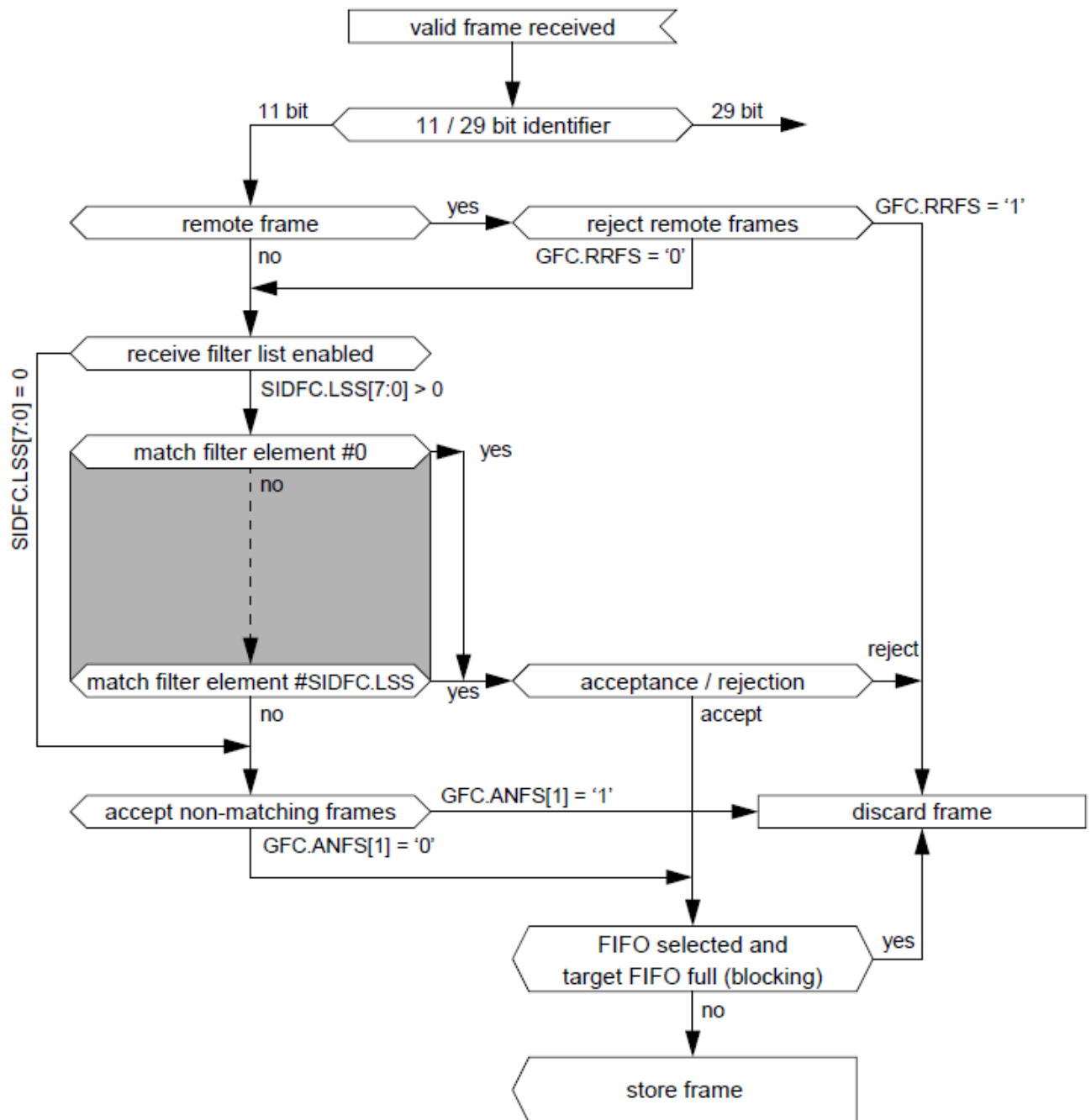


Figure 28.4 Standard Message ID Filter Path

28.2.4.1.5 Extended Message ID Filtering

Figure 28.5 below shows the flow for extended Message ID (29-bit Identifier) filtering. The Extended Message ID Filter element is described in Chapter 28.3.3.6.

Controlled by the Global Filter Configuration **GFC** and the Extended ID Filter Configuration **XIDFC** Message ID, Remote Transmission Request bit (RTR), and the Identifier Extension bit (IDE) of received frames are compared against the list of configured filter elements.

The Extended ID AND Mask **XIDAM** is ANDed with the received identifier before the filter list is executed.

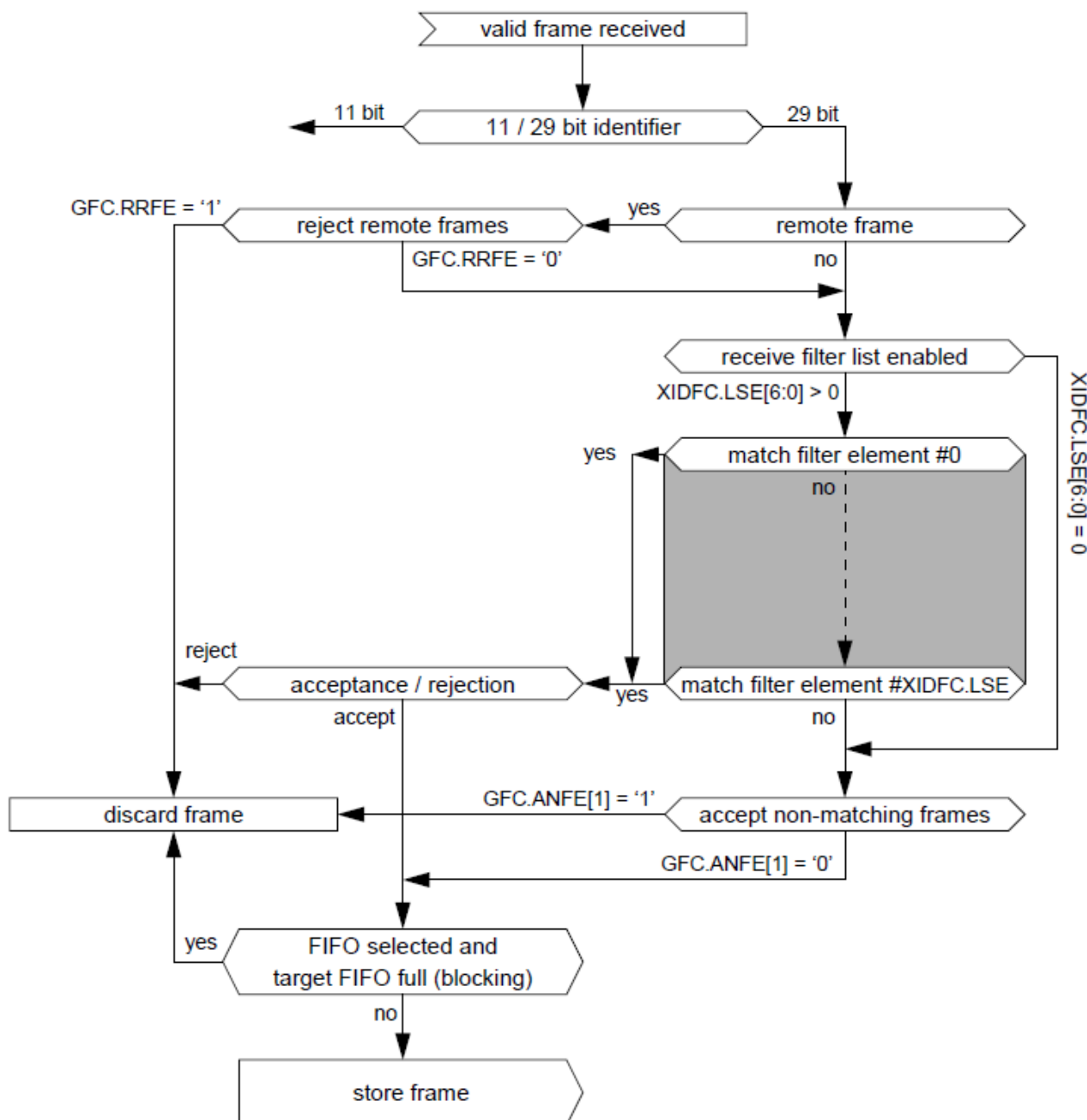


Figure 28.5 Extended Message ID Filter Path

28.2.4.2 Rx FIFOs

Rx FIFO 0 and Rx FIFO 1 can be configured to hold up to 64 elements each. Configuration of the two Rx FIFOs is done via registers **RXF0C** and **RXF1C**.

Received messages that passed acceptance filtering are transferred to the Rx FIFO as configured by the matching filter element. For a description of the filter mechanisms available for Rx FIFO 0 and Rx FIFO 1 see Chapter 28.2.4.1 . The Rx FIFO element is described in Chapter 28.3.3.2 .

To avoid an Rx FIFO overflow, the Rx FIFO watermark can be used. When the Rx FIFO fill level reaches the Rx FIFO watermark configured by **RXFnC.FnWM**, interrupt flag **IR.RFnW** is set. When the Rx FIFO Put Index reaches the Rx FIFO Get Index an Rx FIFO Full condition is signaled by **RXFnC.FnF**. In addition interrupt flag **IR.RFnF** is set.

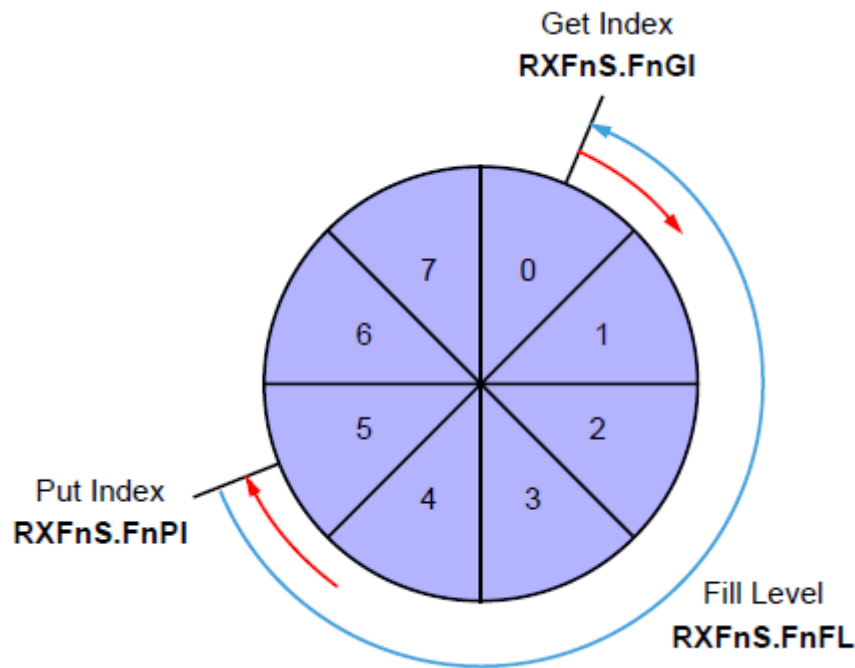


Figure 28.6 Rx FIFO Status

When reading from an Rx FIFO, the Rx FIFO gets Index **RXFnS.FnGI**.

- FIFO Element Size has to be added to the corresponding Rx FIFO start address **RXFnC.FnSA**.

Table 28.2 Rx Buffer / FIFO Element Size

RXESC.RBDS[2:0] RXESC.FnDS[2:0]	Data Field [bytes]	FIFO Element Size [RAM words]
000	8	4
001	12	5
010	16	6
011	20	7
100	24	8
101	32	10
110	48	14
111	64	18

28.2.4.2.1 Rx FIFO Blocking Mode

The Rx FIFO blocking mode is configured by **RXFnC.FnOM** = '0'. This is the default operation mode for the Rx FIFOs. When an Rx FIFO full condition is reached (**RXFnS.FnPI** = **RXFnS.FnGI**), no further messages are written to the corresponding Rx FIFO until at least one message has been read out and the Rx FIFO Get Index has been incremented. An Rx FIFO full condition is signaled by **RXFnS.FnF** = '1'. In addition interrupt flag **IR.RFnF** is set. In case a message is received while the corresponding Rx FIFO is full, this message is discarded and the message lost condition is signaled by **RXFnS.RFnL** = '1'. In addition interrupt flag **IR.RFnL** is set.

28.2.4.2.2 Rx FIFO Overwrite Mode

The Rx FIFO overwrite mode is configured by **RXFnC.FnOM** = '1'. When an Rx FIFO full condition (**RXFnS.FnPI** = **RXFnS.FnGI**) is signaled by **RXFnS.FnF** = '1', the next message accepted for the FIFO will overwrite the oldest FIFO message. Put and get index are both incremented by one. When an Rx FIFO is operated in overwrite mode and an Rx FIFO full condition is signaled, reading of the Rx FIFO elements should start at least at get index + 1. The reason for that is, that it might happen, that a received message is written to the Message RAM (put index) while the CPU is reading from the Message RAM (get index). In this case inconsistent data may be read from the respective Rx FIFO element. Adding an offset to the get index when reading from the Rx FIFO avoids this problem. The offset depends on how fast the CPU accesses the Rx FIFO. Figure 28.7 shows an offset of two with respect to the get index when reading the Rx FIFO. In this case the two messages stored in element 1 and 2 are lost.

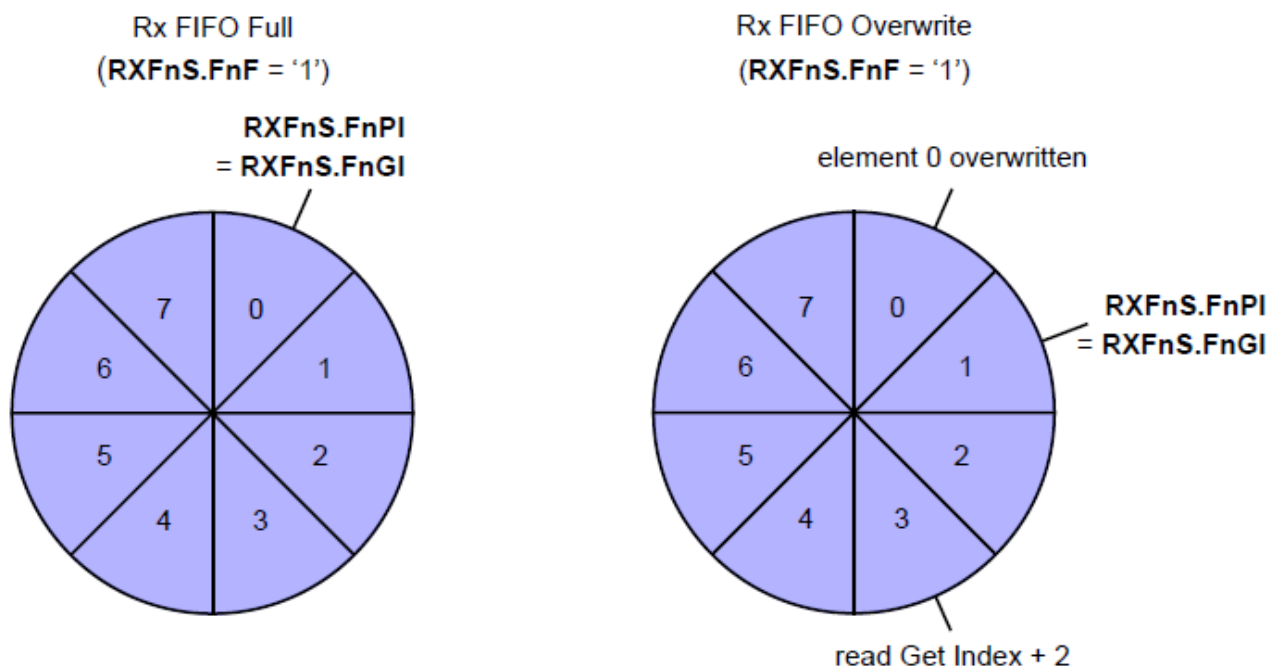


Figure 28.7 Rx FIFO Overflow Handling

After reading from the Rx FIFO, the number of the last element read has to be written to the Rx FIFO Acknowledge Index **RXFnA.FnA**. This increments the get index to that element number. In case that the put index has not been incremented to this Rx FIFO element, the Rx FIFO full condition is reset (**RXFnS.FnF** = '0').

28.2.4.3 Dedicated Rx Buffers

The M_CAN supports up to 64 dedicated Rx Buffers. The start address of the dedicated Rx Buffer section is configured via **RXBC.RBSA**.

For each Rx Buffer a Standard or Extended Message ID Filter Element with **SFEC / EFEC** = "111" and **SFID2 / EFID2[10:9]** = "00" has to be configured (See Chapter 28.3.3.5 and Chapter 28.3.3.6).

After a received message has been accepted by a filter element, the message is stored into the Rx Buffer in the Message RAM referenced by the filter element. The format is the same as for an Rx FIFO element. In addition, the flag **IR.DRX** (Message stored in Dedicated Rx Buffer) in the interrupt register is set.

Table 28.3 Example Filter Configuration for Rx Buffers

Filter Element	SFID1[10:0] EFID1[28:0]	SFID2[10:9] EFID2[10:9]	SFID2[5:0] EFID2[5:0]
0	ID message 1	00	00 0000
1	ID message 2	00	00 0001
2	ID message 3	00	00 0010

After the last word of a matching received message has been written to the Message RAM, the respective New Data flag in register NDAT1, 2 is set. As long as the New Data flag is set, the respective Rx Buffer is locked against updates from received matching frames. The New Data flags have to be reset by the Host by writing '1' to the respective bit position. While an Rx Buffer's New Data flag is set, a Message ID Filter Element referencing this specific Rx Buffer will not match, causing the acceptance filtering to continue. Following Message ID Filter Elements may cause the received message to be stored into another Rx Buffer, or into an Rx FIFO, or the message may be rejected, depending on filter configuration.

28.2.4.3.1 Rx Buffer Handling

- Reset interrupt flag **IR.DRX**
- Read New Data registers
- Read messages from Message RAM
- Reset New Data flags of processed messages

28.2.4.4 Debug on CAN Support

Debug messages are stored into Rx Buffers. For debug handling three consecutive Rx buffers (e.g. #61, #62, #63) have to be used for storage of debug messages A, B, and C. The format is same as for an Rx Buffer or an Rx FIFO element (Refer

to Chapter 28.3.3.2).

Advantage: Fixed start address for the DMA transfers (relative to **RXBC.RBSA**), no additional configuration required. For filtering of debug messages Standard / Extended Filter Elements with **SFEC / EFEC = "111"** have to be set up. Messages matching these filter elements are stored into the Rx Buffers addressed by **SFID2 / EFID2[5:0]**. After message C has been stored, the DMA request output **m_can_dma_req** is activated and the three messages can be read from the Message RAM under DMA control. The RAM words holding the debug messages will not change by the M_CAN while **m_can_dma_req** is activated. The behavior is similar to that of an Rx Buffers with its New Data flag set. After the DMA has completed the DMA unit sets **m_can_dma_ack**. This resets **m_can_dma_req**. Now the M_CAN is prepared to receive the next set of debug messages.

28.2.4.4.1 Filtering for Debug Message

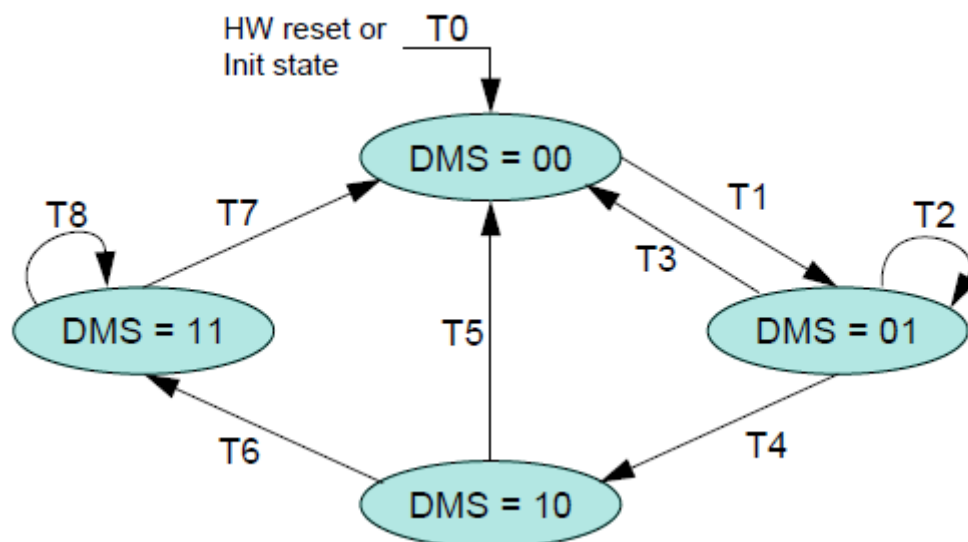
Filtering for debug messages is done by configuring one Standard / Extended Message ID Filter Element for each of the three debug messages. To enable a filter element to filter for debug messages **SFEC / EFEC** has to be programmed to "111". In this case fields **SFID1 / SFID2** and **EFID1 / EFID2** have a different meaning (See Chapter 28.2.4.1.4 and chapter 28.2.4.1.5). While **SFID2 / EFID2[10:9]** controls the debug message handling state machine, **SFID2 / EFID2[5:0]** controls the location for storage of a received debug message. When a debug message is stored, neither the respective New Data flag nor **IR.DRX** are set. The reception of debug messages can be monitored via **RXF1S.DMS**.

Table 28.4 Example Filter Configuration for Debug Messages

Filter Element	SFID1[10:0] EFID1[28:0]	SFID2[10:9] EFID2[10:9]	SFID2[5:0] EFID2[5:0]
0	ID debug message A	01	11 1101
1	ID debug message B	10	11 1110
2	ID debug message C	11	11 1111

28.2.4.4.2 Debug Message Handling

The debug message handling state machine assures that debug messages are stored to three consecutive Rx Buffers in correct order. In case of missing messages the process restarts. The DMA request is activated only when all three debug messages A, B, C have been received in correct order. The status of the debug message handling state machine is signaled via **RXF1S.DMS**.



- T0: reset m_can_dma_req output, enable reception of debug messages A, B, and C
 T1: reception of debug message A
 T2: reception of debug message A
 T3: reception of debug message C
 T4: reception of debug message B
 T5: reception of debug messages A, B
 T6: reception of debug message C
 T7: DMA transfer completed
 T8: reception of debug message A,B,C (message rejected)

Figure 28.8 Debug Message Handling State Machine

28.2.5 Tx Handling

The Tx Handler handles transmission requests for the dedicated Tx Buffers, the Tx FIFO, and the Tx Queue. It controls the transfer of transmit messages to the CAN Core, the Put and Get Indices, and the Tx Event FIFO. Up to 32 Tx Buffers can be set up for message transmission. The CAN mode for transmission (Classic CAN or CAN FD) can be configured separately for each Tx Buffer element. The Tx Buffer element is described in Chapter 28.3.3.3. Table 28.5 below describes the possible configurations for frame transmission.

Table 28.5 Possible Configuration for Frame Transmission

CCCR		Tx Buffer Element		Frame Transmission
BRSE	FDOE	FDF	BRS	
Ignored	0	Ignored	Ignored	Classic CAN
0	1	0	Ignored	Classic CAN
0	1	1	Ignored	FD without bit rate switching
1	1	0	Ignored	Classic CAN
1	1	1	0	FD without bit rate switching
1	1	1	1	FD with bit rate switching

Note: AUTOSAR requires at least three Tx Queue Buffers and support of transmit cancellation. The Tx Handler starts a Tx scan to check for the highest priority pending Tx request (Tx Buffer with the lowest Message ID) when the Tx Buffer Request Pending register TXBRP is updated, or when a transmission starts.

28.2.5.1 Transmit Pause

The transmit pause feature is intended for use in CAN systems where the CAN message identifiers are (permanently) specified to specific values and cannot easily be changed. These message identifiers may have a higher CAN arbitration priority than other defined messages, while in a specific application their relative arbitration priority should be inverse. This may lead to a case where one ECU sends a burst of CAN messages that cause another ECU's CAN messages to be delayed because that other messages have a lower CAN arbitration priority.

e.g. CAN ECU-1 has the transmit pause feature enabled and is requested by its application software to transmit four messages, it will, after the first successful message transmission, wait for two CAN bit times of bus idle before it is allowed to start the next requested message. If there are other ECUs with pending messages, those messages are started in the idle time, they would not need to arbitrate with the next message of ECU-1. After having received a message, ECU-1 is allowed to start its next transmission as soon as the received message releases the CAN bus.

The transmit pause feature is controlled by bit **CCCR.TXP**. If the bit is set, the M_CAN will, each time it has successfully transmitted a message, pause for two CAN bit times before starting the next transmission. This enables other CAN nodes in the network to transmit messages even if their messages have lower prior identifiers. Default is transmit pause disabled (**CCCR.TXP** = '0').

This feature loses up burst transmissions coming from a single node and it protects against "babbling idiot" scenarios where the application program erroneously requests too many transmissions.

28.2.5.2 Dedicated Tx Buffers

Dedicated Tx Buffers are intended for message transmission under complete control of the Host CPU. Each Dedicated Tx Buffer is configured with a specific Message ID. In case that multiple Tx Buffers are configured with the same Message ID, the Tx Buffer with the lowest buffer number is transmitted first.

If the data section has been updated, a transmission is requested by an Add Request via **TXBAR.ARn**. The requested messages arbitrate internally with messages from an optional Tx FIFO or Tx Queue and externally with messages on the CAN bus, and are sent out according to their Message ID.

A dedicated Tx Buffer allocates Element Size 32-bit words in the Message RAM See Table 28.6). Therefore, the start address of a dedicated Tx Buffer in the Message RAM is calculated by adding transmit buffer index (0...31)

- Element Size to the Tx Buffer Start Address **TXBC.TBSA**.

Table 28.6 Tx Buffer / FIFO / Queue Element Size

TXESC.TBDS[2:0]	Data Field [bytes]	Element Size [RAM words]
000	8	4
001	12	5
010	16	6
011	20	7
100	24	8
101	32	10
110	48	14
111	64	18

28.2.5.3 Tx FIFO

Tx FIFO operation is configured by programming **TXBC.TFQM** to '0'. Messages stored in the Tx FIFO are transmitted starting with the message referenced by the Get Index **TXFQS.TFGI**. After each transmission the Get Index is incremented cyclically until the Tx FIFO is empty. The Tx FIFO enables transmission of messages with the same Message ID from different Tx Buffers in the order these messages have been written to the Tx FIFO. The M_CAN calculates the Tx FIFO Free Level **TXFQS.TFFL** as difference between Get and Put Index. It indicates the number of available (free) Tx FIFO elements. New transmit messages have to be written to the Tx FIFO starting with the Tx Buffer referenced by the Put Index **TXFQS.TFQPI**. An Add Request increments the Put Index to the next free Tx FIFO element. When the Put Index reaches the Get Index, Tx FIFO Full (**TXFQS.TFQF** = '1') is signaled. In this case no further messages should be written to the Tx FIFO until the next message has been transmitted and the Get Index has been incremented.

When a single message is added to the Tx FIFO, the transmission is requested by writing '1' to the **TXBAR** bit related to the Tx Buffer referenced by the Tx FIFO's Put Index.

When multiple (n) messages are added to the Tx FIFO, they are written to n consecutive Tx Buffers starting with the Put Index. The transmissions are then requested via **TXBAR**. The Put Index is then cyclically incremented by n. The number of requested Tx buffers should not exceed the number of free Tx Buffers as indicated by the Tx FIFO Free Level.

When a transmission request for the Tx Buffer referenced by the Get Index is cancelled, the Get Index is incremented to the next Tx Buffer with pending transmission request and the Tx FIFO Free Level is recalculated. When transmission cancellation is applied to any other Tx Buffer, the Get Index and the FIFO Free Level remain unchanged.

The Tx FIFO element allocates Element Size 32-bit words in the Message RAM (See Table 28.6). Therefore, the start address of the next available (free) Tx FIFO Buffer is calculated by adding Tx FIFO/Queue Put Index **TXFQS.TFQPI** (0...31)

- Element Size to the Tx Buffer Start Address **TXBC.TBSA**.

28.2.5.4 Tx Queue

Tx Queue operation is configured by programming **TXBC.TFQM** to '1'. Messages stored in the Tx Queue are transmitted starting with the message with the lowest Message ID (highest priority). In case that multiple Queue Buffers are configured with the same Message ID, the Queue Buffer with the lowest buffer number is transmitted first.

New messages have to be written to the Tx Buffer referenced by the Put Index **TXFQS.TFQPI**. An Add Request cyclically increments the Put Index to the next free Tx Buffer. In case that the Tx Queue is full (**TXFQS.TFQF** = '1'), the Put Index is not valid and no further message should be written to the Tx Queue until at least one of the requested messages has been sent out or a pending transmission request has been cancelled.

The application may use register **TXBRP** instead of the Put Index and may place messages to any Tx Buffer without pending transmission request.

The Tx Queue Buffer allocates Element Size 32-bit words in the Message RAM (See Table 28.6). Therefore, the start address of the next available (free) Tx Queue Buffer is calculated by adding Tx FIFO/Queue Put Index **TXFQS.TFQPI** (0...31)

- Element Size to the Tx Buffer Start Address **TXBC.TBSA**.

28.2.5.5 Mixed Dedicated Tx Buffers / Tx FIFO

In this case the Tx Buffers section in the Message RAM is subdivided into a set of Dedicated Tx Buffers and a Tx FIFO. The number of Dedicated Tx Buffers is configured by **TXBC.NDTB**. The number of Tx Buffers assigned to the Tx FIFO is configured by **TXBC.TFQS**.

In case that **TXBC.TFQS** is programmed to *zero*, only Dedicated Tx Buffers are used.

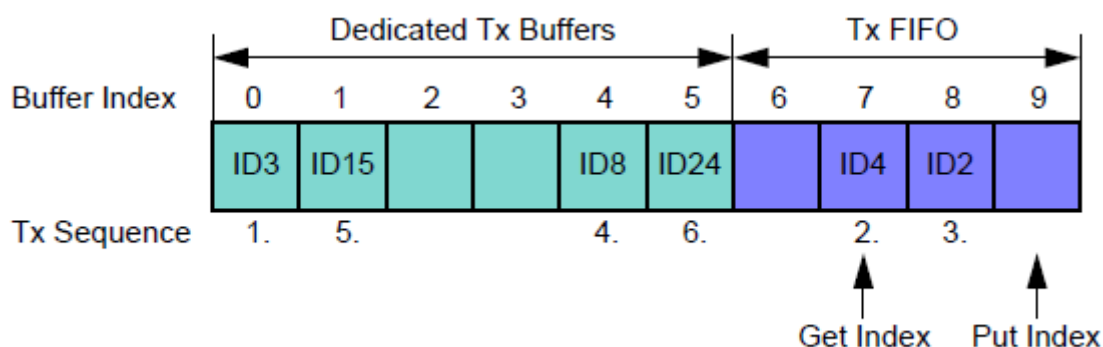


Figure 28.9 Example of mixed Configuration Dedicated Tx Buffers / Tx FIFO

Tx prioritization:

- Scan Dedicated Tx Buffers and oldest pending Tx FIFO Buffer (referenced by **TXFS.TFGI**)
- Buffer with the lowest Message ID gets the highest priority and is transmitted next

28.2.5.6 Mixed Dedicated Tx Buffers / Tx Queue

In this case, the Tx Buffers section in the Message RAM is subdivided into a set of Dedicated Tx Buffers and a Tx Queue. The number of Dedicated Tx Buffers is configured by **TXBC.NDTB**. The number of Tx Queue Buffers is configured by **TXBC.TFQS**. In case that **TXBC.TFQS** is programmed to *zero*, only Dedicated Tx Buffers are used.

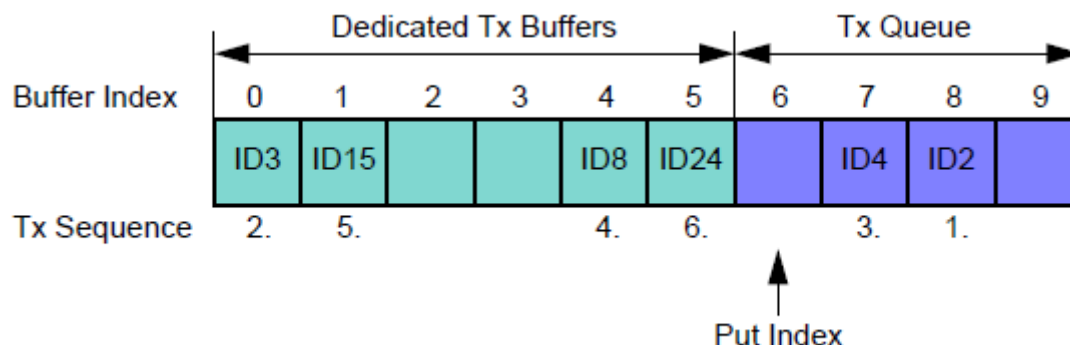


Figure 28.10 Example of mixed Configuration Dedicated Tx Buffers / Tx Queue

Tx prioritization:

- Scan all Tx Buffers with activated transmission request
- Tx Buffer with the lowest Message ID gets the highest priority and is transmitted next

28.2.5.7 Transmit Cancellation

The M_CAN supports transmit cancellation. This feature is especially intended for gateway applications and AUTOSAR based applications. To cancel a requested transmission from a dedicated Tx Buffer or a Tx Queue Buffer the Host has to write a '1' to the corresponding bit position (=number of Tx Buffer) of register **TXBCR**. Transmit cancellation is not intended for Tx FIFO operation.

Successful cancellation is signaled by setting the corresponding bit of register **TXBCF** to '1'.

In case a transmit cancellation is requested while a transmission from a Tx Buffer is already ongoing, the corresponding **TXBRP** bit remains set as long as the transmission is in progress. If the transmission was successful, the corresponding **TXBTO** and **TXBCF** bits are set. If the transmission was not successful, it is not repeated and only the corresponding **TXBCF** bit is set.

Note: In case that a pending transmission is cancelled immediately before this transmission could have been started, there follows a short time window where no transmission is started even if another message is also pending in this node. This may enable another node to transmit a message which may have a lower priority than the second message in this node.

28.2.5.8 Tx Event Handling

To support Tx Event handling, the M_CAN has implemented a Tx Event FIFO. After the M_CAN has transmitted a message on the CAN bus, Message ID and timestamp are stored in a Tx Event FIFO element. To link a Tx event to a Tx Event FIFO element, the Message Marker from the transmitted Tx Buffer is copied into the Tx Event FIFO element.

The Tx Event FIFO can be configured to a maximum of 32 elements. The Tx Event FIFO element is described in Chapter 28.3.3.4.

The purpose of the Tx Event FIFO is to decouple handling transmit status information from transmit message handling i.e. a Tx Buffer holds only the message to be transmitted, while the transmit status is stored separately in the Tx Event FIFO. This has the advantage, especially when operating a dynamically managed transmit queue, that a Tx Buffer can be used for a new message immediately after successful transmission. There is no need to save transmit status information from a Tx Buffer before overwriting that Tx Buffer.

When a Tx Event FIFO full condition is signaled by **IR.TEFF**, no further elements are written to the Tx Event FIFO until at least one element has been read out and the Tx Event FIFO Get Index has been incremented. In case a Tx event occurs while the Tx Event FIFO is full, this event is discarded and interrupt flag **IR.TEFL** is set.

To avoid a Tx Event FIFO overflow, the Tx Event FIFO watermark can be used. When the Tx Event FIFO fill level reaches the Tx Event FIFO watermark configured by **TXEFC.EFWM**, interrupt flag **IR.TEFW** is set.

When reading from the Tx Event FIFO, two times the Tx Event FIFO Get Index **TXEFS.EFGI** has to be added to the Tx Event FIFO start address **TXEFC.EFSA**.

28.2.6 FIFO Acknowledge Handling

The Get Indices of Rx FIFO 0, Rx FIFO 1, and the Tx Event FIFO are controlled by writing to the corresponding FIFO Acknowledge Index (See **RXF0A**, **RXF1A**, and **TXEFA**). Writing to the FIFO Acknowledge Index will set the FIFO Get Index to the FIFO Acknowledge Index plus *one* and thereby updates the FIFO Fill Level. There are two use cases:

When only a single element has been read from the FIFO (the one being pointed to by the Get Index), this Get Index value is written to the FIFO Acknowledge Index.

When a sequence of elements has been read from the FIFO, it is sufficient to write the FIFO Acknowledge Index only once at the end of that read sequence (value: Index of the last element read), to update the FIFO's Get Index.

Due to the fact that the CPU has free access to the M_CAN's Message RAM, special care has to be taken when reading FIFO elements in an arbitrary order (Get Index not considered). This might be useful when reading a High Priority Message from one of the two Rx FIFOs. In this case the FIFO's Acknowledge Index should not be written because this would set the Get Index to a wrong position and also alters the FIFO's Fill Level. In this case some of the older FIFO elements would be lost.

Note: The application has to ensure that a valid value is written to the FIFO Acknowledge Index. The M_CAN does not check for erroneous values.

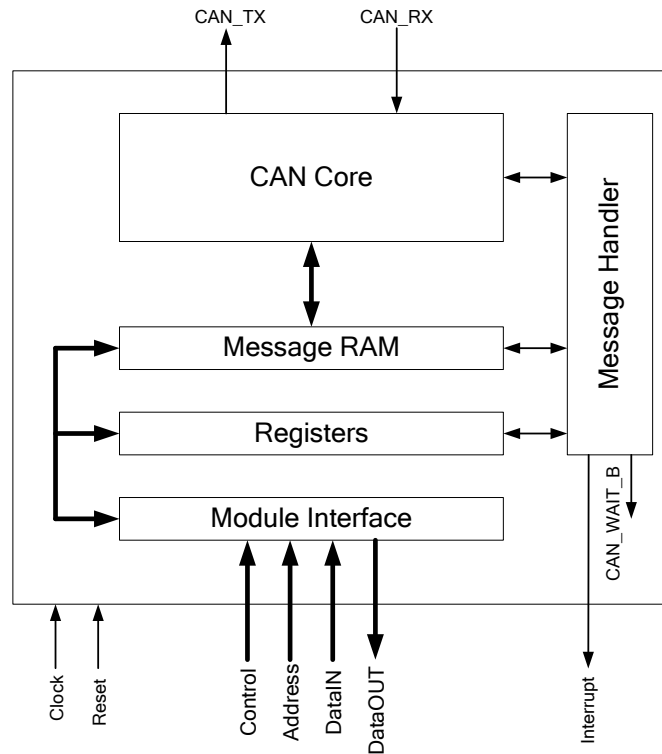


Figure 28.11 Block Diagram of the CAN Controller

- CAN Core: CAN Protocol Controller and Rx/Tx Shift Register for serial/parallel conversion of messages.
- Message RAM: Stores Message Objects and Identifier Masks.
- Registers: All registers are used to control and to configure the CAN module.
- Message Handler: State Machine that controls the data transfer between the Rx/Tx Shift Register of the CAN Core and the Message RAM as well as the generation of interrupts as programmed in the Control and Configuration Registers.
- Module Interface: A 16-bit interfaces to the AMBA APB bus from ARM.

28.3 Programmer's Model

28.3.1 Register Memory Map

Table 28.7 CAN Register Map (Base Address = 0x80007000 / 0x80008000)

Register	Address	R/W	Reset value	Description
CREL	0x00	R	0x00000000	Core Release Register
ENDN	0x04	R	0x87654321	Endian Register
CUST	0x08	R	For debug	Customer Register
DBTP	0x0C	RP	0x00000A33	Data Bit Timing & Prescaler Register
TEST	0x10	R/W	0x00000000	Test Register
RWD	0x14	R/W	0x00000000	RAM Watchdog
CCCR	0x18	R/W	0x00000001	CC Control Register
NBTP	0x1C	R/W	0x06000A03	Nominal Bit Timing & Prescaler Register
TSCC	0x20	R/W	0x00000000	Timestamp Counter Configuration
TSCV	0x24	R/W	0x00000000	Timestamp Counter Value
TOCC	0x28	R/W	0xFFFF0000	Timeout Counter Configuration
TOCV	0x2C	R/W	0x0000FFFF	Timeout Counter Value
	0x30-3C	R/W	0x00000000	Reserved
ECR	0x40	R/W	0x00000000	Error Counter Register
PSR	0x44	R/W	0x00000707	Protocol Status Register
TDCR	0x48	R/W	0x00000000	Transmitter Delay Compensation Register
	0x4C	R/W	0x00000000	Reserved
IR	0x50	R/W	0x00000000	Interrupt Register
IE	0x54	R/W	0x00000000	Interrupt Enable
ILS	0x58	R/W	0x00000000	Interrupt Line Select
ILE	0x5C	R/W	0x00000000	Interrupt Line Enable
	0x60-7C		0x00000000	Reserved
GFC	0x80	RP	0x00000000	Global Filter Configuration
SDIFC	0x84	RP	0x00000000	Standard ID Filter Configuration
XIDFC	0x88	RP	0x00000000	Extended ID Filter Configuration
	0x8C	R	0x00000000	Reserved
XIDAM	0x90	RP	0x00000000	Extended ID AND Mask
HPMS	0x94	R	0x00000000	High Priority Message Status
NDAT1	0x98	RW	0x00000000	New Data 1
NDAT2	0x9C	RW	0x00000000	New Data 2
RXF0C	0xA0	RP	0x00000000	Rx FIFO 0 Configuration
RXF0S	0xA4	R	0x00000000	Rx FIFO 0 Status
RXF0A	0xA8	RW	0x00000000	Rx FIFO 0 Acknowledge
RXBC	0xAC	RP	0x00000000	Rx Buffer Configuration
RXF1C	0xB0	RP	0x00000000	Rx FIFO 1 Configuration
RXF1S	0xB4	R	0x00000000	Rx FIFO 1 Status
RXF1A	0xB8	RW	0x00000000	Rx FIFO 1 Acknowledge
RXESC	0xBC	RP	0x00000000	Rx Buffer / FIFO Element Size Configuration
TXBC	0xC0	RP	0x00000000	Tx Buffer Configuration
TXFQS	0xC4	R	0x00000000	Tx FIFO/Queue Status
TXESC	0xC8	RP	0x00000000	Tx Buffer Element Size Configuration
TXBRP	0xCC	R	0x00000000	Tx Buffer Request Pending
TXBAR	0xD0	R/W	0x00000000	Tx Buffer Add Request
TXBCR	0xD4	R/W	0x00000000	Tx Buffer Cancellation Request
TXBTO	0xD8	R	0x00000000	Tx Buffer Transmission Occurred
TXBCF	0xDC	R	0x00000000	Tx Buffer Cancellation Finished
TXBTIE	0xE0	R/W	0x00000000	Tx Buffer Transmission Interrupt Enable
TXBCIE	0xE4	R/W	0x00000000	Tx Buffer Cancellation Finished Interrupt Enable
	0xE8-0EC	R	0x00000000	Reserved
TXEFC	0xF0	RP	0x00000000	Tx Event FIFO Configuration
TXEFS	0xF4	R	0x00000000	Tx Event FIFO Status
TXEFA	0xF8	R/W	0x00000000	Tx Event FIFO Acknowledge
	0xFC	R	0x00000000	Reserved
TTTMC	0x100	RP	0x00000000	TT Trigger Memory Configuration
TTRMC	0x104	RP	0x00000000	TT Reference Message Configuration
TTOCF	0x108	RP	0x00000000	TT Operation Configuration
TTMLM	0x10C	RP	0x00000000	TT Matrix Limits

Register	Address	R/W	Reset value	Description
TURCF	0x110	RP	0x00000000	TUR Configuration
TTOCN	0x114	R/W	0x00000000	TT Operation Control
TTGTP	0x118	R/W	0x00000000	TT Global Time Preset
TTTMK	0x11C	R/W	0x00000000	TT Time Mark
TTIR	0x120	R/W	0x00000000	TT Interrupt Register
TTIE	0x124	R/W	0x00000000	TT Interrupt Enable
TTILS	0x128	R/W	0x00000000	TT Interrupt Line Select
TTOST	0x12C	R	0x00000000	TT Operation Status
TURNA	0x130	R	0x00000000	TUR Numerator Actual
TTLGT	0x134	R	0x00000000	TT Local & Global Time
TTCTC	0x138	R	0x00000000	TT Cycle Time & Count
TTCPT	0x13C	R	0x00000000	TT Capture Time
TTCSM	0x140	R	0x00000000	TT Cycle Sync Mark
	0x144-1FC	R	0x00000000	Reserved
CFG0	0x80009000	R/W	0x00000000	CFG_CH_DIS_MORD, STS_CH_CLK_STOP_ACK CFG_CH_CLK_STOP_REQ
CFG1	0x80009004	R/w	0x00000000	CFG_CH0_BASE_ADDR
CFG2	0x80009008	R/W	0x00000000	CFG_CH1_BASE_ADDR
CFG3	0x8000900C	R/W	0x00000000	Reserved
CFG4	0x80009010	R/W	0x00000000	CFG_CH0_EXT_TS
CFG5	0x80009014	R/W	0x00000000	CFG_CH1_EXT_TS
CFG6	0x80009018	R/W	0x00000000	Reserved
CFG7	0x8000901C	R	0x00000000	CAN_CH_IRQ[1:0]

28.3.2 Register Description

Core Release Register (CREL)

0x80007000 / 0x80008000 + 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REL[3:0]				STEP[3:0]				SUBSTEP[3:0]				TEAR[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MON[7:0]								DAY[7:0]							

Field	Name	RW	Reset	Description
31:28	REL	R/W	0	Core Release One digit, BCD-coded
27:24	STEP	R/W	0	Step of Core Release One digit, BCD-coded
23:20	SUBSTEP	R/W	0	Sub-step of Core Release One digit, BCD-coded
19:16	YEAR	R/W	0	Time Stamp Year One digit, BCD-coded.
15:8	MON	R/W	0	Time Stamp Month One digit, BCD-coded
7:0	DAY	R/W	0	Time Stamp Day One digit, BCD-coded

Endian Register (ENDN)

0x80007000 / 0x80008000 + 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ETV[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETV[15:0]															

Field	Name	RW	Reset	Description
31:0	ETV	R/W	0x87654321	The endianness test value is 0x87654321

Note: Address 0x08 is reserved for an optional 32 bit customer-specific register. The Customer Register is intended to hold customer-specific configuration, control, and status bits. The functionality is not described in this document.

Data Bit Timing & Prescaler Register (DBTP)

0x80007000 / 0x80008000 + 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-								TDC	-		DBRP[4:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-				DTSEG1[4:0]				DTSEG2[3:0]				DSJW[3:0]			

This register is only writable if bits **CCCR.CCE** and **CCCR.INIT** are set. The CAN bit time may be programmed in the range of 4 to 49 time quanta. The CAN time quantum may be programmed in the range of 1 to 32 **m_ttcan_cclk** periods. $tq = (DBRP + 1) mtq$.

DTSEG1 is the sum of Prop_Seg and Phase_Seg1.

DTSEG2 is Phase_Seg2.

Therefore, the length of the bit time is (programmed values) **[DTSEG1 + DTSEG2 + 3] tq** or (functional values) **[Sync_Seg + Prop_Seg + Phase_Seg1 + Phase_Seg2] tq**.

The Information Processing Time (IPT) is *zero*, meaning the data for the next bit is available at the first clock edge after the sample point.

Field	Name	RW	Reset	Description
23	TDC	R/W	0x0	Transmitter Delay Compensation 0= Transmitter Delay Compensation disabled 1= Transmitter Delay Compensation enabled
20:16	DBRP	R/W	0x0	Data Bit Rate Prescaler The value by which the oscillator frequency is divided for generating the bit time quanta. The bit time is built up from a multiple of this quanta. Valid values for the Bit Rate Prescaler are 0 to 31. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.
12:8	DTSEG1	R/W	0xA	Data time segment before sample point Valid values are 0 to 31. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.
7:4	DTSEG2	R/W	0x3	Data time segment after sample point Valid values are 0 to 15. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.
3:0	DSJW	R/W	0x3	Data (Re)Synchronization Jump Width Valid values are 0 to 15. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.

Note: With a CAN clock (**m_ttcan_cclk**) of 8 MHz, the reset value of 0x00000A33 configures the **M_TTCAN** for a data phase bit rate of 500 kBit/s.

Note: The bit rate configured for the CAN FD data phase via DBTP must be higher or equal to the bit rate configured for the arbitration phase via NBTP.

Test Register (TEST)

0x80007000 / 0x80008000 + 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-								RX	TX[1:0]		LBCK	CAT	CAM	TAT	TAM

Field	Name	RW	Reset	Description
7	RX	R/W	Undef	Receive Pin Monitors the actual value of pin m_ttcn_rx 0= The CAN bus is dominant (m_ttcn_rx = '0') 1= The CAN bus is recessive (m_ttcn_rx = '1')
6:5	TX[1:0]	RP	0x0	TX[1:0] : Control of Transmit Pin 00 Reset value, m_ttcn_tx controlled by the CAN Core, updated at the end of the CAN bit time 01 Sample Point can be monitored at pin m_ttcn_tx 10 Dominant ('0') level at pin m_ttcn_tx 11 Recessive ('1') at pin m_ttcn_tx
4	LBCK	RP	0x0	Loop Back Mode 0= Reset value, Loop Back Mode is disabled 1= Loop Back Mode is enabled (See Chapter 28.2.1.8)
3	CAT	R	0x0	Check ASC Transmit Control Monitors level at output pin m_ttcn_asct . 0= Output pin m_ttcn_asct = '0' 1= Output pin m_ttcn_asct = '1'
2	CAM	R	0x0	Check ASC Multiplexer Control Monitors level at output pin m_ttcn_ascm . 0= Output pin m_ttcn_ascm = '0' 1= Output pin m_ttcn_ascm = '1'
1	TAT	RP	0x0	Test ASC Transmit Control Controls output pin m_ttcn_asct in test mode, ORed with the signal from the FSE 0= Level at pin m_ttcn_asct controlled by FSE 1= Level at pin m_ttcn_asct = '1'
0	TAM	RP	0x0	Test ASC Multiplexer Control Controls output pin m_ttcn_ascm in test mode, ORed with the signal from the FSE 0= Level at pin m_ttcn_ascm controlled by FSE 1= Level at pin m_ttcn_ascm = '1'

RAM Watchdog (RWD)

0x80007000 / 0x80008000 + 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDV[7:0]								WDC[7:0]							

The RAM Watchdog monitors the READY output of the Message RAM (**m_ttcn_aeim_ready**). A Message RAM access via the M_TTCAN's Generic Master Interface (**m_ttcn_aeim_sel** active) starts the Message RAM Watchdog Counter with the value configured by **RWD.WDC**. The counter is reloaded with **RWD.WDC** when the Message RAM signals successful completion by activating its READY output. In case that there is no response from the Message RAM until the counter has counted down to zero, the counter stops and interrupt flag **IR.WDI** is set. The RAM Watchdog Counter is clocked by the Host clock (**m_ttcn_hclk**)

Field	Name	RW	Reset	Description
15-8	WDV[7:0]	R	0x0	Watchdog Value Actual Message RAM Watchdog Counter Value.
7:0	WDC[7:0]	RP	0x0	Watchdog Configuration Start value of the Message RAM Watchdog Counter. With the reset value of "00" the counter is disabled.

CC Control Register (CCCR)

0x80007000 / 0x80008000 + 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NISO	TXP	EFBI	PXHD	-		BRSE	FDOE	TEST	DAR	MON	CSR	CSA	ASM	CCE	INIT

Field	Name	RW	Reset	Description
15	NISO	RP	0x0	Non ISO Operation If this bit is set, the M_TTCAN uses the CAN FD frame format as specified by the Bosch CAN FD Specification V1.0. 0= CAN FD frame format according to ISO11898-1 1= CAN FD frame format according to Bosch CAN FD Specification V1.0
14	TXP	RP	0x0	Transmit Pause If this bit is set, the M_TTCAN pauses for two CAN bit times before starting the next transmission after itself has successfully transmitted a frame (See Chapter 28.2.5.1). 0= Transmit pause disabled 1= Transmit pause enabled
13	EFBI	RP	0x0	Edge Filtering during Bus Integration 0= Edge filtering disabled 1= Two consecutive dominant tq required to detect an edge for hard synchronization
12	PXHD	RP	0x0	Protocol Exception Handling Disable 0= Protocol exception handling enabled 1= Protocol exception handling disabled
9	BRSE	RP	0x0	Bit Rate Switch Enable 0= Bit rate switching for transmissions disabled 1= Bit rate switching for transmissions enabled
8	FDOE	RP	0x0	FD Operation Enable 0= FD operation disabled 1= FD operation enabled
7	TEST	RP	0x0	Test Mode Enable 0= Normal operation, register TEST holds reset values 1= Test Mode, write access to register TEST enabled
6	DAR	RP	0x0	Disable Automatic Retransmission 0= Automatic retransmission of messages not transmitted successfully enabled 1= Automatic retransmission disabled
5	MON	RP	0x0	Bus Monitoring Mode Bit MON can only be set by the Host when both CCE and INIT are set to '1'. The bit can be reset by the Host at any time. 0= Bus Monitoring Mode is disabled 1= Bus Monitoring Mode is enabled
4	CSR	RW	0x0	Clock Stop Request 0= No clock stop is requested 1= Clock stop requested. When clock stop is requested, first INIT and then CSA will be set after all pending transfer requests are completed and the CAN bus reached <i>idle</i> .
3	CSA	R	0x0	Clock Stop Acknowledge 0= No clock stop acknowledged 1= M_TTCAN may be set in power down by stopping m_ttcan_hclk and m_ttcan_cclk
2	ASM	RP	0x0	Restricted Operation Mode Bit ASM can only be set by the Host when both CCE and INIT are set to '1'. The bit can be reset by the Host at any time. For a description of the Restricted Operation Mode. See Chapter 28.2.1.5. 0= Normal CAN operation 1= Restricted Operation Mode active

1	CCE	RP	0x0	Restricted Operation Mode Bit ASM can only be set by the Host when both CCE and INIT are set to '1'. The bit can be reset by the Host at any time. For a description of the Restricted Operation Mode. See Chapter 28.2.1.5.. 0= Normal CAN operation 1= Restricted Operation Mode active
0	INIT	RW	0x0	Initialization 0= Normal Operation 1= Initialization starts

Nominal Bit Timing & Prescaler Register (NBTP)**0x80007000 / 0x80008000 + 0x1C**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NSJW[6:0]								NBRP[8:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NTSEG1[7:0]								-	NTSEG2[6:0]						

Field	Name	RW	Reset	Description
31:25	NSJW	RP	0x3	Nominal (Re)Synchronization Jump Width Valid values are 0 to 127. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.
24:16	NBRP	RP	0x0	Nominal Bit Rate Prescaler 0x000-0x1FF The value by which the oscillator frequency is divided for generating the bit time quanta. The bit time is built up from a multiple of this quanta. Valid values for the Bit Rate Prescaler are 0 to 511. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.
15:8	NTSEG1	RP	0xA	Nominal Time segment before sample point 0x01-0xFF Valid values are 1 to 255. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.
6:0	NTSEG2	RP	0x3	Nominal Time segment after sample point 0x00-0x7F Valid values are 0 to 127. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.

Note: With a CAN clock (m_tcan_cclk) of 8 MHz, the reset value of 0x06000A03 configures the M_TTCAN for a bit rate of 500 kBit/s.

Timestamp Counter Configuration (TSCC)

0x80007000 / 0x80008000 + 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-												TCP[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-														TSS[1:0]	

Field	Name	RW	Reset	Description
19:16	TCP[3:0]	RP	0x0	Timestamp Counter Prescaler Configures the timestamp and timeout counters time unit in multiples of CAN bit times [1...16]. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used. In CAN FD mode the internal timestamp counter TCP does not provide a constant time base due to the different CAN bit times between arbitration phase and data phase.
1:0	TSS[1:0]	RP	0x0	Timestamp Select 00= Timestamp counter value always 0x0000 01=Timestamp counter value incremented according to TCP 10= Reserved 11= Same as "00"

Timestamp Counter Value (TSCV)

0x80007000 / 0x80008000 + 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSC[15:0]															

Field	Name	RW	Reset	Description
15:0	TSC	R	0x0	Timestamp Counter The internal/external Timestamp Counter value is captured on start of frame (Both Rx and Tx). When TSCC.TSS = "01", the Timestamp Counter is incremented in multiples of CAN bit times [1...16] depending on the configuration of TSCC.TCP. A wrap around sets interrupt flag IR.TSW. Write access resets the counter to zero. When TSCC.TSS = "10", TSC reflects the external Timestamp Counter value. A write access has no impact.

Timeout Counter Configuration (TOCC)

0x80007000 / 0x80008000 + 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TOP[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-													TOS[1:0]		ETOC

Field	Name	RW	Reset	Description
31:16	TOP[15:0]	RP	0xFFFF	Timeout Period Start value of the Timeout Counter (down-counter). Configures the Timeout Period.
2:1	TOS[1:0]	RP	0x0	Timeout Select When operating in Continuous mode, a write to TOCV presets the counter to the value configured by TOCC.TOP and continues down-counting. When the Timeout Counter is controlled by one of the FIFOs, an empty FIFO presets the counter to the value configured by TOCC.TOP. Down-counting starts when the first FIFO element is stored. 00= Continuous operation 01= Timeout controlled by Tx Event FIFO 10= Timeout controlled by Rx FIFO 0 11= Timeout controlled by Rx FIFO 1
0	ETOC	RP	0x0	Enable Timeout Counter 0= Timeout Counter disabled 1= Timeout Counter enabled

Timeout Counter Value (TOCV)

0x80007000 / 0x80008000 + 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TOC[15:0]															

Field	Name	RW	Reset	Description
15-0	TOC	R/W	0x0	Timeout Counter The Timeout Counter is decremented in multiples of CAN bit times [1...16] depending on the configuration of TSCC.TCP. When decremented to zero, interrupt flag IR.TOO is set and the Timeout Counter stops. Start and reset/restart conditions are configured via TOCC.TOS.

Error Counter Register (ECR)

0x80007000 / 0x80008000 + 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-								CEL[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RP	REC[6:0]							TEC[7:0]							

Field	Name	RW	Reset	Description
23:16	CEL	X	0x0	CAN Error Logging The counter is incremented each time when a CAN protocol error causes the Transmit Error Counter or the Receive Error Counter to be incremented. It is reset by read access to CEL. The counter stops at 0xFF; the next increment of TEC or REC sets interrupt flag IR.ELO.
15	RP	R	0x0	Receive Error Passive 0= The Receive Error Counter is below the <i>error passive</i> level of 128 1= The Receive Error Counter has reached the <i>error passive</i> level of 128
14:8	REC	R	0x0	Receive Error Counter Actual state of the Receive Error Counter, values between 0 and 127
7:0	TEC	R	0x0	Transmit Error Counter Actual state of the Transmit Error Counter, values between 0 and 255

Note: When CCCR.ASM is set, the CAN protocol controller does not increment TEC and REC when a CAN protocol error is detected, but CEL is still incremented. This enables monitoring of collisions between CAN frames and ASC frames.

Protocol Status Register (PSR)

0x80007000 / 0x80008000 + 0x44

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								TDCV							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-	PXE	RFDF	RBRS	RESI	DLEC[2:0]			BO	EW	EP	ACT[1:0]		LEC[2:0]		

Field	Name	RW	Reset	Description
22:16	TDCV	R	0x0	Transmitter Delay Compensation Value 0x00-0x7F Position of the secondary sample point, defined by the sum of the measured delay from m_ttcn_tx to m_ttcn_rx and TDCR.TDCO. The SSP position is, in the data phase, the number of mtq between the start of the transmitted bit and the secondary sample point Valid values are 0 to 127 mtq.
14	PXE	X	0x0	Protocol Exception Event 0= No protocol exception event occurred since last read access 1= Protocol exception event occurred
13	RFDF	X	0x0	Received a CAN FD Message This bit is set independent of acceptance filtering. 0= Since this bit was reset by the CPU, no CAN FD message has been received 1= Message in CAN FD format with FDF flag set has been received
12	RBRS	X	0x0	BRS flag of last received CAN FD Message This bit is set together with RFDF, independent of acceptance filtering. 0= Last received CAN FD message did not have its BRS flag set 1= Last received CAN FD message had its BRS flag set
11	RESI	X	0x0	ESI flag of last received CAN FD Message This bit is set together with RFDF, independent of

				acceptance filtering. 0= Last received CAN FD message did not have its ESI flag set 1= Last received CAN FD message had its ESI flag set
10:8	DLEC	R	0x7	Data Phase Last Error Code Type of last error that occurred in the data phase of a CAN FD format frame with its BRS flag set. Coding is the same as for LEC. This field will be cleared to zero when a CAN FD format frame with its BRS flag set has been transferred (reception or transmission) without error.
7	BO	R	0x0	Bus_Off Status 0= The M_TTCAN is not Bus_Off 1= The M_TTCAN is in Bus_Off state
6	EW	R	0x0	Warning Status 0= Both error counters are below the Error_Warning limit of 96 1= At least one of error counter has reached the Error_Warning limit of 96
5	EP	R	0x0	Error Passive 0= The M_TTCAN is in the Error_Active state. It normally takes part in bus communication and sends an active error flag when an error has been detected 1= The M_TTCAN is in the Error_Passive state
4:3	ACT	R	0x0	Activity Monitors the module's CAN communication state. 00= Synchronizing - node is synchronizing on CAN communication 01= Idle - node is neither receiver nor transmitter 10= Receiver - node is operating as receiver 11= Transmitter - node is operating as transmitter
2:0	LEC	R	0x7	Last Error Code The LEC indicates the type of the last error to occur on the CAN bus. This field will be cleared to '0' when a message has been transferred (reception or transmission) without error. 0= No Error: No error occurred since LEC has been reset by successful reception or transmission. 1= Stuff Error: More than 5 equal bits in a sequence have occurred in a part of a received message where this is not allowed. 2= Form Error: A fixed format part of a received frame has the wrong format. 3= AckError: The message transmitted by the M_TTCAN was not acknowledged by another node. 4= Bit1Error: During the transmission of a message (with the exception of the arbitration field), the device wanted to send a <i>recessive</i> level (bit of logical value '1'), but the monitored bus value was <i>dominant</i>. 5= Bit0Error: During the transmission of a message (or acknowledge bit, or active error flag, or overload flag), the device wanted to send a <i>dominant</i> level (data or identifier bit logical value '0'), but the monitored bus value was <i>recessive</i>. During <i>Bus_Off</i> recovery this status is set each time a sequence of 11 <i>recessive</i> bits has been monitored. This enables the CPU to monitor the proceeding of the <i>Bus_Off</i> recovery sequence (indicating the bus is not stuck at <i>dominant</i> or continuously disturbed). 6= CRCErr: The CRC check sum of a received message was incorrect. The CRC of an incoming message does not match with the CRC calculated from the received data.

				7= NoChange: Any read access to the Protocol Status Register re-initializes the LEC to '7'. When the LEC shows the value '7', no CAN bus event was detected since the last CPU read access to the Protocol Status Register.
--	--	--	--	--

Note: When a frame in CAN FD format has reached the data phase with BRS flag set, the next CAN event (error or valid frame) will be shown in DLEC instead of LEC. An error in a fixed stuff bit of a CAN FD CRC sequence will be shown as a Form Error, not Stuff Error.

Note: The Bus_Off recovery sequence (See CAN Specification Rev. 2.0 or ISO11898-1) cannot be shortened by setting or resetting CCCR.INIT. If the device goes Bus_Off, it will set CCCR.INIT of its own accord, stopping all bus activities. Once CCCR.INIT is cleared by the CPU, the device will then wait for 129 occurrences of Bus Idle (129 * 11 consecutive recessive bits) before resuming normal operation. At the end of the Bus_Off recovery sequence, the Error Management Counters will be reset. During the waiting time after the resetting of CCCR.INIT, each time a sequence of 11 recessive bits has been monitored, a Bit0 Error code is written to PSR.LEC, enabling the CPU to readily check up whether the CAN bus is stuck at dominant or continuously disturbed and to monitor the Bus_Off recovery sequence. ECR.REC is used to count these sequences.

Transmitter Delay Compensation Register (TDCR)

0x80007000 / 0x80008000 + 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-	TDCO[6:0]							-	TDCF						

Field	Name	RW	Reset	Description
14:8	TDCO	RP	0x0	Transmitter Delay Compensation Offset 0x00-0x7F Offset value defining the distance between the measured delay from m_ttcn_tx to m_ttcn_rx and the secondary sample point. Valid values are 0 to 127 mtq.
6:0	TDCF	RP	0x0	Transmitter Delay Compensation Filter Window Length 0x00-0x7F Defines the minimum value for the SSP position, dominant edges on m_ttcn_rx that would result in an earlier SSP position are ignored for transmitter delay measure

Interrupt Register (IR)

0x80007000 / 0x80008000 + 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
		ARA	PED	PEA	WDI	BO	EW	EP	ELO	BEU	BEC	DRX	TOO	MRAF	TSW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TEFL	TEFF	TEFW	TEFN	TFE	TCF	TC	HPM	RF1L	RF1F	RF1W	RF1N	RF0L	RF0F	RF0W	RF0N

Field	Name	RW	Reset	Description
29	ARA	R/W	0x0	Access to Reserved Address 0= No access to reserved address occurred 1= Access to reserved address occurred
28	PED	R/W	0x0	Protocol Error in Data Phase (Data Bit Time is used) 0= No protocol error in data phase 1= Protocol error in data phase detected (PSR.DLEC ≠ 0,7)
27	PEA	R/W	0x0	Protocol Error in Arbitration Phase (Nominal Bit Time is used) 0= No protocol error in arbitration phase 1= Protocol error in arbitration phase detected (PSR.LEC ≠ 0,7)
26	WDI	R/W	0x0	Watchdog Interrupt 0= No Message RAM Watchdog event occurred 1= Message RAM Watchdog event due to missing READY
25	BO	R/W	0x0	Bus_Off Status 0= Bus_Off status unchanged 1= Bus_Off status changed
24	EW	R/W	0x0	Warning Status 0= Error_Warning status unchanged 1= Error_Warning status changed
23	EP	R/W	0x0	Error Passive 0= Error_Passive status unchanged 1= Error_Passive status changed
22	ELO	R/W	0x0	Error Logging Overflow 0= CAN Error Logging Counter did not overflow 1= Overflow of CAN Error Logging Counter occurred
21	BEU	R/W	0x0	Bit Error Uncorrected Message RAM bit error detected, uncorrected. Controlled by input signal m_ttcn_aeim_berr[1] generated by an optional external parity / ECC logic attached to the Message RAM. An uncorrected Message RAM bit error sets CCCR.INIT to '1'. This is done to avoid transmission of corrupted data. 0= No bit error detected when reading from Message RAM 1= Bit error detected, uncorrected (e.g. parity logic)

20	BEC	R/W	0x0	Bit Error Corrected Message RAM bit error detected and corrected. Controlled by input signal m_ttcn_aeim_berr[0] generated by an optional external parity / ECC logic attached to the Message RAM. 0= No bit error detected when reading from Message RAM 1= Bit error detected and corrected (e.g. ECC)
19	DRX	R/W	0x0	Message stored to Dedicated Rx Buffer The flag is set whenever a received message has been stored into a dedicated Rx Buffer. 0= No Rx Buffer updated 1= At least one received message stored into a Rx Buffer
18	TOO	R/W	0x0	Timeout Occurred 0= No timeout 1= Timeout reached
17	MRAF	R/W	0x0	Message RAM Access Failure The flag is set, when the Rx Handler - has not completed acceptance filtering or storage of an accepted message until the arbitration field of the following message has been received. In this case acceptance filtering or message storage is aborted and the Rx Handler starts processing of the following message. - was not able to write a message to the Message RAM. In this case message storage is aborted. In both cases the FIFO put index is not updated respectively. The New Data flag for a dedicated Rx Buffer is not set, a partly stored message is overwritten when the next message is stored to this location. The flag is also set when the Tx Handler was not able to read a message from the Message RAM in time. In this case message transmission is aborted. In case of a Tx Handler access failure the M_TTCAN is switched into Restricted Operation Mode (See Chapter 28.2.1.5). To leave Restricted Operation Mode, the Host CPU has to reset CCCR.ASM . 0= No Message RAM access failure occurred 1= Message RAM access failure occurred
16	TSW	R/W	0x0	Timestamp Wraparound 0= No timestamp counter wrap-around 1= Timestamp counter wrapped around
15	TEFL	R/W	0x0	Tx Event FIFO Element Lost 0= No Tx Event FIFO element lost 1= Tx Event FIFO element lost, also set after write attempt to Tx Event FIFO of size zero
14	TEFF	R/W	0x0	Tx Event FIFO Full 0= Tx Event FIFO not full 1= Tx Event FIFO full
13	TEFW	R/W	0x0	Tx Event FIFO Watermark Reached 0= Tx Event FIFO fill level below watermark 1= Tx Event FIFO fill level reached watermark
12	TEFN	R/W	0x0	Tx Event FIFO New Entry 0= Tx Event FIFO unchanged 1= Tx Handler wrote Tx Event FIFO element
11	TFE	R/W	0x0	Tx FIFO Empty 0= Tx FIFO non-empty 1= Tx FIFO empty
10	TCF	R/W	0x0	Transmission Cancellation Finished 0= No transmission cancellation finished 1= Transmission cancellation finished
9	TC	R/W	0x0	Transmission Completed 0= No transmission completed 1= Transmission completed

8	HPM	R/W	0x0	High Priority Message 0= No high priority message received 1= High priority message received
7	RF1L	R/W	0x0	Rx FIFO 1 Message Lost 0= No Rx FIFO 1 message lost 1= Rx FIFO 1 message lost, also set after write attempt to Rx FIFO 1 of size zero
6	RF1F	R/W	0x0	Rx FIFO 1 Full 0= Rx FIFO 1 not full 1= Rx FIFO 1 full
5	RF1W	R/W	0x0	Rx FIFO 1 Watermark Reached 0= Rx FIFO 1 fill level below watermark 1= Rx FIFO 1 fill level reached watermark
4	RF1N	R/W	0x0	Rx FIFO 1 New Message 0= No new message written to Rx FIFO 1 1= New message written to Rx FIFO 1
3	RF0L	R/W	0x0	Rx FIFO 0 Message Lost 0= No Rx FIFO 0 message lost 1= Rx FIFO 0 message lost, also set after write attempt to Rx FIFO 0 of size zero
2	RF0F	R/W	0x0	Rx FIFO 0 Full 0= Rx FIFO 0 Not full 1= Rx FIFO 0 full
1	RF0W	R/W	0x0	Rx FIFO 0 Watermark Reached 0= Rx FIFO 0 fill level below watermark 1= Rx FIFO 0 fill level reached watermark
0	RF0N	R/W	0x0	Rx FIFO 0 New Message 0= No new message written to Rx FIFO 0 1= New message written to Rx FIFO 0

Interrupt Enable (IE)

0x80007000 / 0x80008000 + 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
		ARAE	PEDE	PEAE	WDIE	BOE	EWE	EPE	ELOE	BEUE	BECE	DRXE	TOOE	MRAFE	TSWE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TEFLE	TEFFE	TEFWE	TEFNE	TFEE	TCFE	TCE	HPME	RF1LE	RF1FE	RF1WE	RF1NE	RF0LE	RF0FE	RF0WE	RF0NE

Field	Name	RW	Reset	Description
29	ARAE	R/W	0x0	Access to Reserved Address Enable
28	PEDE	R/W	0x0	Protocol Error in Data Phase Enable
27	PEAE	R/W	0x0	Protocol Error in Arbitration Phase Enable
26	WDIE	R/W	0x0	Watchdog Interrupt Enable
25	BOE	R/W	0x0	Bus_Off Status Interrupt Enable
24	EWE	R/W	0x0	Warning Status Interrupt Enable
23	EPE	R/W	0x0	Error Passive Interrupt Enable
22	ELOE	R/W	0x0	Error Logging Overflow Interrupt Enable
21	BEUE	R/W	0x0	Bit Error Uncorrected Interrupt Enable
20	BECE	R/W	0x0	Bit Error Corrected Interrupt Enable
19	DRXE	R/W	0x0	Message stored to Dedicated Rx Buffer Interrupt Enable
18	TOOE	R/W	0x0	Timeout Occurred Interrupt Enable
17	MRAFE	R/W	0x0	Message RAM Access Failure Interrupt Enable
16	TSWE	R/W	0x0	Timestamp Wraparound Interrupt Enable
15	TEFLE	R/W	0x0	Tx Event FIFO Event Lost Interrupt Enable
14	TEFFE	R/W	0x0	Tx Event FIFO Full Interrupt Enable
13	TEFWE	R/W	0x0	Tx Event FIFO Watermark Reached Interrupt Enable
12	TEFNE	R/W	0x0	Tx Event FIFO New Entry Interrupt Enable
11	TFEE	R/W	0x0	Tx FIFO Empty Interrupt Enable
10	TCFE	R/W	0x0	Transmission Cancellation Finished Interrupt Enable
9	TCE	R/W	0x0	Transmission Completed Interrupt Enable
8	HPME	R/W	0x0	High Priority Message Interrupt Enable
7	RF1LE	R/W	0x0	Rx FIFO 1 Message Lost Interrupt Enable
6	RF1FE	R/W	0x0	Rx FIFO 1 Full Interrupt Enable
5	RF1WE	R/W	0x0	Rx FIFO 1 Watermark Reached Interrupt Enable
4	RF1NE	R/W	0x0	Rx FIFO 1 New Message Interrupt Enable
3	RF0LE	R/W	0x0	Rx FIFO 0 Message Lost Interrupt Enable
2	RF0FE	R/W	0x0	Rx FIFO 0 Full Interrupt Enable
1	RF0WE	R/W	0x0	Rx FIFO 0 Watermark Reached Interrupt Enable
0	RF0NE	R/W	0x0	Rx FIFO 0 New Message Interrupt Enable

Interrupt Line Select (ILS)

0x80007000 / 0x80008000 + 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
		ARAL	PEDL	PEAL	WDIL	BOL	EWL	EPL	ELOL	BEUL	BECL	DRXL	TOOL	MRAFL	TSWL
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TEFLL	TEFFL	TEFWL	TEFNL	TFEL	TCFL	TCL	HPML	RF1LL	RF1FL	RF1WL	RF1NL	RF0LL	RF0FL	RF0WL	RF0NL

Field	Name	RW	Reset	Description
29	ARAL	R/W	0x0	Access to Reserved Address Line
28	PEDL	R/W	0x0	Protocol Error in Data Phase Line
27	PEAL	R/W	0x0	Protocol Error in Arbitration Phase Line
26	WDIL	R/W	0x0	Watchdog Interrupt Line
25	BOL	R/W	0x0	Bus_Off Status Interrupt Line
24	EWL	R/W	0x0	Warning Status Interrupt Line
23	EPL	R/W	0x0	Error Passive Interrupt Line
22	ELOL	R/W	0x0	Error Logging Overflow Interrupt Line
21	BEUL	R/W	0x0	Bit Error Uncorrected Interrupt Line
20	BECL	R/W	0x0	Bit Error Corrected Interrupt Line
19	DRXL	R/W	0x0	Message stored to Dedicated Rx Buffer Interrupt Line
18	TOOL	R/W	0x0	Timeout Occurred Interrupt Line
17	MRAFL	R/W	0x0	Message RAM Access Failure Interrupt Line
16	TSWL	R/W	0x0	Timestamp Wraparound Interrupt Line
15	TEFLL	R/W	0x0	Tx Event FIFO Event Lost Interrupt Line
14	TEFFL	R/W	0x0	Tx Event FIFO Full Interrupt Line
13	TEFWL	R/W	0x0	Tx Event FIFO Watermark Reached Interrupt Line
12	TEFNL	R/W	0x0	Tx Event FIFO New Entry Interrupt Line
11	TFEL	R/W	0x0	Tx FIFO Empty Interrupt Line
10	TCFL	R/W	0x0	Transmission Cancellation Finished Interrupt Line
9	TCL	R/W	0x0	Transmission Completed Interrupt Line
8	HPML	R/W	0x0	High Priority Message Interrupt Line
7	RF1LL	R/W	0x0	Rx FIFO 1 Message Lost Interrupt Line
6	RF1FL	R/W	0x0	Rx FIFO 1 Full Interrupt Line
5	RF1WL	R/W	0x0	Rx FIFO 1 Watermark Reached Interrupt Line
4	RF1NL	R/W	0x0	Rx FIFO 1 New Message Interrupt Line
3	RF0LL	R/W	0x0	Rx FIFO 0 Message Lost Interrupt Line
2	RF0FL	R/W	0x0	Rx FIFO 0 Full Interrupt Line
1	RF0WL	R/W	0x0	Rx FIFO 0 Watermark Reached Interrupt Line
0	RF0NL	R/W	0x0	Rx FIFO 0 New Message Interrupt Line

Interrupt Line Enable (ILE)

0x80007000 / 0x80008000 + 0x5C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
														EINT1	EINT0

Field	Name	RW	Reset	Description
1	EINT1	R/W	0x0	Enable Interrupt Line 1 0= Interrupt line m_ttcan_int1 disabled 1= Interrupt line m_ttcan_int1 enabled
0	EINT0	R/W	0x0	Enable Interrupt Line 0 0= Interrupt line m_ttcan_int0 disabled 1= Interrupt line m_ttcan_int0 enabled

Global Filter Configuration (GFC)

0x80007000 / 0x80008000 + 0x80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-										ANFS[1:0]		ANFE[1:0]		RRFS	RRFE

Field	Name	RW	Reset	Description
5:4	ANFS	RP	0x0	Accept Non-matching Frames Standard Defines how received messages with 11-bit IDs that do not match any element of the filter list are treated. 00= Accept in Rx FIFO 0 01= Accept in Rx FIFO 1 10= Reject 11= Reject
3:2	ANFE	RP	0x0	Accept Non-matching Frames Extended Defines how received messages with 29-bit IDs that do not match any element of the filter list are treated. 00= Accept in Rx FIFO 0 01= Accept in Rx FIFO 1 10= Reject 11= Reject
1	RRFS	RP	0x0	Reject Remote Frames Standard 0= Filter remote frames with 11-bit standard IDs 1= Reject all remote frames with 11-bit standard IDs
0	RRFE	RP	0x0	Reject Remote Frames Extended 0= Filter remote frames with 29-bit extended IDs 1= Reject all remote frames with 29-bit extended IDs

Standard ID Filter Configuration (SIDFC)

0x80007000 / 0x80008000 + 0x84

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-								LSS[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLSSA[15:2]														-	

Field	Name	RW	Reset	Description
23:16	LSS	RP	0x0	List Size Standard 0= No standard Message ID filter 1-128= Number of standard Message ID filter elements >128= Values greater than 128 are interpreted as 128
15:2	FLSSA	RP	0x0	Filter List Standard Start Address Start address of standard Message ID filter list (32-bit word address, see Figure 28.4).

Extended ID Filter Configuration (XIDFC)

0x80007000 / 0x80008000 + 0x88

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-								LSE[6:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLESA[15:2]														-	

Field	Name	RW	Reset	Description
22:16	LSE	RP	0x0	List Size Extended 0= No extended Message ID filter 1-64= Number of extended Message ID filter elements >64= Values greater than 64 are interpreted as 64
15:2	FLESA	RP	0x0	Filter List Extended Start Address Start address of extended Message ID filter list (32-bit word address, see Figure 28.5).

Extended ID AND Mask (XIDAM)

0x80007000 / 0x80008000 + 0x90

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-			EIDM[28:16]												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EIDM[15:0]															

Field	Name	RW	Reset	Description
28:0	EIDM	RP	0x1FFF	Extended ID Mask For acceptance filtering of extended frames the Extended ID AND Mask is ANDed with the Message ID of a received frame. Intended for masking of 29-bit IDs in SAE J1939. With the reset value of all bits set to one the mask is not active.

High Priority Message Status (HPMS)

0x80007000 / 0x80008000 + 0x94

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLST	FIDX[6:0]							MSI[1:0]		BIDX[5:0]					

Field	Name	RW	Reset	Description
15	FLST	R	0x0	Filter List Indicates the filter list of the matching filter element. 0= Standard Filter List 1= Extended Filter List.
14:8	FIDX	R	0x0	Filter Index Index of matching filter element. Range is 0 to SIDFC.LSS - 1 resp. XIDFC.LSE - 1.
7:6	MSI	R	0x0	Message Storage Indicator 00= No FIFO selected 01= FIFO message lost 10= Message stored in FIFO 0 11= Message stored in FIFO 1
5:0	BIDX	R	0x0	Buffer Index Index of Rx FIFO element to which the message is stored. Only valid when MSI[1] = '1'.

New Data (NDAT1)

0x80007000 / 0x80008000 + 0x98

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ND31	ND30	ND29	ND28	ND27	ND26	ND25	ND24	ND23	ND22	ND21	ND20	ND19	ND18	ND17	ND16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ND15	ND14	ND13	ND12	ND11	ND10	ND9	ND8	ND7	ND6	ND5	ND4	ND3	ND2	ND1	ND0

Field	Name	RW	Reset	Description
31:0	ND[31:0]	R/W	0x0	New Data The register holds the New Data flags of Rx Buffers 0 to 31. The flags are set when the respective Rx Buffer has been updated from a received frame. The flags remain set until the Host clears them. A flag is cleared by writing a '1' to the corresponding bit position. Writing a '0' has no effect. A hard reset will clear the register. 0= Rx Buffer Not updated 1= Rx Buffer updated from new message

New Data (NDAT2)

0x80007000 / 0x80008000 + 0x9C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ND63	ND62	ND61	ND60	ND59	ND58	ND57	ND56	ND55	ND54	ND53	ND52	ND51	ND50	ND49	ND48
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ND47	ND46	ND45	ND44	ND43	ND42	ND41	ND40	ND39	ND38	ND37	ND36	ND35	ND34	ND33	ND32

Field	Name	RW	Reset	Description
31:0	ND[63:32]	RW	0x0	New Data The register holds the New Data flags of Rx Buffers 32 to 63. The flags are set when the respective Rx Buffer has been updated from a received frame. The flags remain set until the Host clears them. A flag is cleared by writing '1' to the corresponding bit position. Writing '0' has no effect. A hard reset will clear the register. 0= Rx Buffer not updated 1= Rx Buffer updated from new message

Rx FIFO 0 Configuration (RXF0C)

0x80007000 / 0x80008000 + 0xA0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
F0OM	F0WM[6:0]							-	F0S[6:0]						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F0SA[15:2]													-		

Field	Name	RW	Reset	Description
31	F0OM	RP	0x0	FIFO 0 Operation Mode FIFO 0 can be operated in blocking or in overwrite mode (See Chapter 28.2.4.2). 0= FIFO 0 blocking mode 1= FIFO 0 overwrite mode
30:24	F0WM	RP	0x0	Rx FIFO 0 Watermark 0= Watermark interrupt disabled 1-64= Level for Rx FIFO 0 watermark interrupt (IR.RF0W) >64= Watermark interrupt disabled
22:16	F0S	RP	0x0	Rx FIFO 0 Size 0= No Rx FIFO 0 1-64= Number of Rx FIFO 0 elements >64= Values greater than 64 are interpreted as 64 The Rx FIFO 0 elements are indexed from 0 to F0S-1
15:2	F0SA	RP	0x0	Rx FIFO 0 Start Address Start address of Rx FIFO 0 in Message RAM (32-bit word address, see Figure 28.6)

Rx FIFO 0 Status (RXF0S)

0x80007000 / 0x80008000 + 0xA4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-						RF0L	F0F	-			F0PI[5:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-		F0GI[5:0]						-		F0FL[6:0]					

Field	Name	RW	Reset	Description
25	RF0L	R	0x0	Rx FIFO 0 Message Lost This bit is a copy of interrupt flag IR.RF0L . When IR.RF0L is reset, this bit is also reset. 0= No Rx FIFO 0 message lost 1= Rx FIFO 0 message lost, also set after write attempt to Rx FIFO 0 of size zero
24	F0F	R	0x0	Rx FIFO 0 Full 0= Rx FIFO 0 Not full 1= Rx FIFO 0 full
21:16	F0PI	R	0x0	Rx FIFO 0 Put Index Rx FIFO 0 write index pointer, range 0 to 63.
13:8	F0GI	R	0x0	Rx FIFO 0 Get Index Rx FIFO 0 read index pointer, range 0 to 63.
6:0	F0FL	R	0x0	Rx FIFO 0 Fill Level Number of elements stored in Rx FIFO 0, range 0 to 64.

Rx FIFO 0 Acknowledge (RXF0A)

0x80007000 / 0x80008000 + 0xA8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
										F0AI[5:0]					

Field	Name	RW	Reset	Description
5:0	F0AI	R/W	0x0	Rx FIFO 0 Acknowledge Index After the Host has read a message or a sequence of messages from Rx FIFO 0 it has to write the buffer index of the last element read from Rx FIFO 0 to F0AI . This will set the Rx FIFO 0 Get Index RXF0S.F0GI to F0AI + 1 and update the FIFO 0 Fill Level RXF0S.F0FL .

Rx Buffer Configuration (RXBC)

0x80007000 / 0x80008000 + 0xAC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RBSA[15:2]														-	

Field	Name	RW	Reset	Description
15:2	RBSA[15:2]	RP	0x0	Rx Buffer Start Address Configures the start address of the Rx Buffers section in the Message RAM (32-bit word address). Also used to reference debug messages A,B,C..

Rx FIFO 1 Configuration (RXF1C)

0x80007000 / 0x80008000 + 0xB0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
F1OM	F1WM[6:0]							-	F1S[6:0]						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F1SA[15:2]													-		

Field	Name	RW	Reset	Description
31	F1OM	RP	0x0	FIFO 1 Operation Mode FIFO 1 can operate in blocking or in overwrite mode (See Chapter 28.2.4.2). 0= FIFO 1 blocking mode 1= FIFO 1 overwrite mode
30:24	F1WM	RP	0x0	Rx FIFO 1 Watermark 0= Watermark interrupt disabled 1-64= Level for Rx FIFO 1 watermark interrupt (IR.RF1W) >64= Watermark interrupt disabled
22:16	F1S	RP	0x0	Rx FIFO 1 Size 0= No Rx FIFO 1 1-64= Number of Rx FIFO 1 elements >64= Values greater than 64 are interpreted as 64 The Rx FIFO 1 elements are indexed from 0 to F1S - 1.
15:2	F1SA	RP	0x0	Rx FIFO 1 Start Address Start address of Rx FIFO 1 in Message RAM (32-bit word address, see Figure 28.6).

Rx FIFO 1 Status (RXF1S)

0x80007000 / 0x80008000 + 0xB4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMS[1:0]		-				RF1L	F1F	F1P[5:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-		F1GI[5:0]							-	F1FL[6:0]					

Field	Name	RW	Reset	Description
31:30	DMS	R	0x0	Debug Message Status 00= Idle state, wait for reception of debug messages, DMA request is cleared 01= Debug message A received 10= Debug messages A, B received 11= Debug messages A, B, C received, DMA request is set
25	RF1L	R	0x0	Rx FIFO 1 Message Lost This bit is a copy of interrupt flag IR.RF1L . When IR.RF1L is reset, this bit is also reset. 0= No Rx FIFO 1 message lost 1= Rx FIFO 1 message lost, also set after write attempt to Rx FIFO 1 of size zero
24	F1L	R	0x0	Rx FIFO 1 Full 0= Rx FIFO 1 not full 1= Rx FIFO 1 full
21:16	F1P	R	0x0	Rx FIFO 1 Put Index Rx FIFO 1 write index pointer, range 0 to 63.
13:8	F1GI	R	0x0	Rx FIFO 1 Get Index Rx FIFO 1 read index pointer, range 0 to 63
6:0	F1FL	R	0x0	Rx FIFO 1 Fill Level Number of elements stored in Rx FIFO 1, range 0 to 64.

Rx FIFO 1 Acknowledge (RXF1A)

0x80007000 / 0x80008000 + 0xB8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-										F1AI					

Field	Name	RW	Reset	Description
5:0	F1AI	R/W	0x0	Rx FIFO 1 Acknowledge Index After the Host has read a message or a sequence of messages from Rx FIFO 1 it has to write the buffer index of the last element read from Rx FIFO 1 to F1AI . This will set the Rx FIFO 1 Get Index RXF1S.F1GI to F1AI + 1 and update the FIFO 1 Fill Level RXF1S.F1FL .

Rx Buffer / FIFO Element Size Configuration (RXESC)

0x80007000 / 0x80008000 + 0xBC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					RBDS			-	F1DS			-	F0DS		

Field	Name	RW	Reset	Description
10:8	RBDS[2:0]	RP	0x0	Rx Buffer Data Field Size 000= 8 byte data field 001= 12 byte data field 010= 16 byte data field 011= 20 byte data field 100= 24 byte data field 101= 32 byte data field 110= 48 byte data field 111= 64 byte data field
6:4	F1DS[2:0]	RP	0x0	Rx FIFO 1 Data Field Size 000= 8 byte data field 001= 12 byte data field 010= 16 byte data field 011= 20 byte data field 100= 24 byte data field 101= 32 byte data field 110= 48 byte data field 111= 64 byte data field
2:0	F0DS	RP	0x0	Rx FIFO 0 Data Field Size 000= 8 byte data field 001= 12 byte data field 010= 16 byte data field 011= 20 byte data field 100= 24 byte data field 101= 32 byte data field 110= 48 byte data field 111= 64 byte data field

Note: In case that the data field size of an accepted CAN frame exceeds the data field size configured for the matching Rx Buffer or Rx FIFO, only the number of bytes as configured by RXESC are stored to the Rx Buffer resp. Rx FIFO element. The rest of the frame's data field is ignored.

Tx Buffer Configuration (TXBC)

0x80007000 / 0x80008000 + 0xC0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-	TFQM	TFQS[5:0]						NDTB[5:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TBSA[15:2]														-	

Field	Name	RW	Reset	Description
30	TFQM	RP	0x0	Tx FIFO/Queue Mode 0 = TX FIFO operation 1 = Tx Queue operation
29:24	TFQS[5:0]	RP	0x0	Transmit FIFO/Queue Size 0= No Tx FIFO/Queue 1-32= Number of Tx Buffers used for Tx FIFO/Queue >32= Values greater than 32 are interpreted as 32
21:16	NDTB[5:0]	RP	0x0	Number of Dedicated Transmit Buffers 0= No Dedicated Tx Buffers 1-32= Number of Dedicated Tx Buffers >32= Values greater than 32 are interpreted as 32
15:2	TBSA[15:2]	RP	0x0	Tx Buffers Start Address Start address of Tx Buffers section in Message RAM (32-bit word address).

Note: The sum of TFQS and NDTB may be not greater than 32. There is no check for erroneous configurations. The Tx Buffers section in the Message RAM starts with the dedicated Tx Buffers.

Tx FIFO/Queue Status (TXFQS)

0x80007000 / 0x80008000 + 0xC4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
										TFQF	TFQPI[4:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-		TFGI[4:0]						-		TFFL[5:0]					

Field	Name	RW	Reset	Description
21	TFQF	R	0x0	Tx FIFO/Queue Full 0= Tx FIFO/Queue not full 1= Tx FIFO/Queue full
20:16	TFQPI[4:0]	R	0x0	Tx FIFO/Queue Put Index Tx FIFO/Queue write index pointer, range 0 to 31
13:8	TFGI[4:0]	R	0x0	x FIFO Get Index Tx FIFO read index pointer, range 0 to 31. Read as zero when Tx Queue operation is configured (TXBC.TFQM = '1')
5:0	TFFL[5:0]	R	0x0	Tx FIFO Free Level Number of consecutive free Tx FIFO elements starting from TFGI, range 0 to 32. Read as zero when Tx Queue operation is configured (TXBC.TFQM = '1')

Note: In case that mixed configurations where dedicated Tx Buffers are combined with a Tx FIFO or a Tx Queue, the Put and Get Indices indicate the number of the Tx Buffer starting with the first dedicated Tx Buffers.

Example: For a configuration of 12 dedicated Tx Buffers and a Tx FIFO of 20 Buffers, a Put Index of 15 points to the fourth buffer of the Tx FIFO.

Tx Buffer Element Size Configuration (TXESC)

0x80007000 / 0x80008000 + 0xC8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-												TBDS[2:0]			

Field	Name	RW	Reset	Description
2:0	TBDS[2:0]	RP	0x0	Tx Buffer Data Field Size 000= 8 byte data field 001= 12 byte data field 010= 16 byte data field 011= 20 byte data field 100= 24 byte data field 101= 32 byte data field 110= 48 byte data field 111= 64 byte data field

Note: In case that the data length code DLC of a Tx Buffer element is configured to a value higher than the Tx Buffer data field size TXESC.TBDS, the bytes not defined by the Tx Buffer are transmitted as "0xCC" (padding bytes).

Tx Buffer Request Pending (TXBRP)

0x80007000 / 0x80008000 + 0xCC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TRP31	TRP30	TRP29	TRP28	TRP27	TRP26	TRP25	TRP24	TRP23	TRP22	TRP21	TRP20	TRP19	TRP18	TRP17	TRP16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRP15	TRP14	TRP13	TRP12	TRP11	TRP10	TRP9	TRP8	TRP7	TRP6	TRP5	TRP4	TRP3	TRP2	TRP1	TRP0

Field	Name	RW	Reset	Description
31:0	TRP[31:0]	R	0x0	Transmission Request Pending Each Tx Buffer has its own Transmission Request Pending bit. The bits are set via register TXBAR . The bits are reset after a requested transmission has completed or has been cancelled via register TXBCR . TXBRP bits are set only for those Tx Buffers configured via TXBC . After a TXBRP bit has been set, a Tx scan (See Chapter 28.2.5) starts to check for the pending Tx request with the highest priority (Tx Buffer with the lowest Message ID). A cancellation request resets the corresponding transmission request pending bit of register TXBRP . In case that a transmission has already started when a cancellation is requested, this is done at the end of the transmission, regardless whether the transmission was successful or not. The cancellation request bits are reset directly after the corresponding TXBRP bit has been reset. After a cancellation has been requested, a finished cancellation is signaled via TXBCF • after successful transmission together with the corresponding TXBTO bit • when the transmission has not yet been started at the point of cancellation • when the transmission has been aborted due to lost arbitration • when an error occurred during frame transmission In DAR mode all transmissions are automatically cancelled if they are not successful. The corresponding TXBCF bit is set for all unsuccessful transmissions. 0= No transmission request pending 1= Transmission request pending

Note: TXBRP bits which are set while a Tx scan is in progress are not considered during this particular Tx scan. In case that cancellation is requested for such a Tx Buffer, this Add Request is cancelled immediately, and the corresponding TXBRP bit is reset.

Tx Buffer Add Request(TXBAR)

0x80007000 / 0x80008000 + 0xD0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AR31	AR30	AR29	AR27	AR27	AR26	AR25	AR24	AR23	AR22	AR21	AR20	AR19	AR18	AR17	AR16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AR15	AR14	AR13	AR12	AR11	AR10	AR9	AR8	AR7	AR6	AR5	AR4	AR3	AR2	AR1	AR0

Field	Name	RW	Reset	Description
31:0	AR[31:0]	RW	0x0	Add Request Each Tx Buffer has its own Add Request bit. Writing '1' will set the corresponding Add Request bit; writing '0' has no impact. This enables the Host to set transmission requests for multiple Tx Buffers with one write to TXBAR. TXBAR bits are set only for those Tx Buffers configured via TXBC. When no Tx scan is running, the bits are reset immediately, else the bits remain set until the Tx scan process has completed. 0= No transmission request added 1= Transmission requested added

Note: If an add request is applied for a Tx Buffer with pending transmission request (corresponding TXBRP bit is already set), this add request is ignored.

Tx Buffer Cancellation Request (TXBCR)

0x80007000 / 0x80008000 + 0xD4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CR31	CR30	CR29	CR27	CR27	CR26	CR25	CR24	CR23	CR22	CR21	CR20	CR19	CR18	CR17	CR16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CR15	CR14	CR13	CR12	CR11	CR10	CR9	CR8	CR7	CR6	CR5	CR4	CR3	CR2	CR1	CR0

Field	Name	RW	Reset	Description
31:0	CR[31:0]	RW	0x0	Cancellation Request Each Tx Buffer has its own Cancellation Request bit. Writing '1' will set the corresponding Cancellation Request bit; writing '0' has no impact. This enables the Host to set cancellation requests for multiple Tx Buffers with one write to TXBCR. TXBCR bits are set only for those Tx Buffers configured via TXBC. The bits remain set until the corresponding bit of TXBRP is reset. 0= No cancellation pending 1= Cancellation pending

Tx Buffer Transmission Occurred (TXBTO)

0x80007000 / 0x80008000 + 0xD8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TO31	TO30	TO29	TO27	TO27	TO26	TO25	TO24	TO23	TO22	TO21	TO20	TO19	TO18	TO17	TO16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TO5	TO14	TO13	TO12	TO11	TO10	TO9	TO8	TO7	TO6	TO5	TO4	TO3	TO2	TO1	TO0

Field	Name	RW	Reset	Description
31:0	TO[31:0]	R	0x0	Transmission Occurred Each Tx Buffer has its own Transmission Occurred bit. The bits are set when the corresponding TXBRP bit is cleared after a successful transmission. The bits are reset when a new transmission is requested by writing '1' to the corresponding bit of register TXBAR. 0= No transmission occurred 1= Transmission occurred

Tx Buffer Cancellation Finished (TXBCF)

0x80007000 / 0x80008000 + 0xDC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CF31	CF30	CF29	CF27	CF27	CF26	CF25	CF24	CF23	CF22	CF21	CF20	CF19	CF18	CF17	CF16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CF5	CF14	CF13	CF12	CF11	CF10	CF9	CF8	CF7	CF6	CF5	CF4	CF3	CF2	CF1	CF0

Field	Name	RW	Reset	Description
31:0	CF[31:0]	R	0x0	Cancellation Finished Each Tx Buffer has its own Cancellation Finished bit. The bits are set when the corresponding TXBRP bit is cleared after a cancellation is requested via TXBCR . In case that the corresponding TXBRP bit is not set at the point of cancellation, CF is set immediately. The bits are reset when a new transmission is requested by writing '1' to the corresponding bit of register TXBAR . 0= No transmit buffer cancellation 1= Transmit buffer cancellation finished

Tx Buffer Transmission Interrupt Enable (TXBTIE)

0x80007000 / 0x80008000 + 0xE0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TIE31	TIE30	TIE29	TIE27	TIE27	TIE26	TIE25	TIE24	TIE23	TIE22	TIE21	TIE20	TIE19	TIE18	TIE17	TIE16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIE5	TIE14	TIE13	TIE12	TIE11	TIE10	TIE9	TIE8	TIE7	TIE6	TIE5	TIE4	TIE3	TIE2	TIE1	TIE0

Field	Name	RW	Reset	Description
31:0	TIE[31:0]	R/W	0x0	Transmission Interrupt Enable Each Tx Buffer has its own Transmission Interrupt Enable bit. 0= Transmission interrupt disabled 1= Transmission interrupt enabled

Tx Buffer Cancellation Finished Interrupt Enable (TXBCIE)

0x80007000 / 0x80008000 + 0xE4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CFIE31	CFIE30	CFIE29	CFIE27	CFIE27	CFIE26	CFIE25	CFIE24	CFIE23	CFIE22	CFIE21	CFIE20	CFIE19	CFIE18	CFIE17	CFIE16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFIE15	CFIE14	CFIE13	CFIE12	CFIE11	CFIE10	CFIE9	CFIE8	CFIE7	CFIE6	CFIE5	CFIE4	CFIE3	CFIE2	CFIE1	CFIE0

Field	Name	RW	Reset	Description
31:0	CFIE[31:0]	R/W	0x0	Cancellation Finished Interrupt Enable Each Tx Buffer has its own Cancellation Finished Interrupt Enable bit. 0= Cancellation finished interrupt disabled 1= Cancellation finished interrupt enabled

Tx Event FIFO Configuration (TXEFC)

0x80007000 / 0x80008000 + 0xF0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-	-	-	-	-	EFWM[5:0]	-	-	-	-	-	-	-	-	-	-
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
EFSA[15:2]															-

Field	Name	RW	Reset	Description
29:24	EFWM[5:0]	RP	0x0	Event FIFO Watermark 0= Watermark interrupt disabled 1-32= Level for Tx Event FIFO watermark interrupt (IR.TEFW) >32= Watermark interrupt disabled
21:16	EFS[5:0]	RP	0x0	Event FIFO Size 0= Tx Event FIFO disabled 1-32= Number of Tx Event FIFO elements >32= Values greater than 32 are interpreted as 32 The Tx Event FIFO elements are indexed from 0 to EFS-1
15:2	EFSA[15:2]	RP	0x0	Event FIFO Start Address Start address of Tx Event FIFO in Message RAM (32-bit word address).

Tx Event FIFO Status (TXEFS)

0x80007000 / 0x80008000 + 0xF4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-	-	-	-	-	-	TEFL	EFF	-	-	-	-	-	-	-	-
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
EFGI[4:0]						-					EFFL[5:0]				

Field	Name	RW	Reset	Description
25	TEFL	R	0x0	Tx Event FIFO Element Lost This bit is a copy of interrupt flag IR.TEFL . When IR.TEFL is reset, this bit is also reset. 0= No Tx Event FIFO element lost 1= Tx Event FIFO element lost, also set after write attempt to Tx Event FIFO of size zero.
24	EFF	R	0x0	Event FIFO Full 0= Tx Event FIFO not full 1= Tx Event FIFO full
20:16	EFPI[4:0]	R	0x0	Event FIFO Put Index Tx Event FIFO write index pointer, range 0 to 31.
12:8	EFGI[4:0]	R	0x0	Event FIFO Get Index Tx Event FIFO read index pointer, range 0 to 31.
5:0	EFFL[5:0]	R	0x0	Event FIFO Fill Level Number of elements stored in Tx Event FIFO, range 0 to 32.

Tx Buffer Transmission Interrupt Enable(TXBIE)

0x80007000 / 0x80008000 + 0xF8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
-												EFAI[4:0]			

Field	Name	RW	Reset	Description
4:0	EFAI[4:0]	R/W	0x0	Event FIFO Acknowledge Index After the Host has read an element or a sequence of elements from the Tx Event FIFO it has to write the index of the last element read from Tx Event FIFO to EFAI . This will set the Tx Event FIFO GetIndex TXEFS.EFGI to EFAI + 1 and update the Event FIFO Fill Level TXEFS.EFFL .

CFG0

0x80009000 + 0x0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-						CH1_DIS_MORD	CH0_DIS_MORD	-		CH1_CLK_STOP_ACK	CH0_CLK_STOP_ACK	-		CH1_CLK_STOP_REQ	CH0_CLK_STOP_REQ

Field	Name	RW	Reset	Description
9	CH1_DIS_MORD	R/W	0x0	* for debug - This field is always set to '0'
8	CH0_DIS_MORD	R/W	0x0	* for debug - This field is always set to '0'
5	CH1_CLK_STOP_ACK	R/W	0x0	* for debug - This field is always set to '0'
4	CH0_CLK_STOP_ACK	R/W	0x0	* for debug - This field is always set to '0'
1	CH1_CLK_STOP_REQ	R/W	0x0	* for debug - This field is always set to '0'
0	CH0_CLK_STOP_REQ	R/W	0x0	* for debug - This field is always set to '0'

CFG1

0x80009000 + 0x4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFG_CH0_BASE_ADDR															

Field	Name	RW	Reset	Description
15:0	CFG_CH0_BASE_ADDR	R/W	0x0	CH0 RAM BASE ADDR [31:16] - upper 16bit base address of Message RAM. Note: In TCC8132, do Not use 0x1000 (internal SRAM area) in this field. If you want to use internal SRAM area, you can use 0x3000 (mirror address).

CFG2

0x80009000 + 0x8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFG_CH1_BASE_ADDR															

Field	Name	RW	Reset	Description
15:0	CFG_CH1_BASE_ADDR	R/W	0x0	CH1 RAM BASE ADDR [31:16] - upper 16bit base address of Message RAM. Note: In TCC8132, do Not use 0x1000 (internal SRAM area) in this field. If you want to use internal SRAM area, you can use 0x3000 (mirror address).

CFG3

0x80009000 + 0xC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFG_CH2_BASE_ADDR															

Field	Name	RW	Reset	Description
15:0	CFG_CH2_BASE_ADDR	R/W	0x0	Not used

CFG4

0x80009000 + 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
													TCLR	TEN	

Field	Name	RW	Reset	Description
1	TCLR	R/W	0x0	Clear timer count
0	TEN	R/W	0x0	Enable External Timer. (TTCAN Timer)

CFG5

0x80009000 + 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EXT_TIMER_COMP_CLEAR															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXT_TIMER_DIV_RATIO															

Field	Name	RW	Reset	Description
31:16	EXT_TIMER_COMP_CLEAR	R/W	0x0	Timer clear value
15:0	EXT_TIMER_DIV_RATIO	R/W	0x0	Timer Divide ratio

CFG6

0x80009000 + 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Timer Count															

Field	Name	RW	Reset	Description
15:0	Timer Count	R/W	0x0	Timer Counter

CFG7

0x80009000 + 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
										CH2 IRQ		CH1 IRQ		CH0 IRQ	

Field	Name	RW	Reset	Description
5:4	CAN_CH2_IRQ	R	0x0	Not used
3:2	CAN_CH1_IRQ	R	0x0	CAN CH1 IRQ
1:0	CAN_CH0_IRQ	R	0x0	CAN CH0 IRQ

28.3.3 Message RAM

28.3.3.1 Message RAM Configuration

The Message RAM has a width of 32 bits. In case that the parity checking or ECC is used a respective number of bits has to be added to each word. The M_CAN module can be configured to allocate up to 4352 words in the Message RAM. It is not necessary to configure each of the sections listed in Figure 28.12 nor there is any restriction with respect to the sequence of the sections.

When operated in CAN FD mode the required Message RAM size depends on the element size configured for Rx FIFO0, Rx FIFO1, Rx Buffers, and Tx Buffers via **RXESC.F0DS**, **RXESC.F1DS**, **RXESC.RBDS**, and **TXESC.TBDS**.

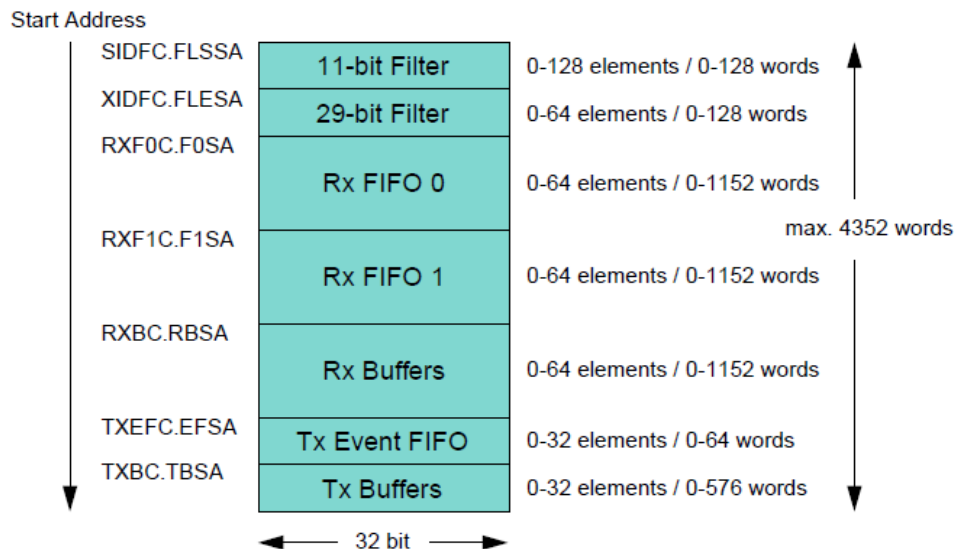


Figure 28.12 CAN Message RAM Configuration

When the M_CAN addresses the Message RAM, it addresses 32-bit words, not single bytes. The configurable start addresses are 32-bit word addresses, i.e. only bits 15 to 2 are evaluated, the two least significant bits are ignored.

Note: The M_CAN does not check for erroneous configuration of the Message RAM. Especially the configuration of the start addresses of the different sections and the number of elements of each section has to be done carefully to avoid falsification or loss of data.

28.3.3.2 Rx Buffer and FIFO Element

Up to 64 Rx Buffers and two Rx FIFOs can be configured in the Message RAM. Each Rx FIFO section can be configured to store up to 64 received messages. The structure of a Rx Buffer / FIFO element is shown in Figure 28.13 below. The element size can be configured for storage of CAN FD messages with up to 64 bytes data field via register RXESC.

	31			24	23	16			15	8	7	0
R0	ESI	XTD	RTR	ID[28:0]								
R1	ANMF	FIDX[6:0]			res	FDF	BRS	DLC[3:0]	RXTS[15:0]			
R2	DB3[7:0]			DB2[7:0]			DB1[7:0]			DB0[7:0]		
R3	DB7[7:0]			DB6[7:0]			DB5[7:0]			DB4[7:0]		
...	...			...			...			...		
Rn	DBm[7:0]			DBm-1[7:0]			DBm-2[7:0]			DBm-3[7:0]		

Figure 28.13 Rx Buffer and FIFO Element

R0

Field	Name	Description
31	ESI	Error State Indicator 0= Transmitting node is error active 1= Transmitting node is error passive
30	XTD	Extended Identifier Signals to the Host whether the received frame has a standard or extended identifier. 0= 11-bit standard identifier 1= 29-bit extended identifier
29	RTR	Remote Transmission Request Signals to the Host whether the received frame is a data frame or a remote frame. 0= Received frame is a data frame 1= Received frame is a remote frame Note: There are no remote frames in CAN FD format. In CAN FD frames (FDF = 1'), the dominant RRS (Remote Request Substitution) bit replaces bit RTR (Remote Transmission Request).
28:0	ID	Identifier. Standard or extended identifier depending on bit XTD . A standard identifier is stored into ID[28:18] .

R1

Field	Name	Description
31	ANMF	Accepted Non-matching Frame Acceptance of non-matching frames may be enabled via GFC.ANFS and GFC.ANFE . 0= Received frame matching filter index FIDX 1= Received frame did not match any Rx filter element
30:24	FIDX	Filter Index 0-127=Index of matching Rx acceptance filter element (invalid if ANMF = '1'). Range is 0 to SIDFC.LSS - 1 resp. XIDFC.LSE - 1.
21	FDF	FD Format 0= Standard frame format 1= CAN FD frame format (new DLC-coding and CRC)
20	BRS	Bit Rate Switch 0= Frame received without bit rate switching 1= Frame received with bit rate switching
19:16	DLC	Data Length Code 0-8= CAN + CAN FD: received frame has 0-8 data bytes 9-15= CAN: received frame has 8 data bytes 9-15= CAN FD: received frame has 12/16/20/24/32/48/64 data bytes
15:0	RXTS	Rx Timestamp Timestamp Counter value captured on start of frame reception. Resolution depending on configuration of the Timestamp Counter Prescaler TSCC.TCP

R2

Field	Name	Description
31:24	DB3	Data Byte 3
23:16	DB2	Data Byte 2
15:8	DB1	Data Byte 1
7:0	DB0	Data Byte 0

R3

Field	Name	Description
31:24	DB7	Data Byte 7
23:16	DB6	Data Byte 6
15:8	DB5	Data Byte 5
7:0	DB4	Data Byte 4

Rn

Field	Name	Description
31:24	DBm	Data Byte m
23:16	DBm-1	Data Byte m-1
15:8	DBm-2	Data Byte m-2
7:0	DBm-3	Data Byte m-3

Note: Depending on the configuration of the element size (RXESC), between two and sixteen 32-bit words (Rn = 3 ..17) are used for storage of a CAN message's data field.

28.3.3.3 Tx Buffer Element

The Tx Buffers section can be configured to hold dedicated Tx Buffers as well as a Tx FIFO / Tx Queue. In case that the Tx Buffer section is shared by dedicated Tx buffers and a Tx FIFO / Tx Queue, the dedicated Tx Buffers start at the beginning of the Tx Buffer section followed by the buffers assigned to the Tx FIFO or Tx Queue. The Tx Handler distinguishes between dedicated Tx Buffers and Tx FIFO / Tx Queue by evaluating the Tx Buffer configuration **TXBC.TFQS** and **TXBC.NDTB**. The element size can be configured for storage of CAN FD messages with up to 64 bytes data field via register TXESC.

	31	24	23	16	15	8	7	0
T0	ESI	XTD	RTR	ID[28:0]				
T1	MM[7:0]			EFC	res	FDF	BRS	DLC[3:0]
T2	DB3[7:0]			DB2[7:0]		DB1[7:0]		DB0[7:0]
T3	DB7[7:0]			DB6[7:0]		DB5[7:0]		DB4[7:0]
...	...			...		...		...
Tn	DBm[7:0]			DBm-1[7:0]		DBm-2[7:0]		DBm-3[7:0]

Figure 28.14 Tx Buffer Element

T0

Field	Name	Description
31	ESI	Error State Indicator 0= ESI bit in CAN FD format depends only on error passive flag 1= ESI bit in CAN FD format transmitted recessive Note: The ESI bit of the transmit buffer is or'ed with the error passive flag to decide the value of the ESI bit in the transmitted FD frame. As required by the CAN FD protocol specification, an error active node may optionally transmit the ESI bit recessive, but an error passive node will always transmit the ESI bit recessive
30	XTD	Extended Identifier 0= 11-bit standard identifier 1= 29-bit extended identifier
29	RTR	Remote Transmission Request 0= Transmit data frame 1= Transmit remote frame Note: When RTR = 1, the M_CAN transmits a remote frame according to ISO 11898-1:2015, even if CCCR.FDOE enables the transmission in CAN FD format.
28:0	ID	Identifier Standard or extended identifier depending on bit XTD . A standard identifier has to be written to ID[28:18] .

T1

Field	Name	Description
31:24	MM	Message Marker Written by CPU during Tx Buffer configuration. Copied into Tx Event FIFO element for identification of Tx message status.
23	EFC	Event FIFO Control 0= Don't store Tx events 1= Store Tx events
21	FDF	FD Format 0= Frame transmitted in Classic CAN format 1= Frame transmitted in CAN FD format
20	BRS	Bit Rate Switching 0= CAN FD frames transmitted without bit rate switching 1= CAN FD frames transmitted with bit rate switching Note: Bits ESI, FDF, and BRS are only evaluated when CAN FD operation is enabled CCCR.FDOE = '1'. Bit BRS is only evaluated when in addition CCCR.BRSE = '1'.
19:16	DLC	Data Length Code 0-8= CAN + CAN FD: transmit frame has 0-8 data bytes 9-15= CAN: transmit frame has 8 data bytes 9-15= CAN FD: transmit frame has 12/16/20/24/32/48/64 data bytes

T2

Field	Name	Description
31:24	DB3	Data Byte 3
23:16	DB2	Data Byte 2
15:8	DB1	Data Byte 1
7:0	DB0	Data Byte 0

T3

Field	Name	Description
31:24	DB7	Data Byte 7
23:16	DB6	Data Byte 6
15:8	DB5	Data Byte 5
7:0	DB4	Data Byte 4

Tn

Field	Name	Description
31:24	DBm	Data Byte m
23:16	DBm-1	Data Byte m-1
15:8	DBm-2	Data Byte m-2
7:0	DBm-3	Data Byte m-3

Note: Depending on the configuration of the element size (TXESC), between two and sixteen 32-bit words (Tn = 3 ..17) are used for storage of a CAN message's data field.

28.3.3.4 Tx Event FIFO Element

Each element stores information about transmitted messages. By reading the Tx Event FIFO the Host CPU gets this information in the order the messages were transmitted. Status information about the Tx Event FIFO can be obtained from register TXEFS.

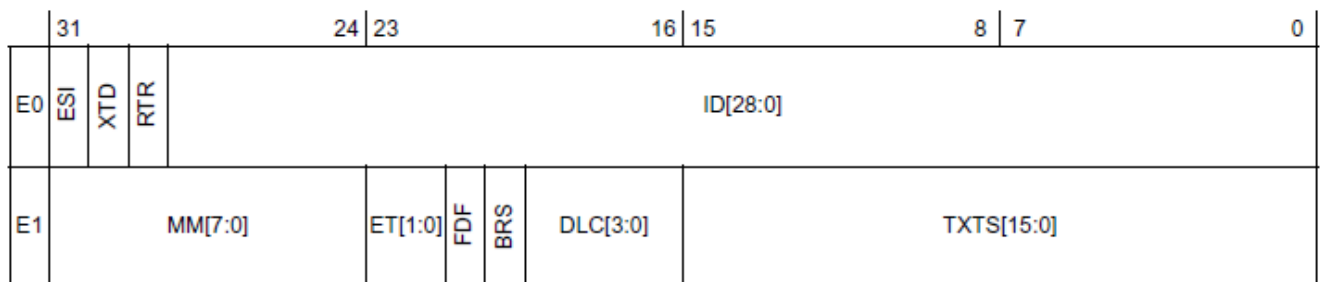


Figure 28.15 Tx Event FIFO Element

E0

Field	Name	Description
31	ESI	Error State Indicator 0= Transmitting node is error active 1= Transmitting node is error passive
30	XTD	Extended Identifier 0= 11-bit standard identifier 1= 29-bit extended identifier
29	RTR	Remote Transmission Request 0= Data frame transmitted 1= Remote frame transmitted
28:0	ID	Identifier Standard or extended identifier depending on bit XTD . A standard identifier is stored into ID[28:18] .

E1

Field	Name	Description
31:24	MM	Message Marker Copied from Tx Buffer into Tx Event FIFO element for identification of Tx message status.
23:22	ET	Event Type 00= Reserved 01= Tx event 10= Transmission in spite of cancellation (always set for transmissions in DAR mode) 11= Reserved
21	FDF	FD Format 0= Standard frame format 1= CAN FD frame format (new DLC-coding and CRC)
20	BRS	Bit Rat Switching 0= Frame transmitted without bit rate switching 1= Frame transmitted with bit rate switching
19:16	DLC	Data Length Code 0-8= CAN + CAN FD: frame with 0-8 data bytes transmitted 9-15= CAN: frame with 8 data bytes transmitted 9-15= CAN FD: frame with 12/16/20/24/32/48/64 data bytes transmitted
15:0	TXTS	Tx Timestamp Timestamp Counter value captured on start of frame transmission. Resolution depending on configuration of the Timestamp Counter Prescaler TSCC.TCP.

28.3.3.5 Standard Message ID Filter Element

Up to 128 filter elements can be configured for 11-bit standard IDs. When accessing a Standard Message ID Filter element, its address is the Filter List Standard Start Address **SIDFC.FLSSA** plus the index of the filter element (0...127).

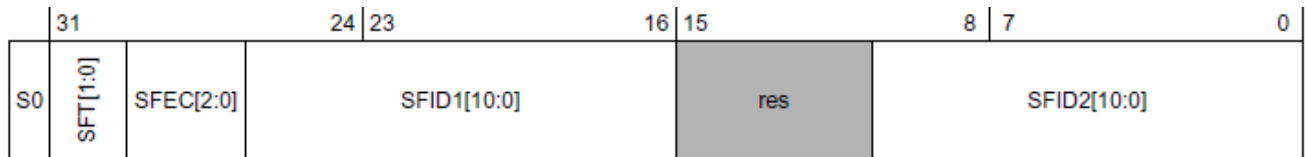


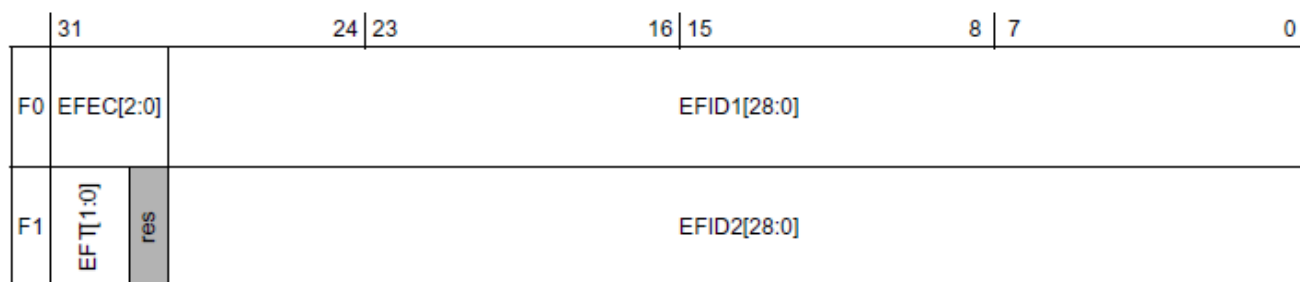
Figure 28.16 Standard Message ID Filter Element

S0

Field	Name	Description
31:30	SFT	Standard Filter Type 00= Range filter from SFID1 to SFID2 (SFID2 $\in$ SFID1) 01= Dual ID filter for SFID1 or SFID2 10= Classic filter: SFID1 = filter, SFID2 = mask 11= Filter element disabled Note: With SFT = "11" the filter element is disabled and the acceptance filtering continues (same behavior as with SFEC = "000")
29:27	SFEC	Standard Filter Element Configuration All enabled filter elements are used for acceptance filtering of standard frames. Acceptance filtering stops at the first matching enabled filter element or when the end of the filter list is reached. If SFEC = "100", "101", or "110" a match sets interrupt flag IR.HPM and, if enabled, an interrupt is generated. In this case register HPMS is updated with the status of the priority match. 000= Disable filter element 001= Store in Rx FIFO 0 if filter matches 010= Store in Rx FIFO 1 if filter matches 011= Reject ID if filter matches 100= Set priority if filter matches 101= Set priority and store in FIFO 0 if filter matches 110= Set priority and store in FIFO 1 if filter matches 111= Store into Rx Buffer or as debug message, configuration of SFT[1:0] ignored
26:16	SFID1	Standard Filter ID 1 First ID of standard ID filter element. When filtering for Rx Buffers or for debug messages this field defines the ID of a standard message to be stored. The received identifiers must match exactly, no masking mechanism is used.
10:0	SFID2	Standard Filter ID 2 This bit field has a different meaning depending on the configuration of SFEC: 1) SFEC = "001"... "110" Second ID of standard ID filter element 2) SFEC = "111" Filter for Rx Buffers or for debug messages SFID2[10:9] decides whether the received message is stored into an Rx Buffer or treated as message A, B, or C of the debug message sequence. 00= Store message into an Rx Buffer 01= Debug Message A 10= Debug Message B 11= Debug Message C SFID2[8:6] is used to control the filter event pins m_can_fe[2:0] at the Extension Interface. '1' at the respective bit position enables generation of a pulse at the related filter event pin with the duration of one m_can_hclk period in case the filter matches. SFID2[5:0] defines the offset to the Rx Buffer Start Address RXBC.RBSA for storage of a matching message.

28.3.3.6 Extended Message ID Filter Element

Up to 64 filter elements can be configured for 29-bit extended IDs. When accessing an Extended Message ID Filter element, its address is the Filter List Extended Start Address **XIDFC.FLESA** plus two times the index of the filter element (0...63).


Figure 28.17 Extended Message ID Filter Element
F0

Field	Name	Description
31:29	EFEC	Extended Filter Element Configuration All enabled filter elements are used for acceptance filtering of extended frames. Acceptance filtering stops at the first matching enabled filter element or when the end of the filter list is reached. If EFEC = "100", "101", or "110" a match sets interrupt flag IR.HPM and, if enabled, an interrupt is generated. In this case register HPMS is updated with the status of the priority match. 000= Disable filter element 001= Store in Rx FIFO 0 if filter matches 010= Store in Rx FIFO 1 if filter matches 011= Reject ID if filter matches 100= Set priority if filter matches 101= Set priority and store in FIFO 0 if filter matches 110= Set priority and store in FIFO 1 if filter matches 111= Store into Rx Buffer or as debug message, configuration of EFT[1:0] ignored
28:0	EFID1	Extended Filter ID 1 First ID of extended ID filter element. When filtering for Rx Buffers or for debug messages this field defines the ID of an extended message to be stored. The received identifiers must match exactly, only XIDAM masking mechanism (See Chapter 28.2.4.1.5) is used.

F1

Field	Name	Description
31:30	EFT	Extended Filter Type 00= Range filter from EFID1 to EFID2 (EFID2 & EFID1) 01= Dual ID filter for EFID1 or EFID2 10= Classic filter: EFID1 = filter, EFID2 = mask 11= Range filter from EFID1 to EFID2 (EFID2 & EFID1), XIDAM mask not applied 111= Store into Rx Buffer or as debug message, configuration of EFT[1:0] ignored
28:0	EFID2	Extended Filter ID 2 This bit field has a different meaning depending on the configuration of EFEC: 1) EFEC = "001"... "110" Second ID of extended ID filter element 2) EFEC = "111" Filter for Rx Buffers or for debug messages EFID2[10:9] decides whether the received message is stored into an Rx Buffer or treated as message A, B, or C of the debug message sequence. 00= Store message into an Rx Buffer 01= Debug Message A 10= Debug Message B 11= Debug Message C EFID2[8:6] is used to control the filter event pins m_can_fe[2:0] at the Extension Interface. '1' at the respective bit position enables generation of a pulse at the related filter event pin with the duration of one m_can_hclk period in case the filter matches. EFID2[5:0] defines the offset to the Rx Buffer Start Address RXBC.RBSA for storage of a matching message.

28.4 Related Blocks

Before accessing the CAN controllers, the registers in the following blocks must be programmed first.

28.4.1 Clock Controller

At reset, the CAN core clocks are all disabled. Clock must be set and enabled by programming the following registers:

- ACLKCAN0 (CAN0 Clock Register)
- ACLKCAN1 (CAN1 Clock Register)
- BCLKCTR1[11:10] (BUS Clock Enable Register bits for CAN1, CAN0 respectively)
- SWRESET1[11:10] (Software Reset Control bits for CAN1, CAN0 respectively)

There is a synchronization logic between fast APB clock domain and the CAN core clock domain. BCLKCTR1[11:10] are clock enable bits for the fast APB clock domain which is synchronous to the CPU. Make sure to enable these bits before accessing the CAN controller(s).

Due to the wait states caused by the clock synchronization logic, the significant degradation in bus performance can happen if the frequency of the CAN controller clock is too slow. Normally, 48MHz or higher frequency is recommended.

28.4.2 GPIO Port

Reset default state for PK[25:22] is GPIO input mode. To enable the signals to/from CAN cores, GPIO Function Select Registers (GPIOPK_FS0 and PMGPIO_FS0) must be programmed as the value shown in the table below. Refer to Chapter 18 GPIO Port for function selection.

Table 28.8 CAN Controller Ports

Signals	Pin Name	GPIO Function Select Register Bits
CAN Controller 0 TXD	PK[22] / C0TXD	GPIOPK_FS0[22] = PMGPIO_FS0[22] = 1
CAN Controller 0 RXD	PK[23] / C0RXD	GPIOPK_FS0[23] = PMGPIO_FS0[23] = 1
CAN Controller 1 TXD	PK[24] / C1TXD	GPIOPK_FS0[24] = PMGPIO_FS0[24] = 1
CAN Controller 1 RXD	PK[25] / C1RXD	GPIOPK_FS0[25] = PMGPIO_FS0[25] = 1

28.4.3 Interrupt Controller

Interrupt signals from the CAN controllers are connected to Interrupt Controller1(B) channel [23:22]. All the corresponding bits of the Interrupt Controller Registers must be programmed with correct values. Note that the interrupt trigger mode default value is "Edge Trigger". For normal operation of CAN cores, the trigger mode should be set as "Level". Refer to TMODE_B and TMODEA_B descriptions in Chapter 11.2.

29 GSB Master/Slave

The TCC8132 provides six General Purpose Serial Bus Controllers, which can be configured as an SPI compatible master/slave. The following figure shows the overall hardware block diagram.

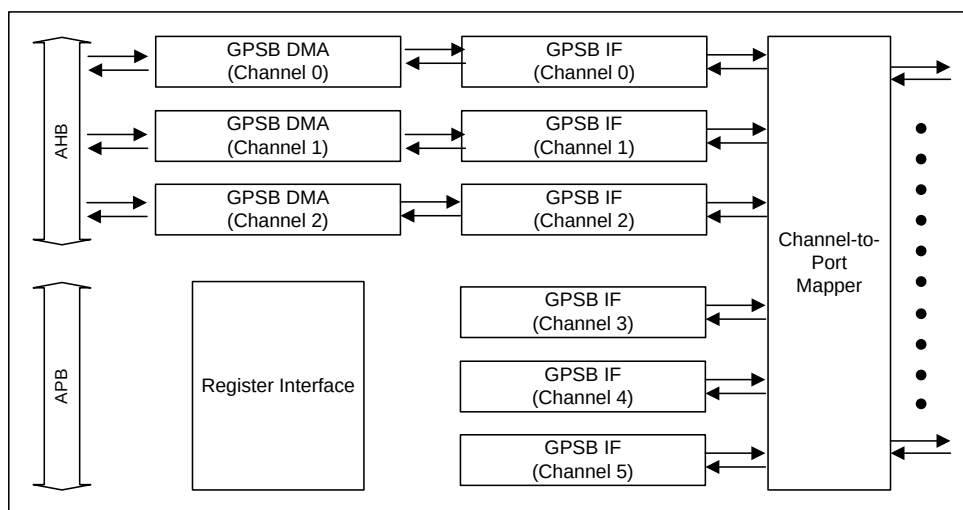


Figure 29.1 Block Diagram

The channel 0,1 and 2 have the dedicated DMA function (GPSB DMA). The GPSB DMA can be used for MPEG-2 TS serial interface with PID matching function.

The “Channel-to-Port Mapper” is for mapping the hardware channel and external ports.

29.1 Features

- Supports Motorola SPI compatible timing
- Supports TI SSP compatible timing
- Programmable Master/Slave configuration
- Programmable timing parameters, data bit width, shift direction, polarity, etc.
- Configurable FIFO depth (either 8x32 or 16x16, determined by bit width)
- Supports Data Swap

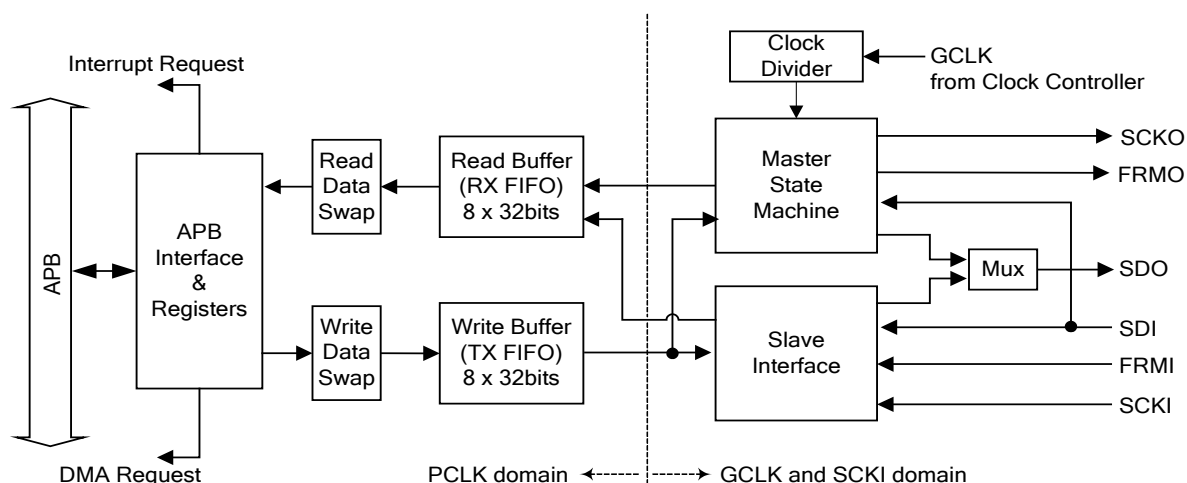


Figure 29.2 GSB Master/Slave Block Diagram

The GSB master/slave block has 6 pins for external serial bus devices.

- SDO Serial Data Output (shared between the master and the slave controller).
- SDI Serial Data Input (shared between the master controller and the slave controller).
- SCKO Serial Clock Output from the master controller
- SCKI Serial Clock Input to the slave controller
- FRMO Frame (Command or Chip Select) Output from the master controller.
- FRMI Frame (Slave Select) Input to slave controller

The frequency of the SCKO pin is determined by the Clock Divider block which uses GCLK as clock source. The GCLK is generated by the Clock Controller Module.

The output enable control is not provided by the core. If a user needs I/O direction control for those output signals, use the GPIO registers.

29.2 GSB Register Description

Table 29.1 GSB Register Map (Base Address = 0x9000Fn00, n=0~5)

Name	Address	Type	Reset	Description
GSBPORT	0x00	R/W	0xFFFFFFFF	GSB Data Port
GSBSTAT	0x04	R/W	0x00000000	GSB Status Register
GSBINTEN	0x08	R/W	0x00000000	GSB Interrupt Enable
GSBMODE	0x0C	R/W	0x00000000	GSB Mode Register
GSBCTRL	0x10	R/W	0x00000000	GSB Control Register
GSBCTRL1	0x14	R/W	0x00000000	GSB Control Register 1

Table 29.2 GSB DMA Register Map (Base Address = 0x9000Fn00, n=0~2)

Name	Address	Type	Reset	Description
TXBASE	0x20	R/W	0xFFFFFFFF	TX Base Address Register
RXBASE	0x24	R/W	0x00000000	RX Base Address Register
PACKET	0x28	R/W	0x00000000	Packet Register
DMA_CTRL	0x2C	R/W	0x00000000	DMA Control Register
DMA_STAT	0x30	R/W	0x00000000	DMA Status Register
DMA_ICR	0x34	R/W	0x00000000	DMA Interrupt Control Register

Table 29.3 GSB Port Selection Register Map (Base Address = 0x9000F800)

Name	Address	Type	Reset	Description
GSB0PORTSEL	0x00	R/W	0x03020100	GSB0 Port Selection Register
GSB1PORTSEL	0x04	R/W	0x07060504	GSB1 Port Selection Register
GSB2PORTSEL	0x08	R/W	0x0B0A0908	GSB2 Port Selection Register
GSB3PORTSEL	0x0C	R/W	0x0F0E0D0C	GSB3 Port Selection Register
GSB4PORTSEL	0x10	R/W	0x13121110	GSB4 Port Selection Register
GSB5PORTSEL	0x14	R/W	0x17161514	GSB5 Port Selection Register

Table 29.4 GSB PID Table Register Map (Base Address = 0x9000FF00)

Name	Address	Type	Reset	Description
PIDTAB	0x00 ~ 0x7F	R/W	-	PID Table

GSB Data Port (GSBPORT)

0x9000Fn00(n=0,1,2,3,4,5)

Bit	Name	R/W	Reset	Description
31:0	DATA	R/W	X	Data Read / Write Port. Any data written to this register is sent to the Write(TX) FIFO. Any data read from this register is from the Read(RX) FIFO. Read data is valid only if RBVCNT is not zero. Although this port can be written even when the core is disabled, no serial bus cycle is generated until the core is enabled with valid parameters.

GSB Status Register (GSBSTAT)

0x9000Fn04(n=0,1,2,3,4,5)

Bit	Name	R/W	Reset	Description
31:29	-	R	0	Reserved
28:24	WBVCNT	R	0	Write(TX) FIFO Valid Entry Count. The maximum value depends on the data bit width.
23:21	-	R	0	Reserved
20:16	RBVCNT	R	0	Read(RX) FIFO Valid Entry Count The maximum value depends on the data bit width.
15:9	-	R	0	Reserved
8	WBOVR	R/W	0	Write FIFO Overrun Error Flag. Cleared by writing "1".
7	RBUDR	R/W	0	Read FIFO Underrun Error Flag. Cleared by writing "1".
6	WBUDR	R/W	0	Write FIFO Underrun Error Flag. Cleared by writing "1".
5	RBOVR	R/W	0	Read FIFO Overrun Error Flag. Cleared by writing "1".
4	RBFULL	R	0	Read FIFO Full Flag
3	WBEMPTY	R	0	Write FIFO Empty Flag
2	RBnEMPTY	R	0	Read FIFO not Empty Flag
1	WBTH	R	0	Write FIFO valid entry count is under threshold ($WBVCNT \leq CFGWTH$)
0	RBTH	R	0	Read FIFO valid entry increased over threshold ($RBVCNT > CFGRTH$)

Among the nine status bits, bit[1:0] are used as conditions for DMA Request signal generation

GSB Interrupt Enable Register (GSBINTEN)

0x9000Fn08(n=0,1,2,3,4,5)

GSBSTAT[8:5] at the end of read cycle.
When this bit is set as "1", GSBSTAT[8:5] are all cleared whenever GSBSTAT register is read.

Bit	Name	R/W	Reset	Description						
31	DRQWB	R/W	0	DMA Request Enable for Write(TX) FIFO. If enabled, the status of WBTH flag is output to the DMA request signal.						
30	DRQRB	R/W	0	DMA Request Enable for Read(RX) FIFO. If enabled, the status of RBTH flag is output to the DMA request signal.						
29:28	-	R	0	Reserved						
27:26	SWAPWD	R/W	0	Swap Enable for Write(TX) Data						
				<table><tr><th>SWAPWD</th><th>Description</th></tr><tr><td>[0]</td><td>Swap byte in half-word { [23:16], [31:24], [7:0], [15:8] }</td></tr><tr><td>[1]</td><td>Swap half-word in word { [15:0], [31:16] }</td></tr></table>	SWAPWD	Description	[0]	Swap byte in half-word { [23:16], [31:24], [7:0], [15:8] }	[1]	Swap half-word in word { [15:0], [31:16] }
				SWAPWD	Description					
				[0]	Swap byte in half-word { [23:16], [31:24], [7:0], [15:8] }					
[1]	Swap half-word in word { [15:0], [31:16] }									
25:24	SWAPRD	R/W	0	Swap Enable for Read(RX) Data						
				<table><tr><th>SWAPRD</th><th>Description</th></tr><tr><td>[0]</td><td>Swap byte in half-word { [23:16], [31:24], [7:0], [15:8] }</td></tr><tr><td>[1]</td><td>Swap half-word in word { [15:0], [31:16] }</td></tr></table>	SWAPRD	Description	[0]	Swap byte in half-word { [23:16], [31:24], [7:0], [15:8] }	[1]	Swap half-word in word { [15:0], [31:16] }
				SWAPRD	Description					
				[0]	Swap byte in half-word { [23:16], [31:24], [7:0], [15:8] }					
[1]	Swap half-word in word { [15:0], [31:16] }									
23:20	CFGWTH	R/W	0	Write(TX) FIFO Threshold for Interrupt/DMA request assertion. (Recommendation: PACKET SIZE's divisor – 1)						
19:16	CFGRTH	R/W	0	Read(RX) FIFO Threshold for Interrupt/DMA request assertion. (Recommendation: PACKET SIZE's divisor – 1)						
15	RCLRSTAT	R/W	0	Clear GSBSTAT[8:5] at the end of read cycle. When this bit is set as "1", GSBSTAT[8:5] are all cleared whenever GSBSTAT register is read.						
14:9	-	R	0	Reserved						
8:0	IROEN	R/W	0	Interrupt Enable for the status bit[8:0] of the GSBSTAT register						

GSB Mode Register (GSBMODE)

0x9000Fn0C(n=0,1,2,3,4,5)

Bit	Name	R/W	Reset	Description									
31:24	DIVLDV	R/W	0	Clock Divider Load Value determines SCKO Frequency (Master Only). $f_{SCKO} = f_{GCLK} / ((DIVLDV + 1) * 2)$									
23	TRECV	R/W	0	Master Recovery Time $t_{RECV} = (TRECV + 1) * (SCKO \text{ period})$									
22	THOLD	R/W	0	Master Hold Time $t_{HOLD} = (THOLD + 1) * (SCKO \text{ period})$									
21	TSETUP	R/W	0	Master Setup Time $t_{SETUP} = (TSETUP + 1) * (SCKO \text{ period})$									
20	POLCS	R/W	0	Polarity Control for Chip Select (FRM). (Master Only) 0: Active Low (Default). Must be 0 for SPI 1: Active High									
19	POLCMD	R/W	0	Polarity Control for Command (FRM) (Master Only) 0: Active High (Default). Must be 0 for SSP 1: Active Low									
18	POLWD	R/W	0	Polarity Control for Write(TX) Data. (Master Only) 0: Data sent at the falling edge of SCK (Default) 1: Data sent at the rising edge of SCK. Must be 1 for SSP									
17	POLRD	R/W	0	Polarity Control for Read(RX) Data. (Master Only) 0: Data received at the rising edge of SCK 1: Data received at the falling edge of SCK Must be 1 for SSP									
16	POLCK	R/W	0	Polarity Control for Serial Clock. Watch out for the different behavior of this bit between Master Mode and Slave Mode.									
				<table><tr><th>POLCK</th><th>Master Mode</th><th>Slave Mode</th></tr><tr><td>0</td><td>SCKO starts from “0”. For SPI Timing 0, 1 and SSP Timing</td><td>SCKI not inverted. For SPI Timing 0 and 3</td></tr><tr><td>1</td><td>SCKO starts from “1” For SPI Timing 2 and 3.</td><td>SCKI inverted For SPI Timing 1, 2, and SSP Timing</td></tr></table>	POLCK	Master Mode	Slave Mode	0	SCKO starts from “0”. For SPI Timing 0, 1 and SSP Timing	SCKI not inverted. For SPI Timing 0 and 3	1	SCKO starts from “1” For SPI Timing 2 and 3.	SCKI inverted For SPI Timing 1, 2, and SSP Timing
				POLCK	Master Mode	Slave Mode							
0	SCKO starts from “0”. For SPI Timing 0, 1 and SSP Timing	SCKI not inverted. For SPI Timing 0 and 3											
1	SCKO starts from “1” For SPI Timing 2 and 3.	SCKI inverted For SPI Timing 1, 2, and SSP Timing											
15:14		R/W	0	Reserved Must have “0” value									
13	-	R	0	Reserved									
12:8	BWS	R/W	0	Bit Width Select. Data bit width is determined by these bits. Data Bit Width = BWS + 1 Valid range for BWS is 7 ~ 31 (8bits ~ 32bits). The FIFOs are configured according to BWS[4] as follows, If BWS[4] == 1, 8 x 32bits. If BWS[4] == 0, 16 x 16bits.									
7	SHIFTDIR	R/W	0	Data Shift Direction Control. 0: Shift Left (MSB First) 1: Shift Right (LSB First)									
6	LOOPBACK	R/W	0	Data Loop Back Enable. SDO is feedback to SDI internally and the incoming data from external I/O is ignored. SDO output is not affected by this bit.									
5	SSDOZ	R/W	0	Slave mode SDO disable. Effective only if GSBCTRL1.SDOCTL == 1.									
4	CONTXFER	R/W	0	Continuous Transfer Mode Enable for SPI and SSP timing. (Master Only) The slave core supports continuous transfer mode regardless of this bit.									
3	ENABLE	R/W	0	CORE ENABLE									
2	CFGSLV	R/W	0	Slave Mode Configuration Enable. 0: GSB Core is in Master Mode 1: GSB Core is in Slave Mode									
1:0	MODE	R/W	0	Mode of operation.									
				<table><tr><th>MODE</th><th>Description</th></tr><tr><td>0</td><td>SPI compatible</td></tr><tr><td>1</td><td>SSP compatible</td></tr><tr><td>2, 3</td><td>Reserved for user defined configurations</td></tr></table>	MODE	Description	0	SPI compatible	1	SSP compatible	2, 3	Reserved for user defined configurations	
				MODE	Description								
				0	SPI compatible								
1	SSP compatible												
2, 3	Reserved for user defined configurations												

The table and the figures below are examples of GSBMODE register values to be programmed for SPI and SSP interface.

Table 29.5 GSB Timing Selection

Timing	CPOL,CPHA	Master	Slave
		POLWD,POLRD,POLCK	POLCK
SPI Timing 0	00	000	0
SPI Timing 1	01	110	1
SPI Timing 2	10	001	1
SPI Timing 3	11	111	0
SSP Timing	NA	110	1

The figures below show the timing of each case listed in the table
DIVLDV = 0 (GCLK / 2), Single Transfer Mode assumed

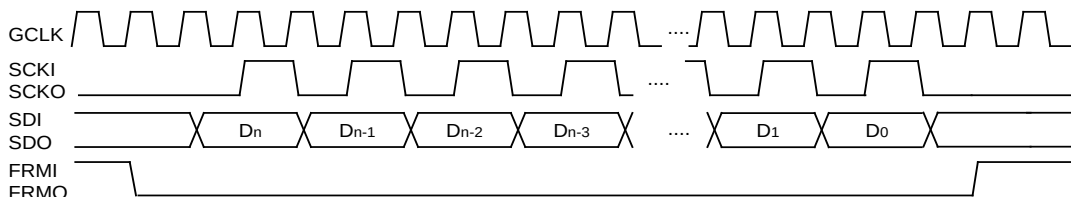


Figure 29.3 SPI Timing 0

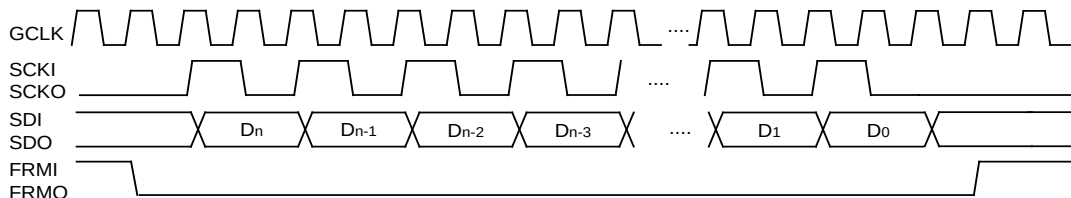


Figure 29.4 SPI Timing 1

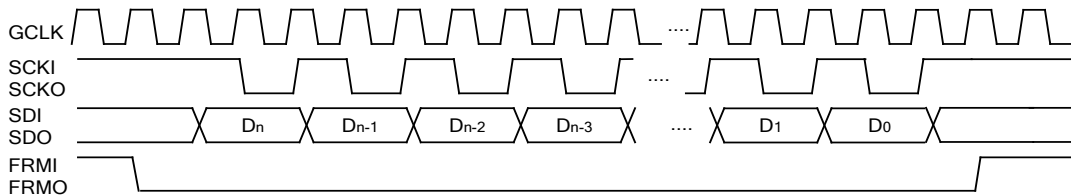


Figure 29.5 SPI Timing 2

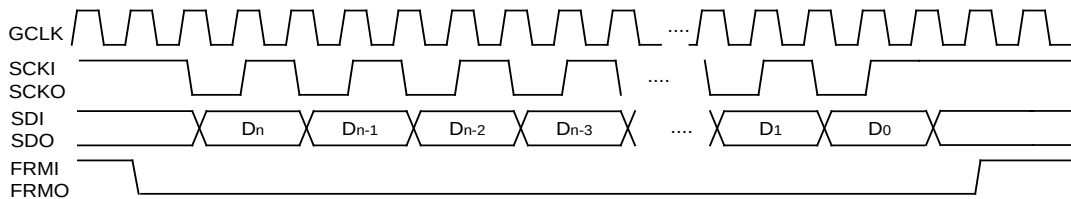


Figure 29.6 SPI Timing 3

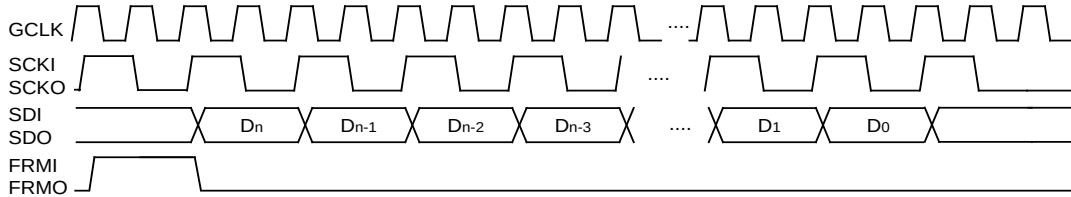


Figure 29.7 SSP Timing

GSB Control Register (GSBCTRL)

0x9000Fn10(n=0,1,2,3,4,5)

Bit	Name	R/W	Reset	Description
31	LASTCKW	R/W	0	Last Clock Disable for Write Cycle.
30	LASTCKR	R/W	0	Last Clock Disable for Read Cycle.
29	-	R/W	0	Reserved
28:24	CMDEND	R/W	0	Command End Position in SCKO clock count.
23:22	-	R	0	Reserved
21	CMDHOLD	R/W	0	Force Command active during Hold time (right after the last SCKO). CMDSTART and CMDEND fields are ignored.
20:16	CMDSTART	R/W	0	Command Start Position in SCKO clock count.
15	CMDFORCE	R/W	0	Force Command Position Control in all modes.
14:13	-	R	0	Reserved
12:8	RDSTART	R/W	0	Read Data Start Position in SCKO clock count.
7	POLWC	R/W	0	Polarity Control for Write Command. (1 = inverted)
6	-	R	0	Reserved
5	FORCEWC	R/W	0	POSWC is ignored and write cycle is forced.
4:0	POSWC	R/W	0	Write Command Position. A bit selected by POSWC is treated as command.

Note: This register is for user defined timing. Do not modify if SPI or SSP timing is required.

GSB Control Register 1 (GSBCTRL1)

0x9000Fn14(n=0,1,2,3,4,5)

Bit	Name	R/W	Reset	Description
31	TXCRX	R/W	0	Enable Counter for Rx operation. When this bit is set to "1" and TXCLDV is written with non-zero value, the write(Tx) buffer is filled with random data until the internal counter is expired.
30	TXCREP	R/W	0	Counter Repeat Enable. The counter is reloaded with TXCLDV.
29	-	R/W	0	Reserved
28	-	R/W	0	Reserved
27	-	R/W	0	Reserved
26	MDLY	R/W	0	Master Output Delay Enable. 0=Disable 1=Enable If enabled, SDO and FRMO signals are delayed by 1/2 GCLK.
25	SDOECTL	R/W	0	SDOE Control Enable.
24	RDPS	R/W	0	Read Data Path Selection.
23:16	-	R	0	Reserved
15:0	TXCLDV	R/W	0	Counter Load Value.

TX Base Register (TXBASE) 0x9000Fn20(n=0,1,2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TX_BASE [31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TX_BASE [15:0]															

Field	Name	RW	Reset	Description
31:0	TX_BASE	R/W	0	TX Base Register

RX Base Register (RXBASE) 0x9000Fn24(n=0,1,2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RX_BASE [31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RX_BASE [15:0]															

Field	Name	RW	Reset	Description
31:0	RX_BASE	R/W	0	RX Base Register

Packet Register (PACKET) 0x9000Fn28(n=0,1,2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-			COUNT												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-			SIZE												

Field	Name	RW	Reset	Description
28:16	COUNT	R/W	0	Packet number information (COUNT + 1)
12:0	SIZE	R/W	0	Packet size information

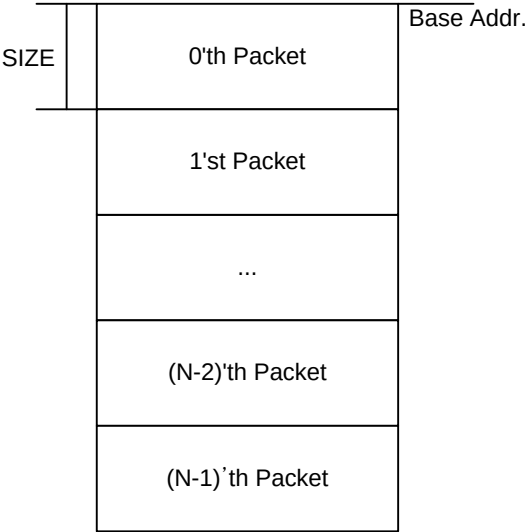


Figure 29.8 Packet Structure

DMA Control Register (DMA_CTRL)

0x9000Fn2C(n=0,1,2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DTE	DRE	CT	END	-								MP	MS	TXAM	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXAM		-								MD		-	PCLR	-	EN

Field	Name	RW	Reset	Description
31	DTE	R/W	0	Transmit DMA request enable '1' for enable
30	DRE	R/W	0	Receive DMA request enable '1' for enable
29	CT	R/W	0	Continuous mode enable (Master) '1' for enable
28	END	R/W	0	Byte endian mode register '0' for little-endian '1' for big-endian If the byte or half-word transfer mode of GPSB mode were used, this field should be '1'. The value of '0' is allowed for word transfer only.
19	MP	R/W	0	PID match mode register '1' for enable (MPEG2-TS mode)
18	MS	R/W	0	Sync byte match control register '1' for enable (MPEG2-TS mode)
17:16	TXAM	R/W	0	TX addressing mode '0' : Multiple Packet '1' : Fixed address (base) '2', '3' : Single Packet
15:14	RXAM	R/W	0	RX addressing mode '0' : Multiple Packet '1' : Fixed address (base) '2', '3' : Single Packet
5:4	MD	R/W	0	DMA mode register 00: normal mode 01: MPEG2-TS mode 1x: Reserved
2	PCLR	W	0	Clear TX/RX packet counter
0	EN	R/W	0	DMA enable register '1' for enable

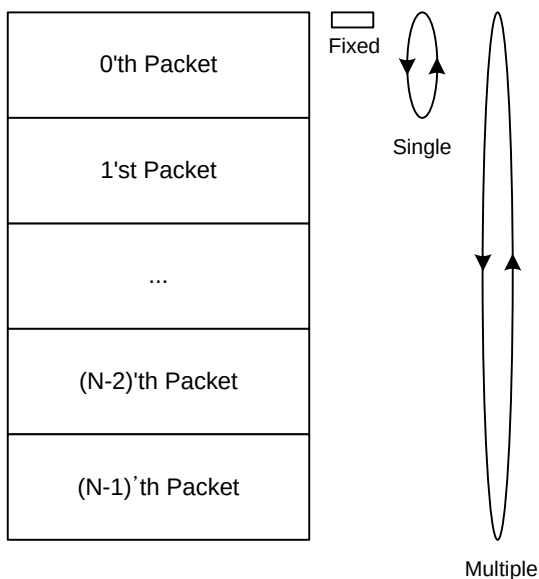


Figure 29.9 TX/RX Addressing Modes

DMA STATUS Register (DMA_STAT)

0x9000Fn30(n=0,1,2)

DMA Channel Register (DMA-CHAN)																
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
0		RXPCNT														0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0			TXPCNT													

Field	Name	RW	Reset	Description
29:17	RXPCNT	R	0	Receive packet count register
12:0	TXPCNT	R	0	Transmit packet count register

DMA IRQ Register (DMAICR)

0x9000Fn34(n=0,1,2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-		ISD	ISP	-							IRQS	-		IED	IEP
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-			IRQPCNT												

Field	Name	RW	Reset	Description
29	ISD	R/C	0	IRQ status for "Done Interrupt" When the number of packets which DMA transfers/receives is equal to the number of packets specified in PACKET register, it is issued. Therefore, it is not valid in continuous mode. When writing 1, it is cleared.
28	ISP	R/C	0	IRQ status for "Packet Interrupt". It is issued every IRQPCNT packets which DMA transfers/receives. When writing 1, it is cleared.
20	IRQS	R/W	0	IRQ select register '0' for receiving '1' for transmitting
17	IED	R/W	0	IRQ enable for "Done Interrupt"
16	IEP	R/W	0	IRQ enable for "Packet Interrupt"
12:0	IRQPCNT	R/W	0	IRQ packet count register

PID Table (PIDTAB)

0x9000FF00 ~ 0x9000FF7C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CH2	CH1	CH0													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-			PID												

Field	Name	RW	Reset	Description
31	CH2	R/W	X	Channel 2 enable
30	CH1	R/W	X	Channel 1 enable
29	CH0	R/W	X	Channel 0 enable
12:0	PID	R/W	X	PID value

Before starting the PID matching, the CH2, CH1 and CH0 fields of all the table entries should be initialized.

29.3 Port Multiplexing

Before the signals are sent to the GPIO block for pin multiplexing, the signals are internally rearranged to have more flexibility in port configuration. The signal name GBUS[191:0] found in GPIO Function Selection Registers (GPIO*_FS* of GPIO block) is the final output of the port multiplexers in this module.

Any GSB signal can be mapped to any bit position of GBUS[191:0] by writing an index number to the registers below in TCC8132. The pre-mapped bit position of GBUS[191:0] is shown the table below.

Table 29.6 Pre-mapped bit position for GSB Port

GBUS	GPIO
GBUS[94:97]	PC[14:17]
GBUS[98:101]	PC[18:21]
GBUS[28:31]	PS[28:31]

Table 29.7 GSB Port Selection Register Map (Base Address = 0x9000F800)

Name	Address	Type	Reset	Description
GSB0PORTSEL	0x00	R/W	0x03020100	GSB0 Port Selection Register
GSB1PORTSEL	0x04	R/W	0x07060504	GSB1 Port Selection Register
GSB2PORTSEL	0x08	R/W	0x0B0A0908	GSB2 Port Selection Register
GSB3PORTSEL	0x0C	R/W	0x0F0E0D0C	GSB3 Port Selection Register
GSB4PORTSEL	0x10	R/W	0x13121110	GSB4 Port Selection Register
GSB5PORTSEL	0x14	R/W	0x17161514	GSB5 Port Selection Register

GSB0 Port Selection Register (GSB0PORTSEL)

0x9000F800

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x03	Port number for SDO
23:16	FRM	R/W	0x02	Port number for FRM
15:8	SCK	R/W	0x01	Port number for SCK
7:0	SDI	R/W	0x00	Port number for SDI

GSB1 Port Selection Register (GSB1PORTSEL)

0x9000F804

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x07	Port number for SDO
23:16	FRM	R/W	0x06	Port number for FRM
15:8	SCK	R/W	0x05	Port number for SCK
7:0	SDI	R/W	0x04	Port number for SDI

GSB2 Port Selection Register (GSB2PORTSEL)

0x9000F808

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x0B	Port number for SDO
23:16	FRM	R/W	0x0A	Port number for FRM
15:8	SCK	R/W	0x09	Port number for SCK
7:0	SDI	R/W	0x08	Port number for SDI

GSB3 Port Selection Register (GSB3PORTSEL)

0x9000F80C

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x0F	Port number for SDO
23:16	FRM	R/W	0x0E	Port number for FRM
15:8	SCK	R/W	0x0D	Port number for SCK
7:0	SDI	R/W	0x0C	Port number for SDI

GSB4 Port Selection Register (GSB4PORTSEL)

0x9000F810

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x13	Port number for SDO
23:16	FRM	R/W	0x12	Port number for FRM
15:8	SCK	R/W	0x11	Port number for SCK
7:0	SDI	R/W	0x10	Port number for SDI

GSB5 Port Selection Register (GSB5PORTSEL)

0x9000F814

Bit	Name	R/W	Reset	Description
31:24	SDO	R/W	0x17	Port number for SDO
23:16	FRM	R/W	0x16	Port number for FRM
15:8	SCK	R/W	0x15	Port number for SCK
7:0	SDI	R/W	0x14	Port number for SDI

29.4 GSB Timing

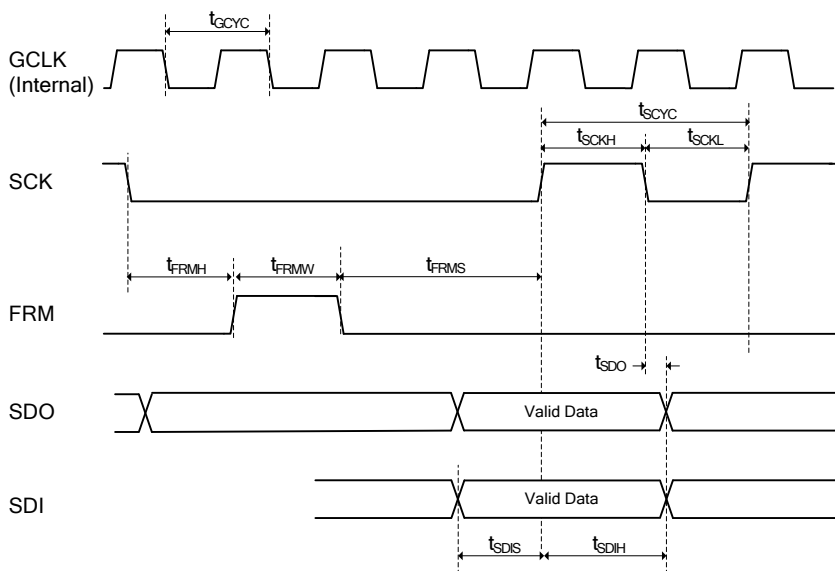


Figure 29.10 GSB Interface Timing (Master)

Table 29.8 GSB Interface Timing Parameters for SPI Timing 0 (Master)

Parameter	Symbol	MIN	MAX	Unit
GCLK cycle time	t_{GCYC}	20	-	ns
SCK cycle time	t_{SCYC}	$2 * t_{GCYC}$	-	ns
SCK pulse width high	t_{SCKH}	$0.45 * t_{SCYC}$	-	ns
SCK pulse width low	t_{SCKL}	$0.45 * t_{SCYC}$	-	ns
SCK rise time	t_{SCKR}	-	$0.05 * t_{SCYC}$	ns
SCK fall time	t_{SCKF}	-	$0.05 * t_{SCYC}$	ns
FRM output low to SCK rising edge	t_{FRMS}	t_{SCYC}	-	ns
FRM output high from SCK falling edge	t_{FRMH}	t_{GCYC}	-	ns
FRM output high pulse width	t_{FRMW}	t_{GCYC}	-	ns
SDO output delay from SCK falling edge	t_{SDO}	-	10	ns
SDI setup time to SCK rising edge	t_{SDIS}	10	-	ns
SDI hold time from SCK rising edge	t_{SDIH}	10	-	ns

Note:

1. CL = 30pF
2. GCLK is an internal signal
3. Parameters are for SPI Timing 0

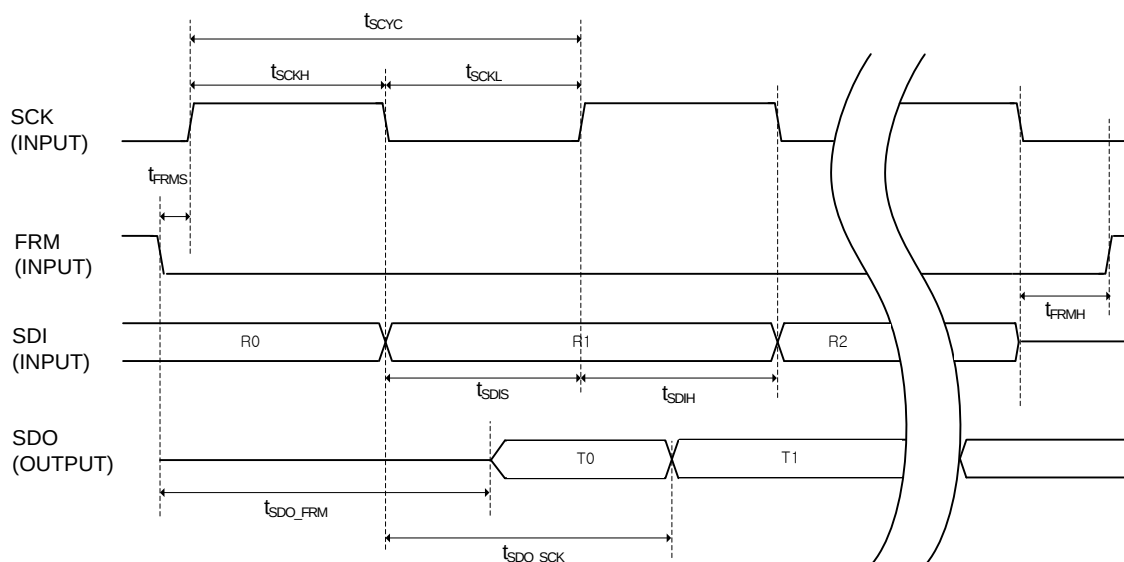


Figure 29.11 GSB Interface Timing (Slave)

Table 29.9 GSB Interface Timing Parameters for SPI Timing 0 (Slave)

Parameter	Symbol	MIN	MAX	Unit
SCK cycle time	t_{SCKC}	20	-	ns
SCK pulse width high	t_{SCKH}	$0.45 * t_{SCKC}$	-	ns
SCK pulse width low	t_{SCKL}	$0.45 * t_{SCKC}$	-	ns
FRM input low setup time to SCK rising edge	t_{FRMS}	2	-	ns
FRM input low hold time from SCK falling edge	t_{FRMH}	11	-	ns
SDO output delay from FRM falling edge	t_{SDO_FRM}	9	16	ns
SDO output delay from SCK falling edge	t_{SDO_SCK}	9	15	ns
SDI setup time to SCK rising edge	t_{SDIS}	8	-	ns
SDI hold time from SCK rising edge	t_{SDIH}	0	-	ns

Note:

CL = 30pF

Parameters are for SPI Timing 0

30 I2C Controller

30.1 Overview

The I2C controller in the TCC8132 supports four masters and two slave controllers. External pins for each controller can be selected by I2C Port Selection Register and GPIO Function Selection Registers.

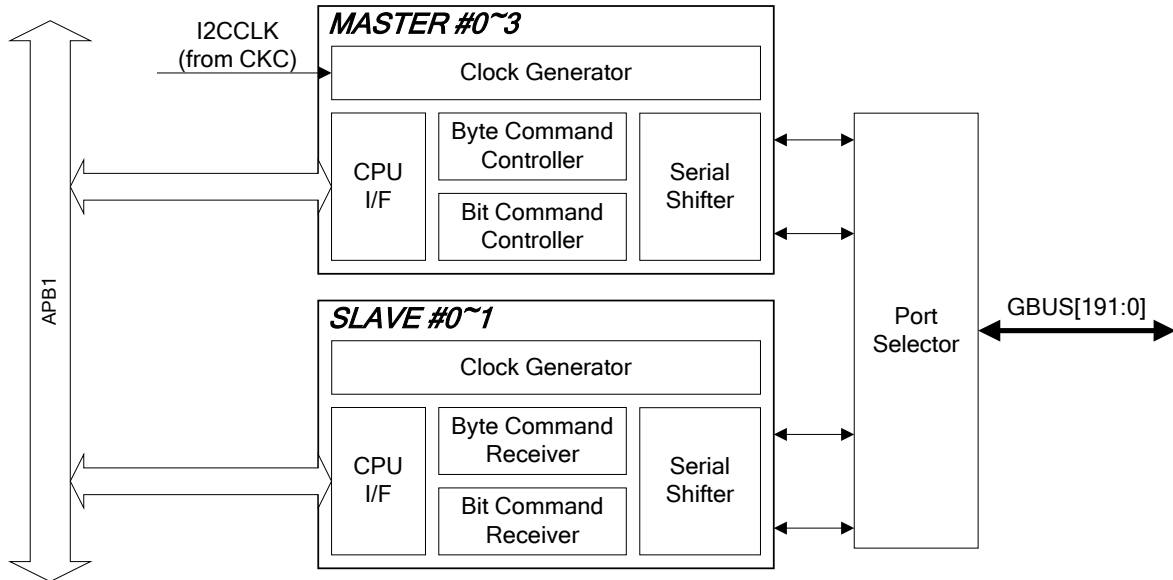


Figure 30.1 I2C Block Diagram

30.2 Related Blocks

After the signals are enabled, I2CCLK (the main clock of I2C) must be enabled and configured to the proper frequency. Refer to chapter 10 for I2CCLK related descriptions.
For internal synchronization, the APB clock frequency must be faster than the I2CCLK frequency.

$$f_{I2CCLK} \leq f_{HCLK} / 4.0$$

30.3 I2C Master Function

30.3.1 Features

- Multi Master Operation
- Supports Fast mode (400Kbps)
- Programmable clock frequency
- Supports wait state by clock stretching

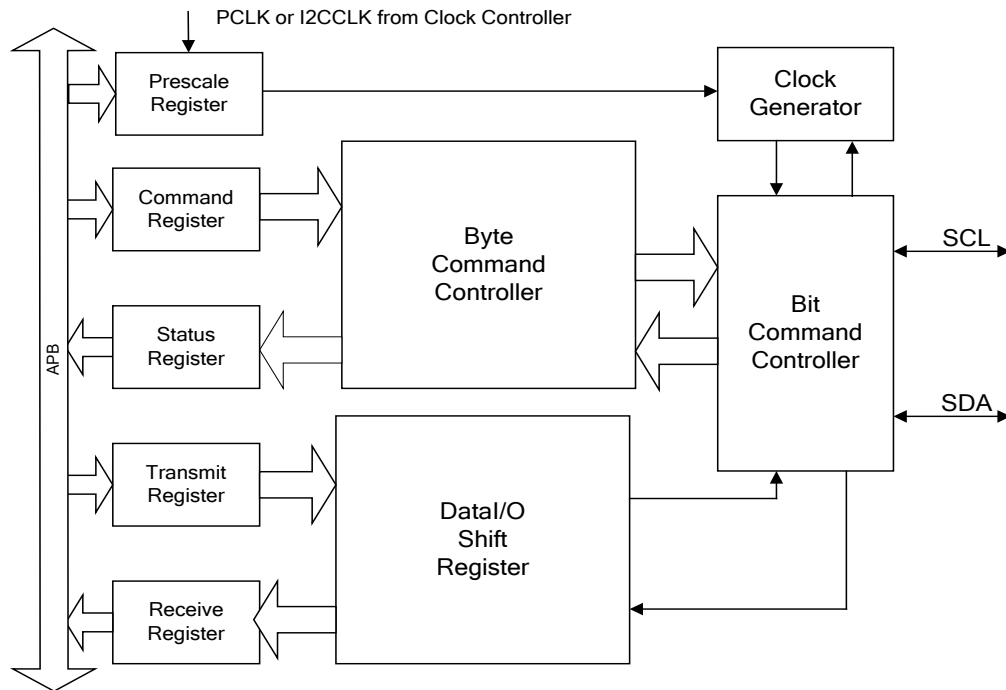


Figure 30.2 I2C Master Block Diagram

30.4 I2C Slave Function

30.4.1 Features

- Supports Fast mode (400kbps)
- 4byte TX FIFO, 4byte RX FIFO and 8byte Buffer

The following figure and table show an example format for accessing Buffers and FIFOs in the I2C slave core.

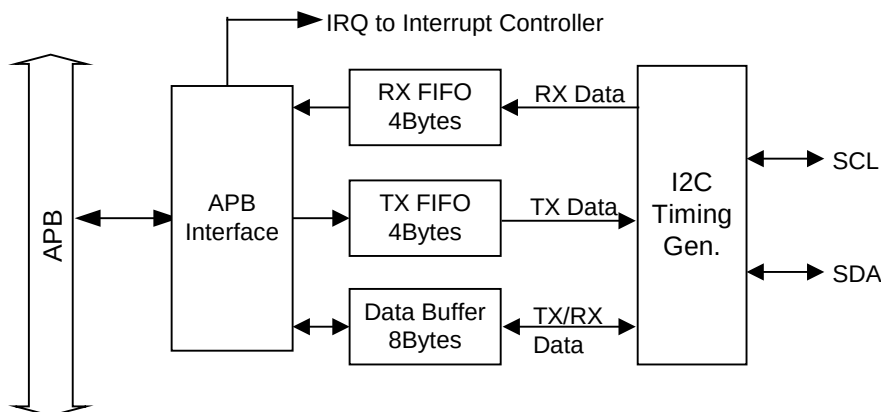


Figure 30.3 I2C Slave Block Diagram

MSB		LSB		MSB		LSB		MSB		LSB									
S	SLAVE ADDRESS		R/W	A	SUB ADDRESS		A	DATA BYTE0		A	DATA BYTE1		A	DATA BYTE2		A	DATA BYTE3		A
7bits			1	1	8bits			8bits											

S: Start Condition

R/W: Read(1) or Write(0)

A: Acknowledge from/to the core

SUB ADDRESS[7:0]	DATA[7:0] Source / Destination	Note
0x00 ~ 0x03	MB0 (Data Buffer 0) Byte 0 ~ 3	
0x04 ~ 0x07	MB1 (Data Buffer 1) Byte 0 ~ 3	
0x08 ~ 0x7F	Reserved	NACK is sent to a master
0x80 ~ 0xFF	RX/TX FIFO	All addresses are directed to the RX/TX FIFO

30.5 Register Description

Table 30.1 I2C Register Map (Base Address = 0x9000B000)

	Name	Address	Type	Reset	Description
MASTER 0	PRES	0x000	R/W	0xFFFF	Clock Prescale register
	CTRL	0x004	R/W	0x0000	Control Register
	TXR	0x008	W	0x0000	Transmit Register
	CMD	0x00C	W	0x0000	Command Register
	RXR	0x010	R	0x0000	Receive Register
	SR	0x014	R	0x0000	Status Register
	TR	0x018	R/W	0x0000	Timing Control Register
MASTER 1	PRES	0x040	R/W	0xFFFF	Clock Prescale register
	CTRL	0x044	R/W	0x0000	Control Register
	TXR	0x048	W	0x0000	Transmit Register
	CMD	0x04C	W	0x0000	Command Register
	RXR	0x050	R	0x0000	Receive Register
	SR	0x054	R	0x0000	Status Register
	TR	0x058	R/W	0x0000	Timing Control Register
SLAVE 0	DPORT	0x080	R/W	-	Data Access port (TX/RX FIFO)
	CTL	0x084	R/W	0x00000000	Control register
	ADDR	0x088	W	0x00000000	Address register
	INT	0x08C	W	0x00000000	Interrupt Enable Register
	STAT	0x090	R	0x00000000	Status Register
	MBF	0x09C	R/W	0x00000000	Buffer Valid Flag
	MB0	0x0A0	R/W	0x00000000	Data Buffer 0 (Byte 3 ~ 0)
	MB1	0x0A4	R/W	0x00000000	Data Buffer 1 (Byte 7 ~ 4)
Status0	IROSTR	0x0C0	R	0x00000000	IRQ Status Register
MASTER 2	PRES	0x100	R/W	0xFFFF	Clock Prescale register
	CTRL	0x104	R/W	0x0000	Control Register
	TXR	0x108	W	0x0000	Transmit Register
	CMD	0x10C	W	0x0000	Command Register
	RXR	0x110	R	0x0000	Receive Register
	SR	0x114	R	0x0000	Status Register
	TR	0x118	R/W	0x0000	Timing Control Register
MASTER 3	PRES	0x140	R/W	0xFFFF	Clock Prescale register
	CTRL	0x144	R/W	0x0000	Control Register
	TXR	0x148	W	0x0000	Transmit Register
	CMD	0x14C	W	0x0000	Command Register
	RXR	0x150	R	0x0000	Receive Register
	SR	0x154	R	0x0000	Status Register
	TR	0x158	R/W	0x0000	Timing Control Register
SLAVE 1	DPORT	0x180	R/W	-	Data Access port (TX/RX FIFO)
	CTL	0x184	R/W	0x00000000	Control register
	ADDR	0x188	W	0x00000000	Address register
	INT	0x18C	W	0x00000000	Interrupt Enable Register
	STAT	0x190	R	0x00000000	Status Register
	MBF	0x19C	R/W	0x00000000	Buffer Valid Flag
	MB0	0x1A0	R/W	0x00000000	Data Buffer 0 (Byte 3 ~ 0)
	MB1	0x1A4	R/W	0x00000000	Data Buffer 1 (Byte 7 ~ 4)
Status1	IROSTR	0x1C0	R	0x00000000	IRQ Status Register
I2CM0/1PORTSEL		0x200	R/W	0xFFFFFFFF	I2C Master 0/1 Port Selection Register
I2CM2/3PORTSEL		0x204	R/W	0xFFFFFFFF	I2C Master 2/3 Port Selection Register
I2CS0/1PORTSEL		0x208	R/W	0xFFFFFFFF	I2C Slave 0/1 Port Selection Register
I2CSWRST		0x210	R/W	0x00000000	Soft Reset Register
I2CSTATUS		0x214	R	0x00000000	I2C Status Register

Prescale Register (PRES)

0x9000B000/040/100/140

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Clock Prescale data															

This register is used to prescale the SCL clock line. Due to the structure of the I2C interface, the core uses a 5*SCL clock internally. The prescale register must be programmed to this 5*SCL frequency (minus 1). Change the value of the prescale register only when 'EN' bit is cleared.

Example:

CLK Input frequency = 8MHz

Desired SCL frequency = 100kHz

Prescale = (8MHz / (5*100kHz)) – 1 = 15

Control Register (CTR)

0x9000B004/044/104/144

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								EN	IEN	MOD	Reserved				

Bit	Name	R/W	Reset	Description
31:8	-	R	0	Reserved
7	EN	R/W	0	I2C Core Enable. The core responds to new commands only when the 'EN' bit is set, and pending commands are finished. Clear the 'EN' bit only when no transfer is in progress, i.e. after a STOP command, or when the command register has the STOP bit set. When halted during a transfer, the core can hang the I ² C bus.
6	IEN	R/W	0	Interrupt Enable.
5	MOD	R/W	0	Data Width 0: 8bit Mode 1: 16bit Mode
4:0	-	R	0	Reserved

Transmit Register (TXR)

0x9000B008/048/108/148

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Transmit Data															

When CTRL[5] is set, the Transmit Data bit width becomes 16bit in case that 16bit mode is selected. Default mode is 8bit mode.

Command Register (CMD)

0x9000B00C/04C/10C/14C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								STA	STO	RD	WR	ACK	Reserved	IACK	

Bit	Name	R/W	Reset	Description
31:8	-	R	0	Reserved
7	STA	R/W	0	Generate Start Condition. Cleared automatically.
6	STO	R/W	0	Generate Stop Condition. Cleared automatically.
5	RD	R/W	0	Read From Slave. Cleared automatically.
4	WR	R/W	0	Write to Slave. Cleared automatically.
3	ACK	R/W	0	When a receiver, sends ACK (ACK = '0') or NACK (ACK = '1')
2:1	-	R	0	Reserved
0	IACK	R/W	0	Interrupt Acknowledge. Writing "1" clears a pending interrupt.

Receive Register (RXR)

0x9000B010/050/110/150

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Receive Data															

When CTRL[5] is set, the Transmit Data bit width becomes 16bit in case of 16bit mode is selected. Default mode is 8bit mode.

Status Register (SR)

0x9000B014/054/114/154

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								RxACK	BUSY	AL	0	0	0	TIP	IF

Bit	Name	R/W	Reset	Description
31:8	-	R	0	Reserved
7	RxACK	R	0	Received Acknowledge from a slave. 0: Acknowledge received 1: No Acknowledge received
6	BUSY	R	0	Bus Busy flag 0: after STOP signal detected 1: after START signal detected
5	AL	R	0	Arbitration Lost flag indicates the core lost arbitration on the bus. Arbitration is lost when: STOP signal is detected, but non requested The master drives SDA high, but SDA is low
4:2	-	R	0	Reserved
1	TIP	R	0	Transfer in progress. "1" indicates that read or write transfer is in progress.
0	IF	R	0	Interrupt Flag indicates interrupt is pending. This bit is set whenever a command is completed or arbitration is lost.

Timing Register (TR)

0x9000B018/058/1B0/158

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RC								0		CKSEL	FC				

Bit	Name	R/W	Reset	Description
31:16	-	R	0	Reserved
15:8	RC[7:0]	R/W	0	Recovery Time Counter Load Value. "0" disables recovery time counter. The recovery time counter is enabled and loaded with RC[7:0] whenever a STOP condition is issued by the core. Execution of a new command written to CMD register is delayed until the counter is expired. Actual wait time = (I2CCLK Period) * PRES[15:0] * 5 * RC[7:0]
7:6	-	R/W	0	Reserved
5	CKSEL	R/W	0	Clock Source Select. 0: I2CCLK from Clock Controller. 1: PCLK (HCLK) divided by 2
4:0	FC	R/W	0	Noise filter counter load value. "0" disables noise filter. SCL and SDA inputs are checked for stability until the counter is expired.

Data Port (DPORT)

0x9000B080/180

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0					TXVC				0				RXVC		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								PORT							

Bit	Name	R/W	Reset	Description
31:27	-	R	0	Reserved
26:24	TXVC	R	0	TX Buffer valid entry count
23:19	-	R	0	Reserved
18:16	RXVC	R	0	RX Buffer valid entry count
15:8	-	R	0	Reserved
7:0	PORT	R/W	X	TX/RX FIFO access port.

Master Write Cycle to RX FIFO:

MSB		LSB		MSB		LSB		MSB		LSB		MSB		LSB	
S	WRITE SLAVE ADDRESS	W(0)	A	WRITE 1XXXXXXXb	A	WRITE BYTE0	A	WRITE BYTE1	A	WRITE BYTE2	A	WRITE BYTE3	A		
7bits		1	1	8bits		8bits									

Master Read Cycle to TX FIFO:

The first byte is always read as 0xFF. The first byte must be discarded by the master.

MSB		LSB		MSB		LSB		MSB		LSB				
S	WRITE SLAVE ADDRESS	R(1)	A	Always Read as 1111111b		A	READ BYTE0	A	READ BYTE1	A	READ BYTE2	A	READ BYTE3	A
7bits		1	1	8bits		8bits								

Control Register (CTL)

0x9000B084/184

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SLV		0				FC				DRQEN					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0		TXTH		0		RXTH		0		RCLR	WS	SDA	CLRFF	0	EN

Bit	Name	R/W	Reset	Description
31:30	SLV			31: Slave 1 30: Slave 0
29:25	-	R	0	Reserved
24:20	FC	R/W	0x4	Filter Counter Load Value
19:16	DRQEN	R/W	0	DMA Request Enable
15:14	-	R	0	Reserved
13:12	TXTH	R/W	0	TX FIFO Threshold for DMA Request
11:10	-	R	0	Reserved
9:8	RXTH	R/W	0	RX FIFO Threshold for DMA Request
7:6	-	R	0	Reserved
5	RCLR	R/W	0	Clear Interrupt Status at read cycle
4	WS	R/W	0	Wait State Control by SCL stretching 0 = Enable 1 = Disable This function is effective only when the TX/RX FIFO is addressed by an external master.
3	SDA	R/W	0	Reserved for test.
2	CLRFF	R/W	0	Clear FIFO. 0 = FIFO enabled 1 = FIFO disabled
1	-	R/W	0	Reserved
0	EN	R/W	0	Enable for this slave core. 0=Disable 1=Enable

Address Register (ADDR)

0x9000B088/188

Address Register (ADDR)															EXCESS BITS/100	
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
0																
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0								ADDR								0

Bit	Name	R/W	Reset	Description
31:8	-	R	0	Reserved
7:1	ADDR	R/W	0	Device Address (Slave Address)
0	-	R	0	Reserved

Interrupt Register (INT)

0x9000B08C/18C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-				IRQSTAT											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-				IRQEN											

Bit	Name	R/W	Reset	Description
31:28	-	R	0	Reserved
27	IRQSTAT[11]	R/W	0	Interrupt Status – All bytes of Data Buffer has been read by a master
26	IRQSTAT[10]	R/W	0	Interrupt Status – All bytes of Data Buffer has been written by a master
25	IRQSTAT[9]	R/W	0	Interrupt Status – Data Buffer has been read by a master
24	IRQSTAT[8]	R/W	0	Interrupt Status – Data Buffer has been written by a master
23	IRQSTAT[7]	R/W	0	Interrupt Status – TX FIFO Under-run
22	IRQSTAT[6]	R/W	0	Interrupt Status – RX FIFO Over-run
21	IRQSTAT[5]	R/W	0	Interrupt Status – TX Bus cycle started with TX FIFO empty.
20	IRQSTAT[4]	R/W	0	Interrupt Status – RX FIFO Full
19	IRQSTAT[3]	R/W	0	Interrupt Status – TX FIFO Empty
18	IRQSTAT[2]	R/W	0	Interrupt Status – RX FIFO Not Empty
17	IRQSTAT[1]	R/W	0	Interrupt Status – TX FIFO Level (TXVC <= TXTH)
16	IRQSTAT[0]	R/W	0	Interrupt Status – RX FIFO Level (RXVC > RXTH)
15:12	-	R	0	Reserved
11:0	IRQEN[11:0]	R/W	0	Interrupt Enable bits for IRQSTAT[11:0]

Note: When RCLR bit of Control Register is “1”, IRQSTAT bits are cleared at the end of read cycle. When RCLR bit of Control Register is “0”, IRQSTAT bits are cleared by writing “1”.

Status Register (STAT)

0x9000B090/190

Status Register (SR17)								SR0000000000000000								
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
0								DDIR	SCBS							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0								SADR								

Bit	Name	R/W	Reset	Description
31:24	-	R	0	Reserved
23	DDIR	R	0	Data Direction 0 = RX 1 = TX
22:16	SCBS	R	0	Slave Core Bus State. SCBS[0] = idle state SCBS[1] = addr0 SCBS[2] = ack0 SCBS[3] = addr1 SCBS[4] = ack1 SCBS[5] = data read or write state SCBS[6] = ack data
15:8	-	R	0	Reserved
7:0	SADR	R	0	Slave Address received from address cycle.

Buffer Valid Flag Register (MBF)

0x9000B09C/19C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								MBFT							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								MBFR							

Bit	Name	R/W	Reset	Description
31:24	-	R	0	Reserved
23:16	MBFT	R	0	Buffer TX Flag for Byte 7 ~ 0. When the Data Buffer 0/1 is read by an external master, the corresponding bit is set. MBFT[3:0] is cleared when the CPU writes to MB0 and MBFT[7:4] is cleared when MB1 is written by the CPU.
15:8	-	R	0	Reserved
7:0	MBFR	R	0	Buffer RX Flag for Byte 7 ~ 0. When each byte is written by an external master, the corresponding bit is set. MBFT[3:0] is cleared when the CPU reads MB0 and MBTF[7:4] is cleared when MB1 is read by the CPU.

Data buffer 0 (MB0)

0x9000B0A0/1A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Data byte 3								Data byte 2							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Data byte 1								Data byte 0							

Data buffer 1 (MB1)

0x9000B0A4, 0x9000B1A4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Data byte 7								Data byte 6							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Data byte 5								Data byte 4							

Note: Deviation from the standard.

The MB0 and MB1 registers cannot be read with the standard I2C read timing sequence. Always TX FIFO is accessed with the standard read timing. To read MB0 or MB1, the sequence below must be followed. If a master is not capable of this non-standard sequence, MB0 and MB1 cannot be read.

MSB		LSB		MSB		LSB		MSB		LSB															
S	WRITE SLAVE ADDRESS			R(1)	A	WRITE 0000XXXXb			A	READ BYTE0			A	READ BYTE1			A	READ BYTE2			A	READ BYTE3			A
7bits				1	1	8bits				8bits															

IRQ Status (IRQSTR)

0x9000B0C0/1C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0													ST2	ST1	ST0

Bit	Name	R/W	Reset	Description
31:3	-	R	0	Reserved
2	ST2	R	0	IRQ Status Flag for I2C Slave Controller 0/1 This is read as '1' when the IRQ status of I2C slave controller is activated.
1	ST1	R	0	IRQ Status Flag for I2C Master Controller of Channel 1. This is read as '1' when the IRQ status of I2C master controller 1/3 is activated.
0	ST0	R	0	IRQ Status Flag for I2C Master Controller of Channel 0. This is read as '1' when the IRQ status of I2C master controller 0/2 is activated.

Table 30.2 I2C Port Selection Register Map (Base Address = 0x9000B200)

Name	Address	Type	Reset	Description
I2CM0/1PORTSEL	0x00	R/W	0xFFFFFFFF	I2C Master 0/1 Port Selection Register
I2CM2/3PORTSEL	0x04	R/W	0xFFFFFFFF	I2C Master 2/3 Port Selection Register
I2CS0/1PORTSEL	0x08	R/W	0xFFFFFFFF	I2C Slave 0/1 Port Selection Register
I2CSWRST	0x10	R/W	0x00000000	Soft Reset Register
I2CSTATUS	0x14	R	0x00000000	I2C Status Register

The signal name GBUS[191:0] found in GPIO Function Selection Registers (GPIO*_FS* of GPIO block) is the final output of the port multiplexers in this module.

Any I2C signal can be mapped to pre-mapped bit position of GBUS[191:0] by writing an index number to the registers below in TCC8132. Pre-mapped bit position of GBUS[191:0] is shown in the table below.

Table 30.3 Pre-mapped bit position for I2C Port

GBUS	GPIO
GBUS[75:76]	PD[19:20]
GBUS[15:16]	PS[15:16]
GBUS[17:18]	PS[17:18]
GBUS[19:20]	PS[19:20]

I2C Master 0/1 Port Selection Register (I2CM0/1PORTSEL)

0x9000B200

Bit	Name	R/W	Reset	Description
31:24	M1SDA	R/W	0xFF	Port number for I2C Master 1 SDA
23:16	M1SCL	R/W	0xFF	Port number for I2C Master 1 SCL
15:8	M0SDA	R/W	0xFF	Port number for I2C Master 0 SDA
7:0	M0SCL	R/W	0xFF	Port number for I2C Master 0 SCL

I2C Master 2/3 Port Selection Register (I2CM2/3PORTSEL)

0x9000B204

Bit	Name	R/W	Reset	Description
31:24	M3SDA	R/W	0xFF	Port number for I2C Master 3 SDA
23:16	M3SCL	R/W	0xFF	Port number for I2C Master 3 SCL
15:8	M2SDA	R/W	0xFF	Port number for I2C Master 2 SDA
7:0	M2SCL	R/W	0xFF	Port number for I2C Master 2 SCL

I2C Slave 0/1 Port Selection Register (I2CS0/1PORTSEL)

0x9000B208

Bit	Name	R/W	Reset	Description
31:24	S1SDA	R/W	0xFF	Port number for I2C Slave 1 SDA
23:16	S1SCL	R/W	0xFF	Port number for I2C Slave 1 SCL
15:8	S0SDA	R/W	0xFF	Port number for I2C Slave 0 SDA
7:0	S0SCL	R/W	0xFF	Port number for I2C Slave 0 SCL

I2C Soft Reset Register (I2CSWRST)**0x9000B210**

Bit	Name	R/W	Reset	Description
31:9	-	R	0	Reserved
8	-	R/W	0	Reserved
7:6	-	R	0	Reserved
5	SWRSTS1	R/W	0	Soft Reset – Slave 1
4	SWRSTM3	R/W	0	Soft Reset – Master 3
3	SWRSTM2	R/W	0	Soft Reset – Master 2
2	SWRSTS0	R/W	0	Soft Reset – Slave 0
1	SWRSTM1	R/W	0	Soft Reset – Master 1
0	SWRSTM0	R/W	0	Soft Reset – Master 0

I2C Status Register (I2CSTATUS)**0x9000B214**

Bit	Name	R/W	Reset	Description
31:9	-	R	0	Reserved
19	DRQTxS1	R	0	DMA Request Tx - Slave 1
18	DRQRxS1	R	0	DMA Request Rx - Slave 1
17	DRQTxS0	R	0	DMA Request Tx - Slave 0
16	DRQRxS0	R	0	DMA Request Rx - Slave 0
15:6	-	R	0	Reserved
5	IRQS1	R	0	Interrupt Request – Slave 1
4	IRQM3	R	0	Interrupt Request – Master 3
3	IRQM2	R	0	Interrupt Request – Master 2
2	IRQS0	R	0	Interrupt Request – Slave 0
1	IRQM1	R	0	Interrupt Request – Master 1
0	IRQM0	R	0	Interrupt Request – Master 0

31 SD/SDIO/MMC Host Controller

31.1 Overview

The SD host controller complies with following versions of specification concerned.

- SD Host Controller Specification Version 3.0
- SDIO Card Specification Version 3.0
- MMC Specification Version 4.41, MMC Plus and MMC Mobile

The TCC8132 supports following SD cards.

- For Capacity of Memory
 - 1) SDSC: Up to and including 2GB
 - 2) SDHC: More than 2GB and up to and including 32GB
 - 3) SDXC: More than 32GB and up to and including 2TB
- For Bus Speed Mode
 - 1) Default Speed: 3.3V signaling
 - 2) High Speed: 3.3V signaling
 - 3) SDR50 mode: 1.8V signaling
 - 4) SDR104 mode: 1.8V signaling

and following MMC cards.

- 1, 4, 8 bit SDR mode and 1, 4, 8 bit DDR mode
- MMC Plus and MMC Mobile
- Normal boot and Alternative boot mode

Cards may be detected or removed by external interrupt trigger. The TCC8132 has two host controllers with one slot. For more effective data transaction, a 2K dual port FIFO is used. You can access the FIFO using inherent SDMA and ADMA. The General DMA access is not available.

31.2 Block Diagram

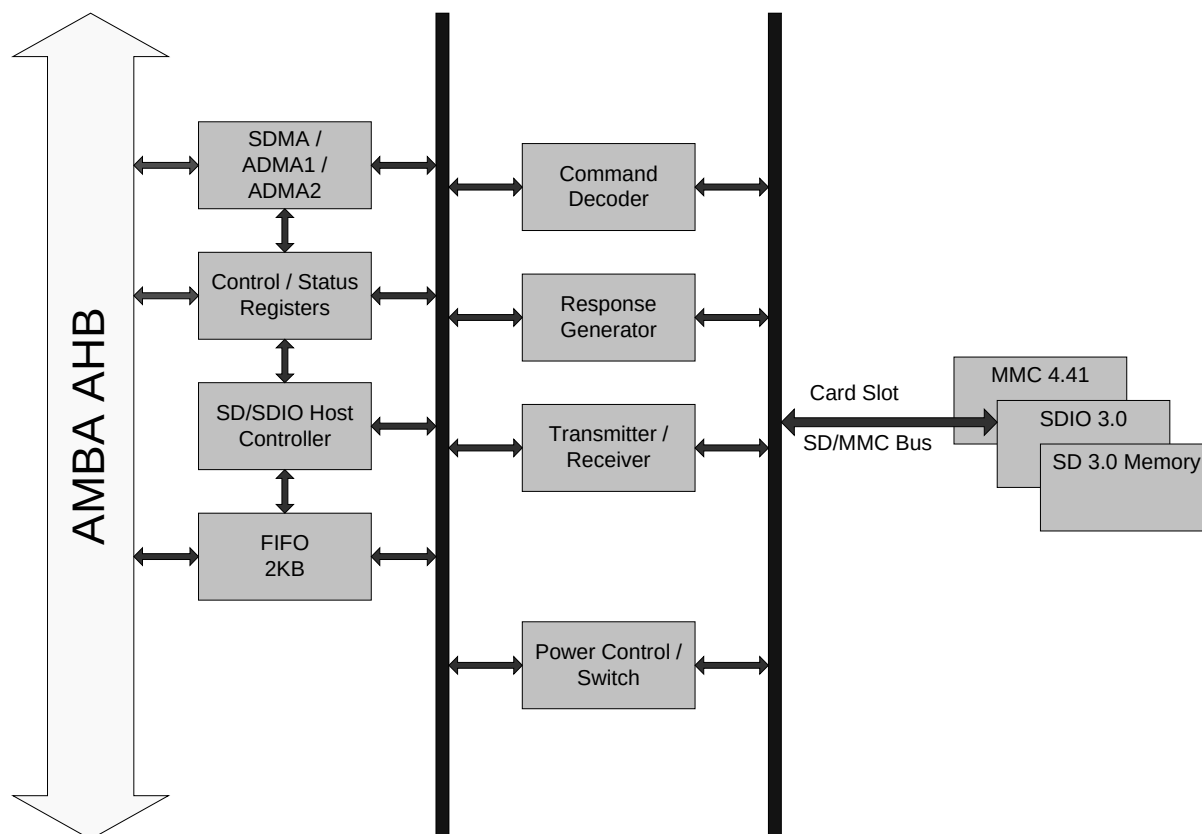


Figure 31.1 SD/SDIO/MMC Block Diagram

31.3 Register Description

Table 31.1 SD Host Register Map (Base Address = 0x8000E000/E200)

Name	Address	Type	Reset	Description
SDMA	0x000	R/W	0x0000	SDMA System Address / Argument 2
BSIZE	0x004	R/W	0x0000	Block Size
BCNT	0x006	R/W	0x0000	Block Count
ARG	0x008	R/W	0x0000	Argument 1
TMODE	0x00C	R/W	0x0000	Transfer Mode
CMD	0x00E	R/W	0x0000	Command
RESP0	0x010	R	0x0000	Response0
RESP1	0x012	R	0x0000	Response1
RESP2	0x014	R	0x0000	Response2
RESP3	0x016	R	0x0000	Response3
RESP4	0x018	R	0x0000	Response4
RESP5	0x01A	R	0x0000	Response5
RESP6	0x01C	R	0x0000	Response6
RESP7	0x01E	R	0x0000	Response7
DATAL	0x020	R/W	-	Buffer Data Port (Low)
DATAH	0x022	R/W	-	Buffer Data Port (High)
STATEL	0x024	R	0x0000	Present State (Low)
STATEH	0x026	R	0x0000	Present State (High)
CONTL	0x028	R/W	0x0000	Power Control / Host Control
CONTH	0x02A	R/W	0x0000	Wakeup Control / Block Gap Control
CLK	0x02C	R/W	0x0000	Clock Control
TIME	0x02E	R/W	0x0000	Software Reset / Timeout Control
STSL	0x030	R	0x0000	Normal Interrupt Status
STSH	0x032	R	0x0000	Error Interrupt Status
STSENL	0x034	R/W	0x0000	Normal Interrupt Status Enable
STSENH	0x036	R/W	0x0000	Error Interrupt Status Enable
INTENL	0x038	R/W	0x0000	Normal Interrupt Signal Enable
INTENH	0x03A	R/W	0x0000	Error Interrupt Signal Enable
CMD12ERR	0x03C	R	0x0000	Auto CMD12 Error Status
CONT2	0x03E	R/W	0x0000	Host Control2
CAPL1	0x040	R	0x0000	Capabilities1(Low)
CAPH1	0x042	R	0x0000	Capabilities1(High)
CAPL2	0x044	R	0x0000	Capabilities2(Low)
CAPH2	0x046	R	0x0000	Capabilities2(High)
CURL	0x048	R	0x0001	Maximum Current Capabilities(Low)
CURH	0x04A	R	0x0000	Maximum Current Capabilities(High)
FORCEL	0x050	W	0x0000	Force event for AutoCmd Error Status
FORCEH	0x052	W	0x0000	Force event for Error Interrupt Status
ADMAERR	0x054	R/W	0x0000	ADMA Error Status
ADDR0	0x058	R/W	0x0000	ADMA Address[15:0]
ADDR1	0x05A	R/W	0x0000	ADMA Address[31:16]
PRESET0	0x060	R	0x0000	Preset Value for Default Speed and Initialization
PRESET1	0x064	R	0x0000	Preset Value for High Speed
	0x068	R	0x0000	Reserved
	0x06C	R	0x0000	Reserved
BTCONTL	0x070	R/W	0x0000	Boot data timeout control(Low)
BTCONTH	0x072	R/W	0x0000	Boot data timeout control(High)
DBGSEL	0x074	R/W	0x0000	Debug Selection
SPIINT	0x0F0	R	0x0000	SPI Interrupt Support
SLOT	0x0FC	R	0x0000	Slot Interrupt Status
VERSION	0x0FE	R	0xa702	Host Controller Version
CFG0	0x100	R/W	0x00000004	Host Controller Configuration 0 register
	0x104	R/W	0x00000000	Reserved
CFG2	0x108	R/W	0x00000000	Host Controller Configuration 2 register
CFG3	0x10C	R/W	0x00000000	Host Controller Configuration 3 register
CFG4	0x110	R/W	0x00000000	Capabilities Preset 0 register
CFG5	0x114	R/W	0x00000000	Capabilities Preset 1 register
CFG6	0x118	R/W	0x00000000	Initialization Preset register
CFG7	0x11C	R/W	0x00000000	Speed Mode Preset 0 register
	0x120	R/W	0x00000000	Reserved

Name	Address	Type	Reset	Description
	0x124	R/W	0x00000000	Reserved
OBSERVABLE	0x128	R	-	Debug Information
MON	0x12C	R	-	Reserved
CFG8	0x140	R/W	0x00000000	DataOE Delay Chain Register
CFG9	0x144	R/W	0x00000000	DATA Delay Chain Register

SDMA System Address / Argument 2

0x8000E000/E200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SDMA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SDMA[15:0]															

SDMA	[31:0]	SDMA System Address / Argument 2
N		This contains system memory address for a SDMA transfer or the second argument for the Auto CMD23. 1) SDMA System Address This should be initialized before starting a SDMA transaction. After SDMA has stopped, the next system address of the next contiguous data position can be read from this register. The SDMA transfer waits at the every boundary specified by the Host SDMA Buffer Boundary in the Block Size register. 2) Argument 2 This register is used with the Auto CMD23 to set a 32-bit block count value to the argument of the CMD23 while executing Auto CMD23.

Block Count and Size

0x8000E004/E204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BCNT[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SDMABUF[2:0]					BSIZE[11:0]									

BCNT	[31:16]	Block Count for Current Transfer
N		This is enabled when Block Count Enable in the Transfer Mode register is set to 1 and is valid only for multiple block transfers. The Host Controller decrements the block count after each block transfer and stops when the count reaches zero. 0000h - Stop Count 0001h - 1 block 0002h - 2 blocks --- --- FFFFh - 65535 blocks

SDMABUF	[14:12]	Host SDMA Buffer Size
N		Specify the size of contiguous buffer in the system memory. The SDMA transfer shall wait at the every boundary specified by these fields and the Host Controller generates the DMA Interrupt to request the Host Driver to update the SDMA System Address register. At the end of transfer, the Host Controller may issue or may not issue DMA Interrupt. In particular, DMA Interrupt shall not be issued after Transfer Complete Interrupt is issued. 000b - 4KB(Detects A11 Carry out) 001b - 8KB(Detects A12 Carry out) 010b - 16KB(Detects A13 Carry out) 011b - 32KB(Detects A14 Carry out) 100b - 64KB(Detects A15 Carry out) 101b - 128KB(Detects A16 Carry out) 110b - 256KB(Detects A17 Carry out) 111b - 512KB(Detects A18 Carry out)

BSIZE	[11:0]	Transfer Block Size
N		The block size for block data transfers for CMD17, CMD18, CMD24, CMD25, and CMD53. Values ranging from 1 up to the maximum buffer size can be set. 0000h - No Data Transfer 0001h - 1 Byte 0002h - 2 Bytes 0003h - 3 Bytes 0004h - 4 Bytes --- --- 01FFh - 511 Bytes 0200h - 512 Bytes --- --- 0800h - 2048 Bytes

Argument 1

0x8000E008/E208

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ARG[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARG[15:0]															

ARG	[31:0]	Command Argument 1
N	This is specified as bit39-8 of Command Format.	

Table 31.2 Command Format

Bit position	47	46	[45:40]	[39:8]	[7:1]	0
Width (bits)	1	1	6	32	7	1
Value	'0'	'1'	x	x	x	'1'
Description	start bit	transmission bit	response index	card status	CRC7	end bit

Command and Transfer Mode

0x8000E00C/E20C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		CMDINDEX[5:0]						CTYPE[1:0]		DATSEL	CICLK	CRCCHK	Reserved	RTYPE[1:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										MS	DIR	ACMDEN		BCNTEN	DMAEN

CMDINDEX	[29:24]	Command Index
N	This bit must be set to the command number (CMD0-63, ACMD0-63) that is specified in bits 45-40 of the Command Format	

CTYPE	[23:22]	Command Type
N	There are three types of special commands. - Abort(11b) : CMD12, CMD52 for writing "I/O Abort" in CCCR - Resume(10b) : CMD52 for writing "Function Select" in CCCR - Suspend(00b) : CMD52 for writing "Bus Suspend" in CCCR - Normal(00b) : All other commands	

DATSEL	[21]	Data Present Select
1	This bit is set to 1 to indicate that data is present and shall be transferred using the DAT line.	
0	No Data Present	

CICLK	[20]	Command Index Check Enable
1	If this bit is set to 1, the HC shall check the index field in the response to see if it has the same value as the command index.	
0	Disable	

CRCCHK	[19]	Command CRC Check Enable
1	If this bit is set to 1, the HC shall check the CRC field in the response.	
0	Disable	

RTYPE	[17:16]	Response Type
N	Response Type Select 00 - No Response 01 - Response length 136 (R2) 10 - Response length 48 (R3, R4, R1, R5, R6, R7) 11 - Response length 48 check Busy after response (R1b, R5b)	

MS	[5]	Multi/Single Block Select
1	This bit is set when issuing multiple-block transfer commands using DAT line. For any other commands, this bit must be set to 0.	
0	Single Block	

DIR	[4]	Data Transfer Direction Select
1	Read (Card to Host)	
0	Write (Host to Card)	

ACMDEN	[3:2]	Auto CMD Enable
N		This field determines use of auto command functions 00b - Auto Command Disabled 01b - Auto CMD12 Enable 10b - Auto CMD23 Enable 11b - Reserved

BCNTEN	[1]	Block Count Enable
1		This bit enables the Block Count register, which is only relevant for multiple block transfers.
0		Disable

DMAEN	[0]	DMA Enable
1		DMA can be enabled only if DMA Support bit in the Capabilities register is set. If this bit is set to 1, a DMA operation must begin when the HD writes to the upper byte of Command register(0x00F).
0		No data transfer or Non-DMA data transfer

Response1/0

0x8000E010/E210

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESPONSE1[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESPONSE0[15:0]															

Response3/2

0x8000E014/E214

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESPONSE3[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESPONSE2[15:0]															

Response5/4

0x8000E018/E218

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESPONSE5[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESPONSE4[15:0]															

Response7/6

0x8000E01C/E21C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESPONSE7[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESPONSE6[15:0]															

RESPONSE	[127:0]	Command Response
N		Command responses from the SD Bus. Refer to the table below.

Table 31.3 Response Register

Kind of Response	Meaning of Response	Response Field	Response Register
R1, R1b(normal response)	Card Status	R[39:8]	RESPONSE[31:0]
R1b(Auto CMD12 response)	Card Status for Auto CMD12	R[39:8]	RESPONSE[127:96]
R2(CID, CSD register)	CID or CSD reg. incl.	R[127:8]	RESPONSE[119:0]
R3(OCR register)	OCR register for memory	R[39:8]	RESPONSE[31:0]
R4(OCR register)	OCR register for I/O etc	R[39:8]	RESPONSE[31:0]
R5, R5b	SDIO response	R[39:8]	RESPONSE[31:0]
R6(Published RCA response)	New published RCA[31:16] etc	R[39:8]	RESPONSE[31:0]

Buffer Data Port

0x8000E020/E220

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DAT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DAT[15:0]															

The Host Controller Buffer can be accessed through this 32-bit Data Port Register.

Present State

0x8000E024/E224

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DAT[7:4]				CMD	DAT[3:0]				SDWP	SDCD	CDST	CDIN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				RDEN	WREN	RDACT	WRACT	Reserved				RTUNE	DATACT	NODAT	NOCMD

DAT	[28:25], [23:20]	Data Line Signal Level
	This status is used to check DAT[7:0] line level to recover from errors, and for debugging.	

CMD	[24]	CMD Line Signal Level
	This status is used to check CMD line level to recover from errors, and for debugging.	

SDWP	[19]	Write Protect Switch Pin Level
1	This bit reflects the SDWP# pin. Write enabled	
0	Write protected	

SDCD	[18]	Card Detect Pin Level
1	This bit reflects the inverse value of the SDCD# pin. Card present.	
0	No Card present.	

CDST	[17]	Card State Stable
1	If this bit is set to 1, it means the Card Detect Pin Level is stable. No Card or Inserted.	
0	Reset of Debouncing.	

CDIN	[16]	Card Inserted
1	This bit indicates whether a card has been inserted.	
0	Reset or Debouncing or No Card.	

RDEN	[11]	Buffer Read Enable
1	This status is used for non-DMA read transfers. If this bit is 1, readable data exists in the buffer.	
0	Read Disable	

WREN	[10]	Buffer Write Enable
1	This status is used for non-DMA read transfers. If this bit is 1, data can be written to the buffer.	
0	Write Disable	

RDACT	[9]	Read Transfer Active
1	This status is used for detecting completion of a read transfer.	
0	No valid data	

WRACT	[8]	Write Transfer Active
1	This status indicates a write transfer is active.	
0	No valid data	

RTUNE	[3]	Re-Tuning Request
1	Sampling clock needs re-tuning	
0	Fixed or well tuned sampling clock	

DATACT	[2]	DAT Line Active
1	This bit indicates whether one of the DAT lines on SD bus is in use.	
0	DAT line inactive.	

NODAT	[1]	Command Inhibit(DAT)
1	This status bit is generated if either the DAT Line Active or the Read transfer Active is set to 1. It cannot issue command which uses the DAT line.	
0	It can issue command which uses the DAT line.	

NOCMD	[0]	Command Inhibit(CMD)
1	Indicates the CMD line is in use.	
0	Indicates the CMD line is not in use and the HC can issue a SD command using the CMD line.	

Wakeup/Block Gap/Power/Host Control

0x8000E028/E228

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					WKOUT	WKIN	WKINT	ACK	ALT	BOOT	SPI	BGINT	RDWAIT	CONREQ	BGSTOP
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			HRST	Reserved			POW	DETSEL	DETCD	SD8	SELDMA[1:0]		HS	SD4	LED

WKOUT	[26]	Wakeup Enable On Card Removal
1		This bit enables wakeup event via Card Removal assertion in the Normal Interrupt Status register. FN_WUS (Wake Up Support) in CIS does not affect this bit.
0		Disable

WKIN	[25]	Wakeup Enable On Card Insertion
1		This bit enables wakeup event via Card Insertion assertion in the Normal Interrupt Status register. FN_WUS (Wake Up Support) in CIS does not affect this bit.
0		Disable

WKINT	[24]	Wakeup Enable On Card Interrupt
1		This bit enables wakeup event via Card Interrupt assertion in the Normal Interrupt Status register. This bit can be set to 1 if FN_WUS (Wake Up Support) in CIS is set to 1
0		Disable

ACK	[23]	Boot Acknowledge
1		Wait for boot ack from eMMC card.
0		Will not wait for boot ack from eMMC card

ALT	[22]	Alternative Boot Mode
1		To start alternate boot mode access
0		To stop alternate boot mode access

BOOT	[21]	Boot Enable
1		To start boot code access
0		To stop boot code access

SPI	[20]	SPI Mode
1		SPI mode
0		SD mode

BGINT	[19]	Interrupt At Block Gap
1		This bit is valid only in 4-bit mode of the SDIO card and selects a sample point in the interrupt cycle. Setting to 1 enables interrupt detection at the block gap for a multiple block transfer.
0		Disable

RDWAIT	[18]	Read Wait Control
1		If the card supports read wait, set this bit to enable use of the read wait protocol to stop read data using DAT[2] line.
0		Disable

CONREQ	[17]	Continue Request
1		This bit is used to restart a transaction, which was stopped using the Stop At Block Gap Request.
0		Stop is ignored

BGSTOP	[16]	Stop at Block Gap Request
1		This bit is used to stop executing a transaction at the next block gap for non-DMA,SDMA and ADMA transfers. The Host Driver shall leave this bit set to 1 until the Transfer Complete is set to 1.
0		Keep transferring

HRST	[12]	Hardware Reset
1		Hardware reset signal is generated for eMMC4.4 card.
0		Hardware reset signal is not generated.

POW	[8]	SD Bus Power
1	Bus power is on.	
0	Bus power is off.	

DETSEL	[7]	Card Detect Signal Selection
1	The card detect test level is selected for test purpose.	
0	SDCD# is selected for normal use.	

DETC	[6]	Card Detect Test Level
1	This bit is enabled while the Card Detect Signal Selection is set to 1 and it indicates card inserted,	
0	No Card	

SD8	[5]	SD 8bit Mode
1	8 bit mode is selected.	
0	Bus-width is selected by Data Transfer Width	

SELDMA	[4:3]	DMA Select
N	One of supported DMA modes can be selected. Use of selected DMA is determined by DMA Enable of the Transfer Mode register. 00 - SDMA is selected. 01 - 32-bit Address ADMA1 is selected. 10 - 32-bit Address ADMA2 is selected. 11 - 64-bit Address ADMA2 is selected.	

HS	[2]	High Speed Enable
1	The Host Controller outputs CMD line and DAT lines at the rising edge of the SD clock (up to 50 MHz).	
0	The Host Controller outputs CMD line and DAT lines at the falling edge of the SD Clock (up to 25MHz).	

SD4	[1]	Data Transfer Width
1	4 bit mode	
0	1 bit mode	

LED	[0]	LED Control
-	This bit is used to caution the user not to remove the card while the SD card is being accessed. This will be supported in the future.	

Software Reset/Timeout/Clock Control

0x8000E02C/E22C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					RSTDAT	RSTCMD	RSTALL	Reserved				TIMEOUT[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SDCLKSEL[7:0]								Reserved				SCKEN		CLKRDY	CLKEN

RSTDAT	[26]	Software Reset for DAT Line
1		Only part of data circuit is reset. The following registers and bits are cleared by this bit: Buffer Data Port Register Buffer is cleared and Initialized. Present State register Buffer read Enable Buffer write Enable Read Transfer Active Write Transfer Active DAT Line Active Command Inhibit (DAT) Block Gap Control register Continue Request Stop At Block Gap Request Normal Interrupt Status register Buffer Read Ready Buffer Write Ready Block Gap Event Transfer Complete
0		Keep working

RSTCMD	[25]	Software Reset For CMD Line
1		Only part of command circuit is reset. The following registers and bits are cleared by this bit: Present State register Command Inhibit (CMD) Normal Interrupt Status register Command Complete
0		Keep working

RSTALL	[24]	Software Reset For All
1		This reset affects the entire HC except for the card detection circuit. If this bit is set to 1, the host driver should issue reset command and reinitialize the SD card.
0		Keep working

TIMEOUT	[19:16]	Data Timeout Counter Value
N		This value determines the interval by which DAT line time-outs are detected. 1111 - Reserved 1110 - $TMCLK * 2^{27}$ ----- 0001 - $TMCLK * 2^{14}$ 0000 - $TMCLK * 2^{13}$

SDCLKSEL	[15:8]	SDCLK Frequency Select
N		This register is used to select the frequency of the SDCLK pin. Only the following settings are allowed. That is 8bit Divide Clock mode. 80h - base clock divided by 256 40h - base clock divided by 128 20h - base clock divided by 64 10h - base clock divided by 32 08h - base clock divided by 16 04h - base clock divided by 8 02h - base clock divided by 4 01h - base clock divided by 2 00h - base clock(10MHz-63MHz) According to the Physical Layer Specification, the maximum SD Clock frequency is 25 MHz in normal speed mode and 50MHz in high speed mode, and must not exceed

	this limit. It is recommended to set 00h under DDR mode to meet duty ratio 50%.
--	--

SDCLKSEL1	[7:6]	SDCLK Frequency Select 1
N		This register is used to extend SDCLK frequency selection.

SCKEN	[2]	SD Clock Enable
1		SD Clock is enabled.
0		The Host Controller shall stop SDCLK when writing this bit to 0. SDCLK Frequency Select can be changed when this bit is 0.

CLKRDY	[1]	Internal Clock Stable
1		This bit is set to 1 when SD clock is stable after writing to Internal Clock Enable in this register to 1. The SD Host Driver shall wait to set SD Clock Enable until this bit is set to 1.
0		Not ready

CLKEN	[0]	Internal Clock Enable
1		Clock starts to oscillate when this bit is set to 1. When clock oscillation is stable, the Host Controller shall set Internal Clock Stable in this register to 1. This bit shall not affect card detection.
0		Stop

Normal Interrupt Status

0x8000E030/E230

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			TR	R		ADMA	ACMD	CLIMIT	DATEND	DATCRC	DATTIME	CINDEX	CMDEND	CMDCRC	CMDTIME
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ERR	BT	BACK	Reserved				CDINT	CDOUT	CDIN	RDRDY	WRRDY	DMA	BLKGAP	TDONE	CDONE

TR	[28]	Target Response Error
1		Occurs when detecting ERROR in m_hresp(dma transaction)
0		No error

ADMA	[25]	ADMA Error
1		This bit is set when the Host Controller detects errors during ADMA based data transfer. The state of the ADMA at an error occurrence is saved in the ADMA Error Status Register.
0		No error

ACMD	[24]	Auto CMD Error
1		Auto CMD12 and Auto CMD23 use this error status. This bit is set when detecting that one of the bits D00-D04 in Auto CMD Error Status register has changed from 0 to 1. In case of Auto CMD12, this bit is set to 1, not only when the errors in Auto CMD12 occur but also when Auto CMD12 is not executed due to the previous command error.
0		No error

CLIMIT	[23]	Current Limit Error
1		Reading 1 means the HC is not supplying power to SD card due to some failure. This will be supported in the future.
0		No error

DATEND	[22]	Data End Bit Error
1		Occurs when detecting 0 at the end bit position of read. Data which uses the DAT line or the end bit position of the CRC status.
0		No error

DATCRC	[21]	Data CRC Error
1		Occurs when detecting CRC error when transferring read data which uses the DAT line or when detecting the Write CRC Status having a value of other than "010".
0		No error

DATTIME	[20]	Data Timeout Error
1		Occurs when detecting one of following timeout conditions. 1) Busy Timeout for R1b, R5b type 2) Busy Timeout after Write CRC status 3) Write CRC status Timeout 4) Read Data Timeout
0		No error

CINDEX	[19]	Command Index Error
1		Occurs if a Command Index error occurs in the Command Response.
0		No error

CMDEND	[18]	Command End Bit Error
1		Occurs when detecting that the end bit of a command response is 0.
0		No error

CMDCRC	[17]	Command CRC Error
1		Command CRC Error is generated in two cases. 1) If a response is returned and the Command Time-out Error is set to 0, this bit is set to 1 when detecting a CRC error in the command response. 2) The Host Controller detects a CMD line conflict by monitoring the CMD line when a command is issued.
0		No error

CMDTIME	[16]	Command Timeout Error
1		This bit is set only if the no response is returned within 64 SDCLK cycles from the end

	bit of the command. If the Host Controller detects a CMD line conflict, this bit shall be set without waiting for 64 SD clock cycles because the command will be aborted by the Host Controller.
0	No error

ERR	[15]	Error Interrupt
1	If any of the bits in the Error Interrupt Status Register is set, then this bit is set. Therefore the HD can test for an error by checking this bit first.	
0	No error	

BT	[14]	Boot Terminate Interrupt
1	This status is set if the boot operation get terminated.	
0	Boot operation is not terminated.	

BACK	[13]	Boot ACK Receive
1	This status is set if the boot acknowledge is received from device.	
0	Boot ack is not received.	

CDINT	[8]	Card Interrupt
1	Card interrupt is generated. In 1-bit mode, the Host Controller shall detect the Card Interrupt without SD Clock to support wakeup. In 4-bit mode, the card interrupt signal is sampled during the interrupt cycle, so there are some sample delays between the interrupt signal from the SD card and the interrupt to the Host System. It is necessary to define how to handle this delay.	
0	No card interrupt	

CDOUT	[7]	Card Removal
1	This status is set if the Card Inserted in the Present State register changes from 1 to 0. When the Host Driver writes this bit to 1 to clear this status, the status of the Card Inserted in the Present State register should be confirmed.	
0	Card State Stable or Debouncing	

CDIN	[6]	Card Insertion
1	This status is set if the Card Inserted in the Present State register changes from 0 to 1. When the Host Driver writes this bit to 1 to clear this status, the status of the Card Inserted in the Present State register should be confirmed.	
0	Card State Stable or Debouncing	

RDRDY	[5]	Buffer Read Ready
1	This status is set if the Buffer Read Enable changes from 0 to 1 and it is ready to read buffer.	
0	Not ready to read buffer.	

WRRDY	[4]	Buffer Write Ready
1	This status is set if the Buffer Write Enable changes from 0 to 1 and it is ready to read buffer.	
0	Not ready to read buffer.	

DMA	[3]	DMA Interrupt
1	This status is set if the Host Controller detects the Host DMA Buffer boundary during transfer. In case of ADMA, by setting Int field in the descriptor table, Host Controller generates this interrupt. This interrupt shall not be generated after the Transfer Complete.	
0	No DMA Interrupt	

BLKGAP	[2]	Block Gap Event
1	If the Stop At Block Gap Request in the Block Gap Control register is set, this bit is set when both a read / write transaction is stopped at a block gap. If Stop At Block Gap Request is not set to 1, this bit is not set to 1. Read Transaction: This bit is set at the falling edge of the DAT Line Active Status (When the transaction stops at SD Bus timing). The Read Wait shall be supported in order to use this function. Write Transaction: This bit is set at the falling edge of Write Transfer Active Status (After getting CRC	

	status at SD Bus timing).
0	No Block Gap Event

TDONE	[1]	Transfer Complete
1	This bit is set when a read / write transaction is completed. Read Transaction: This bit is set at the falling edge of Read Transfer Active Status. This interrupt is generated in two cases. The first is when a data transfer is completed as specified by data length (After the last data has been read to the Host System). The second is when data has stopped at the block gap and completed the data transfer by setting the Stop At Block Gap Request in the Block Gap Control register (After valid data has been read to the Host System). Write Transaction: This bit is set at the falling edge of the DAT Line Active Status. This interrupt is generated in two cases. The first is when the last data is written to the SD card as specified by data length and the busy signal released. The second is when data transfers are stopped at the block gap by setting Stop At Block Gap Request in the Block Gap Control register and data transfers completed. (After valid data is written to the SD card and the busy signal released).	
0	Not Completed	

CDONE	[0]	Command Complete
1	This bit is set when get the end bit of the command response (Except Auto CMD12).	
0	No Command Complete	

Table 31.4 Relation between Transfer Complete and Data Timeout Error

Transfer Complete	Data Timeout Error	Meaning of the status
0	0	Interrupted by another factor
0	1	Timeout occur during transfer
1	Don't Care	Command Execution complete

Table 31.5 Relation between Command Complete and Command Timeout Error

Command Complete	Command Timeout Error	Meaning of the status
0	0	Interrupted by another factor
Don't Care	1	Response not received within 64 SDCLK cycles
1	0	Response received

Normal Interrupt Status Enable

0x8000E034/E234

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			TR	R		ADMA	ACMD	CLIMIT	DATEND	DATCRC	DATTIME	CINDEX	CMDEND	CMDCRC	CMDTIME
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	BT	BACK	Reserved				CDINT	CDOUT	CDIN	RDRDY	WRRDY	DMA	BLKGAP	TDONE	CDONE

Name	Bit	Description (1=Enabled, 0=Masked)
TR	[28]	Target Response Error Status Enable
ADMA	[25]	ADMA Error Status Enable
ACMD	[24]	Auto CMD Error Status Enable
CLIMIT	[23]	Current Limit Error Status Enable
DATEND	[22]	Data End Bit Error Status Enable
DATCRC	[21]	Data CRC Error Status Enable
DATTIME	[20]	Data Timeout Error Status Enable
CINDEX	[19]	Command Index Error Status Enable
CMDEND	[18]	Command End Bit Error Status Enable
CMDCRC	[17]	Command CRC Error Status Enable
CMDTIME	[16]	Command Timeout Error Status Enable
BT	[14]	Boot Terminate Interrupt Status Enable
BACK	[13]	Boot ACK Receive Status Enable
CDINT	[8]	Card Interrupt Status Enable If this bit is set to 0, the Host Controller shall clear interrupt request to the System. The Card Interrupt detection stops when this bit is cleared and restarts when this bit is set to 1. The Host Driver should clear the Card Interrupt Status Enable before servicing the Card Interrupt and should set this bit again after all interrupt requests from the card are cleared to prevent inadvertent interrupts.
CDOUT	[7]	Card Removal Status Enable
CDIN	[6]	Card Insertion Status Enable
RDRDY	[5]	Buffer Read Ready Status Enable
WRRDY	[4]	Buffer Write Ready Status Enable
DMA	[3]	DMA Interrupt Status Enable
BLKGAP	[2]	Block Gap Event Status Enable
TDONE	[1]	Transfer Complete Status Enable
CDONE	[0]	Command Complete Status Enable

Normal Interrupt Signal Enable

0x8000E038/E238

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			TR	R		ADMA	ACMD	CLIMIT	DATEND	DATCRC	DATTIME	CINDEX	CMDEND	CMDCRC	CMDTIME
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	BT	BACK	Reserved				CDINT	CDOUT	CDIN	RDRDY	WRRDY	DMA	BLKGAP	TDONE	CDONE

Name	Bit	Description (1=Enabled, 0=Masked)
TR	[28]	Target Response Error Signal Enable
ADMA	[25]	ADMA Error Signal Enable
ACMD	[24]	Auto CMD Error Signal Enable
CLIMIT	[23]	Current Limit Error Signal Enable
DATEND	[22]	Data End Bit Error Signal Enable
DATCRC	[21]	Data CRC Error Signal Enable
DATTIME	[20]	Data Timeout Error Signal Enable
CMDEND	[18]	Command End Bit Error Signal Enable
CMDCRC	[17]	Command CRC Error Signal Enable
CMDTIME	[16]	Command Timeout Error Signal Enable
BT	[14]	Boot Terminate Interrupt Signal Enable
BACK	[13]	Boot ACK Receive Signal Enable
CDINT	[8]	Card Interrupt Signal Enable
CDOUT	[7]	Card Removal Signal Enable
CDIN	[6]	Card Insertion Signal Enable
RDRDY	[5]	Buffer Read Ready Signal Enable
WRRDY	[4]	Buffer Write Ready Signal Enable
DMA	[3]	DMA Interrupt Signal Enable
BLKGAP	[2]	Block Gap Event Signal Enable
TDONE	[1]	Transfer Complete Signal Enable
CDONE	[0]	Command Complete Signal Enable

Host Control 2 / Auto CMD12 Error Status

0x8000E03C/E23C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PRE	ASYNC	Reserved										DDR			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								NOCMD	Reserved		INDEX	ENDBIT	CRC	TIMEOUT	NORUN

PRE	[31]	Preset Value Enable
1	This bit enables the functions defined in the Preset Value registers. When this bit is set, Automatic Selection by Preset Value is enabled.	
0	SDCLK is controlled by Host Driver	

ASYNC	[30]	Asynchronous Interrupt Enable
1	This bit can be set to 1 if a card support asynchronous interrupt and Asynchronous Interrupt Support is set to 1 in the Capabilities register. When this bit is set, asynchronous interrupt is enabled.	
0	Disabled	

DDR	[18:16]	DDR Mode Select
N	This field is used to select DDR mode for MMC.100b : DDR mode Others : Reserved	

NOCMD	[7]	Command Not Issued
1	Setting this bit to 1 means CMD_wo_DAT is not executed due to an Auto CMD12 error (D04 - D01). This bit is set to 0 when Auto CMD Error is generated by Auto CMD23	
0	No error	

INDEX	[4]	Auto CMD12 Index Error
1	This bit is set if the Command Index error occurs in response to a command.	
0	No error	

ENDBIT	[3]	Auto CMD12 End Bit Error
1	This bit is set when detecting that the end bit of command response is 0.	
0	No error	

CRC	[2]	Auto CMD12 CRC Error
-----	-----	----------------------

1	This bit is set when detecting a CRC error in the command response.
0	No error

TIMEOUT	[1]	Auto CMD12 Timeout Error
1		This bit is set if the no response is returned within 64 SDCLK cycles from the end bit of the command. If this bit is set to 1, the other error status bits (D04-D02) are meaningless.
0		No error

NORUN	[0]	Auto CMD12 Not Executed
1		Setting this bit to 1 means the HC cannot issue Auto CMD12 to stop memory multiple block transfer due to some errors. If this bit is set to 1, other error status bits (D04-D01) are meaningless. This bit is set to 0 when Auto CMD Error is generated by Auto CMD23
0		Executed

Table 31.6 Relation between Auto CMD12 CRC Error and Auto CMD12 Timeout Error

Auto CMD12 CRC Error	Auto CMD12 Timeout Error	Kinds of Error
0	0	No Error
0	1	Response Timeout Error
1	0	Response CRC Error
1	1	CMD line conflict

Capabilities 1

0x8000E040/E240

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STYPE		ASYNC	BUS64	Reserved		V30	V33	RESUME	SDMA	HS	R	ADMA2	EXTBUS	MAXBLK[1:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASECLK[7:0]								TUNIT	R	TIMEOUTCLK[5:0]					

STYPE	[31:30]	Slot Type
N		This field indicates usage of a slot by a specific Host System.00b : Removable Card Slot 01b : Embedded Slot for One Device 10b : Shared Bus Slot 11b : Reserved

ASYNC	[29]	Asynchronous Interrupt Support
1		Asynchronous Interrupt Supported
0		Asynchronous Interrupt Not Supported

BUS64	[28]	64bit System Bus Support
1		Setting 1 to this bit indicates that the Host Controller supports 64-bit address descriptor mode and is connected to 64-bit address system bus.
0		Not supported

V30	[25]	Voltage Support 3.0V
1		Supported
0		Not supported

V33	[24]	Voltage Support 3.3V
1		Supported
0		Not supported

RESUME	[23]	Suspend / Resume Support
1		Supported
0		Not supported

SDMA	[22]	SDMA Support
1		Supported
0		Not supported

HS	[21]	High Speed Support
1		Supported
0		Not supported

ADMA2	[19]	ADMA2 Supported
1		Supported
0		Not supported

EXTBUS	[18]	Extended Media Bus Support
1		Supported
0		Not supported

MAXBLK	[17:16]	Max Block Length
This value indicates the maximum block size that the Host Driver can read and write to the buffer in the Host Controller. The buffer shall transfer this block size without wait cycles. It is noted that transfer block length shall be always 512bytes for SD Memory Cards regardless of this field. 00 - 512 byte 01 - 1024 byte 10 - 2048 byte 11 - 4096 byte		

BASECLK	[15:8]	Base Clock Frequency For SD Clock
1) 6bit Base Clock Frequency This mode is supported by the Host Controller Version 1.00 and 2.00. Upper 2-bit is not effective and always 0. The unit values are 1MHz and the supported clock range		

is 10MHz to 63MHz.
 11xx xxxxb: Not supported
 0011 1111b: 63MHz
 0000 0010b: 2MHz
 0000 0001b: 1MHz
 0000 0000b: Get information via another method

TUNIT	[7]	Timeout Clock Unit
1		This bit shows the unit of base clock frequency used to detect Data Timeout Error. The unit is MHz.
0		The unit is kHz.

TIMEOUTCLK	[5:0]	Timeout Clock Frequency
This bit shows the base clock frequency used to detect Data Timeout Error. The Timeout Clock Unit defines the unit of this field's value. Timeout Clock Unit =0 [kHz] unit: 1kHz to 63kHz Timeout Clock Unit =1 [MHz] unit: 1MHz to 63MHz Not 0 – 1kHz to 63kHz or 1MHz to 63MHz 000000b - Get Information via another method		

Capabilities 2

0x8000E044/E244

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved						SPBLK	SPI	CLKMULT							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

SPBLK	[25]	SPI Block Mode
1		Supported
0		Not Supported

SPI	[24]	SPI Mode
1		Supported
0		Not Supported

CLKMULT	[23:16]	Clock Multiplier
N		This field indicates clock multiplier value of programmable clock generator. Refer to Clock Control register. Setting 00h means that Host Controller does not support programmable clock generator. FFh : Clock Multiplier M = 256 02h : Clock Multiplier M = 3 01h : Clock Multiplier M = 2 00h : Clock Multiplier is not supported

Maximum Current Capabilities

0x8000E048/E248

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MAXCURV30[7:0]								MAXCURV33[7:0]							

Name	Bit	Description
MAXCURV30	[15:8]	Maximum Current for 3.0V. This will be supported in the future.
MAXCURV33	[7:0]	Maximum Current for 3.3V. This will be supported in the future.

Force Event for Error Interrupt / Auto CMD12 Error Status

0x8000E050/E250

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			TR	Reserved		ADMA	ACMD	CLIMIT	DATEND	DATCRC	DATTIME	CINDEX	CMDEND	CMDCRC	CMDTIME
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								NOCMD	Reserved		INDEX	ENDBIT	CRC	TIMEOUT	NORUN

Name	Bit	Description (1=Generate Interrupt, 0=No Interrupt)
TR	[28]	Force Event For Target Response Error
ADMA	[25]	Force Event For ADMA Error
ACMD	[24]	Force Event For Auto CMD Error
CLIMIT	[23]	Force Event For Current Limit Error
DATEND	[22]	Force Event For Data End Bit Error
DATCRC	[21]	Force Event For Data CRC Error
DATTIME	[20]	Force Event For Data Timeout Error
CINDEX	[19]	Force Event For Command Index Error
CMDEND	[18]	Force Event For Command End Bit Error
CMDCRC	[17]	Force Event For Command CRC Error
CMDTIME	[16]	Force Event For Command Timeout Error
NOCMD	[7]	Force Event For Command Not Issued by AutoCMD12 Error
INDEX	[4]	Force Event For Auto CMD12 Index Error
ENDBIT	[3]	Force Event For Auto CMD12 End Bit Error
CRC	[2]	Force Event For Auto CMD12 CRC Error
TIMEOUT	[1]	Force Event For Auto CMD12 Timeout Error
NORUN	[0]	Force Event For Auto CMD12 Not Executed

ADMA Error Status

0x8000E054/E254

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													LEN	ERRSTATE[1:0]	

LEN	[2]	ADMA Length Mismatch Error
1		This error occurs in the following 2 cases. (1) While Block Count Enable being set, the total data length specified by the Descriptor table is different from that specified by the Block Count and Block Length. (2) Total data length cannot be divided by the block length.
0		No error

ERRSTATE	[1:0]	ADMA Error State
N		This field indicates the state of ADMA when error occurs during ADMA data transfer. This field never indicates "10" because ADMA never stops in this state. 00 - ST_STOP (Stop DMA), Points next of the error descriptor 01 - ST_FDS (Fetch Descriptor), Points the error descriptor 10 - Never set this state (Not used) 11 - ST_TFR (Transfer Data), Points the next of the error descriptor

ADMA System Address

0x8000E058/E258

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR[31:16]								8	7	6	5	4	3	2	1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR[15:0]															

ADDR	[31:0]	ADMA System Address
This register holds byte address of executing command of the Descriptor table. 32-bit Address Descriptor uses lower 32-bit of this register. At the start of ADMA, the Host Driver shall set start address of the Descriptor table. The ADMA increments this register address, which points to next line, when every fetching a Descriptor line. When the ADMA Error Interrupt is generated, this register shall hold valid Descriptor address depending on the ADMA state. The Host Driver shall program Descriptor Table on 32-bit boundary and set 32-bit boundary address to this register. ADMA2 ignores lower 2-bit of this register and assumes it to be 00b.		

Refer to the SD Host Controller Standard Specification V2.00 of SD Association to know how to make the descriptor table.
[5]

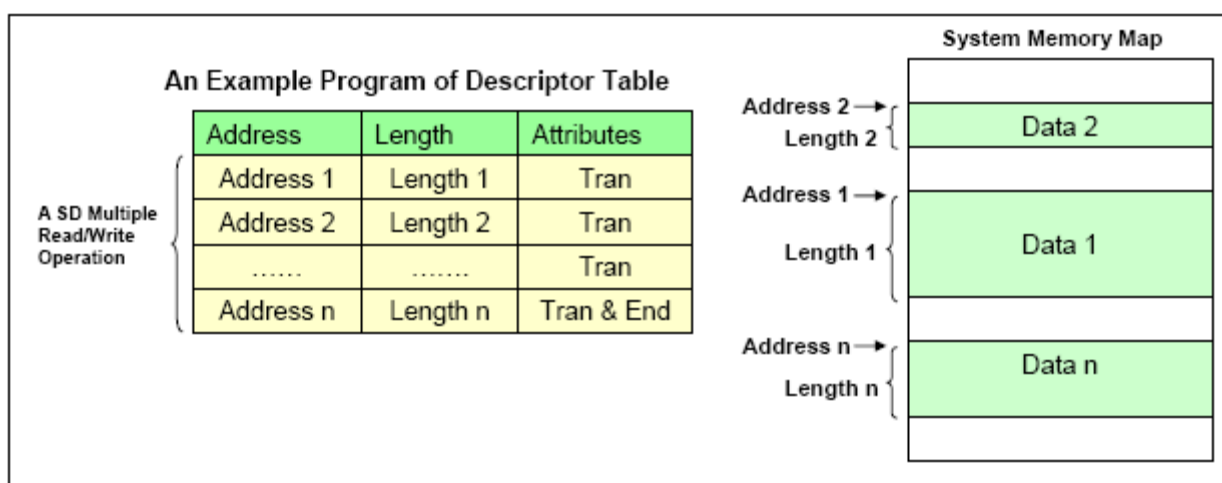


Figure 31.2 Example of ADMA2 Data Transfer

Figure 31.2 shows a typical ADMA2 descriptor program. The data area is sliced in various lengths and each slice is placed somewhere in system memory. The Host Driver describes the Descriptor Table with set of address, length, and attributes. Each sliced data is transferred in turns as programmed in descriptor.

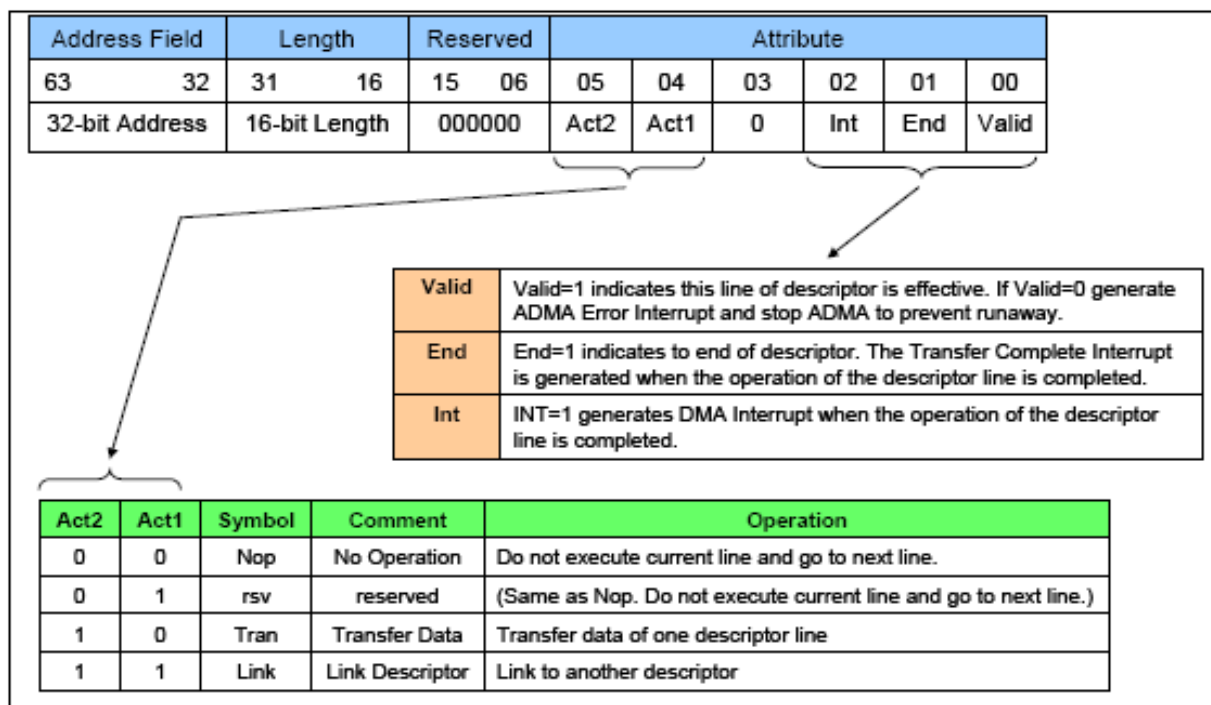


Figure 31.3 32-bit Address Descriptor Table

Preset 0 for Default Speed(0x62) and Initialization(0x60)

0x8000E060/E260

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					CLK	SDCLK									
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CLK	SDCLK									

CLK	[26], [10]	Clock Generator Select Value
1		This bit is effective when Host Controller supports programmable clock generator. When this bit is set, Programmable Clock Generator is selected.
0		Host Controller Ver2.00 Compatible Clock Generator is selected.

SDCLK	[25:16], [9:0]	SDCLK Frequency Select Value
N		10-bit preset value to set SDCLK Frequency Select in the Clock Control Register is described by a host system.

Preset 1 for High Speed(0x64)

0x8000E064/E264

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CLK	SDCLK									

CLK	[10]	Clock Generator Select Value
1		This bit is effective when Host Controller supports programmable clock generator. When this bit is set, Programmable Clock Generator is selected.
0		Host Controller Ver2.00 Compatible Clock Generator is selected.

SDCLK	[9:0]	SDCLK Frequency Select Value
N		10-bit preset value to set SDCLK Frequency Select in the Clock Control Register is described by a host system.

Boot Timeout Control

0x8000E070/E270

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BTOUTCNT[31:16]								8	7	6	5	4	3	2	1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BTOUTCNT[15:0]															

BTOUTCNT	[31:0]	Boot Timeout Count Value
N		This value determines the interval by which DAT line time-outs are detected during boot operation for eMMC4.4 card. The value is in number of sd clock.

Debug Selection

0x8000E074/E274

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														DSEL	

DSEL	[0]	Debug Selection
1		CMD register, Interrupt status, transmitter module, ahb_iface module and clk_sdc card signals are probed out through OBSERVABLE register.
0		Receiver module and fifo_ctrl module signals are probed out through OBSERVABLE register.

SPI Interrupt Support

0x8000E0F0/E2F0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								SPIINT[7:0]							

SPIINT	[7:0]	SPI Interrupt Support
This bit is set to indicate the assertion of interrupts in the SPI mode at any time, irrespective of the status of the card select (CS) line. If this bit is zero, then SDIO card can only assert the interrupt line in the SPI mode when the CS line is asserted.		

Host Controller Version / Slot Interrupt Status

0x8000E0FC/E2FC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								SPEC[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								SLOTINT[7:0]							

SPEC	[23:16]	Specification Version Number
N		This Status indicates the Host Controller Spec Version.00 - SD Host Specification version 1.0 01 - SD Host Specification version 2.00 including only the feature of the Test Register 02 - SD Host Specification version 2.00 including the feature of the Test Register and ADMA Others - Reserved

SLOTINT	[7:0]	Interrupt Signal For Each Slot
N		These status bits indicate the logical OR of Interrupt signal and Wakeup signal for each slot. Bit 00 - Slot 1 Bit 01 - Slot 2 Bit 02 - Slot 3 ----- Bit 07 - Slot 8

CFG0

0x8000E100/E300

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AHB															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										VFPE	VF	CDPE	CD	WPPE	WP

AHB	[31]	AHB HGRANT
1		HGRANT for Host Controller DMA is asserted depending on HBUSREQ.
0		HGRANT for Host Controller DMA is asserted regardless of HBUSREQ.

VFPE	[5]	For debug
VF	[4]	For debug

CDPE	[3]	Write Protection Pin Enable
1		Card Detection is dependent on PS[6]/PS[18] when this bit is set.
0		Card Detection is dependent on CD bit.

CD	[2]	Card Detection
1		Card Detection is asserted when this bit is set and CDPE is cleared.
0		Card Detection is not asserted.

WPPE	[1]	Write Protection Pin Enable
1		Write Protection is dependent on PS[7]/PS[19] when this bit is set.
0		Write Protection is dependent on WP bit.

WP	[0]	Write Protection
1		Write Protection is asserted when this bit is set and WPPE is cleared.
0		Write Protection is not asserted.

CFG2

0x8000E108/E308

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DLYCE				DLYCO				DLYDE				DLYDO			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DLYA	Reserved														

DLYCE	[31:28]	Output Delay for CMD OE
N		Add output delay for CMD Output Enable signal.

DLYCO	[27:24]	Output Delay for CMD
N		Add output delay for CMD Output signal.

DLYDE	[23:20]	Output Delay for DAT OE
N		Add output delay for all DAT Output Enable signal.

DLYDO	[19:16]	Output Delay for DAT
N		Add output delay for all DAT Output signal.

DLYA	[15]	Select Data Output Delay Value
1		Use CFG8, CFG9 Register Value
0		Use DLYDE / DLYDO field value

CFG3

0x8000E10C/E30C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															CONF

CONF	[0]	Disable CMD Conflict
1		Disable the command conflict logic.
0		Enable the command conflict logic.

CFG4

0x8000E110/E310

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CAP[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAP[15:0]															

CAP	[31:0]	Capabilities
N	Capabilities registers reflect these bits	

CFG5

0x8000E114/E314

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												CAP[51:48]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAP[47:32]															

CAP	[51:32]	Capabilities
N	Capabilities registers reflect these bits	

CFG6

0x8000E118/E318

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			INIT												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DEFAULT												

INIT	[28:16]	Initialization Preset
N	Initialization of Preset0 register reflects these bits	

DEFAULT	[12:0]	Default Preset
N	Default Speed of Preset0 register reflects these bits	

CFG7

0x8000E11C/E31C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			HIGH												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

HIGH	[28:16]	High Speed Preset
N	High Speed of Preset1 register reflects these bits	

OBSERVABLE

0x8000E128/E328

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OBSERVE[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OBSERVE[15:0]															

OBSERVE	[31:0]	Observable Signal
N	Debugging use only. Observe signals depending on DSEL of Debug Selection.	

MON

0x8000E12C/E32C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								MON							

MON	[8:0]	Observable Signal
N	Monitor following signals, rst_n, led_on, bus_volt, bus_pow, sig_en_18v, drv_r_str sel.	

CFG8

0x8000E140/E340

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DLYDE7				DLYDE6				DLYDE5				DLYDE4			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DLYDE3				DLYDE2				DLYDE1				DLYDE0			

DLYDE7	[31:28]	Output Delay for DAT OE [7]
N	Add output delay for DAT[7] Output Enable signal.	
DLYDE6	[27:24]	Output Delay for DAT OE [6]
N	Add output delay for DAT[6] Output Enable signal.	
DLYDE5	[23:20]	Output Delay for DAT OE [5]
N	Add output delay for DAT[5] Output Enable signal.	
DLYDE4	[19:16]	Output Delay for DAT OE [4]
N	Add output delay for DAT[4] Output Enable signal.	
DLYDE3	[15:12]	Output Delay for DAT OE [3]
N	Add output delay for DAT[3] Output Enable signal.	
DLYDE2	[11:08]	Output Delay for DAT OE [2]
N	Add output delay for DAT[2] Output Enable signal.	
DLYDE1	[07:04]	Output Delay for DAT OE [1]
N	Add output delay for DAT[1] Output Enable signal.	
DLYDE0	[03:00]	Output Delay for DAT OE [0]
N	Add output delay for DAT[0] Output Enable signal.	

CFG9

0x8000E144/E344

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DLYDO7				DLYDO6				DLYDO5				DLYDO4			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DLYDO3				DLYDO2				DLYDO1				DLYDO0			

DLYDO7	[31:28]	Output Delay for DAT [7]
N	Add output delay for DAT[7] Output Enable signal.	
DLYDO6	[27:24]	Output Delay for DAT [6]
N	Add output delay for DAT[6] Output Enable signal.	
DLYDO5	[23:20]	Output Delay for DAT [5]
N	Add output delay for DAT[5] Output Enable signal.	
DLYDO4	[19:16]	Output Delay for DAT [4]
N	Add output delay for DAT[4] Output Enable signal.	
DLYDO3	[15:12]	Output Delay for DAT [3]
N	Add output delay for DAT[3] Output Enable signal.	
DLYDO2	[11:08]	Output Delay for DAT [2]
N	Add output delay for DAT[2] Output Enable signal.	
DLYDO1	[07:04]	Output Delay for DAT [1]
N	Add output delay for DAT[1] Output Enable signal.	
DLYDO0	[03:00]	Output Delay for DAT [0]
N	Add output delay for DAT[0] Output Enable signal.	

31.4 Command Format

All commands have a fixed code length of 48 bits, needing a transmission time of 1.92µs @ 25MHz and 0.96µs @ 50MHz.

Table 31.7 Command Format

Bit Position	47	46	[45:40]	[39:8]	[7:1]	0
Width (bits)	1	1	6	32	7	1
Value	'0'	'1'	x	x	x	'1'
Description	Start bit	Transmission bit	Command index	Argument	CRC7	End bit

A command always starts with a start bit (always 0), followed by the bit indicating the direction of transmission (host = 1). The next 6 bits indicate the index of the command, this value being interpreted as a binary coded number (between 0 and 63). Some commands need an argument (e.g. an address), which is coded by 32 bits. A value denoted by 'x' in the table above indicates this variable is dependent on the command. All commands are protected by a CRC. Every command codeword is terminated by the end bit (always 1).

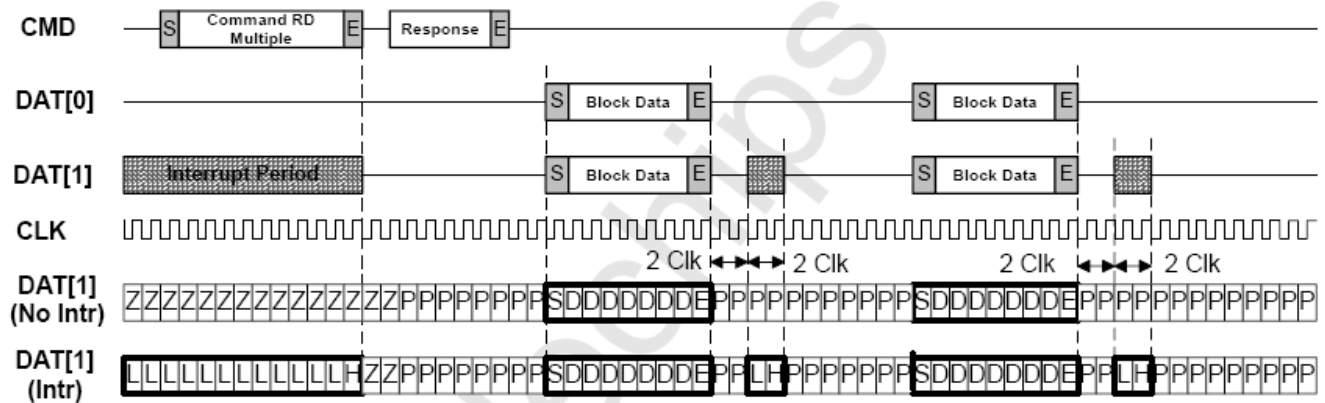


Figure 31.7 SDIO/SD – Multiple Block 4-Bit Read Interrupt Cycle Timing

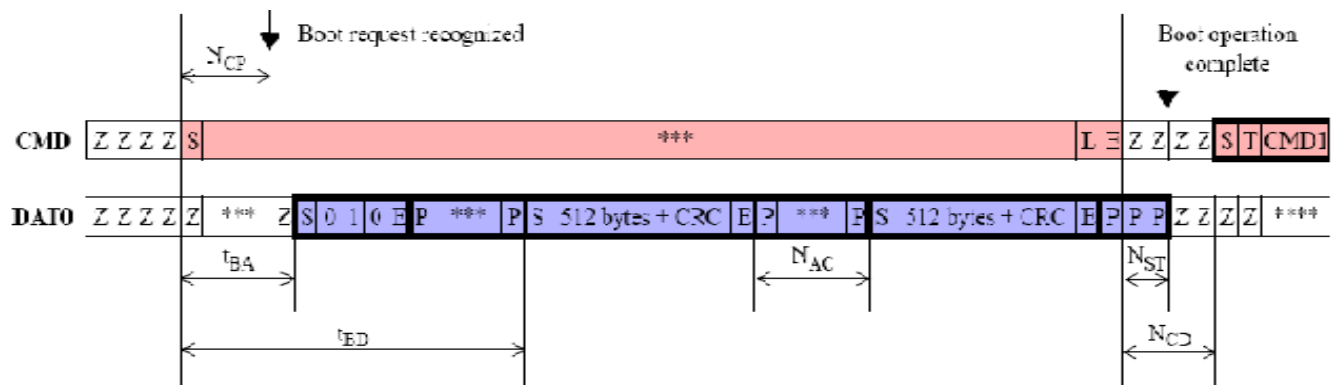


Figure 31.8 Boot Operation Timing with Termination Between Consecutive Data Blocks

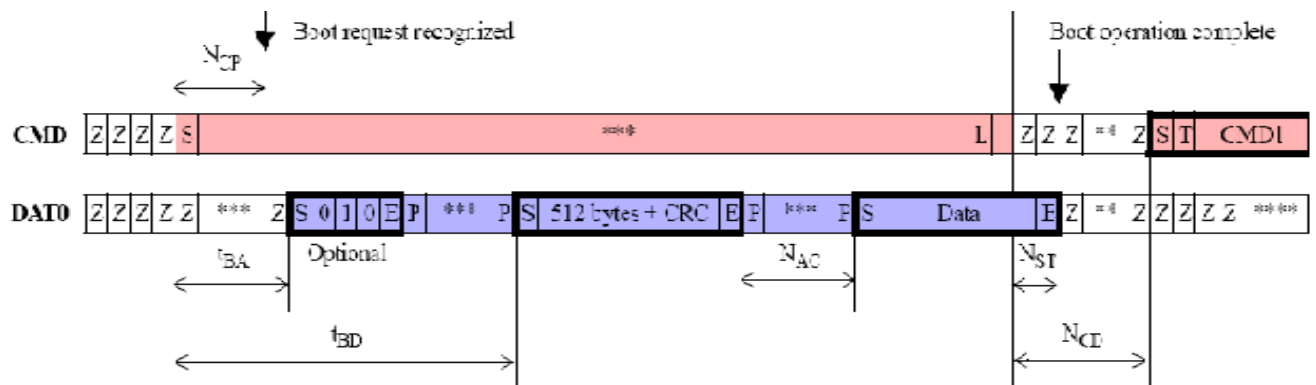
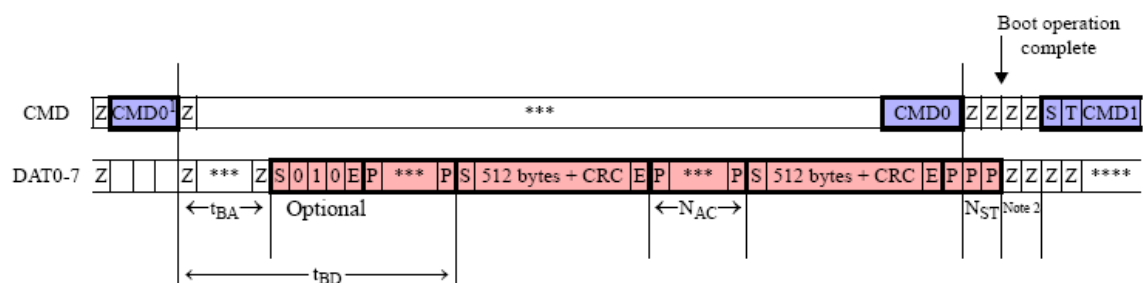
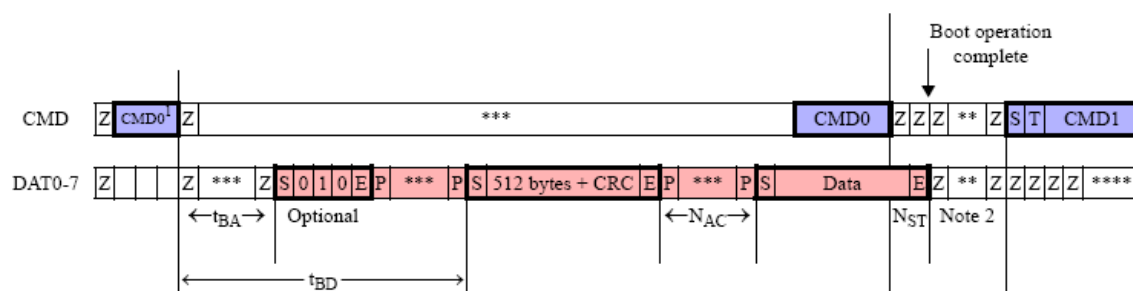


Figure 31.9 Boot Operation Timing with Termination During Transfer



NOTE 1. CMD0 with argument 0xFFFFF0FA.

Figure 31.10 Alternative boot operation, termination between consecutive data blocks



NOTE 1. CMD0 with argument 0xFFFFFFFF.

Figure 31.11 Alternative boot operation, termination during transfer

31.5.2 Timing to Switch Signal Voltage

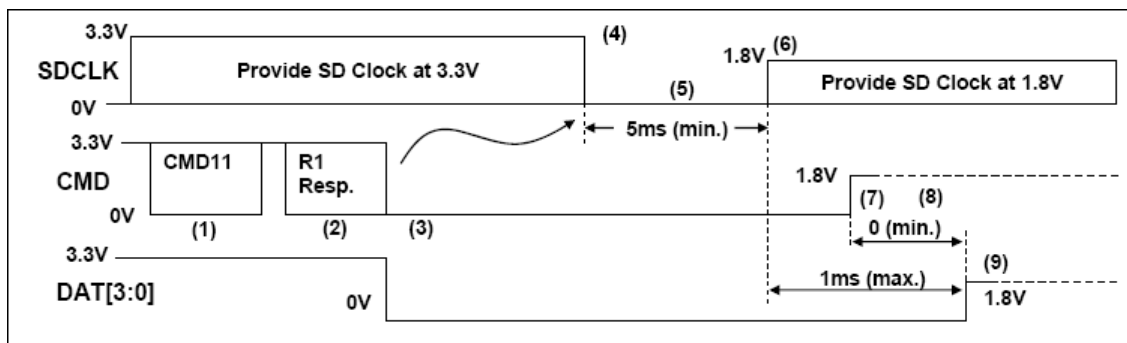


Figure 31.12 Signal Voltage Switching Sequence

- (1) Host issues CMD11 to start voltage switch sequence.
- (2) The card returns R1 response.
- (3) The card drives CMD and DAT[3:0] to low immediately after the response.
- (4) The host stops supplying SDCLK. The card shall start switching voltage after host stops SDCLK. The time to stop SDCLK is not specified. The host can detect whether the sequence starts by checking signal level of either one of CMD, DAT[3:0]. Which signal should be checked depends on ability of the host. If low level is not detected, the host should abort the sequence and execute power cycle.
- (5) 1.8V output of voltage regulator in card shall be stable within 5ms. The host keeps SDCLK low at least 5ms. This means that 5ms is the maximum for the card and the minimum for the host.
- (6) After 5ms from (4) and host voltage regulator is stable, the host starts providing SDCLK at 1.8V. The card can check whether SDCLK voltage is 1.8V.
- (7) By detecting SDCLK, the card drives CMD to high at 1.8V at least one clock and then stop driving (tri-state). CMD is triggered by rising edge of SDCLK (SDR timing).
- (8) The card can check whether the host drives CMD to 1.8V through the host pull-up resistor.
- (9) If switching to 1.8V signaling is completed successfully, the card drives DAT[3:0] to high at 1.8V at least one clock and then stop driving (tri-state). DAT[3:0] is triggered by rising edge of SDCLK (SDR timing). DAT[3:0] shall be high within 1ms from start of providing SDCLK. The host checks whether DAT[3:0] is high after 1ms from supplying SDCLK. This means that 1ms is the maximum for the card and the minimum for the host.

• Power Down and Power Cycle

- When the host shuts down the power, the card VDD shall be lowered to less than 0.5V for a minimum period of 1ms. During power down, DAT, CMD, and CLK should be disconnected or driven to logical 0 by the host to avoid a situation that the operating current is drawn through the signal lines.
- If the host needs to change the operating voltage, a power cycle is required. Power cycle means the power is turned off and supplied again. Power cycle is also needed for accessing cards that are already in Inactive State. To create a power cycle the host shall follow the power down description before power up the card (i.e. the card VDD shall be once lowered to less than 0.5V for a minimum period of 1ms).

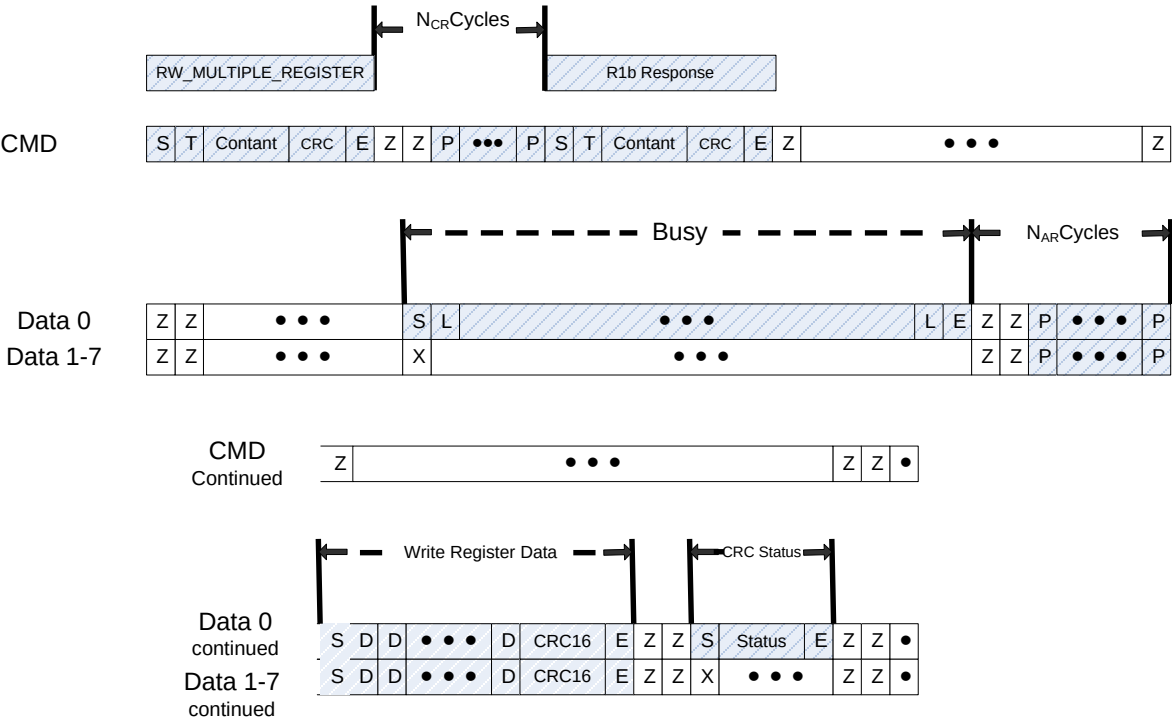


Figure 31.13 CMD60 Write Transaction

31.6 SD/MMC Timing

31.6.1 Default Mode Timing

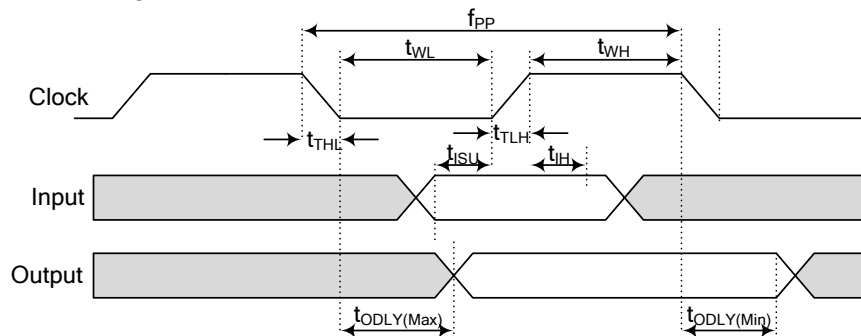


Figure 31.14 SD/MMC Default Mode Timing

Table 31.8 SD/MMC Default Mode Timing Parameters

Parameter	Symbol	MIN	MAX	Unit
Clock frequency (Data Transfer)	f_{PP}	0	25	MHz
Clock frequency (Identification)	f_{OD}	0	400	kHz
Clock low time	t_{WL}	10	-	ns
Clock high time	t_{WH}	10	-	ns
Clock rise time	t_{TLH}	-	10	ns
Clock fall time	t_{THL}	-	10	ns
Input set-up time	t_{ISU}	5	-	ns
Input hold time	t_{IH}	5	-	ns
Output Delay time (Data Transfer)	t_{ODLY}	0	14	ns
Output Delay time (Identification)	t_{ODLY}	0	50	ns

$V_{DDIO} = 3.3V$, $C_L = 30pF$

31.6.2 High-Speed Mode Timing

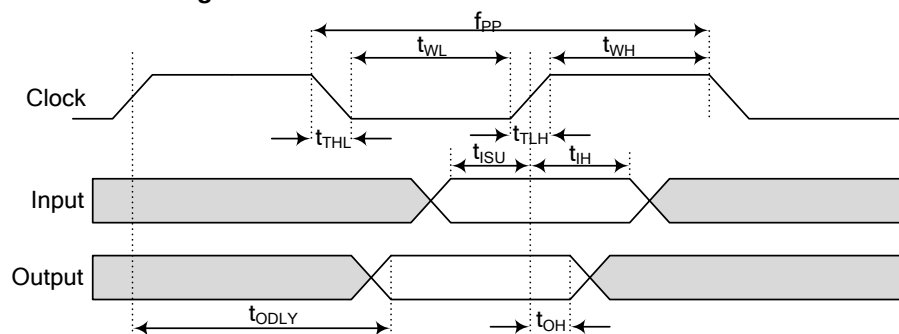


Figure 31.15 SD/MMC High Speed Mode Timing

Table 31.9 SD/MMC High Speed Mode Timing

Parameter	Symbol	MIN	MAX	Unit
Clock frequency (Data Transfer)	f_{PP}	0	50	MHz
Clock low time	t_{WL}	7	-	ns
Clock high time	t_{WH}	7	-	ns
Clock rise time	t_{TLH}	-	3	ns
Clock fall time	t_{THL}	-	3	ns
Input set-up time	t_{ISU}	6	-	ns
Input hold time	t_{IH}	2	-	ns
Output Delay time (Data Transfer)	t_{ODLY}	0	14	ns
Output hold time	t_{OH}	2.5	-	ns

$V_{DDIO} = 3.3V$, $C_L = 30pF$

31.6.3 MMC DDR Mode Timing

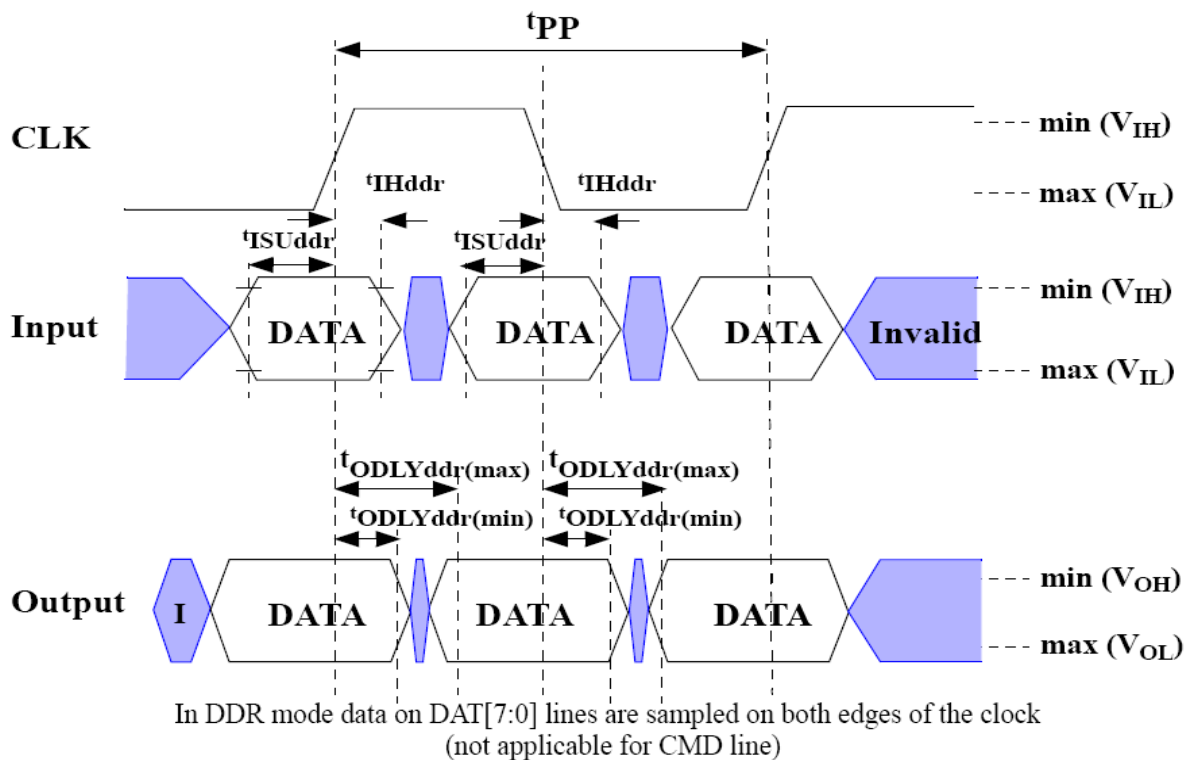


Figure 31.16 MMC DDR Mode Timing

Table 31.10 SD/MMC DDR Mode Timing

Parameter	Symbol	MIN	MAX	Unit
Clock frequency (Data Transfer)	t_{pp}	0	52	MHz
Input set-up time	t_{SUddr}	2.5	-	ns
Input hold time	t_{IHddr}	2.5	-	ns
Output Delay time (Data Transfer)	$t_{ODLYddr}$	1.5	7	ns

$V_{DDIO} = 3.3V$, $C_L = 20pF$

32 Cipher

32.1 Overview

AES and DES are algorithms built in the cipher of the TCC8132 for copy protection.

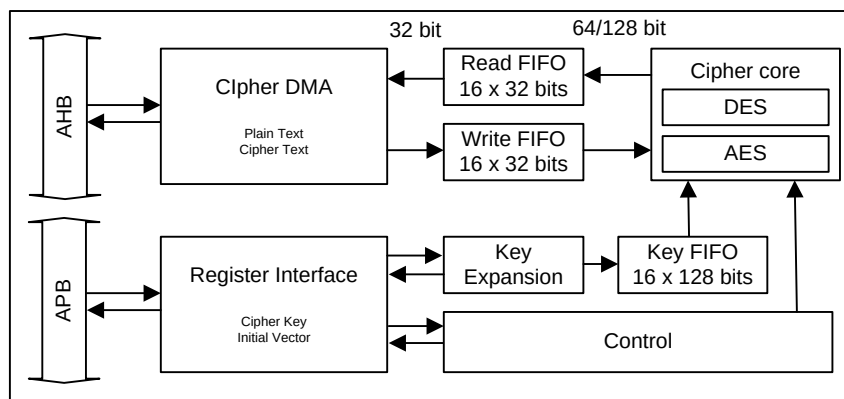


Figure 32.1 Cipher Engine Block Diagram

32.2 Feature

Specifically, AES supports 128/192/256 key mode and DES supports Single DES, Double DES and Triple DES. In addition, Triple DES supports 2 key mode and 3 key mode.

In overall, the cipher engine supports ECB(Electronic Codebook), CBC(Cipher Block Chaining), CFB(Cipher Feedback), OFB(Output Feedback), CTR(Counter) mode in AES and DES.

32.3 Register Descriptions

Table 32.1 Cipher Register Map (Base Address = 0x98018000)

Name	Offset	BITS	RW	Reset	Description
CTRL	0x00	32	R/W	0x00000000	Cipher control register
TXBASE	0x04	32	R/W	0x00000000	TX base address register
RXBASE	0x08	32	R/W	0x00000000	RX base address register
PACKET	0x0C	32	R/W	0x00000000	Packet register
DMACTR	0x10	32	R/W	0x00000000	DMA control register
DMASTR	0x14	32	RO	-	DMA status register
IRQCTR	0x18	32	R/W	0x00000000	Interrupt control register
BLKNUM	0x1C	32	RO	-	Block count register
DATA0	0x30	32	RO	-	Data0 register
DATA1	0x34	32	RO	-	Data1 register
DATA2	0x38	32	RO	-	Data2 register
DATA3	0x3C	32	RO	-	Data3 register
KEY0	0x40	32	WO	-	Key0 register
KEY1	0x44	32	WO	-	Key1 register
KEY2	0x48	32	WO	-	Key2 register
KEY3	0x4C	32	WO	-	Key3 register
KEY4	0x50	32	WO	-	Key4 register
KEY5	0x54	32	WO	-	Key5 register
KEY6	0x58	32	WO	-	Key6 register
KEY7	0x5C	32	WO	-	Key7 register
IV0	0x60	32	WO	-	Initial vector0 register
IV1	0x64	32	WO	-	Initial vector1 register
IV2	0x68	32	WO	-	Initial vector2 register
IV3	0x6C	32	WO	-	Initial vector3 register

32.3.1 Cipher Registers

Control Register (CTRL)

0x98018000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WCLR	RCLR	BCLR	IVLD	KEYLD	KEYLEN		DESMODE		OPMODE			PRT	MODE	SELECT	

Bit	Name	RW	Reset	Description
15	WCLR	WO	0x0	Clear transmit FIFO counter '1' for clear
14	RCLR	WO	0x0	Clear receive FIFO counter '1' for clear
13	BCLR	WO	0x0	Clear Block counter '1' for clear
12	IVLD	WO	0x0	Initial vector load '1' for load
11	KEYLD	WO	0x0	Key data load '1' for load
10:9	KEYLEN	RW	0x0	Select the key length in AES. 2,3: 256 bits key length 1: 192 bits key length 0: 128 bits key length
8:7	DESMODE	RW	0x0	Select the mode in DES. 3: Triple DES 3 key mode 2: Triple DES 2 key mode 1: Double DES mode 0: Single DES mode
6:4	OPMODE	RW	0x0	Select the operation mode. 4,5,6,7: Counter(CTR) 3: Output Feedback(OFB) 2: Cipher Feedback(CFB) 1: Cipher Block Chaining(CBC) 0: Electronic Codebook(ECB)
3	PRT	RW	0x0	Select the parity bit location of the key in DES. 1: MSB is a parity bit. 0: LSB is a parity bit.
2	MODE	RW	0x0	Select Mode 1 : CBCMAC mode 0 : Normal mode
1:0	SELECT	RW	0x0	Select the algorithm 1: DES 0: AES

TX Base Register (TXBASE)

0x98018004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TXBASE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TXBASE															

Bit	Name	RW	Reset	Description
31:0	TXBASE	RW	0x00000000	TX base address register

RX Base Register (RXBASE)

0x98018008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RXBASE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXBASE															

Bit	Name	RW	Reset	Description
31:0	RXBASE	RW	0x00000000	RX base address register

Packet Register (PACKET)

0x9801800C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-	-	-	COUNT												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-	-	-	SIZE												

Bit	Name	RW	Reset	Description
28:16	COUNT	R/W	0x0000	Packet number information (COUNT + 1)
12:0	SIZE	R/W	0x0000	Pack size information

DMA Control Register (DMACTR)

0x98018010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DE	-	-	END	-	-	-	-	-	-	-	-	-	-	TXAM	-
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXAM			-	-	-	-	-	-	-	-	-	-	PCLR	-	EN

Bit	Name	RW	Reset	Description
31	DE	R/W	0x0	DMA request enable 1: Enable 0: Disable
28	END	R/W	0x0	Byte endian mode register 1: Big-endian 0: Little-endian
17:16	TXAM	R/W	0x0	TX addressing mode 2,3: Single packet 1: Fixed address(base) 0: Multiple Packet
15:14	RXAM	R/W	0x0	RX addressing mode 2,3: Single packet 1: Fixed address(base) 0: Multiple Packet
2	PCLR	WO	0x0	Clear TX/RX packet counter '1' for clear
0	EN	R/W	0x0	DMA enable register 1: Enable 0: Disable

DMA Status Register (DMASTS)

0x98018014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-	-	-	RXPCNT												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-	-	-	TXPCNT												

Bit	Name	RW	Reset	Description
28:16	RXPCNT	RO	0x0000	Receive packet count register
12:0	TXPCNT	RO	0x0000	Transmit packet count register

IRQ Control Register (IRQCTR)

0x98018018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-	-	IDS	ISP	-	-	-	-	-	-	-	IRQS	-	-	IED	IEP
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-			IRQPCNT												

Bit	Name	RW	Reset	Description
29	ISD	R/C	0x0	IRQ status for "Done interrupt" When the number of packets which DMA transfers/receives is equal to the number of packets specified in PACKET register, it is issued. When writing 1, it is cleared.
28	ISP	R/C	0x0	IRQ status for "Packet interrupt" It is issued every IRQPCNT packets which DMA transfers/receives When writing 1, it is cleared.
20	IRQS	R/W	0x0	IRQ select register 1: Receiving 0: Transmitting
17	IED	R/W	0x0	IRQ enable for "Done interrupt" 1: Enable 0: Disable
16	IEP	R/W	0x0	IRQ enable for "Packet interrupt" 1: Enable 0: Disable
12:0	IRQPCNT	R/W	0x0000	IRQ packet count register

Blocknum Count Register (DMACTR)

0x9801801C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BLKNUM															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BLKNUM															

Bit	Name	RW	Reset	Description
31:0	BLKNUM	RO	-	Check the number of the current block count.

DATA0-3 Register (DATA0-3)

0x98018030 – 0x981083C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA															

Bit	Name	RW	Reset	Description
31:0	DATA	RO	-	Data Register

When the CM register(In Control register) is set to 1, Data register can read cipher operation.

KEY0-7 Register (KEY0-7)

0x98018040 – 0x981085C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
KEY															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEY															

Bit	Name	RW	Reset	Description
31:0	KEY	WO	-	Key Register

There are 8 key registers and their key sizes vary depending on the selected algorithm. If key setting is completed for these registers, 1 should be set for key load in the control register in order to generate a proper key corresponding to each algorithm. Key load is automatically cleared to 0 after one time key generation. The registers depending on the algorithms are as below.

Table 32.2 Key Map Depending on Cipher Engine

Table 6-21 Key Map Depending on Cipher Engine									
CIPHER	MODE	KEY0	KEY1	KEY2	KEY3	KEY4	KEY5	KEY6	KEY7
AES	128 bits key	KEY[127:0]				Not used			
	192 bits key	KEY[191:0]						Not used	
	256 bits key	KEY[255:0]							
DES	1 des	KEY[63:0]		Not used					
	2 des	KEY[127:0]				Not used			
	3 des 2 key	KEY[127:0]				Not used			
	3 des 3 key	KEY[191:0]						Not used	

IV0-3 Register (IV0-3)

0x98018060 – 0x9801806C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IV															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IV															

Bit	Name	RW	Reset	Description
31:0	IV	WO	-	Initial vector register.

There are four initial vector registers in total and the initial vector size varies from the selected algorithm. Only in AES, all 4 registers are used. In DES, only IV0 and IV1 are used. The initial vector registers are all used when OPMODE of control register is not ECB. Once IV setting is completed, IV load of control register should be set to 1 in order for the initial vector registers to operate properly.

33 VIOC (Video Input/Output Controller)

33.1 Overview

The Video Input/Output Controller (VIOC) is used to send out image data from system memory to various display devices or to receive image data from external video input and write to system memory. The display devices are TFT-LCD (RGB interface type) and the video inputs are camera, TV-input, and etc. The VIOC has one display timing controller and two independent video input controllers

The features of the VIOC are as follows.

Table 33.1 Brief Features

Type	Descriptions
Video Input	• Supports up to 16-bit interface (two channels) • YUV422_16BIT (BT.601, BT.1120, SMPTE274M) • YUV422_8BIT (BT.656, SMPTE125M, BT.1302, SMPTE267M) • YUVK4444_16BIT (BT.799, SMPTE RP-715, BT.1303) • RGBK4444_16BIT • Supports Time-Multiplexing Mode (TW2867 Device in Techwell Inc.)
Video Output	• Supports Thin Film Transistor(TFT) color displays with 8-bit, 16-bit, 18-bit, 24-bit interface • 16, 18, 24 bpp true-color non-palletized color displays for color TFT • Resolution programmable up to 4096 * 4096 • Programmable timing for different display panels • Supports color lookup table • Supports the overlay and alpha blending

The following figure shows the entire block diagram of the VIOC.

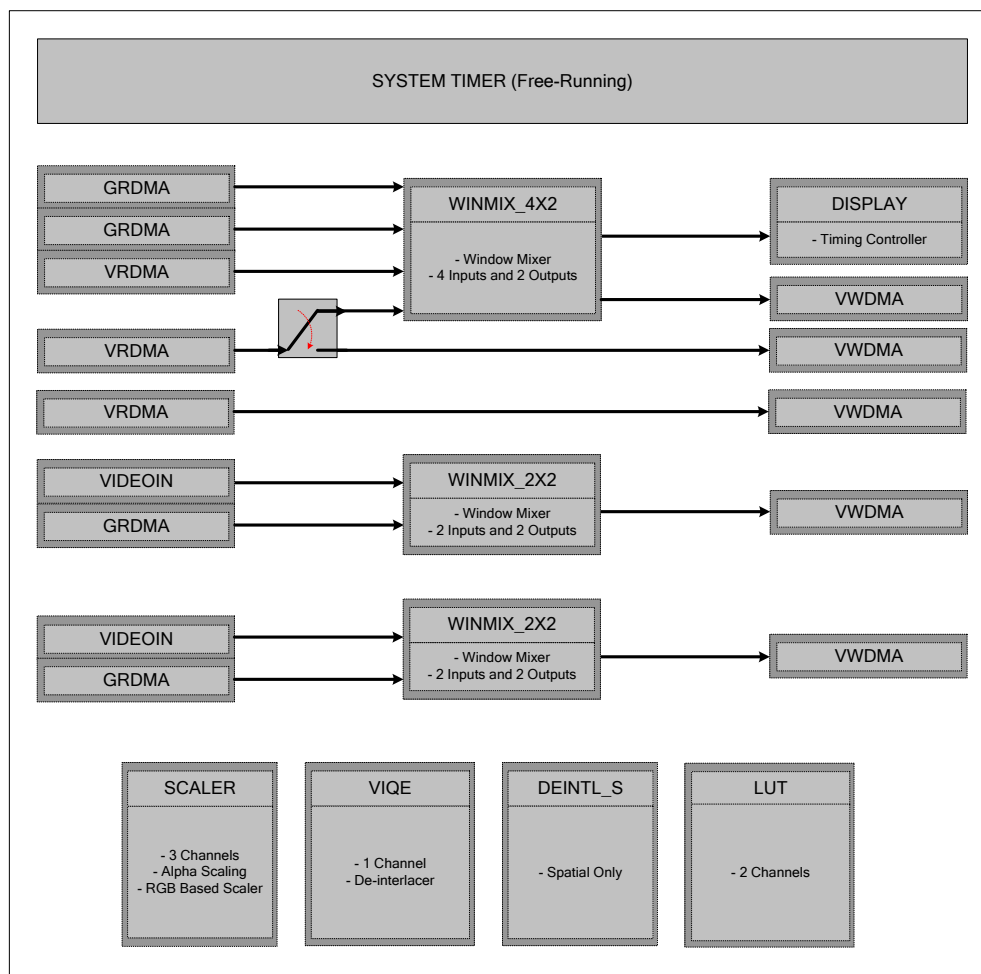


Figure 33.1 VIOC Block Diagram

For more information on each component and interface in the above figure, refer to the detailed explanation below.

The VIOC is divided into component and interface. The component means each hardware block and an interface is the information of channels transmitting data between each hardware block.

Each hardware component of the VIOC is composed of "SYSTEM TIMER", "GRDMA", "VRDMA", "VWDMA", "DISPLAY", "VIDEO IN", "SCALER", "VIQE (DEINTL_temporal)", "DEINTL_S", "LUT", "WINMIX_4X2", "WINMIX_2X2".

The arrows between each hardware block are defined as Interface. The 4 components in the figure above which are not connected with an arrow are supposed to function properly since they are inserted to each interface.

- "SYSTEM TIMER"

It makes system time for PTS (Presentation Time Stamp) in order for streaming control and in "SYSTEM TIMER", an interrupt output is available.

- "GRDMA" and "VRDMA"

It reads Image Source from the memory. For "GRDMA", 1 plane format is supported. For "VRDMA", 2 plane and 3 plane format are supported. Similarly,

- "GWDMA" and "VWDMA"

They are blocks for saving image data to the memory. 1 plane is supported for "GWDMA" and up to 3 planes are supported for "VWDMA". (However, the VIOC does not support GWDMA.)

- "DISPLAY"

It includes Timing Controller for video output.

- "VIDEO IN"

It can receive an image as input from outside.

- "SCALER"

It changes the size of an image.

- "VIQE"

Is a block for enhancing video image quality and can control image quality for the main image channel.

- "DEINTL_S"

It is a block handling simple deinterlacer and is used for sub-channel.

- "LUT"

It changes color mapping of RGB color based on the look-up table and can be used as a palette for 8bpp format.

- "WINMIX_4X2" and "WINMIX_2X2"

They are used as overlay mixer and windowing is available for each input. Simultaneous outputs of two interfaces are available for the overlay mixer.

33.2 Related Blocks

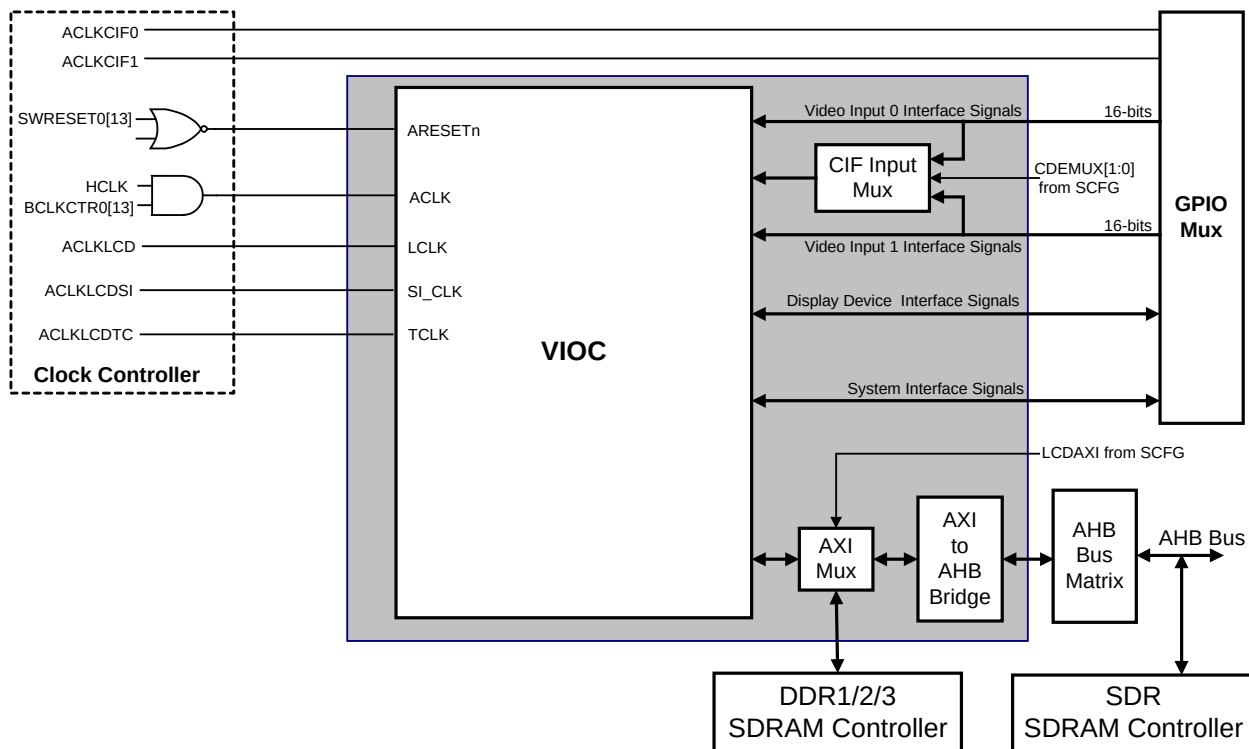


Figure 33.2 Clock/Reset/Bus Control for VIOC

33.2.1 Clock Controller

At reset, the core clocks are all disabled. Clocks must be set and enabled by programming the following registers:

- ACLKCIF0 (CIF0 Clock Register)
- ACLKCIF1 (CIF1 Clock Register)
- ACLKLCD (LCD Clock Register)
- ACLKLCDSI (LCDSI Clock Register)
- ACLKLCDTC (LCDC Timer Clock Register)
- BCLKCTR0[13] (BUS Clock Enable Register bit for VIOC)
- SWRESET0[13] (Software Reset Control bit for VIOC)

33.2.2 Interrupt Controller

Interrupt signal from the VIOC is connected to Interrupt Controller0 (A) channel [15]. Note that the default value of interrupt trigger mode is "Edge Trigger". For normal operation of the VIOC, the trigger mode should be set as "Level". Refer to TMODE_A and TMODEA_A descriptions in Chapter 11.2.

33.2.3 GPIO Controller

The following Function Selection Registers should be programmed appropriately before normal operation starts.

- GPIOLC_FS0, GPIOLC_FS1, GPIOLD_FS0, GPIOLD_FS1
- GPIOPC_FS0, GPIOPC_FS1
- GPIOPS_FS1, GPIOPS_FS2

33.2.4 SCFG (System Configuration Register)

LCDSI Access Enable (LCDSIAE)

0x98013024

Bit	Name	Type	Default	Description
10	CDEMUX[1]	R/W	0	VIOC CIF DEMUX Enable.
9	CDEMUX[0]	R/W	0	VIOC CIF DEMUX Input Selection. 0=CIF0 1=CIF1
8	LCDAXI	R/W	0	VIOC Master Bus Selection.
				0 AHB. If SDR SDRAM is used, this bit must be set as 0.
				1 AXI. If DDR SDRAM is used, this bit must be set as 1.

33.3 Terminology

33.3.1 Frame and Image Organization

The definition of frame window and image window is shown in the following figure. The frame window means that the raw image source and the image window is the window area on the frame window to be displayed. The “FP” stands for frame window pixel and “WP” stands for window pixel. The “FP(0,N)” is the pixel at the end of first line. All the pixels of the frame window should be on the linear and continuous address space.

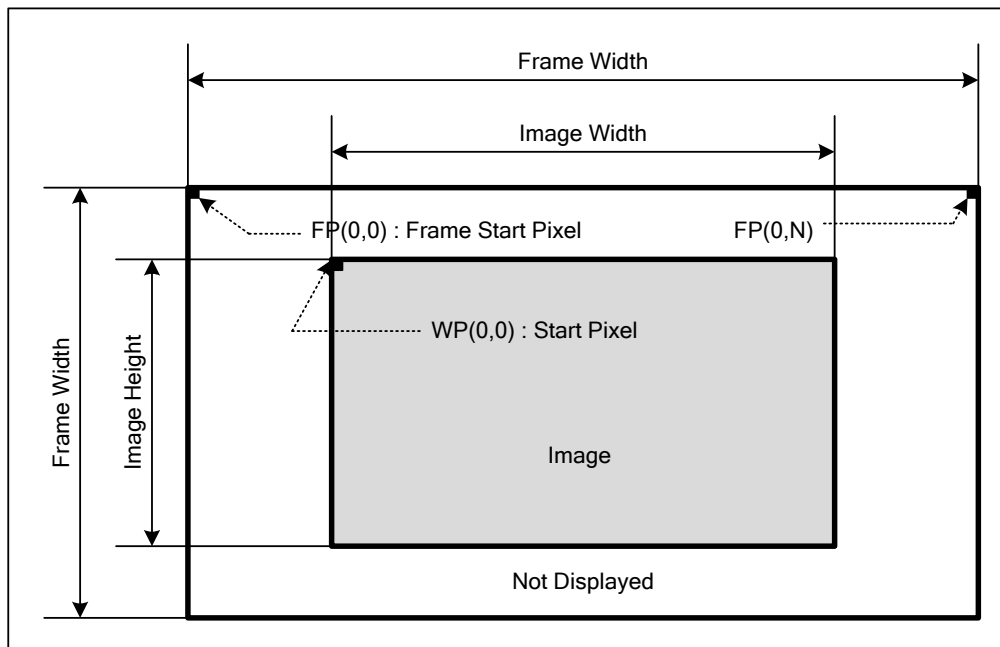


Figure 33.3 Definitions of Frame Window and Display Window

33.3.2 Image Cropping

Image cropping means cutting an image from the frame or cutting another smaller image from the image. The figure below shows the smaller image is cropped from the image.

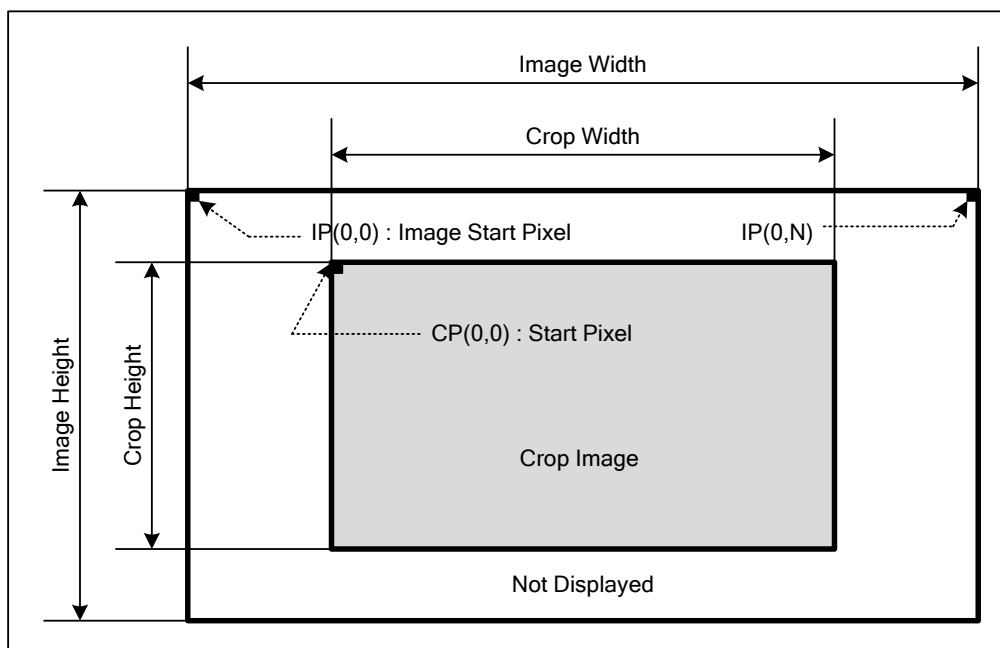


Figure 33.4 Definitions of Image Window and Crop Image

In case of cropping an image, X/Y coordinate of the Image Start Pixel should be 0 or larger than 0. Position X + Crop Width should be same as the Image Width or smaller than the Image Width.

33.3.3 Image Windowing

The windowing function of VIOC refers to both image cropping and image layout.

It is used to place an image inside the window defined as below. In other words, the width and height of the window and X/Y position of an image and the width and height of the image are defined. In this case, X/Y position should be more than 0 and smaller than the window width/height.

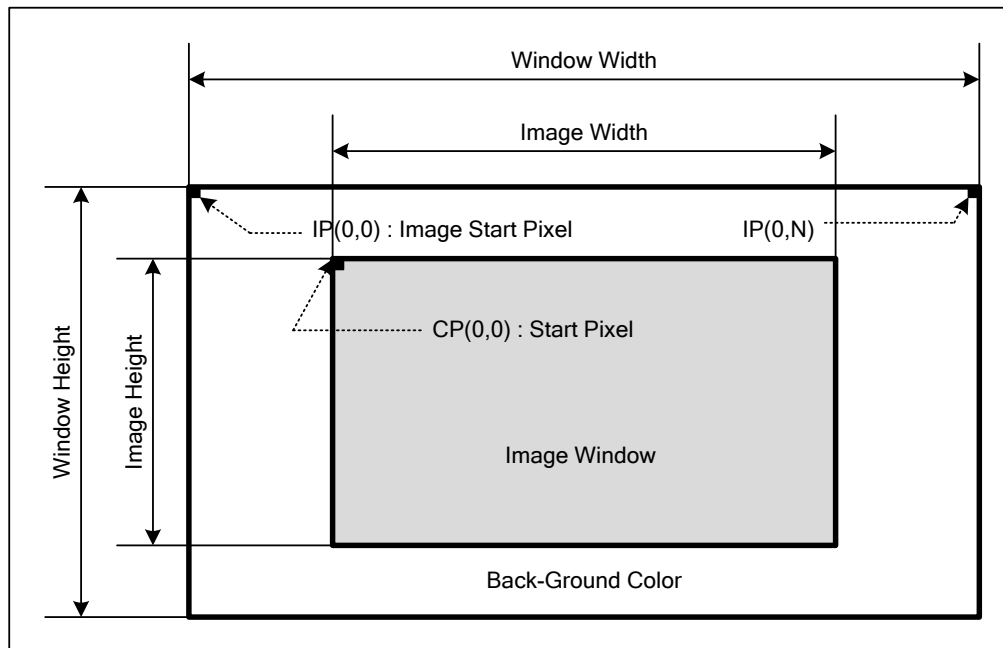


Figure 33.5 Definitions of Image and Window

As the figure above when an image is placed to an imaginary window defined with the above window width and window height, if the image is smaller than the size of the window, it is placed on the above position. The area which is not filled with the image is filled with a background color.

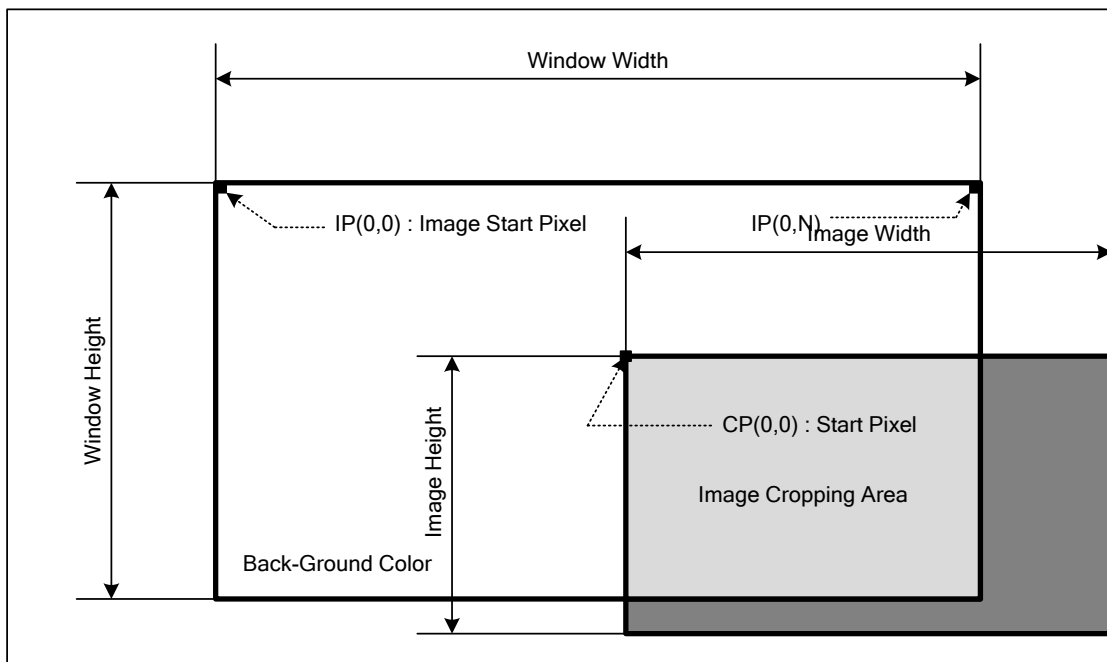


Figure 33.6 Definitions of Image and Window 1

If an image is outside window, the cropping area image above should be cropped and placed inside of the window. In this case, the dark gray color area should be abandoned. Note that since data for the dark gray needs to be transmitted if possible, unnecessary data transmission can occur. In addition, underrun may occur in the display controller, if the abandoned area in Y axis is large.

33.3.4 Images and Layers

For processing the raw image sources, the VIOC defines a virtual display. It is shown in Window size, which consists of "Window Width" and "Window Height" as in Figure 33.7.

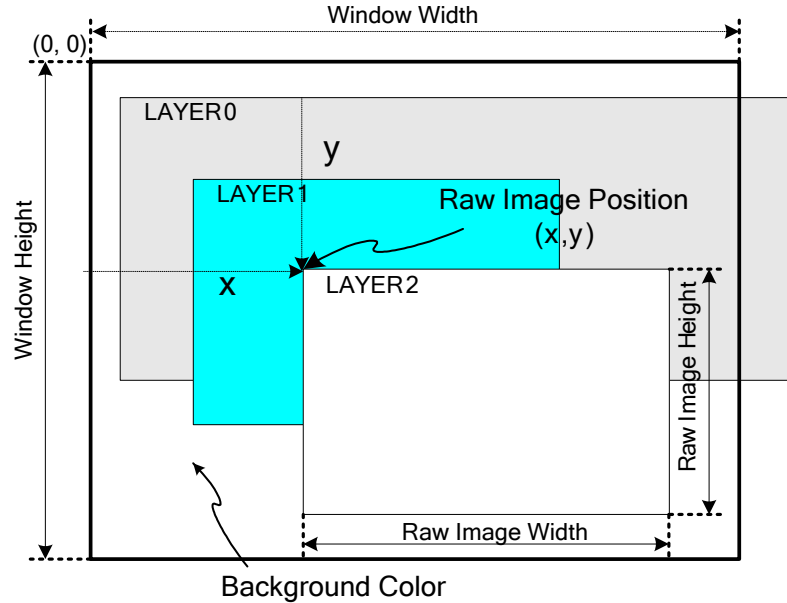


Figure 33.7 Virtual Display in the LCDC

The WINMIX_4X2 can have up to 4 raw image sources, which are named as Image 0, Image 1, Image 2, and Image 3. It can display simultaneously up to 4 layers, which are named as LAYER0, LAYER1, LAYER 2, and LAYER3. LAYER3 is the top layer. If image source was disabled, the corresponding layer is discarded. The area which is not overlapped by layers is filled with the background color. It is determined by the MBG register. Also, the WINMIX_2X2 can have up to 2 raw image sources and 2 layers.

33.3.5 Supported Image Formats

The properties of a layer are determined by those of the corresponding raw image source; size, pixel data format, base address, position in the virtual display, and etc.

33.3.5.1 Overall Supported Image Format

Table 33.2 Overall Supported Image Format

FMT [4:0]	Image Format
0	1bpp indexed color
1	2bpp indexed color
2	4bpp indexed color
3	8bpp indexed color
4 ~ 7	Reserved for future use
8	RGB332 – 1 bytes aligned : R[7:5],G[4:2],B[1:0]
9	RGB444 – 2 bytes aligned : A[15:12],R[11:8],G[7:4],B[3:0]
10	RGB565 – 2 bytes aligned : R[15:11],G[10:5],B[4:0]
11	RGB555 – 2 bytes aligned : A[15],R[14:10],G[9:5],B[4:0]
12	RGB888 – 4 bytes aligned : A[31:24],R[23:16],G[15:8],B[7:0]
13	RGB666 – 4 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]
14	RGB888 – 3 bytes aligned : B1[31:24],R0[23:16],G0[15:8],B0[7:0]
15	ARGB6666 – 3 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]
16	Reserved
17 ~ 20	Reserved for future use
21	4:4:4 separate format – <u>valid for only VRDMA and VWDMA</u> If Y2R is enabled, this can be used for YUV format, otherwise for RGB separated format.
22	YCbCr 4:2:2 sequential format - LSB [U/Y/V/Y] MSB
23	YCbCr 4:2:2 sequential format - LSB [V/Y/U/Y] MSB
24	YCbCr 4:2:0 separated format – <u>valid for only VRDMA and VWDMA</u>
25	YCbCr 4:2:2 separated format – <u>valid for only VRDMA and VWDMA</u>
26	YCbCr 4:2:2 sequential format - LSB [Y/U/Y/V] MSB
27	YCbCr 4:2:2 sequential format - LSB [Y/V/Y/U] MSB
28	YCbCr 4:2:0 interleaved type 0 format – <u>valid for only VRDMA and VWDMA</u>
29	YCbCr 4:2:0 interleaved type 1 format – <u>valid for only VRDMA and VWDMA</u>
30	YCbCr 4:2:2 interleaved type 0 format – <u>valid for only VRDMA and VWDMA</u>
31	YCbCr 4:2:2 interleaved type 1 format – <u>valid for only VRDMA and VWDMA</u>

The VIOC supports formats from 0 to 31 above. #21, #24, #25, #28, #29, #30 and #31 are 3 plan format and they are not supported in GRDMA but supported in VRDMA only.

33.3.5.2 Bit Organization of Each Format

BR=1	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
1BPP	P31	P30	P29	P28	P27	P26	P25	P24	P23	P22	P21	P20	P19	P18	P17	P16
2BPP	P15		P14		P13		P12		P11		P10		P9		P8	
4BPP	P7				P6				P5				P4			
8BPP	P3								P2							
BR=1	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1BPP	P15	P14	P13	P12	P11	P10	P9	P8	P7	P6	P5	P4	P3	P2	P1	P0
2BPP	P7		P6		P5		P4		P3		P2		P1		P0	
4BPP	P3				P2				P1				P0			
8BPP	P1								P0							

(a) The bitmap for 1bpp, 2bpp, 4bpp, 8bpp when BR is '1'

BR=0	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
1BPP	P24	P25	P26	P27	P28	P29	P30	P31	P16	P17	P18	P19	P20	P21	P22	P23
2BPP	P12		P13		P14		P15		P8		P9		P10		P11	
4BPP	P6				P7				P4				P5			
8BPP	P3								P2							
BR=0	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1BPP	P8	P9	P10	P11	P12	P13	P14	P15	P0	P1	P2	P3	P4	P5	P6	P7
2BPP	P4		P5		P6		P7		P0		P1		P2		P3	
4BPP	P2				P3				P0				P1			
8BPP	P1								P0							

(b) The bitmap for 1bpp, 2bpp, 4bpp, 8bpp when BR is '0'

RGB	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RGB332	R1			G1			B1		R0			G0			B0	
ARGB4444	A0				R0				G0				B0			
RGB565	R0					G0					B0					
ARGB1555	A0	R0				G0				B0						

(c) The bitmap for RGB332, ARGB4444, RGB565, and ARGB1555 formats.

RGB	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ARGB6666	IGNORED								A0						R0	
ARGB8888	A0								R0							
RGB	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARGB6666	R0				G0						B0					
ARGB8888	G0								B0							

(d) The bitmap for ARGB6666 and ARGB8888 formats.

RGB	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ARGB6666	G1[1:0]		R1						A0						R0	
RGB888	B1								R0							
RGB	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARGB6666	R0				G0				B0				B0			
RGB888	G0								B0							

(e) The bitmap for ARGB6666 and RGB888 formats. One pixel is mapped by 3 bytes.

Sequential	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CbYCrY	Y1								Cr0							
CrYCbY	Y1								Cb0							
YCbYCr	Cr0								Y1							
YCrYCb	Cb0								Y1							
Sequential	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CbYCrY	Y0								Cb0							
CrYCbY	Y0								Cr0							
YCbYCr	Cb0								Y0							
YCrYCb	Cr0								Y0							

(f) The bitmap for CbYCrY, CrYCbY, YCbYCr, YCrYCb sequential format. The Cb and Cr affect the Y0 and Y1.

Separated	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Y	Y1								Y0							
Cb	Cb1								Cb0							
Cr	Cr1								Cr0							

(g) The bitmap for YCbCr 4:2:0 and 4:2:2 separated format. The Cb0 and Cr0 affect to the Y0 and. The Cb1 and Cr1 affect the Y2s and Y3s.

Interleaved	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Y	Y1								Y0							
C(type 0)	Cr0								Cb0							
C(type 1)	Cb0								Cr0							

(h) The bitmap for YCbCr 4:2:0 and 4:2:2 interleaved format.

Figure 33.8 Supported Pixel Data Format

The pixel-data formats supported in the VIOC are shown in Figure 33.8. The 1bpp, 2bpp, and 4bpp have no color information and can be arranged in big-endian pixel order as well as little-endian pixel order. Others have color information and should be arranged in little-endian pixel order only.

Image 1 and Image 2 do not support YCbCr4:2:0 and YCbCr4:2:2, but support "sequential YCbCr4:2:2". If the pixel data format of a raw image source is YCbCr4:2:0 or YCbCr4:2:2, it needs three base addresses, which are specified by the RDMABASE0, RDMABASE1, and RDMABASE2 register. Otherwise, it needs one base address, which is specified by the RDMABASE0 register. Thus, when the pixel data format of Image 0 is YCbCr4:2:0 or YCbCr4:2:2, the base address of Y, Cb, and Cr image should be set to RDMABASE0, RDMABASE1, and RDMABASE2 register

Though the raw image source is not RGB888 format in RGB color space, it is internally converted to RGB888 as appending additional bits. When this conversion occurs, the PD bits of RDMACTRL determine whether the additional bits are filled with zero or MSB. The difference between them is shown in Figure 33.9.

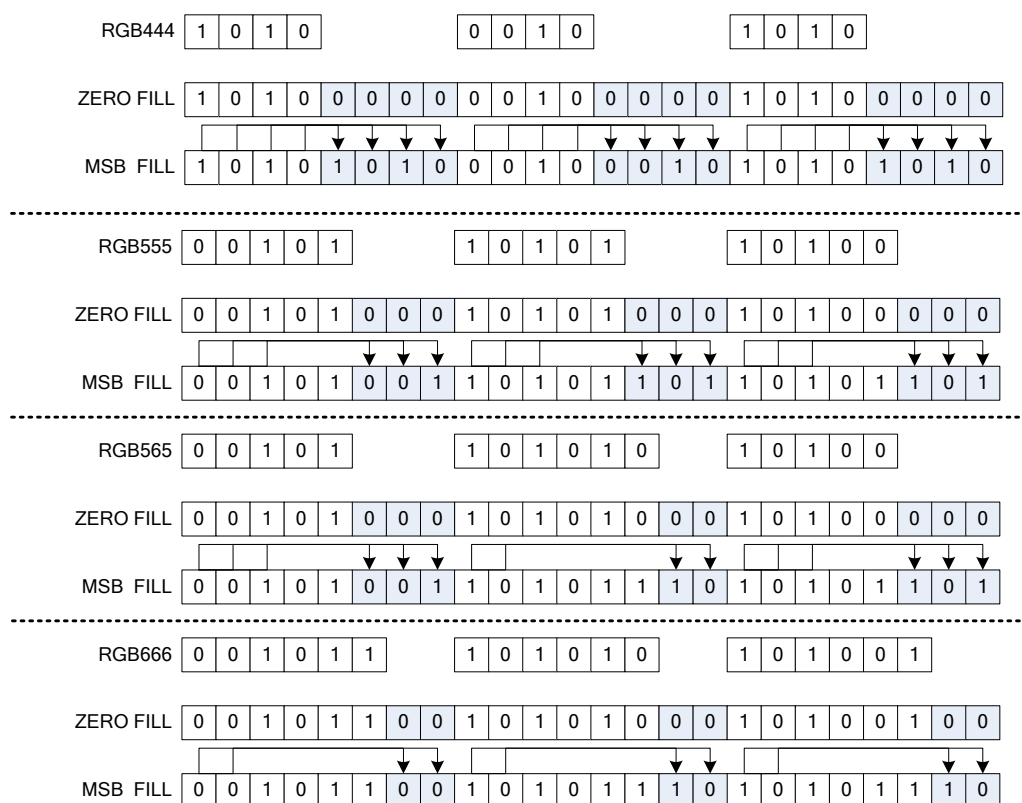


Figure 33.9 RGB2RGB888 Conversion

The origin of the virtual display is the top-left corner ((0, 0) in Figure 33.7) and "Raw Image Size", which consists of "Raw Image Width" and "Raw Image Height", is determined by the RDMASIZE registers. Though the raw image sources are bigger than the size of the virtual display and/or the part of them exits the outside of the virtual display, the VIOC automatically truncates an overlong part.

33.3.6 Supported Display Output Formats

33.3.6.1 TFT-LCD

The VIOC supports 16, 18, 24 bpp true-color non-palletized color displays for color TFT LCD. It generates the control signals for Display Device driver such as VSYNC, HSYNC, PXCLK, PXDEN (ACBIAS), and PXDATA. The timing diagram of TFT mode is shown in Figure 33.10.

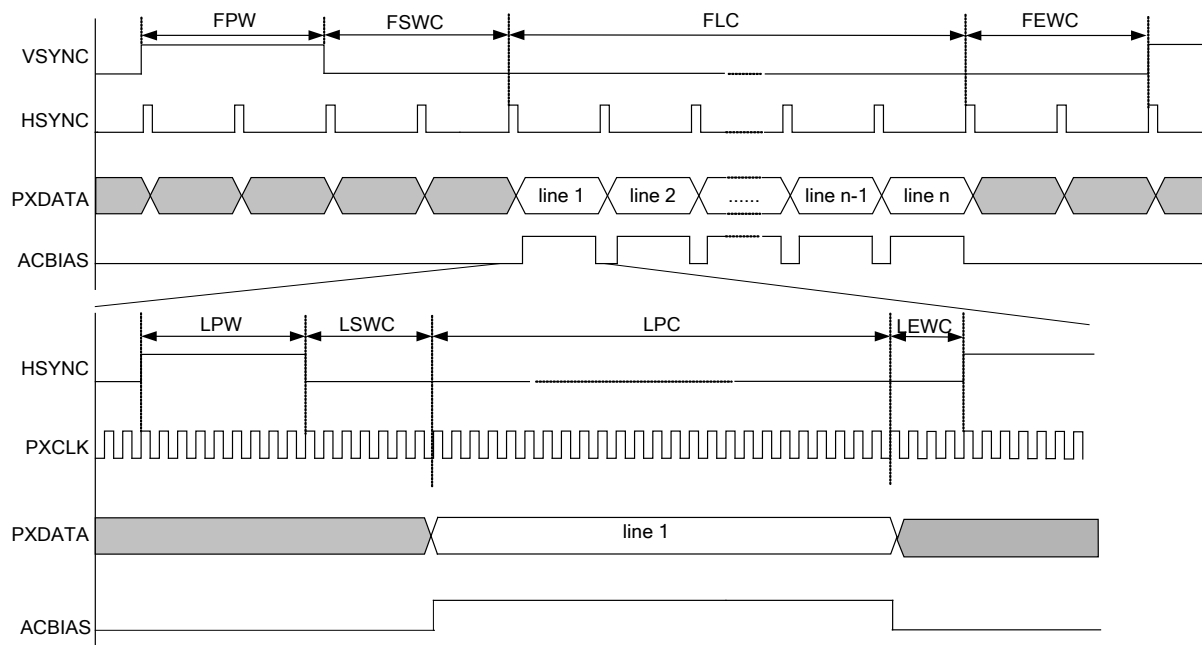


Figure 33.10 TFT Mode Timing Diagram

The VSYNC and HSYNC frequency is controlled by the LPC and FLC field.

$$LPC = (HorizontalDisplaySize) - 1$$

$$FLC = (VerticalDisplaySize) - 1$$

And PXCLK frequency is determined by the CLKDIV value.

$$f_{PXCLK} = f_{LCLK} / (2 \times CLKDIV) \quad (1)$$

The frequency of VSYNC signal is the frame rate. So the frame rate can be calculated as follows:

$$f_{HSYNC} = \frac{f_{PXCLK}}{\{(LPW + 1) + (LPC + 1) + (LSWC + 1) + (LEWC + 1)\}}$$

$$f_{VSYNC} = \frac{f_{HSYNC}}{\{(FPW + 1) + (FLC + 1) + (FSWC + 1) + (FEWC + 1)\}}$$

Therefore, if FR is the required refresh rate in TFT mode, f_{PXCLK_REQ} , which is the required PXCLK, is as following.

$$f_{PXCLK_REQ} = FR \times \{(FPW + 1) + (FLC + 1) + (FSWC + 1) + (FEWC + 1)\} \times \{(LPW + 1) + (LPC + 1) + (LSWC + 1) + (LEWC + 1)\} \quad (2)$$

Example:

For TFT LCD(RGB565), if LCLK = 80MHz and Refresh rate = 60Hz,

PXDW* = 0x4, YCbCr* = 0, BPP* = 0x4, DP* = 0, NI* = 1, TV* = 0, TFT* = 1, STN* = 0
LSWC* = LEWC* = LPW* = 3 (TFT LCD dependent)
FSWC* = FEWC* = FPW* = 1 (TFT LCD dependent)

Table 33.3 TFT LCD (RGB565) Example

Width (pixel)	Height (pixel)	LPC*	FLC*	DHSIZE*	DVSIZE*	F _{PXCLK_REQ} **	CLKDIV*	f _{PXCLK} ***
176	220	175	219	176	220	2.55	15	2.67
240	320	239	319	240	320	4.93	8	5
640	480	639	479	640	480	19.01	2	20

*) control registers to be programmed. **) Refer to expression (2). ***) Refer to expression (1).

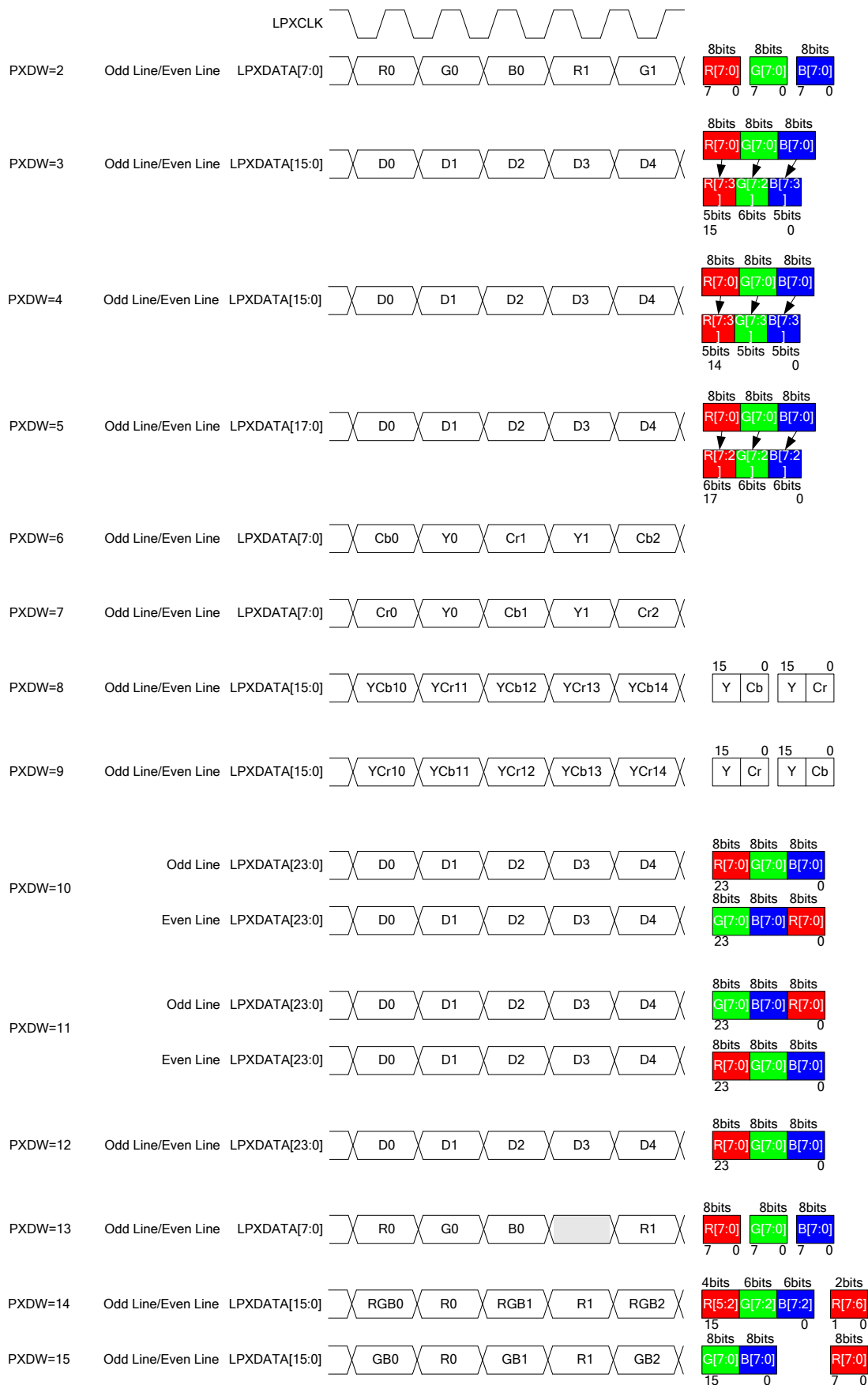


Figure 33.11 Output Pixel Data Format

33.3.7 Supported Video Input Formats

Image formats supported in Video Input (hereinafter VIN) block are as follows.

33.3.7.1 YUV422_16BIT

The following Figure 33.12, Figure 33.13, and Figure 33.14 show the timing diagram of YUV422_16BIT. The following 3 types are the concerned specification.

- BT.601 [SD interlaced]
- BT.1120, SMPTE274M (Include BT.709, SMPTE274M H signal) [HD Interlaced]
- SMPTE274M (Include SMPTE296M H signal) [HD Progressive]

The following figure is BT.601 interface. In case of 525 line and 720 active pixel, it is 27MHz clock frequency. The values in the parenthesis refer to 625 line system.

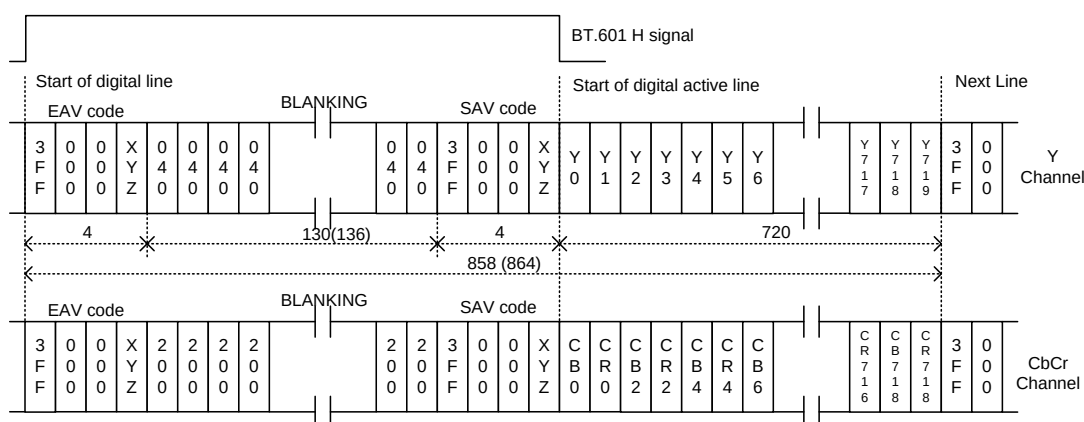


Figure 33.12 BT.601 in YUV422_16BIT (Interlaced)

The following figure is BT.1120 and SMPTE274M interface. In case of 1125 line, it is 29.97Hz, 30Hz, 59.94Hz, and 60 Hz system. In case of 1920 active pixel, it is 74.176MHz, 74.25MHz, 148.35MHz, 148.5MHz clock frequency. The values in the parenthesis refer to 25Hz and 50Hz system.

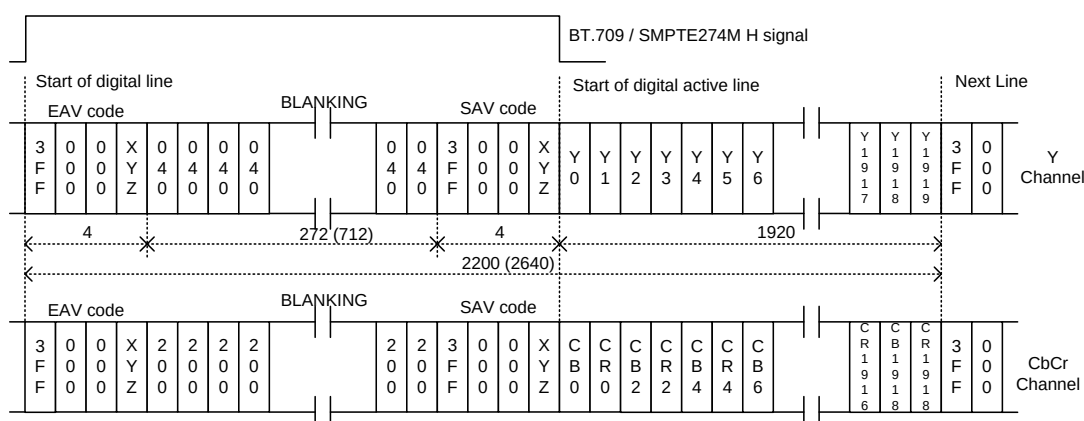


Figure 33.13 BT.1120 and SMPTE274M in YUV422_16BIT (Interlaced)

The following figure is SMPTE274M interface. In case of 750 line, it is 59.94Hz and 60Hz system. In case of 1280 active pixel, it is 74.176MHz and 74.25MHz clock frequency. The values in the parenthesis refer to 50Hz system.

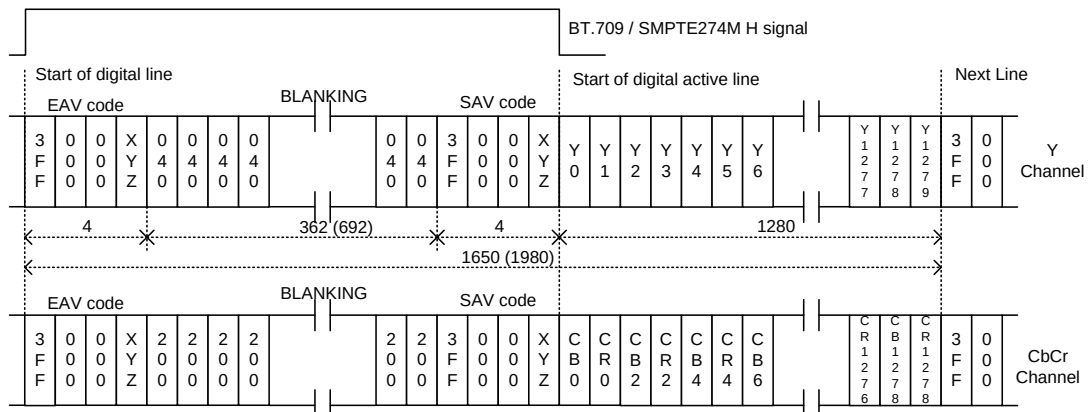


Figure 33.14 SMPTE274M in YUV422_16BIT (Progressive)

33.3.7.2 YUV422_8BIT

The following Figure 33.15 and Figure 33.16 refer to the timing diagram of YUV422_8BIT. The following 2 types are the concerned specification.

- BT.656, SMPTE125M (include BT.601 H signal) [SD interlaced]
- BT.1302, SMPTE267M (include BT.601 H signal) [SD interlaced]

The following figure is BT.656 and SMPTE125M interface. In case of 525 line and 720 active pixel, it is 27MHz clock frequency. The values in the parenthesis refer to 625 line.

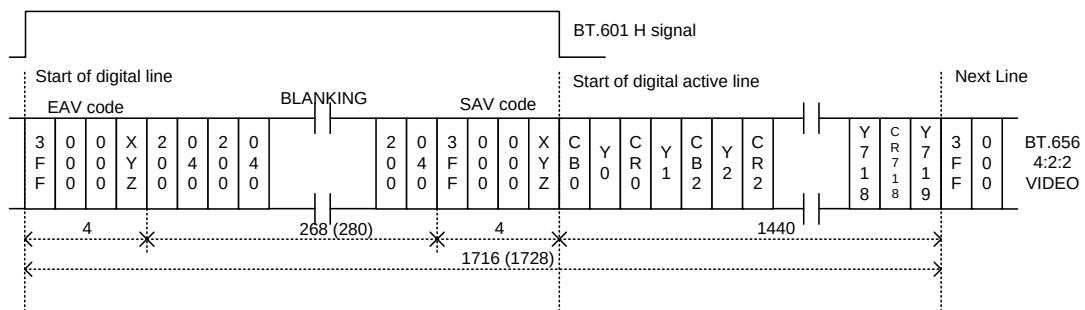


Figure 33.15 BT.656 and SMPTE125M in YUV422_8BIT (Interlaced)

The following figure is BT.1302 and SMPTE267M interface. In case of 525 line and 960 active pixel, it is 36MHz clock frequency. The values in the parenthesis refer to 625 line.

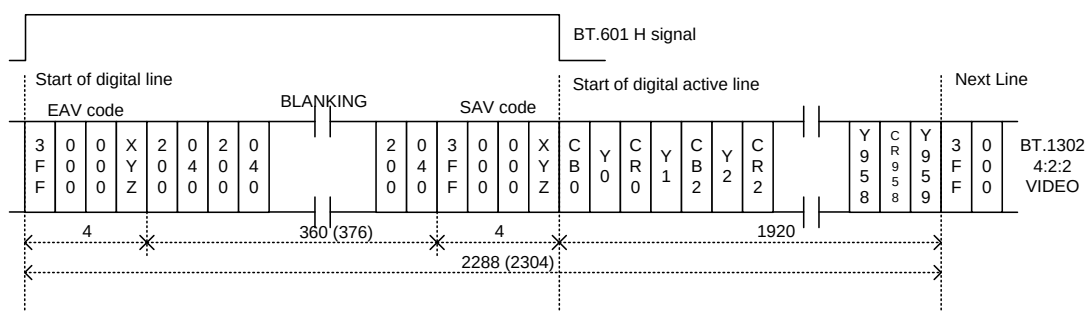


Figure 33.16 BT.1302 and SMPTE267M in YUV422_8BIT (Interlaced)

33.3.7.3 YUVK4444_16BIT

The following Figure 33.17 and Figure 33.18 refer to the timing diagram of YUVK4444_16BIT. The following 2 types are the concerned specification.

- BT.799, SMPTE RP-715 (include BT.601 H signal) [SD interlaced]
- BT.1303 (include BT.601 H signal) [SD interlaced]

The following figure is BT.799 and SMPTE RP-715 interface. In case of 525 line and 720 active pixel, it is 27MHz clock frequency. The values in the parenthesis refer to 625 line.

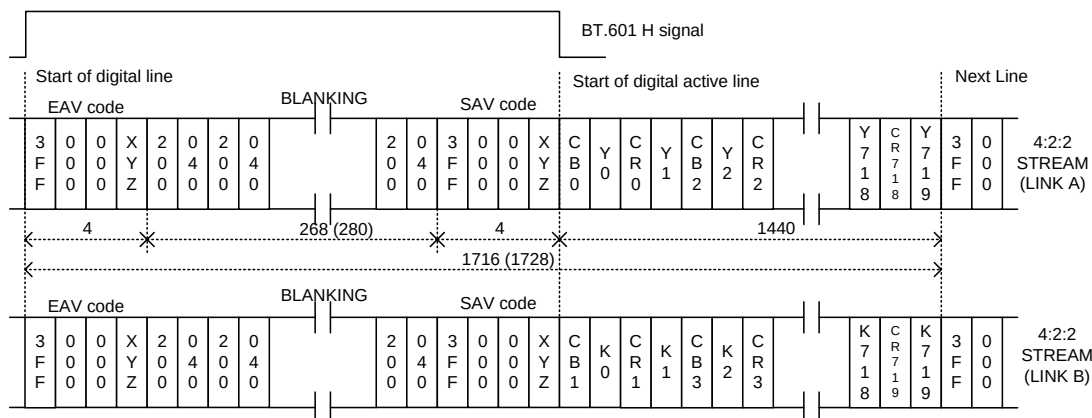


Figure 33.17 BT.799 and SMPTE RP-715 in YUVK4444_16BIT (Interlaced)

The following figure is BT.1303 interface. In case of 525 line and 960 active pixel, it is 36MHz clock frequency. The values in the parenthesis refer to 625 line.

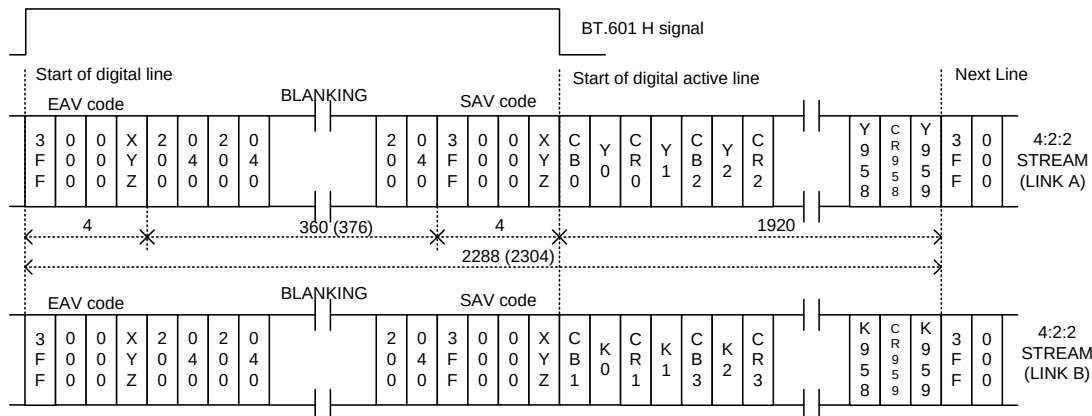


Figure 33.18 BT.1303 in YUVK4444_16BIT (interlaced)

33.3.7.4 RGBK4444_16BIT

The interface type is same between RGBK4444_16BIT format and YUVK4444_16BIT format. The only difference is whether the channel of a color is YUV base or RGB base. For the interface, refer to Chapter 33.3.7.3 YUVK4444_16BIT. For color mapping, refer to Figure 33.19.

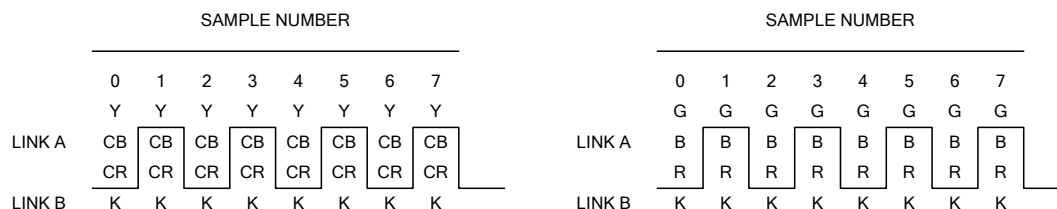


Figure 33.19 Link Content Representation for YCbCrK/RGBK Video Signal

33.3.7.5 Time-Multiplex Mode (TW2867 device for Techwell Inc.)

Time-Multiplex Mode is a format supported by the TW2867 device of Techwell Inc. In this mode, 2 or 4 frame image data based on CCIR656 format is interfaced via one data bus. Normally, this mode is frequently used in a surveillance camera but not a standard format.

a) Two Channel CCIR656 time-multiplexed format with 27 / 54MHz

In two channel mode, 27MHz and 54MHz are supported. In 27MHz mode, data is sent to both positive edges and negative edges of this clock as Figure 33.20. In other words, A-frame is synchronized on a positive edge and B-frame is synchronized on a negative edge.

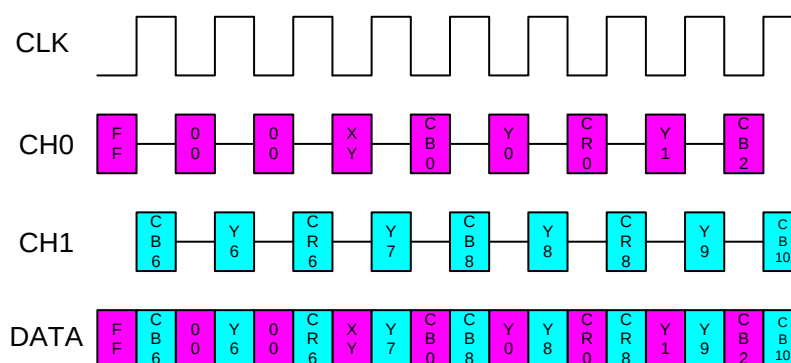


Figure 33.20 Timing diagram of 2 channel Time multiplex in 27MHz

On the other hand, in 54MHz mode, data is sent only at a positive edge of the clock as Figure 33.21. Therefore, 54MHz clock, which is two times faster than 27MHz master clock of CCIR656 format, is used.

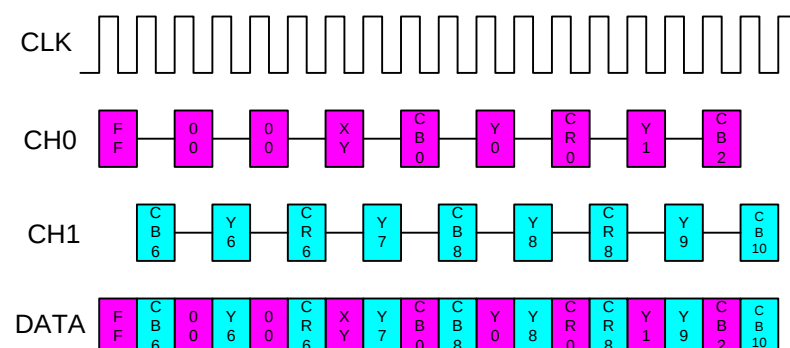


Figure 33.21 Timing diagram of 2 channel Time multiplex in 54MHz

b) Four Channel CCIR656 time-multiplexed format with 54 / 108MHz

Note: 4-channel DEMUX is not supported in the TCC8132.

c) The channel ID code for special format

In time multiplex format, since multiple image data is sent via one data-bus, each frame is assorted by detecting the channel ID. Channel ID is included in embedded sync code for transmission. Table 33.4 shows the channel IDs.

Table 33.4 Time Multiplex format of channel ID code

Condition			656 FVH			SAV – EAV Code						
Field	Vtime	Htime	F	V	H	1st	2nd	3rd	4 th			
									CH1	CH2	CH3	CH4
EVEN	BLANK	EAV	1	1	1	0xFF	0x00	0x00	0xF0	0xF1	0xF2	0xF3
EVEN	BLANK	SAV	1	1	0	0xFF	0x00	0x00	0xE0	0xE1	0xE2	0xE3
EVEN	ACTIVE	EAC	1	0	1	0xFF	0x00	0x00	0xD0	0xD1	0xD2	0xD3
EVEN	ACTIVE	SAV	1	0	0	0xFF	0x00	0x00	0xC0	0xC1	0xC2	0xC3
ODD	BLANK	EAV	0	1	1	0xFF	0x00	0x00	0xB0	0xB1	0xB2	0xB3
ODD	BLANK	SAV	0	1	0	0xFF	0x00	0x00	0xA0	0xA1	0xA2	0xA3
ODD	ACTIVE	EAV	0	0	1	0xFF	0x00	0x00	0x90	0x91	0x92	0x93
ODD	ACTIVE	SAV	0	0	0	0xFF	0x00	0x00	0x80	0x81	0x82	0x83

33.3.8 Component & Interface

The VIOC is composed of a couple of components. Processing that the user wants can be done by the way these components are composed. For more details, refer to Chapter 33.5 Hardware Overview.

33.3.8.1 RDMA

RDMA reads image data saved in the memory and there are GRDMA and VRDMA. The GRDMA supports image formats except for separate format and interleaved format (in other words, only one channel is supported.) and the VRDMA supports all image formats (in other words, three channels are supported.). For more details on format, refer to Chapter 33.3.5 Supported Image Formats.

33.3.8.2 VIDEOIN

VIDEOIN receives image data from an external image input device (i.e. Camera sensor, TV decoder, and etc). For the supported format, refer to Chapter 33.3.7 Supported Video Input Formats.

33.3.8.3 WDMA

WDMA re-saves the processed image data to the memory and there are GWDMA and VWDMA. The GWDMA supports image formats except for separate format and interleaved format (in other words, only one channel is supported.) and the VWDMA supports all image format (in other words, 3 channels are supported.). For more details on format, refer to Chapter 33.3.5 Supported Image Formats.

33.3.8.4 DISPLAY Device

DISPLAY Device makes an interface to display an image on an external image output device (i.e. TFT-LCD, STN-LCD, TV-Encoder, and etc). For more details, refer to Chapter 33.3.6 Supported Display Output Formats

33.3.8.5 WMIX

WMIX handles Overlay, Alpha Blending and Chroma-key processing for inputted images (Functions are described in Chapter 33.4 Basic Image Processing Functions). The WMIX processes four or two images. For output, always two image data are outputted.

33.3.8.6 VIQE

VIQE handles de-interlace (3D) of an image.

The de-interlace converts an inputted image from interlace to progressive. There are 2D and 3D styles. In VIQE, 3D style is used and one progressive image is made by using the previous 4-fields image. The following DEINTLS is 2D style and the previous line and the current line of the current field are interpolated to make a progressive image.

33.3.8.7 SCALER

SCALER enlarges or reduces the size of an image. As Figure 33.22, it can enlarge and reduce the entire area or a certain area. Up to 4 times of enlargement (actual image size) and reduction to 1/64 are available.

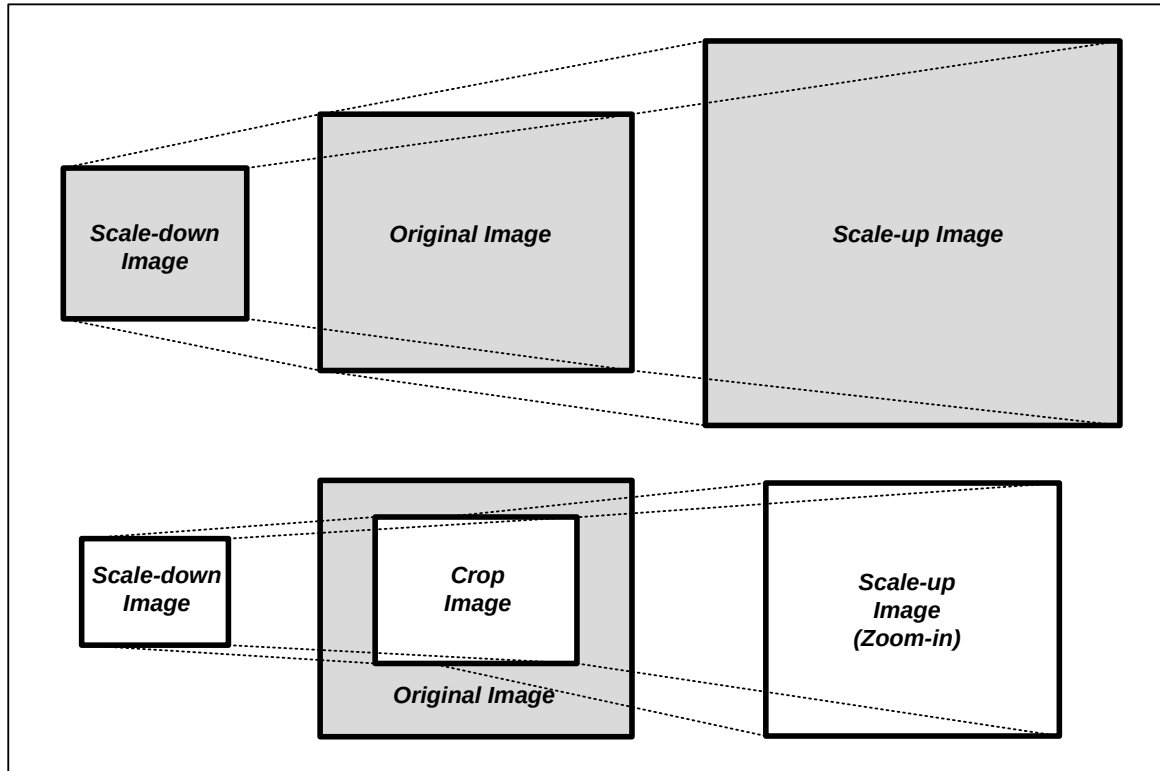


Figure 33.22 Scale Up/Down for SCALER component

33.3.8.8 DEINTL_S

DEINTL_S is a simple de-interlacer and it does interpolation and de-interlacing for data of each line. Unlike VIQE, processing is done only in the spatial domain. For more details on de-interlace, refer to Chapter 33.3.8.6 VIQE.

33.3.8.9 LUT (Look-up Table)

LUT is mainly used to adjust a color of an image. For example, if a pixel value of an inputted image is adjusted by using a certain numerical formula, a user can save the results for each pixel value after using the numerical formula and the inputted pixel value can be mapped to a desired pixel value.

33.4 Basic Image Processing Functions

33.4.1 Format Converting

Image formats processed inside the VIOC are all 4:4:4 format (except for VIQE). For this reason, 4:2:2 / 4:2:0 read from DMA or VIDEOIN need to be converted to 4:4:4 format. (Conversion 4:4:4 to 4:2:2 is available.) As the figure below, when converting 4:2:2 / 4:2:0 to 4:4:4, the empty pixels of the chroma channel interpolate with the previous pixel and the next pixel. For a line, the previous line and the next line are interpolated when converting to 4:4:4 format. When converting 4:4:4 format to 4:2:2 / 4:2:0, only even pixels and even lines are removed.

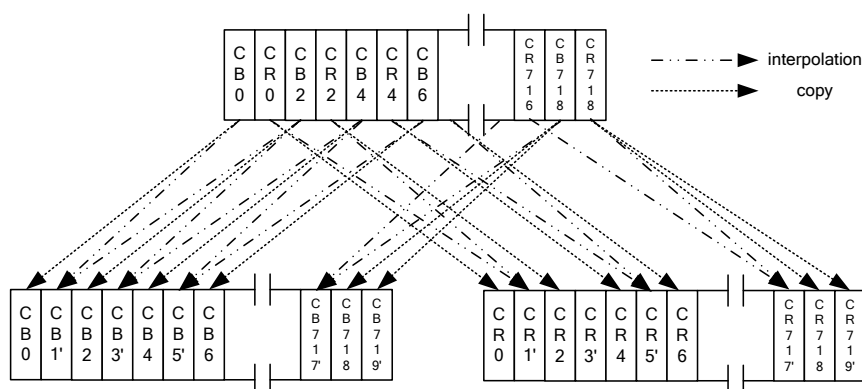


Figure 33.23 Format Converting Method of Chroma channel

33.4.2 YUV-to-RGB Converting

The internal operation in the VIOC is designed for RGB color space. So, if the image source has the YCbCr color space, the source image should be converted to RGB color space. If YCbCr color space is needed by display device, the internal graphic image should be converted to YCbCr color space.

The nominal ranges of “YCbCr” and “RGB” can be one of “0 ~ 255” or “16 ~ 235”. It is called “**Computer System Color**” and “**Studio Color**” for the 0 ~ 255 nominal range and 16 ~ 235 nominal range correspondingly.

The equations for color conversion can be dependent on the color types of source image and destination image.

33.4.3 RGB-to-YUV Converting

The color space of VIOC is RGB but the color space of VIQE is YCbCr. This is because internal calculation and algorithm are optimized for YUV420. Therefore, in case that VIQE is used, the RGB color space needs to be converted to YCbCr color space.

The color range and color space type are the same with YUV2RGB conversion of Chapter 33.4.2 YUV-to-RGB Converting.

33.4.4 Contrast, Brightness, and Hue Functions

Output_value = MAX(MIN((input_value – 128) x contrast + 128 + brightness, 255), 0)

33.4.5 LUT (Look-Up-Table)

The LCDc has the “24 bits x 256 entries” look-up table and it can be connected to one of DISP0, DISP1, and DISP2 or Plug-in type in VIOC. Inputs of this table are RGB888 or YCbCr444 format. Each input pixel data addresses the table entry and the new pixel value is generated from the corresponding entry. Its entry consists of three 8-bit color values. The overall structure of Look-Up Table is shown in Figure 33.24. This table is controlled by LUT_CTRL LUT_DEVk_CFG register. [0x80029000/9004]

The table entries should be initialized before the VIOC or DISP uses it. Their address space is from 0x80029000.

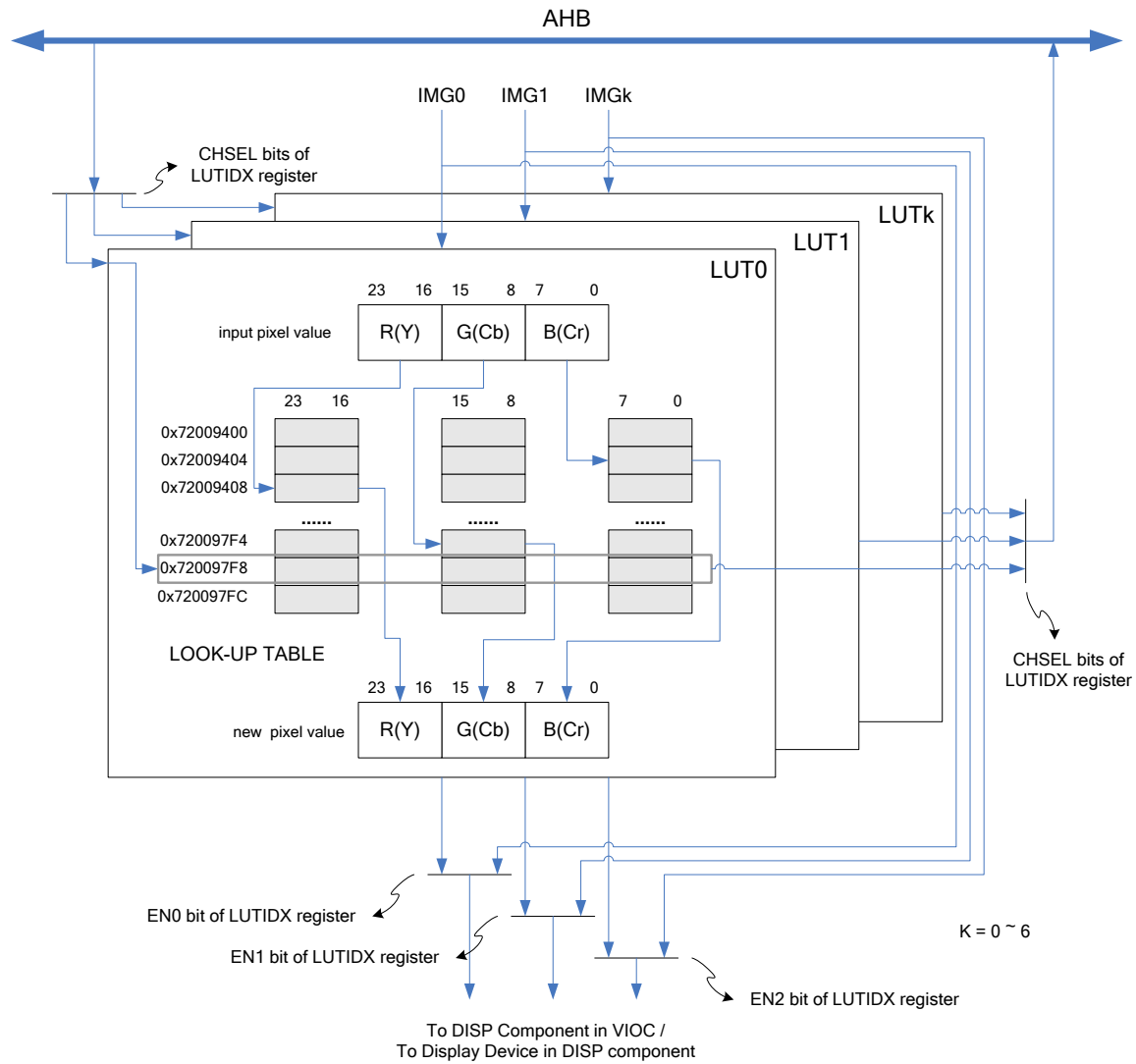


Figure 33.24 Lookup Table Structure

33.4.6 Overlay Mixer

To process the overlapped region between layers, the VIOC supports the alpha-blending and the chroma-keying. When the alpha-blending and the chroma-keying are disabled, the top layer image is only shown in the overlapped region. Thus, LAYER1 is only shown in the overlapped region between the LAYER0 and the LAYER1.

33.4.7 Overlay Mixer (Alpha-blending)

For the window mixer, 2-to-1 mixer is basic. It has input data for 2 channels and these data are mixed and outputted. In 4-channel window mixer, cascade type 2-channel mixer is configured.

Alpha-blending function is decided by MACONk and MCONk register among window mixer related registers. Registers starting with CCON are related with color alphablending function. Registers starting with ACON define the alphablending function for an alpha value.

Basic operations are decided by ACON0, ACON1, CCON0 and CCON1 registers. The ACON0 and ACON1 represent the alphablending function for an alpha value. The CCON0 and CCON1 are used for an alphablending function for color values.

“_00”, “_01”, “_10”, “_11” in the concerned registers are defined as below.

- * “_00” – Region A : The area where Region B, Region C, and Region D are excluded from the entire region.
- * “_10” – Region B : The area where Region C is excluded from Top Layer
- * “_01” – Region D : The area where Region C is excluded from Bottom Layer
- * “_11” – Region C : The common region between Top and Bottom Layer

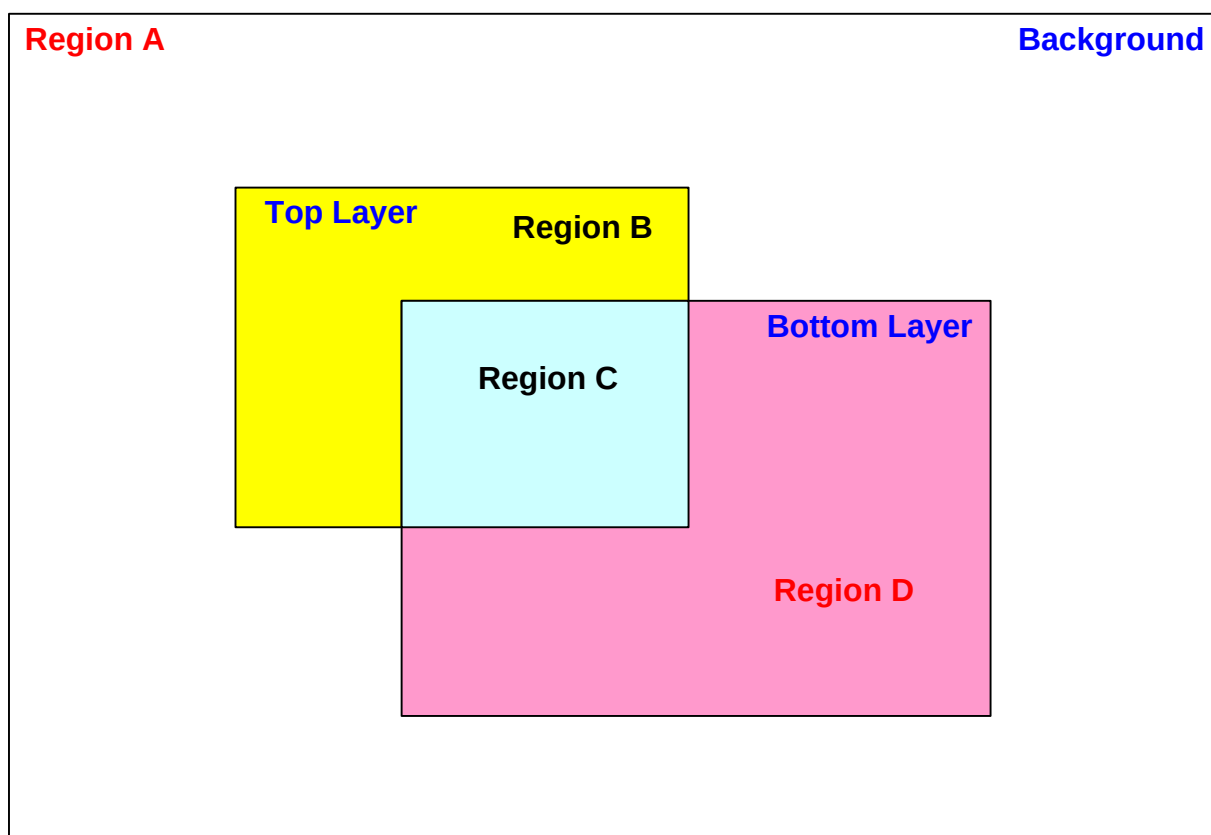


Figure 33.25 Region Definition for Alphablending

The following figure shows how to manipulate the overlaid image in the next overlay mixer. As shown, the superset of the region B, C, D will be valid area for lower layer.

In the figure above, the alpha and color value of the region A have the background value determined by the register. But, in the following figure, the region A has the alpha and color value calculated at the previous overlay mixer.

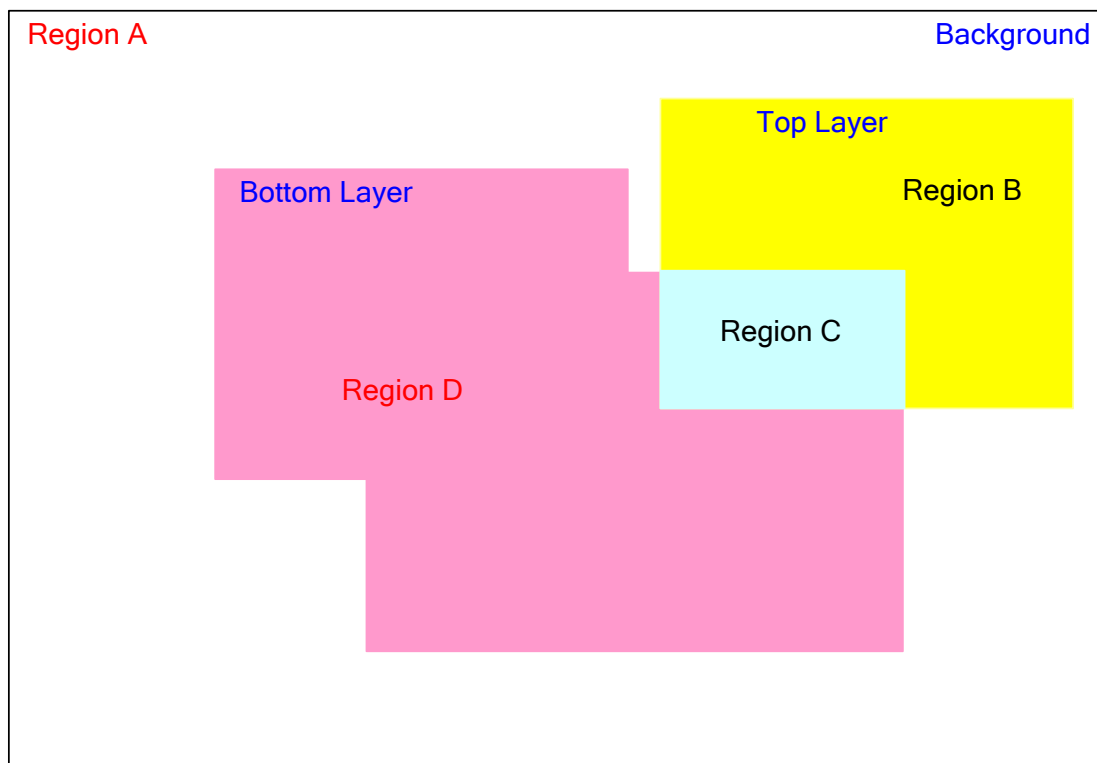


Figure 33.26 Region Definition for Alphablending (Another Layer)

The ACON0, ACON1, CCON0 and CCON1 register are set for each region. The table below shows register values. The ACON0, ACON1, CCON0 and CCON1 are decided by the definitions above. For example, the ACON0 is ACON0_10 for Region B. The ACON1 is ACON1_10, the CCON0 is CCON0_10, and the CCON1 is CCON1_10.

Table 33.5 ACON0, ACON1, CCON0, CCON1 Register Selection

Region	Control Data	Candidate Data
Region A	ACON0	ACON0_00
	ACON1	ACON1_00
	CCON0	CCON0_00
	CCON1	CCON1_00
Region B	ACON0	ACON0_10
	ACON1	ACON1_10
	CCON0	CCON0_10
	CCON1	CCON1_10
Region C	ACON0	ACON0_11
	ACON1	ACON1_11
	CCON0	CCON0_11
	CCON1	CCON1_11
Region D	ACON0	ACON0_01
	ACON1	ACON1_01
	CCON0	CCON0_01
	CCON1	CCON1_01

By using the regions above and ACON0, ACON1, CCON0, CCON1 data, the following numerical formulas are decided.

The AlphaA and AlphaB in the following table are determined by the ROPMODE register and can be either one of the pixel alpha value of input image or ALPHA0 and ALPHA1 determined by the register.

Table 33.6 AlphaA and AlphaB corresponding to each region

Region	Alpha	ROPMODE[4:3]	Candidate Data
Region A	AlphaA	2	ALPHA0 ¹
		3	Image Alpha for A channel ²
	AlphaB	2	ALPHA1 ¹
		3	Image Alpha for B channel ²
Region B	AlphaA	2	ALPHA0 ¹
		3	Image Alpha for A channel
	AlphaB	2	ALPHA1 ¹
		3	Image Alpha for B channel ²
Region C	AlphaA	2	ALPHA0 ¹
		3	Image Alpha for A channel
	AlphaB	2	ALPHA1 ¹
		3	Image Alpha for B channel
Region D	AlphaA	2	ALPHA0 ¹
		3	Image Alpha for A channel ²
	AlphaB	2	ALPHA1 ¹
		3	Image Alpha for B channel

Table 33.7 PixelDataA and PixelDataB corresponding to each region

Region	PixelData	ROPMODE[4:3]	Candidate Data
Region A	PixelDataA	2	BG ³
		3	Image Alpha for A channel ⁴
	PixelDataB	2	BG ³
		3	Image Alpha for B channel ⁴
Region B	PixelDataA	2	BG ³
		3	Image Alpha for A channel
	PixelDataB	2	BG ³
		3	Image Alpha for B channel ⁴
Region C	PixelDataA	2	BG ³
		3	Image Alpha for A channel
	PixelDataB	2	BG ³
		3	Image Alpha for B channel
Region D	PixelDataA	2	BG ³
		3	Image Alpha for A channel ⁴
	PixelDataB	2	BG ³
		3	Image Alpha for B channel

Table 33.8 Alphablending Function for Color Pixel Data

Region	Control Data	Candidate Data
CCON0[1:0]	0	SEL0_Out = ALPHA0
	1	SEL0_Out = ALPHA1
	2	SEL0_Out = 256 - ALPHA0
	3	SEL0_Out = 256 - ALPHA1
CCON0[3:2]	0	Result_A = PixelDataA * SEL0_Out
	1	Result_A = PixelDataA * 256
	2	Result_A = PixelDataA * 128
	3	Result_A = 0
CCON1[1:0]	0	SEL1_Out = ALPHA0
	1	SEL1_Out = ALPHA1
	2	SEL1_Out = 256 - ALPHA0
	3	SEL1_Out = 256 - ALPHA1
CCON1[3:2]	0	Result_B = PixelDataB * SEL1_Out
	1	Result_B = PixelDataB * 256
	2	Result_B = PixelDataB * 128
	3	Result_B = 0
Result	PixelDataOut = (Result_A + Result_B + 128) / 256	

¹ ALPHA0 or ALPHA1 value in the MROPCK register² It would be applied to background alpha value defined in MBG register for the lowest layer. For another layer, the alpha value calculated in the previous layer would be applied.³ Background color defined at the MBG register.⁴ It would be applied to background color data defined in MBG for the lowest layer. For another layer, the color data calculated in the previous layer would be applied.

Table 33.9 Alphablending Function for Alpha Pixel Data

Region	Control Data	Candidate Data
ACON0[0]	0	SEL0_Out = ALPHA1
	1	SEL0_Out = 256 – ALPHA1
ACON0[2:1]	0	Result_A = ALPHA0 * SEL0_Out
	1	Result_A = ALPHA0 * 256
	2	Result_A = ALPHA0 * 128
	3	Result_A = 0
ACON1[0]	0	SEL1_Out = ALPHA0
	1	SEL1_Out = 256 – ALPHA0
ACON1[2:1]	0	Result_B = ALPHA1 * SEL1_Out
	1	Result_B = ALPHA1 * 256
	2	Result_B = ALPHA1 * 128
	3	Result_B = 0
Result	AlphaDataOut = (Result_A + Result_B + 128) / 256	

33.4.8 Layer (Overlay Priority)

For processing the raw image sources, the VIOC defines a virtual display as shown in Figure 33.27. It consists of “Display Width” and “Display Height”, determined by MSIZE register. Actually, it becomes the active display size of the connected LCD or NTSC/PAL encoder.

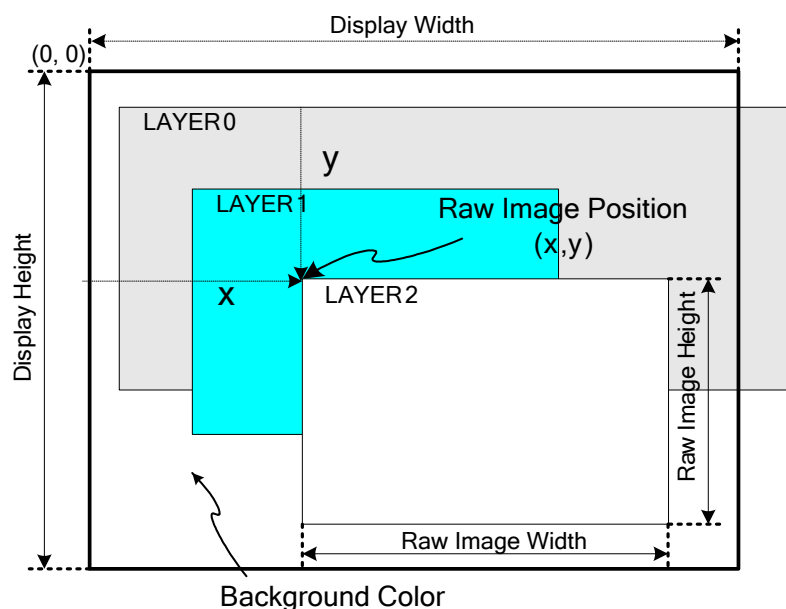


Figure 33.27 Virtual Display in the VIOC

The WMIX can have up to 4 or 2 raw image sources. Each image channel can be enabled by the connected RDMA. The overlay priority between layers can be determined by the OVP field in the MCTRL register. The following table shows the overlay priority information.

Table 33.10 Relationship between Layers and OVP field

OVP	LAYER3	LAYER2	LAYER1	LAYER0
0	Image 3	Image 0	Image 1	Image 2
1	Image 3	Image 1	Image 0	Image 2
2	Image 3	Image 0	Image 2	Image 1
3	Image 3	Image 2	Image 0	Image 1
4	Image 3	Image 1	Image 2	Image 0
5	Image 3	Image 2	Image 1	Image 0
8	Image 2	Image 0	Image 1	Image 3
9	Image 2	Image 1	Image 0	Image 3
10	Image 2	Image 0	Image 3	Image 1
11	Image 2	Image 3	Image 0	Image 1
12	Image 2	Image 1	Image 3	Image 0
13	Image 2	Image 3	Image 1	Image 0
16	Image 1	Image 0	Image 2	Image 3
17	Image 1	Image 2	Image 0	Image 3
18	Image 1	Image 0	Image 3	Image 2
19	Image 1	Image 3	Image 0	Image 2
20	Image 1	Image 2	Image 3	Image 0
21	Image 1	Image 3	Image 2	Image 0
24	Image 0	Image 1	Image 2	Image 3
25	Image 0	Image 2	Image 1	Image 3
26	Image 0	Image 1	Image 3	Image 2
27	Image 0	Image 3	Image 1	Image 2
28	Image 0	Image 2	Image 3	Image 1
29	Image 0	Image 3	Image 2	Image 1

Note: OVP = 5 (Default)

33.4.9 Chroma-Key Function

Usually, a raw image source is the rectangular shape. Chroma-keying is used for clipping the arbitrary shape from a raw image source. In other words, chroma-keying disables the overlay process for selected pixels. They are identified by control registers, which are MKEYk0 (k=0~3). The k=1 is for the LAYER1 and the k=2 is for the LAYER2.

The overlay disabled pixel becomes the transparent pixel; the pixel of background layer is shown in the display.

For LAYER1, it depends on whether a pixel is overlaid or not.

$$\begin{aligned} \text{OverlayDisabledPixel} = & ((\text{Pixel}(\text{RorY}) \& \text{MKEY11.MKR1}) == (\text{MKEY10.KR1} \& \text{MKEY11.MKR1})) \& \\ & ((\text{Pixel}(\text{GorU}) \& \text{MKEY11.MKG1}) == (\text{MKEY10.KG1} \& \text{MKEY11.MKG1})) \& \\ & ((\text{Pixel}(\text{BorV}) \& \text{MKEY11.MKB1}) == (\text{MKEY10.KB1} \& \text{MKEY11.MKB1})) \end{aligned}$$

$$\begin{aligned} \text{OverlayEnabledPixel} = & ((\text{Pixel}(\text{RorY}) \& \text{MKEY11.MKR1}) != (\text{MKEY10.KR1} \& \text{MKEY11.MKR1})) | \\ & ((\text{Pixel}(\text{GorU}) \& \text{MKEY11.MKG1}) != (\text{MKEY10.KG1} \& \text{MKEY11.MKG1})) | \\ & ((\text{Pixel}(\text{BorV}) \& \text{MKEY11.MKB1}) != (\text{MKEY10.KB1} \& \text{MKEY11.MKB1})) \end{aligned}$$

For LAYER2, it depends on whether a pixel is overlaid or not

$$\begin{aligned} \text{OverlayDisabledPixel} = & ((\text{Pixel}(\text{RorY}) \& \text{MKEY21.MKR1}) == (\text{MKEY20.KR1} \& \text{MKEY21.MKR1})) \& \\ & ((\text{Pixel}(\text{GorU}) \& \text{MKEY21.MKG1}) == (\text{MKEY20.KG1} \& \text{MKEY21.MKG1})) \& \\ & ((\text{Pixel}(\text{BorV}) \& \text{MKEY21.MKB1}) == (\text{MKEY20.KB1} \& \text{MKEY21.MKB1})) \end{aligned}$$

$$\begin{aligned} \text{OverlayEnabledPixel} = & ((\text{Pixel}(\text{RorY}) \& \text{MKEY21.MKR1}) != (\text{MKEY20.KR1} \& \text{MKEY21.MKR1})) | \\ & ((\text{Pixel}(\text{GorU}) \& \text{MKEY21.MKG1}) != (\text{MKEY20.KG1} \& \text{MKEY21.MKG1})) | \\ & ((\text{Pixel}(\text{BorV}) \& \text{MKEY21.MKB1}) != (\text{MKEY20.KB1} \& \text{MKEY21.MKB1})) \end{aligned}$$

33.4.10 Image Windowing

Refer to Chapter 33.3.3 Image Windowing.

33.4.11 Image Cropping

Refer to Chapter 33.3.2 Image Cropping.

33.4.12 Dithering

In case that the DISP Components output format is RGB555, RGB565 or RGB666, the RGB Ditherer can be used. RGB Ditherer is based on "Pattern Dithering Algorithm" and pattern matrix can be configurable with register setting.

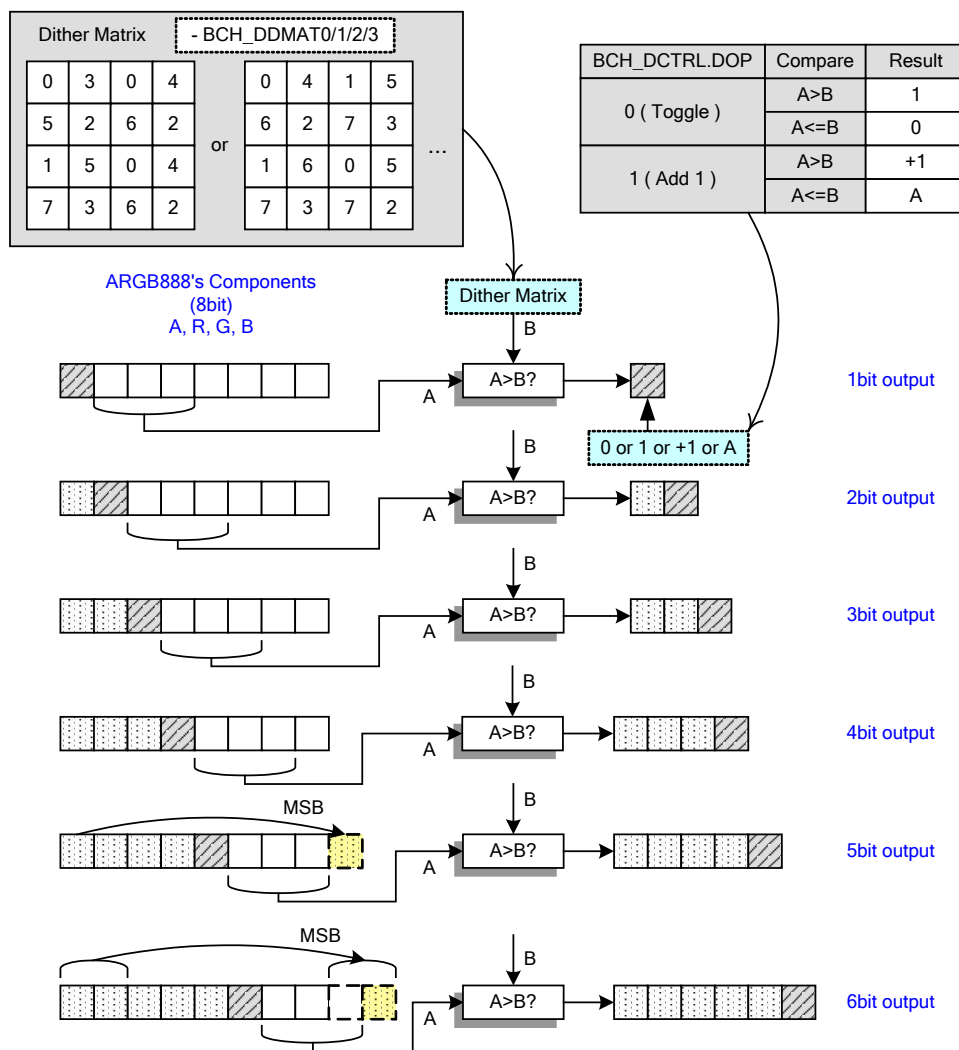


Figure 33.28 LCD RGB Dithering Operation

33.5 Hardware Overview

33.5.1 Detailed Hardware Overview

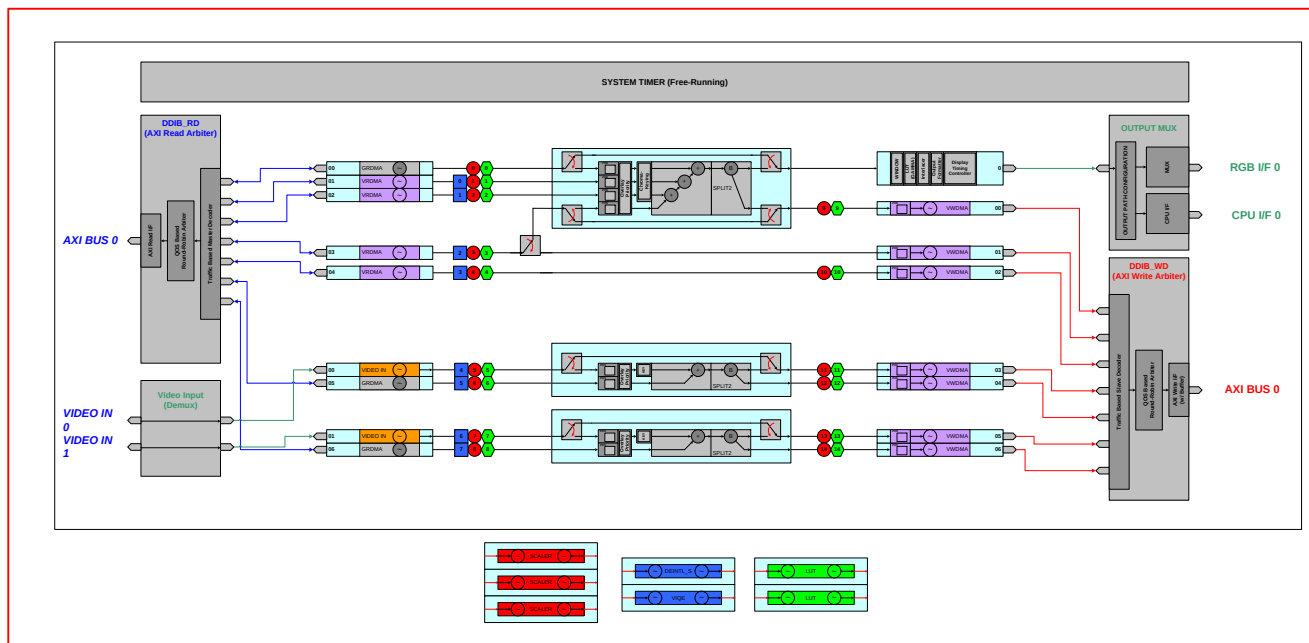


Figure 33.29 VIOC components Structure

Depending on plug-in / plug-out of each component in VIOC, the related component is in use or not in use. Figure 33.1 shows each component within VIOC. The connection interface format of each component is ARGB (or AYUV) 4:4:4:4 format. Image processing of each component is based on 4:4:4:4 format. (However, processing of VIQE is based on YUV4:2:0 format.)

The components relating to VIOC input are RDMA and VIDEOIN. The components relating to output are WDMA, DISPLAY Device. The rest components are for image processing and are used for Image Mixing, Scaling and Enhancing. The connection of these is as Figure 33.29. Plug-in / plug-out are available for each component and a user can easily plug in / out any component.

The number of each component is as Table 33.11.

Table 33.11 Hardware Features

Components	Descriptions
RDMA	● GRDMA : 3 channels ● VRDMA : 4 channels
WDMA	● GWDMA : 0 ● VRDMA : 7 channels
DISPLAY DEVICE	● 1 channel
VIDEOIN	● 2 channels (buffer size : 256 x 64)
WMIX	● WMIX_4-to-2 : 1 channel ● WMIX_2-to-2 : 2 channels
SCALER	● 3 channels (buffer size : 1024)
VIQE	● 1 channel (Note that it cannot be used with DEINTL_S components.)
DEINTL_S	● 1 channel (Note that it cannot be used with VIQE components.)
LUT	● 3 channels (Component of VIQC) ● 1 channel (LUT inside Display Device component)

33.5.2 Display Device Components

Display Device Components makes an interface for the related device in order for display on a final display device such as TFT-LCD, STN-LCD, TV-Encoder, and etc. The VOIC has one Display Device Component, which means triple display is available. Figure 33.30 shows the block of Display Device component. (Upper right of Figure 33.29 is for enlarging a component and 'X' refers to DMA.)

As the following figure, Display Device Components are composed of sub-blocks of WINDOW, LUT, INTERLACER, FORMATTER, DISPLAY_TIMING_CTRL. WINDOW block is for windowing process for an inputted image. LUT is for adjusting GAMMA. When LUT adjusts GAMMA of an image, GAMMA value of each pixel is inputted to LUT in advance and once image data is inputted, the concerned GAMMA value is automatically inputted. INTERLACER block converts an image inputted from progressive to interlace. Formatter outputs an image in the same format with that of an interface supported by an external Display Device. DISPLAY_TIMING_CTRL block makes each type of sync signals.

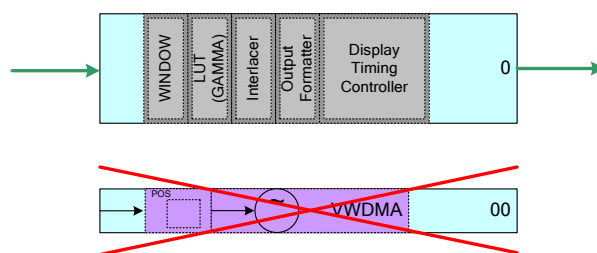


Figure 33.30 Display Device Components

33.5.3 GRDMA & VRDMA Components

GRDMA and VRDMA components read image data from the memory. Figure 33.31 shows the extracted and enlarged RDMA of Figure 33.29. As can be seen below, there are GRDMA and VRDMA. The separate and interleave mode are not supported because GRDMA has only 1 channel. The VRDMA has 3 channels, so it supports all formats including separate and interleave mode.

Processing in VIOC is based on 4:4:4 format. In case of an output in RDMA, the format is converted to 4:4:4 format. For converting, in case of 4:2:2 - to - 4:4:4 a new pixel is made by interpolating chroma channel. In case of 4:2:0 - to - 4:4:4, a new pixel and line are made by interpolating both line and pixel.

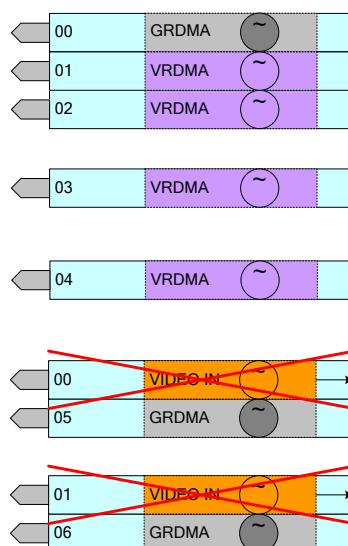


Figure 33.31 GRDMA & VRDMA Components

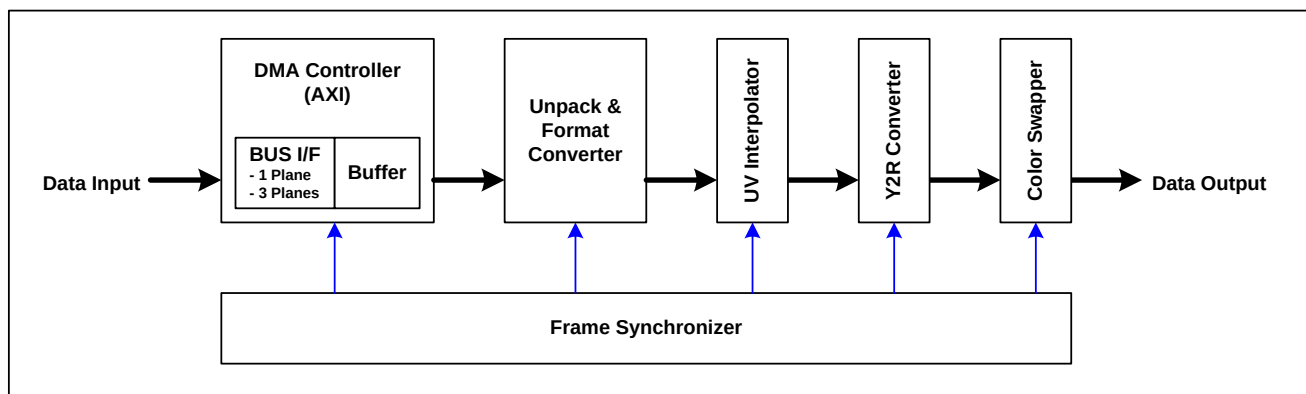


Figure 33.32 Datapath of GRDMA & VRDMA Components

Figure 33.32 shows internal datapath of RDMA. Data read from the data by DMA controller is transmitted to Unpack & Format Converter. Next, in case of YUV format, it is transmitted to UV interpolator and converted to ARGB space. Lastly, color can be changed with Color Swapper. If DMA controller supports only 1 plane, it means GRDMA. If DMA controller supports up to 3 planes, it means VRDMA.

Because ARGB format is used inside of VIOC, YUV Data should be converted to ARGB color via Y2R converter. However, If a block requiring YUV Color processing is connected after RDMA color conversion, conversion to ARGB color should not be done.

33.5.4 GWDMA & VWDMA Components

GWDMA and VWDMA components save processed image data to the memory. Figure 33.33 shows the extracted and enlarged WDMA of Figure 33.29. (The part marked as 'X' is Display Device Component.) Originally, there are GWDMA and VWDMA. As briefly explained before, since GWDMA has only 1 channel, separate and interleave mode are not supported. The VRDMA has 3 channels, so all formats including separate and interleave mode are supported. When data is saved by using WDMA, if conversion to 4:2:2 format is done, even chroma pixels are dropped. If conversion to 4:2:0 format is done, even chroma pixels and lines are dropped and the rest of data are saved.

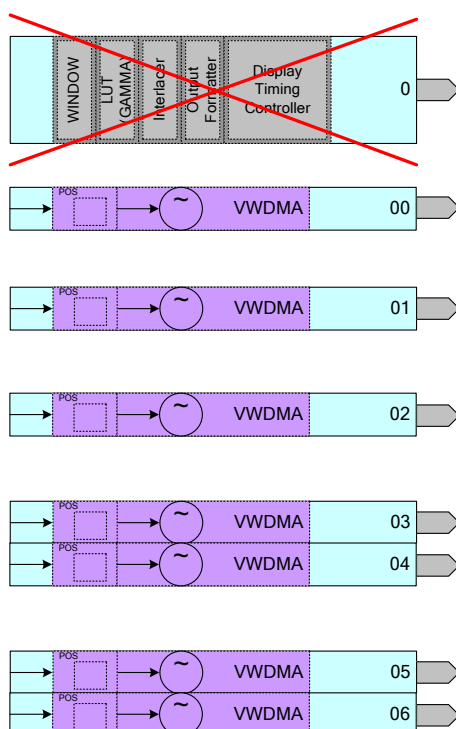


Figure 33.33 VWDMA Components

33.5.5 Scaler Components

Scaler Components reduce / enlarge an image size. Scaling method is briefly explained in Chapter 33.3.8.7 SCALER. Scaler Components are located between RDMA and WMIX and between WMIX and WDMA in Figure 33.29. (The left side of Figure 33.34 shows components located between RDMA and WMIX and the middle one shows components between WMIX and WDMA.) As you can see, SCALER, VIQE/DEINTLS, and LUT are located in order as in Figure 33.34.

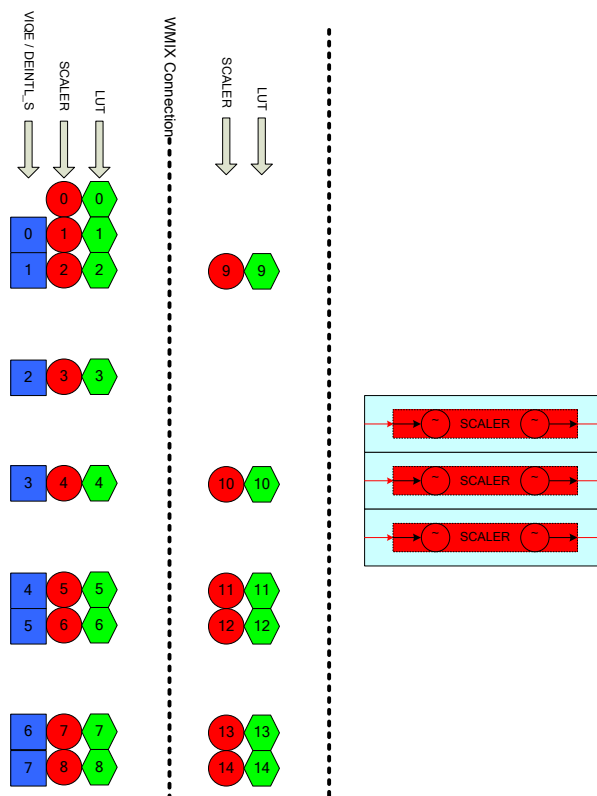


Figure 33.34 SCALER Components

The operation of scaler is defined with source size, destination size, output position and output size. The source size is the size of input image which cannot be defined in the scaler related registers. The destination size, output position and output size can be configured in the scaler related registers as well as these can determine scale-ratio and output image.

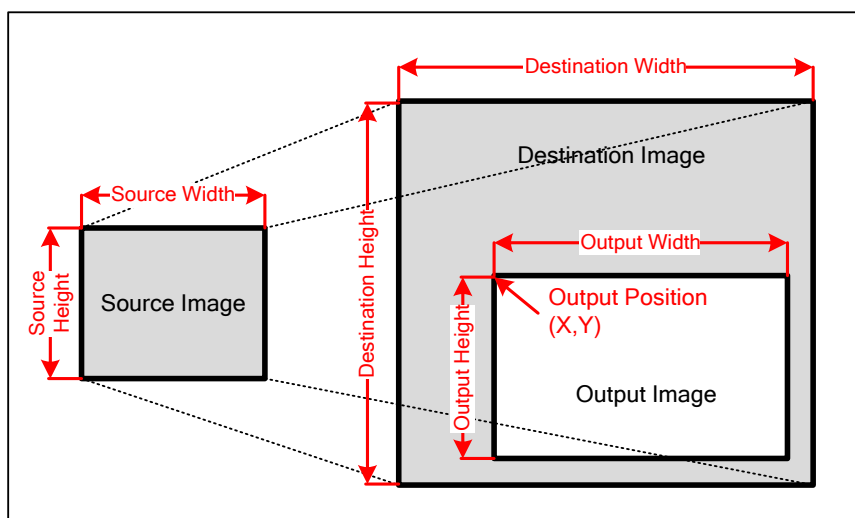


Figure 33.35 Relation between Source Size, Destination Size, Output Position and Output Size

As shown in the figure above, the source image is already defined as the size of input image from the source component.

The destination size, the output position, and the output size can be defined in the scaler related registers.

Basically, the scaler changes the source image resolution to the destination image resolution, which can be up-scale or down-scale. In the output stage, the output image can be transferred to the next component defined by output position and output size. So, the scaler takes time to scale up or down to the destination image size. If the output size is much smaller than the destination size, the hidden region can make the scaler take more time, which is unnecessary.

Specially, in case that the scaler output is connected to the display device, the followings should be concerned. The following case can make the display controller under-run.

1. The down-scale ratio is too much larger to display at time.
The source image size should be reduced by reducing the vertical size of source image and stretching the line offset.
2. The output size is too much smaller than the destination size to display at time.
The destination should have similar size for the output size.

33.5.6 Simple De-interlacer Components

Simple De-interlacer Component does de-interlacing by using interpolation as Figure 33.36. This component is located between SCALER component and RDMA component as Figure 33.29. Both TOP field and BOTTOM field do interpolation. The last line of TOP field and the first line of BOTTOM field simply do copying.

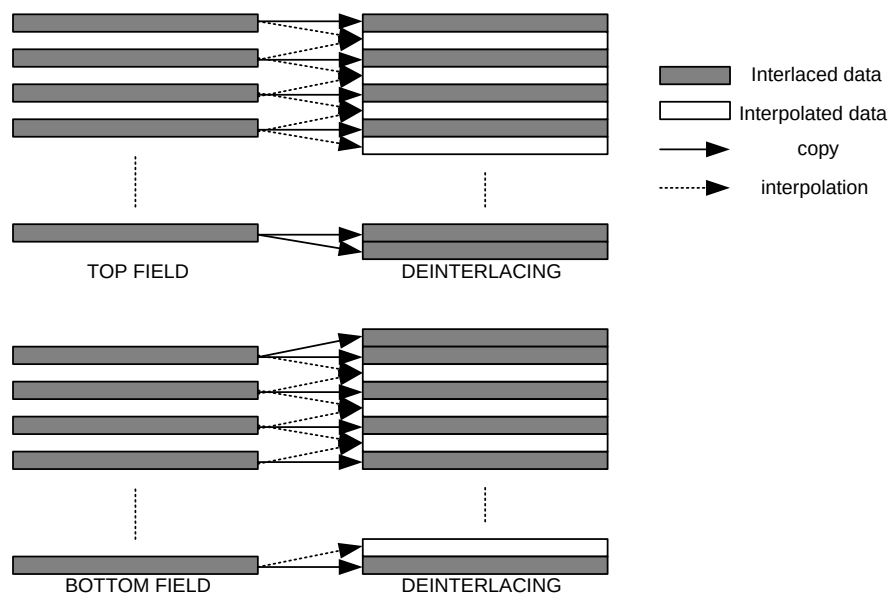


Figure 33.36 Method of Simple De-interlacer

33.5.7 VIQE Components

The VIQE (Video and Image Quality Enhancer) block is high-resolution video/image quality enhancing hardware that facilitates lots of video/image processing features. Aimed primarily at digital flat high-resolution display, it provides advanced and efficient video improvement solutions. In addition to the high quality, the pipeline architecture with specific and characteristic structure can provide minimized memory bandwidth requirement and dynamic module interoperations.

The main functions of the VIQE are listed below.

- De-interlacing
- Edge-based/Motion-adaptive
- Film-mode processing
- Judder Cancellation

The interlaced video sources are spreading very widely, mainly for broadcasting, film sources and DVD video markets. With the need of interlaced contents to be different, nowadays the main display interface is flat panel display, so the conversion into the progressive content is indispensable. The de-interlacing hardware of VIQE has advanced and efficient features, all of which support pixel-wise processing and are composed of several algorithms corresponded to very various situations making the artifacts. The film mode detection becomes an essential feature in interlaced-to-progressive conversion as a usual de-interlacing for film mode contents tends to make the output to be contaminated and to have reduced resolution spatially and temporally. The film mode detector in the VIQE supports reverse 3:2 pull-down and also compensates pixel-by-pixel the bad-edit problem and the additional text problem having not film frame rate.

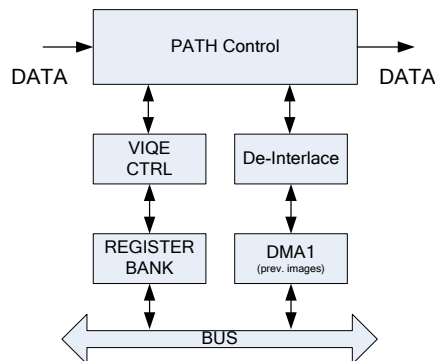


Figure 33.37 VIQE Block Diagram

The VIQE is composed of major two modules, DMA ports and de-Interlacer. Each module can share the hardware resources and operates independently. The enhancing procedure is processed in the VIQE by any combination.

Each module is designed for the hardware resources with other modules to the highest degree, and has the flexibility to select between performance and quality.

The DMA channels have specific services for transferring data. It means the DMA channel (for de-interlacer) for fetching 3 previous field data and current field data. These DMAs are used by the algorithm of frame compression / decompression for reducing memory bandwidth and allocating memory size.

33.5.8 Window Mixer (4-to-2) Components

Window Mixer Components (4-to-2) are handling overlay, alpha blending, chroma-keying processing for 4 image data. The processed image data outputs two images and each of them is connected to VWDMA and Display Device components respectively. This is for displaying an image and saving it to the memory for another processing simultaneously. WMIX 4-to-2 Components are located in upper middle in Figure 33.29 and WMIX 4-to-2 is enlarged in Figure 33.38. Details relating to overlay are explained in Chapter 33.3.4 Images and Layers.

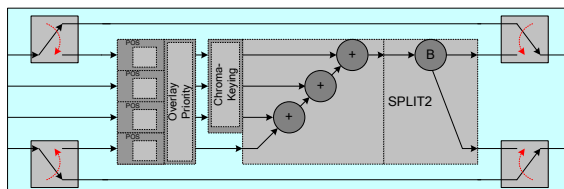


Figure 33.38 WMIX 4-to-2 Components

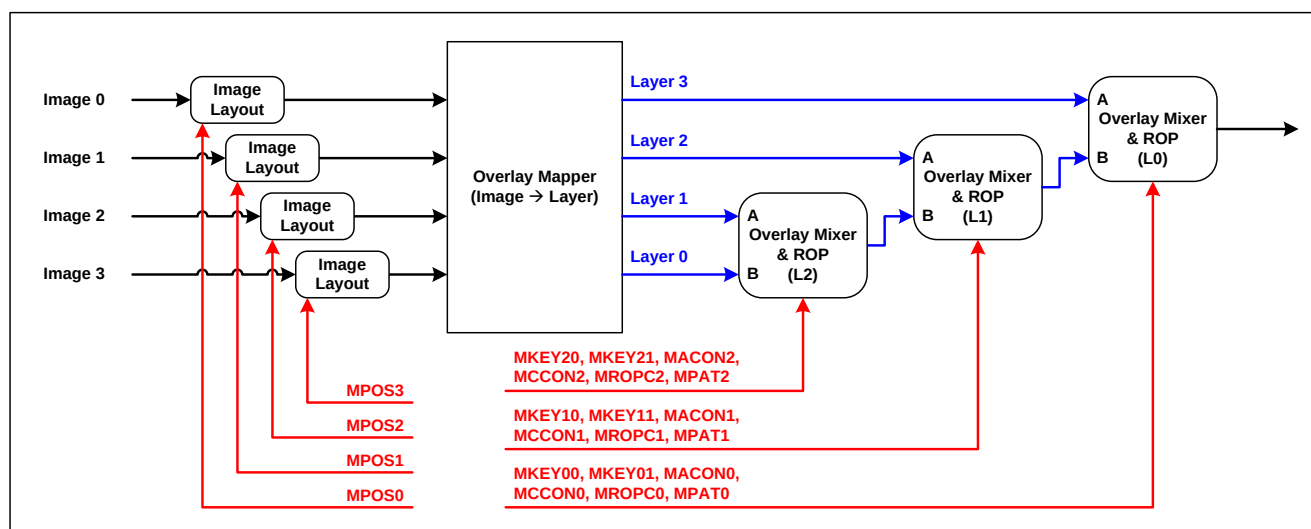


Figure 33.39 Overlay Mixer and ROP with 4 Channel Inputs

The figure above shows the overlay mixer and ROP function with 4 channel inputs. As can be seen in Figure 33.39, 4 input channels are placed to the window of the mixer by using position information. Next, by changing priority with overlay mapper, image data changes to layer data. Layer0 and Layer1 are inputted to L2 block and the output signal is made. L1 makes mixing output by using outputs from Layer 2 and L2. L0 makes a final output by using output signals from Layer3 and L1. For image layer, MPOS0 ~ MPOS3 are used for information of each image channel. MKEYk0, MKEYk1, MACONk, MCONk, MROPk, and MPATk are used in each Lk. (K is 0,1, or 2)

33.5.9 Window Mixer (2-to-2) Components

Window Mixer Components (2-to-2) are handling overlay, alpha blending, chroma-keying processing for 2 image data. The processed image data outputs two images and each of them are connected to VWDMA, GWDMA, and Display Device components respectively. This is for displaying an image and saving it to the memory for another processing simultaneously. Window Mixer (2-to-2) Components are located in upper middle in Figure 33.29 and WMIX 2-to-2 is enlarged in Figure 33.40. Details relating to overlay are explained in Chapter 33.3.4 Images and Layers.

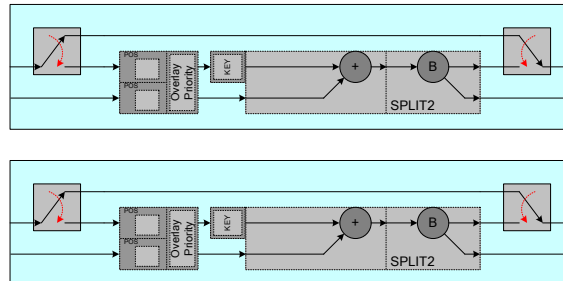


Figure 33.40 WMIX 2-to-2 Components

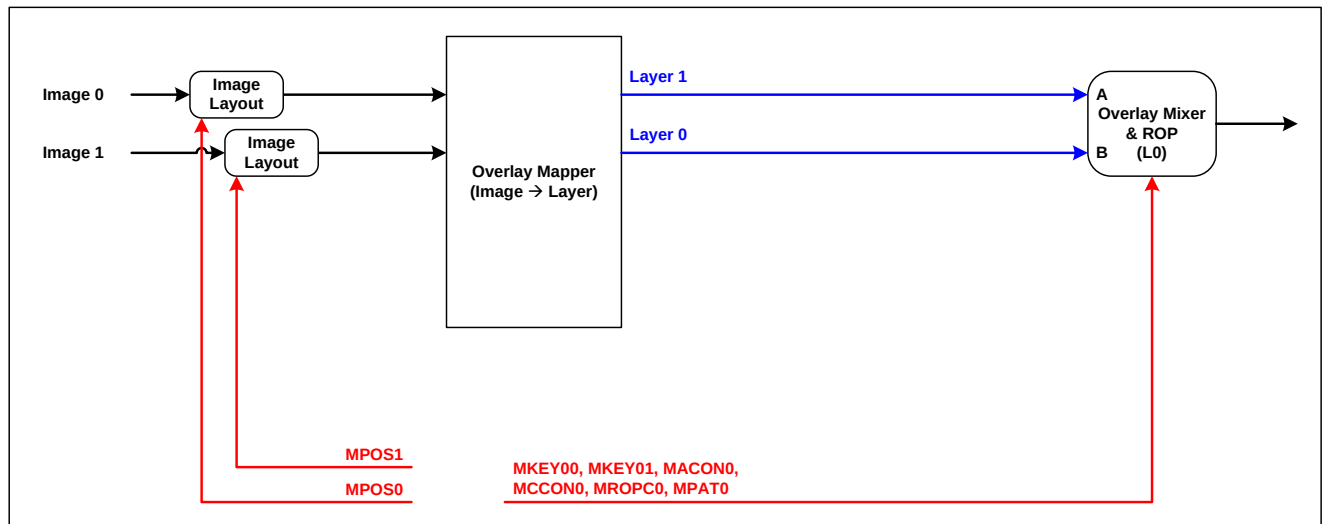


Figure 33.41 Overlay Mixer and ROP with 2 Channel Inputs

The figure above shows the overlay mixer and ROP function with 2 channel inputs. As can be seen in Figure 33.41, 2 input channels are placed to the window of the mixer by using position information. Next, by changing priority with overlay mapper, image data changes to layer data. L0 makes a final output by using output signals from Layer0 and L1.

For image layer, MPOS0 ~ MPOS1 are used for information of each image channel. MKEYk0, MKEYk1, MACONk, MCCONk, MROPCK, and MPATk are used in each Lk. (k is 0.)

33.5.10 VIDEOIN Components

In VIOC, VIDEOIN Components are used in order to receive image data from outside. Also, there are two VIDEOIN Components playing the same role.

VIDEOIN Components receive and save image data from TV-Decoder or Camera module. For an interface, BT.XX or SMPTE.XX parallel interface is used. If image data from TV-Decoder is inputted, it is SD~1080@60. If image data is inputted from the camera module, up to 4M pixel is supported.

- YUV422_16BIT
 - BT.601 [SD interlaced]
 - BT.1120, SMPTE274M (include BT.709, SMPTE274M H signal) [HD Interlaced]
 - SMPTE274M (include SMPTE296M H signal) [HD Progressive]
- YUV422_8BIT
 - BT.656, SMPTE125M (include BT.601 H signal) [SD interlaced]
 - BT.1302, SMPTE267M (include BT.601 H signal) [SD interlaced]
- YUVK4444_16BIT
 - BT.799, SMPTE RP-715 (include BT.601 H signal) [SD interlaced]
 - BT.1303 (include BT.601 H signal) [SD interlaced]
- RGBK4444_16BIT

If data is inputted from camera module, a format whose interface is matched with the interface of the camera module can be used, (Regardless of interlaced/progressive). VIDEOIN Components converts YUV422 format to YUV444 for processing. At this time, the interpolation is supported for UV and the window mode is supported for Crop. In addition, there is YUV-to-RGB format converter for RGB processing. For color processing, there is a Look-up Table.

Figure 33.42 shows VIDEOIN Component of VIOC block of Figure 33.29. (Bottom left of Figure 33.29)

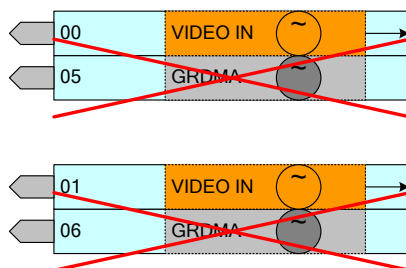


Figure 33.42 VIDEOIN Components

33.5.11 System Timer

The VIOC has an independent built-in Timer. It outputs Timer Counter, Pulse, and 1usec pulse. In order to make two different timer interrupts, two timer counters are included.

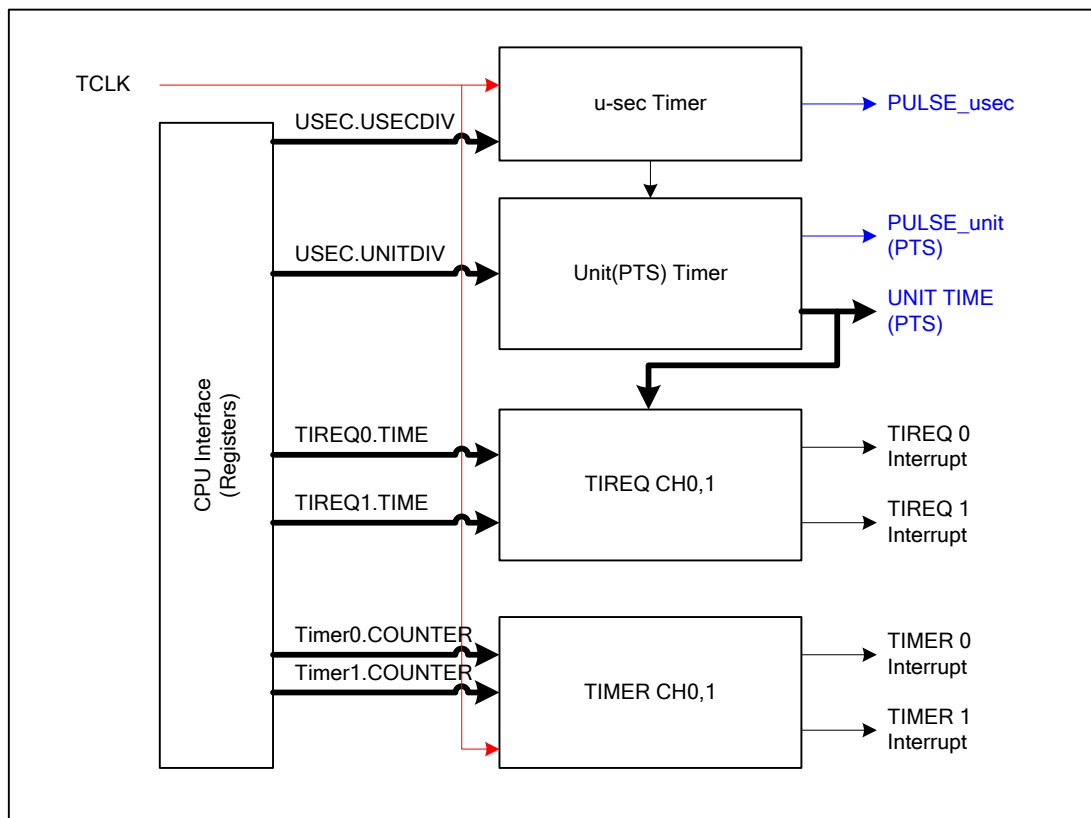


Figure 33.43 VIOC System Timer

The figure above shows System Timer Block Diagram of VIOC. The System Timer uses TCLK as an input clock. The “u-sec Timer” generates pulse every 1 u-second and the “Unit Timer” and independent Timer which has two channels are included. The “Unit Timer” receives 1 usec clock (pulse) as an input and makes 100 usec pulse commonly used in PTS. However, 100 usec is not fixed one. In addition, the “Unit Timer” makes UNIT TIME and this time information is used by blocks which need this time information. Also, it is the source for TIREQ0 and TIREQ1.

The TIREQ0 and TIREQ1 can make interrupt when the unit time get to the time information in the control register.

Two separate Timer Channels are only used to make each independent interrupt.
“PULSE_usec” can be used in order to control the traffic in DMA.

33.5.12 Interrupt Organization

Each block of VIOC makes various interrupts. They receive interrupts from each components and send them to CPU as the figure below.

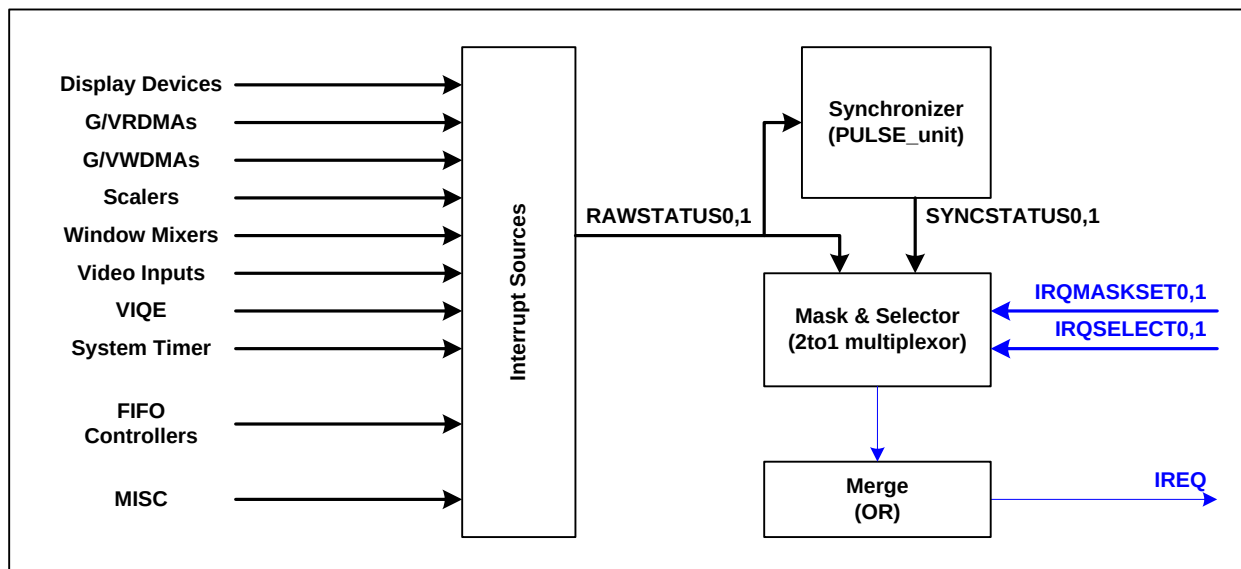


Figure 33.44 VIOC Interrupt Organization

The interrupt inputs from each component are converted to RAWSTATUS. The RAWSTATUS signals are inputted to Synchronizer and the SYNCSTATUS is made. After a valid interrupt is decided by IRQMASKSET and IRQSELECT, this signal is used as IREQ signal transmitted to Main Interrupt Controller.

Synchronizer synchronizes PULSE_unit signals from System Timer. This is in order to prevent the frequent and unnecessary interrupts by interrupt signals occurring for a short time.

The VIOC interrupt is organized by OR operation from various hardware in VIOC. The system interrupt controller should process the VIOC interrupt with the type of level-trigger. Otherwise, the merged interrupt from various hardwares in VIOC cannot make edge trigger and which can make it lost.

33.6 Register Description

33.6.1 Overall Registers

Table 33.12 VIOC Overall Register Map (Base Address = 0x80020000)

Name	Base Address	Description
DISP0	0x00000	Display Device Component in channel 0
RDMA00	0x00400	RDMA(GRDMA) Component in channel 0
RDMA01	0x00500	RDMA(VRDMA) Component in channel 1
RDMA02	0x00600	RDMA(VRDMA) Component in channel 2
RDMA03	0x00700	RDMA(VRDMA) Component in channel 3
RDMA04	0x00800	RDMA(VRDMA) Component in channel 4
RDMA05	0x00900	RDMA(GRDMA) Component in channel 5
RDMA06	0x00A00	RDMA(GRDMA) Component in channel 6
WMIX0	0x01800	WMIX (WMIX 4-to-2) Component in channel 0
WMIX1	0x01900	WMIX (WMIX 2-to-2) Component in channel 1
WMIX2	0x01A00	WMIX (WMIX 2-to-2) Component in channel 2
SC0	0x02000	SCALER Component in channel 0
SC1	0x02100	SCALER Component in channel 1
SC2	0x02200	SCALER Component in channel 2
WDMA00	0x02800	WDMA (VWDMA) Component in channel 0
WDMA01	0x02900	WDMA (VWDMA) Component in channel 1
WDMA02	0x02A00	WDMA (VWDMA) Component in channel 2
WDMA03	0x02B00	WDMA (VWDMA) Component in channel 3
WDMA04	0x02C00	WDMA (VWDMA) Component in channel 4
WDMA05	0x02D00	WDMA (VWDMA) Component in channel 5
WDMA06	0x02E00	WDMA (VWDMA) Component in channel 6
DEINTL_S	0x03800	Simple De-interlacer Component (But this register is Not used.)
VIN00	0x04000	VIDEOIN Component in channel 0
VIN01	0x04400	LUT of VIDEOIN Component in channel 0
VIN10	0x05000	VIDEOIN Component in channel 1
VIN11	0x05400	LUT of VIDEOIN Component in channel 1
LUT	0x09000	LUT Component
LUT_TABLE	0x09400	LUT Table
CONFIG	0x0A000	VIOC Configuration
TIMER	0x0C000	Timer Component
VIQE0	0x0D000	VIQE Components
CPUIF0	0x10000	CPU Interface CTRL of channel 0
OUTCFG	0x10200	Output Configuration

33.6.2 Display Device Components

Table 33.13 Display Device Component Register Map (Base Address = 0x80020000 + (0x100 * n))

Name	Address	Type	Reset	Description
DCTRL	0x000	R/W	0x30400000	Display Device Control Register
DBC	0x004	R/W	0x00000000	Display Device Background Color Register
DCLKDIV	0x008	R/W	0x00000000	Display Device Clock Control Register
DHTIME1	0x00C	R/W	0x00000000	Display Device Horizontal Timing Register 1
DHTIME2	0x010	R/W	0x00000000	Display Device Horizontal Timing Register 2
DVTIME1	0x014	R/W	0x00000000	Display Device Vertical Timing Register 1
DVTIME2	0x018	R/W	0x00000000	Display Device Vertical Timing Register 2
DVTIME3	0x01C	R/W	0x00000000	Display Device Vertical Timing Register 3
DVTIME4	0x020	R/W	0x00000000	Display Device Vertical Timing Register 4
DPOS	0x030	R/W	0x00000000	Display Device Position Register
DDITH	0x040	R/W	0x00000000	Display Device Dithering Register
DCPY	0x044	R/W	0x000000FF	Display Device Luma Clipping Register
DCPC	0x048	R/W	0x000000FF	Display Device Chroma Clipping Register
DDS	0x04C	R/W	0x00000000	Display Device Size Register
DSTATUS	0x050	R/W	0x00000000	Display Device Status Register
DIM	0x054	R/W	0x00000000	Display Device Interrupt Masking Register
DENH	0x070	R/W	0x00000020	Display Device Color Enhancement Register
DADVI	0x074	R/W	0x00000000	Display Device Interlacer Coefficient Register
DDMAT0	0x078	R/W	0x00000000	Display Device Dithering Matrix Register 0
DDMAT1	0x07C	R/W	0x00000000	Display Device Dithering Matrix Register 1

DCTRL (Display Device Control Registers)

0x80020000 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVP	EVS	R2YMD		0	ADVI	0	656	CKG	SREQ	0	0	PXDW<3:0>			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	IV	IH	IP	CLEN	R2Y	DP	NI	TV	SRST	0	0	SWAP			LEN

Bit	Name	RW	Reset	Description																		
31	EVP	R/W	0x0	External VSYNC Polarity Register 0 : Direct Input 1 : Inverted Input																		
30	EVS	R/W	0x0	External VSYNC Enable Register 0 : Disable 1 : Enable																		
29:28	R2YMD	R/W	0x3	RGB-to-YUV Conversion Mode Register																		
				<table><tr><th>R2YMD [29:28]</th><th>RGB to YCbCr Conversion Option</th></tr><tr><td>0</td><td>$* Y = 0.299 * R + 0.587 * G + 0.114 * B$ $* Cb = -0.172 * R - 0.339 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.428 * G - 0.083 * B + 128$ The range for “<i>RGB</i>” is 16 ~ 235, “<i>Studio Color</i>”. The result is “<i>Studio Color</i>” – Normally SDTV.</td></tr><tr><td>1</td><td>$* Y = 0.257 * R + 0.504 * G + 0.098 * B + 16$ $* Cb = -0.148 * R - 0.291 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.368 * G - 0.071 * B + 128$ The range for “<i>RGB</i>” is 0 ~ 255, “<i>Computer System Color</i>”. The result is “<i>Studio Color</i>” – Normally SDTV.</td></tr><tr><td>2</td><td>$* Y = 0.213 * R + 0.715 * G + 0.072 * B$ $* Cb = -0.117 * R - 0.394 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.464 * G - 0.047 * B + 128$ The range for “<i>RGB</i>” is 16 ~ 235, “<i>Studio Color</i>”.. The result is “<i>Studio Color</i>” – Normally HDTV.</td></tr><tr><td>3</td><td>$* Y = 0.183 * R + 0.614 * G + 0.062 * B + 16$ $* Cb = -0.101 * R - 0.338 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.399 * G - 0.040 * B + 128$ The range for “<i>RGB</i>” is 0 ~ 255, “<i>Computer System Color</i>”. The result is “<i>Studio Color</i>” – Normally HDTV.</td></tr></table>	R2YMD [29:28]	RGB to YCbCr Conversion Option	0	$* Y = 0.299 * R + 0.587 * G + 0.114 * B$ $* Cb = -0.172 * R - 0.339 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.428 * G - 0.083 * B + 128$ The range for “ <i>RGB</i> ” is 16 ~ 235, “ <i>Studio Color</i> ”. The result is “ <i>Studio Color</i> ” – Normally SDTV.	1	$* Y = 0.257 * R + 0.504 * G + 0.098 * B + 16$ $* Cb = -0.148 * R - 0.291 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.368 * G - 0.071 * B + 128$ The range for “ <i>RGB</i> ” is 0 ~ 255, “ <i>Computer System Color</i> ”. The result is “ <i>Studio Color</i> ” – Normally SDTV.	2	$* Y = 0.213 * R + 0.715 * G + 0.072 * B$ $* Cb = -0.117 * R - 0.394 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.464 * G - 0.047 * B + 128$ The range for “ <i>RGB</i> ” is 16 ~ 235, “ <i>Studio Color</i> ”.. The result is “ <i>Studio Color</i> ” – Normally HDTV.	3	$* Y = 0.183 * R + 0.614 * G + 0.062 * B + 16$ $* Cb = -0.101 * R - 0.338 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.399 * G - 0.040 * B + 128$ The range for “ <i>RGB</i> ” is 0 ~ 255, “ <i>Computer System Color</i> ”. The result is “ <i>Studio Color</i> ” – Normally HDTV.								
				R2YMD [29:28]	RGB to YCbCr Conversion Option																	
				0	$* Y = 0.299 * R + 0.587 * G + 0.114 * B$ $* Cb = -0.172 * R - 0.339 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.428 * G - 0.083 * B + 128$ The range for “ <i>RGB</i> ” is 16 ~ 235, “ <i>Studio Color</i> ”. The result is “ <i>Studio Color</i> ” – Normally SDTV.																	
				1	$* Y = 0.257 * R + 0.504 * G + 0.098 * B + 16$ $* Cb = -0.148 * R - 0.291 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.368 * G - 0.071 * B + 128$ The range for “ <i>RGB</i> ” is 0 ~ 255, “ <i>Computer System Color</i> ”. The result is “ <i>Studio Color</i> ” – Normally SDTV.																	
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Note: The coefficients of the table above are approximated for fixed point calculation.																						
26	ADVI	R/W	0x0	Advanced Interlaced Mode																		
24	656	R/W	0x0	CCIR 656 Mode 0 : Disable 1 : Enable																		
23	CKG	R/W	0x0	Clock Mask Enable in Under-run Situation 0 : Disable 1 : Enable																		
22	SREQ	R/W	0x1	Stop Request 0 : None 1: Request Caution: You must set this bit to '0'.																		
19:16	PXDW	R/W	0x0	PXDW of Output Format																		
				<table><tr><th>PXDW [19:16]</th><th>#Bits</th><th>Description</th></tr><tr><td>0</td><td>4 bits</td><td>Only for STN LCD</td></tr><tr><td>1</td><td>8 bits</td><td>Only for STN LCD</td></tr><tr><td>2</td><td>8 bits</td><td>RGB Stripe Type * Odd/Even Line : R → G → B → R → G → B ...</td></tr><tr><td>3</td><td>16bits</td><td>RGB565 Format</td></tr><tr><td>4</td><td>16bits</td><td>RGB555 Format</td></tr></table>	PXDW [19:16]	#Bits	Description	0	4 bits	Only for STN LCD	1	8 bits	Only for STN LCD	2	8 bits	RGB Stripe Type * Odd/Even Line : R → G → B → R → G → B ...	3	16bits	RGB565 Format	4	16bits	RGB555 Format
				PXDW [19:16]	#Bits	Description																
				0	4 bits	Only for STN LCD																
				1	8 bits	Only for STN LCD																
				2	8 bits	RGB Stripe Type * Odd/Even Line : R → G → B → R → G → B ...																
3	16bits	RGB565 Format																				
4	16bits	RGB555 Format																				

Bit	Name	RW	Reset	Description
				5 18bits RGB666 Format
				6 8 bits YCbCr Format * Cb → Y → Cr → Y
				7 8 bits YCbCr Format * Cr → Y → Cb → Y
				8 16 bits YCbCr Format * {Y,Cb} → {Y,Cr}
				9 16 bits YCbCr Format * {Y,Cr} → {Y,Cb}
				10 8 bits RGB Delta Type * Odd(1 st) Line : R → G → B → R → G → B ... * Even(2 nd) Line : G → B → R → G → B → R ...
				11 8 bits RGB Delta Type * Odd(1 st) Line : G → B → R → G → B → R ... * Even(2 nd) Line : R → G → B → R → G → B ...
				12 24 bits RGB888 Format
				13 8 bits RGB Dummy Format * Odd/Even Line : R→G→B→Dummy→ ...
				14 16bits RGB666 Format * R[17:12], G[11:6], B[5:0] * [15:0] : First Data * [17:16] : Second Data
				15 16bits RGB888 Format * R[23:16], G[15:8], B[7:0] * [15:0] : First Data * [23:16] : Second Data
15	ID	R/W	0x0	Pixel DE(Data Enable) Output Polarity Register 0 : Active High 1 : Active Low
14	IV	R/W	0x0	Pixel VSYNC Output Polarity Register 0 : Active High 1 : Active Low
13	IH	R/W	0x0	Pixel HSYNC Output Polarity Register 0 : Active High 1 : Active Low
12	IP	R/W	0x0	Pixel Clock Output Polarity Register 0 : Rising Edge 1 : Falling Edge
11	CLEN	R/W	0x0	Output Clipping Enable Register 0 : Disable 1 : Enable
10	R2Y	R/W	0x0	RGB-to-YUV Converter Enable Register 0 : Disable 1 : Enable
9	DP	R/W	0x0	Double Pixel Clock Output Configuration Register 0 : Normal 1 : Double Pixel Clock
8	NI	R/W	0x0	Non-interlaced Information 0 : Interlaced Mode 1 : Non-interlaced Mode
7	TV	R/W	0x0	Display Device Type Configuration Register 0 : TFT or STN LCD 1 : TV
6	SRST	R/W	0x0	Display Device Reset 0 : Normal 1 : Reset
3:1	SWAP	R/W	0x0	Output RGB Swap Register 0 : Normal 1 : Swap
0	LEN	R/W	0x0	Display Enable Control Register 0 : Disable 1 : Enable
				Note: At least one RDMA must be turned on.

DBC (Display Device Background Color)

0x80020004 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BG3<7:0>								BG2<7:0>							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BG1<7:0>								BG0<7:0>							

Bit	Name	RW	Reset	Description
31:24	BG3	R/W	0x0	Background Color 3 (Alpha)
23:16	BG2	R/W	0x0	Background Color 2 (Y/B)
15:8	BG1	R/W	0x0	Background Color 1 (Cb/G)
7:0	BG0	R/W	0x0	Background Color 0 (Cr/R)

DCLKDIV (Display Device Clock Divider Register)

0x80020008 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								ACDIV<7:0>							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								PXCLKDIV<7:0>							

Bit	Name	RW	Reset	Description
23:16	ACDIV	R/W	0x0	AC bias clock divisor (STN only) The number of line clock cycle to count between each toggle of AC_BIAS pin
7:0	PXCLKDIV	R/W	0x0	Pixel clock divider. Note that programming CLKDIV less than 3 is illegal for STN LCD. PXCLK = LCLK / (2*PXCLKDIV) (PXCLKDIV=0~255)

DHTIME1 (Display Device Horizontal Timing Register 1)

0x8002000C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								LPW<8:0>							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								LPC<13:0>							

Bit	Name	RW	Reset	Description
24:16	LPW	R/W	0x0	Line pulse width
13:0	LPC	R/W	0x0	Line pulse count is the number of pixel clock cycles in each line minus 1 on the screen. TFT/NTSC(16bit)/PAL(16bit) : active horizontal pixel - 1 Color STN : (3 * Horizontal display size / pixel width) Mono STN : (Horizontal display size / pixel width) - 1

DHTIME2 (Display Device Horizontal Timing Register 2)

0x80020010 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								LSWC<8:0>							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								LEWC<8:0>							

Bit	Name	RW	Reset	Description
24:16	LSWC	R/W	0x0	Line start wait clock is the number of dummy pixel clock cycles minus 1 to be inserted from the start of each horizontal line of pixels.
8:0	LEWC	R/W	0x0	Line end wait clock is the number of dummy pixel clock cycles minus 1 to be inserted before the end of each horizontal line of pixels

DVTIME1 (Display Device Vertical Timing Register 1)

0x80020014 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VDB<4:0>					0	VDF<3:0>					FPW<5:0>				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0					FLC<13:0>										

Bit	Name	RW	Reset	Description
31:27	VDB	R/W	0x0	Back porch VSYNC delay Delay cycle is -10 to 10cycle delay by PXLCLK. When TV mode, VDB value is equal to VDF. Ex: if VD=5, VSYNC delay is 5 cycle delay by HSYNC.
25:22	VDF	R/W	0x0	Front porch VSYNC delay Delay cycle is 0 to 10 cycle delay by PXLCLK. Ex: if VD=5, VSYNC delay is 5 cycle delay by HSYNC.
21:16	FPW	R/W	0x0	TFT/TV : Frame pulse width is the pulse width of frame clock (VSYNC). STN: None
13:0	FLC	R/W	0x0	Frame line count is the number of lines in each frame on the screen.

DVTIME2 (Display Device Vertical Timing Register 2)

0x80020018 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0							FSWC<8:0>								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0							FEWC<8:0>								

Bit	Name	RW	Reset	Description
24:16	FSWC	R/W	0x0	TFT/TV: Frame start wait cycle is the number of lines to be inserted at the end of each frame. STN : FSWC is N/A. If FSWC[0] is set, VSYNC signal starts on negative falling edge of HSYNC.
8:0	FEWC	R/W	0x0	TFT/TV: Frame end wait cycle is the number of lines to be inserted at the beginning of each frame. STN extra dummy lines between the end and beginning of frame.

DVTIME3 (Display Device Vertical Timing Register 3)

0x8002001C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0					0						FPW<5:0>				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0					FLC<13:0>										

Bit	Name	RW	Reset	Description
21:16	FPW	R/W	0x0	TFT/TV : Frame pulse width is the pulse width of frame clock (VSYNC). STN: None
13:0	FLC	R/W	0x0	Frame line count is the number of lines in each frame on the screen.

If NI of DCTRL is 0, DVTIME3 and DVTIME4 are for even field. Otherwise, DVTIME3 and DVTIME4 must be equal to DVTIME1 and DVTIME2, respectively.

DVTIME4 (Display Device Vertical Timing Register 4)

0x80020020 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0							FSWC<8:0>								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0							FEWC<8:0>								

Bit	Name	RW	Reset	Description
24:16	FSWC	R/W	0x0	TFT/TV: Frame start wait cycle is the number of lines to be inserted at the end of each frame. STN : FSWC is N/A. If FSWC[0] is set, VSYNC signal starts on negative falling edge of HSYNC.
8:0	FEWC	R/W	0x0	TFT/TV: Frame end wait cycle is the number of lines to be inserted at the beginning of each frame. STN extra dummy lines between the end and beginning of frame.

If NI of DCTRL is 0, DVTIME3 and DVTIME4 are for even field. Otherwise, DVTIME3 and DVTIME4 must be equal to DVTIME1 and DVTIME2, respectively.

DPOS (Display Device Position Register)

0x80020030 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			YPOS												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
			XPOS												

Bit	Name	RW	Reset	Description
28:16	YPOS	R/W	0x0	Window Position Y
12:0	XPOS	R/W	0x0	Window Position X

DDITH (Display Device Dithering Control Register)

0x80020040 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DEN	DSEL														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bit	Name	RW	Reset	Description
31	DEN	R/W	0	Dither Enable Register 0 : Dithering Disable 1 : Dithering Enable
30	DSEL	R/W	0	Dither Selection Register 0 : Normal (LSB toggle mode) 1 : Adder Mode

DCPY (Display Device Clipping Register Y)

0x80020044 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								CLPL							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								CLPH							

Bit	Name	RW	Reset	Description
23:16	CLPL	R/W	0x0	Clipping Y/R below this value
7:0	CLPH	R/W	0xFF	Clipping Y/R upper this value

DCPC (Display Device Clipping Register C)

0x80020048 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								CLPL							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0								CLPH							

Bit	Name	RW	Reset	Description
23:16	CLPL	R/W	0x0	Clipping Chroma/G/B below this value
7:0	CLPH	R/W	0xFF	Clipping Chroma/G/B upper this value

DCPY and DCPC can be used not to interpret pixel data as embedded sync signal in the CCIR-656 interface.

DDS (Display Device CD Display Size Register)

0x8002004C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0			VSIZE<12:0>												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0			HSIZE<12:0>												

Bit	Name	RW	Reset	Description
28:16	VSIZE	R/W	0x0	Horizontal size : number of active pixel in a line
12:0	HSIZE	R/W	0x0	Vertical size : number of active lines

Note: VSIZE = 32~4096, HSIZE = 32~4096

DSTATUS (Display Device Status Register)

0x80020050 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VS	BUSY	TFIELD	DEOF	0											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

STATUS[15:0]

Bit	Name	RW	Reset	Description	
31	VS	R	-	Monitoring vertical sync It has the same logic level with LVSYNC port.	
30	BUSY	R	-	Display Device Busy Status 0 : Idle 1 : Busy	
29	TFIELD	R	-	Device Top Field Indicator 0 : Bottom Field 1 : Top Field	
28	DEOF	R	-	Device EOF Indicator 0 : Normal 1 : End of Frame	
15:0	STATUS	R/W	0x0	Status registers The following flag should be cleared by writing '1' to the corresponding bit.	
				FIELD	Description
				FU[0]	LCD output fifo under-run flag.
				VSR[1]	VS rising flag
				VSF[2]	VS falling flag
				RU[3]	Register update flag It indicates that all registers programmed are applied to current frame data.
				DD[4]	Disable done If LEN is disabled, it will be set after current frame is displayed.
				SREQ[5]	Device Stop Request Signal Generated

DIM (Display Device Interrupt Masking Register)

0x80020054 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

MASK[5:0]

Bit	Name	RW	Reset	Description														
5:0	MASK	R/W	0x0	Interrupt Mask Enable Register														
				<table><tr><th>MASK[5:0]</th><th>Description</th></tr><tr><td>FU[0]</td><td>LCD output fifo under-run interrupt mask</td></tr><tr><td>VSR[1]</td><td>VS rising interrupt mask</td></tr><tr><td>VSF[2]</td><td>VS falling interrupt mask</td></tr><tr><td>RU[3]</td><td>Register update interrupt mask</td></tr><tr><td>DD[4]</td><td>Disable done interrupt mask</td></tr><tr><td>SREQ[5]</td><td>Device Stop Request Interrupt Mask</td></tr></table>	MASK[5:0]	Description	FU[0]	LCD output fifo under-run interrupt mask	VSR[1]	VS rising interrupt mask	VSF[2]	VS falling interrupt mask	RU[3]	Register update interrupt mask	DD[4]	Disable done interrupt mask	SREQ[5]	Device Stop Request Interrupt Mask
				MASK[5:0]	Description													
				FU[0]	LCD output fifo under-run interrupt mask													
				VSR[1]	VS rising interrupt mask													
				VSF[2]	VS falling interrupt mask													
				RU[3]	Register update interrupt mask													
				DD[4]	Disable done interrupt mask													
SREQ[5]	Device Stop Request Interrupt Mask																	

DCENH (Display Device Color Enhancement Register)

0x80020070 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0								HUE							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BRIGHTNESS								CONTRAST							

Bit	Name	RW	Reset	Description
23:16	HUE	R/W	0x0	Hue Calibration Register – 2's complement signed value * -30 ~ 30 degree * 0x80 for -30 degree * 0x00 for 0 degree for default value * 0x7F for about 30 degree
15:8	BRIGHTNESS	R/W	0x0	Brightness Calibration Register – 2's complement signed value * -128 ~ 128 value * 0x80 for -128 offset * 0x00 for 0 offset * 0x7F for 127 offset
7:0	CONTRAST	R/W	0x20	Contrast Calibration Register – 2's complement signed value * -4 ~ 4 * 0x80 for -4.0 value * 0xFF for -1.0 value * 0x20 for 1.0 value * 0x7F for about 4.0 value

DADVI (Display Device Advanced Interlacer Coefficient Register)

0x80020074 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								COEFF1				COEFF0			

Bit	Name	RW	Reset	Description
7:4	COEFF1	R/W	0xC	Advanced Interlacer Coefficient 1
3:0	COEFF0	R/W	0x2	Advanced Interlacer Coefficient 0

DDMAT0 (Display Device Dither Matrix Register 0)

0x80020078 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DITH13				DITH12				DITH11				DITH10			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DITH03				DITH02				DITH01				DITH00			

Bit	Name	RW	Reset	Description
30:28	DITH13	R/W	0x0	Dithering Pattern Matrix (1,3)
26:24	DITH12	R/W	0x0	Dithering Pattern Matrix (1,2)
22:20	DITH11	R/W	0x0	Dithering Pattern Matrix (1,1)
18:16	DITH10	R/W	0x0	Dithering Pattern Matrix (1,0)
14:12	DITH03	R/W	0x0	Dithering Pattern Matrix (0,3)
10:8	DITH02	R/W	0x0	Dithering Pattern Matrix (0,2)
6:4	DITH01	R/W	0x0	Dithering Pattern Matrix (0,1)
2:0	DITH00	R/W	0x0	Dithering Pattern Matrix (0,0)

DDMAT1 (Display Device Dither Matrix Register 1)

0x8002007C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
		DITH33				DITH32				DITH31				DITH30	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		DITH23				DITH22				DITH21				DITH20	

Bit	Name	RW	Reset	Description
30:28	DITH33	R/W	0x0	Dithering Pattern Matrix (3,3)
26:24	DITH32	R/W	0x0	Dithering Pattern Matrix (3,2)
22:20	DITH31	R/W	0x0	Dithering Pattern Matrix (3,1)
18:16	DITH30	R/W	0x0	Dithering Pattern Matrix (3,0)
14:12	DITH23	R/W	0x0	Dithering Pattern Matrix (2,3)
10:8	DITH22	R/W	0x0	Dithering Pattern Matrix (2,2)
6:4	DITH21	R/W	0x0	Dithering Pattern Matrix (2,1)
2:0	DITH20	R/W	0x0	Dithering Pattern Matrix (2,0)

For Dithering Matrix, refer to Chapter 33.4.12 Dithering.

33.6.3 Video Input Related

Table 33.14 VIDEOIN Register Map (Base Address = 0x80024000 + (0x1000 * n))

Name	Address	Type	Reset	Description
VIN_CTRL	0x000	R/W	0x00000000	VIN Control Register
VIN_MISC	0x004	R/W	0x00000000	VIN Misc. Register
VIN_SYNC_M0	0x008	R/W	0x00030CB9	VIN Sync Misc. 0 Register
VIN_SYNC_M1	0x00C	R/W	0x000000FF	VIN Sync Misc. 1 Register
VIN_SIZE	0x010	R/W	0x00000000	VIN Size Register
VIN_OFFS	0x014	R/W	0x00000000	VIN Offset Register
VIN_OFFS_INTL	0x018	R/W	0x00000000	VIN Offset Interlaced Register
VIN_INT	0x060	R/W	0x00000000	VIN Interrupt Register
VIN_LUT_C0	Base + 0x1000 ~ 0x10FC	R/W	0x00000000	VIN Look-up Table Set in Channel 0
VIN_LUT_C1	Base + 0x1100 ~ 0x11FC	R/W	0x00000000	VIN Look-up Table Set in Channel 1
VIN_LUT_C2	Base + 0x1200 ~ 0x12FC	R/W	0x00000000	VIN Look-up Table Set in Channel 2

VIN_CTRL (VIN Control Register)

0x80024000 + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CP															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	SE	GFEN	DEAL	FOL	VAL	HAL	PXP		VM	FLUSH	HDCE	INTPLEN	INTEN	CONV	EN

Bit	Name	RW	Reset	Description
31	CP	R/W	0x0	0 : Normal 1 : Capture This bit is always set to 'LOW'. If you want the capture mode, WDMA's operating mode is set to frame-by-frame mode.
27:24	SKIP	R/W	0x0	Skip frame number. If set to '0', the frame skip does not operate.
22:20	DO	R/W	0x0	Input Data Order 0 : RGB – VYU 1 : RBG – VUY 2 : GRB – YVU 3 : GBR – YUV 4 : BRG – UYV 5 : BGR – UYV
19:16	FMT	R/W	0x0	Input Data Format 0x0 : YUV422_16BIT 0x1 : YUV422_8BIT 0x2 : YUVK4444_16BIT 0x3 : Reserved 0x8 : RGBK4444_16BIT 0x9 : Reserved Others : Not used This format is referred to Chapter 33.3.7 Supported Video Input Formats.
14	SE	R/W	0x0	For Debugging Do not set this bit
13	GFEN	R/W	0x0	0 : Generate Field Signal 1 : Use Input Field Signal Field Signal Mode
12	DEAL	R/W	0x0	0 : DE signal – Active High 1 : DE signal – Active Low
11	FOL	R/W	0x0	0 : Field signal – ODD Field High 1 : Field signal – ODD Field Low
10	VAL	R/W	0x0	0 : VS signal – Active High 1 : VS signal – Active Low
9	HAL	R/W	0x0	0 : HS signal – Active High 1 : VS signal – Active Low
8	PXP	R/W	0x0	0 : Normal 1 : Change PXCLK polarity
6	VM	R/W	0x0	0 : Normal 1 : Masking DE signal at vertical blanking area.
5	FLUSH	R/W	0x0	0 : Normal 1 : Buffer Flush
4	HDCE	R/W	0x0	0 : Normal

				1 : Connect DE signal from HS signal.
3	INTPLEN	R/W	0x0	0 : Disable 1 : Enable for interpolating chrominance channels. Converting from format 4:2:2 to 4:4:4, module interpolates the chrominance channel
2	INTLEN	R/W	0x0	0 : Progressive mode 1 : Interlaced mode
1	CONV	R/W	0x0	0 : Normal 1 : Detecting Sync signal from Databus (BT.656)
0	EN	R/W	0x0	0 : Normal 1 : Enable VIDEOIN operating enable bit

VIN_MISC (VIN Misc. Register)

0x80024004 + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
										Y2RM	Y2REN	LUTIF	LUTEN		

Bit	Name	RW	Reset	Description
6:5	Y2RM	R/W	0x0	Refer to table below.
				Y2RMD [6:5]
				YCbCr to RGB Conversion Option
				0 $*R = Y + 1.371 * (Cr - 128)$ $*G = Y - 0.336 * (Cb - 128) - 0.698 * (Cr - 128)$ $*B = Y + 1.732 * (Cb - 128)$ The range for "YCbCr" is 16 ~ 235, "Studio Color". The result is "Studio Color" – Normally SDTV.
				1 $*R = 1.164 * (Y - 16) + 1.596 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.391 * (Cb - 128) - 0.813 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.018 * (Cb - 128)$ The range for "YCbCr" is 16 ~ 235, "Studio Color". The result is "Computer System Color" – Normally SDTV.
4	Y2REN	R/W	0x0	2 $*R = Y + 1.540 * (Cr - 128)$ $*G = Y - 0.183 * (Cb - 128) - 0.459 * (Cr - 128)$ $*B = Y + 1.816 * (Cb - 128)$ The range for "YCbCr" is 16 ~ 235, "Studio Color". The result is "Studio Color" – Normally HDTV.
				3 $*R = 1.164 * (Y - 16) + 1.793 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.213 * (Cb - 128) - 0.534 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.115 * (Cb - 128)$ The range for "YCbCr" is 16 ~ 235, "Studio Color". The result is "Computer System Color" – Normally HDTV.
4	Y2REN	R/W	0x0	0 : Disable 1 : Enable This field is "YUV to RGB format Conv. Enable". If this bit is set to 'HIGH', YUV to RGB converter is enabled.
3	LUTIF	R/W	0x0	0 : Access Look-up Table using VIN module. 1 : Access Look-up Table using Slave port. If you want to initialize Look-up Table. You must set this bit to 'HIGH' and this bit initialize tables. After completing Look-up table, set to 'LOW' for accessing from VIN module. The method of setting Look-up table is referred to VIN_LUT registers.
2:0	LUTEN	R/W	0x0	0 : Not use Look-up Table. 1 : Use Look-up Table. Bit 2 : Channel 2 Bit 1 : Channel 1 Bit 0 : channel 0

VIN_SYNC_M0 (VIN Sync Misc. 0 Register)

0x80024008 + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
												SB		PSL	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FP								VB				HB			

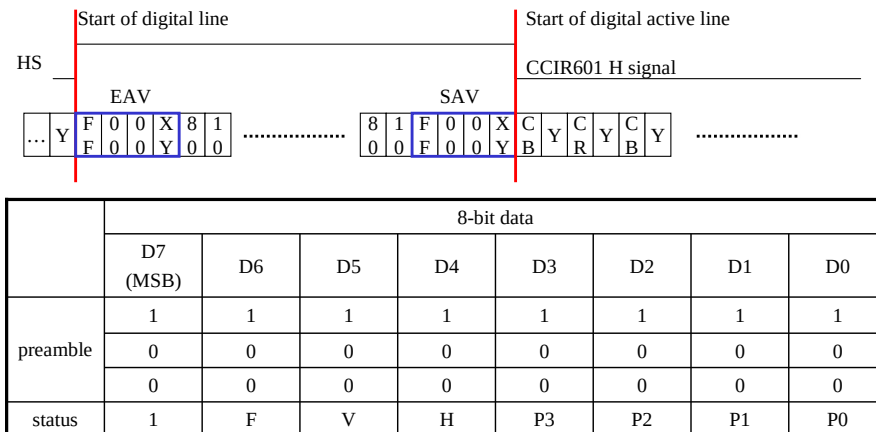
Bit	Name	RW	Reset	Description
19:18	SB	R/W	0x0	0x0 : using databus[07:00] for detecting sync bytes. 0x1 : using databus[15:08] for detecting sync bytes 0x2 : using databus[23:16] for detecting sync bytes Others : Not used This field is "SyncByte" for BT.656 format. If the input format is BT.656 mode, the module must detect Sync Signal using databus. So this bit is decided by the location of Sync Byte.
17:16	PSL	R/W	0x3	0x0 : The status word is located in the first byte of EAV & SAV 0x1 : The status word is located in the second byte of EAV & SAV 0x2 : The status word is located in the third byte of EAV & SAV 0x3 : The status word is located in the fourth byte of EAV & SAV (Refer to Figure 33.45 of the VIN_SYNC_M1 Register)
11:8	FP	R/W	0xC	Field information In status word, it refers to the location of 'F' and F value at field information. The MSB 3 bit means the location of 'F', the other bit means value at field information.
7:4	VB	R/W	0xB	Horizontal blank In status word, it refers to the location of 'H' and H value at blanking. The MSB 3 bit means the location of 'H', the other bit means value at blanking.
3:0	HB	R/W	0x9	Vertical blank In status word, it refers to the location of 'V' and V value at blanking. The MSB 3 bit means the location of 'V', the other bit means value at blanking.

VIN_SYNC_M1 (VIN Sync Misc. 1 Register)

0x8002400C + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								PT							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PS								PF							

Bit	Name	RW	Reset	Description
23:16	PT	R/W	0x0	Third preamble value Define the third preamble value.
15:8	PS	R/W	0x0	Second preamble value Define the second preamble value.
7:0	PF	R/W	0xFF	First preamble value Define the first preamble value.



•Status word define

- F='0' for field 1, '1' for field 2 (interlace mode. If progressive, this value is '0')
- V='1' during vertical blanking
- H='0' at SAV, '1' at EAV

•Protection bits

- P3=V xor H
- P2=F xor H
- P1=F xor V
- P0=F xor V xor H

Figure 33.45 CCIR.656 Stream Structure

VIN_SIZE (VIN Size Register)

0x80024010 + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HEIGHT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WIDTH															

Bit	Name	RW	Reset	Description
31:16	HEIGHT	R/W	0x0	Image Height by pixel.
15:0	WIDTH	R/W	0x0	Image Width by pixel.

VIN_OFFS (VIN Offset Register)

0x80024014 + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OFS_HEIGHT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OFS_WIDTH															

Bit	Name	RW	Reset	Description
31:16	OFS_HEIGHT	R/W	0x0	Offset position in image height
15:0	OFS_WIDTH	R/W	0x0	Offset position In image width

VIN_OFFS_INTL (VIN Offset in Interlaced Register)

0x80024018 + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OFS_HEIGHT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bit	Name	RW	Reset	Description
31:16	OFS_HEIGHT	R/W	0x0	Offset position in image height at interlaced mode.

VIN_INT (VIN Interrupt Register)

0x80024060 + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN												MINVS	MVS	MEOF	MUPD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
				FS								INVS	VS	EOF	UPD

Bit	Name	RW	Reset	Description
31	INTEN	R/W	0x0	0 : Disable 1 : Enable Interrupt Enable
19	MINVS	R/W	0x0	0 : Disable 1 : INVS interrupt enable
18	MVS	R/W	0x0	0 : Disable 1 : VS interrupt enable
17	MEOF	R/W	0x0	0 : Disable 1 : EOF interrupt enable
16	MUPD	R/W	0x0	0 : Disable 1 : Update interrupt enable
11	FS	R	-	0 : Odd field 1 : Even field
3	INVS	R/W	0x0	0 : Normal 1 : Interrupt (start point in blanking area) If write to 'HIGH', interrupt is cleared.
2	VS	R/W	0x0	0 : Normal 1 : Interrupt (start point in active area) If write to 'HIGH', interrupt is cleared.
1	EOF	R/W	0x0	0 : Normal 1 : Interrupt (EOF signal) If write to 'HIGH', interrupt is cleared
0	UPD	R/W	0x0	0 : Normal 1 : Interrupt (Update signal) If write to 'HIGH', interrupt is cleared

VIN_LUTk_C0 (VIN Look-up Table k in Ch.0 Register)

0x80024400 + (0x4*k) + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VALUE _(4n+3)								VALUE _(4n+2)							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALUE _(4n+1)								VALUE _{4n}							

Bit	Name	RW	Reset	Description
31:24	VALUE _(4n+3)	R/W	0x0	The input pixel data (4n+3) is mapped to this value.
23:16	VALUE _(4n+2)	R/W	0x0	The input pixel data (4n+2) is mapped to this value.
15:8	VALUE _(4n+1)	R/W	0x0	The input pixel data (4n+1) is mapped to this value.
7:0	VALUE _(4n)	R/W	0x0	The input pixel data (4n) is mapped to this value.

The range of k is 0 ≤ k ≤ 63. If you want to initialize the Look-up Table, you must set the LUTIF field [bit 3] of VIN_MISC register [0x80024004 / 0x80026004] before initializing the Look-up Tables. The channel of register is mapped to channel 0.

VIN_LUTk_C1 (VIN Look-up Table k in Ch.1 Register)

0x80024500 + (0x4*k) + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VALUE _(4n+3)								VALUE _(4n+2)							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALUE _(4n+1)								VALUE _{4n}							

Bit	Name	RW	Reset	Description
31:24	VALUE _(4n+3)	R/W	0x0	The input pixel data (4n+3) is mapped to this value.
23:16	VALUE _(4n+2)	R/W	0x0	The input pixel data (4n+2) is mapped to this value.
15:8	VALUE _(4n+1)	R/W	0x0	The input pixel data (4n+1) is mapped to this value.
7:0	VALUE _(4n)	R/W	0x0	The input pixel data (4n) is mapped to this value.

The range of k is 0 ≤ k ≤ 63. If you want to initialize the Look-up Table, you must set the LUTIF field [bit 3] of VIN_MISC register [0x80024004 / 0x80026004] before initializing the Look-up Tables. The channel of register is mapped to channel 1.

VIN_LUTk_C2 (VIN Look-up Table k in Ch.2 Register)

0x80024600+(0x4*n) + (0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VALUE_(4n+3)								VALUE_(4n+2)							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALUE_(4n+1)								VALUE_4n							

Bit	Name	RW	Reset	Description
31:24	VALUE_(4n+3)	R/W	0x0	The input pixel data (4n+3) is mapped to this value.
23:16	VALUE_(4n+2)	R/W	0x0	The input pixel data (4n+2) is mapped to this value.
15:8	VALUE_(4n+1)	R/W	0x0	The input pixel data (4n+1) is mapped to this value.
7:0	VALUE_(4n)	R/W	0x0	The input pixel data (4n) is mapped to this value.

The range of k is $0 \leq k \leq 63$. If you want to initialize the Look-up Table, you must set the LUTIF field [bit 3] of VIN_MISC register [0x80024004 / 0x80026004] before initializing the Look-up Tables. The channel of register is mapped to channel 2.

33.6.4 GRDMA and VRDMA

Table 33.15 GRDMA/VRDMA Register Map (Base Address = 0x80020400 + (0x100 * n))

Name	Address	Type	Reset	Description
RDMACtrl	0x00	R/W	0x00000400	RDMA Control Register
RDMAPTS	0x04	R/W	0x00000000	RDMA Image PTS Register
RDMASIZE	0x08	R/W	0x00000000	RDMA Image Size Information Register
RDMABASE0	0x0C	R/W	0x00000000	RDMA Base0 Address for Each Images Register
RDMACADDR	0x10	R/W	0x00000000	RDMA Current Address for Each Images Register
RDMABASE1	0x14	R/W	0x00000000	RDMA Base1 Address for Each Images Register
RDMABASE2	0x18	R/W	0x00000000	RDMA Base2 Address for Each Images Register
RDMAOFFS	0x1C	R/W	0x00000000	RDMA Offset Information for Each Images Register
RDMASCALE	0x20	R/W	0x00000000	RDMA Scale for Each Images Register
RDMAALPHA	0x24	R/W	0x00000000	RDMA Alpha Information for Each Images Register
RDMASTAT	0x28	R/W	0x00000000	RDMA Status Register
RDMAIRQMSK	0x2C	R/W	0x0000007F	RDMA Interrupt Mask Register
RDMASBASE0	0x30	R/W	0x00000000	RDMA Sync Base Address

RDMACTRL (RDMA Control Registers)

0x80020400 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTL	BFMD	BF	IEN	STRM	0		ASEL	UVI	NUVIH	0		0			UPD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PD	SWAP			AEN	Y2RMD		Y2R	BR	0		FMT				

Bit	Name	RW	Reset	Description						
31	INTL	R/W	0x0	Interlaced Image Indication Register 0 : Progressive Data Fetch 1 : Interlace Data Fetch * Line Order is 1,2,3,4,5,6 ... * Line Order of odd frame : 1,3,5,... * Line Order of even frame : 2,4,6,...						
30	BFMD	R/W	0x0	BFIELD Mode Register 0 : Use field signal from image (with BF field [BIT29] of this register) 1 : Use field signal from Display Device.						
29	BF	R/W	0x0	BFIELD Indication Register 0 : Top Field 1 : Bottom Field						
28	IEN	R/W	0x0	Image Enable Register 0 : Disable 1 : Enable						
27	STRM	R/W	0x0	Streaming Mode Enable Register 0 : Normal Mode 1 : Streaming Mode						
24	ASEL	R/W	0x0	Pixel Alpha Select Register 0 : A0 bits and A1 are used as alpha value (A0 and A1 field is located by RDMAALPHA Reg.) 1 : Alpha bits in the pixel are used as alpha value.						
23	UVI	R/W	0x0	UV interpolation Register (Vertical & horizontal) 0 : Normal 1 : Interpolation						
22	NUVIH	R/W	0x0	No UV interpolation in Horizontal Direction 0 : Interpolation in horizontal 1 : No interpolation in horizontal						
16	UPD	R/W	0x0	Information Update Register 0 : Update disable 1 : Immediately Update						
15	PD	R/W	0x0	Padding Option Register 0 : No Padding 1 : Padding (Refer to Figure 33.46 below)						
14:12	SWAP	R/W	0x0	RGB Swap Register 3'b000 : R-G-B 3'b001 : R-B-G 3'b010 : G-R-B 3'b011 : G-B-R 3'b100 : B-R-G 3'b101 : B-G-R						
11	AEN	R/W	0x0	Alpha Enable Register 0 : Disable 1 : Enable						
10:9	Y2RMD	R/W	0x2	YUV-to-RGB Converter Mode Register <table><tr><th>Y2RMD [10:9]</th><th>YCbCr to RGB Conversion Option</th></tr><tr><td>0</td><td>$*R = Y + 1.371 * (Cr - 128)$ $*G = Y - 0.336 * (Cb - 128) - 0.698 * (Cr - 128)$ $*B = Y + 1.732 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Studio Color” – Normally SDTV.</td></tr><tr><td>1</td><td>$*R = 1.164 * (Y - 16) + 1.596 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.391 * (Cb - 128) - 0.813 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.018 * (Cb - 128)$</td></tr></table>	Y2RMD [10:9]	YCbCr to RGB Conversion Option	0	$*R = Y + 1.371 * (Cr - 128)$ $*G = Y - 0.336 * (Cb - 128) - 0.698 * (Cr - 128)$ $*B = Y + 1.732 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Studio Color” – Normally SDTV.	1	$*R = 1.164 * (Y - 16) + 1.596 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.391 * (Cb - 128) - 0.813 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.018 * (Cb - 128)$
Y2RMD [10:9]	YCbCr to RGB Conversion Option									
0	$*R = Y + 1.371 * (Cr - 128)$ $*G = Y - 0.336 * (Cb - 128) - 0.698 * (Cr - 128)$ $*B = Y + 1.732 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Studio Color” – Normally SDTV.									
1	$*R = 1.164 * (Y - 16) + 1.596 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.391 * (Cb - 128) - 0.813 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.018 * (Cb - 128)$									

					The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Computer System Color” – Normally SDTV.																																																						
			2		$*R = Y + 1.540 * (Cr - 128)$ $*G = Y - 0.183 * (Cb - 128) - 0.459 * (Cr - 128)$ $*B = Y + 1.816 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Studio Color” – Normally HDTV.																																																						
			3		$*R = 1.164 * (Y - 16) + 1.793 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.213 * (Cb - 128) - 0.534 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.115 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Computer System Color” – Normally HDTV.																																																						
8	Y2R	R/W	0x0		YUV-to-RGB Converter Enable Register 0 : Disable 1 : Enable																																																						
7	BR	R/W	0x0		Bit Reverse in Byte 0 : Normal (Little-endian) 1 : Reverse (Big-endian) BR is only used when BPP is 1, 2, or 4 bpp																																																						
4:0	FMT	R/W			Image Format Register <table><tr><th>FMT [4:0]</th><th>Image Format</th></tr><tr><td>0</td><td>1bpp indexed color</td></tr><tr><td>1</td><td>2bpp indexed color</td></tr><tr><td>2</td><td>4bpp indexed color</td></tr><tr><td>3</td><td>8bpp indexed color</td></tr><tr><td>4 ~ 7</td><td>Reserved for future use</td></tr><tr><td>8</td><td>RGB332 – 1 bytes aligned : R[7:5],G[4:2],B[1:0]</td></tr><tr><td>9</td><td>RGB444 – 2 bytes aligned : A[15:12],R[11:8],G[7:4],B[3:0]</td></tr><tr><td>10</td><td>RGB565 – 2 bytes aligned : R[15:11],G[10:5],B[4:0]</td></tr><tr><td>11</td><td>RGB555 – 2 bytes aligned : A[15],R[14:10],G[9:5],B[4:0]</td></tr><tr><td>12</td><td>RGB888 – 4 bytes aligned : A[31:24],R[23:16],G[15:8],B[7:0]</td></tr><tr><td>13</td><td>RGB666 – 4 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]</td></tr><tr><td>14</td><td>RGB888 – 3 bytes aligned : B1[31:24],R0[23:16],G0[15:8],B0[7:0]</td></tr><tr><td>15</td><td>ARGB6666 – 3 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]</td></tr><tr><td>16</td><td>Reserved</td></tr><tr><td>17 ~ 20</td><td>Reserved for future use</td></tr><tr><td>21</td><td>4:4:4 separate format - <u>valid for VRDMA only</u> If Y2R is enabled, this can be used for YUV format, otherwise for RGB separated format.</td></tr><tr><td>22</td><td>YCbCr 4:2:2 sequential format LSB [U/Y/V/Y] MSB</td></tr><tr><td>23</td><td>YCbCr 4:2:2 sequential format LSB [V/Y/U/Y] MSB</td></tr><tr><td>24</td><td>YCbCr 4:2:0 separated format - <u>valid for VRDMA only</u></td></tr><tr><td>25</td><td>YCbCr 4:2:2 separated format - <u>valid for VRDMA only</u></td></tr><tr><td>26</td><td>YCbCr 4:2:2 sequential format LSB [Y/U/Y/V] MSB</td></tr><tr><td>27</td><td>YCbCr 4:2:2 sequential format LSB [Y/V/Y/U] MSB</td></tr><tr><td>28</td><td>YCbCr 4:2:0 interleaved type 0 format - <u>valid for VRDMA only</u></td></tr><tr><td>29</td><td>YCbCr 4:2:0 interleaved type 1 format - <u>valid for VRDMA only</u></td></tr><tr><td>30</td><td>YCbCr 4:2:2 interleaved type 0 format - <u>valid for VRDMA only</u></td></tr><tr><td>31</td><td>YCbCr 4:2:2 interleaved type 1 format - <u>valid for VRDMA only</u></td></tr></table> The “A”, “R”, “G” and “B” stand for alpha value, red, green and blue color correspondingly. The alpha value to be applied is determined by ASEL bit. If ASEL is zero, the alpha value equals to “A” with extra bit extended. (Refer to Figure 33.8)	FMT [4:0]	Image Format	0	1bpp indexed color	1	2bpp indexed color	2	4bpp indexed color	3	8bpp indexed color	4 ~ 7	Reserved for future use	8	RGB332 – 1 bytes aligned : R[7:5],G[4:2],B[1:0]	9	RGB444 – 2 bytes aligned : A[15:12],R[11:8],G[7:4],B[3:0]	10	RGB565 – 2 bytes aligned : R[15:11],G[10:5],B[4:0]	11	RGB555 – 2 bytes aligned : A[15],R[14:10],G[9:5],B[4:0]	12	RGB888 – 4 bytes aligned : A[31:24],R[23:16],G[15:8],B[7:0]	13	RGB666 – 4 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]	14	RGB888 – 3 bytes aligned : B1[31:24],R0[23:16],G0[15:8],B0[7:0]	15	ARGB6666 – 3 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]	16	Reserved	17 ~ 20	Reserved for future use	21	4:4:4 separate format - <u>valid for VRDMA only</u> If Y2R is enabled, this can be used for YUV format, otherwise for RGB separated format.	22	YCbCr 4:2:2 sequential format LSB [U/Y/V/Y] MSB	23	YCbCr 4:2:2 sequential format LSB [V/Y/U/Y] MSB	24	YCbCr 4:2:0 separated format - <u>valid for VRDMA only</u>	25	YCbCr 4:2:2 separated format - <u>valid for VRDMA only</u>	26	YCbCr 4:2:2 sequential format LSB [Y/U/Y/V] MSB	27	YCbCr 4:2:2 sequential format LSB [Y/V/Y/U] MSB	28	YCbCr 4:2:0 interleaved type 0 format - <u>valid for VRDMA only</u>	29	YCbCr 4:2:0 interleaved type 1 format - <u>valid for VRDMA only</u>	30	YCbCr 4:2:2 interleaved type 0 format - <u>valid for VRDMA only</u>	31	YCbCr 4:2:2 interleaved type 1 format - <u>valid for VRDMA only</u>
FMT [4:0]	Image Format																																																										
0	1bpp indexed color																																																										
1	2bpp indexed color																																																										
2	4bpp indexed color																																																										
3	8bpp indexed color																																																										
4 ~ 7	Reserved for future use																																																										
8	RGB332 – 1 bytes aligned : R[7:5],G[4:2],B[1:0]																																																										
9	RGB444 – 2 bytes aligned : A[15:12],R[11:8],G[7:4],B[3:0]																																																										
10	RGB565 – 2 bytes aligned : R[15:11],G[10:5],B[4:0]																																																										
11	RGB555 – 2 bytes aligned : A[15],R[14:10],G[9:5],B[4:0]																																																										
12	RGB888 – 4 bytes aligned : A[31:24],R[23:16],G[15:8],B[7:0]																																																										
13	RGB666 – 4 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]																																																										
14	RGB888 – 3 bytes aligned : B1[31:24],R0[23:16],G0[15:8],B0[7:0]																																																										
15	ARGB6666 – 3 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]																																																										
16	Reserved																																																										
17 ~ 20	Reserved for future use																																																										
21	4:4:4 separate format - <u>valid for VRDMA only</u> If Y2R is enabled, this can be used for YUV format, otherwise for RGB separated format.																																																										
22	YCbCr 4:2:2 sequential format LSB [U/Y/V/Y] MSB																																																										
23	YCbCr 4:2:2 sequential format LSB [V/Y/U/Y] MSB																																																										
24	YCbCr 4:2:0 separated format - <u>valid for VRDMA only</u>																																																										
25	YCbCr 4:2:2 separated format - <u>valid for VRDMA only</u>																																																										
26	YCbCr 4:2:2 sequential format LSB [Y/U/Y/V] MSB																																																										
27	YCbCr 4:2:2 sequential format LSB [Y/V/Y/U] MSB																																																										
28	YCbCr 4:2:0 interleaved type 0 format - <u>valid for VRDMA only</u>																																																										
29	YCbCr 4:2:0 interleaved type 1 format - <u>valid for VRDMA only</u>																																																										
30	YCbCr 4:2:2 interleaved type 0 format - <u>valid for VRDMA only</u>																																																										
31	YCbCr 4:2:2 interleaved type 1 format - <u>valid for VRDMA only</u>																																																										

Though raw image source is not RGB888 format, it is internally converted to RGB888 as appending additional bits. When this conversion occurs, these padding bits are determined by PD bit.

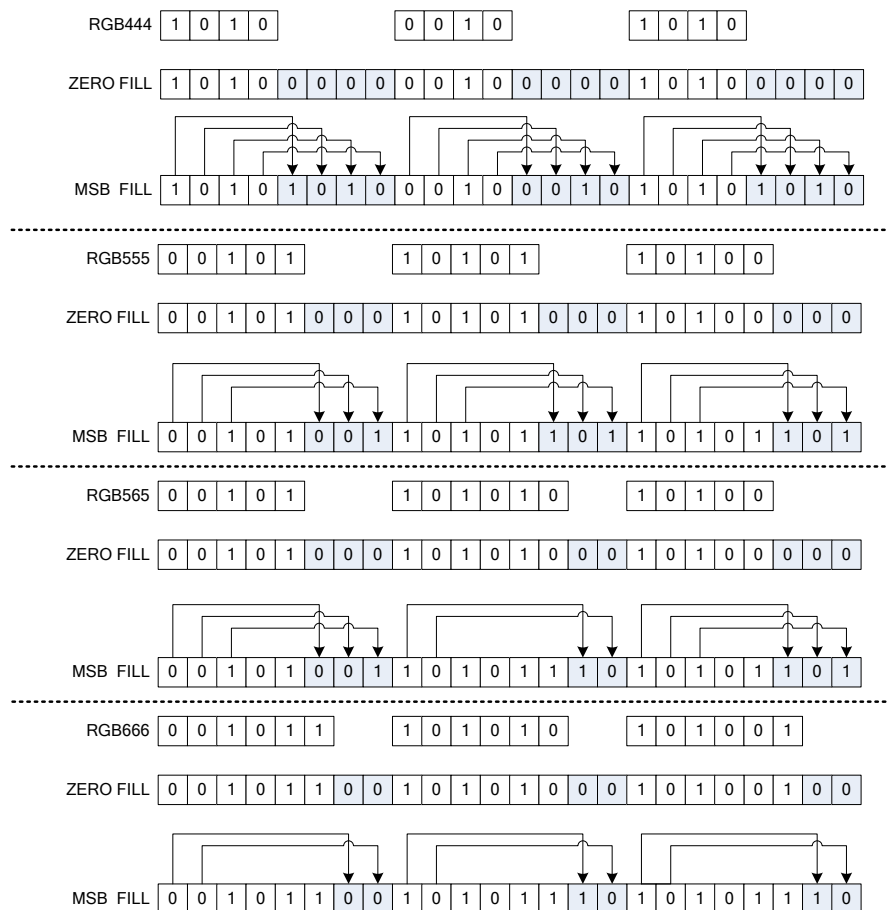


Figure 33.46 Bit Padding

RDMAPTS (RDMA Image PTS Register)

0x80020404 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BOTPTS															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TOPPTS															

Bit	Name	RW	Reset	Description
31:16	BOTPTS	R/W	0x0	Bottom Field Presentation Time Stamp Register
15:0	TOPPTS	R/W	0x0	Top Field Presentation Time Stamp Register

RDMASIZE (Image Size Information Register)

0x80020408 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0			HEIGHT<12:0>												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0			WIDTH<12:0>												

Bit	Name	RW	Reset	Description
28:16	HEIGHT	R/W	0x0	Height Information Register
12:0	WIDTH	R/W	0x0	Width Information Register

RDMABASE0 (Base Address for Each Images)

0x8002040C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE0<31:16>															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE0<15:0>															

Bit	Name	RW	Reset	Description
31:0	BASE0	R/W	0x0	The 1 st Base Address for each image If the image format is YUV separated type, it is the base address of Y.

RDMACADDR (Current Address for Each Images)

0x80020410 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CURR<31:16>															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CURR<15:0>															

Bit	Name	RW	Reset	Description
31:0	CURR	R	-	The working address for base address

RDMABASE1 (2nd Base Address for Each Images)

0x80020414 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE1<31:16>															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE1<15:0>															

Bit	Name	RW	Reset	Description
31:0	BASE1	R/W	0x0	The 2 nd base address for each image. If the image format is YUV separated / interleaved type, it is base address for U / UV, otherwise it is not used.

RDMABASE2 (3rd Base Address for Each Images)

0x80020418 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE2<31:16>															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE2<15:0>															

Bit	Name	RW	Reset	Description
31:0	BASE2	R/W	0x0	The 3 rd base address for each image. If the image format is YUV separated / interleaved type, it is base address for V, otherwise it is not used.

RDMAOFFS (Offset Information Registers for Each Image)

0x8002041C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OFFSET1<15:0>															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OFFSET0<15:0>															

Bit	Name	RW	Reset	Description
31:16	OFFSET1	R/W	0x0	The 2 nd offset information for each image. It is the address offset in U or V channel of FIFO (FIFO1,2) Invalid for Image 1 and Image 2
15:0	OFFSET0	R/W	0x0	The 1 st offset information for each image. It is the address offset in Y or RGB channel of FIFO (FIFO0).

RDMASCALE (SCALE Registers for Each Image)

0x80020420 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
													YSCALE		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
													XSCALE		

Bit	Name	RW	Reset	Description
18:16	YSCALE	R/W	0x0	Image Scaling in Vertical 3'b001 : SCALE2 3'b010 : SCALE3 3'b011 : SCALE4 3'b111 : SCALE8
2:0	XSCALE	R/W	0x0	Image scaling in Horizontal 3'b001 : SCALE2 3'b010 : SCALE3 3'b011 : SCALE4 3'b111 : SCALE8

Scale should be set prior to enabling RDMA. It should not change in the middle of the operation.

RDMAALPHA (Alpha Information Registers for Each Image)

0x80020424 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								A1							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								A0							

Bit	Name	RW	Reset	Description
23:16	A1	R/W	0x0	These bits should be same as value of "A0"
7:0	A0	R/W	0x0	These are used as alpha value for each image when ASEL bit is '0'.

RDMASTAT (RDMA Status Register)

0x80020428 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SFDLY											SBF	SDEOF	SEOFW	SBOTR	STOPR
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
									BOTR	TOPR	EOFW	UPDD	EOFF	EOFR	CFG

Bit	Name	RW	Reset	Description
31:28	SFDLY	R	-	Status of Delayed frame number
20	SBF	R	-	Status of BFIELD
19	SDEOF	R	-	Status of Device EOF
18	SEOFW	R	-	Status of RDMA EOF-WAIT
17	SBOTR	R	-	Status of BOTTOM FIELD READY
16	STOPR	R	-	Status of TOP FIELD READY
6	BOTR	R/W	0x0	Bottom Ready If write '1', this field is cleared.
5	TOPR	R/W	0x0	TOP Ready If write '1', this field is cleared.
4	EOFW	R/W	0x0	EOF-WAIT Rising (it means Frame End) If write '1', this field is cleared
3	UPDD	R/W	0x0	Register Update Done If write '1', this field is cleared.
2	EOFF	R/W	0x0	Device EOF Rising If write '1', this field is cleared.
1	EOFR	R/W	0x0	Device EOF Falling If write '1', this field is cleared.
0	CFG	R/W	0x0	Configuration Update If write '1', this field is cleared.

RDMAIRQMSK (RDMA Interrupt Mask Register)

0x8002042C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
									IBOTR	ITOPR	IEOFW	IUPDD	IEOFF	IEOFR	ICFG

Bit	Name	RW	Reset	Description
6	IBOTR	R/W	0x1	Bottom Ready IRQ Mask 0 : Enable 1 : Disable
5	ITOPR	R/W	0x1	TOP Ready IRQ MASK 0 : Enable 1 : Disable
4	IEOFW	R/W	0x1	EOF-WAIT Rising IRQ MASK(it means Frame End) 0 : Enable 1 : Disable
3	IUPDD	R/W	0x1	Register Update Done IRQ MASK 0 : Enable 1 : Disable
2	IEOFF	R/W	0x1	Device EOF Rising IRQ MASK 0 : Enable 1 : Disable
1	IEOFR	R/W	0x1	Device EOF Falling IRQ MASK 0 : Enable 1 : Disable
0	ICFG	R/W	0x1	Configuration Update IRQ MASK 0 : Enable 1 : Disable

RDMASBASE0 (RDMA Sync Base Address 0)

0x80020430 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SBADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SBADDR[15:00]															

Bit	Name	RW	Reset	Description
31:00	SBADDR	R/W	0x0	Sync Base Address

33.6.5 GWDMA and VWDMA

Table 33.16 GWDMA/VWDMA Register Map (Base Address = 0x80022800 + (0x100 * n))

Name	Address	Type	Reset	Description
WDMACTRL	0x00	R/W	0x00000400	WDMA Control Register
WDMARATE	0x04	R/W	0x00000000	WDMA Rate Control Register
WDMASIZE	0x08	R/W	0x00000000	WDMA Size Register
WDMABASE0	0x0C	R/W	0x00000000	WDMA Base Address 0 Register
WDMACADDR	0x10	R/W	0x00000000	WDMA Current Address Register
WDMABASE1	0x14	R/W	0x00000000	WDMA Base Address 1 Register
WDMABASE2	0x18	R/W	0x00000000	WDMA Base Address 2 Register
WDMAOFFS	0x1C	R/W	0x00000000	WDMA Offset Register
WDMABG	0x24	R/W	0x00000000	WDMA Back Ground Color Register
WDMAPTS	0x28	R/W	0x00000000	WDMA PTS Register
WDMADMAT0	0x2C	R/W	0x00000000	WDMA Dither Matrix 0 Register
WDMADMAT1	0x30	R/W	0x00000000	WDMA Dither Matrix 1 Register
WDMAROLL	0x38	R/W	0x00000000	WDMA Rolling Control Register
WDMASBASE	0x3C	R/W	0x00000000	WDMA Synchronized Base Address Register
WDMAIRQSTS	0x40	R/W	0x00000000	WDMA Interrupt Status Register
WDMAIRQMSK	0x44	R/W	0x000001FF	WDMA Interrupt Mask Register

WDMACtrl (WDMA Control Registers)

0x80022800 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTL		FU	IEN	DITHS	0		DITHE	CONT	SREQ	0					UPD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0				0		R2YMD		R2Y	BR	0		FMT			

Bit	Name	RW	Reset	Description										
31	INTL	R/W	0x0	Interlaced Image Indication Register 0 : Normal 1 : Interlaced Image										
29	FU	R/W	0x0	Field Update Enable 0 : Disable 1 : Enable										
28	IEN	R/W	0x0	Image Enable Register 0 : Disable 1 : Enable										
27	DITHS	R/W	0x0	Dither Select Register 0 : Normal (LSB toggle mode) 1 : Adder Mode										
24	DITHE	R/W	0x0	Dither Enable Register 0 : Disable 1 : Enable										
23	CONT	R/W	0x0	Continuous Mode Enable Register 0 : Disable (Frame-by-Frame mode) 1 : Enable (Continuous mode)										
22	SREQ	RW	0x0	Stop Request Enable Register 0 : Disable 1 : Enable										
16	UPD	R/W	0x0	Information Update Register 0 : Normal 1 : Information Update										
14:12	SWAP	R/W	0x0	RGB Swap Mode 3'b000 : R-G-B 3'b001 : R-B-G 3'b010 : G-R-B 3'b011 : G-B-R 3'b100 : B-R-G 3'b101 : B-G-R										
10:9	R2YMD	R/W	0x2	YUV-to-RGB Converter Mode Register <table><tr><th>R2YMD [10:9]</th><th>RGB to YCbCr Conversion Option</th></tr><tr><td>0</td><td>$* Y = 0.299 * R + 0.587 * G + 0.114 * B$ $* Cb = -0.172 * R - 0.339 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.428 * G - 0.083 * B + 128$ The range for “RGB” is 16 ~ 235, “Studio Color”. The result is “Studio Color” – Normally SDTV.</td></tr><tr><td>1</td><td>$* Y = 0.257 * R + 0.504 * G + 0.098 * B + 16$ $* Cb = -0.148 * R - 0.291 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.368 * G - 0.071 * B + 128$ The range for “RGB” is 0 ~ 255, “Computer System Color”. The result is “Studio Color” – Normally SDTV.</td></tr><tr><td>2</td><td>$* Y = 0.213 * R + 0.715 * G + 0.072 * B$ $* Cb = -0.117 * R - 0.394 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.464 * G - 0.047 * B + 128$ The range for “RGB” is 16 ~ 235, “Studio Color”.. The result is “Studio Color” – Normally HDTV.</td></tr><tr><td>3</td><td>$* Y = 0.183 * R + 0.614 * G + 0.062 * B + 16$ $* Cb = -0.101 * R - 0.338 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.399 * G - 0.040 * B + 128$ The range for “RGB” is 0 ~ 255, “Computer System Color”. The result is “Studio Color” – Normally HDTV.</td></tr></table>	R2YMD [10:9]	RGB to YCbCr Conversion Option	0	$* Y = 0.299 * R + 0.587 * G + 0.114 * B$ $* Cb = -0.172 * R - 0.339 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.428 * G - 0.083 * B + 128$ The range for “ RGB ” is 16 ~ 235, “ Studio Color ”. The result is “ Studio Color ” – Normally SDTV.	1	$* Y = 0.257 * R + 0.504 * G + 0.098 * B + 16$ $* Cb = -0.148 * R - 0.291 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.368 * G - 0.071 * B + 128$ The range for “ RGB ” is 0 ~ 255, “ Computer System Color ”. The result is “ Studio Color ” – Normally SDTV.	2	$* Y = 0.213 * R + 0.715 * G + 0.072 * B$ $* Cb = -0.117 * R - 0.394 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.464 * G - 0.047 * B + 128$ The range for “ RGB ” is 16 ~ 235, “ Studio Color ”.. The result is “ Studio Color ” – Normally HDTV.	3	$* Y = 0.183 * R + 0.614 * G + 0.062 * B + 16$ $* Cb = -0.101 * R - 0.338 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.399 * G - 0.040 * B + 128$ The range for “ RGB ” is 0 ~ 255, “ Computer System Color ”. The result is “ Studio Color ” – Normally HDTV.
R2YMD [10:9]	RGB to YCbCr Conversion Option													
0	$* Y = 0.299 * R + 0.587 * G + 0.114 * B$ $* Cb = -0.172 * R - 0.339 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.428 * G - 0.083 * B + 128$ The range for “ RGB ” is 16 ~ 235, “ Studio Color ”. The result is “ Studio Color ” – Normally SDTV.													
1	$* Y = 0.257 * R + 0.504 * G + 0.098 * B + 16$ $* Cb = -0.148 * R - 0.291 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.368 * G - 0.071 * B + 128$ The range for “ RGB ” is 0 ~ 255, “ Computer System Color ”. The result is “ Studio Color ” – Normally SDTV.													
2	$* Y = 0.213 * R + 0.715 * G + 0.072 * B$ $* Cb = -0.117 * R - 0.394 * G + 0.511 * B + 128$ $* Cr = 0.511 * R - 0.464 * G - 0.047 * B + 128$ The range for “ RGB ” is 16 ~ 235, “ Studio Color ”.. The result is “ Studio Color ” – Normally HDTV.													
3	$* Y = 0.183 * R + 0.614 * G + 0.062 * B + 16$ $* Cb = -0.101 * R - 0.338 * G + 0.439 * B + 128$ $* Cr = 0.439 * R - 0.399 * G - 0.040 * B + 128$ The range for “ RGB ” is 0 ~ 255, “ Computer System Color ”. The result is “ Studio Color ” – Normally HDTV.													
8	R2Y	R/W	0x0	YUV-to-RGB Converter Enable Register										

				0 : Disable 1 : Enable																																																						
7	BR	R/W	0x0	Bit-Reverse in Byte 0 : Little-endian 1 : Big-endian BR is only used when BPP is 1, 2, or 4 bpp																																																						
4:0	FMT	R/W	0x0	Image Format Register																																																						
				<table><tr><th>FMT [4:0]</th><th>Image Format</th></tr><tr><td>0</td><td>1bpp indexed color</td></tr><tr><td>1</td><td>2bpp indexed color</td></tr><tr><td>2</td><td>4bpp indexed color</td></tr><tr><td>3</td><td>8bpp indexed color</td></tr><tr><td>4 ~ 7</td><td>Reserved for future use</td></tr><tr><td>8</td><td>RGB332 – 1 bytes aligned : R[7:5],G[4:2],B[1:0]</td></tr><tr><td>9</td><td>RGB444 – 2 bytes aligned : A[15:12],R[11:8],G[7:4],B[3:0]</td></tr><tr><td>10</td><td>RGB565 – 2 bytes aligned : R[15:11],G[10:5],B[4:0]</td></tr><tr><td>11</td><td>RGB555 – 2 bytes aligned : A[15],R[14:10],G[9:5],B[4:0]</td></tr><tr><td>12</td><td>RGB888 – 4 bytes aligned : A[31:24],R[23:16],G[15:8],B[7:0]</td></tr><tr><td>13</td><td>RGB666 – 4 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]</td></tr><tr><td>14</td><td>RGB888 – 3 bytes aligned : B1[31:24],R0[23:16],G0[15:8],B0[7:0]</td></tr><tr><td>15</td><td>ARGB6666 – 3 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]</td></tr><tr><td>16</td><td>RESERVED</td></tr><tr><td>17 ~ 20</td><td>Reserved for future use</td></tr><tr><td>21</td><td>4:4:4 separate format - <u>valid for VWDMA only</u> If Y2R is enabled, this can be used for YUV format, otherwise for RGB separated format.</td></tr><tr><td>22</td><td>YCbCr 4:2:2 sequential format LSB [U/Y/V/Y] MSB</td></tr><tr><td>23</td><td>YCbCr 4:2:2 sequential format LSB [V/Y/U/Y] MSB</td></tr><tr><td>24</td><td>YCbCr 4:2:0 separated format - <u>valid for VWDMA only</u></td></tr><tr><td>25</td><td>YCbCr 4:2:2 separated format - <u>valid for VWDMA only</u></td></tr><tr><td>26</td><td>YCbCr 4:2:2 sequential format LSB [Y/U/Y/V] MSB</td></tr><tr><td>27</td><td>YCbCr 4:2:2 sequential format LSB [Y/V/Y/U] MSB</td></tr><tr><td>28</td><td>YCbCr 4:2:0 interleaved type 0 format - <u>valid for VWDMA only</u></td></tr><tr><td>29</td><td>YCbCr 4:2:0 interleaved type 1 format - <u>valid for VWDMA only</u></td></tr><tr><td>30</td><td>YCbCr 4:2:2 interleaved type 0 format - <u>valid for VWDMA only</u></td></tr><tr><td>31</td><td>YCbCr 4:2:2 interleaved type 1 format - <u>valid for VWDMA only</u></td></tr></table>	FMT [4:0]	Image Format	0	1bpp indexed color	1	2bpp indexed color	2	4bpp indexed color	3	8bpp indexed color	4 ~ 7	Reserved for future use	8	RGB332 – 1 bytes aligned : R[7:5],G[4:2],B[1:0]	9	RGB444 – 2 bytes aligned : A[15:12],R[11:8],G[7:4],B[3:0]	10	RGB565 – 2 bytes aligned : R[15:11],G[10:5],B[4:0]	11	RGB555 – 2 bytes aligned : A[15],R[14:10],G[9:5],B[4:0]	12	RGB888 – 4 bytes aligned : A[31:24],R[23:16],G[15:8],B[7:0]	13	RGB666 – 4 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]	14	RGB888 – 3 bytes aligned : B1[31:24],R0[23:16],G0[15:8],B0[7:0]	15	ARGB6666 – 3 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]	16	RESERVED	17 ~ 20	Reserved for future use	21	4:4:4 separate format - <u>valid for VWDMA only</u> If Y2R is enabled, this can be used for YUV format, otherwise for RGB separated format.	22	YCbCr 4:2:2 sequential format LSB [U/Y/V/Y] MSB	23	YCbCr 4:2:2 sequential format LSB [V/Y/U/Y] MSB	24	YCbCr 4:2:0 separated format - <u>valid for VWDMA only</u>	25	YCbCr 4:2:2 separated format - <u>valid for VWDMA only</u>	26	YCbCr 4:2:2 sequential format LSB [Y/U/Y/V] MSB	27	YCbCr 4:2:2 sequential format LSB [Y/V/Y/U] MSB	28	YCbCr 4:2:0 interleaved type 0 format - <u>valid for VWDMA only</u>	29	YCbCr 4:2:0 interleaved type 1 format - <u>valid for VWDMA only</u>	30	YCbCr 4:2:2 interleaved type 0 format - <u>valid for VWDMA only</u>	31	YCbCr 4:2:2 interleaved type 1 format - <u>valid for VWDMA only</u>
				FMT [4:0]	Image Format																																																					
				0	1bpp indexed color																																																					
				1	2bpp indexed color																																																					
				2	4bpp indexed color																																																					
				3	8bpp indexed color																																																					
				4 ~ 7	Reserved for future use																																																					
				8	RGB332 – 1 bytes aligned : R[7:5],G[4:2],B[1:0]																																																					
				9	RGB444 – 2 bytes aligned : A[15:12],R[11:8],G[7:4],B[3:0]																																																					
				10	RGB565 – 2 bytes aligned : R[15:11],G[10:5],B[4:0]																																																					
				11	RGB555 – 2 bytes aligned : A[15],R[14:10],G[9:5],B[4:0]																																																					
				12	RGB888 – 4 bytes aligned : A[31:24],R[23:16],G[15:8],B[7:0]																																																					
				13	RGB666 – 4 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]																																																					
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				15	ARGB6666 – 3 bytes aligned : A[23:18],R[17:12],G[11:6],B[5:0]																																																					
				16	RESERVED																																																					
				17 ~ 20	Reserved for future use																																																					
				21	4:4:4 separate format - <u>valid for VWDMA only</u> If Y2R is enabled, this can be used for YUV format, otherwise for RGB separated format.																																																					
				22	YCbCr 4:2:2 sequential format LSB [U/Y/V/Y] MSB																																																					
				23	YCbCr 4:2:2 sequential format LSB [V/Y/U/Y] MSB																																																					
				24	YCbCr 4:2:0 separated format - <u>valid for VWDMA only</u>																																																					
				25	YCbCr 4:2:2 separated format - <u>valid for VWDMA only</u>																																																					
				26	YCbCr 4:2:2 sequential format LSB [Y/U/Y/V] MSB																																																					
				27	YCbCr 4:2:2 sequential format LSB [Y/V/Y/U] MSB																																																					
				28	YCbCr 4:2:0 interleaved type 0 format - <u>valid for VWDMA only</u>																																																					
				29	YCbCr 4:2:0 interleaved type 1 format - <u>valid for VWDMA only</u>																																																					
				30	YCbCr 4:2:2 interleaved type 0 format - <u>valid for VWDMA only</u>																																																					
				31	YCbCr 4:2:2 interleaved type 1 format - <u>valid for VWDMA only</u>																																																					
				The “A”, “R”, “G” and “B” stand for the alpha value, red, green and blue color correspondingly. The alpha value to be applied is determined by ASEL bit. If ASEL is zero, the alpha value equals to “A” with extra bit extended. (Refer to Figure 33.8)																																																						

WDMARATE (WDMA Rate Control Registers)

0x80022804 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REN				0											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
			0				SYNCCMD	SEN							SYNCSSEL

Bit	Name	RW	Reset	Description
31	REN	R/W	0x0	Rate Control Enable 0 : Disable 1 : Enable
23:16	MAXRATE	R/W	0x0	Maximum Pixel Rate (per micro second)
11:9	SYNCCMD	R/W	0x0	For debug
8	SEN	R/W	0x0	RDMA Sync Enable 0 : Disable 1 : Enable
7:0	SYNCSSEL	R/W	0x0	RDMA Select for Sync

WDMASIZE (WDMA Size Registers)

0x80022808 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								HEIGHT							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								WIDTH							

Bit	Name	RW	Reset	Description
28:16	HEIGHT	R/W	0x0	Height Register
12:0	WIDTH	R/W	0x0	Width Register

WDMABASE0 (WDMA Base Address 0 Registers)

0x8002280C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE0															

Bit	Name	RW	Reset	Description
31:0	BASE0	R/W	0x0	The 1 st Base Address for each image If the image format is YUV separated type, it is the base address of Y.

WDMACADDR (WDMA Current Address 0 Registers)

0x80022810 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CADDR															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CADDR															

Bit	Name	RW	Reset	Description
31:0	CADDR	R	-	The working address for base address

WDMABASE1 (WDMA Base Address 1 Registers)

0x80022814 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE1															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE1															

Bit	Name	RW	Reset	Description
31:0	BASE1	R/W	0x0	The 2 nd base address for each image. If the image format is YUV separated / interleaved type, it is base address for U / UV, otherwise it is not used,.

WDMABASE2 (WDMA Base Address 2 Registers)

0x80022818 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE2															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE2															

Bit	Name	RW	Reset	Description
31:0	BASE2	R/W	0x0	The 3 rd base address for each image. If the image format is YUV separated / interleaved type, it is base address for V, otherwise it is not used,.

WDMAOFFS (WDMA Offset Registers)

0x8002281C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OFFSET1								OFFSET0							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bit	Name	RW	Reset	Description
31:16	OFFSET1	R/W	0x0	The 2 nd offset information for each image. It is the address offset in U or V channel of FIFO (FIFO1,2). Invalid for Image 1 and Image 2
15:0	OFFSET0	R/W	0x0	The 1 st offset information for each image. It is the address offset in Y or RGB channel of FIFO (FIFO0).

WDMABG (WDMA BackGround Color Registers)

0x80022824 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BG3								BG2							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BG1								BG0							

Bit	Name	RW	Reset	Description
31:24	BG3	R/W	0x0	Background Color 3 (Alpha)
23:16	BG2	R/W	0x0	Background Color 2 (Y/B)
15:8	BG1	R/W	0x0	Background Color 1 (Cb/G)
7:0	BG0	R/W	0x0	Background Color 0 (Cr/R)

WDMAPTS (WDMA PTS Registers)

0x80022828 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PTS															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bit	Name	RW	Reset	Description
15:0	PTS	R/W	0x0	Presentation Time Stamp Register

WDMADMAT0 (Dither Matrix 0)

0x8002282C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DITH13				DITH12				DITH11				DITH10			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DITH03				DITH02				DITH01				DITH00			

Bit	Name	RW	Reset	Description
30:28	DITH13	R/W	0x0	Dithering Pattern Matrix (1,3)
26:24	DITH12	R/W	0x0	Dithering Pattern Matrix (1,2)
22:20	DITH11	R/W	0x0	Dithering Pattern Matrix (1,1)
18:16	DITH10	R/W	0x0	Dithering Pattern Matrix (1,0)
14:12	DITH03	R/W	0x0	Dithering Pattern Matrix (0,3)
10:8	DITH02	R/W	0x0	Dithering Pattern Matrix (0,2)
6:4	DITH01	R/W	0x0	Dithering Pattern Matrix (0,1)
2:0	DITH00	R/W	0x0	Dithering Pattern Matrix (0,0)

WDMADMAT1 (Dither Matrix 1)

0x80022830 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DITH33				DITH32				DITH31				DITH30			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DITH23				DITH22				DITH21				DITH20			

Bit	Name	RW	Reset	Description
30:28	DITH33	R/W	0x0	Dithering Pattern Matrix (3,3)
26:24	DITH32	R/W	0x0	Dithering Pattern Matrix (3,2)
22:20	DITH31	R/W	0x0	Dithering Pattern Matrix (3,1)
18:16	DITH30	R/W	0x0	Dithering Pattern Matrix (3,0)
14:12	DITH23	R/W	0x0	Dithering Pattern Matrix (2,3)
10:8	DITH22	R/W	0x0	Dithering Pattern Matrix (2,2)
6:4	DITH21	R/W	0x0	Dithering Pattern Matrix (2,1)
2:0	DITH20	R/W	0x0	Dithering Pattern Matrix (2,0)

WDMAROLL (WDMA Rolling Control Register)

0x80022838 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ROL															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ROLLCNT															

Bit	Name	RW	Reset	Description
31	ROL	R/W	0x0	Rolling Enable Register 0 : Disable 1 : Enable
15:0	ROLLCNT	R/W	0x0	Rolling Count Register

WDMASBASE (WDMA Synchronized Base Address)

0x8002283C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SBASE0															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SBASE0															

Bit	Name	RW	Reset	Description
31: 0	SBASE0	R/W	0x0	Synchronized Base Address

WDMAIRQSTS (WDMA Interrupt Status Register)

0x80022840 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ST_EOF	ST_BF	ST_SEN													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
							SEOFF	SEOFr	EOFF	EOFR	ENF	ENR	ROLL	SREQ	UPD

Bit	Name	RW	Reset	Description
31	ST_EOF	R	-	Status of EOF 0 : Normal 1 : EOF (End-of-Frame)
30	ST_BF	R-	-	Status of Bottom Field 0 : Top Field 1 : Bottom Field
29	ST_SEN	R-	-	Status of Synchronized Enabled 0 : Normal 1 : Enable
8	SEOFF	RW	0x0	Falling the Sync EOF 0 : Normal 1 : Falling the Sync EOF If write '1', this field is cleared.
7	SEOFr	RW	0x0	Rising the Sync EOF 0 : Normal 1 : Rising the Sync EOF If write '1', this field is cleared.
6	EOFF	RW	0x0	Falling the EOF 0 : Normal 1 : Falling the EOF If write '1', this field is cleared.
5	EOFR	RW	0x0	Rising the EOF 0 : Normal 1 : Rising the EOF If write '1', this field is cleared.
4	ENF	RW	0x0	Falling the Frame Synchronized Enable 0 : Normal 1 : Falling the Frame Synchronized Enable If write '1', this field is cleared.
3	ENR	RW	0x0	Rising the Frame Synchronized Enable 0 : Normal 1 : Rising the Frame Synchronized Enable If write '1', this field is cleared.
2	ROLL	RW	0x0	Roll Interrupt 0 : Normal 1 : ROLL interrupt If write '1', this field is cleared.
1	SREQ	RW	0x0	STOP Request If write '1', this field is cleared.
0	UPD	RW	0x0	Register Update Done 0 : Normal 1 : Updated Register.

WDMAIRQMSK (WDMA Interrupt Mask Register)

0x80022844 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
							SEOFF	SEOFR	EOFF	EOFR	ENF	ENR	ROL	SREQ	UPD

Bit	Name	RW	Reset	Description
8	SEOFF	R/W	0x0	Sync EOF Falling Interrupt Masked 0 : Enable 1 : Disable
7	EOFR	R/W	0x0	Sync EOF Rising Interrupt Masked 0 : Enable 1 : Disable
6	EOFF	R/W	0x0	EOF Falling Interrupt Masked 0 : Enable 1 : Disable
5	EOFR	R/W	0x0	EOF Rising Interrupt Masked 0 : Enable 1 : Disable
4	ENF	R/W	0x0	Synchronized Enable Falling Interrupt Masked 0 : Enable 1 : Disable
3	ENR	R/W	0x0	Synchronized Enable Rising Interrupt Masked 0 : Enable 1 : Disable
2	ROL	R/W	0x0	Rolling Interrupt Masked 0 : Enable 1 : Disable
1	SREQ	R/W	0x0	Stop Request Interrupt Masked 0 : Enable 1 : Disable
0	UPD	R/W	0x0	Register Update Interrupt Masked 0 : Enable 1 : Disable

33.6.6 Scaler

Table 33.17 Scaler Register Map (Base Address = 0x80022000 + (0x100 * n))

Name	Address	Type	Reset	Description
SCCTRL	0x00	R/W	0x00000000	Scaler Control Register
SCSSIZE	0x08	R	-	Scaler Source Size Register
SCDSIZE	0x0C	R/W	0x00000000	Scaler Destination Size Register
SCOPOS	0x10	R/W	0x00000000	Scaler Output Position Register
SCOSIZE	0x14	R/W	0x00000000	Scaler Output Size Register
SCIRQSTS	0x18	R/W	0x00000000	Scaler Interrupt Status Register
SCIRQMSK	0x1C	R/W	0x0000000F	Scaler Interrupt Mask Register

SCCTRL (Scaler Control Register)

0x80022000 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
															UPD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
															BP

Bit	Name	RW	Reset	Description
16	UPD	R/W	0x0	Register Update
0	BP	R/W	0x0	Bypass (Same Size)

SCSSIZE (Scaler Source Size Register)

0x80022008 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HEIGHT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WIDTH															

Bit	Name	RW	Reset	Description
28:16	HEIGHT	R	-	Source Height Register
12:0	WIDTH	R	-	Source Width Register

The size information comes from RDMA or WMIX (at plug-in location of the Scaler Component).
If SCSSIZE is changed by RDMA or WMIX, you must set UPD field of SCCTRL register. If you do not set this bit, the changed size is not reflected.

SCDSIZE (Scaler Destination Size Register)

0x8002200C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HEIGHT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WIDTH															

Bit	Name	RW	Reset	Description
28:16	HEIGHT	R/W	0x0	Destination Height Register
12:0	WIDTH	R/W	0x0	Destination Width Register

SCOPOS (Scaler Output Position Register)

0x80022010 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
YPOS															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XPOS															

Bit	Name	RW	Reset	Description
28:16	YPOS	R/W	0x0	Y-Position in output image
12:0	XPOS	R/W	0x0	X-Position in output image

SCOSIZE (Scaler Output Size Register)

0x80022014 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HEIGHT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WIDTH															

Bit	Name	RW	Reset	Description
28:16	HEIGHT	R/W	0x0	Output Height Register
12:0	WIDTH	R/W	0x0	Output Width Register

SCIRQSTS (Scaler Interrupt Status Register)

0x80022018 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
												ERR	EOFF	EOFR	UPD

Bit	Name	RW	Reset	Description
3	ERR	R/W	0x0	Status of Scaler Error 0 : Normal 1 : Error If write '1', this field is cleared.
2	EOFF	R/W	0x0	Status of Falling EOF 0 : Normal 1 : Falling EOF If write '1', this field is cleared.
1	EOFR	R/W	0x0	Status of Rising EOF 0 : Normal 1 : Rising EOF If write '1', this field is cleared.
0	UPD	R/W	0x0	Status of Register Updated 0 : Normal 1 : Register updated.

SCIRQMSK (Scaler Interrupt Mask Register)

0x8002201C + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
												MERR	MEOFF	MEOFR	MUPD

Bit	Name	RW	Reset	Description
3	MERR	R/W	0x1	Scaler Error Interrupt Masked 0 : Enable 1 : Disable
2	MEOFF	R/W	0x1	Falling EOF Interrupt Masked 0 : Enable 1 : Disable
1	MEOFR	R/W	0x1	Rising EOF Interrupt Masked 0 : Enable 1 : Disable
0	MUPD	R/W	0x1	Status of Register Updated 0 : Enable 1 : Disable

33.6.7 WMIX (Overlay Mixer)

Table 33.18 WMIX Register Map (Base Address = 0x80021800 + (0x100 * n))

Name	Address	Type	Reset	Description
MCTRL	0x00	R/W	0x00000005	WMIX Control Register
MBG	0x04	R/W	0x00000000	WMIX Background Color Register
MSIZE	0x08	R/W	0x00000000	WMIX Size Register
MPOS0	0x10	R/W	0x00000000	WMIX Position 0 Register
MPOS1	0x14	R/W	0x00000000	WMIX Position 1 Register
MPOS2	0x18	R/W	0x00000000	WMIX Position 2 Register
MPOS3	0x1C	R/W	0x00000000	WMIX Position 3 Register
MKEY00	0x20	R/W	0x00000000	WMIX Chroma-Key 0 & Image 0 Register
MKEY01	0x24	R/W	0x00000000	WMIX Chroma-Key 1 & Image 0 Register
MKEY10	0x28	R/W	0x00000000	WMIX Chroma-Key 0 & Image 1 Register
MKEY11	0x2C	R/W	0x00000000	WMIX Chroma-Key 1 & Image 1 Register
MKEY20	0x30	R/W	0x00000000	WMIX Chroma-Key 0 & Image 2 Register
MKEY21	0x34	R/W	0x00000000	WMIX Chroma-Key 1 & Image 2 Register
MSTS	0x38	R/W	0x00000000	WMIX Status Register
MACON0	0x40	R/W	0x10622662	WMIX Alpha Control Register in layer 0
MCCON0	0x44	R/W	0x20C44CC4	WMIX Color Control Register in layer 0
MPORC0	0x48	R/W	0x00003018	WMIX ROP Control Register in layer 0
MPAT0	0x4C	R/W	0x00000000	WMIX ROP Pattern Register in layer 0
MACON1	0x50	R/W	0x10622662	WMIX Alpha Control Register in layer 1
MCCON1	0x54	R/W	0x20C44CC4	WMIX Color Control Register in layer 1
MPORC1	0x58	R/W	0x00003018	WMIX ROP Control Register in layer 1
MPAT1	0x5C	R/W	0x00000000	WMIX ROP Pattern Register in layer 1
MACON2	0x60	R/W	0x10622662	WMIX Alpha Control Register in layer 2
MCCON2	0x64	R/W	0x20C44CC4	WMIX Color Control Register in layer 2
MPORC2	0x68	R/W	0x00003018	WMIX ROP Control Register in layer 2
MPAT2	0x6C	R/W	0x00000000	WMIX ROP Pattern Register in layer 2

MCTRL (WMIX Control Register)

0x80021800 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
															UPD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
															OVP

Bit	Name	RW	Reset	Description																																																																																																																													
16	UPD	R/W	0x0	Update Register 0 : Normal 1 : Updated Register																																																																																																																													
4:0	OVP	R/W	0x5	Overlay Priority <table border="1"> <thead> <tr> <th>OVP</th><th>LAYER3</th><th>LAYER2</th><th>LAYER1</th><th>LAYER0</th></tr> </thead> <tbody> <tr><td>0</td><td>Image 3</td><td>Image 0</td><td>Image 1</td><td>Image 2</td></tr> <tr><td>1</td><td>Image 3</td><td>Image 1</td><td>Image 0</td><td>Image 2</td></tr> <tr><td>2</td><td>Image 3</td><td>Image 0</td><td>Image 2</td><td>Image 1</td></tr> <tr><td>3</td><td>Image 3</td><td>Image 2</td><td>Image 0</td><td>Image 1</td></tr> <tr><td>4</td><td>Image 3</td><td>Image 1</td><td>Image 2</td><td>Image 0</td></tr> <tr><td>5</td><td>Image 3</td><td>Image 2</td><td>Image 1</td><td>Image 0</td></tr> <tr><td>8</td><td>Image 2</td><td>Image 0</td><td>Image 1</td><td>Image 3</td></tr> <tr><td>9</td><td>Image 2</td><td>Image 1</td><td>Image 0</td><td>Image 3</td></tr> <tr><td>10</td><td>Image 2</td><td>Image 0</td><td>Image 3</td><td>Image 1</td></tr> <tr><td>11</td><td>Image 2</td><td>Image 3</td><td>Image 0</td><td>Image 1</td></tr> <tr><td>12</td><td>Image 2</td><td>Image 1</td><td>Image 3</td><td>Image 0</td></tr> <tr><td>13</td><td>Image 2</td><td>Image 3</td><td>Image 1</td><td>Image 0</td></tr> <tr><td>16</td><td>Image 1</td><td>Image 0</td><td>Image 2</td><td>Image 3</td></tr> <tr><td>17</td><td>Image 1</td><td>Image 2</td><td>Image 0</td><td>Image 3</td></tr> <tr><td>18</td><td>Image 1</td><td>Image 0</td><td>Image 3</td><td>Image 2</td></tr> <tr><td>19</td><td>Image 1</td><td>Image 3</td><td>Image 0</td><td>Image 2</td></tr> <tr><td>20</td><td>Image 1</td><td>Image 2</td><td>Image 3</td><td>Image 0</td></tr> <tr><td>21</td><td>Image 1</td><td>Image 3</td><td>Image 2</td><td>Image 0</td></tr> <tr><td>24</td><td>Image 0</td><td>Image 1</td><td>Image 2</td><td>Image 3</td></tr> <tr><td>25</td><td>Image 0</td><td>Image 2</td><td>Image 1</td><td>Image 3</td></tr> <tr><td>26</td><td>Image 0</td><td>Image 1</td><td>Image 3</td><td>Image 2</td></tr> <tr><td>27</td><td>Image 0</td><td>Image 3</td><td>Image 1</td><td>Image 2</td></tr> <tr><td>28</td><td>Image 0</td><td>Image 2</td><td>Image 3</td><td>Image 1</td></tr> <tr><td>29</td><td>Image 0</td><td>Image 3</td><td>Image 2</td><td>Image 1</td></tr> </tbody> </table>	OVP	LAYER3	LAYER2	LAYER1	LAYER0	0	Image 3	Image 0	Image 1	Image 2	1	Image 3	Image 1	Image 0	Image 2	2	Image 3	Image 0	Image 2	Image 1	3	Image 3	Image 2	Image 0	Image 1	4	Image 3	Image 1	Image 2	Image 0	5	Image 3	Image 2	Image 1	Image 0	8	Image 2	Image 0	Image 1	Image 3	9	Image 2	Image 1	Image 0	Image 3	10	Image 2	Image 0	Image 3	Image 1	11	Image 2	Image 3	Image 0	Image 1	12	Image 2	Image 1	Image 3	Image 0	13	Image 2	Image 3	Image 1	Image 0	16	Image 1	Image 0	Image 2	Image 3	17	Image 1	Image 2	Image 0	Image 3	18	Image 1	Image 0	Image 3	Image 2	19	Image 1	Image 3	Image 0	Image 2	20	Image 1	Image 2	Image 3	Image 0	21	Image 1	Image 3	Image 2	Image 0	24	Image 0	Image 1	Image 2	Image 3	25	Image 0	Image 2	Image 1	Image 3	26	Image 0	Image 1	Image 3	Image 2	27	Image 0	Image 3	Image 1	Image 2	28	Image 0	Image 2	Image 3	Image 1	29	Image 0	Image 3	Image 2	Image 1
OVP	LAYER3	LAYER2	LAYER1	LAYER0																																																																																																																													
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12	Image 2	Image 1	Image 3	Image 0																																																																																																																													
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29	Image 0	Image 3	Image 2	Image 1																																																																																																																													

MBG (WMIX Background Color Register)

0x80021804 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BG3								BG2							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BG1								BG0							

Bit	Name	RW	Reset	Description
31:24	BG3	R/W	0x0	Background Color 3 (Alpha)
23:16	BG2	R/W	0x0	Background Color 2 (Y/B)
15:8	BG1	R/W	0x0	Background Color 1 (Cb/G)
7:0	BG0	R/W	0x0	Background Color 0 (Cr/R)

MSIZE (WMIX Size Register)

0x80021808 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HEIGHT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WIDTH															

Bit	Name	RW	Reset	Description
28:16	HEIGHT	R/W	0x0	Image Height
12:0	WIDTH	R/W	0x0	Image Width

MPOS_k (WMIX Image k Position Register)

0x80021810 + (0x4 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			YPOS												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
			XPOS												

Bit	Name	RW	Reset	Description
28:16	YPOS	R/W	0x0	Y-Position in Image k
12:0	XPOS	R/W	0x0	X-Position in Image k

The range of k is 0≤k≤3. The k means image number. If n = 4 or 5 (WMIX_2to2 Component), the range of k is 0≤k≤1.

MKEY_{k0} (WMIX Image k Chroma-key Register)

0x80021820 + (0x8 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
KEN								KEYR							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KRYG								KEYB							

Bit	Name	RW	Reset	Description
31	KEN	R/W	0x0	Chroma-Key Enable 0 : Disable 1 : Enable
23:16	KEYR	R/W	0x0	Chroma-Key value of R channel
15: 8	KEYG	R/W	0x0	Chroma-Key value of G channel
7:0	KEYB	R/W	0x0	Chroma-Key value of B channel

The range of k is 0≤k≤2. The k means image number.

MKEY_{k1} (WMIX Image k Masked Chroma-key Register)

0x80021824 + (0x8 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								MKEYR							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MKRYG								MKEYB							

Bit	Name	RW	Reset	Description
23:16	MKEYR	R/W	0x0	Masked Chroma-Key value of R channel
15:8	MKEYG	R/W	0x0	Masked Chroma-Key value of G channel
7:0	MKEYB	R/W	0x0	Masked Chroma-Key value of B channel

The range of k is 0≤k≤2. The k means image number.

MSTS (WMIX Status Register)

0x80021838 + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DINTL	DBF	DEN	DEOFW	UINTL	UUPD	UEN	UEOF								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
											EOFF	EOFR	EOFWR	EOFWF	UPD

Bit	Name	RW	Reset	Description
31	DINTL	R	-	Download Interface Status 0 : Non-interlace 1 : Interlace
30	DBF	R	-	Download BFIELD Status 0 : TFIELD 1 : BFIELD
29	DEN	R	-	Download Enable Status 0 : Disable 1 : Enable
28	DEOFW	R	-	Download EOF_WAIT Status 0 : Normal 1 : EOF_WAIT
27	UINTL	R	-	Upload Interface Status 0 : Non-interlace 1 : Interlace
26	UUPD	R	-	Upload Update Status 0 : Normal 1 : Updated
25	UEN	R	-	Upload Enable Status 0 : Disable 1 : Enable
24	UEOF	R	-	Upload EOF Status 0 : Normal 1 : EOF (End-of-Frame)
4	EOFF	R/W	0x0	Falling EOF Status 0 : Normal 1 : Falling EOF Status If write '1', this field is cleared.
3	EOFR	R/W	0x0	Rising EOF Status 0 : Normal 1 : Rising EOF Status If write '1', this field is cleared.
2	EOFWR	R/W	0x0	Rising EOF_WAIT Status 0 : Normal 1 : Rising EOF_WAIT Status If write '1', this field is cleared.
1	EOFWF	R/W	0x0	Falling EOF_WAIT Status 0 : Normal 1 : Falling EOF_WAIT Status If write '1', this field is cleared.
0	UPD	R/W	0x0	Register Update Done 0 : Normal 1 : Updated If write '1', this field is cleared.

MACONk (WMIX Alpha Control Register)

0x80021840 + (0x10 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ACON1_11				ACON0_11				ACON1_10				ACON0_10			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ACON1_01				ACON0_01				ACON1_00				ACON0_00			

Field	Name	RW	Reset	Description
30:28	ACON1_11	R/W	0x1	ACON1 for Region C
26:24	ACON0_11	R/W	0x0	ACON0 for Region C
22:20	ACON1_10	R/W	0x6	ACON1 for Region B
18:16	ACON0_10	R/W	0x2	ACON0 for Region B
14:12	ACON1_01	R/W	0x2	ACON1 for Region D
10:8	ACON0_01	R/W	0x6	ACON0 for Region D (You can set to '0' or '1'. The other values are not meaningful.)
6:4	ACON1_00	R/W	0x6	ACON1 for Region A
2:0	ACON0_00	R/W	0x2	ACON0 for Region A This value is always 0 by hardware. (If you set to other values, the hardware operates by ACON0_00 == 0)

The range of k is 0≤k≤2. The k means mixer layer number.

- * "_00" – Region A : The area where Region B, Region C and Region D are excluded from the entire region.
- * "_10" – Region B : The area where Region C is excluded from Top Layer
- * "_01" – Region D : The area where Region C is excluded from Bottom Layer
- * "_11" – Region C : The common region between Top and Bottom Layer

Refer to Chapter 33.4.7 Overlay Mixer (Alpha-blending).

MCCONk (WMIX Color Control Register)

0x80021844 + (0x10 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCON1_11				CCON0_11				CCON1_10				CCON0_10			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCON1_01				CCON0_01				CCON1_00				CCON0_00			

Field	Name	RW	Reset	Description
31:28	CCON1_11	R/W	0x2	CCON1 for Region C
27:24	CCON0_11	R/W	0x0	CCON0 for Region C
23:20	CCON1_10	R/W	0xC	CCON1 for Region B
19:16	CCON0_10	R/W	0x4	CCON0 for Region B
15:12	CCON1_01	R/W	0x4	CCON1 for Region D
11:8	CCON0_01	R/W	0xC	CCON0 for Region D
7:4	CCON1_00	R/W	0xC	CCON1 for Region A
3:0	CCON0_00	R/W	0x4	CCON0 for Region A

The range of k is 0≤k≤2. The k means mixer layer number.

- * "_00" – Region A : The area where Region B, Region C and Region D are excluded from the entire region.
- * "_10" – Region B : The area where Region C is excluded from Top Layer
- * "_01" – Region D : The area where Region C is excluded from Bottom Layer
- * "_11" – Region C : The common region between Top and Bottom Layer

Refer to Chapter 33.4.7 Overlay Mixer (Alpha-blending).

MROPck (WMIX ROP Control Register)

0x80021848 + (0x10 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ALPHA1								ALPHA0							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ASEL				ROPMODE											

Field	Name	RW	Reset	Description
31 ~ 24	ALPHA1	R/W	0x0	Alpha Register 1
23 ~ 16	ALPHA0	R/W	0x0	Alpha Register 0
15 ~ 14	ASEL	R/W	0x3	Alpha Selection Register 0 : Image alpha → 0 ~ 255 (0% ~ 99.6%) 1 : Not defined 2 : Image alpha → 1 ~ 256 (0.39% ~ 100%) 3 : Image alpha → 0 ~ 127, 129 ~ 256 (0% ~ 49.6%, 50.3% ~ 100%) Caution: You must set this field to 0
4 ~ 0	ROPMODE	R/W	0x18	ROP Mode Register ROPMODE[4:3] = 2 → Global Alpha ROPMODE[4:3] = 3 → Pixel Alpha ROPMODE[4:3] = 0 → not defined

The range of k is $0 \leq k \leq 2$. The k means mixer layer number.

MPATk (WMIX ROP Pattern Register)

0x8002184C + (0x10 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								RED							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GREEN								BLUE							

Field	Name	RW	Reset	Description
23:16	RED	R/W	0x0	RED Color
15:8	GREEN	R/W	0x0	GREEN Color
7:0	BLUE	R/W	0x0	BLUE Color

The range of k is $0 \leq k \leq 2$. The k means mixer layer number.

33.6.8 Simple De-interlacer

The control register of the Simple De-interlacer does not exist.

33.6.9 VIQE (Video Quality Enhancement)

Table 33.19 VIQE Register Map (Base Address = 0x8002D000)

Name	Address	Type	Reset	Description
VCTRL	0x000	R/W	0x00200000	VIQE General Control Register
VSIZE	0x004	R/W	0x00000000	VIQE SIZE Register
VTIMEGEN	0x008	R/W	0x00000016	VIQE Time Generator Register
VINT	0x00C	R/W	0x00000000	VIQE Interrupt Register
VMISC	0x010	R/W	0x00000000	VIQE Miscellaneous Register

Name	Address	Type	Reset	Description
DI_BASE0	0x080	R/W	0x00000000	De-interlacer Base address 0 Register
DI_BASE1	0x084	R/W	0x00000000	De-interlacer Base address 1 Register
DI_BASE2	0x088	R/W	0x00000000	De-interlacer Base address 2 Register
DI_BASE3	0x08C	R/W	0x00000000	De-interlacer Base address 3 Register
DI_SIZE	0x090	R/W	0x00000000	De-interlacer Size Register
DI_OFFS	0x094	R/W	0x00000000	De-interlacer Offset Register
DI_CTRL	0x098	R/W	0x16000000	De-interlacer Control Register
DI_BASE0A	0x0A0	R/W	0x00000000	De-interlacer Base address 0 Register (A frame)
DI_BASE1A	0x0A4	R/W	0x00000000	De-interlacer Base address 1 Register (A frame)
DI_BASE2A	0x0A8	R/W	0x00000000	De-interlacer Base address 2 Register (A frame)
DI_BASE3A	0x0AC	R/W	0x00000000	De-interlacer Base address 3 Register (A frame)
DI_BASE0B	0x0B0	R/W	0x00000000	De-interlacer Base address 0 Register (B frame)
DI_BASE1B	0x0B4	R/W	0x00000000	De-interlacer Base address 1 Register (B frame)
DI_BASE2B	0x0B8	R/W	0x00000000	De-interlacer Base address 2 Register (B frame)
DI_BASE3B	0x0BC	R/W	0x00000000	De-interlacer Base address 3 Register (B frame)
DI_BASE0C	0x0C0	R/W	0x00000000	De-interlacer Base address 0 Register (C frame)
DI_BASE1C	0x0C4	R/W	0x00000000	De-interlacer Base address 1 Register (C frame)
DI_BASE2C	0x0C8	R/W	0x00000000	De-interlacer Base address 2 Register (C frame)
DI_BASE3C	0x0CC	R/W	0x00000000	De-interlacer Base address 3 Register (C frame)
DI_CUR_BASE0	0x0D0	R	-	De-interlacer Current Base 0 Register
DI_CUR_BASE1	0x0D4	R	-	De-interlacer Current Base 1 Register
DI_CUR_BASE2	0x0D8	R	-	De-interlacer Current Base 2 Register
DI_CUR_BASE3	0x0DC	R	-	De-interlacer Current Base 3 Register
DI_CUR_WADDR	0x0E0	R	-	De-interlacer Current WDMA address Register
DI_CUR_RADDR	0x0E4	R	-	De-interlacer Current RDMA address Register

Name	Address	Type	Reset	Description
DI_DEC0_MISC	0x100	R/W	0x00000304	De-interlacer Decomp. 0 Miscellaneous Register
DI_DEC0_SIZE	0x104	R/W	0x00000000	De-interlacer Decomp. 0 Size Register
DI_DEC0_STS	0x108	R	-	De-interlacer Decomp. 0 Status Register
DI_DEC0_CTRL	0x10C	R/W	0x00000100	De-interlacer Decomp. 0 Control Register
DI_DEC1_MISC	0x120	R/W	0x00000304	De-interlacer Decomp. 1 Miscellaneous Register
DI_DEC1_SIZE	0x124	R/W	0x00000000	De-interlacer Decomp. 1 Size Register
DI_DEC1_STS	0x128	R	-	De-interlacer Decomp. 1 Status Register
DI_DEC1_CTRL	0x12C	R/W	0x00000100	De-interlacer Decomp. 1 Control Register
DI_DEC2_MISC	0x140	R/W	0x00000304	De-interlacer Decomp. 2 Miscellaneous Register
DI_DEC2_SIZE	0x144	R/W	0x00000000	De-interlacer Decomp. 2 Size Register
DI_DEC2_STS	0x148	R	-	De-interlacer Decomp. 2 Status Register
DI_DEC2_CTRL	0x14C	R/W	0x00000100	De-interlacer Decomp. 2 Control Register
DI_COM0_MISC	0x160	R/W	0x00000000	De-interlacer Comp. 0 Miscellaneous Register
DI_COM0_NU	0x164	-	-	De-interlacer Comp. 0 Not used.
DI_COM0_STS	0x168	R	-	De-interlacer Comp. 0 Status Register
DI_COM0_CTRL	0x16C	R/W	0x00000000	De-interlacer Comp. 0 Control Register
DI_COM0_AC	0x170	R/W	0x00232A31	De-interlacer Comp.0 AC-length Register

Name	Address	Type	Reset	Description
DI_CTRL	0x280	R/W	0x00000000	De-interlacer Control Register
DI_ENGINE0	0x284	R/W	0x00000000	De-interlacer Engine 0 Register
DI_ENGINE1	0x288	R/W	0x00000000	De-interlacer Engine 1 Register
DI_ENGINE2	0x28C	R/W	0x00000000	De-interlacer Engine 2 Register
DI_ENGINE3	0x290	R/W	0x00000000	De-interlacer Engine 3 Register
DI_ENGINE4	0x294	R/W	0x00000000	De-interlacer Engine 4 Register
PD_THRES0	0x298	R/W	0x00000000	De-interlacer Pulldown Threshold 0 Register

Name	Address	Type	Reset	Description
PD_THRES1	0x29C	R/W	0x00000000	De-interlacer Pulldown Threshold 1 Register
PD_JUDDER	0x2A0	R/W	0x00000000	De-interlacer Pulldown Judder Register
PD_JUDDER_M	0x2A4	R/W	0x00000000	De-interlacer Pulldown Judder Misc. Register
DI_MISCC	0x2A8	R/W	0x00000000	De-interlacer Miscellaneous Control Register
DI_STATUS	0x2AC	R	-	De-interlacer Status Register
PD_STATUS	0x2B0	R	-	De-interlacer Pulldown Status Register
DI_REGION0	0x2B4	R/W	0x00000000	De-interlacer Region 0 Register
DI_REGION1	0x2B8	R/W	0x00000000	De-interlacer Region 1 Register
DI_INT	0x2BC	R/W	0x00000000	De-interlacer Interrupt Register
PD_SAW	0x2E0	R/W	0x00080510	De-interlacer SAW Register
DI_CSIZE	0x2E4	R/W		De-interlacer Core Size Register
DI_FMT	0x2E8	R/W	0x00000000	De-interlacer Format Register

VCTRL (VIQE Control Register)

0x8002D0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					CGPMD	CGDND	CGDID			CFGUPD	UPD		FCF	FCDUF	FCD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
							NHINTPL				DIEN	DNEN	GMEN	HIEN	HILUT

Bit	Name	RW	Reset	Description
26	CGDPM	R/W	0x0	0 : Enable 1 : Disable This field is "Clock Gate Disable for De-noiser". If this bit is set to 'HIGH', the clock gate blocks does not operate. Do not use this field
25	CGD3D	R/W	0x0	0 : Enable 1 : Disable This field is "Clock Gate Disable for De-noiser". If this bit is set to 'HIGH', the clock gate blocks does not operate. Do not use this field
24	CGDID	R/W	0x0	0 : Enable 1 : Disable This field is "Clock Gate Disable for De-interlacer". If this bit is set to 'HIGH', the clock gate blocks does not operate.
21	CFGUPD	R/W	0x1	0 : Do not set register 1 : Set register This field is "Set Registers" Normally, if a register is set, it is immediately applied to a module. However, there is a time gap when all registers are applied. This field is for reducing this time gap. For example, perhaps you set a certain register, but it is not immediately applied to the module you want. By setting this bit, modules can be simultaneously applied to registers that have been set at the same time.
20	UPD	R/W	0x0	0 : Update using EOF (End-of-Frame) 1 : Always update. This field is the "Update Method" Normally, the time of register update is detected by EOF. If you want to update register all the time, set this bit to 'HIGH'
18	FCF	R/W	0x0	0 : Normal 1 : Flush This field is the 'Format Converter Buffer Flush.'
17	FCDUF	R/W	0x0	0 : Enable 1 : Disable This field is the 'Format Converter Disable Using FMT', If this bit is set to 'HIGH', the 4:2:0 to 4:2:2 format converter depends on the input data format without FCD field [bit 16] of this register. If the input data format is 4:2:0, this converter automatically operates. It does not operate for the other formats. If this bit is 'LOW', you must set FCD field by input data format.
16	FCD	R/W	0x0	0 : Enable 1 : Disable This field is the 'Format Converter Disable' for converting 4:2:0 to 4:2:2 or 4:4:4 formats. If the format of input data is 4:2:2, you must set this field to 'HIGH'. (This field is related FCDUF [bit 17] of this register.)
8	NHINTPL	R/W	0x0	0 : Enable 1 : Disable This field is the 'Not-Horizontal Interpolation'. This bit is set, if you do not want to interpolate chrominance data when converts 4:2:0 or 4:2:2 to 4:4:4 format.
4	DIEN	R/W	0x0	0 : Disable 1 : Enable This bit enables de-interlacer.
3	DNEN	R/W	0x0	0 : Disable 1 : Enable This bit enables de-noiser Do not use this field
2	GMEN	R/W	0x0	0 : Disable 1 : Enable This bit enables gamut mapper. Do not use this field
1	HIEN	R/W	0x0	0 : Disable 1 : Enable This bit enables histogram generator.

				Do not use this field
0	HILUT	R/W	0x0	0 : Disable 1 : Enable Do not use this field This bit enables the main PATH Controller to check the status of Histogram Generator in order to dynamically insert the Histogram Generator into pipeline path. When this bit is only enabled, the interpolation engine or LUT in Histogram Generator can be used.

VSIZ (VIQE Size Register)

0x8002D004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					HEIGHT										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					WIDTH										

Bit	Name	RW	Reset	Description
26:16	HEIGHT	R/W	0x0	Input Image Height by pixel
10:0	WIDTH	R/W	0x0	Input Image Width by pixel

This size register is related to TSDU field [bit 1] of VMISC register [0x8002D010]. If the TSDU field is set to 'HIGH', this register is used by VIQE module. (So, these registers are meaningless.) But set to 'LOW', the information of size comes from VIOC module.

VTIMEGEN (VIQE Time Generator Register)

0x8002D008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					Y2RMD		Y2REN		H2H						

Bit	Name	RW	Reset	Description										
10:9	Y2RMD	R/W	0x0	Refer to the table below.										
				<table><tr><th>Y2RMD [10:9]</th><th>YCbCr to RGB Conversion Option</th></tr><tr><td>0</td><td>$*R = Y + 1.371 * (Cr - 128)$ $*G = Y - 0.336 * (Cb - 128) - 0.698 * (Cr - 128)$ $*B = Y + 1.732 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Studio Color” – Normally SDTV.</td></tr><tr><td>1</td><td>$*R = 1.164 * (Y - 16) + 1.596 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.391 * (Cb - 128) - 0.813 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.018 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Computer System Color” – Normally SDTV.</td></tr><tr><td>2</td><td>$*R = Y + 1.540 * (Cr - 128)$ $*G = Y - 0.183 * (Cb - 128) - 0.459 * (Cr - 128)$ $*B = Y + 1.816 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Studio Color” – Normally HDTV.</td></tr><tr><td>3</td><td>$*R = 1.164 * (Y - 16) + 1.793 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.213 * (Cb - 128) - 0.534 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.115 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “Studio Color”. The result is “Computer System Color” – Normally HDTV.</td></tr></table>	Y2RMD [10:9]	YCbCr to RGB Conversion Option	0	$*R = Y + 1.371 * (Cr - 128)$ $*G = Y - 0.336 * (Cb - 128) - 0.698 * (Cr - 128)$ $*B = Y + 1.732 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “ Studio Color ”. The result is “ Studio Color ” – Normally SDTV.	1	$*R = 1.164 * (Y - 16) + 1.596 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.391 * (Cb - 128) - 0.813 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.018 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “ Studio Color ”. The result is “ Computer System Color ” – Normally SDTV.	2	$*R = Y + 1.540 * (Cr - 128)$ $*G = Y - 0.183 * (Cb - 128) - 0.459 * (Cr - 128)$ $*B = Y + 1.816 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “ Studio Color ”. The result is “ Studio Color ” – Normally HDTV.	3	$*R = 1.164 * (Y - 16) + 1.793 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.213 * (Cb - 128) - 0.534 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.115 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “ Studio Color ”. The result is “ Computer System Color ” – Normally HDTV.
				Y2RMD [10:9]	YCbCr to RGB Conversion Option									
				0	$*R = Y + 1.371 * (Cr - 128)$ $*G = Y - 0.336 * (Cb - 128) - 0.698 * (Cr - 128)$ $*B = Y + 1.732 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “ Studio Color ”. The result is “ Studio Color ” – Normally SDTV.									
				1	$*R = 1.164 * (Y - 16) + 1.596 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.391 * (Cb - 128) - 0.813 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.018 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “ Studio Color ”. The result is “ Computer System Color ” – Normally SDTV.									
2	$*R = Y + 1.540 * (Cr - 128)$ $*G = Y - 0.183 * (Cb - 128) - 0.459 * (Cr - 128)$ $*B = Y + 1.816 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “ Studio Color ”. The result is “ Studio Color ” – Normally HDTV.													
3	$*R = 1.164 * (Y - 16) + 1.793 * (Cr - 128)$ $*G = 1.164 * (Y - 16) - 0.213 * (Cb - 128) - 0.534 * (Cr - 128)$ $*B = 1.164 * (Y - 16) + 2.115 * (Cb - 128)$ The range for “YCbCr” is 16 ~ 235, “ Studio Color ”. The result is “ Computer System Color ” – Normally HDTV.													
8	Y2REN	R/W	0x0	0 : Disable 1 : Enable This field is “YUV2RGB format Conv. Enable”. When this bit is set to ‘HIGH’, YUV toRGB converter is enabled.										
7:0	H2H	R/W	0x16	This field is the pixel count between the ending pixel of the line and the first pixel of the next line.										

VINT (VIQE Interrupt Register)

0x8002D00C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					MPMI	MDNI	MDII						PMI	DNI	DII

Bit	Name	RW	Reset	Description
10	MPMI	R/W	0x0	0 : Interrupt disable 1 : Interrupt enable Pixel-mapper interrupt enable bit (DO NOT USE)
9	MDNI	R/W	0x0	0 : Interrupt disable 1 : Interrupt enable De-noiser interrupt enable bit (DO NOT USE)
8	MDII	R/W	0x0	0 : Interrupt disable 1 : Interrupt enable De-interlacer interrupt enable bit
2	PMI	R/W	0x0	0 : Normal 1 : Interrupt (If write to '1', this field is cleared.) Pixel-mapper interrupt bit. (DO NOT USE)
1	DNI	R/W	0x0	0 : Normal 1 : Interrupt (If write to '1', this field is cleared.) De-noiser interrupt bit. (DO NOT USE)
0	DII	R/W	0x0	0 : Normal 1 : Interrupt (if write to '1', this field is cleared.) De-interlacer interrupt bit

VMISC (VIQE Miscellaneous Register)

0x8002D010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
												GED	SDDU	TSDU	GENDU

Bit	Name	RW	Reset	Description
3	GED	R/W	0x0	0 : Generate EOF of VIQE signal 1 : Don't generate EOF of VIQE signal This field is "Do not generate EOF Signal".
2	SDDU	R/W	0x0	0 : Use the stream information for de-interlacer 1 : Don't use the stream information for de-interlacer This field is "Stream Information. Do not use for de-interlacer". If stream mode (from VIOC), the enable signal of de-interlacer is get from VIOC. If you want to operate de-interlacer, DIEN field of VCTRL register [0x8002D000] and stream information should be set to 'HIGH'. If this bit is set to 'LOW', VIQE is ignored in stream information.
1	TSDU	R/W	0x0	0 : Use 1 : Don't Use This field is "Size Do not use". If this bit set to 'LOW', the size information of VIQE comes from VIOC modules. For the others, the size information is gotten from VSIZE register [0x8002D004].
0	GENDU	R/W	0x0	0 : Use 1 : Don't Use This field is "Global Enable Do not use". If this bit set to 'LOW', the global enable of VIQE comes from VIOC modules. For the others, the global enable of VIQE is related the sub modules of VIQE without the enable of VIOC modules.

DI_BASE n (De-interlacer Base n Register)

0x8002D080 + (0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								BASE[31:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE[15:00]															

Bit	Name	RW	Reset	Description
31:0	BASE	R/W	0x0	De-interlacer Base n Address. (0 ≤ n ≤ 3)

The DMA for de-interlacer has to allocate 4 areas of memory for temporal de-interlace mode. If spatial-only mode, these registers are ignored. (So the enable bit of de-interlacer DMA is not set to 'HIGH').

DI_SIZE (De-interlacer Size Register)

0x8002D090

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					HEIGHT										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					WIDTH										

Bit	Name	RW	Reset	Description
26:16	HEIGHT	R/W	0x0	Input Image Height by pixel
10:00	WIDTH	R/W	0x0	Input Image Width by pixel

This size register is related to TSDU field [bit 0] of DI_CTRL register [0x8002D098]. The TSDU field is set to 'HIGH', this register is used de-interlacer module. (So, these registers are meaningless.) If set to 'LOW', the information of size is from VIOC module.

DI_OFFS (De-interlacer Offset Register)

0x8002D094

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OFFS1															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OFFS0															

Bit	Name	RW	Reset	Description
31:16	OFFS1	R/W	0x0	Address offset in chrominance by pixel
15:00	OFFS0	R/W	0x0	Address offset in luminance by pixel

Address offset means that the start line point to the next start line point.
Do not write any data since this register is not used by de-interlacer DMA.

DI_CTRL (De-interlacer Control Register)

0x8002D098

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
H2H													FRD	CFGUPD	EN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FRMNUM						UVINTPL							SDDU	TSDU	

Bit	Name	RW	Reset	Description
31:24	H2H	R/W	0x16	This field is the pixel count between the ending pixel of the line and the first pixel of the next line.
18	FRD	R/W	0x0	0 : Enable 1 : Disable This field is "Monitoring Frame Rate". De-interlacer operates while the field broken signal occurrence due to mismatch of frame rates between source and display in field information, is monitored.
17	CFGUPD	R/W	0x0	0 : Do not set register 1 : Set register This field is "Set Registers" Normally, if a register is set, it is immediately applied to module. However, there is time gap when each of registers is all applied. This field is for reducing this time gap. For example, even though you set a certain register, the set register is not immediately applied to a module that you want. However, the modules can be simultaneously applied to registers which have been set at the same time by setting this bit.
16	EN	R/W	0x0	0 : Disable 1 : Enable This field is "De-interlacer DMA Enable". If you want to operate temporal de-interlace mode, this bit is set to 'HIGH'. For other modes, set to 'LOW'.
9:8	FRMNUM	R.W	0x0	Operation Frame Number (Maximum 4 Frames) If a value over 2 is set, an address post 2nd address should be set. (DI_BASE n A DI_BASE n B DI_BASE n C register)
2	UVINTPL	R/W	0x0	For Debug. Caution: Do not use this field
1	SDDU	R/W	0x0	0 : Use the stream information for de-interlacer 1 : Do not use the stream information for de-interlacer This field is "Stream Info. Do not use for de-interlacer". If stream mode (from VIOC), the enable signal of de-interlacer comes from VIOC. To operate de-interlacer, DIEN field of VCTRL register [0x8002D000] and stream information is set to 'HIGH'. If this bit is set to 'LOW', VIQE is ignored in stream information.
0	TSDU	R/W	0x0	0 : Use 1 : Do not use This field is "Size Do not use". If this bit is set to 'LOW', the size information of de-interlacer comes from VIOC modules. For the others, the size information is gotten from DI_SIZE register [0x8002D090].

DI_BASE n A (De-interlacer Base n Register (A frame))

0x8002D0A0 + (0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE[15:00]															

Bit	Name	RW	Reset	Description
31:0	BASE	R/W	0x0	De-interlacer Base n Address. (0 ≤ n ≤ 3)

The DMA for de-interlacer has to allocate 4 areas of memory for temporal de-interlace mode. But If spatial-only mode, these registers are ignored. (Therefore, the enable bit of de-interlacer DMA is not set to 'HIGH').

DI_BASE n B (De-interlacer Base n Register (B frame))

0x8002D0B0 + (0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE[15:00]															

Bit	Name	RW	Reset	Description
31:0	BASE	R/W	0x0	De-interlacer Base n Address. (0 ≤ n ≤ 3)

The DMA for de-interlacer has to allocate 4 areas of memory for temporal de-interlace mode. But If spatial-only mode, these registers are ignored. (Therefore, the enable bit of de-interlacer DMA is not set to 'HIGH').

DI_BASE n C (De-interlacer Base n Register (C frame))

0x8002D0C0 + (0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BASE[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE[15:00]															

Bit	Name	RW	Reset	Description
31:0	BASE	R/W	0x0	De-interlacer Base n Address. (0 ≤ n ≤ 3)

The DMA for de-interlacer has to allocate 4 areas of memory for temporal de-interlace mode. But If spatial-only mode, these registers are ignored. (Therefore, the enable bit of de-interlacer DMA is not set to 'HIGH').

DI_CUR_BASE n (De-interlacer Current Base n Register)

0x8002D0D0 + (0x4*n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CUR_BASE[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CUR_BASE[15:00]															

Bit	Name	RW	Reset	Description
31:0	CUR_BASE	R	-	De-interlacer Current Base n Address. (0 ≤ n ≤ 3)

DI_CUR_WADDR (De-interlacer Current WAddress Register)

0x8002D0E0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CUR_WADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CUR_WADDR[15:00]															

Bit	Name	RW	Reset	Description
31:0	CUR_WADDR	R	-	De-interlacer WDMA Current Address.

DI_CUR_RADDR (De-interlacer Current RAddress Register)

0x8002D0E4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CUR_RADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CUR_RADDR[15:00]															

Bit	Name	RW	Reset	Description
31:0	CUR_RADDR	R	-	De-interlacer RDMA Current Address.

DI_DECn_MISC (De-interlacer Decomp. n MISC. Register)

0x8002D100+(0x20*n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FMT				DEC_DIV				SF				ECR	FLUSH	DE	

Bit	Name	RW	Reset	Description
16:12	FMT	R/W	0x0	0 : YUV 4:2:0 format 1 : YUV 4:2:2 format Others : Not used.
9:8	DEC_DIV	R/W	0x3	The divisor value of stream decompressor.
3	SF	R/W	0x0	0 : Normal 1 : Stream size is set to maximum value. If this bit is set to 'HIGH', the stream size of frame-stream is changed minimum value (-1). For the other modes, the size gets from frame-stream information.
2	ECR	R/W	0x1	This bit must set to 'HIGH'. Do not set to 'LOW'.
1	FLUSH	R/W	0x0	0 : Normal 1 : Flush If this bit is set to 'HIGH', the FIFO of frame decompressor is flushed.
0	DE	R/W	0x0	0 : Disable 1 : Enable This field is "Detect EndStream". If this bit is set to 'HIGH', the frame decompressor detects the end-stream in streaming data.

In case of write mode, the frame decompressor does not need FMT information. (This is because frame-stream has FMT information.) However, if start-stream information is not detected in the frame-stream, frame decoding cannot be done. If start-stream information is not detected for a certain amount of time, frame decoding is done with FMT information written here. In read mode, FMT information obtained from frame-stream is read.

DI_DECn_SIZE (De-interlacer Decomp. n Size Register)

0x8002D104+(0x20*n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					HEIGHT										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					WIDTH										

Bit	Name	RW	Reset	Description
26:16	HEIGHT	R/W	0x0	- Write • Image Height divided by 2 - Read • Image width information is gotten from frame stream information. • Image Height divided by 2 (Because the height size of field is one divided by 2 of frame in interlace mode.)
10:00	WIDTH	R/W	0x0	- Write • Image Width - Read • Image width information is gotten from frame stream information.

In case of write mode, the frame decompressor does not need size information. (This is because frame-stream has size information.) However, if start-stream information is not detected in the frame-stream, frame decoding cannot be done. If start-stream information is not detected for a certain amount of time, frame decoding is done with size information written here and FMT field of DI_DECn_MISC (0x8002D100). In read mode, the size information obtained from frame-stream is read.

DI_DECn_STS (De-interlacer Decomp. n Status Register)

0x8002D108+(0x20*n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DEC_STS				DEC_ER	HR_ER	EOFO	EOFI	EMPTY				FULL			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WIDTH_DWALIGN															

Bit	Name	RW	Reset	Description
31:28	DEC_STS	R	-	Status from decoder core
27	DEC_ER	R	-	0 : Normal 1 : Error from decoder core
26	HR_ER	R	-	0 : Normal 1 : Error from decoding header
25	EOFO	R	-	Output EOF form frame decompressor
24	EOFI	R	-	Input EOF form VIOC
23:20	EMPTY	R	-	Empty Status in Buffer
19:16	FULL	R	-	Full Status in Buffer
15:0	WIDTH_DWALIGN	R	-	The width size based double-word aligned. This information is gotten from frame stream.

DI_DECn_CTRL (De-interlacer Decomp. n Control Register)

0x8002D10C+(0x20*n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN								STS							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HRER_EN							ER_CK	SELECT							

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	0 : Disable 1 : Enable When this bit is set to 'HIGH', the frame decompressor operates.
23:16	STS	R	-	The status from the frame decompressor (for debugging)
9	HRER_EN	R/W	0x0	Check error from the stream header. When this bit is set to 'HIGH', the frame decompressor operates in spite of non-detecting start-stream information.
8	ER_CK	R/W	0x0	Error check to decoder core. When this bit is set to 'HIGH', the decoder core operates in spite of generating ERROR.
7:0	SELECT	R/W	0x0	Not used.

DI_COM0_MISC (De-interlacer Comp. 0 Misc. Register)

0x8002D160

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FMT				ENC_DIV				SF				FLUSH		DE	

Bit	Name	RW	Reset	Description
16:12	FMT	R/W	0x0	0 : YUV 4:2:0 format 1 : YUV 4:2:2 format Others : Not used.
9:8	ENC_DIV	R/W	0x0	The divisor value of stream decompressor.
3	SF	R/W	0x0	0 : Normal 1 : Stream size is set to maximum value. If this bit is set to 'HIGH', the stream size of frame-stream changes to maximum value (-1). In other modes, the size comes from frame encoder core.
1	FLUSH	R/W	0x0	0 : Normal 1 : Flush When this bit is set to 'HIGH', the FIFO of frame compressor is flushed.
0	DE	R/W	0x0	Not used

DI_COM0_STS (De-interlacer Comp. 0 Status Register)

0x8002D168

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
						EOFO	EOFI	EMPTY				FULL			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bit	Name	RW	Reset	Description
25	EOFO	R	-	Output EOF form frame compressor
24	EOFI	R	-	Input EOF form VIOC
23:20	EMPTY	R	-	Empty Status in Buffer
19:16	FULL	R	-	Full Status in Buffer

DI_COM0_CTRL (De-interlacer Comp. 0 Control Register)

0x8002D16C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN								STS							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								SELECT							

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	0 : Disable 1 : Enable When this bit is set to 'HIGH', the frame compressor operates.
23:6	STS	R	-	The status from the frame compressor (for debugging)
7:0	SELECT	R/W	0x0	Not used.

DI_COM0_AC (De-interlacer Comp. 0 AC Length Register)

0x8002D170

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
										K2_AC					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
K1_AC								K0_AC							

Bit	Name	RW	Reset	Description
21:16	K2_AC	R/W	0x23	K2 AC Length from frame encoder core
13:8	K1_AC	R/W	0x2A	K1 AC Length from frame encoder core
5:0	K0_AC	R/W	0x31	K0 AC Length from frame encoder core

DI_CTRL (De-interlacer Main Control Register)

0x8002E280

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CLRIR	FLSFF	BYPASS	RESERVED												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PDCLFI				PDPF		PDJUD	PDEN	YDM		MRSP	MRTM		JT	JSP	JS

Bit	Name	RW	Reset	Description
31	CLRIR	R/W	0x0	- Clear Internal Registers 1'b0 = No meaning 1'b1 = Clear Internal Registers in De-interlacer
30	FLSFF	R/W	0x0	- Flush Synchronizer FIFO 1'b0 = No meaning 1'b1 = Flush Synchronizer FIFO
29	BYPASS	R/W	0x0	- Bypass Register 1'b0 = Normal process 1'b1 De-interlacer Bypass Enable. If this bit is set, the input data to the de-interlacer will be intact through hardware block.
15	PDCLFI	R/W	0x0	- Clear Internal Frame Index Register 1'b0 = Normal process 1'b1 = Clear Internal Frame Index Register in Pulldown Detector
11	PDPF	R/W	0x0	- Use Prevention-Flag in Pulldown Detector 1'b0 = Use Prevention-Flag algorithm in Pulldown Detector. The Prevention-Flag algorithm can reduce the possibility of miss-detection in film mode checking. 1'b1 = Do not use Prevention-Flag Algorithm.
9	PDJUD	R/W	0x0	- User Judder Detection in Pulldown Detector 1'b0 = Do not use judder detection in Pulldown detector. 1'b1 = Use judder detection in Pulldown detector. The judder in the reverse pull-down processing means the artifacts from bad-edit or flowing text that be edited additionally at the normal frame rate. These artifacts can be removed using the judder detection algorithm in the VIQE.
8	PDEN	R/W	0x0	- Pulldown Detector Enable 1'b0 = Pulldown detector is disabled. 1'b1 = Pulldown detector is enabled. Pulldown detector can be used for detecting the film source, which is transferred into digital format using 3:2 pull-down method.
7	YDM	R/W	0x0	- YD Mode 1'b0 = Disable 1'b1 = Enable
6	GTHSJ	R/W	0x0	- Generate Jaggy Checker Threshold 1'b0 = Use only pre-defined threshold for jaggy checker. 1'b1 = Use generated threshold for jaggy checker. This is optional feature that the thresholds for jaggy checker can be generated adaptively by the specific hardware. This feature is strongly recommended.
5	MRSP	R/W	0x0	- Pixel Output Range Enable in Spatial Processing 1'b0 = Do not use output range limitation algorithm. 1'b1 = Use output range limitation algorithm. This option can be used only when the spatial de-interlacer only mode(DISP bit in CTRL register) is off.
4	MRTM	R/W	0x0	- Pixel Output Range Enable in Temporal Processing 1'b0 = Do not use output range limitation algorithm. 1'b1 = Use output range limitation algorithm. This option can be used only when the spatial de-interlacer only mode(DISP bit in CTRL register) is off.
2	JT	R/W	0x0	- Temporal Jaggy check 1'b0 = Disable 1'b1 = Enable
1	JSP	R/W	0x0	- Temporal Jaggy check 1'b0 = Disable 1'b1 = Enable
0	JS	R/W	0x0	- Spatio Jaggy check 1'b0 = Disable 1'b1 = Enable

DI_ENGINE0 (De-noiser Engine 0 Register)

0x8002D284

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMTPXLC								DMTSADC							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMTPXL								DMTSAD							

Bit	Name	RW	Reset	Description
31:24	DMTSADC	R/W	0x0	- SAD Threshold in Motion Detection for Chroma Threshold for SAD value in Motion Detection for Chrominance
23:16	DMTPXLC	R/W	0x0	- Pixel Threshold in Motion Detection for Chroma Threshold for Pixel value in Motion Detection for Chrominance
15:8	DMTPXL	R/W	0x0	- Pixel Threshold in Motion Detection Threshold for Pixel value in Motion Detection for Luminance
7:0	DMTSAD	R/W	0x0	- SAD Threshold in Motion Detection Threshold for SAD value in Motion Detection for Luminance

DI_ENGINE1 (De-interlacer Engine 1 Register)

0x8002D288

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
THSJMAX								THSJMIN							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LRT		LRL				LRD		DIRLT				GLD	JDH		

Bit	Name	RW	Reset	Description
31:24	THSJMAX	R/W	0x0	- Maximum Threshold in Jaggy Threshold Generator This value means the maximum boundary for the jaggy threshold generator.
23:16	THSJMIN	R/W	0x0	- Minimum Threshold in Jaggy Threshold Generator This value means the minimum boundary for the jaggy threshold generator.
15:14	LRT	R/W	0x0	- Horizontal Reference Type Normally, it should be zero.
13:9	LRL	R/W	0x2	- Horizontal Reference Length Normally, it should be 2.
8	LRD	R/W	0x0	- Horizontal Reference Disable 1'b0 = Enable 1'b1 = Disable
7:4	DIRLT	R/W	0x0	- Spatial Edge Direction Detector Length Type Spatial Edge Direction Detector Length Type Normally, it should be zero.
3	GLD	R/W	0x0	- Small Angle Detection Disable 1'b0 = Enable 1'b1 = Disable
2	JDH	R/W	0x0	- Jaggy Detection Half Divider 1'b0 = Jaggy Detection Half Divider Off 1'b1 = Jaggy Detection Half Divider On

DI_ENGINE2 (De-interlacer Engine 2 Register)

0x8002D28C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
A_THS												E_THS			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
E_THS								EP2	EP1				OPPD	EARLYD	ADAPD

Bit	Name	RW	Reset	Description
31:20	A_THS	R/W	0x0	- Threshold of Early determination This value indicates threshold value which determines whether the current edge directional result is ignored comparing to vertical direction or not.
19:8	E-THS	R/W	0x0	- Threshold of Adaptive Edge determination This value indicates threshold value which determines whether the current edge directional result is acceptable comparing to other directional cost values or not.
7	EP2	R/W	0x0	- Early determination Suppress Algorithm 2 Disable 1'b0 = Suppress algorithm 2 enable 1'b1 = Suppress algorithm 2 disable
6	EP1	R/W	0x0	- Early determination Suppress Algorithm 1 Disable 1'b0 = Suppress algorithm 1 enable 1'b1 = Suppress algorithm 1 disable
2	OPPD	R/W	0x0	- Edge adaptive direction determination Disable 1'b0 = Enable 1'b1 = Disable
1	EARLYD	R/W	0x0	- Early determination Disable 1'b0 = Enable 1'b1 = Disable
0	ADAPD	R/W	0x0	- Adaptive Edge determination Disable 1'b0 = Enable 1'b1 = Disable

DI_ENGINE3 (De-interlacer Engine 3 Register)

0x8002D290

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STTHW												STTHM			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STTHM								STTHD							

Bit	Name	RW	Reset	Description
31:20	STTHW	R/W	0x0	- Threshold Weight Parameter of Stationary Detection Increasing one bit means 0.5 weight addition.
19:8	STTHM	R/W	0x0	- Threshold Multiplier Parameter of Stationary Detection Increasing one bit means 0.5 multiplier addition
0	STTHD	R/W	0x0	- Adaptive Stationary Checker Disable 1'b0 = Use adaptive stationary checker algorithm. The stationary checker in the VIQE does not determine only whether this pixel is stationary or not, but also analyzes the degree of stationarity quantitatively. This value can be used in the compensating the outputs. These procedures shall process in the adaptive stationary checker hardware. 1'b1 = Do not use adaptive stationary checker algorithm

DI_ENGINE4 (De-interlacer Engine 4 Register)

0x8002D294

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VARLG				VARR1				VARR0				DYNVAR			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HPFVAR				HPFFLD				HPFFRM				DMDC			

Bit	Name	RW	Reset	Description
31:28	VARLG	R/W	0x0	- Variance small stationary range
27:24	VARR1	R/W	0x0	- Variance dynamic threshold ratio 1
23:20	VARR0	R/W	0x0	- Variance dynamic threshold ratio 0
19:16	DYNVAR	R/W	0x0	- Derivative Variance Minimum Threshold
15:12	HPFVAR	R/W	0x0	- Derivative Variance Threshold for HPF output
11:8	HPFFLD	R/W	0x0	- HPF Threshold of Field Difference
7:4	HPFFRM	R/W	0x0	- HPF Threshold of Frame Difference
1	DMDC	R/W	0x0	- Stationary Checker Mode in Chrominance 1'b0 = Use advanced stationary checker algorithm in chrominance processing. 1'b1 = Do not use advanced stationary checker algorithm in chrominance processing
0	DMDL	R/W	0x0	- Stationary Checker Mode in Luminance 1'b0 = Use advanced stationary checker algorithm in luminance processing. 1'b1 = Do not use advanced stationary checker algorithm in luminance processing

Note: All bits in DI_ENGINE4 register are meaningful only when either DMDC or DMDL is zero.

PD_THRES0 (Pulldown Detector Threshold 0 Register)

0x8002D298

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
							ZP	CNTSCO				CO			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UVALDIS							CNTS								

Bit	Name	RW	Reset	Description
24	ZP	R/W	0x0	- Avoidance of zero difference 1'b0 = Do not use the avoidance of zero frame difference. 1'b1 = Use the avoidance of zero frame difference. It is recommended to use.
23:20	CNTSCO	R/W	0x0	- Threshold for counter value of Pulldown Prevention-Flag
17:16	CO	R/W	0x0	- Value of Pulldown Detection type 2'b00 = Not interleaving, but normal de-interlacing process 2'b01 = Reserved 2'b10 = Interleaving fields from the previous fields 2'b11 = Interleaving fields from the next fields Through this field, the user can control manually the method of interleaving in the reverse pull-down process.
15	UVALDIS	R/W	0x0	- Not Use User-defined Threshold Value in Prevention-Flag 1'b0 = Use a user-defined threshold value 1'b1 = Use a hardware-generated threshold value Its range can be controlled by WEIGHT field in PD_THRES0 register.
14:10	WEIGHT	R/W	0x0	- Weight Value for Threshold of Pulldown Prevention-Flag This means the ratio of relative amount between minimum detection value and the second smaller detection value. This bit is meaningless when UVALDIS is '0'.
9:0	CNTS	R/W	0x0	- Threshold for counter value of Pulldown checker If the number of continuous success of pulldown detection is over that threshold, field interleaving operates instead of de-interlacing.

PD_THRES1 (Pulldown Detector Threshold 1 Register)

0x8002D29C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
THRES1								THRES0							

Bit	Name	RW	Reset	Description
23:16	THRES2	R/W	0x0	- Threshold for pixel difference value of the third level This means the threshold value for pixel difference between the current frame and the previous frame.
15:8	THRES1	R/W	0x0	- Threshold for pixel difference value of the second level This means the threshold value for pixel difference between the current frame and the previous frame.
7:0	THRES0	R/W	0x0	- Threshold for pixel difference value of the first level This means the threshold value for pixel difference between the current frame and the previous frame.

In normal configuration, threshold value of a level should be larger than that of lower level. Or, threshold value of the first level is smallest of all, and that of the third level is largest.

PD_JUDDER (Pulldown Detector Judder Register)

0x8002D2A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
THSJDMAX								THSJDMIN							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HORLINE				DNLIN				CNTS							

Bit	Name	RW	Reset	Description
31:24	THSJDMAX	R/W	0x0	- Threshold maximum boundary for Pulldown Judder Detector This means the maximum boundary of hardware-generated threshold for Pulldown Judder Detector
23:16	THSJDMIN	R/W	0x0	- Threshold minimum boundary for Pulldown Judder Detector This means the minimum boundary of hardware-generated threshold for Pulldown Judder Detector
15:12	HORLINE	R/W	0x0	- Horizontal Margin for Judder Elimination Processing This means the left/right margin of the lines detected as having some judder pixels. If its value is larger, the negative effect for judder misdetection is less.
11:8	DNLIN	R/W	0x0	- Downward Margin for Judder Elimination Processor This means the downward margin of the lines detected as having some judder pixels. If its value is larger, the negative effect for judder misdetection is less.
7:0	CNTS	R/W	0x0	- Threshold for counter of judder pixels This value determines whether the current line is judder line or not. If the pixels detected as judder pixels are more of that value, the current line is determined as the judder line.

PD_JUDDER_M (Pulldown Detector Judder Misc. Register)

0x8002D2A4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								MULDMT							NOJDS
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
THSJD2B															JDH

Bit	Name	RW	Reset	Description
23:17	MULDMT	R/W	0x0	Multiplier Stationary Checker Threshold for stationary pixels during Film-mode processing
16	NOJDS	R/W	0x0	- Prevent to count stationary pixels 1'b0 = Count judder pixels to be even stationary 1'b1 = Not count judder pixels to be stationary
15:8	THSJD2B	R/W	0x0	- Threshold Pulldown judder for Two buffers mode.
0	JDH	R/W	0x0	- Judder Detection Half Divider 1'b0 = Judder Detection Half Divider Off 1'b1 = Judder Detection Half Divider On

DI_MISCC (De-interlacer Misc. Register)

0x8002D2A8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				INIT L2L				Y_MST JMP		X_MST END					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDX_Y_MST								IDX_X_MST							

Caution: This register is TEST mode. Do not access this register

DI_STATUS (De-interlacer Status Register)

0x8002D2AC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								POS_Y					POS_X		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
POS_X															BUSY

Bit	Name	RW	Reset	Description
29:19	POS_Y	R	-	- Current Vertical Position This indicates the vertical position in image to be processed currently. This position value is based on luminance resolution.
18:8	POS_X	R	-	- Current Horizontal Position This indicates the horizontal position in image to be processed currently. This position value is based on luminance resolution.
0	BUSY	R	-	- De-interlacer Busy This indicates that the De-interlacer hardware is processing currently.

PD_STATUS (Pulldown Detector Status Register)

0x8002D2B0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								MPOFF			CO		STATE		

Bit	Name	RW	Reset	Description
10:8	MPOFF	R	-	- Minimum Position Offset This indicates the information to know the minimum frame position in current operation. This field is used for hardware debugging.
5:4	CO	R	-	- Value of Pulldown Detection type 2'b00 = Not interleaving, but normal de-interlacing process 2'b01 = Reserved 2'b10 = Interleaving fields from the previous fields 2'b11 = Interleaving field from the next fields
3:0	STATE	R	-	- Current FSM State This indicates the current state of the hardware FSM. This field is used for hardware debugging.

DI_REGION0 (De-interlacer Region 0 Register)

0x8002D2B4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						XSTART									

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	- Region Selection Enable 1'b0 = Normal Process 1'b1 = Only specific region of the whole image will be processed in de-interlacing
25:16	XEND	R/W	0x0	- Last Horizontal Position of Region Selection This means the last horizontal position of region selection.
9:0	XSTART	R/W	0x0	- First Horizontal Position of Region Selection This means the first horizontal position of region selection.

DI_REGION1 (De-interlacer Region 1 Register)

0x8002D2B8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						YSTART									

Bit	Name	RW	Reset	Description
25:16	YEND	R/W	0x0	- Last Vertical Position of Region Selection This means the last vertical position of region selection.
9:0	YSTART	R/W	0x0	- First Vertical Position of Region Selection This means the first vertical position of region selection.

DI_INT (De-interlacer Interrupt Register)

0x8002D2BC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
															MINT
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
															INT

Bit	Name	RW	Reset	Description
16	MINT	R/W	0x0	0 : Disable Interrupt 1 : Enable Interrupt
0	INT	R/W	0x0	0 : Finished one frame processing register 1 : Clear interrupt.

DI_PD_SAW (De-interlacer Pulldown SAW Register)

0x8002D2E0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SAWEN															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FRM_THR								FLD_THR							

Bit	Name	RW	Reset	Description
31	SAWEN	R/W	0x0	For Debugging Do not set this bit
23:16	SAW_DUR	R/W	0x8	For Debugging Do not set this bit
15:8	FRM_THR	R/W	0x5	For Debugging Do not set this bit
7:0	FLD_THR	R/W	0x10	For Debugging Do not set this bit

DI_CSIZ (De-interlacer Core Size Register)

0x8002D2E4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					HEIGHT										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					WIDTH										

Bit	Name	RW	Reset	Description
10:00	WIDTH	R/W	0x0	Input Image Width by pixel
26:16	HEIGHT	R/W	0x0	Input Image Height by pixel

These size register are related to TSDU field [bit 16] of DI_FMT register [0x800202E8]. When the TSDU field is set to 'HIGH', this register is used in de-interlacer module. (So, these registers are meaningless.) If set to 'LOW', the information of size is from VIOC module.

DI_FMT (De-interlacer Format Register)

0x8002D2E8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TFCD															TSDU
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
															F422

Bit	Name	RW	Reset	Description
31	TFCD	R/W	0x0	Do not set this bit (always 'LOW')
16	TSDU	R/W	0x0	0 : Use 1 : Do not use This field is "Size Do not Use". If This bit set to 'LOW', the size information of De-interlacer comes from VIOC modules. For the others, the size information is from DI_CSIZ register [0x800202E4].
0	F422	R/W	0x0	0 : 4:2:0 format 1 : 4:2:2 format

33.6.10 CPU Interface

Table 33.20 CPU Interface Register Map (Base Address = 0x80030000 + (0x100 * n))

Name	Address	Type	Reset	Description
CS0_CMD0_CTRL	0x00	R/W	0xA0229011	CS0 CMD0 (XA[1]==0) Control Register
CS0_DAT0_CTRL	0x04	R/W	0xA0429021	CS0 DAT0 (XA[1]==0) Control Register
CS0_CMD1_CTRL	0x08	R/W	0xA0229011	CS0 CMD1 (XA[1]==1) Control Register
CS0_DAT1_CTRL	0x0C	R/W	0xA0429021	CS0 DAT1 (XA[1]==1) Control Register
CS1_CMD0_CTRL	0x10	R/W	0xA0229011	CS1 CMD0 (XA[1]==0) Control Register
CS1_DAT0_CTRL	0x14	R/W	0xA0429021	CS1 DAT0 (XA[1]==0) Control Register
CS1_CMD1_CTRL	0x18	R/W	0xA0229011	CS1 CMD1 (XA[1]==1) Control Register
CS1_DAT1_CTRL	0x1C	R/W	0xA0429021	CS1 DAT1 (XA[1]==1) Control Register
CS0_BSWAP0	0x20	R/W	0xE4E4E4E4	CS0 Byte Swap 0 Register
CS0_BSWAP1	0x24	R/W	0xE4E4E4E4	CS0 Byte Swap 1 Register
CS1_BSWAP0	0x28	R/W	0xE4E4E4E4	CS1 Byte Swap 0 Register
CS1_BSWAP1	0x2C	R/W	0xE4E4E4E4	CS1 Byte Swap 1 Register
TYPE	0x38	R/W	0x00000000	CPUIF Type Register
CS0_CMD0	0x40 ~ 0x4C	R/W	0x00000000	If this register is read or written, reading or writing operations are generated on CS0 & CMD0 (XA[1:0] == 2'b00)
CS0_DAT0	0x50 ~ 0x5C	R/W	0x00000000	If this register is read or written, reading or writing operations are generated on CS0 & DAT0 (XA[1:0] == 2'b01)
CS0_CMD1	0x60 ~ 0x6C	R/W	0x00000000	If this register is read or written, reading or writing operations are generated on CS0 & CMD1 (XA[1:0] == 2'b10)
CS0_DAT1	0x70 ~ 0x7C	R/W	0x00000000	If this register is read or written, reading or writing operations are generated on CS0 & DAT1 (XA[1:0] == 2'b11)
CS1_CMD0	0x80 ~ 0x8C	R/W	0x00000000	If this register is read or written, reading or writing operations are generated on CS1 & CMD0 (XA[1:0] == 2'b00)
CS1_DAT0	0x90 ~ 0x9C	R/W	0x00000000	If this register is read or written, reading or writing operations are generated on CS1 & DAT0 (XA[1:0] == 2'b01)
CS1_CMD1	0xA0 ~ 0xAC	R/W	0x00000000	If this register is read or written, reading or writing operations are generated on CS1 & CMD1 (XA[1:0] == 2'b10)
CS1_DAT1	0xB0 ~ 0xBC	R/W	0x00000000	If this register is read or written, reading or writing operations are generated on CS1 & DAT1 (XA[1:0] == 2'b11)

The CMD and DAT is mapped to XA[1:0]. The CMD0, DAT0, CMD1, and DAT1 mean (XA[1:0] == 2'b00), (XA[1:0] == 2'b01), (XA[1:0] == 2'b10), and (XA[1:0] == 2'b11).

CSk_CMDm_CTRL(CSk CMDm Control Register)

0x80030000 + (0x10 * k) + (0x8 * m) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BW1			WSTP					WPW						WHLD	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BW0			RSTP					RPW						RHLD	

Bit	Name	RW	Reset	Description
31, 15	{BW1,BW0}	R/W	0x0	Data Bus Bit Width 2'b00 : 8 bit 2'b01 : 16 bit 2'b10 : Not used (Do not set this field) 2'b11 : 32 bit
30:28	WSTP	R/W	0x0	Setup Time for Writing N cycles are issued between the falling edge of LCSN and the falling edge of LWEN (Writing operation) or LOEN (Reading operation).
27:19	WPW	R/W	0x0	Pulse Width Time for Writing (N+1) cycles are issued between the falling edge of LWEN (or LOEN) and the rising edge of LWEN (or OEN).
18:16	WHLD	R/W	0x0	Hold Time for Writing N cycles are issued between the rising edge of LWEN (or LOEN) and the rising edge of LCSN.
14:12	RSTP	R/W	0x0	Setup Time for Reading N cycles are issued between the falling edge of LCSN and the falling edge of LWEN (Writing operation) or LOEN (Reading operation).
11:3	RPW	R/W	0x0	Pulse Width Time for Reading (N+1) cycles are issued between the falling edge of LWEN (or LOEN) and the rising edge of LWEN (or LOEN).
2:0	RHLD	R/W	0x0	Hold Time for Reading N cycles are issued between the rising edge of LWEN (or OEN) and the rising edge of LCSN.

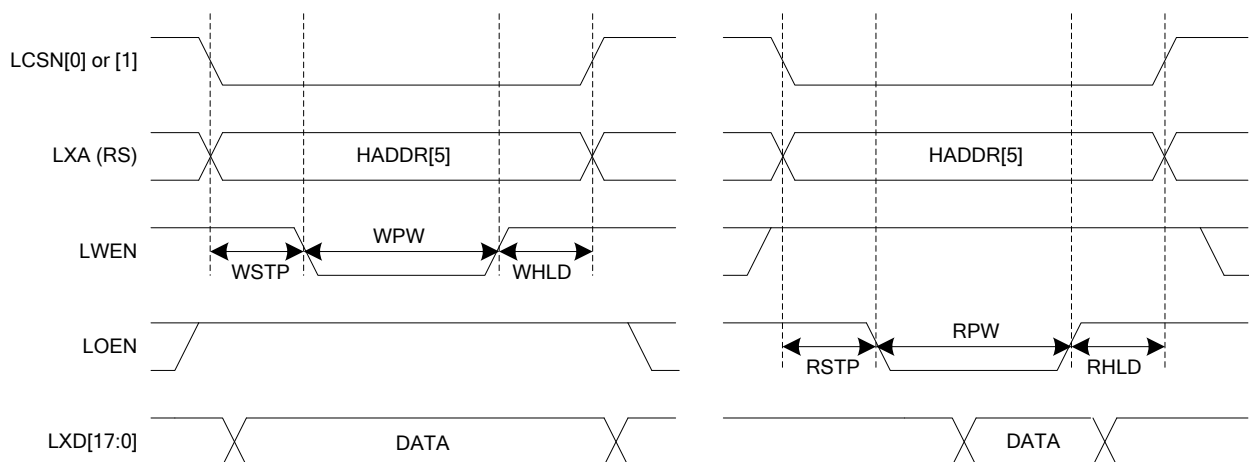


Figure 33.47 Timing Configuration of CPU interface Output Signals for the on-chip CPU Access

CSk_DATm_CTRL(CSk DATm Control Register)

0x80030004 + (0x10 * k) + (0x8 * m) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BW1	WSTP			WPW				WHLD				RSTP			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BW0	RSTP			RPW				RHL D							

Bit	Name	RW	Reset	Description
31, 15	{BW1,BW0}	R/W	0x0	Data Bus Bit Width 2'b00 : 8 bit 2'b01 : 16 bit 2'b10 : Not used (Do not set this field) 2'b11 : 32 bit
30:28	WSTP	R/W	0x0	Setup Time for Writing N cycles are issued between the falling edge of LCSN and the falling edge of LWEN (Writing operation) or LOEN (Reading operation).
27:19	WPW	R/W	0x0	Pulse Width Time for Writing (N+1) cycles are issued between the falling edge of LWEN (or LOEN) and the rising edge of LWEN (or OEN).
18:16	WHLD	R/W	0x0	Hold Time for Writing N cycles are issued between the rising edge of LWEN (or LOEN) and the rising edge of LCSN.
14:12	RSTP	R/W	0x0	Setup Time for Reading N cycles are issued between the falling edge of LCSN and the falling edge of LWEN (Writing operation) or LOEN (Reading operation).
11:3	RPW	R/W	0x0	Pulse Width Time for Reading (N+1) cycles are issued between the falling edge of LWEN (or LOEN) and the rising edge of LWEN (or LOEN).
2:0	RHL D	R/W	0x0	Hold Time for Reading N cycles are issued between the rising edge of LWEN (or OEN) and the rising edge of LCSN.

CSk_BSWAP0 (CSk Byte Swap 0 Register)

0x80030020 + (0x8 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
XA01_WR_BSWAP								XA01_RD_BSWAP							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XA00_WR_BSWAP								XA00_RD_BSWAP							

Bit	Name	RW	Reset	Description
31:24	XA01_WR_BSWAP	R/W	0xE4	Data Bus Byte Swap in XA01_WR
23:16	XA01_RD_BSWAP	R/W	0xE4	Data Bus Byte Swap in XA01_RD
15:8	XA00_WR_BSWAP	R/W	0xE4	Data Bus Byte Swap in XA00_WR
7:0	XA00_RD_BSWAP	R/W	0xE4	Data Bus Byte Swap in XA00_RD

CSk_BSWAP1 (CSk Byte Swap 1 Register)

0x80030024 + (0x8 * k) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
XA11_WR_BSWAP								XA11_RD_BSWAP							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XA10_WR_BSWAP								XA10_RD_BSWAP							

Bit	Name	RW	Reset	Description
31:24	XA11_WR_BSWAP	R/W	0xE4	Data Bus Byte Swap in XA11_WR
23:16	XA11_RD_BSWAP	R/W	0xE4	Data Bus Byte Swap in XA11_RD
15:8	XA10_WR_BSWAP	R/W	0xE4	Data Bus Byte Swap in XA10_WR
7:0	XA10_RD_BSWAP	R/W	0xE4	Data Bus Byte Swap in XA10_RD

TYPE (CPUIF Type Register)

0x80030038+ (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
															T68

Bit	Name	RW	Reset	Description
0	T68	R/W	0x0	CPUIF Mode 0 : 80 Mode 1 : 68 Mode

CSk_CMDm (CSk CMDm Register)

0x80030040 + (0x10 * k) + (0x8 * m) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CSk_CMDm[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CSk_CMDm[15:00]															

Bit	Name	RW	Reset	Description
31:0	CSk_CMDm	W	-	If this register is read or written, reading or writing operations are generated on CSk & CMDm The range of n is 0≤k≤1, 0≤m≤1. XA[1:0] is 2'b00 or 2'b01.

CSk_DATm (CSk DATm Register)

0x80030044 + (0x10 * k) + (0x8 * m) + (0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CSk_DATm[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CSk_DATm[15:00]															

Bit	Name	RW	Reset	Description
31:0	CSk_DATm	W	-	If this register is read or written, reading or writing operations are generated on CSk & DATm The range of n is 0≤k≤1, 0≤m≤1. XA[1:0] is 2'b10 or 2'b11.

33.6.11 Look-Up-Table

Table 33.21 LUT Register Map (Base Address = 0x80029000)

Name	Address	Type	Reset	Description
LUT_CTRL	0x00	R/W	0x00000000	LUT Configuration Register
LUT_DEV0_CFG	0x04	R/W	0x00000000	LUT in DEV0 Configuration Register
LUT_DEV1_CFG	0x08	R/W	0x00000000	Reserved
LUT_DEV2_CFG	0x0C	R/W	0x00000000	Reserved
LUT_VIOC0_CFG	0x10	R/W	0x00000000	LUT in VIOC0 Configuration Register
LUT_VIOC1_CFG	0x14	R/W	0x00000000	LUT in VIOC1 Configuration Register
LUT_VIOC2_CFG	0x18	R/W	0x00000000	Reserved
LUT_VIOC3_CFG	0x1C	R/W	0x00000000	Reserved
LUT_TABLE	0x400 ~0x7FC	R/W	-	LUT for Initializing Register

LUT_CTRL (LUT Configuration Register)

0x80029000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
												EN			

Bit	Name	RW	Reset	Description
3:0	SEL	R/W	0x0	Lookup Table Select for initializing value. 4'h0 : LUT in DEV0 4'h1 : Reserved 4'h2 : Reserved 4'h3 : LUT in VIOC0 4'h4 : LUT in VIOC1 4'h5 : Reserved 4'h6 : Reserved When initializing LUT, set LUT to this field and write LUT value to 0x80029400.

LUT_DEVk_CFG (LUT in DEVk Configuration Register)

0x80029004 + (0x4 * k)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								SEL							

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	LUT in DEVk Enable 1'h0 : Disable 1'h1 : Enable

LUT_VIOCK_CFG (LUT in VIOCK Configuration Register)

0x80029010 + (0x4 * k)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								SEL							

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	Plug-in Enable 1'h0 : Disable 1'h1 : Enable
7:0	SEL	R/W	0x0	Location of Plug-in 8'h00 : GRDMA00 – WMIX0 path 8'h01 : VRDMA01 – WMIX0 path 8'h02 : VRDMA02 – WMIX0 path 8'h03 : VRDMA03 – VWDMA1 path 8'h04 : VRDMA04 – VWDMA2 path (front) 8'h05 : VIDEOIN0 – WMIX1 path 8'h06 : GRDMA05 – WMIX1 path 8'h07 : VIDEOIN1 – WMIX2 path 8'h08 : GRDMA06 – WMIX2 path 8'h09 : WMIX0 – VWDMA0 path 8'h0A : VRDMA04 – VWDMA2 path (post) 8'h0B : WMIX1 – VWDMA3 path 8'h0C : WMIX1 – VWDMA4 path 8'h0D : WMIX2 – VWDMA5 path 8'h0E : WMIX2 – VWDMA6 path This value is same as the number of LUT location of Figure 33.29.

LUT_TABLE (LUT Initialize Register)

0x80029400 + (0x4 * k)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								Rk							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Gk								Bk							

Bit	Name	RW	Reset	Description
23:16	Rk	R/W	0x0	Lookup-Table initialize value in R channel
15:8	Gk	R/W	0x0	Lookup-Table initialize value in G channel
7:0	Bk	R/W	0x0	Lookup-Table initialize value in B channel

In order to initialize the look-up table, select LUT to be initialized from SEL field of LUT_CTRL register and initialize LUT value in this register. (Refer to Figure 33.24.)

33.6.12 VIN_DEMUX

Table 33.22 VIN_DEMUX Register Map (Base Address = 0x8002A800)

Name	Address	Type	Reset	Description
VD_CTRL	0x00	R/W	0x32100000	VIN_DEMUX Control Register
VD_BLANK0	0x04	R/W	0x00030CB9	VIN_DEMUX Blank Set 0 Register
VD_BLNAK1	0x08	R/W	0x000000FF	VIN_DEMUX Blank Set 1 Register
VD_MISC	0x0C	R/W	0x00004444	VIN_DEMUX Miscellaneous Register
VD_STAT	0x10	R	-	VIN_DEMUX Status Register

VD_CTRL (VIN_DEMUX Control Register)

0x8002A800

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
SEL3				SEL2				SEL1				SEL0				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
					CM						BS					EN

Field	Name	RW	Reset	Description
30 - 28	SEL3	R/W	0x3	DEMUX Output Port 3 Connection 3'h0~3 : External Port 3 3'h4 : Multiplex Port 0 3'h5 : Multiplex Port 1 3'h6 : Multiplex Port 2 3'h7 : Multiplex Port 3
26 - 24	SEL2	R/W	0x2	DEMUX Output Port 2 Connection 3'h0~3 : External Port 2 3'h4 : Multiplex Port 0 3'h5 : Multiplex Port 1 3'h6 : Multiplex Port 2 3'h7 : Multiplex Port 3
22 - 20	SEL1	R/W	0x1	DEMUX Output Port 1 Connection 3'h0~3 : External Port 1 3'h4 : Multiplex Port 0 3'h5 : Multiplex Port 1 3'h6 : Multiplex Port 2 3'h7 : Multiplex Port 3
18 - 16	SEL0	R/W	0x0	DEMUX Output Port 0 Connection 3'h0~3 : External Port 0 3'h4 : Multiplex Port 0 3'h5 : Multiplex Port 1 3'h6 : Multiplex Port 2 3'h7 : Multiplex Port 3
10 - 8	CM	R/W	0x0	DEMUX Clock Mode 3'h0 : Using Double Edge (positive & negative edge) 3'h1 : Using Double Frequency 3'h2 : Using Double Edge and Frequency 3'h3 : Using 4-times Frequency
5 - 4	BS	R/W	0x0	Byte Select of Data Bus. 2'h0 : Using Byte0 Lane (bit[07:00]) 2'h1 : Using Byte1 Lane (bit[15:08]) 2'h2 : Using Byte2 Lane (bit[23:16]) Others : Not used
0	EN	R/W	0x0	DE-MUX Enable 0 : Disable 1 : Enable

VD_BLANK0 (VIN_DEMUX BLANK 0 Register)

0x8002A804

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
												SB		PSL	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
				FP				VB				HB			

Field	Name	RW	Reset	Description
19-18	SB	R/W	0x0	0x0 : Using databus[07:00] for detecting sync bytes. 0x1 : Using databus[15:08] for detecting sync bytes 0x2 : Using databus[23:16] for detecting sync bytes Others : Not used This field is "SyncByte" for BT.656 format. If the input format is BT.656 mode, the module must detect Sync Signal using data bus. So this bit is decided by the location of Sync Byte.
17-16	PSL	R/W	0x3	0x0 : The status word is located in the first byte of EAV & SAV 0x1 : The status word is located in the second byte of EAV & SAV 0x2 : The status word is located in the third byte of EAV & SAV 0x3 : The status word is located in the fourth byte of EAV & SAV (Refer to Figure 33.45)
11-8	FP	R/W	0xC	Field information In status word, it refers to the location of 'F' and F value at field information. The MSB 3 bit means the location of 'F', the other bit means value at field information.
7-4	VB	R/W	0xB	Horizontal blank In status word, it refers to the location of 'H' and H value at blanking. The MSB 3 bit means the location of 'H', the other bit means value at blanking.
3-0	HB	R/W	0x9	Vertical blank In status word, it refers to the location of 'V' and V value at blanking. The MSB 3 bit means the location of 'V', the other bit means value at blanking.

VD_BLANK1 (VIN_DEMUX BLANK 1 Register)

0x8002A808

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
												PT			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PS								PF							

Field	Name	RW	Reset	Description
23-16	PT	R/W	0x0	Third preamble value Define the third preamble value.
15-8	PS	R/W	0x0	Second preamble value Define the second preamble value.
7-0	PF	R/W	0xFF	First preamble value Define the first preamble value.

VD_MISC (VIN_DEMUX Miscellaneous Register)

0x8002A80C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DSEL3				DSEL2				DSEL1				DSEL0			

Field	Name	RW	Reset	Description
14 - 12	DSEL3	R/W	0x4	Not Using Detect Frame Index in decoding port 3 3'h0 : Using clock index port 0 3'h1 : Using clock index port 1 3'h2 : Using clock index port 2 3'h3 : Using clock index port 3 Others : Using detected frame index
10 - 8	DSEL2	R/W	0x4	Not Using Detect Frame Index in decoding port 2 3'h0 : Using clock index port 0 3'h1 : Using clock index port 1 3'h2 : Using clock index port 2 3'h3 : Using clock index port 3 Others : Using detected frame index
6 - 4	DSEL1	R/W	0x4	Not Using Detect Frame Index in decoding port 1 3'h0 : Using clock index port 0 3'h1 : Using clock index port 1 3'h2 : Using clock index port 2 3'h3 : Using clock index port 3 Others : Using detected frame index
2 - 0	DSEL0	R/W	0x4	Not Using Detect Frame Index in decoding port 0 3'h0 : Using clock index port 0 3'h1 : Using clock index port 1 3'h2 : Using clock index port 2 3'h3 : Using clock index port 3 Others : Using detected frame index

VD_STS (VIN_DEMUX Status Register)

0x8002A810

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDX3				IDX2				IDX1				IDX0			

Field	Name	RW	Reset	Description
14 - 12	IDX3	R	-	Detected frame index in decoding port 3
10 - 8	IDX2	R	-	Detected frame index in decoding port 2
6 - 4	IDX1	R	-	Detected frame index in decoding port 1
2 - 0	IDX0	R	-	Detected frame index in decoding port 0

33.6.13 Output Configuration

Table 33.23 Output Configuration Register Map (Base Address = 0x80030200)

Name	Address	Type	Reset	Description
D0CPUIF	0x00	R/W	0x00000008	CPUIF0 Output Configuration Register
D1CPUIF	0x04	R/W	0x00000088	CPUIF1 Output Configuration Register
D2CPUIF	0x08	R/W	0x00000088	CPUIF2 Output Configuration Register
MISC	0x10	R/W	0x00000000	Miscellaneous Register

DkCPUIF(CPUIFk Control Register)

0x80030200 + (0x4 * k)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	IV					EN	SI	CS		XA				FMT	

Bit	Name	RW	Reset	Description
15	ID	R/W	0x0	DE Polarity Register Note: It should be same as the DE polarity control register of the Display Device Component.
14	IV	R/W	0x0	VSYNC Polarity Register Note: It should be inverted value of the VSYNC polarity control register of Display Device Component.
9	EN	R/W	0x0	Enable Register * Note: If enabled, the LCD data transferred to the CPUIF
8	SI	R/W	0x0	Select CPUIF 0 : CPUIF0 1 : CPUIF1
7	CS	R/W	0x0	Setup Time for Reading N cycles are issued between the falling edge of LCSN and the falling edge of LWEN (Writing operation) or LOEN (Reading operation).
6:5	XA	R/W	0x0	XA[1:0] Mapping Register These are transferred to the XA[1:0]
3:0	FMT	R/W	0x8	Hold Time for Reading N cycles are issued between the rising edge of LWEN (or OEN) and the rising edge of LCSN.

MISC(Miscellaneous Register)

0x8003020C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

33.6.14 Configuration & Interrupt

Table 33.24 Configuration & Interrupt Register Map (Base Address = 0x8002A000)

Name	Address	Type	Reset	Description
RAWSTATUS0	0x00	R	-	Interrupt Status 0 Register
RAWSTATUS1	0x04	R	-	Interrupt Status 1 Register
SYNCSTATUS0	0x08	R	-	Synchronized Interrupt Status 0 Register
STNCSTATUS1	0x0C	R	-	Synchronized Interrupt status 1 Register
IRQSELECT0	0x10	R/W	0x00000000	Interrupt Select 0 Register
IRQSELECT1	0x14	R/W	0x00000000	Interrupt Select 1 Register
VECTORID	0x28	R	-	Vector ID Register
CFG_MISC	0x40	R/W	0x00000000	Miscellaneous Register
CFG_PATH_SC0	0x44	R/W	0x00000000	Scaler 0 PATH Configuration Register
CFG_PATH_SC1	0x48	R/W	0x00000000	Scaler 1 PATH Configuration Register
CFG_PATH_SC2	0x4C	R/W	0x00000000	Scaler 2 PATH Configuration Register
CFG_PATH_VIQE	0x54	R/W	0x00000000	VIQE PATH Configuration Register
CFG_PATH_DEINTLS	0x58	R/W	0x00000000	DEINTL_S PATH Configuration Register
ARID	0xC0	R/W	0x00000000	ARID
AWID	0xC4	R/W	0x00000000	AWID
PWR_AUTOPD	0xC8	R/W	0x00003FFF	Auto Power Down Mode Register
PWR_CLKCTRL	0xCC	R/W	0x00000103	Clock Control Register
PWR_BLK_PWDN0	0xD0	R/W	0x00000000	Block Power Down Mode 0 Register
PWR_BLK_PWDN1	0xD4	R/W	0x00000000	Block Power Down Mode 1 Register
PWR_BLK_SWR	0xD8	R/W	0x00000000	Block Software Reset 0 Register
PWR_BLK_SWR	0xDC	R/W	0x00000000	Block Software Reset 1 Register

RAWSTATUS0 (Interrupt Status 0 Register)

0x8002A000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					RD6	RD5	RD4	RD3	RD2	RD1	RD0	TIMER			DEV0

Bit	Name	RW	Reset	Description
10	RD6	R	-	RDMA06 Interrupt Status 0 : Normal 1 : Interrupt
9	RD5	R	-	RDMA05 Interrupt Status 0 : Normal 1 : Interrupt
8	RD4	R	-	RDMA04 Interrupt Status 0 : Normal 1 : Interrupt
7	RD3	R	-	RDMA03 Interrupt Status 0 : Normal 1 : Interrupt
6	RD2	R	-	RDMA02 Interrupt Status 0 : Normal 1 : Interrupt
5	RD1	R	-	RDMA01 Interrupt Status 0 : Normal 1 : Interrupt
4	RD0	R	-	RDMA00 Interrupt Status 0 : Normal 1 : Interrupt
3	TIMER	R	-	TIMER Interrupt Status 0 : Normal 1 : Interrupt
0	DEV0	R	-	Display Device 0 Interrupt Status 0 : Normal 1 : Interrupt

RAWSTATUS1 (Interrupt Status 1 Register)

0x8002A004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	SC2	SC1	SC0	VIQE									WMIX2	WMIX1	WMIX0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
									WD6	WD5	WD4	WD3	WD2	WD1	WD0

Bit	Name	RW	Reset	Description
30	SC2	R	-	Scaler 2 Interrupt Status. 0 : Normal 1 : Interrupt
29	SC1	R	-	Scaler 1 Interrupt Status. 0 : Normal 1 : Interrupt
28	SC0	R	-	Scaler 0 Interrupt Status. 0 : Normal 1 : Interrupt
27	VIQE	R	-	VIQE Interrupt Status. 0 : Normal 1 : Interrupt
18	WMIX2	R	-	Not used
17	WMIX1	R	-	Not used
16	WMIX0	R	-	Not used
6	WD6	R	-	WDMA06 Interrupt Status 0 : Normal 1 : Interrupt
5	WD5	R	-	WDMA05 Interrupt Status 0 : Normal 1 : Interrupt
4	WD4	R	-	WDMA04 Interrupt Status 0 : Normal 1 : Interrupt
3	WD3	R	-	WDMA03 Interrupt Status 0 : Normal 1 : Interrupt
2	WD2	R	-	WDMA02 Interrupt Status 0 : Normal 1 : Interrupt
1	WD1	R	-	WDMA01 Interrupt Status 0 : Normal 1 : Interrupt
0	WD0	R	-	WDMA00 Interrupt Status 0 : Normal 1 : Interrupt

SYNCSTATUS0 (Sync Interrupt Status 0 Register)

0x8002A008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					RD6	RD5	RD4	RD3	RD2	RD1	RD0	TIMER			DEV0

Bit	Name	RW	Reset	Description
10	RD6	R	-	RDMA06 Sync Interrupt Status 0 : Normal 1 : Interrupt
9	RD5	R	-	RDMA05 Sync Interrupt Status 0 : Normal 1 : Interrupt
8	RD4	R	-	RDMA04 Sync Interrupt Status 0 : Normal 1 : Interrupt
7	RD3	R	-	RDMA03 Sync Interrupt Status 0 : Normal 1 : Interrupt
6	RD2	R	-	RDMA02 Sync Interrupt Status 0 : Normal 1 : Interrupt
5	RD1	R	-	RDMA01 Sync Interrupt Status 0 : Normal 1 : Interrupt
4	RD0	R	-	RDMA00 Sync Interrupt Status 0 : Normal 1 : Interrupt
3	TIMER	R	-	TIMER Sync Interrupt Status 0 : Normal 1 : Interrupt
0	DEV0	R	-	Display Device 0 Sync Interrupt Status 0 : Normal 1 : Interrupt

SYNCSTATUS1 (Sync Interrupt Status 1 Register)

0x8002A00C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	SC2	SC1	SC0	VIQE									WMIX2	WMIX1	WMIX0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
									WD6	WD5	WD4	WD3	WD2	WD1	WD0

Bit	Name	RW	Reset	Description
30	SC2	R	-	Scaler2 Sync Interrupt Status. 0 : Normal 1 : Interrupt
29	SC1	R	-	Scaler1 Sync Interrupt Status. 0 : Normal 1 : Interrupt
28	SC0	R	-	Scaler0 Sync Interrupt Status. 0 : Normal 1 : Interrupt
27	VIQE	R	-	VIQE Sync Interrupt Status. 0 : Normal 1 : Interrupt
18	WMIX2	R	-	Not used
17	WMIX1	R	-	Not used
16	WMIX0	R	-	Not used
6	WD6	R	-	WDMA06 Sync Interrupt Status 0 : Normal 1 : Interrupt
5	WD5	R	-	WDMA05 Sync Interrupt Status 0 : Normal 1 : Interrupt
4	WD4	R	-	WDMA04 Sync Interrupt Status 0 : Normal 1 : Interrupt
3	WD3	R	-	WDMA03 Sync Interrupt Status 0 : Normal 1 : Interrupt
2	WD2	R	-	WDMA02 Sync Interrupt Status 0 : Normal 1 : Interrupt
1	WD1	R	-	WDMA01 Sync Interrupt Status 0 : Normal 1 : Interrupt
0	WD0	R	-	WDMA00 Sync Interrupt Status 0 : Normal 1 : Interrupt

IRQSELECT0 (Interrupt Select 0 Register)

0x8002A010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					RD6	RD5	RD4	RD3	RD2	RD1	RD0	TIMER			DEV0

Bit	Name	RW	Reset	Description
10	RD6	R/W	0x0	RDMA06 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
9	RD5	R/W	0x0	RDMA05 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
8	RD4	R/W	0x0	RDMA04 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
7	RD3	R/W	0x0	RDMA03 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
6	RD2	R/W	0x0	RDMA02 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
5	RD1	R/W	0x0	RDMA01 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
4	RD0	R/W	0x0	RDMA00 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
3	TIMER	R/W	0x0	TIMER Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
0	DEV0	R/W	0x0	Display Device 0 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt

IRQSELECT1 (Interrupt Select 1 Register)

0x8002A014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	SC2	SC1	SC0	VIQE									WMIX2	WMIX1	WMIX0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
									WD6	WD5	WD4	WD3	WD2	WD1	WD0

Bit	Name	RW	Reset	Description
30	SC2	R/W	0x0	Scaler2 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
29	SC1	R/W	0x0	Scaler1 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
28	SC0	R/W	0x0	Scaler0 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
27	VIQE	R/W	0x0	VIQE Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
18	WMIX2	R/W	0x0	Not used
17	WMIX1	R/W	0x0	Not used
16	WMIX0	R/W	0x0	Not used
6	WD6	R/W	0x0	WDMA06 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
5	WD5	R/W	0x0	WDMA05 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
4	WD4	R/W	0x0	WDMA04 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
3	WD3	R/W	0x0	WDMA03 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
2	WD2	R/W	0x0	WDMA02 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
1	WD1	R/W	0x0	WDMA02 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt
0	WD0	R/W	0x0	WDMA00 Interrupt Select 0 : Async Interrupt 1 : Sync Interrupt

VECTORID (Vector ID Register)

0x8002A024

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INVALID															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
										INDEX					

Bit	Name	RW	Reset	Description
31	INVALID	R	-	Invalid Interrupt 0 : Generating Interrupt 1 : Normal
5:0	INDEX	R	-	Interrupt Index 6'h00 : Display Device 0 6'h01 : Display Device 1(Not used) 6'h02 : Display Device 2(Not used) 6'h03 : TIMER 6'h04 : RDMA 00 6'h05 : RDMA 01 6'h06 : RDMA 02 6'h07 : RDMA 03 6'h08 : RDMA 04 6'h09 : RDMA 05 6'h0A : RDMA 06 6'h0B : RDMA 07(Not used) 6'h0C : RDMA 08(Not used) 6'h0D : RDMA 09(Not used) 6'h0E : RDMA 10(Not used) 6'h0F : RDMA 11(Not used) 6'h10 : RDMA 12(Not used) 6'h11 : RDMA 13(Not used) 6'h12 : RDMA 14(Not used) 6'h13 : RDMA 15(Not used) 6'h14 : RDMA 16(Not used) 6'h15 : RDMA17(Not used) 6'h16 : Not used 6'h17 : Not used 6'h18 : MMU(Not used) 6'h19 : Not used 6'h1A : Not used 6'h1B : Not used 6'h1C : Async FIFO 0(Not used) 6'h1D : Async FIFO 1(Not used) 6'h1E : Not used 6'h1F : Not used 6'h20 : WDMA 00 6'h21 : WDMA 01 6'h22 : WDMA 02 6'h23 : WDMA 03 6'h24 : WDMA 04 6'h25 : WDMA 05 6'h26 : WDMA 06 6'h27 : WDMA 07(Not used) 6'h28 : WDMA 08(Not used) 6'h29 ~ 6'h2F : Not used 6'h30 : WMIX 00(Not used) 6'h31 : WMIX 01(Not used) 6'h32 : WMIX 02(Not used) 6'h33 : WMIX 03(Not used) 6'h34 : WMIX 04(Not used) 6'h35 : WMIX 05(Not used) 6'h36 ~ 6'h3A : Not used 6'h3B : VIQE 6'h3C : Scaler 0 6'h3D : Scaler 1 6'h3E : Scaler 2 6'h3F : Scaler 3(Not used)

CFG_MISC (Miscellaneous Configuration Register)

0x8002A040

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RD04	RD03									MIX21	MIX20	MIX11	MIX10	MIX01	MIX00
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
													L0_EVS_SEL		

Bit	Name	RW	Reset	Description
31	RD04	R/W	0x0	RDMA04 Path Control (FIX) 0 : RDMA04 PATH 1 : Not used
30	RD03	R/W	0x0	RDMA03 Path Control 0 : WMIX0 PATH 1 : WDMA_01 PATH
21	MIX21	R/W	0x0	WMIX2 Post Path Control 0 : Bypass PATH 1 : Mixing PATH Note: This register is applied to the datapath by the WDMA connected to 0'th output channel. So, if the corresponding WDMA does not work, it should be enabled with dummy operation.
20	MIX20	R/W	0x0	WMIX2 Pre Path Control 0 : Bypass PATH 1 : Mixing PATH Note: This register is applied to the datapath by the WDMA connected to 0'th output channel. So, if the corresponding WDMA does not work, it should be enabled with dummy operation.
19	MIX11	R/W	0x0	WMIX1 Post Path Control 0 : Bypass PATH 1 : Mixing PATH Note: This register is applied to the datapath by the WDMA connected to 0'th output channel. So, if the corresponding WDMA does not work, it should be enabled with dummy operation.
18	MIX10	R/W	0x0	WMIX1 Pre Path Control 0 : Bypass PATH 1 : Mixing PATH Note: This register is applied to the datapath by the WDMA connected to 0'th output channel. So, if the corresponding WDMA does not work, it should be enabled with dummy operation.
17	MIX01	R/W	0x0	WMIX0 Post Path Control 0 : Bypass PATH 1 : Mixing PATH Note: This register is applied to the datapath by the WDMA connected to 0'th output channel. So, if the corresponding WDMA does not work, it should be enabled with dummy operation.
16	MIX00	R/W	0x0	WMIX0 Pre Path Control 0 : Bypass PATH 1 : Mixing PATH Note: This register is applied to the datapath by the WDMA connected to 0'th output channel. So, if the corresponding WDMA does not work, it should be enabled with dummy operation.
2:0	L0_EVS_SEL	R/W	0x0	Select VS signal for Display Device Output Port 0 3'h0 : Port 0 External VS 3'h1 : Port 1 External VS 3'h2 : Port 2 External VS 3'h3 : Device 0 Internal VS 3'h4 : Device 1 Internal VS

				3'h5 : Device 2 Internal VS 3'h6 : VIDEOIN 0 VS 3'h7 : VIDEOIN 1 VS
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CFG_PATH_SCK (Scaler k Path Configuration k Register)

0x8002A044 + (0x4 * k)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN													ERR	STS	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								SEL							

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	Path Enable 0 : Disable 1 : Enable
18	ERR	R	-	Device Error 1'h0 : CONNECTED 1'h1 : BUSY
17:16	STS	R	-	Path Status 2'h0 : DISCONNECTED 2'h1 : CONNECTING 2'h2 : CONNECTED 2'h3 : DISCONNECTING
7:0	SEL	R/W	0x0	Path Selection 8'h00 : RDMA_00 8'h01 : RDMA_01 8'h02 : RDMA_02 8'h03 : RDMA_03 8'h04 : RDMA_04 8'h05 : VIDEOIN_00 8'h06 : RDMA_05 8'h07 : VIDEOIN_01 8'h08 : RDMA_06 8'h09 : WDMA_00 8'h0A : WDMA_03 8'h0B : WDMA_04 8'h0C : WDMA_05 8'h0D : WDMA_06

CFG_PATH_VIQE (VIQE Path Configuration k Register)

0x8002A054

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN													ERR	STS	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								SEL							

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	Path Enable 0 : Disable 1 : Enable
18	ERR	R	-	Device Error 1'h0 : CONNECTED 1'h1 : BUSY
17:16	STS	R	-	Path Status 2'h0 : DISCONNECTED 2'h1 : CONNECTING 2'h2 : CONNECTED 2'h3 : DISCONNECTING
7:0	SEL	R/W	0x0	Path Selection 8'h00 : RDMA_01 8'h01 : RDMA_02 8'h02 : RDMA_03 8'h03 : RDMA_04 8'h04 : VIDEOIN_00 8'h05 : RDMA_05 8'h06 : VIDEOIN_01 8'h07 : RDMA_06

CFG_PATH_DEINTLS (DEINTLS Path Configuration k Register)

0x8002A058

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN													ERR	STS	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								SEL							

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	Path Enable 0 : Disable 1 : Enable
18	ERR	R	-	Device Error 1'h0 : CONNECTED 1'h1 : BUSY
17:16	STS	R	-	Path Status 2'h0 : DISCONNECTED 2'h1 : CONNECTING 2'h2 : CONNECTED 2'h3 : DISCONNECTING
7:0	SEL	R/W	0x0	Path Selection 8'h00 : RDMA_01 8'h01 : RDMA_02 8'h02 : RDMA_03 8'h03 : RDMA_04 8'h04 : VIDEOIN_00 8'h05 : RDMA_05 8'h06 : VIDEOIN_01 8'h07 : RDMA_06

ARID (ARID of DMA Register)

0x8002A0C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				ARID[27:16]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARID[15:0]															

Bit	Name	RW	Reset	Description
27:0	ARID	R/W	0x0	ARID of Master DMA

AWID (AWID of DMA Register)

0x8002A0C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
													AWID[18:16]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AWID[15:0]															

Bit	Name	RW	Reset	Description
18:0	AWID	R/W	0x0	AWID of Master DMA

PWR_AUTOPD (Power Auto Power Down Register)

0x8002A0C8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
			VIN	L2A	CPUIF				DEINTS	VIQE		WDMA	MIX	SC	RDMA

Bit	Name	RW	Reset	Description
12	VIN	R/W	0x1	VIDEOIN Auto Power Down Mode 0 : Normal 1 : Auto PWDN
11	L2A	R/W	0x1	LCD2AHB Auto Power Down Mode 0 : Normal 1 : Auto PWDN
10	CPUIF	R/W	0x1	CPU Interface Auto Power Down Mode 0 : Normal 1 : Auto PWDN
6	DEINTS	R/W	0x1	DEINTL_S Auto Power Down Mode 0 : Normal 1 : Auto PWDN
5	VIQE	R/W	0x1	VIQE Auto Power Down Mode 0 : Normal 1 : Auto PWDN
3	WDMA	R/W	0x1	WDMA Auto Power Down Mode 0 : Normal 1 : Auto PWDN
2	MIX	R/W	0x1	WMIX Auto Power Down Mode 0 : Normal 1 : Auto PWDN
1	SC	R/W	0x1	Scaler Auto Power Down Mode 0 : Normal 1 : Auto PWDN
0	RDMA	R/W	0x1	RDMA Auto Power Down Mode 0 : Normal 1 : Auto PWDN

PWR_CLKCTRL (Power Clock Control Register)

0x8002A0CC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						PFEN	EN							MIN	

Bit	Name	RW	Reset	Description
9	PFEN	R/W	0x0	Clock Profile Enable 0 : Disable 1 : Enable
8	EN	R/W	0x1	Clock Control (Clock Gating) Enable 0 : Disable 1 : Enable
3:0	MIN	R/W	0x3	Clock Disable Minimum size

PWR_BLK_PWDN0 (Power Block Power Down 0 Register)

0x8002A0D0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SC				VIN				RDMA							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDMA															

Bit	Name	RW	Reset	Description
31:28	SC	R/W	0x0	Scaler Block Power Down 0 : Normal 1 : PWDN [Bit 31] : Not used [Bit 30] : Scaler 2 [Bit 29] : Scaler 1 [Bit 28] : Scaler 0
25:24	VIN	R/W	0x0	VIDEOIN Block Power Down 0 : Normal 1 : PWDN [Bit 25] : VIN 1 [Bit 24] : VIN 0
17:0	RDMA	R/W	0x0	RDMA Power Down 0 : Normal 1 : PWDN [Bit #] : RDMA #

PWR_BLK_PWDN1 (Power Block Power Down 1 Register)

0x8002A0D4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								DEV				DINTS			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WMIX								WDMA							

Bit	Name	RW	Reset	Description
22:20	DEV	R/W	0x0	Display Device Power Down 0 : Normal 1 : PWDN [Bit 22] : Not used [Bit 21] : Not used [Bit 20] : DEV 0
17	DINTS	R/W	0x0	DEINTL_S Power Down 0 : Normal 1 : PWDN
16	VIQE	R/W	0x0	VIQE Power Down 0 : Normal 1 : PWDN
15:10	WMIX	R/W	0x0	WMIX Power Down 0 : Normal 1 : PWDN [Bit 15] : Not used [Bit 14] : Not used [Bit 13] : Not used [Bit 12] : WMIX 2 [Bit 11] : WMIX 1 [Bit 10] : WMIX 0
8:0	WDMA	R/W	0x0	WDMA Power Down 0 : Normal 1 : PWDN [Bit #] : WMIX #

PWR_BLK_SWR0 (Power Block SWRESET 0 Register)

0x8002A0D8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SC				VIN				RDMA							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDMA															

Bit	Name	RW	Reset	Description
31:28	SC	R/W	0x0	Scaler Block Reset 0 : Normal 1 : Reset [Bit 31] : Not used [Bit 30] : Scaler 2 [Bit 29] : Scaler 1 [Bit 28] : Scaler 0
25:24	VIN	R/W	0x0	VIDEOIN Block Reset 0 : Normal 1 : Reset [Bit 25] : VIN 1 [Bit 24] : VIN 0
17:0	RDMA	R/W	0x0	RDMA Reset 0 : Normal 1 : Reset [Bit #] : RDMA #

PWR_BLK_SWR1 (Power Block SWRESET 1 Register)

0x8002A0DC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								DEV							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WMIX								WDMA							

Bit	Name	RW	Reset	Description
22:20	DEV	R/W	0x0	Display Device Reset 0 : Normal 1 : Reset [Bit 22] : Not used [Bit 21] : Not used [Bit 20] : DEV 0
17	DINTS	R/W	0x0	DEINTL_S Reset 0 : Normal 1 : Reset
16	VIQE	R/W	0x0	VIQE Reset 0 : Normal 1 : Reset
15:10	WMIX	R/W	0x0	WMIX Reset 0 : Normal 1 : Reset [Bit 15] : Not used [Bit 14] : Not used [Bit 13] : Not used [Bit 12] : WMIX 2 [Bit 11] : WMIX 1 [Bit 10] : WMIX 0
8:0	WDMA	R/W	0x0	WDMA Reset 0 : Normal 1 : Reset [Bit #] : WMIX #

33.6.15 Timer

Table 33.25 Timer Register Map (Base Address = 0x8002C000)

Name	Address	Type	Reset	Description
USEC	0x00	R/W	0x8000630B	Micro second (usec) Register
CURTIME	0x04	R	-	Current Time Register
TIMER0	0x08	R/W	0x00000000	Timer Core 0 Register
TIMER1	0x0C	R/W	0x00000000	Timer Core 1 Register
TIREQ0	0x10	R/W	0x00000000	Time Interrupt Control 0 Register
TIREQ1	0x14	R/W	0x00000000	Time Interrupt Control 1 Register
IRQMASK	0x18	R/W	-	Interrupt Mask Register
IRQSTAT	0x1C	R/W	-	Interrupt Status Register

USEC (Micro-second Control Register)

0x8002C000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UNITDIV								USECDIV							

Bit	Name	RW	Reset	Description
31	EN	R/W	0x1	Timer Enable 0 : Disable 1 : Enable
15:8	UNITDIV	W	0x63	To make 100usec pulse from 1usec (Normally 99) It makes the T_{UNIT} – Unit Time.
7:0	USECDIV	R/W	0x0B	To make 1usec pulse from TCLK (Freq TCLK) / (USECDIV + 1) If TCLK = 12MHz, USECDIV = 0x0B (decimal is 11)

CURTIME (Current Time Register)

0x8002C004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CURTIME[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CURTIME[15:00]															

Bit	Name	RW	Reset	Description
31:0	CURTIME	R	-	Current Time. CURTIME is based on T_{UNIT} - Unit Time. The unit time is 100usec in reset value.

Timer k (Timer Core k Register)

0x8002C008 + (0x4 * k)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COUNTER															

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	Timer Core k Enable 0 : Disable 1 : Enable
15:0	COUNTER	R/W	0x0	Timer Core k Divide Value * $F_{TIMER} = F_{TCLK} / (COUNTER + 1)$ - F_{TIMER} : Frequency of Timer Interrupt - F_{TCLK} : Frequency of Timer Input Clock The interrupt occurs periodically.

TIREQ k (Time Interrupt k Register)

0x8002C010 + (0x4 * k)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EN															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIME															

Bit	Name	RW	Reset	Description
31	EN	R/W	0x0	Time Interrupt k Enable 0 : Disable 1 : Enable
15:0	TIME	R/W	0x0	Time Value based on the unit of Unit Time. Note: The unit of TIME should be same as CURTIME It is automatically disabled after interrupt occurrence.

IRQMASK (Interrupt Mask Register)

0x8002C0018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
												TIREQ1	TIREQ0	TIMER1	TIMER0

Bit	Name	RW	Reset	Description
3	TIREQ1	R/W	0x0	Time Interrupt 1 Masked 0 : Enable 1 : Disable
2	TIREQ0	R/W	0x0	Time Interrupt 0 Masked 0 : Enable 1 : Disable
1	TIMER1	R/W	0x0	Timer Core Interrupt 1 Masked 0 : Enable 1 : Disable
0	TIMER0	R/W	0x0	Timer Core Interrupt 0 Masked 0 : Enable 1 : Disable

IRQSTAT (Interrupt Status Register)

0x8002C001C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
												TIREQ1	TIREQ0	TIMER1	TIMER0

Bit	Name	RW	Reset	Description
3	TIREQ1	R/W	0x0	Time Interrupt 1 If write '1', this field is cleared.
2	TIREQ0	R/W	0x0	Time Interrupt 0 If write '1', this field is cleared.
1	TIMER1	R/W	0x0	Timer Core Interrupt 1 If write '1', this field is cleared.
0	TIMER0	R/W	0x0	Timer Core Interrupt 0 If write '1', this field is cleared.

33.7 Turn On/Off

33.7.1 Turning On/Off Display Device

Turning On/Off can be controlled by depending on LEN field [Bit0] of DCTRL Register [0x80020000] in Display Device Component.

33.7.2 Turning On/Off Video Input

Turning On/Off VIDEOIN Component is controlled by On/Off signal of VWDMA05~VWDMA08. In other words, since VWDMA05 and VWDMA06 are connected to VIDEOIN00 and VIDEOIN01, if the related VWDMA is enabled, the VIDEOIN is automatically enabled (On). A user does not need to set something particularly regarding On/Off.

33.8 Component Path Configurations

Plug-in and out for each component of VIOC is available. This chapter explains path-configuration method of each component. Refer to VIOC Block Diagram of Figure 33.29.

33.8.1 Scaler Components

As explained in overview, plugging-in/out Scaler Component is connected between the path of RDMA-to-WMIX and WMIX-to-WDMA. The number of Scaler Component is 4, and 29 spots can be used for plug-in as shown in VIOC Block Diagram.

The following is how to set.

- ① To set Destination, Output Size (SCDSIZE, SCOSIZE Register [0x80022000] setting of Scaler Component)
- ② Path Configuration Related Setting
 - * CFG_PATH_SCK Register [0x8002A044~50] setting of Configuration & Interrupt Block
 - a) Select one channel to use among Scaler 4 channel.
 - b) Set a path to connect in CFG_PATH_SCK Register relating to the selected Scaler [Bit[7:0]].
 - c) After setting the path, if you set EN field [BIT[31]], setting the path is done.
- ③ Lastly, if you set 1 to UPD field [Bit[16]] of SCCTRL Register[0x80022000] of Scaler Component, the Scaler works.

Prior to connecting a path, check whether a component that you want to use has been already used. After connecting is completely done, the next operation should be followed. This applies to not only Scaler Component but also all components to plug-in/out. The followings are the concerned check-points.

- ① After connection is enabled, check ERR field [Bit[18]] of CFG_PATH_SCK Register [0x8002A044]. If the value is '1', that means the concerned component is being used and another component should be used. If the value is '0', that means the concerned component can be usable.
- ② If you check STS field [Bit[17:16]], if the value is '2', that means connection has been completed, If so, you can proceed the next operation. If not '2' but '1' (CONNECTING) is read, that means connection is being proceeded and you have to wait until '2' is read.

33.8.2 Simple De-interlacer Component

A user can plug-in/out Simple De-interlacer Component on the RDMA-to-WMIX path. Although the number of De-interlacer Component that can be used is 1, 11 spots can be used for Plug-in. Since this component shares a plug-in spot with VIQE Component, two components cannot be used at the same time. (In other words, only one component must be used at a time.)

The setting order is as below.

- ① Path Configuration Related Setting
 - * CFG_PATH_DEINTLS Register [0x8002A058] setting of Configuration & Interrupt Block
 - a) Set a path to connect in DEINTLS related CFG_PATH_DEINTLS Register. [Bit[7:0]]
 - b) After setting the path to connect, set EN field [Bit[31]].
- ② After plug-in, De-interlacer automatically works.

33.8.3 VIQE Component

A user can plug-in/out VIQE Component on the RDMA-to-WMIX path. Although the number of VIQE Component that can be used is 1, 11 spots can be used for Plug-in. Since this component shares a plug-in spot with Simple De-interlacer, two components cannot be used at the same time.

The setting order is as below.

- ① Path Configuration Related Setting
* CFG_PATH_VIQE Register [0x8002A054] setting of Configuration & Interrupt Block
a) Set a path to connect in VIQE related CFG_PATH_VIQE Register. [Bit[7:0]]
b) After setting the path to connect, set EN field [Bit[31]].
- ② If VIQE is enabled under the condition that internal setting of VIQE is done, the VIQE component works normally.

33.8.4 WMIX

WMIX Component is located between RDMA and WDMA and it plays a role of image mix. Right and Left are decided based on this component and the component handles the connection between RDMA and WDMA.

In case that the image size changes due to other components, the image size should be particularly set by the WMIX.

The setting order is as below.

- ① Set an image size to be processed to MSIZE Register [0x80021808] of WMIX Component. If a size has changed by image scanning prior to WMIX, the changed size should be set.
- ② Enable UPD field [bit[16]] of Register[0x80021800].

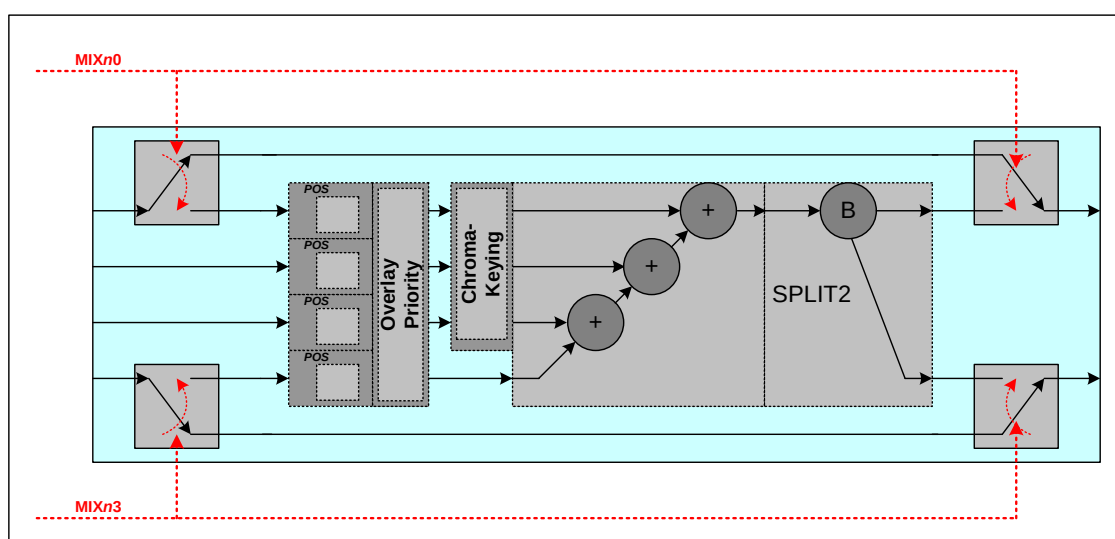


Figure 33.48 4-to-2 Window Mixer Structure

The figure above shows 4-to-2 Window Mixer structure. The MIXn0 of CFG_MISC register decides the data path of 0th input channel of the mixer and MIXn3 of CFG_MISC register decides the data path of 3rd input channel of the mixer. The figure above shows the configuration when the values of MIXn0 and MIXn3 are all 0 (n = 0).

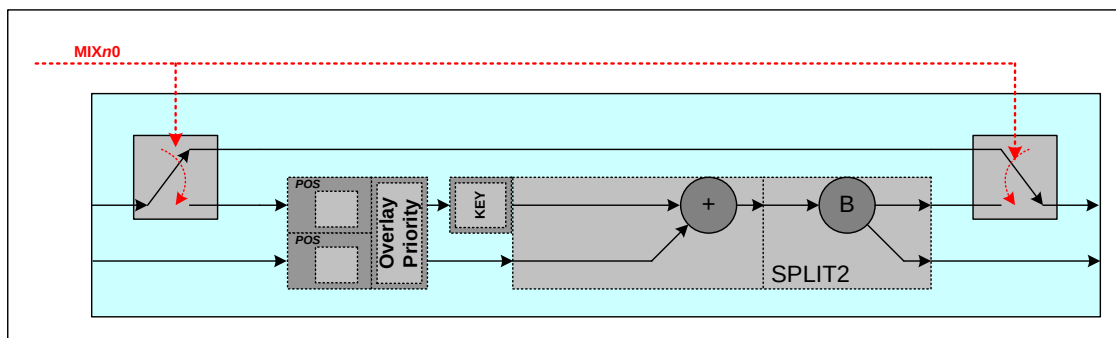


Figure 33.49 2-to-2 Window Mixer Structure

The figure above shows the structure of 2-to-2 Window Mixer. As shown above, the MIXn0 of CFG_MISC register decides the data path of 0th input channel of the mixer. It shows the configuration when MIXn0 is 0 (n = 1, 2).

33.9 Synchronization Between Source and Sink Components

There are three data paths in VIOC.

1. DATAPATH-1
Path starting from RDMA through the mixer to the display device
i.e. the case that data is simultaneously delivered to display device and WDMA via splitter from the 4-to-2 mixer
2. DATAPATH-2
Path starting from VIDEOIN to WDMA via the mixer
i.e. the case that RDMA is used in order to overlay with VIDEOIN
3. DATAPATH-3
Path starting from RDMA to WDMA via the mixer
i.e. the case that the display device and VIDEOIN are excluded from the entire path

The 3 data paths above are basic. In case of DATAPATH-1, the data transmission speed and synchronization are decided by the display device. In the case of DATAPATH-2, these are decided by VIDEOIN. However, in the case of DATAPATH-3, since synchronization is not done, speed cannot be consistent.

In the case that an image made from DATAPATH-3 is used as an input for DATAPATH-1, inter-frame flickering (the phenomenon that the previous frame and the current frame are displayed on one screen) due to each different speed may occur. In order to remove this, at least more than 3 frame buffers are required. Therefore, additional resources and frame buffer management are required.

This chapter explains how to minimize additional resource and frame buffer management. In the following description, Source Component refers to RDMA and Sink Component refers to WDMA.

33.9.1 Synchronous Synchronization Between Source and Sink Component

As above, an output of DATAPATH-3 is used as an input of DATAPATH-1. Synchronous synchronization is a method to remove inter-frame flickering between two components by using only one frame buffer.

Since sink component processes one frame slower than source component, eventually the operation speed of the sink component is the same as that of the source component. For example, when synchronous synchronization is used, if the source component works at 60fps(frame-per-second), the sink component works at 60fps, too.

Synchronous synchronization can be executed by using SYNCEN, SYNCMD and SYNCSEL bits of WDMARATE register. In addition, a special attention should be given to setting the base address of the sink component and the source component

0th bit of SYNCMD[2:0] refers to a mode used to make the speed of the sink component slower than that of the source component. Normally by comparing an address, this method makes the sink component slower than the source component. In the case of 0, an absolute address is used for comparing. In the case of 1, a relative address is used for comparing. Normally, an absolute address is better to be used. However, in order for a couple of different buffers to have the same

speed for synchronization, a relative address can be used.

2nd bit of SYNCMD[2:0] refers to a mode to decide the start timing of the sink component operation. In the case of 0, it means edge sensitive. In the case of 1, it refers to level sensitive. In edge sensitive mode, the sink component starts its operation at the start timing of the source component operation. In level sensitive mode, the sink component starts its operation while the source component is operating.

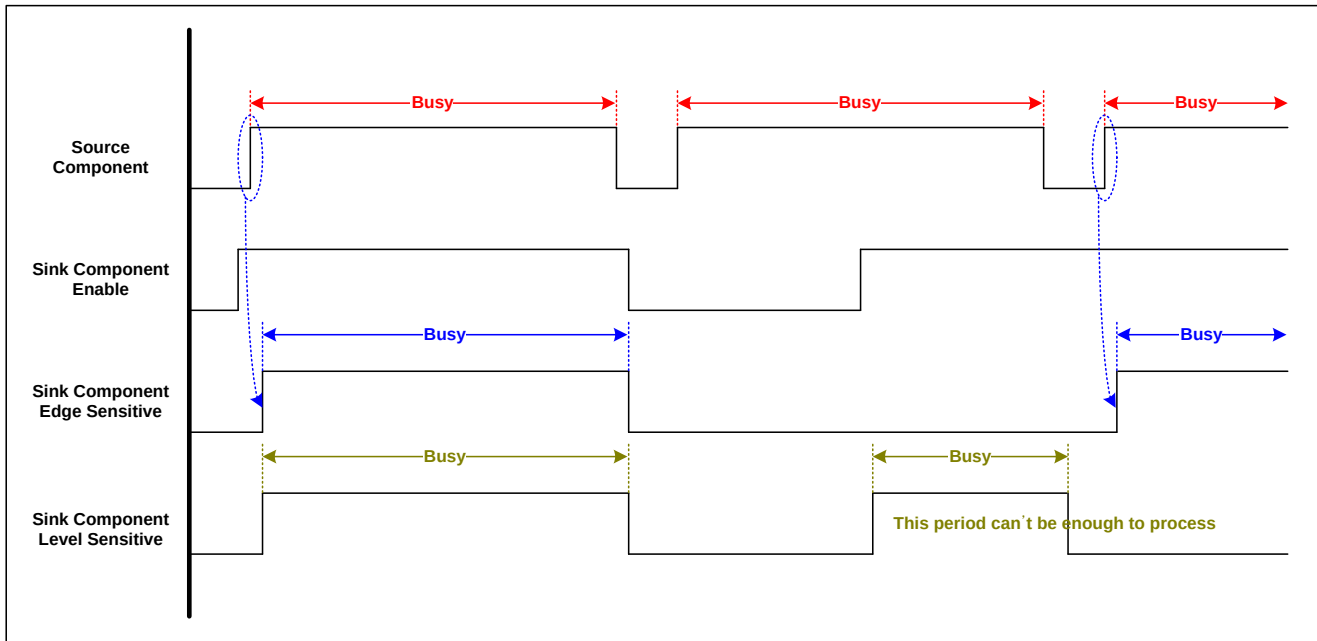


Figure 33.50 Synchronous Synchronization Between Source and Sink Component

The figure above shows the difference between edge-sensitive and level-sensitive modes. The operations of the two modes are same when the sink component is enabled for the first time. However, when the sink component is enabled for the second time, the sink component operates at the start time of the next frame although the source component is operating.. In the case of level-sensitive mode, since the source component is operating when the sink component is enabled, the sink component immediately starts operating. However, if the time until the operation of the source component is completed is short, the sink component may not operate normally in level-sensitive mode. Therefore, it is recommended to use the edge-sensitive mode.

33.9.2 Asynchronous Synchronization Between Source and Sink Component

Asynchronous synchronization is used in order to remove inter-frame flickering occurred when data is transmitted between different DATAPATHs whose frame rates are different. If the same image as an image outputted from display device 1 is intended to be outputted with display device 2, these two display devices have different frame rates. Therefore, the asynchronous synchronization can be used in this case. In addition, it can be used when one video stream is outputted with more than 2 different display devices.

If asynchronous synchronization is used, since there are data paths with different frame rates, frame information should include time information. This is because when contents with the same speed are displayed on two different display devices with different speed, a frame should be either duplicated or skipped.

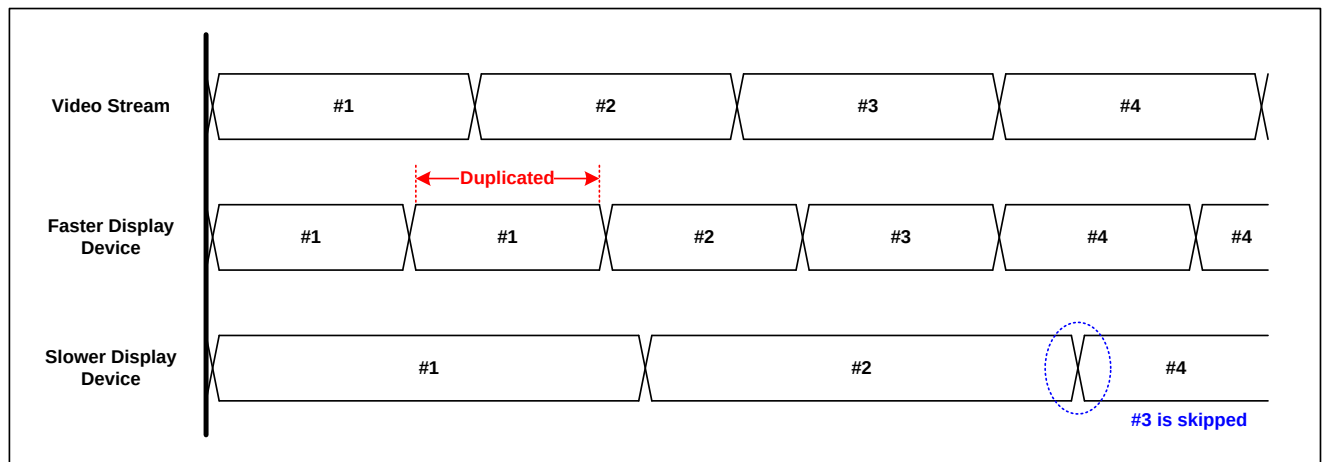


Figure 33.51 Case of Displaying Same Content Through the Two Different Display Devices with Different Frame Rates

The figure above shows frame management where content is outputted with two display devices (Faster and Slower). As can be seen in the figure above, “Video Stream” frames with a certain speed can be outputted with a “Faster Display Device” and a “Slower Display Device”. At this time, either frame duplication or frame skip occurs. If the content is outputted with a faster display device, repeated frame-duplication occurs. If the content is outputted with a slower display device, frame skip (as can be seen where “#3” is missing in the figure above) occurs.

More importantly, the lifetime variance of #1 ~ #4 in the “Video Stream” is too great. Therefore, frame buffer management to remove inter-frame flickering may be difficult.

Normally, asynchronous synchronization connects the buffer controller between the WDMA and RDMA. In other words, an asynchronous FIFO controller should be connected between the WDMA and RDMA. However, since the FIFO controller does not have a frame buffer, frame buffer allocation is definitely required. It does not only have the FIFO structure but also has time information in order to support frame skip. Since either frame-duplication or frame-skip is decided by time information in the VIOC System Timer, time information corresponding to the system timer needs to be inputted.

33.10 Streaming Mode (VRDMA)

Screen is composed of a user interface and video streaming. In case of a user interface, there is no time information for an image itself. In case of a video clip, time information (frame-rate) is included. In some cases, the frame-rate is different from a refresh rate of a display. In this case, buffer management of a video stream should be controlled corresponding to the refresh rate. If a frame-rate of the video stream gets high, the discrepancy between two gets big and this leads to that an image is displayed unnaturally. The followings explain how to process stream mode for a progressive and interlaced image. Note that when stream mode is used, more attention should be paid to managing life time of a frame buffer.

33.10.1 Progressive Images

If a refresh rate of a display device is 60Hz (60 frames-per-second) and a frame rate of a video clip is 24fps, based on the refresh rate of the display device, sampling 24 frames per second is done at 60Hz. The following figure shows the relation between the frame rate and refresh rate in this example.

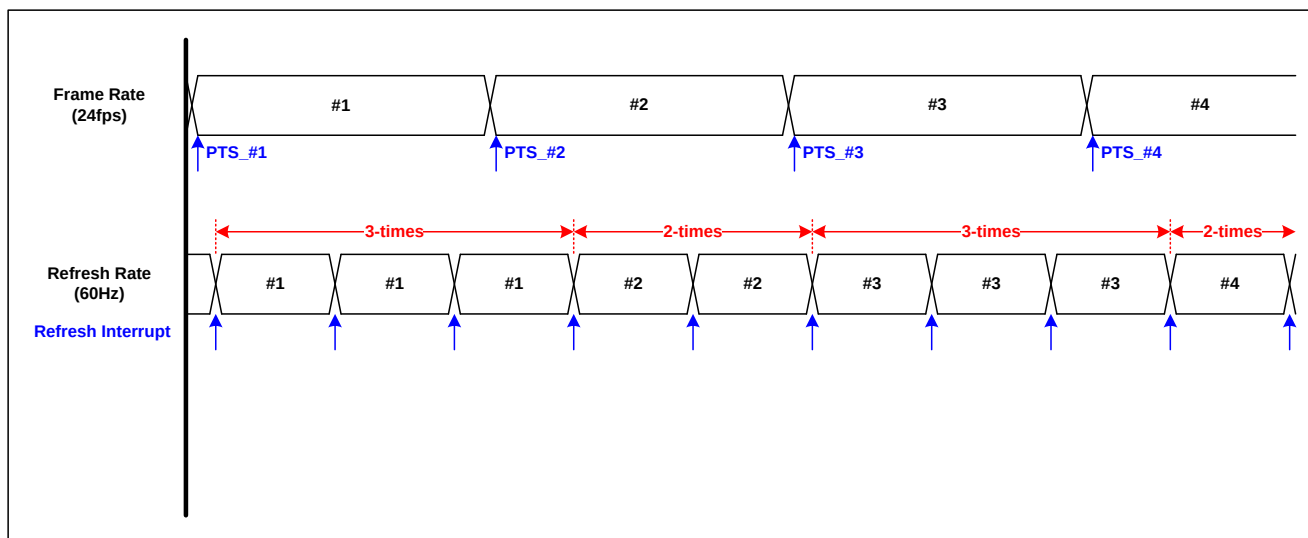


Figure 33.52 Frame Rate vs. Refresh Rate in the Progressive Image

As the figure above, since the refresh rate is higher than the frame rate, one image is repeated with 2 or 3 times repetitive refresh. An interrupt occurs corresponding to the refresh-rate. At this time, the video image buffer should be controlled. However, in stream mode time information such as PTS_#1, PTS_#2 or PTS_#3 is inputted for each image and when the time, which has been set previously, comes hardware automatically updates the above procedure. After updating, an interrupt occurs as soon as display starts. Therefore, a frame buffer can be managed based on an occurred interrupt. However, since this update is executed based on time information, it should be considered.

33.10.2 Interlaced Images

The interlaced stream is more complicated compared to the progressive stream. This is because interlaced frame contains time information for two numbers of timings per one frame. The following figure shows the interlaced stream in progress. “T” refers to the first field, which means it should be displayed first and “B” refers to the second field.

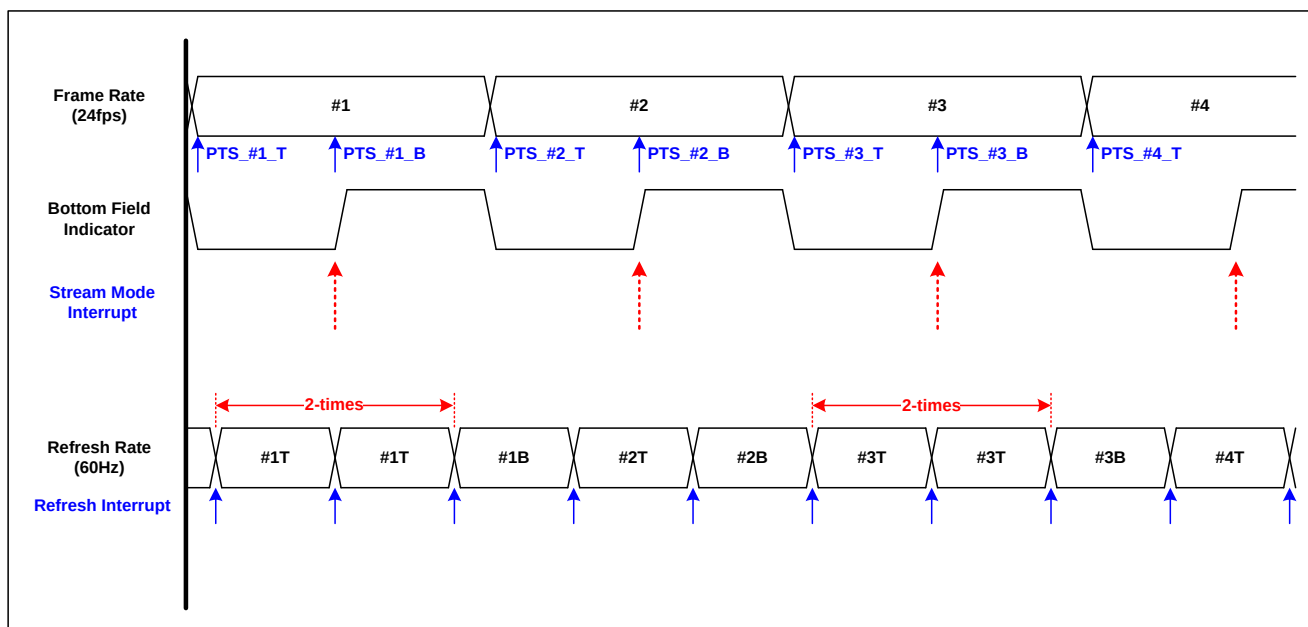


Figure 33.53 Frame Rate vs. Refresh Rate in the Interlaced Image

Since one frame has two numbers of timing information, if there are two frames, repetitions of the vide stream after synchronized with the refresh rate are dramatically reduced unlike progressive. Therefore, a stream is more difficult to be controlled. Software can be controlled corresponding to the refresh interrupt. After being projected to the bottom field, stream mode interrupt occurs. At this time, if information for the next frame is updated, the software can be easily controlled.

33.11 Scaler Related Operation

33.11.1 Definitions of Destination and Output Image

SCSSIZE, SCDSIZE, SCOPOS, and SCOSIZE are the registers defining scaler setting. The SCSSIZE represents the width and height of a source image decided by an input of the scaler. The SCDSIZE means a destination size decided by a ratio of the scaler (Region B in the following figure). The SCOPOS and SCOSIZE registers decide an output image of the scaler and show an image with the size decided by SCOSIZE register (the width and height from the output position decided by SCOPOS register), which refers to Region A in the following figure.

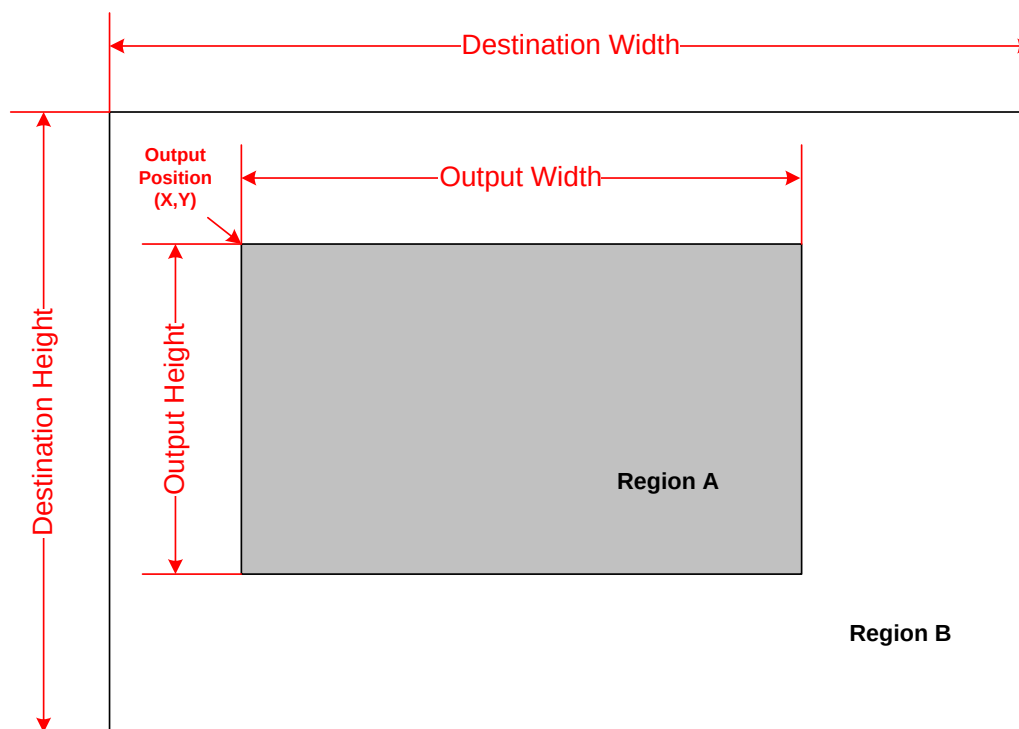


Figure 33.54 Scaler Definitions of Destination and Output Image

Normally, SCOPOS is (0,0) and the value of SCOSIZE is same as that of SCDSIZE. However, when Zoom-In is executed with a fixed output ratio, SCOPOS and SCOSIZE can be used.

33.11.2 Up and Down Scaling on the On-The-Fly Path and Restrictions

In the datapath from memory to memory, there is no limitation according to the scale ratio. However, in case of display device operation, which is an operation with a constant data rate, there may be limitations according to a ratio of the scaler because an image should be outputted within a certain period of time.

In up-scale, if destination image size is very large and output image size is relatively smaller, data processing for the area which is included in the destination image area but not included in the output area is required. At this time, due to unnecessarily wasted time for the data processing, the operation of the scaler is hard to be completed within the certain amount of time.

In down-scale, if a source image is too large and a destination image is too small, it is impossible to finish data processing within a certain amount of time because all data of the source image should be read and scaled-down for output.

In this case, it should be carefully processed not to cause underrun on the application in on-the-fly.

33.12 Video (Camera) Input Related Operation

33.12.1 VIDEOIN Module

Figure 33.55 shows the block diagram of VIDEOIN component consisting of 6 blocks. There are ONOFF, and SYNC_CTRL to process sync and WINDOW block for cropping an inputted image data. In addition, there is BUFFER block for image processing in the entire VIOC module. This block is shared for interpolation of chrominance. Y2R converter for RGB base processing of VOIC module exists and there is also Look-up Table per each channel for gamma, histogram, pixel mapping of color.

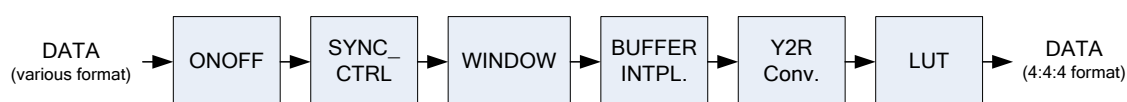


Figure 33.55 VIDEOIN Component

For inputted data, they are 4:2:2 or 4:4:4 format. However, ternal VIOC 4:4:4 format is used for processing so 4:2:2 to 4:4:4 conversion is required. For doing this, interpolation for chrominance channel is processed.

33.13 Video Input DEMUX Operation

33.13.1 VINDEMUX Module

VINDEMUX Module is a module located left bottom of Figure 33.29. It connects the data line of 2-ports with 4 VIDEOIN components inside of VIOC. It also supports Time Multiplex Mode and separates each frame data and connects them to each component. (In TCC8132, the VIOC has 2 VIDEOIN components.)

Figure 33.56 is the block diagram of VIN DEMUX. There are two input ports. These are connected to VIDEOIN component via MUX without passing through the decoder. Since there are two input ports, they are connected to two components out of four components. Since multiple connection is supported, one image data can be saved as 2~4 numbers of the same image at a time. For this MUX setting, set SEL0~4 of VD_CTRL Register.

“Time Multiplex Decoder” block decodes the mode explained in Chapter 33.3.7.5 Time-Multiplex Mode (TW2867 device for Techwell Inc.) and separates an image into 2 or 4 images. Post_MUX block can control whether data inputted from outside will be used or the decoded data will be used. After this, it connects the concerned data to VIDEOIN component inside of VIOC.

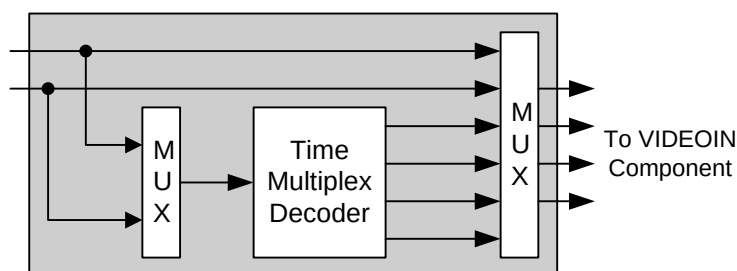


Figure 33.56 VIN DEMUX Block Diagram

33.14 VIQE (Video Image Quality Enhancement) Related Operation

33.14.1 De-interlacer

The objective of de-interlacing is to convert interlaced scanning into progressive scanning, overcoming the intrinsic nature of the interlaced video sampling structure. In interlaced scanning method, if a fast moving part exists between two fields resulting annoying jitters between lines, and an edge especially along horizontal direction exists, that performance degradation of de-interlacing becomes more noticeable. In other words, there is a theoretical limitation in high frequency components temporally or spatially.

The de-interlacing module in the VIQE has the following features.

- ① Pixel-wise processing
- ② Edge directional interpolation
- ③ Motion adaptive processing
- ④ 3:2 pull-down film mode detection
- ⑤ Bad edition/judder detection

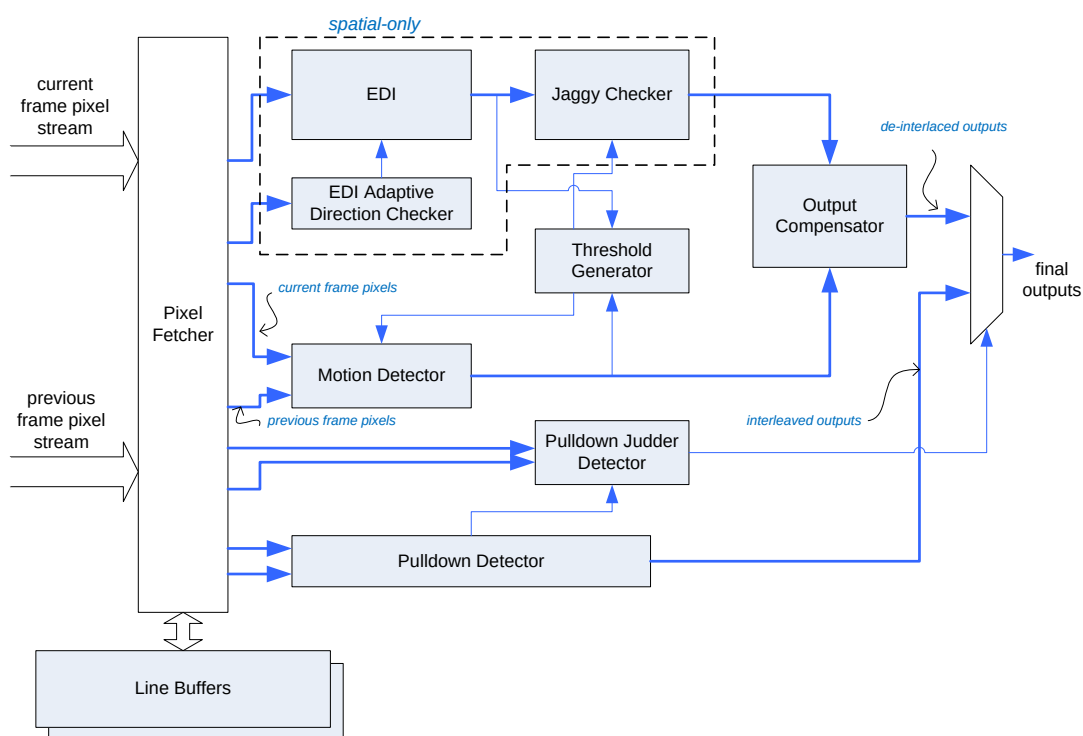


Figure 33.57 De-Interlacer Block Diagram

De-interlacing hardware is composed of several sub modules to implement various algorithms for compensating and coping with the serious negative effects from conversion scanning. The figure above describes the internal block diagram of de-interlacing module. All modules including de-interlacing module in the VIQE are pipelined in inter-module as well as inner-module, and input/output interfaces consist of actual pixel data stream and timing control signals.

The EDI, EDI adaptive direction checker, and jaggy checker have functions for spatial de-interlacing, and are default streaming path if the motion detector determines solid existence of motion or if the spatial-only mode in the control register is selected. The threshold generator block can provide the thresholds or parameter values which can be changed dynamically adapting the neighboring pixels instead of fixing as user-defined values. Motion detector decides whether the current pixel is moving or not through examination of neighboring pixels spatially or temporally. The failure of detecting motion causes serious visual artifacts, so the output compensator would check the reliability of the result of motion adaptive processing as well as spatial adaptive processing.

The pull-down detector and pull-down judder detector are designed to operate reverse 3:2 pull-down procedure, which consists of three steps in the VIQE. The first is to detect whether this stream is film source or not, and the next step is to check the existence of judder caused by a bad edition or addition of texture at normal display rate and the last step is to interleave accurate field line or to operate de-interlacing at the detected judder pixels. The detection of film mode takes advantage of statistical information acquired during some frames.

34 Overlay Mixer

34.1 Overview

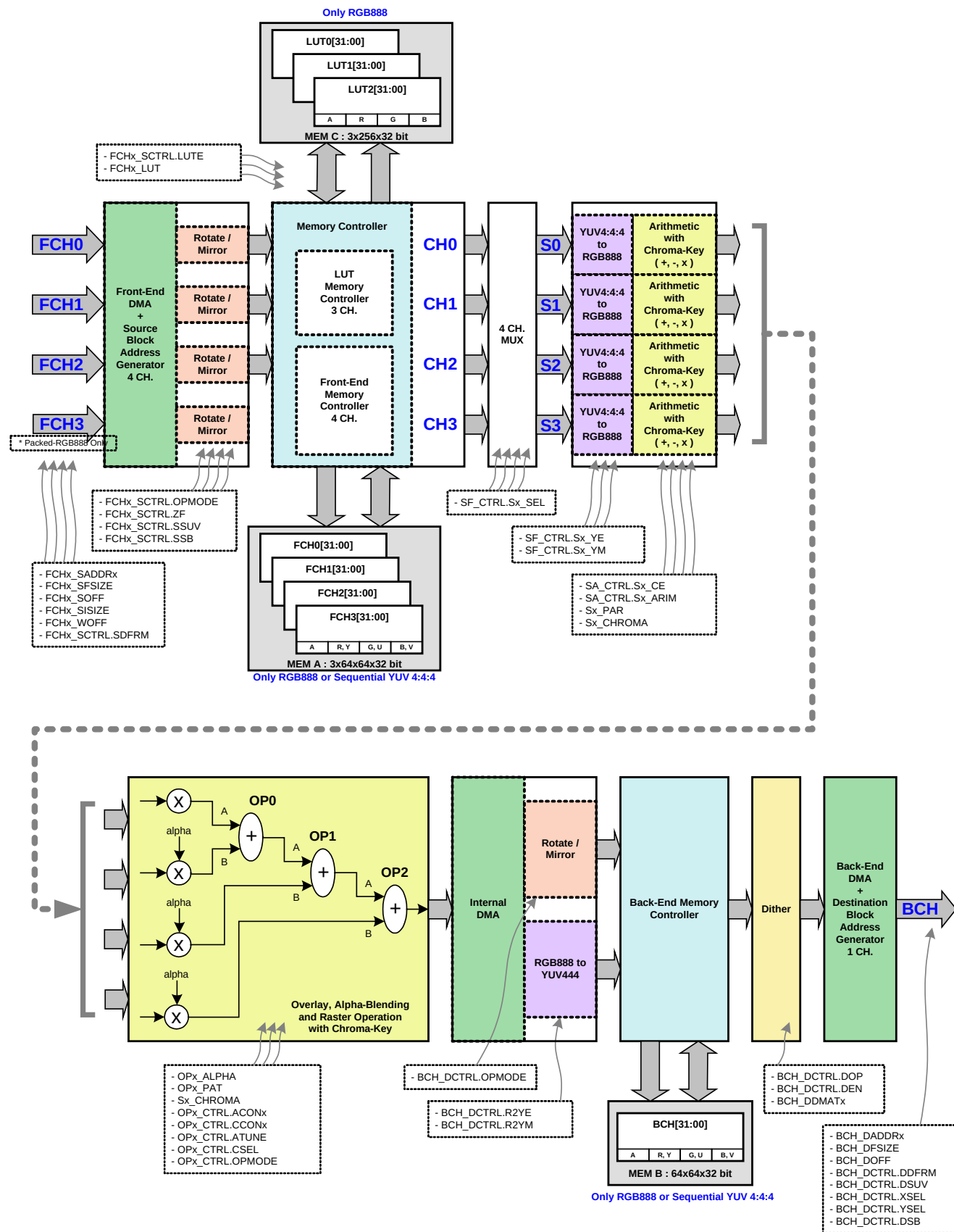
Figure 34.1 shows the whole block diagram of Overlay Mixer and its corresponding registers

As shown in Figure 34.1, the Overlay Mixer supports Rotating/Mirroring image, bitwise ROP, Alpha-Blending, Arithmetic or Format Converting function. These functions can simultaneously work. The Overlay Mixer has four different source channels and one destination channel. One output image can be created with three Input images by using an appropriate function of Overlay Mixer. Moreover, specific local region operation is available since the Overlay Mixer supports Source Image Offset, Destination Image Offset and Window Offset.

34.2 Features

The features of Overlay Mixer are as follows;

- Input / Output Image Format
 - Indexed Color Format : 8bpp
 - RGB Format : RGB332, RGB444, RGB454, RGB555, RGB565, RGB666, RGB888, Packed-RGB888 (Only Input)
 - Alpha-RGB(ARGB) Format : ARGB4444, ARGB3454, ARGB1555, ARGB4666, ARGB6666, ARGB4888, ARGB8888
 - Sequential YUV : YUV4:4:4, YUV4:2:2
 - Separated YUV : YUV4:4:4, YUV4:4:0, YUV4:2:2, YUV4:2:0, YUV4:1:1, YUV4:1:0
 - Interleaved YUV : YUV4:2:2, YUV4:2:0
- Rotate/Mirror Operation
 - 3 Channel Source 90/180/270 Rotate with Window Offset and Source Offset
 - 1 Channel Destination 90/180/270 Rotate with Destination Offset
 - 3 Channel Source Vertical/Horizontal/Vertical & Horizontal Mirror with Window Offset and Source Offset
 - 1 Channel Destination Vertical/Horizontal/Vertical & Horizontal Mirror with Destination Offset
- Arithmetic Operation
 - 3 Channel Arithmetic Operation with Chroma-Key Function
 - ◆ Arithmetic Function : Bypass, Fill, Inversion, Addition, Subtract, Multiplication
- Raster Operation & Alpha-Blending & Overlay
 - 3 Channel to 1 Channel ROP Operation with Chroma-Key & Window Offset
 - 3 Channel to 1 Channel Alpha Blending with Chroma-Key & Window Offset
 - ROP Function : 16 Type
 - Alpha-Blending Operation by Fixed Alpha-Value
 - Alpha-Blending Operation by ARGB Input Image Format
- YUV to RGB Format Converting
 - 3 Independent Source YUV to RGB Format Converting
 - 4 YUV to RGB Format Converting Type
- RGB to YUV Format Converting
 - 1 Destination RGB to YUV Format Converting
 - 4 RGB to YUV Format Converting Type
- Color LUT
 - 3 Channel Source Indexed Color with LUT : 8bpp
 - 3 Channel Source Alpha with LUT : ARGB4444, ARGB3454, ARGB1555, ARGB4666, ARGB6666, ARGB4888, ARGB8888
- RGB Format Dithering
 - 1 Channel Destination Dithering : RGB332, RGB444, ARGB4444, RGB454, ARGB3454, RGB555, ARGB1555, RGB565, RGB666, ARGB4666, ARGB6666, ARGB4888



[TOP Architecture]

Figure 34.1 Overlay Mixer Block Diagram

		31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Indexed Color Format	8bpp	Data3[7:0]								Data2[7:0]								Data1[7:0]								Data0[7:0]							
		R3[2:0]		G3[2:0]		B3[1:0]		R2[2:0]		G2[2:0]		B2[1:0]		R1[2:0]		G1[2:0]		B1[1:0]		R0[2:0]		G0[2:0]		B0[1:0]									
RGB Format	RGB332(8bit)	R3[2:0]		G3[2:0]		B3[1:0]		R2[2:0]		G2[2:0]		B2[1:0]		R1[2:0]		G1[2:0]		B1[1:0]		R0[2:0]		G0[2:0]		B0[1:0]									
	RGB444(12bit)	PAD		R1[3:0]		G1[3:0]		B1[3:0]		PAD		R0[3:0]		G0[3:0]		B0[3:0]																	
	ARGB4444(16bit)	A1[3:0]		R1[3:0]		G1[3:0]		B1[3:0]		A0[3:0]		R0[3:0]		G0[3:0]		B0[3:0]																	
	RGB454(13bit)	PAD		R1[3:0]		G1[4:0]		B1[3:0]		PAD		R0[3:0]		G0[4:0]		B0[3:0]																	
	ARGB3454(16bit)	A1[2:0]		R1[3:0]		G1[4:0]		B1[3:0]		A0[2:0]		R0[3:0]		G0[4:0]		B0[3:0]																	
	RGB555(15bit)	P		R1[4:0]		G1[4:0]		B1[4:0]		P		R0[4:0]		G0[4:0]		B0[4:0]																	
	ARGB1555(16bit)	A1		R1[4:0]		G1[4:0]		B1[4:0]		A0		R0[4:0]		G0[4:0]		B0[4:0]																	
	RGB565(16bit)	R1[4:0]		G1[5:0]		B1[4:0]		R0[4:0]		G0[5:0]		B0[4:0]																					
	RGB666(18bit)	PAD		PAD		PAD		R[5:0]		G[5:0]		B[5:0]																					
	ARGB4666(22bit)	PAD		PAD		A[3:0]		R[5:0]		G[5:0]		B[5:0]																					
	ARGB6666(24bit)	PAD		PAD		A[5:0]		R[5:0]		G[5:0]		B[5:0]																					
	RGB888(24bit)	PAD		PAD		PAD		R[7:0]		G[7:0]		B[7:0]																					
	ARGB4888(28bit)	PAD		A[3:0]		R[7:0]		G[7:0]		B[7:0]																							
	ARGB8888(32bit)	A[7:0]		R[7:0]		G[7:0]		B[7:0]																									
	Packed RGB888 (24bit)	B1[7:0]		G2[7:0]		R3[7:0]		B0[7:0]		G1[7:0]		R2[7:0]																					
Sequential YUV Format	Sequential YUV 4:4:4	PAD		Y[7:0]		U[7:0]		V[7:0]																									
	Sequential YUV 4:2:2	V0[7:0]		Y1[7:0]		U0[7:0]		Y0[7:0]																									
Separated YUV Format	Separated YUV 4:4:4	Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]	
	Separated YUV 4:4:0	U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]		U[7:0]	
	Separated YUV 4:2:2	V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]		V[7:0]	
	Separated YUV 4:1:1																																
Interleaved YUV Format	Interleaved YUV 4:2:2	Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]		Y[7:0]	
	Interleaved YUV 4:2:0 (Swap U and V)	U[7:0]		V[7:0]		Y[7:0]		U[7:0]		V[7:0]		Y[7:0]		U[7:0]		V[7:0]		Y[7:0]		U[7:0]		V[7:0]		Y[7:0]		U[7:0]		V[7:0]		Y[7:0]		U[7:0]	

Figure 34.2 Supported Data Format of Overlay Mixer

Note: Refer to Figure 34.4 for detail information of YUV Format

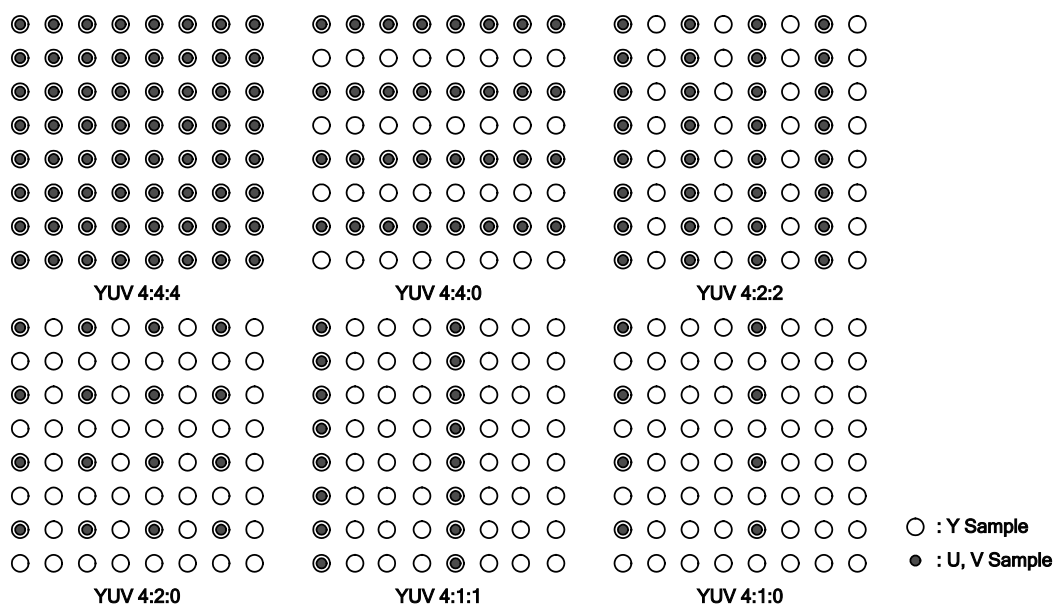


Figure 34.3 Sampling of YUV Format

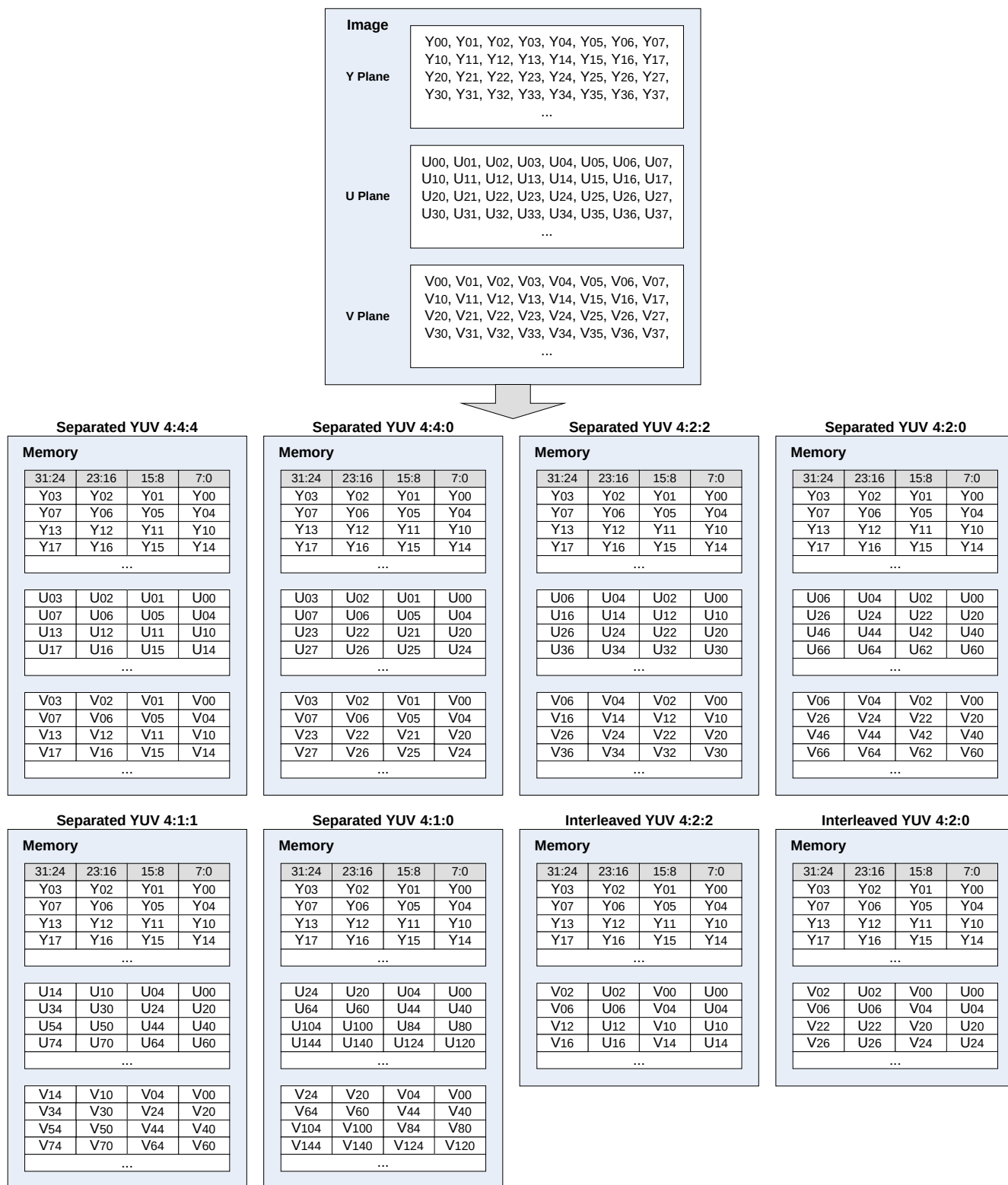


Figure 34.4 Detailed Information of YUV Format

34.3 Register Description

Table 34.1 Overlay Mixer Register Map (Base Address = 0x80001000)

Name	Address	Type	Reset	Description
FCH0_SADDR0	0x00	R/W	0x00000000	Front-End Channel 0 Source Address 0
FCH0_SADDR1	0x04	R/W	0x00000000	Front-End Channel 0 Source Address 1
FCH0_SADDR2	0x08	R/W	0x00000000	Front-End Channel 0 Source Address 2
FCH0_SFSIZE	0x0C	R/W	0x00000000	Front-End Channel 0 Source Frame Pixel Size
FCH0_SOFF	0x10	R/W	0x00000000	Front-End Channel 0 Source Pixel Offset
FCH0_SISIZE	0x14	R/W	0x00000000	Front-End Channel 0 Source Image Pixel Size
FCH0_WOFF	0x18	R/W	0x00000000	Front-End Channel 0 Window Pixel Offset
FCH0_SCTRL	0x1C	R/W	0x00000000	Front-End Channel 0 Control
FCH1_SADDR0	0x20	R/W	0x00000000	Front-End Channel 1 Source Address 0
FCH1_SADDR1	0x24	R/W	0x00000000	Front-End Channel 1 Source Address 1
FCH1_SADDR2	0x28	R/W	0x00000000	Front-End Channel 1 Source Address 2
FCH1_SFSIZE	0x2C	R/W	0x00000000	Front-End Channel 1 Source Frame Pixel Size
FCH1_SOFF	0x30	R/W	0x00000000	Front-End Channel 1 Source Pixel Offset
FCH1_SISIZE	0x34	R/W	0x00000000	Front-End Channel 1 Source Image Pixel Size
FCH1_WOFF	0x38	R/W	0x00000000	Front-End Channel 1 Window Pixel Offset
FCH1_SCTRL	0x3C	R/W	0x00000000	Front-End Channel 1 Control
FCH2_SADDR0	0x40	R/W	0x00000000	Front-End Channel 2 Source Address 0
FCH2_SADDR1	0x44	R/W	0x00000000	Front-End Channel 2 Source Address 1
FCH2_SADDR2	0x48	R/W	0x00000000	Front-End Channel 2 Source Address 2
FCH2_SFSIZE	0x4C	R/W	0x00000000	Front-End Channel 2 Source Frame Pixel Size
FCH2_SOFF	0x50	R/W	0x00000000	Front-End Channel 2 Source Pixel Offset
FCH2_SISIZE	0x54	R/W	0x00000000	Front-End Channel 2 Source Image Pixel Size
FCH2_WOFF	0x58	R/W	0x00000000	Front-End Channel 2 Window Pixel Offset
FCH2_SCTRL	0x5C	R/W	0x00000000	Front-End Channel 2 Control
FCH3_SADDR0	0x60	R/W	0x00000000	Front-End Channel 3 Source Address 0
-	0x64-0x6B	-	-	Reserved
FCH3_SFSIZE	0x6C	R/W	0x00000000	Front-End Channel 3 Source Frame Pixel Size
FCH3_SOFF	0x70	R/W	0x00000000	Front-End Channel 3 Source Pixel Offset
FCH3_SISIZE	0x74	R/W	0x00000000	Front-End Channel 3 Source Image Pixel Size
FCH3_WOFF	0x78	R/W	0x00000000	Front-End Channel 3 Window Pixel Offset
FCH3_SCTRL	0x7C	R/W	0x00000000	Front-End Channel 3 Control
S0_CHROMA	0x80	R/W	0x00000000	Source 0 Chroma-Key Parameter
S0_PAR	0x84	R/W	0x00000000	Source 0 Arithmetic Parameter
S1_CHROMA	0x88	R/W	0x00000000	Source 1 Chroma-Key Parameter
S1_PAR	0x8C	R/W	0x00000000	Source 1 Arithmetic Parameter
S2_CHROMA	0x90	R/W	0x00000000	Source 2 Chroma-Key Parameter
S2_PAR	0x94	R/W	0x00000000	Source 2 Arithmetic Parameter
S3_CHROMA	0x98	R/W	0x00000000	Source 3 Chroma-Key Parameter
S3_PAR	0x9C	R/W	0x00000000	Source 3 Arithmetic Parameter
SF_CTRL	0xA0	R/W	0x00000000	Source Function Control Register
SA_CTRL	0xA4	R/W	0x00000000	Source Arithmetic Control Register
-	0xA8-0xAF	-	-	Reserved
OP0_ALPHA	0xB0	R/W	0x00000000	Source Operator 0 Alpha
OP1_ALPHA	0xB4	R/W	0x00000000	Source Operator 1 Alpha
OP2_ALPHA	0xB8	R/W	0x00000000	Source Operator 2 Alpha
-	0xBC-0xBF	-	-	Reserved
OP0_PAT	0xC0	R/W	0x00000000	Source Operator 0 Pattern
OP1_PAT	0xC4	R/W	0x00000000	Source Operator 1 Pattern
OP2_PAT	0xC8	R/W	0x00000000	Source Operator 2 Pattern
-	0xCC-0xCF	-	-	Reserved
OP0_CTRL	0xD0	R/W	0x42130000	Source Operation 0 Control Register
OP1_CTRL	0xD4	R/W	0x42130000	Source Operation 1 Control Register
OP2_CTRL	0xD8	R/W	0x42130000	Source Operation 2 Control Register
-	0xDC-0xDF	-	-	Reserved
BCH_DADDR0	0xE0	R/W	0x00000000	Back-End Channel Destination Address 0
BCH_DADDR1	0xE4	R/W	0x00000000	Back-End Channel Destination Address 1
BCH_DADDR2	0xE8	R/W	0x00000000	Back-End Channel Destination Address 2
BCH_DFSIZE	0xEC	R/W	0x00000000	Back-End Channel Destination Frame Pixel Size
BCH_DOFF	0xF0	R/W	0x00000000	Back-End Channel Destination Pixel Offset
BCH_DCTRL	0xF4	R/W	0x00000000	Back-End Channel Control

Name	Address	Type	Reset	Description
-	0xF8 – 0xFF	-	-	Reserved
BCH_DDMAT0	0x100	R/W	0x00000000	Back-End Channel Destination Dither Matrix 0
BCH_DDMAT1	0x104	R/W	0x00000000	Back-End Channel Destination Dither Matrix 1
BCH_DDMAT2	0x108	R/W	0x00000000	Back-End Channel Destination Dither Matrix 2
BCH_DDMAT3	0x10C	R/W	0x00000000	Back-End Channel Destination Dither Matrix 3
OM_CTRL	0x110	R/W	0x00000000	Overlay Mixer Control
OM_IREQ	0x114	R/W	0x00000000	Overlay Mixer Interrupt Request
-	0x118 – 0x3FF	-	-	Reserved
FCH0_LUT	0x400 – 0x7FF	R/W	-	Front-End Channel 0 Lookup Table
FCH1_LUT	0x800 – 0xBFF	R/W	-	Front-End Channel 1 Lookup Table
FCH2_LUT	0xC00 – 0xFFF	R/W	-	Front-End Channel 2 Lookup Table

Front-End Channel 0 Source Address 0(FCH0_SADDR0)

0x80001000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR0[15:2]														Reserved	

SADDR0	[31:2]	Source Address 0
N (0 ~ 4095)		It determines the Y image base address of channel 0 image when the data format of channel 0 is YUV format. But, in case the data format of channel 0 is a sequential YUV or an RGB, it determines the base address of the whole image

The data format of FCH0 image is determined by the FCH0_SCTRL register.

Front-End Channel 0 Source Address 1 (FCH0_SADDR1)

0x80001004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR1[15:2]														Reserved	

SADDR1	[31:2]	Source Address 1
N (0 ~ 4095)		It determines the U image base address of front-end channel 0 (FCH0) when the data format of channel 0 is YUV format. But, in case the data format of FCH0 image is a sequential YUV or an RGB, it is not used.

Front-End Channel 0 Source Address 2 (FCH0_SADDR2)

0x80001008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR2[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR2[15:2]														Reserved	

SADDR2	[31:2]	Source Address 2
N (0 ~ 4095)		It determines the V image base address of FCH0 when the data format of FCH0 is YUV format. But, in case the data format of FCH0 image is a sequential YUV or an RGB, it is not used.

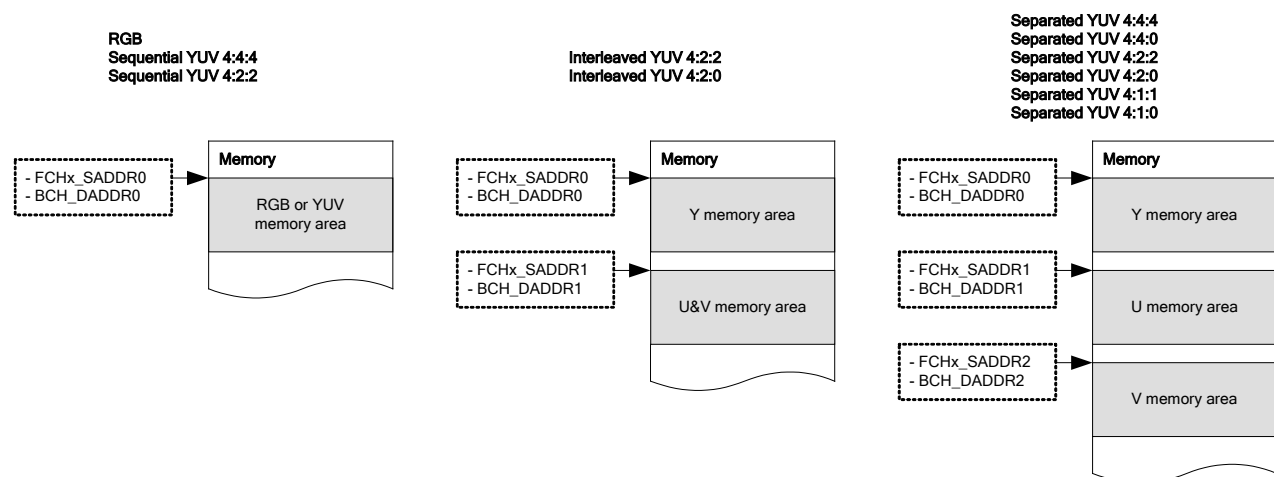


Figure 34.5 Source/Destination Base Address of Data Format

Front-End Channel 0 Source Frame Pixel Size (FCH0_SFSIZE)

0x8000100C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SFSIZE_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SFSIZE_X[11:0]											

SFSIZE_Y	[27:16]	Source Frame Pixel Size Y
N (0 ~ 4095)	height of FCH0 image by pixel units	
SFSIZE_X	[11:0]	Source Frame Pixel Size X
N (0 ~ 4095)	width of FCH0 image by pixel units	

Front-End Channel 0 Source Pixel Offset (FCH0_SOFF)

0x80001010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SOFF_X[11:0]											

SOFF_Y	[27:16]	Source Pixel Offset Y
N (0 ~ 4095)	N =Y Axis Source Pixel Offset	
SOFF_X	[11:0]	Source Pixel Offset X
N (0 ~ 4095)	N =X Axis Source Pixel Offset	

Front-End Channel 0 Source Image Pixel Size (FCH0_SISIZE)

0x80001014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SISIZE_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SISIZE_X[11:0]											

SISIZE_Y	[27:16]	Source Image Pixel Size Y
N (0 ~ 4095)	N =Y Axis Source Image Pixel Size	
SISIZE_X	[11:0]	Source Image Pixel Size X
N (0 ~ 4095)	N =X Axis Source Image Pixel Size	

Front-End Channel 0 Window Pixel Offset (FCH0_WOFF)

0x80001018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WOFF_X[11:0]											

WOFF_Y	[27:16]	Window Pixel Offset Y
N (0 ~ 4095)	N =Y Axis Window Pixel Offset	
WOFF_X	[11:0]	Window Pixel Offset X
N (0 ~ 4095)	N =X Axis Window Pixel Offset	

When FCH0 Image is selected as Source 0 by SF_CTRL.S0SEL register, Window Offset must be '0'.

Front-End Channel 0 Control (FCH0_SCTRL)

0x8000101C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SSB[2:0]			Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MABC	Reserved		LUTE	SSUV	OPMODE[2:0]			0	ZF[1:0]		SDFRM[4:0]				

SSB	[26:24]	Source Swap Byte
000		ARGB
001		ARBG
010		AGRB
011		AGBR
100		ABRG
101		ABGR
110		RABG
111		BGRA

MABC	[15]	AXI Bus Master Address Boundary Control
0		AXI Bus Master Address Boundary = 4KByte
1		AXI Bus Master Address Boundary = 1KByte This bit must be written as "1"

LUTE	[12]	LUT Enable
0		LUT Disable
1		LUT Enable

SSUV	[11]	Source Swap U for V
0		Keep on U and V (LSB = U, MSB = V)
1		Swap U for V (LSB = V, MSB = U)

This register applies only when Data Format is Interleaved YUV Format

OPMODE	[10:8]	Operation Mode
00x		Data Copy
010		Horizontal Mirror
011		Vertical Mirror
100		Vertical & Horizontal Mirror
101		90 Degree Rotate (Clockwise)
110		180 Degree Rotate (Clockwise)
111		270 Degree Rotate (Clockwise)

ZF	[6:5]	Zero Fill
00		MSB Fill Mode Enable
01		Zero Fill Mode Enable
1x		HOB (High-Order Bits) Fill Mode Enable

The content above applies only when Data Format is RGB Format, not RGB888 / ARGB8888. Refer to Figure 34.6 for detailed information.

Color Field Bits	A				R				G				B			
	31	...	24		23	...	16		15	...	8		7	...	0	
RGB444 ARGB4444	0	1	1	0			1	0	1	0			0	0	1	0
ZERO FILL	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0
MSB FILL	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0
HOB FILL	0	1	1	0	0	1	1	0	1	0	0	0	1	0	1	0
RGB454 ARGB3454	1	1	0				0	0	1	0			1	0	1	0
ZERO FILL	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0
MSB FILL	1	1	0	1	1	1	1	1	0	0	1	0	1	1	1	1
HOB FILL	1	1	0	1	1	0	1	1	0	1	0	1	1	0	1	0
RGB555 ARGB1555	1						0	0	1	0	1		1	0	1	0
ZERO FILL	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0
MSB FILL	1	1	1	1	1	1	1	1	0	0	1	0	1	0	0	0
HOB FILL	1	1	1	1	1	1	1	1	0	0	1	0	1	0	0	0
RGB565							0	0	1	0	1		1	0	1	0
ZERO FILL							0	0	1	0	1	0	0	0	0	0
MSB FILL							0	0	1	0	1	0	0	0	0	0
HOB FILL							0	0	1	0	1	0	0	0	0	0
RGB666 ARGB4666 ARGB6666	0	1	1	0	0	0	0		0	0	1	0	1	1	0	
ZERO FILL	0	1	1	0	0	0	0	0	0	0	1	0	1	1	0	0
MSB FILL	0	1	1	0	0	1	0	0	0	0	1	0	1	1	0	0
HOB FILL	0	1	1	0	0	1	0	0	0	0	1	0	1	1	0	0
ARGB4888	1	1	1	0			0	0	1	0	1	1	0	1	1	1
ZERO FILL	1	1	1	0	0	0	0	0	0	0	1	0	1	0	0	0
MSB FILL	1	1	1	0	1	1	1	1	0	0	1	0	1	0	0	0
HOB FILL	1	1	1	0	1	1	1	0	0	0	1	0	1	0	0	0
RGB332							1	0	1				0	1		
ZERO FILL							1	0	1	0	0	0	0	0	0	0
MSB FILL							1	0	1	1	1	1	1	1	1	1
HOB FILL							1	0	1	1	0	1	1	0	1	0

Figure 34.6 Example: Input Source Image Data to RGB888/ARGB8888 Conversion

SDFRM	[4:0]	Source Data Format
00000		Separated YUV 4:4:4
00001		Separated YUV 4:4:0
00010		Separated YUV 4:2:2
00011		Separated YUV 4:2:0
00100		Separated YUV 4:1:1
00101		Separated YUV 4:1:0
00110		Interleaved YUV 4:2:2
00111		Interleaved YUV 4:2:0
01000		Sequential YUV 4:4:4
01001		Sequential YUV 4:2:2
0101x		RGB332 (8bpp)
01100		RGB444
01101		ARGB4444
01110		RGB454
01111		ARGB3454
10000		RGB555
10001		ARGB1555
1001x		RGB565
10100		RGB666
10101		ARGB4666
10110		ARGB6666
10111		RGB888
11000		ARGB4888
11001		ARGB8888
Others		Not used

Front-End Channel 1 Source Address 0 (FCH1_SADDR0) 0x80001020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR0[15:2]														Reserved	

In case that Source Image Format is Separated YUV X:X:X, the value of the address is Source Address of Y region.

In case that Source Image Format is not Separated YUV X:X:X, the value of the address is Source Address.

Front-End Channel 1 Source Address 1 (FCH1_SADDR1) 0x80001024

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR1[15:2]														Reserved	

In case that Source Image Format is Separated YUV X:X:X, the value of the address is Source Address of U region.

In case that Source Image Format is not Separated YUV X:X:X, the value of the address is not used.

Front-End Channel 1 Source Address 2 (FCH1_SADDR2) 0x80001028

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR2[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR2[15:2]														Reserved	

In case that Source Image Format is Separated YUV X:X:X, the value of the address is Source Address of V region.

In case that Source Image Format is not Separated YUV X:X:X, the value of the address is not used.

Front-End Channel 1 Source Frame Pixel Size (FCH1_SFSIZE) 0x8000102C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SFSIZE_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SFSIZE_X[11:0]											

SFSIZE_Y	[27:16]	Source Frame Pixel Size Y
N (0 ~ 4095)	N =Y Axis Source Frame Pixel Size	

SFSIZE_X	[11:0]	Source Frame Pixel Size X
N (0 ~ 4095)	N =X Axis Source Frame Pixel Size	

Front-End Channel 1 Source Pixel Offset (FCH1_SOFF) 0x80001030

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SOFF_X[11:0]											

SOFF_Y	[27:16]	Source Pixel Offset Y
N (0 ~ 4095)	N =Y Axis Source Pixel Offset	

SOFF_X	[11:0]	Source Pixel Offset X
N (0 ~ 4095)	N =X Axis Source Pixel Offset	

Front-End Channel 1 Source Image Pixel Size (FCH1_SISIZE) 0x80001034

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SISIZE_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SISIZE_X[11:0]											

SISIZE_Y	[27:16]	Source Image Pixel Size Y
N (0 ~ 4095)	N =Y Axis Source Image Pixel Size	

SISIZE_X	[11:0]	Source Image Pixel Size X
N (0 ~ 4095)	N =X Axis Source Image Pixel Size	

Front-End Channel 1 Window Pixel Offset (FCH1_WOFF)

0x80001038

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WOFF_X[11:0]											

WOFF_Y	[27:16]	Window Pixel Offset Y
N (0 ~ 4095)		N =Y Axis Window Pixel Offset

WOFF_X	[11:0]	Window Pixel Offset X
N (0 ~ 4095)		N =X Axis Window Pixel Offset

When FCH1 Image is selected as Source 0 by SF_CTRL.S0SEL register, Window Offset must be '0'.

Front-End Channel 1 Control (FCH1_SCTRL)

0x8000103C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SSB[2:0]				Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MABC	Reserved		LUTE	SSUV	OPMODE[2:0]			0	ZF[1:0]		SDFRM[4:0]				

SSB	[26:24]	Source Swap Byte
000		ARGB
001		ARBG
010		AGRB
011		AGBR
100		ABRG
101		ABGR
110		RABG
111		BGRA

MABC	[15]	AXI Bus Master Address Boundary Control
0		AXI Bus Master Address Boundary = 4KByte
1		AXI Bus Master Address Boundary = 1KByte This bit must be written as "1"

LUTE	[12]	LUT Enable
0		LUT Disable
1		LUT Enable

SSUV	[11]	Source Swap U for V
0		Keep on U and V (LSB = U, MSB = V)
1		Swap U for V (LSB = V, MSB = U)

This register applies only when Data Format is Interleaved YUV Format.

OPMODE	[10:8]	Operation Mode
00x		Data Copy
010		Horizontal Mirror
011		Vertical Mirror
100		Vertical & Horizontal Mirror
101		90 Degree Rotate (Clockwise)
110		180 Degree Rotate (Clockwise)
111		270 Degree Rotate (Clockwise)

ZF	[6:5]	Zero Fill
00		MSB Fill Mode Enable
01		Zero Fill Mode Enable
1x		HOB(High-Order Bits) Fill Mode Enable

The content above applies only when Data Format is RGB Format, not RGB888 / ARGB8888. Refer to Figure 34.6 for detailed information.

SDFRM	[4:0]	Source Data Format
00000		Separated YUV 4:4:4
00001		Separated YUV 4:4:0
00010		Separated YUV 4:2:2
00011		Separated YUV 4:2:0
00100		Separated YUV 4:1:1
00101		Separated YUV 4:1:0
00110		Interleaved YUV 4:2:2
00111		Interleaved YUV 4:2:0
01000		Sequential YUV 4:4:4
01001		Sequential YUV 4:2:2
0101x		RGB332 (8bpp)
01100		RGB444
01101		ARGB4444
01110		RGB454
01111		ARGB3454
10000		RGB555
10001		ARGB1555
1001x		RGB565
10100		RGB666
10101		ARGB4666
10110		ARGB6666
10111		RGB888
11000		ARGB4888
11001		ARGB8888
else		Not used

Front-End Channel 2 Source Address 0 (FCH2_SADDR0) 0x80001040

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR0[15:2]														Reserved	

In case that Source Image Format is Separated YUV X:X:X, the value of the address is Source Address of Y region.
In case that Source Image Format is not Separated YUV X:X:X, the value of the address is Source Address.

Front-End Channel 2 Source Address 1 (FCH2_SADDR1) 0x80001044

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR1[15:2]														Reserved	

In case that Source Image Format is Separated YUV X:X:X, the value of the address is Source Address of U region.
In case that Source Image Format is not Separated YUV X:X:X, the value of the address is not used.

Front-End Channel 2 Source Address 2 (FCH2_SADDR2) 0x80001048

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR2[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR2[15:2]														Reserved	

In case that Source Image Format is Separated YUV X:X:X, the value of the address is Source Address of V region.
In case that Source Image Format is not Separated YUV X:X:X, the value of the address is not used.

Front-End Channel 2 Source Frame Pixel Size (FCH2_SFSIZE)

0x8000104C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SFSIZE_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SFSIZE_X[11:0]											

SFSIZE_Y	[27:16]	Source Frame Pixel Size Y
N (0 ~ 4095)	N=Y Axis Source Frame Pixel Size	

SFSIZE_X	[11:0]	Source Frame Pixel Size X
N (0 ~ 4095)	N=X Axis Source Frame Pixel Size	

Front-End Channel 2 Source Pixel Offset (FCH2_SOFF)

0x80001050

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SOFF_X[11:0]											

SOFF_Y	[27:16]	Source Pixel Offset Y
N (0 ~ 4095)	N=Y Axis Source Pixel Offset	

SOFF_X	[11:0]	Source Pixel Offset X
N (0 ~ 4095)	N=X Axis Source Pixel Offset	

Front-End Channel 2 Source Image Pixel Size (FCH2_SISIZE)

0x80001054

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SISIZE_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SISIZE_X[11:0]											

SISIZE_Y	[27:16]	Source Image Pixel Size Y
N (0 ~ 4095)	N=Y Axis Source Image Pixel Size	

SISIZE_X	[11:0]	Source Image Pixel Size X
N (0 ~ 4095)	N=X Axis Source Image Pixel Size	

Front-End Channel 2 Window Pixel Offset (FCH2_WOFF)

0x80001058

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WOFF_X[11:0]											

WOFF_Y	[27:16]	Window Pixel Offset Y
N (0 ~ 4095)	N=Y Axis Window Pixel Offset	

WOFF_X	[11:0]	Window Pixel Offset X
N (0 ~ 4095)	N=X Axis Window Pixel Offset	

When FCH2 Image is selected as Source 0 by SF_CTRL.S0SEL register, Window Offset must be '0'.

Front-End Channel 2 Control (FCH2_SCTRL)

0x8000105C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SSB[2:0]			Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MABC	Reserved		LUTE	SSUV	OPMODE[2:0]			0	ZF[1:0]		SDFRM[4:0]				

SSB	[26:24]	Source Swap Byte
000	ARGB	
001	ARBG	
010	AGRB	
011	AGBR	
100	ABRG	
101	ABGR	
110	RABG	
111	BGRA	

MABC	[15]	AXI Bus Master Address Boundary Control
0		AXI Bus Master Address Boundary = 4KByte
1		AXI Bus Master Address Boundary = 1KByte <i>This bit must be written as "1"</i>

LUTE	[12]	LUT Enable
0		LUT Disable
1		LUT Enable

SSUV	[11]	Source Swap U for V
0		Keep on U and V (LSB = U, MSB = V)
1		Swap U for V (LSB = V, MSB = U)

This register applies only when Data Format is Interleaved YUV Format

OPMODE	[10:8]	Operation Mode
00x		Data Copy
010		Horizontal Mirror
011		Vertical Mirror
100		Vertical & Horizontal Mirror
101		90 Degree Rotate (Clockwise)
110		180 Degree Rotate (Clockwise)
111		270 Degree Rotate (Clockwise)

ZF	[6:5]	Zero Fill
00		MSB Fill Mode Enable
01		Zero Fill Mode Enable
1x		HOB(High-Order Bits) Fill Mode Enable

The content above applies only when Data Format is RGB Format, not RGB888 / ARGB8888. Refer to Figure 34.6 for detailed information.

SDFRM	[4:0]	Source Data Format
00000		Separated YUV 4:4:4
00001		Separated YUV 4:4:0
00010		Separated YUV 4:2:2
00011		Separated YUV 4:2:0
00100		Separated YUV 4:1:1
00101		Separated YUV 4:1:0
00110		Interleaved YUV 4:2:2
00111		Interleaved YUV 4:2:0
01000		Sequential YUV 4:4:4
01001		Sequential YUV 4:2:2
0101x		RGB332 (8bpp)
01100		RGB444
01101		ARGB4444
01110		RGB454
01111		ARGB3454

10000	RGB555
10001	ARGB1555
1001x	RGB565
10100	RGB666
10101	ARGB4666
10110	ARGB6666
10111	RGB888
11000	ARGB4888
11001	ARGB8888
else	Not used

Front-End Channel 3 Source Address 0 (FCH3_SADDR0)

0x80001060

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SADDR0[15:2]														Reserved	

Front-End Channel 3 Source Frame Pixel Size (FCH3_SFSIZE)

0x8000106C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SFSIZE_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SFSIZE_X[11:0]											

SFSIZE_Y	[27:16]	Source Frame Pixel Size Y
N (0 ~ 4095)	N =Y Axis Source Frame Pixel Size	

SFSIZE_X	[11:0]	Source Frame Pixel Size X
N (0 ~ 4095)	N =X Axis Source Frame Pixel Size	

Front-End Channel 3 Source Pixel Offset (FCH3_SOFF)

0x80001070

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SOFF_X[11:0]											

SOFF_Y	[27:16]	Source Pixel Offset Y
N (0 ~ 4095)	N =Y Axis Source Pixel Offset	

SOFF_X	[11:0]	Source Pixel Offset X
N (0 ~ 4095)	N =X Axis Source Pixel Offset	

Front-End Channel 3 Source Image Pixel Size (FCH3_SISIZE)

0x80001074

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SISIZE_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SISIZE_X[11:0]											

SISIZE_Y	[27:16]	Source Image Pixel Size Y
N (0 ~ 4095)	N =Y Axis Source Image Pixel Size	

SISIZE_X	[11:0]	Source Image Pixel Size X
N (0 ~ 4095)	N =X Axis Source Image Pixel Size	

Front-End Channel 3 Window Pixel Offset (FCH3_WOFF)

0x80001078

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WOFF_X[11:0]											

WOFF_Y	[27:16]	Window Pixel Offset Y
N (0 ~ 4095)		N = Y Axis Window Pixel Offset

WOFF_X	[11:0]	Window Pixel Offset X
N (0 ~ 4095)		N = X Axis Window Pixel Offset

When FCH3 Image is selected as Source 0 by SF_CTRL.S0SEL register, Window Offset must be '0'.

Front-End Channel 3 Control (FCH3_SCTRL)

0x8000107C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SSB[2:0]				Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MABC	Reserved				OPMODE[2:0]			0	Reserved						

SSB	[26:24]	Source Swap Byte
000		ARGB
001		ARBG
010		AGRB
011		AGBR
100		ABRG
101		ABGR
110		RABG
111		BGRA

MABC	[15]	AXI Bus Master Address Boundary Control
0		AXI Bus Master Address Boundary = 4KByte
1		AXI Bus Master Address Boundary = 1KByte <i>This bit must be written as "1"</i>

OPMODE	[10:8]	Operation Mode
00x		Data Copy
010		Horizontal Mirror
011		Vertical Mirror
100		Vertical & Horizontal Mirror
101		90 Degree Rotate (Clockwise)
110		180 Degree Rotate (Clockwise)
111		270 Degree Rotate (Clockwise)

Source 0 Chroma-Key Parameter (S0_CHROMA)**0x80001080**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CHROMA_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHROMA_GU[7:0]								CHROMA_BV[7:0]							

CHROMA_RY	[23:16]	Chroma-Key Value
n	R or Y Chroma-Key Value	

CHROMA_GU	[15:8]	Chroma-Key Value
n	G or U Chroma-Key Value	

CHROMA_BV	[7:0]	Chroma-Key Value
n	B or V Chroma-Key Value	

Chroma-Key Value should be set to Sequential YUV4:4:4 or RGB888 irrespective of Input Source Format.

Writing should be done in RGB888 Format when Input Format is RGBxxx. When Input Format is YUVxxx and YUVtoRGB Converter is disabled, YUV444 Format is used for Writing. If input format is YUVxxx and YUVtoRGB Converter is enabled, writing is done in RGB888 Format.

S0_CHROMA Register is effective only when SA_CTRL.S0CE is enabled and OP_CTRL.CSEL0/1 Register is 2bit "01".

Source 0 Arithmetic Parameter (S0_PAR)**0x80001084**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								PAR_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAR_GU[7:0]								PAR_BV[7:0]							

PAR_RY	[23:16]	Parameter Value for Arithmetic Operation
n	R or Y Parameter Value for Arithmetic Operation.	

PAR_GU	[15:8]	Parameter Value for Arithmetic Operation
n	G or U Parameter Value for Arithmetic Operation.	

PAR_BV	[7:0]	Parameter Value for Arithmetic Operation
n	B or V Parameter Value for Arithmetic Operation.	

Parameter Value should be set to Sequential YUV4:4:4 or RGB888 irrespective of Input Source Format.

Writing should be done in RGB888 Format when Input Format is RGBxxx. When Input Format is YUVxxx and YUVtoRGB Converter is disabled, YUV444 Format is used for Writing. If input format is YUVxxx and YUVtoRGB Converter is enabled, the writing is done in RGB888 Format.

S0_PAR Register is effective only when SA_CTRL.S0ARITHMODE is Fill, Add, Sub and Multiply.

Source 1 Chroma-Key Parameter (S1_CHROMA)**0x80001088**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CHROMA_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHROMA_GU[7:0]								CHROMA_BV[7:0]							

CHROMA_RY	[23:16]	Chroma-Key Value
n	R or Y Chroma-Key Value	

CHROMA_GU	[15:8]	Chroma-Key Value
n	G or U Chroma-Key Value	

CHROMA_BV	[7:0]	Chroma-Key Value
n	B or V Chroma-Key Value	

Chroma-Key Value should be set to Sequential YUV4:4:4 or RGB888 irrespective of Input Source Format.

Writing should be done in RGB888 Format when Input Format is RGBxxx. When Input Format is YUVxxx and YUVtoRGB Converter is disabled, YUV444 Format is used for Writing. If input format is YUVxxx and YUVtoRGB Converter is enabled, the writing is done in RGB888 Format.

S1_CHROMA Register is effective only when SA_CTRL.S1CE is enabled and OP_CTRL.CSEL0/1 Register is 2bit "01".

Source 1 Arithmetic Parameter (S1_PAR)

0x8000108C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								PAR_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAR_GU[7:0]								PAR_BV[7:0]							

PAR_RY	[23:16]	Parameter Value for Arithmetic Operation
n		R or Y Parameter Value for Arithmetic Operation.

PAR_GU	[15:8]	Parameter Value for Arithmetic Operation
n		G or U Parameter Value for Arithmetic Operation.

PAR_BV	[7:0]	Parameter Value for Arithmetic Operation
n		B or V Parameter Value for Arithmetic Operation.

Parameter Value should be set to Sequential YUV4:4:4 or RGB888 irrespective of Input Source Format.

Writing should be done in RGB888 Format when Input Format is RGBxxx. When Input Format is YUVxxx and YUVtoRGB Converter is disabled, YUV444 Format is used for Writing. If input format is YUVxxx and YUVtoRGB Converter is enabled, the writing is done in RGB888 Format.

S1_PAR Register is effective only when SA_CTRL.S1ARITHMODE is Fill, Add, Sub and Multiply.

Source 2 Chroma-Key Parameter (S2_CHROMA)

0x80001090

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CHROMA_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHROMA_GU[7:0]								CHROMA_BV[7:0]							

CHROMA_RY	[23:16]	Chroma-Key Value
n		R or Y Chroma-Key Value

CHROMA_GU	[15:8]	Chroma-Key Value
n		G or U Chroma-Key Value

CHROMA_BV	[7:0]	Chroma-Key Value
n		B or V Chroma-Key Value

Chroma-Key Value should be set to Sequential YUV4:4:4 or RGB888 irrespective of Input Source Format.

Writing should be done in RGB888 Format when Input Format is RGBxxx. When Input Format is YUVxxx and YUVtoRGB Converter is disabled, YUV444 Format is used for Writing. If input format is YUVxxx and YUVtoRGB Converter is enabled, the writing is done in RGB888 Format.

S2_CHROMA Register is effective only when SA_CTRL.S2CE is enabled and OP_CTRL.CSEL1 Register is 2bit "01".

Source 2 Arithmetic Parameter (S2_PAR)

0x80001094

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								PAR_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAR_GU[7:0]								PAR_BV[7:0]							

PAR_RY	[23:16]	Parameter Value for Arithmetic Operation
n		R or Y Parameter Value for Arithmetic Operation.

PAR_GU	[15:8]	Parameter Value for Arithmetic Operation
n		G or U Parameter Value for Arithmetic Operation.

PAR_BV	[7:0]	Parameter Value for Arithmetic Operation
n		B or V Parameter Value for Arithmetic Operation.

Parameter Value should be set to Sequential YUV4:4:4 or RGB888 irrespective of Input Source Format.

Writing should be done in RGB888 Format when Input Format is RGBxxx. When Input Format is YUVxxx and YUVtoRGB Converter is disabled, YUV444 Format is used for Writing. If input format is YUVxxx and YUVtoRGB Converter is enabled, the writing is done in RGB888 Format.

S2_PAR Register is effective only when SA_CTRL.S2ARITHMODE is Fill, Add, Sub and Multiply.

Source 3 Chroma-Key Parameter (S3_CHROMA)

0x80001098

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CHROMA_R[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHROMA_G[7:0]								CHROMA_B[7:0]							

CHROMA_R	[23:16]	Chroma-Key Value
n		R Chroma-Key Value

CHROMA_G	[15:8]	Chroma-Key Value
n		G Chroma-Key Value

CHROMA_B	[7:0]	Chroma-Key Value
n		B Chroma-Key Value

Chroma-Key Value should be set to RGB888, irrespective of Input Source Format.

S3_CHROMA Register is effective only when SA_CTRL.S3CE is enabled and OP_CTRL.CSEL1 Register is 2bit "01".

Source 3 Arithmetic Parameter (S3_PAR)

0x8000109C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								PAR_R[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAR_G[7:0]								PAR_B[7:0]							

PAR_R	[23:16]	Parameter Value for Arithmetic Operation
n		R Parameter Value for Arithmetic Operation.

PAR_G	[15:8]	Parameter Value for Arithmetic Operation
n		G Parameter Value for Arithmetic Operation.

PAR_B	[7:0]	Parameter Value for Arithmetic Operation
n		B Parameter Value for Arithmetic Operation.

Parameter Value should be set to RGB888, irrespective of Input Source Format.

S3_PAR Register is effective only when SA_CTRL.S3ARITHMODE is Fill, Add, Sub, and Multiply.

Source Function Control (SF_CTRL)

0x800010A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					S2YE	S1YE	S0YE	Reserved		S2YM[1:0]		S1YM[1:0]		S0YM[1:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								S3SEL[1:0]		S2SEL[1:0]		S1SEL[1:0]		S0SEL[1:0]	

S2YE	[26]	Source 2 YUV to RGB Converter Enable
0		YUV to RGB Converter Disable
1		YUV to RGB Converter Enable

To enable YUV to RGB Converter, Data Format of Source 2 decided by SF_CTRL.S2SEL[1:0] should be YUVxxx.

S1YE	[25]	Source 1 YUV to RGB Converter Enable
0		YUV to RGB Converter Disable
1		YUV to RGB Converter Enable

To enable YUV to RGB Converter, Data Format of Source 1 decided by SF_CTRL.S1SEL[1:0] should be YUVxxx.

S0YE	[24]	Source 0 YUV to RGB Converter Enable
0		YUV to RGB Converter Disable
1		YUV to RGB Converter Enable

To enable YUV to RGB Converter, Data Format of Source 0 decided by SF_CTRL.S0SEL[1:0] should be YUVxxx.

S2YM	[21:20]	Source 2 YUVtoRGB Convert Type
00		YUVtoRGB Converter Type0
01		YUVtoRGB Converter Type1
10		YUVtoRGB Converter Type2
11		YUVtoRGB Converter Type3

S1YM	[19:18]	Source 1 YUVtoRGB Convert Type
00		YUVtoRGB Converter Type0
01		YUVtoRGB Converter Type1
10		YUVtoRGB Converter Type2
11		YUVtoRGB Converter Type3

S0YM	[17:16]	Source 0 YUVtoRGB Convert Type
00		YUVtoRGB Converter Type0
01		YUVtoRGB Converter Type1
10		YUVtoRGB Converter Type2
11		YUVtoRGB Converter Type3

- ☐ YUVtoRGB Format Converter Type 0
 $R = Y + 1.371 (V - 128)$
 $G = Y - 0.336 (U - 128) - 0.698 (V - 128)$
 $B = Y + 1.732 (U - 128)$
- ☐ YUVtoRGB Format Converter Type 1
 $R = 1.164 (Y - 16) + 1.159 (V - 128)$
 $G = 1.164 (Y - 16) - 0.391 (U - 128) - 0.813 (V - 128)$
 $B = 1.164 (Y - 16) + 2.018 (U - 128)$
- ☐ YUVtoRGB Format Converter Type 2
 $R = Y + 1.540 (V - 128)$
 $G = Y - 0.183 (U - 128) - 0.459 (V - 128)$
 $B = Y + 1.816 (U - 128)$
- ☐ YUVtoRGB Format Converter Type 3
 $R = 1.164 (Y - 16) + 1.793 (V - 128)$
 $G = 1.164 (Y - 16) - 0.213 (U - 128) - 0.534 (V - 128)$
 $B = 1.164 (Y - 16) + 2.115 (U - 128)$

Figure 34.7 YUV4:4:4 to RGB888 Converter Type

S3_SEL	[7:6]	Source 3 Selection
00	Source 3 = Front-End Channel 0	
01	Source 3 = Front-End Channel 1	
10	Source 3 = Front-End Channel 2	
11	Source 3 = Front-End Channel 3	

S2_SEL	[5:4]	Source 2 Selection
00	Source 2 = Front-End Channel 0	
01	Source 2 = Front-End Channel 1	
10	Source 2 = Front-End Channel 2	
11	Source 2 = Front-End Channel 3	

S1_SEL	[3:2]	Source 1 Selection
00	Source 1 = Front-End Channel 0	
01	Source 1 = Front-End Channel 1	
10	Source 1 = Front-End Channel 2	
11	Source 1 = Front-End Channel 3	

S0_SEL	[1:0]	Source 0 Selection
00	Source 0 = Front-End Channel 0	
01	Source 0 = Front-End Channel 1	
10	Source 0 = Front-End Channel 2	
11	Source 0 = Front-End Channel 3	

Source 0 should select the image. In other words, the Input Source Image should be selected to Source 0 if there is only one Input Source Image in FCH0, FCH1, FCH2, or FCH3.

If there are more than 2 Input Source Images in two of FCH0, FCH1, FCH2, and FCH3, the larger Input Source Image should be selected to S0 and the others should be selected to either S1,S2, or S3.

Source Arithmetic Control (SA_CTRL)

0x800010A4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												S3CE	S2CE	S1CE	S0CE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	S3ARIM[2:0]			R	S2ARIM[2:0]			R	S1ARIM[2:0]			R	S0ARIM[2:0]		

S3CE	[19]	Source 3 Chroma-Key Enable for Arithmetic
0		Chroma-Key Disable for Arithmetic
1		Chroma-Key Enable for Arithmetic

S2CE	[18]	Source 2 Chroma-Key Enable for Arithmetic
0		Chroma-Key Disable for Arithmetic
1		Chroma-Key Enable for Arithmetic

S1CE	[17]	Source 1 Chroma-Key Enable for Arithmetic
0		Chroma-Key Disable for Arithmetic
1		Chroma-Key Enable for Arithmetic

S0CE	[16]	Source 0 Chroma-Key Enable for Arithmetic
0		Chroma-Key Disable for Arithmetic
1		Chroma-Key Enable for Arithmetic

S3_ARITHMODE	[14:12]	Source 2 Arithmetic Mode
00x		Bypass Mode
010		Fill Mode
011		Inverter Mode
100		Add Mode
101		Substrate Type A Mode
110		Substrate Type B Mode
111		Multiplier Mode

S2_ARITHMODE	[10:8]	Source 2 Arithmetic Mode
00x		Bypass Mode
010		Fill Mode
011		Inverter Mode
100		Add Mode
101		Substrate Type A Mode
110		Substrate Type B Mode
111		Multiplier Mode

S1_ARITHMODE	[6:4]	Source 1 Arithmetic Mode
00x		Bypass Mode
010		Fill Mode
011		Inverter Mode
100		Add Mode
101		Substrate Type A Mode
110		Substrate Type B Mode
111		Multiplier Mode

S0_ARITHMODE	[2:0]	Source 0 Arithmetic Mode
00x		Bypass Mode
010		Fill Mode
011		Inverter Mode
100		Add Mode
101		Substrate Type A Mode
110		Substrate Type B Mode
111		Multiplier Mode

Arithmetic Fnc.	Equation	A	B
BYPASS	$Y = A$	Source Image	Do not use
FILL	$Y = B$	Source Image	S0/1_PAR
INV	$Y = 255 - A$	Source Image	Do not use
ADD	$Y = A + B$	Source Image	S0/1_PAR
SUBA	$Y = A - B$	Source Image	S0/1_PAR
SUBB	$Y = B - A$	Source Image	S0/1_PAR
MUL	$Y = A \times B$ Mantissa = B[7:6] Fraction = B[5:0]	Source Image	S0/1_PAR

Since Arithmetic Operation is done after YUVtoRGB Converting Operation, operation is working in RGB888 or YUV4:4:4 format irrespective of Input Source Format. Therefore, writing should be done in RGB888 or YUV4:4:4 format.

If Chroma-Key Value and Selected Source Data are matched when Chroma-Key is enabled, the BYPASS Operation is carried out.

Source Operator 0 Alpha (OP0_ALPHA)

0x800010B0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ALPHA1[7:0]								ALPHA0[7:0]							

ALPHA1	[15:8]	Source Operator 0 Alpha Value 1
n		Alpha Value for Alpha-Blending Operation of Source Operator 0

ALPHA0	[7:0]	Source Operator 0 Alpha Value 0
n		Alpha Value for Alpha-Blending Operation of Source Operator 0

These registers are effective only when OPCTRL.OP0MODE is Alpha-Blending Type 0.

Source Operator 1 Alpha (OP1_ALPHA)

0x800010B4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ALPHA1[7:0]								ALPHA0[7:0]							

ALPHA1	[15:8]	Source Operator 1 Alpha Value 1
n		Alpha Value for Alpha-Blending Operation of Source Operator 1

ALPHA0	[7:0]	Source Operator 1 Alpha Value 0
n		Alpha Value for Alpha-Blending Operation of Source Operator 1

These registers are effective only when OPCTRL.OP1MODE is Alpha-Blending Type 0.

Source Operator 2 Alpha (OP2_ALPHA)

0x800010B8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ALPHA1[7:0]								ALPHA0[7:0]							

ALPHA1	[15:8]	Source Operator 2 Alpha Value 1
n		Alpha Value for Alpha-Blending Operation of Source Operator 2

ALPHA0	[7:0]	Source Operator 2 Alpha Value 0
n		Alpha Value for Alpha-Blending Operation of Source Operator 2

These registers are effective only when OPCTRL.OP2MODE is Alpha-Blending Type 0.

Source Operator 0 Pattern (OP0_PAT)

0x800010C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								PAT_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAT_GU[7:0]								PAT_BV[7:0]							

PAT_RY	[23:16]	Source Operator 0 Pattern Value
n	R or Y Pattern Value for Raster Operation of Source Operator 0	

PAT_GU	[15:8]	Source Operator 0 Pattern Value
n	G or U Pattern Value for Raster Operation of Source Operator 0	

PAT_BV	[7:0]	Source Operator 0 Pattern Value
n	B or V Pattern Value for Raster Operation of Source Operator 0	

Source Operator 1 Pattern (OP1_PAT)

0x800010C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								PAT_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAT_GU[7:0]								PAT_BV[7:0]							

PAT_RY	[23:16]	Source Operator 1 Pattern Value
n	R or Y Pattern Value for Raster Operation of Source Operator 1	

PAT_GU	[15:8]	Source Operator 1 Pattern Value
n	G or U Pattern Value for Raster Operation of Source Operator 1	

PAT_BV	[7:0]	Source Operator 1 Pattern Value
n	B or V Pattern Value for Raster Operation of Source Operator 1	

Source Operator 2 Pattern (OP1_PAT)

0x800010C8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								PAT_RY[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAT_GU[7:0]								PAT_BV[7:0]							

PAT_RY	[23:16]	Source Operator 2 Pattern Value
n	R or Y Pattern Value for Raster Operation of Source Operator 2	

PAT_GU	[15:8]	Source Operator 2 Pattern Value
n	G or U Pattern Value for Raster Operation of Source Operator 2	

PAT_BV	[7:0]	Source Operator 2 Pattern Value
n	B or V Pattern Value for Raster Operation of Source Operator 2	

Source Operation 0 Control (OP0_CTRL)

0x800010D0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	ACON1[2:0]			R	ACON0[2:0]			CCON1[3:0]			CCON0[3:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		ATUNE[1:0]		Reserved		CSEL[1:0]		Reserved			OPMODE[4:0]				

ACON1	[30:28]	Alpha-Value Control 1 for ARGB Output
000		$\text{ALPHA1} * \text{ALPHA0}$
001		$\text{ALPHA1} * (1 - \text{ALPHA0})$
01x		ALPHA1
10x		$\text{ALPHA1} * (1/2)$
11x		0

Note: '11x' is recommended.

ACON0	[26:24]	Alpha-Value Control 0 for ARGB Output
000		$\text{ALPHA0} * \text{ALPHA1}$
001		$\text{ALPHA0} * (1 - \text{ALPHA1})$
01x		ALPHA0
10x		$\text{ALPHA0} * (1/2)$
11x		0

CCON1	[23:20]	Color Control 1 for Alpha-Blending
0000		$B * \text{ALPHA0}$
0001		$B * \text{ALPHA1}$
0010		$B * (1 - \text{ALPHA0})$
0011		$B * (1 - \text{ALPHA1})$
01xx		B
10xx		$B * (1/2)$
11xx		0

CCON0	[19:16]	Color Control 0 for Alpha-Blending
0000		$A * \text{ALPHA0}$
0001		$A * \text{ALPHA1}$
0010		$A * (1 - \text{ALPHA0})$
0011		$A * (1 - \text{ALPHA1})$
01xx		A
10xx		$A * (1/2)$
11xx		0

ATUNE	[13:12]	Alpha Value Tuning for Alpha-Blending
0x		Alpha Value is 0 ~ 255 (0% ~ 99.6%)
10		Alpha Value is 1 ~ 256 (0.39% ~ 100%)
11		Alpha Value is 0 ~ 127 and 129 ~ 256 (0% ~ 49.6% and 50.39% ~ 100%)

CSEL	[9:8]	Chroma-key Source Selection
00		Chroma-Key Operation Disable of Operator 0
01		Chroma-Key Source = Source 0, Chroma-Key Value = S0_CHROMA
10		Chroma-Key Source = Source 1, Chroma-Key Value = S1_CHROMA
11		Reserved.

OPMODE	[4:0]	Operation Mode
00000		Result = 0
00001		Result = P & A
00010		Result = ~ A B
00011		Result = P
00100		Result = P ^ B
00101		Result = (P ~ A) B
00110		Result = A
00111		Result = A ^ B
01000		Result = A B
01001		Result = A & B
01010		Result = A & ~ B
01011		Result = ~ A

Raster
Operation

01100	Not Source Erase	Result = $\sim (A \mid B)$	
01101	Destination Copy	Result = B	
01110	Destination Inverter	Result = $\sim B$	
01111	Whiteness	Result = 1	
10xxx	Alpha-Blending Type 0	Alpha-Blending with Alpha-Value	Alpha- Blending
11xxx	Alpha-Blending Type 1	Alpha-Blending with Alpha-Value of ARGB	

Source Operation 1 Control (OP1_CTRL)

0x800010D4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	ACON1[2:0]			R	ACON0[2:0]			CCON1[3:0]			CCON0[3:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		ATUNE[1:0]		Reserved		CSEL[1:0]		Reserved			OPMODE[4:0]				

ACON1	[30:28]	Alpha-Value Control 1 for ARGB Output
000		$\text{ALPHA1} * \text{ALPHA0}$
001		$\text{ALPHA1} * (1 - \text{ALPHA0})$
01x		ALPHA1
10x		$\text{ALPHA1} * (1/2)$
11x		0

ACON0	[26:24]	Alpha-Value Control 0 for ARGB Output
000		$\text{ALPHA0} * \text{ALPHA1}$
001		$\text{ALPHA0} * (1 - \text{ALPHA1})$
01x		ALPHA0
10x		$\text{ALPHA0} * (1/2)$
11x		0

CCON1	[23:20]	Color Control 1 for Alpha-Blending
0000		$B * \text{ALPHA0}$
0001		$B * \text{ALPHA1}$
0010		$B * (1 - \text{ALPHA0})$
0011		$B * (1 - \text{ALPHA1})$
01xx		B
10xx		$B * (1/2)$
11xx		0

CCON0	[19:16]	Color Control 0 for Alpha-Blending
0000		$A * \text{ALPHA0}$
0001		$A * \text{ALPHA1}$
0010		$A * (1 - \text{ALPHA0})$
0011		$A * (1 - \text{ALPHA1})$
01xx		A
10xx		$A * (1/2)$
11xx		0

ATUNE	[13:12]	Alpha Value Tuning for Alpha-Blending
0x		Alpha Value is 0 ~ 255 (0% ~ 99.6%)
10		Alpha Value is 1 ~ 256 (0.39% ~ 100%)
11		Alpha Value is 0 ~ 127 and 129 ~ 256 (0% ~ 49.6% and 50.39% ~ 100%)

CSEL	[9:8]	Chroma-key Source Selection
00		Chroma-Key Operation Disable of Operator 1
01		Chroma-Key Source = Source 0, Chroma-Key Value = S0_CHROMA
10		Chroma-Key Source = Source 1, Chroma-Key Value = S1_CHROMA
11		Chroma-Key Source = Source 2, Chroma-Key Value = S2_CHROMA

OPMODE	[4:0]	Operation Mode	Raster Operation
00000		Result = 0	
00001		Result = P & A	
00010		Result = ~ A B	
00011		Result = P	
00100		Result = P ^ B	
00101		Result = (P ~ A) B	
00110		Result = A	
00111		Result = A ^ B	
01000		Result = A B	
01001		Result = A & B	
01010		Result = A & ~ B	
01011		Result = ~ A	
01100		Result = ~ (A B)	
01101		Result = B	

01110	Destination Inverter	Result = $\sim B$	Alpha- Blending
01111	Whiteness	Result = 1	
10xxx	Alpha-Blending Type 0	Alpha-Blending with Alpha-Value	
11xxx	Alpha-Blending Type 1	Alpha-Blending with Alpha-Value of ARGB	

Source Operation 2 Control (OP2_CTRL)

0x800010D8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	ACON1[2:0]			R	ACON0[2:0]			CCON1[3:0]			CCON0[3:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		ATUNE[1:0]		Reserved		CSEL[1:0]		Reserved			OPMODE[4:0]				

ACON1	[30:28]	Alpha-Value Control 1 for ARGB Output
000		$\text{ALPHA1} * \text{ALPHA0}$
001		$\text{ALPHA1} * (1 - \text{ALPHA0})$
01x		ALPHA1
10x		$\text{ALPHA1} * (1/2)$
11x		0

ACON0	[26:24]	Alpha-Value Control 0 for ARGB Output
000		$\text{ALPHA0} * \text{ALPHA1}$
001		$\text{ALPHA0} * (1 - \text{ALPHA1})$
01x		ALPHA0
10x		$\text{ALPHA0} * (1/2)$
11x		0

CCON1	[23:20]	Color Control 1 for Alpha-Blending
0000		$B * \text{ALPHA0}$
0001		$B * \text{ALPHA1}$
0010		$B * (1 - \text{ALPHA0})$
0011		$B * (1 - \text{ALPHA1})$
01xx		B
10xx		$B * (1/2)$
11xx		0

CCON0	[19:16]	Color Control 0 for Alpha-Blending
0000		$A * \text{ALPHA0}$
0001		$A * \text{ALPHA1}$
0010		$A * (1 - \text{ALPHA0})$
0011		$A * (1 - \text{ALPHA1})$
01xx		A
10xx		$A * (1/2)$
11xx		0

ATUNE	[13:12]	Alpha Value Tuning for Alpha-Blending
0x		Alpha Value is 0 ~ 255 (0% ~ 99.6%)
10		Alpha Value is 1 ~ 256 (0.39% ~ 100%)
11		Alpha Value is 0 ~ 127 and 129 ~ 256 (0% ~ 49.6% and 50.39% ~ 100%)

CSEL	[9:8]	Chroma-key Source Selection
00		Chroma-Key Operation Disable of Operator 1
01		Chroma-Key Source = Source 0, Chroma-Key Value = S0_CHROMA
10		Chroma-Key Source = Source 1, Chroma-Key Value = S1_CHROMA
11		Chroma-Key Source = Source 2, Chroma-Key Value = S2_CHROMA

OPMODE	[4:0]	Operation Mode	Raster Operation
00000		Result = 0	
00001		Result = P & A	
00010		Result = ~ A B	
00011		Result = P	
00100		Result = P ^ B	
00101		Result = (P ~ A) B	
00110		Result = A	
00111		Result = A ^ B	
01000		Result = A B	
01001		Result = A & B	
01010		Result = A & ~ B	
01011		Result = ~ A	
01100		Result = ~ (A B)	
01101		Result = B	

01110	Destination Inverter	Result = $\sim B$	Alpha- Blending
01111	Whiteness	Result = 1	
10xxx	Alpha-Blending Type 0	Alpha-Blending with Alpha-Value	
11xxx	Alpha-Blending Type 1	Alpha-Blending with Alpha-Value of ARGB	

Back-End Destination Address 0 (BCH_DADDR0)**0x800010E0**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BCH_DADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BCH_DADDR0[15:2]														Reserved	

In case that Destination Image Format is Separated YUV X:X:X, the value of the address is Source Address of Y region.

In case that Destination Image Format is not Separated YUV X:X:X, the value of the address is Source Address.

Back-End Destination Address 1 (BCH_DADDR1)**0x800010E4**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BCH_DADDR1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BCH_DADDR1[15:2]														Reserved	

In case that Destination Image Format is Separated YUV X:X:X, the value of the address is Source Address of U region.

In case that Destination Image Format is not Separated YUV X:X:X, the value of the address is not used.

Back-End Destination Address 2 (BCH_DADDR2)**0x800010E8**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BCH_DADDR2[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BCH_DADDR2[15:2]														Reserved	

In case that Destination Image Format is Separated YUV X:X:X, the value of the address is Source Address of V region.

In case that Destination Image Format is not Separated YUV X:X:X, the value of the address is not used.

Back-End Channel Destination Frame Pixel Size (BCH_DFSize)**0x800010EC**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				DFSize_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DFSize_X[11:0]											

DFSize_Y	[27:16]	Destination Frame Pixel Size Y
N (0 ~ 4095)	N =Y Axis Destination Frame Pixel Size	

DFSize_X	[11:0]	Destination Frame Pixel Size X
N (0 ~ 4095)	N =X Axis Destination Frame Pixel Size	

Back-End Channel Destination Pixel Offset (BCH_DOFF)

0x800010F0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				DOFF_Y[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DOFF_X[11:0]											

DOFF_Y	[27:16]	Destination Pixel Offset Y
N (0 ~ 4095)	N =Y Axis Destination Pixel Offset	

DOFF_X	[11:0]	Destination Pixel Offset X
N (0 ~ 4095)	N =X Axis Destination Pixel Offset	

BCH_DCTRL.DDFRM	Destination Frame Size / Image Size / Offset Limitation
RGB332(8bpp)	No Limitation
RGB454	No Limitation
RGB565	No Limitation
RGB666	No Limitation
RGB888	No Limitation
Separated YUV 4:4:4	No Limitation
Separated YUV 4:4:0	X axis = No Limitation, Y axis = multiplies of 2
Separated YUV 4:2:2	X axis = multiplies of 2, Y axis = No Limitation
Separated YUV 4:2:0	X axis = multiplies of 2, Y axis = multiplies of 2
Separated YUV 4:1:1	X axis = multiplies of 4, Y axis = No Limitation
Separated YUV 4:1:0	X axis = multiplies of 4, Y axis = multiplies of 2
Interleaved YUV 4:2:2	X axis = multiplies of 2, Y axis = No Limitation
Interleaved YUV 4:2:0	X axis = multiplies of 2, Y axis = multiplies of 2
Sequential YUV 4:4:4	No Limitation
Sequential YUV 4:2:2	X axis = multiplies of 2, Y axis = No Limitation

Destination Image Size is decided by both Source 0 size controlled by S0_CTRL and Back-End Destination Operation Mode controlled by BCH_CTRL.

Back-End Channel Control (BCH_DCTRL)

0x800010F4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					DSB[2:0]			Reserved			MABC	Reserved		YSEL	XSEL[1:0]
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R2YE	R2YM[1:0]		Res.	DSUV	OPMODE[2:0]			0	DOP	DEN	DDFRM[4:0]				

DSB	[26:24]	Destination Swap Byte
000	ARGB	
001	ARBG	
010	AGRB	
011	AGBR	
100	ABRG	
101	ABGR	
110	RABG	
111	BGR	

MABC	[21]	AXI Bus Master Address Boundary Control
0	AXI Bus Master Address Boundary = 4KByte	
1	AXI Bus Master Address Boundary = 1KByte <i>This bit must be written as "1"</i>	

YSEL	[18]	YUV4:4:4 to YUVx:x:x Y Control
n	N = 0 or 1	

XSEL	[17:16]	YUV4:4:4 to YUVx:x:x X Control
n	N = 0 ~ 3	

Since inside Overlay Mixer, only RGB888 and Sequential YUV4:4:4 exist as Data Format, if Destination Data Format is RGBxxx, RGB888 is converted to RGBxxx. If Destination Data Format is Sequential / Separated YUVxxx, Sequential YUV4:4:4 is converted to Sequential / Separated YUVxxx.

In this case, BCH_DCTRL.X/YSEL Register is reflected when Sequential YUV4:4:4 is converted to Sequential/Separated YUVxxx.

R2YE	[15]	Destination Format Converter Enable
0	RGB to YUV Converter Disable	
1	RGB to YUV Converter Enable	

R2YM	[14:13]	RGBtoYUV Converter Type
0	RGBtoYUV Converter Type0	
1	RGBtoYUV Converter Type1	
2	RGBtoYUV Converter Type2	
3	RGBtoYUV Converter Type3	

- ☐ RGBtoYUV Format Converter Type 0
 $Y = 0.299 R + 0.587 G + 0.114 B$
 $U = -0.172 R - 0.339 G + 0.511 B + 128$
 $V = 0.511 R - 0.428 G - 0.083 B + 128$

☐ RGBtoYUV Format Converter Type 1
 $Y = 0.257 R + 0.504 G + 0.098 B + 16$
 $U = -0.148 R - 0.291 G + 0.439 B + 128$
 $V = 0.439 R - 0.368 G - 0.071 B + 128$

☐ RGBtoYUV Format Converter Type 2
 $Y = 0.213 R + 0.715 G + 0.072 B$
 $U = -0.117 R - 0.394 G + 0.511 B + 128$
 $V = 0.511 R - 0.464 G - 0.047 B + 128$

☐ RGBtoYUV Format Converter Type 3
 $Y = 0.183 R + 0.614 G + 0.062 B + 16$
 $U = -0.101 R - 0.338 G + 0.439 B + 128$
 $V = 0.439 R - 0.399 G - 0.040 B + 128$

Figure 34.8 RGB to YUV Converter Type

DSUV	[11]	Destination Swap U for V
0	Keep on U and V (LSB = U, MSB = V)	
1	Swap U for V (LSB = V, MSB = U)	

The content above applies only when Data Format is Interleaved YUV.

OPMODE	[10:8]	Operation Mode of Back-End DMA
00x	Data Copy	
010	Horizontal Mirror	
011	Vertical Mirror	
100	Vertical & Horizontal Mirror	
101	90 Degree Rotate (Clockwise)	
110	180 Degree Rotate (Clockwise)	
111	270 Degree Rotate (Clockwise)	

DOP	[6]	Dithering Operation Type
0	1-Bit Toggle Operation	
1	Add 1 Operation	

The content above applies only when Data Format is RGB, not RGB888 / ARGB8888.

DEN	[5]	Dithering Enable
0	Dithering Disable	
1	Dithering Enable	

The content above applies only when Data Format is RGB, not RGB888 / ARGB8888.

DDFRM	[4:0]	Destination Data Format of Back-End DMA
00000		Separated YUV 4:4:4
00001		Separated YUV 4:4:0
00010		Separated YUV 4:2:2
00011		Separated YUV 4:2:0
00100		Separated YUV 4:1:1
00101		Separated YUV 4:1:0
00110		Interleaved YUV 4:2:2
00111		Interleaved YUV 4:2:0
01000		Sequential YUV 4:4:4
01001		Sequential YUV 4:2:2
0101x		RGB332 (8bpp)
01100		RGB444
01101		ARGB4444
01110		RGB454
01111		ARGB3454
10000		RGB555
10001		ARGB1555
1001x		RGB565
10100		RGB666
10101		ARGB4666
10110		ARGB6666
10111		RGB888
11000		ARGB4888
11001		ARGB8888
Others		Not used

Back-End Channel Destination Dither Matrix 0 (BCH_DDMAT0)

0x80001100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DDMAT03[4:0]				Reserved				DDMAT02[4:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DDMAT01[4:0]				Reserved				DDMAT00[4:0]				

Back-End Channel Destination Dither Matrix 1 (BCH_DDMAT1)

0x80001104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DDMAT07[4:0]				Reserved				DDMAT06[4:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DDMAT05[4:0]				Reserved				DDMAT04[4:0]				

Back-End Channel Destination Dither Matrix 2 (BCH_DDMAT2)

0x80001108

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DDMAT11[4:0]				Reserved				DDMAT10[4:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DDMAT09[4:0]				Reserved				DDMAT08[4:0]				

Back-End Channel Destination Dither Matrix 3 (BCH_DDMAT3)

0x8000110C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DDMAT15[4:0]				Reserved				DDMAT14[4:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DDMAT13[4:0]				Reserved				DDMAT12[4:0]				

These BCH_DDMAT registers are for setting dither matrix values for dithering. The patterned dithering algorithm using 4x4 dither matrix is used for dithering in Overlay Mixer. Each BCH_DDMAT 0/1/2/3 register sets 4 values in one row of the matrix. The below figure shows the relation between 4x4 dither matrix and BCH_DDMAT register and it provides simple setting value examples.

DDMAT 00 [4:0]	DDMAT 01 [4:0]	DDMAT 02 [4:0]	DDMAT 03 [4:0]
DDMAT 04 [4:0]	DDMAT 05 [4:0]	DDMAT 06 [4:0]	DDMAT 07 [4:0]
DDMAT 08 [4:0]	DDMAT 09 [4:0]	DDMAT 10 [4:0]	DDMAT 11 [4:0]
DDMAT 12 [4:0]	DDMAT 13 [4:0]	DDMAT 14 [4:0]	DDMAT 15 [4:0]

(a)

0	7	1	9
11	3	13	5
2	10	0	8
14	6	12	4

0	9	2	11
13	4	15	6
3	12	1	10
16	7	14	5

(b)

Figure 34.9 (a) 16 Level 4x4 Dither Matrix (b) Example of Dither Matrix Setting

Overlay Mixer Control (OM_CTRL)

0x80001110

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															IEN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												EN[3:0]			

IEN	[16]	Overlay Mixer Interrupt Enable
0		Interrupt Disable
1		Interrupt Enable

EN	[3:0]	Overlay Mixer Enable
0000		Overlay Mixer Disable
0001		Overlay Mixer Enable With Front-End Channel 0
0010		Overlay Mixer Enable With Front-End Channel 1
0011		Overlay Mixer Enable With Front-End Channel 2
0100		Overlay Mixer Enable With Front-End Channel 3
0101		Overlay Mixer Enable With Front-End Channel 0,1
0110		Overlay Mixer Enable With Front-End Channel 0,2
0111		Overlay Mixer Enable With Front-End Channel 0,3
1000		Overlay Mixer Enable With Front-End Channel 1,2
1001		Overlay Mixer Enable With Front-End Channel 1,3
1010		Overlay Mixer Enable With Front-End Channel 2,3
1011		Overlay Mixer Enable With Front-End Channel 0,1,2
1100		Overlay Mixer Enable With Front-End Channel 0,1,3
1101		Overlay Mixer Enable With Front-End Channel 0,2,3
1110		Overlay Mixer Enable With Front-End Channel 1,2,3
1111		Overlay Mixer Enable With Front-End Channel 0,1,2,3

After all transfers complete, EN[3:0] Register is automatically cleared to 4bit "0000".

EN Register should be set last after all Control Registers are set.

Overlay Mixer Interrupt Request Control (OM_IREQ)

0x80001114

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															FLG
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															IRQ

FLG	[16]	Overlay Mixer Flag Bit
0		Transfer of Overlay Mixer is not completed.
1		Transfer of Overlay Mixer is completed
1		Automatically set, and cleared by writing '1'

IRQ	[0]	Overlay Mixer Interrupt Request
0		Interrupt Request does not occur.
1		Interrupt Request occurs.
1		Cleared by writing '1'

Front-End Channel 0 Lookup Table (FCH0_LUT)

0x80001400 ~ 0x800017FF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FCH0_LUT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FCH0_LUT[15:0]															

When the Overlay Mixer is operating, CPU cannot write and read this register.

Front-End Channel 1 Lookup Table (FCH1_LUT)

0x80001800 ~ 0x80001BFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FCH0_LUT[31:16]															
9	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FCH0_LUT[15:0]															

When the Overlay Mixer is operating, CPU cannot write and read this register.

Front-End Channel 2 Lookup Table (FCH2_LUT)

0x80001C00 ~ 0x80001FFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FCH0_LUT[31:16]															
9	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FCH0_LUT[15:0]															

When the Overlay Mixer is operating, CPU cannot write and read this register.

FCH_SCTRL.SDFRM	Supported LUT Field (ARGB8888)			
	A	R	G	B
RGB332(8bpp)	Yes	Yes	Yes	Yes
ARGB4444	Yes	No	No	No
ARGB3454	Yes			
ARGB1555	Yes			
ARGB4666	Yes			
ARGB6666	Yes			
ARGB4888	Yes			
ARGB8888	Yes			
RGB444	No	No	No	No
RGB454				
RGB555				
RGB565				
RGB666				
RGB888				
Separated YUV x:x:x				
Interleaved YUV x:x:x				

34.4 Overlay Mixer Operation

34.4.1 Source Rotate / Mirror and Destination Rotate / Mirror Operation

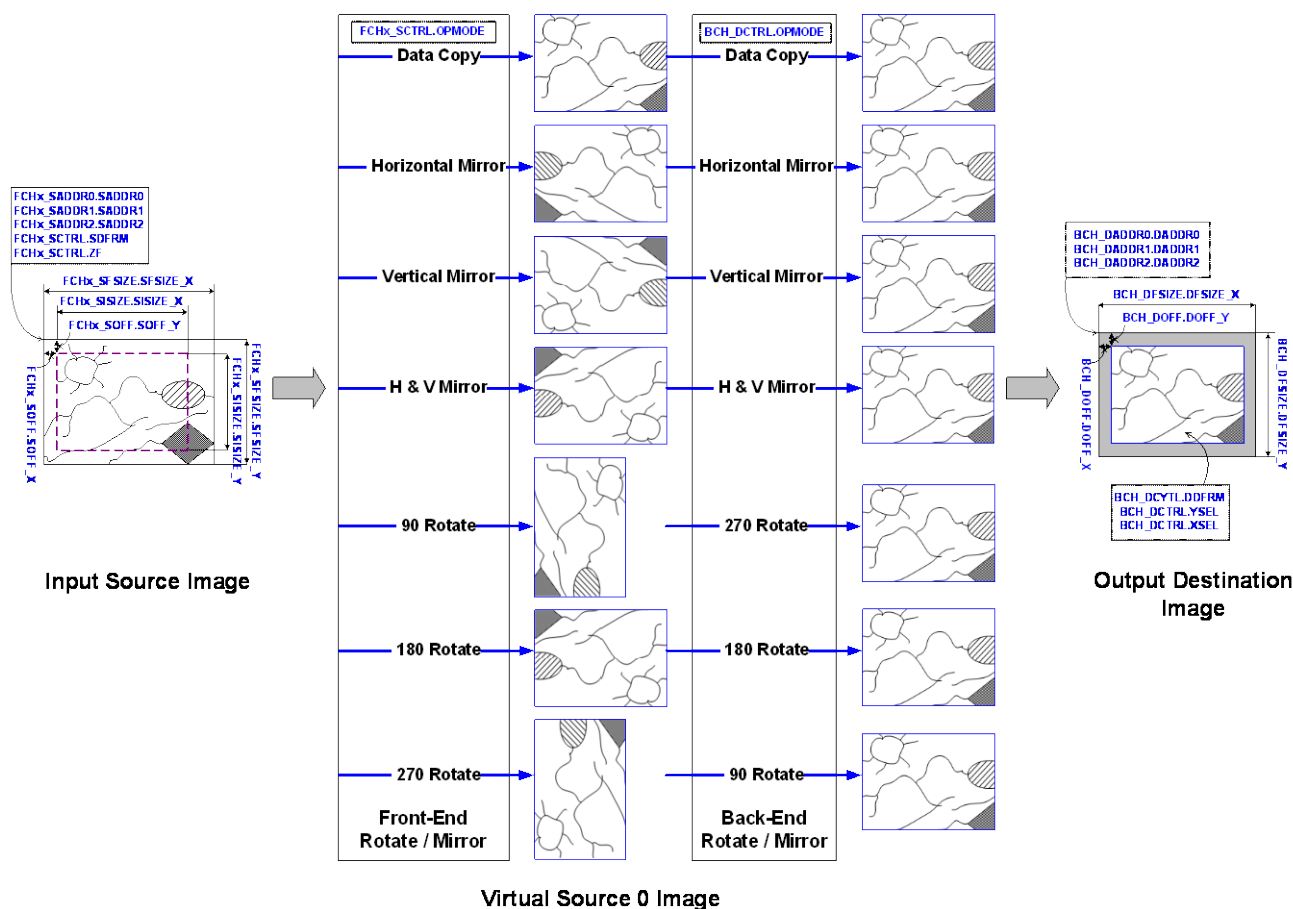


Figure 34.10 Source & Destination Mirror / Rotate Operation

Figure 34.10 illustrates one of functions supported by Overlay Mixer, Rotate / Mirror function. The figure shows Internal Register Setting when there is one Input Source Image and Input Image Source is FCH1.

As explained before, if the number of an image is One, the image should be selected to Source 0. Window Offset of the image which is selected to Source 0 must be '0'. In addition, the OM_CTRL.EN Register should be set last.

- (1) FCH1_SADDR0 / 1 / 2 : The Input Source Image Start Address.
- (2) FCH1_SCTRL.SDFRM, FCH1_SCTRL.ZF : The Input Source Image Data Format
- (3) FCH1_SF.SIZE : The Input Source Image Frame Size
- (4) FCH1_SF.SIZE : The Input Source Image Size
- (5) FCH1_SOFF : The Input Source Image Offset
- (6) FCH1_SCTRL.OPMODE : FCH1 Operation Mod
- (7) FCH1_WOFF : Should be Zero.
- (8) SF_CTRL.S0SEL : Should be Front-End Channel 1
- (9) SA_CTRL.S0ARIM : Should be Bypass Mode.
- (10) SF_CTRL.S0YE, SA_CTRL.S0CE : Should be Disable.
- (11) OP0_CTRL.CSEL, OP1_CTRL.CSEL : Should be Disable.
- (12) OP0_CTRL.OPMODE, OP1_CTRL.OPMODE, OP2_CTRL.OPMODE: Should be Source Copy
- (13) BCH_DCTRL.OPMODE : BCH Operation Mode.
- (14) BCH_DCTRL.R2YE : Should be Disable.
- (15) BCH_DADDR0 / 1 / 2 : The Output Destination Image Start Address.
- (16) BCH_DCTRL.DDFRM, BCH_DCTRL.X / YSEL : The Output Destination Image Data Format
- (17) BCH_DF.SIZE : The Output Destination Image Frame Size
- (18) BCH_DOFF : The Output Destination Image Offset
- (19) OM_CTRL : OM_CTRL.EN = b'0010

34.4.2 YUV to RGB Format Conversion

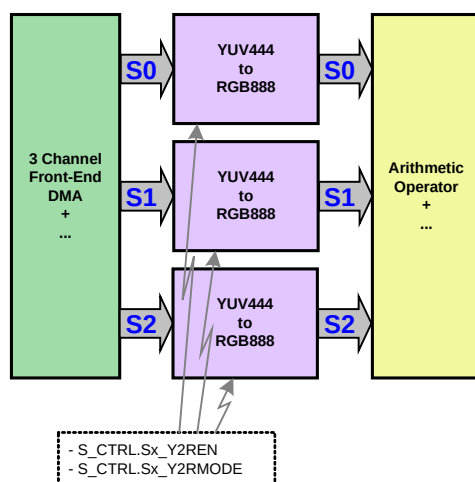


Figure 34.11 Source YUV to RGB Conversion

Figure 34.11 explains one of functions supported by Overlay Mixer, 3-Channel YUV to RGB Format Conversion function. As can be seen, 3-Channel YUV Source Image of Overlay Mixer can be converted to RGB888. S0 and S1 in Figure 34.11 are the images selected by SF_CTRL.SxSEL Register.

Below shows 4 equations in YUV to RGB Conversion per each SF_CTRL.SxYM Register.

- YUVtoRGB Format Converter Type 0
 - $R = Y + 1.371 (V - 128)$
 - $G = Y - 0.336 (U - 128) - 0.698 (V - 128)$
 - $B = Y + 1.732 (U - 128)$
- YUVtoRGB Format Converter Type 1
 - $R = 1.164 (Y - 16) + 1.159 (V - 128)$
 - $G = 1.164 (Y - 16) - 0.391 (U - 128) - 0.813 (V - 128)$
 - $B = 1.164 (Y - 16) + 2.018 (U - 128)$
- YUVtoRGB Format Converter Type 2
 - $R = Y + 1.540 (V - 128)$
 - $G = Y - 0.183 (U - 128) - 0.459 (V - 128)$
 - $B = Y + 1.816 (U - 128)$
- YUVtoRGB Format Converter Type 3
 - $R = 1.164 (Y - 16) + 1.793 (V - 128)$
 - $G = 1.164 (Y - 16) - 0.213 (U - 128) - 0.534 (V - 128)$
 - $B = 1.164 (Y - 16) + 2.115 (U - 128)$

34.4.3 Arithmetic Operation

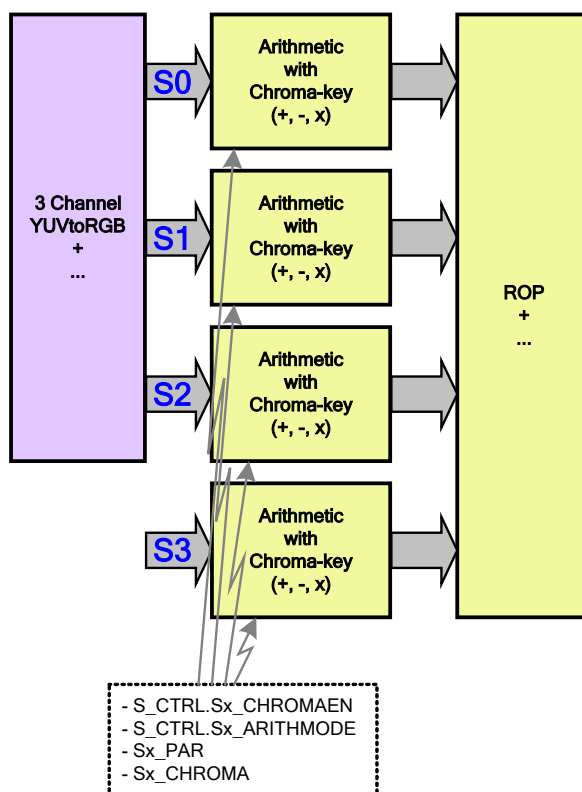


Figure 34.12 Arithmetic Operation

Figure 34.12 explains 4-Channel Arithmetic Operation, one of functions which Overlay Mixer supports. As shown above, the arithmetic operation for 4-Channel Image Source is available in Overlay Mixer. In the figure above, S0, S1, S2, and S3 are the results of performing YUV to RGB Conversion in the images selected by SF_CTRL.SxSEL Register.

The following is about Arithmetic Operation per each SF_CTRL.SxARIM Register.

	A	B	Result	
			A = Sx_CHROMA	A != Sx_CHROMA
BYPASS	Source 0/1/2/3 Image	Do not use	A	Y = A
FILL	Do not Used	Sx_PAR	A	Y = B
INV	Source 0/1/2/3 Image	Do not use	A	Y = 255 - A
ADD	Source 0/1/2/3 Image	Sx_PAR	A	Y = A + B
SUBA	Source 0/1/2/3 Image	Sx_PAR	A	Y = A - B
SUBB	Source 0/1/2/3 Image	Sx_PAR	A	Y = B - A
MUL	Source 0/1/2/3 Image	Sx_PAR	A	Y = A x B Mantissa = B[7:6] Fraction = B[5:0]

As described above, Arithmetic Operation is performed after Source YUV to RGB Conversion.

Therefore, for Arithmetic Operation with Chroma-Key, the value after Source YUV to RGB Conversion should be set as Chroma-Key Value (Sx_CHROMA).

34.4.4 Alpha-Blending and ROP with Window Offset and Chroma-Key

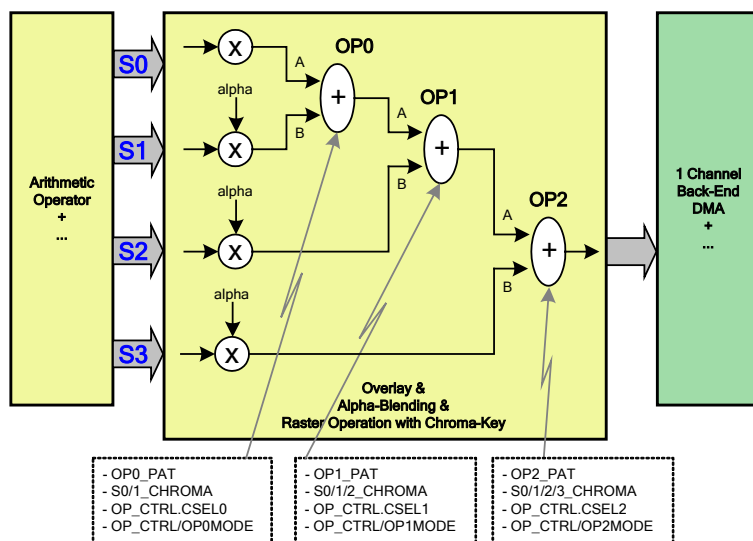


Figure 34.13 ROP / Alpha-Blending Block Diagram

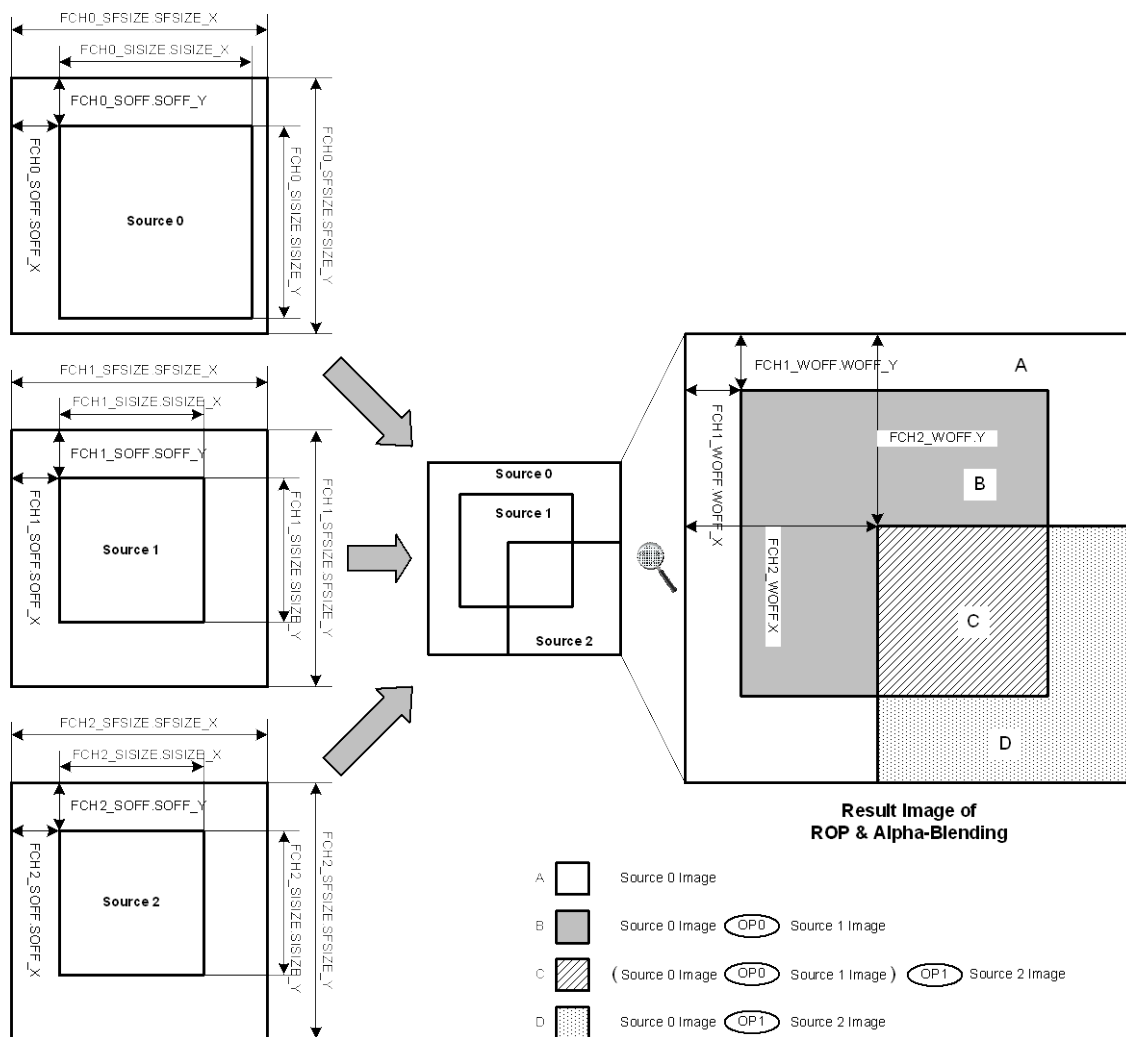


Figure 34.14 ROP and Alpha-Blending Operation Example

Figure 34.14 illustrates ROP and Alpha-Blending Operation. It assumes that FCH0 is Source 0, and FCH1, and FCH2 are Source 1 and Source 2 respectively. As the figure explains, Source 0 Image must include other Source Images.

In addition, in case of Source 0 Image, Window Offset must be '0'. Other than these, there is no requirement for ROP and Alpha-Blending Operation.

The table below shows the function of Operator 0/1/2.

Mode	Result of Operator 0		
	A = Source 0 selected by SF_CTRL.S0SEL		
	B = Source 1 selected by SF_CTRL.S1SEL		
	P = OP0_PAT[23:0]		
	ALPHA0 = OP0_PAT[31:24] / 256		
	ALPHA1 = Alpha-Value of ARGB Format Source 1 Data		
	OP0_CTRL.CSEL	01	10
		A = S0_CHROMA	B = S1_CHROMA
Blackness	$Y = 0$	B	A
Merged Copy	$Y = P \& A$	B	A
Merged Paint	$Y = \sim A B$	B	A
Pattern Copy	$Y = P$	B	A
Pattern Invert	$Y = P \wedge B$	B	A
Pattern Paint	$Y = (P \sim A) B$	B	A
Src Copy	$Y = A$	B	A
Src Inverter	$Y = A \wedge B$	B	A
Srd Paint	$Y = A B$	B	A
Src AND	$Y = A \& B$	B	A
Src Erase	$Y = A \& \sim B$	B	A
Not Src Copy	$Y = \sim A$	B	A
Not Src Erase	$Y = \sim (A B)$	B	A
Dst Copy	$Y = B$	B	A
Dst Inverter	$Y = \sim B$	B	A
Whiteness	$Y = 1$	B	A
Alpha-Blending 0	$Y = ALPHA0 * B + (1 - ALPHA0) * A$	B	A
Alpha-Blending 1	$Y = ALPHA1 * B + (1 - ALPHA1) * A$	B	A

Mode	Result of Operator 1		
	A = Result Image of Operator 0		
	B = Source 2 selected by SF_CTRL.S2SEL		
	P = OP1_PAT[23:0]		
	ALPHA0 = OP1_PAT[31:24] / 256		
	ALPHA1 = Alpha-Value of ARGB Format Source 2 Data		
	OP1_CTRL.CSEL	01	10
		A = S0_CHROMA	B = S1_CHROMA
Blackness	$Y = 0$	B	A
Merged Copy	$Y = P \& A$	B	A
Merged Paint	$Y = \sim A B$	B	A
Pattern Copy	$Y = P$	B	A
Pattern Invert	$Y = P \wedge B$	B	A
Pattern Paint	$Y = (P \sim A) B$	B	A
Src Copy	$Y = A$	B	A
Src Inverter	$Y = A \wedge B$	B	A
Srd Paint	$Y = A B$	B	A
Src AND	$Y = A \& B$	B	A
Src Erase	$Y = A \& \sim B$	B	A
Not Src Copy	$Y = \sim A$	B	A
Not Src Erase	$Y = \sim (A B)$	B	A
Dst Copy	$Y = B$	B	A
Dst Inverter	$Y = \sim B$	B	A
Whiteness	$Y = 1$	B	A
Alpha-Blending 0	$Y = ALPHA0 * B + (1 - ALPHA0) * A$	B	A
Alpha-Blending 1	$Y = ALPHA1 * B + (1 - ALPHA1) * A$	B	A

Mode	Result of Operator 2		
	A = Result Image of Operator 1		
	B = Source 3 selected by SF_CTRL.S3SEL		
	P = OP3_PAT[23:0]		
	ALPHA0 = OP2_PAT[31:24] / 256		
	ALPHA1 = Alpha-Value of ARGB Format Source 3 Data		
	OP2_CTRL.CSEL	01	10
		A = S0_CHROMA	B = S1_CHROMA
Blackness	$Y = 0$	B	A
Merged Copy	$Y = P \& A$	B	A
Merged Paint	$Y = \sim A \mid B$	B	A
Pattern Copy	$Y = P$	B	A
Pattern Invert	$Y = P \wedge B$	B	A
Pattern Paint	$Y = (P \mid \sim A) \mid B$	B	A
Src Copy	$Y = A$	B	A
Src Inverter	$Y = A \wedge B$	B	A
Srd Paint	$Y = A \mid B$	B	A
Src AND	$Y = A \& B$	B	A
Src Erase	$Y = A \& \sim B$	B	A
Not Src Copy	$Y = \sim A$	B	A
Not Src Erase	$Y = \sim (A \mid B)$	B	A
Dst Copy	$Y = B$	B	A
Dst Inverter	$Y = \sim B$	B	A
Whiteness	$Y = 1$	B	A
Alpha-Blending 0	$Y = \text{ALPHA0} * B + (1 - \text{ALPHA0}) * A$	B	A
Alpha-Blending 1	$Y = \text{ALPHA1} * B + (1 - \text{ALPHA1}) * A$	B	A

34.4.5 RGB to YUV Format Conversion

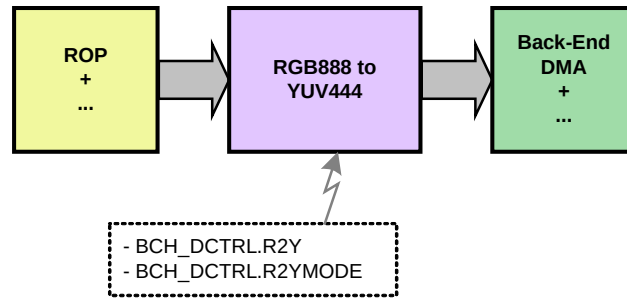


Figure 34.15 RGB to YUV Format Converter

Figure 34.15 shows 1-Channel RGB to YUV Format Conversion function provided by Overlay Mixer.

As shown above, RGB to YUV Format Conversion is performed after ROP and Alpha-Blending Operation.

The following explains the equation of RGB to YUV Conversion per BCH_DCTRL.R2YMODE Register.

- RGBtoYUV Format Converter Type 0
 - $Y = 0.299 R + 0.587 G + 0.114 B$
 - $U = -0.172 R - 0.339 G + 0.511 B + 128$
 - $V = 0.511 R - 0.428 G - 0.083 B + 128$
- RGBtoYUV Format Converter Type 1
 - $Y = 0.257 R + 0.504 G + 0.098 B + 16$
 - $U = -0.148 R - 0.291 G + 0.439 B + 128$
 - $V = 0.439 R - 0.368 G - 0.071 B + 128$
- RGBtoYUV Format Converter Type 2
 - $Y = 0.213 R + 0.715 G + 0.072 B$
 - $U = -0.117 R - 0.394 G + 0.511 B + 128$
 - $V = 0.511 R - 0.464 G - 0.047 B + 128$
- RGBtoYUV Format Converter Type 3
 - $Y = 0.183 R + 0.614 G + 0.062 B + 16$
 - $U = -0.101 R - 0.338 G + 0.439 B + 128$
 - $V = 0.439 R - 0.399 G - 0.040 B + 128$

34.4.6 Destination RGB Dithering

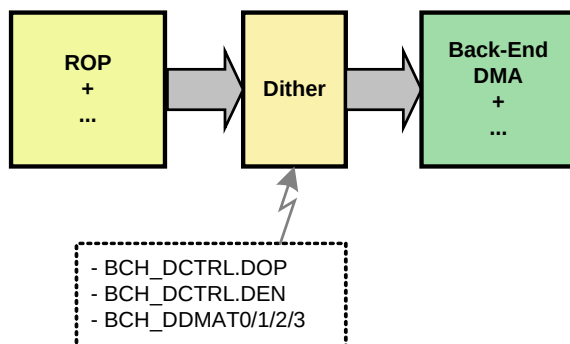


Figure 34.16 Destination RGB Dithering

Figure 34.16 shows 1-Channel RGB dithering function provided by Overlay Mixer. Patterned Dithering Algorithm is used in dithering algorithm of Overlay Mixer.

Figure below illustrates detailed operation of Patterned Dithering.

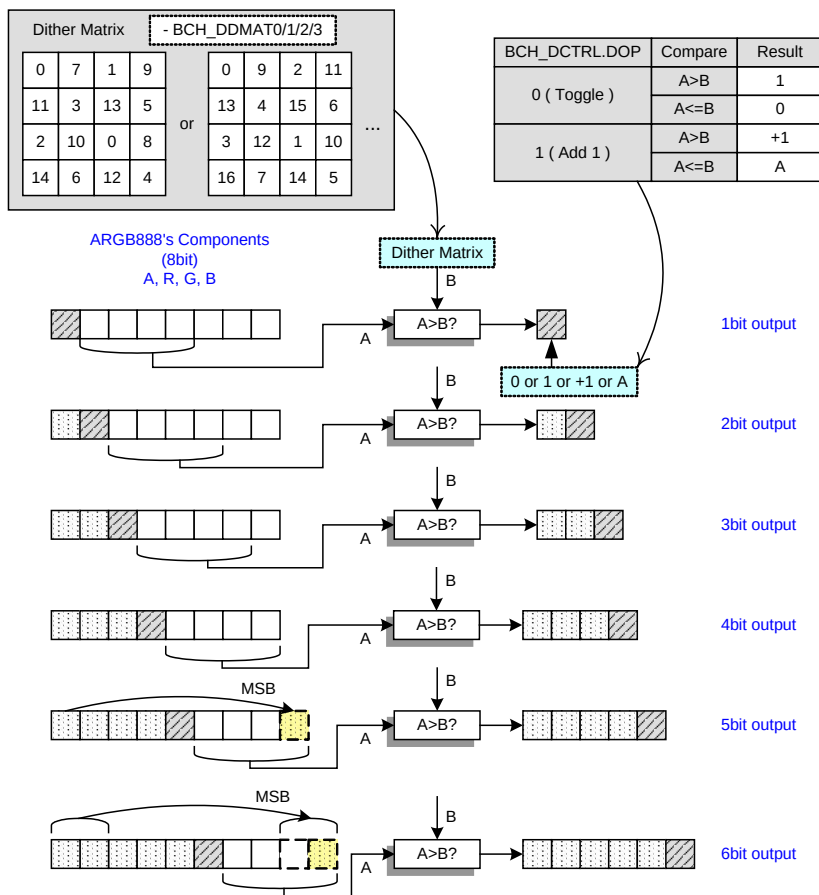


Figure 34.17 Destination RGB Dithering Operation

35 References

- [1] Contact Telechips for ARM TRM Technical Reference Manual version r0p5:
auto_sales@telechips.com/ce_sales@telechips.com
- [2] DesignWare USB 1.1 OHCI Host Controller with AHB/VC1 Databook, Synopsys
- [3] Enhanced Host Controller Interface Specification for Universal Serial Bus, Revision 1.0, Intel Corporation
- [4] Open Host Controller Interface Specification for USB, Release 1.0a, Compaq, Inc., Microsoft Corporation
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- [6] USB Battery Charging 1.2; http://www.usb.org/developers/docs/devclass_docs/USB_Battery_Charging

36 Revision History

Date	Revision	Description
2018-05-10	1.02	* Figure 8.2 is updated * Correct the Treatment of JTAG related pins in chapter 9.3 * Correct the PLL's configuration range in chapter 10.2 – PLL* Configuration Register * Modified Figure 8.11 and Figure 8.12 in SD/eMMC boot
2018-01-01	1.01	* Correct the Package and Ballmap information * Correct the XIWAIT field in (PMGPIO_CTL) Register. * Table 29.8 GPSB Interface Timing Parameters for SPI Timing 0 (master) revised * Figure 29.11 GPSB Interface Timing (slave) added * Table 29.9 GPSB Interface Timing Parameters for SPI Timing 0 (slave) added * Change the Package dimensions (POD)
2017-10-23	1.00	* V1.00 Official Release
2017-09-30	0.04	* Modified CAN description * Modified Electrical Characteristics for General I/O * Fixed some wrong information
2017-06-30	0.03	* Correct the fail-safe information. * Correct the Pull-up / down information of GPIO * Change the name from OTP to USB2.0DH(Device & Host)
2017-05-22	0.02	* 29.2 GSB Register Description revised(GSBMODE) * Remove/modify SFMC OCTA related description * Some mistake of spec. is corrected.
2017-03-13	0.01	* 22.2 NFC Register Description revised(NFC_CTRL) * 29.2 GSB Register Description revised(GSBINTEN)
2017-02-01	0.00	Initial release

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